

Erytra

SERVICING CENTER

Document version: 1.8.0

Software version $\geq 4.6.1$



Original instructions

Read this manual before using the equipment

Retain this manual for future use

GRIFOLS

About the Erytra Servicing Center

Published on: Tuesday, October 7, 2025

Document ID: 000166 - 1.8.0

Software version: 4.6.1

Copyright

Copyright © 2025 Grifols All Rights Reserved.

Contains information owned by Grifols and/or its affiliates. Do not copy, store, transmit or disclose to any third party without prior written permission from Grifols.

Other product and company names may be trademarks or registered trademarks of other companies, and are the property of their owners. They are used only for explanation, without intent to infringe.

Contact address

Grifols International

Avinguda de la Generalitat, 152
Parc empresarial Can Sant Joan
08174 Sant Cugat del Vallès, Barcelona
Spain

Safety notices

This manual uses the following symbols to facilitate the location of specific sections.



DANGER

Any kind of injury, critical or not, affecting a person, which can only be diminished by following the provided instructions.



WARNING

Any kind of injury affecting a person for whom protection systems are provided, but it is convenient the qualified technician is aware of it.



CAUTION

Any kind of situation which may cause damages for the equipment or other materials. This situation can be avoided or moderated by using this indication or with other protections.



NOTE

Used for clarification and complementary or emphasized information.

Contents

Erytra	1
About the Erytra Servicing Center	2
Copyright	2
Contact address	2
Safety notices	2
Contents	3
Related documents	19
Modifications	22
FSE tools	56
 1 Getting started	 58
1.1 Introduction	58
1.2 Users	58
1.3 Security remarks	59
1.4 Functional characteristics	59
1.4.1 Purified water or equivalent for laboratory use	59
1.5 Technical specifications	61
1.6 Erytra modules	62
1.6.1 Overview	63
1.6.2 General module (GEN)	63
1.6.2.1 Embedded computer and touchscreen monitor	64
1.6.2.2 Supply and Communication bus	64
1.6.2.3 Samples management module	65
1.6.2.4 Reagents management module	65
1.6.2.5 Cards management module	65
1.6.2.6 Doors	65
1.6.3 Incubator module (INC)	66
1.6.4 Centrifuge module (CEN)	66
1.6.5 Pipetting robot arm module (PIP)	66
1.6.6 Card transport robot arm module (CAT)	67
1.6.7 Fluidics module (FLU)	68
1.6.8 Dilution cup module (CUP)	69
1.6.9 Reader module (LEC)	69
1.7 Software structure	69
1.7.1 Overview	69

1.7.2 SSD Structure	69
1.7.2.1 Files structure	70
1.7.2.2 Description of log files	75
1.7.3 Windows registry structure	77
1.8 Setting up and starting the instrument	78
1.8.1 Location remarks	78
1.8.2 Unpacking	79
1.8.3 Instrument levelling	85
1.8.4 Installation Qualification (IQ)	87
1.8.5 External drain	87
1.8.6 Installing software for the first time	88
1.8.7 Working with Boot Application	88
1.8.8 Software configurations	91
1.8.8.1 ARCHIVE_ENABLED	91
1.8.8.2 ARCHIVE_INTERVAL	92
1.8.8.3 ARCHIVE_PATH	92
1.8.8.4 AUTO_LIS_QUERY_ENABLED	92
1.8.8.5 AUTO_LIS_QUERY_MODE	92
1.8.8.6 AUTO_LIS_PERIOD_UPDATE	93
1.8.8.7 AUTOMATIC_RESULTS_VERIFICATION	93
1.8.8.8 Barcode reader configuration	93
1.8.8.9 DEFAULT_BC_IS_ISBT	93
1.8.8.10 DELTA_CHECK_INCONSISTENCIES_CONFIGURATION	93
1.8.8.11 DELTA_CHECK_HISTORICAL_FROM	93
1.8.8.12 DIANA_SYSTEM_BATCH_CREATION	94
1.8.8.13 DIRECT_PRINT	94
1.8.8.14 DISABLE_VALIDATION_BUTTON	94
1.8.8.15 DISABLE_EXPORT_BUTTON	94
1.8.8.16 DISABLE_CHANGE_RESULTS_CONFIRMATION	94
1.8.8.17 DOUBLE_TUBE_ENABLED	95
1.8.8.18 DUAL_MODE	95
1.8.8.19 ENABLE_AUTOVERIFICATION_QCPASS	95
1.8.8.20 FLUIDIC_SYSTEM_ALWAYS_READY	95
1.8.8.21 IMPACT_DETECTOR	95
1.8.8.22 INSTRUMENT_NAME	96
1.8.8.23 INSTRUMENT_SERIAL	96
1.8.8.24 INTERPRETATION_FORMAT	96
1.8.8.25 ISBT_ENABLED	96
1.8.8.26 ISBT_LIS_DIGITS	96
1.8.8.27 LANGUAGE	97
1.8.8.28 LIS_ACTIVATED	97
1.8.8.29 LIS_AUTO_QUERY_BY_SAMPLE	98
1.8.8.30 LIS_SHARED_MODE	98

1.8.8.31 NO_SAMPLE_ID_IF_NO_USER_LOGGED	98
1.8.8.32 QC_ENABLED	98
1.8.8.33 QC_HOURS_CADENCE	98
1.8.8.34 QC_NEW_LOT_CARDS_AND_REAGENTS	99
1.8.8.35 QC_NEW_LOT_WARNING_ENABLE	99
1.8.8.36 PASSWORD_REAUTHENTICATION	100
1.8.8.37 PASSWORD_EXPIRATION_DAYS	100
1.8.8.38 RACK_BC_ENABLED	100
1.8.8.39 REAGENTS_ON_BOARD_STABILITY_ENABLED	100
1.8.8.40 RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA	100
1.8.8.41 RESERVE_INCUBATOR_FOR_STAT	100
1.8.8.42 RESULT_CONFIGURATION	101
1.8.8.43 SAMPLE_RACKS_CONFIGURATION	101
1.8.8.44 SEND_PENDING_TO_VALIDATE	103
1.8.8.45 SERVICE_RACK_MODE	103
1.8.8.46 SHOW_CONFIRMATION_WHEN_VALIDATE_MODIFIED_ SAMPLEPROFILES	103
1.8.8.47 WELL_NAME_FORMAT	104
1.8.9 Operational Qualification (OQ)	104
1.8.10 Setting Time Zone, Date and Time	105
1.8.11 Communications	105
1.8.11.1 LIS connection	105
1.8.11.2 Remote connection	106
1.8.12 Installing peripheral devices	116
1.8.13 Decontaminating	117
1.8.14 Uninstalling and shipping	117
1.8.15 Instrument disposal	123
1.8.15.1 Personal data at instrument disposal	123
1.9 Preventive maintenance	124
1.9.1 Items and calendar	124
1.9.2 Adjustment tools	124
1.9.3 Lubrication points	125
1.9.3.1 General module (GEN)	126
1.9.3.2 Pipetting arm module (PIP)	130
1.9.3.3 Card transport arm module (CAT)	132
1.9.3.4 Fluidics module (FLU)	135
1.9.4 How to report	137
2 Modules overview	138
2.1 General module (GEN)	138
2.1.1 Description	138
2.1.1.1 PCA GEN-BCKPL59	138
2.1.1.2 Power supply	140

2.1.1.3 Computer	144
2.1.1.4 Touchscreen monitor	148
2.1.1.5 User tray	148
2.1.1.6 Drawers	149
2.1.1.7 Keyboards	152
2.1.1.8 Status LEDs	152
2.1.1.9 Doors	153
2.1.1.10 Temperature control system	154
2.1.1.11 Buzzer	155
2.1.1.12 Chassis	155
2.1.1.13 Cables and tubes rails	156
2.1.1.14 Covers	156
2.1.2 Mechanical adjustments	160
2.1.2.1 Adjusting upper / intermediate area door	160
2.1.2.2 Adjusting touchscreen drawer lock	161
2.1.2.3 Adjusting user tray drawer lock	163
2.1.2.4 Adjusting safety latch reagents, samples and cards drawers	164
2.1.2.5 Adjusting front panel of reagents, samples and cards drawers	165
2.1.2.6 Adjusting tension of the belt of the reagents drawers	166
2.1.2.7 Adjusting reagents drawers flatness	167
2.1.2.8 Adjusting samples drawers flatness	169
2.1.2.9 Adjusting cards drawers flatness	170
2.1.3 Functional adjustments	171
2.1.3.1 Adjusting touchscreen	171
2.1.3.2 Adjusting touchscreen monitor menu	175
2.1.4 Verification procedures	177
2.1.4.1 Verifying power supply (SN < 059-0001583)	177
2.1.4.2 Verifying PCA GEN-STDBY	178
2.1.4.3 Verifying PCA GEN-BCKPL59	178
2.1.4.4 Verifying computer	179
2.1.4.5 Verifying touchscreen monitor	179
2.1.4.6 Verifying status LEDs	180
2.1.4.7 Verifying ambient light	181
2.1.4.8 Verifying user tray	181
2.1.4.9 Verifying temperature control system	182
2.1.4.10 Verifying buzzer	183
2.1.4.11 Verifying keyboards	183
2.1.4.12 Verifying doors	184
2.1.4.13 Verifying drawers	184
2.1.4.14 Verifying the flatness of reagent drawer	186
2.1.4.15 Verifying the flatness of sample drawer	187
2.1.4.16 Verifying the flatness of card drawer	189
2.1.5 Replacement procedures	191
2.1.5.1 Replacing power supply (SN < 059-0001583)	191

2.1.5.2 Replacing touchscreen monitor	194
2.1.5.3 Replacing reagents and samples drawers keyboard	198
2.1.5.4 Replacing card drawers keyboard	199
2.1.5.5 Replacing PCA RM-ST59	200
2.1.5.6 Replacing RAM memory modules	201
2.1.5.7 Replacing PC motherboard	205
2.1.5.8 Replacing card drawer electric lock	208
2.1.5.9 Replacing samples/reagents drawer electric lock	212
2.1.5.10 Replacing fluidic drawer electric lock	213
2.1.5.11 Replacing user tray	217
2.1.5.12 Replacing upper door	219
2.1.6 Electronics	221
2.1.6.1 Fuses	222
2.1.6.2 LEDs	222
2.1.6.3 Jumpers and test points	223
2.1.6.4 Wiring table	225
2.1.7 Troubleshooting	236
2.2 Incubator module (INC)	256
2.2.1 Description	256
2.2.2 Mechanical adjustments	259
2.2.2.1 Adjusting incubator flatness	259
2.2.3 Functional adjustments	260
2.2.3.1 Adjusting incubator height	260
2.2.3.2 Adjusting temperature	263
2.2.4 Verification procedures	267
2.2.4.1 Verifying temperature	267
2.2.4.2 Verifying movement	268
2.2.4.3 Verifying incubator flatness	269
2.2.5 Replacement procedures	271
2.2.5.1 Replacing incubator module	271
2.2.5.2 Replacing Peltier module	274
2.2.5.3 Replacing PCA INC-BL59	276
2.2.6 Electronics	277
2.2.6.1 Fuses	277
2.2.6.2 LEDs	277
2.2.6.3 Jumpers and test points	278
2.2.6.4 Wiring table	279
2.2.7 Troubleshooting	284
2.3 Centrifuge module (CEN)	295
2.3.1 Description	295
2.3.2 Mechanical adjustments	297
2.3.2.1 Adjusting rotor height	297
2.3.3 Functional adjustments	299

2.3.4 Verification procedures	300
2.3.5 Replacement procedures	301
2.3.5.1 Replacing motor	302
2.3.5.2 Replacing centrifuge rotor	304
2.3.5.3 Removing centrifuge pot	308
2.3.6 Electronics	310
2.3.6.1 Fuses	310
2.3.6.2 LEDs	310
2.3.6.3 Jumpers and test points	311
2.3.6.4 Wiring table	312
2.3.7 Troubleshooting	316
2.4 Pipetting robot arm module (PIP)	324
2.4.1 Description	324
2.4.2 Mechanical adjustments	328
2.4.2.1 Adjusting probes parallelism	329
2.4.2.2 Adjusting X-Axis small belt tension	330
2.4.2.3 Adjusting X-Axis large belt tension	331
2.4.2.4 Adjusting Y-Axis small belt tension	332
2.4.2.5 Adjusting Y-axis large belt tension	333
2.4.2.6 Adjusting Z-axis belt tension	334
2.4.2.7 Adjusting presence detector inclination	335
2.4.3 Functional adjustments	336
2.4.3.1 Adjusting probes offset	336
2.4.3.2 Adjusting reagents drawers	338
2.4.3.3 Adjusting samples drawers	342
2.4.3.4 Adjusting incubators	345
2.4.3.5 Adjusting dilution cup	347
2.4.3.6 Adjusting barcode reader inclination position	348
2.4.3.7 Adjusting horns position	352
2.4.3.8 Adjusting presence detector distance of detection	355
2.4.3.9 Adjusting barcode readers configuration	358
2.4.4 Verification procedures	360
2.4.4.1 Verifying movement	360
2.4.4.2 Verifying home of the horns	361
2.4.4.3 Verifying home of the X-Y-Z1-Z2 axis	362
2.4.4.4 Verifying probes parallelism	362
2.4.4.5 Verifying probes offset	364
2.4.4.6 Verifying reagents drawer	365
2.4.4.7 Verifying samples drawer	366
2.4.4.8 Verifying incubators	367
2.4.4.9 Verifying dilution cup	368
2.4.4.10 Verifying barcode reader	369
2.4.4.11 Verifying presence detector	370

2.4.4.12 Verifying level detection system	372
2.4.4.13 Verifying impact detector system	374
2.4.5 Replacement procedures	375
2.4.5.1 Replacing probes	375
2.4.5.2 Replacing probes washer	377
2.4.5.3 Replacing probes washer fitting	379
2.4.5.4 Replacing level detector PCA	380
2.4.5.5 Replacing leadscrews	382
2.4.5.6 Replacing PIP X axis chain (CH1)	392
2.4.5.7 Replacing PIP Y axis chain (CH2)	396
2.4.5.8 Replacing PIP Z1 axis chain (CH3) / Z2 axis chain (CH4)	400
2.4.5.9 Replacing barcode reader mirror	404
2.4.5.10 Replacing horns chain	406
2.4.5.11 Replacing toothed belt T2.5 245	417
2.4.5.12 Replacing toothed belt T5/2000 (PIP X-axis)	421
2.4.5.13 Replacing toothed belt T5/1200 (PIP Y-axis)	425
2.4.6 Electronics	429
2.4.6.1 Fuses	429
2.4.6.2 LEDs	430
2.4.6.3 Jumpers and test points	430
2.4.6.4 Wiring table	431
2.4.7 Troubleshooting	440
2.5 Card transport module (CAT)	462
2.5.1 Description	462
2.5.2 Mechanical adjustments	464
2.5.2.1 Adjusting X-axis small belt tension	465
2.5.2.2 Adjusting X-axis large belt tension	465
2.5.2.3 Adjusting Y-axis small belt tension	466
2.5.2.4 Adjusting Y-axis large belt tension	467
2.5.2.5 Adjusting Z-axis gripper belt tension	468
2.5.2.6 Adjusting gripper parallelism	469
2.5.2.7 Adjusting the inclination of card presence detector	470
2.5.3 Functional adjustments	471
2.5.3.1 Adjusting gripper in the card drawers area	472
2.5.3.2 Adjusting gripper in the incubator area	475
2.5.3.3 Adjusting gripper in the card waste container area	478
2.5.3.4 Adjusting gripper in the centrifuge area	479
2.5.3.5 Adjusting card presence detector-gripper parallelism	482
2.5.3.6 Adjusting card presence detector height detection	483
2.5.3.7 Adjusting card barcode reader inclination	486
2.5.3.8 Adjusting gripper in the card reader area	487
2.5.4 Verification procedures	490
2.5.4.1 Verifying gripper parallelism	490
2.5.4.2 Verifying card drawers area	492

2.5.4.3 Verifying incubators area	494
2.5.4.4 Verifying card waste container area	496
2.5.4.5 Verifying centrifuges area	497
2.5.4.6 Verifying card presence detector inclination	498
2.5.4.7 Verifying card presence detector-gripper parallelism	499
2.5.4.8 Verifying card presence detector height detection	500
2.5.4.9 Verifying card barcode reader inclination	502
2.5.4.10 Verifying reader area	502
2.5.5 Replacement procedures	504
2.5.5.1 Replacing gripper leadscrew	504
2.5.5.2 Replacing card barcode reader mirror	508
2.5.5.3 Replacing CAT X chain	509
2.5.5.4 Replacing CAT Y chain	512
2.5.5.5 Replacing gripper (Z) chain	515
2.5.5.6 Replacing gripper	517
2.5.5.7 Replacing toothed belt T2.5 245	520
2.5.5.8 Replacing toothed belt T5/2000 (CAT X-axis)	524
2.5.5.9 Replacing toothed belt T5/1200 (CAT Y-axis)	527
2.5.6 Electronics	530
2.5.6.1 Fuses	530
2.5.6.2 LEDs	531
2.5.6.3 Jumpers and test points	531
2.5.6.4 Wiring table	532
2.5.7 Troubleshooting	538
2.6 Fluidics module (FLU)	555
2.6.1 Description	555
2.6.1.1 Parts	555
2.6.2 Fluidic procedures	564
2.6.2.1 Priming	564
2.6.2.2 Rinsing	567
2.6.2.3 Emptying circuits	570
2.6.2.4 Washing	572
2.6.2.5 Managing overpressure	573
2.6.3 Mechanical adjustments	574
2.6.3.1 Adjusting the safety latch of the drawers of the bottles	574
2.6.3.2 Adjusting the cosmetic front panel of the drawers of the bottles	575
2.6.4 Functional adjustments	576
2.6.4.1 Adjusting load cells of the bottle drawers	576
2.6.4.2 Adjusting the load cell of the card waste container	577
2.6.4.3 Adjusting pressure-vacuum sensors	579
2.6.4.4 Adjusting dilutors configuration	581
2.6.5 Verification procedures	582
2.6.5.1 Verifying pressure sensors	582

2.6.5.2 Verifying CPD sensors	583
2.6.5.3 Verifying dispensation pumps and expansion vessels	586
2.6.5.4 Verifying aspiration pump	587
2.6.5.5 Verifying electrovalves	587
2.6.5.6 Verifying manifold	588
2.6.5.7 Verifying load cell	588
2.6.6 Replacement procedures	589
2.6.6.1 Replacing pressure sensors	589
2.6.6.2 Replacing CPD sensors	592
2.6.6.3 Replacing dispensation pumps	594
2.6.6.4 Replacing expansion vessels	596
2.6.6.5 Replacing aspiration pump	597
2.6.6.6 Replacing electrovalves	599
2.6.6.7 Replacing manifold	600
2.6.6.8 Replacing manifold-distribution block	601
2.6.6.9 Replacing load cell	603
2.6.6.10 Replacing fitting of the tubes	607
2.6.6.11 Replacing syringe and valve	608
2.6.6.12 Replacing fluidic drawer linear guide	616
2.6.6.13 Replacing coupling CPC bottles connector	619
2.6.6.14 Replacing coupling CPC bottles connector O-ring	620
2.6.6.15 Replacing bottle CPC connector and internal tube	621
2.6.6.16 Replacing connection block	623
2.6.6.17 Replacing solution A and B filter	625
2.6.6.18 Replacing vacuum filter	627
2.6.6.19 Replacing aspiration pump tubing	629
2.6.6.20 Replacing PCA FLU-CTRL59 board	630
2.6.6.21 Replacing protector plates	632
2.6.7 Electronics	634
2.6.7.1 Fuses	634
2.6.7.2 LEDs	634
2.6.7.3 Jumpers and test points	635
2.6.7.4 Wiring table	636
2.6.8 Troubleshooting	640
2.6.8.1 FLUIDIC SYSTEM ISSUES	664
2.6.8.2 BOTTLES /CARD WASTE DRAWERS ISSUES	670
2.7 Dilution cup module (CUP)	672
2.7.1 Description	672
2.7.2 Mechanical adjustments	673
2.7.2.1 Adjusting dilution cup parallelism	673
2.7.3 Functional adjustments	676
2.7.4 Verification procedures	676
2.7.5 Replacement procedures	678

2.7.5.1 Replacing dilution cup wells assembly	678
2.7.5.2 Replacing dilution cup motor	679
2.7.5.3 Replacing fitting	685
2.7.6 Electronics	686
2.7.6.1 Fuses	686
2.7.6.2 LEDs	686
2.7.6.3 Jumpers and test points	686
2.7.6.4 Wiring table	687
2.7.7 Troubleshooting	690
2.8 Card reader module (LEC)	695
2.8.1 Description	695
2.8.2 Functional adjustments	698
2.8.2.1 Adjusting diaphragm aperture	701
2.8.2.2 Adjusting focus	703
2.8.2.3 Adjusting illumination position	705
2.8.2.4 Adjusting illumination dimensions	707
2.8.2.5 Adjusting illumination rotation	708
2.8.2.6 Adjusting illumination perspective	709
2.8.2.7 Adjusting illumination intensity	711
2.8.2.8 Adjusting illumination color	712
2.8.2.9 Adjusting conversion factor	714
2.8.3 Verification procedures	716
2.8.3.1 Verifying diaphragm aperture	720
2.8.3.2 Verifying focus	722
2.8.3.3 Verifying illumination position	723
2.8.3.4 Verifying illumination dimensions	724
2.8.3.5 Verifying illumination rotation	726
2.8.3.6 Verifying illumination perspective	727
2.8.3.7 Verifying illumination intensity	728
2.8.3.8 Verifying illumination color	729
2.8.3.9 Verifying luminance uniformity	731
2.8.3.10 Verifying color uniformity	732
2.8.3.11 Verifying illumination cleanliness	733
2.8.4 Replacement procedures	734
2.8.4.1 Removing reader module	734
2.8.4.2 Replacing camera	736
2.8.4.3 Replacing reader illumination/PCA RD-IL421	737
2.8.4.4 Replacing reader illumination pattern	739
2.8.4.5 Removing reader cover	740
2.8.5 Electronics	743
2.8.5.1 Fuses	743
2.8.5.2 LEDs	743
2.8.5.3 Jumpers and test points	743
2.8.5.4 Wiring table	744

2.8.6 Troubleshooting	746
3 Technical Service software overview	754
3.1 Introduction	754
3.2 Modular tabs	754
3.2.1 Procedures tab	756
3.2.2 Incubators tab	760
3.2.2.1 Incubator 1, 2, 3 and 4 subtabs	760
3.2.2.2 Home Cycles subtab	765
3.2.3 Centrifuges tab	767
3.2.3.1 Centrifuge 1 and 2 subtabs	767
3.2.3.2 Home Cycles subtab	770
3.2.3.3 Random Position subtab	772
3.2.4 Dilution Cup tab	774
3.2.5 Gripper Arm tab	777
3.2.5.1 Movement subtab	777
3.2.5.2 Presence Detector subtab	782
3.2.5.3 BCR subtab	785
3.2.5.4 Home Cycles subtab	788
3.2.5.5 Random Position subtab	790
3.2.6 Drawers tab	792
3.2.6.1 Card Drawers subtab	792
3.2.6.2 Sample Drawers subtab	794
3.2.6.3 Reagent Drawers subtab	796
3.2.6.4 Fluidic Drawers subtab	800
3.2.6.5 Home Cycles subtab	802
3.2.7 Pip Arm tab	804
3.2.7.1 Movement subtab	804
3.2.7.2 Presence Detector subtab	808
3.2.7.3 Level Detector subtab	814
3.2.7.4 BCR subtab	817
3.2.7.5 CPD subtab	820
3.2.7.6 Home Cycles subtab	822
3.2.7.7 Random Position subtab	824
3.2.8 Fluidics tab	826
3.2.9 General tab	832
3.2.10 Firmware tab	837
3.2.10.1 PCA control subtab	837
3.2.10.2 Eprom control subtab	843
3.2.11 Operations tab	845
3.2.11.1 Dilution Cup Operations subtab	845
3.2.11.2 Aspire and Dispense Operations subtab	848

4 Diagnostic application	851
4.1 Introduction	851
4.2 Overview	851
4.3 Tests	853
4.3.1 Initialization	854
4.3.1.1 Initialization	854
4.3.2 Cards Management	855
4.3.2.1 Robot movements	855
4.3.2.2 BC reading and presence detection	856
4.3.2.3 Card Transport card drawers	856
4.3.2.4 Card transport centrifuges	857
4.3.2.5 Card transport incubators	858
4.3.3 Pipetting	859
4.3.3.1 Robot movements	859
4.3.3.2 Presence detection and BC reading	860
4.3.3.3 Level detection	861
4.3.3.4 Pipetting precision and accuracy	861
4.3.3.5 CPD and PCA level detector	863
4.3.3.6 Impact detection	863
4.3.4 Fluidics	865
4.3.4.1 Drawers and Load cells	865
4.3.4.2 Prime, rinse and empty	866
4.3.4.3 Pressure circuits	867
4.3.4.4 Vacuum circuits	868
4.3.5 Reader	869
4.3.5.1 Initialization	870
4.3.6 Incubators	870
4.3.6.1 Movements	870
4.3.6.2 Incubation	871
4.3.7 Centrifuges	871
4.3.7.1 Movements	871
4.3.7.2 Centrifugation	871
4.3.8 Samples drawers	872
4.3.8.1 Drawers	872
4.3.9 Reagents drawers	872
4.3.9.1 Drawers and stirrers	873
4.3.10 Cards drawers	873
4.3.10.1 Drawers	873
4.3.11 Dilution cup	874
4.3.11.1 Stirring	874
4.3.12 Miscellaneous	874
4.3.12.1 Door, keyboard, LEDs and buzzer	874

4.4 Report	875
5 Importer application	883
5.1 Introduction	883
5.2 Overview	883
5.3 Importer	884
5.3.1 Importing a reagent pack file	885
5.3.2 Importing a reagent language pack file	885
5.3.3 Importing an Erytra language pack file	886
5.4 Profiles	886
5.4.1 Enabling profiles from an imported reagent pack	887
5.4.2 Creating a not unitary profile	888
5.4.3 Editing not unitary profiles	893
5.5 QC kits	893
5.5.1 Enabling QC kits from an imported reagent pack	894
5.6 Special results	895
5.6.1 Configuring special results that are not set by default	896
5.7 Setup	899
5.7.1 Configuring instrument settings	899
5.7.2 Changing the language	900
5.7.3 Configuring QC settings	900
5.7.4 Configuring inconsistencies settings	901
5.7.5 Configuring results settings	902
5.7.6 Configuring other settings	904
5.7.7 Password policies	904
6 Mechanical drawings	906
6.1 Requirements	906
6.2 Drawings reference table	906
6.3 Creo View Express installation	906
6.3.1 Within Grifols company	907
6.3.2 External distributors	907
6.4 Creo View configuration	907
6.5 Models structure and navigation	909
6.5.1 Tree structure	909
6.5.2 Models navigation	910
6.6 Animated assembly procedures	911
6.7 Spare parts searching	913
6.7.1 Finding a drawing part for a spare part list's reference code	913

6.7.2 Finding the spare part list's reference code for a drawing part	914
7 Software information	915
7.1 Software version changes and components versions	915
7.1.1 Erytra Software v4.5.1	915
7.1.1.1 Software changes	915
7.1.1.2 Application and firmware versions	916
7.1.2 Erytra Software v4.6.0	917
7.1.2.1 Software changes	917
7.1.3 Erytra Software v4.6.1	934
7.1.3.1 Software changes	934
7.1.3.2 Application and firmware versions	949
7.1.4 Erytra Software v5.2.0	951
7.1.4.1 Software changes	951
7.1.5 Erytra Software v5.2.1	989
7.1.5.1 Software changes	989
7.1.5.2 Application and firmware versions	1004
7.2 Installation procedures	1006
7.2.1 Downloading Erytra files	1006
7.2.2 Installing or restoring Operating System	1006
7.2.2.1 Restoring from USB pen drive	1007
7.2.2.2 Restoring from Boot Application menu	1016
7.2.2.3 Restoring from Grifols Secure Remote Connectivity tool	1019
7.2.2.4 Verifying OS version	1028
7.2.2.5 Activating Windows License	1029
7.2.2.6 Verifying Licence activation	1038
7.2.3 Installing Erytra software v4.6.1	1039
7.2.3.1 Installing steps in units with OS version v2.3.3 (or v2.3.1+ SU v1.7 or v2.3.2+SU v1.7)	1042
7.2.3.2 Installing steps in units with OS version $\geq$ 2.2.0 and OS $\neq$ 2.3.3 (or 2.3.1+SU, or 2.3.2+SU)	1045
7.2.3.3 Installing steps in units with OS version $<$ 2.2.0	1054
7.2.5.1 Software installer steps	1065
7.2.5.2 Software installation Final Setup	1070
7.2.6 Installing Erytra software (v5.2.0)	1071
7.2.6.1 Installing steps in units with OS version $\geq$ 2.2.0 (v5.2.0)	1073
7.2.6.2 Installing steps in units with OS version $<$ 2.2.0 (v5.2.0)	1077
7.2.6.3 Software installer wizard (v5.2.0)	1085
7.2.6.4 Final Setup (v5.2.0)	1093
7.2.6.5 Verifying software installation (v5.2.0)	1094
7.2.6.6 Verifying software version	1096
7.2.7 Verifying OS and SW versions	1097
7.2.7.1 From Boot Application	1097

7.2.7.2 From ErytraApplication	1098
7.3 Operating system	1098
7.3.1 Return to Boot Application	1098
7.3.2 Windows security packages	1099
7.3.2.1 Windows security update installation	1101
7.3.3 Accessing Windows 10 file explorer	1105
7.3.3.1 Technical service mode	1106
7.3.3.2 Diagnostic application mode	1106
7.3.3.3 User application mode	1108
7.3.3.4 OS shell mode	1109
7.3.4 Sharing and accessing a shared folder or printer	1109
7.3.5 Mapping a network drive	1113
7.3.6 Collecting Windows event logs	1114
7.4 Database and relevant data	1116
7.4.1 Backing up with user interface or OS Shell backup option	1117
7.4.2 Backing up databases and relevant data with boot menu	1117
7.4.2.1 Full backup	1117
7.4.2.2 Customized backup	1118
7.4.3 Backing up relevant data manually	1120
7.4.4 Restoring databases and relevant data with boot menu	1124
7.4.5 Erytra database manager application	1127
7.4.5.1 Installation procedure	1127
7.5 BIOS, Firmware, EEPROM and Drivers procedures	1129
7.5.1 Updating firmware	1129
7.5.1.1 Updating firmware in a valid PCA	1129
7.5.1.2 Updating firmware in a spare part PCA (CEN-SRM, INC-CUP, CAT-CM or PIP MotorControl)	1130
7.5.1.3 Updating firmware in a PCA with non valid firmware (CEN-SRM, INC-CUP, CAT-CM or PIP MotorControl)	1131
7.5.2 BIOS settings	1131
7.5.2.1 Checking BIOS version	1132
7.5.2.2 Configuring BIOS	1134
7.5.2.3 Checking assignment of Windows COM ports	1135
7.5.3 Backing up and restoring EEPROM	1140
7.5.3.1 EEPROM backup	1140
7.5.3.2 EEPROM restore	1141
7.5.4 Installing and configuring touchscreen driver	1141
7.5.4.1 Installing touchscreen driver	1141
7.5.4.2 Configuring touchscreen v2 drivers	1141
7.6 Applications	1143
7.6.1 Applocker application	1143
7.6.1.1 Updating the whitelist	1143
7.6.1.2 Blocked executables	1145

7.6.2 Erytra Macro executor	1146
7.6.3 Support pack	1147
7.6.3.1 Generating Support Pack from user software	1148
7.6.3.2 Generating Support Pack from OS shell	1149
7.7 Additional information	1153
7.7.1 Guidelines on error codes location	1154
7.8 Erytra-LIS connection	1154
7.8.1 DianaServer program	1155
7.8.2 Serial communication mode	1158
7.8.3 Files communication mode	1158
7.8.4 TCP-IP communication mode	1159
7.8.5 HostDemo program	1162
7.8.5.1 Patients Management	1165
7.8.5.2 Test and Templates management	1167
7.8.5.3 Configuration	1167
7.8.5.4 Orders section	1167
7.8.6 Enabling and configuring LIS connection	1168
7.8.7 Erytra-LIS errors	1170
7.8.7.1 Diana System errors related to LIS communication	1171
7.8.7.2 List of errors related to Erytra - LIS communication module	1174
7.8.8 LIS interface specification	1183
Bulletin index	1183
7.9 MTB-0059#0001 Erytra Software version 4.5.1 (Patch 1 for version 4.5.0)	1198
7.10 MTB-0059#0002 Reader module foam removal	1200
7.11 MTB-0059#0003 Erytra Software version 4.6.1 (Patch 1 for version 4.6.0)	1202
7.12 MTB-0059#0004 Erytra Software version 5.2.1 (Patch 1 for version 5.2.0)	1205
8 Spare parts list	1207
Eflexis Revision history table	1362
Erytra Revision history table	1383
DG Reader Net Revision history table	1395
DG Spin Revision history table	1399
DG Therm Revision history table	1404

Related documents

This section lists the documents referenced in this document.



NOTE

All related documents are stored in the **RelatedDocuments/Erytra** folder supplied with this document.

Wiring diagrams

- Electronic Wiring diagram (W5608)
Located in **RelatedDocuments > Erytra**.
- Base module Wiring diagram (W5614)
Located in **RelatedDocuments > Erytra**.
- Modules old Wiring diagram (W56xx-old)
Located in **RelatedDocuments > Erytra**.

2D models

- Fluidics module - Fluidic diagram (p2475V0)
Located in **RelatedDocuments > Erytra**.
- External BCR holder (s2018V0)
Located in **RelatedDocuments > Erytra**.

3D models

- Base (I) (M6007961V0)
Located in **RelatedDocuments > Erytra > 3D models**.
- Base (II) (M6011717V0)
Located in **RelatedDocuments > Erytra > 3D models**.
- Base (III) (M6006516V0)
Located in **RelatedDocuments > Erytra > 3D models**.
- Packaging (M210400V0)
Located in **RelatedDocuments > Erytra > 3D models**.
- General module - user tray (M6018839V0)
Located in **RelatedDocuments > Erytra > 3D models > General module**.
- General module - touchscreen monitor drawer (M6006416V0)
Located in **RelatedDocuments > Erytra > 3D models > General module**.
- General module - chassis (M6006420V0)
Located in **RelatedDocuments > Erytra > 3D models > General module**.

- General module - lower level (M6006421V0)
Located in **RelatedDocuments > Erytra > 3D models > General module.**
- General module - electronics cabinet (M6006424V0)
Located in **RelatedDocuments > Erytra > 3D models > General module.**
- Incubator module (M6006413V0)
Located in **RelatedDocuments > Erytra > 3D models > Incubator module.**
- Centrifuge module (M6006412V0)
Located in **RelatedDocuments > Erytra > 3D models > Centrifuge module.**
- Pipetting module - X axis (M6006419V0)
Located in **RelatedDocuments > Erytra > 3D models > Pipetting module.**
- Pipetting module - Y and Z axis (M6006418V1)
Located in **RelatedDocuments > Erytra > 3D models > Pipetting module.**
- Card transport module - X axis (M6006419V0)
Located in **RelatedDocuments > Erytra > 3D models > Card transport module.**
- Card transport module - Y and Z axis (M6006417V0)
Located in **RelatedDocuments > Erytra > 3D models > Card transport module.**
- Fluidics module - solution bottle (M6006428V0)
Located in **RelatedDocuments > Erytra > 3D models > Fluidics module.**
- Fluidics module - waste bottle (M6006780V0)
Located in **RelatedDocuments > Erytra > 3D models > Fluidics module.**
- Fluidics module - bottle drawer (M6006422V0)
Located in **RelatedDocuments > Erytra > 3D models > Fluidics module.**
- Dilution cup module (M6006411V0)
Located in **RelatedDocuments > Erytra > 3D models > Dilution cup module.**
- Card reader module (M6006414V0)
Located in **RelatedDocuments > Erytra > 3D models > Card reader module.**

Security and data privacy

- Data privacy assessment for Erytra (DG-QA-REP-000233)
Located in **RelatedDocuments > Erytra.**
- Manufacturer Disclosure Statement for Medical Device Security DG-59 (DG-DI-DHF-009117)
Located in **RelatedDocuments > Erytra.**
- Security and privacy, Erytra Hardening guide (DG-DI-DHF-008144)
Located in **RelatedDocuments > Erytra.**

Communications

- Erytra, Erytra Eflexis and DG Reader Net LIS interface specification (DG-PDS-TSM-000001)
Located in **RelatedDocuments > Common**.

Modifications

v4.6.1 - October 2025

MOD050_new_v1.8

MTB-0059#0004 Erytra Software v5.2.1

See "MTB-0059#0004 Erytra Software version 5.2.1 (Patch 1 for version 5.2.0)" on page 1205

MOD049_mod_v1.8

Minor changes and error fixing

CHANGE DESCRIPTION

- SW version 4.6.1 requirements have been updated. See "MTB-0059#0003 Erytra Software version 4.6.1 (Patch 1 for version 4.6.0)" on page 1202
- An incorrect note regarding liquibase date and time has been deleted from See "Installing steps in units with OS version $\geq$ 2.2.0 and OS $\neq$ 2.3.3 (or 2.3.1+SU, or 2.3.2+SU)" on page 1045
- Updated SW v4.6.1 components, see "Erytra Software v4.6.1" on page 934
 - Updated SW v4.6.1 components, Erytra diagnostic, Importer and Technical service version is 4.15.0.
 - Language pack version is 1.33.
- See "Items and calendar" on page 124: A new check to ensure that no other devices are plugged into the UPS, if there is on, has been added to the PM.

v4.6.1 - September 2025

MOD048_new_v1.7

MTB-0059#0003 Erytra Software v4.6.1

See "MTB-0059#0003 Erytra Software version 4.6.1 (Patch 1 for version 4.6.0)" on page 1202

MOD047_mod_v1.7

Minor changes and error fixing

CHANGE DESCRIPTION

- New BIOS version added to "Checking BIOS version" on page 1132
- Added document number and repaired document version in "Installation Qualification (IQ)" on page 87, See "Operational Qualification (OQ)" on page 104 and See "Items and calendar" on page 124 (PM).
- Restored INR-0059#0163 in See "Bulletin index" on page 1183
- Repaired pdf download link.

v4.6.0 - September 2025

MOD046_new_v1.6

New IQ, OQ and PM in HTML format

The Installation Qualification (IQ), Operation Qualification (OQ) and Preventive Maintenance (PM) have been migrated into HTML format and included in the Erytra Servicing Center. Also IQ, OQ and PM are available in English, Spanish and Portuguese.

- Access IQ from "Installation Qualification (IQ)" on page 87.
- Access OQ from "Operational Qualification (OQ)" on page 104.
- Access PM from "Items and calendar" on page 124.



NOTE

If required, empty IQ, OQ and PM check-list in PDF should be generated from the HTML topics in advance.

MOD045_new_v1.6

MTB-0059#0002 Reader module foam removal

See "MTB-0059#0002 Reader module foam removal" on page 1200

MOD044_new_v1.6

MTB-0059#0001 Erytra Software v4.5.1

See "MTB-0059#0001 Erytra Software version 4.5.1 (Patch 1 for version 4.5.0)" on page 1198

MOD043_mod_v1.6

Minor changes and error fixing

CHANGE DESCRIPTION

- Correction in MOD029 (See "MOD029_new_v1.3 FSB-0059#0041 Erytra Software v4.6.0" on page 36) and MOD030 (See "MOD030_new_v1.3 FSB-0059#0042 Erytra Software v5.2.0 (China)" on page 34): no modification on FW CEN-SRM
- New Manufacturer Technical Bulletin (MTB) table in "Bulletin index" on page 1183 (old INR/FSB index).
- "Replacing reader illumination/PCA RD-IL421" on page 737: a note has been added to remind that firmware version should be check and updated if necessary when installing a new PCA.

v4.6.0 - May 2025

MOD042_new_v1.5
FSB-0059#0045 Windows security updates March 2025

A OS image and a standalone installer has been created to update Windows 10 Enterprise 2019 LTSC Operating System of Erytra software version 4.6.0 with Security Updates until March 2025.

TYPE	FSB-0059_0045 Located in RelatedDocuments/Erytra .
SCOPE	All Erytra units: - Running under software version 4.6.0 with Windows 10 Enterprise 2019 LTSC version 2.3.1 - Running under software version 4.6.0 with Windows 10 Enterprise 2019 LTSC version 2.3.2 (Operating System image installed from factory in units with S/N ≥ 059-0001991).
CHARACTER	RECOMMENDED
PRIORITY	ASAP (The installation of the Security Updates is recommended to be done in remote mode or during the next scheduled service action)
REGULATORY IMPACT	The Security Updates March 2025 can be installed in all Erytra instruments.
PROCEDURE	- See "Windows security update installation seccion" (mod) - See "Windows security packages update v1.7.0" section (new)

CHANGE DESCRIPTION

A standalone installer has been created to update Windows 10 Enterprise 2019 LTSC version 2.3.1 and 2.3.2 Operating System (Erytra software version 4.6.0) with Security Updates until March 2025.

A new Operating System image version 2.3.3 with Security Updates until March 2025 has also been created for Erytra software's versions 4.6.0.



NOTE

Windows 10 Enterprise 2019 LTSC version 2.3.1 includes Security Updates until March 2024.

Security Updates standalone installer v1.6.0 includes Security Updates from February 2022 to September 2024.

Windows 10 Enterprise 2019 LTSC version 2.3.2 includes Security Updates until September 2024.

Security Updates standalone installer v1.7.0 includes Security Updates from February 2022 to March 2025.

Windows 10 Enterprise 2019 LTSC version 2.3.3 includes Security Updates until March 2025.

The standalone installer and the new Operating System image include the following Security Updates packages:

ID Sec. Package	Description
KB5005112	Servicing stack update for Windows 10, version 1809: August 10, 2021 This update makes quality improvements to the servicing stack, which is the component that installs Windows updates.
KB5049608	Cumulative Update for .NET Framework 3.5 and 4.7.2 for Windows 10, version 1809 and Windows Server 2019
KB5053596	March 11, 2025 - KB5053596 (OS Build 17763.7009) This update addresses security issues for your Windows operating system. Improvements: • [Daylight saving time (DST)] Updated: DST changes for Paraguay. For more information, see the Daylight Saving Time & Time Zone Blog. • [GB18030-2022] This update adds support for this amendment. • [Temporary files] This update enables system processes to store temporary files in a secure directory "C:\Windows\SystemTemp" via either calling GetTempPath2 API or using .NET's GetTempPath API, thereby reducing the risk of unauthorized access. • [Open Secure Shell (OpenSSH) (known issue)] Fixed: The service fails to start, which stops SSH connections. There is no detailed logging, and you must run the sshd.exe process manually.

FILES DOWNLOAD

Windows Security updates files		
Standalone installer	Installer	w10-1809-ar-secupdates-installer-1.7.0.zip
		SHA-256:
Windows 10 Enterprise 2019 LTSC version 2.3.3	Installer	w10-1809-2.3.3.zip
		SHA-256:
		24e2989955bd0ad059e1919bbc064183760782a0efd61494e4854b0dbf4f2ac4
	Drivers	imb-1003j-drivers-1.2.0.jar
		SHA256:
		8ef8f059df53eac96aaf4be07b5460bc765188cfe24d27c0eec2add824792d79
		imb-150d-drivers-1.1.0.jar
		SHA-256:
		5d42eba4d66ab8ce65c197383f2d4098b26ff0dc6d909bd024fe376462219624

REQUIREMENTS AND CONSIDERATIONS

The Security Updates standalone installer v1.7.0 includes the Security Updates from February 2020 to March 2025.

The Security Updates standalone installer v1.7.0 require that Operating system version is Windows 10 Enterprise 2019 LTSC v2.3.1 (with or without any previous Security Updates installed).

The Security Updates standalone installer v1.7.0 does not require that any previous Security Updates have been installed.

The Security Updates standalone installer v1.7.0 can update Operating System with any previous Security Updates installed.

PROCEDURES

Standalone System update

- Security Updates installation: See "Windows security update installation section"
- Security Updates Verification: See "Windows security packages update v1.7.0" section

Operating System v2.3.3

- Installation: See "Restoring Operating System or full SSD image" section
- Verification: See "Verifying OS and Software version" section

MOD041_new_v1.5

INR-0059#0163 New probe material

In order to improve mechanical and chemical characteristics the probe material has been changed maintaining shape and installation process.

TYPE	INR-0059_0163 Located in Related Documents > Erytra > INRs.
SCOPE	Implemented in factory from s/n ≥ 059-0002055.
P/N UPDATE	CREATED: • 736237 SAMPLES PROBE (Impact detector) v2 MODIFIED: N/A CANCELLED: • 233899 SAMPLES PROBE (Impact detector) COMPATIBILITY: • New P/N are compatible with all Erytra units. • Previous SAMPLES PROBE (Impact detector) model with P/N 233899 has been obsoleted, but remaining stock can still be used in all instruments.
PROCEDURE	N/A

CHANGE DESCRIPTION

Probe material has changed from AISI 304 to AISI 316, which has better physical characteristics and chemical resistance.

MOD040_modf_v1.5

FSB-0059#0041 Erytra Software Version 4.6.0 v2

CHANGE DESCRIPTION

FW PIP version has been modified from 4.10.0 to 4.12.0.

Requirements update about CE and US configuration.

See "MOD029_new_v1.3 FSB-0059#0041 Erytra Software v4.6.0" on page 36

MOD039_modf_v1.5

FSB-0059#0042 Erytra Software Version 5.2.0 (China) v2

CHANGE DESCRIPTION

FW PIP version has been modified from 4.10.0 to 4.12.0.

Requirements update about CE and US configuration.

See "MOD030_new_v1.3 FSB-0059#0042 Erytra Software v5.2.0 (China)" on page 34

MOD038_mod_v1.5

Minor changes and error fixing

CHANGE DESCRIPTION

See "ARCHIVE_PATH" on page 92: A note has been added to help avoid mistakes when configuring the Archive_Path in Erytra properties.

See "Replacing reader illumination/PCA RD-IL421" on page 737: Verify/update firmware step when Replacing PCA RD-IL421.

See "Related documents" on page 19: The link from Modules old Wiring diagram has been changed to the correct one.

MOD037_modf_v1.5

FSB-0059#0044 Windows security updates September 2024 v2

CHANGE DESCRIPTION

The cut number has been added.

See "MOD035_new_v1.4 FSB-0059#0044 Windows security updates September 2024 " on page 28

MOD036_new_v1.5**INR-0059#0162 New 8GB DDR3L RAM module for IMB-150D**

TYPE	INR-0059_0162 Located in RelatedDocuments > Erytra > INRs.
SCOPE	Implemented from factory in Erytrainstruments with S/N ≥ 059-0002047.
P/N UPDATE	CREATED: 735899 RAM DDR3L 8GB SO-DIMM MODIFIED:N/A CANCELLED: 734048 //RAM DDR3 2GB SO-DIMM KINGMAX (2 units) COMPATIBILITY: - P/N is compatible with all Erytra units with MINI-ITX ASROCK IBM-150. - Previous DDR3 2GB SO-DIMM KINGMAX RAM(2 units) model with P/N 734048 has been obsoleted. Remaining stock should not be consumed and new RAM DR3L 8GB SO-DIMM model with P/N 735899 should be used instead.
PROCEDURE	(modified)

CHANGE DESCRIPTION

A new DDR3L 8GB RAM module has been released due to the obsolescence of the DDR3 2GB SO-DIMM KINGMAX RAM.

PROCEDURE

A new RAM module DDR3L 8GB has been released in order to improve IMB-150D efficiency. One module of DDR3L 8GB replaces the previous two modules of 2GB DDR3.

v4.6.0 - November 2024>**MOD035_new_v1.4****FSB-0059#0044 Windows security updates September 2024**

TYPE	FSB-0059_0044 Located in RelatedDocuments > Erytra > FSBs.
SCOPE	Implemented from factory in Erytrainstruments with S/N ≥ 059-0001991.
CHARACTER	RECOMMENDED
PRIORITY	ASAP (The installation of the Security Updates is recommended to be done in remote mode or during the next scheduled service action)

REGULATORY IMPACT

The Security Updates September 2024 can be installed in all Erytra instruments.

PROCEDURE

- See "Windows security update installation seccion" (new)
 - See "Windows security packages update v1.6.0" section (new)
-

CHANGE DESCRIPTION

A standalone installer has been created to update Windows 10 Enterprise 2019 LTSC version 2.3.1 Operating System (Erytra software version 4.6.0) with Security Updates until September 2024.



NOTE

Windows 10 Enterprise 2019 LTSC version 2.3.1 includes Security Updates until March 2024.

Security Updates standalone installer v1.6.0 includes Security Updates from March 2022 to September 2024.

Windows 10 Enterprise 2019 LTSC version 2.3.2 includes Security Updates until September 2024.

This standalone installer includes all the necessary Security Updates to update the Operating System. It does not require that any other Security Updates have been previously installed. The installer can update Operating System with any previous Security Updates installed.

A new Operating System image version 2.3.2 with Security Updates until September 2024 has also been created for Erytra software's versions 4.6.0.

The standalone installer and the new Operating System image include the following Security Updates packages:

ID Sec. Package	Description
KB5005112	Servicing stack update for Windows 10, version 1809: August 10, 2021 This update makes quality improvements to the servicing stack, which is the component that installs Windows updates.
KB5039879	July 9, 2024-KB5039879 Cumulative Update for .NET Framework 3.5 and 4.7.2 for Windows 10, version 1809 and Windows Server 2019. This security update addresses a remote code execution vulnerability detailed in CVE-2024-38081 (.NET Framework Elevation of Privilege Vulnerability).
KB5043050	September 10, 2024-KB5043050 (OS Build 17763.6293) This security update includes improvements. Below is a summary of the key issues that this update addresses when you install this KB. If there are new features, it lists them as well. The bold text within the brackets indicates the item or area of the change we are documenting. • [AppContainer] Printers do not work as you expect when you use them in a restricted setting, like AppContainer. • [BitLocker] You might not be able to decrypt a BitLocker data drive. This occurs when you move that drive from a newer version of Windows to an older version. • [Windows Installer] When it repairs an application, the User Account Control (UAC) does not prompt for your credentials. After you install this update, the UAC will prompt for them. Because of this, you must update your automation scripts. Application owners must add the Shield icon. It indicates that the process requires full administrator access. To turn off the UAC prompt, set the HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\Installer\DisableLUInRepair registry value to 1. • [Windows Server 2019 (known issue)] Some devices slow down, stop responding, and have high CPU usage. This is most likely to occur with Cryptographic Services. A few companixes report that the issue occurs when antivirus software scans the %systemroot%\system32\catroot2 folder for Windows updates.

FILES DOWNLOAD

Windows Security updates files		
Standalone installer	Installer	w10-1809-ar-secupdates-installer-1.6.0.zip
		SHA-256: 1c2f6036c02f224433669db147e3d3a0350421e54bd8bb51f7053d7aa0103e5a
Windows 10 Enterprise 2019 LTSC version 2.3.2	Installer	w10-1809-2.3.2.zip
		SHA-256: 24e2989955bd0ad059e1919bbc064183760782a0efd61494e4854b0dbf4f2ac4
	Drivers	imb-150d-drivers-1.1.0.jar
		SHA-256: 5d42eba4d66ab8ce65c197383f2d4098b26ff0dc6d909bd024fe376462219624

REQUIREMENTS AND CONSIDERATIONS

The Security Updates standalone installer v1.6.0 includes the Security Updates from March 2022 to September 2024.

The Security Updates standalone installer v1.6.0 require that Operating system version is Windows 10 Enterprise 2019 LTSC v2.3.1 (with or without any previous Security Updates installed).

The Security Updates standalone installer v1.6.0 does not require that any previous Security Updates have been installed.

The Security Updates standalone installer v1.6.0 can update Operating System with any previous Security Updates installed.

PROCEDURES

Standalone System update

- Security Updates installation: See "Windows security update installation seccion"
- Security Updates Verification: See "Windows security packages update v1.6.0" section

Operating System v2.3.2

- Installation: See "Restoring Operating System or full SSD image" section
- Verification: See "Verifying OS and Software version" section

MOD034_mod_v1.4

FSB-0059#0043 New illumination pattern.

The design of the 'Illumination Pattern DGGel' has been modified to avoid shadows in the upper area of the card well chamber (just below the meniscus) which could result in false incorrect volumes.

TYPE	FSB-0511_0043 Located in RelatedDocuments > Erytra > FSBs.
SCOPE	All Erytra units with s/n < 511-0002305: Running under software version 3.2.0 with Windows 10 Enterprise 2019 LTSC version 2.3.1 (Operaiong System image installed from factory in Erytra instruments with S/N ≥ 059-0001991.)
CHARACTER	RECOMMENDED(Installing the new illumination pattern model is recommended for those instruments working with Serial Titration techniques available in software version 4.6.0)
PRIORITY	ASAP

P/N UPDATE

CREATED:

- 735851 READER ILLUMINATION SYSTEM v4
- 735852 ILLUMINATION PATTERN DGGel V2

MODIFIED: N/A

CANCELLED: N/A

- 233710 ILLUMINATION PATTERN DGGEL
- 733042 READER ILLUMINATION SYSTEM V3

COMPATIBILITY: New P/N are compatible with all Erytra units.

AFFECTED PROCEDURES N/A

CHANGE DESCRIPTION

The horizontal silk-screened marks (lipemia marks) have been removed. These horizontal marks could sometimes cause an unexpected shadow in the upper area of the card well chamber, just below the meniscus.

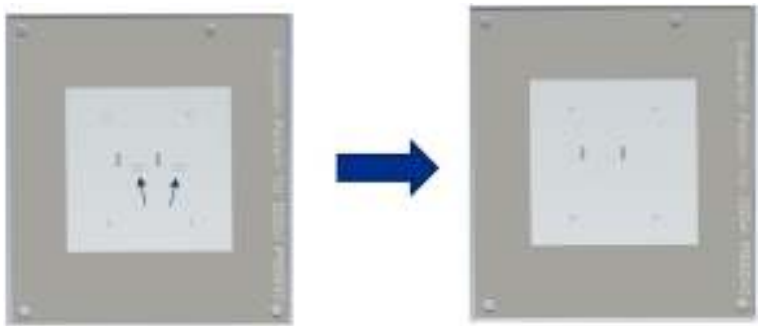


Figure 1 - Illumination pattern's Horizontal silk-screened marks removal

UPDATE PROCEDURE

See "Replacing the reader illumination pattern" section

MOD032_mod_v1.4
Minor changes and error fixing

CHANGE DESCRIPTION

- See "Bulletin index" on page 1183: FSB-059#0041 and FSB-059#0042 release dates have been corrected.
- See "Backing up relevant data manually" on page 1120: A caution note has been added to inform that databases should be updated before backing them up to avoid problems with the user list.

v4.6.0 - v5.2.0 (CN) - October 2024

MOD031_modf_v1.3**INR-0059#0136 v2 Pressure sensors improvement**

Pressure sensor model already used in Erytra Eflexis instrument replaces the one being used in Erytra.

TYPE	INR-0059#0136
SCOPE	Implemented from factory in Erytra instruments from 059-0001955 to 059-0001957 and S/N $\geq$ 059-0001959.
P/N UPDATE	RELATED: • 234210 PRESSURE SENSOR (S3,S4,S5,S6): 1 unit including the corresponding o-ring • 734000 PRESSURE SENSOR CABLE • 733519 O-RING 5.28x1.78 (10 units) CANCELLED: • 233763 PRESSURE SENSOR (S3, S4, S5, S6) COMPATIBILITY: New spare part references detailed above (already used in Erytra Eflexis instruments) are compatible and can be installed in all Erytra instruments See "Spare parts list" on page 1207.
PROCEDURE	See "Replacing pressure sensors" chapter (modified)

CHANGE DESCRIPTION

Pressure sensor model already used in Erytra Eflexis instrument replaces the one being used in Erytra to obtain the necessary pressure and vacuum readings. The sensor component is the same, however the sensor assembly in Eflexis instrument is more robust, minimizing the risk of wire breakage when manipulating it.

Previous pressure sensor was welded to its cable, while the new pressure sensor is composed of the sensor welded to an intermediate PCA with a connector, and the cable as a separate part. The o-ring is the same in both assemblies.

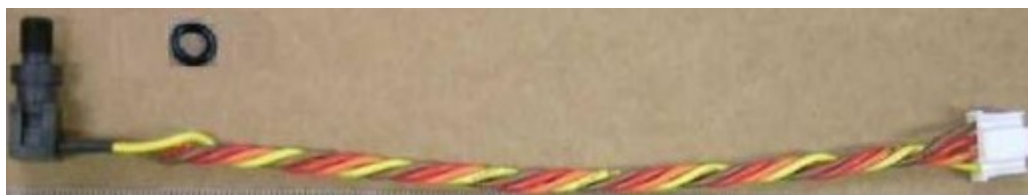


Figure 2 - Previous pressure sensor



Figure 3 - New pressure sensor

MOD030_new_v1.3

FSB-0059#0042 Erytra Software v5.2.0 (China)

TYPE	FSB-0059#0042 (See "Index FSBs and INRs")
SCOPE	All Erytra units that are under National Medical Products Administration (NMPA) regulation (China).
CHARACTER	RECOMMENDED
PRIORITY	ASAP
REQUIREMENTS	Erytra software version 5.2.0 requires the following Reagents Pack versions or higher or a Reagents Pack generated from IH Templates Pack (These packs contain stability on-board information): • CE configuration: IH Reagents Pack v9.0 or higher or IH Reagents Pack AU v2.0 or higher.
REGULATORY IMPACT	Erytra software version 5.2.0 can only be installed in China.
PROCEDURE/DOCUMENT	See "Installing Erytra software (v5.2.0)" chapter (new) See "Verifying software version" chapter (new)

IMPROVEMENTS AND MODIFICATIONS

See "Erytra Software v5.2.0 changes chapter"

VERSION FILES

Download the files needed to install Erytra software version 5.2.0 and Windows 10 version 2.3.1.

Erytra Software version 5.2.0 with OS version 2.3.1

Erytra Software files	File name	ERYTRA-5.2.0..zip
	SHA-256	
OS files	File name	w10-1809-2.3.1.zip
	SHA-256	f9b8f2b63abc075574ac5f2c54e46b7a1e900c01f49188da31cdc983773a0927
Drivers	File name	imb-150d-drivers-1.1.0.jar
	SHA-256	5d42eba4d66ab8ce65c197383f2d4098b26ff0dc6d909bd024fe376462219624
Remote Connection	File name	RemoteConnection.zip
	SHA-256	34072a1035d55d67e2cefdda2afa3930237ed1eb6c282c1f7a5a3375f4d1740f

FIRMWARE VERSIONS

PCA	Firmware version
CEN-SRM	4.5.0
INC-CUP	4.5.0
CAT-CM	4.5.0
PIP	4.12.0
FLU	4.6.2
RD	5.5.0

OS & SW INSTALLATION

See "Installing Erytra software (v5.2.0)" chapter.

VERSION VERIFICATION

See "Verifying software version (v5.2.0)" chapter

MOD029_new_v1.3

FSB-0059#0041 Erytra Software v4.6.0

TYPE	FSB-0059#0041 (See "Index FSBs and INRs")
SCOPE	All Erytra instruments that are not under National Medical Products Administration (NMPA) regulation (China).
CHARACTER	RECOMMENDED
PRIORITY	ASAP
REQUIREMENTS	Erytra software version 4.6.0 requires the following Reagents Pack versions or higher or a Reagents Pack generated from IH Templates Pack (These packs contain stability on-board information): • CE configuration: IH Reagents Pack v9.0 or higher or IH Reagents Pack AU v2.0 or higher. • US configuration: IH Reagents Pack US v2.4 or higher. • CE configuration with stability on-board feature enabled: IH Reagents Pack v9.0 or higher, or IH Reagents Pack AU v2.0 or higher (these packs contain stability on-board information). • CE configuration with stability on-board feature disabled: IH Reagents Pack v9.0 or higher, or IH Reagents Pack AU v2.4. • US configuration: IH Reagents Pack US v2.3 or higher (this pack and the IH Templates Pack v11 do not contain stability on-board information for the US region. The stability on-board information for the US region is included in the IH Reagents Pack US v2.4 and IH Templates Pack v12).
REGULATORY IMPACT	Erytra software version 4.6.0 can be installed in all the countries where the Erytra is registered except China
PROCEDURE / DOCUMENT	See "Installing software" chapter (modified) See "Verifying software installation" chapter (new)

IMPROVEMENTS AND MODIFICATIONS

See "Erytra Software v4.6.0"

VERSION FILES

Download the files needed to install Erytra software version 4.6.0 and Windows 10 version 2.3.1 (See "Downloading Erytra software version" chapter):

Erytra Software version 4.6.0 with OS version 2.3.1

Erytra Software files	File name	ERYTRA-4.6.0.zip
	SHA-256	
OS files	File name	w10-1809-2.3.1.zip
	SHA-256	f9b8f2b63abc075574ac5f2c54e46b7a1e900c01f49188da31cdc983773a0927
Drivers	File name	imb-150d-drivers-1.1.0.jar
	SHA-256	5d42eba4d66ab8ce65c197383f2d4098b26ff0dc6d909bd024fe376462219624
Remote Connection	File name	RemoteConnection.zip
	SHA-256	34072a1035d55d67e2cefdda2afa3930237ed1eb6c282c1f7a5a3375f4d1740f

FIRMWARE VERSIONS

PCA	Firmware version
CEN-SRM	4.5.0
INC-CUP	4.5.0
CAT-CM	4.5.0
PIP	4.12.0
FLU	4.6.2
RD	5.5.0

OS & SW INSTALLATION

See "Installing software" chapter.

VERSION VERIFICATION

See "Verifying software installation" chapter.

MOD028_modf_v1.3**New Spare parts list**

TYPE	topic modification
SCOPE	All Erytra instruments.
P/N UPDATE	CREATED: N/A MODIFIED: N/A CANCELLED: N/A COMPATIBILITY: N/A
PROCEDURE	See "Spare parts list"

CHANGE DESCRIPTION

New spare parts list with commercial regions recommended stock columns.

v4.5.0 - April 2024**MOD027_new_v1.2****New T5 PIP/CAT arm large belt relacelment procedures**

A new procedure has been created to replace large belts on X/Y axis on PIP/CAT arms.

TYPE	New procedure
SCOPE	All Erytra units.
P/N UPDATE	CREATED: CANCELLED: NA COMPATIBILITY: • 233866TOOTHED BELT T5/1200 • 233865TOOTHED BELT T5/2000 See "Spare parts list"
PROCEDURE	See "Replacing toothed belt T5/2000 (CAT X-axis)" on page 524 See "Replacing toothed belt T5/1200 (CAT Y-axis)" on page 527 See "Replacing toothed belt T5/2000 (PIP X-axis)" on page 421 See "Replacing toothed belt T5/1200 (PIP Y-axis)" on page 425 See "Replacing leadscrews" on page 382

CHANGE DESCRIPTION

In order to inform about T5 belt clamping plate assembly considerations, T5 belts replacement procedures have been added.

MOD026_new_v1.2**INR-0059#0161 New PM structure and new format for PM, IQ and OQ documents**

As not all the instruments have the same workload and usage, the Preventive Maintenance (PM) structure has been redefined with the aim of improving execution steps and adapt the frequency of parts replacement to the workload of the instrument.

TYPE	INR-0059#0161
SCOPE	All Erytra units.
P/N UPDATE	CREATED: NA CANCELLED: NA COMPATIBILITY: NA
PROCEDURE	See "Preventive maintenance" on page 124 See "Installation Qualification (IQ)" on page 87 See "Operational Qualification (OQ)" on page 104 See "Replacing coupling CPC bottles connector" on page 619 See "Replacing bottle CPC connector and internal tube" on page 621

CHANGE DESCRIPTION

Instruments have been classified in 3 categories depending on their annual usage and the frequency of the Preventive Maintenance has been adapted for each category.

The new structure of the PM is as follows:

1. Parts replacement (according to usage).
2. Visual / Physical inspections & cleaning.
3. Lubrications (according to usage).
4. Functional checks & adjustments.
5. Qualification.

The parts replaced, lubrications, cleanings and checks have been recalculated and adapted to the current situation of the instruments' parts life expectancy, resulting in a substantial reduction of time and maintenance time and costs for those instruments working below the full potential usage of the instrument. Moreover, the new structure of the PM will result in an improvement on the efficiency of the field service engineer's tasks when executing the different steps.

Additionally, PM, IQ and OQ check list documents have been changed to Excel format, allowing their edition in a PC or laptop during the execution of the corresponding tasks. These excel files will not be able to be saved (so that a blank form is available the next time it's opened) but they can be printed out to a PDF file or sent to the lab printer.

PM OPERATION

Once opening the Excel file, the Preventive Maintenance home page will be prompted. There, the “Annual usage of the instrument” should be selected (40.000, 80.000 or 130.000 tests/year) as well as “Maintenance year of the instrument” (the years that can be selected will be filtered according to the previously selected annual usage). Then, click on the “Go to...” link and the corresponding PM check-list page will be opened.

ERTYRA
PREVENTIVE MAINTENANCE

GRIFOLS
 DG-AT-Y9M-000174
 Form version 1.0



INSTRUCTIONS: Select from the dropdown list and Go to corresponding PM. Remember selecting again the maintenance year when you change the Annual Usage selection in order to ensure a correct PM combination.

Annual Usage of the instrument:	40.000 tests
Maintenance year of the instrument:	Y8

Go to PM: Prescribed: 40.000 tests Y8

Figure 4 - Preventive maintenance home page

Once in the PM check-list, each part replacing procedure can be accessed by clicking on the links provided.

[illegible]

Figure 5 - Preventive maintenance check-list page

NEW PM PROCEDURES

New procedures have been created for bottles CPC records replacement.

v4.5.0 - March 2024

MOD025_modf_v1.0

The number of characters making up the serial number of the instrument in erytra.properties file must match that defined on the product label.

Changing the number of characters in this file has an impact at SW level. Complaint 800011676.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "INSTRUMENT_SERIAL" on page 96.

MOD024_new_v1.0

The incubators temperature adjustment wizard now displays a message indicating whether the incubator is adjusting the temperature or it is already stabilized.

If the temperature is saved while the incubator is still regulating it, this action will be logged in **DEBUG.technical_service** log file.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Adjusting temperature" on page 263.

MOD023_modf_v1.0

Instructions have been added for tightening the syringe against the valve.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Replacing syringe and valve" on page 608.

MOD022_del_v1.0

From Operating system version 2.2.x onwards, to access a shared folder of a remote computer the mapped drives is the only procedure that can be performed. Procedure 1 has been deleted.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Mapping a network drive" on page 1113.

MOD021_new_v1.0

INR-059#0159 New capacitive touchscreen

TYPE	INR-059#0159
SCOPE	Implemented from factory in Erytra instruments with S/N ≥ 059-0001911.
P/N UPDATE	CREATED: 735520TOUCHSCREEN V4 CANCELLED: 732288 TOUCHSCREEN V3 COMPATIBILITY: The new parts can be installed in all Erytra units. See "Spare parts list" on page 1207.
PROCEDURE	See "Replacing touchscreen monitor" on page 194 (procedure has not been modified) See "Adjusting touchscreen" on page 171 (procedure has not been modified)

CHANGE DESCRIPTION

Due to the obsolescence of the resistive touch panel, Instruments' touch screen is improved replacing the previous panel for a capacitive touch panel. The TFT/monitor is the same model as before therefore the configuration buttons did not change.

Both, previous and new, touch screens have a similar appearance but can be differentiated by the model number located at the back of the panel and by how they react when touched. New panels provide multi-point properties and also a better reaction when used.



Figure 6 - Previous and new touchscreen back label

MOD020_modf_v1.0

Spare parts list has been updated

CHANGE DESCRIPTION	
P/N UPDATE	See "Spare parts list" on page 1207.
PROCEDURE	See "Erytra Revision history table" on page 1383

SCOPE	N/A
-------	-----

TYPE

CHANGE DESCRIPTION

Updated spare part list

MOD019_new_v1.0

When starting the installation of Erytra Database Manager, it is automatically checked that BTM is not installed on the computer.

Nowadays the PostgreSQL version and the configuration of both softwares is not compatible.

TYPE	Modified section.
------	-------------------

PROCEDURE / DOCUMENT	See "Erytra database manager application" on page 1127.
----------------------	---

MOD018_modf_v1.0

Leak tightness test has been improved.

CPD values before creating pressure or vacuum will now be checked and the allowed margins have been reviewed and take into account the effects of instrument location (altitude).

TYPE	Modified section.
------	-------------------

PROCEDURE / DOCUMENT	See "Verifying CPD sensors" on page 583.
----------------------	--

MOD017_modf_v1.0

The Pressure circuits test from diagnostic has been improved for instruments configured to use an external drain.

the fluidic section of Solution A container in drawer 4 and Solution B container in drawer 5 are now tested.

TYPE	Modified section.
------	-------------------

PROCEDURE / DOCUMENT	See "Pressure circuits" on page 867.
----------------------	--------------------------------------

MOD016_new_v1.0

New section "Collecting Windows event logs" has been added. This section shows how to collect the Both application and System logs.

These logs are not included in the Support Pack and are usually needed to investigate Software issues.

TYPE	New section.
PROCEDURE / DOCUMENT	See "Collecting Windows event logs" on page 1114.

MOD015_modf_v1.0

DSC.def and DSC_Custom.def files will now be included in the backup and restored automatically when the restoring and baking up process is performed from the boot application.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Backing up databases and relevant data with boot menu" on page 1117. See "Restoring databases and relevant data with boot menu" on page 1124.

MOD014_new_v1.0

Added information on water recommended for use with Grifols Wash Solutions or DG Fluids to wash the fluidic system of Grifols immunohematology instruments.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Functional characteristics" on page 59.

MOD013_modf_v1.0

The software installation procedure has been updated.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Installing Erytra software v4.6.1" on page 1039.

MOD012_modf_v1.0

Assembly drawings have been updated.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Related documents" on page 19.

MOD011_modf_v1.0

The Importer log files will now be included in the Support Pack.

New folder named Importer in Support Pack Erytra\Log folder.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Support pack" on page 1147.

MOD010_modf_v1.0

Logs updated according to SW upgrade from 4.5.0 and above.

The system will now post in a log the CPD values measured during aspirations. The new log is named PipettingProfile.log and it will be created in E:\log folder.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Description of log files" on page 75.

MOD009_new_v1.0

Added Danish (Denmark), Slovak (Slovakia), Estonian (Estonia), Hungarian (Hungary), Lithuanian (Lithuania), Latvian (Latvia), Russian (Russia) and Slovenian (Slovenia) languages.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "LANGUAGE" on page 97.

MOD008_modf_v1.0

The Windows Security packages has been updated according to SO 2.2.2.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Windows security packages" on page 1099.

MOD007_modf_v1.0

Uninstalling and shipping information has been updated with a requirement to use theader sealant on the illumination pattern screws, should the instrument need to be transported.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Uninstalling and shipping" on page 117.

MOD006_new_v1.0

Hard disc disposal instructions has been added in order to safekeeping their containing personal data.

TYPE	Modified section.
PROCEDURE / DOCUMENT	See "Instrument disposal" on page 123.

MOD005_del_v1.0

INR-0059#0156 Instruments with serial number $\geq$ 059-0001903 do not include the factory pre-installed SW.

Instrument logistics will be simplified by removing the software version dependency for the manufactured analyzers.

The instruments include the operating system and it is necessary to install the latest Software version on the instrument during its installation.

TYPE	INR-0059#0156
SCOPE	Software will not be pre-installed anymore from factory in Erytra instruments with S/N $\geq$ 059-0001903.
P/N UPDATE	CREATED: NA COMPATIBILITY:NA
PROCEDURE	See "Installing software for the first time" on page 88.

CHANGE DESCRIPTION

Erytra software will not be pre-installed from factory for instruments with SN $\geq$ 059-0001903, so the necessary software version should be installed at the moment of installing the instrument. Operating system will still be installed in all manufactured instruments.

MOD004_modf_v1.0

INR-0059#0155 The walls of the PIP robotic head horns have been redesigned to improve the robustness of the assembly. The new design will also allow a new mechanical adjustment system for the barcode reader devices, as well as integrating the mirror support in the walls.

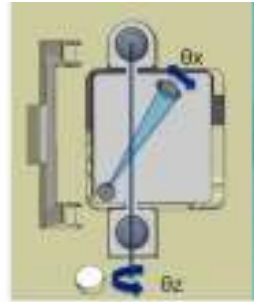
TYPE	INR-0059#0155
SCOPE	Implemented from factory for instruments with SN $\geq$ 059-0001887.
P/N UPDATE	CREATED: • 735329HORNS LATERAL v3 (RIGHT). • 735327HORNS LATERAL v3 (LEFT). • 735328HORNS-FRONTAL v2. COMPATIBILITY: The new spare part number can be installed in all Erytra instruments with SN $\geq$ 059-0001887. If the new horns walls are to be installed in Erytra units below the above-indicated SN, the whole horns structure, including horns holder plate must be replaced by parts of the new model. Thus, spare parts for the horns lateral walls (P/N 732508HORNS LATERAL v2 (RIGHT) and P/N 732509HORNS LATERAL v2 (LEFT)) and the horns frontal (P/N 233835) are still available in case they need to be replaced in the instruments with SN $<$ 059-0001887. See "Spare parts list" on page 1207.
PROCEDURE	See "Adjusting barcode reader inclination position" on page 348. See "Replacing horns chain" on page 406.

CHANGE DESCRIPTION

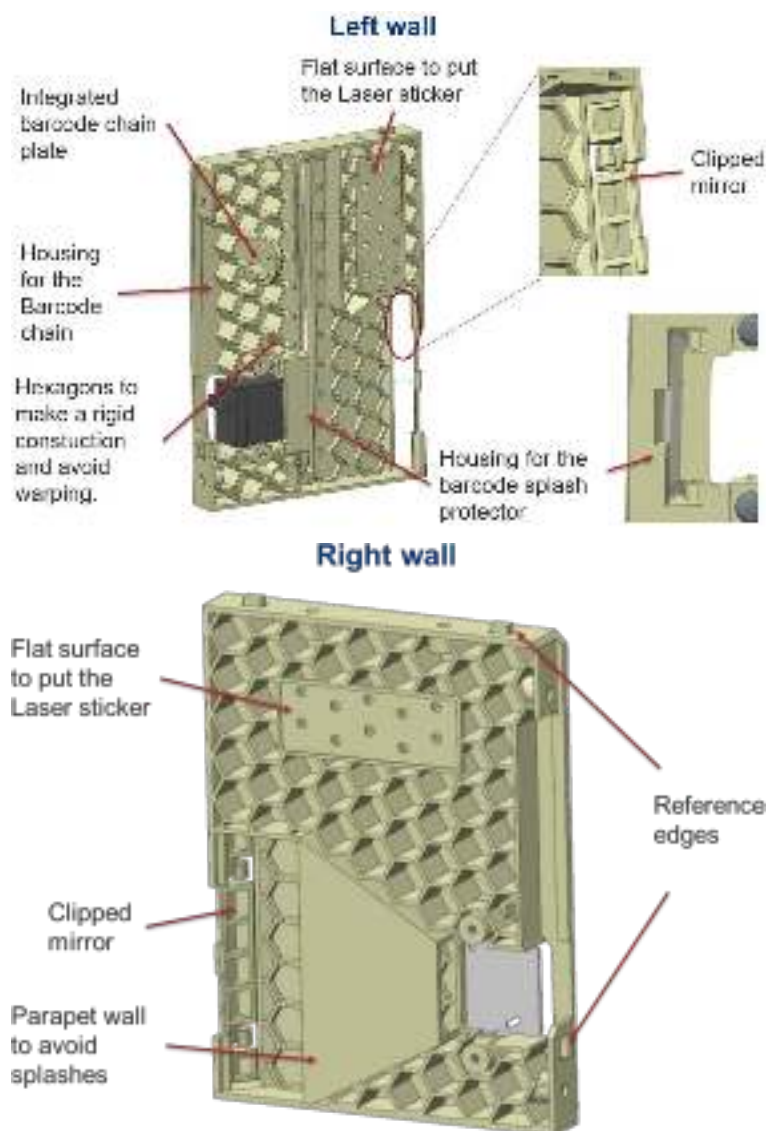
The new horns design improves the robustness of the PIP robotic head assembly of Erytra, solving issues related to the geometry of the horn walls seen during the manufacturing process.

Additionally, this new design simplifies the barcode reader adjustment. Now, this adjustment can be performed through a metallic plate holding the barcode reader device, which allows to adjust the beam position in the Z axis. The angle of the plate can be modified by applying a small amount of pressure onto the plate. When the desired position is reached the plate will maintain that position.

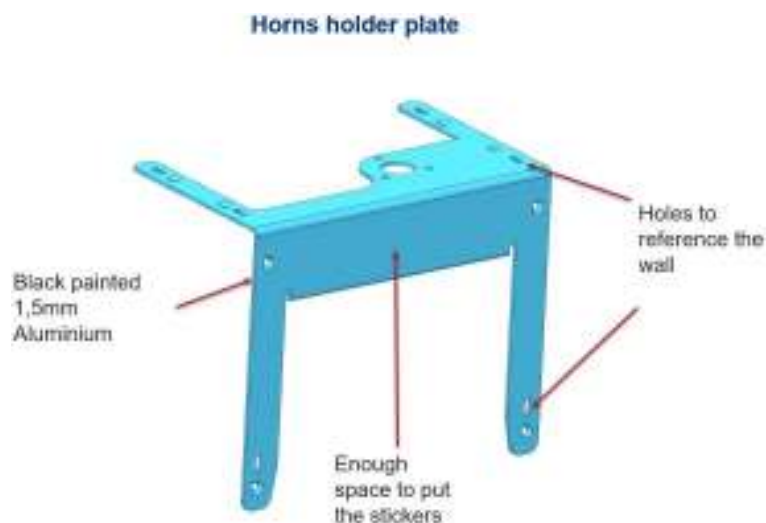
X angle adjustment (Slotted hole)
Z angle adjustment (deformable plate)



The current horn walls were made by LTP printing technology which has been replaced by 3D printing. The new structure simplifies the assembly integrating the wall elements like the barcode reader plate and the BCR chain plate. The mirrors are held in place by clips as the splash protector.



The structure of the new horn holder plate, now with L shape, has tiny perforations that match the wall references, ensuring a correct perpendicularity between the two parts.

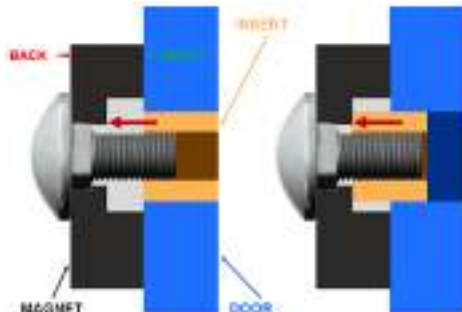
**MOD003_modf_v1.0**

INR-0059#0151 The assembly of the magnetic sensor of the upper door has been improved, adapting it to the geometry of the component.

TYPE	INR-0059#0151
SCOPE	Implemented from factory for instruments with SN $\geq$ 059-0001859.
P/N UPDATE	CREATED: • 735084UPPER DOOR v2. CANCELLED: • 233825 UPPER DOOR. COMPATIBILITY: New 735084UPPER DOOR v2 is compatible with the existing spare part 233807DOOR SENSOR and can be installed in all Erytra instruments using the screws of the original sensor being replaced. See "Spare parts list" on page 1207.
PROCEDURE	See "Replacing upper door" on page 219.

CHANGE DESCRIPTION

The assembly procedure of the magnetic sensor in charge of detecting the status of the upper door has been improved, adapting it to the geometry of the component. When the magnet is screwed to the door, the applied force may cause the insert to move out of position due to the corkscrew effect.



To solve the problem, the position of the sensor has been reversed:



The new orientation of the sensor makes it necessary to change the position of the screws in the door.



Additionally, the current M3x10 screw is replaced by a M3x8. The screw head now fits inside the sensor housing.

MOD002_new_v1.0

FSB-0059#0039 Filters have been added in the path between solution containers and the manifold to prevent undesired residues from reaching the latter.

TYPE	FSB-0059#0039
SCOPE	Implemented from factory for instruments with SN ≥ 059-0001879.
P/N UPDATE	CREATED: 735206SOLUTION LINES FILTERS SET.  NOTE 735206SOLUTION LINES FILTERS SET contains 4 x 50-micron filters. One full set can be installed in an Erytra configured as Dual Mode or in 2 Erytra configured as Single Mode. COMPATIBILITY: New spare part 735206SOLUTION LINES FILTERS SET can be installed in all Erytra units. See "Spare parts list" on page 1207.
PROCEDURE	See "Replacing solution A and B filter" on page 625.

CHANGE DESCRIPTION

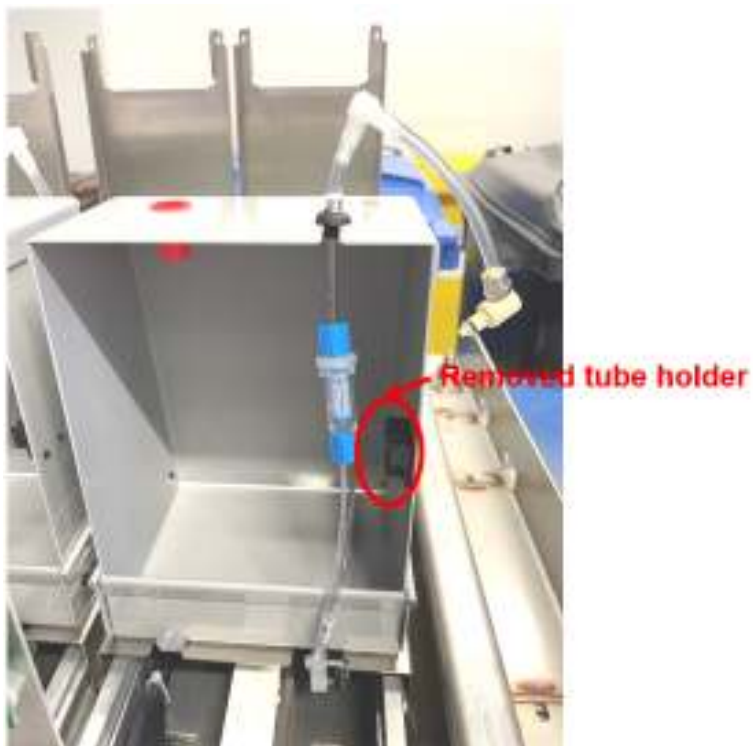
50-micron filters are added between the containers of solution A and B and the manifold of the instrument. The selected filters are the same that are used for the vacuum lines.

The main objective is to prevent solid particles coming from solution bottles could reach the fluidic circuit, causing obstruction in electrovalves.

**NOTE**

In the upcoming version of Erytra PM, a step will be added to check the status of the Solution A and B circuits' filters.

Since after the installation of the filter, the tube section behind the container basis will not fit into the tube holder, Erytra units with the new solution filters from factory will not include the tube holder.

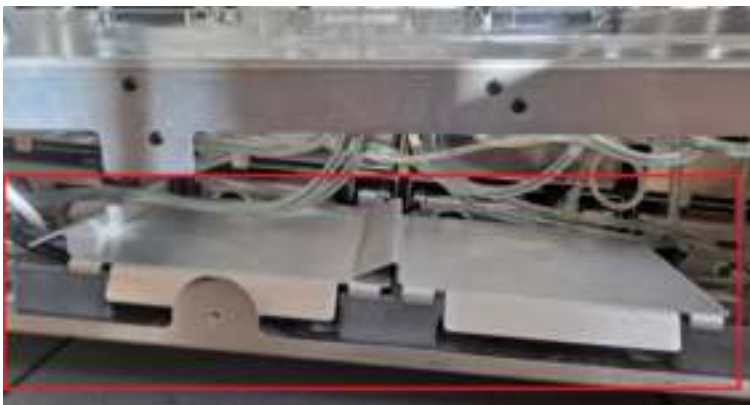
**MOD001_new_v1.0**

INR-0059#0149 Two metallic plates have been included beneath the manifold in order to protect the fluidic connections in that area against possible spillages.

TYPE	INR-0059#0149
SCOPE	Implemented from factory for instruments with SN $\geq$ 059-0001851.
P/N UPDATE	CREATED: • 735052PROTECTOR PLATES (FLU). COMPATIBILITY: The new spare part number can be installed in all Erytra units. See "Spare parts list" on page 1207.
PROCEDURE	See "Replacing protector plates" on page 632.

CHANGE DESCRIPTION

Two metallic plates have been included beneath the manifold in order to protect the fluidic connections in that area against possible spillages from the manifold.



The metallic plates have two front legs which are inserted into a snapping foot to fix their position and two rear legs for support. Both plates are overlapped to prevent leakage between the space that may be left between them.



The snapping feet have been designed to be inserted into the multi tube holders of the lower tray.



The geometry of the parts restricts the evacuation of the liquid in only one direction.



FSE tools

All the procedures in this document require a standard set of tools. The recommended standard tools are listed in the table below.

Table 1 - Standard tools

TOOL	DESCRIPTION
SET OF STANDARD ALLEN WRENCHES	SET OF SIX HEX KEYS (1.5, 2, 2.5, 3, 4, and 5 MM).
10" ADJUSTABLE WRENCH	WIDTH ADJUSTABLE JAW WRENCH.
SCREWDRIVER SET	
NUT DRIVER SET	8.0 MM, 7.0 MM, 5.5 MM, and 4.5 MM.
TUBE CUTTER	TUBING CUTTER FROM 4 MM - 6 MM.
WIRE CUTTERS	
FLASHLIGHT	
ESD WRIST STRAP	

Some procedures also require special adjustment tools. The **Adjustment tools** list at the top of every procedure shows the tools that are needed for that procedure. The full list of adjustment tools is listed below.

Table 2 - Adjustment tools

ADJUSTMENT TOOL	MODULE	PN
PROBES ADJUSTMENT TOOL (Ver. 4)	PIP	234213
OMNIFIT TOOL (for tube connection Ver 2)	PIP	233898
DILUTION CUP/PROBE ADJUSTMENT TOOL	PIP	234178
PROBE POSITIONING VERIFICATION TOOL	PIP	234216
CARDS DRAWER ADJUSTMENT TOOL (Ver. 2)	CAT	233878
CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL (Ver. 3)	CAT	733056
LOWER CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL v3	CAT	733068
CENTRIFUGE ROTOR ADJUSTMENT TOOL	CEN	233821
DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V	INC	230903
TEMP. PROBE PT-100 FOR CARDS - MiniDIN 4v	INC	230905

ADJUSTMENT TOOL	MODULE	PN
READER ADJUSTMENT TOOL-SUPPORT (Ver. 2)	LEC	234212
READER ADJUSTMENT TOOL-PATTERN v2	LEC	731398
USB 2.0 CABLE 1.8 M (TOUCHSCREEN, ELECTRONICS CONNECTORS COVER AND PCA RD IL421)	LEC	233443
FIREWIRE READER CABLE TOOL	LEC	233822
PRESSURE SENSOR ADJUSTMENT TOOL	FLU	233817
DIGIT. GAUGE DIGITRON MOD.2023	FLU	247032

1 Getting started

Before you start using Erytra, get familiar with information about the main modules, software and technical specifications.

1.1 Introduction

Configurable as a standalone instrument or interfaced to the customer's Laboratory Information System (LIS), Erytra automates many of the test processing functions and data management requirements using gel cards and digital image processing.

Samples contained in standard immunohaematology laboratory tubes are introduced into the instrument and are processed using cards and reagents also introduced, according to the instructions for use. Results are automatically read, interpreted and registered. Results can be reviewed, printed or exported upon user request.

This manual is for a qualified technician and contains the necessary information for proper and secure use of the instrument.

Read carefully all the information in this manual before you start using this instrument. It is important to read the information concerning maintenance, correct use and liquids management.



NOTE

All the operations described in this manual must be carried out only by a qualified technician.

1.2 Users

Supervisor	Individual or group responsible for the use and maintenance of the equipment, and for ensuring that operators are adequately trained.
Operator	Person operating the equipment for its intended purpose. The operator should have received the appropriate training for this purpose.
Qualified technician	Person responsible for the installation, reparation and special maintenance of the equipment, who has received specific training for this purpose.

All restrictions referred to the supervisor are valid for the operator.

1.3 Security remarks

All the information concerning security, marking and regulatory information of the instrument is described in the Instructions for Use manual.

1.4 Functional characteristics

Functional characteristics of Erytra can be found in the Instructions for Use manual.

1.4.1 Purified water or equivalent for laboratory use

It is hereby informed that as “purified water or equivalent for laboratory use”, any water that has been treated for any of the following methods: distillation, deionization or reverse osmosis, is appropriate for its use with Grifols Wash Solutions or DG Fluids to wash the fluidic system of Grifols immunohematology instruments as long as it meets the following specifications:

Parameters	European Pharmacopeia 11.0 Monograph 04/2018:0008	USP Chapters 1231, 643 and 645	ISO 3696:1987			CLSI C3-A4
	Water, purified	Purified Water	Water Grade 1	Water Grade 2	Water Grade 3	Clinical Laboratory Reagent Water
pH @25°C	Not specified	5.0 - 7.0 (Specified for conductivity tests. 0.3ml per 100 ml of potassium chloride)	Not Applicable (Because of difficulties associated with pH value of high purity water)	Not Applicable (Because of difficulties associated with pH value of high purity water) 4	5.0 - 7.5	Not specified
Conductivity @25°C (µS/cm)	5.1	2.1 (Stage 2)	0.1	1.0	5.0	0.1 Min. theoretical: 0.05 Min. air exposed: 1.0
TOC or oxidisable substances	0.5 mg/l	0.5 mg/l	Not specified	Not specified		500 ng/g (0.5 mg/l)
Microbial count (UFC/ml)	100	100	Not specified	Not specified	Not specified	10
Heavy metals (ppm)	Not specified	0.1 (some considerations given)	Not specified	Not specified	Not specified	Not specified
Nitrates (ppm)	0.2	Not specified	Not specified	Not specified	Not specified	Not specified
Absorbance @254 nm, 1 cm op Path	Not specified	Not specified	0.001	0.01	Not specified	Not specified

Parameters	European Pharmacopeia 11.0 Monograph 04/2018:0008	USP Chapters 1231, 643 and 645	ISO 3696:1987			CLSI C3-A4
	Water, purified	Purified Water	Water Grade 1	Water Grade 2	Water Grade 3	Clinical Laboratory Reagent Water
Res. after evaporation mg/kg	Not specified	Not specified	Not applicable (See ISO 3696)	1	2	Not specified
Silica mg/l	Not specified	Not specified	0.01	0.02	Not specified	Not specified
Filtrado	Not specified	Not specified	Not specified	Not specified	Not specified	0.22 µm

1.5 Technical specifications

The following table summarizes the main technical specifications of Erytra.

Table 3 - Technical specifications

Model	Erytra	
Supply	Voltage	100-240 V AC
	Frequency	50-60 Hz
	Maximum input power	600 W
	Standby power	30 VA
	Fuses	2 x T6.3 A L 250 V, 5x20 mm
Protection against electric shock	Class I	
Installation category	Overvoltage category II	

Dimensions	700 mm (depth) x 1060 mm (width) x 1785 mm (height) 725 mm (depth, with door handles installed) x 1060 mm (width) x 1785 mm (height) Packaging: 800 mm (depth) x 1200 mm (width) x 1800 mm (height)	
Weight	Approximately 350 kg (with washing and waste containers full)	
Working conditions	Indoors use	
	Temperature	15 °C to 30 °C
	Maximum relative humidity	85 % (non condensing)
	Maximum altitude	3000 m (9842 ft)
	Maximum fluctuations of the power supply	±10 % of the nominal power supply
	Pollution degree range	2
Storage and transport conditions	Temperature	-29 °C to 60 °C
	Maximum relative humidity	85 % (non condensing)
Sound level	58,9 dBA	
Heat dispersion (per hour)*	Standby	0,005 kWh / 18 kJ / 4,3 kcal / 17,1 BTU
	Start Up / Operation	0,171 kWh / 615,6 kJ / 147,1 kcal / 583,5 BTU
	Ready for Workload	0,138 kWh / 496,8 kJ / 118,7 kcal / 470,9 BTU
Heat dispersion (per hour)**	Standby	0,006 kWh / 21,6 kJ / 5,2 kcal / 20,5 BTU
	Start Up / Operation	0,126 kWh / 453,6 kJ / 108,4 kcal / 429,9 BTU
	Ready for Workload	0,121 kWh / 435,6 kJ / 104,1 kcal / 412,9 BTU

* with old electronics cabinet

** with new electronics cabinet (S/N ≥ 059-0001583)

1.6 Erytra modules

In this chapter an overall description of all the modules of the instrument is presented.

- Status LEDs: allow the operator to know the instrument status without looking at the software screen. The instrument contains status LEDs in the sample drawers, in the pushbutton and on the top of the instrument (General LED).
- Temperature control system: system to check that the internal temperature does not exceed defined values.
- Buzzer: it alerts acoustically of any incidence to the operator.
- Structure of the instrument: chassis, cables, tubes rails and external covers. In the base of the instrument there are four rotating supporting wheels, one in each corner, to allow moving the instrument.

1.6.2.1 Embedded computer and touchscreen monitor

An embedded computer and a touchscreen monitor are integrated in the instrument, thus there is no need to use an external computer to control the instrument.

There are some ports available to allow the user to connect external peripherals and to connect the instrument to a Laboratory Information System (LIS). These ports are:

- Two USB ports located at the front part of the instrument.
- Four USB ports located at the rear part of the instrument.
- One RS-232 port located at the rear part of the instrument.
- One ETHERNET/LAN connection located at the rear part of the instrument.

The screen is installed in a drawer located under samples and reagents area. This drawer allows hiding completely the screen inside the instrument when it is not in use. Also the drawer allows the operator to adjust its position/inclination.

1.6.2.2 Supply and Communication bus

This is a system that provides the standby power supply to the computer in order to, when the pushbutton is pressed to switch it on, activate the ATX control of the computer and allow the system to turn on. Once the instrument is switched on, main supply voltages (24 V and 5 V) in Erytra are generated by the power supply system.

Regarding the communication system, the instrument has three different types of communication buses: RS-232, RS-485 and I2C. The different parts of the instrument that use each communication bus are described in the following chapters:

- "General module (GEN)" on page 138.
- "Incubator module (INC)" on page 256.
- "Centrifuge module (CEN)" on page 295.
- "Pipetting robot arm module (PIP)" on page 324.
- "Card transport module (CAT)" on page 462.
- "Fluidics module (FLU)" on page 555.
- "Dilution cup module (CUP)" on page 69.
- "Card reader module (LEC)" on page 695.

1.6.2.3 Samples management module

This module is located in the upper area of the instrument. The instrument has four sample drawers that are labelled as 3, 4, 5 and 6 in front of the instrument. Each drawer has a safety latch (controlled electronically) to monitor open/close status and assure closed/working position. A button is located just below each drawer to activate the latch, opening the drawer.

Each sample drawer allocates two samples racks. The correct centering of different diameter tubes in their lodgings is ensured by centering clips. The racks have an opening window to allow reading the barcode of the samples tubes.

1.6.2.4 Reagents management module

This module is also located in the upper area of the instrument. The instrument has two reagents drawers, labelled as 1 and 2, in front of the instrument. Each drawer has a safety latch (controlled electronically) to monitor open/close status and assure closed/working position. As well as the samples drawers, a button is located just below each drawer to activate the latch, opening the drawer.

Each reagents drawer allocates two racks (18 vial rack, six vials and three diluents rack). The reagents racks have a window to allow reading the barcode of the vials/bottles. All vial lodgings have a spring clip to fix its position against the reading window. The spring clips avoid any movement of the vial in its lodging. All the vials lodgings place the vial/bottle with a certain inclination to minimise dead volumes.

Both drawers include a rotating mechanism in order to stir the vials of the 18 vials rack all together.

1.6.2.5 Cards management module

This module is located in the middle area of the instrument and includes four drawers, labelled as 1, 2, 3 and 4, in front of the instrument. Each drawer can allocate four card holders, orientated in landscape mode.

Also four safety latches (controlled electronically) are included, one per each drawer, that ensure the drawer is closed in the working position and allows close/open status monitoring.

Drawers are opened by means of a keyboard including buttons for each drawer. This keyboard is located under the drawers.

1.6.2.6 Doors

An upper area door gives to the technician / operator the access to the samples and reagents management modules. Once opened, it keeps the position and does not close down by its own weight. The door does not have a latch, it has a sensor that monitors open/close status.

An intermediate area door gives to the technician the access to the cards management modules. The door does not have a latch, it has a sensor that monitors open/close status. In order to prevent the access to the user to the cards management area, the door is closed by means of a screw located in the sensor area.

1.6.3 Incubator module (INC)

There are four independent incubator blocks. Each block has capacity for 12 cards which are positioned equidistantly in the block. A Peltier module unit (formed by Peltier cell, fan and heatsink) attached to incubator block is used as reversible heat/cool pump to warm up or cool the block to the desired temperature.

The other function of the Incubator module is to transport the cards between the upper and intermediate level moving vertically the incubator block. The incubator movement is performed by a servocontrolled motor, with a home position in the lower position of the incubator detected by means of a light barrier. The higher position of the incubator is determined by an adjustment.

1.6.4 Centrifuge module (CEN)

The instrument has two identical centrifuges. Each centrifuge has a capacity of 12 cards. Main parts of each centrifuge are the rotor with the card holders, the motor and the case.

The motors are DC brushless motors (one per centrifuge) with a home position detected by means of a Hall effect sensor. The motors are isolated with silentblocks to reduce a possible transmission of vibration to the rest of the instrument.

The rotor is the rotating part of the centrifuge and is attached to the motor axis. It holds the card holders that contain the cards to be centrifuged.

Centrifuge design ensures that any even number of cards can be introduced in a way that the centrifuge is not unbalanced. To balance the centrifuge, when an odd number of cards need to be centrifuged, an empty card is used. There is no specific card to balance the centrifuge, it will be used a card waiting to be used in card drawers. There is not unbalance detector.

1.6.5 Pipetting robot arm module (PIP)

The PIP module is allocated in the upper area of the instrument and is responsible for pipetting samples and reagents into cards (with the appropriate controls and level detection) and identification and presence detection of samples and reagents.

PIP module is composed of the following submodules:

- X-axis: it allows the movement along the X-axis (left to right in the instrument).
- Y-axis: it allows the movement along the Y-axis (back to front in the instrument).
- Robot head: it contains the following items:
 - Probes including probes washer, liquid level detector system as well as related mechanics and electronics.
 - Barcode readers and related mechanics and electronics.
 - Presence detector and related mechanics and electronics.
- Correct pipetting detectors (CPDs) located at the rear part of the robot arm, and digital pumps (dilutors) are explained in the "Fluidics module (FLU)" on page 555.

This module can read the barcodes of each sample tube, vial or bottle. Samples and reagents identification is done automatically and samples can also be identified manually. There are two barcode readers allowing reading the barcodes from each side of the rack. They are located in such a way that ensures that when one reader is centered with a sample tube the other one is not focused to any other tube.

The identification includes two main tasks:

- Presence detection: detects the presence of tubes and vials/bottles in the drawers.
- Barcode reading: reading of the barcode of each tube/vial/bottle.

In order to perform the pipetting actions, the robot arm includes two probes. Using the level detection system (capacitive sensing, based on frequency variations detection), the instrument calculates the available amount of reagent in each vial in order to give the operator feedback of the available and required reagent volume. This estimation is performed before each aspiration and in each reagent drawer identification. For this purpose, a double detection is performed. The system monitors the position of both detections and raises an error if they are different. If impact detector system is installed, it is possible to detect and protect probes from undesired impacts which are frequently caused by improper user operation.

Probe internal wash is required to remove any trace of aspirated liquid (extra volume of liquid aspirated and rests of this liquid in the tubing) in order to minimise the risk of carry over. Internal probe wash is performed with Solution A. Probe external wash is required to remove any trace of aspirated liquid in the probe external surface in order to minimise the risk of carry over. External wash is performed with Solution B.

1.6.6 Card transport robot arm module (CAT)

The Card transport robot arm module (CAT) is allocated in the intermediate floor of the instrument and is responsible for card transportation between cards drawers, incubators, centrifuges and reader, as well as release cards to waste container and card inventory (mapping and identification).

Card transport is required as a link between functionalities:

- From card load/unload to card identification.
- From card identification to card revision.
- From card revision to card pipetting and incubation (includes card lift).
- From card incubation to card centrifugation (includes card lift).
- From card centrifugation to card reading.
- From card reading to card return.

The CAT module is composed of the following submodules:

- X-axis: it allows the movement along the X-axis (left to right).
- Y-axis: it allows the movement along the Y-axis (back to front).

- Robot head: it contains the following items:
 - Gripper (and related mechanics and electronics).
 - Barcode reader (and related mechanics and electronics).
 - Presence detector (and related mechanics and electronics).

Cards identification includes two main tasks:

- Presence detection: detection of cards in any location.
- BC reading: reading of the barcode of each card. The reader is in a fixed position and the card is placed in front of it by the Card Transport system.

After being processed, cards are normally disposed to a waste container except if the result obtained requires a review from the operator or there are some wells not used. In these cases the cards are left in a special area (service rack) and they can be recovered by the operator.

1.6.7 Fluidics module (FLU)

The fluidics module is composed of the following submodules:

- Manifold: acrylic block (PMMA) where fluidic components are integrated. These components are electrovalves, dispensation pumps, expansion vessels, sensors and bottles configuration selector.
- Aspiration Pump: responsible for removing waste solution from probes washer and dilution cup.
- Electronics: provide interface between fluidic components (electrovalves, pumps, sensors) and the instrument control. Additionally, it provides interface with lockers and with bottle weighting system.
- Tubing: provide connection between fluidic components.
- Module support: integrates module components and gives integration with the rest of the instrument.
- Interconnection manifolds: small manifolds used to interconnect tubing in order to allow mechanical module union. There are three units, one to join tubing that comes from fluidics module to the tubing of X-axis module, one to join tubing that comes from X-axis module to the tubing of Y-axis module and one to join tubing that comes from Y-axis module to the tubing of Z-axis module.
- Bottles and cards disposal container drawers: there is one drawer per each bottle and card disposal container, located in the lower part of the instrument.
- Dilutors: responsible for aspirating and dispensing samples, reagents, diluents and dilutions at the required accuracy.
- Correct Pipetting Detectors (CPD): sensors to check possible obstructions in the system.

The dilution cup (see "Dilution cup module (CUP)" on page 672) and probes washer (see "Pipetting robot arm module (PIP)" on page 324) are fluidic components but are considered as independent module or included in other modules.

1.6.8 Dilution cup module (CUP)

The dilution cup has two independent and identical wells, so two independent dilutions can be prepared simultaneously, or a single dilution can be processed in any of the two wells. To homogenize the dilution, a stirring system is provided.

Each well has two fittings for the tubes required to wash and drain. For each well, one fitting (located at the upper part of the well) allows filling the well with Solution A or B, while the other one (located at the bottom part of the well) allows emptying to the waste bottle.

1.6.9 Reader module (LEC)

Reading module for gel cards tests, including image capturing and processing in order to obtain the level of agglutination of every single microtube, as well as applying the necessary safeguards to ensure proper results.

1.7 Software structure

This chapter provides an overview of the software structure of Erytra: general overview, computer SSD structure, a description of the files structure related to the software and a register structure description.

1.7.1 Overview

Erytra is provided with an embedded computer that contains all the software applications installed. Installation processes are required in update operations or during some corrective maintenance procedures.

1.7.2 SSD Structure

The instrument contains one SSD of 256 GB (SSD) divided in seven volumes. The following table summarizes the main characteristics of each drive.

Table 4 - SSD structure

Drive	Name	Size	Contents
C:\	SYSTEM	69.4 GB	Windows 10
D:\	APPS	1 GB	Internal applications data
E:\	DATA	159.9 GB	Software of the instrument
R:\	DGRECOVERY	15 GB	OS Image Windows 10 v2.2.1 (only allowed in maintenance mode)

Drive	Name	Size	Contents
--	RECOVERY	499 MB	Internal Recovery feature
--	EFI PARTITION	100 MB	Boot loader feature
V:\	DGFILESMGMT	7.1 GB	Default unit to save reports (Virtual unit)

Operating system (O.S.) of the instrument is Windows 10 Enterprise 2019 LTSC version 2.1.1. To check Windows 10 version is installed in the instrument, press the grid () button during system startup, log in as `adminUser` (case sensitive, admin password) and the Boot Application options will be displayed. Once in there, press the arrow in the right side of the screen to switch to page 2/2 of the menu, then select the **Information** icon among the Boot Application options.



Figure 8 - Information option

After switching on the instrument, the OS is loaded and login is automatically performed. Default username and password for the Erytra computer are (small letters):

- Username: `application`
- Password: `root`

Administrator username and password:

- Username: `admin`
- Password: `dgroot`

When the instrument is switched on, the Erytra application is automatically launched.

1.7.2.1 Files structure

Main application files (program files, data and logs) are found on the **E:** drive.

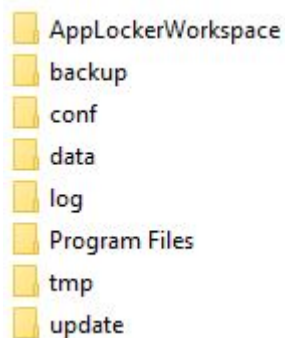


Figure 9 - E drive directories

- Folder **E:\AppLockerWorkspace**: contains AppLocker information files.
- Folder **E:\backup**: default folder where the backups are stored.

**NOTE**

The backup destination folder can be changed by indicating the desired path in the **Backup.conf** file in **E:\Program Files\System Tools**.

- Folder **E:\Conf**: contains the **erytra.properties** file and logo picture (**logo.jpg**, up to 200x45 pixels), if any. It also contains the **erytra.properties.backup** file, used by the system when the **erytra.properties** file doesn't exist or can't be read.
- Folder **E:\data** contains:

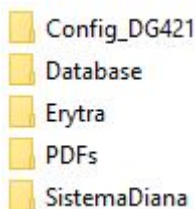


Figure 10 - E:\data subdirectories

- **E:\data\Config_DG421**: contains Reader module configuration files.
- **E:\data\Database**: contains Erytra databases files and data (samples,...).
- **E:\data\Erytra**: contains Erytra data (configuration files, pictures, logs,...) and **staterepository.xml** file.

- **E:\data\PDFs**: contains the PDF report files requested through the user application.

The name of the files is **PAC00XY.pdf**, where X is the report type (1 for Traceability report, 2 for Reading report and 3 for Results List report) and Y is a 13-digit timestamp in milliseconds. PDF report files older than three hours will be deleted when a new report is requested.

- **E:\data\SistemaDiana**: contains DianaSystem files (Working directory, DianaServer).

- **Folder E:\Log** contains:

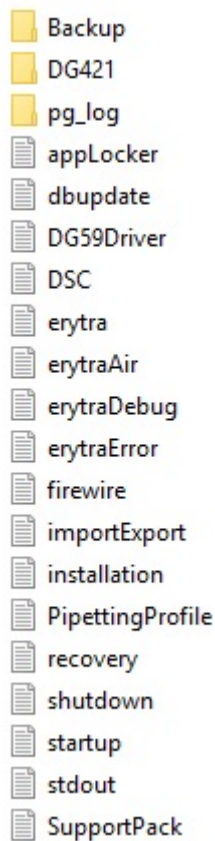


Figure 11 - E:\Log subdirectories

- **E:\Log:** contains log files of the instrument. See "Description of log files" on page 75
- **E:\Log\Backup:** contains backup of log files.
- **E:\Log\DG421:** contains reader log files.
- **E:\Log\pg_log** contains Database engine (PostgreSQL) log files.
- **Folder E:\Program Files** contains:

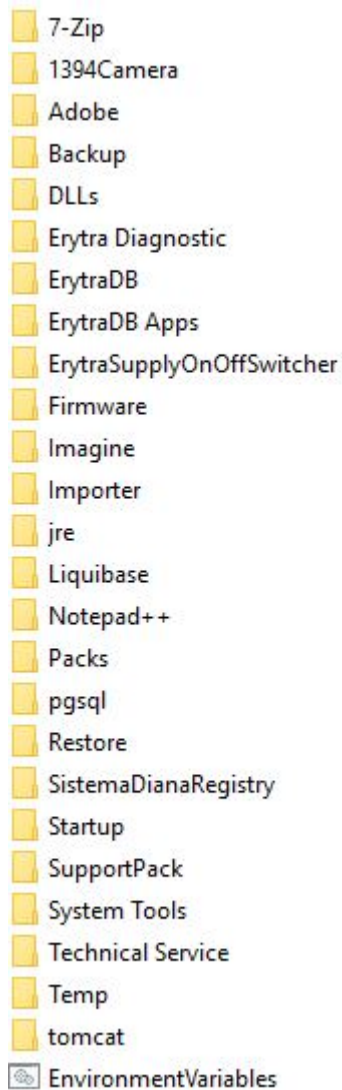


Figure 12 - E:\Program Files subdirectories

- **E:\Program Files\7-Zip:** contains the 7-Zip application to compress and decompress files with a high compression ratio.
- **E:\Program Files\1394Camera:** contains the camera drivers.
- **E:\Program Files\Adobe:** contains the Adobe application to view pdf files.
- **E:\Program Files\Backup:** includes the backup tool.
- **E:\Program Files\DLLs:** contains Erytra application libraries.
- **E:\Program Files\Erytra Diagnostic:** contains Erytra Diagnostic application.
- **E:\Program Files\ErytraDB:** contains Erytra (factory) databases backup files.
- **E:\Program Files\ErytraDB Apps:** includes the **Restore_DataBase** tool.

- **E:\Program Files\ErytraSupplyOnOffSwitcher**: includes the ErytraSupplyOnOffSwitcher application to switch the electronics ON and OFF.
- **E:\Program Files\Firmwares DG-59**: contains all Erytra modules firmwares and related kernels.
- **E:\Program Files\Imagine**: includes the Imagine application for viewing RAW images.
- **E:\Program Files\Importer**: contains Erytra database importer (new tests,...).
- **E:\Program Files\jre**: contains java runtime environment files.
- **E:\Program Files\Liquibase**: includes Liquibase application.
- **E:\Program Files\Notepad++**: includes the Notepad++ application for displaying log files.
- **E:\Program Files\Packs**: includes languagepack, reagentlanguagepack and reagentpack files.
- **E:\Program Files\pgsql**: contains PostgreSQL files.
- **E:\Program Files\Restore**: contains batch files to restore the required data.
- **E:\Program Files\SistemaDianaRegistry**: includes the necessary files to set Windows registry key configuration for Sistema Diana.
- **E:\Program Files\Startup**: contains Erytra application start up.
- **E:\Program Files\SupportPack**: contains batch files to generate the support pack.
- **E:\Program Files\System Tools**: contains batch files to perform automatic backups.
- **E:\Program Files\Technical Service**: contains Technical Service application files.
- **E:\Program Files\Temp**: temporary folder.
- **E:\Program Files\tomcat**: contains Web server (Tomcat 8.5), related application files and Erytra web applications.
- **EnvironmentVariables.bat**: add environment variables to the OS.
- Folder **E:\update**: any software installer copied to this folder will be automatically executed on reboot.
- Folder **E:\data\Erytra** contains:

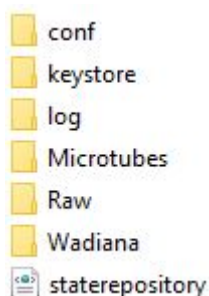


Figure 13 - E:\data\Erytra subdirectories

- **staterepository.xml** file: the file where the disabled modules are stored.
- **E:\data\Erytra\conf**: contains instrument configuration files: geometry, adjustments (**PIPCalibrations.txt** and **CATCalibrations.txt** files), etc. and **testcounter.txt** file registering the number of executed (completed or cancelled) tests in the current database.
- **E:\data\Erytra\log**: not in use (software versions ≥ 2.00).
- **E:\data\Erytra\Microtubes**: contains (individual) microtubes jpg and bmp pictures.
- Microtubes jpg files are organized into different folders as follows:
 - First folder level: card type, card barcode digits 1 to 4.
 - Second folder level: card lot, card barcode digits 9 to 11.
 - Third folder level: card lot, card barcode digits 12 to 13.
 - Fourth folder level: last two digits of card serial number, card barcode digits 17 to 18.
- **E:\data\Erytra\Raw**: contains reader raw images acquired last 14 days (two wells per picture, pixels information as they were acquired by the sensor -8 bits).
- **E:\data\Erytra\Wadiana**: contains Diana System Connectivity definition and messages files.
- Folder **E:\data\SistemaDiana** contains:

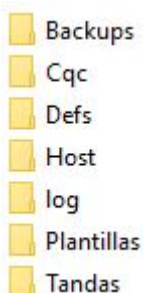


Figure 14 - E:\data\SistemaDiana subdirectories

- **E:\data\SistemaDiana\CQC**: contains Diana System quality control files.
- **E:\data\SistemaDiana\defs**: contains Diana System definition files (cards, wells,...).
- **E:\data\SistemaDiana\Host**: contains Diana System LIS database.
- **E:\data\SistemaDiana\log**: contains Diana System programs log and logs backups.
- **E:\data\SistemaDiana\Plantillas**: contains Diana System templates.
- **E:\data\SistemaDiana\Tandas**: contains Diana System batches files (*.tan and *.bmp).

1.7.2.2 Description of log files

Log files of the instrument are stored in the **E:\Log** folder. Different log files are the following ones:

- **E:\Log:** contains log files of the instrument:
 - **appLocker.log:** registers events related to the AppLocker application.
 - **dbupdate.log:** registers events related to database updates (software installation, patches).
 - **DG59Driver.log:** registers Driver events and it is set to **INFO** level when registering events from the user application, and set to **DEBUG** level when registering events from the Technical Service software. Once it reaches 100 MB, it is compressed, moved to the **E:\Log\Backup** folder and renamed to **DG59Driver_yyyymmdd_hhmmss.zip**, using the date and time of the last entry.
 - **DSC.log:** Diana System Connectivity log file and it is set to **DEBUG** level. Once it reaches 1 MB, it is moved to the **E:\Log\Backup** folder and renamed to **DSC_YYYYMMDD_hhmmss.log**, using the date and time of the last entry.
 - **erytra.log:** registers user interface events and is set to **INFO** level. Once it reaches 100 MB, a compressed backup file is created in the **E:\Log** folder and renamed to **erytra_yyyymmdd_hhmmss.zip**, using the date and time of the last entry.
 - **erytraAir.log:** Registers application events at the AIR runtime level. Once it reaches 100 MB, it is compressed, moved to **E:\Log\Backup** folder and renamed to **erytraAir_yyyymmdd_hhmmss.zip**, using the date and time of the last entry.
 - **erytraDebug.log:** registers user interface events and it is set to **DEBUG** level. Once it reaches 100 MB, a backup file is created. Only the last 20 backup files are kept. This log does not contain patient demographic data.
 - **erytraError.log:** registers user interface events and it is set to **ERROR** level. Once it reaches 100 MB, a compressed backup file is created in the **E:\Log** folder and renamed to **erytraError_yyyymmdd_hhmmss.zip**, using the date and time of the last entry.
 - **firewire.log:** registers events related to reader module initialization.
 - **PipettingProfile.log:** registers the parameters of all the aspirations performed, including the CPD values during aspiration.
 - **stdout.log** file: registers current day web application (Tomcat) console events. When the day is over it is compressed, moved to the **E:\Log\YYYY-MM** folder and renamed to **stdout.log.YYYY-MM-DD-hh-mm-ss.zip**, using the date and time of the last entry.
 - **SupportPack.log:** registers events related to Support Pack application. As for example when the process starts and ends, saved files, ...
 - **System.log:** information about Boot Application and Operating System. Registers events of access to Windows File Explorer and also registers entries to the system when it is in unprotected mode.
- **E:\Log\DG421:** contains reader active log file (**DioneExpert.log**). Once it reaches 1,7 MB it is compressed and moved to **E:\Log\DG421\Backup** and renamed to **DioneExpert_YYYYMMDD_hhmmss.zip** using the date and time of the last entry.
- **E:\Log\Backup:** contains backup of log files.

- **E:\Log\pg_log**: contains the database engine (PostgreSQL) log. The log file name is **postgresql-YYYY-MM-DD_HHmm.log**, where YYYY=year, MM=Month, DD=Day, HH=hour and mm=minutes when PostgreSQL started.

The log files of previous days are compressed in a zip file at system start up in files named **postgresql_YYY-MM-DD.zip**, where YYY=year, MM=Month and DD=Day.

The logs older than 14 days are deleted.

- **E:\Program Files\tomcat\logs**: contains web server application logs:
 - **catalina.YYYY-MM-DD.log** (where YYYY=year, MM=month and DD=day): Web Server servlet container log.
 - **localhost.YYYY1-MM-DD.log** (where YYYY=year, MM=month and DD=day).

If running Diana System / LIS connection in Erytra PC (default configuration):

- **E:\data\SistemaDiana\log**:
 - **DianaASTM.log**: it registers DianaServer (LIS communication) events.
 - **HostDemo.log**: it registers HostDemo (LIS communication) events.
- **E:\Program Files\Backup\logs**:
 - **ihExport.log**: it registers the backup process. Once this file reaches 2MB it compressed (gnu zip) and renamed to **ihExport-backup-YYYYMM-DD.log**, where YYYY is the year, MM the month and DD the day.
 - **ihImport.log**: it registers the restore process. Once this file reaches 2MB it compressed (gnu zip) and renamed to **ihImport-backup-YYYYMM-DD.log**, where YYYY is the year, MM the month and DD the day.

For the logs related to Windows events or applications, See "Collecting Windows event logs" on page 1114.

1.7.3 Windows registry structure

In case it is necessary to modify the Diana System working directory and/or the language of Diana System, run the **E:\data\SistemaDiana\Config.exe** file as administrator (before running it, the rest of applications must be closed). In the main tab it is possible to define both parameters, and the selection is stored in the Windows registry:

- Working directory: **HKEY_LOCAL_MACHINE > SOFTWARE > WOW6432Node > Grifols > SistemaDiana > General > DirBase** parameter.
- Diana System language: **HKEY_LOCAL_MACHINE > SOFTWARE > WOW6432Node > Grifols > SistemaDiana > General > NumIdioma** parameter.

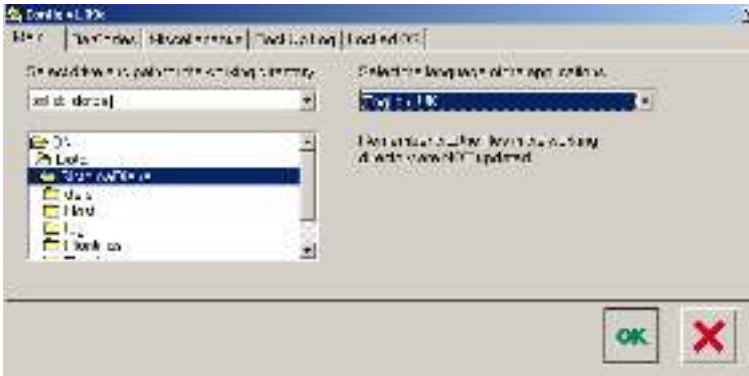


Figure 15 - Config Diana System software

1.8 Setting up and starting the instrument

This section contains information about unpacking the instrument and setting up the needed software.

1.8.1 Location remarks

Before placing the instrument in the installation site, the following requirements should be taken into account:

- The installation process must be carried out by a qualified technician.
- Space requirements (with the transportation pallet):
 - Size: 1210 mm wide x 2010 mm height x 800 mm deep.
 - Weight: 500 kg (approximately).
- Place the instrument so that it is over a non inclined surface.
- Do not place the instrument out of doors.
- Do not place the instrument over a flammable material surface.
- Do not place the instrument so that it is difficult to access for maintenance or servicing.
- Do not block air openings nor place the instrument on a soft surface which might block them.
- Do not place the instrument where it could get wet.
- Do not position the instrument so that it is difficult to operate the mains switch.
- Do not let the instrument or its flexible cord come into contact with surfaces which are too hot.
- Do not place anything on the top of the instrument.
- Do not locate the instrument in an area of direct sunlight.
- Locate the instrument near a fluid drain access.
- Do not place the instrument over any electric or electronic equipment since failures in the

fluidic circuit may damage them.

- Appliance coupler or plug shall be readily and easy accessible.

**CAUTION**

Disregard may lead to malfunction of and cause damage to the Erytra.

1.8.2 Unpacking

The instrument is dispatched in a wooden box on a transportation pallet. Before unpacking the instrument, the pallet should be placed on the floor, as close as possible to the installation site.

In order to unpack the instrument, a minimum of two people is necessary, one of which must be a qualified technician. The packaging should be removed by following the instructions below.

Tools:

- Open end spanner number 12 (for arms shipping clamps) and number 19 (for feet).
- Electric screwdriver with Phillips screw head.

Procedure

1. Place the packaging as close as possible to the installation site.
2. Remove the upper part of the wooden packaging.

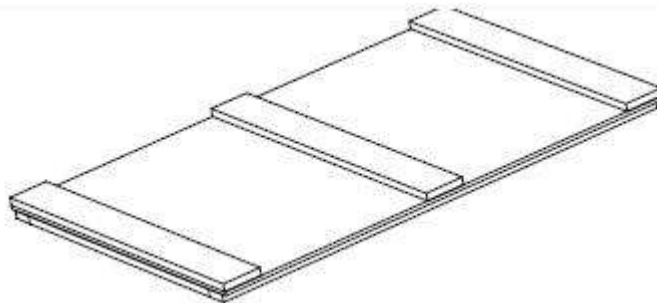


Figure 16 - Upper part of the wooden packaging

3. Remove the planar lateral (narrow lateral without wooden strips) of the packaging and place it attached to the pallet in order to act as a ramp. Place the upper part of the wooden packaging under the ramp.

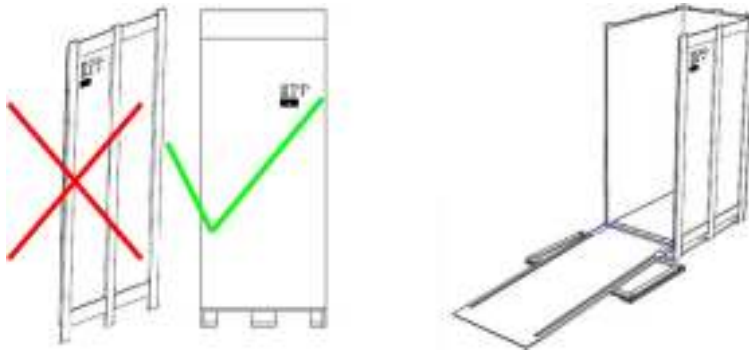


Figure 17 - Packing assembly of the Erytra

4. Fix the ramp with the two hook-and-loop fasteners.



Figure 18 - Hook-and-loop fasteners

5. Dismantle the rest of the laterals of the wooden box.
6. Remove the wooden small ramp included inside the packaging (above the instrument) and place it at the end of the main ramp. Stick it to the main ramp with adhesive tape.



Figure 19 - Small ramp attached to the main ramp of the packaging

7. Remove the door handles that are also located above the instrument.
8. Remove the aluminium anti-humidity cover that is wrapped around the instrument.
9. Remove the retractile film and protective foams located at each corner of the Erytra.

10. Remove the adhesive tape of the feet.



Figure 20 - Adhesive tape in a foot of the Erytra

11. Move up the instrument turning in counterclockwise direction the nut of each foot with an open end spanner number 19. Move down the feet in a crossed way.



Figure 21 - Location of the nut in a foot of Erytra



Figure 22 - Instructions to screw / unscrew the feet of the Erytra

12. Once the instrument is lifted over its feet, remove the foam and the wooden board located underneath the Erytra.

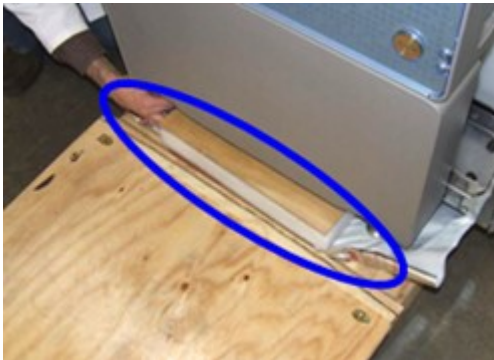


Figure 23 - Foam and wooden protection of the packaging

13. Move down the Erytra by means of the nut of the feet (turning them in clockwise direction, use the open end spanner number 19) until wheels rest on the pallet.
14. Move carefully by the ramp the instrument to the floor, with minimum two people avoiding that the instrument moves by itself for the ramp.



DANGER

Weight of the instrument is quite considerable and it is recommended to strictly follow instructions in this chapter.

15. Place the instrument in the installation site.
16. Remove the frontal foams located from the user tray and the touchscreen monitor.
17. Remove the protective adhesive tape of the frontal USB connectors.
18. Install the door handles. The screws for install them are located in a bag attached to the door handles themselves. To install the door handle of the intermediate door (remove its fixing screw), the support of the celduc must be removed. Once the door handle is installed, place again the support of the celduc attached to the door and the door fixing screw as well.



Figure 24 - Door handles installed on the doors

19. Open all the fluidic drawers (push the lever located in the front of the drawer, at the bottom area) and remove the foams located on both sides of the card disposal container and in the handles of the bottles. Remove the contents of the card disposal container: antihumidity bags, power cords, decontamination tool, disposable bags for card waste and documentation. Remove the protective transparent adhesive of identification labels in fluidic drawers surface and solution A/B and waste bottles (lids and bottles). Close the drawers.
20. Open the upper door and remove the box plug between the RM comb and SM comb and the X-axis support arm.



Figure 25 - Location of the box plug

21. Remove the SM comb between the RM comb and the supplement comb lateral.
22. Remove the external barcode reader box and then the RM comb.



Figure 26 - Location of the RM comb

23. Remove the supplement comb lateral.



Figure 27 - Location of the supplement comb lateral

24. Remove the shipping clamp of the PIP arm (use an open end spanner number 12) which prevents the arm to move.



Figure 28 - Location of the shipping clamp of the PIP arm

25. Remove the adhesive tape of the dilution cup wells and remove its fixing foams.
26. Remove the decontamination bottle, two samples racks, two diluents racks and two reagents racks from the internal packaging tray. Remove the protective transparent adhesive from the decontamination bottle identification labels (lid and bottle).
27. Remove the internal packaging tray and the plug of the tray.



Figure 29 - Location of the internal packaging tray (left) and plug of the tray (right)

28. Open user tray and remove the screw of the intermediate door. Open the intermediate door and remove the shipping clamp of the CAT arm (use an open end spanner number 12) which prevents the arm to move. Check also inside of the instrument for any foreign objects. Close the intermediate door and put its fixing screw.



Figure 30 - Location of the shipping clamp of the CAT arm

29. Remove the protective transparent adhesive from the surface of the reagents-samples keyboard and card keyboard.
30. Move down the frontal feet of the instrument until they rest on the ground.

1.8.3 Instrument levelling

Instrument levelling is recommended to be performed once the instrument has been set in its corresponding position in the laboratory: during its initial installation or when an instrument in use has been moved from its physical location due to a technical intervention.

Instrument levelling can be done by means of the front feet of the instrument:

1. Switch on the instrument and access the Boot application menu.
2. Make sure the rear feet of the instrument are resting on the ground.
3. Open the Technical Service software and perform a home of the PIP and CAT arms. See "Pip Arm tab" on page 804.
4. Go to the **Procedures** tab (see "Procedures tab" on page 756) and perform a presence detector verification check routine (see "Verifying presence detector" on page 370) using P/N 234213 PROBES ADJUSTMENT TOOL (Ver. 4). See "Spare parts list" on page 1207
5. Activate the PIP presence detector (**PIP Arm** tab > **Presence Detector** subtab, then select **Set on**).
6. Check that the laser is over the hole of the tool.



Figure 31 - Laser PIP alignment

7. Perform a gripper parallelism check routine (see "Verifying card presence detector-gripper parallelism" on page 499) using the P/N 233878 CARDS DRAWER ADJUSTMENT TOOL (Ver. 2). See "Spare parts list" on page 1207.
8. Activate the CAT presence detector. Go to **Gripper Arm** tab > **Presence Detector** subtab, then select **Set on**. See "Gripper Arm tab" on page 777.

9. Check that the laser is over the central slot of the tool.



Figure 32 - Laser CAT alignment

Should the check for the laser position for the PIP or CAT modules fail, then it is needed to level the instrument:

10. With both lasers on over the corresponding adjustment tools, access the upper nut of the left or right front instrument feet.
11. Loosen the upper nut by turning it upwards (counter clockwise).



Figure 33 - Upper nut of the foot

12. Access the lower nut of the foot, turning it clockwise or counter clockwise (depending on the correction needed for each instrument) until the laser beams fit in the corresponding adjustment tools for each module.
 - Turning clockwise left front foot: the lasers move to the left.
 - Turning clockwise right front foot: the lasers move to the right.



Figure 34 - Lower nut of the foot

13. Switch off the PIP and CAT presence detector (the **Set off** button).
14. Remove the adjustment tools from the instrument.
15. Tighten back the upper nut.

1.8.4 Installation Qualification (IQ)

During the Installation Qualification (IQ) phase, all of the activities involved in the installation of the instrument at the users' site are documented. The IQ procedure should be executed during the installation of the instrument at customer site.

Installation qualification



NOTE

To view the Installation Qualification (IQ) procedure refer to the Erytra. If an empty IQ check-list is needed, it should be prepared in advance by printing it as a PDF from the HTML check-list.

Once the IQ procedure has successfully finished, the Operational Qualification (OQ) procedure should be performed. See "Operational Qualification (OQ)" on page 104.

1.8.5 External drain

The specifications for the tube to be used for external drain connection are the following:

- Recommended PVC 6x4 mm.
- Tube connection with no loops.

- Maximum tube length: 5 m.
- Maximum drain height: 1 m above the external drain connector.
- Minimum drain height: 0.55 m below the external drain connector.

1.8.6 Installing software for the first time

Operating System is installed on Erytra instruments from factory by default. Once the instrument has been unpacked and placed in its final position in the lab, the software installation should be done.

Follow the installation procedure in "Installing Erytra software v4.6.1" on page 1039 to perform the first installation.

1.8.7 Working with Boot Application

When switching on the instrument, the boot application is launched, showing the Erytra logo and bottom corner buttons on screen.



Figure 35 - Boot Application initial window

From there, the protected/administrative options can be accessed by pressing the bottom-left grid () button and entering the corresponding credentials. Additionally, the Diagnostic menu can be accessed by pressing the bottom-right diagnostic () button. The Diagnostic menu includes options which are mainly focused on the end user.

**NOTE**

There is a limited time of five seconds to select the desired menu before the application is launched. At this point, the grid and diagnostic buttons will not be available and the user application login screen will be shown.

**NOTE**

A file named **BootApp_Extended_time.vsb** (E:\Program Files\System Tools) can be executed to increase the timeout to enter Boot application to 1 hour only on next reboot.



By pressing the grid () button, the **Boot Application** login window will be displayed.



Figure 36 - Boot Application login window

Login should be done with the following credentials (case sensitive):

- User: adminUser
- Password: admin

**NOTE**

Only Grifols authorized service personnel (FSE or TAS users) is intended to enter the protected/administrative options within the Boot Application. Thus, the above login credentials cannot be changed and should not be provided to users other than Grifols authorized service personnel.

After logging in, the Boot Application options will be displayed in two screens.



Figure 37 - Boot Application protected/administrative options

The available options are listed below:

- **Start Erytra:** starts running the Erytra Software program.
- **Importer:** opens the Importer application. See "Importer application" on page 883.
- **Technical Service:** opens the Technical Service application. See "Technical Service software overview" on page 754.
- **OS Shell:** opens the OS Shell mode.
- **Maintenance Mode:** opens Maintenance mode.
- **Update Software:** updates the SW version in the instrument. The system will automatically check if there is a removable device connected with a folder named **installers** inside. If so, the system will execute the software installer within that folder. See "Installing Erytra software v4.6.1" on page 1039.
- **Recovery system:** this button allows to perform Operating System Full Recovery or OS Recovery from instrument SSD and Boot Application; without the need of connecting a USB pen drive. This new feature allows to update or recover the OS remotely. See "Installing or restoring Operating System" on page 1006.
- **Diagnostic:** opens the Diagnostic application. See "Diagnostic application" on page 851.
- **Backup:** starts the Backup application. See "Backing up databases and relevant data with boot menu" on page 1117.
- **Restore:** starts the Restore application. See "Restoring databases and relevant data with boot menu" on page 1124.

- **Information:** shows the information window of the current OS and SW version.



Figure 38 - Information window

- **Backwards:** arrow located on the top left corner of the screen. When pressed, it goes back to the **Boot Application** initial window.
- **Shutdown:** when pressing the button located in the top right corner of the screen, a confirmation message appears informing that the system will be shut down. Press **Yes** to confirm the action.

1.8.8 Software configurations

Before performing OQ procedure, it is recommended to define some software parameters such as language, Erytra serial number, sample drawers configuration, LIS connection configuration, etc. These configurations are defined in the **erytra.properties** file located in **E:\conf**.



NOTE

When modifying the default configuration, check that spelling of the written parameters is correct. Otherwise the default value for the wrongly spelt parameter will be used without being changed in **erytra.properties** file.



NOTE

If a properties parameter is deleted while modifying the default configuration, the default value for that parameter will be used and written in the file.



NOTE

Before modifying any parameter, check that software is not running and the MSDOS command windows are closed.

1.8.8.1 ARCHIVE_ENABLED

Parameter to activate the archive option. This option moves the contents of **E:\data\Erytra\Microtubes** and **E:\Log** folders to the folder defined in the **ARCHIVE_PATH** parameter.

- Options: `true/false`.
- Default option: `false`.

1.8.8.2 ARCHIVE_INTERVAL

Parameter to define the number of days that files to be archived remain in the corresponding default folder of the instrument (recommended value ≤ 14). After the number of days defined (checked by the software every hour), the files are moved to the folder defined in the ARCHIVE_PATH parameter. If the parameter is defined as 0, the files are continuously moved to the corresponding folder. Negative values are not allowed.

- Options: any positive integer number or NONE.
- Default option: NONE.

1.8.8.3 ARCHIVE_PATH

Parameter to define the path where the above-mentioned files will be moved.

- Options: Any valid path. Any '\' symbol must be written as '\\' (for instance, E:\\ or C:\\Program Files\\XXXX). There is no limitation regarding the number of introduced characters. Note that '\' symbol would be added automatically in the defined path by the software before the ':' symbol when the software is launched.
- Default option: NONE.



NOTE

Recommended path destination format is NTFS.



NOTE

Ensure that the path does not end with a \ symbol (e.g., use "E:\\ARCHIVO" instead of "E:\\ARCHIVO\\").

e.g: ARCHIVE_PATH=E:\\Erytra_XXXX

1.8.8.4 AUTO_LIS_QUERY_ENABLED

It allows enabling and disabling the periodic automatic query to LIS.

Values allowed:

- true: Auto LIS query feature is enabled.
- false: Auto LIS query feature is disabled. Default value.

1.8.8.5 AUTO_LIS_QUERY_MODE

Parameter to set the query mode to Query All or to Query by Sample (when AUTO_LIS_QUERY_ENABLED=true).

Values allowed:

- QALL: Query All will be sent to LIS (when AUTO_LIS_QUERY_ENABLED=true).
- QBSAMPLE: Q: Query by Sample will be sent to LIS (when AUTO_LIS_QUERY_ENABLED=true). Default value.

1.8.8.6 AUTO_LIS_PERIOD_UPDATE

Parameter to define the amount of time in minutes that will elapse between each periodic automatic query (when `AUTO_LIS_QUERY_ENABLED=true`).

Values allowed: integers from 1 to 60.

Default value: 5.

Default value will be applied if a value out of the allowed range is found.

1.8.8.7 AUTOMATIC_RESULTS_VERIFICATION

Parameter to define if SampleProfiles without incidences or special results, and with QC status exported or QC disabled, are automatically validated and exported.

- Options: `true/false`.
- Default option: `false`.

1.8.8.8 Barcode reader configuration

PIP barcode readers should be configured/checked to match the sample tube barcode characteristics used in each site (barcode type, checksum, etc.). While most of the barcode types currently used to label sample tubes are enabled by default, the default configuration can be modified. See "Adjusting barcode readers configuration" on page 358.

1.8.8.9 DEFAULT_BC_IS_ISBT

Available options are:

- `true`: any manually introduced barcode is considered as ISBT type (16 characters expected).
- `false`: any manually introduced barcode is considered as non-ISBT type (default value).

When manually identifying a sample tube barcode, the manual identification window prompted by the software includes a checkbox allowing to indicate whether the barcode is ISBT type or not (ticked by default when the `DEFAULT_BC_IS_ISBT` parameter is enabled and vice versa).

1.8.8.10 DELTA_CHECK_INCONSISTENCIES_CONFIGURATION

Parameter to define the inconsistencies on results to be checked. Available options are:

- `None`: no inconsistencies are checked.
- `Sample`: inconsistencies are checked on Sample tube BC basis.
- `Patient`: inconsistencies are checked on Patient ID basis.

1.8.8.11 DELTA_CHECK_HISTORICAL_FROM

Parameter to define the time interval where Inconsistencies are checked within exported SampleProfiles.

Available options are:

- `ALL`: inconsistencies are checked within all data stored in database.
- `yyyy/mm/dd`: inconsistencies are checked within all SampleProfiles processed from defined date onwards.
- `any number`: inconsistencies are checked within all SampleProfiles processed within the defined number of days prior the current date. Default option is 0.

1.8.8.12 DIANA_SYSTEM_BATCH_CREATION

Parameter to define if the system creates Diana System batches (.tan files and related bmp files) or not.

- `TRUE`: the tan (batch) file creation is activated.
- `FALSE`: the tan (batch) file creation is not activated. This is the default option.

1.8.8.13 DIRECT_PRINT



NOTE

Direct print functionality requires the following specifications to be met by the printer device being used:

- At least 128 MB memory.
- Compatibility with PCL5, PCL6 or PostScript languages.

Available options are:

- `true`: any report requested by the operator will be sent directly to the printer without being displayed on screen.
- `false`: any report requested by the operator will be displayed on screen in PDF format (default value).

1.8.8.14 DISABLE_VALIDATION_BUTTON

Parameter to define if the Validation button in the Validation list is disabled.

- Options: `true/false`.
- Default option: `false`.

1.8.8.15 DISABLE_EXPORT_BUTTON

Parameter to define if the Export button in the Validation list is disabled.

- Options: `true/false`.
- Default option: `false`.

1.8.8.16 DISABLE_CHANGE_RESULTS_CONFIRMATION

Parameter to define if the confirmation message when the user modifies results is disabled.

- Options: `true/false`.
- Default option: `false`.

1.8.8.17 DOUBLE_TUBE_ENABLED

Parameter to define if double sample tubes (one with serum and one with RBC's) are used.

- Options: `true/false`.
- Default option: `false`.

1.8.8.18 DUAL_MODE

Parameter to configure the usage of external drain. When activated, waste 1 position is used as solution A and waste 2 position is used as solution B. An external draining system must be connected to the instrument when activating this parameter, and also a mechanical configuration must be applied to the manifold. See "Parts" on page 555.

- Options: `true/false`.
- Default option: `false`.

1.8.8.19 ENABLE_AUTOVERIFICATION_QCPASS

Parameter to define if a QC Pass protocol is automatically validated and exported.

- Options: `NONE, ONLY_VALIDATE, VALIDATE AND EXPORT`.
- Default option: `ONLY_VALIDATE`.

1.8.8.20 FLUIDIC_SYSTEM_ALWAYS_READY

This parameter is used to select the fluidic procedure to be performed after one hour of fluidic inactivity. At that moment, when the parameter is set to `true`, the system will perform a quick internal and external wash of the probe and dilution cup, repeating these actions every hour for 24 consecutive hours as long as the fluidic inactivity condition is met. Otherwise (when set to `false`) a full rinsing procedure will be performed and no further fluidic procedures will take place afterwards until required by the operator. This rinsing procedure will also be executed when the parameter is set to `true` and the system has performed 24 consecutive inactivity washes.

Available options are:

- `true`: a quick washing procedure will be performed after each hour of fluidic inactivity for 24 consecutive hours (default value).
- `false`: a rinsing procedure will be performed after one hour of fluidic inactivity.

When this parameter is set to `true` and the last performed fluidic action has been an inactivity wash, the system will empty and then prime the probe circuit before executing an action required by the operator which involves any fluidic procedure.

1.8.8.21 IMPACT_DETECTOR

Parameter to define if impact detector is installed in the system.

- Options: `true/false`.
- Default option: `true`.

1.8.8.22 INSTRUMENT_NAME

Parameter to define the desired instrument name (maximum 10 characters). This parameter can also be defined by means of Importer application. See "Importer application" on page 883.

1.8.8.23 INSTRUMENT_SERIAL

This parameter defines the serial number of the instrument (maximum seven characters). By default it is set in a 000xxxx format.



CAUTION

The number of characters of this parameter must match the **last seven characters** of the Product label.

Other settings, including a different number of characters may cause Software errors.

For example if the serial number of the instrument is 059-0001500, this parameter must be 0001500.

This parameter can also be defined by means of the Technical Service software **Run** menu > **Edit serial number** option). See "Modular tabs" on page 754.

1.8.8.24 INTERPRETATION_FORMAT

Defines the standard used to designate interpretation results.

- **TRADITIONAL**: uses the traditional nomenclature. This is the default option.
- **ISBT**: uses the ISBT nomenclature.
- **TRADITIONAL_EXTENDED**: it uses an extended traditional nomenclature.

1.8.8.25 ISBT_ENABLED

Parameter to define if ISBT 128 DIN barcodes are detected and interpreted by the system.

- Options: `true/false`.
- Default option: `false`.

1.8.8.26 ISBT_LIS_DIGITS

Parameter to define how ISBT BCs are shown in Worksheet, IC-Samples, Validation list, Validation detail, Database list and Database detail screens and in Reports, and, if LIS is enabled, received from LIS or sent to LIS. Options are:

- **DIN**: system communication to/from LIS consists of 13-character ISBT codes.
- **FULL**: system communication to/from LIS consists of 16-character ISBT codes (default value, includes all the available characters).

1.8.8.27 LANGUAGE

Parameter to define User Software language.

LANGUAGE=xx_XX where xx_XX is the defined language. Available languages are:

- ca_ES: Catalan language.
- cs_CZ: Czech language.
- cs_DK: Danish language.
- de_DE: German language.
- el_GR: Greek language.
- en_GB: English (UK) language (default in CE configuration).
- en_US: English (US) language (default in US configuration).
- es_ES: Spanish language.
- et_ET: Estonian language.
- fi_FI: Finnish language.
- fr_FR: French language.
- hu_HU: Hungarian language.
- it_IT: Italian language.
- ja_JP: Japanese language.
- lt_LT: Lithuanian language.
- lv_LV: Latvian language.
- nb_NO: Norwegian language.
- pl_PL: Polish language.
- pt_BR: Portuguese (Brazil) language.
- pt_PT: Portuguese (Portugal) language.
- ru_RU: Russian language.
- sk_SK: Slovak language.
- sl_SI: Slovenian language.
- sv_SE: Swedish language.
- zh_SG: Chinese language.

This parameter can also be defined by means of the Importer application.

1.8.8.28 LIS_ACTIVATED

Defines whether the system connects to LIS or not.

- TRUE: the LIS connection is activated if the system is connected to LIS, the samples shall be exported to LIS.
- FALSE: the LIS connection is not activated. This is the default option.

1.8.8.29 LIS_AUTO_QUERY_BY_SAMPLE

Defines if a workload query to LIS is executed after samples drawer recognition. Available options are:

- `true`: once a sample drawer is identified or a sample tube is manually identified, automatically request to LIS all pending work for every identified sample inside the instrument.
- `false`: no automatic request to LIS will be performed after a sample drawer is identified or a sample tube is manually identified (default value).

1.8.8.30 LIS_SHARED_MODE

Defines the behaviour of the instrument when receiving a request for a sample that is not present. The options are:

- `false`: sample requests with sample missing will be hold waiting for sample (default value).
- `true`: sample request with missing sample will be sent back to LIS as cancelled.
Recommended option when there are several instruments connected together to the LIS.

When several instruments share the same LIS connection, `LIS_SHARED_MODE` parameter must be set to `TRUE`. Otherwise, all the workload would be sent to only one Erytra and not received by the others.

In case there is only one Erytra connected to LIS, it is recommended to set `LIS_SHARED_MODE` parameter to `FALSE`.

1.8.8.31 NO_SAMPLE_ID_IF_NO_USER_LOGGED

The values allowed are:

- `true`: sample drawers will not be opened when there is no user logged in.
If a sample drawer that was already opened is closed when there is no user logged in, the samples in that drawer will not be identified.
- `false`: sample drawers can be opened when there is no user logged in. The samples in the sample drawers closed when there is no user logged in will be identified and available in the system. Default value.

1.8.8.32 QC_ENABLED

This parameter defines whether quality controls (QCs) are used on the system.

- Options: `true/false`.
- Default option: `true`.

1.8.8.33 QC_HOURS_CADENCE

Parameter to define the periodicity in hours for the QC.

- Options: any integer value.
- Default option: 8.

1.8.8.34 QC_NEW_LOT_CARDS_AND_REAGENTS

Parameter to define if the system shows a warning when a reagent or card newly detected lot has not passed any QC protocol.

Options:

- **false** (Default option): the system will not check if a QC has been passed, when new cards or reagents lots are identified.
- **warning**: when the 'QC_ENABLED' parameter is set to true and the card or reagent is used in a technique that has QC, the system will check if a QC has been passed and show a warning in case it has not.



Figure 39 - Warning configuration message

- **mandatory**: when the 'QC_ENABLED' parameter is set to true and the card or reagent is used in a technique that has QC, the system will check if a QC has been passed. In case it has not, the system will show a warning and will not use that lot of cards or reagents until it has been used in a QC test with a pass result.



Figure 40 - Mandatory configuration message

Affected cards or reagents will not be considered to during the system planning regardless of technique and the system will stop the queued techniques and activate the corresponding missings.



NOTE

This parameter has replaced QC_NEW_LOT_WARNING_ENABLED from Erytra v4.6.1

1.8.8.35 QC_NEW_LOT_WARNING_ENABLE

Parameter to define if the system shows a warning when a reagent or card newly detected lot has not passed any QC protocol.

- Options: `true/false`.
- Default option: `false`.

**NOTE**

This parameter has been replaced QC_NEW_LOT_CARDS_AND_REAGENTS from Erytra v4.6.1

1.8.8.36 PASSWORD_REAUTHENTICATION

Parameter to allow enabling or disabling the authentication request (user/password) for some of the actions performed by the operator.

- Options: `true/false`.
- Default option: `true`.

1.8.8.37 PASSWORD_EXPIRATION_DAYS

Parameter to define after how many days the user passwords will expire. Default is 90 days.

1.8.8.38 RACK_BC_ENABLED

This parameter is used to configure sample holders barcode requirement. Available options are:

- `true`: all sample holders must have a barcode meeting one of those defined in **erytra.properties** in order to be able to use the samples in that holder (default value, recommended).
- `false`: barcode is not required for any holder. If a holder barcode is found, the corresponding tube settings will be applied. If no holder barcode is found, `DEFAULT_RACK_CONFIGURATION` will be applied. See "SAMPLE_RACKS_CONFIGURATION" on page 101.

1.8.8.39 REAGENTS_ON_BOARD_STABILITY_ENABLED

The available options are:

- `true`: stability on board feature is enabled.
- `false`: stability on board feature is disabled (default value).

1.8.8.40 RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA

Allows defining if sample demographic data sent from LIS is to be stored in the system and shown to the operator.

- `TRUE`: to receive and show the demographic data.
- `FALSE`: to not show the demographic data (default option).

1.8.8.41 RESERVE_INCUBATOR_FOR_STAT

Parameter to configure whether there is one incubator reserved for STAT or not.

- Options: `true/false`.
- Default option: `true`.

1.8.8.42 RESULT_CONFIGURATION

Parameter to define the microtube's results that shall be checked by the operator before validation. That is, the cards will be left at the service rack so the operator can visually check the results. Multiple options shall be selected by entering them separated by commas.

Available options are:

- **SRW:** weak well results (only for CE).
- **SRDL:** doubtful ("?",) results.
- **SRDP:** double population (DP) results.
- **SROD:** operator defined special results. This special results are defined through the importer application.
- **RI:** results with reading incidences.
- **PI:** results with process incidences.
- **CSI:** results with sample or card incidences.
- **SRI:** interpretation special results (NI or DISCREPANT or AUTOCONTROL NOT NEG or DAT NOT NEG results).
- **ALL:** all processed cards.

By default, all the partial cards are loaded on the Service Rack. The default option is:

- **CE:** SRW, SRDP and SRDL.
- **US:** SRDP and SRDL.

1.8.8.43 SAMPLE_RACKS_CONFIGURATION

In order to define the tubes contained in a sample holder, different configurations can be used depending on the barcode labels attached to the sample holders.

It is possible to configure seven different barcode labels. There are four configurable parameters for each barcode label:

BCv=w, x, y, z where,

- **v:** number from 1 to 7.
- **w:** identifies the sample tube type, related to level detection sensitivity. Available options are:
 - NORMAL
 - PLUNGER
 - PEDRIATIC
- **x:** sample holder barcode.

- **y**: sample tube external diameter (limited from 9 to 16 mm). It may be used to ignore the tube diameter readings given by the presence detector and use a configured value instead.
Available options are:
 - If set to 0, the system considers tube diameters given by the presence detector.
 - Otherwise, all of the tube diameters for the holder will be set to the configured value (in mm).
- **z**: sample tube bottom height (limited from -2 to 100 mm). It indicates the height offset in mm (number without decimals) added to the position set by the probe adjustment.

**NOTE**

A negative height offset means that the probe will be moved down further than the position set by the adjustment. A positive height offset means that the probe movement will be stopped before reaching the position set by the adjustment.

Also there is another parameter to define the default holder configuration (grey label):

DEFAULT_RACK_CONFIGURATION= w, x, y, z where,

- **w**: identifies the sample tube type, related to level detection sensitivity. Available options are:
 - NORMAL
 - PLUNGER
 - PEDRIATIC
- **x**: sample holder barcode. Default value is GY.
- **y**: sample tube external diameter (limited from 9 to 16 mm). It may be used to ignore the tube diameter readings given by the presence detector and use a configured value instead.
Available options are:
 - If set to 0, the system considers tube diameters given by the presence detector.
 - Otherwise, all of the tube diameters for the holder will be set to the configured value (in mm).
- **z**: sample tube bottom height (limited from -2 to 100 mm). It indicates the height offset in mm (number without decimals) added to the position set by the probe adjustment.

**NOTE**

A negative height offset means that the probe will be moved down further than the position set by the adjustment. A positive height offset means that the probe movement will be stopped before reaching the position set by the adjustment.

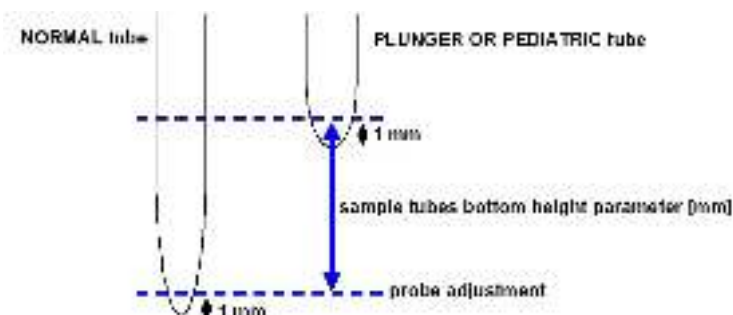


Figure 41 - Samples drawers configuration

DEFAULT_RACK_CONFIGURATION will be applied to those holders with the barcode configured in this parameter (by default GY, matching grey label provided with the instrument). If rack barcodes configuration has been set to not required. See "RACK_BC_ENABLED" on page 100. DEFAULT_RACK_CONFIGURATION settings will be applied also to those holders without barcode or with a barcode not successfully decoded.

1.8.8.44 SEND_PENDING_TO_VALIDATE

Parameter to inform LIS that the execution of the requested test has been completed. The status P (Preliminary results) in the Results Status field will be transmitted to LIS when this feature is enabled.

This parameter will be automatically added and set to disabled in **erytra.properties** when any application starts (user application, Technical Service,...), if it is not found (no manual action is required to add it).

- Options: `true/false`.
- Default option: `false`.

1.8.8.45 SERVICE_RACK_MODE

Defines the number of racks used as service rack. It is possible to work with the following configurations:

- **NORMAL**: there are two service racks in position rack 4 of the card drawer 3 and rack 4 of the card drawer 4. This is the default option.
- **HALF**: there are eight service racks placed in position racks 1,2,3 and 4 of card drawers 3 and 4.
- **NONE**: there is no service rack is defined, all cards will be discarded directly in the waste cards container.

1.8.8.46 SHOW_CONFIRMATION_WHEN_VALIDATE_MODIFIED_SAMPLEPROFILES

Parameter to define if the system shows a confirmation window (not requiring the username and password of the user) whenever the user validates any modified result.

- Options: `true/false`.
- Default option: `true`.

1.8.8.47 WELL_NAME_FORMAT

Defines the standard used to designate well names.

- `TRADITIONAL`: uses the traditional nomenclature. This is the default option.
- `ISBT`: uses the ISBT nomenclature.

1.8.9 Operational Qualification (OQ)

Operational Qualification (OQ) testing is performed to verify that the instrument or instrument modules operate as intended. OQ procedure should be performed in the following cases:

- After IQ procedure. See "Installation Qualification (IQ)" on page 87.
- After PM procedure. See "Items and calendar" on page 124.
- After a corrective maintenance, executing the part related to the module involved in the corrective action and the functional checking.

Operational qualification



NOTE

To view the Operation Qualification (OQ) procedure refer to the Erytra. If an empty OQ check-list is needed, it should be prepared in advance by printing it as a PDF from the HTML check-list.

First of all, ensure that the mains switch, which is located at the back side of the instrument, is turned off. Then plug the mains cable, supplied with the instrument, to Erytra first and to the mains later.



DANGER

Electromagnetic compatibility and electrical safety tests have been performed using the cord supplied by the manufacturer. Using cables different to the one supplied may degrade performance of the instrument concerning electromagnetic compatibility and safety.



NOTE

If the instrument has been stored under different environmental conditions than operation conditions, wait for at least one hour under operation conditions before turning the instrument on.

Before turning on the instrument, check that at least one empty waste bottle is connected and both wash solution bottles are filled with the expected solutions and connected. Check also that card waste container is properly placed.

Then, turn the mains switch on and press the push button located in the front part of the instrument. The LED of this button changes to green colour and after that, Erytraapplication is loaded. Finally, run the Technical Service software.

1.8.10 Setting Time Zone, Date and Time

Follow these steps to set the date and time shown by the OS:

1. Enter **Maintenance Mode** (See "Working with Boot Application" section)
2. Go to **Start > Windows System > Control Panel**.
3. Set the appropriate Time Zone in the **Clock and Region > Date and Time > Change Time zone...** option. If it is available for the selected time zone, the **Automatically adjust clock for daylight saving changes** option is enabled by default.
4. If it is displayed and enabled, disable the **Notify me when the clock changes** option in the **Date and Time** window.
5. Set the appropriate date and time in the **Clock and Region > Date and Time > Change Date and Time...** option (password for admin user will be required).
6. Switch off the instrument.

1.8.11 Communications

This section contains a brief description of the LIS and remote connections that can be performed in Erytra. Regarding the physical connection ports, one RS-232 port, one ETHERNET/LAN connection and six USB connections are available.



NOTE

Use only shielded cables to ensure the proper performance of the equipment.

1.8.11.1 LIS connection

Communication of Erytra with LIS is performed using DianaServer software. See "DianaServer program" on page 1155.

Diana System working directory must be checked to confirm that it is properly configured (default value is **E:\data\SistemaDiana**, run **E:\data\SistemaDiana\Config.exe**, main tab).

1. Copy **RemoteConnection.zip** in any folder of the computer where the remote connection is required to be installed.
2. Unzip **RemoteConnection.zip**.
3. Run **\RemoteConnection\Erytra\Installer_RemoteConnection.bat** with admin privileges.
4. Once the installation has finished, press **Enter** to end the installation process.



Figure 43 - Remote connection installation

5. Check that **ERYTRA_REMOTE.bat** has been copied into the **C:\Grifols\Manager** path.
6. Check that **erytra_remote.txt** has been copied into the **C:\Grifols\Manager** path.

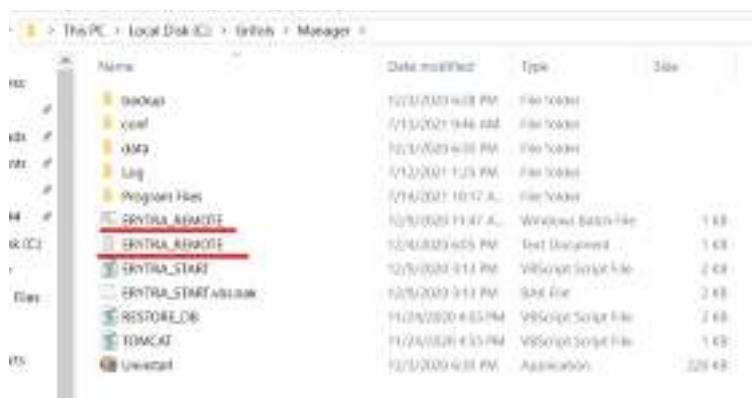


Figure 44 - Remote connection final checkings

7. Open **C:\Grifols\Manager\erytra_remote.txt** and edit the IP address of the instrument (same IP address as the one that was previously used when connecting using the old browser).

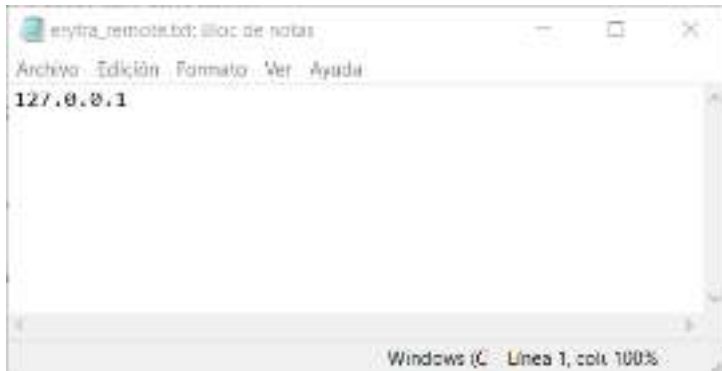


Figure 45 - IP edition for remote connection

The remote connection functionality using the alternative browser can be started by running **C:\Grifols\Manager\ERYTRA_REMOTE.bat** and logging in to the application with the corresponding credentials.

For the first time that the remote connection is started, the following steps should be performed:

1. Click **Advanced settings** and click **Proceed to URL (unsafe)**.



Figure 46 - Starting remote connection step 1

2. Click **Get Flash** and then click **Allow** to allow Flash to run on this site.



Figure 47 - Starting remote connection step 2

3. Log in to the application as usual.



Figure 48 - Log in to the application

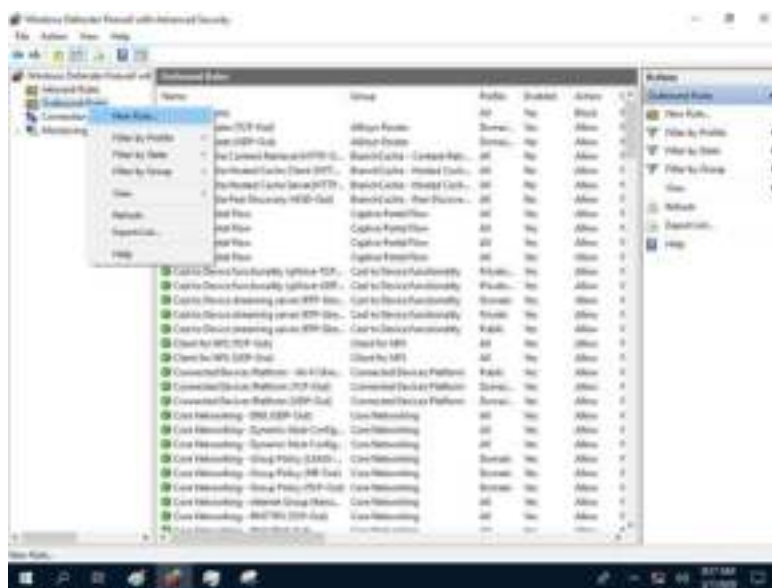
1.8.11.2 Windows firewall configuration

Once the software has been installed, it is recommended to block outbound connections of the alternative browser by means of Windows firewall configuration in the remote computer. With this, access from the alternative browser to IP addresses different than the one configured for the analyzer will not be allowed, increasing the security of the system by blocking unwanted connections. To do so, the following steps should be executed:

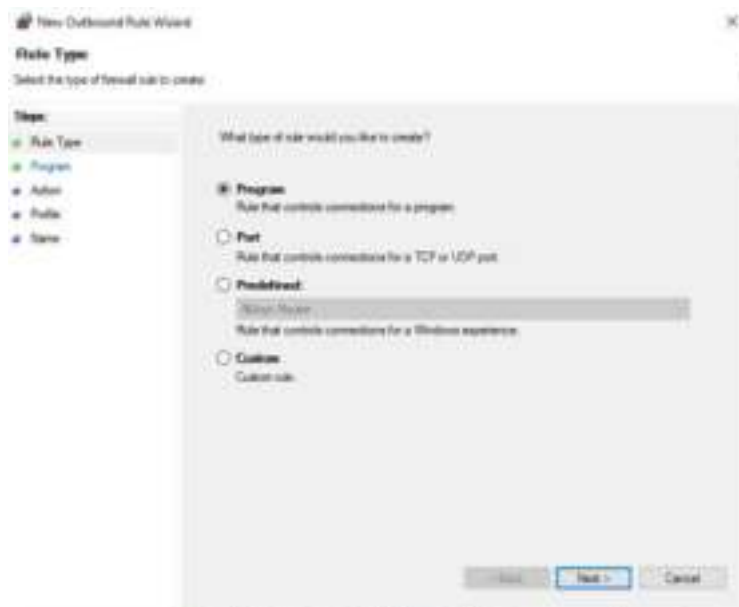
1. Go to **Windows Menu / Windows Administrative tools / Windows Defender Firewall with Advanced Security**. Right click and select the **Run as administrator** option (credentials may be requested).



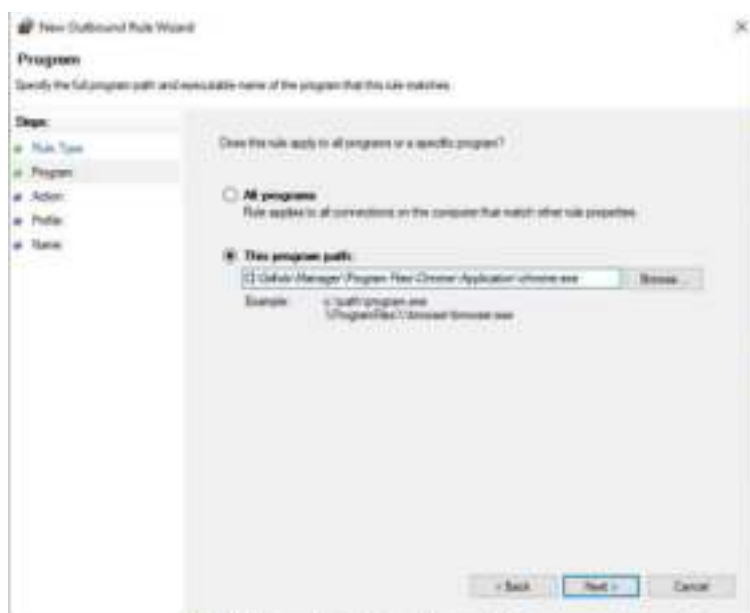
2. Right click on **Outbound Rules** and select **New Rule**.



3. Select **Program** (selected by default) and click on **Next**.

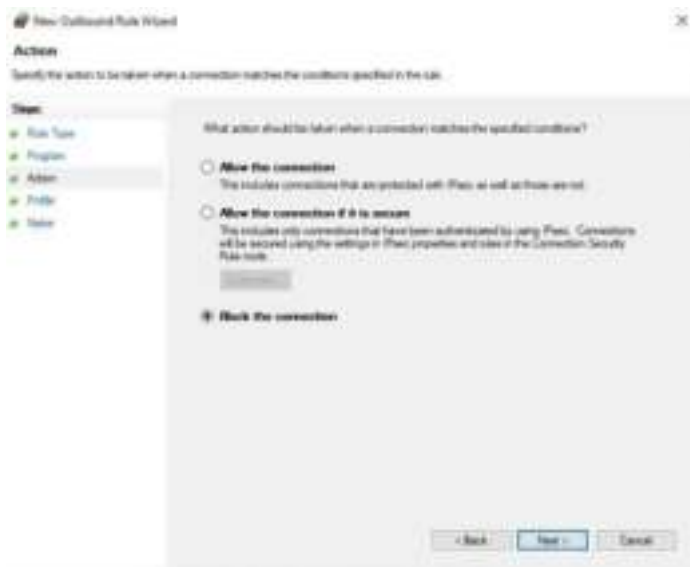


4. Select **This program path:** and enter the path of the Chrome application: **C:\Grifols\Manager\Program Files\Chrome\Application\chrome.exe**.



5. Click **Next**.

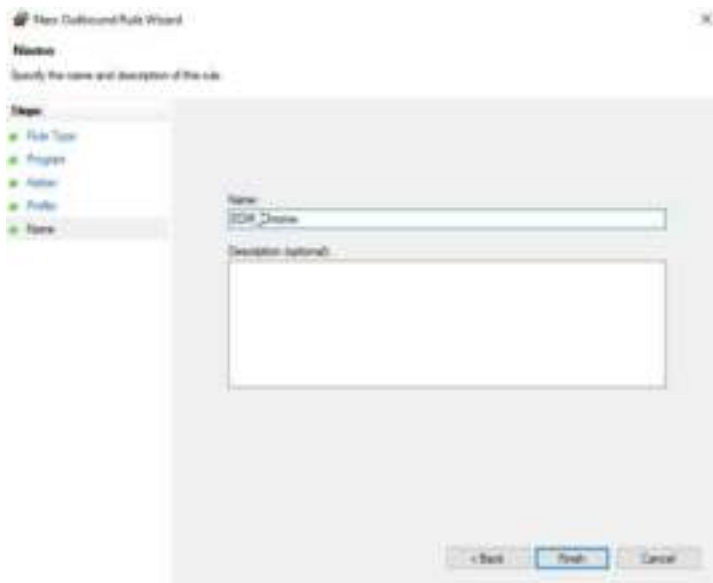
6. Select **Block the connection** option and click **Next**.



7. Do not change the default settings and click **Next**.

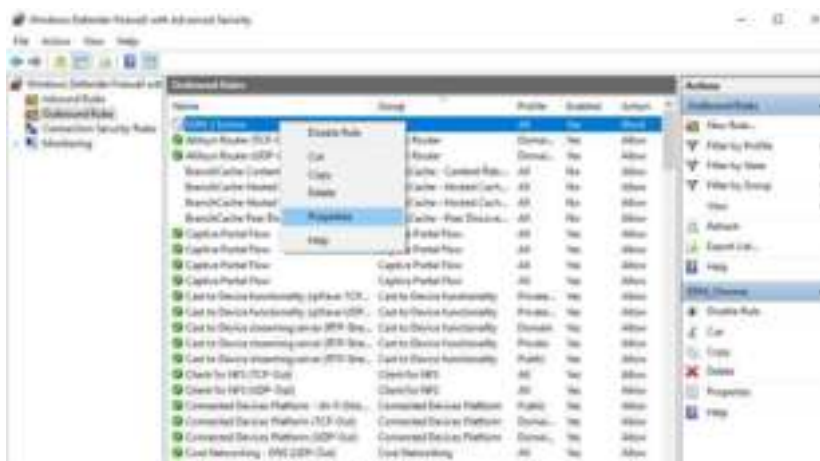


8. Give a name to the created rule. For instance, `EDM_Chrome`.



9. Click **Finish**. The created rule will then be shown among the available Outbound rules.

10. Right click on the created rule and select **Properties**.



11. Go to the **Scope** tab and select **These IP addresses** on the **Remote IP address** section. Click **Add...**



12. Select the **This IP address** range option and add the following IP addresses:
- From: **1.1.1.1**
 - To: IP configured in **C:\Grifols\Manager\erytra_remote.txt** -1. For instance, if the configured IP is 192.168.1.50, type **192.168.1.49**.



13. Click **OK**.
14. Click **Add...** in the **Remote IP addresses** section once again.

15. Select the **This IP address** range option and add the following IP addresses:
- From: IP configured in **C:\Grifols\Manager\erytra_remote.txt** +1. For instance, if the configured IP is 192.168.1.50, type 192.168.1.51.
 - To: 255.255.255.255



16. Click **OK**, then **Apply** and **OK** again in the **EDM Chrome Properties** window.



To avoid the warnings related to SSL certificate when using the remote connection functionality, the following steps must be taken:

1. Open the **C:\Grifols\Manager\ERYTRA_REMOTE.bat** file in edition mode by means of Notepad++ or similar software.

2. Edit line 15 by adding `-ignore-certificate-errors` after `"%RUNTIME_BASE%\Program Files\Chrome\Application\chrome.exe"`. Keep the rest of the line 15 as it is:

```
"%RUNTIME_BASE%\Program Files\Chrome\Application\chrome.exe" --  
ignore-certificate-errors --user-data-dir="% RUNTIME_BASE%\Program  
Files\Chrome\Application\userData" --app=https://%ip_  
erytra%:8443/IHManager
```

3. Save the **ERYTRA_REMOTE.bat** file and close the edition software.

1.8.12 Installing peripheral devices

To install a peripheral device, system volumes shall be unprotected. If connected, it is also recommended to disconnect the instrument from the network during the installation procedure.

Unprotect:

1. Switch on the instrument and press the grid () button during the system startup. Login credentials will be required. See "Working with Boot Application" on page 88.
2. Select the **Maintenance Mode** icon among the Boot Application options.
3. The PC will be rebooted and Maintenance mode will be accessed in the subsequent startup.



CAUTION

Once the computer has been restarted, drives are not protected.

4. Once the desktop is available, connect a mouse and a keyboard.
5. Press the Windows **Start** button and type `Command Prompt`. Right click at the top result and select **Run as administrator**, then type the administrator password `dgroot`.
6. Enter the command: `applocker bu` (Begin Update). The AppLocker Manager graphical interface will be opened in a separate window.
7. Wait for 10 seconds so that the change to the update mode is completed.
8. Install the drivers of the peripheral device.
9. At this point, consider reviewing the **Advanced sharing settings** in **Control Panel > Network and Internet > Network and Sharing Center**. Note that file and printer sharing option is turned off by default.

Protect:

1. Type `eu` (End Update) in the AppLocker Manager graphical interface window.
2. Wait for 10 seconds so that the change to the protected mode is completed.
3. Shut down the system.
4. The volumes will be protected and the system will shut down.

1.8.13 Decontaminating

Before manipulating the instrument for any maintenance purpose or before preparing it for storing and packaging, it is necessary to decontaminate the instrument. All procedures for this purpose are described in Instructions for Use manual.

1.8.14 Uninstalling and shipping

If the instrument needs to be stored for a long time, it is recommended to decontaminate the fluidic circuit and package it for transport. For the necessary space requirements, see "Location remarks" on page 78.

**WARNING**

The instrument must be decontaminated before transport or storage.

**WARNING**

Liquids need to be removed from the fluidic circuit when uninstalling the instrument.

**CAUTION**

Use only the original packaging to transport the instrument.

When the software is closed, the instrument automatically performs a wash of the fluidic system and empties the liquid of the circuit. Nevertheless, if the software is not closed properly or a liquid remains in the fluidic circuit, it is possible to empty the circuit with the Technical Service software. To do so, open the Technical Service software, start the **Driver** and press the **fluidic empty** button in the **Fluidics** tab. See "Fluidics tab" on page 826.

Once the circuit is empty, remove the liquids from solution A and B bottles and from waste bottles (if used). Empty the card waste container if necessary.

Before packing the instrument to be transported the screws that fix the illumination pattern to the reader must be fixed with a thread sealant (such as Loctite 5331). These screws can come loose during the transport.

**WARNING**

Anaerobic sealant such as Loctite 222 or Loctite 577 must not be used as they are incompatible with methacrylate and can damage it severely.

To apply the sealant, follow the first four steps of "Replacing reader illumination pattern" on page 739, and once the illumination pattern is accessible, unscrew one by one the screws, apply a bit of sealant in the thread and screw it back.

**CAUTION**

At least 2 screws needs to remain in position to avoid any adjustment alterations.

Two people (three recommended) are required to pack the instrument and at least one of them must be a qualified technician. To pack the instrument, follow the instructions below:

1. Open all the fluidic drawers and place the foams in the card disposal container and in the handles of the bottles.



Figure 49 - Foams in the card disposal container and in the bottles

2. Place inside the card disposal container two antihumidity bags, power cords, the decontamination tool, disposable bags for card waste and the documentation related to the instrument (manuals, etc.). Close the fluidic drawers.



Figure 50 - Decontamination tool

3. Move the PIP arm to the right of the instrument. Place the internal packaging tray in the upper floor, over the PIP platform and in the left position.



Figure 51 - Location of the internal packaging tray

4. Place the plug of the tray between the internal packaging tray and the right lateral.



Figure 52 - Location of the plug of the tray

5. Place the decontamination bottle, two samples racks, two diluents racks and two reagents racks in the packaging tray.



Figure 53 - Decontamination bottle and racks in packaging tray

6. Cover with some adhesive tape the dilution cup wells and place the fixing foams of the dilution cup.

7. Move the PIP robotic head to the front of the instrument and remove the two samples racks from the drawer labelled as 6 of the upper area to place the supplement comb lateral as shown in the following picture.



Figure 54 - Positioning of the supplement comb lateral

8. Place again the samples racks in the drawer 6 of the upper area and place the RM comb so it touches the drawers frontal and the lateral of PIP platform.



Figure 55 - Positioning of the RM comb

9. Place the external barcode reader box in its position in the RM comb. Place one antihumidity bag at each side of the box.
10. Place the SM comb between the RM comb and the supplement comb lateral.



Figure 56 - Positioning of the SM comb

11. Place the box plug between the RM comb and SM comb and the X-axis support arm.



Figure 57 - Positioning of the box plug

12. Place the frontal foams located in the user tray and in the touchscreen monitor.
13. Place the two shipping clamps (use an open end spanner number 12) which prevent the arms from moving. Check that the shipping clamps are blocking the arm and the robot head at the same time (there is only one position of the robotic head that matches with a hole in the arm to block both at the same time).
14. Remove the door handles and place the screws in a bag. Attach the bag to the door handles.
15. Move up manually the feet in order to avoid touching the surface.
16. Place the pallet of the packaging as close as possible to where the instrument is located.
17. Place the planar lateral (narrow lateral without wooden strips) of the packaging attached to the pallet in order for it to act as a ramp. Place the upper part of the wooden packaging under the ramp.
18. Fix the ramp with the two hook-and-loop fasteners.
19. Place the wooden small ramp included inside the packaging at the end of the main ramp. Stick it with adhesive tape.
20. Push the Erytra through the ramp until it is over the pallet.

21. Put the instrument in the position where the feet of the instrument are over the marks that are located in the pallet.

**NOTE**

It is very important that the instrument is centered as much as possible on the pallet.

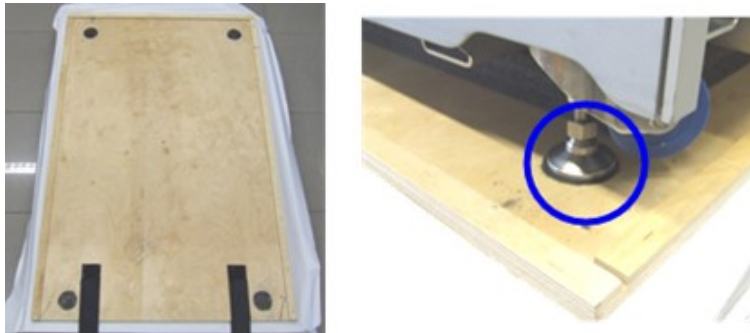


Figure 58 - Marks for the feet located in the pallet

22. Move up the instrument turning in the counterclockwise direction the nut of each feet with an open end spanner number 19. Move down the foot in a crossed way.
23. Place the foam and the wooden board included in the packaging underneath the instrument.
24. Move down the Erytra by means of the nut of the feet, turning them in the clockwise direction (use the open end spanner number 19) until the instrument rests on the wooden board over the foam.
25. Put adhesive tape on the feet to prevent the instrument from falling during the shipment.
26. Place the protective foams at each corner of the Erytra. Fix them with retractile film.

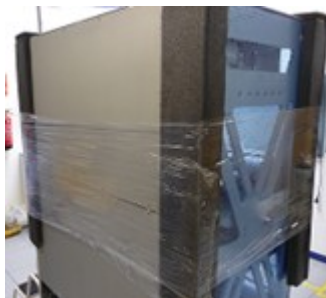


Figure 59 - Retractable film in the packaging

27. Place the aluminium anti-humidity cover.
28. Place the door handles and the small ramp inside their protective packaging and place them over the instrument.

29. Assemble the wooden box. There are two considerations in this step:
- The laterals of the packaging must be levelled with the basis of the pallet in the rear part of the packaging (the rear part is the opposite where is the main ramp).
 - The flat side of the upper part of the packaging must be faced with the frontal part of the packaging (the one where the main ramp is).

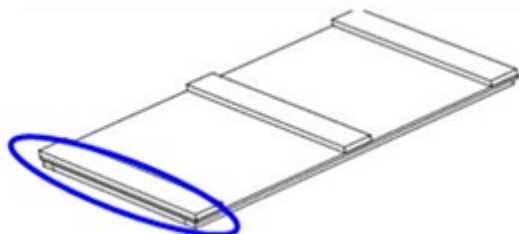


Figure 60 - Upper part of the packaging

1.8.15 Instrument disposal

If the instrument is to be disposed of, all possible residues of reagents and samples must be removed and it must be cleaned and totally decontaminated. Then, it must be sent to an authorized centre for such purpose.



CAUTION

When the equipment is at the end of its lifetime, the user must take into account all local regulations to dispose of it.



NOTE

When disposing of the instrument, Erytra should be considered an electronic residue.

1.8.15.1 Personal data at instrument disposal



CAUTION

As the device has accessed, processed, and stored potentially confidential and valuable data, consideration must be taken before its disposal.

If the data container (hard drive, SSD,...) can be extracted from the instrument, it can be released to the instrument owner, for safekeeping and to apply their local policies on personal data protection and product disposal. Otherwise, data contained in its software/hardware that is no longer used, must be securely deleted and must be unreadable and unrecoverable before device disposal. Under no circumstances, a device storing customer valuable data shall be custodied by a Grifols representative.

1.9 Preventive maintenance

Description of instrument maintenance.

1.9.1 Items and calendar

Preventive Maintenance (PM) form defines the procedures that must be carried out periodically on Erytra instruments in order to ensure that it is running under proper operating conditions. The period of the preventive maintenance plan (PMP) is defined according to the number of samples processed annually. Three categories of instruments have been defined according to its annual usage: 40.000 test/year, 80.000 test/year and 130.000 test/year.



NOTE

The number of executed tests of the current database can be found in the file (**testcounter.txt** in **E:\data\Erytra\conf**). The number of executed test is the addition of all the lines in the file.

The PM related to this Servicing Center can be found in the following form and must always be executed before the corresponding Operational Qualification (OQ) procedure. See "Operational Qualification (OQ)" on page 104

[Preventive maintenance](#)



NOTE

To view the Preventive Maintenance (PM) procedure refer to the Erytra. If an empty PM check-list is needed, it should be prepared in advance by printing it as a PDF from the HTML check-list.

In this PM form, category of use (number of test/year) and maintenance year should be selected in order to configure the PM Check-list which defines the items and procedures that should be performed according to the selected parameters. The parts to be replaced are decided taking into account the number of tests carried out during each period.

1.9.2 Adjustment tools

All the adjustment tools are detailed in each section where an adjustment is explained. In the following table there are all of them:

Table 5 - Erytra adjustment tools

ADJUSTMENT TOOL	MODULE	PN
PROBES_WAGON ADJUSTMENT TOOL (Ver. 2)	PIP	233901
PROBES ADJUSTMENT TOOL (Ver. 4)	PIP	234213
HORNS ADJUSTMENT TOOL (BCR)	PIP	233819
OMNIFIT TOOL (for tube connection Ver 2)	PIP	233898
CARDS DRAWER ADJUSTMENT TOOL (Ver. 2)	CAT	233878
CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL (Ver. 3)	CAT	733056
LOWER CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL v3	CAT	733068
DILUTION CUP ALIGNMENT TOOL (Ver. 2)	CUP	233938
CENTRIFUGE ROTOR ADJUSTMENT TOOL	CEN	233821
DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V	INC	230903
TEMP. PROBE PT-100 FOR CARDS - MiniDIN 4v	INC	230905
READER ADJUSTMENT TOOL-SUPPORT (Ver. 2)	LEC	234212
READER ADJUSTMENT TOOL-PATTERN v2	LEC	731398
USB 2.0 CABLE 1.8 M (TOUCHSCREEN, ELECTRONICS CONNECTORS COVER AND PCA RD IL421)	LEC	233443
FIREWIRE READER CABLE TOOL	LEC	233822
PRESSURE SENSOR ADJUSTMENT TOOL	FLU	233817
DIGIT. GAUGE DIGITRON MOD.2023	FLU	247032
DRAWER FRONT PANEL ALIGNMENT TOOL	GEN	233824

1.9.3 Lubrication points

During some preventive maintenance actions or even after some corrective maintenance actions, lubrication procedures are required to apply to some parts of the equipment. Recommended lubrication products are:

- Grease: P/N 232697 SUPER LUBE GREASE LUBRICANT 85 g (5 units).
- Oil: P/N 232696 SUPER LUBE OIL LUBRICANT 7 ml (5 units).

For more information about spare parts, see "Spare parts list" on page 1207.

**NOTE**

Deenergize the motors before performing lubrication procedures in order to distribute the grease or oil. When necessary, perform a home of the module and move it from edge to edge to distribute the grease or oil. Do not apply grease/oil in excess.

1.9.3.1 General module (GEN)

Touchscreen drawer

- Lubricate with grease the tilting latch lock.



Figure 61 - Lubrication point in touchscreen drawer

- Lubricate with grease the guides of the touchscreen monitor.

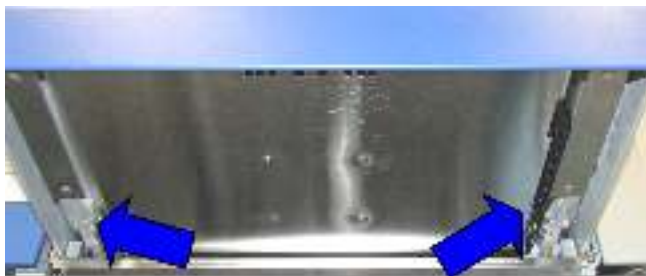


Figure 62 - Lubrication points in touchscreen drawer

User tray

**NOTE**

Not necessary for instruments with SN $\geq$ 059-0001675 or any Erytra unit where P/N 733671 USER TRAY (Ver. 4) is assembled.

- Lubricate with grease over the protection caps and the guiding edge.



Figure 63 - Lubrication points in user tray

Upper door

- Lubricate with a drop of oil in the joints of the hinges.

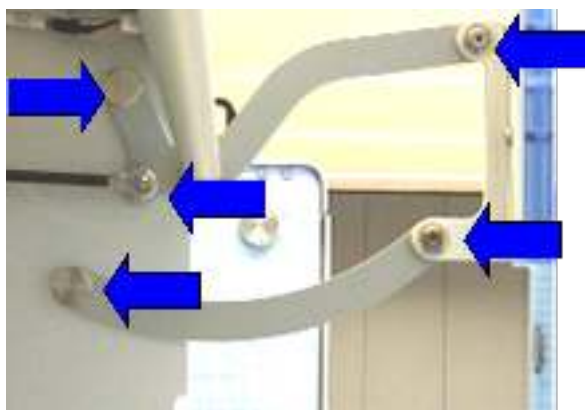


Figure 64 - Lubrication points in upper door

- Lubricate with grease the surface of the hinges.



Figure 65 - Lubrication points in intermediate door

Reagent drawers

- Lubricate the end of the latch with grease.



Figure 66 - Lubrication point in reagent drawer (latch)

- Lubricate the linear guides with grease (use a brush).



Figure 67 - Lubrication point in reagent drawer (linear guides)

- Lubricate with grease underneath the foot of the stirred reagents rack platform.



Figure 68 - Lubrication point in reagent drawer (stirred reagents platform)

Samples drawers

- Lubricate with grease the end of the latch.



Figure 69 - Lubrication point in sample drawer (latch)

- Lubricate the linear guides with grease (use a brush).



Figure 70 - Lubrication point in sample drawer (linear guide)

Cards drawers

- Lubricate with grease the end of the latch.



Figure 71 - Lubrication point in card drawer (latch)

- Lubricate the linear guides with grease (use a brush).



Figure 72 - Lubrication point in card drawer (linear guides)

1.9.3.2 Pipetting arm module (PIP)

Horns structure

- Lubricate with grease.



Figure 73 - Lubrication point in PIP module (horns)

X-axis linear guide

- Lubricate with oil.



Figure 74 - Lubrication point in PIP module (X-axis linear guide)

Y-axis linear guide

- Lubricate with oil.



Figure 75 - Lubrication point in PIP module (Y-axis linear guide)

Leadscrews

- Lubricate with oil the probes leadscrews and horns leadscrew.

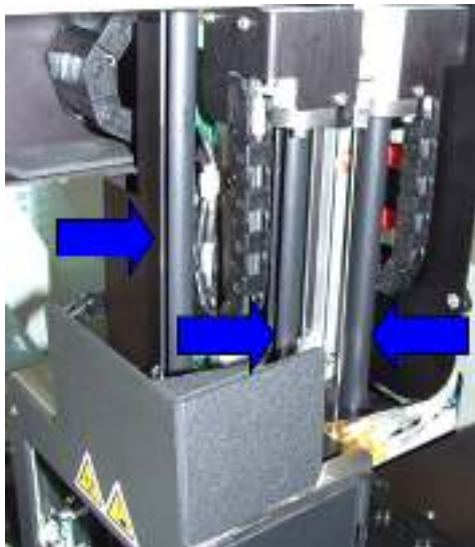


Figure 76 - Lubrication points in PIP module (leadscrews)

1.9.3.3 Card transport arm module (CAT)

X-axis linear guide

- Lubricate with oil.



Figure 77 - Lubrication point in CAT module (X-axis linear guide)

Y-axis linear guide

- Lubricate with oil.



Figure 78 - Lubrication point in CAT module (Y-axis linear guide)

Leadscrew

- Lubricate with oil.

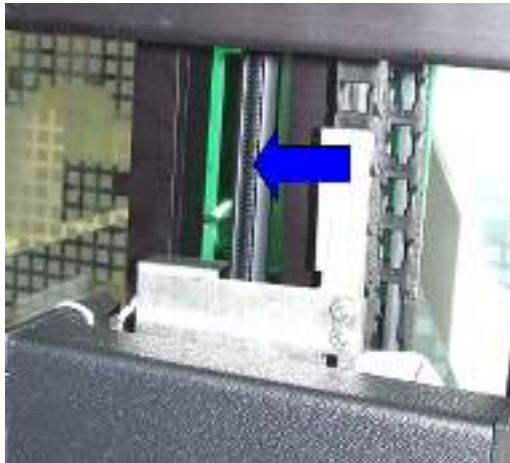


Figure 79 - Lubrication point in CAT module (leadscrew)

Gripper linear guide

- Lubricate with oil.



Figure 80 - Lubrication point in CAT module (gripper linear guide)

Gripper



NOTE

Consider removing only the lower part of the gripper for lubrication purposes.

In case of complete removal of the gripper, it is necessary to carry out all adjustments of the gripper again. Including the gripper parallelism.

In order to disassemble the lower part of the gripper follow the 8 first steps described in "Replacing gripper (Z) chain" on page 515. Then follow these steps:

1. Remove the two lower screws.



Figure 81 - Gripper two lower screws

2. Remove the screw holding the PCA and disassemble it.



Figure 82 - Screw holding PCA

3. Completely unscrew the two screws that hold the lower part to the upper part of the gripper. A flat screwdriver can be used to prise the two parts apart.



Figure 83 - Two screws attaching lower and upper parts of gripper

4. Once the two parts are separated, lubricate with a small drop of oil the four gripper pins.

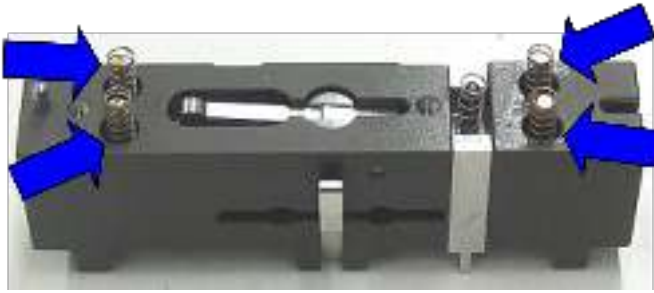


Figure 84 - Lubrication point in CAT module (gripper pins)

5. Lubricate with grease the surface of the centering system in contact with the lower part of the gripper.

**NOTE**

Make sure the spring is placed over the centering part when assembling back all the parts.



Figure 85 - Lubrication point in CAT module (gripper centering system)

**NOTE**

Threadlocking adhesive (such as Loctite 222) must be applied to the two removed screws when assembling the gripper in the instrument.

1.9.3.4 Fluidics module (FLU)

Bottle drawers / Cards disposal tray drawer

- Put some grease in load cell ramp.



Figure 86 - Lubrication point in fluidics drawer (load cell)

- Lubricate with grease the tilting base joint.



Figure 87 - Lubrication point in fluidics drawer (tilting base joints)

- Put grease at the end of the latch.



Figure 88 - Lubrication point in fluidics drawer (latch)

- If the linear guide of the drawer is cleaned, put some grease in the front and rear part.

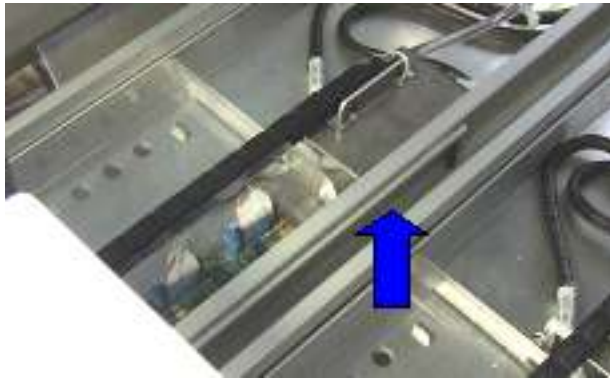


Figure 89 - Lubrication point in fluidics drawer (linear guide)

1.9.4 How to report

The necessary information needed to analyze different incidences occurring during the daily routine can be compiled using the Support Pack feature, which allows the creation of a zipped file with all the data required for support. The generated data will not contain patient data in order to meet the General Data Protection Regulation (GDPR) compliance, and will be required to be stored in a USB drive, using 7-Zip software for its compression.

For further information on how to create a Support Pack and the generated contents, see "Support pack" on page 1147.

2 Modules overview

Description of modules and procedures related to them.

2.1 General module (GEN)

In this chapter, General module is described. It includes an overall description of the module, adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, electronic schematics and a troubleshooting section.

2.1.1 Description

An overall description of the parts included in the general module is explained in this chapter.

2.1.1.1 PCA GEN-BCKPL59

Main PCA of the instrument, located at the upper part of the backpanel. This PCA has the following functionalities:

- Convert between RS-232 (computer) and RS-485 (PCAs internal bus) protocol.
- Digital pumps RS-485 interface (convert three digital I/O signals of PCA PIP control into RS-485 two-wire bus).
- Support all PCA MotorControl. They control specific modules of the instrument and some of these PCAs can control two or more modules.
- Provide a connection for PCA Fluidics Control.
- Incorporate all necessary electronic interfaces between PCAs MotorControl and the rest of PCAs, including connection.
- Distribute power supply (with the appropriate protections) to PCAs MotorControl, and the rest of PCAs interfaces and PCAs.

2.1.1.1.1 RS-232 (PC-side) to 4-wire RS-485 (machine-side) converter

PCA GEN-BCKPL59 has a RS-232 to 4-wire RS-485 converter. This converter is optically isolated and only converts TxD and RxD signals of the RS-232 port, with a maximum baudrate of 115200 bauds. PC-side is supplied from +5 VPC input supply, and machine-side is supplied from +5 V supply rail.

PC-side connection (TxD, RxD and GND) is routed to a standard DB-9 shielded female connector, in such a way that a pin to pin DB-9 cable is required to connect PCA GEN-BCKPL59 to a PC. Configuration of the pins is as follows:

- Transmission to PC : Pin 2.
- Reception from PC : Pin 3.
- GND : Pin 5.
- Chassis : Connector shield.

Machine-side 4-wire RS-485 signals are routed to the RS-485 pins of all PCA MotorControl slots and also to the RS-485 pins of the PCA Fluidics Control connector.

2.1.1.1.2 PCA MotorControl support

PCA GEN-BCKPL59 provides four DIN connectors (slots) to connect four PCA MotorControl units. There are two protections against connection of a PCA MotorControl in a wrong slot:

- Two PCA MotorControl digital inputs of each slot are reserved for slot-addressing purposes.
- These two slot-addressing PCA MotorControl digital inputs are the same at all slots.

Addressing is as follows:

- PCA CEN-SRM Control: Address 0 ("00") - SLOT 0.
- PCA INC-CUP Control: Address 1 ("01") - SLOT 1.
- PCA CAT-CM Control: Address 2 ("10") - SLOT 2.
- PCA PIP Control: Address 3 ("11") - SLOT 3.



Figure 90 - Location of the different PCA MotorControl in PCA GEN-BCKPL59

2.1.1.1.3 I2C Bus

Communication between PCA GEN-BCKPL59 and some of the different modules of the instrument is performed through I2C buses. This bus only requires two bus lines and each device connected to the bus is software-addressable by a unique address.

The different I2C buses of the instrument are the following ones:

- SM safety latches + RM safety latches + GEN SM&RM floor Temperature Sensor.
- CM safety latches + GEN CM floor Temperature Sensor.
- PIP (Probe1 Level Detector activation + Probe2 Level Detector activation + Probe1 Correct Pipetting Detector (activation and signal) + Probe2 Correct Pipetting Detector (activation and signal) + Barcode Readers activation + Barcode readers axis home sensor activation and current limit sensing + Presence Detector activation).
- Incubator 1 (Peltier Cell + Peltier Fan + two x Temperature Sensors).

- Incubator 2 (Peltier Cell + Peltier Fan + two x Temperature Sensors).
- Incubator 3 (Peltier Cell + Peltier Fan + two x Temperature Sensors).
- Incubator 4 (Peltier Cell + Peltier Fan + two x Temperature Sensors).
- FLU safety latches.

2.1.1.1.4 Power Supply input connection

PCA GEN-BCKPL59 has two supply-input connectors:

- One connector to input +24 V and +5 V power supplies, coming directly from the Power Supply system. Both supplies share ground connection (common ground plane).
- One connector to input +5 VPC, that comes from PCA GEN-STDBY511 +5 VPC output.

A block diagram of the PCA GEN-BCKPL59 is shown below:

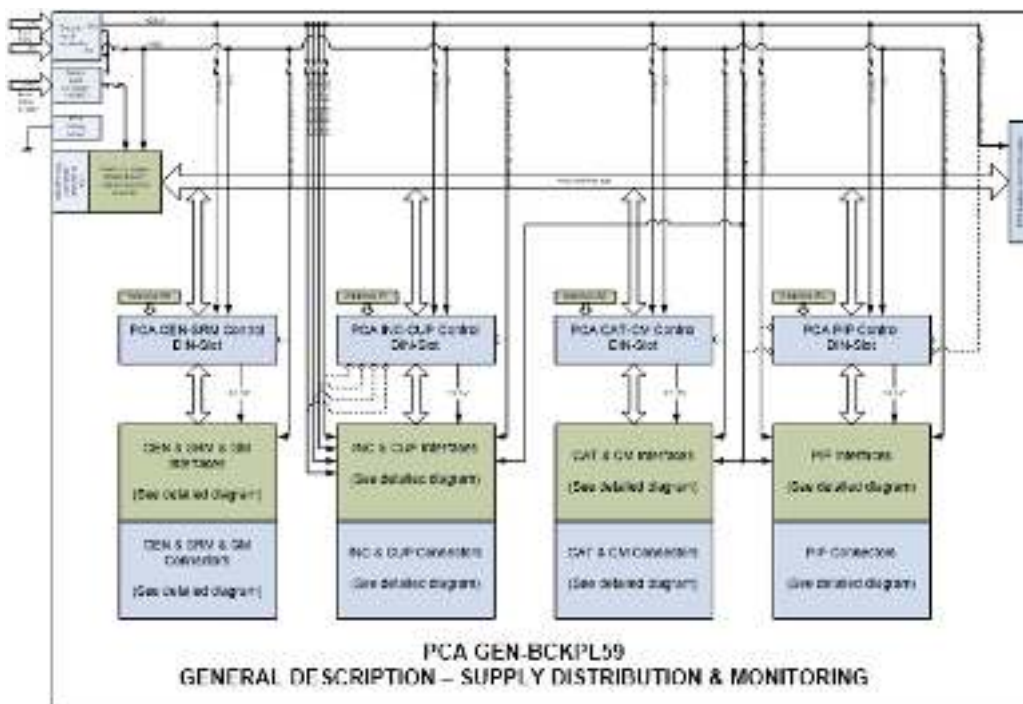


Figure 91 - Block diagram of PCA GEN-BCKPL59

2.1.1.2 Power supply

Located at the upper part of the instrument, it connects Erytra to the mains AC voltage.

It is a switched power supply system which automatically adapts to the mains supply voltage, thus does not require any type of manual switching. It has a standby output voltage in order to supply standby-mode electronics that is kept on as long as mains voltage is applied at the power supply input.

Internal AC power cabling is composed of three cables (Ground, Line and Neutral). Each of these three cables can be identified by its color and has a specific position when connected to the mains filter (position identified/marked in the mains filter label).

1. Brown: Line (P).
2. Light blue: Neutral (N).
3. Yellow-green: Ground (PE).



Figure 92 - Internal AC power cabling

The power supply system consists of two individual power supply units: one providing 24 VDC and another one providing 12 VDC. Additionally, a voltage converter PCA provides a 5 VDC output signal using a 24 VDC input.

The whole power supply system is assembled in an independent plate, which is attached to the top of the electronics box (above the PC) by means of two hand screws.

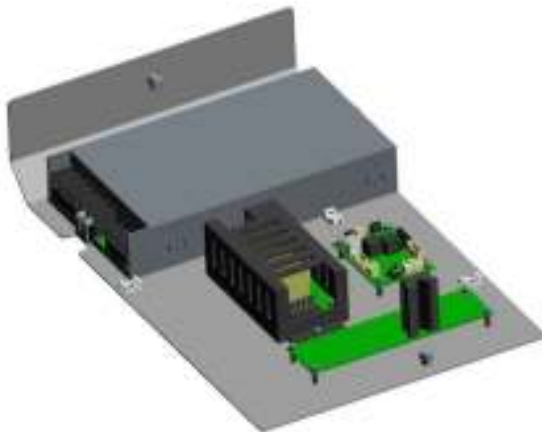


Figure 93 - Power supply system assembled in an independent plate

Specifications of the power supply:

- Input range: 85 to 264 VAC, 47 to 64 Hz.
- Power Factor Correction.

- Output voltages:
 - +24 VDC $\pm 2\%$ (23,52 V to 24,48 V): Motors, syringes, solenoids, electrovalves, pumps and Peltier cells.
 - +5 VDC $\pm 4\%$ (4,8 V to 5,2 V): Control electronics logic.
 - +12 V $\pm 2\%$ (11,76 V to 12,24 V): PC supply.
 - -12 V $\pm 2\%$ (-11,76 V to -12,24 V): PC supply.
 - +5 V $\pm 4\%$ (4,8 V to 5,2 V): PC supply.
 - +3.3 V $\pm 4\%$ (3,17 V to 3,43 V): PC supply.
 - +5 V $\pm 4\%$ (4,8 V to 5,2 V): Standby power.

2.1.1.2.1 Erytra On/Off Switcher

24 V and 5 V power supply outputs can be activated / deactivated by the Technical Service software.



NOTE

For units with S/N < 059-0001583, jumper JP3 of PCA GEN-STDBY59 must be shortcircuited to use this function.

1. In the Technical Service software, go to **Run > Erytra On/Off Switcher** (see "Modular tabs" on page 754). A new window will appear on screen.

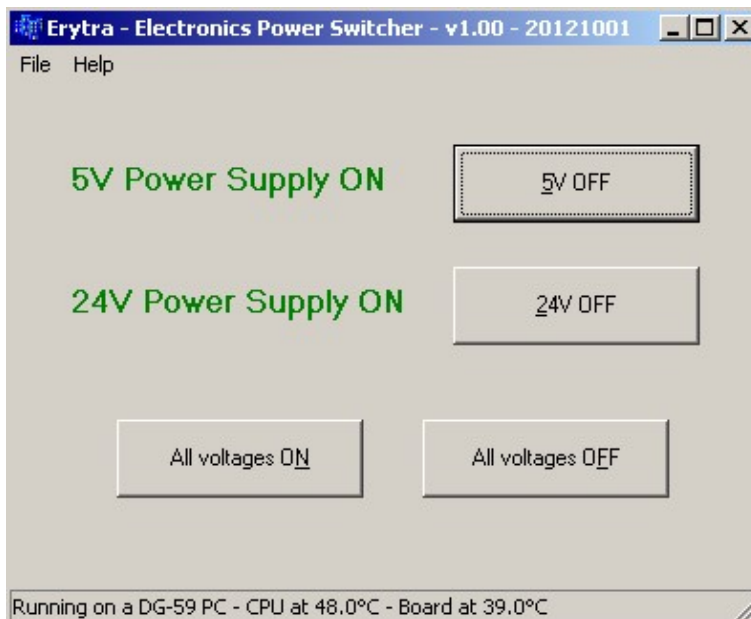


Figure 94 - Erytra On/Off Switcher

2. To switch off the 5V and 24V power supply outputs, press the **All voltages OFF** button. This action will not switch off the computer. Then in order to switch on the 5V and 24V power supply outputs, press the **All voltages ON** button.

3. For units with S/N < 059-0001583, also it is possible to switch off only the 5V or 24V power supply output pressing the **5V OFF** or the **24V OFF** button, respectively. Then in order to switch them on press the **5V ON** or the **24V ON** button, respectively.

2.1.1.3 Computer

The computer contains the OS (Windows 10 Embedded (WES10) v2.1.0) and Erytra software. It is located in the upper part of the instrument, under the power supply system.

The computer is switched on by means of the pushbutton that allows switching on the instrument and setting it to standby mode. It is located between the upper floor and the intermediate one. It includes a green LED that is used as a status LED:

- Off: instrument switched off.
- On: instrument PC-system (PC, screen, ports) is switched on.



Figure 95 - Pushbutton

2.1.1.3.1 PC EMBEDDED v3, v4 and v5 (PC ASRock)

Connectors of the computer are the following ones:

- Eight USB ports:
 - USB 3.0: USB 1, USB 3, USB 6 and USB 7 (related to the USB ports available at the rear part of the instrument).
 - USB 2.0: USB 0, USB 2, USB 4 and USB 5 (related to internal communications and USB ports available at the front part of the instrument).
- ATX POWER.
- COM 1 and COM 3.
- VGA.
- FW (FireWire/IEEE1394).

- LAN1.
- PC ATX POWER.
- PC FEATURE.

For a detailed wiring list, see "Wiring table" on page 225.



Figure 96 - Computer connectors for PC EMBEDDED v3. Front view

Connections for PC EMBEDDED v4 and PC EMBEDDED v5 should be done in the same way as indicated in the picture above for PC EMBEDDED v3. The only difference is that two ports of the built-in firewire PCIe are available in PC EMBEDDED v4 and PC EMBEDDED v5 (one remains unused), instead of the three ports available in PC EMBEDDED v3 (two of them remain unused).



Figure 97 - Firewire ports in PC EMBEDDED v4/v5. Front view

2.1.1.3.2 PC EMBEDDED v2 / PC EMBEDDED - USA (PC Kontron)

Connectors of the computer are the following ones:

- Eight USB 2.0 ports:
 - USB 1, USB 3, USB 6 and USB 7 (related to the USB ports available at the rear part of the instrument).
 - USB 0, USB 2, USB 4 and USB 5 (related to internal communications and USB ports available at the front part of the instrument).
- ATX POWER.
- COM 1 and COM 3.
- VGA.
- FW (FireWire/IEEE1394).
- LAN1.
- PC ATX POWER.
- PC FEATURE.

For a detailed wiring list, see "Wiring table" on page 225.



Figure 98 - Computer connectors for PC EMBEDDED v2 / PC EMBEDDED - USA. Front view (left) and lateral view (right)

Both PC models have a direct connection with the reader module by means of a FireWire protocol for the camera control and with a USB protocol for the reader backlight and for the communication emulating a serial port.

**NOTE**

If the PC battery needs to be replaced for any of the above described models, the new battery shall be a 3 V CR2032-type, with a maximum operating temperature $\geq 80^{\circ}\text{C}$.

2.1.1.4 Touchscreen monitor

Located in a drawer under the loading area of samples, it is the main operator interface. The mechanism of the drawer allows the operator to tilt the screen and to hide it completely inside the instrument when it is not in use. In order to extract the touchscreen monitor, the frontal panel must be pushed once and then pulled from the frontal panel. When the touchscreen monitor is outside completely of the instrument, it can be tilted.

The monitor is TFT LCD 15" with a resolution of 1024 x 768 pixels that incorporates a touchscreen. The monitor is connected to the computer using a VGA protocol and the touchscreen is controlled by an internal USB port that is not accessible externally.



Figure 99 - Touchscreen monitor

2.1.1.5 User tray

Located under the reagents loading area, it helps the operator to handle samples and reagents. It has a usable surface of 340 mm width x 200 mm depth. The mechanism of the drawer allows the operator to hide it completely inside the instrument when it is not in use.

In order to extract the user tray, the frontal panel must be pushed once and then pulled from the frontal panel.



Figure 100 - User tray

2.1.1.6 Drawers

There are four types of drawers in the Erytra: reagents, samples, cards and bottles drawers. In this chapter, reagents, samples and cards drawers are explained. For an explanation of the bottles drawers, see "Fluidics module (FLU)" on page 555.

Samples and reagents drawers are controlled with an I2C bus. In this bus is also attached the temperature sensors of the upper floor. The same communication bus applies to the cards drawers, attached as well to the temperature sensors of the intermediate floor.

The system assigns one address to each reagent / samples drawer. PCA GEN-BCKPL59 provides a '000' binary address at SRM-LCK interface connector, as the address of the first PCA GEN-LCK59 connected to the safety latch bus. The output of each PCA GEN-LCK59 is input address+1 and this is the address provided to the next PCA GEN-LCK59 and so on. The same applies to the cards drawers.

2.1.1.6.1 Reagent drawers

The structure of the module is formed by the following elements:

- Moving part: it supports the reagent racks (located between centerers) and it allows the operator to load / unload reagent racks. It includes a cosmetic front panel and the mechanics and electronics to perform the required reagent stirring.
- Fixed part: it is responsible for allowing the movement of the moving part outside the instrument by means of a linear guide and to attach the module to it. It also includes the cable that connects the moving part to the electronics.
- Safety latch: it controls open / close status, it allows opening the drawer electronically and it assures closed position repeatability.

There are two reagents drawers located over the user tray, on the left of the instrument and labelled as 1 and 2. Each drawer allocates two reagents racks, oriented in a way that the reagents are located in two rows parallel to drawer movement. The racks are manufactured from plastic and are washable in Laboratory Glassware Washer.



Figure 101 - Top view of reagents racks

In each drawer, the frontal rack allows to an orbital stirring of the reagents. Each rack for stirred reagents allocates 18 vials and each rack for non-stirred reagents allocates six vials and three diluent bottles. There is a Hall effect sensor in order to determine the home position of the stirring platform.

Racks centerers are placed on the drawers in order to do not place incorrectly the reagents racks.

Reagents drawers incorporate a magnetic braking system in order to avoid the possibility during its movement of a splashing process of the reagents.

Each reagent / diluent position in the rack has a window to allow reading (vertical) bar code of the vials / bottles.

There are clips in order to avoid any movement of the vials / bottles on its lodgings. All the vials lodgings place the vial / bottle with an inclination to minimise dead volumes.

A safety latch (controlled electronically), located at the rear part of the drawer, assures drawer is closed in the working position and allows open / close status monitoring. The latch is attached to the support, while the bolt is attached to the moving part. The bolt position is adjustable in order to adjust the cosmetic front panel.

2.1.1.6.2 Sample drawers

The structure of the module is formed by the following elements:

- Moving part: it supports the sample racks and allows the operator to load / unload samples racks. It includes a cosmetic front panel.
- Fixed part: it is responsible for allowing the movement of the moving part outside the instrument by means of a linear guide and to attach the module to it.
- Safety latch: it controls open / close status, it allows opening the drawer electronically and it assures closed position repeatability.

There are four samples drawers located over the touchscreen monitor, on the right of the instrument and labelled as 3, 4, 5 and 6. Each drawer allocates two samples racks oriented in a way that samples are located in two rows parallel to drawer movement. The racks are manufactured from injected plastic and are washable in Laboratory Glassware Washer.



Figure 102 - Sample rack

To ensure the correct position of the racks in the drawer, there are positioning devices that hold the rack by their four corners. Due to the rack symmetry, a 180° rotation in the drawer is allowed.

Each rack allows placing 12 samples tubes. The distance between the center of the tubes in the rack is 25 mm in X and Y axis. The correct centering of tubes with different diameter in their lodging is assured by centering clips. Allowed dimensions of the tubes (maximum tube height: 100 mm) are:

- Glass tubes (external diameter 9 to slightly smaller than 17 mm).
- Plastic tubes (external diameter 9 to slightly smaller than 17 mm).



NOTE

To load sample tubes with an external diameter larger than 16 mm and slightly smaller than 17 mm, P/N 210715 SAMPLES HOLDER >16 mm needs to be used.

For more information about spare parts, see "Spare parts list" on page 1207.

Samples drawers incorporate a magnetic braking system in order to avoid the possibility, during its movement, of a splashing process, aerosolization, bubble formation, breakage or remixing of a centrifuged specimen.

A safety latch (controlled electronically), located at the rear part of the drawer, assures drawer is closed in the working position and allows open / close status monitoring. The latch is attached to the support, while the bolt is attached to the moving part. The bolt position is adjustable in order to adjust the cosmetic front panel.

2.1.1.6.3 Card drawers

The structure of the module is formed by the following elements:

- Moving part: it supports the card holders and allows the operator to load / unload cards holders.

- Fixed part: they are responsible for allowing the movement of the moving part outside the instrument and to attach the module to it.
- Safety latch: it controls open / close status, it allows opening the drawer electronically and it assures closed position repeatability.

There are four card drawers located on the intermediate floor and labelled as 1, 2, 3 and 4 in the keyboard below. Each drawer can allocate four 25-card holders, which can be properly placed on the drawer by means of centerers attached to its surface. Additionally, card holders are retained by means of a plastic bar attached to the vertical linear guide on the left side of each drawer and a plastic clip on the opposite side (one for each rack). Should a holder be positioned with its card bar codes facing the right side of the instrument, the software will warn the user.

A safety latch (controlled electronically) located at the rear part of the drawer, assures the drawer position when closed and allows the open / close status monitoring. The latch is located in the rear support, while the bolt is attached to the moving part of the drawer. The bolt position can be adjusted in order to modify the depth of the drawer inside the instrument.

2.1.1.7 Keyboards

Keyboards allow the operator to open the drawers of the samples, reagents and cards by means of a button located under each drawer. It is possible also to know samples drawers status by means of a LED located at the right side of each opening button. There is a button on the right side of the keyboard to request the opening of a samples drawer in order to introduce a STAT sample.

Samples/Reagents keyboard is connected directly to the PCA GEN-BCKPL59 with a cable that transports each button signal/line in parallel. Communication between PCA GEN-BCKPL59 and the PCA CEN-SRM Control is performed with an I2C bus. Identical communication system is used for the Cards keyboard (between PCA GEN-BCKPL59 and PCA CAT-CM Control).



Figure 103 - Samples / Reagents and Cards drawers keyboards

2.1.1.8 Status LEDs

Status LEDs allow the operator to know the instrument status without looking at the software screen. The instrument contains the following status LEDs:

- Samples drawers: the availability of each drawer is highlighted in blue color.

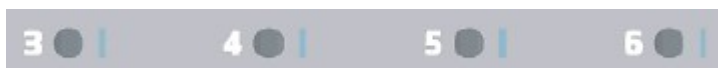


Figure 104 - Samples drawers keyboard

Samples drawers status LEDs are connected directly to the PCA GEN-BCKPL59 with the same cable that transports information of the Samples/Reagents keyboard. Communication between PCA GEN-BCKPL59 and the PCA CEN-SRM Control is performed with an I2C bus.

- General LED: located at the top right of the instrument, it shows by means of three different colors (green, yellow and red) if there is any alarm in the system allowing to check the instrument status without the need to open screen drawer.



Figure 105 - General LED

General LED is connected directly with a cable to the PCA GEN-BCKPL59. Communication between PCA GEN-BCKPL59 and the PCA CEN-SRM Control is performed with an I2C bus.

2.1.1.9 Doors

There are two doors that give technician access to the instrument for servicing purposes:

- Upper area door: located where pipetting is performed. It opens manually upwards and keeps its position until it is closed down by the operator/technician. The door does not have a latch.



Figure 106 - Upper area door

- Intermediate area door: located where card transport is performed. It opens downwards manually by its own weight with a cushioned movement. The door does not have a latch, it is closed by means of a screw located in the sensor area. To open the door, the screw must be removed and the user tray must be opened.



Figure 107 - Intermediate area door

Status of the doors is monitored by means of one sensor (Celduc sensor) for each door. This Hall effect sensor is split in two parts: one attached to the door and the other one attached to the instrument, so when the door is closed the part attached to the door is in contact with the fixed part.



Figure 108 - Celduc sensors in upper area door (left) and in intermediate area door (right)

2.1.1.10 Temperature control system

There are three areas in the Erytra where the temperature is monitored. Two of them are located in the lower part of each floor of the instrument, where there are located a PCA with two temperature sensors (PCA GEN_TEMP59). The other area is located in the electronics cabinet, where there is a temperature sensor on the PCA GEN-BKCPL59 (U21), close to the RM_2 connector. Sensors are LM92 with an accuracy of $\pm 0.33^{\circ}\text{C}$.



Figure 109 - Temperature sensors in upper (left) and intermediate (right) areas

There is a forced air extraction system in the electronics area activated by electronics cabinet temperature (30 °C) while instrument is working. This extraction system consists of two fans that extract the heat outside the instrument, one located in the right side of the electronics cabinet (looking from the rear side of the instrument) and another one in the central part, assembled on a plate providing rigidity of the whole cabinet.

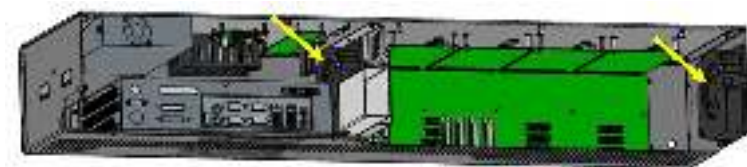


Figure 110 - Fans in the electronics cabinet

Communication buses of the temperature control system of the upper and intermediate area are through I2C buses, sharing same bus with the reagents/samples drawers (upper area) and cards drawers (intermediate area).

2.1.1.11 Buzzer

The instrument incorporates a buzzer to alert of any incidence to the operator. It is placed at the PCA GEN-BCKPL59, under slot number 2.



Figure 111 - Location of the buzzer in PCA GEN-BCKPL59

2.1.1.12 Chassis

It is formed by two laterals (made of aluminium) held together with crossbars (made of aluminium) and the X axis of the upper and intermediate floors. It is mounted on the instrument base.



Figure 112 - Chassis of Erytra

2.1.1.13 Cables and tubes rails

The rails route the tubes / cables in order to distribute them throughout the instrument. They are located at the rear part of the instrument, attached to the chassis crossbeams. There are three rails, one at each side of the Erytra and another one centered.

In each tie wrap of the rails there are labels indicating the codes of the cables / tubes in order to facilitate the identification.



Figure 113 - Cables and tubes rails

2.1.1.14 Covers

Aesthetical covers of the instrument. There are the following covers:

- Upper and intermediate area laterals: covered by a transparent lateral attached to the chassis.
- Lower area laterals: covered by an opaque lateral attached to the chassis.
- Top cover: it protects the instrument from spills.

- Frontal covers: include all frontal areas that are not covered by the frontal of the drawers and doors. One of the frontal covers includes the pushbutton and two USB ports.

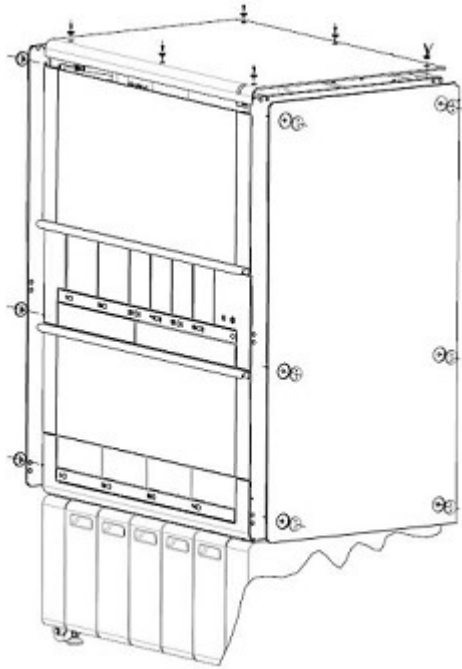


Figure 114 - Frontal covers

- Back covers: there are three back covers, one per each area (upper, intermediate and lower). Upper and lower covers have a matrix of holes in order to allow heat extraction.

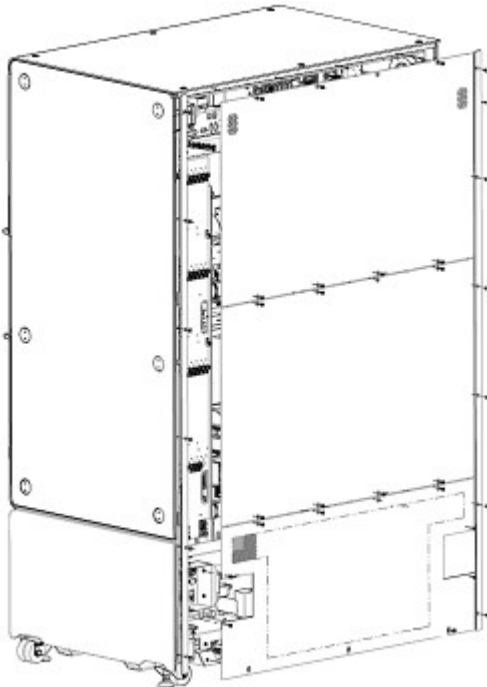


Figure 115 - Back covers

- Connectors covers: they are attached to instrument laterals and are independent from lower back covers. Connector covers allow room for external connected cables, therefore when those are connected they do not jut out the instrument footprint. In the lower area, there are two specific covers:
 - Fluidics: includes a fitting to connect the fluidic system to the laboratory drain.



Figure 116 - Fluidic connector cover

- Electronics: to allow mains connection and external available ports (four USB ports, one Ethernet connector and one serial port).



Figure 117 - Electronics connectors cov

A general block diagram of the General module is shown in the following figure:

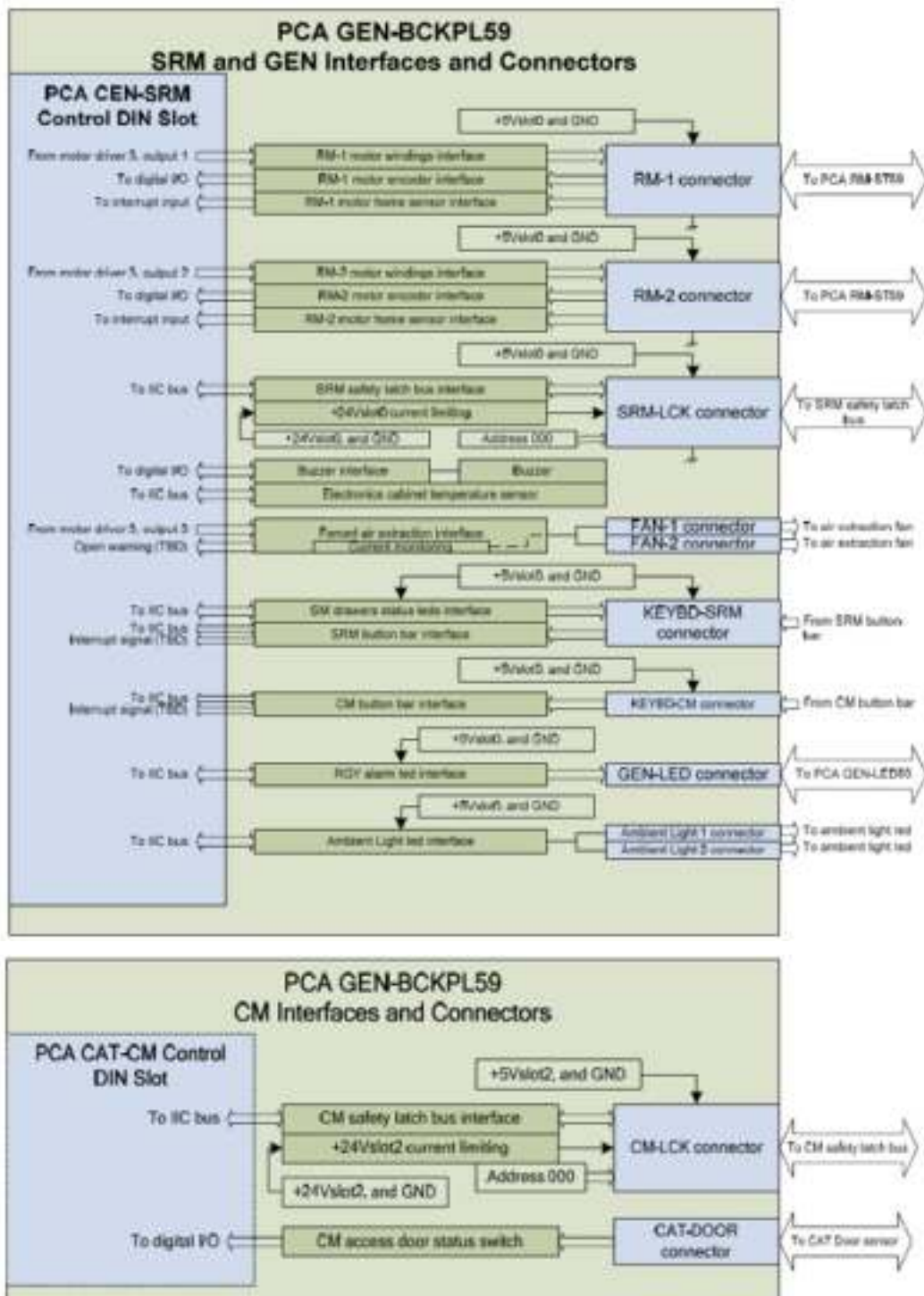


Figure 118 - Block diagram of General module

PCAs related to General module are:

- **PCA GEN-BCKPL59:** see "PCA GEN-BCKPL59" on page 138.
- **PCA GEN-SRM Control:** located at the back panel (slot 0), controls all the signals used in the centrifuge module and samples/reagents management module. It contains the firmware of both modules.
- **PCA CAT-CM Control:** located at the back panel (slot 2), controls all the signals used in the card transport module and card management module. It contains the firmware of both modules.
- **PCA GEN-STDBY511:** it contains the standby power supply system and interfaces the Power Supply system to the computer (PC) and PCA GEN-BCKPL59. It is able to activate / deactivate the power supply of the PC and electronics control unit (power and control).
- **PCA GEN-LED59:** it contains a high-power LED and by means of current fixing resistors, it is possible to obtain three different colors (red, green and blue). One additional color, yellow, is obtained with a combination of red and green. There are three PCAs on the instrument:
 - One PCA located at the top right of the instrument (Status LED).
 - One PCA located under the electronics drawer (Ambient light).
 - One PCA located under the intermediate floor (Ambient light).
- **PCA GEN-FRNT59:** it supports and interfaces the standby switch button and two USB Type A ports located at the front of the instrument. This PCA is located under the cover of the dilution cup, at the front of the instrument.
- **PCA GEN-TEMP59:** it contains the temperature sensors to monitor temperature at both instrument floors. There are two PCA located at:
 - Between the upper and the intermediate floor, at the rear part of the instrument, under the dilution cup area.
 - At the rear part of the card drawer number 4, in the intermediate floor.
- **PCA GEN-LCK59:** it supports latch electronic control. It contains the safety latch interface (activate / deactivate) and it monitors its status (opened / closed). This PCA is located at the rear part of the reagents / samples / cards drawers.
- **PCA RM-ST59:** it interfaces the stirrer motor and the home detector of the stirrer. It is located under the reagents drawers.
- **PCA RM-CON59:** it interconnects the PCA RM-ST59 with the PCA GEN-BCKPL59.

2.1.2 Mechanical adjustments

Mechanical adjustments related to the General (GEN) module are detailed in the following chapters.

2.1.2.1 Adjusting upper / intermediate area door

This adjustment is required to set the reagents-samples and cards door front panel depth into the instrument in order to align it with the rest of the instrument frontal.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. To adjust the parallelism of the door, loosen the screws that attach the magnets to the Erytra chassis.



Figure 119 - Magnet of the doors

2. Adjust the position of the magnets by means of the slotted holes in order to place the reagents and samples door parallel to the chassis once it is closed.

2.1.2.2 Adjusting touchscreen drawer lock

This adjustment is required to set the touchscreen monitor housing front panel depth into the instrument in order to align it with the rest of the instrument front.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. To adjust the position of the touchscreen monitor inside its drawer, loose the two screws of the lock and adjust the position by means of its slotted holes. Access to the lock screws is reached through CAT area.

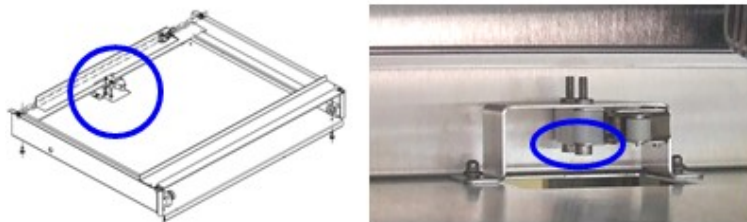


Figure 120 - Lock of the touchscreen door

2. Insert the screen support plate into the instrument. Check that the two flags at the end of the screen support plate are inserted into the two slots.



Figure 121 - Slots to insert the flags

3. Check that touchscreen monitor is aligned with the frontal of the instrument in closed position.

2.1.2.3 Adjusting user tray drawer lock

This adjustment is required to set the user tray front panel depth into the instrument in order to align it with the rest of the instrument frontal.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Extract the user tray from the instrument. Remove the screws located under the reagents / samples keyboard and extract the user tray.
2. Loosen the two screws of the lock located at the rear part of the user tray and adjust the position by means of its slotted holes.



Figure 122 - Lock of the user tray

3. Insert the user tray into the instrument. Check that the two pins at the end of the user tray are inserted into the two slots.



Figure 123 - Slots to insert the pins of the user tray

4. Tighten the screws that attach the user tray to the instrument.
5. Check that the user tray is aligned with the front of the instrument in closed position.

2.1.2.4 Adjusting safety latch reagents, samples and cards drawers

This adjustment is required to set the depth of the drawer front panel into the instrument in order to align it with the rest of the drawers and the instrument front.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233824 DRAWER FRONT PANEL ALIGNMENT TOOL.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the drawer to be adjusted. The safety latch is located in the rear part of the drawer. Open the upper door (reagent / samples drawer) or the intermediate door (cards drawer) to have better access.
2. Loosen the nut attached to the safety latch, and loosen or screw the safety latch in order to align the drawer.

3. Check if the drawer is aligned, place the alignment tool and close the drawer to check if the front panel is aligned with the front of the instrument.
4. Once the drawer is aligned, tighten the nut.



Figure 124 - Safety latch adjustment with drawers front panel alignment tool



Figure 125 - Latches of reagents / samples drawers (left) and cards drawers (right)

5. Close the drawer and check its alignment.
6. Perform probes adjustment in reagents / samples drawers (see "Adjusting reagents drawers" on page 338) or gripper adjustment in cards drawers (see "Adjusting gripper in the card drawers area" on page 472) depending on which safety latch has been adjusted.

2.1.2.5 Adjusting front panel of reagents, samples and cards drawers

This adjustment is required to set the front panel of the drawers in order to align it with the rest of the drawers and the instrument front.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233824 DRAWER FRONT PANEL ALIGNMENT TOOL.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the drawer to be adjusted.
2. Remove the front panel cover by means of the screws attached to each side.
3. Loosen the four screws that are attaching the front panel to the drawer.



Figure 126 - Cosmetic front panels screws. Samples / reagents (left) and cards (right)

4. Place the alignment tool and move in Z and X axis on the front panel in order to align it. Once the frontal panel is aligned, tighten the four screws.



Figure 127 - Front panel adjustment with drawers front panel alignment tool

5. Place the frontal panel cover and tighten the screws.
6. Close the drawer and check its alignment.

2.1.2.6 Adjusting tension of the belt of the reagents drawers

The purpose of this adjustment is to adjust the tension of the belt of the motors located under the reagents drawers.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the reagents drawer to be adjusted.
2. Loosen the screws of the motor.
3. Move the motor to adjust the belt tension and tighten the screw of the slotted hole. Then tighten the other screw.
4. Verify reagents stirring process. See "Verifying drawers" on page 184.

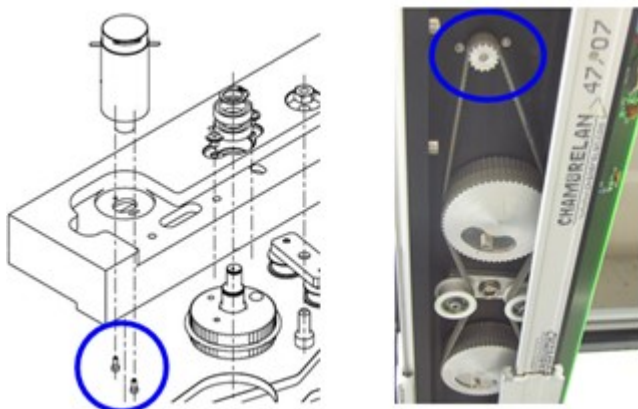


Figure 128 - Reagents motor screws

2.1.2.7 Adjusting reagents drawers flatness

The purpose of this procedure is to adjust the flatness of the samples drawers.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the reagent drawer.
2. Remove the plate of the magnetic brake of the reagent drawer.



Figure 129 - Magnetic brake screws

3. Loosen the four screws of the vertical linear guide and then modify manually the position of the drawer (there are slotted holes) to adjust the flatness and finally tighten the screws. Difference in counts between the maximum and the minimum encoder counts in four different points should be ≤ 1047 . See "Verifying the flatness of reagent drawer" on page 186.



Figure 130 - Vertical linear guide screws - reagents drawers

4. Install the plate of the magnetic brake.

5. Perform probes adjustment in the reagent drawer. See "Adjusting reagents drawers" on page 338.

2.1.2.8 Adjusting samples drawers flatness

The purpose of this procedure is to adjust the flatness of the samples drawers.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the sample drawer to be verified and remove the front panel cover by means of its four screws to get access to the screws of the vertical linear guide.
2. Loosen the four screws of the vertical linear guide, modify manually the position of the drawer (there are slotted holes) to adjust the flatness and finally tighten the screws. Difference in encoder counts between left and right part of the previous positions should be ≤ 1047 . See "Verifying the flatness of sample drawer" on page 187.



Figure 131 - Vertical linear guide screws - samples drawers

3. Install the front panel cover.

4. Perform probes adjustment in the samples drawer. See "Adjusting samples drawers" on page 342.

2.1.2.9 Adjusting cards drawers flatness

The purpose of this procedure is to adjust the flatness of the cards drawers.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the card drawer.
2. Remove the front panel cover by means of its two screws.
3. Loosen the four screws that attach the vertical linear guide to the drawer.



Figure 132 - Vertical linear guide screws - cards drawers

4. Modify manually the position of the drawer due to the slotted holes of the cover and tighten the screws of the linear guide to adjust the flatness. Difference in encoder counts between left and right part of the same rack (1 and 4) is ≤ 600 . See "Verifying the flatness of card drawer" on page 189.



Figure 133 - Card drawer flatness adjustment

5. Install the front panel cover.
6. Perform gripper adjustment in the drawer. See "Adjusting gripper in the card drawers area" on page 472.

2.1.3 Functional adjustments

Functional adjustments related to the General (GEN) module are detailed in the following chapters.

2.1.3.1 Adjusting touchscreen

This procedure is needed to calibrate the touchscreen, in order to make sure that the point touched on the screen matches with the point where the cursor of the OS is placed.



NOTE

The touchscreen calibration process must be done in Maintenance mode of the software. Once the calibration is finished, shut the system down so that the next time it can be started in the desired mode.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

2.1.3.1.1 Touchscreen v2 (SN < 059-0001515)

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the **Maintenance Mode** icon among the Boot Application options.
2. Once the PC reboots and the desktop is available, press the **Windows Start** button and select **MicroTouch MT 7** program, then click on **Control Panel** among the drop down options.

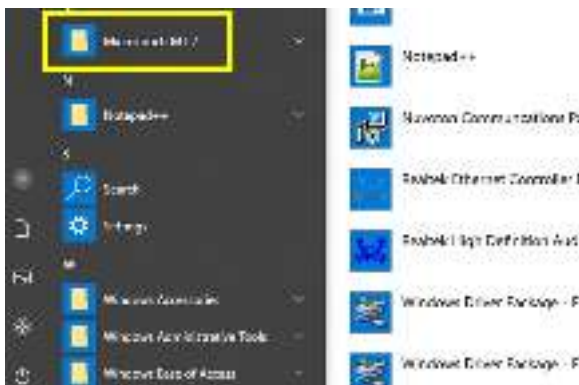


Figure 134 - MicroTouch MT 7 program

3. Select the **Main** tab and click the **Calibrate** button. The calibration target () button will be displayed.
4. Touch the target centre for several seconds and then lift your finger off (the target will change to red colour). Repeat this step for the next two calibration points.
5. Once you have calibrated the touchscreen, a dialog box will appear giving the option to test, accept or cancel the performed calibration. Finish the calibration process by pressing the **Accept** button (do not test).
6. Verify the calibration accuracy. If it is not accurate enough then repeat the calibration and verify again.

7. If suitable accuracy cannot be achieved, perform a 'Linearization':
 - a. In Micro Touch MT 7 Control Panel, select the **Tools** tab and click **Advanced Options**.
 - b. Check the **Enable manual linearization** option and click **OK**.
 - c. Select the **Controller** tab and click the **Linearize** button.
 - d. Twenty five calibration targets () (one at a time) will appear. Press with a pointer over the centre for several seconds until changes to red and remove the pointer.
 - e. Finish the process by pressing **Accept** (do not test).
 - f. Verify calibration accuracy and, if considered necessary, perform touchscreen calibration again.
8. Switch off the instrument.

2.1.3.1.2 Touchscreen v3 (SN ≥ 059-0001515)

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the **Maintenance Mode** icon among the Boot Application options.
2. Once the PC reboots and the desktop is available, go to **Start > Windows System > Control Panel**.
3. Select **Hardware and Sound > Tablet PC Settings > Calibrate the screen for pen or touch input** option.

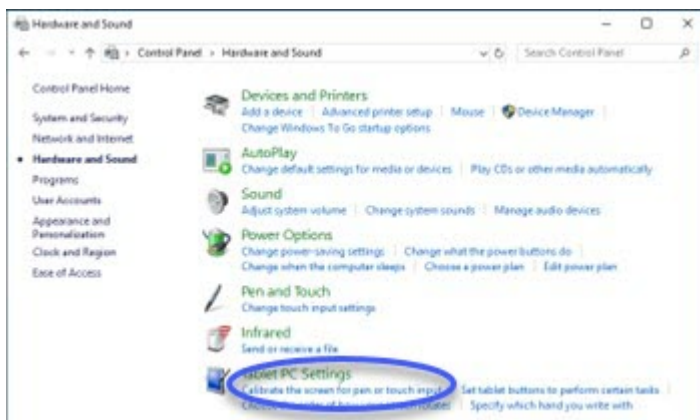


Figure 135 - Tablet PC Settings

- The Tablet PC Settings window will be displayed. Select **Calibrate** option in the **Display** tab and enter the admin user password in the **User Account control** window (if displayed).



Figure 136 - Tablet PC Settings window

- The calibration process will start, requesting to tap the crosshair each time that it appears on screen.



Figure 137 - Calibration screen

- It will display 16 crosshairs (four in each corner) the first time the touchscreen is calibrated (or when the **Tablet PC Settings > Reset** option has been selected) and just four crosshairs the next time (only inner lines are displayed).

- Once the calibration process finishes, a new **Digitizer Calibration Tool** message window asking whether data should be saved will be displayed. Select **Yes**.

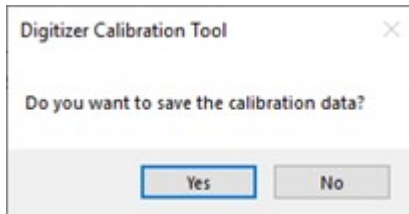


Figure 138 - Save calibration data

- Click the **Tablet PC Settings > OK** button to close this window.
- Close the **Hardware and Sound** window.
- Switch off the instrument.

2.1.3.2 Adjusting touchscreen monitor menu

The holes in the rear part of the touchscreen drawer allow accessing the monitor menu buttons, so that settings such as brightness, contrast, etc. can be customized. Default brightness values are set to 100 for the monitor with touchscreen v2 and 50 for the one with touchscreen v3.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

2.1.3.2.1 Touchscreen v2 (SN < 059-0001515)

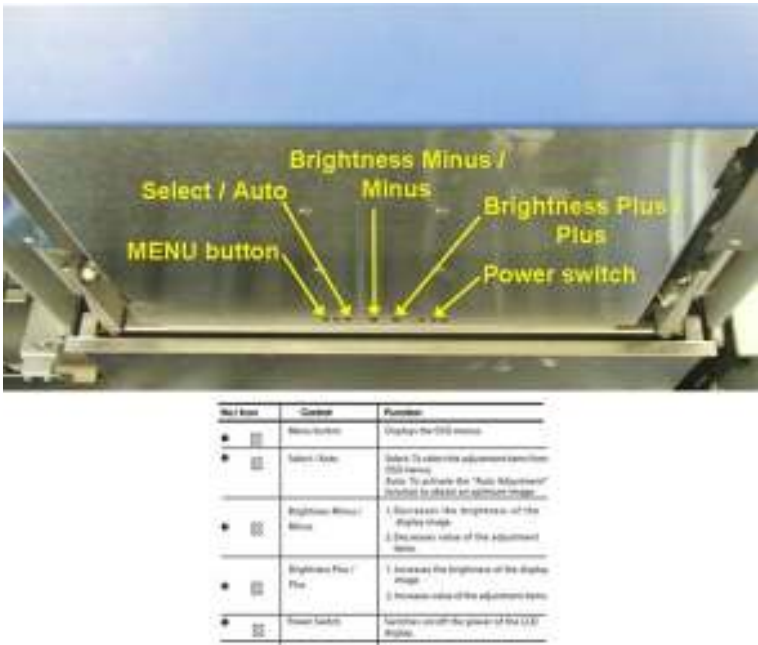
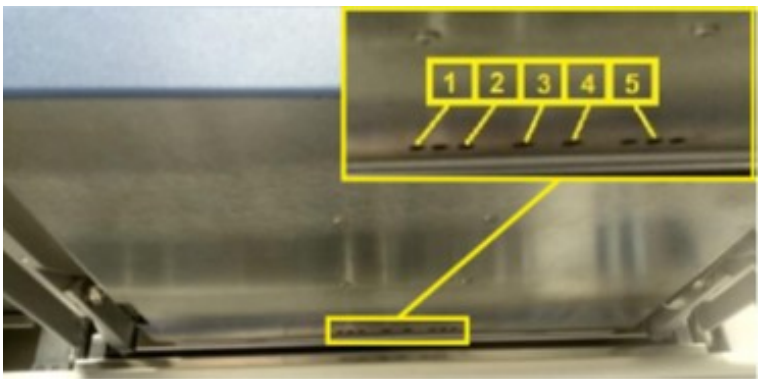


Figure 139 - Touchscreen v2 buttons

2.1.3.2.2 Touchscreen v3 (SN ≥ 059-0001515)



Button number	Function	Description
1	MENU	Activate menu / Select menu
2	SELECT	Menu exit / Auto adjust
3	DOWN	Cursor control down / Display monitor information
4	UP	Cursor control up / Source change (VGA, DVI, Auto select)
5	POWER	Power on / off

Figure 140 - Touchscreen v3 buttons

2.1.4 Verification procedures

In this section the verification of parts of the General (GEN) module are detailed.

2.1.4.1 Verifying power supply (SN < 059-0001583)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check fuses of the mains connector.
2. Check that all the cables are correctly connected.
3. Connect Erytra to the mains and switch on the button from mains connector.
4. Switch on the instrument and check voltages at the output of power supply with a calibrated multimeter:
 - +24 VDC $\pm$ 2% (23,52 V to 24,48 V): motors, syringes, solenoids, electrovalves, pumps and Peltier cells.
 - +5 VDC $\pm$ 4% (4,8 V to 5,2 V): control electronics logic.
 - +12 V $\pm$ 2% (11,76 V to 12,24 V): PC supply.
 - -12 V $\pm$ 2% (-11,76 V to -12,24 V): PC supply.
 - +5 V $\pm$ 4% (4,8 V to 5,2 V): PC supply.
 - +3.3 V $\pm$ 4% (3,17 V to 3,43 V): PC supply.
 - +5 V $\pm$ 4% (4,8 V to 5,2 V): standby power.

2.1.4.2 Verifying PCA GEN-STDBY



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check that all the cables are correctly connected.
2. Connect the instrument to the mains and switch on the instrument.
3. Check that the LED is on.

2.1.4.3 Verifying PCA GEN-BCKPL59



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check that all the cables are correctly connected.
2. Check that all fuses are present.
3. Connect Erytra to the mains and switch on the instrument.
4. Check status of fuses by means of their LEDs.

2.1.4.4 Verifying computer



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check that the computer is switched on when the push button is pressed. Once pressed, the green LED of the push button is lit.
2. Check that the OS is loaded correctly and that there are no problems in the hardware (**Start menu > Control Panel > System and Security > Device Manager**).

2.1.4.5 Verifying touchscreen monitor



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check that the mechanism of the touchscreen monitor opens smoothly.
2. Check that it is possible to tilt the touchscreen monitor when it is extracted completely from its drawer.
3. Check that when the touchscreen monitor is tilted is not possible to enter it in its drawer.
4. Check that the lock system works properly.
5. Check that the point where you put the pointer on the screen is the same that the point where the cursor is placed.

2.1.4.6 Verifying status LEDs



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the Technical Service software, go to the **General** tab (see "General tab" on page 832) and switch on the different colours of the upper general LED. Then switch off the light.
2. In the same tab, check in the same way the LEDs of the keyboard of the samples drawers.

2.1.4.7 Verifying ambient light



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. In the Technical Service software, go to the **General** tab (see "General tab" on page 832) and switch on the ambient lights. Check visually that they are on.
2. Switch off the lights, checking visually that they are off.

2.1.4.8 Verifying user tray



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check that the user tray opens smoothly.
2. Check that the lock system works properly.

2.1.4.9 Verifying temperature control system



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the Technical Service software, go to the **General** tab (see "General tab" on page 832) and press the **general fans ON** button. Check that the fans are switched on.
2. Press the **general fans OFF** button and check that they are switched off.
3. Check that the temperature in the electronics cabinet, reagents and samples floor and cards floor returned by the software is updated and correct.

2.1.4.10 Verifying buzzer



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

- Open the Technical Service software, go to the **General** tab (see "General tab" on page 832) and activate the buzzer (the **Buzzer Mode** section) with the **MODE1 ... MODE7** options checking its sound.

2.1.4.11 Verifying keyboards



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

- Open the Technical Service software, go to the **General** tab (see "General tab" on page 832) and press the buttons of the keyboard and check that its associated key (under the **Keyboard** section) is activated. Check that the drawer is opened (before pressing different buttons wait for two seconds).

2.1.4.12 Verifying doors



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the Technical Service software, go to **General** tab (see "General tab" on page 832) and check that the status of the upper area door is correctly detected (the **SRM Door** section).
2. Open the Technical Service software, go to **General** tab and check that the status of the intermediate area door is correctly detected (the **CM Door** section).

2.1.4.13 Verifying drawers



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

2.1.4.13.1 OPEN / CLOSE

1. In the Technical Service software, perform a home of the PIP arm, by going to **Pip Arm > Movement > home** (see "Pip Arm tab" on page 804) and then selecting **all axes**.
2. In the **Procedures** tab, press the **verify** button of **Drawers Module** located in the **Checks** section.
3. It will start a wizard in order to check if the reagents / samples / cards drawers are detected in their opened / closed positions.
4. Close all the reagents / samples / cards drawers and press the **Next** button.
5. Open all the reagents / samples / cards drawers and press the **Next** button.
6. Close all drawers and press the **Next** button.
7. A screen appears showing the cards detection status. In case status detection has been performed correctly, pass result is shown.

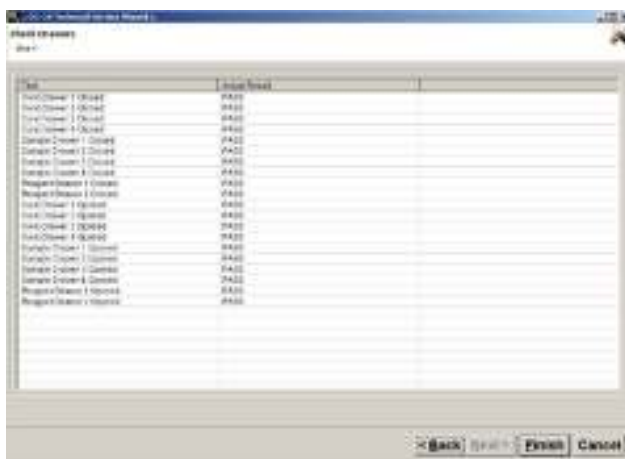


Figure 141 - Verification drawers wizard

8. Press **Finish** and **OK** in the information window.
9. Press the **OK** button in the confirmation window that indicates that home movements will be performed.
10. Press the **OK** button to finish the verification procedure.

2.1.4.13.2 REAGENTS STIRRING

1. Open the Technical Service software, go to the **Home Cycles** subtab in the **Drawers** tab. See "Drawers tab" on page 792.
2. Press the **start** button for **Reagent Drawer 1** and / or **Reagent Drawer 2** in order to start cycles of home detection of the reagent racks. Check that no errors appear.
3. In the Technical Service software, go to the **Reagents Drawers** subtab in the **Drawers** tab.
4. Select **REAGENT_DRAWER_1** and / or **REAGENT_DRAWER_2** with **PROFILE_0** and press **Shake** to start a stirring process. Check that no errors appear.

2.1.4.14 Verifying the flatness of reagent drawer



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the reagents racks of the drawer to be verified.
2. Remove the reagents racks platforms from the cover of the drawer.
3. Perform a home of probe 1 using the **home** button in the Technical Service software, go to **Pip Arm > Movement > Position Actions** (see "Pip Arm tab" on page 804) and then selecting **z1** axis.
4. De-energize the PIP robotic head the using the **unlock / stop** button in the Technical Service software, go to **Pip Arm > Movement > Position Actions** and then selecting **z1** axis.
5. Move manually the robotic head over the drawer to be checked.

6. Lower probe 1 manually (by means of its belt) until the probe touches gently the following positions, checking in all of them Z1 encoder counts.
 - a. Position 4 of rack 1.
 - b. Between positions 5 and 6 of rack 1.
 - c. Position 9 of rack 2.
 - d. Position 10 of rack 2.

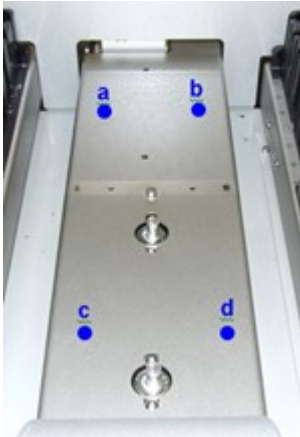


Figure 142 - Flatness verification points in reagent drawer

7. Check that the difference between the maximum and the minimum encoder counts of the previous four points is ≤ 1047 . Otherwise the reagent drawer flatness should be adjusted. See "Adjusting reagents drawers flatness" on page 167.

2.1.4.15 Verifying the flatness of sample drawer



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the samples racks of the drawer to be verified.
2. Perform a home of probe 1 by using the **home** button in the Technical Service software, go to **Pip Arm > Movement > Position Actions** (see "Pip Arm tab" on page 804) and then selecting **z1** axis.
3. De-energize the PIP robotic head by using the **unlock / stop** button in the Technical Service software, go to **Pip Arm > Movement > Position Actions** and then selecting **z1** axis.
4. Move manually the robotic head over the drawer to be checked.
5. Lower probe 1 manually (by means of its belt) until the probe touches gently the position corresponding to sample tube 1 in rack 1.



Figure 143 - Left verification in rack 1 of sample drawer

6. Check Z1 encoder counts in the Technical Service software.
7. Repeat the same procedure for position of sample tube 2 in rack 1. Check that the difference in counts between the two positions is ≤ 1047 encoder counts.

8. Finally compare if the Z1 encoder counts between positions of samples tubes 11 and 12 in rack 2 is also ≤ 1047 encoder counts by means of lowering probe 1 manually (by means of its belt) until the probe touches gently these positions.



Figure 144 - Flatness verification points in sample drawer

9. In case the difference of any of the two comparisons between left and right side is > 1047 encoder counts, the flatness of the drawer should be adjusted. See "Adjusting samples drawers flatness" on page 169.

2.1.4.16 Verifying the flatness of card drawer



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the card racks of the drawer to be verified if any.
2. Perform a home of gripper using the **home** button in the Technical Service software, go to **Gripper Arm > Movement > Position Actions** (see "Gripper Arm tab" on page 777) and then selecting **Z** axis.
3. De-energize CAT robotic head using the **unlock / stop** button in the Technical Service software, go to **Gripper Arm > Movement > Position Actions**.
4. Move manually the robotic head over the drawer to be checked.
5. Lower the gripper manually (by means of its belt) until the gripper rear main surface touches gently the left part of the rear centre of rack 1 of the selected drawer.



Figure 145 - Rear left part verification in rack 1

6. Check the Z encoder counts in the Technical Service software.
7. Perform a home of the gripper, unlock the axes by means of the Technical Service software and then lower the gripper manually (by means of its belt) until the gripper rear main surface touches gently the right part of the rear centre of rack 1 of the selected drawer.



Figure 146 - Rear right part verification in rack 1

8. Check the Z encoder counts in the Technical Service software.

9. Repeat previous steps (from 4 to 7) in the frontal center of rack 4 of selected drawer.



Figure 147 - Frontal right part (right picture) and frontal left part (left picture) verification in rack 4

10. Check if the difference in counts between left and right part of the same rack is ≤ 600 . Otherwise adjust the flatness of the drawer.
See "Adjusting cards drawers flatness" on page 170.

2.1.5 Replacement procedures

This section details replacement procedures for different parts of the General module.

2.1.5.1 Replacing power supply (SN < 059-0001583)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233756 POWER SUPPLY.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

The procedure to replace the power supply is the following:

1. Switch off the instrument and disconnect it from the mains.
2. Take out the rear upper cover to get access to the electronic drawer and disconnect the output cables from the power supply.
3. Remove the input connectors cover of the PCA GEN-STDBY59 by means of its two screws.

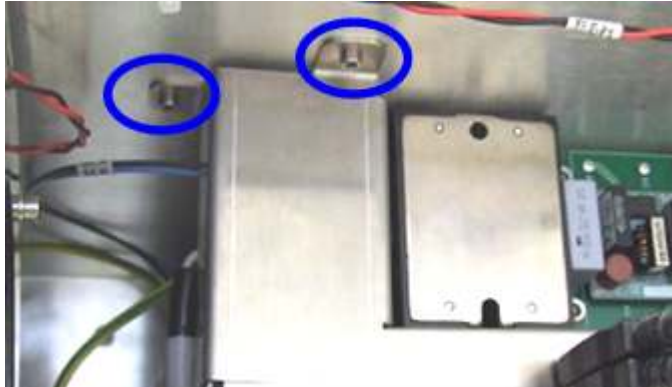


Figure 148 - Input connectors cover of the PCA GEN-STDBY59

4. Disconnect the ELE-AC cable from the PCA GEN-STDBY59.
5. Unscrew the PCA GEN-STDBY59 that is attached to the power supply.
6. Remove top cover of the instrument by means of its four screws.
7. Remove the four screws that hold the power supply to the upper part of the instrument. Be careful when the four screws are removed because the power supply can fall over the computer.
8. Remove from the instrument the power supply. Before to remove completely, disconnect the ground cable and the ELE-AC cable from the input of the power supply
9. Place the new power supply and connect all the cables.
10. Repeat steps 1 to 7 in reverse mode.

Table 6 - List of power supply cables

CABLE	COLOR	OUTPUT	VOLTAGE
ELE-V1	Orange	V1 +	+5V
	Blue	V1 -	
ELE-V1_EARTH	Yellow / Green	V1 -	-

CABLE	COLOR	OUTPUT	VOLTAGE
ELE-V2-	Black	V2 -	+24V
	Black	V2 -	
	Black	V2 -	
ELE-V2+	Red	V2 +	
	Red	V2 +	
	Red	V2 +	
ELE-V2_EARTH	Yellow / Green	V2 -	-
ELE-V3+	Red	V3 +	+5V
	Red	V3 +	
ELE-V3-	Black	V3 -	
	Black	V3 -	
ELE-V4	Orange	V4 +	+3V3
	Black	V4 -	
ELE-V5	Yellow	V5 +	+12V
	Black	V5 -	
ELE-V6	Black	V6 +	-12V
	Blue	V6 -	

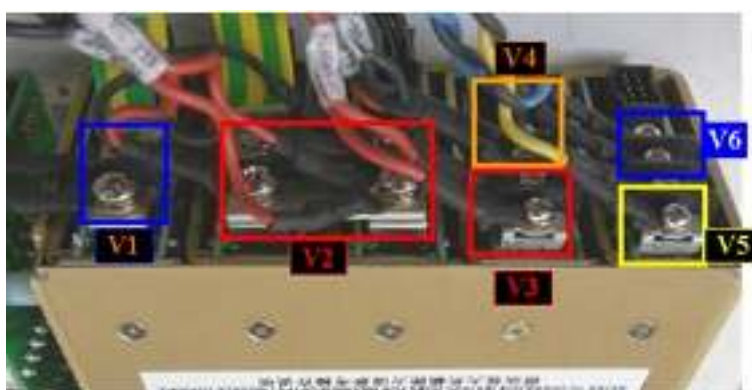


Figure 149 - Position of the cables in the power supply as installed on the instrument

2.1.5.2 Replacing touchscreen monitor



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- The touchscreen V2 spare part no longer exists.
- [[[Undefined variable spare-parts-number.732288]]] [[[Undefined variable spare-parts-description.732288]]].

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

The procedure to replace the touchscreen monitor is listed below for each touchscreen model:

(Touchscreen v2 (SN < 059-0001515))

1. Make sure that the instrument is switched off and disconnect it from the mains.
2. Extract the touchscreen drawer from the instrument by pushing it and then pulling it outwards.
3. Remove the six screws attaching the monitor cover to the touchscreen drawer (three at each side, underneath the drawer).
4. Remove the four screws attaching the monitor to the touchscreen drawer.

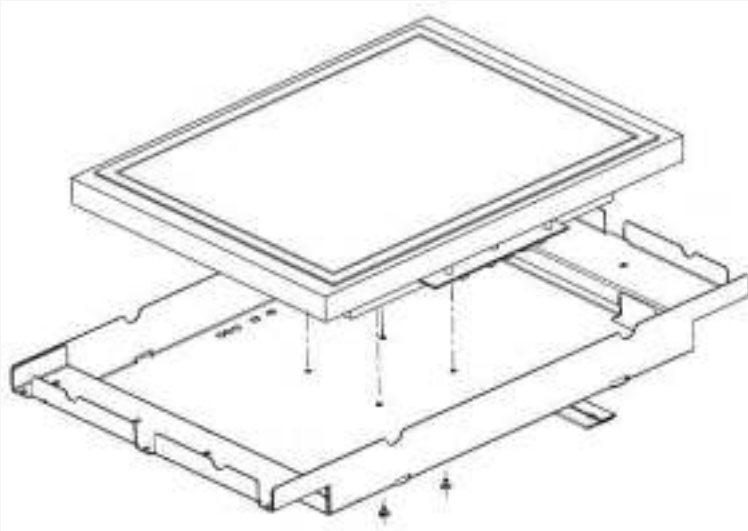


Figure 150 - Screws attaching the monitor to the touchscreen drawer

5. Carefully lift up the monitor and disconnect the VGA, USB and power cables.
6. Remove the USB PCA located in the replaced touchscreen monitor and take the two hexagonal spacers. Note that one of the spacers is larger than the other one.



Figure 151 - Hexagonal spacers

7. Remove the USB PCA from the new touchscreen monitor and the plate attached.



Figure 152 - USB PCA in new touchscreen monitor

8. Install the USB PCA in the new touchscreen monitor in the same place than the replaced touchscreen monitor using the hexagonal spacers. The larger spacer is the one acting as a stopper.

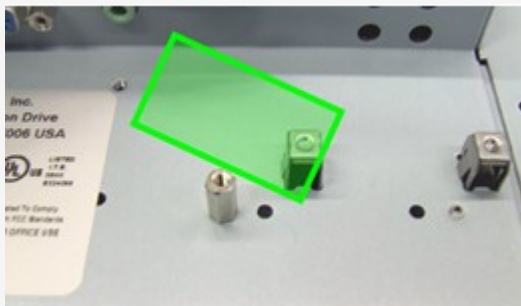


Figure 153 - USB PCA installation place

9. Place the new touchscreen monitor, connect the cables and repeat steps 1 to 4 in reverse order.
10. Perform driver installation. See "Software information" on page 915.
11. If necessary, execute adjustment procedure. See "Functional adjustments" on page 171.

Touchscreen v3 (SN ≥ 059-0001515)

1. Make sure that the instrument is switched off and disconnect it from the mains.
2. Extract the touchscreen drawer from the instrument by pushing it and then pulling it outwards.
3. Remove the six screws attaching the monitor cover to the touchscreen drawer (three at each side, underneath the drawer).
4. Remove the four screws attaching the monitor to the touchscreen drawer.

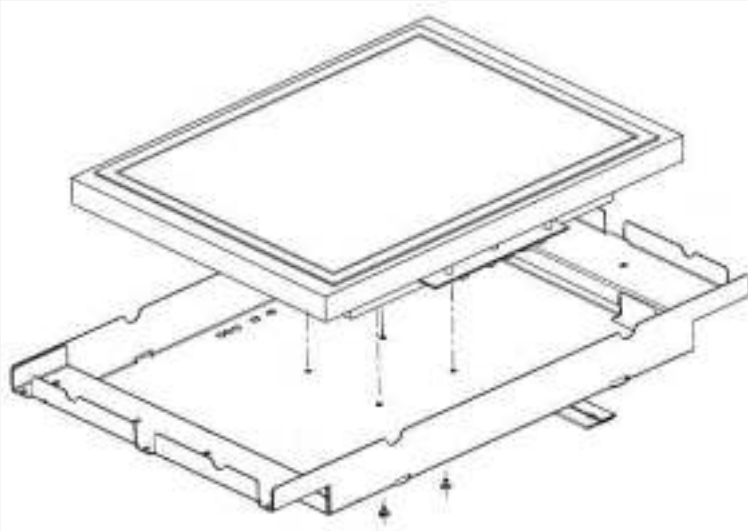


Figure 154 - Screws attaching the monitor to the touchscreen drawer

5. Carefully lift up the monitor and disconnect the VGA, USB and power cables.
6. Remove the monitor, placing the new one in its place. Make sure that the PCA including the monitor configuration buttons is on the inner part of the drawer.
7. Carefully lift up the monitor and connect the VGA, USB and power cables, using the adapter cable for the last one. Consider using a cable tie to hold the three cables together in the space underneath the lower side of the monitor.



Figure 155 - From left to right: Power (adapter), VGA, USB cables

8. Attach the monitor to the touchscreen drawer using the four previously removed screws.

9. Place the monitor cover back in its place, using the previously removed six screws.
10. Perform driver installation. See "Software information" on page 915.
11. If necessary, execute adjustment procedure. See "Functional adjustments" on page 171.

2.1.5.3 Replacing reagents and samples drawers keyboard



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233964 KEYBOARD REAG/SAMPLE DRAWER VER.2.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

The procedure to replace the keyboard of the reagents/samples drawers is the following:

1. Extract the user tray from the instrument removing the two screws that attach it to the handrail.
2. Disconnect the VGA, power and USB cables from the computer.
3. Remove the two frontal screws of the touchscreen drawer and the two frontal set screws located rear them.

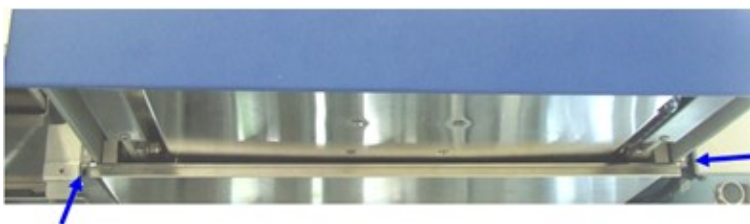


Figure 156 - Touchscreen drawer assembly

4. Remove the two set screws located at the rear part of the drawer. To get a better access, it is recommended to remove the blue right lateral cover of the instrument and the PCA GEN-LED59 (ambient light) of the card drawers area.



Figure 157 - Rear set screws in touchscreen drawer

5. Extract as much as possible the touchscreen drawer from the instrument.
6. Remove the keyboard cable protective cover by means of loosening the four screws and moving the cover to the right and pulling it.
7. Remove the four nuts that attach the screws of the keyboard to the handrail.
8. Before removing completely the keyboard from the instrument, disconnect the keyboard cable.
9. Place the new keyboard and repeat steps 1 to 4 in reverse mode.

2.1.5.4 Replacing card drawers keyboard



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233965 KEYBOARD CARDS DRAWER VER.2.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

The procedure to replace the keyboard of the card drawers is the following:

1. Open completely all the card drawers in order to get a better access to the screws.
2. Remove the keyboard cable protective cover located under card drawer four by means of loosening the four screws and moving the cover to the right and pulling it.



Figure 158 - Position of the protective cover in the card drawers keyboard

3. Remove the four nuts that attach the screws of the keyboard to the handrail. In order to remove the nut close to the drawer 3, it is recommended to take out the pot of the centrifuge 2.

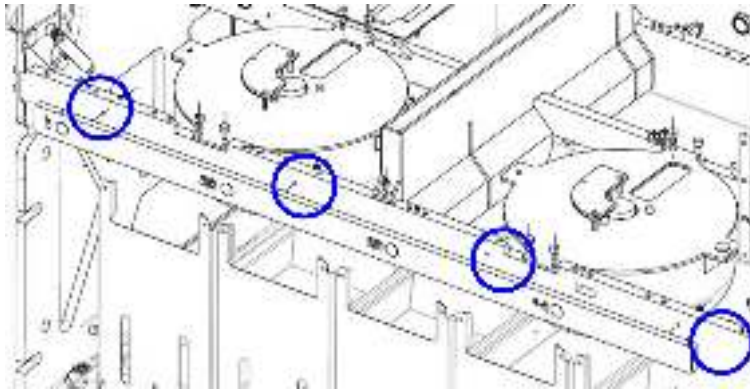


Figure 159 - Position of the nuts of the card drawers keyboard

4. Before removing completely the keyboard from the instrument, disconnect the keyboard cable.
5. Place the new keyboard and repeat steps 1 to 4 in reverse mode.

2.1.5.5 Replacing PCA RM-ST59



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233713 PCA RM-ST59.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

The procedure to replace the PCA RM-ST59, located under the reagent drawer, is the following:

1. Open the reagents drawer that contains the PCA RM-ST59 to be replaced.
2. Remove the reagents racks.
3. Remove both reagents platforms.
4. Take out the cosmetic cover of the motor.
5. Remove carefully the basis of the drawer by means of its screws.
6. Disconnect the cable from the motor (power and encoder) and the cable located at the end of the drawer.
7. Replace the PCA removing its four screws and repeat the steps 1 to 6 in reverse mode.
8. Verify the performance of the reagents stirring function.

2.1.5.6 Replacing RAM memory modules



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- SN < 059-0000987: //732489 //PC KONTRON RAM DDR2 2 GB v3 (2 units).
- SN < 059-0000987 and the PC has been updated: //734048 //RAM DDR3 2GB SO-DIMM KINGMAX (2 units).
- SN ≥ 059-0000987: //734048 //RAM DDR3 2GB SO-DIMM KINGMAX (2 units).
- SN ≥ 059-000XXXX: 735899RAM DDR3L 8GB SO-DIMM.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace RAM memories, part number 735899 RAM DR3L 8GB SO-DIMM for IMB-150D, follow the next steps:

1. Switch off the instrument and disconnect it from the mains.
2. Remove the screws that attach the instrument upper rear cover, then remove it.
3. For instruments with SN < 059-0001583, refer to INR-0059#0111: NEW ELECTRONICS CABINET (see "Bulletin index" on page 1183), remove two screws of the metallic plate.



Figure 160 - Metallic plate screws

4. For instruments with SN $\geq$ 059-0001583, to access the screws on the right side of the computer cabinet, remove the fan and its support by removing two screws.
5. Disconnect the fan cable.



Figure 161 - Fan cable disconnection

6. Remove the four screws that attach ASROCK computer cabinet (rear side of the instrument).



Figure 162 - ASROCK computer fitting screws

7. Disconnect all ASROCK computer cabling and remove the computer cabinet.



Figure 163 - ASROCK computer

8. Remove the six screws of the cover of the ASROCK computer cabinet to have access to the two RAM memories, pull apart the cover carefully as there is one cable connected.

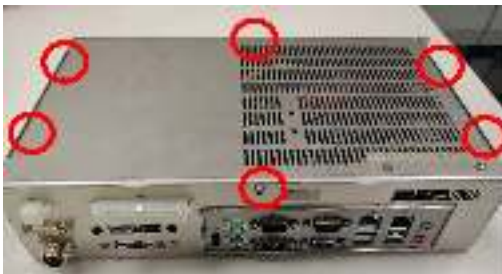


Figure 164 - ASROCK computer cover screws

9. Remove the RAM memories from their slots in the ASROCK computer.



Figure 165 - RAM memory location

10. Install the new RAM memories and reassemble all the parts again.

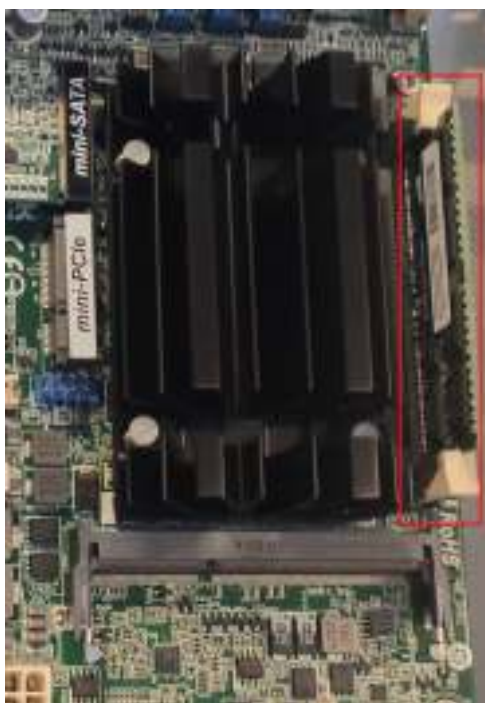


Figure 166 - RAM memory location



NOTE

Only one 8GB RAM module is to be installed on the primary slot. Otherwise the instrument would not start.

11. Finally, switch on the instrument, enter the boot application menu and choose Maintenance Mode. Then check in **Control Panel > System and Security > System** that 8GB of RAM are detected.

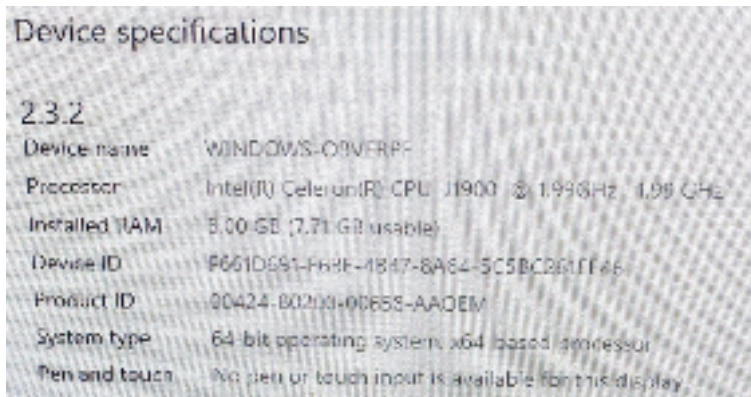


Figure 167 - Installed memory detected in Control Panel

2.1.5.7 Replacing PC motherboard



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 734235 MOTHERBOARD v3 (MINI-ITX ASROCK IBM-150).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the PC motherboard, follow these steps:

1. Switch off the instrument and disconnect it from the mains.
2. Remove the screws that attach the instrument upper rear cover, then remove it.

3. For instruments with SN < 059-000158, refer to INR-0059#0111: NEW ELECTRONICS CABINET (see "Bulletin index" on page 1183) remove two screws of the metallic plate.



Figure 168 - Metallic plate screws

4. For instruments with SN ≥ 059-0001583, to access the screws on the right side of the computer cabinet, remove the fan and its support by removing two screws. Disconnect the fan cable.

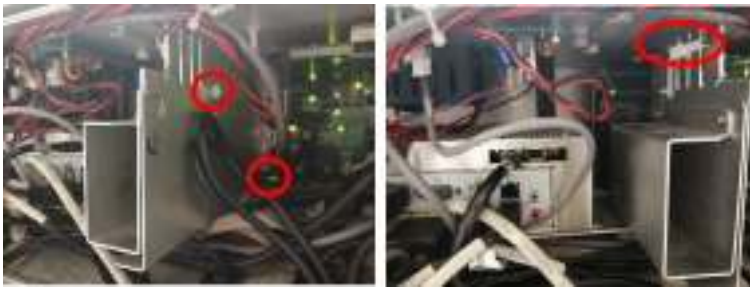


Figure 169 - Fan cable disconnection

5. Remove the four screws that attach ASROCK computer cabinet (rear side of the instrument).



Figure 170 - ASROCK computer fitting screws

6. Disconnect all ASROCK computer cabling and remove the computer cabinet.



Figure 171 - ASROCK computer

7. Remove the six screws of the cover of the ASROCK computer cabinet to have access to the PC motherboard, pull apart the cover carefully as there is one cable connected.



Figure 172 - ASROCK computer cover screws

8. Disconnect all the cables connected to the ASROCK computer and remove the board by unscrewing the four screws attaching it to its base plate in the computer cabinet.



Figure 173 - Motherboard location

9. Install the new PC motherboard and reassemble all the parts again.
10. Connect an external keyboard, switch on the instrument and press Del/Supr key during the initialization of the computer to enter in the BIOS/UEFI menu.



NOTE

A password to access the BIOS has been set. When requested enter password: TSDGIH.

11. Once the BIOS software is opened, go to the **Main** tab and check in the **UEFI version** parameter that the BIOS firmware version is IMB-150 P2.20L.
12. If a new SSD has been installed in the new motherboard, proceed with a Full Recovery procedure. See "Installing or restoring Operating System" on page 1006
13. If the SSD from the replaced motherboard will be used with the new one, proceed with an Operating System Recovery procedure. See "Installing or restoring Operating System" on page 1006.

2.1.5.8 Replacing card drawer electric lock



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 734238 ELECTRIC LOCK v2.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Previous considerations

When assembling P/N 734238 ELECTRIC LOCK v2, the following considerations should be considered in order to ensure a correct assembly of the system:

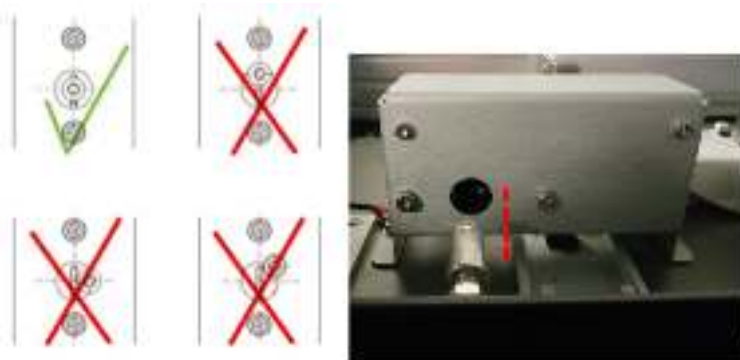
- Ensure a parallel positioning of the electric lock when assembled.



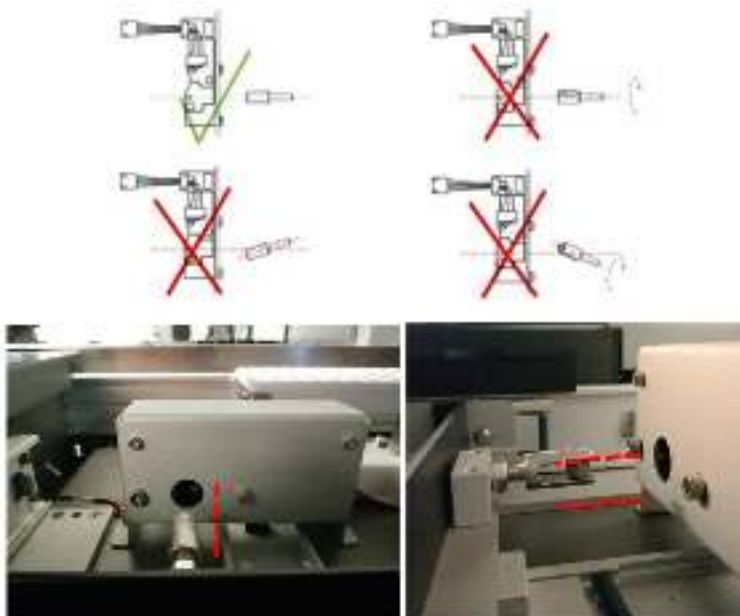
- Initially, the screws of the electric lock should be lightly tightened by hand, so the latch will help to define electric lock final position once inserted.



- Electric lock opening and latch must be centered in relation to each other.



- Screw the latch nut securing the latch against rotations. Parallelism between electric lock opening and the latch must be ensured. Make sure the latch is not rotated and its bolard is in vertical position.



- Before tightening completely the screws of the electric lock, gently close the door/drawer for which the lock is being replaced. The closing force is directly related to the correct alignment and parallelism of the components. If necessary, loosen the nut and adjust the thread length of the latch. Then lock its final position with the nut, taking into account the above-indicated instructions.



- Open the door/ drawer and tighten completely the screws of the electric lock.



Procedure

To replace the card drawer electric lock follow the next steps:

1. Using the Technical Service software (see "Drawers tab" on page 792), open the drawer corresponding to the electric lock to be replaced and move the incubator located behind the opened drawer to the upper position.
2. Switch off the instrument and disconnect it from the mains.
3. Pull out the user tray and the screen, then remove the screw of the intermediate door and open intermediate door.

4. Remove the two screws that assemble the electric lock cover and the two screws that assemble the electric lock.



Figure 174 - Electric lock cover and lock

5. Pull apart the electric lock cover carefully. Disconnect electric lock cable and remove it.



Figure 175 - Electric lock cable

6. Assemble the new electric lock (do it without completely tightening the screws, then close the drawer placing the locker in a position where the drawer latch can be properly inserted) then tighten its screws and reassemble the rest of the parts.
7. Once all the parts are assembled make sure that the cabling in the electric lock area does not interfere with the incubator movement.
8. Check that the drawer closes/opens correctly.
9. Adjust the CAT arm (gripper) on the drawer corresponding to the electric lock that has been replaced.

See "Adjusting gripper in the card drawers area" on page 472.

2.1.5.9 Replacing samples/reagents drawer electric lock



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 734238 ELECTRIC LOCK v2.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- See "Previous considerations" on page 208 before replacing the electric lock.

Procedure

To replace the samples/reagents drawer electric lock follow the next steps:

1. Switch off the instrument and disconnect it from the mains.
2. Open manually the drawer by pushing the lever upwards.



Figure 176 - Electric lock lever

3. Remove the two screws that assemble electric lock.



Figure 177 - Electric lock assembly screws

4. Disconnect the electric lock cable and remove it.



Figure 178 - Electric lock cable

5. Assemble the new electric lock and reassemble the rest of the parts. When doing so, make sure that the electric lock cable does not interfere with the path of the incubator located behind.
6. Once all the parts are assembled check that the drawer closes/opens correctly.
7. Adjust the PIP arm (probe) on the drawer corresponding to the electric lock that has been replaced. See "Adjusting reagents drawers" on page 338 and see "Adjusting samples drawers" on page 342.

2.1.5.10 Replacing fluidic drawer electric lock



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 734238 ELECTRIC LOCK v2.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- See "Previous considerations" on page 208 before replacing the electric lock.

Procedure

To replace the fluidic drawer electric lock follow the next steps:

1. With the instrument switched off, open manually the drawer corresponding to the electric lock to be replaced by pushing the lever located under the bottle drawer.

2. Disconnect and remove the bottle from the drawer, then remove the front cover of the drawer by removing its three screws.



Figure 179 - Three screws holding the front cover

3. Pull apart the container tube and remove the container basis.



Figure 180 - Container basis

4. Loosen the two screws of the electric lock.



Figure 181 - Two screws holding the electric lock

5. Disengage one slot of the chain so that the plastic cover can be easily removed afterwards.



Figure 182 - Disengaging the chain

6. Carefully remove the plastic cover.



Figure 183 - Removing the plastic cover

7. Disconnect the electric lock cable and remove the electric lock.

8. Assemble the new electric lock.

**NOTE**

When assembling the new lock, consider all the tips listed in "Previous considerations" on page 208

**CAUTION**

Make sure that the lever of the drawer metallic plate is inserted in the hole for the manual emergency opening of the electric lock.



Figure 184 - Lever of the drawer metallic plate inserted in the hole

9. Check that the bottle drawer closes/opens correctly manually through the emergency opening (an external latch can be used to test it) .



Figure 185 - External latch

10. Remove the external latch.
11. Reassemble the rest of the parts.
12. Once all the parts are assembled check that the bottle drawer closes/opens correctly.
13. Check that the load cell adjustment of the bottle drawer is correct. See "Verifying load cell" on page 588.

2.1.5.11 Replacing user tray



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 733671 USER TRAY (Ver. 4).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the User tray follow the next steps:

1. Switch off the instrument and disconnect it from the mains.
2. Open the Intermediate area door.
3. Pull the user tray out, and remove the cosmetic front panel, unscrewing the four screws on its back side.



Figure 186 - Cosmetic front panel screws

4. Remove the two allen screws attaching the user tray to the instrument.



Figure 187 - User tray attachment screws

5. Pull the whole module out of the instrument.



Figure 188 - User tray

6. Insert the new user tray making sure that the rear flanges enter the corresponding slots in the instrument.

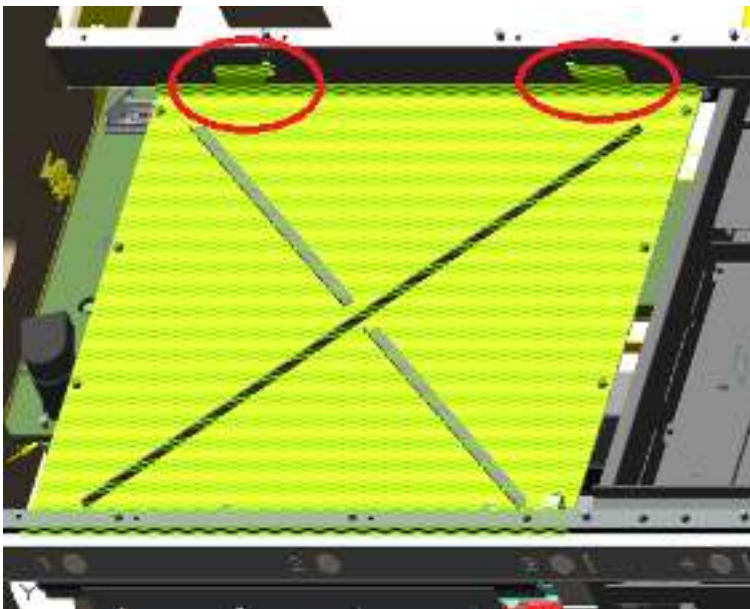


Figure 189 - User tray rear flanges

7. Screw the user tray to the instrument, using the two previously removed screws.

8. Attach the cosmetic front panel, using the four previously removed screws.

2.1.5.12 Replacing upper door



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To install the upper door in the instrument:

1. Open manually the main door through the top.
2. Remove the six screws attaching the upper door to the hinge assembly.

**NOTE**

The door must be held firmly while removing the screws to prevent it from falling down.

3. Disassemble the magnet.
4. Assemble the magnet in the new upper door.

5. Assemble the new upper door in the hinge assembly by means of the six screws previously removed.



6. Check that the sensor installed in the new door is in contact with the sensor magnet attached to the instrument. If necessary, adjust the position of the magnets by means of the slotted holes in order to place the door parallel to the chassis once it is closed.



7. Open the Technical Service software, go to the **General** tab and check that the status of the upper area door is correctly detected (SRM Door section). See "General tab" on page 832.

2.1.6 Electronics

This section covers the electronics related to the General module.

2.1.6.1 Fuses

Fuses related to General module are detailed in following table:

Table 7 - List of fuses of the General module

PCA	FUSE	DESCRIPTION
PCA GEN-BCKPL59	FUSGEN1	MST250 0034.6620 3.15 A / 250 V
	FUSGEN2	MST250 0034.6615 1 A / 250 V
	FUSGEN4	MST250 0034.6615 1 A / 250 V
	FUS01	MST250 0034.6615 1 A / 250 V
	FUS02	MST250 0034.6620 3.15 A / 250 V
	FUS21	MST250 0034.6615 1 A / 250 V
	FUS22	MST250 0034.6620 3.15 A / 250 V
PCA CEN-SRM Control	FUS1	MSF250 0034.6013 0.8 A / 250 V
PCA CAT-CM Control	FUS1	MSF250 0034.6013 0.8 A / 250 V
PCA GEN-STDBY	FUS1	MSF250 0034.6018 F2.5 A / 250 V
PCA GEN-LED59	NA	NA
PCA GEN-FRNT59	NA	NA
PCA GEN-TEMP59	NA	NA
PCA GEN-LCK59	NA	NA
PCA RM-ST59	NA	NA
MAINS CONNECTOR	NA	2xT 6,3 A L 250 V, 5x20 mm

2.1.6.2 LEDs

LEDs related to the General module are detailed in following table:

Table 8 - List of LEDs of General module

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED1	Green 5 V (FUSGEN1)
	LED2	Green 5 V (FUSGEN2)
	LED4	Green 24 V (FUSGEN4)
	LED5	Green 5 V (FUS01)
	LED6	Green 24 V (FUS02)
	LED13	Green 5 V (FUS21)
	LED14	Green 24 V (FUS22)
PCA CEN-SR Control	LED1	Yellow When the instrument is switched on, LED is on and a few seconds later is off. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is ≥ 12 V.
PCA CAT-CM Control	LED1	Yellow When the instrument is switched on, LED is on and a few seconds later is off. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is ≥ 12 V.
PCA GEN-STDBY	LED1	Green It indicates that there is AC voltage at the input.
PCA GEN-LED59	NA	NA
PCA GEN-FRNT59	NA	NA
PCA GEN-TEMP59	NA	NA
PCA GEN-LCK59	NA	NA
PCA RM-ST59	NA	NA

2.1.6.3 Jumpers and test points

Jumpers and test points related to the General module are detailed in the following table:

Table 9 - List of jumpers and test points of General module

PCA	JUMPER	TEST POINT
PCA GEN-STDBY (SN < 059-0001583)	JP1	Not present: standby button controls the switch on of the power supply.
		Present: power supply is always receiving the mains voltage. Instrument is switched on by means of the switch located in the mains connector. Computer is not switched on.
	JP3	Not present: electronic power is activated when the push button is pressed.
		Present: electronic power is controlled by software.

2.1.6.4 Wiring table

Cables related to the General module are detailed in the following table and their connection are shown in the wiring diagram. To view the wiring diagram, see "Related documents" on page 19 or refer to the Erytra Online Help.

Table 10 - List of wiring of General module

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
ELE-F1	H	FAN ELEC	FAN1	PCA GEN-BCKPL59
ELE-F2 + ELE-EXT	H	FAN PC	FAN2	PCA GEN-BCKPL59
ELE_+5V	H14	PCA GEN-STDBY	H2	PCA GEN-BCKPL59
ELE-V1 , ELE-V2-, ELE-V2+	V1+, V2-, V2+	POWER SUPPLY	H1	PCA GEN-BCKPL59
ELE-AC	FASE+NEUTRAL	POWER SUPPLY	H2	PCA GEN-STDBY59
ELE-GND	EARTH	POWER SUPPLY	-	CHASSIS
B-AC PWR	OUTPUT	POWER LINE FILTER	H1-PH+N	PCA GEN-STDBY59
B-FIL_AC	PH+N	MAINS	INPUT	POWER LINE FILTER
B-FIL_EA	EARTH	MAINS	INPUT	POWER LINE FILTER
B-EARTH	EARTH	MAINS	H1-EARTH	PCA GEN-STDBY59
ELE-V4,ELE-V3+,ELE-V3-	V3+,V3-,V4+	POWER SUPPLY	H3	PCA GEN-STDBY
ELE-V5,ELE-V6	V5+,V5-,V6+,V6-	POWER SUPPLY	H4	PCA GEN-STDBY

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
ELE-ATX	ATX POWER	COMPUTER	H5	PCA GEN-STDBY
ELE-V1s	V1s	POWER SUPPLY	H9	PCA GEN-STDBY
ELE-V2s	V2s	POWER SUPPLY	H10	PCA GEN-STDBY
ELE-V3s/V4s	V3/4s	POWER SUPPLY	H11	PCA GEN-STDBY
ELE-FEAT	Pc Feature	COMPUTER	H12	PCA GEN-STDBY
ELE-COM	COM1	COMPUTER	H3	PCA GEN-BCKPL59
B52_BUTTON	-	PUSHBUTTON	H4	PCA GEN-FRNT59
B-PWR_FR	ATX	COMPUTER	H5	PCA GEN-FRNT59
B-USB1_FR	USB4	COMPUTER	H3	PCA GEN-FRNT59
B-USB2_FR	USB5	COMPUTER	H2	PCA GEN-FRNT59
B-TFT_USB	USB0	COMPUTER	USB	TFT SCREEN
B-TFT_VGA	VGA	COMPUTER	ADAPTOR DB15	NA
TFT-VGA	VGA	TFT SCREEN	ADAPTOR DB15	NA
B-TFT_PWR	H13	PCA GEN-STDBY59	DC In +12V	TFT SCREEN
B-LAN_RE	LAN1	COMPUTER	LAN	ELECTRONIC COVER
B-COM_RE	COM2	COMPUTER	COM	ELECTRONIC COVER

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-USB2_RE	USB1	COMPUTER	USB1	ELECTRONIC COVER
B-USB1_RE	USB3	COMPUTER	USB3	ELECTRONIC COVER
B-USB4_RE	USB6	COMPUTER	USB6	ELECTRONIC COVER
B-USB3_RE	USB7	COMPUTER	USB7	ELECTRONIC COVER
B-LED_SRM	H1	PCA GEN-LED59 SRM	GEN_LED SRM	PCA GEN-BCKPL59
B-LED_CM	H1	PCA GEN-LED59 CM	GEN_LED CM	PCA GEN-BCKPL59
B-LED	H1	PCA GEN-LED59	GEN_LED	PCA GEN-BCKPL59
B-KCM	-	KEYBOARD CM	KEYBD_CM	PCA GEN-BCKPL59
B-KSRM	-	KEYBOARD SRM	KEYBD_SRM	PCA GEN-BCKPL59
B-CM_LOCK	H1 FROM MASTER	PCA GEN-LCK59 CARD DRAWER 1	CM_LCK	PCA GEN-BCKPL59
B-CM1/2	H2 TO SLAVE	PCA GEN-LCK59 CARD DRAWER 1	H1 FROM MASTER	PCA GEN-LCK59 CARD DRAWER 2
NA	H3	PCA GEN-LCK59 CARD DRAWER 1	H	SAFETY LOCK
B-CM2/3	H2 TO SLAVE	PCA GEN-LCK59 CARD DRAWER 2	H1 FROM MASTER	PCA GEN-LCK59 CARD DRAWER 3

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
NA	H3	PCA GEN-LCK59 CARD DRAWER 2	H	SAFETY LOCK
B-CM3/4	H2	PCA GEN-LCK59 CARD DRAWER 3	H1	PCA GEN-LCK59 CARD DRAWER 4
NA	H3	PCA GEN-LCK59 CARD DRAWER 3	H	SAFETY LOCK
B-CM4/TEMP	H2	PCA GEN-LCK59 CARD DRAWER 4	H1	PCA GEN-TEMP59
NA	H3	PCA GEN-LCK59 CARD DRAWER 4	H	SAFETY LOCK
B_SRM_LOCK	H1 FROM MASTER	PCA GEN-LCK59 REAGENT DRAWER 1	SRM_LCK	PCA GEN-BCKPL59
B-RM1/2	H2 TO SLAVE	PCA GEN-LCK59 REAGENT DRAWER 1	H1 FROM MASTER	PCA GEN-LCK59 REAGENT DRAWER 2
B-STIR1	RM_1	PCA GEN-BCKPL59	H1	PCA RM-CON59 REAGENT DRAWER 1
NA	H2	PCA RM-CON59 REAGENT DRAWER 1	H1	PCA RM-ST59 REAGENT DRAWER 1
NA	H2	PCA RM-ST59 REAGENT DRAWER 1	ENCODER	REAGENT MOTOR

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
NA	H3	PCA RM-ST59 REAGENT DRAWER 1	POWER	REAGENT MOTOR
B-STIR2	RM_2	PCA GEN-BCKPL59	H1	PCA RM-CON59 REAGENT DRAWER 2
NA	H2	PCA RM-CON59 REAGENT DRAWER 2	H1	PCA RM-ST59 REAGENT DRAWER 2
NA	H2	PCA RM-ST59 REAGENT DRAWER 2	ENCODER	REAGENT MOTOR
NA	H3	PCA RM-ST59 REAGENT DRAWER 2	POWER	REAGENT MOTOR
B-SRM_LOCK	SRM_LCK	PCA GEN-BCKPL59	H1	PCA GEN-LCK59 REAGENT DRAWER 1
B-RM1/2	H2	PCA GEN-LCK59 REAGENT DRAWER 1	H1	PCA GEN-LCK59 REAGENT DRAWER 2
NA	H3	PCA GEN-LCK59 REAGENT DRAWER 1	H	SAFETY LOCK
B-RM2/SM1	H2	PCA GEN-LCK59 REAGENT DRAWER 2	H1	PCA GEN-LCK59 SAMPLE DRAWER 1
NA	H3	PCA GEN-LCK59 REAGENT DRAWER 2	H	SAFETY LOCK

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-SM1/2	H2	PCA GEN-LCK59 SAMPLE DRAWER 1	H1	PCA GEN-LCK59 SAMPLE DRAWER 2
NA	H3	PCA GEN-LCK59 SAMPLE DRAWER 1	H	SAFETY LOCK
B-SM2/3	H2	PCA GEN-LCK59 SAMPLE DRAWER 2	H1	PCA GEN-LCK59 SAMPLE DRAWER 3
NA	H3	PCA GEN-LCK59 SAMPLE DRAWER 2	H	SAFETY LOCK
B-SM3/4	H2	PCA GEN-LCK59 SAMPLE DRAWER 3	H1	PCA GEN-LCK59 SAMPLE DRAWER 4
NA	H3	PCA GEN-LCK59 SAMPLE DRAWER 3	H	SAFETY LOCK
B-SM4/TEMP	H2	PCA GEN-LCK59 SAMPLE DRAWER 4	H1	PCA GEN-TEMP59
NA	H3	PCA GEN-LCK59 SAMPLE DRAWER 4	H	SAFETY LOCK
B-DOOR_SRM	NA	SRM DOOR SENSOR	PIP_DOOR	PCA GEN-BCKPL59
B-DOOR_CM	NA	CM DOOR SENSOR	CAT_DOOR	PCA GEN-BCKPL59

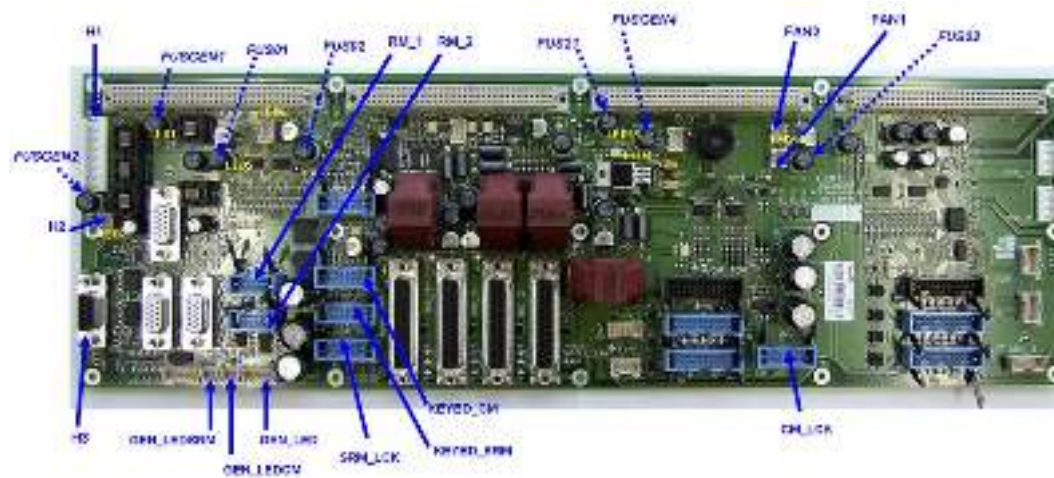


Figure 190 - Connectors, fuses and LEDs of PCA GEN-BCKPL59 related to General module

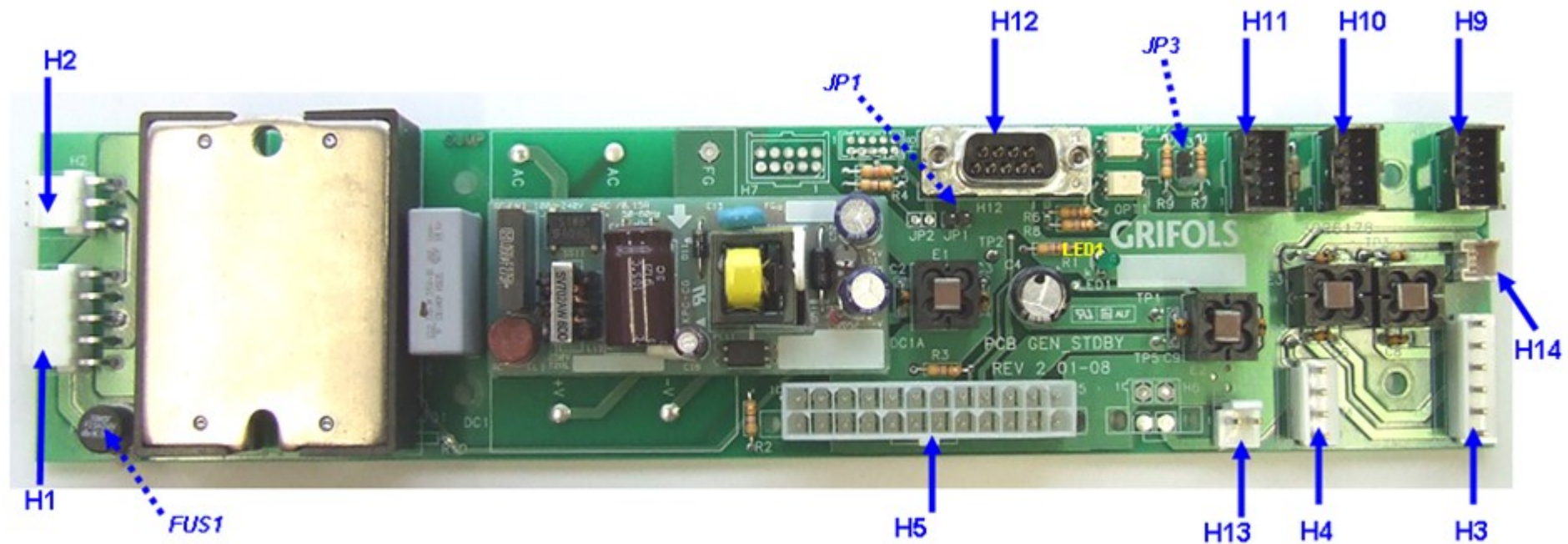


Figure 191 - Connectors, fuses and LEDs of PCA GEN-STDBY (units with S/N < 059-0001583)

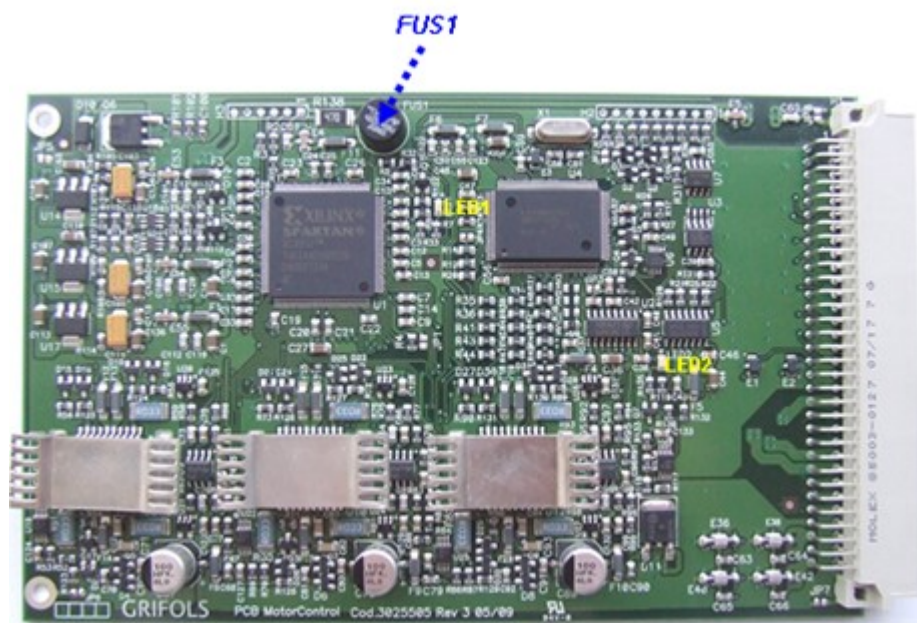


Figure 192 - Location of the fuse and LEDs in CEN-SRM / CAT-CM Control



Figure 193 - Connectors of PCA RM-ST59



Figure 194 - Connectors of PCA GEN-FRNT59



Figure 195 - Connectors of PCA GEN-TEMP59

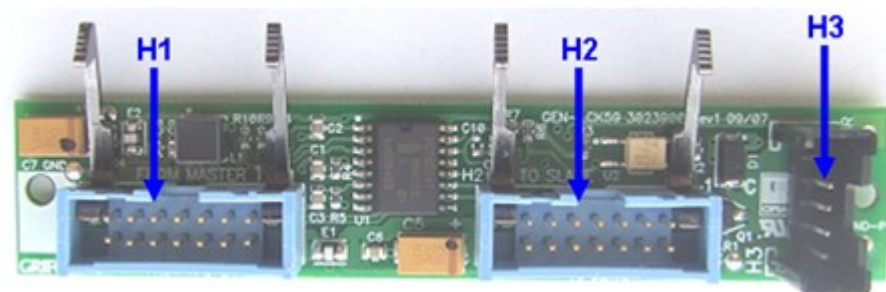


Figure 196 - Connectors of PCA GEN-LCK59

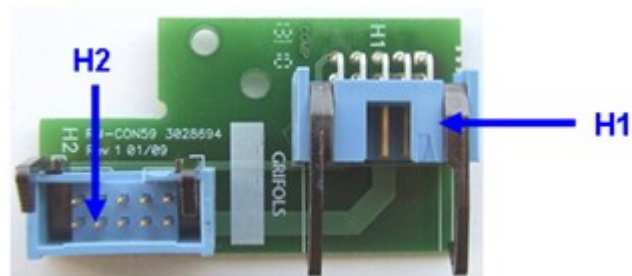


Figure 197 - Connectors of PCA RM-CON59



Figure 198 - Connectors of PCA GEN-LED59

2.1.7 Troubleshooting

Table 11 - List of errors related to General module

Module	Device	Error code	Error	Cause	Text	Options
SRM (Samples And Reagents Management Module) (09)	Samples and reagents lockers	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Unrecoverable error. Please press OK to RESTART.	OK
	(01) Drawer 1	005000	ERRORCODE_ARG	Command has wrong arguments		
	(02) Drawer 2					
	(03) Drawer 3					
	(04) Drawer 4	039000	SWITCH_ERR	Locker error detected		
	(05) Drawer 5					
	(06) Drawer 6					
		040000	ERRORCODE_ DRAWER_ UNKNOWN_ERROR	Drawer unknown error		

Module	Device	Error code	Error	Cause	Text	Options
SRM (Samples And Reagents Management Module) (09)	Samples and reagents lockers	006000	ERRORCODE_ BUSY	Command is addressed to a busy device	NA	NA
	(01) Drawer 1	038000	ERRORCODE_ DRAWER_ OPEN_SYNC_ TIMEOUT	Drawer has been busy for more than its timeout margin		
	(02) Drawer 2					
	(03) Drawer 3					
	(04) Drawer 4	088000	ERRORCODE_ PRECONDITION_ FAILED	A precondition is not reach to perform the action		
	(05) Drawer 5					
	(06) Drawer 6					

Module	Device	Error code	Error	Cause	Text	Options
SRM (Samples And Reagents Management Module) (09)	Reagents stirrer motor (16) Drawer 1 (17) Drawer 2	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Error during reagents stirring process or homogenization in dilution cup. Please choose one of the following options:	RETRY
		005000	ERRORCODE_ARG	Command has wrong arguments		CANCEL Works (it leads to: Reagent drawer has been disabled. Please reboot the system as soon as possible. If problem persists, contact Technical Service.)
		006000	ERRORCODE_BUSY	Command is addressed to a busy device		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device		
		012000	ERRORCODE_SPINBADPROFILE	Command includes a spin profile that is invalid for the addressed device		

Module	Device	Error code	Error	Cause	Text	Options
SRM (Samples And Reagents Management Module) (09)	Reagents stirrer motor (16) Drawer 1 (17) Drawer 2	013000	ERRORCODE_ NOTSPINNING	The addressed device is not performing the requested spin	Error during reagents stirring process or homogenization in dilution cup. Please choose one of the following options:	RETRY
		032000	CURRENT_LIMIT	Current limit at servo device		CANCEL Works (it leads to: Reagent drawer has been disabled. Please reboot the system as soon as possible. If problem persists, contact Technical Service.)
		033000	POSITION_ERROR	A position error has been produced at a servo device		
		034000	MOVEMENT_ ERROR	A movement error has been produced at a servo device		
		035000	HOME_TIMEOUT	A home timeout has been produced at a servo device		

Module	Device	Error code	Error	Cause	Text	Options	
General module (06)	Keyboard (01)	info	KEYBD1_DRIVER_ERR	Keyboard driver error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK	
		info	KEYBD2_DRIVER_ERR	Keyboard driver error			
	Lights (03)	info	AMBIENT_DRIVER_ERR	Ambient lights driver error			
		info	DRIVER_ERR	Driver error			
	Backpanel Temp sensors & fans (04)	info	SENSOR_ERR	Error detected in the value measured by the sensor - electronics cabinet			
		info	DRIVER_ERR	Electronics cabinet fan is in Circuit open, Driver error			
		info	CURRENT_LIMIT	Electronics cabinet fan is in Circuit open, Current limit error			
	Instrument temp sensors (04)	info	SENSOR_ERROR	Error detected in the value measured by the sensor			

Module	Device	Error code	Error	Cause	Text	Options
CM (Cards Management Module) (01)	Cards lockers (00) Drawer 1 (01) Drawer 2 (02) Drawer 3 (03) Drawer 4	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Unrecoverable error. Please press OK to RESTART.	OK
		005000	ERRORCODE_ARG	Command has wrong arguments		
		039000	SWITCH_ERR	Locker error detected		
		040000	ERRORCODE_DRAWER_UNKNOWN_ERROR	Drawer unknown error		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device	NA	NA
		038000	ERROR_CODE_DRAWER_OPEN_SYNC_TIMEOUT	Drawer has been busy for more than its timeout margin		
		088000	ERRORCODE_PRECONDITION_FAILED	A precondition is not reach to perform the action		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received a command with incorrect length		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device		
		005000	ERRORCODE_ARG	Command has wrong arguments		
		014000	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW		
		015000	ERRORCODE_HW_HWFAIL	There is a problem with PCA HW		
		016000	ERRORCODE_HW_HWNOTSET	PCA setup error		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	000000	ERRORCODE_UNKNOWN	Unknown error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_NOTSAFE	Movement not safe		
		017000	ERRORCODE_FW_UNKNOWN_ERROR	Unknown error		
		061000	ERRORCODE_INVALID_ARGUMENTS	Invalid arguments		
		062000	ERRORCODE_FWCOMM_TIMEOUT	Communication timeout		
		063000	ERRORCODE_FWCOMM_CONNECTION_LOST	Communication lost		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	064000	ERRORCODE_FWCOMM_COMM_ERROR	Communication error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		088000	ERRORCODE_PRECONDITION_FAILED	Previous condition for the action is not met		
		089000	ERRORCODE_STARTUP_ERROR	Driver startup error		
		091000	ERRORCODE_ALREADY_STARTED	Another instance of Driver is already opened		
0	240	090000	ERRORCODE_UNEXPECTED_ERROR	A Driver unexpected error has been produced		

Table 12 - Power supply issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Instrument does not switch on	AC cable disconnected	Connect AC cable to mains connector
	Switch of mains connector is in OFF position	Switch on mains connector
	Fuses of mains connector are blown	Replace fuses
	Cables of mains connector disconnected / damaged	Check / replace cables
	Pushbutton is damaged	Check / replace pushbutton
	Fuse of PCA GEN-STDBY59 blown (S/N < 059-0001583)	Check / replace FUS1 of PCA GEN-STDBY59
	PCA GEN-FRNT59 damaged	Check / replace PCA GEN-FRNT59
	PCA GEN-STDBY59 damaged (S/N < 059-0001583)	Check / replace PCA GEN-STDBY59
	Power supply damaged	Replace power supply
Output voltages do not match with expected ones	Power supply damaged	Replace power supply

Table 13 - Computer/Software issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Not possible to run the computer	Power supply failure	Check / replace power supply
		Check / replace PCA GEN-STDBY59 (S/N < 059-0001583)
	Hardware error	Replace computer
		Replace SSD
Computer is switched off automatically	Fan of the computer is blocked	Check internal cables of the computer
	Electronic cabinet fans are not working	Check/replace electronic cabinet fans
O.S. errors	Software error	Install default image of SSD
	Wrong configurations	Check applied configurations
USB connector does not work	Hardware error	Check/replace wiring
	Software error	Install default image of SSD
	Computer USB plug does not work	Replace computer
Low SSD space	No space in SSD	Backup and delete files
Adobe Air installer reports that installer is corrupted	Software error	Check configuration of OS date
		Check that ErytraLogin is not running in the background when installing Adobe Air

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Java error when starting the Driver in the Technical Service software or login into application	Adjustment files are not found	Check that PIPCalibrations.txt and CATCalibrations.txt are present in E:\data\Erytra\conf and they are not read only.
java.lang.NullPointerException when starting Erytra software	Corrupted startup.check file	Delete E:\data\Erytra\startup.check file
Server not available in Erytra Login	Incorrect erytra.properties configuration	Check erytra.properties configuration
	Erytra database is not found	Install Erytra database
	Erytra database folder is corrupted	Rename E:\data\Database folder and run E:\Program Files\Liquibase\createdb.bat to create new database structure. Then restore a database backup.
	Erytra software files corrupted	Reinstall OS and Erytra software

Table 14 - Touchscreen issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Flickering	VGA cable is not connected properly	Check VGA connection
	VGA cable is damaged	Replace VGA cable
	Touchscreen is damaged	Replace touchscreen monitor

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Image has a red/green color	VGA cable is not connected properly	Check VGA connection
	VGA cable is damaged	Replace VGA cable
	Touchscreen is damaged	Replace touchscreen monitor
Cursor on the screen does not match where it is pointed on the screen	Screws of the touchscreen cover are too tight	Loose a little bit touchscreen cover screws
	Incorrect drivers installed	Check/Install proper drivers
	Touchscreen is not calibrated	Calibrate touchscreen
	Touchscreen is damaged	Replace touchscreen monitor
Hard to insert / extract touchscreen monitor into/from the instrument	Drawer lock is not correctly placed	Check/adjust drawer lock position
	Touchscreen cover is not correctly placed	Check cover position and tighten/loose its screws
	Cables of the touchscreen monitor are not correctly placed	Check cables position
Impossible to insert touchscreen monitor into the instrument	Touchscreen monitor can not be placed flat	Check that touchscreen monitor can be placed completely flat before to insert it into the instrument
		Lubricate tilting latch lock

Table 15 - Upper/Intermediate area door issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Door status not detected	Both parts of the celduc sensor are not in contact when door is closed	Check assembly of celduc sensor
	Celduc sensor is damaged	Replace celduc sensor
Noise when opening/closing the door	Lack of lubrication	Lubricate the door
Door is not aligned	Incorrect assembly of the magnets of the doors	Adjust the position of the door

Table 16 - User tray issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Hard to insert / extract user tray into/from the instrument	User tray is not lubricated	Lubricate user tray
	Drawer lock is not correctly placed	Check/adjust drawer lock position
User tray exists completely from the drawer	Stopper broken	Replace user tray

Table 17 - Status LED issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Status LED does not switch on	PCA GEN-LED59 damaged	Replace PCA GEN-LED59
	Cable damaged	Replace cable (B-LED)
A color is not activated	PCA GEN-LED59 damaged	Replace PCA GEN-LED59

Table 18 - Ambient lights issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Upper/Intermediate area ambient light does not switch on	PCA GEN-LED59 damaged	Replace PCA GEN-LED59
	Cable damaged	Replace cable (B-LED_SRM or B-LED_CM)

Table 19 - Sample drawers issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Drawer not opened	Keyboard damaged	Check/replace keyboard
SD3: 0903006000 SD3: 0903038000 SD3: 0903039000 SD4: 0904006000 SD4: 0904038000 SD4: 0904039000 SD5: 0905006000 SD5: 0905038000 SD5: 0905039000 SD6: 0906006000 SD6: 0906038000 SD6: 0906039000	Keyboard cable damaged	Check/replace B-KSRM cable
	Cable between locks damaged	Check / replace cable between locks
	Lid electric lock damaged	Check/replace lid electric lock
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Incorrect drawer opened	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace keyboard cable
	Cable between locks damaged	Check / replace cable between locks
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
	PCA GEN-BCKPL59 damaged	Check/replace PCA GEN-BCKPL59
Sample rack is too loose / too tight over the drawer	Centerers of the drawer are not correctly placed	Adjust centerers position and then readjust probe in the drawer
Cosmetic front panel is not aligned	Cosmetic front panel screws are loose	Align cosmetic front panel
Drawer is not aligned	Safety latch depth position is not correct	Adjust safety latch position and then readjust probe in the drawer
Hard to open / close the drawer	Lack of lubrication	Lubricate
	Linear guides damaged	Replace linear guides
Sample tube tilted	Sample holder of rack damaged	Replace sample rack

Table 20 - Card drawers issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Drawer not opened	Keyboard damaged	Check/replace keyboard
CD1: 0100006000 CD1: 0100038000 CD1: 0100039000	Keyboard cable damaged	Check/replace B-KCM cable
CD2: 0101006000 CD2: 0101038000 CD2: 0101039000	Cable between locks damaged	Check / replace cable between locks
CD3: 0102006000 CD3: 0102038000 CD3: 0102039000	Lid electric lock damaged	Check/replace lid electric lock
CD4: 0103006000 CD4: 0103038000 CD4: 0103039000	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
Incorrect drawer opened	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace B-KCM cable
	Cable between locks damaged	Check / replace cable between locks
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
	PCA GEN-BCKPL59 damaged	Check/replace PCA GEN-BCKPL59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Card rack is not correctly placed over the drawer	Centerers of the drawer are not correctly placed	Check centerers position
Cosmetic front panel is not aligned	Cosmetic front panel screws are loose	Align cosmetic front panel
Drawer is not aligned	Safety latch depth position is not correct	Adjust safety latch position and then readjust probe in the drawer
Hard to open / close the drawer	Lack of lubrication	Lubricate
	Linear guides damaged	Replace linear guides

Table 21 - Reagent drawers issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / Current limit error RD1: 0916032000 RD1: 0916034000 RD2: 0917032000 RD2: 0917034000	Fuse blown	Check / replace FUS01 or FUS02
	Dirt under reagent rack platform	Clean under reagent rack platform
	Feet of the rack platform are not lubricated properly	Lubricate with grease underneath the feet of rack platform
	Motor belt is too tight	Check/adjust motor belt tension
	Motor belt is damaged	Replace motor belt
	Motor/encoder damaged	Check/replace motor
	Ball bearing damaged	Check/replace ball bearing
	PCA RM-ST59 failure	Check/replace PCA RM-ST59
	Cable between PCA RM-ST59 and PCA RM-CON59 damaged	Check/replace cable
	PCA CEN-SRM Control failure	Check/replace PCA CEN-SRM Control
Home position is not detected RD1: 0916035000 RD2: 0917035000	Hall effect sensor tilted	Check position of Hall effect sensor in PCA RM-ST59
	Cable between PCA RM-ST59 and PCA RM-CON59 damaged	Check/replace cable
	PCA RM-ST59 failure	Check/replace PCA RM-ST59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Drawer not opened RD1: 0901006000 RD1: 0901038000 RD1: 0901039000 RD2: 0902006000 RD2: 0902038000 RD2: 0902039000	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace B-KSRM cable
	Cable between locks damaged	Check / replace cable between locks
	Lid electric lock damaged	Check/replace lid electric lock
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
Incorrect drawer opened	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace keyboard cable
	Cable between locks damaged	Check / replace cable between locks
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
	PCA GEN-BCKPL59 damaged	Check/replace PCA GEN-BCKPL59
Cosmetic front panel is not aligned	Cosmetic front panel screws are loose	Align cosmetic front panel
Drawer is not aligned	Safety latch depth position is not correct	Adjust safety latch position and then readjust probe in the drawer
Hard to open / close the drawer	Lack of lubrication	Lubricate
	Linear guides damaged	Replace linear guides

2.2 Incubator module (INC)

In this chapter, the Incubator module is described. It includes an overall description of the functions, as well as the corresponding adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, and a troubleshooting section.

2.2.1 Description

There are two main functions of this module:

- Incubate the cards to the defined temperature.
- Move the cards from the intermediate level to the upper level and vice-versa (lift), retaining the incubation block/card position during the operations performed on each level.

There are four independent incubator blocks located at the rear part of the instrument. Each block has a capacity for 12 cards positioned equidistantly.

The edges of the metallic plates where the incubators move upwards and downwards along the linear guides have been hemmed in order to avoid injuries to the field engineer when accessing this area. However there are available edge protectors to be installed on instruments without hemmed edges to mitigate this risk.

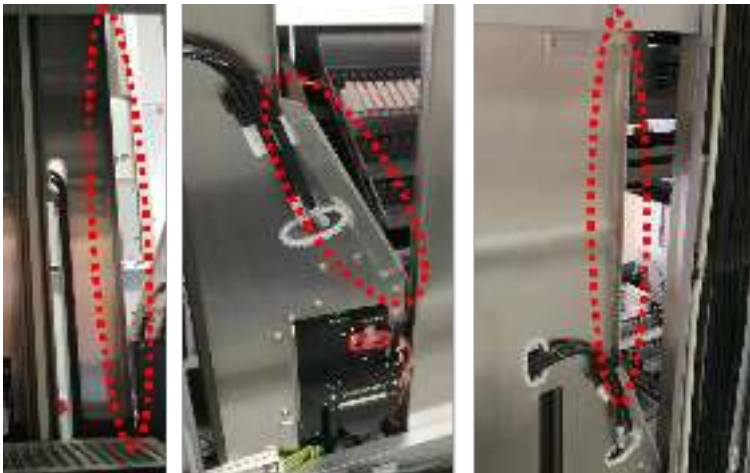


Figure 199 - Edge protectors on the incubator area

The incubation procedure has two parameters: temperature and time. Temperature is controlled by firmware, while time is controlled by software.

Two temperatures are allowed: RT (room temperature), that is $24\text{ }^{\circ}\text{C} \pm 3\text{ }^{\circ}\text{C}$, and $37\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$. Temperature is modified by a Peltier module unit (consisting of a Peltier cell, fan and heat sink; 24 V DC, 3.3 A) attached to the incubator block (bottom). This module unit is used as a reversible heat pump to warm up or cool down the block to the desired temperature. Temperature is monitored by two temperature sensors in order to protect against a single fail condition. These sensors are

assembled in PCA INC-BL59 attached to the right side of the incubator, and protected by a bumper from unexpected impacts with the CAT arm. Incubator built-in isolation material all around and the heat transmission U shape.

The incubator block target temperature shall be reached within 20 minutes at any of the allowed environmental conditions, and taking under consideration that the internal instrument temperature may be different from the external temperature.

A servocontrolled system is used in order to achieve required movement speed and positioning precision. This system ensures that the incubator position is kept without powering the motor up to 70 N. The motor transmits the movement to the guide by means of a coupling.

Encoder resolution for the incubator axis is as follows:

Table 22 - INC axis encoder resolution

INC Axis	Encoder counts / mm
Z	851.851

There is an optical home sensor located at the right-bottom end of PCA INC-CON59 that determines the lower position of the incubator (card floor) and that is used as a home reference. A small L-shaped plate assembled in the guide wagon is used as a home flag. The Upper level/pipetting floor position of the incubator is determined with the incubator height adjustment.

Cards are retained in the incubator slot by means of the frontal probes platform (PIP floor), including a window per each incubator with a width lower than card width so it retains for both ends. This platform also has a hole per each incubator that is used by the PIP arm presence detector to check if the incubator is at the pipetting floor during user software initialization.

The Peltier module and temperature sensor are controlled/signals between PCA INC-CON59 and PCA GEN-BCKPL59 are transmitted through I2C bus.

The general block diagram of the module is shown in the following figure:

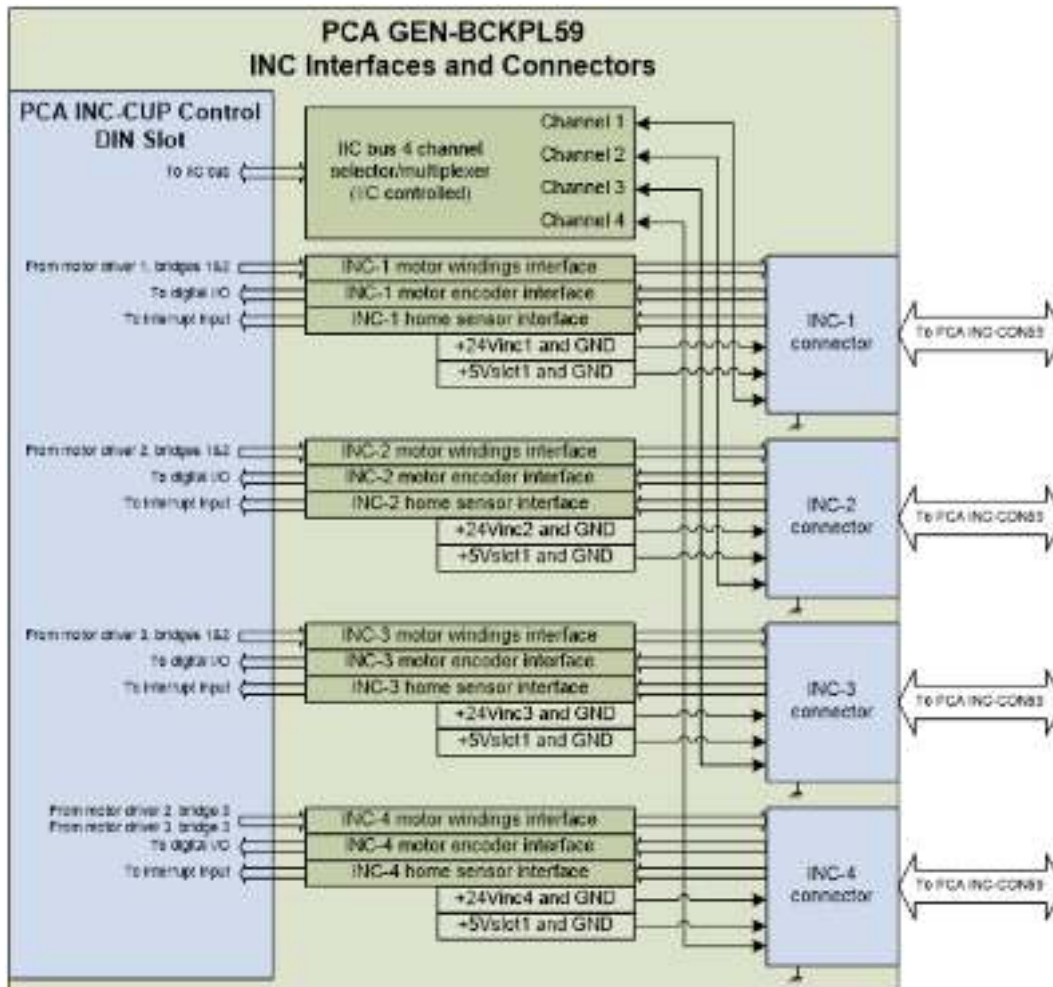


Figure 200 - Incubator module block diagram

PCAs related to the Incubator module are:

- **PCA GEN-BCKPL59:** main PCA of the instrument, located in the backpanel.
- **PCA INC-CUP Control:** located in the backpanel (slot 1), attached to PCA GEN-BCKPL59 (slot 1). It controls signals used in the Incubator module and in the Dilution cup module. It contains the firmware of both modules.
- **PCA INC-CON59:** one PCA for each incubator, located in the incubator panel and protected by a cover. It contains the Peltier module driver, Peltier module connections, lift home sensor, motor connections and INC-BL59 connections.
- **PCA INC-BL59:** there is one PCA for each incubator, attached to the incubator block. It contains two temperature sensors (LM92 sensors).

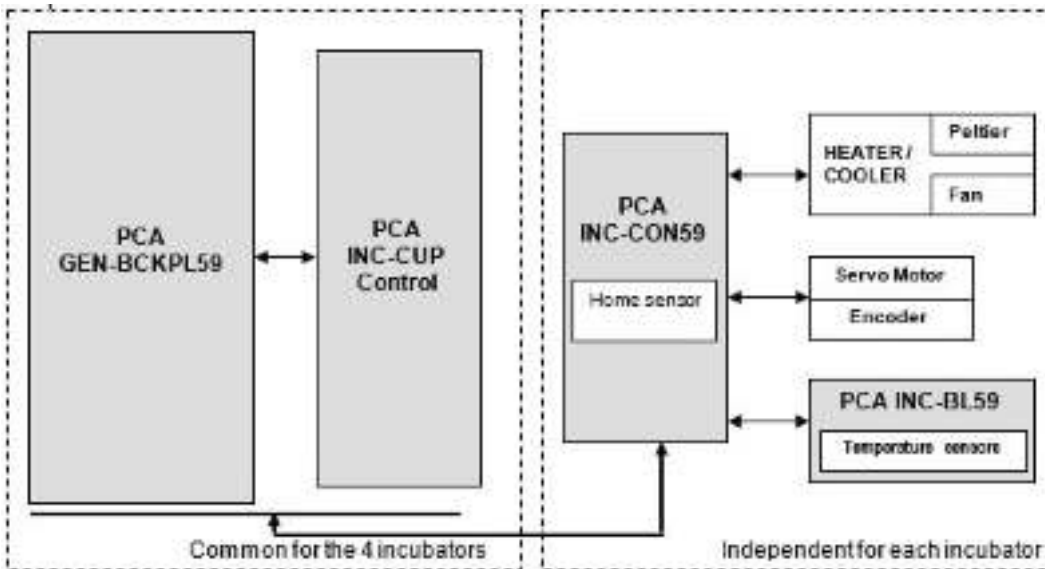


Figure 201 - Main blocks of the Incubator module

2.2.2 Mechanical adjustments

Mechanical adjustments related to the Incubator module are detailed in this section.

2.2.2.1 Adjusting incubator flatness



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Loosen the screws of the incubator laterals and modify manually the position of the incubator, making sure that the distance between the incubator surface and the gripper is ≤ 0.4 mm over slots 1 and 12 of the incubator to be adjusted. See "Verifying incubator flatness" on page 269.
2. Retighten the screws.



Figure 202 - Location of the screws of the laterals of the incubators

3. Loosen the screws of the incubator laterals and modify manually the position of the incubator, making sure that the difference in the Z gripper encoder counts between slots 1 and 12 of the incubator to be adjusted is ≤ 600 . See "Verifying incubator flatness" on page 269.
4. Retighten the screws.
5. Loosen the screws of the incubator laterals and modify manually the position of the incubator, making sure that the difference between the maximum and the minimum encoder counts of probe 1 over the defined points in the incubator to be adjusted is ≤ 1047 . See "Verifying incubator flatness" on page 269.
6. Retighten the screws.
7. In case flatness of the incubator is adjusted, check and perform the incubator height adjustment (see "Adjusting incubator height" on page 260), probes adjustment in the incubator and gripper adjustment in the incubator area (see "Adjusting gripper in the incubator area" on page 475).

2.2.3 Functional adjustments

Incubator height and temperature adjustments are detailed in this section.

2.2.3.1 Adjusting incubator height

This procedure is needed to adjust the incubator position at the upper/pipetting floor (maximum height of each incubator). Adjustment is then performed with the incubator in the upper module.



CAUTION

Check the CAT arm position to avoid it interfering with the movement of the incubator and leading to a crash.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Incubators homed.

Procedure

1. Open the intermediate door (remove the screw to open it) and place manually three cards in the incubator to be adjusted, in slots 1, 6 and 12.
2. Go to the **Procedures** tab in the Technical Service software > **PIP floor Modules** section > **incubators height** button (see "Procedures tab" on page 756).

3. An adjustment wizard is displayed:
 - a. Select the incubator to be adjusted and press the **Next** button.
 - b. Move the CAT arm so it does not interfere with the incubator movement.
 - c. Move up the incubator with the joystick buttons by the amount of counts selected and indicated by the software.

It is possible to modify the amount of counts by sliding the bar.

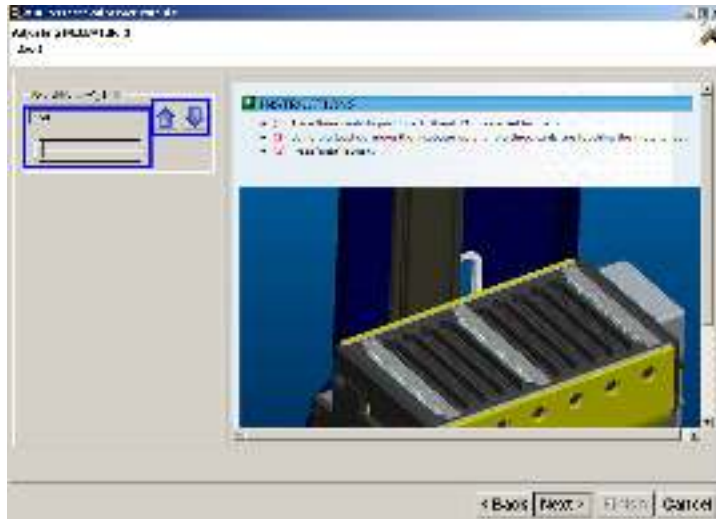


Figure 203 - Joystick buttons and counts sliding bar

Correct adjustment position of the incubator is achieved when the cards placed in the incubator are just touching the surface of the platform.



CAUTION

Check that the incubator block does not crash with the platform, avoiding deforming the platform or forcing the incubator motor.



Figure 204 - Correct adjustment incubator position

If the cards are not touching at the same time, check the assembly of the platform flatness or the incubator flatness.

- d. Once the incubator is in the correct position, press the **Next** button.
A screen appears showing the parameters to be stored.

- e. Press the **Finish** button and press **OK** in the confirmation window.
- f. Once the adjustment is stored, press **OK** in the information window.
- g. Press **OK** in the confirmation window that indicates that home movements will be performed.
- h. Press **OK** to finish the adjustment procedure.

**NOTE**

The adjustment is stored in `E:\data\Erytra\conf\PIPCalibrations.txt`.

2.2.3.2 Adjusting temperature

This procedure is needed to adjust the temperatures (24 °C and 37 °C) for each Incubator module. The incubators can be adjusted at any environmental temperature between 15 °C and 30 °C.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- 230903 DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V.
- 230905 TEMP. PROBE PT-100 FOR CARDS - MiniDIN 4v

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- Upper and intermediate area doors closed.
- All covers placed.
- Incubators homed.

Procedure

1. Place the card probe in slot 6 of the incubator that is going to be adjusted.
2. In the Technical Service software go to the **Procedures** tab > **Adjustments** > **Others** > **temperature** (see "Procedures tab" on page 756) in order to start the adjustment wizard.

3. Select the incubator to adjust and press the **Next** button.

The instrument automatically activates the temperature regulation in RT mode.

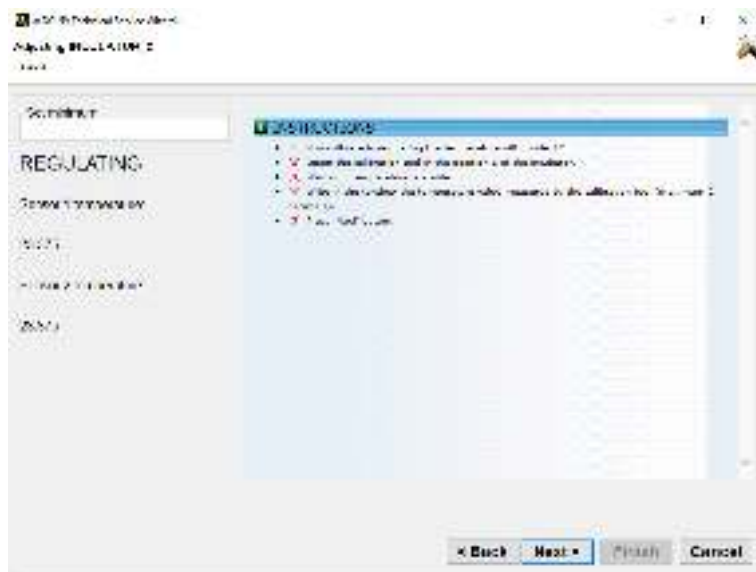


Figure 205 - Incubator adjustment wizard (adjustment mode RT)

4. Follow the instructions of the adjustment wizard:
 - a. Wait until the temperature has stabilized. The wizard displays a message indicating when the temperature is stable.



Figure 206 - Temperature stabilized (adjustment mode RT)



NOTE

The temperature of the incubator is considered as stabilized when it does not vary more than 0.4°C in 3 minutes.

- b. In the **Set minimum** textbox, enter the measured temperature as displayed by the calibrated thermometer (maximum two decimals separated by dot).
- c. Press the **Next** button.

The instrument automatically activates the temperature regulation in the 37 °C mode.

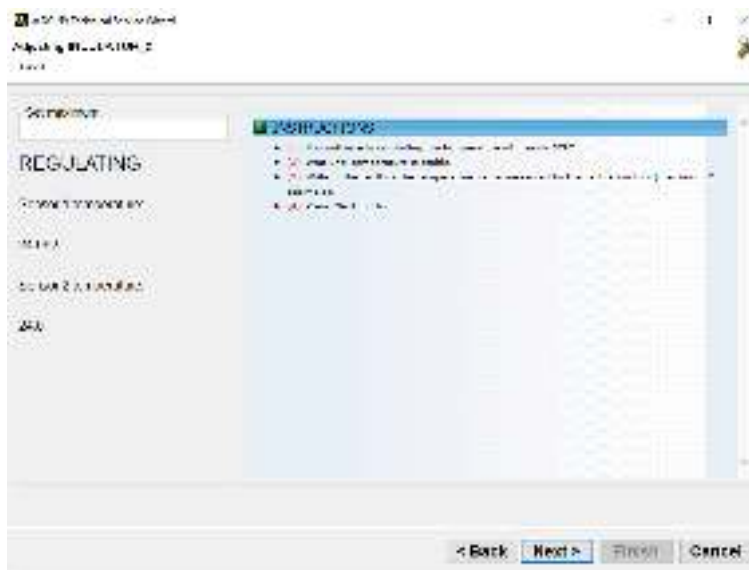


Figure 207 - Incubator adjustment wizard (adjustment mode 37)

5. Follow the instructions of the adjustment wizard:
 - a. Wait until the temperature has stabilized. The wizard displays a message indicating when the temperature is stable.



Figure 208 - Temperature stabilized (adjustment mode 37)



NOTE

The temperature of the incubator is considered as stabilized when its temperature does not vary more than 0.2°C in 5 minutes.

- b. In the **Set maximum** textbox, enter the measured temperature as displayed by the calibrated thermometer.
- c. Press the **Next** button.

A screen appears showing the parameters to be stored.

6. Press the **Finish** button and press the **OK** button in the information window.
7. Press the **OK** button in the confirmation window that indicates that home movements will be performed.

The incubator performs a home movement.

8. Press the **OK** button to finish the adjustment procedure.



NOTE

The adjustment is stored in PCA INC-CUP Control. In case of replacing this PCA, the adjustment must be restored (if a previous backup was performed). Otherwise the adjustment must be performed.

9. If desired, verify the temperature adjustment (See "Verifying temperature" chapter).

If the temperature is introduced and the **Next** button is selected to save it when the temperature is not considered stabilized, a warning message will be displayed:

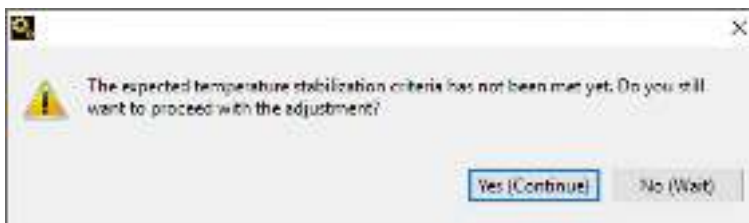


Figure 209 - Warning not stable temperature

- When **Yes (Continue)** option is selected, the action will be logged in DEBUG.technical_ service log file with the temperature values that caused the incubator's temperature not to be stabilized.
- When **No (Wait)** option is selected, the warning window will close, and current step temperature regulation will continue. The temperature values that were previously entered in the textbox will be erased.

2.2.4 Verification procedures

In this section, the verification procedures for the Incubator module are detailed.

2.2.4.1 Verifying temperature



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Place a temperature card probe in slot 6 of the incubator to verify.
2. Move the incubator to verify to the PIP area, checking that the movement does not interfere with the CAT arm.
3. Check that the upper and intermediate area doors are closed and all covers are placed.
4. In the Technical Service software, go to the **Incubators** tab (see "Incubators tab" on page 760) and select the incubator tab to verify.

5. In **Regulate Actions**, activate the **On** option and select 24 °C temperature (**MODE RT**).

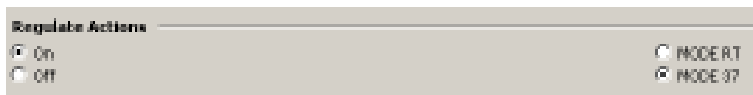


Figure 210 - Regulate Actions section

6. Wait until the temperature has stabilized (maximum 20 minutes, checking that the **Out of Margin** flag is deactivated) and check that the measured temperature displayed by the calibrated thermometer and the temperature in the software are 24 °C ± 3 °C.
7. Change temperature regulation to 37 °C (**MODE 37**) in the Technical Service software.
8. Wait until the temperature has stabilized (maximum 20 minutes checking that the **Out of Margin** flag is deactivated) and check that the measured temperature displayed by the calibrated thermometer and the temperature in the software are 37 °C ± 1 °C.
9. Change temperature regulation to 24 °C (**MODE RT**) in the Technical Service software.
10. Wait until the temperature has stabilized (maximum 20 minutes checking that the **Out of Margin** flag is deactivated) and check that the temperature is 24 °C ± 3 °C.
11. Deactivate temperature regulation (**Incubator** tab > **Regulate Actions**, select the **Off** option).

2.2.4.2 Verifying movement



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Make a home of the CAT arm axes. Click the **Home** button in the **Gripper Arm** tab > **Movement** subtab > **Position Actions** area. See "Gripper Arm tab" on page 777.

2. In the Technical Service software, go to the **Incubators** tab (see "Incubators tab" on page 760) and select the subtab of the incubator to verify.
3. Press the **home** button and check that the **Homed** indicator is activated when the incubator reaches the home position.

**CAUTION**

Check the CAT arm position to avoid it interfering with the incubator movement and leading to a crash.

4. In the Technical Service software, go to the **Incubators** tab > **Home Cycles** tab. See "Incubators tab" on page 760.
5. Press the **start** button of the incubator to verify.
6. Perform home cycles for 5 minutes, checking that no errors appear during the movement.

2.2.4.3 Verifying incubator flatness

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Perform a home of the CAT robotic head and a home of the incubator to verify via the Technical Service software, checking that movements between them do not interfere.
2. Move the CAT robotic head to slot 1 of the incubator to verify via the Technical Service software, **Gripper Arm** tab > **Movement** subtab > **Move Actions** section. See "Gripper Arm tab" on page 777.
3. De-energize the Z axis using the **unlock/stop** button in the Technical Service software, **Gripper Arm** tab > **Movement** subtab > **Position Actions** section. See "Gripper Arm tab" on page 777.

4. Lower the gripper manually (by means of its belt) until the gripper touches gently the surface of the incubator.
5. Check that the distance between the gripper and incubator surface is ≤ 0.4 mm.



Figure 211 - Distance verification between gripper and incubator

6. Perform a home of the Z-axis and repeat the previous procedure in slot 12, checking that the distance between the gripper and incubator surface is ≤ 0.4 mm.
7. Perform a home of the Z-axis and move the CAT robotic head to slot 1 of the incubator to be verified via the Technical Service software, **Gripper Arm** tab > **Movement** subtab > **Move Actions** section. See "Gripper Arm tab" on page 777.
8. Unlock the Z-axis using the **unlock/stop** button in the Technical Service software, **Gripper Arm** tab > **Movement** subtab > **Position Actions** section. See "Gripper Arm tab" on page 777.
9. Lower the gripper manually (by means of its belt) until the gripper touches gently the surface of the incubator.
10. Check Z encoder counts in the Technical Service software.
11. Perform a home of the Z-axis and repeat steps 8 to 11 in slot 12 of the incubator.
12. Check that the difference in Z encoder counts between slot 1 and slot 12 is ≤ 600 .
13. Move the incubator to the PIP area (checking where the gripper arm is located in order to avoid any contact between the incubator and gripper arm) via the Technical Service software, **Gripper Arm** tab > **Movement** subtab > **Gripper Actions** section > **up** button. See "Gripper Arm tab" on page 777.
14. Perform a home of the PIP robotic head.
15. Unlock the PIP robotic head using the **unlock/stop** button in the Technical Service software, **Pip Arm** tab > **Movement** subtab > **Position Actions** section. See "Pip Arm tab" on page 804.

16. Manually move the PIP robotic head and lower probe 1 in the following points, checking Z1 encoder counts:



Figure 212 - Location of the verification points of the incubator flatness in PIP area

17. Check that the difference between Z1 encoder counts in the previous two steps is ≤ 1047 .

2.2.5 Replacement procedures

In this section the replacement procedures related to the Incubator module are described.

For some replacement procedures, it is recommended to disassemble the whole Incubator module from the instrument for a convenient access to all the parts. See "Replacing incubator module" on page 271.

2.2.5.1 Replacing incubator module



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 733620 DG-59 INCUBATOR MODULE.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to disassemble the desired incubator module from the instrument, follow these steps:

1. Switch off the instrument and disconnect it from the mains.
2. Remove the upper and middle rear covers.
3. Remove the screw that attaches the incubator module to the frontal probes platform, located in the PIP floor.



Figure 213 - Incubator module and frontal probes platform

4. Unplug the DB25 connector from PCA INC-CON59 (cable from PCA INC-CON59 to PCA GEN-BCKPL59).
5. Disconnect the ground cable that it is located between the motor and the incubator chassis from the motor side.



Figure 214 - Incubator module ground cable and motor fixing screws

6. Loosen the four screws that attach the motor with the motor flange and move the motor to the right, so the incubator is free to move by the linear guide.



CAUTION

Hold the incubator in order to not allow it to move freely by the linear guide.

7. Move up manually the incubator block until it is placed in the middle of the linear guide (to gain access to the two bottom screws fixing the module to the instrument chassis).
8. Move the motor to the left to couple the system again and fix the motor position by tightening the four screws that attach it to the motor flange.
9. Remove the two bottom screws that attach the incubator module to the chassis of the Erytra.

10. Loosen again the four screws that attach the motor with the motor flange and move the motor to the right, so the incubator is free to move by the linear guide.

**CAUTION**

Hold the incubator in order to not allow it to move freely by the linear guide.

11. Remove the motor-linear guide coupling (it is not fixed and can easily fall from its location when motor and linear guide are not coupled).



Figure 215 - Incubator motor-linear guide coupling

12. Remove the four screws that attach the motor with motor flange and remove the flange.
13. Remove the two upper screws that attach the incubator module to the chassis of the Erytra.

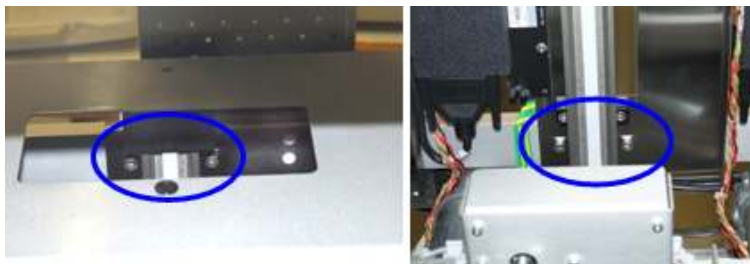


Figure 216 - Location of the screws that attach incubator module to Erytra chassis

14. Unplug the motor (motor power and encoder) from PCA INC-CON59 and remove the motor.
15. Remove the whole module by the CAT floor. It is advisable to first lay the module horizontally on the cards drawers and then extract it by handling the module from instrument front side (intermediate floor door).
16. In order to assemble the new module, repeat steps from 1 to 15 in reverse. Couple the motor-linear guide so the resulting incubator position ensures that the home plate is located higher than the home optical barrier.
17. Check that all incubator module cables and any other cable that may have been affected by the removal/assembly are properly routed and fixed. Pay special attention to the SRM and CM locks ribbon cables and the cables routed through instrument laterals.
18. Check/perform the incubator flatness (see "Adjusting incubator flatness" on page 259), incubator height adjustment (see "Adjusting incubator height" on page 260), probe and gripper adjustment in the incubator.

2.2.5.2 Replacing Peltier module



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233907 PELTIER MODULE (Ver. 2).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to replace a Peltier module, it is not necessary to remove the whole Incubator module, just the incubator block. Do the following:

1. Open the Technical Service software, start the **Driver** (see "Modular tabs" on page 754) and perform a home of the incubator from which the Peltier module is going to be replaced.
2. Move the Incubator module 150000 encoder counts.
3. Switch off the instrument and disconnect it from the mains.
4. Disconnect the Peltier cable.

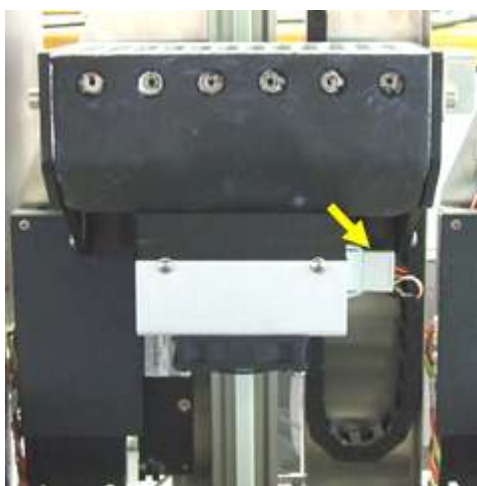


Figure 217 - Peltier module cable

5. Remove the two upper screws fixing the left lateral incubator support to the Incubator module.



Figure 218 - Upper screws fixing left lateral incubator support

6. Remove grounding connection and bumper (two upper screws fixing right lateral incubator) from the right side of the Incubator module.



Figure 219 - Upper screws bumper and grounding connection

7. Disconnect the cable from PCA INC-BL59.
8. Remove the three screws fixing the right lateral incubator support to the linear guide platform.
9. Extract carefully the Incubator module from its position.

10. Remove the four screws fixing the Peltier module to the incubator and remove the module.



Figure 220 - Peltier module assembly

11. Replace the thermal silicon from the incubator U-shape Peltier module area.
12. Assemble the new Peltier module, following the previous steps in reverse order.
13. Check/perform incubator flatness, incubator height adjustment, probe and gripper adjustment in the incubator and the incubator temperature adjustment. See "Procedures tab" on page 756.

2.2.5.3 Replacing PCA INC-BL59



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233718 PCA INC-BL59.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

Once the incubator block has been removed from the instrument (check steps 1-8 in "Replacing Peltier module" on page 274), do the following:

1. Remove the two screws that attach PCA INC-BL59 to the incubator block and remove the PCA.



Figure 221 - PCA INC-BL59 assembly

2. Assemble the new PCA INC-BL59.
3. Install the incubator block in the instrument and check/perform incubator flatness , incubator height adjustment, probe and gripper adjustment in the incubator and the incubator temperature adjustment. See "Procedures tab" on page 756.

2.2.6 Electronics

This section covers the electronics related to the Incubator module.

2.2.6.1 Fuses

Fuses related to the Incubator module are detailed in the following table:

Table 23 - List of Incubator module fuses

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS11	MST250 0034.6615 1A/250V
	FUS12	MST250 0034.6620 3.15A/250V
	FUSINC-1	MST250 0034.6620 3.15A/250V
	FUSINC-2	MST250 0034.6620 3.15A/250V
	FUSINC-3	MST250 0034.6620 3.15A/250V
	FUSINC-4	MST250 0034.6620 3.15A/250V
PCA INC-CUP Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA INC-CON59	NA	NA
PCA INC-BL59	NA	NA

2.2.6.2 LEDs

LEDs related to the Incubator module are detailed in the following table:

Table 24 - List of Incubator module LEDs

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED7	Green 5V (FUS11)
	LED8	Green 24V (FUS12)
	LED9	Green 24V (FUSINC-1)
	LED10	Green 24V (FUSINC-2)
	LED11	Green 24V (FUSINC-3)
	LED12	Green 24V (FUSINC-4)
PCA INC-CUP Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$
PCA INC-CON59	NA	NA
PCA INC-BL59	NA	NA

2.2.6.3 Jumpers and test points

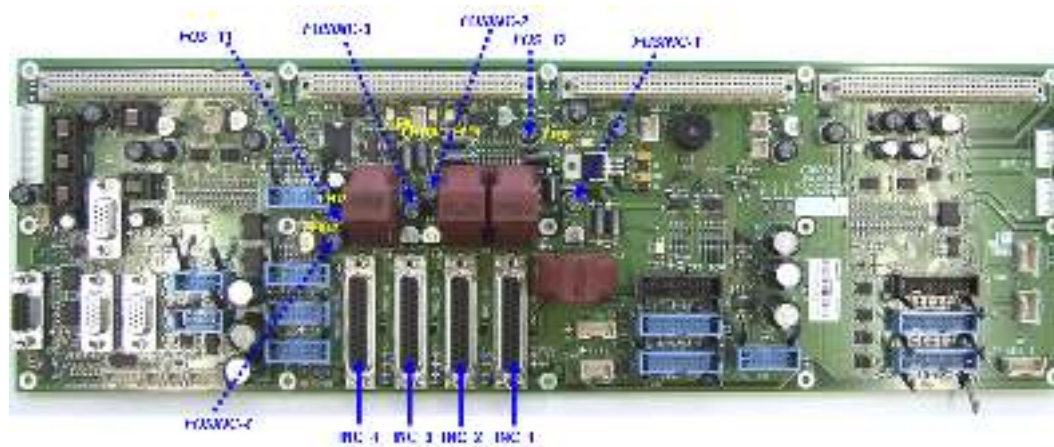
Neither jumpers nor test points are related to the Incubator module.

2.2.6.4 Wiring table

Cables related to the Incubator module are detailed in the following table and their connection are shown in the wiring diagram. To view the wiring diagram, see "Related documents" on page 19 or refer to the Erytra Online Help.

Table 25 - List of Incubator module wiring

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-INC1	H1	PCA INC-CON59	INC_1	PCA GEN-BCKPL59
B-INC2	H1	PCA INC-CON59	INC_2	PCA GEN-BCKPL59
B-INC3	H1	PCA INC-CON59	INC_3	PCA GEN-BCKPL59
B-INC4	H1	PCA INC-CON59	INC_4	PCA GEN-BCKPL59
INC-ENC	H2	PCA INC-CON59	ENCODER	INCUBATOR MOTOR
INC-MOT	H3	PCA INC-CON59	POWER	INCUBATOR MOTOR
INC-PELT	H4	PCA INC-CON59	PELTIER + FAN	PELTIER MODULE
INC-TEMP	H5	PCA INC-CON59	H1	PCA INC-BL59



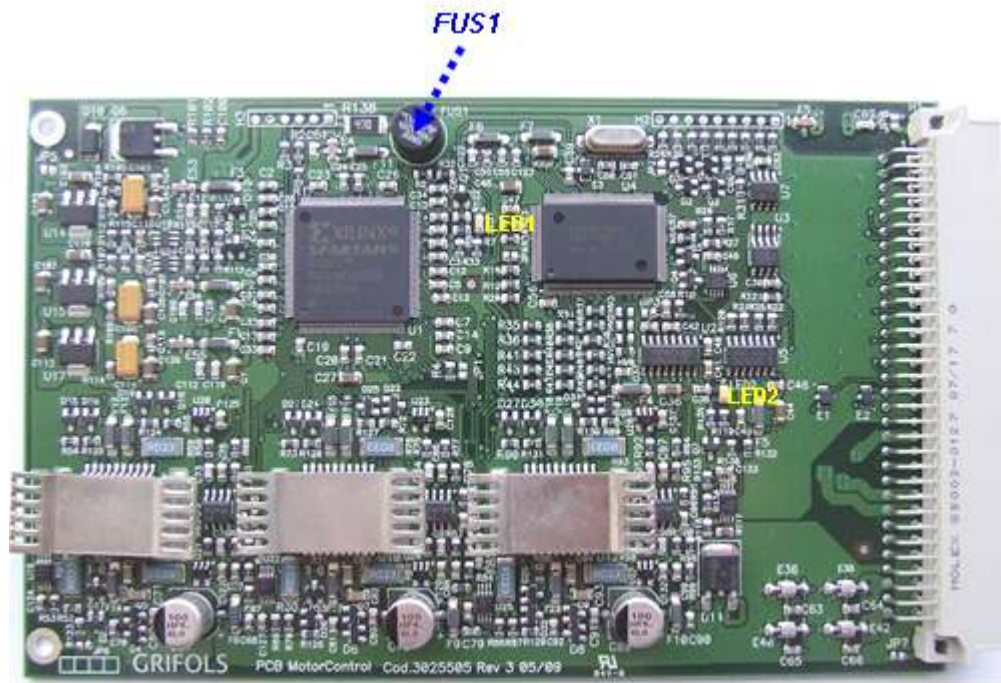


Figure 223 - Location of the fuse and LEDs in PCA MotorControl

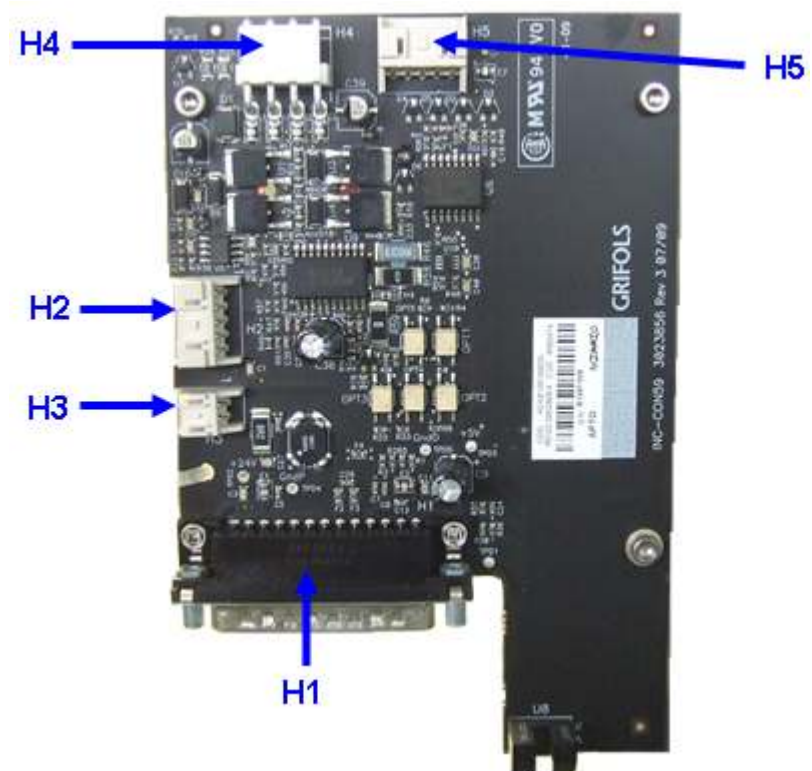


Figure 224 - Location of the connectors in PCA INC-CON59

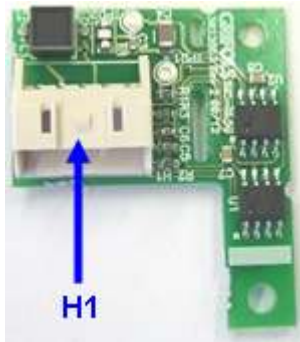


Figure 225 - Location of the connector in PCA INC-BL59

2.2.7 Troubleshooting

Table 26 - List of errors related to the Incubator module

Module	Device	Error code	Error	Cause	Text	Options
INC (07)	Incubator motor (20) Incub 1 (36) Incub 2 (52) Incub 3 (68) Incub 4	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Error when trying to move an incubator up/down. Please choose one of the following options:	RETRY CANCEL Works (it leads to: Incubator {1, 2, 3 or 4} has been disabled)
		005000	ERRORCODE_ARG	Command has wrong arguments		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device		
		032000	CURRENT_LIMIT	Current limit at servo device		
		033000	POSITION_ERROR	A position error has been produced at a servo device		

Module	Device	Error code	Error	Cause	Text	Options
INC (07)	Incubator motor (20) Incub 1 (36) Incub 2 (52) Incub 3 (68) Incub 4	034000	MOVEMENT_ERROR	A movement error has been produced at a servo device	Error when trying to move an incubator up/down. Please choose one of the following options:	RETRY CANCEL Works (it leads to: Incubator {1, 2, 3 or 4} has been disabled)
		035000	HOME_TIMEOUT	A home timeout has been produced at a servo device		
		036000	ERRORCODE_SRV_SYNC_TIMEOUT	A synchronization error has been produced at a servo device		
		037000	ERRORCODE_SRV_UNKNOWN_ERROR	An unknown error has been produced at a servo device		

Module	Device	Error code	Error	Cause	Text	Options
INC (07)	Temperature regulator (16) Incub 1 (32) Incub 2 (48) Incub 3 (64) Incub 4	info	FUSE_FAIL	Incubator temperature regulator is at Fuse fail, Fan fail, Channel error, Driver Error, Sensor 1 error or Sensor 2 error	Incubator {1, 2, 3 or 4} has been disabled	OK
		info	FAN_FAIL			
		info	DRIVER_ERR			
		info	PWM_ERR			
		info	SENSOR_2_ERR			
		info	SENSOR_1_ERR			
		info	DELTA_ERR			
		info	CHANNEL_ERR			
		info	OUT_OF_MARGIN (if Ramping=false)	Incubator temperature is out of margins		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_ CMD	Unknown command code received	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received incorrect command length		
		004000	ERRORCODE_ DEV	Command addressed to an incorrect device		
		005000	ERRORCODE_ ARG	Command has wrong arguments		
		014000	ERRORCODE_ HW_DEV	Addressed device has a problem with its related HW		
		015000	ERRORCODE_ HW_HWFAIL	There is a problem with PCA HW		
		016000	ERRORCODE_ HW_HWNOTSET	PCA setup error		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx) 0	Non firmware errors (xx) 240	000000	ERRORCODE_UNKNOWN	Unknown error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_NOTSAFE	Movement not safe		
		017000	ERRORCODE_FW_UNKNOWN_ERROR	Unknown error		
		061000	ERRORCODE_INVALID_ARGUMENTS	Invalid arguments		
		062000	ERRORCODE_FWCOMM_TIMEOUT	Communication timeout		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx) 0	Non firmware errors (xx) 240	063000	ERRORCODE_ FWCOMM_ CONNECTION_ LOST	Communication lost	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		064000	ERRORCODE_ FWCOMM_COMM_ ERROR	Communication error		
		088000	ERRORCODE_ PRECONDITION_ FAILED	Previous condition for the action is not met		
		089000	ERRORCODE_ STARTUP_ERROR	Driver startup error		
		091000	ERRORCODE_ ALREADY_ STARTED	Another instance of Driver is already opened		
		090000	ERRORCODE_ UNEXPECTED_ ERROR	A Driver unexpected error has been produced		

Table 27 - Incubator module troubleshooting

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Incubator disabled when running tests	No heating / cooling	Check no heating / cooling section in this troubleshooting
	Temperature sensors error	Check temperature sensors error section in this troubleshooting
	Incorrect temperature regulation	Check Incorrect temperature regulation section in this troubleshooting
Incubator never used when running tests	Reserved for STAT samples	Check RESERVE_INCUBATOR_FOR_STAT parameter in the E:\conf\erytra.properties file
	No heating / cooling	Check no heating / cooling section in this troubleshooting
	Temperature sensors error	Check temperature sensors error section in this troubleshooting
	Incorrect temperature regulation	Check Incorrect temperature regulation section in this troubleshooting

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
No heating / cooling	Fuse blown	Check / replace FUSINC-x
	Peltier cell failure	Check / replace Peltier module
	Peltier fan failure	
	Cable damaged	Check / replace incubator chain
	PCA INC-CON59 failure	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Replace PCA GEN-BCKPL59
Temperature sensors error	Temperature sensor/s damaged	Check / replace PCA INC-BL59
	Cable damaged	Check / replace incubator chain
	PCA INC-CON59 failure	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Incorrect temperature regulation	Temperature of the room out of specifications	Check room conditions
	Wrong adjustment of incubator temperature	Check / readjust incubator temperature
	Temperature sensor/s damaged	Check / replace PCA INC-BL59
	Cable damaged	Check / replace incubator chain
	PCA INC-CON59 failure	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / Current limit error INC1: 0720032000 INC2: 0736032000 INC3: 0752032000 INC4: 0768032000 INC1: 0720034000 INC2: 0736034000 INC3: 0752034000 INC4: 0768034000	Temperature of the room out of specifications	Check room conditions
	Incubator movement is interfered	Check if something is interfering in incubator movement
	Incubator crashes with the probes platform	Adjust incubator height
		Check probes platform position
	Fuse blown	Check / replace FUS11 or FUS12
	Motor and linear guide are not correctly coupled	Check coupling between motor and linear guide
	Motor/encoder damaged	Check / replace motor
	PCA INC-CON59 failure	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Home position is not detected INC1: 0720007000 INC2: 0736007000 INC3: 0752007000 INC4: 0768007000 INC1: 0720035000 INC2: 0736035000 INC3: 0752035000 INC4: 0768035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Incubator position is lower than optical barrier	Move manually incubator over optical barrier position
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

2.3 Centrifuge module (CEN)

In this chapter, the Centrifuge module is described. It includes an overall description of the functions, as well as the corresponding adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, and a troubleshooting section.

2.3.1 Description

The main function of this module is the centrifugation of the cards. There are two centrifuges located in the lower part of the instrument, under card drawers. Each centrifuge has a capacity of 12 cards.

The structure of the centrifuge is formed by the following elements:

- Top cover: it holds the whole centrifuge module to the instrument and supports the centrifuge motor structure and the casing. It has an opening to allow the gripper of the CAT arm to load / unload cards.
- Motor structure: formed by a motor, a holder and a set of silentblocks.
- Rotor and card holders: the rotor is the rotating part of the centrifuge and it is attached to the motor shaft. Holders are located in the rotor and they are responsible for allocating the cards to be centrifuged.
- Casing: attached to the base, it covers the complete structure in order to improve operator safety and power consumption.
- Electronics: provides interface with the instrument control.

Centrifuge motors are servo controlled and have a reference position, which is detected by the Hall effect sensors. The home process allows the motors to move to the desired position once the known reference is found. The spinning direction is clockwise. The motor is isolated dynamically with silentblocks to reduce possible transmission of vibration to the rest of the instrument.

The encoder resolution for centrifuge motor is as follows:

Table 28 - CEN encoder resolution

CEN	Encoder counts / turn
Rotation	4000 counts / turn

There is no unbalance detector, thus a non pipetted card is used to balance the centrifuge when an odd number of cards are centrifuged. There is no specific card to balance the centrifuge, it is used a card located in any of the card drawers.

The centrifugation process has four parameters: acceleration, speed, brake cycle and time and all of them are controlled by firmware. Values of the centrifugation are the following:

- Centrifuging speed: 1206 rpm $\pm$ 36 rpm.
- Centrifugation duration: 550 s $\pm$ 15 s (including braking time).

Time monitoring of the centrifuge is controlled by the computer clock. Once the firmware sends the command to start a centrifugation, the software stores initial time. When the firmware sends the command to stop the centrifugation, the software compares the final time with the initial one and it checks if there is any deviation.

All card holders are suspended and unbalanced to assure correct position for centrifuging the cards when the centrifuge is rotating, and assure that CAT arm can grab the card when the centrifuge is stopped. When a centrifugation is running, centrifugal force is applied aligned to microtube axis due to holders rotation. Positioning is forced by rotation itself, no external device is used to position the centrifuge holder.

A general block diagram of the module is shown in the following figure.

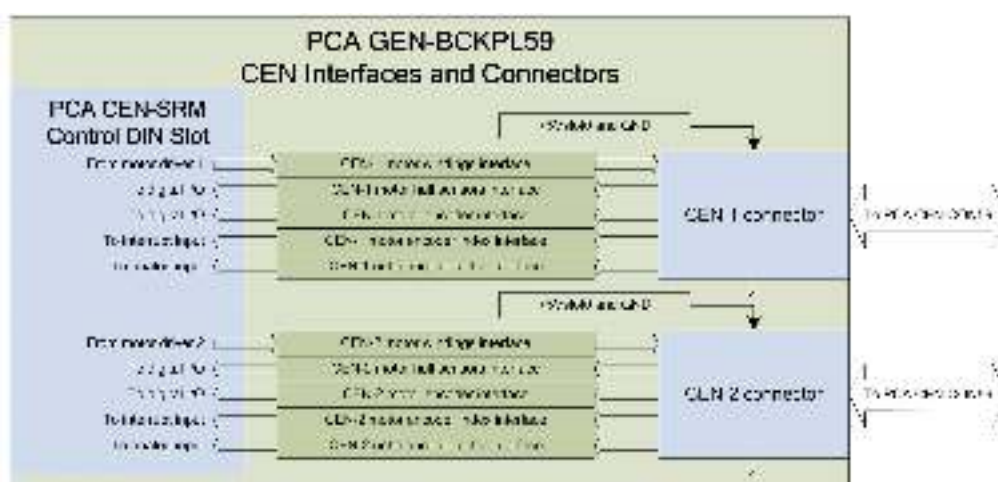


Figure 226 - Centrifuge module block diagram

PCAs related to the centrifuge module are:

- PCA GEN-BCKPL59: main PCA of the instrument, located in the backpanel.
- PCA CEN-SRM Control: located in the backpanel (slot 0), attached to PCA GEN-BCKPL59. It controls signals used in the Centrifuge module and samples / reagents drawers management module. It contains the firmware of both modules.
- PCA CEN-CON59: located over the centrifuge, this PCA contains the centrifuge motor and encoder interface.

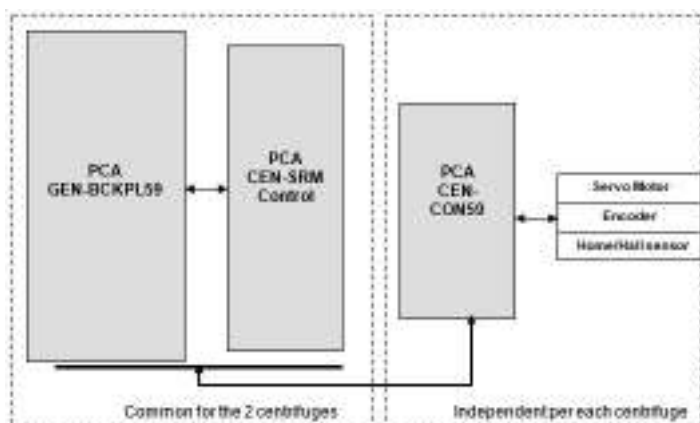


Figure 227 - Main blocks of the Centrifuge module

2.3.2 Mechanical adjustments

Mechanical adjustments related to the Centrifuge module are detailed in this section.

2.3.2.1 Adjusting rotor height

Rotor height mechanical adjustment is required to set the rotor vertical position in the motor shaft in order to allow the card transport robot arm to grab cards from the centrifuge.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233821 CENTRIFUGE ROTOR ADJUSTMENT TOOL

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Extract the centrifuge from the instrument.

To extract the centrifuge from the instrument, follow the next steps:

1. Open completely solution A and solution B bottle drawers (for centrifuge 1) or Waste 1 and Waste 2 bottle drawers (for centrifuge 2) using the software or manually. See "Drawers tab" on page 792.

2. Remove the bottles and close bottle drawers.
3. Open the intermediate door, opening first the user tray and then removing the screw of the intermediate door.
4. Open completely card drawers 1 and 2 (for centrifuge 1) or card drawers 3 and 4 (for centrifuge 2).

**WARNING**

It is recommended to turn off the instrument at this point.

5. Switch off the instrument. Disconnect cable B-CEN1 (for centrifuge 1) or B-CEN2 (for centrifuge 2), located at the rear part of the centrifuge module.
6. Remove the four screws that attach the centrifuge to the chassis.



Figure 228 - Centrifuge removal

7. Close the card drawers.
8. Open bottles drawers in order to have access to the centrifuge module and rotate the centrifuge in a clockwise direction until it is free from its position. Remove it from the instrument by the bottles drawers area.
9. Remove the centrifuge cover by means of the lateral flap located at one side of the cover and the screw located at the other side.
10. Place the centrifuge cover over the pot with the motor axis outside the pot.
11. Remove all the centrifuge card holders.
12. Loosen the nut and extract the rotor.

Procedure

To adjust the rotor height, follow the next steps:

1. Place the centrifuge rotor height adjustment tools over the centrifuge cover at each side and place the rotor over them.

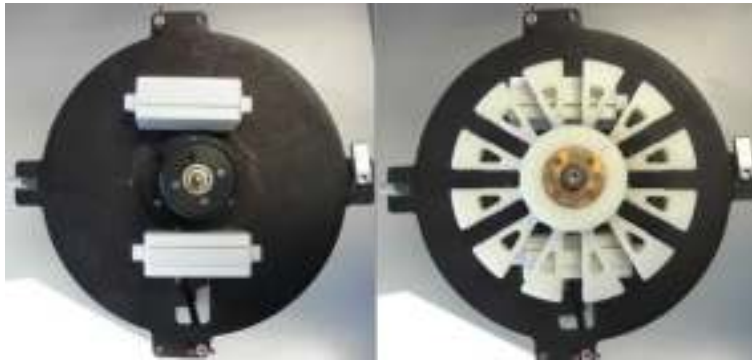


Figure 229 - Rotor height adjustment tools

2. Screw the nut to the centrifuge motor axis. Remove the CENTRIFUGE ROTOR ADJUSTMENT TOOL and check that distance between rotor and basis of the cover is $81.3 \text{ mm} \pm 0.5 \text{ mm}$.

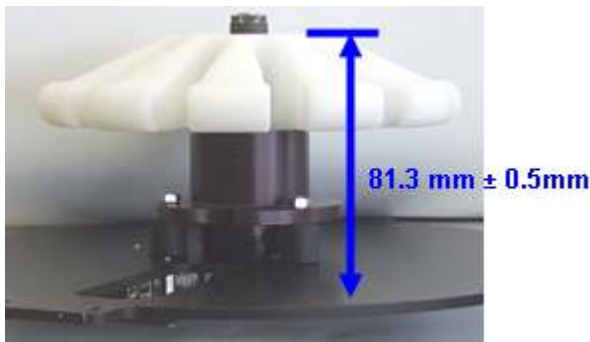


Figure 230 - Distance between the rotor and the basis of the cover

3. Remove the cover from the pot and place the centrifuge card holders in the rotor.
4. Place correctly the centrifuge cover in the pot (with the rotor inside the pot) and lock them with the lateral flap and the screw.
5. Install the centrifuge module in the instrument, placing the four screws that attach the centrifuge to the structure of the instrument.
6. Connect cable B-CEN1 (for centrifuge 1) or B-CEN2 (for centrifuge 2).
7. Perform gripper adjustment in the centrifuge. See "Procedures tab" on page 756.

2.3.3 Functional adjustments

There are no functional adjustments in the Centrifuge module. Functional adjustments related to the gripper inside the centrifuge are detailed in "Card transport module (CAT)" on page 462.

2.3.4 Verification procedures



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. In the Technical Service software, go to the **Centrifuges** tab. See "Centrifuges tab" on page 767. Then select the **Centrifuge 1** tab or **Centrifuge 2** tab and press the **home** button. Wait until the rotor stops its movement and check that the **Homed** flag is activated.



Figure 231 - Centrifuge motor section in the Technical Service software

2. Place a reflective adhesive in the rotor between two consecutive holders.



Figure 232 - Reflective adhesive in the rotor of the centrifuge

- Then, in **Centrifuge 1** tab or **Centrifuge 2** tab, select **PROFILE_0** in the **Move centrifuge** to section, and press the **start** button.



Figure 233 - Move Actions section in the Technical Service software

- Place a tachometer vertically over the slot of the centrifuge in order to measure the rpm. Check that the centrifugation speed is $1206 \text{ rpm} \pm 36 \text{ rpm}$ with duration of $550 \text{ s} \pm 15 \text{ s}$ (including braking time, so measure the time between centrifuge encoder counts start to change until they stop). Check that there are no errors in the software.



Figure 234 - Slot of the centrifuge (left) and measurement of the centrifugation speed with a tachometer (right)

- Remove the reflective adhesive placed in the rotor.



NOTE

The use of any tachometer is possible as long as it complies with following specifications:

Type: optical.

Scale division (rpm): $\leq 1 \text{ rpm}$.

Range (rpm): 1 - 99999 rpm.

Maximum uncertainty permitted: $\pm 3 \text{ rpm}$.

Minimum detection distance: 5 cm with visible white light.



NOTE

The use of any chronometer is possible as long as it complies with having at least a precision about 1 second in 10 minutes.

2.3.5 Replacement procedures

Replacement procedures related to the Centrifuge module are detailed in this section.

2.3.5.1 Replacing motor



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233727 MOTOR CENTRIFUGE.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the centrifuge module from the instrument and centrifuge rotor. See "Adjusting rotor height" on page 297.
2. Remove the cover of PCA CEN-CON59.
3. Disconnect cables CEN-1, CEN-2 and CEN-3 from PCA CEN-CON59.

4. Remove the three nuts that attach the motor cover to the silentblocks (blocking the set screw with an Allen key and then loosening the nut).



Figure 235 - Removing the motor cover of the centrifuge

5. Remove the motor with the cover.
6. Remove the four screws that attach the cover to the motor.
7. Replace the motor.
8. Attach the cover to the new motor by means of the four screws aligning motor cables with cover flat section.



Figure 236 - Motor cover-motor alignment

- Place the motor with the cover in the silentblocks and screw the nuts to the setscrews (block the set screw with an Allen key and screw the nut). Check that the set screw is 10 mm above the basis of the silentblock. Pay attention that the cover is not symmetrical; the rounded slot must be positioned in front of the cover slot.



Figure 237 - Height of the set screw in the silentblocks and motor cover alignment

- Connect cables CEN-1, CEN-2 and CEN-3 to PCA CEN-CON59.
- Place the cover of the PCA CEN-CON59 and install the whole module in the instrument.
- Perform gripper adjustment in the centrifuge. See "Procedures tab" on page 756.

2.3.5.2 Replacing centrifuge rotor



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233821 CENTRIFUGE ROTOR ADJUSTMENT TOOL.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 733965 CENTRIFUGE ROTOR V2.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the centrifuge rotor, follow these next steps:

- Open completely solution A and solution B bottle drawers (for centrifuge 1) or Waste 1 and Waste 2 bottle drawers. See "Drawers tab" on page 792.
- Remove the bottles and close the bottle drawers.

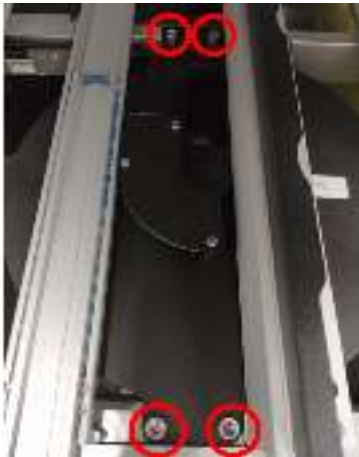
3. Open the intermediate door, opening first the user tray and then removing the screw of the intermediate door.
4. Using the Technical Service software, open completely card drawers 1 and 2 (for centrifuge 1) or card drawers 3 and 4 (for centrifuge 2).



5. Switch off the instrument. Disconnect cable B-CEN1 (for centrifuge 1) or B-CEN2 (for centrifuge 2), located at the rear part of the centrifuge module.



6. Remove the four screws that attach the centrifuge to the chassis.



7. Close the card drawers.
8. Open bottle drawers in order to have access to the centrifuge module and rotate the centrifuge in a clockwise direction until it is free from its position. Carefully remove it from the instrument by the bottles drawers area.



9. Remove the centrifuge cover by means of the lateral flap located at one side of the cover and the screw located on the other side.



10. Place the centrifuge cover over the pot with the motor axis outside the pot.
11. Remove all the centrifuge card holders.

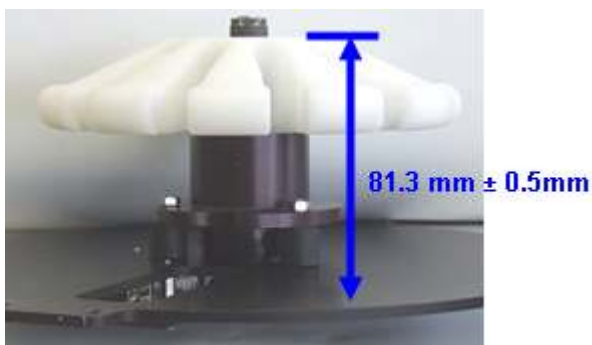
12. Loosen the nut and extract the rotor.



13. Place the CENTRIFUGE ROTOR ADJUSTMENT TOOL over the centrifuge cover at each side and place the new centrifuge rotor over them.



14. Screw the nut to the centrifuge motor axis. Remove the CENTRIFUGE ROTOR ADJUSTMENT TOOL and check that distance between rotor and basis of the cover is $81.3 \text{ mm} \pm 0.5 \text{ mm}$.



15. Remove the cover from the pot and place the centrifuge card holders in the rotor.
16. At this point, it is recommended to manually spin the new rotor to ensure that holders can freely move up and down once placed in the rotor.

17. Place the centrifuge cover in the pot (with the rotor inside the pot) and lock them with the lateral flap and the screw.
18. Install the centrifuge module in the instrument, placing the four screws that attach the centrifuge to the structure of the instrument.
19. Connect cable B-CEN1 (for centrifuge 1) or B-CEN2 (for centrifuge 2).
20. Perform gripper adjustment in the centrifuge which rotor has been replaced. See "Procedures tab" on page 756.

2.3.5.3 Removing centrifuge pot



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to remove the pot of centrifuge 1, follow the next steps:

1. Open completely solution A and solution B, and remove the two solutions.
2. Loosen the screw (at the left side), and pull down the lever (on the right side).

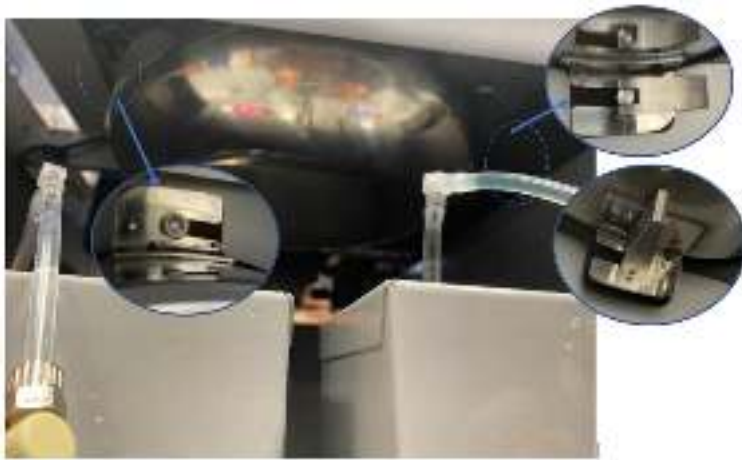


Figure 238 - Screws and lever of the centrifuge pot

3. Turn the centrifuge pot clockwise while pressing slightly upwards until it is released.



Figure 239 - Turning the centrifuge pot

4. Once the pot has been released, it can be removed from the instrument through the opening above the fluidic drawers.



Figure 240 - Removing the centrifuge pot

To remove the pot of centrifuge 2, the same procedure must be carried out by opening the other three fluidic drawers.

2.3.6 Electronics

This section covers the electronics related to the Centrifuge module.

2.3.6.1 Fuses

Fuses related to the Centrifuge module are detailed in the following table.

Table 29 - List of fuses of Centrifuge module

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS01 FUS02	MST250 0034.6615 1A/250V MST250 0034.6620 3.15A/250V
PCA CEN-SRM Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA CEN-CON59	NA	NA

2.3.6.2 LEDs

LEDs related to Centrifuge module are detailed in the following table.

Table 30 - List of LEDs of Centrifuge module

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED5	Green 5V (FUS01)
	LED6	Green 24V (FUS02)
PCA CEN-SRM Control	LED1	Yellow When the instrument is switched on, LED is on and a few seconds later is off. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$.
PCA CEN-CON59	NA	NA

2.3.6.3 Jumpers and test points

Neither jumpers nor test points are related to the Centrifuge module.

2.3.6.4 Wiring table

Cables related to the Centrifuge module are detailed in the following table and their connection are shown in the wiring diagram. To view the wiring diagram, see "Related documents" on page 19 or refer to the Erytra Online Help.

Table 31 - List of wiring of Centrifuge module

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-CEN1	H3	PCA CEN-CON59	CEN1	PCA GEN-BCKPL59
B-CEN2	H3	PCA CEN-CON59	CEN2	PCA GEN-BCKPL59
CEN-MOT	H1	PCA CEN-CON59	NA	MOTOR POWER
CEN-HALL	H2	PCA CEN-CON59	NA	MOTOR HALL SENSORS
CEN-ENC	H4	PCA CEN-CON59	NA	MOTOR ENCODER

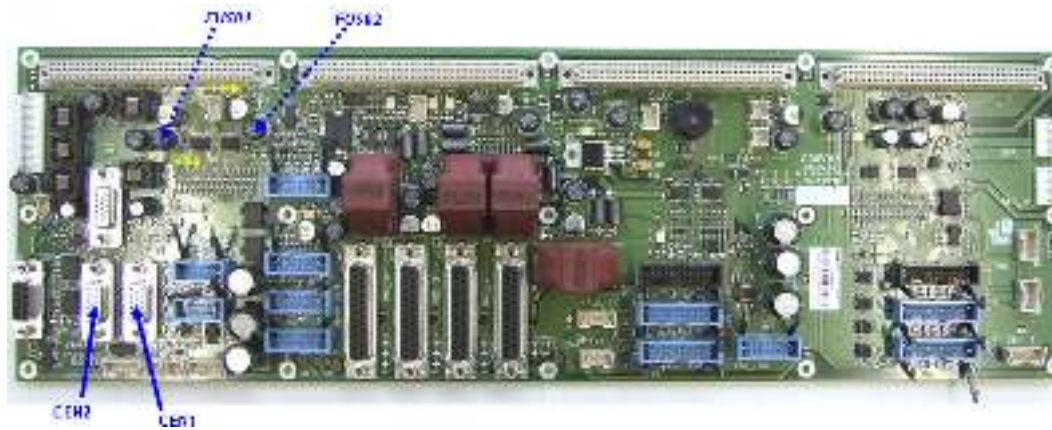


Figure 241 - Location of the connectors, fuses and LEDs in PCA GEN-BCKPL59 related to Centrifuge module

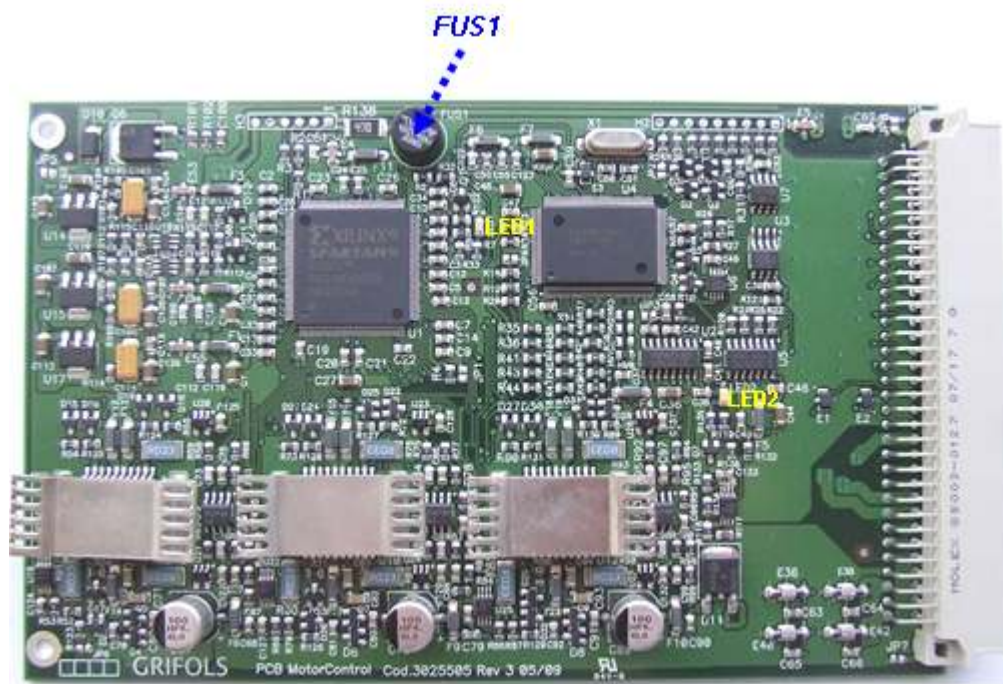


Figure 242 - Location of the fuse and LED in PCA CEN-SRM Control

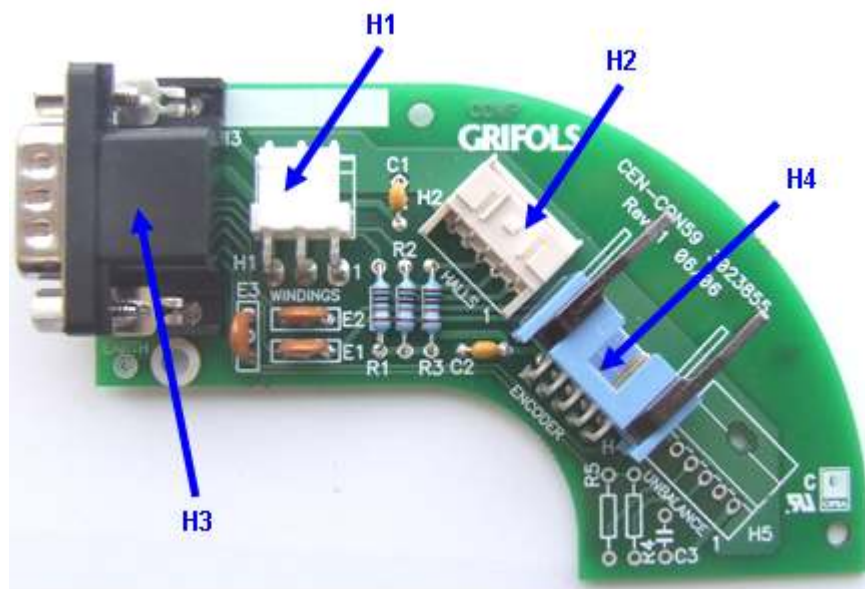


Figure 243 - Location of the connectors in PCA CEN-CON59

2.3.7 Troubleshooting

Table 32 - List of errors related to Centrifuge module

Module	Device	Error code	Error	Cause	Text	Options
Centrifuge (03)	Centrifuge motor (00) Centr 1 (01) Centr 2	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Centrifuge {1 or 2} has been disabled	OK
		005000	ERRORCODE_ARG	Command has wrong arguments		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device		
		012000	ERRORCODE_SPINBADPROFILE	Command includes a spin profile that is invalid for the addressed device		

Module	Device	Error code	Error	Cause	Text	Options
Centrifuge (03)	Centrifuge motor (00) Centr 1 (01) Centr 2	013000	ERRORCODE_NOTSPINNING	The addressed device is not performing the requested spin	Centrifuge {1 or 2} has been disabled	OK
		033000	POSITION_ERROR	A position error has been produced at a servo device		
		035000	HOME_TIMEOUT	A home timeout has been produced at a servo device		
		036000	ERRORCODE_SRV_SYNC_TIMEOUT	A synchronization error has been produced at a servo device		
		037000	ERRORCODE_SRV_UNKNOWN_ERROR	An unknown error has been produced at a servo device		
Centrifuge (03)	Centrifuge motor (00) Centr 1 (01) Centr 2	032000	CURRENT_LIMIT	Current limit at servo device	When loading/unloading a card:	RETRY
					Error during centrifuge positioning.	CANCEL Works (it leads to: Centrifuge {1 or 2} has been disabled)
					During centrifugation:	OK
					Centrifuge {1 or 2} has been disabled	

Module	Device	Error code	Error	Cause	Text	Options
Centrifuge (03)	Centrifuge motor (00) Centr 1 (01) Centr 2	034000	MOVEMENT_ERROR	A movement error has been produced at a servo device	When loading/unloading a card:	RETRY
					Error during centrifuge positioning.	CANCEL Works (it leads to: Centrifuge {1 or 2} has been disabled)
					During centrifugation:	OK
					Centrifuge {1 or 2} has been disabled	

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received a command with incorrect length		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device		
		005000	ERRORCODE_ARG	Command has wrong arguments		
		014000	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW		
		015000	ERRORCODE_HW_HWFAIL	There is a problem with PCA HW		
		016000	ERRORCODE_HW_HWNOTSET	PCA setup error		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	000000	ERRORCODE_ UNKNOWN	Unknown error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_ NOTSAFE	Movement not safe		
		017000	ERRORCODE_FW_ UNKNOWN_ERROR	Unknown error		
		061000	ERRORCODE_ INVALID_ ARGUMENTS	Invalid arguments		
		062000	ERRORCODE_ FWCOMM_TIMEOUT	Communication timeout		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	063000	ERRORCODE_ FWCOMM_ CONNECTION_LOST	Communication lost	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		064000	ERRORCODE_ FWCOMM_COMM_ ERROR	Communication error		
		088000	ERRORCODE_ PRECONDITION_ FAILED	Previous condition for the action is not met		
		089000	ERRORCODE_ STARTUP_ERROR	Driver startup error		
		091000	ERRORCODE_ ALREADY_STARTED	Another instance of Driver is already opened		
0	240	090000	ERRORCODE_ UNEXPECTED_ ERROR	A Driver unexpected error has been produced	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Table 33 - Centrifuge module troubleshooting.

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Centrifuge disabled	Centrifuge disabled was already disabled in software	Check centrifuge status in Erytra software
	Movement error / Current limit error	Check Movement error / Current limit error section in this troubleshooting
	Home position is not detected	Check Home position is not detected section in this troubleshooting
Movement error / Current limit error	Fuse blown	Check / replace FUS01 or FUS02
CEN1: 0300032000 CEN2: 0301032000 CEN1: 0300034000 CEN2: 0301034000	Rotor is not correctly assembled	Check rotor assembly
	Motor/encoder damaged	Check / replace motor
	PCA CEN-CON59 failure	Check / replace PCA CEN-CON59
	Check wiring	Check / replace wiring
	PCA CEN-SRM Control failure	Check / replace PCA CEN-SRM Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Home position is not detected CEN1: 0300007000 CEN2: 0301007000 CEN1: 0300035000 CEN2: 0301035000	Fuse blown	Check / replace FUS01 or FUS02
	Failure in home detector	Check / replace motor
	PCA CEN-CON59 failure	Check / replace PCA CEN-CON59
	Check wiring	Check / replace wiring
	PCA CEN-SRM Control failure	Check / replace PCA CEN-SRM Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
Centrifuging speed / time is not the correct one	Motor/encoder damaged	Check / replace motor
	Check wiring	Check / replace wiring
	PCA CEN-SRM Control failure	Check / replace PCA CEN-SRM Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

2.4 Pipetting robot arm module (PIP)

In this chapter, the Pipetting robot arm module is described. It includes an overall description of the functions, as well as the corresponding adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, and a troubleshooting section.

2.4.1 Description

The functions of this module are:

- Presence detection and identification of samples and reagents.
- Pipetting samples and reagents into cards, including appropriate controls and level detection.

To do so, this module has a five-axis robotic head: X, Y, Z1 (Probe 1), Z2 (Probe 2) and horns (involved in barcode reading).

Robotic head X-axis structure is formed by a mechanized aluminium plate screwed to the instrument laterals. A linear guide screwed to the plate allows movement of the X-axis. The X-axis is moved by a servo-controlled motor assembled in the left area. This motor turns and provokes that a small belt moves. This belt transmits the movement by means of a gear to a large belt attached to the X wagon (assembled on the linear guide). Large belt tension is adjusted by modifying the X end pulley position. Home position is defined at the left and using an L-shaped part assembled in the X wagon that will get in an optical barrier. PIP module and CAT module are built in the same X-axis.

Y-axis is moved by a servo-controlled motor assembled in the back axis area. This motor turns and provokes that the small belt moves. This belt transmits the movement by means of a gear to a large belt attached to the Y wagon (assembled on a linear guide). Small belt tension can be adjusted by modifying the motor position (slotted hole). Large belt tension is adjusted by modifying the Y end pulley position. Home position is defined at the back and using an L-shaped part assembled in the Y wagon that will get in an optical barrier.

Z axes (composed of probe 1 axis, probe 2 axis and horns axis) are moved by servo-controlled motors assembled in robotic head bottom. Motor turns and provokes that a small belt moves. This belt provokes that a pulley fixed to a leadscrew with a set screw turns and this one that the probes or horns move. Small belt tension can be adjusted modifying motor position (slotted hole).

Encoders resolution for the PIP axes is as follows:

Table 34 - PIP axes encoder resolution

PIP Axis	Encoder counts / mm
X	285.714 counts / mm
Y	285.714 counts / mm
Z1	2095.2 counts / mm
Z2	2095.2 counts / mm
Horns	2095.2 counts / mm

Horns home position is defined at Z axis top and using an L-shaped part assembled in leadscrew nut that will get in an optical barrier placed in PCA PIP-TOW59. Once horns activate the optical barrier, they move up. Probes home positions are defined at the top Z axis. Using L-shaped parts assembled in the probe wagons, they get in optical barriers placed in PCA PIP-TOW59. Once probes activate the optical barrier, they move to the position adjusted in the probes offset adjustment.

Level detection system is based on a capacitive system (RC oscillator) where the probe is part of the capacitor. When probe touches the surface of liquid, its capacitance is modified and a decrease in the oscillation frequency is obtained. There are two different PCA, one per each probe, sharing PCB and all components except those related to generate the oscillation frequency, in order to have a different frequency at each probe and to avoid any coupling. The level detector system is used also to detect if card foil is present when first pipetting into a well. If impact detector system is installed, it is possible to detect and protect probes from undesired impacts which are frequently caused by improper user operation.

Presence detection of samples and reagents is performed by means of a laser detection system. The laser is placed in a fixed position in the robot head, no movement is required. The robot arm passes over the racks with samples and reagents and with the detection of the reflected light of the laser, the system is able to detect whether is located a tube, vial or bottle in the scanned rack. Also the reflected light of the laser gives information of the sample tubes diameter. To scan the reagents racks, X-movements of the PIP arm are performed whereas to scan the samples racks, Y-movements of the PIP arm are performed.

Identification of samples and reagents is performed by means of two barcode readers assembled at each side of the horns located in the robotic head. Barcode reader reference is Symbol SE-955 (S/N < 059-0001335) or Zebra SE-965 (S/N ≥ 059-0001335) and the communication is through RS-232. The robot arm passes over the rack and the horns move down, activating the barcode reader to read the barcode labels. The tubes, vials and bottles must be faced with the barcode label outside the rack in order to let the system read it.

Allowed types of barcodes are the following ones:

- Code 39 (check digit enabled).
- Interleaved 2 of 5 (check digit enabled).
- Code 128.
- ISBT 128.

- GS1 - 128 (formerly UCC/EAN-128).
- Codabar.
- Others upon barcode readers configuration.

Sample barcode specifications (* based on code type 128 barcode) are the following ones:

- Minimum resolution*: 0.15 mm.
- Position*: ≥ 3 mm from top / bottom.
- Quiet zone*: ≥ 2 mm.
- Maximum length*: 45 mm.
- Minimum width*: 8 mm.
- Slope*: $\leq 20^\circ$
- Maximum number of characters: 24.

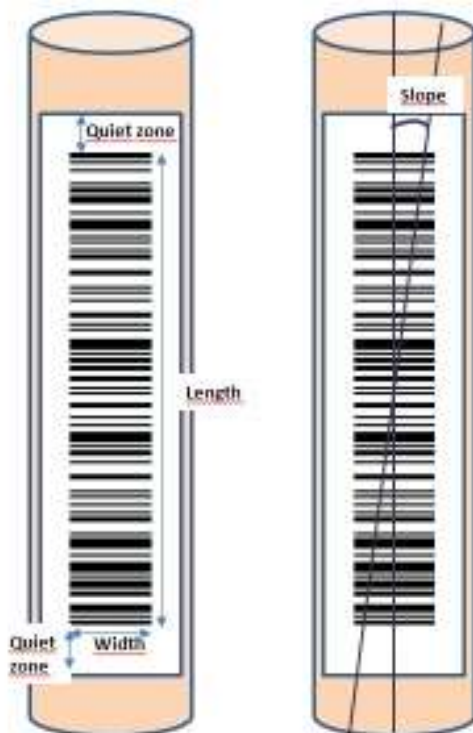


Figure 244 - Sample barcode specifications

The number of PIP barcodes reading retries for sample tubes and reagents and the total number of samples and reagents barcodes read in the current session are logged at instrument Driver (DG59Driver) log when the shutdown process is performed.

A block diagram of the module is shown in following figure:

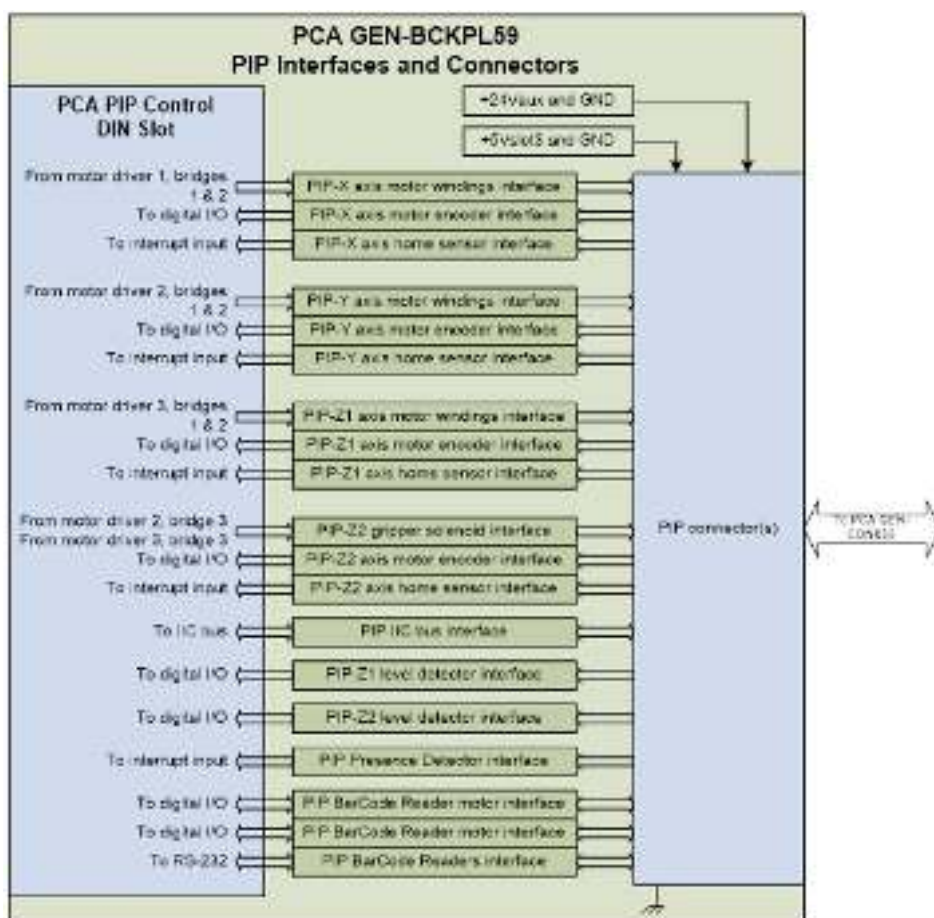


Figure 245 - Pipetting robot arm module block diagram

PCAs related to PIP module are:

- **PCA GEN-BCKPL59:** the main PCA of the instrument, located at the backpanel.
- **PCA PIP Control:** located at the backpanel (slot 3), attached to PCA GEN-BCKPL59. It controls signals used in the pipetting module. It contains the firmware of this module.
- **PCA GEN-CONX59:** this PCA is located at the X axis left side (looking from the front of the instrument). It contains the X axis home sensor and X axis motor powering and encoder.
- **PCA PIP-CONY59:** located at the Y axis back (looking from the front of the instrument). It contains the Y axis home optical barrier (power and signal), Y axis motor powering and encoder. Other parts are detailed in the fluidics module.
- **PCA PIP-TOW59:** located inside the robotic head. It contains probes and horns home optical barriers (power and signal), probes axis motor powering and encoder, horns axis motor powering and encoder and barcode reader signals.
- **PCA PIP-P1OSC59:** located inside the robotic head, over Z1 probe. It contains a level detector oscillator and sends oscillator signal to PCA PIP-TOW59. Oscillation frequency for the Z1 probe is different than for the Z2 probe.

- **PCA PIP-P2OSC59:** located inside the robotic head, over the Z2 probe. It contains a level detector oscillator and sends oscillator signal to PCA PIP-TOW59.
- **PCA PIP-BCR59:** located inside the robotic head, it connects barcode readers interfaces with PCA PIP-TOW59.

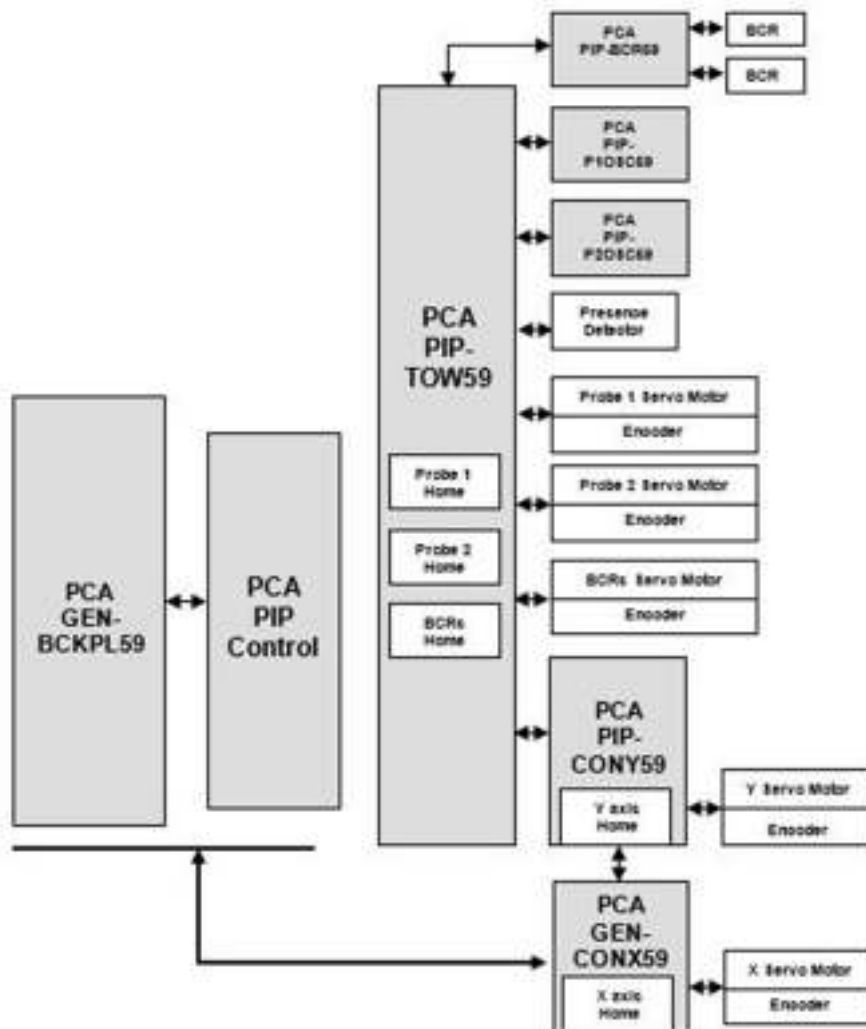


Figure 246 - Main blocks of PIP module

This module uses a bus I2C for Probe1 Level Detector activation + Probe2 Level Detector activation + Probe1 Correct Pipetting Detector (activation and signal) + Probe2 Correct Pipetting Detector (activation and signal) + Barcode Readers activation + Barcode readers axis home sensor activation and current limit sensing + Presence Detector activation.

Communication with both barcode readers is through a RS-232 bus.

2.4.2 Mechanical adjustments

Mechanical adjustments related to the Pipetting robot arm module are detailed in this section.

2.4.2.1 Adjusting probes parallelism

This adjustment is performed to ensure the parallelism between both probes. It must be performed if:

- Probe leadscrew is replaced.
- Probe wagon is replaced.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233901 PROBES_WAGON ADJUSTMENT TOOL (Ver. 2).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Loosen the two Allen screws of the probes washer and the three Allen screws that attach each probe wagon to the leadscrew.



Figure 247 - Allen screws of the probe wagon - leadscrew

2. Place the tool between the upper part of the two leadscrews, checking also that both probe wagons are located at the top of the robotic head.

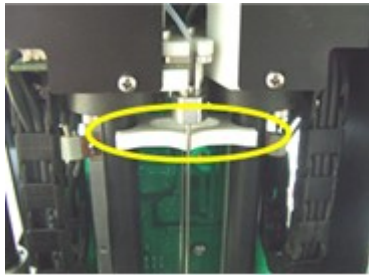


Figure 248 - Parallelism checking at the top of the robotic head

3. Adjust the position of both probe wagons until both probes touch the external edges of the tool. Then tighten the screws of the probe wagon - leadscrew and remove the tool.



Figure 249 - External edges of the probes / wagon adjustment tool (Ver. 2)

4. Move down manually (by means of their belts) both probes and adjust the position of the probes washer, checking that both probes are parallel in the X-Y axes. Tighten the screws of the probes washer. In order to verify the parallelism, see "Verifying probes parallelism" on page 362.

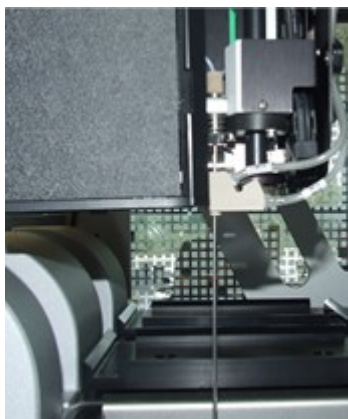


Figure 250 - Parallelism checking at the bottom of the robotic head

5. Check alignment between home flag and home sensor.

2.4.2.2 Adjusting X-Axis small belt tension

No manual tension adjustment is required for the X-axis small belt. Assembly and silentblocks ensure a proper belt tension.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.



Figure 251 - X-axis motor small belt

2.4.2.3 Adjusting X-Axis large belt tension

Adjustment performed to set large belt tension of the PIP arm X-axis.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Screw the pulley tightening screw to allow pulley movement.
2. Loose both pulley fixing screws.
3. Tighten the belt by unscrewing the pulley tightening screw, and fix its position by tightening both fixing screws.

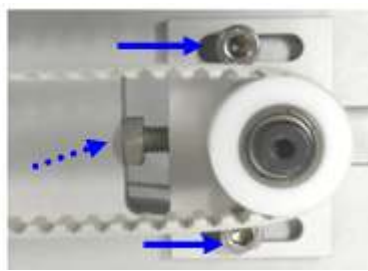
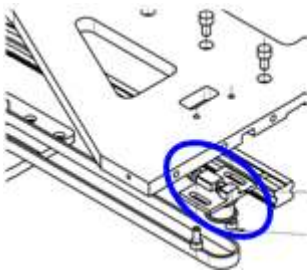


Figure 252 - X-axis motor large belt

2.4.2.4 Adjusting Y-Axis small belt tension

This adjustment is performed to set small belt tension of PIP arm Y-axis.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the Y arm motor cover (remove four fixing screws).
2. Unplug the Y arm ribbon cables from PIP-CONY59 board.
3. Remove the screws fixing the Y motor-belt assembly to the Y arm and remove the assembly.
4. Loosen the Y motor slotted hole screw.



Figure 253 - Y-axis motor small belt

5. Move the Y motor to adjust the belt tension.
6. Fix motor position tightening slotted hole screw.
7. Assemble the Y motor-belt assembly to the Y arm by tightening the three screws.
8. Connect the Y-arm ribbon cables to PCA PIP-CONY59.
9. Assemble the Y motor cover.

2.4.2.5 Adjusting Y-axis large belt tension

This adjustment is performed to adjust tension of the large belt of the Y-axis of the PIP arm.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Screw the pulley tightening screw to allow pulley movement.
2. Loose both pulley fixing screws.
3. Tighten the belt by unscrewing the pulley tightening the screw, and fix its position by tightening both fixing screws.

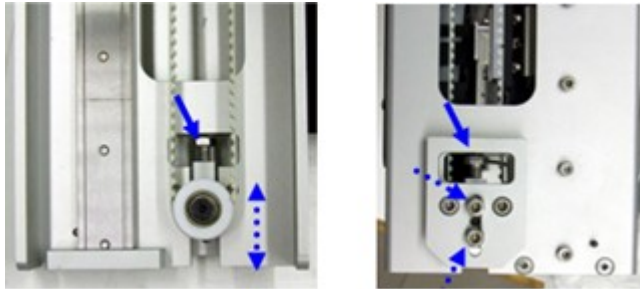


Figure 254 - Y-axis motor large belt

2.4.2.6 Adjusting Z-axis belt tension

The adjustment method applies for the three Z-axis motors (Z1 probe, Z2 probe and Z horns). It is performed to adjust tension of the belt of any of the Z-axis of the PIP arm.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Loose the two screws that are attaching the motor to the PIP robot bottom base.
2. Push the motor pinion in order to adjust the tension of the motor belt and tighten the screw of the slotted hole. Then tighten the other screw.



Figure 255 - Z-axis motors belts

2.4.2.7 Adjusting presence detector inclination

Adjustment of the inclination of the presence detector system. The adjustment must be performed if:

- Presence detector is replaced.
- Wrong detections are observed.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Loosen the nut of the screw that attaches the presence detector to the PIP robot head.
2. Move the presence detector so the pin of the presence detector is inserted in its location.

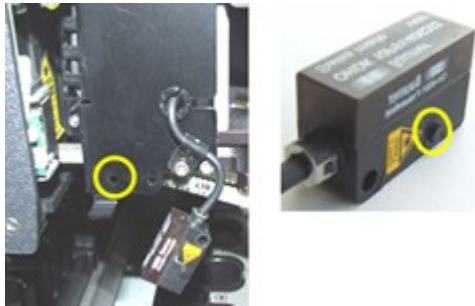


Figure 256 - Presence detector assembly

3. Tighten the nut of the screw that attaches the presence detector to the presence detector holder plate.
4. Verify the position of the presence detector. See "Verifying presence detector" on page 370.

2.4.3 Functional adjustments

Functional adjustments related to the PIP module are detailed in this section.

2.4.3.1 Adjusting probes offset

This adjustment is required to define the exact position of the probes washer in respect of the probes and to compensate for a possible difference in the Z position of both probes. The adjustment must be performed / restored if:

- The probe wagon is replaced.
- The leadscrew of a probe is replaced.
- PCA PIP-TOW59 is replaced.
- PCA PIP Control is replaced.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234213 PROBES ADJUSTMENT TOOL (Ver. 4).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Fluidic circuit is empty.
- PIP homed.

Procedure

1. In the Technical Service software, go to the **Procedures** tab. Then select **probes offset** in the **PIP floor Modules** section (see "Procedures tab" on page 756). A precondition message is shown on the screen, advising that the fluidic circuit must be empty (**Fluidics** tab > **fluidic empty**).
2. After accepting the message, an adjustment wizard window is opened.



Figure 257 - Probes offset wizard in the Technical Service software

3. Place the adjustment tool (holding it with your hand) as indicated in the following picture, checking that the adjustment tool holes match with the probes washer exit holes and that the adjustment tool is completely flat when inserted in this position.



Figure 258 - Adjustment tool position in probes offset adjustment

4. Press the **Next** button. A warning message informing that the instrument will perform movements / settings appears on the screen. Press the **OK** button. Probes will move down automatically until they detect tool surface.
5. Once the process is finished, a screen appears showing the parameters to be stored. Remove the tool from the probes washer, press the **Finish** button and press **OK** on the information window.
6. Press **OK** in the confirmation window that indicates that home movements will be performed.
7. Press **OK** to finish the adjustment procedure.
8. Verify the adjustment (See "Verifying probes offset" chapter), if desired.

**NOTE**

Probes offset adjustment is stored in PCA PIP Control. In case of replacing this PCA, the adjustment can be restored (if a previous backup was performed). Otherwise the adjustment must be performed.

2.4.3.2 Adjusting reagents drawers

Adjustment of the probes position on reagent drawers. The adjustment must be performed if:

- A reagent drawer is removed from the instrument.
- The PIP robotic head is removed.
- The probe wagon or the leadscrew is replaced.
- The position of the latch of the reagent drawer is modified.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- 234213 PROBES ADJUSTMENT TOOL (Ver. 4).

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- Offset between probes adjustment performed. See "Adjusting probes offset" on page 336.
- Reagent drawer flatness verified. See "Verifying the flatness of reagent drawer" on page 186.
- PIP homed.

Procedure

1. In the Technical Service software go to the **Procedures** tab > **PIP floor Modules** section > **reagent drawers** button. See "Procedures tab" on page 756. An adjustment wizard is opened.
2. Select the reagent drawer to adjust and press the **Next** button.



Figure 259 - Reagent drawer selection in the Technical Service software

3. A warning message informing that the instrument will perform movements / settings appears on the screen. Press the **OK** button. Rack 2 of the selected drawer will perform a home movement. Insert the adjustment tool over positions 1 and 5 in rack 2 of the selected drawer. Place manually the PIP arm over the adjustment tool and with the help of the screen joysticks (or manually), move the PIP arm until the Z1 probe is above the adjustment area and close to the upper left corner (not touching tool surface and not further than 6 mm from tool surface or adjustment edges), as it is shown in following pictures.

**NOTE**

The Z1 probe must not touch the tool surface.

**NOTE**

Check that the adjustment area of the tool is over holder 1 in reagent rack.

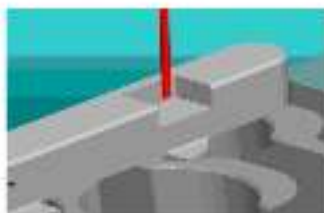
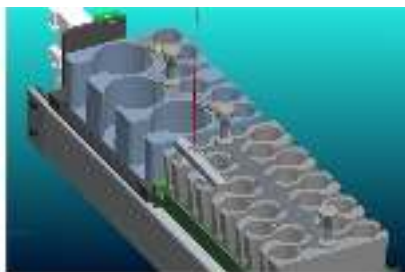


Figure 260 - Adjustment of the PIP arm on the reagents drawers - step

4. Press the **Next** button. The Z1 probe is moved automatically to find the adjustment position (upper left corner on the adjustment area). First, probe 1 is moved automatically in the Z axis and, by means of level detection, it is detected when probe 1 touches the tool surface. Then probe 1 is moved automatically in the X axis to detect, by means of level detection, left wall of the adjustment tool. Finally, probe 1 is moved automatically in the Y axis to detect the upper wall of the adjustment tool. When procedure finishes, probe 1 is homed.

5. Then place adjustment tool over positions 4 and 6 in rack 1 of the selected drawer, pushing the tool to the right in its location. Place manually the PIP arm over the adjustment tool and with the help of the screen joysticks (or manually), move the PIP arm until the Z1 probe is above the adjustment area (not touching tool surface) avoiding the hole, as it is shown in following picture.

**NOTE**

Only the Z coordinate will be adjusted in this step. The adjustment tool is free in the X and Y axes when placed into adjustment position. Check that adjustment tool remains flat over adjustment position.

**NOTE**

The Z1 probe must not touch surface of the tool.

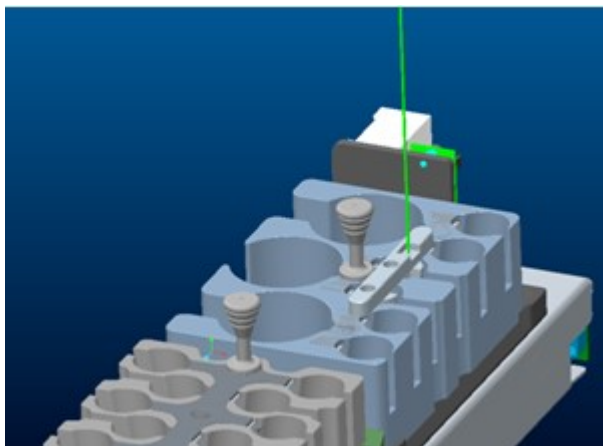


Figure 261 - Adjustment of PIP arm on reagents drawers - step 2

6. Press the **Next** button. The Z1 probe is moved automatically to find the adjustment position (surface on the adjustment area). Probe 1 is moved automatically in the Z axis and, by means of level detection, it is detected when probe 1 touches the adjustment area tool surface. When procedure finishes, probe 1 is homed.
7. Once the process is finished, a screen appears showing the parameters to be stored. Press the **Finish** button and press the **OK** button on the confirmation window. When the adjustments are stored, press the **OK** button in the information window.
8. Press the **OK** button in the confirmation window that indicates that home movements will be performed.
9. Press the **OK** button to finish the adjustment procedure.

**NOTE**

The adjustment is stored in **E:\data\Erytra\conf\PIPCalibrations.txt**.

10. Repeat the same procedure for the other reagent drawer if necessary.



CAUTION

Remove the adjustment tool from the instrument when adjustment procedure finishes.

11. Verify the adjustment (See "Verifying reagents drawer" chapter), if desired.

2.4.3.3 Adjusting samples drawers

Adjustment of the position of the probes in samples drawers. The adjustment must be performed if:

- A sample drawer is removed from the instrument.
- The PIP robotic head is removed.
- The probe wagon or the leadscrew is replaced.
- The position of the latch of the samples drawer is modified.
- The position of the centers of the samples drawer is modified.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234213 PROBES ADJUSTMENT TOOL (Ver. 4).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Offset between probes adjustment performed. See "Adjusting probes offset" on page 336.
- Sample drawer flatness verified. See "Verifying the flatness of sample drawer" on page 187.
- Home of the PIP arm: X, Y, Z1, Z2 and horns.

Procedure

1. In the Technical Service software, go to the **Procedures** tab > **PIP floor Modules** section > **sample drawers** button. See "Procedures tab" on page 756. An adjustment wizard is opened.
2. Select the sample drawer to adjust and press the **Next** button.

3. Insert the adjustment tool over positions 2 and 4 in rack 1 of the selected drawer. Place manually the PIP arm over the adjustment tool and with the help of the screen joysticks (or manually), move the PIP arm until the Z1 probe is above the adjustment area and close to the upper left corner (not touching tool surface and not further than 6 mm from tool surface or adjustment edges), as it is shown in the following pictures.

**NOTE**

Z1 probe must not touch the tool surface.

**NOTE**

Check that the adjustment area of the tool is over holder 2 in the sample rack.

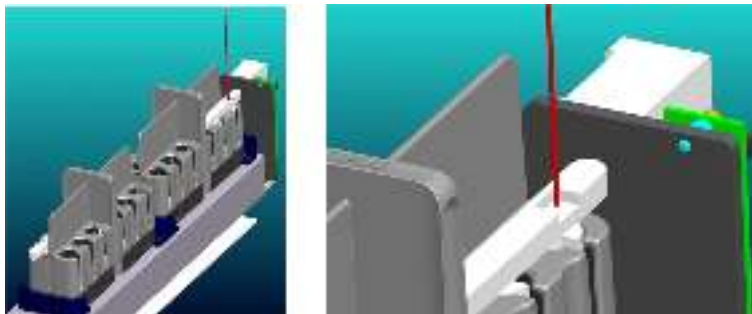


Figure 262 - Adjustment of PIP arm over rear part of samples drawer

4. Press the **Next** button. A warning message informing that the instrument will perform movements / settings appears on the screen. Press the **OK** button. The Z1 probe is moved automatically to find the adjustment position (upper left corner on the adjustment area) by means of level detection. First, probe 1 is moved automatically in the Z axis until it is detected that probe 1 touches the tool surface. Then probe 1 is moved automatically in the X axis to detect the left wall of the adjustment tool. Finally, probe 1 is moved automatically in the Y axis to detect upper wall of the adjustment tool. When procedure finishes, probe 1 is homed.

- Then insert the adjustment tool over positions 9 and 11 in the rack 2 of the selected drawer. Place manually the PIP arm over the adjustment tool and with the help of the screen joysticks (or manually), move the PIP arm until the Z1 probe is above the adjustment area and close to the lower right corner, as it is shown in the following pictures.

**NOTE**

Z1 probe must not touch surface of the tool.

**NOTE**

Check that the adjustment area of the tool is over holder 11 in sample rack.

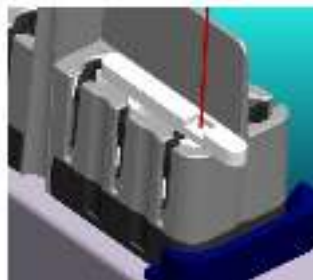
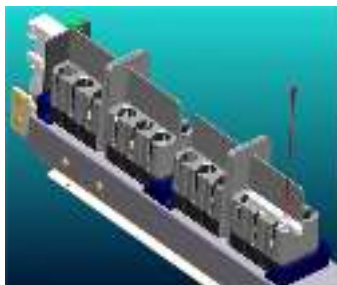


Figure 263 - Adjustment of PIP arm over frontal part of samples drawer

- Press the **Next** button. The Z1 probe is moved automatically to find the adjustment position (lower right corner on the adjustment area) by means of level detection. First, the probe is moved automatically in the Z axis until it is detected that the probe touches the surface of the tool. Then the probe is moved automatically in the X axis to detect the right wall of the adjustment tool. Finally, the probe is moved automatically in the Y axis to detect the lower wall of the adjustment tool.
- Once the process is finished, a screen appears showing the parameters to be stored. Press the **Finish** button and press the **OK** button in the confirmation window. When the adjustments are stored, press the **OK** button in the information window.
- Press the **OK** button in the confirmation window that indicates that home movements will be performed.
- Press the **OK** button to finish the adjustment procedure.

**NOTE**

The adjustment is stored in `E:\data\Erytra\conf\PIPCalibrations.txt`.

- Repeat the same procedure for other samples drawers if necessary.

**CAUTION**

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

- Verify the adjustment (See "Verifying samples drawer" chapter), if desired.

2.4.3.4 Adjusting incubators

Adjustment of the position of the probes on each incubator. The adjustment must be performed if:

- An incubator is removed from the instrument.
- The PIP robotic head is removed.
- The probe wagon or the leadscrew is replaced.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234213 PROBES ADJUSTMENT TOOL (Ver. 4).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Offset between probes adjustment performed. See "Adjusting probes offset" on page 336.
- Incubator flatness verification. See "Verifying incubator flatness" on page 269.
- Adjustment of the height of the incubator performed. See "Adjusting incubator height" on page 260.
- Incubator in home position.
- PIP homed.

Procedure

1. In the Technical Service software, go to the **Procedures** tab > **PIP floor Modules** section > **incubators** button. See "Procedures tab" on page 756. A precondition message on screen appears, indicating the requirement of the incubator height adjustment. If the **OK** button is pressed, a wizard adjustment is opened.

2. Select the incubator to adjust. A message appears on screen informing that it is necessary to place the adjustment tool in the sixth slot of the selected incubator and that when the **Next** button is pressed, the selected incubator will move up. Place the tool in the sixth slot (once in this position, it is recommended to push it as far as possible to the rear part of the slot) and then press the **OK** button and then press the **Next** button. The incubator will move to the intermediate area.

**CAUTION**

Check that the CAT arm is not over the incubator to adjust (interfering with its movement).

3. Place manually the PIP arm over the adjustment tool and with the help of the screen joysticks (or manually), move the PIP arm until the Z1 probe is above the adjustment area and close to the upper right corner (not touching tool surface and not further than 6 mm from tool surface or adjustment edges), as it is shown in the following pictures.

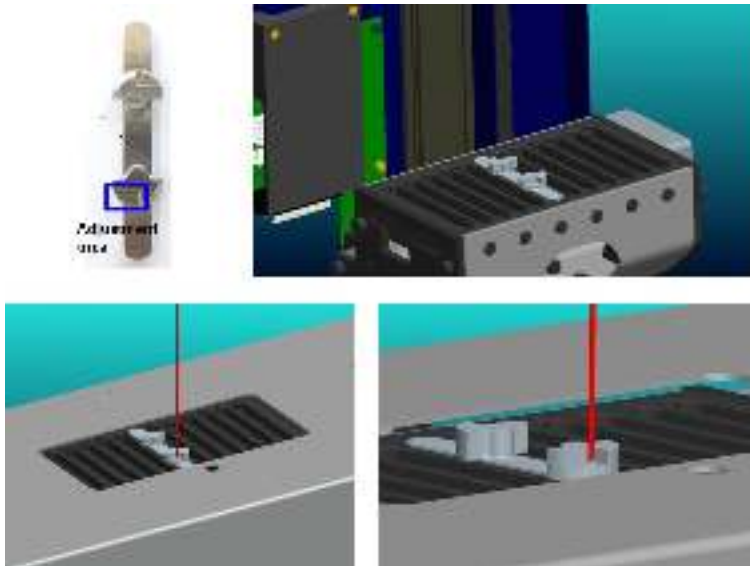


Figure 264 - Adjustment of PIP arm on incubators

4. Press the **Next** button. The Z1 probe is moved automatically to find the adjustment position (upper right corner on the adjustment area) by means of level detection. First, the probe is moved automatically in the Z axis until it is detected that the probe touches the surface of the tool. Then the probe is moved automatically in the X axis to detect the right wall of the adjustment tool. Finally, the probe is moved automatically in the Y axis to detect the upper wall of the adjustment tool.
5. After locating the adjustment area, the Z1 probe is placed automatically over well 1 position of slot 6 and then the probe is moved automatically in the Z axis until it is detected that probe touches the surface of the tool. Then the Z1 probe is placed automatically over well 8 position of slot 6 and then the probe is moved automatically in the Z axis until it is detected that probe touches the surface of the tool.

6. Once the process is finished, a screen appears showing the parameters to be stored. Press the **Finish** button and press the **OK** button on the confirmation window. When the adjustments are stored, press the **OK** button on the information window.
7. Press the **OK** button in the confirmation window that indicates that home movements will be performed. The incubator will automatically move to the home position (CAT area).
8. Press the **OK** button to finish the adjustment procedure.

**NOTE**

The adjustment is stored in **E:\data\Erytra\conf\PIPCalibrations.txt**.

9. Repeat the same procedure for other incubators if necessary.

**CAUTION**

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

10. Verify the adjustment (See "Verifying incubators" chapter), if desired.

2.4.3.5 Adjusting dilution cup

Adjustment of the position of the probes in the dilution cup. The adjustment must be performed if:

- The dilution cup module is removed.
- The PIP robotic head is removed.
- The probe wagon or the leadscrew is replaced.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Offset between probes adjustment performed. See "Adjusting probes offset" on page 336.
- Dilution cup parallelism adjustment performed. See "Adjusting dilution cup parallelism" on page 673.
- PIP homed.

Procedure

1. In the Technical Service software, press the **Procedures** tab > **PIP floor Modules** section > **dilution cup** button. See "Procedures tab" on page 756. It will open a wizard adjustment. Press the **Next** button. A warning message informing that the instrument will perform movements / settings appears on the screen. Press the **OK** button.
2. Place manually the PIP arm over the dilution cup and with the help of the screen joysticks (or manually), lower the Z1 probe until it is inserted in the centre notch located on the surface of the dilution cup, touching the bottom of the hole but without moving the cup.



Figure 265 - Position of the probe in dilution cup adjustment

3. Press the **Next** button. A screen appears showing the parameters to be stored. Press the **Finish** button and press the **OK** button in the confirmation window. When the adjustments are stored, press the **OK** button in the information window.
4. Press the **OK** button in the confirmation window that indicates that home movements will be performed.
5. Press the **OK** button to finish the adjustment procedure.



NOTE

The adjustment is stored in **E:\data\Erytra\conf\PIPCalibrations.txt**.

6. Verify the adjustment (See "Verifying dilution cup" chapter), if desired.

2.4.3.6 Adjusting barcode reader inclination position

Adjustment of the laser beam position of the barcode reader. The adjustment must be performed if a barcode reader is removed.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

Depending on the instrument serial number follow the described steps:

(SN < 059-0001887)

1. Loosen the two screws that attach the barcode reader to the horns.
2. Activate the barcode reader. To do so, open the Technical Service software and press the **Pip Arm** tab > **BCR** tab > **set on** button (see "BCR subtab" on page 817), activating the desired barcode reader.

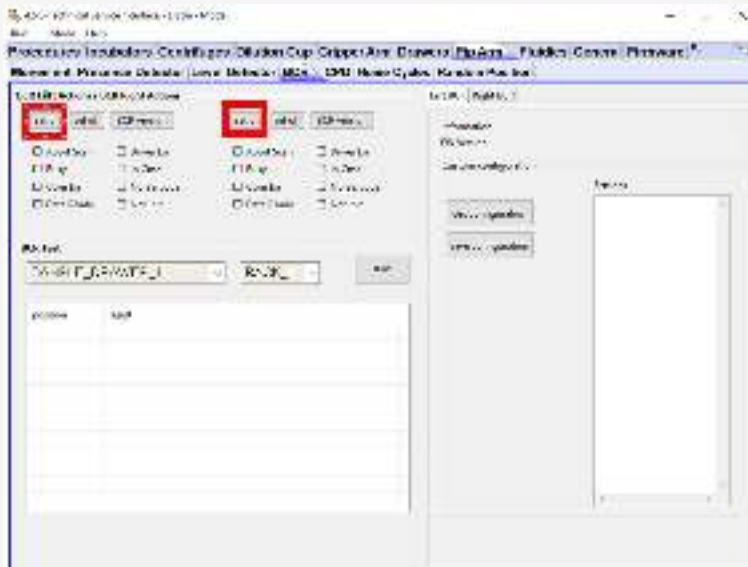


Figure 266 - Switching ON the barcode reader

3. Move manually the barcode reader in order to centre the laser light in the mirror, and then tighten both screws.



Figure 267 - Correct position of the barcode reader

4. Deactivate the barcode reader by means of the **set off** button of the desired barcode reader (go to the Technical Service software, **Pip Arm** tab > **BCR** tab).
5. Verify the adjustment (See "Verifying barcode reader" chapter), if desired.

(SN ≥ 059-0001887)

1. Activate the barcode reader. To do so, open the Technical Service software and press the **Pip Arm** tab > **BCR** tab > **set on** button (see "BCR subtab" on page 817), activating the desired barcode reader.

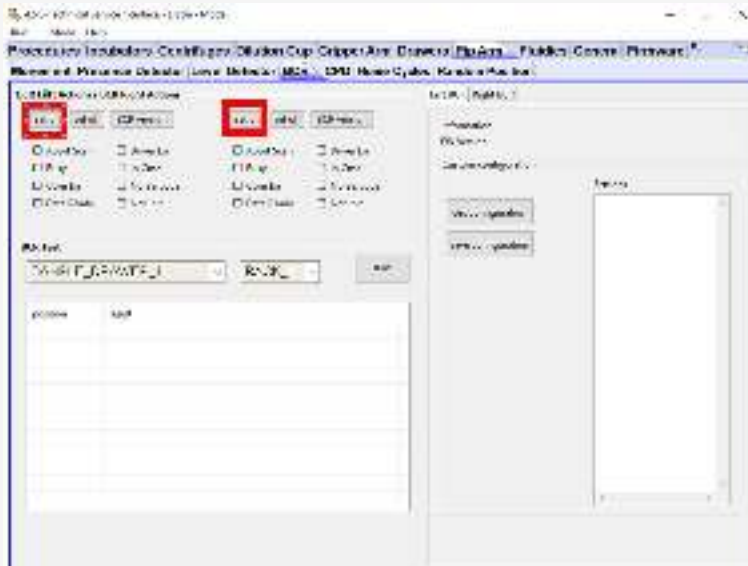


Figure 268 - Switching ON the barcode reader

2. Manually move the barcode reader in order to center the height of the laser light in the mirror, and then tighten both screws.

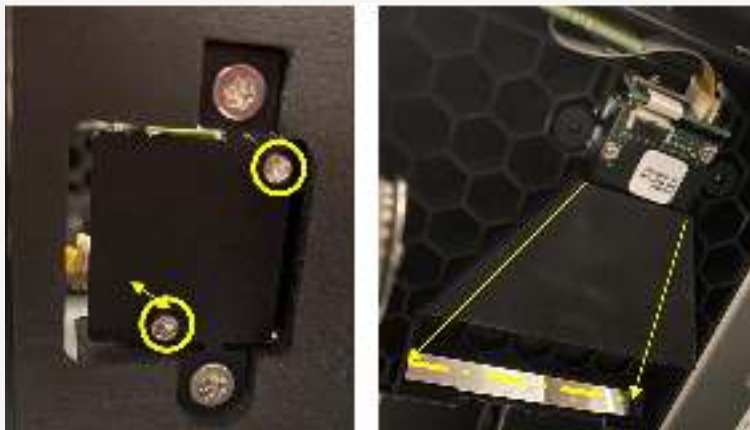


Figure 269 - Centering the height of the laser light in the mirror

3. By pressing slightly the barcode plate with the finger. Move the laser extremes along the mirror width until it is completely centered.



Figure 270 - Correct width position of barcode reader

4. Deactivate the barcode reader by means of **set off** button of the desired barcode reader (Technical Service software, **Pip Arm tab** → **BCR tab**).
5. Verify the adjustment (See "Verifying barcode reader" chapter), if desired.

2.4.3.7 Adjusting horns position

Adjustment of the position of barcode reading in the drawers. The adjustment must be performed if:

- The PIP robotic head is removed.
- The horns wagon or the horns leadscrew is replaced.
- The position of the latch of the reagent drawer is modified.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Adjustment of the barcode reader inclination position. See "Adjusting barcode reader inclination position" on page 348.
- Adjustment of the reagents and samples drawers. See "Adjusting reagents drawers" on page 338 and see "Adjusting samples drawers" on page 342.
- PIP homed.

Procedure

1. In the Technical Service software, press the **Procedures** tab > **PIP floor Modules** section > **horns** button. See "Procedures tab" on page 756. It will open a wizard adjustment.
2. Check that the barcode readers are ON and then press the **Next** button.
3. Adjustment of the horns X and Z axes: place a diluent bottle in diluent position 4 of the reagent rack 1 of drawer 1, with barcode in the middle of the reading window. Move by means of the joystick the buttons in the software or manually the PIP arm in the X axis until the horns are centered in reagents rack 1 of drawer 1. Move down the horns until the laser beam of the barcode reader illuminates the whole barcode label. Then press the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button.



Figure 271 - Position of the horns in first step of the adjustment

4. Adjustment in the Y axis of left barcode reader: move the PIP arm in the Y axis until the laser beam of the left barcode reader illuminates the centre of the frontal handle of reagent drawer 1 - rack 2. Then press the **Next** button. A warning message informing that the instrument will perform movements / settings appears on the screen. Press the **OK** button.

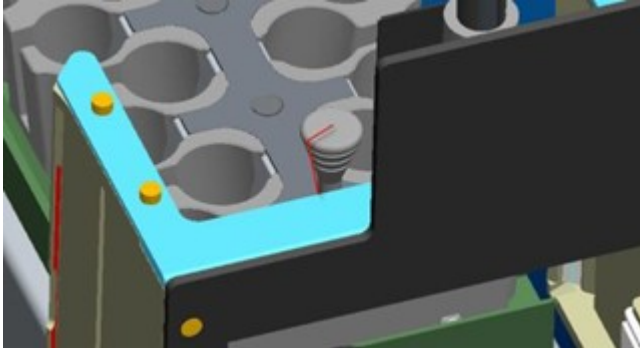


Figure 272 - Adjustment in Y axis of left barcode reader

5. Adjustment in the Y axis of the right barcode reader: move the PIP arm in the Y axis until the laser beam of the right barcode reader illuminates the centre of the frontal handle of reagent drawer 1 - rack 2. Then press the **Next** button. A warning message informing that the instrument will perform movements / settings appears on the screen. Press the **OK** button.

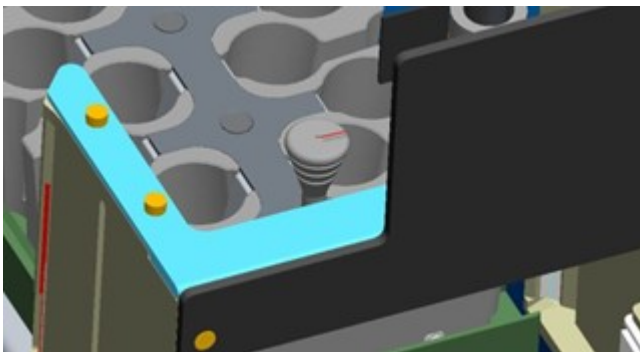


Figure 273 - Adjustment in Y axis of right barcode reader

- Adjustment in the X axis of samples drawer 1: Move by means of the joystick the buttons in the software or manually the PIP arm in the X axis until the horns are centered in samples drawer 1 - rack 1. Then press the **Next** button. A warning message informing that the instrument will perform movements / settings appears on the screen. Press the **OK** button.

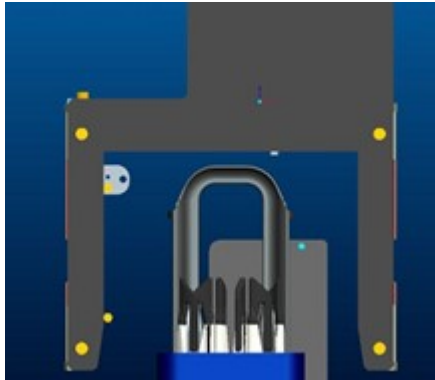


Figure 274 - Adjustment in samples drawer of the barcode readers

- Once the process is finished, a screen appears showing the parameters to be stored. Press the **Finish** button and press the **OK** button in the confirmation window. When the adjustments are stored, press the **OK** button in the information window.
- Press the **OK** button in the confirmation window that indicates that home movements will be performed.
- Press the **OK** button to finish the adjustment procedure.



NOTE

The adjustment is stored in **E:\data\Erytra\conf\PIPCalibrations.txt**.

2.4.3.8 Adjusting presence detector distance of detection

This section details the adjustment of the detection distance for the presence detector system. The adjustment must be performed after replacing the presence detector.



NOTE

The PIP presence detector device comprises two LEDs:

- The green LED should always be ON during the scanning routines, indicating that presence detector device is activated.
- The orange LED will only light (ON mode) when presence is detected.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234213 PROBES ADJUSTMENT TOOL (Ver. 4).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Adjustment of presence detector alignment. See "Adjusting presence detector inclination" on page 335.

Procedure

1. De-energize the PIP arm. In the Technical Service software, go to the **PIP Arm** tab > **Movement** subtab (see "Movement subtab" on page 804), and select the **Unlock** button from the **Position Actions** tab in the **Actions** section.
2. Place the adjustment tool over a reagent rack.



Figure 275 - Adjustment tool position on reagent rack

3. Activate the presence detector (in the Technical Service software, go to the **PIP Arm** tab > **Presence Detector** subtab (see "Presence Detector subtab" on page 808), and select the **On Presence Detector** button from the **On/Off** actions tab in the **Actions** section).

4. Turn slowly the presence detector screw counter-clockwise (remove the sealant if any) until a 'click' is heard.

The potentiometer is located in one side of the presence detector. In order to get a better access, it is recommended to move up the horns.



Figure 276 - Access to presence detector potentiometer

5. Move manually the PIP arm until the laser beam of the presence detector illuminates the higher adjustment surface of the adjustment area of the tool (avoiding the hole).



Figure 277 - Location of the adjustment area of detection of presence detector

6. Turn the presence detector screw slowly clockwise until the point where presence is detected (orange LED is ON and not blinking).
7. Manually move the PIP arm until the laser beam of the presence detector illuminates the black surface of the reagent rack.

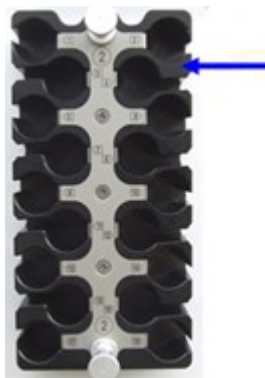


Figure 278 - Location of the adjustment area of no detection of the presence detector

8. Check that presence is not detected. This is indicated by means of the LEDs of the presence detector:
 - The green LED: ON. The presence detector system is ON.
 - The orange LED: OFF. The system has not detected presence.
9. Move manually the PIP arm alternating between the higher adjustment surface of the tool and the black surface of the reagent rack, checking that the orange LED is continuously ON and OFF, respectively. Should this not be the case, repeat the previous procedure.
10. Deactivate the laser of the presence detector (in the Technical Service software, go to **PIP Arm** tab > **Presence Detector** subtab, and select the **Off Presence Detector** button from the **On/Off** actions tab in the **Actions** section).
11. Consider applying a sealant to the presence detector screw.

2.4.3.9 Adjusting barcode readers configuration

Barcode reader devices installed in the PIP module use a default configuration which is loaded every time that the instrument is restarted. The enabled barcode types by default are:

- Code 39 (check digit enabled).
- Interleaved 2 of 5 (check digit enabled).
- Code 128.
- ISBT 128.
- GS1 - 128 (formerly UCC/EAN-128).
- Codabar.

In order to adapt the system to a specific lab configuration where different barcode types are used, the default configuration can be modified by means of the Technical Service software. All the introduced changes will be applied for each barcode reader device after the default configuration has been loaded, modifying it accordingly.



NOTE

The same barcode reader configuration is applied to read sample and reagent barcodes.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to program a customized barcode reader configuration, follow the steps listed below:

1. In the Technical Service software, go to **PIP Arm** tab > **Barcode Reader** tab. See "BCR subtab" on page 817.
2. Select the device where the custom configuration is going to be applied (**Left BCR** or **Right BCR**).
3. Press the **Get configuration** button so that the current customized configuration (if any) is displayed in the **Settings** box.
4. Use the **Settings** box to write or delete the desired modifications to be applied with respect to the default configuration.



NOTE

New settings should be introduced by writing the Parameter number corresponding to the desired barcode type / feature, followed by a code representing the action to be implemented (enable or disable). Parameter number for each barcode type / feature can be checked in the Symbol SE955/SE965 Integration Guide barcode reader manual.

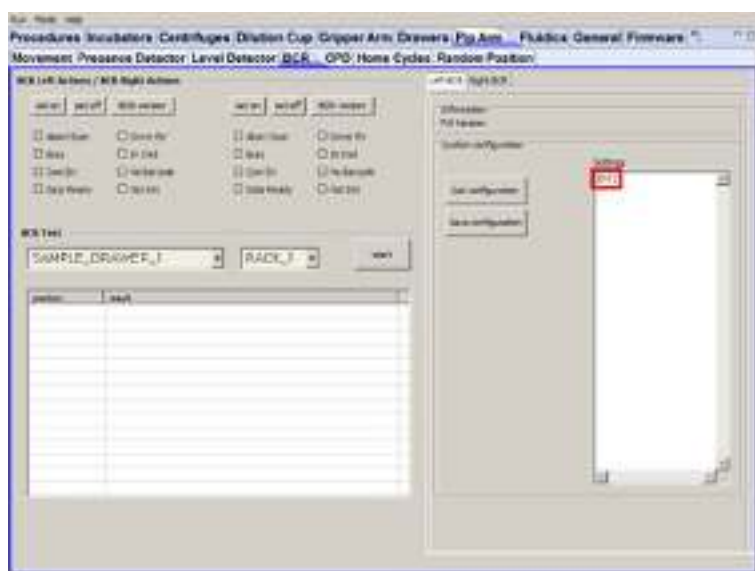


Figure 279 - Barcode readers custom configuration

- Click the **Save configuration** button to save the introduced changes.

**NOTE**

Changes applied with respect to the default configuration are saved in **E:\data\Erytra\conf** using independent files for each barcode reader device. These files are named **LeftBCCustom.cfg** and **RightBCCustom.cfg**, and apply for the left and right barcode reader device (respectively). These files are included in the backup functionality available in the user software.

- Press the **Get configuration** button and check that the new barcode reader configuration is properly read and displayed in the **Settings** box.
- Repeat the previous steps 3 to 6 for the other barcode reader device.

**NOTE**

The same configuration should be applied for both left and right barcode reader devices.

- Perform sample barcode readings (the **Barcode Detection** bar in the **Barcode Reader** subtab) to verify the applied configuration. .

2.4.4 Verification procedures

In this section the verifications of the different parts of the PIP arm are detailed.

2.4.4.1 Verifying movement

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. In the Technical Service software, go to the **Pip Arm** tab > **Movement** subtab > **Position Actions** section. See "Movement subtab" on page 804.
2. Press the **home** button and select **all axes**.
3. Press the **start** button in the **Pip Arm** tab > **Random Position** and perform random movements during 5 minutes, checking that no errors appear. See "Random Position subtab" on page 824.
4. Press the **home** button (the **Pip Arm** tab > **Movement** subtab > **Position Actions** section) and select **all axes**.
5. Press the **unlock/stop** button selecting all axes and move manually PIP head over a reagent drawer, assuring that the horns are centered over the drawer.
6. Press the **horns down** and **horns up** button 10 times checking that no errors appear.

2.4.4.2 Verifying home of the horns



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. In the Technical Service software, go to the **Pip Arm** tab > **Movement** subtab. See "Movement subtab" on page 804.
2. Press the **Home sensor on** button to activate the home sensor.
3. Move up and down manually the horns by means of its belt checking that when the home flag cuts the home sensor, its Home sensor flag is activated.
4. Press the **Home** button selecting horns option and check that home movement and detection are performed correctly, without errors.
5. Repeat step 4, 10 times.

2.4.4.3 Verifying home of the X-Y-Z1-Z2 axis



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. In the Technical Service software, go to **Pip Arm** tab > **Movement** subtab. See "Movement subtab" on page 804.
2. Press the **Home sensor on** button to activate the home sensors.
3. Move manually the X and Y PIP axes checking that when the home flag cuts the home sensor, its related Home sensor flag is activated.
4. In the Technical Service software, go to **Pip Arm** tab > **Home Cycles** subtab. See "Home Cycles subtab" on page 822.
5. Press the **start** button and perform homes within five minutes. Check that home movement and detection for all axes are performed correctly, without errors.

2.4.4.4 Verifying probes parallelism



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234213 PROBES ADJUSTMENT TOOL (Ver. 4).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Place the PROBES ADJUSTMENT TOOL (Ver. 4) in a sample drawer, rack 2, between positions 9 and 11, checking that the hole of the tool is over position 11.



Figure 280 - First step of the probes parallelism verification

2. With the Technical Service software, move Pip Arm over the sample drawer where the tool is placed, rack 2, position 11, probe 1 (**Pip Arm** tab > **Movement** subtab > **Move Actions** section). See "Movement subtab" on page 804.
3. Unlock the probe 1 motor (the **unlock/stop** button > **z1 axis** in the Technical Service software) and lower manually, by means of its belt, probe 1 checking that it fits with the hole.
4. Perform a home of probe 1 by means of the Technical Service software.

5. Place the PROBES ADJUSTMENT TOOL (Ver. 4) in the same sample drawer, rack 2, between positions 10 to 12 checking that the hole of the tool is over position 10.



Figure 281 - Second step of the probes parallelism verification

6. With the Technical Service software, move the Pip Arm over the sample drawer where the tool is placed, rack 2, position 9, probe 1 (**Pip Arm** tab > **Movement** subtab > **Move Actions** section).
7. Unlock the probe 2 motor (**unlock/stop** button > **Z2 axis** in the Technical Service software) and lower manually, by means of its belt, probe 2 checking that it fits with the hole.
8. Perform a home of probe 2 by means of the Technical Service software.

2.4.4.5 Verifying probes offset



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Perform a home of the PIP axes: X, Y, Z1, Z2 and horns.
2. Unlock the PIP axes (**Pip Arm** tab > **Movement** tab > **unlock/stop** button and select **all axes**). See "Movement subtab" on page 804.
3. Move manually the PIP robotic head to the front of the instrument.
4. Press the **probes offset** button in the Technical Service software > **Procedures** tab > **Checks** section. See "Procedures tab" on page 756.
5. A precondition message informing that probes will move down 31300 encoder counts appears on screen. If the **OK** button is pressed, probes move down to that position.
6. After the automatic movement, the following message is shown on screen:

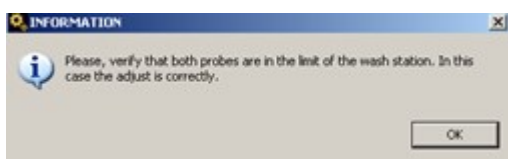


Figure 282 - Information window in the probes offset verification

7. Check visually that both probes are levelled with the probes washer exit holes and then press the **OK** button. The probes are homed automatically.

2.4.4.6 Verifying reagents drawer



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Perform a home of the PIP axes: X, Y, Z1, Z2 and horns.
2. Place vials in position 2 of rack 1 and position 17 of rack 2 of the reagent drawer to verify.

3. In the Technical Service software, go to the **Pip Arm** tab > **Movement** tab (see "Movement subtab" on page 804), and select the reagent drawer to verify, position rack 1, pos_2, probe_1 and press the **move to** button.
4. Once the robot is placed over the indicated position, write 150000 encoder counts in Move servo to encoder count box in the Z1 axis Motor and press the **Move** button.
5. Check that the Z1 probe is centered in the Y axis with the vial place.
6. Perform a home of probe Z1 (press the **Pip Arm** tab > **Movement** tab > **home** button and select the Z1 axis).
7. Repeat steps from 1 to 5 for rack 2, pos_17.
8. Perform a home of PIP axes: X, Y, Z1, Z2 and horns.

2.4.4.7 Verifying samples drawer



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Perform a home of all axes of the PIP arm: X, Y, Z1, Z2 and horns.
2. Place samples tubes in position 1 of rack 1 and position 11 of rack 2 of the sample drawer to verify.
3. In the Technical Service software, go to the **Pip Arm** tab > **Movement** tab (see "Movement subtab" on page 804), and select the sample drawer to verify, position rack 1, pos_1, probe_1 and press the **move to** button.
4. Once the robot is placed over the indicated position, deenergize the Z1 axis (**unlock/stop** button and select the Z1 axis) and move down manually the probe by means of its belt until enters into the tube.

5. Check that Z1 probe is centered in the Y axis with the sample tube placed and 0.5 mm from the center of the tube to the right in the X axis.
6. Perform a home of probe Z1 (go to the **Pip Arm** tab > **Movement** tab > **home** button and select the Z1 axis).
7. Repeat steps from 1 to 5 for the rack 2, pos_11.
8. Perform a home of all axes of the PIP arm: X, Y, Z1, Z2 and horns.

2.4.4.8 Verifying incubators



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Perform a home of the incubator.



CAUTION

Check that the CAT arm is not interfering with the incubator movement.

2. Place two cards in slots 1 and 3 of the incubator and move the incubator to the PIP floor.
3. In the Technical Service software, go to the **Pip Arm** tab > **Movement** tab (see "Movement subtab" on page 804), and select the incubator to verify, position **SLOT_1**, **WELL_1**, **PROBE_1** and press the **move to** button.
4. Press the **down probe 1** button and the **down probe 2** button in order to move both probes until dispensation position inside the cards. Both probes should perforate the cards film.
5. Perform a home of both probes (the **Pip Arm** tab > the **home** button and select the Z1 and Z2 axes).

6. In the Technical Service software, go to the **Pip Arm** tab > **Movement** tab, and select the incubator to verify, position **SLOT_1**, **WELL_8**, **PROBE_1** and press the **move to** button.
7. Press the **down probe 1** button and the **down probe 2** button in order to move both probes until dispensation position inside the cards. Both probes should perforate the cards film.
8. Perform a home of both probes (the **Pip Arm** tab > the **home** button and select the Z1 and Z2 axes).
9. Check that holes in the film card are centered in the X axis and shifted to the rear part of the instrument in the Y axis 1 mm as maximum.
10. Perform a home of the incubator. Be careful with the position of the CAT arm in order not to crash with the incubator.
11. Remove the cards from the incubator.

2.4.4.9 Verifying dilution cup



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. In the Technical Service software, perform a home of the PIP arm (**Pip Arm** tab > **Movement** tab > **home** button and then select all axes). See "Movement subtab" on page 804.
2. Move PIP over dilution cup: go to the **Pip Arm** tab > **Movement** tab, select **DILUTION_CUPS**, **PROBE_1**, **ASPIRE**, and press the **move to** button.
3. Press the **down probe 1** button and the **down probe 2** button.
4. In the Technical Service software, go to the **Dilution Cup** tab (see "Dilution Cup tab" on page 774), and press the **Spin** button (check that **PROFILE_0** is selected). Check that walls of the wells are not touching the probes.
5. Once stirring process finishes, go to the **Pip Arm** tab and perform a home of the probes.

6. Then in the **Movement** subtab, select **DILUTION_CUPS**, **probe_1**, **DISPENSE** and press the **move to** button.
7. Press the **down probe 1** button and **down probe 2** button.
8. Check that position of both probes is close to the rear part of the wells of the dilution cup in the Y axis and centered in the X axis. Check that probes do not touch the walls of the wells.

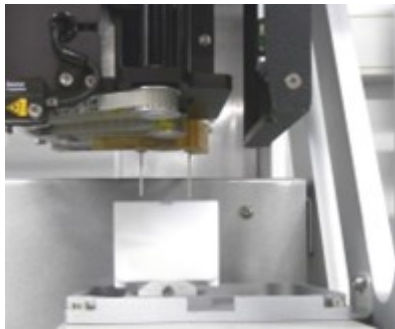


Figure 283 - Probes position on dilution cup

9. In the Technical Service software, go to the **Dilution Cup** tab, and press the **Spin** button (check that **PROFILE_0** is selected). Check that probes do not touch the walls of the wells.
10. Perform a home of both probes (press the **Pip Arm** tab > **Movement** tab > **home** button and select the Z1 and Z2 axes).

2.4.4.10 Verifying barcode reader



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

Reagents drawer:

1. Place some vials in both racks of a reagents drawer, with barcodes facing the reading windows.
2. In the Technical Service software, go to the **Pip Arm** tab > **BCR** tab (see "BCR subtab" on page 817), and select the reagent drawer to verify and rack 1. Then press the **start** button.
3. Once reading process is finished, check that all the barcodes are correctly read.
4. In the Technical Service software, go to the **Pip Arm** tab > **BCR** tab, and select the reagents drawer to verify and rack 2. Then press the **start** button.
5. Once the reading process is finished, check that all the barcodes are correctly read.

Samples drawer:

1. Place some allowed tubes in both samples racks of a samples drawer, with barcodes faced outside to reading windows.
2. In the Technical Service software, go to the **Pip Arm** tab > **BCR** tab, and select the samples drawer to verify and rack 1. Then press the **start** button.
3. Once reading process is finished, check that all barcodes are correctly read.
4. In the Technical Service software, go to the **Pip Arm** tab > **BCR** tab, and select the samples drawer to verify and rack 2. Then press the **start** button.
5. Once the reading process is finished, check that all barcodes are correctly read.

2.4.4.11 Verifying presence detector



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234213 PROBES ADJUSTMENT TOOL (Ver. 4).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Perform a home of the PIP axes.
2. Place the PROBES ADJUSTMENT TOOL (Ver. 4) in position 9 and 11 of rack 2 of samples drawer 1.
3. In the Technical Service software, go to the **Procedures** tab > **Checks** section (see "Procedures tab" on page 756), and press the **pip laser alignment** button. Press the **OK** button.
4. The PIP arm is moved over samples drawer 1 - rack 2 and presence detector laser is switched on. Check that laser fits inside the hole of the adjustment tool.



Figure 284 - Presence detector alignment verification

2.4.4.11.1 Reagents drawer

1. Check that no vials / bottles are present in the racks of a reagent drawer.
2. In the Technical Service software, go to the **Pip Arm** tab > **Presence Detector** tab > **Reagent drawers** tab (see "Reagent drawers subtab" on page 808), and select the reagents drawer to verify and rack 1. Then press the **start** button.
3. Once reading process is finished, select rack 2 and press the **start** button.
4. Once reading process is finished, check that no vials are detected.
5. Place some vials in both racks of a reagent drawer, with distilled water.
6. In the Technical Service software, go to the **Pip Arm** tab > **Presence Detector** tab > **Reagent Drawers** tab, and select the reagents drawer to verify and rack 1. Then press the **start** button.
7. Once reading process is finished, select rack 2 and press the **start** button.
8. Once reading process is finished, check that vials are correctly detected.



NOTE

Check that surface of reagent rack is dry in order to avoid false detections by the presence of liquid.

2.4.4.11.2 Samples drawer

1. Check that no tubes are present in the racks of a sample drawer.
2. In the Technical Service software, go to the **Pip Arm** tab > **Presence Detector** tab > **Sample Drawers** tab (see "Sample drawers subtab" on page 810), and select the samples drawer to verify and rack 1. Then press the **start** button.
3. Once reading process is finished, select rack 2 and press the **start** button.
4. Once reading process is finished, check that no tubes are detected.
5. Place some allowed tubes in both racks of samples drawer to verify, with distilled water.
6. In the Technical Service software, go to the **Pip Arm** tab > **Presence Detector** tab > **Sample Drawers** tab, and select the samples drawer to verify and rack 1. Then press the **start** button.
7. Once reading process is finished, select rack 2 and press the **start** button.
8. Once reading process is finished, check that tubes are correctly detected (0, -2 mm):
 - Diameter value shown on screen is 1 mm smaller than the detected one. This value cannot be equal or bigger than the real one.
 - If the diameter value shown on screen is smaller than the real one, it can be accepted until up to 2 mm smaller than the real one.

2.4.4.12 Verifying level detection system



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Empty the fluidic circuit from the dilutor to the probe (syringe, valve, tubing and probe). Use the fluidic empty button in the **Fluidics** tab in the Technical Service software. See "Fluidics tab" on page 826.

2. Place both probes in home position (go to the **Pip Arm** tab > **Movement** subtab (see "Movement subtab" on page 804) > **home** button and select **Z1** and **Z2**) and go to the **Pip Arm** tab > **Level Detector** subtab.
3. Select **PROBE_1** in the probe selection box and press the **on** button.
4. Select **PROBE_2** in the probe selection box and press the **on** button.
5. Check that level detection system values are the following ones:

Table 35 - Level detection system values in dry situation

	Probe 1	Probe 2
Probes not installed	6300 ± 300	8400 ± 500
Probes installed	6700 ± 300	9000 ± 500

6. If the probes are not installed, move up-down three times the probes holder assembly to simulate impacts, then check that level detection values match with the ones of previous table (probes not installed case).
7. Prime the fluidic circuit (in the Technical Service software, select the **Fluidics** tab > **initializing fluidic** option and then select the **fluidic prime** option).
8. Prime probe 1 and probe 2 fluidic circuits (the **Fluidics** tab > **prime probe 1** button and the **prime probe 2** button).
9. Check that level detection system values are the following ones:

Table 36 - Level detection system values in prime situation

Probe 1	Probe 2
7600 ± 300	10400 ± 500

10. Perform a home of reagent racks two of both reagent drawers.
11. Place a vial with saline solution in position 1 of rack 2 of both reagents drawers.
12. In the Technical Service software, go to the **Pip Arm** tab > **Level Detector** subtab, see "Level Detector subtab" on page 814. Introduce five in the **iterations** field, select **REAGENT_DRAWER_1**, **PROBE_1**, **rack 2**, **first tube 1**, **last tube 1**. Press the **start** button and check that all the detections are ok.
13. Repeat previous procedure for reagent drawer 2 and probe 2 (select **REAGENT_DRAWER_2**, **PROBE_2**, **rack 2**, **first tube 1**, **last tube 1**).
14. Place a sample tube with saline solution in position 1 of rack 2 in the 4 samples drawers.
15. In the Technical Service software, go to the **Pip Arm** tab > **Level Detector** subtab. Introduce five in the **iterations** field, select **SAMPLE_DRAWER_1**, **PROBE_1**, **rack 2**, **first tube 1**, **last tube 1** and **tube type NORMAL** (in case of using a pediatric tube, select **PEDIATRIC** and then define the difference between the maximum depth of the pediatric tube and a normal one). Press the **start** button and check that all the detections are ok.

16. Repeat the previous procedure for sample drawer 2 and probe 2 (select **SAMPLE_DRAWER_2, PROBE_2, rack 2, first tube 1, last tube 1** and the tube type).
17. Repeat the previous procedure for sample drawer 3 and probe 1 (select **SAMPLE_DRAWER_3, PROBE_1, rack 2, first tube 1, last tube 1** and the tube type).
18. Repeat the previous procedure for sample drawer 4 and probe 2 (select **SAMPLE_DRAWER_4, PROBE_2, rack 2, first tube 1, last tube 1** and the tube type).

2.4.4.13 Verifying impact detector system



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In case the impact detector system is installed on the instrument (installed from factory in instruments with S/N $\geq$ 059-0000611), verification procedure is as follows:

1. Remove probe 1 and check the value of the level detection system in the **Pip Arm** tab > **Level Detector** subtab. See "Level Detector subtab" on page 814. Move up-down three times the probes holder assembly, checking that the probes holder rests over the pad of the PCA and that the final value of level detection system is the same as the initial one $\pm$ 30 counts. Install probe 1 again.
2. Remove probe 2 and check the value of the level detection system in the **Pip Arm** tab > **Level Detector** subtab. Move up-down three times the probes holder assembly, checking that the probes holder rests over the pad of the PCA and that the final value of the level detection system is the same as the initial one $\pm$ 30 counts. Install probe 2 again.
3. Check the level detection system. See "Verifying level detection system" on page 372.
4. Place a diluent with a cap in reagent drawer 1 - rack 1 - pos 4.

5. Perform a home of the PIP arm (in the Technical Service software, click the **Pip Arm** tab > **Movement** subtab > **home** button and select **all axes**). See "Pip Arm tab" on page 804.
6. Go to the **Pip Arm** tab > **Level Detector** subtab. Write 3 in iterations field, select **REAGENT_DRAWER_1, PROBE_1, rack 1, first tube 4** and **last tube 4**. Press the **start** button and check that impact is detected three times (in the graph of the window or in log execution section).
7. Place a diluent with cap in reagent drawer 2 - rack 1 - pos 4.
8. Perform a home of the PIP arm (in the Technical Service software, click the **Pip Arm** tab > **Movement** subtab > **home** button and select **all axes**). .
9. Go to the **Pip Arm** tab > **Level Detector** subtab. Write 3 in the iterations field, select **REAGENT_DRAWER_2, PROBE_2, rack 1, first tube 4** and **last tube 4**. Press the **start** button and check that impact is detected three times (in the graph of the window or in log execution section).

2.4.5 Replacement procedures

Replacement procedures related to the Pipetting robot arm are detailed in this section.

2.4.5.1 Replacing probes



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233898 OMNIFIT TOOL (for tube connection Ver 2).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- //233748 //SAMPLES PROBE.
- For instruments with impact detector installed from factory with S/N $\geq$ 059-0000611: //233899 //SAMPLES PROBE (Impact detector).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Before replacing a probe, follow these steps:

1. Switch off the instrument and disconnect it from the mains.
2. Open one of the samples drawer:
 - In the Technical Service software, go to the **Drawers** tab > **Sample Drawers** subtab (see "Drawers tab" on page 792) and press the **Open** button in the selected drawer.
 - Or manually, with the emergency open located in the lock at the rear part of the drawer.



Figure 285 - Manual opening of the drawer

3. Remove the racks of the opened drawer and then move manually the PIP arm over the drawer at the frontal part of the instrument to get a better access.
4. Move down the horns:
 - With the Technical Service software, go to **Pip Arm** tab > **Movement** tab (see "Movement subtab" on page 804) and press **horns down** button.
 - Or manually, by means of rotating the horns belt located under the PIP arm.

2.4.5.1.1 Z1 probe

5. Move down the Z2 probe to get a better access to the Z1 probe:
 - In the Technical Service software, move down the Z2 probe to 250000 encoder counts. Write this counts number in Move servo to encoder count box from the Z2 axis (Motor) and then press the **Move** button.
 - Manually, rotating the Z2 probe belt located under the PIP arm.
6. Move down the Z1 probe to 220000 encoder counts.
 - In the Technical Service software, write this encoder counts number in Move servo to encoder count box from the Z1 axis (Motor) and then press the **Move** button.
 - Manually, rotating the Z1 probe belt located under the PIP arm.

7. Unscrew the fitting - it is possible to use OMNIFIT TOOL (for tube connection Ver 2) for unscrewing it - and remove the probe from its position.

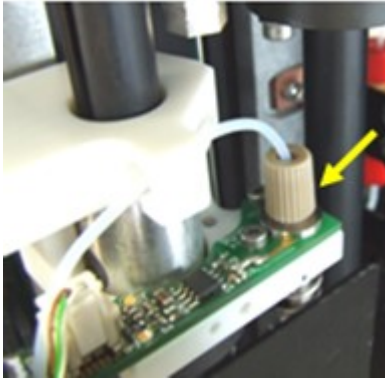


Figure 286 - Probe fitting

8. Insert the new probe first in the probe holder and then in its hole of the probes washer. The upper part of the probe has a flat area that must be faced properly inside the probe holder. Once the probe is correctly placed, tighten the probe fitting to the maximum. It is possible to use the OMNIFIT TOOL (for tube connection Ver 2) for tightening it.
9. Perform a home of all the axes of the PIP arm.

2.4.5.1.2 Z2 probe

10. Move down the Z2 probe to 220000 encoder counts.
 - In the Technical Service software, write this encoder counts number in Move servo to encoder count box from the Z2 axis (Motor) and then press the **Move** button.
 - Manually, rotating the Z2 probe belt located under the PIP arm.
11. Unscrew the fitting and remove the probe from its position.
12. Insert the new probe first in the probe holder and then in its hole of the probes washer. The upper part of the probe has a flat area that must be faced properly inside the probe holder. Once the probe is correctly placed, tighten the probe fitting to the maximum. It is possible to use the OMNIFIT TOOL (for tube connection Ver 2) for tightening it.
13. Perform a home of all the axes of the PIP arm.

2.4.5.2 Replacing probes washer



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233924 PROBES WASHER (Ver. 2).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the two screws that attach the probes washer to the PIP robotic head. To get a better access, it is recommended to move up the horns.



Figure 287 - Probes washer screws

2. Disconnect all tubes from the probes washer.
3. Replace the probes washer. Check the status of the end of the tubes and connect the tubes to the fittings of the washer. Make sure the tubes routing of C22 and C24 is as in the image below.



Figure 288 - Probes washer tubes C22 and C24 routing

4. Slightly tighten the two screws of the probes washer.

5. Move down manually both probes (by means of their belts) and check that both probes are aligned.
6. Tighten the two screws of the probes washer to the maximum and then $\frac{1}{4}$ of a turn more.

**WARNING**

Avoid the possibility of pricking yourself with the probes when tightening the screws of the probes washer.

7. Move up manually both probes (by means of their belts).

2.4.5.3 Replacing probes washer fitting

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- 233844 TUBE CONNECT. DISTRIB. (for 2,3x3,6 tube).

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

Probes washer fittings are sealed with acetoxysilicone, recommended for use on plastic pipe threaded fittings (such as Loctite 5331). To replace a fitting, follow these steps:

1. Check that the fluidic system is empty.
2. Disconnect the tube from the fitting to be replaced.
3. Unscrew the fitting from the probes washer and clean its allocation.

**CAUTION**

Check that no seal product gets into the probes washer during cleaning process.

4. Place the new fitting in its allocation only tightening it a little bit so most of the thread is out of its allocation.

5. Put the sealant product around the fitting thread. The sealant product should be spread only on the fitting thread as uniformly as possible from the middle of fitting thread toward the union point with nut. To ensure a better and easier spreading, use a pointed-end object.

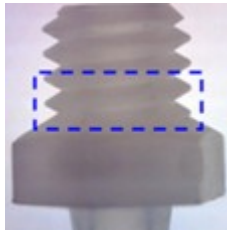


Figure 289 - Sealant product application area

6. Tighten the fitting manually to the maximum and then $\frac{1}{4}$ of a turn more in its allocation.



CAUTION

Check that the seal product does not block the connection between the fitting and the probes washer.

2.4.5.4 Replacing level detector PCA



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- //233751 //PCA PIP-P2OSC59.
- //233752 //PCA PIP-P1OSC59.
- For instruments with impact detector installed from factory with S/N $\geq$ 059-0000611:
 - 233894 PCA PIP-P1OSC59 (Impact detector).
 - 233895 PCA PIP-P2OSC59 (Impact detector).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Remove the three screws of the cover of the wagon probe and carefully move it apart it from its position as much as possible.



Figure 290 - Wagon probe cover for chain CH4 (left) and CH3 (right)

3. In order to remove the Teflon tube from its position, unscrew the fitting from the PCA.

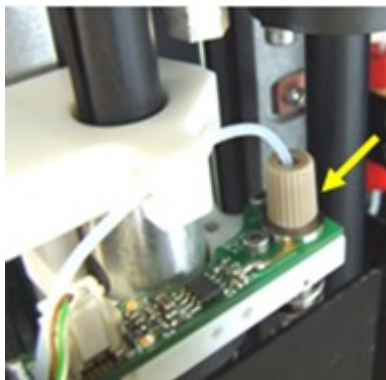


Figure 291 - Teflon tube of the probe

4. Disconnect carefully the cable (level detection signals) of the chain to be replaced, connected to the level detection PCA (PCA PIP-P1OSC59 or PCA PIP-P2OSC59).



Figure 292 - Location of the connector in the level detection PCA

5. Remove the PCA from its position:

- Unscrew the nut of the probe holder and push the probe holder to the top. Then remove the two screws of the PCA.

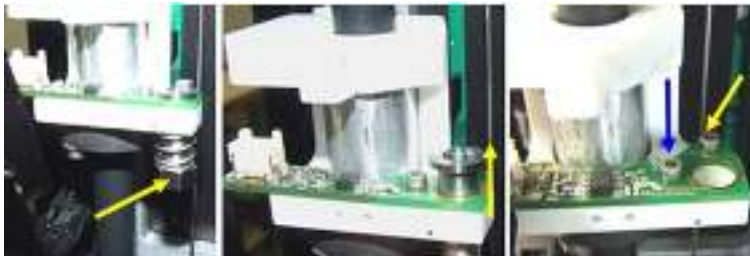


Figure 293 - Level detector PCA replacement steps

- Place the new PCA over the wagon, checking that the hole of the PCA is aligned with the hole of the wagon.
- Insert the probes holder through the PCA hole, and then place the spring around the probes holder. Tighten manually the nut of the probes holder assembly.
- Move up-down three times the probes holder assembly, checking that the probes holder rests over the pad of the PCA.
- Connect the level detection cable, install the probe, and screw the Teflon tube fitting. Finally, place the wagon probe cover.

6. Verify the level detection system. See "Verifying level detection system" on page 372.

2.4.5.5 Replacing leadscrews



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233819 HORNS ADJUSTMENT TOOL (BCR)

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233749 LEAD SCREW (Samples).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

2.4.5.5.1 Set up

To replace the leadscrews it is necessary to remove the PIP robotic head from the instrument.

1. Switch off the instrument, disconnect it from the mains, remove the upper rear cover of the instrument and open the upper area door.
2. Remove both probes.
3. Free from its clamps the fluidic tubing from the probes washer to the Y chain.



Figure 294 - Location of the clamps in the PIP robotic head



NOTE

Make sure that the position of the black tubing clamp is not modified while removing or inserting the fluidic tubes (see image above).

4. Disconnect, from the interconnection manifold located at the rear part of the PIP robotic head, the tubes of the probes that come from Y-chain.



Figure 295 - Interconnection manifold in the robotic head

5. Remove the two screws of the probes washer.



Figure 296 - Probes washer in the robotic head

6. Remove the two screws from the Y arm lateral cover.



Figure 297 - Lateral cover screws of the Y arm

7. From the rear part of the instrument remove the two screws that attach Y-chain to PIP robotic head.



Figure 298 - Location of the screws that attach Y chain to Z tower / robotic head

8. Disconnect Y chain ribbon cables from PCA PIP-TOW59.

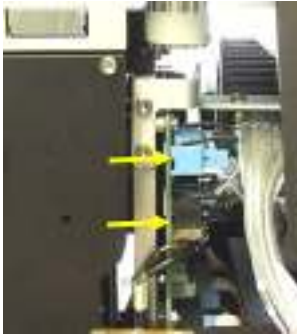


Figure 299 - Location of the ribbon cables connectors at PCA PIP-TOW59

9. From the front of the instrument remove Y-arm linear guide front end stopper (remove two screws).

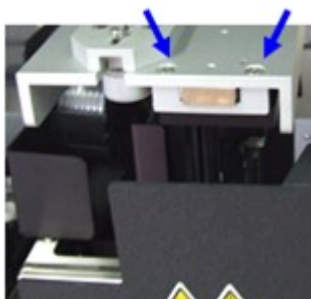


Figure 300 - Y arm guide front end stopper screws

10. Take out Y-belt from its position. First remove the two fixing screws of the tensor assembly and then free the belt from the rear gear.



Figure 301 - Rear gear Y belt (left) and fixing screws of the tensor assembly (right)



CAUTION

Be careful to do not damage Y axis home barrier when placing the belt.

11. Remove carefully PIP robotic head by the front of the instrument, opening a drawer if necessary.



Figure 302 - Removing the PIP robot head

2.4.5.5.2 Horns leadscrew

In order to remove the horns leadscrew from PIP robotic head follow the next steps:

1. Remove the belt and loose the set screw from the horns leadscrew pulley and remove it. Horns leadscrew pulley and belt are the ones close to the presence detector.

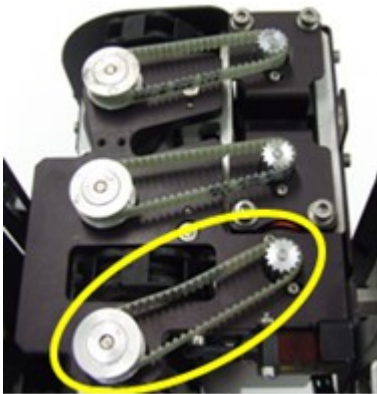


Figure 303 - Belt and motor pinion of the horns

2. Remove Y axis frontal robotic head stopper in order to have access to the leadscrew holder. Screws are located under the robotic head stopper.

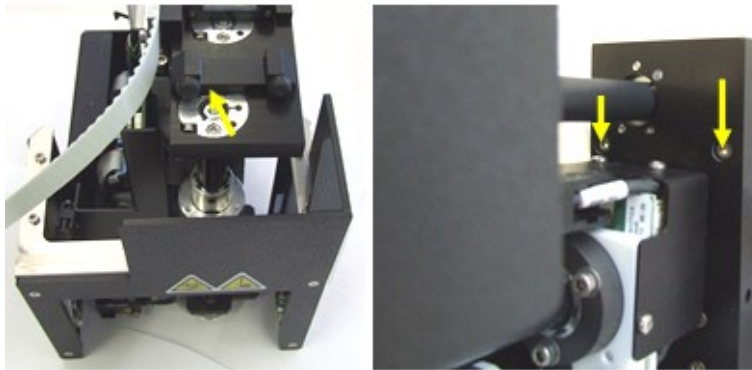


Figure 304 - Frontal robotic head stopper location

3. Remove the screws that attach the leadscrew holder to the robotic head and the three screws that attach the leadscrew to the home flag ring and the horns structure.
4. Remove the leadscrew upwards until the leadscrew holder is free from its position. If it is hard to remove, hit smoothly from the bottom upwards with a rubber hammer.
5. Remove the leadscrew holder from the leadscrew by means of loosening the set screw.

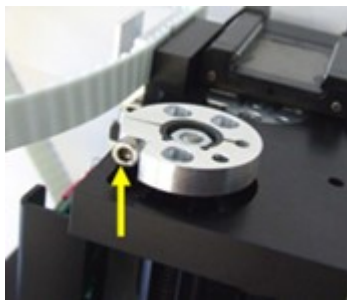


Figure 305 - Set screw of the leadscrew holder

6. Finally, take out the leadscrew from the horns removing the home flag ring.

Install new horns leadscrew into the PIP robotic head following previous steps 1-6 in reverse mode. Some considerations must be taken into account:

1. Place the leadscrew between the home flag ring and barcode readers structure plate. Use the HORNS ADJUSTMENT TOOL (BCR) in order to align it.

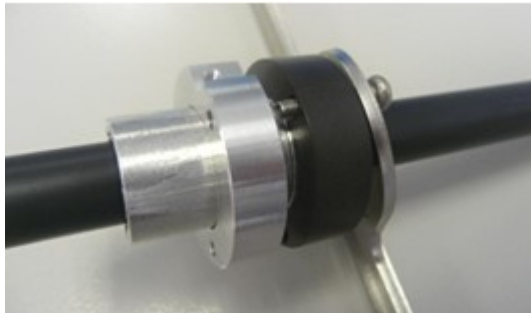


Figure 306 - Position of the home flag ring and barcode reader structure plate in the leadscrew

2. When the leadscrew holder is placed in the leadscrew, it must be placed at the same level of the leadscrew.



Figure 307 - Leadscrew holder

3. If it is hard to place the leadscrew holder in its position on the PIP robotic head, hit smoothly from the top downwards with a rubber hammer facing correctly the other side of the leadscrew.



Figure 308 - Leadscrew install

4. Place the horns leadscrew pulley at the same level than the leadscrew axis. Then tighten the set screw of the pulley in the flat side of the leadscrew axis.

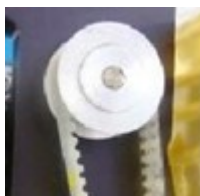


Figure 309 - Pulley of the motor of the horns

5. Place the belt and adjust its tension. See "Adjusting Z-axis belt tension" on page 334.
6. Check the probes offset. See "Verifying probes offset" on page 364.
7. Check probes parallelism. See "Verifying probes parallelism" on page 362.
8. Perform probes adjustment in all positions: reagents drawers (see "Adjusting reagents drawers" on page 338), samples drawers (see "Adjusting samples drawers" on page 342), incubators (see "Adjusting incubators" on page 345), and dilution cup (see "Adjusting dilution cup" on page 347).
9. Perform adjustments verifications related to the PIP module. See "Verification procedures" on page 360.

2.4.5.5.3 Probes leadscrew

1. Remove the belt of the probe and loosen the set screw from the probe leadscrew pulley motor and remove it.



Figure 310 - Probes belts

2. Move down the two probes wagons manually and remove the Y axis robotic head rear stopper in order to have full access to the leadscrew holder in case replacing probe 1 leadscrew.



WARNING

Be careful to do not prick yourself with the probes, if assembled.

3. Remove the four screws that attach the Y axis linear guide wagon to the PIP robotic head.



Figure 311 - Location of the screws that attach Y linear guide wagon to the PIP robotic head

4. Loosen (if present) the set screw that fixes the position of the ball bearing wagon. To get access to the set screw, remove the plate that holds the Y-axis belt by means of its two screws. Then loose the set screw and remove the ball bearing wagon. Before removing the ball bearing wagon, mark with a pen its orientation in order to have a reference when it will be installed again.



Figure 312 - Location of the set screw to fix the linear guide wagon

5. Remove the three screws (depending on the probe to be replaced) that attach the leadscrew holder to the robotic head and the three screws that attach the leadscrew to the probe wagon.

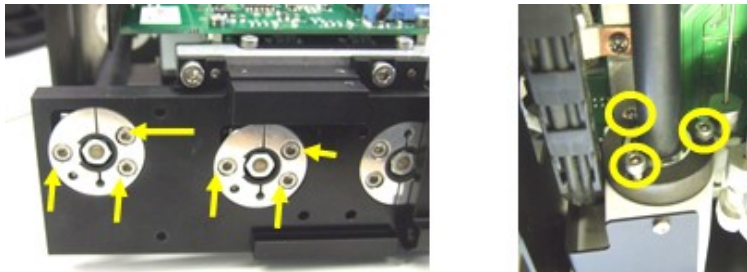


Figure 313 - Robotic head and leadscrew holders

6. Remove the leadscrew upwards until the leadscrew holder is free from its position. If it is hard to remove, hit smoothly from the bottom upwards with a rubber hammer.
7. Remove the leadscrew holder from the leadscrew by loosening its set screw.
8. Finally, take out the leadscrew downwards.

Install the probe leadscrew into the PIP robotic head by following the previous steps in reverse. Some considerations must be taken into account:

- When the leadscrew holder is placed in the leadscrew, it must be placed at the same level of the leadscrew.
- If it is hard to place the leadscrew holder in its position on the PIP robotic head, hit smoothly from the top downwards with a rubber hammer facing correctly the other side of the leadscrew.

- When placing the linear guide wagon, and before placing the screws that attach it to the PIP robotic head, fix the position of the linear guide wagon with the set screw included in the leadscrew spare part. To get access to the set screw allocation, remove the plate that holds the Y-axis belt by means of its two screws and then tighten the set screw against the linear guide wagon. Place the screws that attach the Y linear guide wagon to the PIP robotic head, tighten them, and finally remove the set screw. Install the Y-axis belt.



Figure 314 - Location of the set screw to fix the linear guide wagon



NOTE

When tightening the fixing part make sure to tension the belt outwards in order to eliminate any possible play that could affect the belt tension in the future.



Figure 315 - Belt tension direction when fixing the clamping plate (this view is hidden in the instrument)

- Place the motor pulley at the same level as the leadscrew axis. Then tighten the set screw of the motor pulley to block it.
- Place the belt and adjust its tension. See "Adjusting Z-axis belt tension" on page 334.
- Check the probes offset. See "Verifying probes offset" on page 364.
- Check the probes parallelism. See "Verifying probes parallelism" on page 362.
- Perform probes adjustment in all positions: reagents drawers (see "Adjusting reagents drawers" on page 338), samples drawers (see "Adjusting samples drawers" on page 342), incubators (see "Adjusting incubators" on page 345), and dilution cup (see "Adjusting dilution cup" on page 347).
- Perform adjustments verifications related to the PIP module. See "Verification procedures" on page 360.

2.4.5.6 Replacing PIP X axis chain (CH1)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233788 X WIRING/CHAIN (PIP).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open upper door.
3. Disconnect tubing from the connection block and open the clamp.



Figure 316 - Tubing

4. Disengage the last slot of the chain.



Figure 317 - Last chain slot

5. Disconnect the three flat cables from PCA PIP-CONY59 and remove the ties.



Figure 318 - Flat cables

6. Remove the instrument left lateral methacrylate cover.



Figure 319 - Lateral removal

7. Disconnect the four tubes from connection block.



Figure 320 - Tubing

8. Disconnect the three flat cables from PCA GEN-CONX59.



Figure 321 - Flat cables

9. Remove the two screws that attach the chain.



Figure 322 - Chain screws

10. Disconnect the Teflon tubes from the valves.



Figure 323 - Teflon tubes

11. Remove the screws of the blocks over the dilutors (side of the Teflon tube) to be able to pull apart the Teflon tubes.



Figure 324 - Teflon tubes under the blocks

12. Remove the PIP X chain.
13. Place the new PIP X chain and assemble back all the parts.



CAUTION

Do not swap the ribbon cables in PCA GEN-CONX59 or PCA PIP-CONY59 when connecting them (it could damage different parts of the instrument), specially the cables labelled as XB-PIP_XY and XB-PIP_Z that both have a blue connector. The cable labelled as XB-PIP_TOW has a black connector. Cables should match the connector colour when connected to the PCA.

14. Perform a Prime to check the fluidic system. See "Fluidics tab" on page 826.
15. Perform the Leak-tightness test. See "Leak-tightness test" on page 583.

2.4.5.7 Replacing PIP Y axis chain (CH2)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233789 Y WIRING/CHAIN (PIP).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Free from its clamps the fluidic tubing from the probes washer to the Y axis chain.



Figure 325 - Location of the clamps in the PIP robotic head



NOTE

Make sure that the position of the tubing clamp is not modified while removing or inserting the fluidic tubes (see image above).

3. Remove the two screws of the probes washer, pull it apart it from its position and disconnect all the tubes from their fittings.
4. Disconnect, from the interconnection manifold located in the robotic head, the tubes of the probes that come from the Y axis chain.



Figure 326 - Interconnection manifold in the robotic head

5. Remove the two screws from the lateral cover of the Y arm.



Figure 327 - Lateral cover screws of the Y arm

6. Remove the screws that attach the Y chain to the Z tower / robotic head.



Figure 328 - Location of the screws that attach the Y chain to the Z tower / robotic head

7. Disconnect the Y chain ribbon cables from PCA PIP-TOW59.

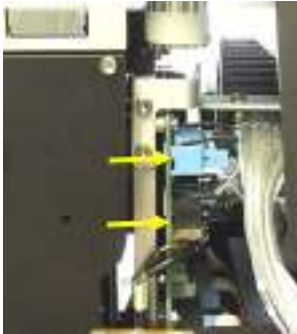


Figure 329 - Location of the ribbon cables connectors at PCA PIP-TOW59

8. Disconnect the tubes that come from the PIP Y axis chain to the interconnection manifolds located at the rear part of the PIP arm.

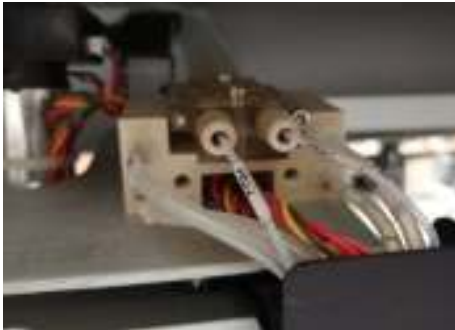


Figure 330 - Interconnection manifolds at the rear part of PIP arm

9. Open the tie-wraps attached to the lateral cover of the Y arm in order to release the ribbon cables and tubing from the Y axis chain.

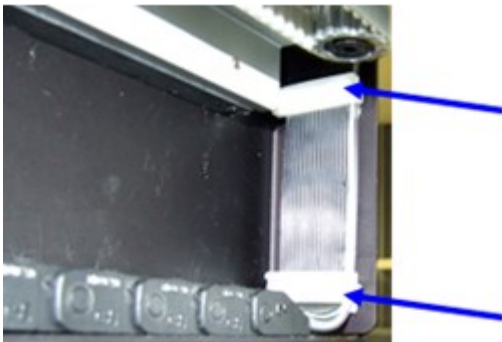


Figure 331 - Tie-wraps in the lateral cover of the Y arm

10. Remove the screws that attach the PIP Y axis chain to the lateral cover of the PIP arm.



Figure 332 - Location of the screws that attach the Y chain to the lateral cover of the PIP arm

11. Remove the screws of the rear cover of the PIP arm and apart the cover from its position.



Figure 333 - Location of the screws of the rear cover of the PIP arm

12. Open the tie-wrap in order to release the ribbon cables that come from the Y axis chain to the PCA PIP-CONY59, and disconnect the cables from the PCA.



Figure 334 - Tie-wrap under the rear cover of the PIP arm (left) and the connectors in PCA PIP-CONY59 (right)

13. Remove completely the chain from the instrument. Then remove the plate that attaches the Y chain to the Z tower/robotic head from the chain in order to place it in the new Y chain.



Figure 335 - Plate attaching the Y chain to the Z tower / robotic head

14. Repeat previous steps in reverse mode in order to install the new chain.



CAUTION

Do not swap the ribbon cables in PCA PIP-CONY59 or PCA PIP-TOW59 when connecting them (it could damage different parts of the instrument).

Cables should match the connector colour when connected to the PCA: cable labelled as PIP-TOW black and PIP-Z blue.

15. Perform a Prime to check the fluidic system. See "Fluidics tab" on page 826.
16. Perform the Leak-tightness test. See "Leak-tightness test" on page 583.

2.4.5.8 Replacing PIP Z1 axis chain (CH3) / Z2 axis chain (CH4)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- //233790 //LEVEL DETECTOR WIRING/CHAIN (PIP).
- For instruments with impact detector installed from factory with S/N $\geq$ 059-0000611: 233897 LEVEL DETECTOR WIRING/CHAIN (PIP) (Impact detector)

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Disconnect, from the interconnection manifold located in the robotic head, the tube of the probe that comes from the chain to be replaced.



Figure 336 - Interconnection manifold in the robotic head

3. Move down manually (by means of its belt) the horns and remove the lateral cover of the PIP robotic head.



Figure 337 - Lateral cover of the PIP robotic head

4. In case that the chain to be replaced is the PIP Z1 axis chain (CH3), remove also the rear cover of the PIP robotic head.



Figure 338 - Rear cover of the PIP robotic head

5. Remove the two screws from the lateral cover of the Y arm.



Figure 339 - Lateral cover screws of the Y arm

6. Disconnect level detection cable of the chain to be replaced from PCA PIP-TOW59.



Figure 340 - Connector location in PCA PIP-TOW59 for chain CH4 (left) and CH3 (right)

7. Cut the two tie-wraps located at PCA PIP-TOW59 fixing the position of the tubes.
8. Remove the probe related with the chain to be replaced.

9. Remove the three screws of the cover of the wagon probe and carefully apart it from its position as much as possible.



Figure 341 - Wagon probe cover for chain CH4 (left) and CH3 (right)

10. Unscrew the fitting from the PCA in order to remove the Teflon tube from its position.



Figure 342 - Teflon tube of the probe

11. Disconnect carefully the cable (level detection signals) of the chain to be replaced connected to the level detection PCA (PCA PIP-P1OSC59 or PCA PIP-P2OSC59).



Figure 343 - Connector location in the level detection PCA

12. Unscrew the ground connection of the chain to be replaced.



Figure 344 - Ground connection

13. Remove the chain to be replaced from the instrument.
14. Remove the cover of the probe wagon from the chain and install it in the new chain.
15. Place the new chain and assemble back all the parts.



NOTE

When installing the ground connection clamp, make sure that the level detection cable and the Teflon tube are tightened without squeezing them (place the Teflon tube at the back of the clamp) but also making sure that no movement is allowed by pulling them slightly up and down.



NOTE

Once the ground connection is done, consider placing adhesive tape around the remaining visible shielded area of the Teflon tube and the level detection cable to avoid undesired contact of this shielded area with other parts of the instrument.

16. Perform a fluidic prime to check that the Teflon tube is not squeezed.
17. Perform the Leak-tightness test. See "Leak-tightness test" on page 583.
18. Check the level detection. See "Verifying level detection system" on page 372.

2.4.5.9 Replacing barcode reader mirror



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233777 BCR MIRROR (ERY PIP&CAT / EFL CAT).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the mirror by pressing the plastic clipping stoppers

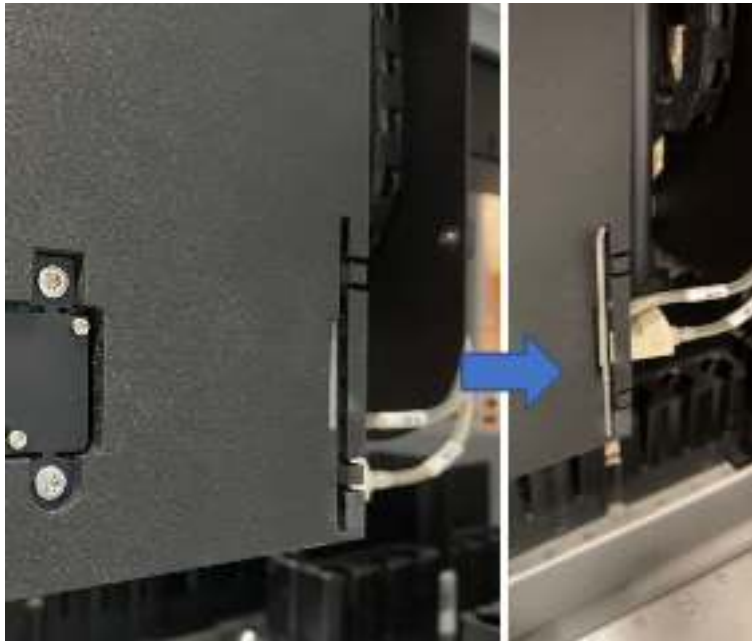


Figure 345 - Removing the mirror

2. In order to place the mirror in the correct position it is necessary to know which is its first face. Touch the mirror with an object. If the tip of the object seems to touch the reflected image, this side of the mirror is the first face.



Figure 346 - The mirror first face (left) and the mirror second face (right)

3. Insert the mirror in the slot. Place the mirror facing the first face to the barcode reader.

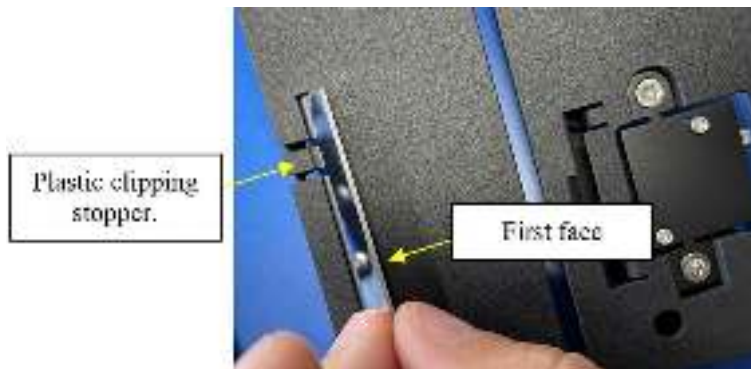


Figure 347 - Placing the mirror



NOTE

Check that the mirror is securely fixed, making sure that the plastic clipping stopper locks it in place.

2.4.5.10 Replacing horns chain



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233791 BCR WIRING/CHAIN (PIP).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

Depending on the instrument serial number follow the described steps:

(SN < 059-0001887)

1. Switch off the instrument and disconnect it from the mains.
2. Open the upper door.
3. Remove the PIP module left cover.

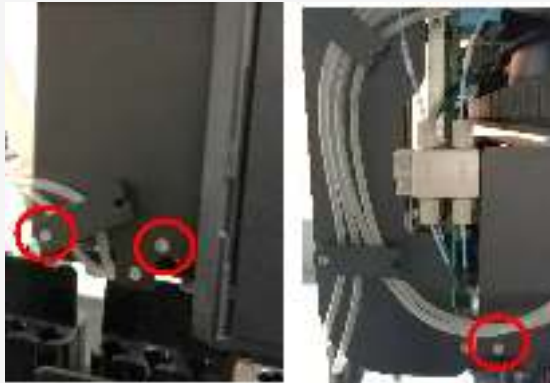


Figure 348 - Left cover screws

4. Disconnect the BCR cable, then remove the tie and the screw attaching the chain.



Figure 349 - Cable, tie and chain screw

5. Remove the other screw attaching the chain.



Figure 350 - Chain screw

6. Disconnect the BCR cable and the ground cable.



Figure 351 - Horns cabling

7. Remove the horns chain.
8. Place the new chain.
9. Assemble back all the parts.

10. Place the tie ensuring that the cable does not interfere with any part of the instrument when moving upwards/downwards.



Figure 352 - Final assembly

11. Verify the card barcode reading performance. See "Verifying barcode reader" on page 369.

(SN ≥ 059-0001887)

1. Switch off the instrument and disconnect it from the mains.
2. Open the upper door.
3. Remove the PIP module left cover.

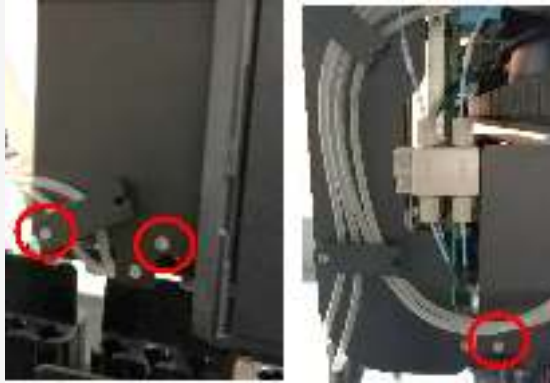


Figure 353 - Left cover screws

4. Disconnect the chain cable from the PCA and remove the zip tie that holds it.



Figure 354 - Disconnecting the barcode chain cable

5. Detach the left horn wall by removing the four screws holding it.

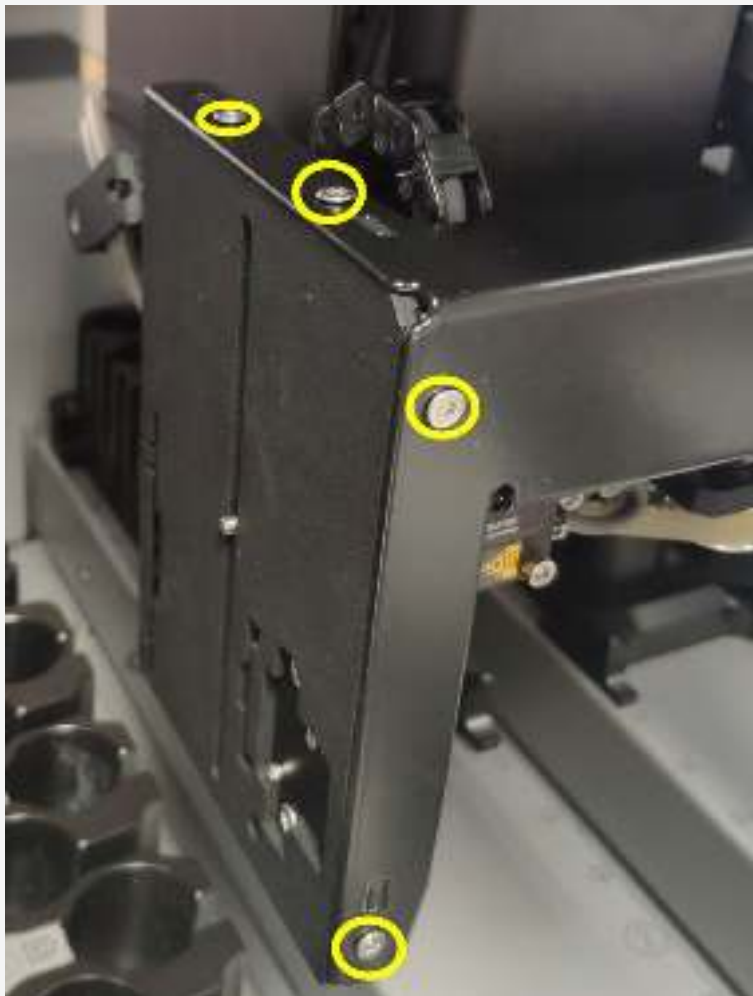


Figure 355 - Detaching the left horn

**NOTE**

Caution should be taken when removing the horn wall, as the barcode flat cable is connected and prevents it from detaching completely

6. Disconnect the BCR flat cable from the barcode reader. Make sure that the connector flap is opened before disconnecting the cables.

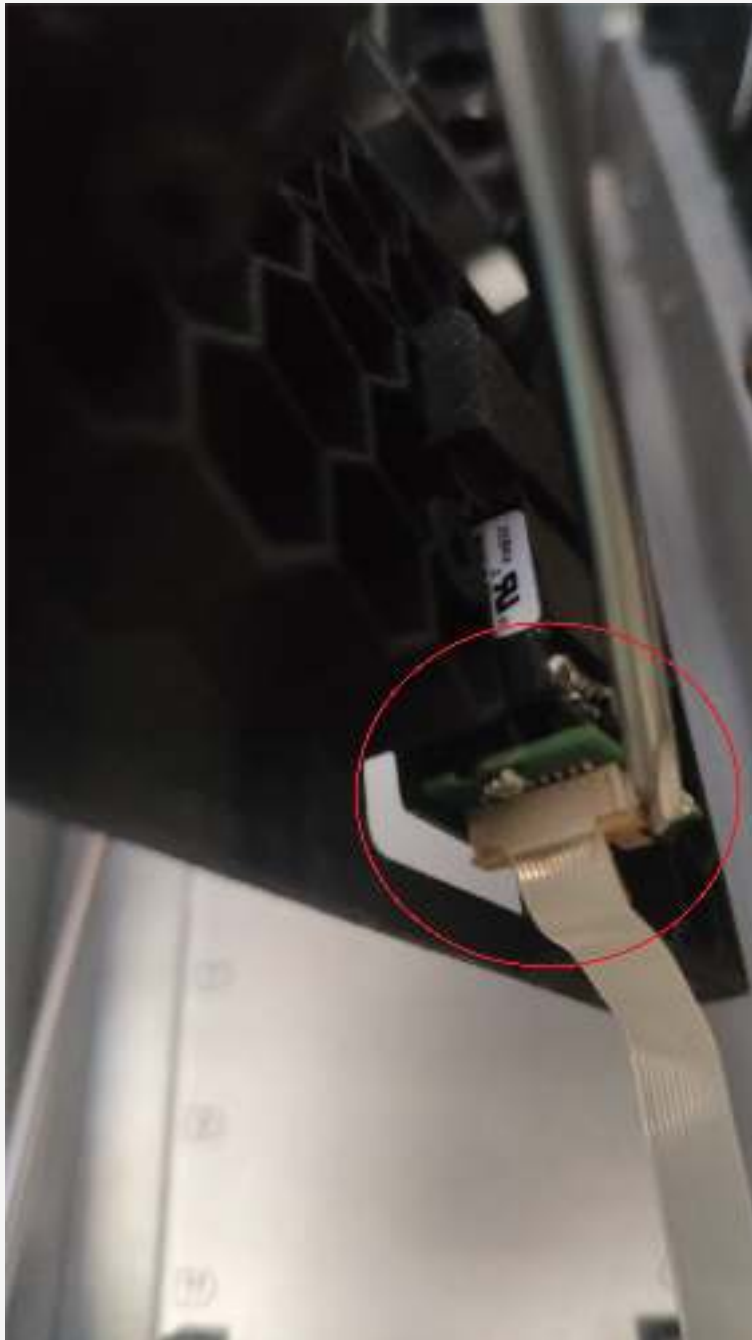


Figure 356 - Disconnecting the BCR flat cable

7. Remove the screw attaching the chain to the wall.

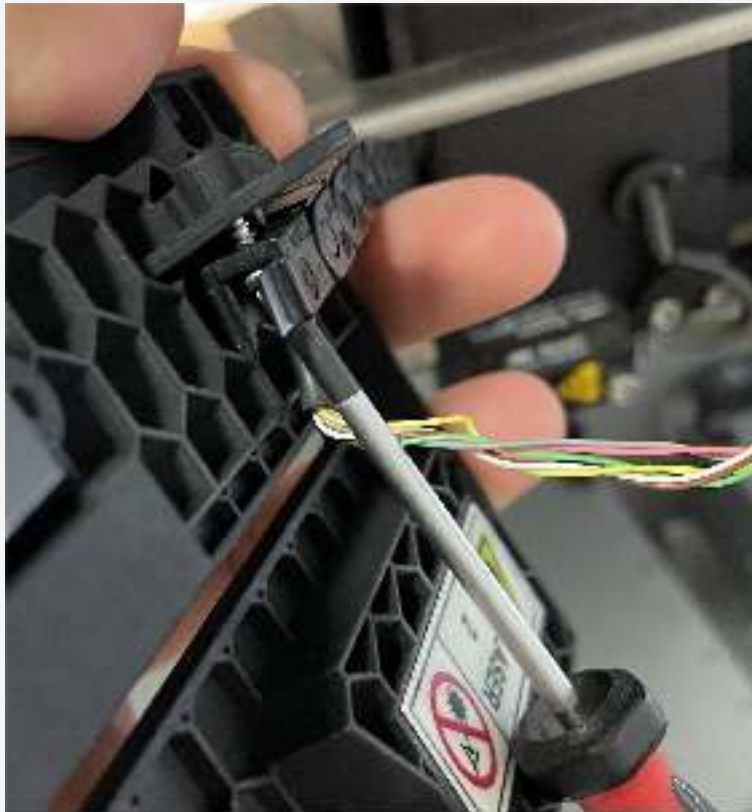


Figure 357 - First chain screw

8. Remove the other screw attaching the chain.

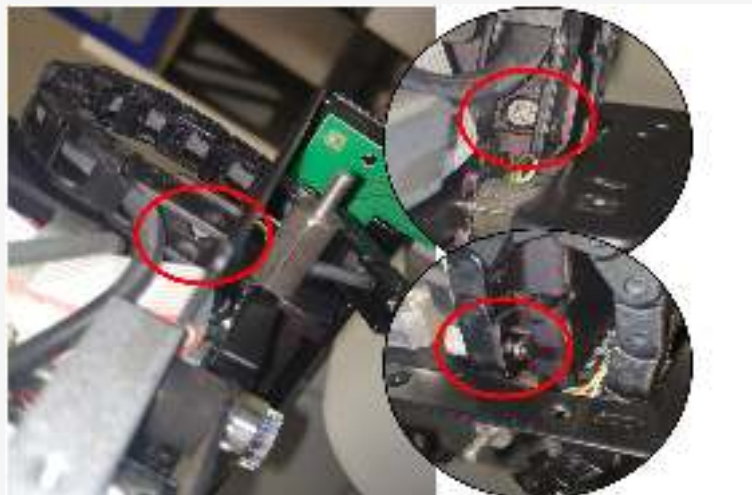


Figure 358 - Second chain screw

9. Disconnect the BCR cable and the ground cable.



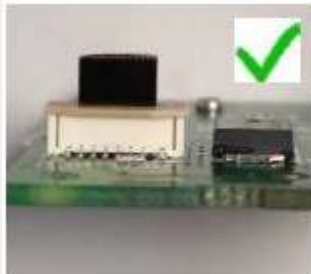
Figure 359 - Horns cabling

10. Remove the horns chain.
11. Place the new chain.
12. Clean the barcode mirror/s and the splash protector.
13. Place the horns near its position and connect the flat cable to the barcode reader.



NOTE

Make sure that the pins are correctly aligned, and the connector flap completely



closed.



14. Assemble back the left horn wall.
15. Connect the chain cable to the PCA.

16. Place a new zip tie to hold the barcode chain cable, ensuring that the cable does not interfere with any part of the instrument when moving upwards/downwards.

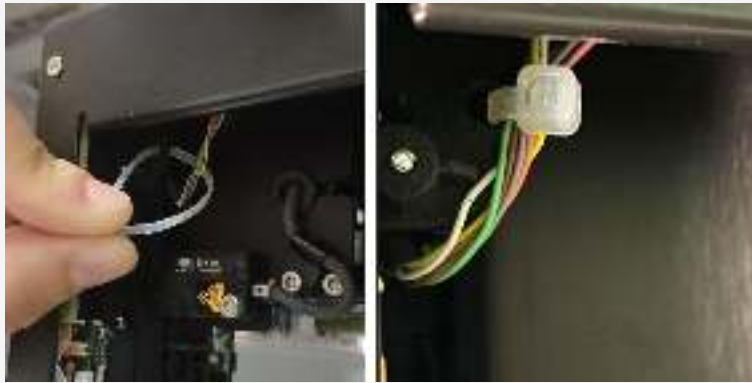


Figure 360 - Placing the barcode chain cable holding tie

17. Route the flat barcode cables as shown in the image:



Figure 361 - Flat cable routing

18. Assemble back the rest of parts.
19. Switch on the instrument.
20. Up and down the horns using the Technical Service application and verify the flat barcode reader cable is moving smoothly upwards/downwards. See "Movement subtab" on page 804.
21. Verify the card barcode reading performance. See "Verifying barcode reader" on page 369.

2.4.5.11 Replacing toothed belt T2.5 245

Replacement procedures related to the T2.5 245 Toothed belt in PIP and CAT arms:

- See "Replacing toothed belt T5/2000 (CAT X-axis)" on page 524
- See "Replacing toothed belt T5/1200 (CAT Y-axis)" on page 527
- See "Replacing toothed belt T5/2000 (PIP X-axis)" on page 421
- See "Replacing toothed belt T5/1200 (PIP Y-axis)" on page 425

2.4.5.11.1 Replacing toothed belt T2.5/245 (PIP X-axis)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 231889 TOOTHED BELT T2,5/245.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the upper door.

3. Remove the small belt.



Figure 362 - Small belt

4. Place the new small belt.

2.4.5.11.2 Replacing toothed belt T2.5/245 (PIP Y-axis)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 231889 TOOTHED BELT T2,5/245.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the upper door.

3. From the front of the instrument, take out the Y-belt from its position by removing the fixing screw of the pulley.



Figure 363 - PIP large belt

4. Remove the large belt from the pinion. The belt will be hang on the PIP robotic head.



Figure 364 - Large belt (pinion)

5. Remove the upper rear cover.
6. Remove the screws of the rear cover of PIP arm.



Figure 365 - Location of the screws of the rear cover of the PIP arm

7. Disconnect the tubing from the Y chain and pull apart the cover.
8. Remove the motor support screws.



Figure 366 - Support screws

9. Disconnect the two flat cables, open the cable clamp, and pull apart the cables.



Figure 367 - Cabling and cable clamp

10. Remove the motor screw (slotted hole).



Figure 368 - Motor screw

11. Move the motor to be able to remove the small belt.

12. Place the new small and large belts and assemble back all the parts.

**NOTE**

If the pulley tightening screw is in the same position before disassembly, the belt tension will remain unchanged, else belt tension must be adjusted by modifying the end pulley position.



Figure 369 - Tension screw

**NOTE**

Any variation of the belt tension may modify probe adjustments in all positions.

13. Adjust the tension of the small belt.
14. Perform the probe positioning check. See "Procedures tab" on page 756.

2.4.5.12 Replacing toothed belt T5/2000 (PIP X-axis)

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- 233865 TOOTHED BELT T5/2000

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the upper door and remove X-chain fairing by removing its 4 screws.



Figure 370 - X chain fairing



Figure 371 - X chain fairing fixing screws

3. Removing the fixing screw of the tension pulley.



Figure 372 - Tension pulley of the large belt (x axis)

4. Remove the large belt from the pinion. The belt will be left hanging from the PIP robotic head.
5. Loosen the belt clamp plate (without removing) and remove the belt.

- Put the new belt in the clam plate and tighten the screws tensioning the belt outwards.



Figure 373 - Large belt clamping plate and belt tension direction needed at the plate fixing moment.



NOTE

When tightening the fixing part make sure to tension the belt outwards in order to eliminate any possible play that could affect the belt tension in the future.



Figure 374 - Belt tension direction when fixing the clamping plate (this view is hidden in the instrument)

- Check that the home barrier do not collide with the sensor by hand-moving the arm (x-axis case).

8. Put the belt around the pulleys and fix the tension pulley again.

**NOTE**

If the pulley tightening screw is in the same position before disassembly, the belt tension will remain unchanged, else belt tension must be adjusted by modifying the end pulley position.



Figure 375 - Tension screw (tension pulley is under these parts)

**NOTE**

Any variation of the belt tension may modify probe adjustments in all positions.

9. Mount the fairing plate back in place.
10. Perform the probe positioning check. See "Procedures tab" on page 756.

2.4.5.13 Replacing toothed belt T5/1200 (PIP Y-axis)

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- 233866TOOTHED BELT T5/1200

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the upper door and remove the lateral cover by removing its screws.



Figure 376 - PIP arm lateral cover

3. Removing the fixing screw of the tension pulley.



Figure 377 - Tension pulley of the large belt (PIP y axis)

4. Remove the large belt from the pinion. The belt will be left hanging from the PIP robotic head.



Figure 378 - Large belt (pinion)

5. Loosen the belt clamp plate (without removing) and remove the belt.

6. Put the new belt in the clam plate and tighten the screws tensioning the belt outwards.

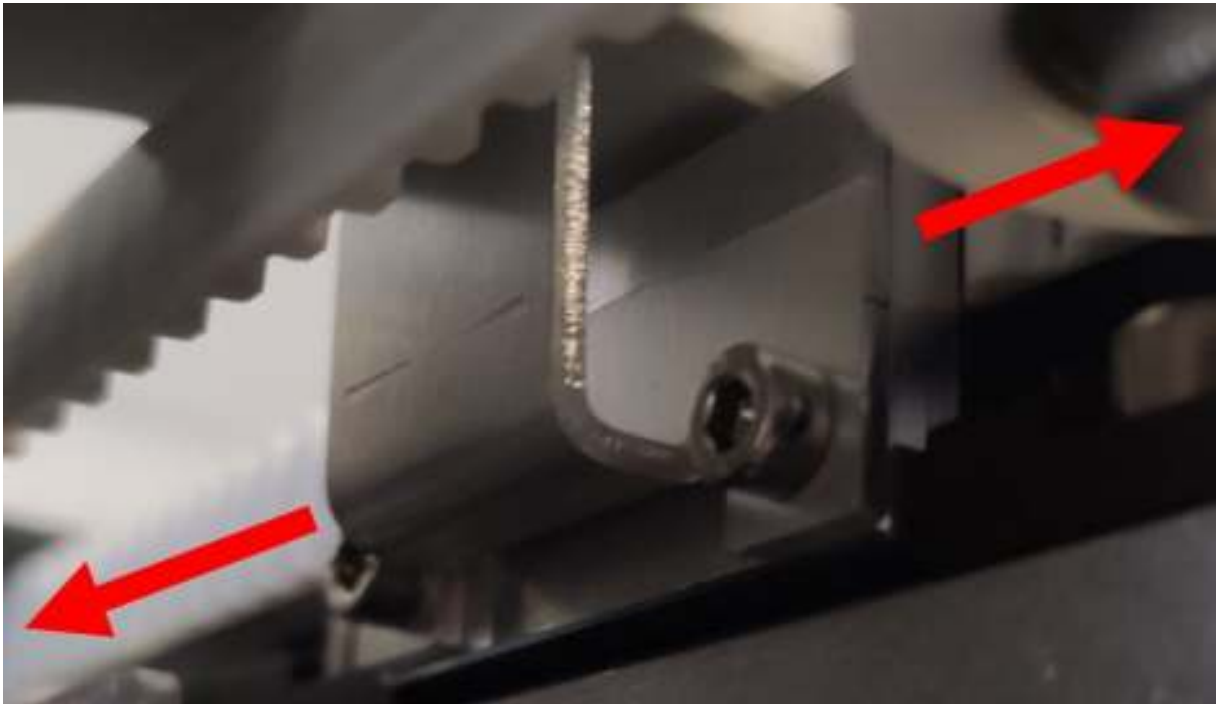


Figure 379 - Large belt clamping plate and belt tension direction needed at the plate fixing moment.



NOTE

When tightening the fixing part make sure to tension the belt outwards in order to eliminate any possible play that could affect the belt tension in the future.



Figure 380 - Belt tension direction when fixing the clamping plate (this view is hidden in the instrument)

7. Check that the home barrier do not collide with the sensor by hand-moving the tower (y-axis case).

8. Put the belt around the pulleys and fix the tension pulley again.

**NOTE**

If the pulley tightening screw is in the same position before disassembly, the belt tension will remain unchanged, else belt tension must be adjusted by modifying the end pulley position.



Figure 381 - Tension screw (tension pulley is under these parts)

**NOTE**

Any variation of the belt tension may modify probe adjustments in all positions.

9. Mount the cover back in place.
10. Perform the probe positioning check. See "Procedures tab" on page 756.

2.4.6 Electronics

This section covers the electronics related to the Pipetting robot arm module.

2.4.6.1 Fuses

Fuses related to the PIP module are detailed in the following table:

Table 37 - List of the PIP module fuses

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS31 FUS32	MST250 0034.6615 1A/250V MST250 0034.6620 3.15A/250V3
PCA PIP Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA PIP-P1OSC59	NA	NA

PCA	FUSE NAME	DESCRIPTION
PCA PIP-P2OSC59	NA	NA
PCA PIP-TOW59	NA	NA
PCA PIP-BCR59	NA	NA
PCA PIP-CONY59	NA	NA
PCA GEN-CONX59	NA	NA

2.4.6.2 LEDs

LEDs related to the PIP module are detailed in the following table:

Table 38 - List of the PIP module LEDs

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED15 LED16	Green 5V (FUS31) Green 24V (FUS32)
PCA PIP Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$
PCA PIP-P1OSC59	NA	NA
PCA PIP-P2OSC59	NA	NA
PCA PIP-TOW59	NA	NA
PCA PIP-BCR59	NA	NA
PCA PIP-CONY59	NA	NA
PCA GEN-CONX59	NA	NA

2.4.6.3 Jumpers and test points

Neither jumpers nor test points are related to the PIP module.

2.4.6.4 Wiring table

Cables related to the Pipetting module are detailed in the following table and their connection are shown in the wiring diagram. To view the wiring diagram, see "Related documents" on page 19 or refer to the Erytra Online Help.

Table 39 - List of the PIP module wiring

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
BRX-EXT + BRX-MotX	H1 MotX	PCA GEN-CONX59	Power motor	X AXIS MOTOR
BRX-EncX	H2 EncX	PCA GEN-CONX59	Encoder motor	X AXIS MOTOR
B-PIP_XY	H3 In CAT/PIP-XY	PCA GEN-CONX59	PIP_XY	PCA GEN-BCKPL59
B-PIP_Z	H4 In CAT/PIP-Z	PCA GEN-CONX59	PIP_Z	PCA GEN-BCKPL59
B-PIP_TOW	H5 In CAT/PIP-TOW	PCA GEN-CONX59	PIP_TOW	PCA GEN-BCKPL59
XB-PIP_TOW	H8 Out CAT/PIP-TOW	PCA GEN-CONX59	H4	PCA PIP-CONY59
XB-PIP_Z	H7 Out CAT/PIP-Z	PCA GEN-CONX59	H3	PCA PIP-CONY59
XB-PIP_XY	H6 Out CAT/PIP-XY	PCA GEN-CONX59	H5	PCA PIP-CONY59
PIP-EncY	H8 MotY	PCA PIP-CONY59	Encoder motor	Y AXIS MOTOR
PIP-MotY	H9 EncY	PCA PIP-CONY59	Power motor	Y AXIS MOTOR
PIP-Z	H1 Out PIP-Z	PCA PIP-CONY59	H2 PIP-Z	PCA PIP-TOW59
PIP-TOW	H2 Out PIP-TOW	PCA PIP-CONY59	H1 PIP-TOW	PCA PIP-TOW59

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
PIP-Z1	H4 PIP-Z1	PCA PIP-TOW59	H1	PCA PIP-P1OSC59
PIP-Z2	H6 PIP-Z2	PCA PIP-TOW59	H1	PCA PIP-P2OSC59
PIP-BCR	H7 BCR	PCA PIP-TOW59	H2	PCA PIP-BCR59
PIP-PresDet	H8 PRES_DET	PCA PIP-TOW59	NA	LASER DETECT
PIP-EncZ1	H9 ENC_Z1	PCA PIP-TOW59	Encoder motor	P1 AXIS MOTOR
PIP-MotZ1	H10 MOT_Z1	PCA PIP-TOW59	Power motor	P1 AXIS MOTOR
PIP-EncZ2	H11 ENC-Z2	PCA PIP-TOW59	Encoder motor	P2 AXIS MOTOR
PIP-MotZ2	H12 MOT-Z2	PCA PIP-TOW59	Power motor	P2 AXIS MOTOR
PIP-EncBCR	H13 ENC_BCR	PCA PIP-TOW59	Encoder motor	BCR AXIS MOTOR
PIP-MotBCR	H14 MOT_BCR	PCA PIP-TOW59	Power motor	BCR AXIS MOTOR
NA	H1	PCA PIP-BCR59	NA	BARCODE READER
NA	H3	PCA PIP-BCR59	NA	BARCODE READER



Figure 382 - Connectors, fuses and LEDs of PCA GEN-BCKPL59 related to PIP module

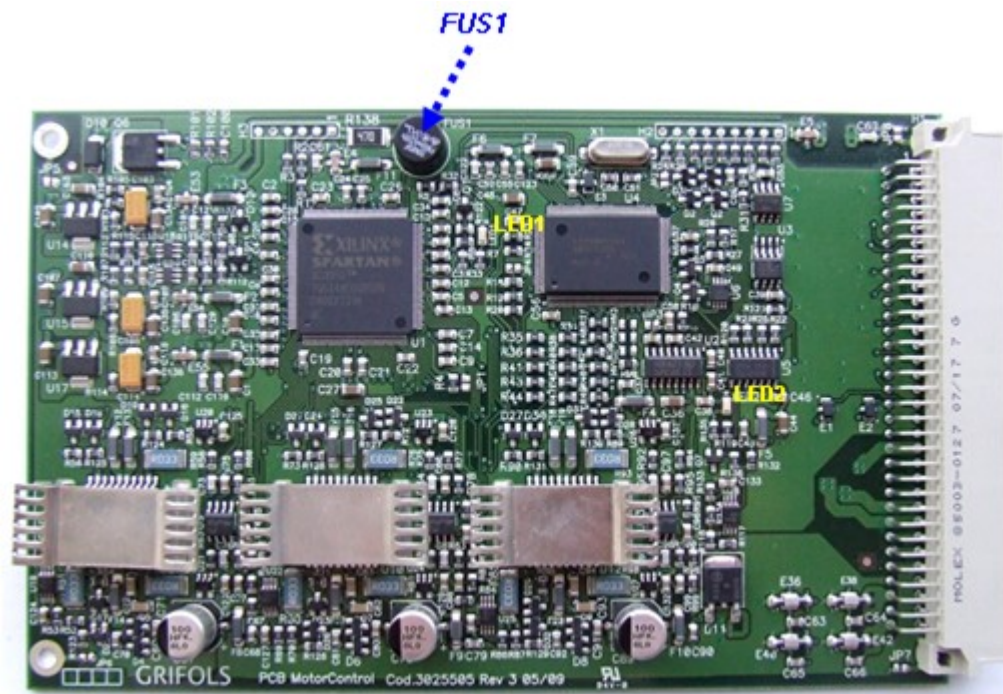


Figure 383 - Location of the fuse and LEDs in PCA PIP Control

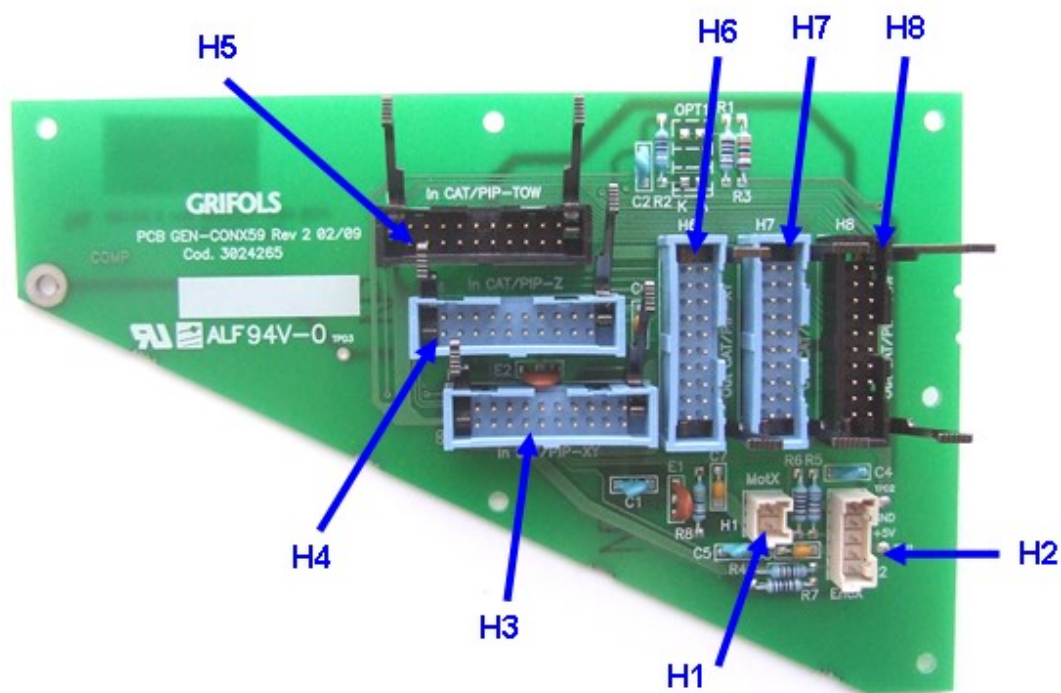


Figure 384 - Location of the connectors in PCA GEN-CONX59

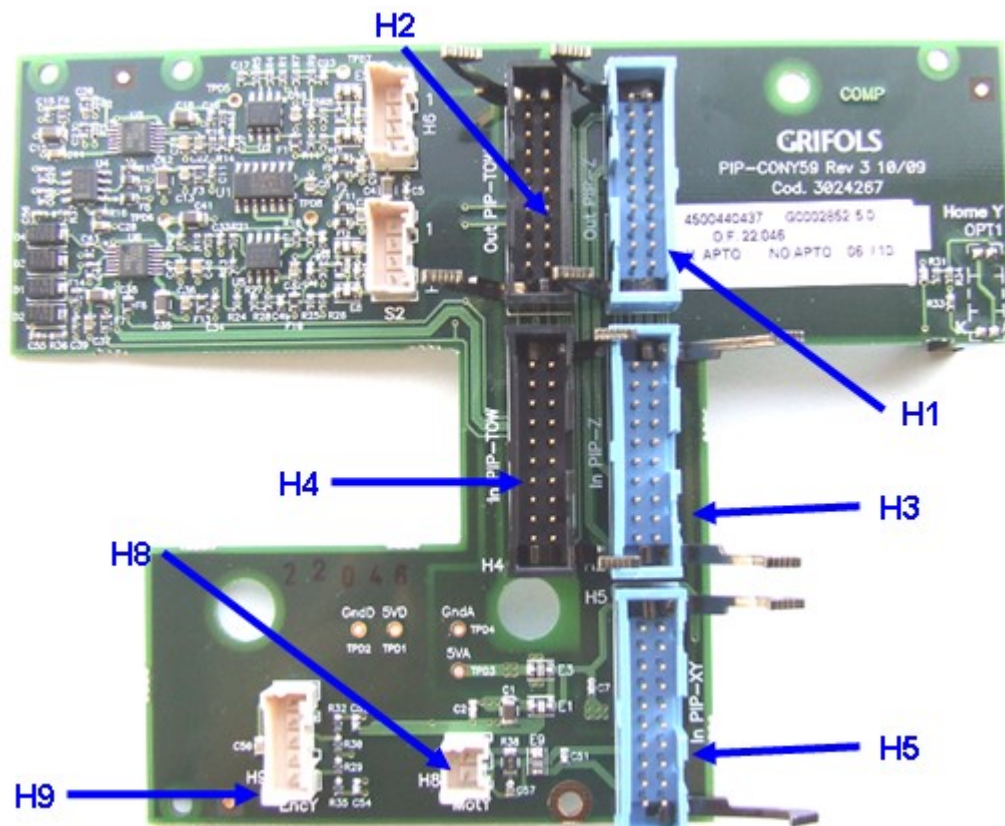


Figure 385 - Location of the connectors in PCA PIP-CONY59 related to PIP module

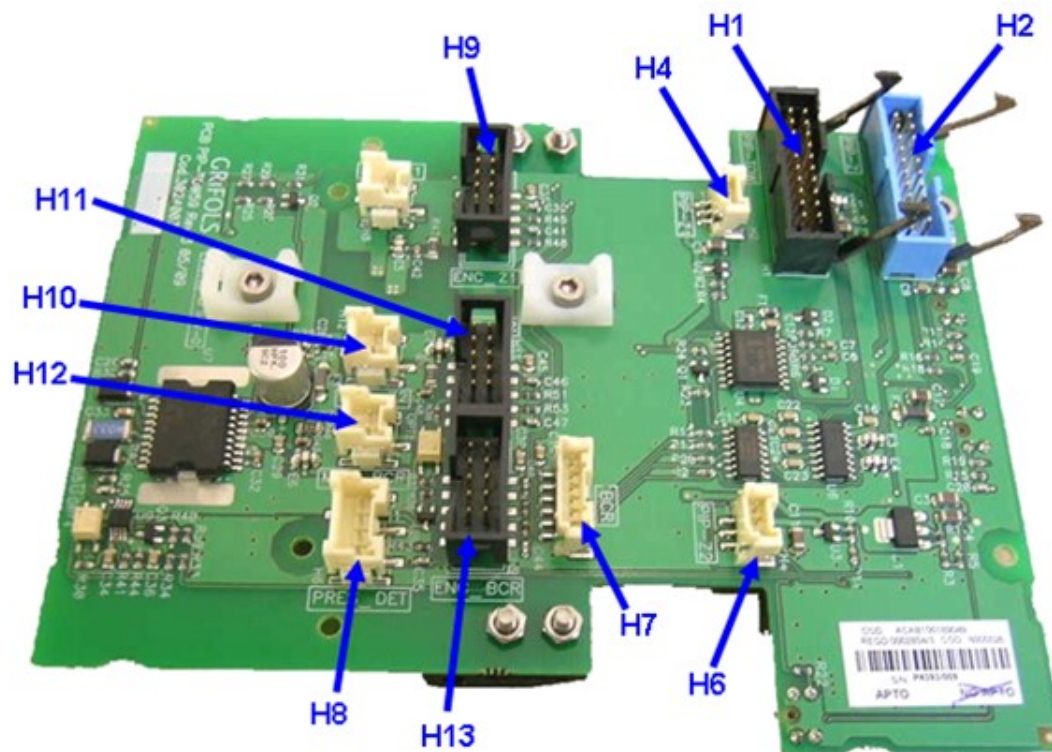


Figure 386 - Location of the connectors in PCA PIP-TOW59

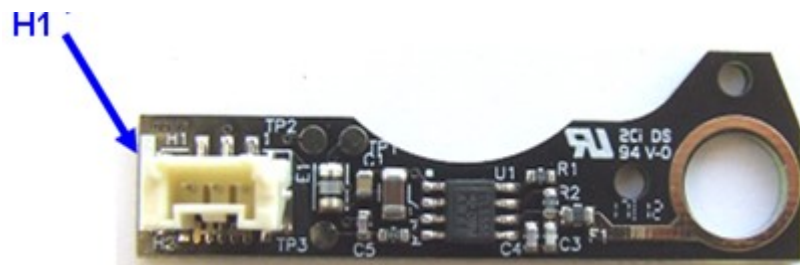


Figure 387 - Location of the connector in PCA PIP-P1OSC59 (Impact detector)



Figure 388 - Location of the connector in PCA PIP-P2OSC59 (Impact detector)

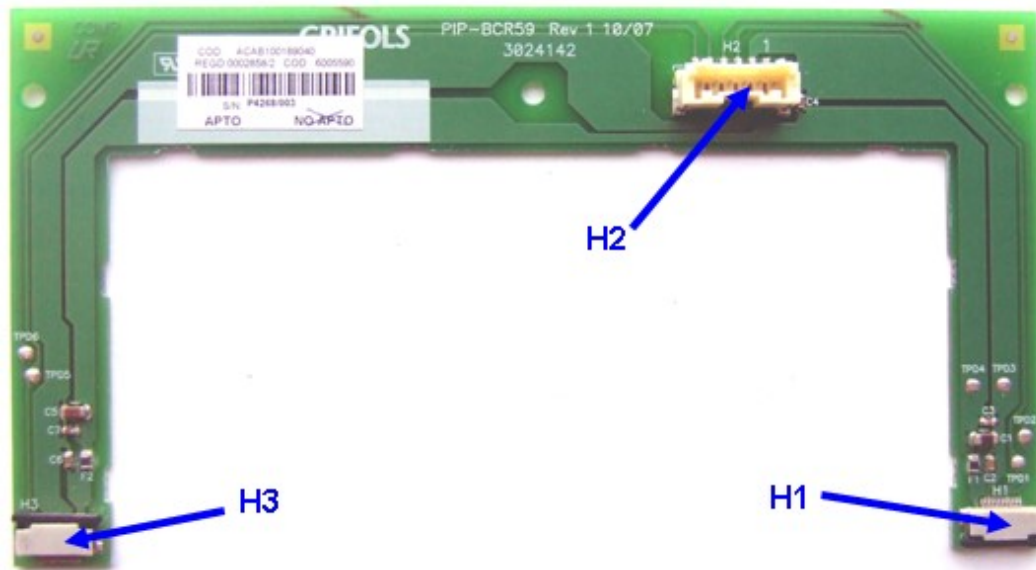


Figure 389 - Location of the connectors in PCA PIP-BCR59

2.4.7 Troubleshooting

Table 40 - List of errors related to the PIP module

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	XY Motors (01) X axis (02) Y axis (11) Presence detector during X movements (05) Presence detector during Y movements	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY
		005000	ERRORCODE_ARG	Command has wrong arguments.		CANCEL Works (it does not show any further error. When queueing new samples, The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test is shown)
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	XY Motors (01) X axis (02) Y axis (11) Presence detector during X movements (05) Presence detector during Y movements	008000	ERRORCODE_ NOTSAFE	Command is not safe for the instrument.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY CANCEL Works (it does not show any further error. When queueing new samples, The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test is shown)
		032000	CURRENT_LIMIT	Current limit at servo device.		
		033000	ERRORCODE_ SRV_POSITION_ ERROR	A position error has been produced at a servo device.		
		034000	MOVEMENT_ ERROR	A movement error has been produced at a servo device.		
		035000	HOME_TIMEOUT	A home timeout has been produced at a servo device.		
		036000	ERRORCODE_ SRV_SYNC_ TIMEOUT	A synchronization error has been produced at a servo device.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	XY Motors (01) X axis (02) Y axis (11) Presence detector during X movements (05) Presence detector during Y movements	037000	ERRORCODE_SRV_UNKNOWN_ERROR	An unknown error has been produced at a servo device.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY
		041000	ERRORCODE_PD_SYNC_TIMEOUT	Presence detector has been busy for more than its timeout margin.		CANCEL Works (it does not show any further error. When queueing new samples, The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test is shown)
		042000	ERRORCODE_PD_UNKNOWN_ERROR	Unknown error produced during a presence detector related operation.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Probe / BCR Motors (03) Probe 1 (04) Probe 2 (08) BCR	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY CANCEL WORKS (it does not show any further error. When queueing new samples, The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test is shown. In case of BCR motors errors, no error is shown upon cancel error.)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device.		
		032000	CURRENT_LIMIT	Current limit at servo device.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Probe / BCR Motors (03) Probe 1 (04) Probe 2 (08) BCR	033000	POSITION_ERROR	Position error at a servo device.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY CANCEL WORKS (it does not show any further error. When queueing new samples, The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test is shown. In case of BCR motors errors, no error is shown upon cancel error.)
		034000	MOVEMENT_ERROR	Movement error at a servo device.		
		035000	HOME_TIMEOUT	Home timeout at a servo device.		
		036000	ERRORCODE_SRV_SYNC_TIMEOUT	A synchronization error has been produced at a servo device.		
		037000	ERRORCODE_SRV_UNKNOWN_ERROR	An unknown error has been produced at a servo device.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	BCR (06) BCR 1 (07) BCR 2	003000	ERRORCODE_LEN	FW received incorrect length command.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY CANCEL WORKS (it does not show any further error. When queueing new samples, The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test is shown. In case of BCR motors errors, no error is shown upon cancel error.)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		011000	ERRORCODE_DATANOTREADY	The data requested is not ready.		
		048000	DRIVER_ERR	Barcode reader driver error.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	BCR (06) BCR 1 (07) BCR 2	050000	ERRORCODE_ BCR_COM_ERR	Unable to communicate with a barcode reader.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY CANCEL WORKS (it does not show any further error. When queueing new samples, The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test is shown. In case of BCR motors errors, no error is shown upon cancel error.)
		051000	ERRORCODE_ BCR_NO_ BARCODE	No barcode is read.		
		052000	ERRORCODE_ BCR_SYNC_ TIMEOUT	Timeout.		
		053000	ERRORCODE_ BCR_UNKNOWN_ ERROR	Unknown error.		
		049000	NOT_INIT	Unable to initialize barcode reader.	An error in a barcode reader has been detected. Please, RESTART the system.	OK

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Level detector (09) Probe 1 (10) Probe 2	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY
		005000	ERRORCODE_ARG	Command has wrong arguments.		RE-RUN
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		Open (option not available if error occurs in Dilution Cup)
		054000	ERRORCODE_LD_IMPRECISE_LEVEL	Level detector is unable to determine a tube or vial level.	Level detection error during pipetting. Please check there is enough volume and the absence of bubbles.	RETRY
		055000	ERRORCODE_LD_INSUF_LEVEL	Not enough level detected to perform an aspire command.		RE-RUN Open

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Level detector (09) Probe 1 (10) Probe 2	056000	ERRORCODE_LD_ DRIVER_ERR	Level detector driver error.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY RE-RUN Open (option not available if error occurs in Dilution Cup)
		057000	ERRORCODE_LD_ NOT_LEVEL	Not level detected in a tube or vial.	Level detection error during pipetting. Please check there is enough volume and the absence of bubbles.	RETRY RE-RUN Open

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Level detector (09) Probe 1 (10) Probe 2	058000	ERRORCODE_LD_ SYNC_TIMEOUT	Timeout.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY RE-RUN Open (option not available if error occurs in Dilution Cup)
		059000	ERRORCODE_LD_ IMPACT	Impact detected.	Probe of upper robotic arm has found an obstacle. Please check: - Samples and reagents suitability. - Samples and reagents racks positioning.	RETRY CANCEL Works Open
		060000	ERRORCODE_LD_ UNKNOWN_ ERROR	Unknown error.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY RE-RUN Open (option not available if error occurs in Dilution Cup)

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_ CMD	Unknown command code received.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	Incorrect length command received.		
		004000	ERRORCODE_ DEV	Command is addressed to an incorrect device.		
		005000	ERRORCODE_ ARG	Command has wrong arguments.		
		014000	ERRORCODE_ HW_DEV	Addressed device has a problem with its related HW.		
		015000	ERRORCODE_ HW_HWFAIL	There is a problem with PCA HW.		
		016000	ERRORCODE_ HW_HWNOTSET	PCA setup error.		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	000000	ERRORCODE_UNKNOWN	Unknown error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_NOTSAFE	Movement not safe		
		017000	ERRORCODE_FW_UNKNOWN_ERROR	Unknown error		
		061000	ERRORCODE_INVALID_ARGUMENTS	Invalid arguments		
		062000	ERRORCODE_FWCOMM_TIMEOUT	Communication timeout		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	063000	ERRORCODE_FWCOMM_CONNECTION_LOST	Communication lost	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		064000	ERRORCODE_FWCOMM_COMM_ERROR	Communication error		
		088000	ERRORCODE_PRECONDITION_FAILED	Previous condition for the action is not met		
		089000	ERRORCODE_STARTUP_ERROR	Driver startup error		
		091000	ERRORCODE_ALREADY_STARTED	Another instance of Driver is already opened		
0	240	090000	ERRORCODE_UNEXPECTED_ERROR	A Driver unexpected error has been produced	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Table 41 - PIP module troubleshooting

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / current limit error (X axis) 0801032000 0801034000	Fuse blown	Check / replace FUS31 and FUS32
	Large belt too tight	Check tightness of belt / Replace belt
	Incorrect X motor assembly	Check / replace X motor silentblocks
	Lack of lubrication in X linear guide	Lubricate linear guide
	X linear guide damaged	Check / replace linear guide
	Motor damaged	Check / replace motor
	PIP X-Y flat cable damaged	Check / replace PIP X-Y cable
Movement error / current limit error (Y axis) 0802032000 0802034000	Fuse blown	Check / replace FUS31 and FUS32
	Small belt too tight	Check tightness of belt / Replace belt
	Large belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in Y linear guide	Lubricate linear guide
	Y linear guide damaged	Check / replace linear guide
	Motor damaged	Check / replace motor
	PIP X-Y flat cable damaged	Check / replace PIP X-Y cable / X PIP chain

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / current limit error (Z1/Z2 axis) Z1: 0803032000 Z1: 0803034000 Z2: 0804032000 Z2: 0804034000	Fuse blown	Check / replace FUS31 and FUS32
	Belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in leadscrew	Lubricate leadscrew
	Leadscrew damaged	Check / replace leadscrew
	Motor damaged	Check / replace motor
	PIP-Z flat cable damaged	Check / replace X - Y PIP chain
Movement error / current limit error (Horns axis) 0808032000 0808034000	Fuse blown	Check / replace FUS31, FUS32, FUSGEN4
	Belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in leadscrew	Lubricate leadscrew
	Leadscrew damaged	Check / replace leadscrew
	Motor damaged	Check / replace motor
	PIP-TOW flat cable damaged	Check / replace X / Y PIP chain
X axis home is not detected 0801035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA GEN-CONX59
	PIP X-Y flat cable damaged	Check / replace PIP X-Y cable

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Y axis home is not detected 0802035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA PIP-CONY59
	PIP X-Y flat cable damaged	Check / replace X chain
Z1 / Z2 axis home is not detected Z1:0803035000 Z2: 0804035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA PIP-TOW59
	PIP-Z flat cable damaged	Check / replace X/ Y PIP chain
Horns axis home is not detected 0808035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA PIP-TOW59
	PIP-TOW flat cable damaged	Check / replace X / Y PIP chain

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Samples tubes/reagents vials are not detected / incorrectly detected	Fuse blown	Check / replace FUSGEN4
	Tube / vial is not placed correctly	Place correctly tube / vial in the rack
	Sample / reagent rack is not placed correctly over the drawer	Place correctly sample / reagent rack in the drawer
	Drawer is not flat	Check sample / reagent drawer flatness
	Adjustment of the PIP in the drawer is not correct	Check / perform PIP adjustment in the drawer
	Presence detector is not correctly assembled	Check presence detector assembly / PIP laser alignment
	Presence detector adjustment is not correct	Perform presence detector adjustment
	Presence detector damaged	Replace presence detector
	PCA PIP-TOW59 damaged	Replace PCA PIP-TOW59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Diameter of the samples tubes is not detected correctly	Sample tube is not placed correctly	Place correctly sample tube in the rack
	Sample rack is not placed correctly over the drawer	Place correctly sample rack in the drawer
	Sample tube shape provokes this effect	Check / set the erytra.properties configuration
	Drawer is not flat	Check sample / reagent drawer flatness
	Adjustment of the PIP in the drawer is not correct	Check / perform PIP adjustment in the drawer
	Presence detector is not correctly assembled	Check presence detector assembly
	Presence detector adjustment is not correct	Perform presence detector adjustment
	Presence detector damaged	Replace presence detector
	PCA PIP-TOW59 is damaged	Replace PCA PIP-TOW59
Presence detector is always ON	PCA PIP-TOW59 is damaged	Replace PCA PIP-TOW59
	Presence detector damaged	Replace presence detector

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
BCRs are not switched on BC1: 0806050000 BC1: 0806052000 BC1: 0806053000 BC1: 0806049000 BC2: 0807050000 BC2: 0807052000 BC2: 0807053000 BC2: 0807049000	Fuse blown	Check / replace FUS31
	BCRs cables damaged / disconnected	Check / replace BCRs cables
	BCRs are not working	Check / replace BCRs
	PCA PIP-BCR59 damaged	Check / replace PCA PIP-BCR59
	Horns chain damaged	Check / replace horns chain
	PCA PIP-TOW59 damaged	Replace PCA PIP-TOW59
	PIP-TOW / PIP-Z cable damaged	Check / replace X / Y PIP chains
Barcodes are not read BC1: 0806051000 BC2: 0807051000	Sample tube/vial is not placed correctly	Place correctly sample tube / vial in the rack
	Barcode label in bad conditions	Check barcode label
	Mirrors are dirty	Clean mirrors
	Drawer is not flat	Check drawer flatness
	Adjustment of the horns is not correct	Check / perform horns adjustment
	BCR is not properly assembled	Check the assembly of the BCR
	BCR is not working	Check / replace BCR
	BCR cable damaged / disconnected	Replace the BCR cable
	PCA PIP-TOW59 damaged	Replace PCA PIP-TOW59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Level is not detected properly with probe 1 or probe 2 Z1: 0809054000 Z1: 0809057000 Z2: 0810054000 Z2: 0810057000	Level Detection PCA damaged	Check / replace PCA PIP-P1OSC59 (probe 1) or PCA PIP-P2OSC59 (probe 2)
	Dirt in connection pad of level detection PCA	Clean connection pad of PCA PIP-P1OSC59 (probe 1) or PCA PIP-P2OSC59 (probe 2)
	Incorrect assembly between level detection PCA and probes holder	Check assembly between PCA PIP-P1OSC59 (probe 1) or PCA PIP-P2OSC59 (probe 2) and its probes holder
	Probes washer leakage	Check/clean/replace probes washer
		Check probes washer aspiration circuit
	Probes holder damaged	Check/replace probes holder
	Probe chain damaged	Check / replace probe chain
	Bad ground connection of level detection cables	Check ground connection of level detection cables
	Bubbles on the liquid	Ensure there are no bubbles on the liquid.
	Drops in the tube walls	Ensure there are no drops at the tube walls
Level detected too early	Bent probe	Replace probe
	Probably due to probe/s being wet.	Clean probe. Check for leakages
	Possible leak in probes washer	Check tubing and fittings

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Probe is not correctly placed over tube / vial / incubator / dilution cup	Reagent / Sample rack is not placed correctly over the drawer	Place correctly reagent / sample rack in the drawer
	Reagent vial / Sample tube is not placed correctly	Place correctly reagent vial / sample tube in the rack
	Probes are not correctly adjusted	Perform probes adjustment
Probe crashes with the bottom of sample tubes Z1: 0809059000 Z2: 0810059000	Rack not correctly placed	Place correctly reagent rack in the drawer
	Drawer is not flat	Check sample drawer flatness
	Probe adjustment is not performed correctly	Perform probe adjustment in sample drawers position
	Incorrect configuration	Check configuration defined in the erytra.properties file
Probe crashes with the bottom of reagent/diluent vials Z1: 0809059000 Z2:0810059000	Rack not correctly placed	Place correctly reagent rack in the drawer
	Rack platform is not correctly assembled	Check rack platform assembly
	Drawer is not flat	Check reagent drawer flatness
	Probe adjustment is not performed correctly	Perform probe adjustment in reagent drawers position
Horns crash with drawer during barcode reading	Horns are not correctly adjusted	Perform horns adjustment

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Liquid leaking on top of card foil	Some bubbles are considered normal due to pressurised cards.	Replace probe/s
	Incubator is not flat	Check incubator flatness
	Incubator height adjustment is not performed correctly	Perform incubator height adjustment
	Probe adjustment is not performed correctly	Perform probe adjustment in incubator position

2.5 Card transport module (CAT)

In this chapter, the Card transport robot arm module is described. It includes an overall description of the functions as well as the corresponding adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, and a troubleshooting section.

2.5.1 Description

The functions of this module are:

- Card inventory:
 - Card presence detection/location in drawers.
 - Card identification (type, expiration date, ...).
- Card transportation (card rack, incubator, centrifuge, ...).

To do so, this module consists of a three-axis robotic head. Card transportation is achieved by means of a gripper. This gripper is found in the robotic head and moves along the Z axis. The gripper includes a centering system to grab cards from the card holder that consists of an H-shaped retractile part. The cards are grabbed by two fingers that are opened/closed by means of an electromagnet. Four pins with springs contribute to card release/horizontal position. Card grabbing detection is also implemented by means of an optical barrier that detects one of the pins (up) position.

Card presence detection/location is achieved by means of a laser reflection sensor assembled in the robotic head. The robotic head scans the rack from left to right.

Card identification is achieved by means of a barcode reader assembled in the robotic head, and it is performed when the gripper raises the card to upper position. The barcode reader reference is Symbol SE-955 (S/N < 059-0001335) or Zebra SE-965 (S/N ≥ 059-0001335) and the communication is through RS-232.

The X-axis is moved by a servo-controlled motor assembled in the left area. This motor turns and provokes that a small belt moves. This belt transmits the movement by means of a gear to a large belt attached to the X wagon (assembled on a linear guide). The small belt tension is automatically set by assembly and silentblocks. The large belt tension is adjusted by modifying the X end pulley position. Home position is defined at the left and using an L-shaped part assembled in the X wagon that will get in an optical barrier. The CAT module and the PIP module are built in the same X-axis.

The Y-axis is moved by a servo-controlled motor assembled in the back axis area. This motor turns and provokes that a small belt moves. This belt transmits the movement by means of a gear to a large belt attached to the Y wagon (assembled on a linear guide). The small belt tension can be adjusted by modifying the motor position (slotted hole). The large belt tension is adjusted by modifying the Y end pulley position. Home position is defined at the back and using an L-shaped part assembled in the Y wagon that will get in an optical barrier.

The Z-axis is moved by a servo-controlled motor assembled in the robotic head bottom. This motor turns and provokes that a small belt moves. This belt provokes that a pulley fixed to a leadscrew with a set screw turns and this one that the gripper moves. A linear guide is also assembled to assure proper Z movement. The small belt tension can be adjusted by modifying the motor position (slotted hole). Home position is defined at the Z-axis top and using an L-shaped part assembled in the leadscrew nut that will get in an optical barrier.

Encoders resolutions for the CAT axes are listed below:

Table 42 - CAT axes encoder resolution

CAT Axis	Encoder counts / mm
X	285.714 counts/mm
Y	285.714 counts/mm
Z	1257.1 counts/mm

General block diagram of the module is shown in the following figure:

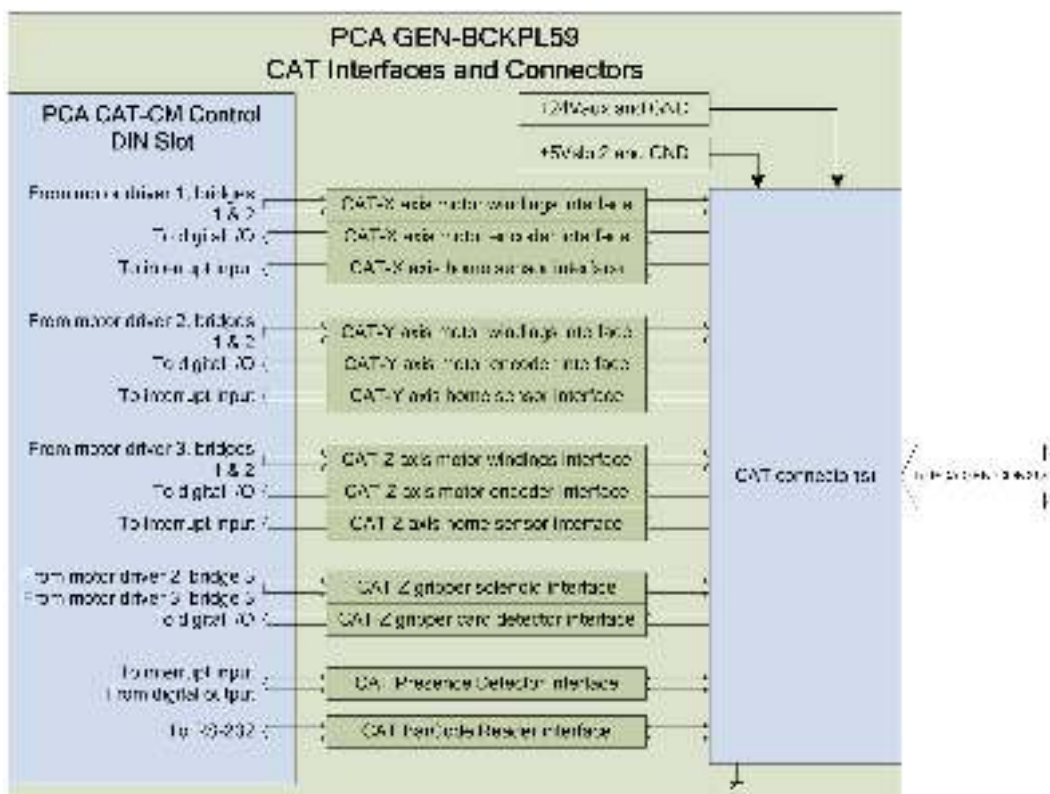


Figure 390 - CAT module block diagram

PCAs related to the CAT module are:

- **PCA GEN-BCKPL59:** main PCA of the instrument, located in the backpanel.
- **PCA CAT-CM Control:** located in the backpanel (slot 2), attached to the PCA GEN-BCKPL59.

- **PCA GEN-CONX59:** located at the X axis left side (looking from the front). It contains the X axis home optical barrier (power and signal), the X axis motor powering and encoder and the three ribbon cables (intermediate) connectors: CAT XY (X and Y motor power and encoder and homes), CATZ (Z motor powering and encoder, electromagnet activation and card presence optical barrier powering and signal) and CAT-TOW (card presence detector and barcode reader signals).
- **PCA CAT-CONY59:** located at the Y axis back (looking from the front). It contains the Y axis home optical barrier (power and signal), the Y axis motor powering and encoder and two ribbon cables (intermediate) connection: CATZ (Z motor powering and encoder, electromagnet activation and card presence optical barrier powering and signal) and CAT-TOW (Presence detector and barcode reader signals). CATXY ribbon cable transmits the Y axis home sensor powering and signal and the Y motor powering and encoder signal.
- **PCA CAT-TOW59:** located inside the robotic head. It contains the Z/gripper axis home sensor, Z motor powering and encoder and card presence detector and barcode reader signals.
- **PCA PINZA57:** located at the top of the gripper. It contains the card presence detector and electromagnet powering (open/close fingers).

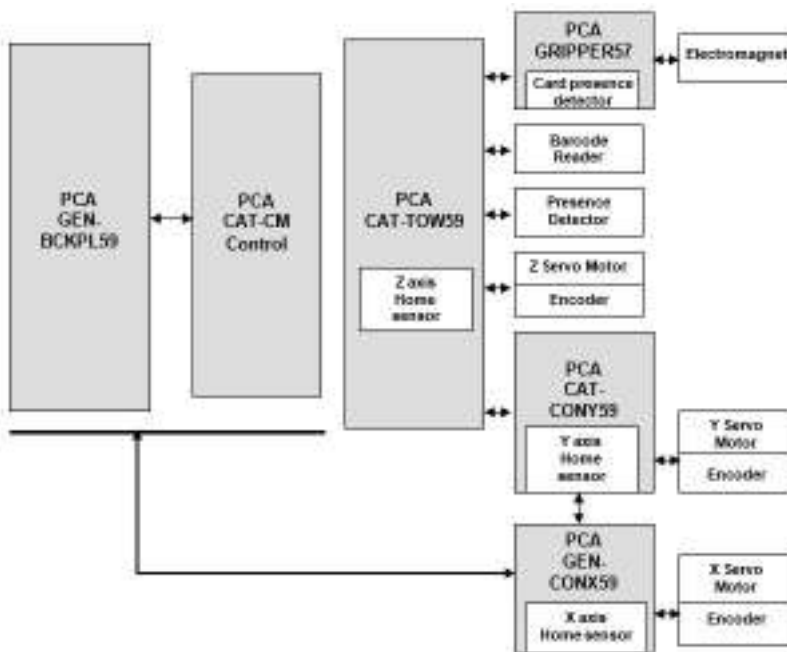


Figure 391 - Main blocks of the CAT module

Barcode reader communication is performed through RS-232.

2.5.2 Mechanical adjustments

Mechanical adjustments related to the CAT module are detailed in this section.

2.5.2.1 Adjusting X-axis small belt tension

No manual X axis belt tension adjustment is required. The assembly and silentblocks ensure proper belt tension.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

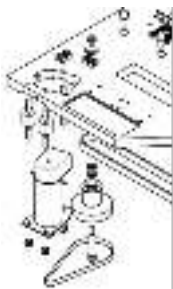


Figure 392 - X-axis motor small belt

2.5.2.2 Adjusting X-axis large belt tension

This adjustment is performed to adjust the tension of the CAT arm X-axis large belt.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Screw the pulley tightening the screw to allow pulley movement.
2. Loosen both pulley fixing screws.
3. Tighten the belt by unscrewing the pulley tightening the screw, and fix its position by tightening both fixing screws.

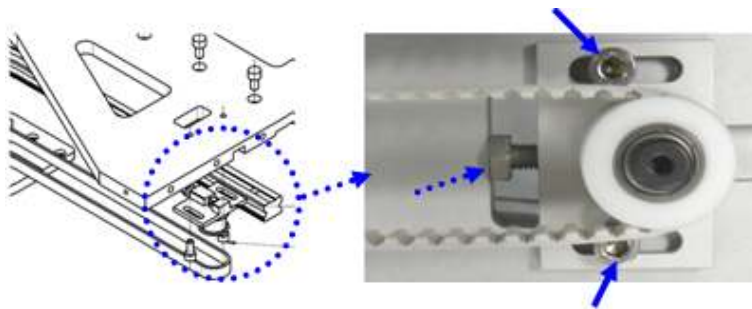


Figure 393 - X-axis motor large belt

2.5.2.3 Adjusting Y-axis small belt tension

This adjustment is performed to adjust the tension of the small belt of the Y-axis of the CAT arm.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the Y arm motor cover (remove two fixing screws).
2. Unplug the Y arm ribbon cables from PCA CAT-CONY59.
3. Remove the screws fixing the Y motor-belt assembly to the Y arm and remove the assembly.
4. Loosen the Y motor slotted hole screw.



Figure 394 - Y-axis motor small belt

5. Move the Y motor to adjust the belt tension.
6. Fix the motor position by tightening the slotted hole screw.
7. Assemble the Y motor-belt assembly to the Y arm by tightening the tree screws.
8. Connect the Y-arm ribbon cables to PCA CAT-CONY59.
9. Assemble the Y motor cover.

2.5.2.4 Adjusting Y-axis large belt tension

This adjustment is performed to adjust the tension of the large belt of the Y-axis of the CAT arm.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Screw the pulley tightening the screw to allow pulley movement.
2. Loosen both pulley fixing screws.
3. Tighten the belt by unscrewing the pulley tightening screw, and fix its position by tightening both fixing screws.

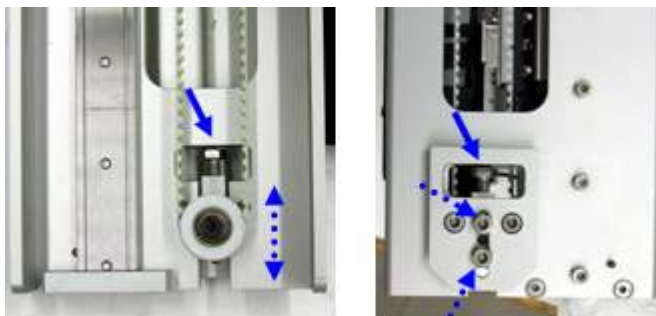


Figure 395 - Y-axis motor large belt



NOTE

Excessively low belt tension can lead to an offset from the intended position in the CAT Z-axis.

2.5.2.5 Adjusting Z-axis gripper belt tension

This adjustment is performed to adjust the tension of the gripper belt.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Loosen the gripper motor screws.
2. Move the gripper motor to adjust the gripper belt tension.
3. Set the gripper motor position/belt tension by tightening both gripper motor screws.

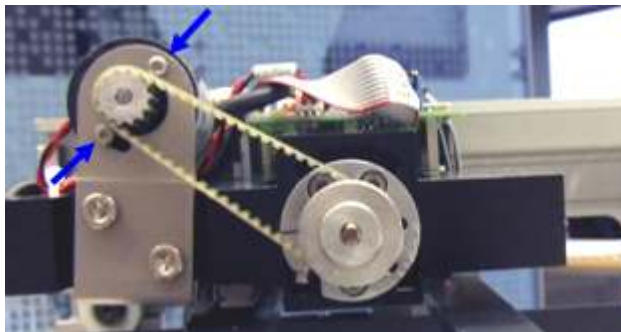


Figure 396 - Z-axis motor belt

2.5.2.6 Adjusting gripper parallelism

This adjustment must be performed when:

- The gripper (whole) is disassembled.
- The robotic head is disassembled.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Unlock the CAT robotic head by using the **unlock/stop** button in the **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Position Actions** area and selecting all axes.
2. Open card drawer 2.
3. Lower the gripper until the gripper edges gently touch the centrifuge cover surface.



Figure 397 - Gripper parallelism adjustment

4. Check that the gripper edges (left or right) touch the centrifuge cover surface at the same time. Should this not be the case, loosen the four screws that fix the gripper, align the gripper, and tighten the screws again.
5. Apply a thread locking adhesive (such as Loctite 222) to the four screws that fix the gripper.



NOTE

No more than one screw should be removed at a time when applying the sealant. Remove the first screw, apply the adhesive, screw it back in and continue with the next screw.

6. Verify the adjustment (See "Verifying gripper parallelism" chapter), if desired.

2.5.2.7 Adjusting the inclination of card presence detector

This adjustment must be performed when the card presence detector is disassembled / replaced.

The presence detector inclination is set using two points: a screw and a presence detector pin. Inclination is assured by assembly if the presence detector pin is correctly inserted while screwing. The presence detector bottom is aligned with the holder bottom.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Loosen the screw attaching the presence detector to the presence detector holder.
2. Move the presence detector until the pin is properly inserted in the detector holder and the detector bottom is aligned with the detector holder.

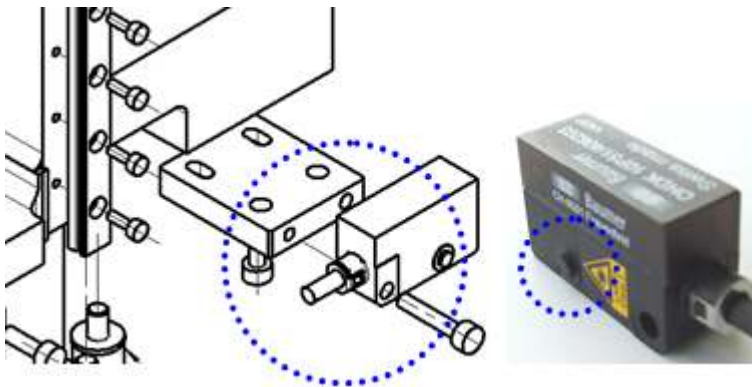


Figure 398 - Card presence detector inclination

3. Tighten the screw.
4. Verify the adjustment (See "Verifying card presence detector inclination" chapter), if desired.

2.5.3 Functional adjustments

In this section, the functional adjustments related to the Card transport module are explained.

All the adjustments determining a given set of coordinates are stored in the **CATCalibrations.txt** file found in **E:\data\Erytra\conf**.

It is assumed in all the adjustment procedures that the proper conditions for movement are met, that the gripper arm is homed and their movements can be completed without risk of collision.

To home the gripper arm, click the **Home** button in the **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) and perform a home of all gripper arm axes.

**CAUTION**

Check the incubators position to avoid them interfering with the CAT movement and leading to a crash.

2.5.3.1 Adjusting gripper in the card drawers area

This adjustment will allow the software to determine coordinates of two card drawer known positions by attaching the card drawer adjustment tool to these two drawer positions and inserting the gripper into the tool.

This adjustment must be performed when:

- The gripper is disassembled.
- The CAT robotic head is disassembled.
- The position of the latch of the card drawer is modified.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- 233878 CARDS DRAWER ADJUSTMENT TOOL (Ver. 2).

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- CAT module homed.
- Gripper parallelism verification. See "Verifying gripper parallelism" on page 490.
- Card drawer flatness verification. See "Verifying the flatness of card drawer" on page 189.

Procedure

1. Click the **drawers** button found in the Technical Service software **Procedures** tab > **CAT floor Modules** section. See "Procedures tab" on page 756. The following window is displayed.



Figure 399 - Card drawer selection in the Technical Service software

2. Select a card drawer to adjust and click **Next**.
3. Insert the cards drawer adjustment tool in rear rack position.
4. Move the gripper manually or using the cursor buttons to fit it with adjustment tool (pins inserted, gripper bottom lays on adjustment tool). When placing the gripper over the tool, gripper fingers are not involved so they do not grip the tool. When placing the gripper over the tool, be careful to not raise robot (the gripper just rests over the tool).

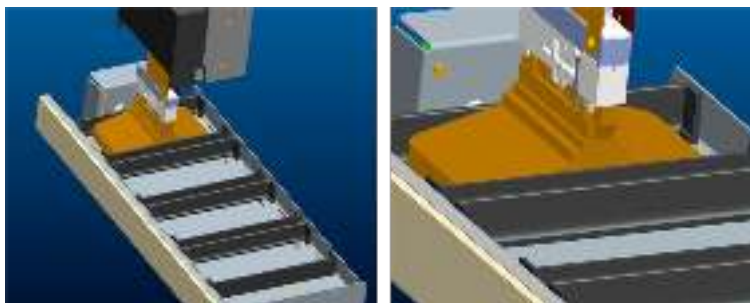


Figure 400 - Rear position of card drawer adjustment tool in the drawer

5. Start the automatic adjustment by pressing the **Next** button on the screen. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button. The automatic adjustment is as follows:
 - a. The gripper moves in the Z axis until the position is detected where the gripper stops and no movement is detected (in encoder counts) if the gripper is asked to move further down. The system stores this first Z adjustment in order to allow the system to correctly adjust the X and Y axes.
 - b. Then the gripper is moved in the X and Y axes until the position is detected where the affected axis stops when the gripper is discentered and then moved down to fit the adjustment tool.
 - c. Finally, the Z axis is adjusted automatically again using the X and Y values from the previous step.
6. The system repeats the previous procedure at least twice (maximum six executions). The obtained values are compared, and if they are smaller than the established threshold, it is considered that the adjustment is completed. Otherwise the procedure is repeated and the results are compared with the previous execution.
7. Once the process finishes, the gripper performs a home in the Z axis. Remove the cards drawer adjustment tool from the rear position and place it in the front rack position.
8. Move the gripper manually or using the cursor buttons to fit it with the adjustment tool (pins inserted, gripper bottom lays on adjustment tool). When placing the gripper over the tool, gripper fingers are not involved so they do not grip the tool. When placing the gripper over the tool, be careful to not raise the robot (the gripper just rests over the tool).

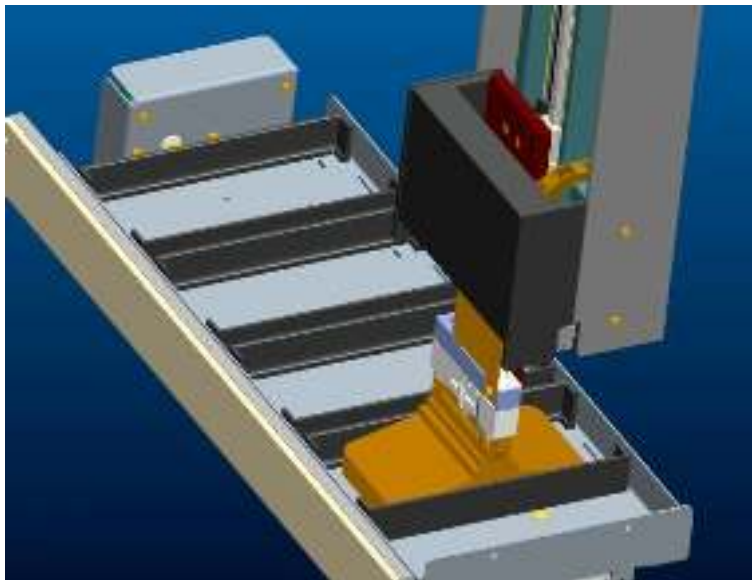


Figure 401 - Front position of the cards drawer adjustment tool in the drawer

9. Start the automatic adjustment by pressing the **Next** button on the screen. The automatic adjustment is performed again.

10. Once the process is finished, a screen appears showing the parameters to be stored. Press the **Finish** button and press **OK** in the confirmation window. When the adjustments are stored, press **OK** in the information window.
11. Press **OK** in the confirmation window that indicates that home movements will be performed. The gripper will perform a home.
12. Press **OK** to finish the adjustment procedure.
13. Verify the adjustment (See "Verifying card drawers area" chapter), if desired.

**NOTE**

The adjustment is stored in **E:\data\Erytra\conf\CATCallibrations.txt**.

**CAUTION**

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

2.5.3.2 Adjusting gripper in the incubator area

This adjustment allows the software to determine the coordinates of a known position by inserting the tool to the incubator slot 6 and matching the gripper with the adjustment tool.

This adjustment must be performed when:

- The gripper is disassembled.
- The CAT robotic head is disassembled.
- The Incubator module/block is disassembled.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- 733068 LOWER CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL v3.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- CAT module homed.
- Gripper parallelism verification. See "Verifying gripper parallelism" on page 490.
- Incubator flatness verification. See "Verifying incubator flatness" on page 269.
- Incubator homed.

Procedure

1. Click the **incubators** button found in the **Technical Service > Procedures** tab > **CAT floor Modules** section. See "Procedures tab" on page 756. The following window is displayed:



Figure 402 - Incubator adjustment selection in the Technical Service software

2. Select the incubator to adjust and click the **Next** button. A warning message informing that the instrument will perform a home of the selected incubator appears on the screen. Press the **OK** button and then the **Next** button.



CAUTION

Check the CAT arm position to avoid it interfering with the incubator movement and leading to a crash.

3. Insert the incubator section of the centrifuge/incubator adjustment tool in incubator position 6.

4. Move the gripper manually or using the cursor buttons to fit it with the adjustment tool (pins inserted, gripper bottom lays on adjustment tool. The tool has been designed so gripper fingers are not used). Be careful not to raise the robot.

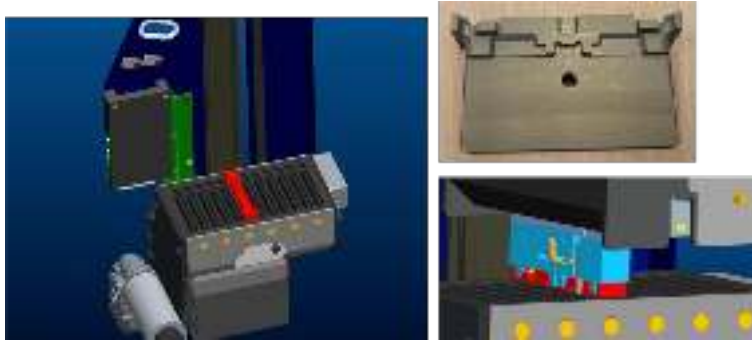


Figure 403 - Incubator adjustment in the Technical Service software

5. Start the automatic adjustment by pressing the **Next** button on the screen. The automatic adjustment is as follows:
 - a. The gripper moves in the Z axis until the position is detected where the gripper stops and no movement is detected (in encoder counts) if the gripper is asked to move further down. The system stores this first Z adjustment in order to allow the system to correctly adjust the X and Y axes.
 - b. Then the gripper is moved in the X and Y axes until the position is detected where the affected axis stops when the gripper is discentered and then moved down to fit the adjustment tool.
 - c. Finally, the Z axis is adjusted automatically again using the X and Y values from the previous step.
6. The system repeats the previous procedure at least twice (maximum six executions). The obtained values are compared and, if they are smaller than the established threshold, it is considered that the adjustment is completed. Otherwise the procedure is repeated and the results are compared with the previous execution.
7. Once the process is finished, a screen appears showing the parameters to be stored. Press the **Finish** button and press **OK** in the confirmation window. When the adjustments are stored, press **OK** in the information window.
8. Press **OK** in the confirmation window that indicates that home movements will be performed. The gripper will perform a home.
9. Press **OK** to finish the adjustment procedure.
10. Verify the adjustment (See "Verifying incubators area" chapter), if desired.



NOTE

The adjustment is stored in **E:\data\Erytra\conf\CATCalibrations.txt**.



CAUTION

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

2.5.3.3 Adjusting gripper in the card waste container area

This adjustment allows the software to determine coordinates of the card waste container once the gripper has been placed in a known position.

This adjustment must be performed when:

- The gripper is disassembled.
- The CAT robotic head is disassembled.
- The card disposal guide/nozzle is disassembled.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- CAT module homed.
- Gripper parallelism verification. See "Verifying gripper parallelism" on page 490.

Procedure

1. Click the **waste tray** button found in the Technical Service **Procedures** tab > **CAT floor Modules** section. See "Procedures tab" on page 756. It will open the wizard adjustment. Press the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button.

2. Move the gripper manually or using the keypad, following the steps listed below:
 - a. Move the gripper arm so that the rear part of the gripper leans against the top surface of the rear wall of the waste tray.
 - b. Move the gripper along the X axis as far to the right as possible, until it touches the card waste right wall.



Figure 404 - Gripper Adjustment in Card waste container (removed cover)

3. Press the **Next** button. A screen appears showing the parameters to be stored. Press the **Finish** button and press **OK** in the confirmation window. When the adjustments are stored, press **OK** in the information window.
4. Press **OK** in the confirmation window that indicates that home movements will be performed. The gripper will perform a home.
5. Press **OK** to finish the adjustment procedure.
6. Verify the adjustment (See "Verifying card waste container area" chapter), if desired.



NOTE

The adjustment is stored in **E:\data\Erytra\conf\CATCalibrations.txt**.

2.5.3.4 Adjusting gripper in the centrifuge area

This adjustment allows the software to determine coordinates of centrifuge position 1 (gripper and centrifuge).

This adjustment must be performed when:

- The gripper is disassembled.
- The CAT robotic head is disassembled.
- The centrifuge is disassembled.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 733056 CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL (Ver. 3).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- CAT module homed.
- Centrifuges homed.
- Gripper parallelism verification. See "Verifying gripper parallelism" on page 490.
- Centrifuge rotor height adjustment. See "Adjusting rotor height" on page 297.
- Centrifuge with all the holders.

Procedure

1. Click the **centrifuges** button found in the Technical Service software **Procedures** tab (see "Procedures tab" on page 756) > **CAT floor Modules** section.

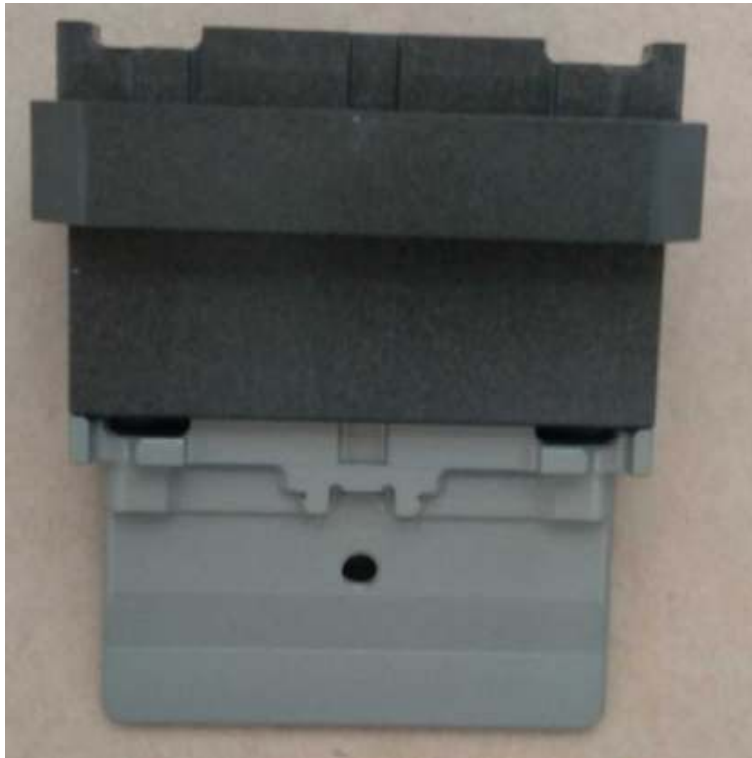


Figure 405 - Centrifuge / incubator adjustment tool



Figure 406 - Centrifuge adjustment tab in the Technical Service software

2. Select a centrifuge to adjust and click **Next**. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button. The centrifuge will perform a home.
3. The centrifuge will move slot 1 to load/unload position. Insert manually the whole/assembled centrifuge/incubator adjustment tool set in the more centred slot.

4. Move the gripper manually or using the cursor buttons to fit it with the adjustment tool. When placing the gripper over the tool, gripper fingers are not involved so they do not grip the tool. Click the **Next** button.

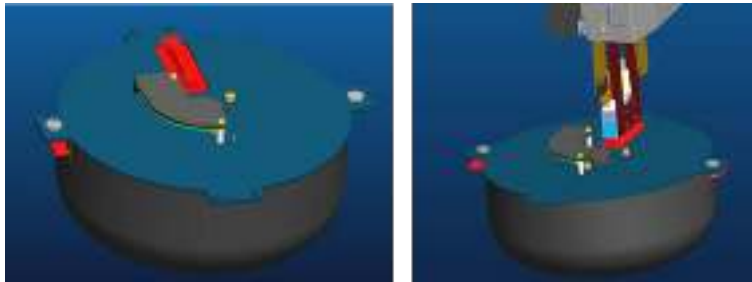


Figure 407 - Gripper wizard adjustment in centrifuge in the Technical Service software

5. A screen appears showing the parameters to be stored. Press the **Finish** button and press **OK** in the confirmation window. When the adjustments are stored, press **OK** in the information window.
6. Press **OK** in the confirmation window that indicates that home movements will be performed. The gripper will perform a home.
7. Press **OK** to finish the adjustment procedure.
8. Verify the adjustment (See "Verifying centrifuges area" chapter), if desired.



NOTE

The adjustment is stored in `E:\data\Erytra\conf\CATCalibrations.txt`.



CAUTION

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

2.5.3.5 Adjusting card presence detector-gripper parallelism



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233878 CARDS DRAWER ADJUSTMENT TOOL (Ver. 2).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Gripper parallelism verification. See "Verifying gripper parallelism" on page 490.

Procedure

1. Insert the cards drawer adjustment tool in the front position of any card drawer.

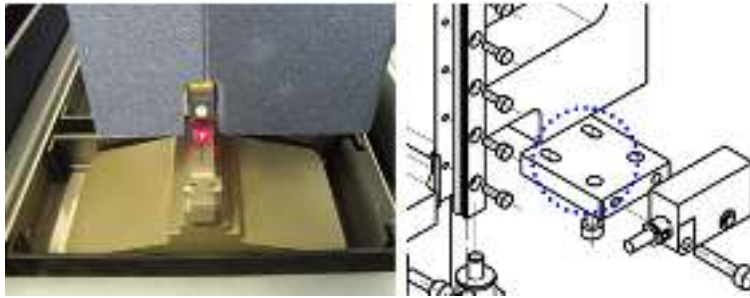


Figure 408 - Card presence detector parallelism adjustment

2. Move the gripper manually to fit it with the adjustment tool (pins inserted, gripper bottom lays on adjustment tool).
3. Perform a home of the Z axis in the Technical Service software, energize all the CAT axes and then deenergize just the Y axis.
4. Move manually the robotic head in the Y axis so presence detector light spot illuminates (click the **set on** button in the **Gripper Arm** tab > **Presence detector** subtab (see "Presence Detector subtab" on page 782) > **presence detector** area) the section of cards drawer adjustment tool that contains the central slot.
5. Loosen the two screw fixing the presence detector holder to the Gripper robotic head and move the holder backwards/forward until the presence detector beam spot illuminates the card drawer gripper adjustment tool central slot.
6. Tighten the two screws fixing the presence detector holder.



CAUTION

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

2.5.3.6 Adjusting card presence detector height detection

This adjustment must be performed when:

- Card presence detector is disassembled.
- CAT robotic head is disassembled.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233878 CARDS DRAWER ADJUSTMENT TOOL (Ver. 2).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Card presence detection inclination adjustment. See "Adjusting the inclination of card presence detector" on page 470.

Procedure



NOTE

The card presence detector device comprises two LEDs:

- Green LED should always be ON as long as the drivers are started, indicating that the presence detector device is activated.
- Orange LED will only light (ON mode) when presence is detected.

1. Insert the cards drawer gripper adjustment tool in the front position of any card drawer.
2. Activate the presence detector by clicking the **set on** button in **Presence detector** area of **Presence detector** subtab (see "Presence Detector subtab" on page 782) of the **Gripper Arm** tab.
3. Access the presence detector screw and turn it slowly clockwise (remove the sealant if any) until a 'click' is heard.

4. De-energize the gripper arm (**Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Position Actions** area > **unlock/stop** button and select all axes) and move it so that the presence detector laser beam spot illuminates adjustment area 1 (see picture below).

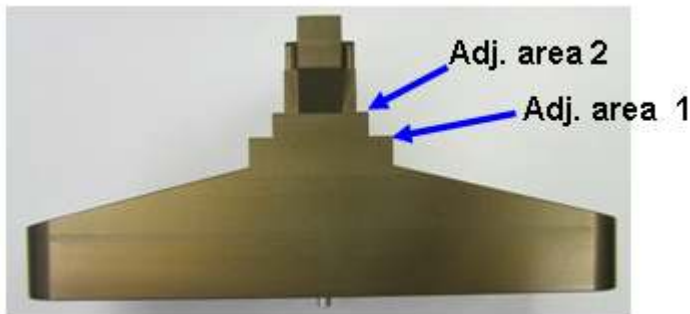


Figure 409 - Card presence detector height adjustment areas

5. Turn slowly the presence detector screw counter-clockwise until the orange LED is permanently OFF (presence is not detected).
6. Move the gripper arm so that the presence detector laser beam illuminates adjustment area 2, checking that presence is detected (orange LED lights uniformly and no blinking is observed). Otherwise, go back to step 3 to start over with the adjustment procedure.

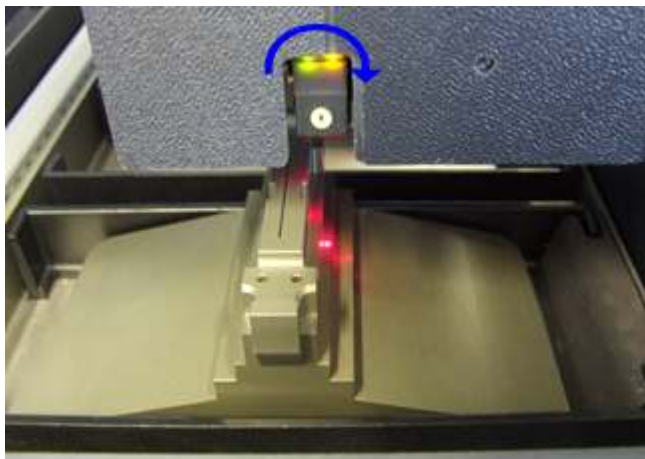


Figure 410 - Presence detector over adjustment area 2

7. Move alternately the CAT's laser beam between adjustment areas 1 and 2, checking that the orange LED is OFF over adjustment area 1 and ON over adjustment area 2.
8. Deactivate the presence detector by clicking the **set off** button in the **Presence detector** area of the **Presence detector** subtab (see "Presence Detector subtab" on page 782) of the **Gripper Arm** tab.
9. Consider applying a sealant to the potentiometer.
10. Verify the adjustment (See "Verifying card presence detector height detection" chapter), if desired.

**CAUTION**

Remove the adjustment tool from the instrument once the adjustment procedure is finished.

2.5.3.7 Adjusting card barcode reader inclination

This adjustment must be performed when:

- The card barcode reader is disassembled.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the right CAT robotic head cover (fixed by six screws, three per side).
2. Detach the barcode reader assembly (removing carefully the left CAT robotic head cover) to gain access to the barcode reader. Keep the barcode reader connected to PCA CAT-TOW59.
3. Activate the barcode reader: in the Technical Service software, click the **set on** button in the **Gripper Arm** tab > **BCR** subtab. See "BCR subtab" on page 785.

4. Loosen the two screws fixing the barcode reader position and move the barcode reader so the laser beam is focused on the center of the reflection mirror.

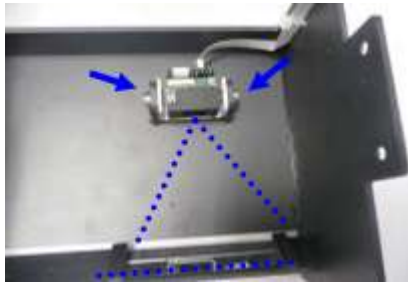


Table 43 - Barcode reader inclination adjustment

5. Tighten the two screws and deactivate the barcode reader.
6. Assemble carefully the left CAT robotic head cover and then the right CAT robotic head cover.
7. Verify the adjustment (See "Verifying card barcode reader inclination" chapter), if desired.

2.5.3.8 Adjusting gripper in the card reader area

This adjustment must be performed when:

- The gripper is disassembled.
- The reader module is disassembled.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 731398 READER ADJUSTMENT TOOL-PATTERN v2.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- CAT homed.
- Incubators homed.
- Gripper parallelism verification. See "Verifying gripper parallelism" on page 490.
- Reader adjusted.

Procedure

1. Select the **Procedures** tab in the Technical Service software and in the **CAT floor Modules** section, click the **acquisition (central) position** button. See "Procedures tab" on page 756. The following window will be displayed.

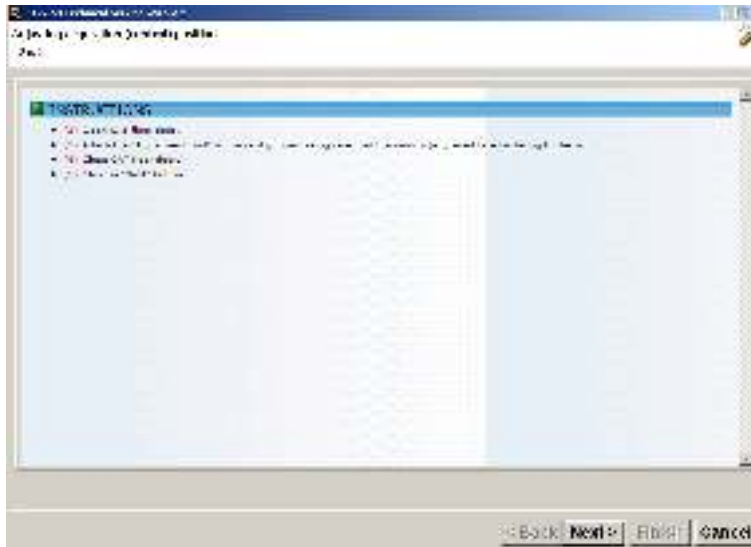


Figure 411 - Acquisition (central) position wizard - screen 1

2. De-energize the gripper arm motors by clicking the **unlock/stop** button in the **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Position Actions** and selecting **all axes**.
3. Open the CAT floor door (opening first the user tray and then removing the screw fixing the intermediate door).

4. Attach manually the reader adjustment tool-pattern to the gripper as indicated in the picture (pattern side looking at the left, screws heads looking at right).

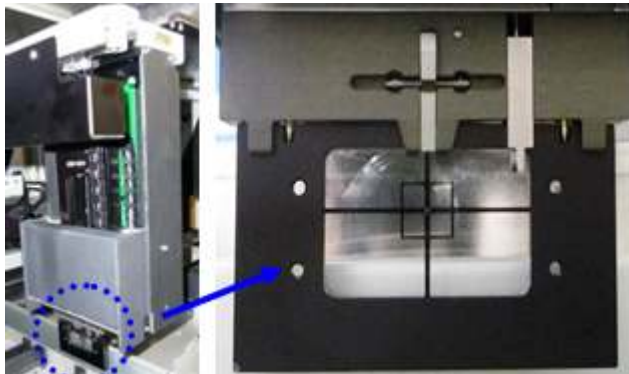


Figure 412 - Gripper adjustment in card reader area



CAUTION

In the next step, the robot head will move without performing automatically a gripper home.

5. Perform a home of the gripper (**Gripper Arm** tab > **home** button > **z axis**). Click the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button. The adjustment is automatic and consists in determining gripper coordinates that set the pattern in the desired position (current pixels / mm conversion factor matches the adjustment one). Wait for the adjustment completion.
6. At the adjustment completion, the following window appears:



Figure 413 - Acquisition (central) position wizard - screen 2

7. Remove the reader adjustment tool-pattern from the gripper by activating the electromagnet (manually or using the Technical Service software). Click the **Next** button.

8. A screen appears showing the parameters to be stored. Press the **Finish** button and press **OK** on the confirmation window. When the adjustments are stored, press **OK** on the information window.
9. Press **OK** in the confirmation window that indicates that home movements will be performed.
10. Press **OK** to finish the adjustment procedure.
11. Verify the adjustment (See "Verifying reader area" chapter), if desired.

**NOTE**

The adjustment is stored in **E:\data\Erytra\conf\CATCalibrations.txt**.

2.5.4 Verification procedures

It is assumed in all the verification procedures that the proper conditions for movement are met, that is the gripper arm is homed and their movements can be completed without risk of collision.

To home the gripper arm, click the **home** button in **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) and perform a home of all gripper arm axes.

**CAUTION**

Check the incubators position to avoid them interfering with the CAT movement and leading to a crash.

2.5.4.1 Verifying gripper parallelism

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. De-energize the CAT robotic head by using the **unlock/stop** button in the **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Move Actions** and selecting all axes.
2. Open card drawer 2.
3. Lower the gripper until the gripper edges (left or right) gently touch the centrifuge cover surface. Check that the gripper edges (left or right) touch the centrifuge cover surface at the same time.



Figure 414 - Gripper parallelism verification

4. Should this not be the case, loosen the four screws that fix the gripper (see picture below), align the gripper and tighten the screws again.

- Once the alignment is correct, a thread locking adhesive (such as Loctite 222) must be applied to the for screws.

**NOTE**

No more than one screw should be removed at a time when applying the sealant. Remove the first screw, apply the adhesive, screw it back in and continue with the next screw.



Figure 415 - Four screws fixing gripper

2.5.4.2 Verifying card drawers area

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Insert a card rack with all the cards in the lower floor in the selected card drawer rack 1 position.
2. Scan the selected card drawer - rack 1 (**Gripper Arm** tab > **Presence Detector** subtab). See "Presence Detector subtab" on page 782.
3. Select the desired card drawer, rack 1 and card 1 in the **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Move Actions** fields and click the **move to** button. Press **Yes** in the confirmation window.
4. Click the **get card** button to take the card.
5. Select **CAMERA, READ and WELLS_1_2** in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** section and click the **Move to** button. Press **Yes** in confirmation window.
6. Lower gripper selecting down button in the **Gripper Arm** tab > **Movement** subtab > **Position Actions** area.
7. Click on the real time image button in the **Procedures** tab (see "Procedures tab" on page 756) > **Checks** section > **Reader Module** area to obtain the camera image.



Figure 416 - Gripper adjustment in card drawers verification

8. Check that the gripper pin is centered in the midpoint between well 1 and 2. Should this not be the case, readjust the gripper in that card drawer. Press **Cancel** in the real time image window to close it and then **OK** in the confirmation window.
9. Remove the card from the reader by selecting the **up** button in **Gripper Arm** tab > **Movement** subtab > **Position Actions** area.

10. Bring the card back to the rack by selecting the desired card drawer, rack 1 and card 1 in the **Gripper Arm** tab > **Movement** subtab > **move Actions** field, clicking the **move to** button, **Yes** in the confirmation window, and then clicking the **leave card** button.
11. Move the card rack to rack 4 position of the selected drawer and repeat steps 2 to 10, selecting rack 4 instead of rack 1 and card 23 instead of card 1.

2.5.4.3 Verifying incubators area



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Select the **Incubators** tab and the subtab related to the incubator area to check. See "Incubator 1, 2, 3 and 4 subtabs" on page 760.
2. Select the **Home** button in **Movement Area**.



CAUTION

Check the CAT arm position to avoid it interfering with the incubator movement and leading to a crash.

3. Insert a card rack with all the cards in the lower floor in the selected card drawer rack 1 position.
4. Scan the card drawer 1 - rack 1 (**Gripper Arm** tab > **Presence Detector** subtab). See "Presence Detector subtab" on page 782.
5. Take a card from the rack selecting card drawer 1, rack 1 and desired card position in the **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then clicking the **get card** button.

6. Insert the card in the incubator slot 1, selecting the incubator to check, and in slot 1 in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then clicking the **leave card** button. Check that the card is properly left.
7. Take another card from the rack, selecting card drawer 1, rack 1, and selecting the desired card position in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then clicking the **get card** button.
8. Insert the card in the incubator slot 12, selecting the incubator to check, and in slot 12 in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then clicking the **leave card** button. Check that the card is properly left.
9. Select the incubator to check and slot 1 in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields and click the **move to** button. Press **Yes** in the confirmation window.
10. Click the **get card** button to grip the card.
11. Select **CAMERA, READ and WELLS_1_2** in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields and click the **move to** button. Press **Yes** in the confirmation window.
12. Lower the gripper by selecting the **down** button in the **Gripper Arm** tab > **Movement** subtab > **Position Actions** area.
13. Click the **real time image** button in the **Procedures** tab (see "Procedures tab" on page 756) > **Checks** section > **Reader Module** area to obtain the camera image.



Figure 417 - Gripper adjustment in incubator verification

14. Check that the gripper pin is centered in the midpoint between well 1 and 2. Press **Cancel** in the real time image window to close it and then **OK** in the confirmation window.
15. Remove the card from the reader selecting the **up** button in the **Gripper Arm** tab > **Movement** subtab > **Position Actions** area.
16. Bring the card back to the rack by selecting card drawer 1, rack 1, and selecting the desired card position in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then the **leave card** button.

17. Repeat steps 9 to 16 taking the card from incubator slot 12 instead of 1, and leaving the card in a free card rack position.

2.5.4.4 Verifying card waste container area



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Insert a card rack with all the cards in the lower floor in the selected card drawer rack 1 position.
2. Scan the desired rack of the card drawer (**Gripper Arm** tab > **Presence Detector** subtab). See "Presence Detector subtab" on page 782.
3. Use the robot to get and dispose of the cards in the waste container position and check proper performance:
 - a. Select the desired card drawer, rack position and card position and click **move to**. Press **Yes** in the confirmation window.
 - b. Click the **get card** button to take the card.
 - c. Select the **WASTE_TRAY** position in the **Movement** area and click the **move to** button. Press **Yes** in the confirmation window.
 - d. Dispose of the card by clicking the **leave card** button.
 - e. Repeat the previous steps until all cards have been disposed of.

2.5.4.5 Verifying centrifuges area



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **home** button in the **Centrifuges** tab > **Centrifuge 1/2** subtab (see "Centrifuge 1 and 2 subtabs" on page 767) and perform a home of the centrifuge.
2. Insert a card rack with all the cards in the lower floor in the selected card drawer rack 1 position.
3. Scan the card drawer 1 - rack 4 (**Gripper Arm** tab > **Presence Detector** subtab). See "Presence Detector subtab" on page 782.
4. Take a card from the rack by selecting card drawer 1, rack 4, and selecting a desired card position in the **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then clicking the **get card** button.
5. Insert the card in centrifuge slot 1 by selecting the centrifuge to check and slot 1 in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then clicking the **leave card** button. Check that card is properly left.
6. Move centrifuge slot 6 to the loading/unloading position by selecting the centrifuge to check and slot 6 in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields and clicking the **move to** button. Press **Yes** in the confirmation window.
7. Remove the card from the centrifuge slot 1 by selecting the centrifuge to check and slot 1 in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then clicking the **get card** button.

8. Move the card to the reader position by selecting '**Camera**', '**Read**' and '**Wells 1_2**' in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and lower gripper selecting the **down** button in **Gripper Arm** tab > **Movement** subtab > **Position Actions** area.
9. Click the **real time image** button in the **Procedures** tab (see "Procedures tab" on page 756) > **Checks** section > **Reader Module** area to obtain the camera image.
10. Check that the gripper pin is centered in the midpoint between well 1 and 2. Press **Cancel** in the real time image window to close it and then **OK** in the confirmation window.



Figure 418 - Gripper adjustment in centrifuge verification

11. Remove the card from the reader by selecting the **up** button in the **Gripper Arm** tab > **Movement** subtab > **Position Actions** area.
12. Bring the card back to the rack by selecting card drawer 1, rack 4 and the desired card position in the **Gripper Arm** tab > **Movement** subtab > **Move Actions** fields, clicking the **move to** button, pressing **Yes** in the confirmation window and then the **leave card** button.

2.5.4.6 Verifying card presence detector inclination



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

- Check that the presence detector bottom and the presence detector holder bottom are aligned.

2.5.4.7 Verifying card presence detector-gripper parallelism



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233878 CARDS DRAWER ADJUSTMENT TOOL (Ver. 2).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Insert the cards drawer gripper adjustment tool in the front position of any card drawer.

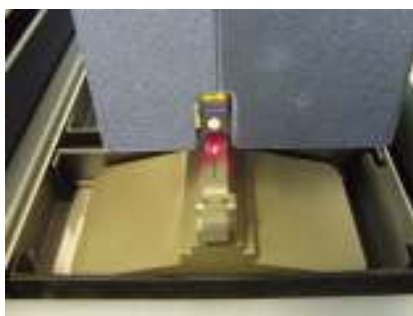


Figure 419 - Card presence detector parallelism checking

2. Move the gripper manually to fit it with the adjustment tool (pins inserted, gripper bottom lays on adjustment tool).

3. Perform a home of the Z axis in the Technical Service software, energize all the CAT axes and then deenergize just the Y axis. See "Movement subtab" on page 777.
4. Move manually the robotic head in the Y axis so the presence detector light spot illuminates (click the **set on** button in the **Gripper Arm** tab > **Presence detector** subtab (see "Presence Detector subtab" on page 782) > **presence detector** area) the section of the cards drawer adjustment tool that contains the central slot.
5. Check that the presence detector beam spot illuminates the card drawer gripper adjustment tool central slot.

**CAUTION**

Remove the adjustment tool from the instrument when the verification procedure finishes.

2.5.4.8 Verifying card presence detector height detection

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- 233878 CARDS DRAWER ADJUSTMENT TOOL (Ver. 2).

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

**NOTE**

When the presence detector is activated, the green LED lights. The orange LED lights for a moment and:

- If no detection occurs, the light goes out.
- If detection occurs, the light stays on.

1. Insert the cards drawers gripper adjustment tool in the front position of any card drawer.
2. Activate the presence detector by clicking the **set on** button in the **Gripper Arm** tab > **Presence detector** subtab (see "Presence Detector subtab" on page 782) > **Presence detector** area.
3. De-energize the CAT arm. In the Technical Service software, press the **unlock/stop** button in the **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Position Actions** area and select all axes. Then move it so the presence detector laser beam spot illuminates adjustment area 2 (see picture below).

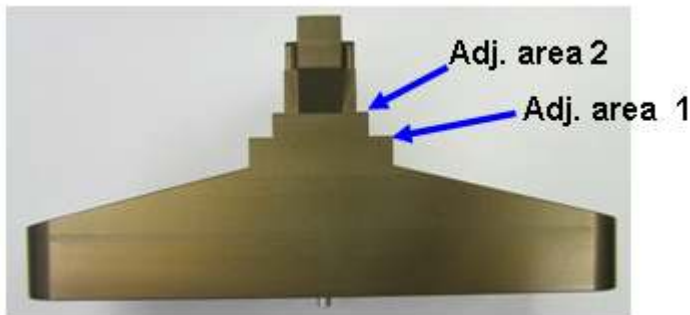


Figure 420 - Card presence detector height adjustment

4. Check that presence is detected (orange LED lights).

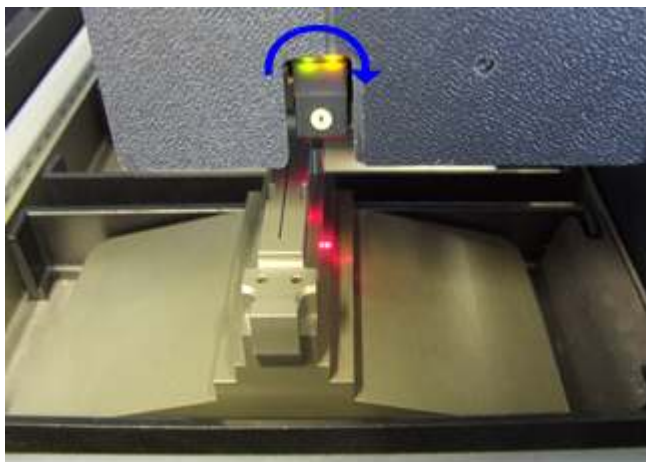


Figure 421 - Presence detected over adjustment area 2

5. Move the gripper arm so the presence detector laser beam spot illuminates adjustment area 1.
6. Check that no presence is detected (orange LED does not light).



CAUTION

Remove the adjustment tool from the instrument when the verification procedure finishes.

2.5.4.9 Verifying card barcode reader inclination



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Activate the barcode reader: in the Technical Service software, go to the **Gripper Arm** tab > **BCR** subtab (see "BCR subtab" on page 785) and click the **set on** button.
2. Check that the barcode reader laser beam is focused on the center of the reflection mirror.

2.5.4.10 Verifying reader area



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 731398 READER ADJUSTMENT TOOL-PATTERN v2.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Perform a home of the CAT module.
2. Move the incubators to the home position by clicking the **Home** button in the **Incubators** tab > **Incubator X** subtab. See "Incubator 1, 2, 3 and 4 subtabs" on page 760.



CAUTION

Check the CAT arm position to avoid it interfering with incubator movement and leading to a crash.

3. De-energize the gripper arm motors by clicking the **unlock/stop** button in **Gripper Arm** tab > **Movement** subtab (see "Movement subtab" on page 777) > **Move Actions** and selecting all axes.
4. Open the CAT floor door (opening first the user tray and then removing the screw fixing the intermediate door).
5. Attach the reader adjustment tool-pattern to the gripper as indicated in the picture (pattern side looking at the left, screws heads looking at right).

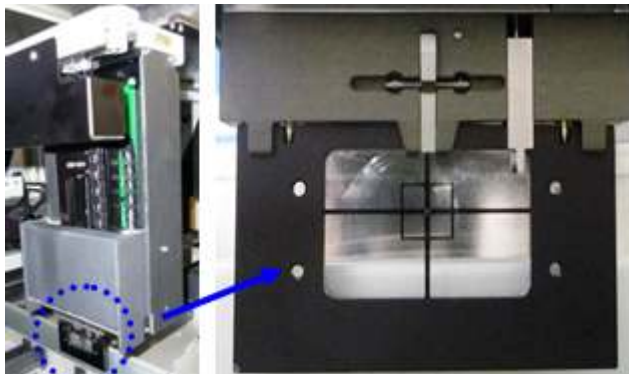


Figure 422 - Gripper adjustment in card reader area



CAUTION

In the next step, the robot head will move without performing a gripper home. Check gripper Z position to avoid a potential crash with any intermediate floor element.

6. Click the **reference check** button in the **Procedures** tab (see "Procedures tab" on page 756) > **Checks** area > **Reader Module** section. The instrument will show on-screen information of the three axes (X, Y and Z) of the adjustment. In case any of the axes return a result of false, perform a gripper adjustment in the reader area.
7. Remove the reader adjustment tool-pattern from the gripper by activating the electromagnet (manually or using the Technical Service software). Click **Next**.

8. Click the **read/check cards test** button in the **Procedures** tab > **Checks** area > **Reader Module** section. The following window will be displayed.

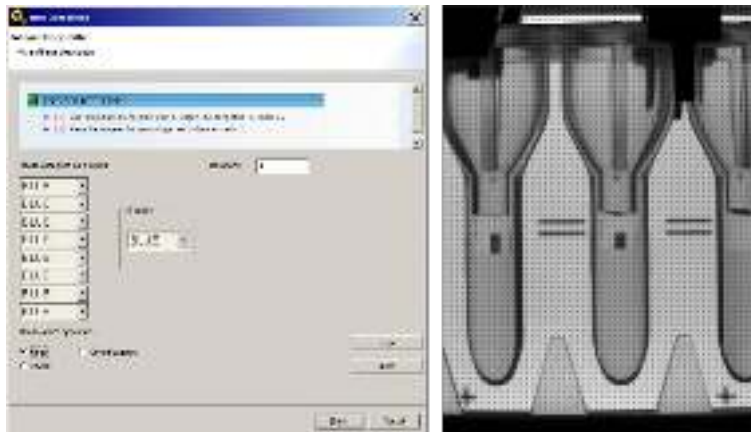


Figure 423 - Read/Check card test configuration window and wells picture

9. Insert a rack with several cards in card drawer 1, rack 1 position.
10. Configure microtubes color and the number of iterations.
11. Select the **Check** option and click the **start** button. The cards will be moved to reader for initial check positions. The obtained pictures will be displayed/saved. Check proper pattern location inside the wells.

2.5.5 Replacement procedures

Replacement procedures related to the CAT module are detailed in this section.

2.5.5.1 Replacing gripper leadscrew



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233722 LEAD SCREW (gripper).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure



CAUTION

Do not handle (or gently handle) the Y arm by the card reader barcode assembly, as the stress induced in this way may break the card barcode mirror. The barcode reader mirror is fragile.

In order to replace the gripper leadscrew, it is necessary to remove the robotic head from the instrument. To do so, follow these steps:

1. Remove the CAT robotic head right cover (fixed by six screws, three per side).
2. Detach the barcode reader assembly from the robotic head (four screws) to gain access to the barcode reader. Unplug the barcode reader from PCA CAT-TOW59. Be careful with the connector flap/flange.
3. Remove the screws that attach the Y chain to the Z tower/robotic head.

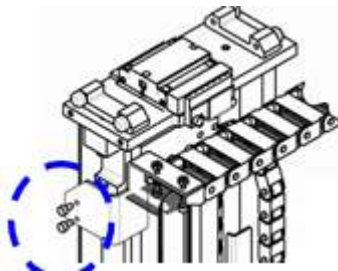


Figure 424 - Location of the screws that attach Y chain to Z tower/robotic head

4. Disconnect the Y chain ribbon cables from PCA CAT-TOW59.



Figure 425 - Location of Y chain ribbon cables at PCA CAT-TOW59

5. Remove the Y arm guide front end stopper (remove two screws, see dotted arrows).

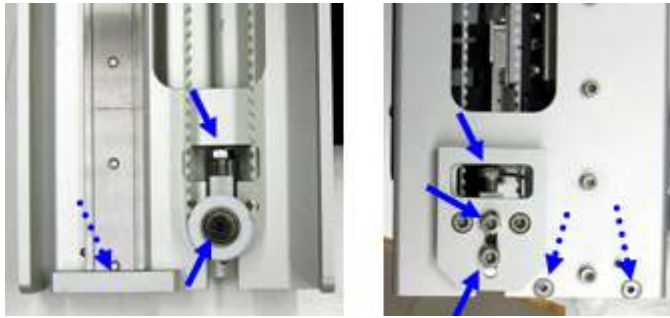


Figure 426 - Y arm guide

6. Remove the Y belt pulley by removing its fixing screw. To ease removal, loosen the belt tension screw and both tensor assembly fixing screws (solid arrows).
7. Remove the robot head.

2.5.5.1.1 Replace leadscrew

1. Remove the two screws fixing the gripper guide assembly to the leadscrew nut flange.



Figure 427 - Gripper guide assembly and leadscrew nut flange

2. Remove the leadscrew pinion set screw and remove the pinion.
3. Remove the three screws fixing the leadscrew holder to the robotic head bulk. The leadscrew is now free.
4. Extract the leadscrew and the leadscrew holder from their location, loosen the holder screw and remove the holder. A rubber hammer may be used to ease the set release.



Figure 428 - Leadscrew removing process

5. Remove the leadscrew. The gripper guide assembly (lineal guide wagon) can be removed from the guide. The wagon has wires to prevent the wagon bearings falling.



Figure 429 - Leadscrew

6. Remove the three screws fixing the leadscrew nut flange to the leadscrew nut and remove the flange.
7. Dispose of the leadscrew.
8. Insert the leadscrew nut flange into the new leadscrew, checking proper orientation. Flange built-in gripper axis home flag.
9. Assemble the flange by means of the three screws (check proper screws insertion direction). Do not tighten the screws completely.
10. Insert the leadscrew into the robotic head.
11. Insert the leadscrew holder into the leadscrew, aligning the holder notches and leadscrew and levelling both part external edges (see following picture).



Figure 430 - Adjustment of the leadscrew in the leadscrew holder

12. Tighten the holder screw.
13. Insert the leadscrew and holder in their location. The holder tightening screw head must be located on the side closest to the gripper. A rubber hammer may be used to help to insert the set.
14. Fix the holder to the robotic head.
15. Insert the pinion into the leadscrew and level it with the leadscrew end. Fix it by tightening the set screw to the flat side.
16. Attach the gripper assembly to the leadscrew: insert and tighten the two screws.
17. Tighten the screws attaching the leadscrew nut flange to the nut.
18. Assemble the Y arm/robot head back, following the removal procedure steps in reverse order.

2.5.5.2 Replacing card barcode reader mirror



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233777 BCR MIRROR (ERY PIP&CAT / EFL CAT).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the gripper arm robotic head right lid (fixed by six screws, three per side).
2. Detach the barcode reader assembly from the robotic head (four screws) to gain access to the barcode reader. Unplug the barcode reader from PCA CAT-TOW. Be careful with the connector flap/flange.
3. Remove the CAT mirror wedge with pliers (if installed).
4. Remove the mirror through the slot, pulling it up and then to the right.



Figure 431 - Barcode reader mirror

5. Insert the mirror with the reflective coating side/first side facing the barcode reader. The reflective coating side/first side can be easily identified: touch the mirror with an object. If the tip of the object seems to touch the reflected image, that is the mirror coating side/first side.



Figure 432 - First face of the mirror (left) and second face of the mirror (right)

6. Place the CAT mirror wedge (if available) with pliers, checking that it exits from the surface of the cover about 3-4 mm.
7. Place the left robotic head lid with its screws and connect carefully the barcode reader cable to PCA CAT-TOW59.
8. Finally, place the right robotic head lid with its screws.

2.5.5.3 Replacing CAT X chain



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233793 X WIRING/CHAIN (CAT).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Remove the middle rear instrument cover.
3. Open the intermediate door.

4. Remove the interconnection cover (X-Y chains).

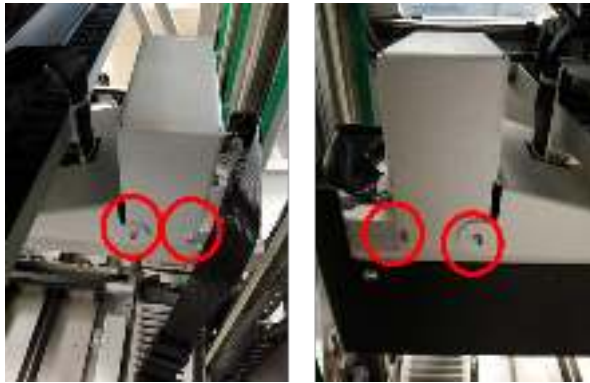


Figure 433 - Interconnection cover

5. Disconnect the three flat cables from PCA CAT-CONY59 and remove the cable ties.



Figure 434 - Flat cables and ties

6. Disengage the X chain from the last chain slot.



Figure 435 - X chain slot

7. Open the cable clamp and disconnect the three flat cables from PCA GEN-CONX59. If needed, remove the left lateral methacrylate cover and the user tray to have better access to this area. See "Replacing user tray" on page 217.



Figure 436 - Flat cables and cable clamp

8. Remove the two screws that attach the X chain.



Figure 437 - X chain screws

9. Remove the X chain.
10. Place the new X chain and assemble back all the parts.



CAUTION

Do not swap ribbon cables in PCA GEN-CONX59 or PCA CAT-CONY59 when connecting them (it could damage different parts of the instrument), specially cables labelled as XB-CAT_XY and XB-CAT_Z that have both a blue connector. The cable labelled as XB-CAT_TOW has a black connector. Cables should match the connector colour when connected to the PCA.

2.5.5.4 Replacing CAT Y chain



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233794 Y WIRING/CHAIN (CAT).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Remove the middle rear instrument cover.
3. Open the intermediate door.
4. Remove the right cover of the CAT module (eight screws).

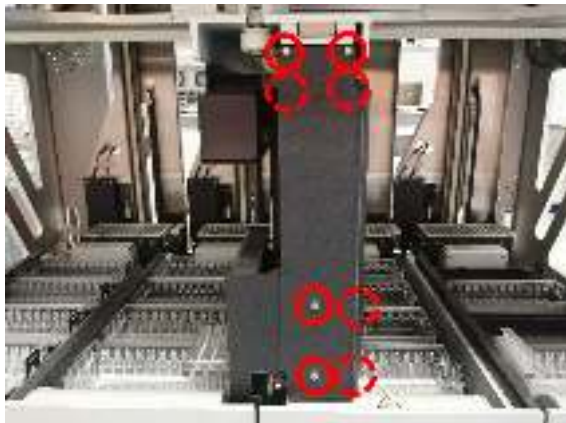


Figure 438 - Right cover of the CAT module

5. Disconnect the two flat cables from PCA CAT-TOW59 of the CAT module.



Figure 439 - Flat cables

6. Disengage the Y chain from the last chain slot.



Figure 440 - Y chain slot

7. Remove the two screws from the Y chain and open the cable clamp.

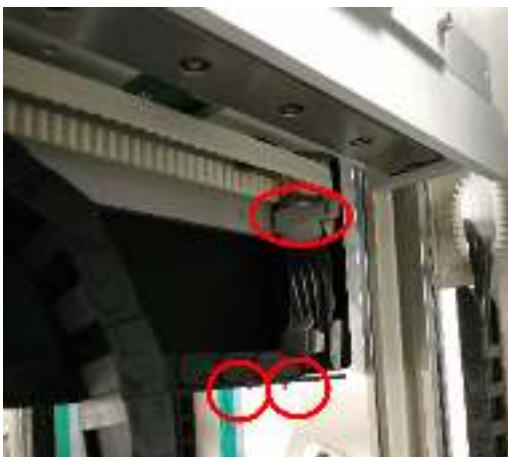


Figure 441 - Y chain screws and cable clamp

8. Remove the interconnection cover (X-Y chains).

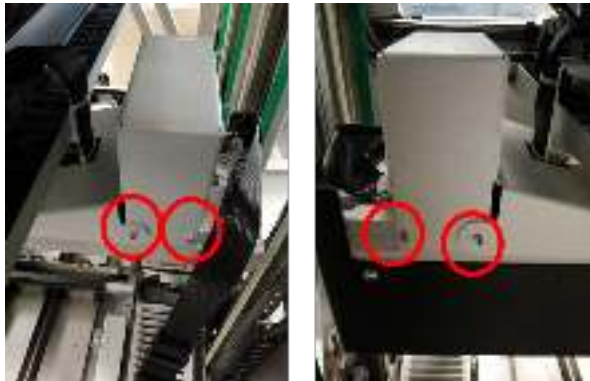


Figure 442 - Interconnection cover screws

9. Disconnect the two flat cables from PCA CAT-CONY59 and open the cable clamp.



Figure 443 - Flat cables and cable clamp

10. Remove the Y chain.
11. Place the new Y chain and assemble back all the parts.



CAUTION

Do not swap ribbon cables in PCA CAT-CONY59 or PCA CAT-TOW59 when connecting them (it could damage different parts of the instrument).

Cables should match the connector colour when connected to the PCA: cable labelled as CAT-TOW black and CAT-Z blue.

2.5.5.5 Replacing gripper (Z) chain



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233795 GRIPPER WIRING_CHAIN.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Remove the middle rear instrument cover.
3. Open the intermediate door.
4. Remove the right cover of the CAT module (eight screws).

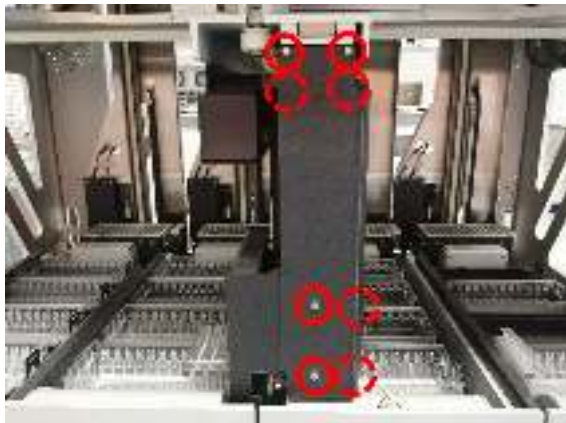


Figure 444 - Right cover of the CAT module

5. Remove the left cover of the CAT module (four screws)



Figure 445 - Left cover of the CAT module

6. Pull it apart carefully, not damaging the barcode reader cable that is attached to this cover.



Figure 446 - Barcode reader cable

7. Remove the screw of the metallic plate



Figure 447 - Metallic plate screw

8. Disconnect the cables.



Figure 448 - Gripper cabling (1)

9. Disconnect cables.



Figure 449 - Gripper cabling (2)

10. Remove the gripper Z chain.
11. Place the new Z chain and assemble back all the parts.
12. Verify gripper performance.

2.5.5.6 Replacing gripper



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233721 GRIPPER.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the intermediate door.
3. Open card drawer 1.
4. Remove the right cover of the CAT module (eight screws)

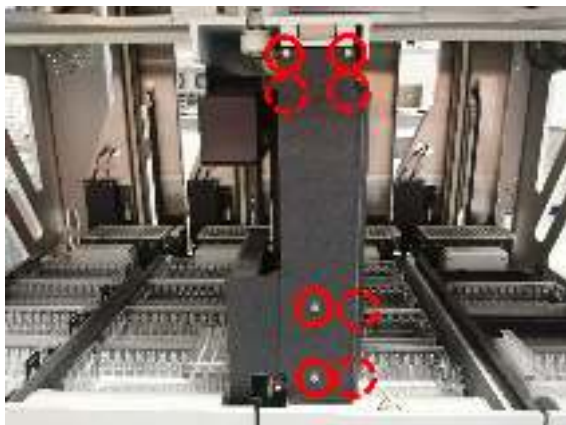


Figure 450 - Right cover of the CAT module

5. Pull it apart carefully, not damaging the barcode reader cable that is attached to this cover.



Figure 451 - Barcode reader cable

6. Move down the gripper manually by means of the gripper motor pinion.
7. Remove the screw of the metallic plate.



Figure 452 - Metallic plate screw

8. Disconnect the cables.



Figure 453 - Gripper cabling

9. Remove the four screws that attach the gripper to the CAT Module.
10. Remove the gripper together with its PCA.
11. Remove the PCA from the gripper.
12. Attach the PCA to the new gripper.

13. Place the new gripper and its PCA to the CAT module and tighten the four screws of the gripper. Consider applying a sealant to the four screws.

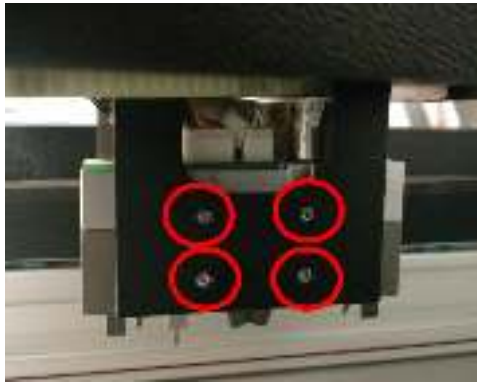


Figure 454 - Gripper screws

14. Assemble back all the parts and reconnect the cabling.
15. Perform Gripper parallelism verification. See "Verifying gripper parallelism" on page 490.
16. Perform Gripper adjustments in all positions. See "Adjusting gripper parallelism" on page 469.
17. Verify gripper performance.

2.5.5.7 Replacing toothed belt T2.5 245

Replacement procedures related to the T2.5 245 Toothed belt in PIP and CAT arms:

- See "Replacing toothed belt T5/2000 (CAT X-axis)" on page 524
- See "Replacing toothed belt T5/1200 (CAT Y-axis)" on page 527
- See "Replacing toothed belt T5/2000 (PIP X-axis)" on page 421
- See "Replacing toothed belt T5/1200 (PIP Y-axis)" on page 425

2.5.5.7.1 Replacing toothed belt T2.5/245 (CAT X-axis)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 231889 TOOTHED BELT T2,5/245.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the intermediate door.
3. Remove the small belt.



Figure 455 - Small belt

4. Place the new small belt.

2.5.5.7.2 Replacing toothed belt T2.5/245 (CAT Y-axis)



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 231889 TOOTHED BELT T2,5/245.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the intermediate door.
3. Remove the middle rear cover.
4. From the front of the instrument, take out the Y-belt from its position by removing the fixing screw of the pulley.

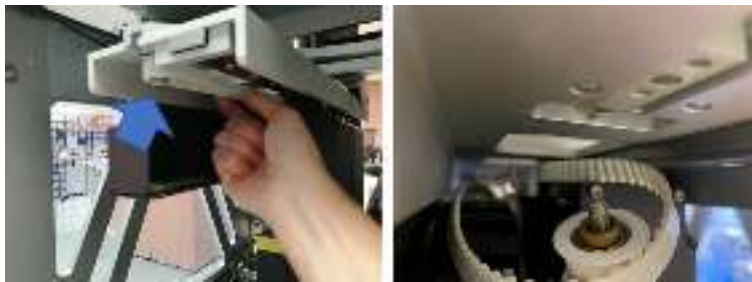


Figure 456 - CAT large belt

5. Remove the large belt from the pinion. The belt will hang on the CAT robotic head.



Figure 457 - Large belt (pinion)

6. Remove the interconnection cover (X-Y chains).

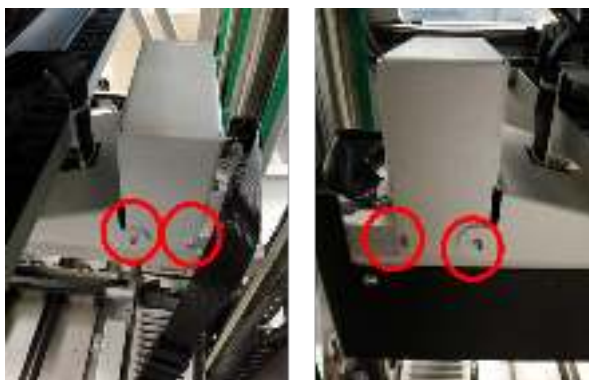


Figure 458 - Interconnection cover screws

7. Remove the motor support screws.



Figure 459 - Support screws

8. Disconnect the two flat cables, open the cable clamp and pull apart the cables.



Figure 460 - Cabling and cable clamp

9. Remove the motor screw (slotted hole).



Figure 461 - Motor screw

10. Move the motor to be able to remove the small belt.
11. Place the new small and large belts and assemble back all the parts.

**NOTE**

If the pulley tightening screw is in the same position before disassembly, the belt tension will remain unchanged, else belt tension must be adjusted by modifying the end pulley position.



Figure 462 - Pulley tightening screw

**NOTE**

Any variation of the belt tension may modify probe adjustments in all positions.

12. Adjust the tension of the small belt. See "Adjusting Y-axis small belt tension" on page 466.
13. Perform a card transport check.

2.5.5.8 Replacing toothed belt T5/2000 (CAT X-axis)

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233865 TOOTHED BELT T5/2000

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the lower door.
3. Removing the fixing screw of the tension pulley.



Figure 463 - Tension pulley of the large belt (PIP y axis)

4. Remove the large belt from the pinion. The belt will be left hanging.
5. Loosen the belt clamp plate (without removing) and remove the belt.

- Put the new belt in the clam plate and tighten the screws tensioning the belt outwards.



Figure 464 - Large belt clamping plate and belt tension direction needed at the plate fixing moment.



NOTE

When tightening the fixing part make sure to tension the belt outwards in order to eliminate any possible play that could affect the belt tension in the future.



Figure 465 - Belt tension direction when fixing the clamping plate (this view is hidden in the instrument)

- Check that the home barrier do not collide with the sensor by hand-moving the arm (x-axis case).

8. Put the belt around the pulleys and fix the tension pulley again.

**NOTE**

If the pulley tightening screw is in the same position before disassembly, the belt tension will remain unchanged, else belt tension must be adjusted by modifying the end pulley position.



Figure 466 - Tension screw (tension pulley is under these parts)

**NOTE**

Any variation of the belt tension may modify probe adjustments in all positions.

9. Mount the fairing plate back in place.
10. Perform the probe positioning check. See "Procedures tab" on page 756.

2.5.5.9 Replacing toothed belt T5/1200 (CAT Y-axis)

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- 233866TOOTHED BELT T5/1200

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Open the upper door and remove the lateral cover by removing its screws.



Figure 467 - CAT arm lateral cover

3. Removing the fixing screw of the tension pulley.



Figure 468 - Tension pulley of the large belt (PIP y axis)

4. Remove the large belt from the pinion. The belt will be left hanging.
5. Loosen the belt clamp plate (without removing) and remove the belt.

6. Put the new belt in the clam plate and tighten the screws tensioning the belt outwards.

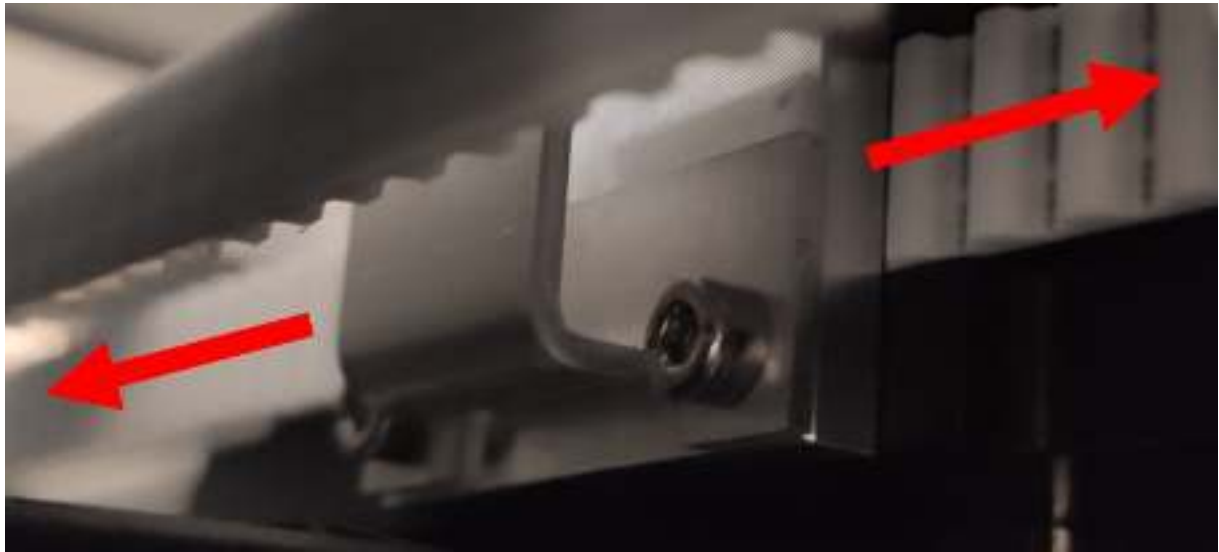


Figure 469 - Large belt clamping plate and belt tension direction needed at the plate fixing moment.



NOTE

When tightening the fixing part make sure to tension the belt outwards in order to eliminate any possible play that could affect the belt tension in the future.



Figure 470 - Belt tension direction when fixing the clamping plate (this view is hidden in the instrument)

7. Check that the home barrier do not collide with the sensor by hand-moving the tower (y-axis case).

8. Put the belt around the pulleys and fix the tension pulley again.



NOTE

If the pulley tightening screw is in the same position before disassembly, the belt tension will remain unchanged, else belt tension must be adjusted by modifying the end pulley position.



Figure 471 - Tension screw (tension pulley is under these parts)



NOTE

Any variation of the belt tension may modify probe adjustments in all positions.

9. Mount the cover back in place.
10. Perform the probe positioning check. See "Procedures tab" on page 756.

2.5.6 Electronics

This section covers the electronics related to the CAT module.

2.5.6.1 Fuses

Fuses related to the CAT module are detailed in the following table:

Table 44 - List of CAT module fuses

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS21	MST250 0034.6615 1A/250V
	FUS22	MST250 0034.6620 3.15A/250V
PCA CAT-CM CONTROL	FUS1	MSF250 0034.6013 0.8A/250V

PCA	FUSE NAME	DESCRIPTION
PCA GEN-CONX59	NA	NA
PCA CAT-CONY59	NA	NA
PCA CAT-TOW59	NA	NA
PCA PINZA57	NA	NA

2.5.6.2 LEDs

LEDs related to the CAT module are detailed in the following table:

Table 45 - List of CAT module LEDs

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED13	Green 5V (FUS21)
	LED14	Green 24V (FUS22)
PCA CAT-CM CONTROL	LED1	Yellow, When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green, VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$.
PCA GEN-CONX59	NA	NA
PCA CAT-CONY59	NA	NA
PCA CAT-TOW59	NA	NA
PCA PINZA57	NA	NA

2.5.6.3 Jumpers and test points

Neither jumpers nor test points are related to the CAT module.

2.5.6.4 Wiring table

Cables related to the Card transport module are detailed in the following table and their connection are shown in the wiring diagram. To view the wiring diagram, see "Related documents" on page 19 or refer to the Erytra Online Help.

Table 46 - List of CAT module wiring

ID	FROM CONNECTOR	FROM PCA/ COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-CAT_TOW	CAT_TOW	GEN-BCKPL59	H5 In CAT/PIP-TOW	PCA GEN-CONX59
B-CAT_Z	CAT_Z	GEN-BCKPL59	H4 In CAT/PIP-Z	PCA GEN-CONX59
B-CAT_XY	CAT_XY	GEN-BCKPL59	H3 In CAT/PIP-XY	PCA GEN-CONX59
XB-CAT_TOW	H8 Out CAT/PIP-TOW	PCA GEN-CONX59	H5	PCA CAT-CONY59
XB-CAT_Z	H7 Out CAT/PIP-Z	PCA GEN-CONX59	H4	PCA CAT-CONY59
XB-CAT_XY	H6 Out CAT/PIP-XY	PCA GEN-CONX59	H3	PCA CAT-CONY59
BRX-EncX	H2 EncX	PCA GEN-CONX59	Encoder	X Axis motor
BRX-EXT + BRX-MotX	H1 MotX	PCA GEN-CONX59	Motor powering	X Axis motor
CAT-MotY	H1 MotY	PCA CAT-CONY59	Motor powering	Y Axis Motor
CAT-EncY	H2 EncY	PCA CAT-CONY59	Encoder	Y Axis Motor
CAT-Z	H6 Out CAT-Z	PCA CAT-CONY59	H7 CAT-Z	PCA CAT-TOW59
CAT-TOW	H7 Out CAT-TOW	PCA CAT-CONY59	H6 CAT-TOW	PCA CAT-TOW59

ID	FROM CONNECTOR	FROM PCA/ COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
CAT-PRES	H1 PresDet	PCA CAT-TOW59	Presence detector	Presence detector.
CAT-BCR	H8 BarCode	PCA CAT-TOW59	Barcode reader	Barcode reader
CAT-MotZ	H5 MotZ	PCA CAT-TOW59	Motor powering	Gripper motor
CAT-EncZ	H2 EncZ	PCA CAT-TOW59	Encoder	Gripper motor
CAT-CARD	H3 CardDet	PCA CAT-TOW59	J2	PCA PINZA57
CAT-SOL	H4 Sol	PCA CAT-TOW59	J1	PCA PINZA57
NA	Solenoid	Solenoid	J3	PCA PINZA57

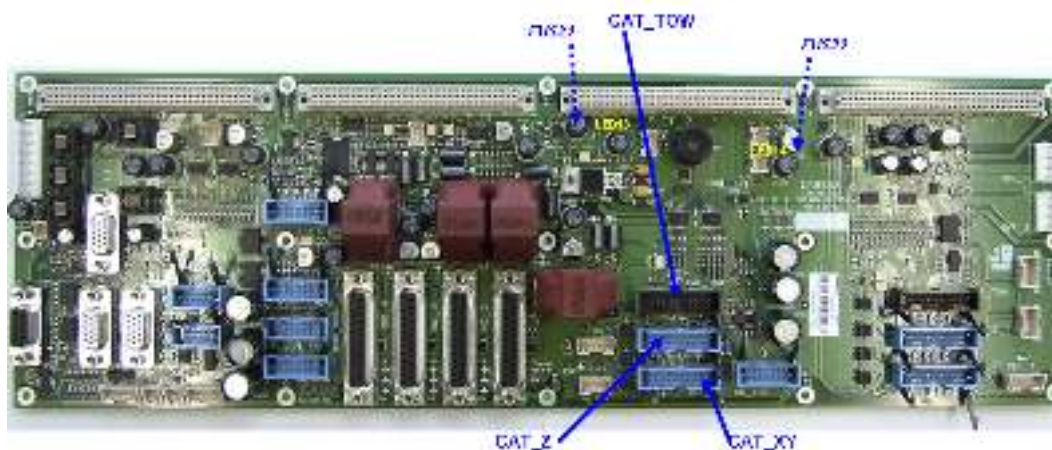


Figure 472 - Connectors, fuses and LEDs of PCA GEN-BCKPL59 related to CAT module

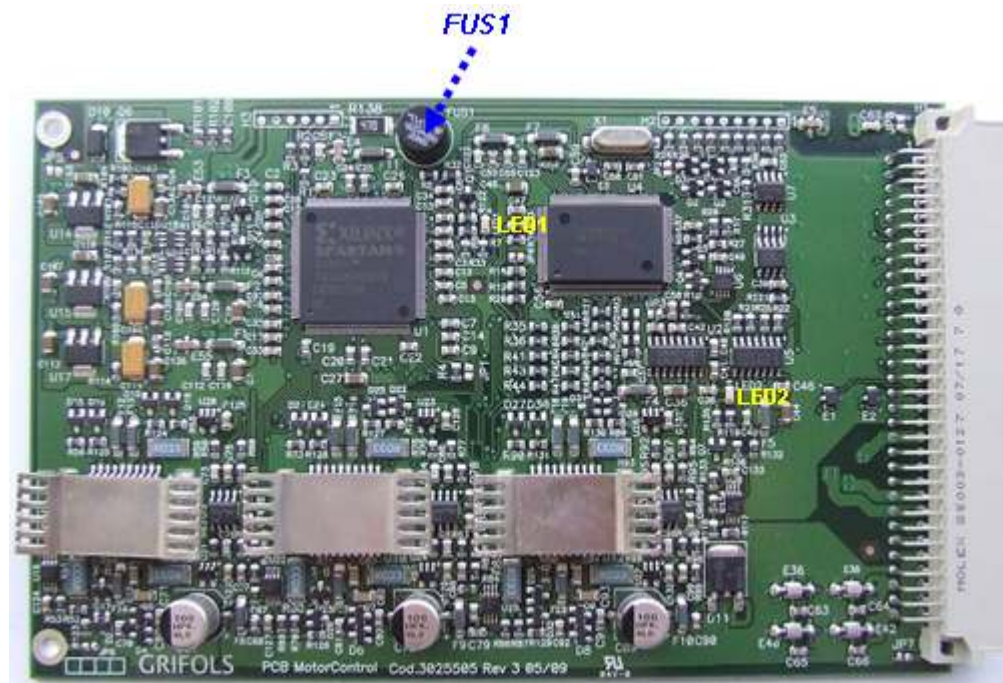


Figure 473 - Location of the fuse and LEDs in PCA CAT-CM Control

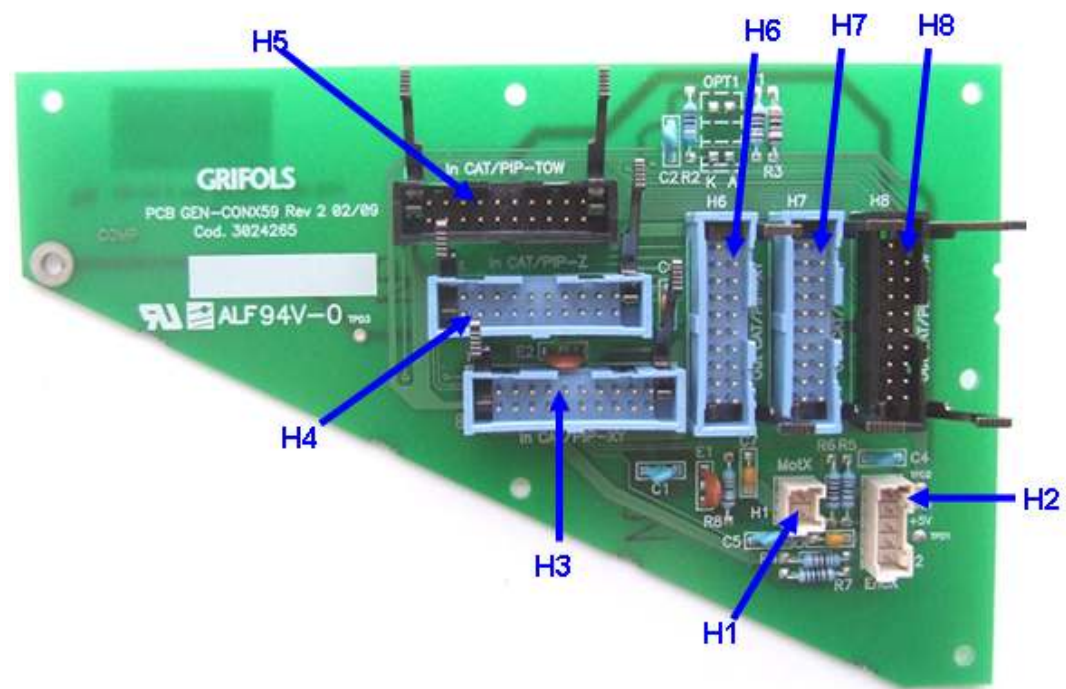
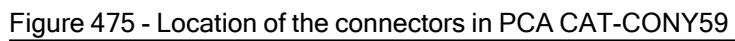


Figure 474 - Location of the connectors in PCA GEN-CONX59



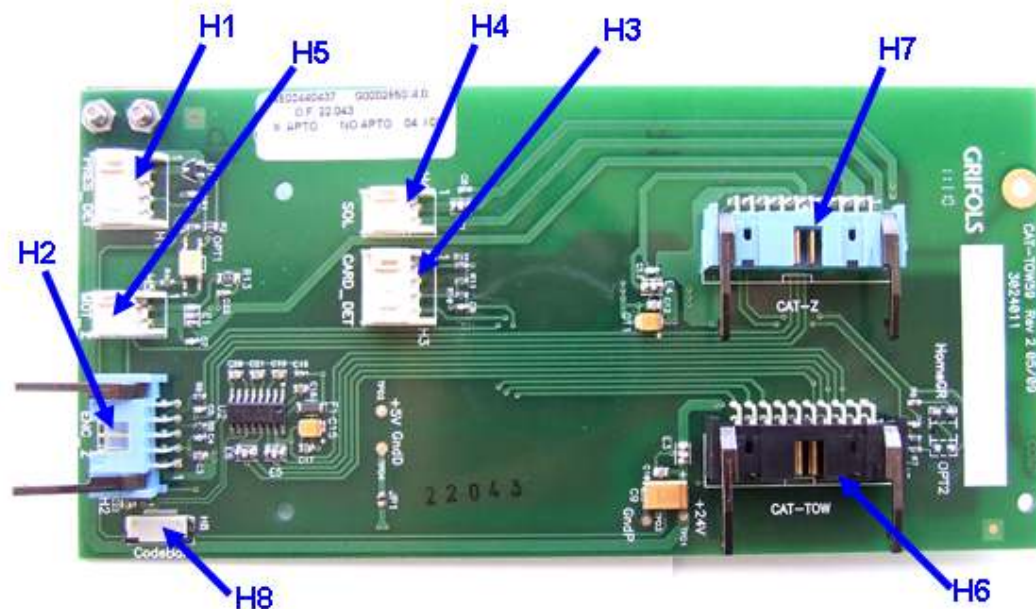


Figure 476 - Location of the connectors in PCA CAT-TOW59

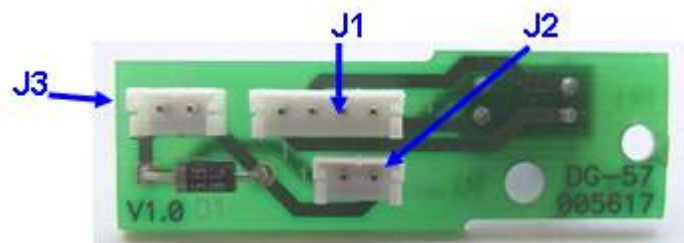


Figure 477 - Location of the connectors in PCA PINZA57

2.5.7 Troubleshooting

Table 47 - List of errors related to CAT module

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	XYZ Motors	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during the lower robotic arm movement. Please, inspect the instrument middle level. Please choose one of the following options:	RETRY CANCEL Works (it leads to: Gripper disabled)
	(01) X axis					
	(02) Y axis					
	(04) Z axis					
	(03) Gripper	005000	ERRORCODE_ARG	Command has wrong arguments.		
	(05) Presence detector during X movements	006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
	(07) Presence detector during Y movements	007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device.		
		008000	ERRORCODE_NOTSAFE	Command is not safe for the instrument.		

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	XYZ Motors (01) X axis (02) Y axis (04) Z axis (03) Gripper (05) Presence detector during X movements (07) Presence detector during Y movements	011000	ERRORCODE_DATANOTREADY	The data requested is not ready.	Error during the lower robotic arm movement. Please, inspect the instrument middle level. Please choose one of the following options:	RETRY
		032000	CURRENT_LIMIT	Current limit at servo device.		CANCEL
		033000	POSITION_ERROR	A position error has been produced at a servo device.		Works
		034000	MOVEMENT_ERROR	Movement error produced at servo device.		(it leads to: Gripper disabled)
		035000	HOME_TIMEOUT	Home timeout has been produced at a servo device.		

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	XYZ Motors (01) X axis (02) Y axis (04) Z axis (03) Gripper (05) Presence detector during X movements (07) Presence detector during Y movements	036000	ERRORCODE_SRV_SYNC_TIMEOUT	Synchronization error has been produced at a servo device.	Error during the lower robotic arm movement. Please, inspect the instrument middle level. Please choose one of the following options:	RETRY CANCEL Works (it leads to: Gripper disabled)
		037000	ERRORCODE_SRV_UNKNOWN_ERROR	Unknown error has been produced at a servo device.		
		041000	ERRORCODE_PD_SYNC_TIMEOUT	Presence detector has been busy for more than its timeout margin.		
		042000	ERRORCODE_PD_UNKNOWN_ERROR	Unknown error produced during a presence detector related operation.		
		047000	ERRORCODE_GRP_UNKWON_ERROR	Unknown error.		

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	XYZ Motors (01) X axis (02) Y axis (04) Z axis (03) Gripper (05) Presence detector during X movements (07) Presence detector during Y movements	009000	ERRORCODE_CARDPRESENT	Card present in the gripper.	No error reported when reading BCs from cards racks.	NA
					If error is detected in incubator or centrifuge, reported error is	RETRY
					Error in the lower robotic arm when taking a card. A card is already present.	CANCEL Works
		010000	ERRORCODE_NOCARDPRESENT	There is no card present in the gripper.	No error reported when reading BCs from cards racks.	NA
					If error is detected in incubator or centrifuge, following error is reported.	RETRY
					Error in the lower robotic arm when releasing a card. Please inspect middle level for lost card.	CANCEL Works
		044000	CARD_LOST	The gripper has lost a card.	No error reported when reading BCs from cards racks.	NA
					If error is detected in incubator or centrifuge, reported error is	RETRY
					Error in the lower robotic arm when taking a card. Please inspect middle level for lost card.	CANCEL Works
					If error is detected in reader, reported error is	
					Card lost while reading. Please select one of the following options:	

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	XYZ Motors (01) X axis (02) Y axis (04) Z axis (03) Gripper (05) Presence detector during X movements (07) Presence detector during Y movements	045000	CARD_NOTLEFT	Card not left during a left card operation.	Error in the lower robotic arm when releasing a card. Please inspect middle level for lost card.	RETRY
						CANCEL Works
						Release card
		046000	CARD_NOTFOUND	Card not found during a take card operation.	No error is reported when reading BCs from cards racks.	NA
					If error is detected in incubator or centrifuge, reported error is	RETRY
					Error in the lower robotic arm when taking a card.	CANCEL Works

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	BCR (06)	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	No error is reported when reading BCs from card racks. If the error is detected in incubator or centrifuge, following error is reported if card has not been previously cancelled: Card barcode not expected. Card will be discarded and samples cancelled.	RETRY CANCEL Works (it leads to: Gripper disabled)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		011000	ERRORCODE_DATANOTREADY	Data requested is not ready.		
		048000	ERRORCODE_BCR_DRIVER_ERROR	Driver error.		

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	BCR (06)	049000	ERRORCODE_BCR_NOT_INIT	Barcode reader is not initialized	No error is reported when reading BCs from card racks. If the error is detected in incubator or centrifuge, following error is reported if card has not been previously cancelled:	RETRY
		050000	ERRORCODE_BCR_COM_ERR	Barcode reader communication error.		CANCEL Works (it leads to: Gripper disabled)
		051000	ERRORCODE_BCR_NO_BARCODE	No barcode is returned by the reader.	Card barcode not expected. Card will be discarded and samples cancelled.	
		052000	ERRORCODE_BCR_SYNC_TIMEOUT	Barcode reader has been busy more than its timeout margin.		
		053000	ERRORCODE_BCR_UNKNOWN_ERROR	Barcode reader unknown error.		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received a command with incorrect length.		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device.		
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		014000	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW.		
		015000	ERRORCODE_HW_HWFAIL	There is a problem with PCA HW.		
		016000	ERRORCODE_HW_HWNOTSET	PCA setup error.		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	000000	ERRORCODE_ UNKNOWN	Unknown error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_ NOTSAFE	Movement not safe.		
		017000	ERRORCODE_FW_ UNKNOWN_ERROR	Unknown error.		
		061000	ERRORCODE_ INVALID_ ARGUMENTS	Invalid arguments.		
		062000	ERRORCODE_ FWCOMM_TIMEOUT	Communication timeout.		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	063000	ERRORCODE_ FWCOMM_ CONNECTION_ LOST	Communication lost.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		064000	ERRORCODE_ FWCOMM_COMM_ ERROR	Communication error.		
		088000	ERRORCODE_ PRECONDITION_ FAILED	Previous condition for the action is not met.		
		089000	ERRORCODE_ STARTUP_ERROR	Driver startup error.		
		091000	ERRORCODE_ ALREADY_ STARTED	Another instance of Driver is already opened.		
0	240	090000	ERRORCODE_ UNEXPECTED_ ERROR	A Driver unexpected error has been produced.		

Table 48 - CAT module troubleshooting

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / current limit error (X axis) 0201032000 0201034000	Fuse blown	Check / replace FUS21 or FUS22
	Large belt too tight	Check tightness of belt / Replace belt
	Incorrect X motor assembly	Check/replace X motor silentblocks
	Lack of lubrication in X linear guide	Lubricate linear guide
	Motor damaged	Check / replace motor
	CAT X-Y flat cable damaged	Check / replace CAT X-Y cable
	X linear guide damaged	Check / replace linear guide
Movement error / current limit error (Y axis) 0202032000 0202034000	Fuse blown	Check / replace FUS21 or FUS22
	Small belt too tight	Check tightness of belt / Replace belt
	Large belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in Y linear guide	Lubricate linear guide
	Motor damaged	Check / replace motor
	CAT X-Y flat cable damaged	Check / replace CAT X-Y cable / X CAT chain
	Y linear guide damaged	Check / replace linear guide

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / current limit error (Z axis) 0204032000 0204034000	Fuse blown	Check / replace FUS21 or FUS22
	Belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in Z linear guide	Lubricate linear guide
	Lack of lubrication in leadscrew	Lubricate leadscrew
	Motor damaged	Check / replace motor
	CAT-TOW flat cable damaged	Check / replace CAT-TOW flat cable
	CAT-Z flat cable damaged	Check / replace X- Y CAT chain
	Z linear guide damaged	Check / replace linear guide
	Leadscrew damaged	Check / replace leadscrew
X axis home is not detected 0201035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Faulty belt	Check / replace small / large belt
	Failure in optical barrier	Check / replace PCA GEN-CONX59
	Flat cable damaged	Check / replace CAT X-Y cable

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Y axis home is not detected 0202035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Faulty belt	Check / replace belt
	Failure in optical barrier	Check / replace PCA CAT-CONY59
	Flat cable damaged	Check / replace X CAT chain (CAT X-Y cable)
Z axis home is not detected 0204035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Faulty belt	Check / replace belt
	Failure in optical barrier	Check / replace PCA CAT-TOW59
	Flat cable damaged	Check / replace Z CAT chain (CAT-Z cable)

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Cards are not taken from card drawer/rack	Card rack is not placed correctly	Place correctly card rack in the drawer
	Incorrect card rack detection	Check / readjust / fix CAT presence detector
		Check card drawer flatness
		Check / perform CAT adjustment in card drawer
	Gripper is not parallel	Check gripper parallelism
	Card drawer is not flat	Check card drawer flatness
	Lack of lubrication in gripper	Lubricate gripper
Cards are not taken from incubator	Wrong adjustment	Check / perform CAT adjustment in card drawer
	Gripper is not parallel	Check gripper parallelism
	Incubator is not flat	Check incubator flatness
	Lack of lubrication in gripper	Lubricate gripper
	Wrong adjustment	Check / perform CAT adjustment in incubator

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Cards are not taken from centrifuge	Gripper is not parallel	Check gripper parallelism
	Card holder defective	Check / replace card holders
	Rotor is not correctly assembled	Check rotor assembly
	Lack of lubrication in gripper	Lubricate gripper
	Wrong adjustment	Check / perform CAT adjustment in centrifuge
When a card is taken, it is not detected	Home detector or connections broken	Replace PCA PINZA57
0204046000	Pin bent	Replace gripper pin
Card is lost during transport	Wrong CAT arm adjustment	Check/perform CAT arm adjustment
0204044000	Gripper pins bent or springs not equal	Replace pins / springs
Cards tilted in the gripper	Gripper is not parallel	Check gripper parallelism
	Wrong CAT arm adjustment	Check/perform CAT arm adjustment

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Cards are not detected / incorrectly detected	Fuse blown	Check / replace FUSGEN4
	Card rack is not placed correctly	Place correctly card rack in the drawer
	Presence detector is not correctly assembled	Check presence detector assembly
	Adjustment of the presence detector is not correct	Perform presence detector adjustment
	Wrong CAT arm adjustment	Check/perform CAT arm adjustment
	Presence detector is damaged	Replace presence detector
	PCA CAT-TOW59 is damaged	Replace PCA CAT-TOW59
Barcodes are not read 0206051000	Card rack / card is not placed correctly	Place correctly card rack / rack
	Barcode label in bad conditions	Check barcode label
	Mirror is dirty	Clean mirror
	BCR is not properly assembled	Check BCR assembly
	BCR is not working	Check / replace BCR
	BCR cable damaged / disconnected	Check / replace BCR cable
	PCA CAT-TOW59 damaged	Replace PCA CAT-TOW59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
BCR is not switched on	Fuse blown	Check / replace FUS21 or FUS22
0206049000	BCR cables damaged / disconnected	Check / replace BCR cables
0206050000	BCR is not working	Check / replace BCR
0206052000	PCA CAT-TOW59 is damaged	Replace PCA CAT-TOW59
0206053000	CAT-TOW / CAT-Z cable damaged	Check / replace X / Y CAT chains

2.6 Fluidics module (FLU)

In this chapter, the Fluidics module is described. It includes an overall description of the functions as well as the corresponding adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, and a troubleshooting section.

2.6.1 Description

In this section a description of all the parts related to the fluidics module and a description of the fluidics procedures performed by the Erytra are detailed.

2.6.1.1 Parts

The fluidics module is decomposed by the following submodules:

- **Manifold:** acrylic block (PMMA) where fluidic components are integrated. These components are electrovalves, expansion vessels, dispensation pumps, sensors and bottles configuration selector.
- **Electrovalves:** there are 18 electrovalves that allow to pass or to block the liquid by the fluidic system. Block mounting body. Nominal voltage: 24 VDC. Electrovalves involved in the dispensing part are closed by firmware 30 seconds after its activation in order to avoid undesired overflows.
- EV1, EV2, EV3, EV4, EV5, EV6, EV7, EV8, EV9, EV10, EV13, EV14, EV15, EV16, EV17, EV18, EV19 and EV20.
- **Dispensation pumps:** there are two dispensation pumps to drive solution A and B to the fluidic system. The type of the pump is diaphragm for liquids. The motor of the pump is a DC motor with a nominal voltage of 24 V. The pump is screwed to a support that holds it in the correct position. The pump is screwed to the manifold by means of clamp. Clamp and seals used are capable to guarantee watertight union in working conditions. PM1 and PM3.



Figure 478 - Dispensation pumps

- **Expansion vessels:** pressure is regulated by means of the dispensation pumps. As the pump head is membrane type, the pressure given to the system is not constant but pulsed. In order to absorb these pulses, an expansion vessel is placed at each pump outlet.



Figure 479 - Expansion vessels

- **Pressure / vacuum sensors:** they incorporate an overpressure characteristic that, in case of an electronic failure, it controls pressure levels over nominal range without damage. Working pressure range: ± 15 psi (~ 1034 mbar). S3, S4, S5 and S6.

- **Bottles configuration selector:** change manually the use of two bottles of the Erytra between waste solution (bottles 1 and 2) and wash solution (solution A and solution B).



Figure 480 - Bottles configuration selector

In order to configure the bottles configuration selector if the external drain option is activated (see "DUAL_MODE" on page 95), follow next steps:

1. Perform instrument decontamination.
2. Empty the fluidic system and then disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the bottom rear cover of the instrument in order to get access to the manifold.
5. Remove the screw of the bottles configuration selector.
6. Remove the bottles configuration selector with its silicone seal.
7. Dry, if necessary, the location of the bottles configuration selector.
8. Rotate it 90° in counter clockwise direction and place it on its position in the manifold again.



Figure 481 - Bottles configuration selector positions - external drain not activated (left), external drain activated (right)

9. Place the screw of the bottles configuration selector and tighten it to the maximum and then a ¼ turn more.
10. Place the bottom rear cover of the instrument.
11. Connect properly the bottles, taking into account that previous waste solution 1 and 2 drawers are considered now as additional solution A and solution B drawers.

12. Connect to the external drain connector of the instrument a tube with a fitting to connect the instrument to an external drain.

**CAUTION**

The specifications for the tube to be used for the external drain are described in See "External drain" on page 87

13. Perform instrument decontamination after configuring DUAL_MODE parameter.

- **Aspiration pump:** to remove waste liquids from probes washer and dilution cup. The type of the pump is diaphragm for liquids. It has two independent heads in order to be associated to each probe circuit. Its flow rate avoids liquid overflow in dilution cup and drops in probes washer. The motor of the pump is a DC motor with a nominal voltage of 24 V; an EMI filter is incorporated. Pump is attached to the support by means of four silentblocks. PM2.



Figure 482 - Aspiration pump

- **Bottles:** there are four bottles in the instrument with a capacity of 8 litres, located at the bottom of the instrument. Are allowed two possible configurations (from left to right):
 - First configuration: Solution A bottle - Solution B bottle - Waste 1 - Waste 2.
 - Second configuration: Solution A bottle - Solution B bottle - Solution A bottle - Solution B bottle, when the system is directly disposing waste liquids to the laboratory drain.

The bottles contain one tube connector to connect it to the fluidic system and it can be manually disconnected by the operator for filling/emptying/removing/replacing purposes. Caps of the bottles are also the exhaust for the fluidic system (bottles are not pressurized).

To prevent solid particles coming from solution bottles could reach the fluidic circuit causing obstruction in electrovalves, 50-micron filters are placed between the containers of solutions A and B and the manifold of the instrument.

- **Cards disposal tray:** there is a container located at the bottom of the instrument where the instrument throws the cards that are not necessary for the system. It has a capacity for 425 cards.

- **Bottles and cards disposal container drawers:** there is one drawer per each bottle and card disposal container. The structure of each drawer is formed by the following elements:

- **Moving part:** it supports the bottle or cards disposal container and allows the operator to load/unload them. It moves by sliding movement and it includes a cosmetic front panel.

The edges of the moving part of each of the fluidic drawers have been hemmed in order to avoid injuries to the field engineer when manipulating it. However there are available edge protectors to be installed in instruments without hemmed edges to mitigate this risk.



Figure 483 - Fluidic drawer edge protectors

- **Fixed part:** it is responsible for allowing the movement of the moving part outside the instrument and to attach the drawer to the base. It also contains the load cell.
- **Safety latch:** it monitors open/close status, allows opening the drawer electronically and it assures closed position repeatability. The latch is attached to the support, while the bolt is attached to the moving part. The bolt position is adjustable in order to adjust the cosmetic front panel position.

Bottles and card disposal container can be removed when the drawers are fully opened. The amount of liquid in bottles or cards in cards disposal container is monitored by a load cell, located at the rear part of the drawer.

- **Digital pumps:** they are responsible for aspirating and dispensing samples, reagents, diluents and dilutions at the required accuracy. There are two digital pumps, one for each probe, located in the upper floor on the left and with a syringe of 1 ml. Communication with each dilutor is through 2-wire RS-485.

- **Correct Pipetting Detector (CPD):** there are two pressure sensors located in the interconnection manifold of the Y-axis of the PIP arm that connect each dilutor with its probe. They monitor pressure changes during aspiration / dispensation to detect obstruction in the fluidic system.



Figure 484 - CPDs in the interconnection manifold

- **Electronics:** provide interface between the fluidic components (electrovalves, pumps, sensors) and the instrument control. Additionally, they provide interface with the lockers and with the bottle weighting system. It is located rear the instrument, on the lower floor.

- **Interconnection manifolds:** they interconnect tubing in order to allow mechanical module union. There are three units in the Erytra:
 - To join the tubing that comes from the fluidics module to the tubing of the X-axis module. Located at the rear part of the digital pumps. BK2.
 - To join the tubing that comes from the X-axis module to the tubing of the Y-axis module. Located at the rear part of the Y-axis of the PIP arm. BK3.



Figure 485 - Location of the interconnection manifolds, BK2 (left) and BK3 (right)

- To join the tubing that comes from the Y-axis module to the probes. Located at the rear part of the head of the PIP arm. BK6.



Figure 486 - Location of the interconnection manifold (c)

- **Tubing:** provide connection between the fluidic components. It is made with the following types (description, dimensions IDxOD):
 - PVC, 4x6 mm.
 - Multilumen, 2.3x3.6 mm.
 - Silicone, 3x5 mm (not present in fluidic system including expansion vessels).
 - PTFE, 1.1x1.7 mm.
 - Shielded PTFE tubing.

Multilumen tubing is a set of eight tubes extruded together in ribbon shape. It is used in chains tubing and when many lines go across the instrument together. The number of tubes of the set varies along the instrument: the multilumen inside the chains has eight tubes, the multilumen of the dilution cup has four tubes and the multilumen that goes to the interconnection manifold located in the X-axis has three tubes related to probes washer solution B.

- **Module support:** this support integrates the module components and gives integration with the rest of the instrument. It is screwed to the base module by means of four silentblocks.



Figure 487 - Fluidic module support with all the components

- **Fluidic metallic plates:** these metal plates protect the manifold from possible spillages from the manifold.



A general block diagram of the module is shown in the following figure:

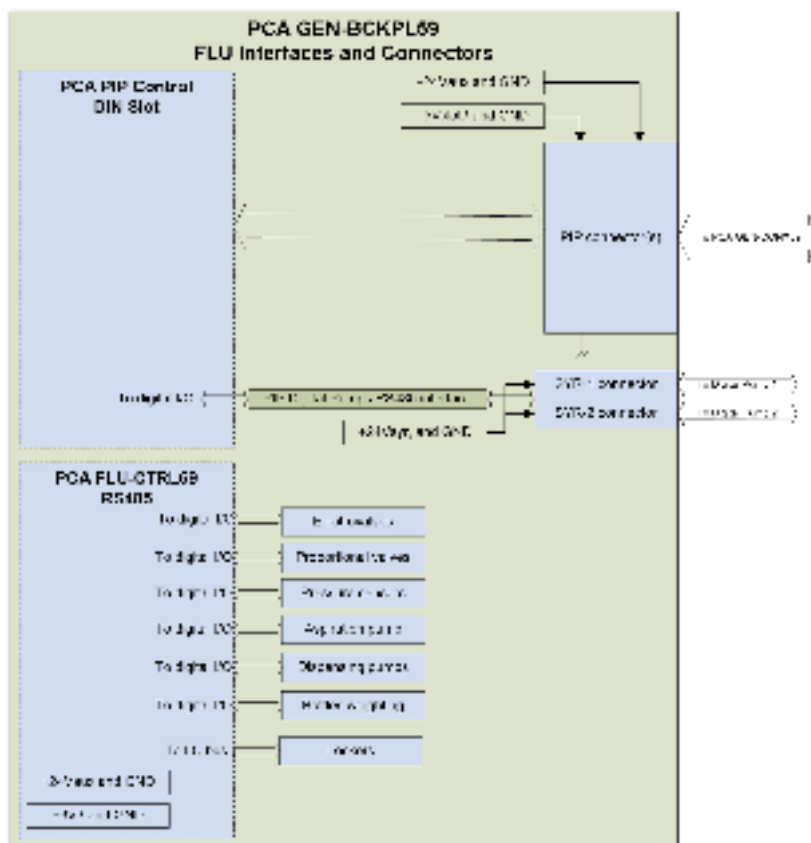


Figure 488 - Fluidic block diagram

The PCAs related to the fluidics module are:

- **PCA GEN-BCKPL59:** main PCA of the instrument, located in the backpanel. It includes an interface to convert three Digital I/O signals of PCA PIP Control (transmission, reception and transmission/reception enable) into a RS-485 2-wire bus. This interface supports 38400 bauds communications speed minimum. This bus is used to communicate with both digital pumps. There are two connectors, SYR-1 and SYR-2, to connect both dilutors and it interfaces +24V and GND supply rails to SYR-1 and SYR-2 connectors.
- **PCA FLU-CTRL59:** located at the module support part, close to the manifold. It controls the fluidic system and load cells. It includes a HD15 connector type to connect the module with PCA GEN-BCKPL59.
- **PCA GEN-LCK59:** it supports the latch electronic control. It contains the safety latch interface (activate / deactivate) and it monitors its status (opened/closed). This PCA is located inside the drawer at the frontal part and it is protected from liquid spills by means of a cover.
- **PCA PIP-CONY59:** it contains the sensor and interface to control the CPD of the probes. It is located at the Y-axis back (looking from the front of the instrument). Other parts are detailed on the PIP arm module.

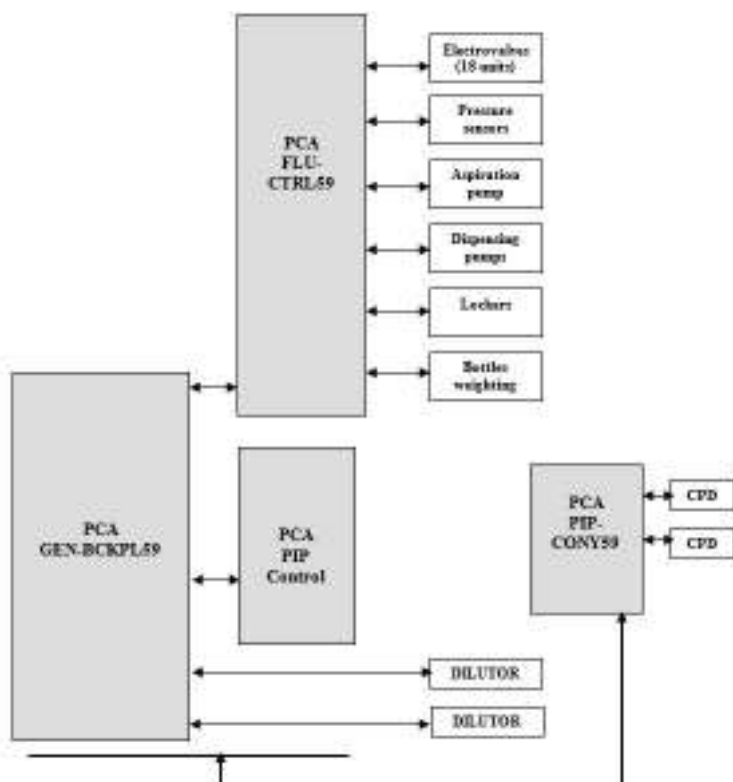


Figure 489 - Main blocks of the Fluidics module

2.6.2 Fluidic procedures

In this chapter, the different fluidics procedures are explained.

2.6.2.1 Priming

Priming processes can be divided in two main processes: low priming (manifold) and high priming (from manifold to probes, dilution cups, etc.). High priming can also be divided in some smaller processes: high priming with solution B, high priming with solution A, dilution cup priming and syringes priming.

In each fluidic step where S3 and S4 sensors are monitored, a previous checking of all the sensors is performed to monitor their status.

2.6.2.1.1 Low priming

The general aim of the low priming is to prime the circuit from bottles to manifold and it can be done for solution A and solution B.

1. (no external drain) Check waste levels.
2. (no external drain) Select waste bottle:
 - In case of using waste bottle 1: clear EV17 and EV18.
 - In case of using waste bottle 2: set EV17 and EV18.

3. (no external drain/external drain) Check waste circuit.
4. (external drain) Clear EV16.
5. (no external drain/external drain) Fill with solution B bottle drawer 2, path between EV15 and EV13 and waste bottle/external drain:
 - a. Switch on PM1, EV13, EV19 and EV20. Wait 40 s.
 - b. Check pressure $S3 = 200 \text{ mbar} \pm 150 \text{ mbar}$.
 - c. Clear EV13, EV19 and EV20.
 - d. Clear PM1.
6. (no external drain/external drain) Fill with solution B bottle drawer 2 path between solution B bottle and waste bottle/external drain (fast prime solution B):
 - a. Set PM3 and EV19. Wait 40 s.
 - b. Check pressure $S4 = 180 \text{ mbar} \pm 100 \text{ mbar}$.
 - c. Clear EV19.
 - d. (external drain) Clear EV14 and EV16.
 - e. Clear PM3.
7. (external drain) Fill with solution B path between solution B bottle drawer 5 and external drain (fast prime solution B):
 - a. Set EV16.
 - b. Set PM3 and EV19. Wait 40 s.
 - c. Check pressure $S4 = 180 \text{ mbar} \pm 100 \text{ mbar}$.
 - d. Clear EV19, EV14 and EV16.
 - e. Clear PM3.
8. (no external drain/external drain) Fill with solution A path between solution A bottle drawer 1 and waste bottle/external drain (fast prime solution A):
 - a. Set PM1 and EV20. Wait 40 s.
 - b. Check pressure $S3 = 180 \text{ mbar} \pm 100 \text{ mbar}$.
 - c. Clear EV20.
 - d. (external drain) Clear EV14 and EV16.
9. (external drain) Fill with solution A path between solution A bottle drawer 3 and external drain (fast prime solution A):
 - a. Set EV14.
 - b. Set PM1 and EV20. Wait 40 s.
 - c. Check pressure $S3 = 180 \text{ mbar} \pm 100 \text{ mbar}$.
 - d. Clear EV20, EV14 and EV16.

2.6.2.1.2 High priming solution B (probes washer)

The general aim of the high priming solution B is to prime the circuit from manifold to probes washer.

1. (no external drain) Check waste levels.
2. (no external drain) Select waste bottle:
 - In case of using waste bottle 1: clear EV17 and EV18.
 - In case of using waste bottle 2: set EV17 and EV18.
3. (external drain) Check bottle levels.
4. (external drain) Select solution B drawer. By default, solution B drawer 2 is used. In case of using solution B drawer 5, set EV16.
5. (no external drain/external drain) Set PM2 and PM3.
6. (no external drain/external drain) Check pressure/vacuum sensors:
 - $S4 = 180 \text{ mbar} \pm 30 \text{ mbar}$.
 - $S5 < -500 \text{ mbar}$, $S6 < -500 \text{ mbar}$.
7. (no external drain/external drain) Store S5 and S6 values.
8. (no external drain/external drain) Set EV5 and EV10.
9. (no external drain/external drain) Check that vacuum is stable (S5 and S6 values variation is less than 10 mbar).
10. (no external drain/external drain) Check that there is a difference in vacuum before and after opening aspiration electrovalves (difference at least of 40% between current S5 and S6 values and the stored ones).
11. (no external drain/external drain) Store S5 and S6 values.
12. (no external drain/external drain) Set EV1 and EV6. Wait 50 s.
13. (no external drain/external drain) Check that vacuum increases (S5 and S6 values are lower than the stored ones).
14. (no external drain/external drain) Check that liquid arrives to the probes washer (there is a difference at least of 10% between current S5 and S6 values and the stored ones).
15. (no external drain/external drain) Clear EV1, EV6, EV5 and EV10.

2.6.2.1.3 High priming solution A (dilution cup)

The general aim of the high priming solution A is to prime the circuit with solution A from manifold to dilution cup.

1. (external drain) Check bottle levels.
2. (external drain) Select solution A drawer. By default, solution A drawer 1 is used. In case of using solution A drawer 4, set EV14.
3. (no external drain/external drain) Set PM1 and PM2.
4. (no external drain/external drain) Check pressure/vacuum sensors:
 - $S3 = 180 \text{ mbar} \pm 10 \text{ mbar}$.
 - $S5 < -400 \text{ mbar}$, $S6 < -400 \text{ mbar}$.
5. (no external drain/external drain) Store S5 and S6 values.
6. (no external drain/external drain) Set EV4 and EV9.

7. (no external drain/external drain) Check that vacuum is stable (S5 and S6 values variation is less than 10 mbar).
8. (no external drain/external drain) Check that there is a difference in vacuum before and after opening aspiration electrovalves (there is a difference at least of 40% between current S5 and S6 values and the stored ones).
9. (no external drain/external drain) Store S5 and S6 values.
10. (no external drain/external drain) Set EV3 and EV8. Wait 20 s.
11. (no external drain/external drain) Check that vacuum increases (S5 and S6 values are lower than the stored ones).
12. (no external drain/external drain) Check that liquid arrives to the dilution cup (there is a difference at least of 10% between current S5 and S6 values and the stored ones).
13. (no external drain/external drain) Clear EV3, EV8, EV4 and EV9.

2.6.2.1.4 Syringe priming

The general aim of the syringe priming is to prime the probes and their related fluidic circuit.

1. Perform one obstruction test:
 - a. Set dilutor valves to input position and dilutor parameters for PRIMING/RINSE ASPIRATION.
 - b. Set PM2.
 - c. Set EV5 and EV10.
 - d. (external drain) In case of taking solution A from solution A drawer 4, set EV14.
 - e. Make an aspiration movement with both dilutors.
 - f. Make a dispensation movement with both dilutors.
 - g. Check obstruction in both probes (DCP).
2. 12 cycles of aspiration and dispensation.
3. Five cycles of aspiration and dispensation. Check that the mean of S5 and S6 sensors when the cycles have finished is > 10% than the mean of S5 and S6 sensors before the five cycles started.
4. Clear EV5 and EV10. Clear EV14 if activated.

2.6.2.2 Rinsing

There are four different rinse procedures in the system: low rinse, high rinse solution B, dilution cup rinse and syringe rinse.

In each fluidic step where S3 and S4 sensors are monitored, a previous checking of all the sensors is performed to monitor their status.

2.6.2.2.1 Low rinse

The general aim of this procedure is to rinse the circuit from bottles to manifold.

1. (no external drain) Check waste levels.
2. (no external drain) Select waste bottle:
 - In case of using waste bottle 1: clear EV17 and EV18.
 - In case of using waste bottle 2: set EV17 and EV18.
3. (no external drain) Check waste circuit.
4. (no external drain/external drain) Fill with solution B bottle drawer 2, path between EV15 and EV13 and waste bottle/external drain:
 - Switch on PM1, EV13, EV19 and EV20. Wait 40 s.
 - Check pressure S3 = 200 mbar $\pm$ 150 mbar.
 - Clear EV13, EV19 and EV20.
 - Clear PM1.
5. (no external drain/external drain) Fill with solution B bottle drawer 2 path between solution B bottle and waste bottle/external drain (fast prime solution B)
 - a. Set PM3 and EV19. Wait 40 s.
 - b. Check pressure S4 = 180 mbar $\pm$ 100 mbar.
 - c. Clear EV19.
 - d. Clear PM3.
6. (no external drain/external drain) Fill with solution B bottle drawer 2, path between EV15 and EV13 and waste bottle/external drain:
 - a. Switch on PM1, EV13 and EV20. Wait 40 s.
 - b. Check pressure S3 = 200 mbar $\pm$ 150 mbar.
 - c. Clear EV13 and EV20.

2.6.2.2.2 High rinse probes washer

The general aim of this procedure is to rinse the circuit from manifold to probes washer with solution B.

1. (external drain) Check bottle levels.
2. (external drain) Select solution B drawer. By default, solution B drawer 2 is used. In case of using solution B drawer 5, set EV16.
3. (no external drain/external drain) Set PM2 and PM3.
4. (no external drain/external drain) Check pressure/vacuum sensors:
 - S4 = 180 mbar $\pm$ 10 mbar.
 - S5 < -500 mbar, S6 < -500 mbar.
5. (no external drain/external drain) Store S5 and S6 values.
6. (no external drain/external drain) Set EV5 and EV10.

7. (no external drain/external drain) Check that vacuum is stable (S5 and S6 values variation is less than 10 mbar).
8. (no external drain/external drain) Check that there is a difference in vacuum before and after opening aspiration electrovalves (there is a difference at least of 40% between current S5 and S6 values and the stored ones).
9. (no external drain/external drain) Store S5 and S6 values.
10. (no external drain/external drain) Set EV1 and EV6. Wait 50 s.
11. (no external drain/external drain) Check that vacuum increases (S5 and S6 values are lower than the stored ones).
12. (no external drain/external drain) Check that liquid arrives to the probes washer (there is a difference at least of 10% between current S5 and S6 values and the stored ones).
13. (no external drain/external drain) Clear EV1, EV6, EV5 and EV10.

2.6.2.2.3 High rinse dilution cup

The general aim of this procedure is to rinse the circuit from manifold to dilution cup with solution B.

1. (external drain) Check bottle levels.
2. (external drain) Select solution B drawer. By default, solution B drawer 2 is used. In case of using solution B drawer 5, set EV16.
3. (no external drain/external drain) Set PM1, PM2 and EV13.
4. (no external drain/external drain) Check pressure/vacuum sensors:
 - $S3 = 180 \text{ mbar} \pm 10 \text{ mbar}$.
 - $S5 < -400 \text{ mbar}$, $S6 < -400 \text{ mbar}$.
5. (no external drain/external drain) Store S5 and S6 values.
6. (no external drain/external drain) Set EV4 and EV9.
7. (no external drain/external drain) Check that vacuum is stable (S5 and S6 values variation is less than 10 mbar).
8. (no external drain/external drain) Check that there is a difference in vacuum before and after opening aspiration electrovalves (there is a difference at least of 40% between current S5 and S6 values and the stored ones).
9. (no external drain/external drain) Store S5 and S6 values.
10. (no external drain/external drain) Set EV3 and EV8. Wait 20 s.
11. (no external drain/external drain) Check that vacuum increases (S5 and S6 values are lower than the stored ones).
12. (no external drain/external drain) Check that liquid arrives to the dilution cup (there is a difference at least of 10% between current S5 and S6 values and the stored ones).
13. (no external drain/external drain) Clear EV3, EV8, EV4, EV9 and EV13.

2.6.2.2.4 Syringe rinse

The general aim of this procedure is to rinse the syringes and their related circuit.

1. Perform one obstruction test:
 - a. Set dilutor valves to input position and dilutor parameters for PRIMING/RINSE ASPIRATION.
 - b. Set PM2.
 - c. Set EV13. Set EV5 and EV10.
 - d. (external drain) In case of taking solution B from solution B drawer 5, set EV16.
 - e. Make an aspiration movement with both dilutors.
 - f. Make a dispensation movement with both dilutors.
 - g. Check obstruction in both probes (DCP).
2. 12 cycles of aspiration and dispensation.
3. Five cycles of aspiration and dispensation. Check that the mean of S5 and S6 sensors when the cycles have finished is > 10% than the mean of S5 and S6 sensors before the 5 cycles started.
4. Clear EV5 and EV10.
5. Clear EV13.

2.6.2.3 Emptying circuits

The general aim of this procedure is to empty the fluidic circuit when a shutting down order is given, after the final wash.

2.6.2.3.1 EXPANSION VESSELS

1. (no external drain) Check waste level.
2. (no external drain) Select waste bottle:
 - a. In case of using waste bottle 1: clear EV17 and EV18.
 - b. In case of using waste bottle 2: set EV17 and EV18.
3. (no external drain/external drain) Check waste circuit.
4. (no external drain/external drain) Set PM2, EV15, EV2 and EV13.
5. (no external drain/external drain) Set EV3 and wait 15 s.
6. (no external drain/external drain) Check pressure level S3 > -250 mbar.
7. (no external drain/external drain) Clear EV3.
8. (no external drain/external drain) Set EV8 and wait 15 s.
9. (no external drain/external drain) Check pressure level S3 > -250 mbar.
10. (no external drain/external drain) Clear EV8.
11. (no external drain/external drain) Set PM2, EV15, EV2, EV13 and EV7.
12. (no external drain/external drain) Set EV1 and wait 15 s.
13. (no external drain/external drain) Check pressure level S4 > -250 mbar.
14. (no external drain/external drain) Clear EV1.

15. (no external drain/external drain) Set EV6 and wait 15 s.
16. (no external drain/external drain) Check pressure level S4 > -250 mbar.
17. (no external drain/external drain) Clear EV6, EV15, EV2, EV13 and EV7.
18. (no external drain/external drain) Set PM2 and EV5 and wait 1 s.
19. (no external drain/external drain) Clear EV5.
20. (no external drain/external drain) Set EV10 and wait 1s.
21. (no external drain/external drain) Clear EV10.
22. (no external drain/external drain) Set PM2 and EV4 and wait 1 s.
23. (no external drain/external drain) Clear EV4.
24. (no external drain/external drain) Set EV9 and wait 1s.
25. (no external drain/external drain) Clear EV9.
26. (no external drain/external drain) Set EV13, EV15, EV1 and EV3.
27. (no external drain/external drain) Check pressure sensors:
 - S3 > -50 mbar.
 - S4 > -50 mbar.
28. (no external drain/external drain) Clear EV1 and EV3.
29. (no external drain/external drain) Set PM1, PM3, EV13, EV15, EV4 and EV5.
30. (no external drain/external drain) Check pressure sensors:
 - S3 = 180 ± 30 mbar.
 - S4 = 180 ± 30 mbar.
31. (no external drain/external drain) Set EV19 and EV20.
32. (no external drain/external drain) Check pressure/vacuum sensors:
 - S3 < 150 mbar.
 - S4 < 150 mbar.
 - S5 < 150 mbar.
33. (no external drain/external drain) Clear EV19, EV20, PM1, PM3, EV4, EV5, EV13 and EV15.
34. (no external drain/external drain) Set dilutor 1 valve to bypass position.
35. (no external drain/external drain) Set PM2, EV15 and EV13 and wait 10 s.
36. (no external drain/external drain) Clear PM2, EV15 and EV13.
37. (no external drain/external drain) Set dilutor 1 valve to probe position.
38. (no external drain/external drain) Set dilutor 2 valve to bypass position.
39. (no external drain/external drain) Set PM2, EV15 and EV13 and wait 10 s.
40. (no external drain/external drain) Clear PM2, EV15 and EV13.
41. (no external drain/external drain) Set dilutor 2 valve to probe position.

2.6.2.4 Washing

There are five different washing procedures in the system: internal washing, fast/slow external washing, probe end washing and dilution cup washing.

2.6.2.4.1 Internal washing

The general aim of the internal washing is to wash internally the probes with a specific volume and cycles determined by the software.

1. (no external drain/external drain) Check bottle levels.
2. (no external drain/external drain) Switch on PM2.
3. (no external drain/external drain) Set EV5 and EV10.
4. (external drain) If using solution A drawer 4, set EV14.
5. (no external drain/external drain) Check vacuum sensors: S5 or S6 between -450 mbar and -50 mbar.
6. (no external drain/external drain) Move probes up to high position.
7. (no external drain/external drain) Aspirate and dispense with both dilutors the number or cycles and volume determined by the scheduler.
8. (no external drain/external drain) Clear EV5 and EV10.

2.6.2.4.2 Fast/slow external washing

The general aim of this washing is to wash externally the probes

1. (no external drain/external drain) Check bottle levels.
2. (no external drain) Check waste level.
3. (no external drain/external drain) Switch on PM2 and PM3. Set EV5 and EV10.
4. (no external drain/external drain) Check vacuum sensors S5 or S6 between -450 mbar and -100 mbar. Timeout = 5 s.
5. (no external drain/external drain) Check average pressure $S4 = 160 \text{ mbar} \pm 50 \text{ mbar}$.
6. (no external drain/external drain) Set EV1 and/or EV6.
7. (external drain) If using solution B drawer 5, set EV16.
8. (no external drain/external drain) Set washing probe speed.
9. (no external drain/external drain) Move probes up.
10. (no external drain/external drain) Clear EV1 or EV6.

2.6.2.4.3 Probe end washing

The general aim of this washing is to wash internally the end of the probes with a specific number of cycles determined by the software.

1. (no external drain/external drain) Check bottle levels.
2. (no external drain/external drain) Switch on PM2. Set EV5 and EV10. Set EV1 or EV6. Switch on PM3.

3. (external drain) In case of external drain connected and using solution B drawer 5, set EV16.
4. (no external drain/external drain) Check average pressure S4 (160 mbar $\pm$ 50 mbar).
5. (no external drain/external drain) Check vacuum sensors S5 or S6 between -450 mbar and -100 mbar. Timeout = 5 s.
6. (no external drain/external drain) Set washing probe speed.
7. (no external drain/external drain) Move probes up and down the number of cycles determined by the scheduler.
8. (no external drain/external drain) Clear EV1 or EV6.
9. (no external drain/external drain) Clear EV5 and EV10.

2.6.2.4.4 Dilution cup washing

The general aim of this washing is to wash during routine the dilution cup with a specific number of cycles determined by the scheduler.

1. (no external drain/external drain) Check bottle levels.
2. (external drain) In case of external drain connected and using solution A drawer 4, set EV14.
3. (no external drain) Check waste level.
4. (no external drain/external drain) Switch on PM2. Set EV4 and/or EV9.
5. (no external drain/external drain) Check vacuum sensors S5 or S6 between -450 mbar and -50 mbar.
6. (no external drain/external drain) Clear EV4 and/or EV9.
7. (no external drain/external drain) Switch on PM1.
8. (no external drain/external drain) Check pressure S3 = 160 mbar $\pm$ 50 mbar.
9. (no external drain/external drain) Set EV3 and/or EV8.
10. (no external drain/external drain) Clear EV3 and/or EV8.
11. (no external drain/external drain) Switch on PM2. Set EV4 and/or EV9.
12. (no external drain/external drain) Check vacuum sensors S5 or S6 between -450 mbar and -50 mbar.
13. (no external drain/external drain) Clear EV4 or EV9.
14. (no external drain/external drain) If there are more cycles programmed by the scheduler, repeat steps 4 to 11 this number of cycles.

2.6.2.5 Managing overpressure

Before starting any fluid process, a system status check is carried out.

In the event of a pump being activated in a situation where pressure is already high (should have descended in the previous step but did not achieve it) the fluidic procedure may fail and the software will keep re-trying, but as long as pressure is not released, the error will keep appearing. So, in cases where all EVs related to an active dispensation pump are closed and pressure is > 25 mbar above threshold, it:

- Opens EV20 during 50 ms (related to a high pressure in S3 or Dispensation Pump PM1), or:
- Opens EV19 during 50 ms (related to a high pressure in S4 or Dispensation Pump PM3)

This algorithm is repeated up to three times per second in order to allow pressure to descend. It remains in this state until the dispensation pump is closed, or any other EV associated to the sensor (thus dispensation pump) is open.

**NOTE**

This process implies that sounds of constant electrovalve activation / deactivation might be heard during this operation.

2.6.3 Mechanical adjustments

Mechanical adjustments of the fluidic module are detailed.

2.6.3.1 Adjusting the safety latch of the drawers of the bottles

This adjustment is required to set the depth of the drawer front into the instrument in order to align it with the rest of the drawers and the instrument frontal.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the drawer to be adjusted and remove the bottle/cards disposal container. The safety latch is located underneath the drawer.



Figure 490 - Location of the latch in the bottles / cards disposal container drawers

2. Loosen the nut attached to the safety latch, and loosen or screw the safety latch in order to align the drawer. Once the drawer is aligned, tighten the nut.

2.6.3.2 Adjusting the cosmetic front panel of the drawers of the bottles

This adjustment is required to set the cosmetic frontal panel of the drawers in order to align it with the rest of the drawers and the instrument frontal.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the drawer to be adjusted and remove the bottle/cards disposal container.
2. Loosen the three screws that are attaching the front panel to the drawer. Move in Z and X axis the front panel in order to align it. Once the frontal panel is aligned, tighten the three screws and check its alignment.



Figure 491 - Location of the screws that attach the front panel to the drawer

2.6.4 Functional adjustments

Functional adjustments of the fluidic module are detailed in this section.

2.6.4.1 Adjusting load cells of the bottle drawers

This adjustment is required to set the full scale and the offset of the load cell.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Drawers closed.

Procedure

1. Start the adjustment wizard pressing the **gauge** button in the Technical Service software **Procedures** tab > **Adjustments** section > **FLU floor Modules** area. See "Procedures tab" on page 756.
2. Select the drawer to be adjusted (solution A, solution B, waste 1 or waste 2). Press the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button. The selected drawer will be automatically opened.
3. Place an empty bottle in the drawer.
4. Close the drawer and wait until a stable measure is obtained in the software main interface, **Fluidics** tab. See "Fluidics tab" on page 826. Then press the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button.
5. After drawer opening, place a bottle filled with 8 litres in the drawer.
6. Close the drawer and wait until a stable measure is obtained in Technical Service software main interface, **Fluidics** tab. Then press the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button.



NOTE

When adjusting the load cells with the filled bottle (8 litres), wait to press the **Next** button until the liquid does not move to assure a stable measurement.

7. After drawer opening, remove the full bottle and close the drawer.
8. A screen appears showing the parameters to be stored. Press the **Finish** button and **OK** on the information window.
9. Press **OK** in the confirmation window that indicates that home movements will be performed. The drawer is automatically opened to place its original bottle.
10. Press **OK** to finish the adjustment procedure.



NOTE

Load cells adjustments are stored in the PCA FLU-CTRL59. In case of replacing this PCA, new adjustments must be performed. In case of restoring the adjustments from a previous backup, check whether the obtained readings are correct.

11. Verify the adjustment (See "Verifying load cell" chapter), if desired.

2.6.4.2 Adjusting the load cell of the card waste container

This adjustment is required to set the full scale and the offset of the load cell.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Drawer closed.

Procedure

1. Start the adjustment wizard pressing gauge button in the Technical Service software **Procedures** tab > **Adjustments** section > **FLU floor Modules** area. See "Procedures tab" on page 756.
2. Select the card waste tray drawer from the pull-down menu (**WASTE_TRAY**). Press the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button. The card waste tray drawer is automatically opened.
3. Place an empty card waste container in the drawer.
4. Close the drawer and wait until a stable measure is obtained in the software main interface, **Fluidics** tab. See "Fluidics tab" on page 826. Then press the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button.
5. Once the drawer is open, place a 2 kg weight into the container by either pouring 2 litres of water (recommended) or placing a known 2 kg weight in the center of the container.
6. Close the drawer and wait until a stable measure is obtained in the Technical Service software main interface, **Fluidics** tab. Then press the **Next** button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press the **OK** button.
7. Once the drawer is open, empty the container and close the drawer.
8. A screen appears showing the parameters to be stored. Press the **Finish** button and **OK** on the information window.
9. Press **OK** in the confirmation window that indicates that home movements will be performed. The drawer is automatically opened. Close the drawer.

10. Press **OK** to finish the adjustment procedure.

**NOTE**

Load cells adjustments are stored in the PCA FLU-CTRL59. In case of replacing this PCA, new adjustments must be performed. In case of restoring the adjustments from a previous backup, check whether the obtained readings are correct.

11. Verify the adjustment (See "Verifying load cell" chapter), if desired.

2.6.4.3 Adjusting pressure-vacuum sensors

This adjustment is required to set the full scale and the offset of the pressure/vacuum sensors located in the manifold.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- 247032 DIGIT. GAUGE DIGITRON MOD.2023.
- 233817 PRESSURE SENSOR ADJUSTMENT TOOL.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- Fluidic system is empty.
- Bottles disconnected.

Procedure

**NOTE**

Be careful when manipulating a sensor already connected to PCA FLU-CTRL59. Avoid that the welded connections of the PCA assembled in the sensor touch any metallic part of the instrument (for example the metallic support plate where the manifold is mounted) as it may damage PCA FLU-CTRL59.

1. Start the adjustment wizard pressing sensors button in the Technical Service software **Procedure** tab > **Adjustments** section > **FLU floor Modules** area. See "Procedures tab" on page 756.
2. Select the sensor to be adjusted (S3, S4, S5 or S6) and press the **Next** button.
3. Disconnect the sensor from the PCA FLU-CTRL59 (to do not damage cables when unscrewing it) and then unscrew the selected sensor (with the O-ring) from the manifold and press the **Next** button in the software.
4. Connect the sensor (with the O-ring) and calibrated manometer to pressure sensor adjustment tool set
5. Connect the sensor to the PCA FLU-CTRL59. Press the **Next** button in the software.



Figure 492 - Setup of the adjustment of the pressure sensors

6. Apply pressure to the syringe until the manometer measures a pressure of $500 \text{ mbar} \pm 50 \text{ mbar}$. Press the **Next** button in the software.
7. Write in the text box the measured pressure in the manometer (without decimals). Press the **Next** button in the software.
8. Disconnect the syringe from the 'T' connector and check that the manometer measures $0 \text{ mbar} \pm 10 \text{ mbar}$. Press the **Next** button in the software.
9. Write in the textbox the measured pressure in the manometer (without decimals). Press the **Next** button in the software.
10. A screen appears showing the parameters to be stored. Press the **Finish** button and **OK** on the information window.
11. Press **OK** in the confirmation window that indicates that home movements will be performed.
12. Press **OK** to finish the adjustment procedure.
13. Disconnect the sensor from the PCA FLU-CTRL59.
14. Assemble carefully the sensor (with the O-ring) to the manifold. Screw it as aligned as possible with the thread and manually to the maximum, and then an extra $\frac{1}{4}$ turn.



CAUTION

Applying too much force can break the sensor.

15. Connect the sensor again to the PCA FLU-CTRL59.



NOTE

The pressure/vacuum sensors are stored in PCA FLU-CTRL59. In case of replacing this PCA, the adjustment must be performed. In case restoring the adjustment from a previous backup, then check if the readings are correct.

16. Verify the adjustment (See "Verifying pressure sensors" chapter), if desired.

2.6.4.4 Adjusting dilutors configuration

This adjustment is required to set the correct address and configuration of the dilutors. Address selector and configuration microswitches are located at the rear part of the dilutor.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

Configure dilutor as detailed in the following table:

Table 49 - Configuration of the dilutors

Dilutor	Label	Address selector	Microswitch	Jumpers	CONFIGURATION
#1	DIL1	1	3, 7, 8: ON 1, 2, 4, 5, 6: OFF	5-6, 7-8 (factory implemented)	
#2	DIL2	2			

Dilutor #1 corresponds to the dilutor associated to the probe #1. It is the dilutor on the left, looking at the instrument from the front. Dilutor #2 corresponds to the dilutor associated to the probe #2. It is the dilutor on the right looking at the instrument from the front.

2.6.5 Verification procedures

Verifications of the fluidic module are detailed in this section.

2.6.5.1 Verifying pressure sensors



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233817 PRESSURE SENSOR ADJUSTMENT TOOL.
- 247032 DIGIT. GAUGE DIGITRON MOD.2023.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure



CAUTION

Be careful when manipulating a sensor already connected to PCA FLU-CTRL59. Avoid that the welded connections of the PCA assembled in the sensor touch any metallic part of the instrument (for example the metallic support plate where the manifold is mounted) as it may damage PCA FLU-CTRL59.

1. Disconnect the sensor from the PCA FLU-CTRL59 (to not damage cables when unscrewing it).
2. Unscrew the selected sensor (with the O-ring) from the manifold
3. Connect sensor to the PCA FLU-CTRL59.
4. Connect the pressure sensor (with the O-ring) and calibrated manometer to the pressure sensor adjustment tool set.

5. Apply pressure to the syringe and check that manometer reading and sensor reading (see "Fluidics tab" on page 826) does not differ more than the following values:
 - Pressure: 0 mbar $\pm$ 10 mbar → Reading: 0 mbar $\pm$ 10 mbar.
 - Pressure: 500 mbar $\pm$ 10 mbar → Reading: 500 mbar $\pm$ 10 mbar.
6. Disconnect sensor from the PCA FLU-CTRL59 and from pressure sensor adjustment tool.
7. Assemble carefully the sensor (with the O-ring) to the manifold. Screw it as aligned as possible with the thread and manually to the maximum, and then an extra $\frac{1}{4}$ turn.

**CAUTION**

Applying too much force can break the sensor.

8. Connect the sensor to the PCA FLU-CTRL59.

2.6.5.2 Verifying CPD sensors

There are two tests to check the circuit between the dilutors and the probes by means of the CPD sensors.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- 232572 SILICONE TUBE 4x0,76 mm (15 cm).

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

2.6.5.2.1 Leak-tightness test

1. Empty the fluidic circuit (in the Technical Service software > **Fluidics** tab > **initializing fluidic** option (see "Fluidics tab" on page 826) and then **fluidic empty** option).
2. Remove the racks of sample drawer 1.

3. Run the verification in the Technical Service software, **Procedures** tab > **Checks** section > **Others** subsection (see "Procedures tab" on page 756) > **leak-tightness** button. A wizard window is opened:

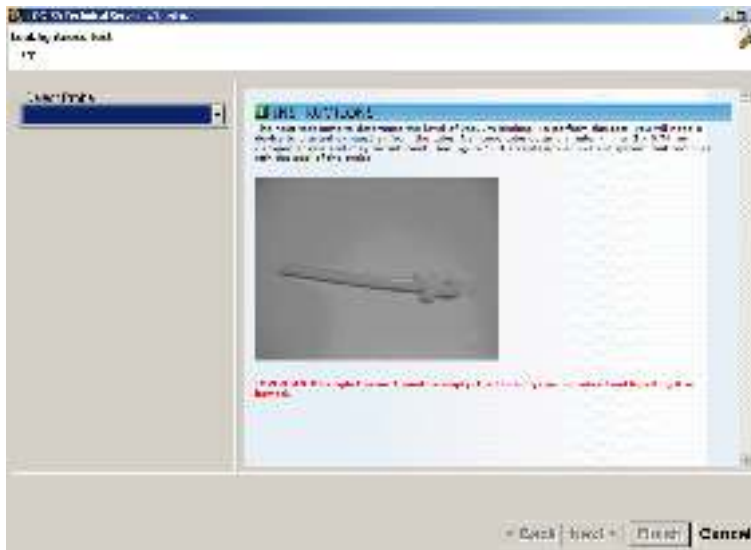


Figure 493 - Leak-tightness wizard

4. Select the probe of the circuit to be checked and press the **Next** button.
5. Press the **Next** button and then **OK** in the warning message informing that the instrument will perform movements/settings to empty the fluidic circuit between the dilutor and the probe selected. The PIP robot will move to rack 2 position of sample drawer 1, it will move down the probe and it will perform aspiration/dispensation movements with the dilutor. Then it will check if the CPD value is between 17000 and 21000 counts.
6. If the above condition is met, block the end of the probe so that air cannot enter into the probe. It is recommended to use a piece of silicone tube of 3x0.74 mm (external / internal diameter) tied by one of its sides for this purpose. Place the tube in the probe and make sure that there is no play between the probe and the tube.



Figure 494 - Blocking the end of a tube

7. Press the **Next** button to start the verification. The instrument will perform an aspiration of 1ml from the probe, so that vacuum is created inside the circuit. The system will check whether CPD value is between 6000 and 13000 counts and that the first CPD value minus the current one is less than -6000 counts. Otherwise an error will be shown on screen.
 - If the differences is more than 220 counts but less than 376 counts, it shall be considered a warning.
 - If this test fails, fluidic circuit between dilutor and the probe should be checked.
 - If the test is passed, a new step in the adjustment wizard appears on the screen requesting to remove the blockage from the probe. Remove it and press the **Next** button.
8. The instrument will then perform movements/settings corresponding to an empty procedure of the fluidic circuit between the dilutor and the probe. At this point, the CPD value should be between 17000 and 21000 counts.
9. If the above condition is met, an aspiration of 1ml(valve input position) will be made and the software will ask for the silicon tube to be placed again in the probe end so that air cannot enter into the probe. Place the tube in the probe and make sure that there is no play between the probe and the tube.
10. Press the **Next** button in order to start the verification. The instrument will perform a dispensation (valve output position), so a pressure is created inside the circuit. The system will check whether CPD value is between 26000 and 38000 counts and that the value of the first CPD value minus the current one is greater than 5700 counts. Otherwise an error will be shown on screen.
11. If CPD counts are within margins, the system will check the CPD value on that moment and 30 seconds later. Then it will calculate the difference between these 2 values, which should be smaller than 220 counts.
 - If the differences is more than 220 counts but less than 376 counts, it shall be considered a warning.
 - If this test fails, fluidic circuit between dilutor and the probe should be checked.
 - If the test is passed, a new step in the adjustment wizard appears on the screen requesting to remove the blockage from the probe. Remove it and press the **Next** button.
12. A screen appears showing the parameters obtained during the test. Press the **Finish** button and **OK** on the information window (probe will be homed).
13. Press **OK** in the confirmation window to finish the verification procedure.

2.6.5.2.2 Prime test

Perform the following steps in order to make sure that the circuit from the dilutor to the probe can be properly primed:

1. Prime the fluidic circuit (in the Technical Service software > **Fluidics** tab > **initializing fluidic** option (see "Fluidics tab" on page 826) and then the **fluidic prime** option)
2. Prime the pipetting circuit from the dilutor to the probe (in the Technical Service software > **Fluidics** tab > **prime probe 1** or **prime probe 2** option (see "Fluidics tab" on page 826).
3. Place the probe in the home position.

4. With the pressure/vacuum pumps switched off, check that the CPD value is 17700 ± 700 counts.
5. If the CPD value is out of the previous value:
 - a. If the CPD value > 18400 , the circuit contains air.
 - b. If the CPD value < 17000 , the circuit contains a possible obstruction.

2.6.5.3 Verifying dispensation pumps and expansion vessels



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to verify the performance of the dispensation pump-expansion vessel set, follow the next steps:

1. Check that waste bottles or external drain is connected and there is solution A and B liquid.
2. Empty the fluidic circuit (in the Technical Service software > **Fluidics** tab > **initializing fluidic** option (see "Fluidics tab" on page 826) and then **fluidic empty** option).
3. Check that aspiration electrovalves are closed (activate and deactivate EV5 and EV10 in the Technical Service software).
4. Activate PM1 in the Technical Service software and check that S3 sensor = $160 \text{ mbar} \pm 10 \text{ mbar}$. Deactivate PM1.
5. Activate PM3 in the Technical Service software and check that S4 sensor = $160 \text{ mbar} \pm 10 \text{ mbar}$. Deactivate PM3.
6. Perform a prime of the fluidic circuit (in the Technical Service software > **Fluidics** tab > **fluidic prime** option).
7. Perform a rinse of the fluidic circuit (in the Technical Service software > **Fluidics** tab > **fluidic rinse** option).

2.6.5.4 Verifying aspiration pump



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To verify the performance of the aspiration pump, follow the next steps:

1. Check that aspiration electrovalves are closed (activate and deactivate EV4, EV5, EV9 and EV10 in the Technical Service software). See "Fluidics tab" on page 826.
2. To check the maximum value of vacuum, check that S5 and S6 vacuum values are higher than -600 mbar when it is activated only PM2.
3. To check the minimum value of vacuum, check that S5 and S6 vacuum values are higher than -100 mbar when it is activated PM2 and EV5 and EV10.

2.6.5.5 Verifying electrovalves



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To check the performance of the electrovalves follow the next steps:

1. Activate the EV to be checked in the Technical Service software, **Fluidics** tab. See "Fluidics tab" on page 826. Check that a sound must be heard when the EV is activated / deactivated.
2. Check that the LED of the PCA FLU-CTRL59 is activated when the EV is activated in the software.

2.6.5.6 Verifying manifold



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

- Check that no leakages or scratches are observed.

2.6.5.7 Verifying load cell



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to verify the performance of a load cell, follow the next steps:

Bottle drawers:

1. Place a bottle with a known volume (4 litres recommended) in the different positions and check that the value shown by the software is the expected one.
2. Repeat the process with an empty bottle and with a one filled with 8 litres.
3. Remove the bottle and check that the software detects that it is not present.

Card waste drawer:

1. Place a 2 kg weight into the container by either pouring 2 litres of water (recommended) or placing a known 2 kg weight in the center of the container, checking that the value shown by the software is close to 410 cards.
2. Repeat the process with an empty card container, checking that the value shown by the software is close to 0 cards.

2.6.6 Replacement procedures

This section describes the replacement procedures for several parts of the fluidic system: pressure and CPD sensors, dispensation and aspiration pumps, electrovalves, manifold, load cell, fitting of the tubes, syringe and valve.

2.6.6.1 Replacing pressure sensors



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 234210 PRESSURE SENSOR (S3,S4,S5,S6).
- 734000 PRESSURE SENSOR CABLE.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To install a new pressure sensor and the corresponding cable, follow next steps:



NOTE

There are four pressure sensors in Erytra instrument.



CAUTION

Be careful when manipulating a sensor already connected to the PCA FLU-CTRL59. Avoid that the welded connections of the PCA assembled in the sensor touch any metallic part of the instrument (for example the metallic support plate where the manifold is mounted) as it may damage the PCA FLU-CTRL59.

1. Switch on the instrument and enter the **Boot Application** menu > **Technical Service** option. See "Technical Service software overview" on page 754.
2. Perform an empty of the fluidic system and disconnect all the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the rear bottom cover of the instrument in order to get access to the manifold.

5. Disconnect the pressure sensor from the PCA FLU-CTRL59 and remove the sensor (with the O-ring) from the manifold.



Figure 495 - Pressure sensors

6. Connect the new cable to the new sensor.
7. Connect the new cable and sensor to the PCA FLU-CTRL59. Do not place the sensor in the manifold yet.



Figure 496 - New cable and sensor connected to the PCA FLU-CTRL59

8. Connect the instrument to the mains, switch it on and enter the **Boot Application** menu > Technical Service option.
9. Perform the pressure sensor adjustment of the new sensor. See "Adjusting pressure-vacuum sensors" on page 579.
10. Disconnect the new sensor from the new cable.

11. Assemble carefully the new sensor (with the O-ring) to the manifold. Screw it as aligned as possible with the thread and manually to the maximum, and then an extra ¼ turn.

**CAUTION**

Applying too much force can break the sensor.

12. Connect back the new sensor to the cable.



Figure 497 - New sensor connected to new cable

13. Place the lower cover of the rear part.
14. Connect the bottles to the fluidic circuit.
15. Connect the instrument to the mains.
16. Check pressure sensors. See "Verifying pressure sensors" on page 582.

2.6.6.2 Replacing CPD sensors

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- 2x 233800 CPD SENSOR (S1,S2).
- 233799 CONNECTION BLOCK CPD SENSOR (BK4 - BK5).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Remove the middle rear cover of the instrument in order to get access to the rear part of the PIP arm.
3. Remove the screws of the rear cover of the PIP arm to get a better access to the parts.
4. Disconnect the cables of the sensors from the PCA PIP-CONY59.
5. Disconnect the tube fittings from the interconnection manifold.
6. Remove carefully the interconnection manifold in order to do not damage its cables.
7. Remove the two screws that attach both parts of the interconnection manifold and separate them smoothly.

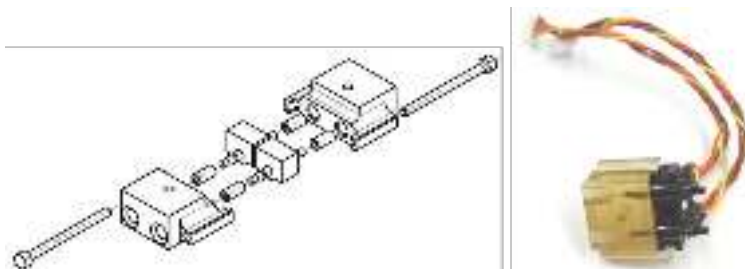


Figure 498 - CPD sensors

8. Insert silicone tubes (length 8 mm) on both new CPD sensor sides and replace it. Check that the cables of the sensor are correctly placed.
9. Join both parts of the interconnection manifold and tighten its screws. Tighten the screws to the maximum, and then a $\frac{1}{4}$ turn more.

10. Place it carefully on its position over the distributor, inserting the cables of the sensors through the distributor and making sure of the CPD orientation is as shown in below picture.



Figure 499 - CPD sensors position inside a block



NOTE

Route carefully the CPD sensors cable through the distributor.

11. Connect the tube fittings to the interconnection manifold, screwing them until a 'click' is heard.
12. Connect the cables of the sensors to the PCA PIP-CONY59.
13. Place the rear cover of the PIP arm.
14. Place the central rear cover of the instrument.
15. Perform Prime test (CPD performance). See "Verifying CPD sensors" on page 583.
16. Perform Leak-tightness test. See "Verifying CPD sensors" on page 583.

2.6.6.3 Replacing dispensation pumps



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233766 LIQUID PUMP (PM1,PM3).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. By means of the Technical Service software perform an empty of the fluidic system. See "Fluidics tab" on page 826.
2. Disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the lower cover of the rear part of the Erytra in order to get access to the manifold.
5. Disconnect the dispensation pump from the PCA FLU-CTRL59.
6. Loose the two screws of the dispensation pumps support and remove it. Take out the dispensation pump.



Figure 500 - Dispensation pump screws

7. Replace the two O-rings that are placed in the manifold.



Figure 501 - O-rings in dispensation pump support

8. Place the new dispensation pump, checking that the input and output ports of the dispensation pump match with the lodgings of the manifold. Place the dispensation pumps support and tighten the screws to the maximum, and then a ¼ turn more.
9. Connect the dispensation pump to the PCA FLU-CTRL59.

10. Connect the instrument to the mains and switch it on.
11. Check the connection of the bottles.
12. Verify the performance of the dispensation pump. See "Verifying dispensation pumps and expansion vessels" on page 586.
13. Place the lower cover of the rear part.

2.6.6.4 Replacing expansion vessels



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233916 EXPANSION VESSEL.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check that the fluidic system is empty and disconnect the bottles.
2. Switch off the instrument and disconnect it from the mains.
3. Remove the lower cover of the rear part of the instrument in order to get access to the manifold.

4. Remove the two screws of the expansion vessel and remove it.



Figure 502 - Expansion vessels screws

5. Assemble the new expansion vessel. Tighten the screws to the maximum, and then a $\frac{1}{4}$ turn more.
6. Place the lower cover of the rear part.
7. Check status of the bottles.
8. Connect the instrument to the mains and switch it on.

2.6.6.5 Replacing aspiration pump



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233960 WASTE PUMP (Liquid) (Ver. 2) (PM2).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. By means of the Technical Service software perform an empty of the fluidic system. See "Fluidics tab" on page 826.
2. Disconnect the bottles.

3. Switch off the instrument and disconnect it from the mains.
4. Remove the lower cover of the rear part of the instrument in order to get access to the manifold.
5. Disconnect the aspiration pump from the PCA FLU-CTRL59.



Figure 503 - Aspiration pump cable connection

6. Disconnect the ground cable of the aspiration pump.
7. Disconnect the fluidic tubes from the aspiration pump.



Figure 504 - Tubes and ground cable

8. Unscrew the aspiration pump and plate assembly from the basis of the fluidics and remove the assembly.



Figure 505 - Plate assembly

9. Unscrew the silentblocks of the aspiration pump and place the new pump.
10. Assemble the pump and plate to the basis of the fluidics.
11. Connect the ground cable to the aspiration pump.
12. Connect the fluidics tubes to the pump.
13. Connect the aspiration pump to the PCA FLU-CTRL59.
14. Connect the bottles.
15. Place the lower cover of the rear part.
16. Connect the instrument to the mains and switch it on.
17. Verify the performance of the pump. See "Verifying aspiration pump" on page 587.

2.6.6.6 Replacing electrovalves



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233225 ELECTROVALVE 3 WAYS.
- 233226 ELECTROVALVE 2 WAYS.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. By means of the Technical Service software perform an empty of the fluidic system. See "Fluidics tab" on page 826.
2. Disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the rear lower cover of the instrument in order to get access to the manifold.
5. Disconnect the electrovalve from the PCA FLU-CTRL59.
6. Remove the electrovalve from the manifold by means of the two screws.



Figure 506 - Electrovalve screws

7. Place the new electrovalve in its position. Tighten the screws to the maximum, and then a $\frac{1}{4}$ turn more.
8. Connect the electrovalve to the PCA FLU-CTRL59.
9. Place the lower cover of the rear part.
10. Connect the bottles.
11. Verify the performance of the electrovalve. See "Verifying electrovalves" on page 587.

2.6.6.7 Replacing manifold



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233870 MANIFOLD ASSEMBLY.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check that the fluidic system is empty and disconnect the bottles.
2. Remove the rear lower cover of the instrument in order to get access to the manifold.
3. Switch off the instrument and disconnect it from the mains.
4. Disconnect all the parts of the manifold from the PCA FLU-CTRL59.
5. Remove the two screws that attach the manifold to the fluidic basis.
6. Move up carefully the manifold and disconnect all the tubes that are located at the bottom of the manifold.
7. Place the new manifold, connecting all the tubes and all the parts to the PCA FLU-CTRL59.



NOTE

The sealant product already present in the fittings might break due to the fittings manipulation when connecting back the tubes.

8. Put a second layer of sealant product around the fittings thread once the tubes have been inserted.
9. Tighten the screws that attach the manifold to the fluidic basis to the maximum, and then a ¼ turn more.
10. Connect the instrument to the mains and switch it on.
11. Verify the performance of the manifold. See "Verifying manifold" on page 588.
12. Place the lower cover of the rear part.

2.6.6.8 Replacing manifold-distribution block

Manifold /distribution block (BK2, BK3) fittings are sealed with acetoxysilicone, recommended for use on plastic pipe threaded fittings (such as Loctite 5331).



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 733466 DISTRIBUTION BLOCK BK2 (Ver. 3).
- 733465 DISTRIBUTION BLOCK BK3 (Ver. 3).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace a fitting, follow the next steps:

1. Check that the fluidic system is empty.
2. Switch off the instrument and disconnect it from the mains.
3. Disconnect the tube from the fitting to be replaced.
4. Unscrew the fitting from the manifold/distribution block (BK2, BK3) and clean its allocation.



CAUTION

Check that no seal product gets into the manifold/distribution block (BK2, BK3) during cleaning process.

5. Place the new fitting in its allocation only tightening it a little bit, so most of the thread is out of its allocation.

6. Put the sealant product around the fitting thread. The sealant product should be spread only on the fitting thread as uniformly as possible from the middle of fitting thread toward the union point with nut. To ensure a better and ease spreading use a pointed end object.

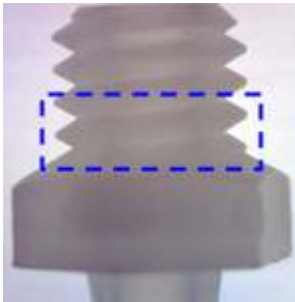


Figure 507 - Sealant product application area

7. Tighten the fitting manually to the maximum and then $\frac{1}{4}$ turn more in its allocation.



CAUTION

Check that the seal product does not block the connection between the fitting and the manifold/distribution block (BK2, BK3).

8. Connect the tube to the fitting.



NOTE

The sealant product already present in the fitting might break due to the fitting manipulation when connecting back the tube.

9. Put a second layer of the sealant product around the fitting thread once the tube has been inserted.

2.6.6.9 Replacing load cell



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233736 LOAD CELL.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the load cell, follow these steps:

1. Remove the lower cover of the rear part of the instrument in order to get access to the fluidics area, by removing the 11 screws.



Figure 508 - Screws holding the lower rear cover

2. Disconnect from the PCA FLU-CTRL59 the cable of the load cell to replace.



Figure 509 - Load cells connections to PCA FLU-CTRL59

3. Using a number 8 open end spanner, remove the four nuts that attach the manifold basis to the instrument.



Figure 510 - Nuts attaching the manifold basis to the instrument.



CAUTION

Be careful with the grounding connection when handling the manifold basis.

4. Rotate the manifold basis to access the rear part of the PCA FLU-CTRL59 and unplug the cable of the load cell of its clamps.



Figure 511 - Clamps holding load cell cable.

5. Unplug the cable from all the clamps and connectors below the manifold basis.



Figure 512 - Connector and clamp below the manifold basis

6. Remove the three screws that attach the drawer frame to the basis of the instrument.



Figure 513 - Location of screws attaching the drawer frame to the basis of the instrument.

7. Carefully rotate the basis of the drawer to get access to the screws holding the load cell underneath the basis and remove these two screws.

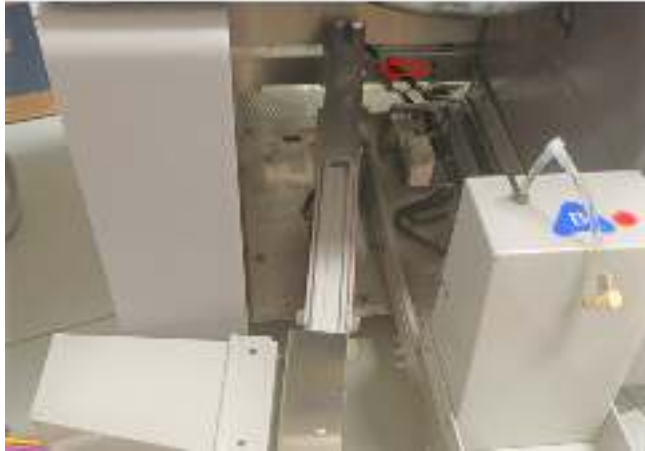


Figure 514 - Screws holding the load cell

8. Extract the load cell from the front of the instrument.

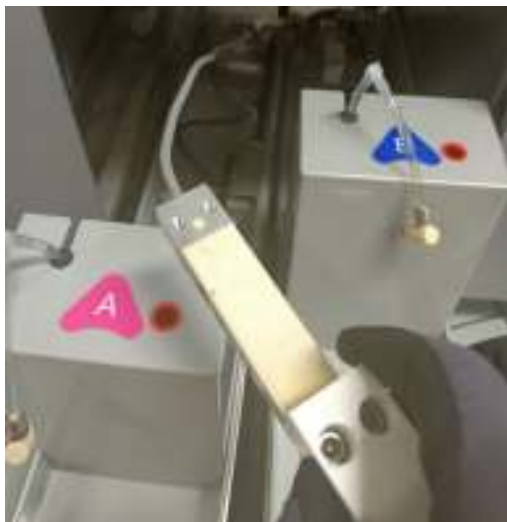


Figure 515 - Load cell removal

9. Position the new load cell on the drawer pushing it to the right to the two reference points marked on the image.

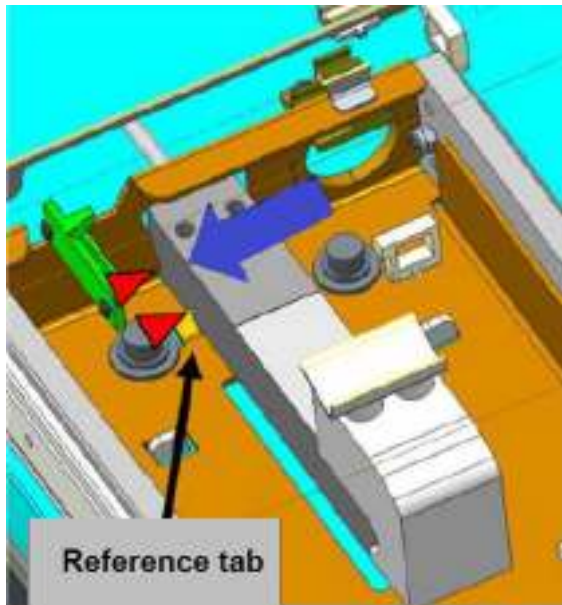


Figure 516 - Positioning of the new load cell

10. Attach the load cell to the drawer frame using the two previously removed screws.
11. Execute steps 1 to 6 in reverse order to assemble the drawer.
12. Perform the adjustment of the load cell.

2.6.6.10 Replacing fitting of the tubes



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233900 TEFLON TUBE CONNECTION (seal nut+ferrule) (Ver. 2).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect it from the mains.
2. Cut the end of the tube square.
3. Fit an Omnifit fitting nut onto the tube. Slide an Omni-lok lock ring and ferrule onto the cut tube end. Ensure the lip of the lock ring and ferrule is facing towards the tube end.



Figure 517 - Lock ring (left) and ferrule (right) tube assembly

4. Place lock ring and ferrule between two pliers and close them until the cone and case are located together but a gap between the lip of the lock ring and the lip of the ferrule remains (see below picture).
5. Close slowly both pliers until half of the gap between the lip of the lock ring and the lip of the ferrule is closed.
6. Rotate 180° the tube and close slowly the pliers until the lip of the lock ring and the lip of the ferrule are between 0 and 0.2 mm apart. Do not deform the lip of the ferrule. The assembly is now ready for use.



Figure 518 - Lock ring and ferrule assembly

7. Perform a Prime to check the fluidic system. See "Fluidics tab" on page 826.
8. Perform Leak-tightness test. See "Verifying CPD sensors" on page 583.

2.6.6.11 Replacing syringe and valve



NOTE

In case of replacing only one of the two parts it is necessary to follow the same steps, but replacing only the required part.

**NOTE**

If the instrument has the syringe locking system mounted on a MK syringe, it should be removed and discarded.

P/N [[[Undefined variable spare-parts-number.734834]]] [[[Undefined variable spare-parts-description.734834]]] and P/N 734835 MK SYRINGE 1ml assemblies do not require the syringe locking mechanism.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch on Erytra and click the bottom-left corner grid () button . The Boot application login window will be displayed.
2. Log in as `adminUser` (it is case sensitive). Boot Application options will be displayed.
3. Select the Technical Service () option .
4. Once the Technical Service software is opened, start the Driver (select **Run > Start**).
5. Move the PIP arm to its top-right position in order to obtain better access to the dilutor module.

- Initialize the dilutor by means of the Technical Service software (**Fluidics** tab, select **INIT** from the dilutor section for the syringe that is going to be replaced. Then press the **Send** button).



Figure 519 - INIT button in the Fluidics tab

- Move the syringe plunger drive to the center of the stroke by means of the Technical Service software. Go to the **Fluidics** tab and type the command **1A1500R** in the dilutor section for the syringe that is going to be replaced. Press the **send** button.



Figure 520 - Dilutor controller in the Fluidics tab

- Loosen the thumbscrew of the plunger.



Figure 521 - Plunger thumbscrew

9. Remove the syringe and valve assembly from the dilutor by loosening the two screws attaching it to the dilutor.

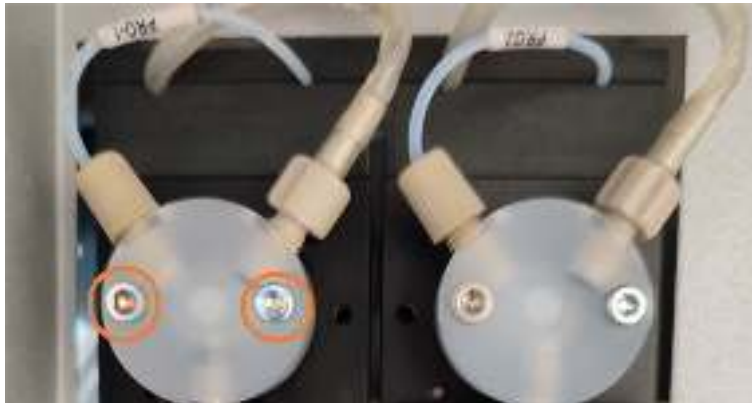


Figure 522 - Two screws attaching syringe and valve assembly to dilutor

10. Remove the output tube (PRO) first and then the input tube (SYR) while rotating the valve.

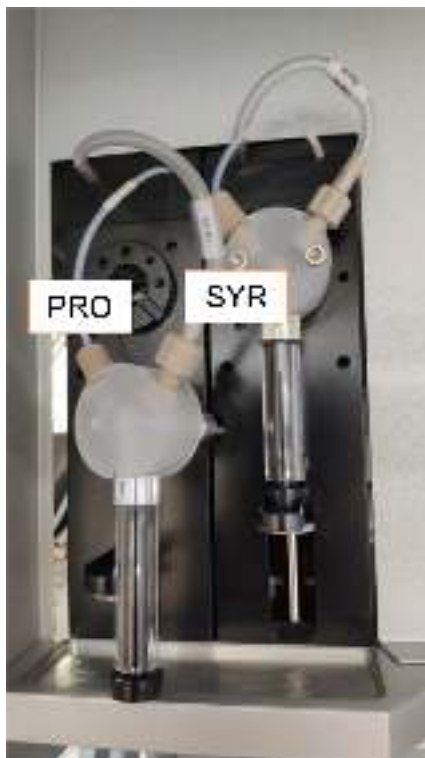


Figure 523 - Output tube (PRO) and input tube (SYR)

11. Insert the upper end of the new syringe into the valve and rotate until tight.

**NOTE**

Screw the syringe into the valve until it just touches the bottom of the valve, making sure that it is properly aligned with the valve thread. Then tighten the syringe an extra $\frac{1}{4}$ turn.



Figure 524 - Output tube (PRO) and input tube (SYR)

12. Place the new O-ring into the new valve input tubing area.



Figure 525 - New O-ring

13. Screw the input tube (SYR), rotating the valve and then the output tube (PRO) to the new valve (for output tubing is necessary to screw until a 'click' is heard).



Figure 526 - Input tube (SYR), output tube (PRO) and new valve

14. Install the new assembly by aligning the coupler on the valve stem with the hole in the encoder and fix the valve using the new screws. Do not tighten the screws completely (a small play should be present when executing the next steps).

**NOTE**

A manual valve axis rotation might be needed in order to align the coupling system between the valve system and the hole in the encoder.

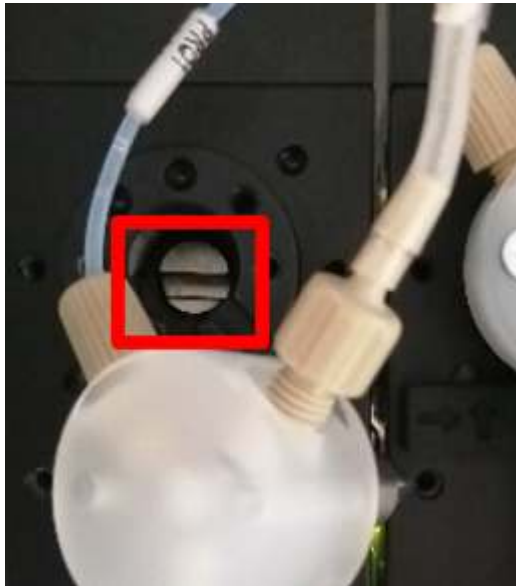


Figure 527 - Valve stem coupler and hole in encoder

15. Pull the plunger down until it reaches the plunger drive of the dilutor.
16. Position the plunger so that the plunger button sets into the channel of the drive system.



Figure 528 - Plunger button position

17. Hold the plunger and tighten the thumbscrew into the plunger. Do not tighten it completely yet.



Figure 529 - Plunger thumbscrew

18. Using the Technical Service software, bring back the plunger to initial position with an **INIT** command (see step 6 above) and then position the plunger to the end of the stroke: go to the **Fluidics** tab and select **ASPIRATE_1ML_FROM_PROBE** from the dilutor section for which the syringe is going to be replaced. Press the **Send** button.

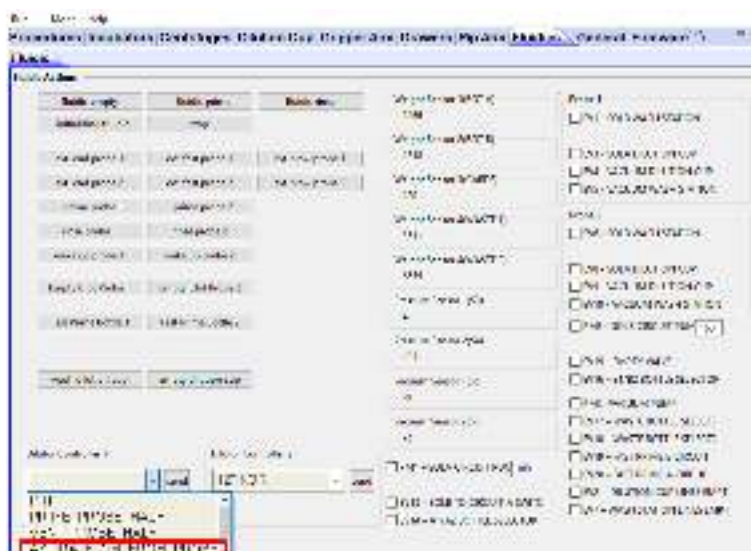


Figure 530 - ASPIRATE_1ml_FROM_PROBE button in Fluidics tab

19. Dispense the aspirated air using the **DISPENSE_TO_PROBE** command from the dilutor section for which the syringe is going to be replaced. Press the **Send** button.



Figure 531 - ASPIRATE_1ml_FROM_PROBE button in Fluidics tab

20. Tighten the thumbscrew of the plunger that was not completely tightened previously.
21. Tighten the two screws attaching the valve that were not completely tightened previously.
22. Re-tighten the syringe into the valve.
23. Initialize again the dilutor by means of the Technical Service software (**Fluidics** tab, select **INIT** from the dilutor section for the syringe that is going to be replaced. Then press the **Send** button).



Figure 532 - INIT button in the Fluidics tab

24. Repeat step 6 to 21step 2.6.6.11 above for the syringe related to probe/dilutor 2 circuit.

25. Perform a Prime of the fluidic circuit. See "Fluidics tab" on page 826.

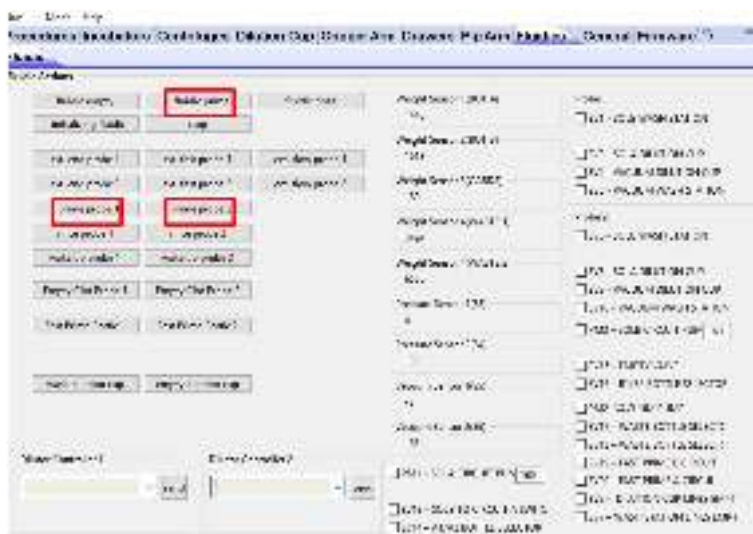


Figure 533 - Prime buttons in the Fluidics tab

26. Go to the **Procedures** tab > **Checks** section > **Others** subsection (See "Procedures tab" on page 756) and run leak-tightness test on both circuits to verify that there are no leakages and that the new valve and syringe are correctly tightened.



CAUTION

If a MK valve already installed in an instrument needs to be replaced, the valve axis may be disengaged while removing the valve from the dilutor.



2.6.6.12 Replacing fluidic drawer linear guide



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233704 LINEAR GUIDE (Cards_Bottles drawer) 500 mm.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open fluidic drawers and remove the four fluidic bottles and the card waste container.
2. Switch off the instrument and disconnect it from the mains.
3. Remove the three Allen screws that fix the fluidic drawer where is needed to replace the linear guide to lower basis.

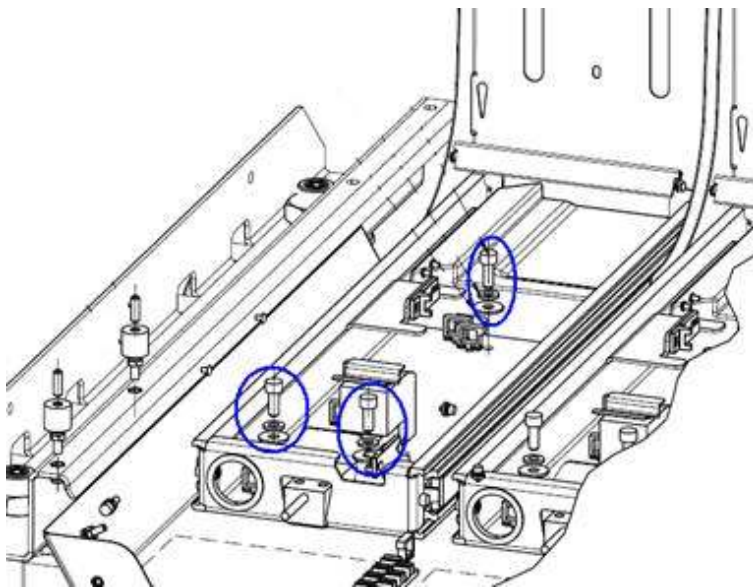


Figure 534 - Screws fixing fluidic drawer to lower basis

4. Rotate carefully the drawer depending on the linear guide (left or right) you want to replace.

5. Remove the three Allen screw of the linear guide attached to fluidic drawer lateral.

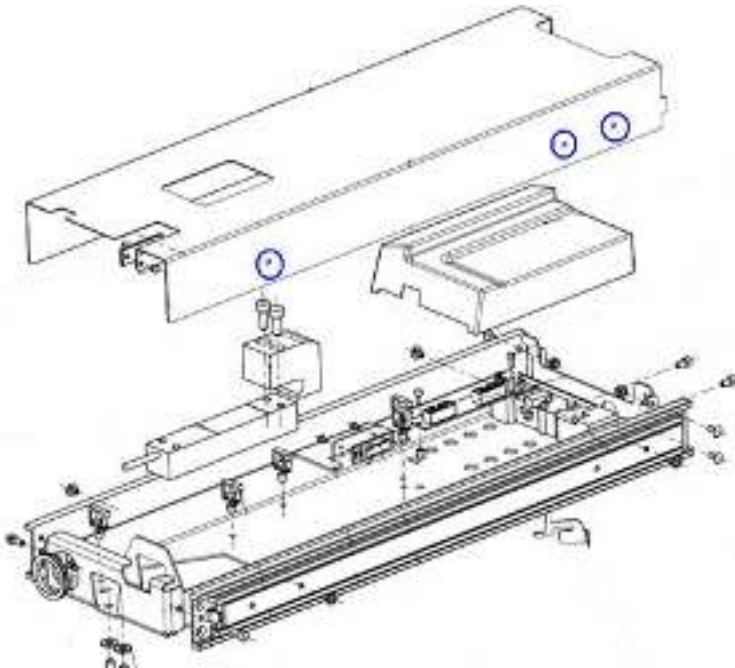


Figure 535 - Linear guide screws on fluidic drawer lateral

6. Remove the two frontal Allen screws (including nuts) and the rear one (including nut) holding the linear guide to main lateral part of the drawer.

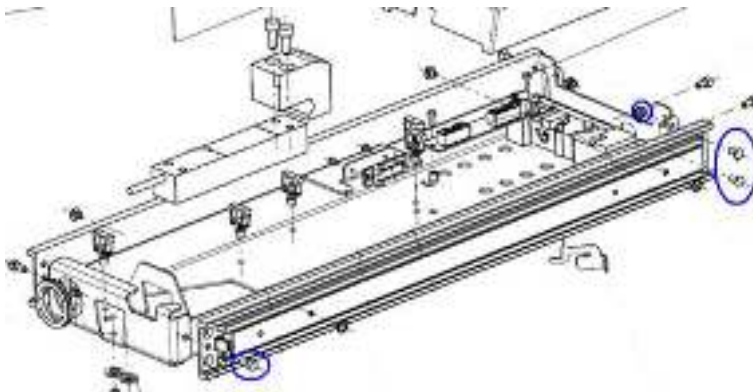


Figure 536 - Linear guide screws on drawer main lateral part

7. Remove the linear guide and install the new one, lubricating it properly.
8. After assembling back everything, check load cell adjustment. See "Verifying load cell" on page 588.

2.6.6.13 Replacing coupling CPC bottles connector



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 231894 TUBE CONNECTION CPC M (right angle)

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the fluidic drawers.
2. Disconnect the four fluidic bottles by pressing the lever and pulling the connector.



Figure 537 - CPC bottles connector lever

3. Remove the coupling CPC bottles connector unscrewing the thread and pulling the connector off the tube.



Figure 538 - CPC bottles connector replacement

4. Place the new coupling CPC bottles connector and screw the thread hand tight.
5. Connect again the four fluidic bottles.
6. Close the fluidic drawers.

2.6.6.14 Replacing coupling CPC bottles connector O-ring



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 232806 TUBE CONNECTION CPC O-RING (20 units).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open the fluidic drawers.
2. Disconnect the four fluidic bottles by pressing the lever and pulling the connector.



Figure 539 - CPC bottles connector lever

3. Remove the coupling CPC bottles connector O-ring.



Figure 540 - CPC bottles connector O-ring

4. Place the new O-ring to the coupling CPC bottles connector.
5. Connect again the four fluidic bottles.
6. Close the fluidic drawers.

2.6.6.15 Replacing bottle CPC connector and internal tube



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233920 SOL. CONTAINERS INTERNAL TUBE (w. filter)

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. If necessary, disconnect and take the bottle out of the instrument.
2. Unscrew the racord of the bottle in order to extract the internal tub.



Figure 541 - Unscrewed bottle CPC racord

3. Remove the CPC racord with the internal tube and replace it for the new one.
4. Screw the new part hand tight in the bottle.
5. Put the bottle back in place.

2.6.6.16 Replacing connection block



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233799 CONNECTION BLOCK CPD SENSOR (BK4 - BK5).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the connection block (BK4, BK5) where the CPD sensors are placed follow the next steps:

1. Open the upper door.
2. Check that the fluidic system is empty.
3. Switch off the instrument and disconnect it from the mains.
4. Disconnect the Teflon tubes on both sides of the connection blocks.

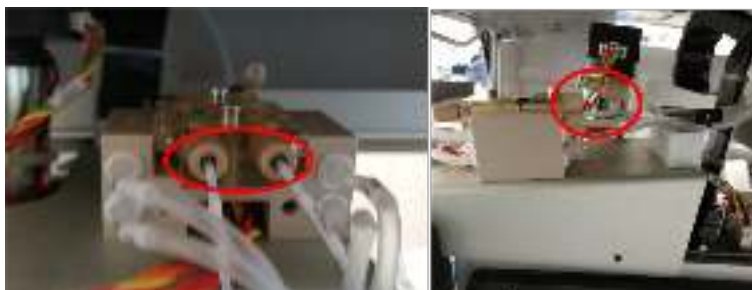


Figure 542 - Tubing

5. Remove the two screws that attach both blocks to each other (BK4, BK5).

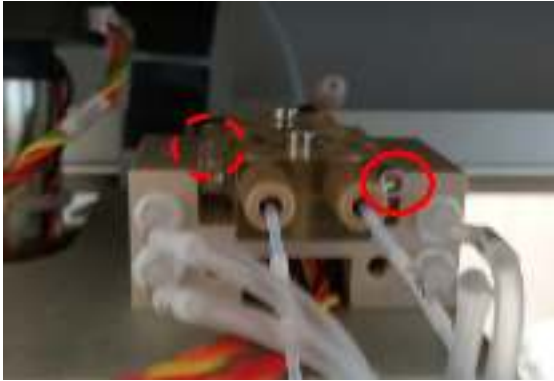


Figure 543 - Screws attaching the two blocks together

6. Remove the screw that attach BK4 to the distributor and also the remove screw that attach BK5 to the distributor.

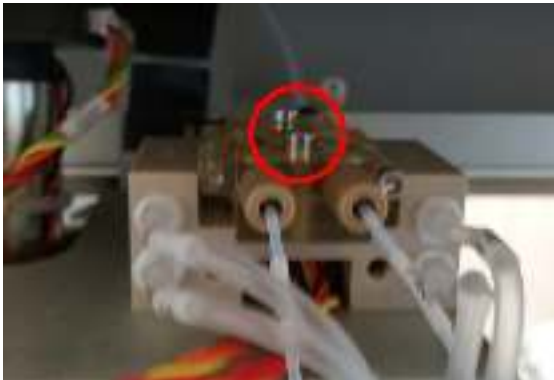


Figure 544 - Screws attaching the two blocks to the distributor

7. Carefully pull apart both CDP sensors from inside BK4 and BK5 blocks and remove BK4 and BK5 blocks.

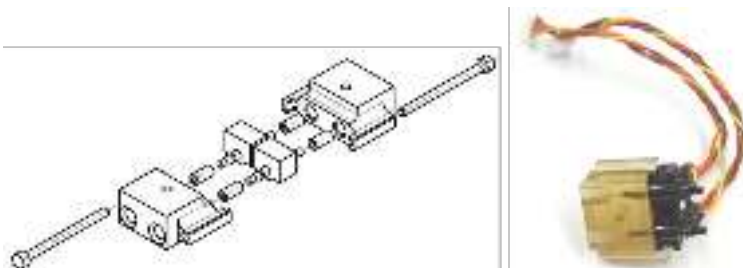


Figure 545 - CPD sensors

8. Place the new connection blocks (BK4, BK5).

9. Place the CPD sensors inside the blocks.



Figure 546 - CPD sensors position inside a block



NOTE

Route carefully the CPD sensors cable through the distributor.

10. Join both parts of the interconnection manifold and tighten its screws. Tighten the screws to the maximum, and then a ¼ turn more.
11. Assemble back all the parts.
12. Verify the performance of the CPD sensor. See "Verifying CPD sensors" on page 583.

2.6.6.17 Replacing solution A and B filter



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the solution A and B filter:

1. With the instrument switched off, open manually the electric lock of the corresponding solution drawer by pushing the lever located under the bottle drawer.
2. Disconnect and remove the bottle.



Figure 547 - Removing bottle

3. Open completely the drawer and then replace vacuum filter. Make sure the filter is correctly oriented (filtering liquid towards the manifold).



Figure 548 - Vacuum filter



NOTE

Filters must be installed in all solution A and B container lines. In case of dual mode, install the filters in all the 4 lines.

4. Perform a rinse, empty and prime to check the fluidic system. See "Fluidics tab" on page 826.

2.6.6.18 Replacing vacuum filter



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233934 VACUUM FILTERS SET.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the vacuum filter follow the next steps:

1. By means of the Technical Service software perform an empty of the fluidic system. See "Fluidics tab" on page 826.
2. Disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the lower rear cover.
5. Disconnect the vacuum filters and remove them. There are four vacuum filters (tubes C20 / C22 / C24 / C26).



Figure 549 - Vacuum filters

6. Place the new vacuum filter and assemble back all the parts. Make sure the filter is correctly oriented (filtering liquid towards aspiration pump).

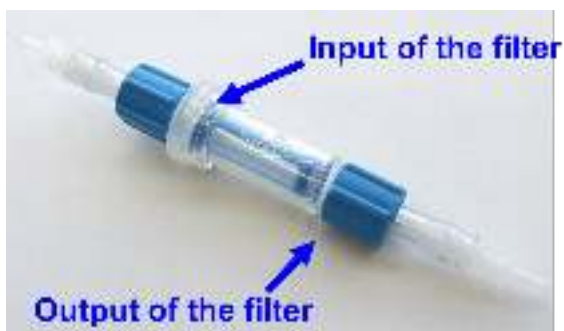


Figure 550 - Vacuum filters orientation

7. Perform a Prime to check the fluidic system. See "Fluidics tab" on page 826.

2.6.6.19 Replacing aspiration pump tubing



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 231915 TUBE PVC 6x4 (50 m.).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the aspiration pump tubing follow the next steps:

1. By means of the Technical Service software perform an empty of the fluidic system. See "Fluidics tab" on page 826.
2. Disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the lower rear cover.
5. Disconnect the tubing from the aspiration pump (C29, C31).



Figure 551 - Aspiration pump tubing

6. Remove the two screws that attach the manifold to the fluidic basis. If needed disconnect cabling from the PCA FLU-CTRL59.



Figure 552 - Manifold screws

7. Disconnect the tubing from the manifold (C29,C31).



Figure 553 - Aspiration pump tubing in the manifold

8. Place the new tubing and assemble back all the parts.



NOTE

The sealant product already present in the fitting might break due to the fitting manipulation when connecting back the tube.

9. Put a second layer of sealant product around the fitting thread once the tube has been inserted.
10. Perform a Prime to check the fluidic system. See "Fluidics tab" on page 826.
11. Check aspiration pump performance. See "Verifying aspiration pump" on page 587.

2.6.6.20 Replacing PCA FLU-CTRL59 board



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233956 PCA FLU-CTRL59 (Ver. 2).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the PCA FLU-CTRL59 follow the next steps. First a backup of the adjustments stored in the PCA FLU-CTRL59 EPROM is performed:

1. Switch on the instrument and enter the **Boot application** menu
2. Enter the Technical Service software.
3. Go to the **Firmware** tab. See "Firmware tab" on page 837.
4. Go to the **EPROM control** tab and select **FLU**.
5. Click the **Read** button.
6. Click the **Backup** button to save the EPROM values with the adjustments of the fluidic module in a file.
7. Close the Technical Service software.

Then the PCA FLU-CTRL59 is replaced:

1. Switch off the instrument and disconnect it from the mains.
2. Remove the lower rear cover.
3. Remove all cabling from the PCA FLU-CTRL59.
4. Remove the PCA FLU-CTRL59 by removing its six screws.



5. Place the new PCA FLU-CTRL59 board.
6. Reconnect all the cabling.

Finally the adjustments backup previously performed is restored to the new PCA FLU-CTRL59 EPROM:

1. Switch on the instrument and enter **Boot application** menu.
2. Enter the Technical Service software
3. Go to the **Firmware** tab. See "Firmware tab" on page 837.
4. Go to the **EPROM control** and select **FLU**.
5. Click on the **Read** button.
6. Click on the **Restore** button and select the previous backup file to restore the EPROM values with the adjustments of the fluidic module.
7. Click the **Write** button to save the adjustments to the EPROM.
8. Go to the **PCA control** tab.
9. Select **FLU** and click on the **Reset** button.
10. Check all fluidic and load cells adjustments. See "Verifying load cell" on page 588.

2.6.6.21 Replacing protector plates



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace the protector plates:

1. Remove the lower cover of the rear part of the instrument in order to get access to the fluidics area, by removing 11 screws.



Figure 554 - Screws holding the lower rear cover

2. Using a number 8 open end spanner, remove four nuts that attach the manifold basis to the instrument.



Figure 555 - Nuts attaching the manifold basis to the instrument



CAUTION

Be careful with the grounding connection when handling the manifold basis.

3. Rotate the manifold basis to access the rear part of the lower tray of the instrument.
4. Replace the two plates by inserting the front legs into each foot.



Figure 556 - Inserting cover plates

5. Return the manifold to its position.



CAUTION

Be careful with the grounding connection.



CAUTION

Make sure that all the tubes are properly routed, not pinched, and avoiding curves with small radius.



Figure 557 - Fluidic protector plates

6. Place back the lower cover of the rear part of the instrument.

2.6.7 Electronics

2.6.7.1 Fuses

Fuses related to Fluidics module are detailed in the following table.

Table 50 - List of fuses of Fluidics module

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUSGEN3 FUSGEN5	MST250 0034.6621 T4A/250V MST250 0034.6620 3.15A/250V
PCA FLU-CTRL59	FUS1	MST250 0034.6621 T4A/250V
PCA PIP-CONY 59	NA	NA

2.6.7.2 LEDs

LEDs related to Fluidics module are detailed in the following table.

Table 51 - List of LEDs of Fluidics module

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED3	Green 24V (FUSGEN3)
	LED17	Green 24V (FUSGEN5)
PCA FLU-CTRL59	LED1	EV9 activated (Yellow)
	LED2	EV10 activated (Yellow)
	LED3	EV4 activated (Yellow)
	LED4	EV5 activated (Yellow)
	LED5	EV7 activated (Yellow)
	LED6	EV8 activated (Yellow)
	LED7	EV2 activated (Yellow)
	LED8	EV3 activated (Yellow)
	LED9	EV13 activated (Yellow)
	LED10	EV14 activated (Yellow)
	LED11	EV20 activated (Yellow)
	LED12	EV17 activated (Yellow)
	LED13	EV18 activated (Yellow)
	LED14	EV19 activated (Yellow)
	LED15	NA
	LED16	EV16 activated (Yellow)
	LED17	PM1 activated (Yellow)
	LED18	PM3 activated (Yellow)
	LED19	EV15 activated (Yellow)
	LED20	EV11 activated (Yellow)
	LED21	EV12 activated (Yellow)
	LED22	EV1 activated (Yellow)
	LED23	EV6 activated (Yellow)
	LED24	PM2 activated (Yellow)
	LED25	Blinks (Yellow) during communication.
	LED26	Problem in the voltage or there is not communication with the lockers (Red).
	LED27	Blinks (Green) : - during normal operation : 1 Hz freq - error in the sensors : 5 Hz freq
PCA PIP-CONY59	NA	NA

2.6.7.3 Jumpers and test points

Neither jumpers nor test points are related to Fluidics module.

2.6.7.4 Wiring table

Cables related to the Fluidics module are detailed in the following table and their connection are shown in the wiring diagram. To view the wiring diagram, see "Related documents" on page 19 or refer to the Erytra Online Help.

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
INF-SOL.A + INF-SOL.A_EXT	H7	PCA FLU-CTRL59	NA	LOAD CELL A SOLUTION DRAWER
INF-SOL.B + INF-SOL.B_EXT	H8	PCA FLU-CTRL59	NA	LOAD CELL B SOLUTION DRAWER
INF-CARD + INF-CARD_EXT	H9	PCA FLU-CTRL59	NA	LOAD CELL CARD DRAWER
INF-WAS1 + INF-WAS1_EXT	H10	PCA FLU-CTRL59	NA	LOAD CELL WASTE 1 DRAWER
INF-WAS2 + INF-WAS2_EXT	H11	PCA FLU-CTRL59	NA	LOAD CELL WASTE 2 DRAWER
B-FLU	H13	PCA FLU-CTRL59	FLU	PCA GEN-BCKPL59
INF-LOCK + BOT-IN	H14	PCA FLU-CTRL59	H1 From Master	PCA GEN-LCK59 A SOLUTION DRAWER
NA	H3	PCA GEN-LCK59 A SOLUTION DRAWER	H	SAFETY LOCK
BOT-OUT + BOT-IN	H2 To Slave	PCA GEN-LCK59 A SOLUTION DRAWER	H1 From Master	PCA GEN-LCK59 B SOLUTION DRAWER

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
NA	H3	PCA GEN-LCK59 B SOLUTION DRAWER	H	SAFETY LOCK
BOT-OUT + BOT-IN	H2 To Slave	PCA GEN-LCK59 A SOLUTION DRAWER	H1 From Master	PCA GEN-LCK59 CARD DRAWER
NA	H3	PCA GEN-LCK59 CARD DRAWER	H	SAFETY LOCK
BOT-OUT + BOT-IN	H2 To Slave	PCA GEN-LCK59 CARD SOLUTION DRAWER	H1 From Master	PCA GEN-LCK59 WASTE 1 DRAWER
NA	H3	PCA GEN-LCK59 WASTE 1 DRAWER	H	SAFETY LOCK
BOT-OUT + BOT-IN	H2 To Slave	PCA GEN-LCK59 WASTE 1 DRAWER	H1 From Master	PCA GEN-LCK59 WASTE 2 DRAWER
NA	H3	PCA GEN-LCK59 WASTE 2 DRAWER	H	SAFETY LOCK
B-SYR1	H	DILUTOR 1	SYR_1	PCA GEN-BCKPL59
B-SYR2	H	DILUTOR 2	SYR_2	PCA GEN-BCKPL59
PIP-S1	H6 S1	PCA PIP-CONY59	S1	PRESSURE SENSOR
PIP-S2	H7 S2	PCA PIP-CONY59	S2	PRESSURE SENSOR
GROUND CABLE	ASPIRATION PUMP	ASPIRATION PUMP	FLUIDIC BASIS	FLUIDIC BASIS
GROUND CABLE	INSTRUMENT BASIS	INSTRUMENT BASIS	FLUIDIC BASIS	FLUIDIC BASIS



Figure 558 - Location of the connectors, fuses and LEDs in PCA GEN-BCKPL59 related to Fluidics module

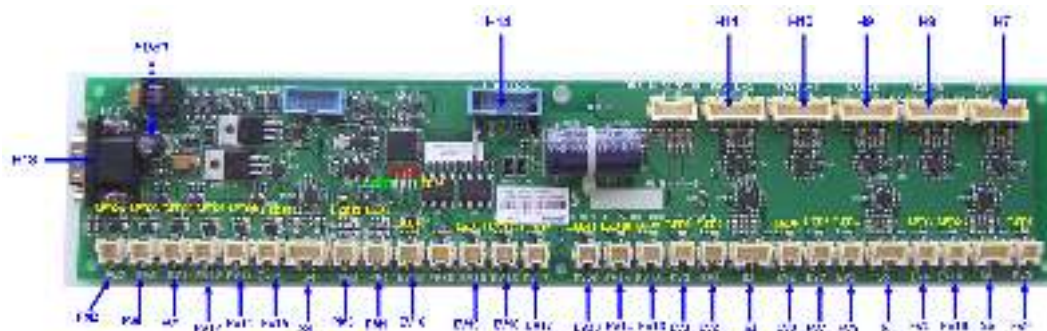


Figure 559 - Location of the connectors, fuse and LEDs in PCA FLU-CTRL59

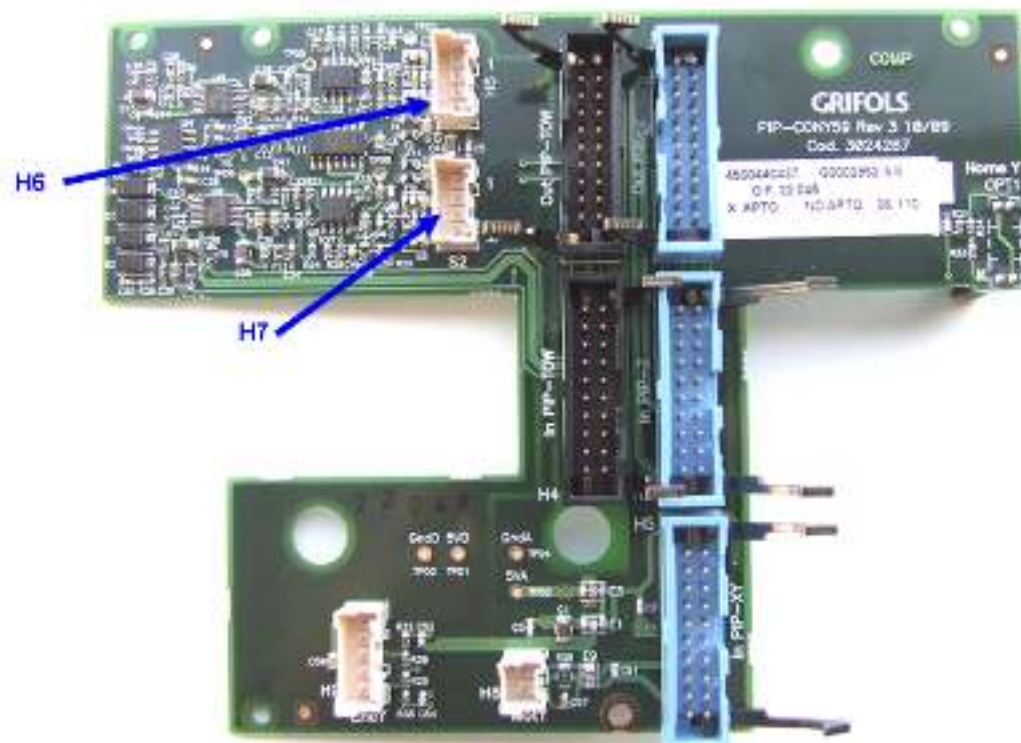


Figure 560 - Location of the connectors in PCA PIP-CONY59 related to Fluidics module

2.6.8 Troubleshooting

In fluidics module, last 3 digits errors can be 000 or a different number than 000 (it will indicate the fluidic step where the error has been found. See "Last 3 digits error code" on page 660 table).

Table 52 - List of errors related to Fluidics module

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	PCA FLU (89)	info	FLG_POWER_VOLTAGE_KO	No power supply at Fluidic PCA.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		info	FLG_SENSOR_VOLTAGE_KO	Fluidic sensors voltage error.		
		info	FLG_I2C_BUS_KO	Error I2C bus.		

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Fluidic drawers lockers (01) Drawer 1 (02)	info	FLG_LOCKER_VOLTAGE_KO	Fluidic lockers voltage error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
	Drawer 2 (03) Drawer 3 (04) Drawer 4 (05) Drawer 5	info	FLG_ACCELEROMETER_KO	NA		
FLU (05)	Sol A Lock (01)	039xxx	FLG_ERR_LOCK1	Locker error detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
	Sol A Lock (01)	077xxx	ERRORCODE_FLU_BOTTLE_A_DRAWER_IS_OPEN	Drawer A is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL
FLU (05)	Sol A Load cell (68)	075xxx	FLG_ERR_GAL1	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Sol B Lock (02)	039xxx	FLG_ERR_LOCK2	Locker error detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		078xxx	ERRORCODE_FLU_BOTTLE_B_DRAWER_IS_OPEN	Drawer B is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL
FLU (05)	Sol B Load cell (69)	075xxx	FLG_ERR_GAL2	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
	Waste tray Lock (03)	039xxx	FLG_ERR_LOCK3	Locker error detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	
FLU (05)	Waste tray Lock (03)	081xxx	ERRORCODE_FLU_WASTE_TRAY_DRAWER_IS_OPEN	Cards waste tray is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL
FLU (05)	Waste tray Load cell (70)	075xxx	FLG_ERR_GAL3	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Waste 1 Lock (04)	039xxx	FLG_ERR_LOCK4	Locker error detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		079xxx	ERRORCODE_FLU_WASTE_1_DRAWER_IS_OPEN	Waste 1 drawer is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL
FLU (05)	Waste 1 Load cell (71)	075xxx	FLG_ERR_GAL4	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
FLU (05)	Waste 1 Load cell (71)	083xxx	ERRORCODE_FLU_WASTE_1_CAPACITY_OUT_OF_MARGINS	Waste 1 bottle full.	If there are running works: System has been stopped caused by a container problem and there are pending works. Please, select one of the following options: If there are not running works: System has been stopped caused by a container problem. Please, select one of the following options:	RETRY RE-RUN OPEN

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Waste 2 Lock (05)	039xxx	FLG_ERR_LOCK5	Locker error detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		080xxx	ERRORCODE_FLU_WASTE_2_DRAWER_IS_OPEN	Waste 2 drawer is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL
FLU (05)	Waste 2 Load cell (72)	075xxx	FLG_ERR_GAL5	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
FLU (05)	Waste 2 Load cell (72)	084xxx	ERRORCODE_FLU_WASTE_2_CAPACITY_OUT_OF_MARGINS	Waste 2 bottle full.	If there are running works: System has been stopped caused by a container problem and there are pending works. Please, select one of the following options: If there are not running works: System has been stopped caused by a container problem. Please, select one of the following options:	RETRY RE-RUN OPEN

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Pressure sensor 3 (06)	069xxx	ERRORCODE_FLU_ PRESSURE_FAIL_A	Pressure error detected on solution A fluidic circuit.	If external drain is not activated:	RETRY
					Pressure error on solution A circuit. Please check:	CANCEL Works
					- Containers are properly connected. - There is no spilling in the upper level.	OPEN solution A drawer
					If external drain is activated:	RETRY
FLU (05)	Pressure sensor 3 (06)	074xxx	FLG_ERR_ PRESSURE_SENSOR_ 1	Pressure sensor malfunction detected.	Pressure error on solution A circuit. Please check:	CANCEL Works
					- Containers are properly connected. - There is no spilling in the upper level.	OPEN solution A drawer 1
						OPEN solution A drawer 2
					The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Pressure sensor 4 (07)	070xxx	ERRORCODE_FLU_ PRESSURE_FAIL_B	Pressure error detected on solution B fluidic circuit.	If external drain is not activated:	RETRY
					Pressure error on solution B circuit. Please check:	CANCEL Works
					- Containers are properly connected. - There is no spilling in the upper level.	OPEN solution B drawer
					If external drain is activated:	RETRY
FLU (05)	Pressure sensor 4 (07)	074xxx	FLG_ERR_ PRESSURE_SENSOR_ 2	Pressure sensor malfunction detected.	Pressure error on solution B circuit. Please check:	CANCEL Works
					- Containers are properly connected. - There is no spilling in the upper level.	OPEN solution B drawer 1
						OPEN solution B drawer 2
					The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Pressure sensor 5 (08)	071xxx	ERRORCODE_FLU_VACUUM_FAIL_1	Vacuum error detected on vacuum 1 fluidic circuit.	If external drain is not activated:	RETRY
					Vacuum error. Please check:	CANCEL Works
					- Containers are properly connected. - There is no spilling in the upper level.	OPEN Waste 1 drawer
						OPEN Waste 2 drawer
FLU (05)	Pressure sensor 5 (08)	Info	FLG_ERR_VACUUM_SENSOR_1	Pressure sensor malfunction detected.	If external drain is activated:	RETRY
					Vacuum error. Please check:	CANCEL Works
					- Lab drain tube and connection. - There is no spilling in the upper level.	
FLU (05)	Pressure sensor 5 (08)	Info	FLG_ERR_VACUUM_SENSOR_1	Pressure sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Pressure sensor 6 (09)	072xxx	ERRORCODE_FLU_VACUUM_FAIL_2	Vacuum error detected on vacuum 2 fluidic circuit.	If external drain is not activated:	RETRY
					Vacuum error. Please check:	CANCEL Works
					- Containers are properly connected. - There is no spilling in the upper level.	OPEN Waste 1 drawer
						OPEN Waste 2 drawer
FLU (05)	Pressure sensor 6 (09)	Info	FLG_ERR_VACUUM_SENSOR_2	Pressure sensor malfunction detected.	If external drain is activated:	RETRY
					Vacuum error. Please check:	CANCEL Works
					- Lab drain tube and connection. - There is no spilling in the upper level.	
FLU (05)	Pressure sensor 6 (09)	Info	FLG_ERR_VACUUM_SENSOR_2	Pressure sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	CPD (17) CPD 1 (18) CPD 2	085xxx	ERRORCODE_FLU_UNABLE_TO_READ_DILUTOR_1_PLUNGER_POSITION	Not possible to read dilutor 1 plunger position. FW aborts the syringe process, due to an overpressure error for example, and this causes this dilutor reading error in the driver.	NA (error is the Technical Service software).	NA
		086xxx	ERRORCODE_FLU_UNABLE_TO_READ_DILUTOR_2_PLUNGER_POSITION	Not possible to read dilutor 2 plunger position. FW aborts the syringe process, due to an overpressure error for example, and this causes this dilutor reading error in the driver.		
PIP (08)	CPD (17) CPD 1 (18) CPD 2	003xxx	ERRORCODE_FW_LEN	FW has received a command with incorrect length.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		005xxx	ERRORCODE_FW_ARG	Command has wrong arguments.		
		027xxx	ERRORCODE_DIL_DCP_DRIVER_ERROR	A DCP FW driver error has been produced.		
PIP (08)	CPD (17) CPD 1 (18) CPD 2	028xxx	ERRORCODE_DIL_DCP_OVER_PRESSURE	Pressure value beyond limits.	Obstruction detected in a probe during pipetting.	RETRY RE-RUN OPEN (option not available if error occurs in Dilution Cup)

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	CPD (17) CPD 1 (18) CPD 2	029xxx	ERRORCODE_DIL_ DCP_TIMEOUT_ ERROR	Timeout (waiting for DCP pressure sensor to return to a non-clot value) has expired.	Obstruction detected during pipetting sample(s) x located in position(s) x. Please Open the drawer and remove the obstruction.	RETRY RE-RUN OPEN
		030xxx	ERRORCODE_DIL_ DCP_CLOT	DCP value is above threshold.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		Info	ERRORCODE_ DATANOTREADY	NA	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
PIP (08)	Dilutors (17) Dilutor 1 (18) Dilutor 2	003xxx	ERRORCODE_FW_ LEN	FW has received a command with incorrect length.		
		005xxx	ERRORCODE_FW_ ARG	Command has wrong arguments.		
		006xxx	ERRORCODE_FW_ BUSY	Command is addressed to a busy device.		
		018xxx	ERRORCODE_DIL_ CORRUPTED_ CONFIGURATION_ FILE	The dilutor configuration file is unreadable.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Dilutors (17) Dilutor 1 (18) Dilutor 2	019xxx	ERRORCODE_DIL_ CONFIGURATION_ FILE_NOT_FOUND	The dilutor configuration file is unable to be opened.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		020xxx	ERRORCODE_DIL_ CANNOT_LOAD_ CONFIGURATION_ FILE	There is a problem during the dilutors configuration file reading.		
		021xxx	ERRORCODE_DIL_ UNKNOWN_ COMMAND	Dilutor has received an unknown command.		
		022xxx	ERRORCODE_DIL_ COMMAND_TOO_ LONG	A command received by the dilutor is too long.		
		023xxx	ERRORCODE_DIL_ COM_ERR	Dilutor communication error.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Dilutors (17) Dilutor 1 (18) Dilutor 2	024xxx	ERRORCODE_DIL_ COMMAND_NOT_ SENT	FW has received a command addressed to a dilutor, but the command has not been send to the dilutor.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		025xxx	ERRORCODE_DIL_ SYNC_TIMEOUT	Dilutor has been busy for more than its timeout margin.		
		026xxx	ERRORCODE_DIL_ OEM_ERROR	A dilutor internal error has been produced.		
		031xxx	ERRORCODE_DIL_ UNKNOWN_ERROR	Unknown dilutor error has been produced.		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002xxx	ERRORCODE_CMD	Unknown command code received.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003xxx	ERRORCODE_LEN	FW has received a command with incorrect length.		
		004xxx	ERRORCODE_DEV	Command is addressed to an incorrect device.		
		005xxx	ERRORCODE_ARG	Command has wrong arguments.		
		014xxx	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW.		
		015xxx	ERRORCODE_HW_HWFAIL	There is a problem with PCA HW.		
		016xxx	ERRORCODE_HW_HWNOTSET	PCA setup error.		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	000xxx	ERRORCODE_ UNKNOWN	Unknown error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008xxx	ERRORCODE_FW_ NOTSAFE	Movement not safe.		
		017xxx	ERRORCODE_FW_ UNKNOWN_ERROR	Unknown error.		
		061xxx	ERRORCODE_ INVALID_ARGUMENTS	Invalid arguments.		
		062xxx	ERRORCODE_ FWCOMM_TIMEOUT	Communication timeout.		
		063xxx	ERRORCODE_ FWCOMM_ CONNECTION_LOST	Communication lost.		
		064xxx	ERRORCODE_ FWCOMM_COMM_ ERROR	Communication error.		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	065xxx	ERRORCODE_FLU_ NOT_ENOUGH_SOL_A	Not enough solution A to execute a fluidic operation.	If there are running works:	RETRY
					System has been stopped caused by a container problem and there are pending works. Please, select one of the following options:	RE-RUN Open
					If there are not running works: System has been stopped caused by a container problem. Please, select one of the following options:	
Non firmware errors (xx)	Non firmware errors (xx)	066xxx	ERRORCODE_FLU_ NOT_ENOUGH_SOL_B	Not enough solution B to execute a fluidic operation.	If there are running works:	RETRY
					System has been stopped caused by a container problem and there are pending works. Please, select one of the following options:	RE-RUN Open
					If there are not running works: System has been stopped caused by a container problem. Please, select one of the following options:	

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	067xxx	ERRORCODE_FLU_NOT_ENOUGH_WASTE_CAPACITY	Not enough waste capacity to execute fluidic operations.	If there are running works: System has been stopped caused by a container problem and there are pending works. Please, select one of the following options: If there are not running works: System has been stopped caused by a container problem. Please, select one of the following options:	RETRY RE-RUN Open
		Non firmware errors (xx)	ERRORCODE_FLU_WASTE_TRAY_FULL	The card waste disposal tray is full.	Waste container is full. Please open the drawer and empty it.	OK
		073xxx	ERRORCODE_FLU_ADJUST_ERROR	Impossible to set adjustment.	NA (error is the Technical Service software)	NA
		082xxx	ERRORCODE_WASTE_CIRCUIT_ERROR	Error in waste circuit.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	087xxx	ERRORCODE_FLU_UNKNOWN_ERROR	Unknown error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		088xxx	ERRORCODE_PRECONDITION_FAILED	Previous condition for the action is not met.		
		089xxx	ERRORCODE_STARTUP_ERROR	Driver startup error.		
		091xxx	ERRORCODE_ALREADY_STARTED	Another instance of Driver is already opened.		
Non firmware errors (xx)	Non firmware errors (xx)	093001	ERRORCODE_FLU_LOW_PRIMED_SOL_A	Solution A in drawer 1 not primed.	There is a problem in the fluidic system. Please, reintroduce the containers and ensure that they are properly connected and select RETRY. Select CANCEL if the problem persists only.	CANCEL Works CONTINUE
		093004		Solution A in drawer 4 not primed.		
		094002	ERRORCODE_FLU_LOW_PRIMED_SOL_B	Solution B in drawer 2 not primed.		
		094005		Solution B in drawer 5 not primed.		
0	240	090xxx	ERRORCODE_UNEXPECTED_ERROR	A Driver unexpected error has been produced.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
None	None	None	None	Possible dripping detected at PRESSURE_SENSOR_x (where x is the affected sensor).	If error occurs over a sample / reagent drawer: Dripping on drawer error If error occurs over dilution cups or incubator: Dripping on instrument	CANCEL Works CONTINUE

Due to the fluidic procedures consist on different steps, in order to focus properly the troubleshooting it is necessary to know in which step the error appears. This can be done in two different ways:

- By means of the last three digits of the fluidic error (if they are different than 000).
- By means of DG59Driver.log, the error message is stored with the following structure:

[ERROR] ERROR Error Code: (xxyyzzzsss)

Fluidic system error

Description: _____

where **xxyyzzzsss** is the error code that can be found in the error lists of the Servicing Center. Example (please note that, between these lines of the log file, it is possible that some others lines are registered):

```
[WARN ] FLU   Starting Initial Low Prime
[INFO ] FLU   Resetting fluidic device: PUMP_ASP (PM2)
[INFO ] FLU   Resetting fluidic device: PUMP_DISP_AB (PM1)
[INFO ] FLU   Resetting fluidic device: PUMP_DISP_B (PM3)
[INFO ] FLU   Setting fluidic device: PUMP_DISP_AB (PM1)
[INFO ] FLU   Setting fluidic device: EV_FAST_PRIME_B (EV19)
[INFO ] FLU   Setting fluidic device: EV_FAST_PRIME_AB (EV20)
[INFO ] FLU   Setting fluidic device: EV_CH_SOL (EV13)
[INFO ] FLU   PRIME_LOW_INITIAL_2 - Delay (sec.): 40
[INFO ] FLU   PRIME_LOW_INITIAL_3
[INFO ] FLU   Verify Margins failed. Sensor: 3 - Instant value: 48 - Min: 50 Max: 350
[ERROR] FLU   Initial low Prime Aborted!!
[INFO ] FLU   Sensor PRESSURE_SENSOR_3 set to 160 milibars
[INFO ] FLU   Sensor PRESSURE_SENSOR_4 set to 160 milibars
[INFO ] FLU   Resetting all Pumps and Valves and terminating Dilutors
[WARN ] PIP   PROBE_1 dilutor is in mode IDLE and will not be terminated because there is no instrument flow risk
[WARN ] PIP   PROBE_2 dilutor is in mode IDLE and will not be terminated because there is no instrument flow risk
[ERROR] FLUSR Service FLUIDIC_SYSTEM_SERVICE with mode FAST_PRIME failed
[ERROR] ERROR Error Code: (0506069053)
Fluidic system error
Description: Pressure fail on circuit A,
```

Once the error is located in the log file, it is necessary to check the specific step of the fluidic procedure where this error has appeared. Just checking a few lines above the error, it can be found the step and the parts activated in that moment. Further information in all the fluidic procedures can be found in this Servicing Center, Fluidics (FLU) Module chapter.

In the log file, the components that are activated are registered with the Setting fluidic device command. Components that are deactivated / returned to its original position are registered with Resetting fluidic device command.

Once the fluidic step is known (and the parts that are activated), in the Technical Service software is possible to simulate this step activating the components in Fluidics tab and checking the value of the fluidic sensors (S3, S4, S5 or S6).

**NOTE**

Empty the fluidic circuit every time a simulation of a specific step of the fluidic procedures is performed activating in the Technical Service software the different related components.

Table 53 - List of errors (3 last digits) related to Fluidics module

Last 3 digits error code	Procedure	Fluidic step see "Fluidic procedures" on page 564
001	Check sol. A level - internal drain	Error during check bottle levels step in fluidic procedures
002	Check sol. B level - internal drain	Error during check bottle levels step in fluidic procedures
003	Check waste level - internal drain	Error during check bottle levels step in fluidic procedures
004	Check sol. A level - external drain	Error during check bottle levels step in fluidic procedures
005	Check sol. B level - external drain	Error during check bottle levels step in fluidic procedures
006	Dilution cup fill	Error during step 8 of dilution cup wash procedure
007	Dilution cup drain	Error during step 5 / 12 dilution cup wash procedure

Last 3 digits error code	Procedure	Fluidic step see "Fluidic procedures" on page 564
008	Probe Prime	Error during step 3 of syringe prime procedure
009	Probe Rinse	Error during step 3 of syringe rinse procedure
010	Probe Empty	Error during steps 18/ 21 / 22 / 25 of empty procedure
026	Internal wash	Error during step 5 of internal wash procedure
027	Internal wash with sol. B	Error during step 5 of internal wash procedure
028	Slow external wash	Error during step 4 / 5 of slow external wash procedure
029	Fast external wash	Error during step 4 / 5 of fast external wash procedure
052	Low Prime	Error during sensors status monitoring before step 5 of low prime procedure Error during sensors status monitoring before step 4 of low rinse procedure
053	Low Prime	Error during step 5 of low prime procedure Error during step 4 of low rinse procedure
102	Low Prime	Error during sensors status monitoring before step 6/7 of low prime procedure Error during sensors status monitoring before step 5 of low rinse procedure
103	Low Prime	Error during step 6 / 7 of low prime procedure Error during step 5 of low rinse procedure
152	Low Prime	Error during sensors status monitoring before step 8/9 of low prime procedure
153	Low Prime	Error during step 8 / 9 of low prime procedure
252	Probes washer High Prime	Error during step 6 of high prime solution B (probes washer) procedure

Last 3 digits error code	Procedure	Fluidic step see "Fluidic procedures" on page 564
254	Probes washer High Prime	Error during step 10 of high prime solution B (probes washer) procedure
256	Probes washer High Prime	Error during sensors status monitoring before step 13 of probes washer prime procedure
257	Probes washer High Prime	Error during step 14 of high prime solution B (probes washer) procedure
260	Dilution cup High Prime	Error during step 4 of high prime solution A (dilution cup) procedure
262	Dilution cup High Prime	Error during step 8 of high prime solution A (dilution cup) procedure
264	Dilution cup High Prime	Error during sensors status monitoring before step 11 of dilution cup prime procedure
265	Dilution cup High Prime	Error during step 12 of high prime solution A (dilution cup) procedure
302	Low Rinse	Error during sensors status monitoring before step 6 of low rinse procedure
303	Low Rinse	Error during step 6 of low rinse procedure
306	Probes washer High Rinse	Error during step 4 of high rinse (probes washer) procedure
308	Probes washer High Rinse	Error during step 8 of high rinse (probes washer) procedure
310	Probes washer High Rinse	Error during sensors status monitoring in step 10 of high rinse (probes washer) procedure
311	Probes washer High Rinse	Error during step 12 of high rinse (probes washer) procedure
314	Dilution cup High Rinse	Error during step 4 of high rinse (dilution cup) procedure

Last 3 digits error code	Procedure	Fluidic step see "Fluidic procedures" on page 564
316	Dilution cup High Rinse	Error during step 8 of high rinse (dilution cup) procedure
318	Dilution cup High Rinse	Error during sensors status monitoring in step 10 of high rinse (dilution cup) procedure
319	Dilution cup High Rinse	Error during step 12 of high rinse (dilution cup) procedure
352	Empty	Error during sensors status monitoring before step 6/9 of empty procedure Error during step 6/9 of empty procedure
355	Empty	Error during sensors status monitoring before step 13/16 of empty procedure Error during step 13/16 of empty procedure
358	Empty	Error during sensors status monitoring before step 19/21 of empty procedure
359	Empty	Error during sensors status monitoring before step 23/25 of empty procedure
360	Empty	Error during sensors status monitoring before step 27 of empty procedure Error during step 27 of empty procedure
361	Empty	Error during step 30 of empty procedure
362	Empty	Error during sensors status monitoring before step 32 of empty procedure Error during step 32 of empty procedure
363	Empty	Error during sensors status monitoring before step 36 of empty procedure
364	Empty	Error during sensors status monitoring before step 40 of empty procedure
402	Check Waste	Error during check waste step in fluidic procedures
404	Check Waste	Error during check waste step in fluidic procedures

2.6.8.1 FLUIDIC SYSTEM ISSUES

Table 54 - Fluidics module troubleshooting - fluidic system issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
The system is not primed / rinsed	Dilutor/s not initialised	Check that dilutor is initialised
	Dilutor/s valve/s failure	Check / replace dilutor/s valve
	Leak in tube or fitting	Check tubing and fittings
	Dispensation circuit is not working	Check that there is enough liquid in the bottles and they are properly connected Check in DG59Driver log file, the prime / rinse step where the error appears (the fluidic parts activated are detailed in the log file)
The system is not primed / rinsed	Aspiration circuit is not working	Check that there is enough room in the waste bottles and they are properly connected
		Check in DG59Driver log file, the prime / rinse step where the error appears (the fluidic parts activated are detailed in the log file)

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
The system is not emptied	Dilutor/s not initialized	Check that dilutor is initialised
	Dilutor/s valve/s failure	Check / replace dilutor/s valve
	Aspiration circuit is not working	Check that there is enough room in the waste bottles and they are properly connected Check in DG59Driver log file, the empty step where the error appears (the fluidic parts activated are detailed in the log file)
Leak in probes washer when washing probes	Not enough vacuum arrives to the probes washer	Check waste bottles
		Check aspiration pump (PM2)
		Check electrovalves EV5 / EV10
		Check/adjust/replace vacuum sensors (S5 / S6)
	Tubes/fittings obstructed by dirt / squeezed	Replace tubing, fittings
	Home position of probe/s too low	Check probe home position. Perform probes offset adjustment.
	Probe/s bent	Replace probe/s

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Leak in probes washer when washing probes	Too much pressure/liquid arrives to the probes washer	Check solution bottles
		Check/replace dispensation pump (PM3)
		Check electrovalves EV1 / EV6
		Check/adjust/replace pressure sensors (S3 / S4)
	Clot/obstruction in probes washer	Remove clot/obstruction
	Probes washer scratched	Replace probes washer
	Probe/s partially blocked	Replace probe/s
Dirt on probes washer	Probably salt from saline solution	Clean it with distilled water
Electrovalve is not working	Electrovalve broken	Check / replace electrovalve
	PCA FLU-CTRL59 damaged	Check / replace PCA FLU-CTRL59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Bubbles in sample line	Syringe tube connector defective or wrongly assembled	Replace/fix tube connector
	O-ring (inside syringe valve) lost or defective	Replace O-ring
	Syringe valve defective	Replace dilutor valve
	Syringe plunger seal worn out	Replace plunger
	Syringe glass worn out	Replace glass
	Syringe not properly installed	Install correctly the syringe
	Tubes connected wrongly.	Check tubes connection
	Probe assembly defective	Check probe assembly
Bubbles in sample line	Intermediate manifold BK2, BK4, BK5 and/or BK6 tube connector defective or not properly connected	Replace/fix tube connector/connection
	CPDs assembly defective or incorrectly assembled	Check CPDs assembly
	CDP defective	Replace CDP
	Intermediate manifold BK2, BK4, BK5 and/or BK6 defective	Replace defective interconnection manifold
	Syringe to probe Teflon tube defective	Replace defective Teflon tube / chain
	Bubbles coming from solution A/B bottles	Check tubing/parts from solution A/B bottle/s to dilutor valves

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Overflow in dilution cup	Clot expelled during pipetting in the bottom of the cup/obstruction	Remove clot / obstruction
	Tube blocked or squeezed	Replace / fix tube
	Leak from probes washer	Refer to Leak in probes washer section in this table
	Too much pressure/liquid arrives to dilution cup	Check solution bottles
		Check/replace dispensation pump (PM1)
		Check / replace electrovalves EV3 / EV8
		Check/adjust/replace pressure sensors (S3 / S4)
	Not enough vacuum arrives to dilution cup	Check waste bottles
		Check aspiration pump (PM2)
		Check / replace EV4 / EV9
		Check/adjust/replace vaccumm sensors (S5 / S6)

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Wells not dispensed/incorrect (too low) volume	Incorrect tube diameter detection	Check / readjust / fix PIP presence detector
	Aspiration valves of dilution cup defective	Check / replace EV4 / EV9
	Level detection failure	Check/fix level detection
	Sufficient volume determination failure	Check / readjust probe maximum depth
		Check / readjust / fix PIP presence detector
	Bubbles in sample line	Refer to bubbles in sample line in this table
	Incorrect probe positioning in dilution cup	Check / adjust probes in dilution cup
Wells not dispensed/incorrect (too low) volume	Incorrect probe positioning in samples drawers	Check / adjust probes in samples drawers
	Incorrect probe positioning in reagent drawers	Check / adjust probes in reagent drawers
	Incorrect probe positioning in incubators	Check / adjust incubator flatness
		Check / adjust incubator height
		Check / adjust probes in incubators
	Dilutor failure	Replace dilutor
Wells with incorrect (too high) volume	Bubbles in sample line	Refer to bubbles in sample line in this table
	Dilutor failure	Replace dilutor
	Leak from probes washer	Refer to Leak in probes washer section in this table

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
CPD leakage	CPD sensor is broken	Check / replace CPD sensor (S1, S2)
	Incorrect CPD assembly into interconnection manifold	Check CPD assembly
Plunger overload in dilutor module	Dilutor failure	Replace dilutor
	Syringe failure	Replace syringe
	Input/output valve tubing obstructed	Check/replace related tubing

2.6.8.2 BOTTLES /CARD WASTE DRAWERS ISSUES

Table 55 - Fluidics module troubleshooting - bottles/card waste drawers issues

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Drawer is not opened	Lid electric lock damaged	Check / replace lid electric lock
	PCA GEN-LCK59 wiring damaged	Check / replace cables
	PCA GEN-LCK59 damaged	Check / replace PCA GEN-LCK59
	PCA FLU-CTRL59 damaged	Check / replace FLU-CTRL59
Cosmetic front panel is not aligned	Cosmetic front panel screws are loose	Align cosmetic front panel
Drawer is not aligned	Latch depth position is not correct	Adjust latch position

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Hard to open / close the drawer	Bad status of linear guides	Replace linear guides
	Incorrect PCA GEN-LCK59 protective cover assembly	Check PCA GEN-LCK59 assembly
	Lack of lubrication	Lubricate
Weight measured is not correct	Drawer not closed properly	Check fluidic drawer
	Incorrect load cell adjustment	Check / perform load cell adjustment
No weight is measured	Load cell damaged	Check / replace load cell
	Load cell wiring damaged	Check / replace load cell wiring
	PCA FLU-CTRL59 damaged	Check / replace PCA FLU-CTRL59

2.7 Dilution cup module (CUP)

In this chapter, the Dilution cup module is described. It includes an overall description of the functions as well as the corresponding adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, and a troubleshooting section.

2.7.1 Description

The dilution cup has two independent Teflon wells so two independent dilutions can be prepared simultaneously or in any of the two wells. In order to homogenize the dilution, one stirring system is provided.

The structure of the module is formed by the following elements:

- Wells: where the liquids are dispensed to make the dilution.
- Stirring system: is the mechanism responsible for homogenize the dilution.
- Supporting structure is responsible for holding the stirring system and the wells, attaching them to the instrument structure. Isolate the dilution cup vibration to the rest of the modules.

Each well has two fluidic connectors to connect the tubes required to wash and drain each well. For each well, one connector (located at the upper part of the well) allows filling the well with solution A or B, while the other one (located at the bottom part of the well) allows emptying to the waste bottle. The wells are oriented in a way that allows the two probes to access the wells at the same time.

The stirring is accomplished using a motor with a counterweight attached to its axis, and with a silentblock acting as an elastic element to allow the orbital movement. The silentblock is placed between the motor, located at the bottom part of the structure, and the wells located in the upper part of the structure, in order to achieve a proper stirring.

The motor, a brushless type, rotates in only one direction. The control of the velocity is performed by the hall sensors set of the motor and it is overcurrent monitored. Speed and time are determined by the software.

A cover attached to the main structure prevents any spill from reaching the motor or the PCA. The cover has an evacuation channel for the liquid that may come from a well overflow.

A general block diagram of the module is shown in the following figure:

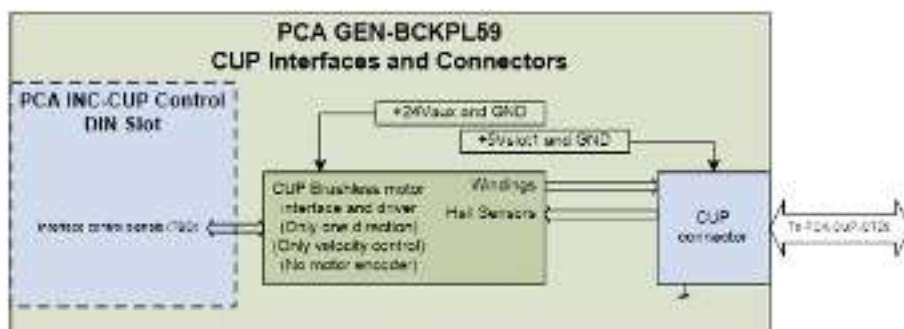


Figure 561 - Block diagram of the Dilution cup module

The PCAs related to the CUP module are:

- **PCA GEN-BCKPL59:** main PCA of the instrument, located at the backpanel.
- **PCA INC-CUP Control:** located at the backpanel (slot 1), attached to PCA GEN-BCKPL59. It includes one brushless motor interface and driver to manage dilution cup motor.
- **PCA CUP-ST59:** located under the dilution cup. It contains the connectors to match with the stirring motor.

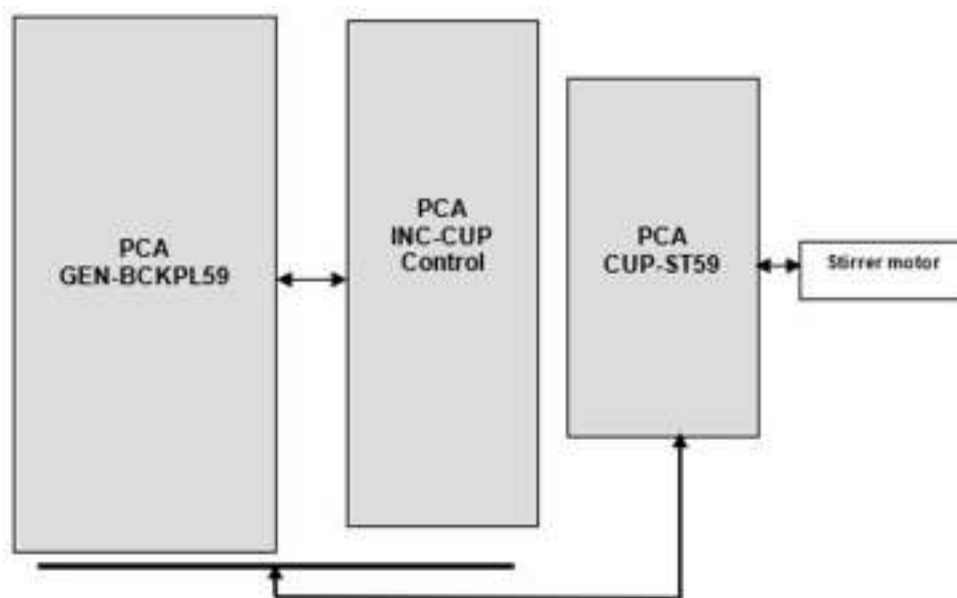


Figure 562 - Main blocks of the Dilution cup module

2.7.2 Mechanical adjustments

Mechanical adjustments of the dilution cup module are detailed.

2.7.2.1 Adjusting dilution cup parallelism

This adjustment is needed to place parallel the dilution cup wells with the probes in order to allow that the probes enter into the dilution cup wells.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 233938 DILUTION CUP ALIGNMENT TOOL (Ver. 2).

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- Remove the dilution cup from the instrument.

In order to remove the dilution cup from the instrument, the next steps should be followed:

1. Remove the four screws of the dilution cup bath.
2. Remove the screws of the cover and take out the cover by the front of the instrument.
3. Remove the two screws that are attaching the dilution cup module to the Erytra chassis.
4. Disconnect the flat cable from the PCA CUP-ST59.
5. Remove the dilution cup wells by means of the two screws that attach it to the dilution cup structure. In order to get a better access to the rear screw, it is recommended to rotate a little bit the dilution cup module.
6. Remove the dilution cup bath.
7. Remove the whole module.

Procedure

1. Block the set screw with an Allen key and loose the nut.
2. Check that the silentblock is touching the bottom surface of the dilution cup support and the set screw is tightened until is going out from the dilution cup support 6 mm, as it is shown in the following picture,



Figure 563 - set screw inserted in the silentblock

3. Place the spacer in the set screw and then place the dilution cup parallelism Y-axis tool inside the dilution cup structure. Use the screws of the dilution cup wells to fix it.



Figure 564 - Dilution cup alignment tool fixed into dilution cup structure

4. The set screw is fixed by means of an Allen key and the tool, as it is shown in the picture.

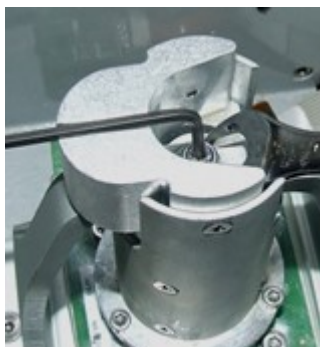


Figure 565 - Parallelism adjustment of the dilution cup module

5. Tighten the nut (check that it is not too tight or or too loose), remove Allen key and the tool and check that the bar of the shaking support is centered in the shaking tube.



Figure 566 - Checking of the correct adjustment of the dilution cup

2.7.3 Functional adjustments

There are no functional adjustments in the Dilution cup module.

2.7.4 Verification procedures



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to verify that the Dilution cup module is working properly, follow the next steps:

1. Remove the four screws of the dilution cup bath and take out the cover by the front of the instrument.

2. Remove the dilution cup wells by means of the two screws that attach it to the dilution cup structure and then remove the dilution cup bath.
3. Place the dilution cup wells and tighten its two screws.
4. Stick a reflective adhesive to the basis of the motor and use a tachometer to read the rpm during a stirring. Check that the adhesive does not interfere with anything during the rotation of the motor. In order to start a stirring process, open the Technical Service software and go to the **Dilution Cup** tab, select **PROFILE_0** in the **Stirrer Actions** submenu (see "Dilution Cup tab" on page 774) and press the **Spin** button. The expected value of the tachometer measurement is $800 \text{ rpm} \pm 80 \text{ rpm}$.

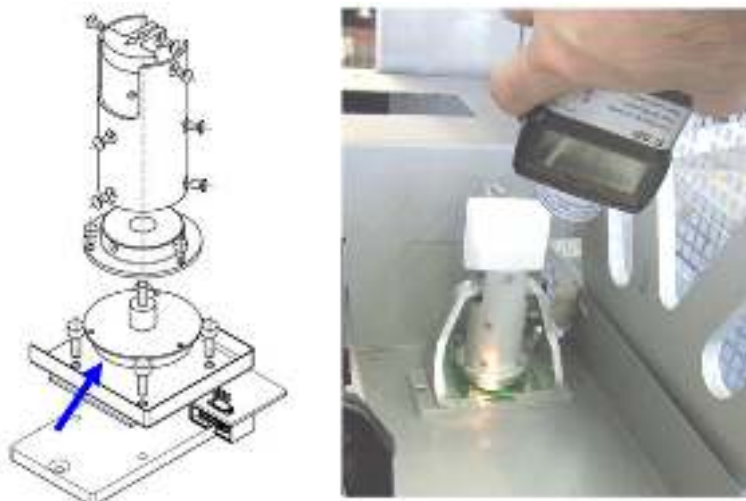


Figure 567 - Location of the basis of the motor (left) and rpm measurement with a tachometer during stirring (right)



NOTE

The use of any tachometer is possible as long as it complies with the following specifications:

- Type: optical.
- Scale division (rpm): $\leq 1 \text{ rpm}$.
- Range (rpm): 1 - 99999 rpm.
- Maximum uncertainty permitted: $\pm 3 \text{ rpm}$.

5. Remove the reflective adhesive.
6. Check the status of the PCA plastic cover.
7. Remove the dilution cup wells and place the cover with its screws.
8. Place the dilution cup bath and tighten its four screws.
9. Place the dilution cup wells and tighten its two screws.

10. Check that the dilution cup bath is centred in the dilution cup structure to avoid that the dilution cup wells touch the dilution cup bath during the stirring movement.
11. Perform a prime and a wash of the dilution cup in order to check that liquid is dispensed into the wells and it is properly aspirated. To do so, select **fluidic prime** and **wash dilution cup** (selecting desired number of iterations) options in **Fluidics** tab of Technical Service software. See "Fluidics tab" on page 826.

2.7.5 Replacement procedures

Replacement procedures related to the dilution cup module are detailed in this section.

2.7.5.1 Replacing dilution cup wells assembly



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 733966 DILUTION STATION CUP V2.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Switch off the instrument and disconnect from mains.
2. Open the upper door.

3. Remove the dilution cup wells by means of the two screws that attach it to the dilution cup structure. In order to get a better access to the rear screw, it is recommended to rotate a little bit the dilution cup module.



Figure 568 - Dilution cup screws

4. Place the new dilution cup wells. After connecting back the tubes, make sure they don't modify/force the dilution cup wells resting position.
5. Perform a Prime to check the fluidic system. See "Fluidics tab" on page 826.
6. Check that the probes don't impact against the dilution cup in both aspiration and dispensing positions when the dilution cup is moving.

2.7.5.2 Replacing dilution cup motor



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 733699 MOTOR STIRRER DILUTION (Ver. 2).
- 733700 SHOCK ABSORBER DILUTION CUP (Ver. 3).
- 733698 PCA CUP-ST59 v2.
- 730606 SCREW M4x10 DIN 912 (100 units).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to replace the motor of the dilution cup follow the next steps:

1. Remove the dilution cup wells unscrewing the two screws, one on the front and the other one on the back, attaching it to dilution cup structure.

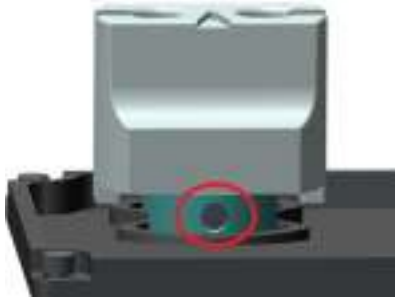


Figure 569 - Dilution cup screw

2. Disconnect the four tubes connected to the rear part of the dilution cup.
3. Remove the four screws of the dilution cup bath.



Figure 570 - Dilution cup bath screws

4. Unscrew the five screws attaching the cover and remove it by the front of the instrument.



Figure 571 - Screws attaching the cover

5. Remove the two screws that are attaching the dilution cup module to the upper floor, disconnect the flat cable from PCA CUP-ST59 and extract the whole module outside the instrument.



Figure 572 - Screws attaching the dilution cup module

6. Carefully disconnect the motor cable from PCA CUP-ST59.
7. Remove PCA CUP-ST59 and the plastic protective cover by means of the four screws that are attaching them to base of the dilution cup module. Dispose of the plastic cover, PCA and the four screws.

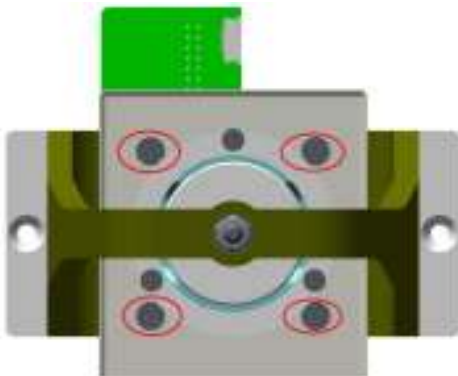


Figure 573 - Screws attaching the PCA CUP-ST59 and plastic protective cover

8. Unscrew the silentblock from the dilution cup support.



Figure 574 - Unscrewing the silentblock

9. Remove the dilution cup structure from the dilution cup support.



Figure 575 - Dilution cup structure and support

10. Remove the three screws that are located in the lower part of the dilution cup structure fixing the motor assembly and remove it.



Figure 576 - Lower screws of the dilution cup structure

11. Remove the three screws that are located in the upper part of the dilution cup structure and remove the shock absorber assembly.



Figure 577 - Upper screws of the dilution cup structure

12. Put the new motor with MOTOR STIRRER DILUTION (Ver. 2) in the same place as the one previously removed, in the lower part of the dilution cup structure. Orientation of the cable must match the one in the following picture:



Figure 578 - Orientation of the cable

13. Put the SHOCK ABSORBER DILUTION CUP (Ver. 3) in the same place as the one previously removed, in the upper part of the dilution cup structure.
14. Screw the PCA CUP-ST59 v2 to the dilution cup support on any of its sides, as shown on the following image:



Figure 579 - New PCA attached to the dilution cup support

15. Attach the dilution cup structure to the dilution cup support and base using SCREW M4x10 DIN 912 (100 units), with the motor cable orientated to PCA H1 connector.



Figure 580 - Dilution cup structure attached to the dilution cup support (left) and cable orientation (right)



NOTE

At this point, the dilution cup parallelism adjustment must be performed. See "Adjusting dilution cup parallelism" on page 673.

16. Put back the dilution cup bath in its place, then connect the tubes again on the back on the dilution cup wells and attach this one to the structure, using the previously removed screws.
17. Connect the flat cable to PCA CUP-ST59 v2, then screw back the dilution cup module on its place in the upper floor using the two previously removed screws. The PCA must be facing the right lateral cover of the instrument.

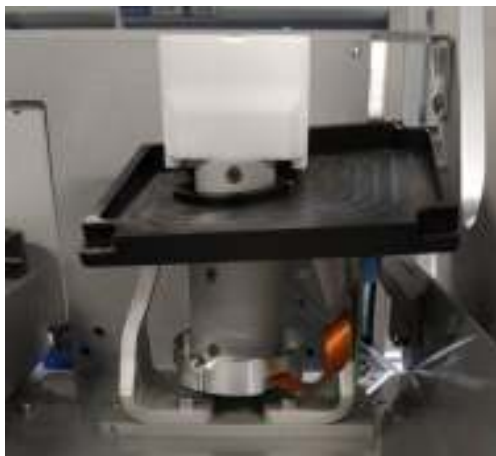


Figure 581 - Dilution cup module in the upper floor

18. Place back the cover on its place, using the previously removed screws.



NOTE

At this point, the probes adjustment over the dilution cup should be performed.

19. Check that the probes don't impact against the dilution cup in both aspiration and dispensing positions when the dilution cup is moving.

20. Check the RPM of the dilution cup. See "Verification procedures" on page 676.

2.7.5.3 Replacing fitting



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233844 TUBE CONNECT. DISTRIB. (for 2,3x3,6 tube).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

To replace a fitting of the dilution cup wells, follow next steps:

1. Remove dilution cup wells by means of the two screws that attach it to dilution cup structure.
2. Disconnect tubing from dilution cup wells.
3. Remove the fitting to be replaced.
4. Place the new fitting. Tighten it manually to the maximum and then $\frac{1}{4}$ turn more with an open end spanner number six.
5. Connect tubing, checking the label attached to the tubes and the reference label attached to the rear part of the dilution cup wells.

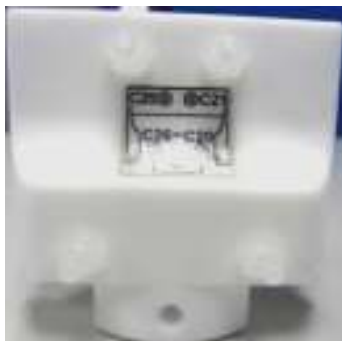


Figure 582 - Rear part of dilution cup wells

6. Place dilution cup wells into the structure and tighten its two screws.
7. Perform a Prime to check the fluidic system. See "Fluidics tab" on page 826.

2.7.6 Electronics

2.7.6.1 Fuses

Fuses related to the Dilution cup module are detailed in the following table:

Table 56 - List of fuses of the Dilution cup module

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS11	MST250 0034.6615 1A/250V
	FUS12	MST250 0034.6620 3.15A/250V
PCA INC-CUP Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA CUP-ST59	NA	NA

2.7.6.2 LEDs

LEDs related to the Dilution cup module are detailed in the following table:

Table 57 - List of LEDs of the Dilution cup module

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED7	Green 5V (FUS11)
	LED8	Green 24V (FUS12)
PCA INC-CUP Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$
PCA CUP-ST59	NA	NA

2.7.6.3 Jumpers and test points

Neither jumpers nor test points are related to the Dilution cup module.

2.7.6.4 Wiring table

Cables related to the Dilution cup module are detailed in the following table and their connection are shown in the wiring diagram. To view the wiring diagram, see "Related documents" on page 19 or refer to the Erytra Online Help.

Table 58 - List of wiring of Dilution cup module

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-CUP	H2	PCA CUP-ST59	CUP	PCA GEN-BCKPL59
NA	H1	PCA CUP-ST59	MOTOR	CUP MOTOR

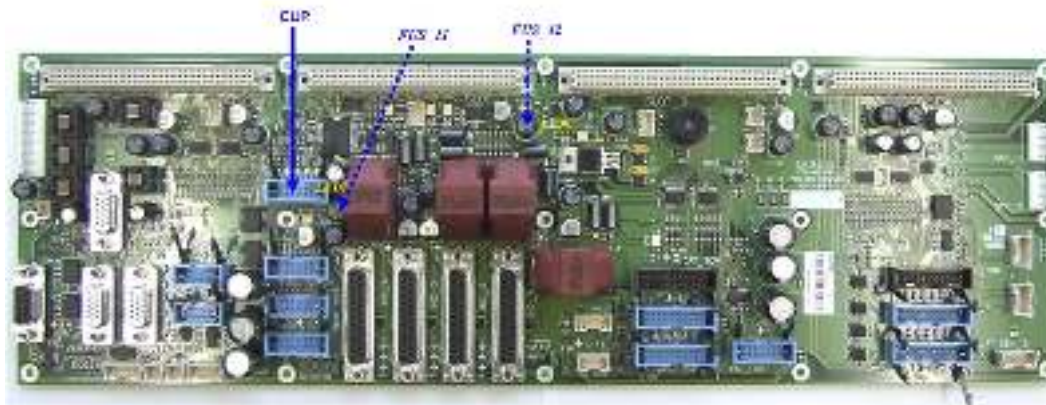


Figure 583 - Location of the connector, fuses and LEDs in PCA GEN-BCKPL59 related to Dilution cup module

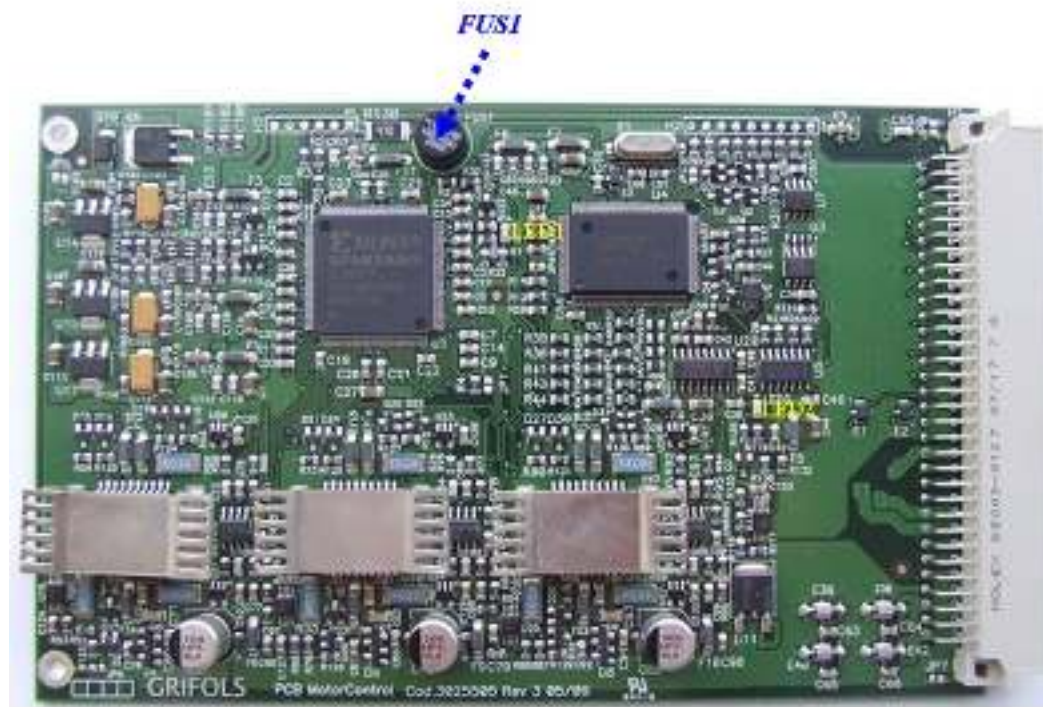


Figure 584 - Location of the fuse and LEDs in PCA MotorControl

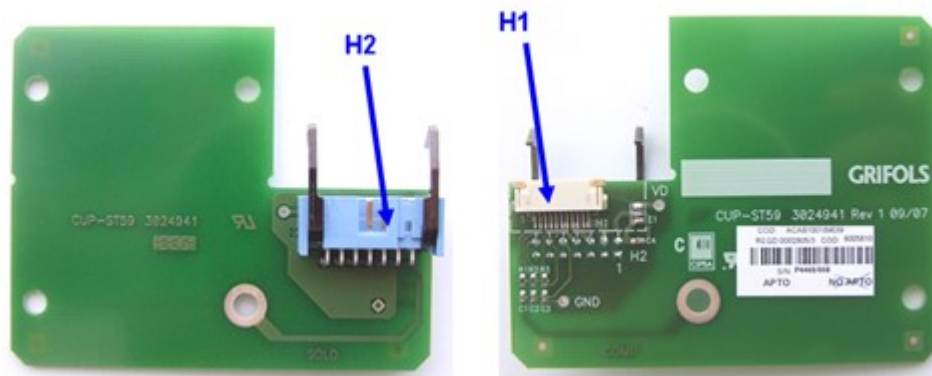


Figure 585 - Location of the connectors in PCA CUP-ST59

2.7.7 Troubleshooting

Table 59 - List of errors related to the Dilution cup module

Module	Device	Error code	Error	Cause	Text	Options
CUP (04)	Dilution cup (01)	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during reagents stirring process or homogenization in dilution cup. Please choose one of the following options:	RETRY
		005000	ERRORCODE_ARG	Command has wrong arguments.		CANCEL Works
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		012000	ERRORCODE_SPINBADPROFILE	Command includes a spin profile that is invalid for the addressed device.		
		013000	ERRORCODE_NOTSPINNING	The addressed device is not performing the requested spin.		
		032000	CURRENT_LIMIT	Current limit at servo device.		
		034000	MOVEMENT_ERROR	A movement error has been produced at a servo device.		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received a command with incorrect length.		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device.		
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		014000	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW.		
		015000	ERRORCODE_HW_HWFAIL	There is a problem with PCA HW.		
		016000	ERRORCODE_HW_HWNOTSET	PCA setup error.		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	000000	ERRORCODE_ UNKNOWN	Unknown error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_ NOTSAFE	Movement not safe.		
		017000	ERRORCODE_FW_ UNKNOWN_ERROR	Unknown error.		
		061000	ERRORCODE_ INVALID_ ARGUMENTS	Invalid arguments.		
		062000	ERRORCODE_ FWCOMM_TIMEOUT	Communication timeout.		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	063000	ERRORCODE_ FWCOMM_ CONNECTION_LOST	Communication lost.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		064000	ERRORCODE_ FWCOMM_COMM_ ERROR	Communication error.		
		088000	ERRORCODE_ PRECONDITION_ FAILED	Previous condition for the action is not met.		
		089000	ERRORCODE_ STARTUP_ERROR	Driver startup error.		
		091000	ERRORCODE_ ALREADY_STARTED	Another instance of Driver is already opened.		
0	240	090000	ERRORCODE_ UNEXPECTED_ ERROR	A Driver unexpected error has been produced.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Table 60 - Dilution cup module troubleshooting

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Stirring is not performed 0401013000 0401032000 0401034000	Fuse blown	Check / replace FUS11, FUSGEN4
	Motor does not turn	Check / replace motor
	Dilution cup is not stirred but the motor turns	Check assembly of dilution cup
	Cable B-CUP damaged	Check / replace B-CUP cable
	PCA CUP-ST59 failure	Check / replace PCA CUP-ST59
	PCA INC-CUP MotorControl failure	Check / replace PCA INC-CUP MotorControl
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
Stirring at different speed than defined one 0401013000 0401032000 0401034000	Improper assembly	Check assembly of dilution cup
	Motor damaged	Check / replace the motor
	PCA CUP-ST59 damaged	Check / replace PCA CUP-ST59
	Wrong firmware	Check / install INC-CUP firmware
	PCA INC-CUP MotorControl failure	Check / replace PCA INC-CUP MotorControl
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
Dilution cup is not parallel with probes	Dilution cup incorrectly assembled	Check assembly of dilution cup
Noise when stirring	Dilution cup bath incorrectly assembled	Check assembly of dilution cup bath
	Dilution cup incorrectly assembled	Check assembly of dilution cup

2.8 Card reader module (LEC)

In this chapter, the Card reader module is described. It includes an overall description of the functions as well as the corresponding adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, and a troubleshooting section.

2.8.1 Description

The functions of this module are the card integrity check and results reading:

- Card integrity check and initial identification:
 - Initial total volume measurement.
 - Gel volume measurement.
 - Microtube proper condition and artifact detection.
- Results reading:
 - Total volume measurement.
 - Reaction grading.
 - Microtube warnings and incidences: lipemic, haemolysis, artifacts, etc.

To do so, this module consists of a FireWire color camera with optics, a pattern and LED illumination panel controlled by a PCA RD-IL421. The camera resolution is 656 (H) x 494 (V).

The reader pattern has four crosses, two vertical lines and two sets of two horizontal lines. The crosses at each corner are used to determine pattern location and orientation (adjustment and checking routines). Vertical lines are used in the microtube integrity check and initial check to ease gel volume and total volume calculation (supernatant-air meniscus and gel-supernatant meniscus detection). Horizontal line sets are used in the microtube reading to detect turbidity in the non-gel section.

The illumination panel is made of two independent arrays of LEDs. One array is assembled at the top and the other at the bottom. Its light intensity is controlled by the software setting the current. Erytra automatically performs light intensity verification and adjustment at the user software start-up. It is also performed in the illumination intensity adjustment step of the Technical Service software.

The reader takes a picture of two microtubes at a time and then four steps and images are required to complete a card processing. Raw images are 640x480 images that contain the pixel information as acquired by the sensor, 8 bits. Pixels are green, red or blue according to their position on the sensor and the Bayer filter of the camera.



Figure 586 - Reader raw image and microtube image

Raw images are saved (images older than 14 days are automatically deleted) in **E:\data\Erytra\Raw** directory with the file name:

BBBBBBBBBBBBBBBBBBBB_CCCCCCCC_MXMY_CXCY_PPP_Q.PGM

where:

BBBBBBBBBBBBBBBBBBBB: card barcode.

CCCCCCCC: internal unique identification of a test execution performed by the instrument.

MX and MY: microtube numbers associated to that picture, counting from left to right from the non-barcode card side.

CX and CY: microtube MX and MY color.

PPP: card type (DGG for DG Gel cards).

Q: indicates if the picture belongs to the initial/integrity check (Q = C) or to results reading (Q = R).

Raw images are used to obtain each microtube picture when grading reactions. A mask is applied; no reference image exists. Two files, a bmp and a jpg file, per microtube will be generated and saved in the **E:\data\Erytra\Microtubes** directory. The jpg reference file will be included in the instrument database, upon automatic or manual validation, and reports. These files names will be:

BBBBBBBBBBBBBBBBBBBB_CCCCCCCC_MX_MC_PPP.bmp/jpg

where:

BBBBBBBBBBBBBBBBBBBB: card barcode.

CCCCCCCC: internal unique identification of a test execution performed by the instrument.

MX: microtube numbers associated to that picture, counting from left to right from the non-barcode card side.

MC: microtube MX color.

PPP: card type (DGG for DG Gel cards).

A general block diagram of the module is shown in the following figure:

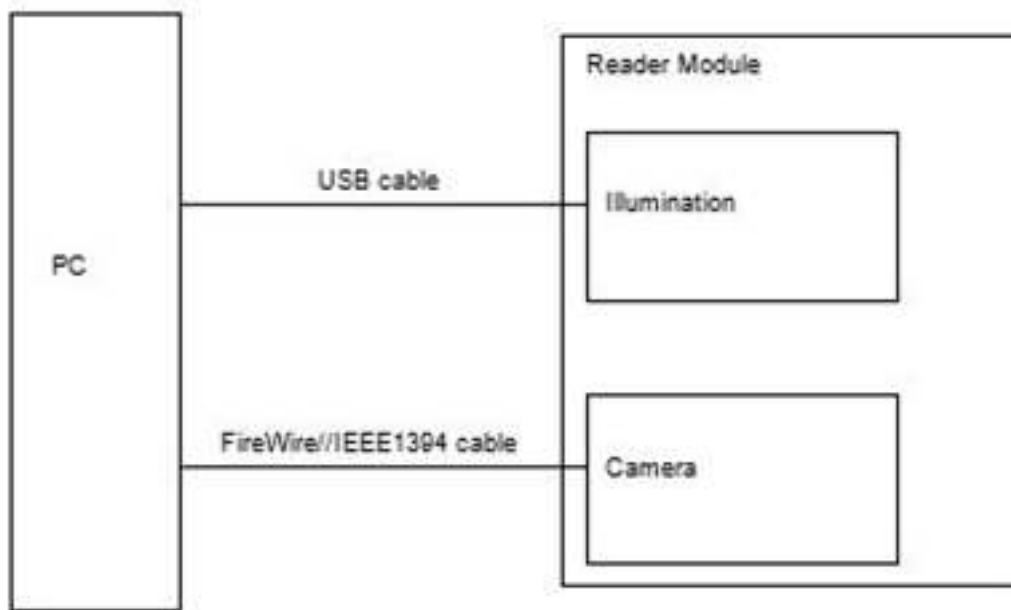


Figure 587 - LEC module block diagram

PC - Camera communication is through FireWire/IEEE1394.

PC - PCA RD-IL421 (illumination panel) is through USB.

The PCA related to the LEC module is:

- PCA RD-IL421: located in the reader, attached to the reader connectors side. It controls reader illumination through the PC-RD-IL421 USB cable.

2.8.2 Functional adjustments

All adjustments are included in a single wizard in a row. Diaphragm and focus adjustments require access to the camera iris and focus and then it is advisable to perform them with the reader module disassembled from the instrument, or assembled in the instrument but with the reader lateral cover removed. All other adjustments (illumination and integration adjustments) can be easily completed with the reader assembled inside the instrument.



NOTE

The reader adjustment tool-support shall be inserted in the reader in all the steps of adjustment or verification whenever performed with the reader module not assembled inside the Erytra, to avoid external light contribution.



NOTE

The reader adjustment tool-support shall be inserted in the reader together with the reader adjustment tool-pattern in all the steps of adjustment or verification that require the reader adjustment tool-pattern.

Illumination intensity adjustment must be performed whenever the reader module is re-assembled in the instrument.

The adjustment procedure is described in the following lines assuming that adjustment is performed outside the Erytra (Reader adjustment tool-support is inserted in the reader in all the steps). Should this not be the case, take into account that the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) should only be inserted when the software requires the READER ADJUSTMENT TOOL-PATTERN v2.

Whenever an adjustment procedure is performed, the last image acquired during the adjustment routine will be saved in PGM and BMP formats in **E:\Log\DG421\Images** as **LastAdjImage.PGM** and **LastAdjImage.BMP**. Additionally, a log file will be created, containing the values for each parameter, illumination and software version, camera model and serial number. This file is stored in the **E:\Logs\DG421** directory with the name: **Adjustment - DD_MM_YYYY - hh_mm_ss.log**, where:

DD: day.

MM: month.

YYYY: year.

hh: hours.

mm: minutes.

ss: seconds.

Summary of adjustments:

Table 61 - Summary of adjustments of the LEC module

Adjustment	Reader adjustment tool-support (module removed)	Reader adjustment tool-pattern	Automatic / manual	Reader lateral cover	Allowed range
Optics Adjustments					
"Adjusting diaphragm aperture" on page 701.	X	-	Manual. Diaphragm adjustment	Removed	160 to 180
"Adjusting focus" on page 703.	X	X	Manual. Focus adjustment	Removed	65 to 100
Illumination Adjustments					
"Adjusting illumination position" on page 705.	X	-	Automatic	Assembled	Hor: 16 to 79 Ver: 16 to 89
"Adjusting illumination dimensions" on page 707.	X	-	Automatic	Assembled	Width: 371 to 399 Height: 500 to 523
"Adjusting illumination rotation" on page 708.	X	-	Automatic	Assembled	Rotation*100: -129 to 129
"Adjusting illumination perspective" on page 709.	X	-	Automatic	Assembled	HorPersp(%): -83 to 83 VerPersp(%): -80 to 80

Adjustment	Reader adjustment tool-support (module removed)	Reader adjustment tool-pattern	Automatic / manual	Reader lateral cover	Allowed range
"Adjusting illumination intensity" on page 711.	X	-	Automatic	Assembled	No range; 170
"Adjusting illumination color" on page 712.	X	-	Automatic	Assembled	Red factor*100: 120 to 229 Blue factor*100: 69 to 160
Integration Adjustments					
"Adjusting conversion factor" on page 714.	X	X	Automatic	Assembled	Pix/mm*100: 2050 to 2150

2.8.2.1 Adjusting diaphragm aperture

An estimation of the diaphragm aperture is obtained by means of an image process (i.e. mean of the luminance layer) supposing the incident light is constant. To perform this adjustment, the illumination panel light intensity is set to the default (current) value.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **reader** button found in the Technical Service software > **Procedures** tab (see "Procedures tab" on page 756) > **Adjustments** area > **CAT floor Modules** area.
2. Attach the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) to the reader without the READER ADJUSTMENT TOOL-PATTERN v2.
3. Remove the reader lateral cover.

4. Loosen the screw fixing the diaphragm to unlock it. The ring is the furthest one from the camera.



Figure 588 - Fixing screw of the diaphragm

- Click the **Start Adjustment** button. Images are acquired and displayed together with maximum, minimum, current light intensity/aperture values and the area used for measurement.

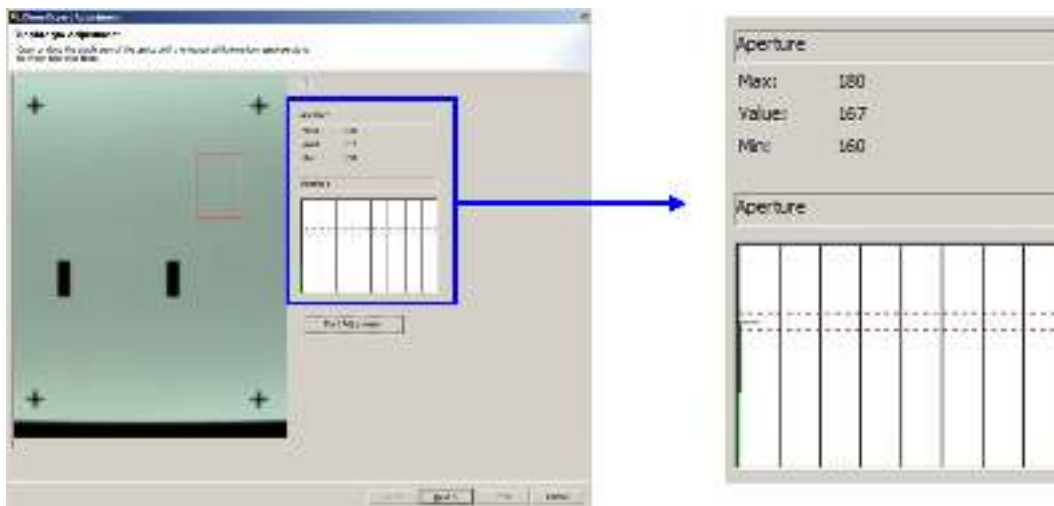


Figure 589 - Diaphragm adjustment screen in the Technical Service software

- Adjust the diaphragm position so the aperture/current light intensity value is within the allowed range of 180 to 160, and the green line is within the dotted red lines. The target value is 170. Then, the **Next** button is enabled.
- Tighten the diaphragm fixing screw to lock it, checking that the measured aperture value remains within the allowed margin.
- Click the **Next** button.



NOTE

Verify the adjustment (See "Verification procedures" chapter), if desired.

2.8.2.2 Adjusting focus

The READER ADJUSTMENT TOOL-PATTERN v2 provides a reference to measure the level of focus of the card/microtubes plane. The contrast of the pattern vertical line is measured.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. A message box indicating that the reader adjustment tool-pattern is required for this adjustment is displayed. Click the **Ok** button.
2. Remove the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2).

3. Attach the READER ADJUSTMENT TOOL-PATTERN v2 to the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2), as shown in the picture, so that the screw heads are in the illumination panel side.



Figure 590 - READER ADJUSTMENT TOOL-PATTERN v2 attached to the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2)

4. If the reader lateral cover is assembled, remove it.
5. Loosen the screw fixing the focus to unlock it. The ring is the closest one to the camera.



Figure 591 - Fixing screw of the focus

- Click the **Start Adjustment** button. Images are acquired and displayed together with the area where the pattern vertical line is searched and maximum, minimum and current contrast value for the pattern vertical line.

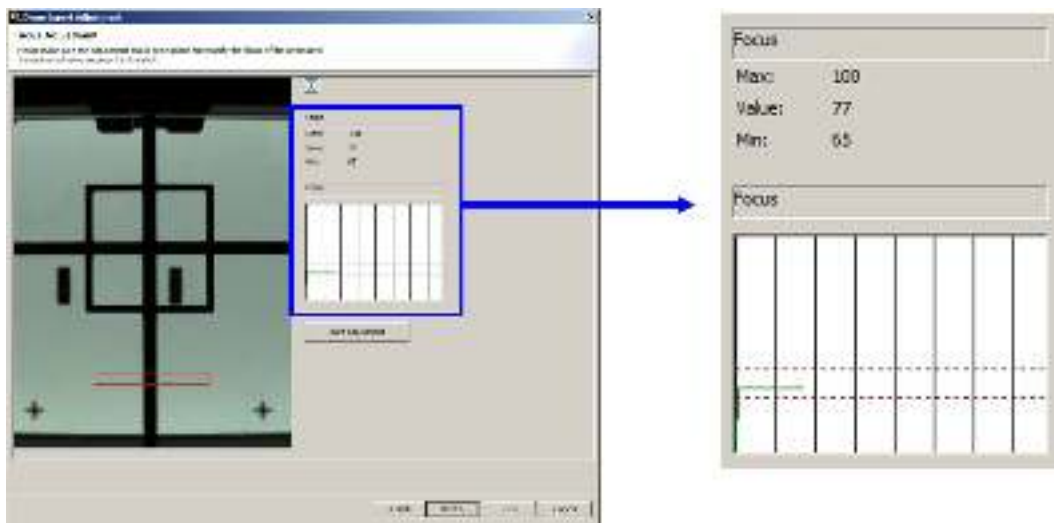


Figure 592 - Focus adjustment screen in the Technical Service software

- Adjust the focus position so that the focus current value is the maximum value in the graph and it is within the allowed range (green line is within dotted red lines). The **Next** button is enabled.
- Tighten the focus fixing screw to lock the focus, checking that the measured focus value remains within the allowed margin.
- Assemble the reader lateral cover.
- Click the **Next** button.



NOTE

Verify the adjustment (See "Verification procedures" chapter), if desired.

2.8.2.3 Adjusting illumination position

The software determines the illumination position by looking for the illumination pattern top-left corner cross within the allowed area (red square). The detected cross positions are displayed in red in the picture and the **Value** fields are populated with its coordinates.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. A message box requests **READER ADJUSTMENT TOOL-PATTERN v2removal**, so remove it. Click the **Ok** button.
2. Attach the **READER ADJUSTMENT TOOL-SUPPORT (Ver. 2)** to the reader.
3. Click the **Start Adjustment** button. Images are acquired and the top-left corner cross is searched. The acquired image, searching area (red square) and detected cross position (red cross) are displayed together with the maximum, minimum and current position values for vertical and horizontal cross positions.

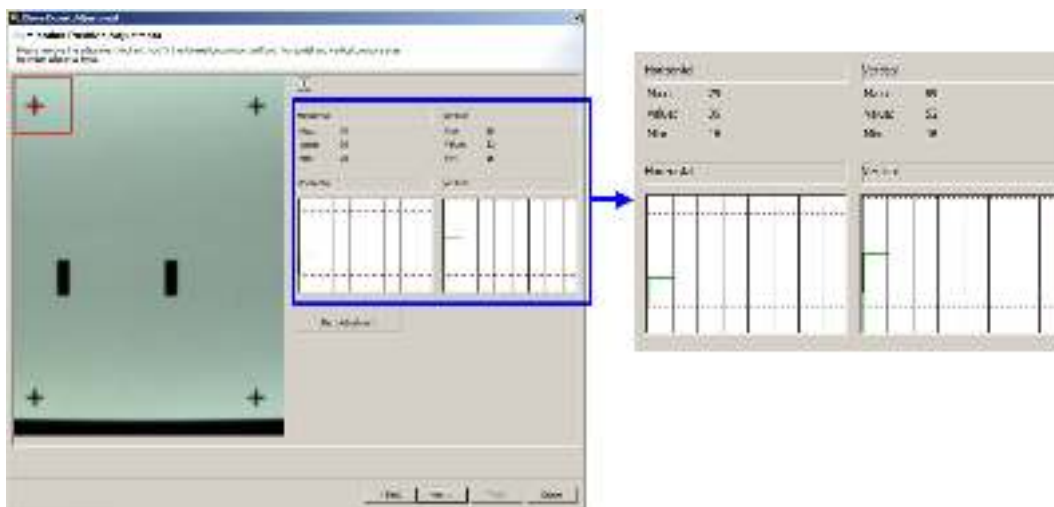


Figure 593 - Illumination position adjustment screen in the Technical Service software

The **Next** button is enabled if both position values are within the allowed range.

4. Click the **Next** button.



NOTE

Verify the adjustment (See "Verification procedures" chapter), if desired.

2.8.2.4 Adjusting illumination dimensions

The software determines the illumination panel dimensions/camera-illumination panel distance by looking for the four illumination pattern corner crosses within the allowed areas (red squares). The detected cross positions are displayed in red in the picture and the obtained coordinates are used to calculate the illumination pattern horizontal and vertical dimensions. The **Value** fields are populated with the obtained dimensions.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) is attached to the reader without the READER ADJUSTMENT TOOL-PATTERN v2 and that the reader lateral cover is assembled.

- Click the **Start Adjustment** button. Images are acquired and displayed together with cross searching areas (red squares), located cross positions (red crosses) and maximum, minimum and current height and width values.

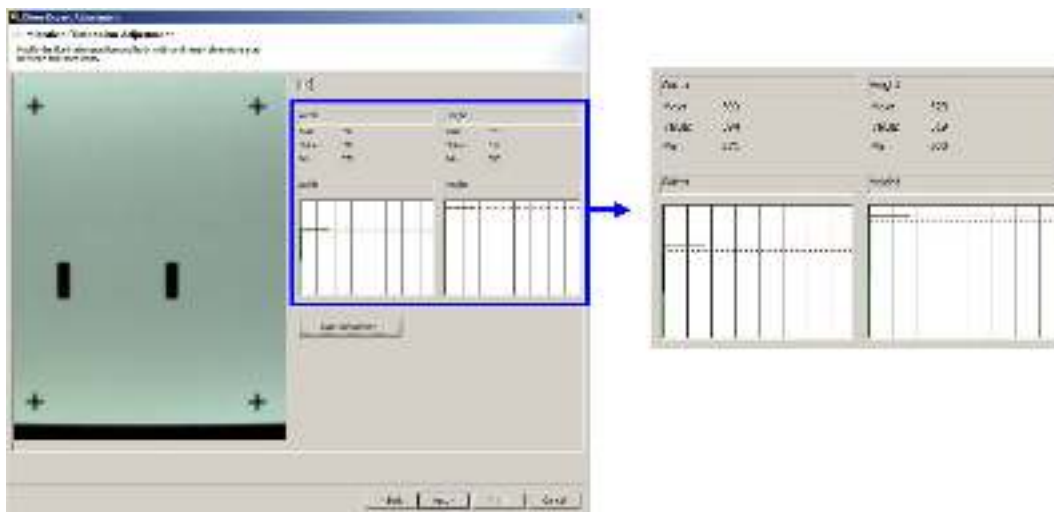


Figure 594 - Illumination dimensions adjustment screen in the Technical Service software

The **Next** button is enabled when both dimension values are within the allowed range.

- Click the **Next** button.



NOTE

Verify the adjustment (See "Verification procedures" chapter), if desired.

2.8.2.5 Adjusting illumination rotation

The software determines the illumination panel rotation by comparing vertical and horizontal lines obtained using corner crosses with absolute/complete vertical and horizontal lines. The software looks for the four illumination pattern corner crosses within the allowed areas (red squares) and displays obtained cross positions and lines. The **Value** field is populated with the obtained rotation.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) is attached to the reader without the READER ADJUSTMENT TOOL-PATTERN v2 and that the reader lateral cover is assembled.
2. Click the **Start Adjustment** button. Images are acquired and displayed together with cross searching area, located cross positions, maximum and minimum and current absolute angle value multiplied by 100.

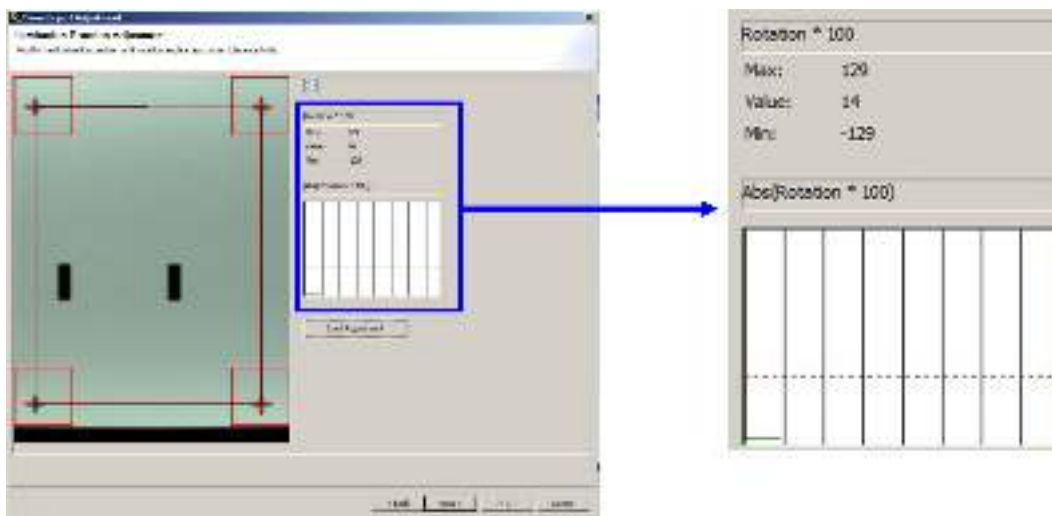


Figure 595 - Illumination rotation verification screen in the Technical Service software

The **Next** button is enabled when the measured angle value is within the allowed range.

3. Click the **Next** button.



NOTE

Verify the adjustment (See "Verification procedures" chapter), if desired.

2.8.2.6 Adjusting illumination perspective

The software determines the illumination panel vertical/horizontal perspective by comparing both vertical and horizontal line lengths obtained using the detected corner cross positions. The software looks for the four illumination pattern corner crosses within the allowed areas (red squares) and displays obtained cross positions and lines. The **Value** fields are populated with the obtained dimensions.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) is attached to the reader without the READER ADJUSTMENT TOOL-PATTERN v2 and that the reader lateral cover is assembled.

- Click the **Start Adjustment** button. Images are acquired and displayed together with maximum, minimum and current horizontal and vertical perspective values, expressed as a percentage and multiplied by 100.

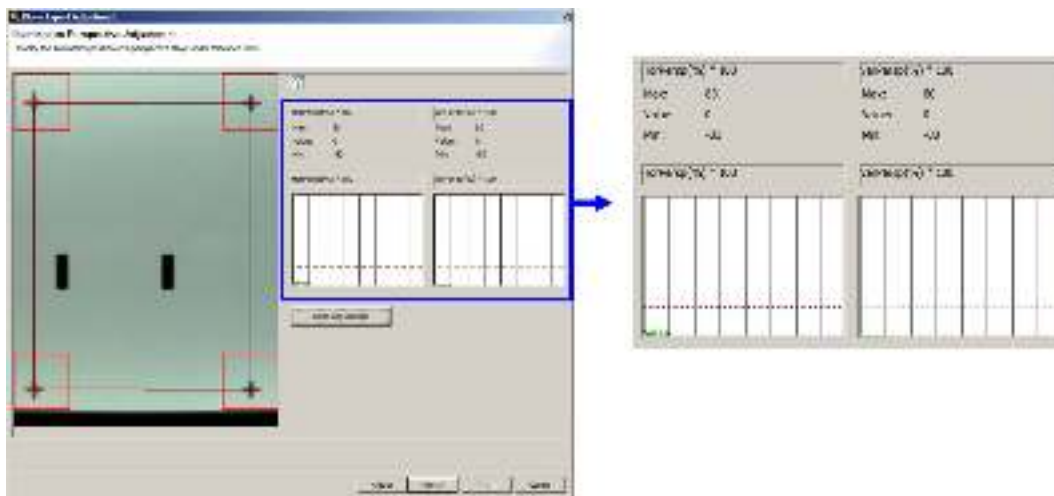


Figure 596 - Illumination perspective adjustment screen in the Technical Service software

0% will indicate no perspective distortion.

The **Next** button is enabled when both perspective values are within the allowed range.

- Click the **Next** button.



NOTE

Verify the adjustment (See "Verification procedures" chapter), if desired.

2.8.2.7 Adjusting illumination intensity

The software measures the amount of light reaching from a certain area (red rectangle) and automatically adjusts the illumination panel light intensity (current) to obtain the target value of 170.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) is attached to the reader without the READER ADJUSTMENT TOOL-PATTERN v2 and that the reader lateral cover is assembled.
2. Click the **Start Adjustment** button. Images are acquired and displayed together with measured area (red rectangle), maximum, minimum and current measured intensity value.

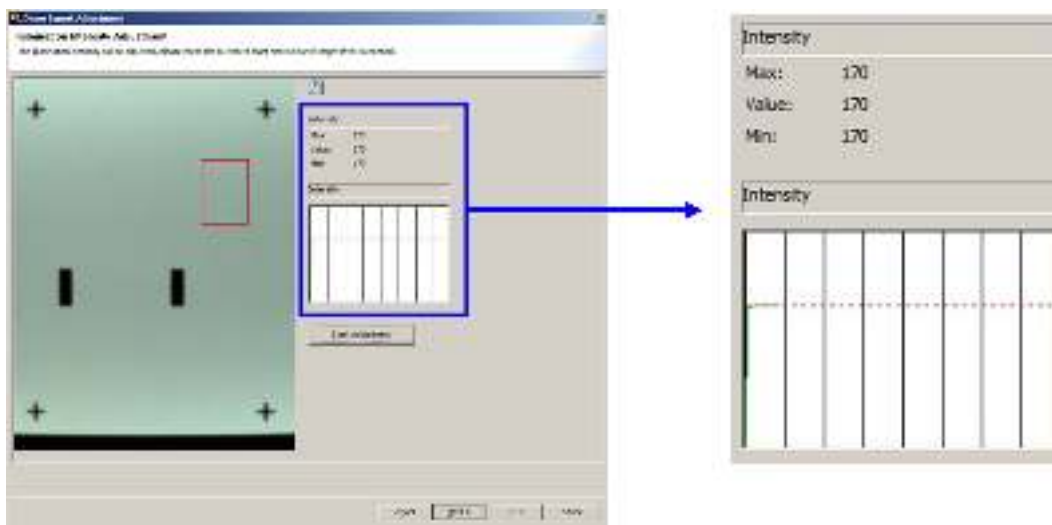


Figure 597 - Illumination intensity adjustment screen in the Technical Service software

The software automatically modifies the illumination intensity so the target value of 170 is obtained. At that time, the **Next** button is enabled.

3. Click the **Next** button.



NOTE

Verify the adjustment (See "Verification procedures" chapter), if desired.

2.8.2.8 Adjusting illumination color

The software measures the amount of light reaching from a certain area (red rectangle area) and calculates red and blue factors (color and white balance).



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Check the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) is attached to the reader without the READER ADJUSTMENT TOOL-PATTERN v2 and that the reader lateral cover is assembled.
2. Click the **Start Adjustment** button. Images are acquired and displayed together with the area used for measurement, maximum, minimum and current measured factors multiplied by 100 for red and blue.

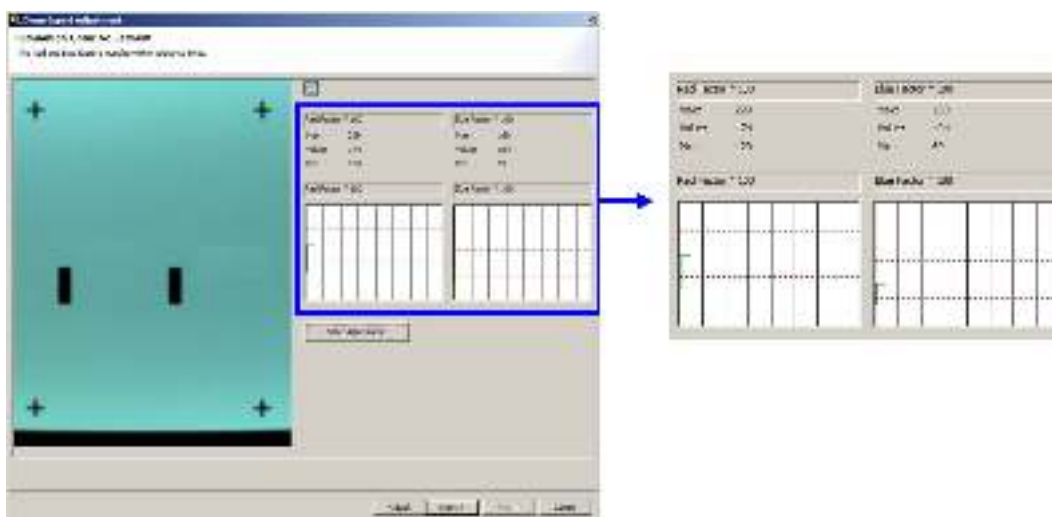


Figure 598 - Illumination color adjustment screen in the Technical Service software

The **Next** button is enabled if both factor values are within the allowed range.

3. Click the **Next** button.

**NOTE**

Verify the adjustment (See "Verification procedures" chapter), if desired.

2.8.2.9 Adjusting conversion factor

The software determines the pixels/mm conversion factor by obtaining the length in pixels of an already known length in millimetres. The length used is square side (10 mm). The software looks for the internal edge of the square lower horizontal line and external edge of the square higher horizontal line within the allowed areas (red rectangles). The detected positions are displayed in green in the picture and used to calculate the conversion factor. The **Value** field is populated with the obtained conversion factor.

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Attach the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) with the READER ADJUSTMENT TOOL-PATTERN v2 to the reader.
2. Check that the reader lateral cover is assembled.

- Click the **Start Adjustment** button. Images are acquired and displayed together with maximum, minimum and current measured conversion factor multiplied by 100.

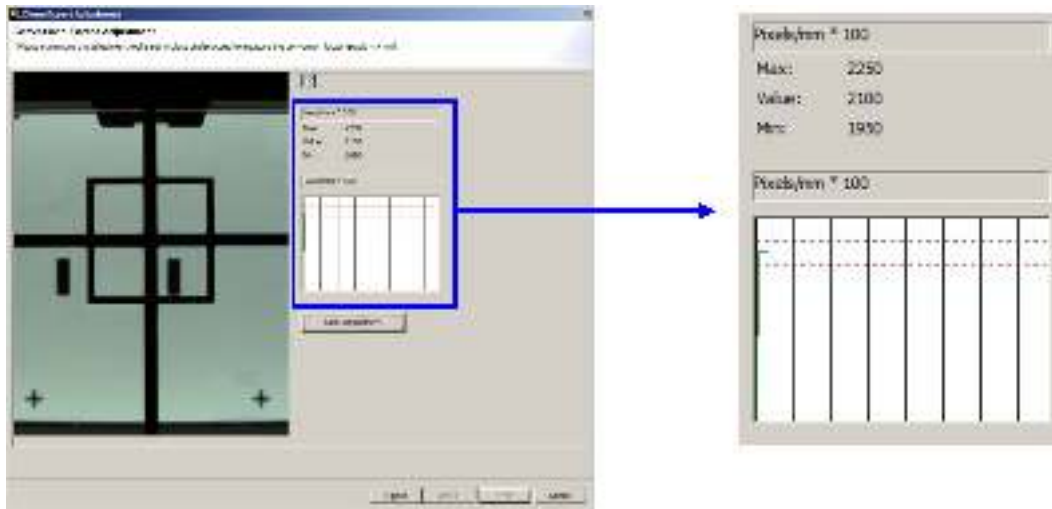


Figure 599 - Conversion factor adjustment screen in the Technical Service software

If the obtained conversion factor is within the allowed limit, a message indicating that the module has been successfully adjusted will be displayed and the **Finish** button will be enabled.

- Click the **Finish** button.
- Check/perform a gripper adjustment in the reader area. See "Adjusting gripper in the card reader area" on page 487.

2.8.3 Verification procedures

The following verifications can be performed in a row:

Optics adjustment checks:

- Diaphragm aperture check.
- Focus check (requires the reader adjustment tool-pattern).

Illumination checks:

- Illumination position check.
- Illumination dimensions check.
- Illumination rotation check.
- Illumination perspective check.
- Illumination intensity check.
- Illumination color check.
- Luminance uniformity check.
- Color uniformity check.
- Illumination cleanliness check.



NOTE

The reader adjustment tool-support must be inserted into the reader in all the steps of verification whenever performed with the reader module not assembled inside the Erytra to avoid external light contribution.

**NOTE**

The reader adjustment tool-support must be inserted into the reader together with the reader adjustment tool-pattern in all the steps of verification that require the reader adjustment tool-pattern.

Summary of verifications:

Table 62 - Summary of verifications

Check	Reader adjustment tool-support (module removed)	Reader adjustment tool-pattern	Allowed range
Optics adjustment checks			
"Verifying diaphragm aperture" on page 720.	X	-	158 to 182
"Verifying focus" on page 722.	X	X	60 to 100
Illumination checks			
"Verifying illumination position" on page 723.	X	-	Horizontal: 15 to 80 Vertical: 15 to 90
"Verifying illumination dimensions" on page 724.	X	-	Width: 370 to 400 Height: 499 to 524

Check	Reader adjustment tool-support (module removed)	Reader adjustment tool-pattern	Allowed range
"Verifying illumination rotation" on page 726.	X	-	Rotation*100: -135 to 135
"Verifying illumination perspective" on page 727.	X	-	Horizontal*100: -110 to 110 Vertical*100: -100 to 100
"Verifying illumination intensity" on page 728.	X	-	165 to 175
"Verifying illumination color" on page 729.	X	-	Red factor*100: 119 to 219 Blue factor*100: 59 to 159
"Verifying luminance uniformity" on page 731.	X	-	Horizontal: 80 to 100 Vertical: 70 to 100
"Verifying color uniformity" on page 732.	X	-	Horizontal: 90 to 100 Vertical: 90 to 100
"Verifying illumination cleanliness" on page 733.	X	-	240 to 0

The verification procedure is described assuming that the reader module is assembled inside the Erytra. Should this not be the case, take into account the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) requirements stated above.

A log file is created whenever a verification procedure is performed. A log file contains the values for each parameter, illumination and software version, camera model and serial number. This file is stored in the **E:\Logs\DG421** directory with the name: **Check - DD_MM_YYYY - hh_mm_ss.log**, where:
DD: day, MM: month, YYYY: year, hh: hours, mm: minutes, ss: seconds.

2.8.3.1 Verifying diaphragm aperture

An estimation of the diaphragm aperture can be obtained by means of an image process (i.e. mean of the luminance layer) supposing the incident light is constant.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **verify** button found in the Technical Service software **Procedures** tab (see "Procedures tab" on page 756) > **Checks** section > **Reader Module** area. The following window will be displayed:

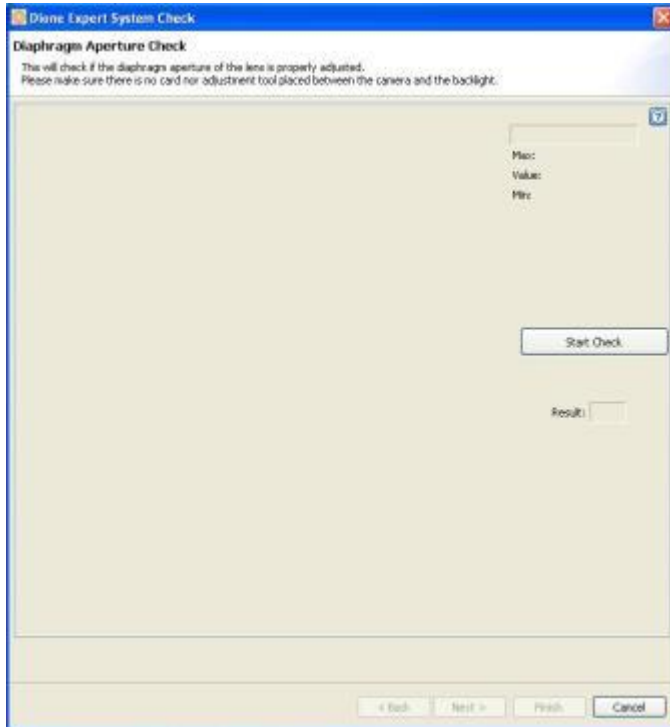


Figure 600 - Diaphragm aperture check initial screen in the Technical Service software

- Click the **Start Check** button. Images will be acquired and displayed together with maximum, minimum, current light intensity/aperture values, area used for measurement (red rectangle) and check result.

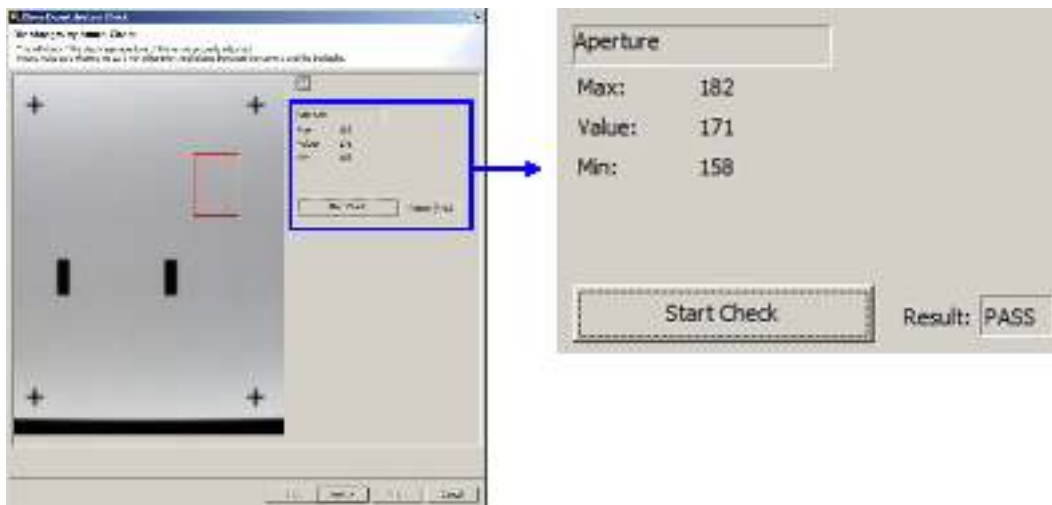


Figure 601 - Diaphragm aperture check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected.

The **Next** button is enabled if the test result is Pass.

- Click the **Next** or **Cancel** button.

2.8.3.2 Verifying focus

The reader adjustment tool-pattern is used to provide a reference to measure the level of focus of the card/microtubes plane. The contrast of pattern vertical line is measured.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2).
- 731398 READER ADJUSTMENT TOOL-PATTERN v2.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Attach the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) with the READER ADJUSTMENT TOOL-PATTERN v2 to the reader.
2. Click the **Start Check** button. Images will be acquired and displayed together with the maximum, minimum, current light intensity/focus values, area used for measurement and check result.

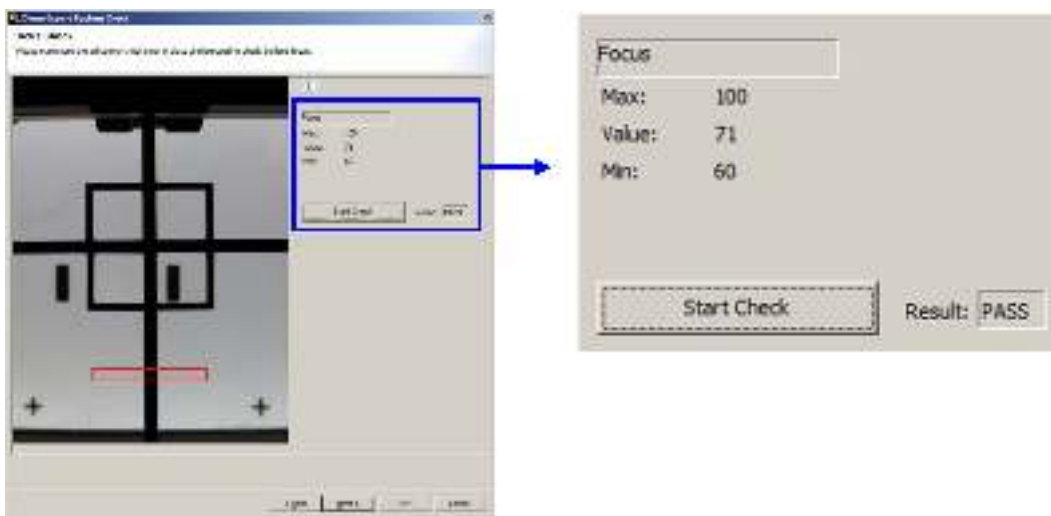


Figure 602 - Optical focus check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

3. Click the **Next** or **Cancel** button.

2.8.3.3 Verifying illumination position

The software will determine the illumination position by looking for the illumination pattern top-left corner cross within the allowed area (red square). The detected cross position will be displayed in red in the picture and the **Value** fields will be populated with its coordinates.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- 234212 READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) .
- 731398 READER ADJUSTMENT TOOL-PATTERN v2 .

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the READER ADJUSTMENT TOOL-SUPPORT (Ver. 2) and the READER ADJUSTMENT TOOL-PATTERN v2 from the reader.
2. Click the **Start Check** button. Images will be acquired and the top-left corner cross will be searched. The acquired image, searching area (red square) and detected cross position (red cross) will be displayed together with the maximum, minimum, current position values for vertical and horizontal cross position values and check result.

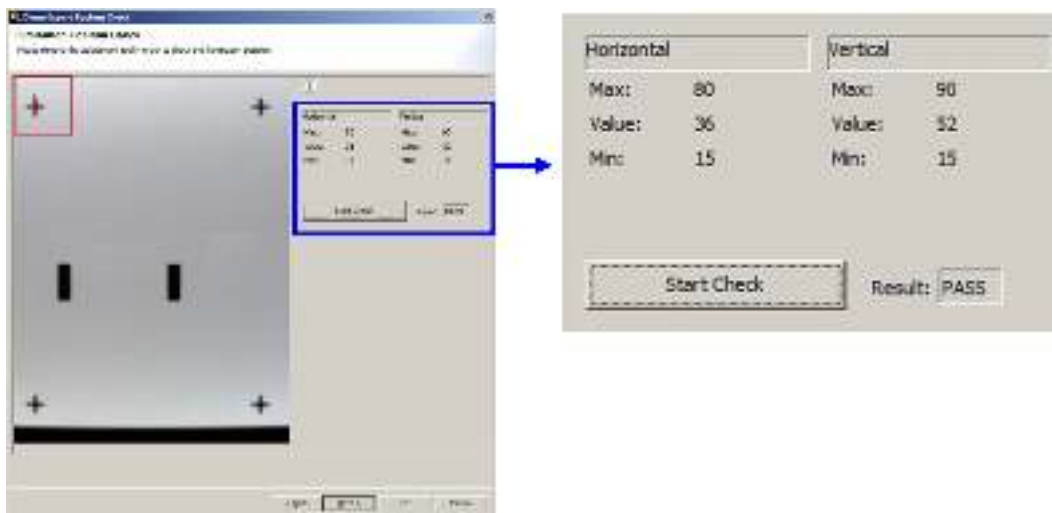


Figure 603 - Illumination position check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

3. Click the **Next** or **Cancel** button.

2.8.3.4 Verifying illumination dimensions

The software will determine the illumination panel dimensions and camera-illumination panel distance, looking for the four illumination pattern corner crosses within allowed areas (red squares). The detected cross positions will be displayed in red in the picture and the obtained coordinates will

be used to calculate the illumination pattern horizontal and vertical dimensions. The **Value** fields will be populated with the obtained dimensions.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **Start Check** button. Images will be acquired and displayed together with the maximum, minimum, current height and width values and check result.

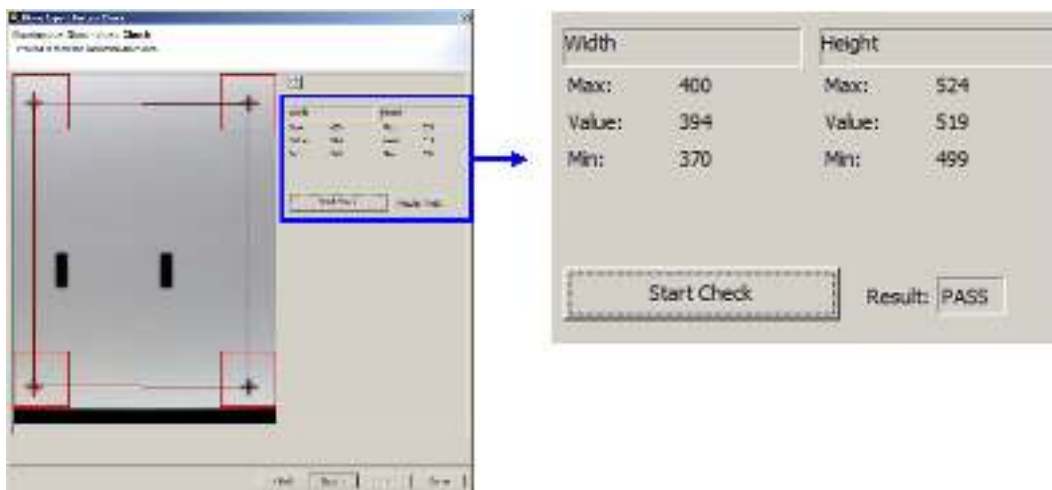


Figure 604 - Illumination dimensions check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

2. Click the **Next** or **Cancel** button.

2.8.3.5 Verifying illumination rotation

The software will determine the illumination panel rotation by comparing vertical and horizontal lines obtained using corner crosses with absolute/complete vertical and horizontal lines. The software will look for the four illumination pattern corner crosses within allowed areas (red squares) and display the obtained cross positions and lines. The **Value** field will be populated with the obtained rotation.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **Start Check** button. Images will be acquired and displayed together with the crosses searching area, located crosses position and maximum, minimum, current angle value multiplied by 100 and check result.

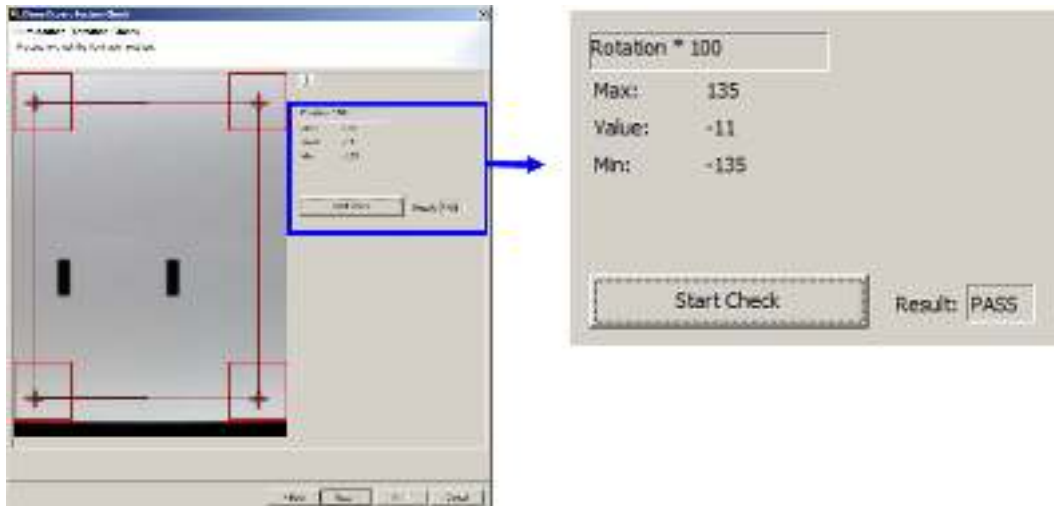


Figure 605 - Illumination rotation check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

2. Click the **Next** or **Cancel** button.

2.8.3.6 Verifying illumination perspective

The software will determine the illumination panel vertical and horizontal perspective by comparing both vertical and horizontal line lengths obtained using detected corner cross positions. The software will look for the four illumination pattern corner crosses within allowed areas (red squares) and display the obtained cross positions and lines. The **Value** fields will be populated with the obtained dimensions.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **Start Check** button. Images will be acquired and displayed together with the maximum, minimum, current horizontal and vertical perspective values expressed as percentages and multiplied by 100 and check result.

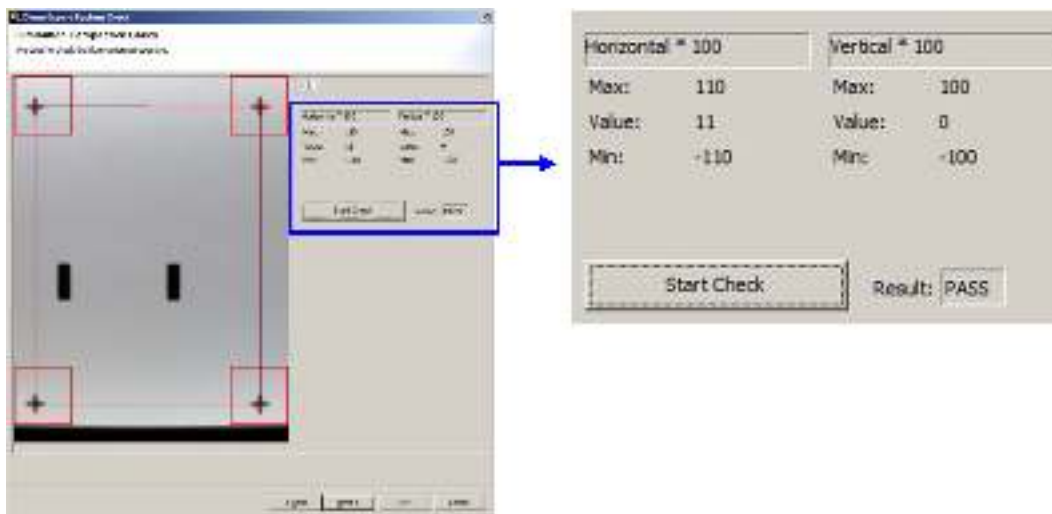


Figure 606 - Illumination perspective check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

2. Click the **Next** or **Cancel** button.

2.8.3.7 Verifying illumination intensity

The software will measure the amount of light reaching from a certain area (red rectangle area) and will automatically adjust the light intensity to obtain the target value of 170.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **Start Check** button. Images will be acquired and displayed together with the maximum, minimum, current measured intensity value and check result.

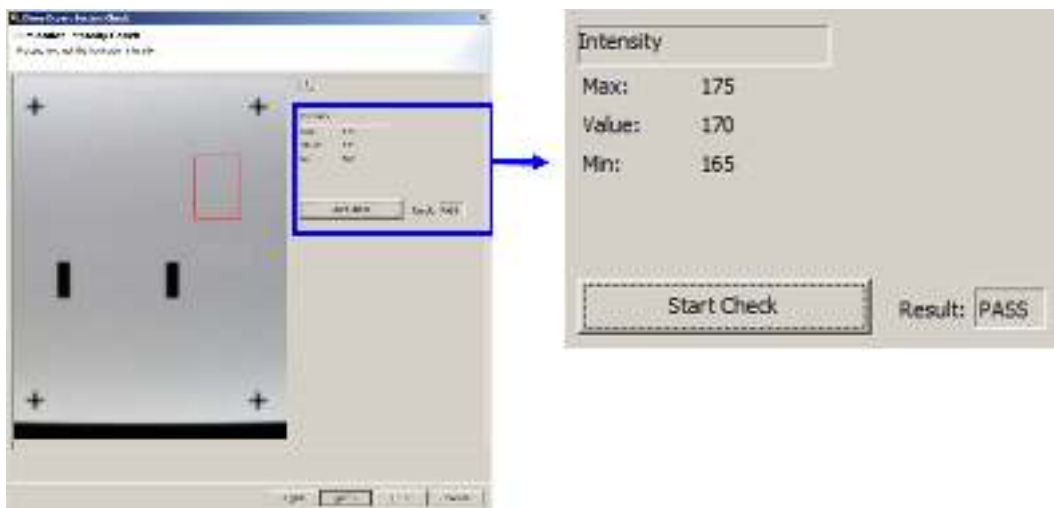


Figure 607 - Illumination intensity check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

2. Click the **Next** or **Cancel** button.

2.8.3.8 Verifying illumination color

The software will measure the amount of light reaching from a certain area (red rectangle), calculate red and blue factors, color and white balance, and compare them with the adjustment ones.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **Start Check** button. Images will be acquired and displayed together with the maximum, minimum, current measured factors variation multiplied by 100 for red and blue and check result.

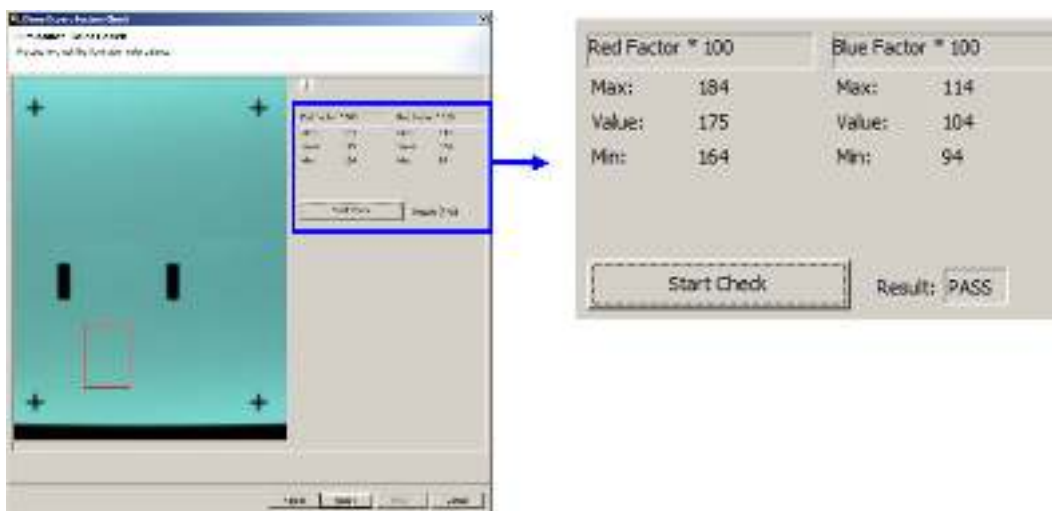


Figure 608 - Illumination color check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

2. Click the **Next** or **Cancel** button.

2.8.3.9 Verifying luminance uniformity



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **Start Check** button. Images will be acquired and displayed together with the maximum, minimum, current measured factors multiplied by 100 for red and blue and check result.

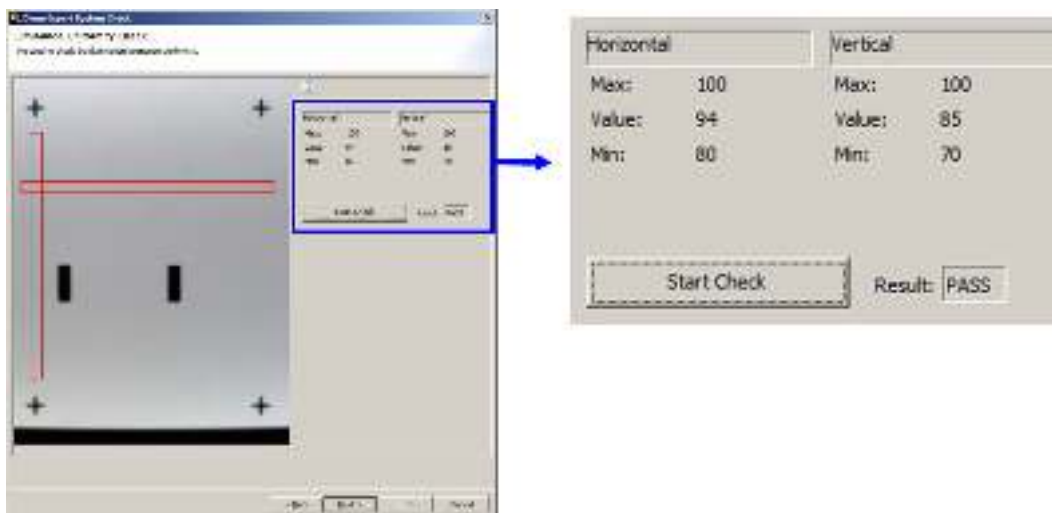


Figure 609 - Illumination uniformity check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

2. Click the **Next** or **Cancel** button.

2.8.3.10 Verifying color uniformity



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **Start Check** button. Images will be acquired and displayed together with the maximum, minimum, current measured factors multiplied by 100 for red and blue and check result.

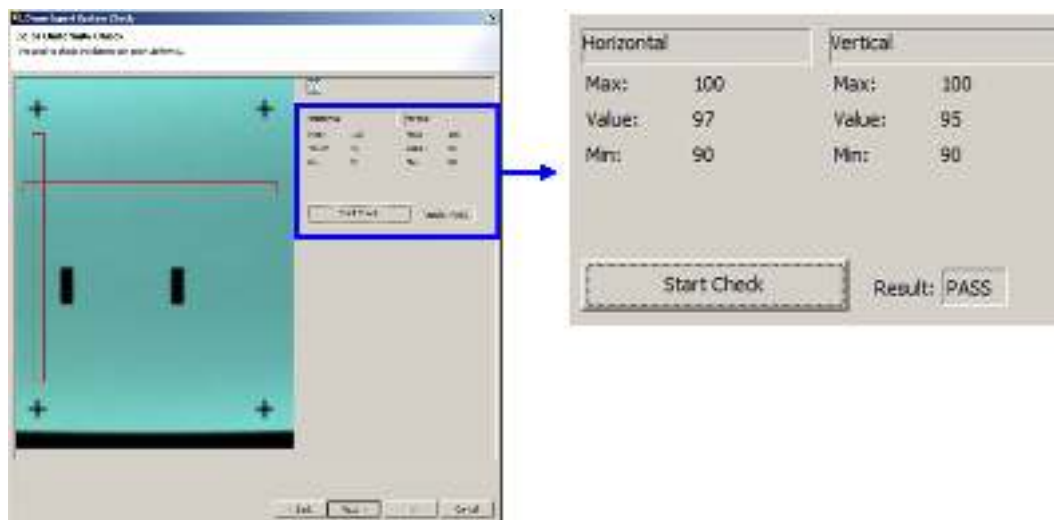


Figure 610 - Color uniformity check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

2. Click the **Next** or **Cancel** button.

2.8.3.11 Verifying illumination cleanliness

The software will look for unexpected elements, artifacts, or dirt in the illumination and illumination pattern.



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Click the **Start Check** button. Images will be acquired and displayed together with the maximum, minimum, current measured factors multiplied by 100 for red and blue and check result.

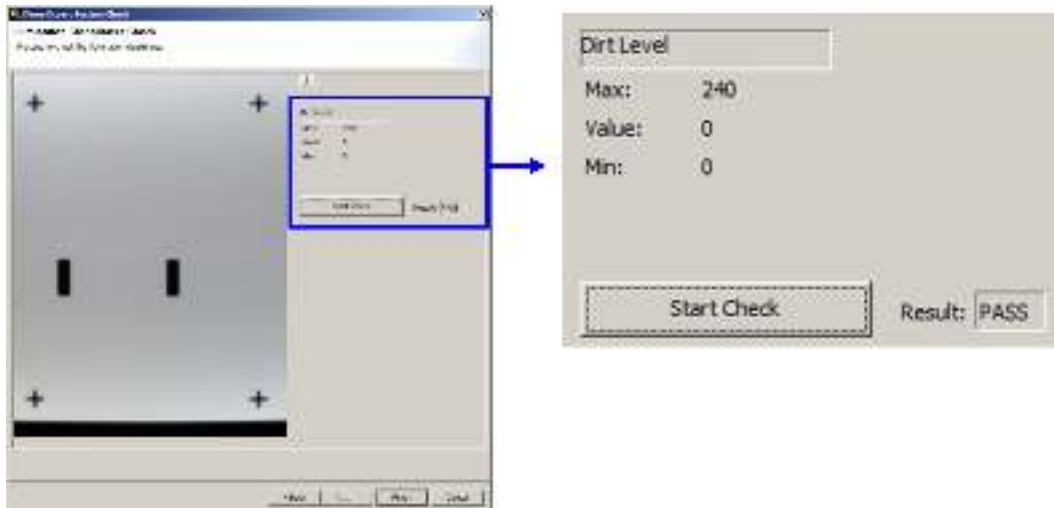


Figure 611 - Illumination cleanliness check screen in the Technical Service software

The check result can be Pass or Fail. If the check result is Fail, the verification process can not go on and the **Cancel** button must be selected. The **Next** button is enabled if the test result is Pass.

A message box with **System Check passed** will be displayed if the test result is Pass.

2.8.4 Replacement procedures

Replacement procedures related to the Card reader module are detailed in this section.

2.8.4.1 Removing reader module



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Open card drawer number 1 and number 2 to have access to the reader module fixing screws and wiring.
2. Unplug the camera cable (FireWire/IEEE1394) and illumination screen cable (USB) from the reader module (illumination panel side).



Figure 612 - USB and FireWire cables. Reader and grounding fixing screws

3. Remove the screw fixing the grounding cable to the reader and remove the grounding cable.
4. Remove the two screws fixing the reader module to the instrument. To ease the removal of the screw, closer to the back of the instrument, the card drawer electric lock might be unscrewed from the instrument.
5. Open solution A and B drawers and remove solution A and B bottles.
6. Partially close both container drawers, slip the reader to the right so it gets out of its location and lower/place it in the solution A and B drawers.
7. Open the drawers and remove the reader.
8. Place solution A and B bottles back in the drawers.



NOTE

Once the reader is assembled back:

- Perform a reader verification. See "Verification procedures" on page 716.
- Check/perform a gripper adjustment in reader area. See "Adjusting gripper in the card reader area" on page 487.

2.8.4.2 Replacing camera



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 233735 CAMERA (w. optics).

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the reader module. See "Removing reader module" on page 734.
2. Remove the eight screws fixing the reader cover to the reader bulk (two screws per side) and remove the cover.
3. Unplug the FireWire/IEEE1394 cable from the camera.

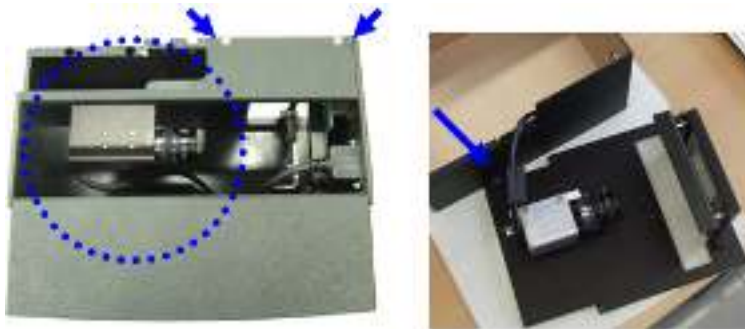


Figure 613 - FireWire/IEEE1394 camera cable

4. Remove the two screws fixing the camera case, and then remove the four screws fixing the camera to its support plate so that the camera can be removed.



Figure 614 - Camera assembly

5. Place the new camera so it rests on the holder's front and bottom edges of the camera shape area and fix it with the four screws.



Figure 615 - Camera and optics set screws assembly

6. Make sure that the optic rings set screws (iris and focus) are assembled facing the camera holder side and as far as possible from the reader basis, to ease handling during adjustment.
7. Assemble the camera cover.
8. Plug the FireWire/IEEE1394 cable into the camera.
9. Assemble the reader cover.
10. Perform the reader adjustment (). Check that the four crosses lay within the expected square area displayed in the adjustment/verification routines. Should this not be the case, loosen the four screws fixing the camera and align the camera.
11. Perform the reader verification. See "Verification procedures" on page 716.
12. Check/perform a gripper adjustment in the reader area. See "Adjusting gripper in the card reader area" on page 487.

2.8.4.3 Replacing reader illumination/PCA RD-IL421



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- 732911 PCA RD-IL421 v3.

For more information about spare parts, see "Spare parts list" on page 1207.



PRECONDITIONS

The following preconditions must be met before starting this procedure:

- None.

Procedure

The reader illumination is part of PCA RD-IL421 and then whole assembly replacement is required. To replace the PCA RD-IL421 proceed as follows:

1. Switch off the instrument and disconnect it from the mains power supply.
2. Remove the reader module. See "Removing reader module" on page 734.
3. Remove the eight screws fixing the reader cover to the reader bulk (two screws per side) and remove the cover.
4. Unplug the FireWire/IEEE1394 cable from the camera.



Figure 616 - PCA RD-IL421 v3 assembly

5. Remove the four screws fixing the PCA RD-IL421 to the reader bulk (two screws per side) and remove the PCA RD-IL421.
6. Assemble the new PCA RD-IL421 using the four screws and checking proper positioning (see previous figure).

7. Assemble the reader cover.

**NOTE**

Verify PCA RD-IL421 firmware version. See "Check firmware version".
Apply latest relevant firmware update if applicable.

8. Verify current firmware version in TSS. Apply latest relevant firmware update if applicable. See "Updating firmware" on page 1129
9. Perform a reader adjustment. See "Functional adjustments" on page 698.
10. Perform a reader verification. See "Verification procedures" on page 716.
11. Check/perform a gripper adjustment in the reader area. See "Adjusting gripper in the card reader area" on page 487.

2.8.4.4 Replacing reader illumination pattern

**ADJUSTMENT TOOLS**

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.

**SPARE PARTS**

The following spare parts are referenced in this procedure:

- //233710 //ILLUMINATION PATTERN DGCEL.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

1. Remove the reader module. See "Removing reader module" on page 734.
2. Remove the eight screws fixing the reader cover to the reader bulk (two screws per side) and remove the cover.
3. Unplug the FireWire/IEEE1394 cable from the camera.

4. Remove the two screws fixing the reader splitter to the reader bulk and remove the splitter.

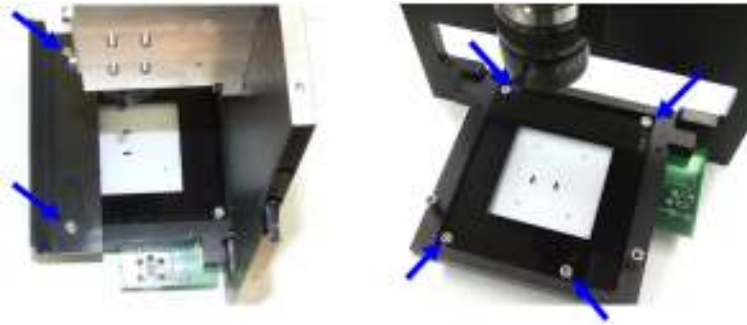


Figure 617 - Reader illumination pattern assembly

5. Remove the four screws fixing the reader illumination pattern to the reader bulk and remove the illumination pattern.
6. Remove the new illumination pattern protective foil and insert and fix it to the reader bulk, avoiding fingerprints. The screw positions assure proper insertion and assembly (serigraphy side facing camera).
7. Clean the illumination pattern with an optics cleaner.
8. Assemble the reader splitter.
9. Connect the FireWire/IEEE 1394 cable from the camera.
10. Assemble the reader cover.
11. Perform a reader adjustment. See "Functional adjustments" on page 698.
12. Perform a reader verification. See "Verification procedures" on page 716.
13. Check/perform a gripper adjustment in the reader area. See "Adjusting gripper in the card reader area" on page 487.

2.8.4.5 Removing reader cover



ADJUSTMENT TOOLS

The following adjustment tools are needed for this procedure:

- None.

For more information about tools, see "FSE tools" on page 56.



SPARE PARTS

The following spare parts are referenced in this procedure:

- None.

For more information about spare parts, see "Spare parts list" on page 1207.

**PRECONDITIONS**

The following preconditions must be met before starting this procedure:

- None.

Procedure

In order to access the reader easily, follow the steps below:

1. Open fluidic drawers A and B and remove the two solutions.
2. Slide down the front cover of the reader, using the side guides until it is completely removed.

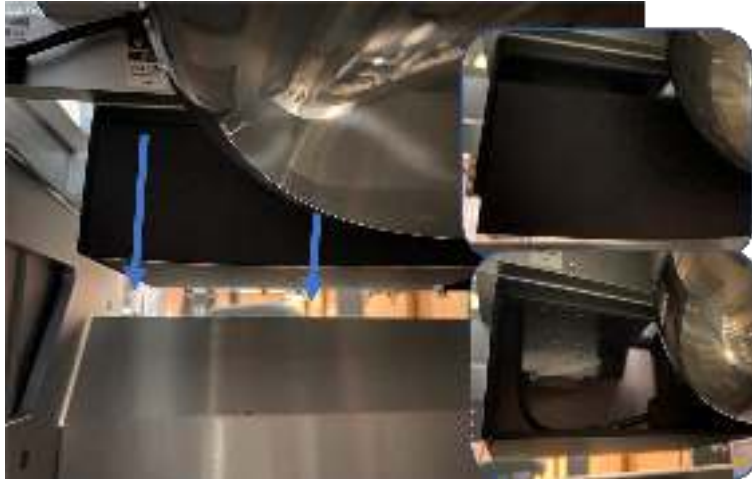


Figure 618 - Reader front cover

It is possible to remove the centrifuge pot for a better access.



Figure 619 - Camera optics, reader and reader foam



NOTE

If the centrifuge pot has been removed, check that all the card holders are correctly fitted before placing the pot back.

2.8.5 Electronics

2.8.5.1 Fuses

Fuses related to the Card reader module are detailed in the following table.

Table 63 - List of fuses of the Card reader module

PCA	Fuse name	Description
PCA RD_IL421	NA	NA

2.8.5.2 LEDs

LEDs related to the Card reader module are detailed in the following table.

Table 64 - List of LEDs of the Card reader module

PCA	LED	Description
PCA RD_IL421	NA	NA

2.8.5.3 Jumpers and test points

Neither jumpers nor test points are related to the Card reader module.

2.8.5.4 Wiring table

Cables related to the Card reader module are detailed in the following table and their connection are shown in the wiring diagram. To view the wiring diagram, see "Related documents" on page 19 or refer to the Erytra Online Help.

Table 65 - List of wiring of the Card reader module

ID	From connector	From PCA/component	To connector	To PCA/component
B-READ_FW	PC.FW (FW1)	PC	NA	Reader box
READ_FW	NA	Reader	Firewire	Camera
READ_GND	NA	Chassis	Earth	Camera
B-READ_USB	PC.USB2	PC	H2	PCA RD-IL421



Figure 620 - Location of the connectors in PCA RD-IL421 v3

2.8.6 Troubleshooting

Table 66 - List of errors related to the reader module

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17409	CAMERA_INITIALIZATION_ERROR	Unable to initialize acquisition system, camera, FireWire bus,...(hardware error).	Error during Reader initialization. Please RESTART the instrument.	Reboot
		17410	ILLUMINATION_INITIALIZATION_ERROR	Unable to initialize the communication with illumination module (hardware error).		
		17413	VERSIONS_DO_NOT_MATCH	Version of the Acquisition Structure stored in EEPROM does not match the software version (versions error).		

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17441	ACQUISITION_ HARDWARE_ ERROR	Error acquiring image (hardware error).	No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		17442	ILLUMINATION_ COMM_ERROR	Error sending command to illumination module (hardware error).	Instrument initialisation step:	Reboot
					Error during Reader initialization. Please RESTART the instrument.	
					Integrity check/reading step:	NA
					No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17443	ILLUMINATION_ HARDWARE_ ERROR	Error executing command in the illumination module or an error flag of the status word is on (hardware error).	Instrument initialisation step:	Reboot
					Error during Reader initialization. Please RESTART the instrument.	
					Integrity check/reading step:	NA
					No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	
		17457	UNABLE_TO_ ADJUST_LIGHT_ INTENSITY	Unable to reach intensity target when adjusting illumination current.	Instrument initialisation step:	Reboot
					Error during Reader initialization. Please RESTART the instrument.	
					Integrity check/reading step:	NA
					No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17458	STARTUP_CHECK_FAILED	Start up check failed. One or more of the checks performed failed.	Error during Reader initialization. Please RESTART the instrument.	Reboot
		17459	STARTUP_CHECK_REQUIRED	A Start up check is required in order to guarantee that the system is within working parameters.		
		17460	NOT_ADJUSTED	Adjustment parameter currently being adjusted is still out of the acceptance range.	NA (only it appears in the Technical Service software)	NA

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17473	DISK_ACCESS_ERROR	Error opening, reading or writing a file (xml, image, log, ...) to disk.	Instrument initialisation step:	RESTART
					Error during Reader initialization. Please RESTART the instrument.	
					Integrity check/reading step:	NA
					No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	
		17474	MEMORY_ALLOCATION_ERROR	Error allocating memory (possible memory leak somewhere).	Instrument initialisation step:	RESTART
					Error during Reader initialization. Please RESTART the instrument.	
					Integrity check/reading step:	NA
					No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17489	UNEXPECTED_ERROR	Unexpected error has been found.	Instrument initialisation step:	RESTART
					Error during Reader initialization. Please RESTART the instrument.	
					Integrity check/reading step:	NA
					No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	
		17490	INCORRECT_USE	A method has been called before initializing the module.	No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
Reader	Reader	17491	WRONG_ARGUMENT	Wrong argument passed to a method.	Instrument initialisation step:	RESTART
					Error during Reader initialization. Please restart the instrument.	
					Integrity check/reading step:	NA
					No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	
Reader	Reader	17505	INVALID_INPUT_IMAGE	Image being processed is not what was expected.	No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		NA	NA	Reader module does not respond in 15 seconds.	System has an unrecoverable error. Please restart the instrument as soon as possible. If problem persists, please contact your Technical Service.	OK

Table 67 - Reader module troubleshooting

Symptom	Possible causes	Recommended actions
System is not able to initialise Reader module / error when software is opened	Software was not closed properly	Switch off the instrument and disconnect from the mains power supply
	Camera cable damaged	Check / replace FireWire cable
	PCA illumination cable damaged	Check / replace USB cable
	Unable to reach illumination intensity	Check / perform reader adjustment
	PCA RD-IL421 failure	Check / replace PCA RD-IL421
	Camera failure	Check / replace camera
Cards are rejected during initial checking 17489	Cards are not in good conditions	Check status of the cards
	Incorrect adjustment of CAT in card drawer	Check adjustment of the gripper in the drawer where the card has been taken
	Camera not correctly assembled	Check camera assembly
	Incorrect adjustment of reader	Check / perform adjustment of the reader
	Incorrect adjustment of CAT in reader	Check / perform adjustment of the gripper in the reader

Symptom	Possible causes	Recommended actions
Cards are not read during final reading 17489	Incorrect adjustment of CAT in centrifuge	Check / perform adjustment of the gripper in the centrifuge where the card has been taken
	Camera not correctly assembled	Check camera assembly
	Incorrect adjustment of reader	Check / perform adjustment of the reader
	Incorrect adjustment of CAT in reader	Check / perform adjustment of the gripper in the reader

3 Technical Service software overview

The Technical Service software allows a qualified technician to adjust and verify the modules.

3.1 Introduction

This chapter contains an overview of the Technical Service software that allows the QUALIFIED TECHNICIAN to adjust and verify all the modules of the instrument.

In order to access the Technical Service software, the boot menu must be accessed. For that, switch on the PC, wait for the Erytra logo to appear on screen and then press the grid () button. Log in as `adminUser` (case sensitive, admin password) and the Boot Application options will be displayed. Once in there, select the Technical Service icon among the Boot Application options.



Figure 621 - Technical Service software icon

Once the Technical Service software is opened, start the **Driver** (select **Run > Start**). See "Modular tabs" on page 754.

3.2 Modular tabs

There is a menu bar at the top of the application, which manages the configuration of the application.



Figure 622 - Menu bar of the Technical Service software

- **Run:**
 - **Start:** it starts the Driver application, enables serial port communication.
 - **Stop:** it stops the Driver application, disables serial port communication.
 - **Edit serial number:** option to enter the serial number of the instrument (maximum seven characters). It is stored in **erytra.properties** file (E:\conf).

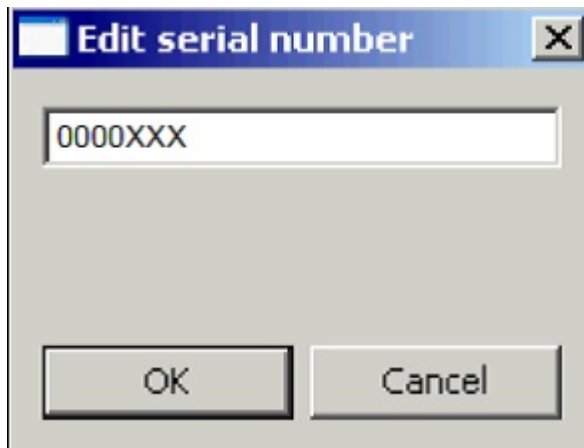


Figure 623 - Edit serial number window

- **Erytra MacroExecutor:** it opens the graphical interface designed to run macros. See "Erytra Macro executor" on page 1146.
- **Erytra On/Off switcher:** it runs the application to control by software the outputs of the power supply. See "Erytra On/Off Switcher" on page 142.
- **Start Windows Keyboard:** it opens the Windows keyboard.
- **Exit:** it ends the program execution.
- **Mode: Driver:** it enables serial port communication.
- **Help: Software Version:** it shows software version.

3.2.1 Procedures tab

The **Procedures** tab includes verifications and adjustments of the different modules of the Erytra. Adjustment of each module is done through a wizard to facilitate the qualified technician to monitor the different steps of the adjustment process.



Figure 624 - Procedures tab in the Technical Service software

Table 68 - List of options in the Procedures tab

Option	Description
Checks	
Reader Module	
verify	Verification of the camera / reader adjustment.
read/check cards test	Perform card initial check and results reading position (microtubes color can be selected).
reference check	Check adjustment of the gripper inside reader module.
backlight usage	Check time [hours] that reader backlight has been switched on.
real time image	It shows real time image of reader module.
Drawers Module	
verify	Verification of the drawers status (samples, reagents and cards).
PIP Module	
pip laser alignment	Check alignment of presence detector laser of PIP arm.
probes offset	Check the adjustment of the probes offset.
pre-analytical laser alignment	Check alignment of barcode reader laser of PIP arm.
Others	
leak-tightness	Check fluidic circuit between dilutors and probes.

Option	Description
Adjustments	
Modules CAT floor	
drawers	Adjustment wizard of card drawers (gripper adjustment in drawers).
centrifuges	Adjustment wizard of centrifuges (gripper adjustment in centrifuges).
incubators	Adjustment wizard of incubators (gripper adjustment in incubators).
reader	Adjustment wizard of the camera / reader module.
acquisition (central) position	Adjustment wizard of the gripper inside reader module.
waste tray	Adjustment wizard of card waste tray (gripper adjustment in card waste tray).
PIP floor Modules	
sample drawers	Adjustment wizard of samples drawers (probes adjustment in samples drawers).
reagent drawers	Adjustment wizard of reagents drawers (probes adjustment in reagents drawers).
incubators	Adjustment wizard of incubators (probes adjustment in incubators).
dilution cup	Adjustment wizard of dilution cup (probes adjustment in dilution cup).
horns	Adjustment wizard of the horns (horns adjustment in reagents / samples drawers).
horns: pre-analytil	Adjustment wizard of the horns for pre-analytical samples racks.
incubators height	Adjustment wizard of the maximum height of the incubators.

Option	Description
probes offset	Adjustment wizard to set both probes at same level in Z-axis.
FLU floor Modules	
gauge	Adjustment wizard of the load cells (bottles of solution A and B, cards waste container and bottles of waste).
sensors	Adjustment wizard of the pressure / vaccum sensors located in the manifold (S3, S4, S5 and S6).
Others	
temperature	Adjustment wizard of the incubators temperature.

3.2.2 Incubators tab

The **Incubators** tab includes all the parameters and functions related to the incubators module. It contains five subtabs: one per each incubator and **Home Cycles**.

3.2.2.1 Incubator 1, 2, 3 and 4 subtabs

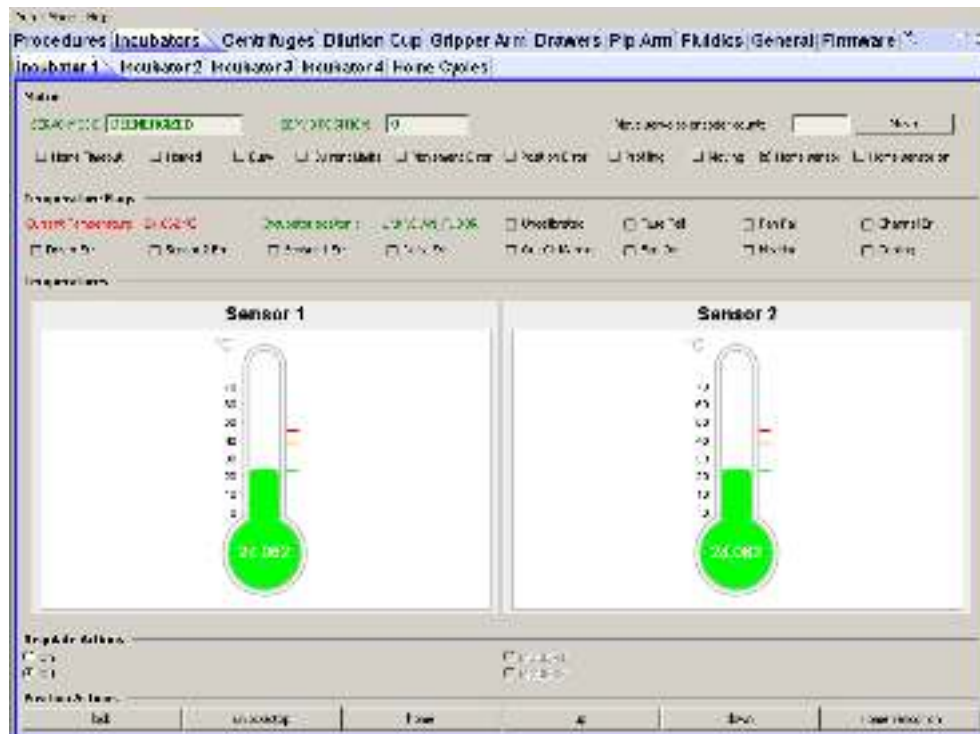


Figure 625 - Incubator 1 - Incubators tab in the Technical Service software

Table 69 - List of options in the Incubator 1, 2, 3 and 4 - Incubators tab

Option	Description
Motor	
SERVO MODE	Describes motor movement. There are four possible status: • Position Control: motor energized, maintaining the actual position. • Deenergized: motor deenergized. • Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. • Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Move	Incubator moves to the number of counts written in the Move servo to encoder count text box.
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.
Homed	Home position has been detected.
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.

Option	Description
Profiling	Servo is executing trapezoidal movement or integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.
Temperature Flags	
Present Temperature	Average temperature of the two temperature sensors.
Incubator position	Information about where is located the incubator. There are three possibilities: • Probes: incubator placed in the upper area. • Gripper: incubator placed in the intermediate area. • UNKNOWN_FLOOR: if a home process has not been performed.
Uncalibrated	Indicates that the incubator is not calibrated.
Fuse Fail	Indicates that temperature regulator fuse is blown.
Fan Fail	Indicates that an error happened when communicating with the Peltier cell fan.
Channel Err	Indicates that an error happened when communicating with incubator multiplexer peripheral.
Driver Err	Indicates an error happened when communicating with the temperature regulator auxiliary peripheral.
Sensor 2 Err	Indicates that an error happened when communicating with the temperature regulator sensor 2.
Sensor 1 Err	Indicates that an error happened when communicating with the temperature regulator sensor 1.

Option	Description
Delta Err	Indicates that temperature difference between sensors 1 and 2 exceeds 1 °C.
Out Of Margin	Indicates that the Incubator temperature is out of regulation margin (only for regulation modes): 3 °C for RT and 1 °C for 37 °C.
Temperature Flags	
Fan On	Indicates that Peltier cell fan is on.
Heating	Indicates that Peltier cell is in heating mode (increasing regulator temperature).
Cooling	Indicates that Peltier cell is in cooling mode (decreasing regulator temperature).
Temperatures	
Sensor 1	Indicates the temperature measured by the sensor 1 in real time.
Sensor 2	Indicates the temperature measured by the sensor 2 in real time.
Regulate Actions	
On	Activates the temperature regulation system.
Off	Deactivates the temperature regulation system.
Mode RT	When the temperature regulation system is on, the temperature is regulated to 24 °C.
Mode 37	When the temperature regulation system is on, the temperature is regulated to 37 °C.
Position Actions	
lock	Lock the servo motor.

Option	Description
unlock/stop	Unlock the servo motor or to stop the current movement.
home	Perform a home movement.
up	Move the incubator to the upper position (samples area).
down	Move the incubator to the lower position (cards area).
Home sensor on	Activate home optical barrier.

3.2.2.2 Home Cycles subtab

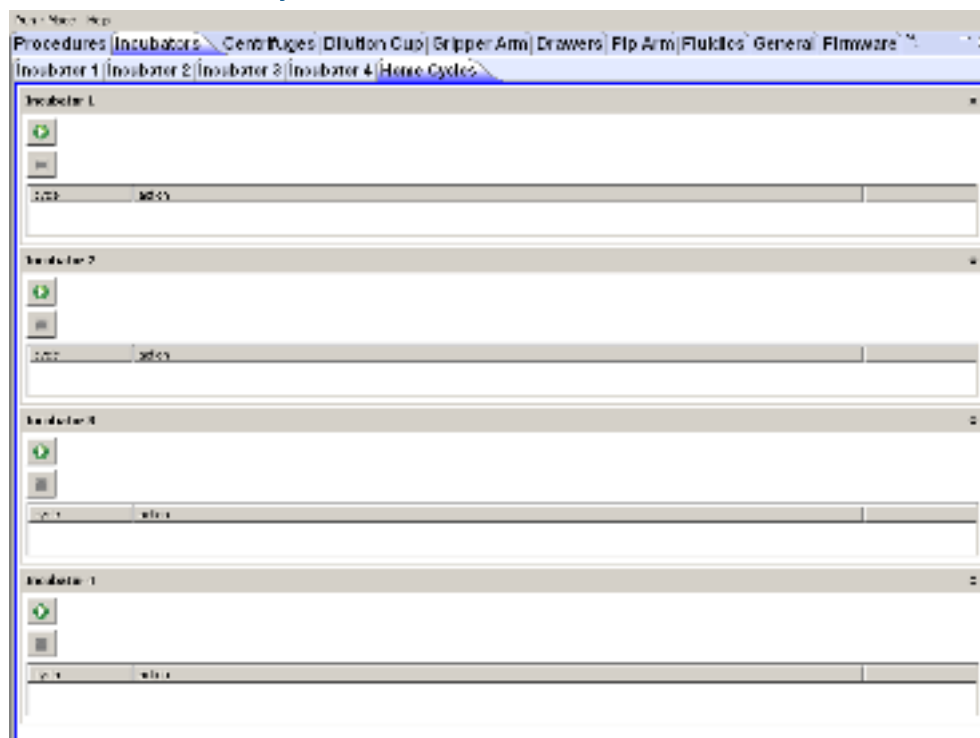


Figure 626 - Home Cycles - Incubators tab in the Technical Service software

Table 70 - List of options in the Home Cycles - Incubators tab

Option	Description
Incubator 1, Incubator 2, Incubator 3 and Incubator 4	
	Perform continuous home movements of the selected incubator.
	Stop the continuous home movements of the selected incubator.

3.2.3 Centrifuges tab

The **Centrifuges** tab includes all the parameters and functions related to the centrifuges module. It contains four subtabs: one per each centrifuge, **Home Cycles** and **Random Position**.

3.2.3.1 Centrifuge 1 and 2 subtabs

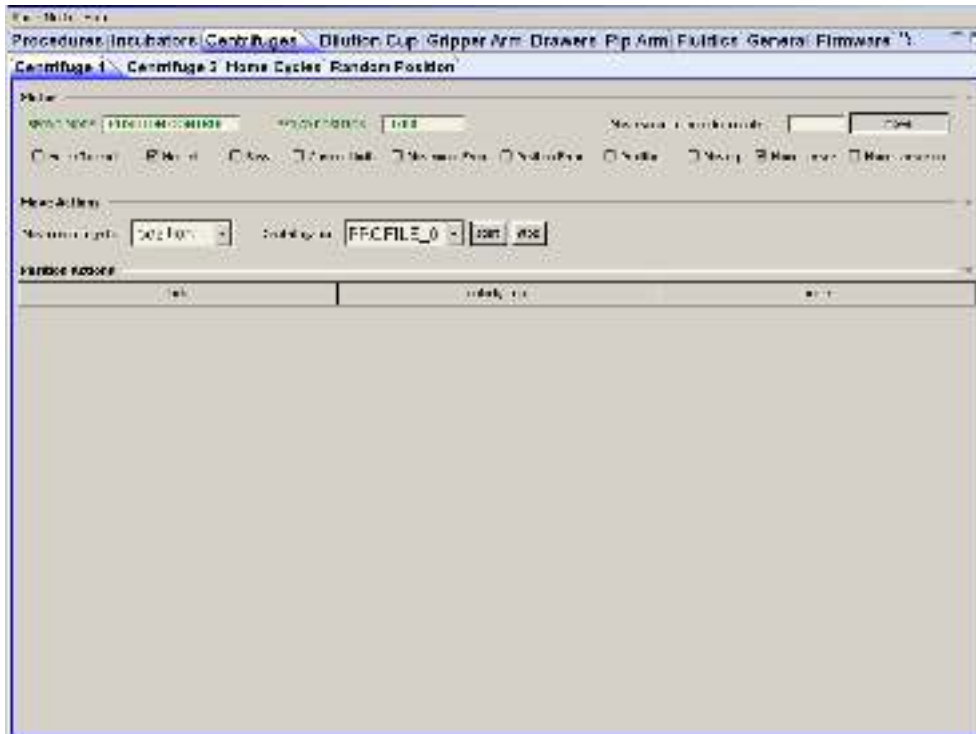


Figure 627 - Centrifuge x - Centrifuges tab in the Technical Service software

Table 71 - List of options in the Centrifuge x - Centrifuges tab

Option	Description
Motor	
SERVO MODE	Describes motor movement. There are four possible status: • Position Control: motor energized, maintaining the actual position. • Deenergized: motor deenergized. • Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. • Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Move	Centrifuge moves to the number of counts written in the Move servo to encoder count text box.
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.
Homed	Home position has been detected.
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.

Option	Description
Profiling	Servo is executing trapezoidal movement or integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.
Move Actions	
Move centrifuge to	Move centrifuge to any of the selected slots (1 to 12) to the entry / exit card position.
Centrifugation	Selection list to choose programmed parameters depending on the Profile selected: • Profile_0: 1206 rpm, 540 s. • Profile_1: 1206 rpm, 120 s. • Profile_2: Not used. • Profile_3: Not used.
start	Initiate the centrifugation with the selected profile.
stop	Cancel / stop current centrifugation.
Position Actions	
lock	Lock the servo motor.
unlock/stop	Unlock the servo motor or to stop the current movement.
home	Perform a home movement.

3.2.3.2 Home Cycles subtab

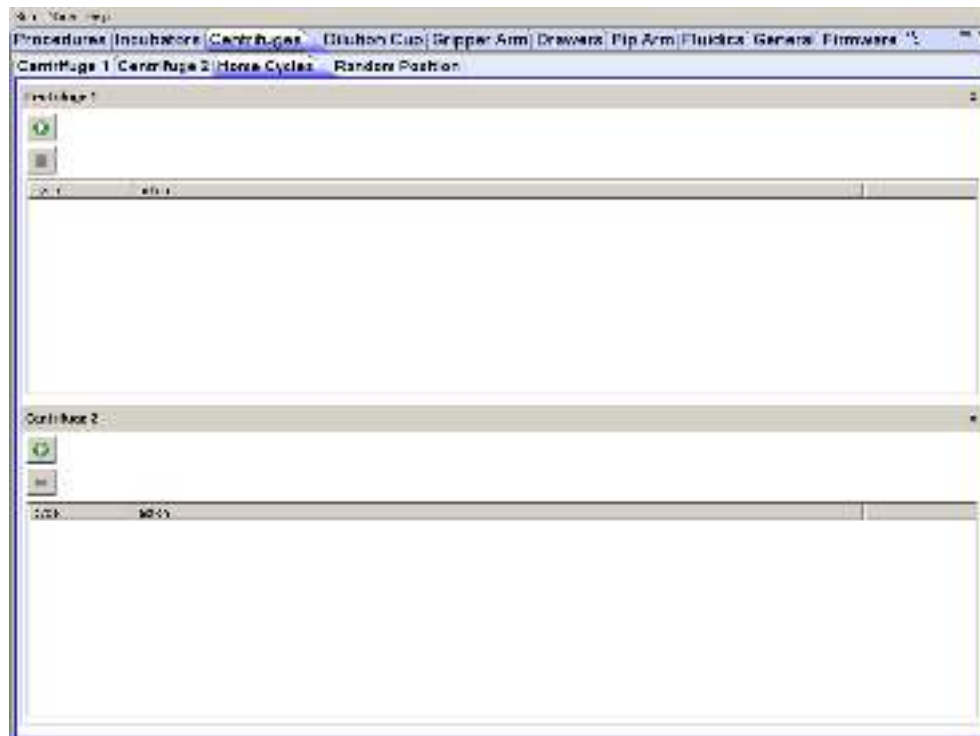


Figure 628 - Home Cycles - Centrifuges tab in the Technical Service software

Table 72 - List of options in the Home Cycles - Centrifuges tab

Option	Description
Centrifuge 1 and Centrifuge 2	
	Perform continuous home movements of the selected centrifuge.
	Stop the continuous home movements of the selected centrifuge.

3.2.3.3 Random Position subtab

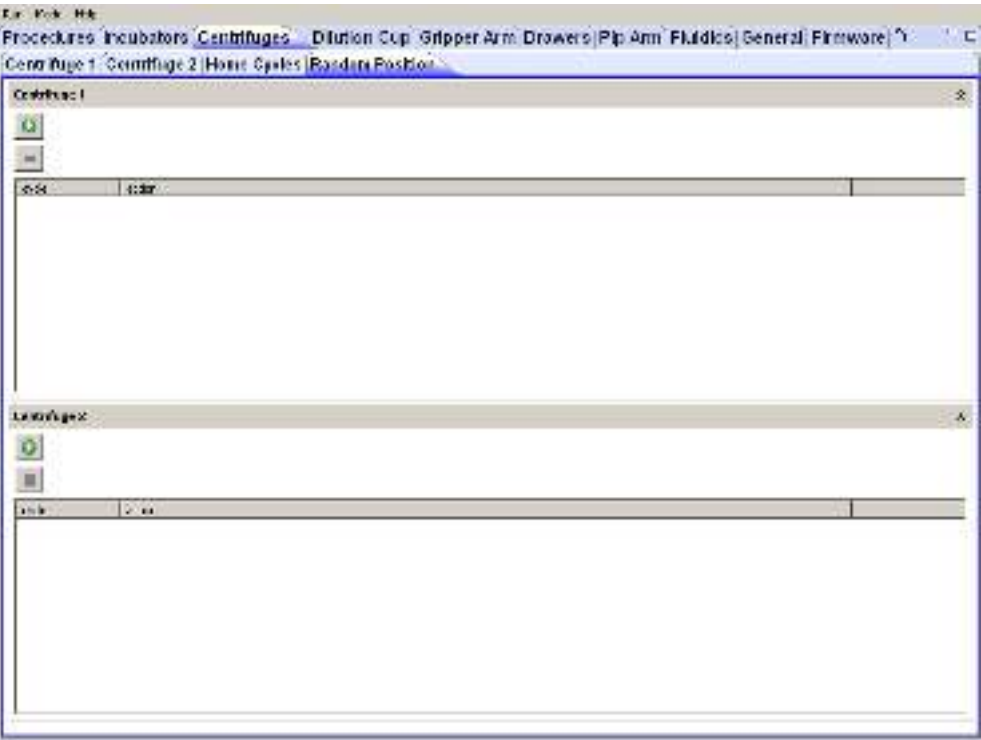


Figure 629 - Random Position - Centrifuges tab in the Technical Service software

Table 73 - List of options in the Random Position - Centrifuges tab

Option	Description
Centrifuge 1 and Centrifuge 2	
	Perform random movements of the selected centrifuge.
	Stop the random movements of the selected centrifuge.

3.2.4 Dilution Cup tab

The **Dilution Cup** tab includes all the parameters and functions related to the dilution cup system.

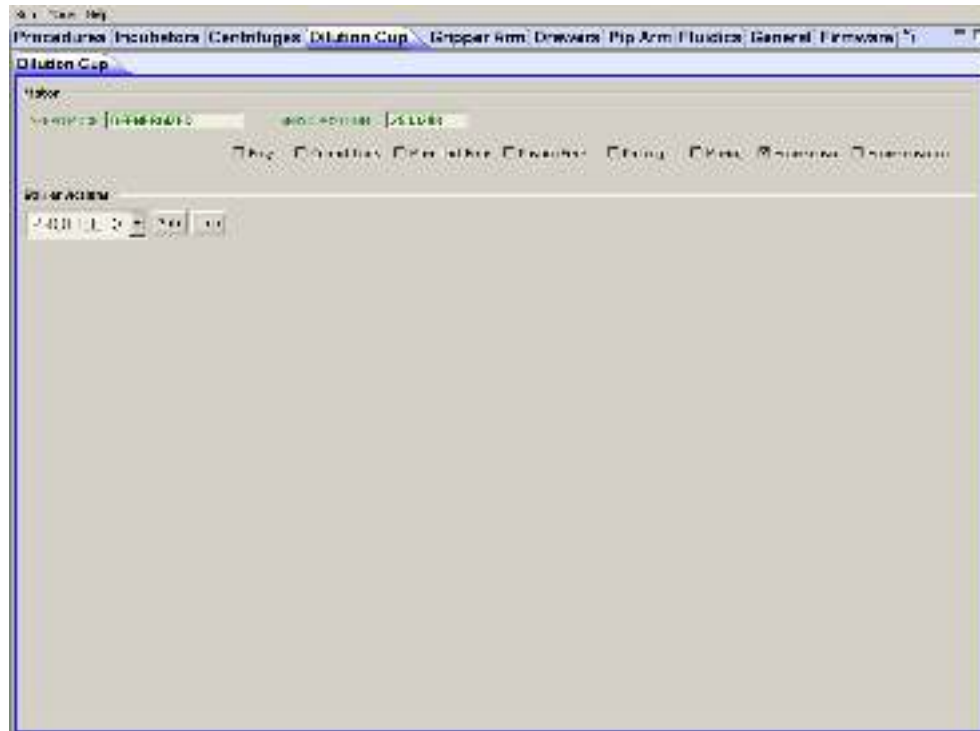


Figure 630 - Dilution Cup tab in the Technical Service software

Table 74 - List of options in the Dilution Cup tab

Option	Description
Motor	
SERVO MODE	Describes motor movement. There are four possible statuses: • Position Control: motor energized, maintaining the actual position. • Deenergized: motor deenergized. • Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. • Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.
Profiling	Servo is executing trapezoidal movement or integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.

Option	Description
Stirrer Actions	It is possible to select four different stirring profiles: • Profile_0: 800 rpm, 9 s. • Profile_1: Not used. • Profile_2: Not used. • Profile_3: Not used.
Spin	Initiate the stirring process of the dilution cup.
stop	Stop the stirring process of the dilution cup.

3.2.5 Gripper Arm tab

The **Gripper Arm** tab includes all the parameters and functions related to the CAT arm. It contains five subtabs: **Movement**, **Presence Detector**, **BCR**, **Home Cycles** and **Random Position**.

3.2.5.1 Movement subtab

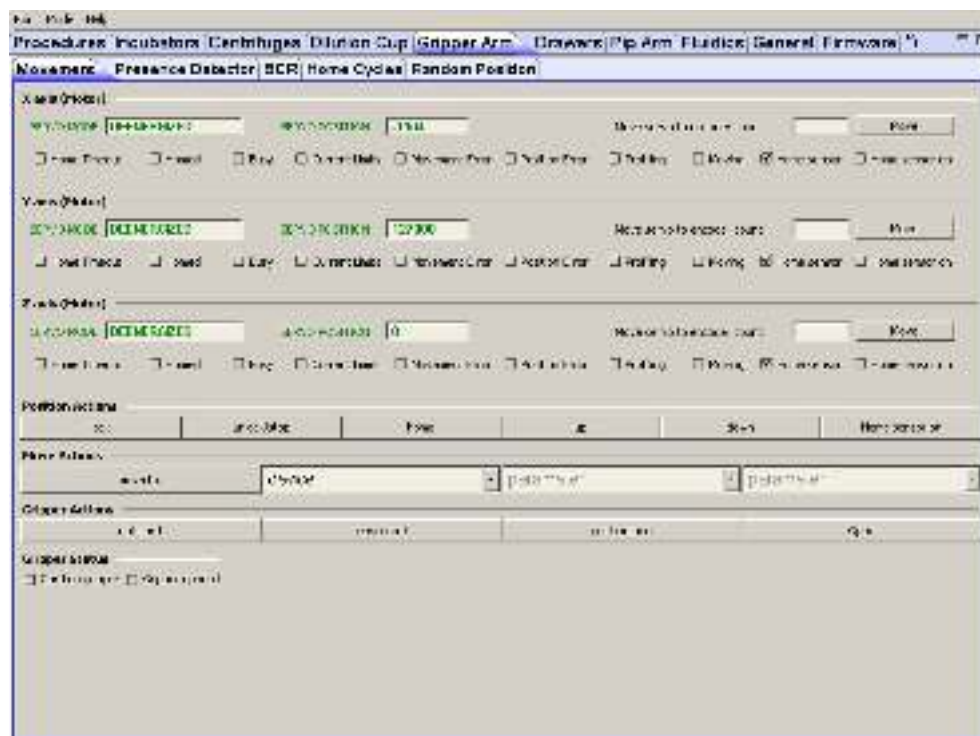


Figure 631 - Movement - Gripper Arm tab in the Technical Service software

Table 75 - List of options in the Movement - the Gripper Arm tab

Option	Description
X axis (Motor) , Y axis (Motor)and Z axis (Motor)	
SERVO MODE	Describes motor movement. There are four possible status: • Position Control: motor energized, maintaining the actual position. • Deenergized: motor deenergized. • Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. • Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Move	Selected axis moves to the number of counts introduced in the Move servo to encoder count text box.
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.
Homed	Home position has been detected.
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.

Option	Description
Profiling	Servo is executing trapezoidal movement or integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.
Position Actions	
lock	Lock a servo motor (X, Y, Z or all of them).
unlock/stop	Unlock a servo motor (X, Y, Z or all of them) or to stop the current movement.
home	Perform a home movement (X, Y, Z or all of them).
up	Move up the gripper.
down	Move down the gripper when CAT is over reader area.
Home sensor on	Activate home optical barriers.
Move Actions	

Option	Description		
move to:	Button to move the gripper arm to any of the following allowed positions:		
	First combo box	Second combo box	Third combo box
	INCUBATOR_1 INCUBATOR_2 INCUBATOR_3 INCUBATOR_4	SLOT_1 ... SLOT_12	INACTIVE
	CENTRIFUGE_1 CENTRIFUGE_2	SLOT_1 ... SLOT_12	INACTIVE
	CARD_DRAWER_1 CARD_DRAWER_2 CARD_DRAWER_3 CARD_DRAWER_4	RACK_1 ... RACK_4	CARD_1 ... CARD_25
	CAMERA	INTEGRITY READ	WELLS_1_2 WELLS_3_4 WELLS_5_6 WELLS_7_8
	WASTE_TRAY	INACTIVE	INACTIVE

Gripper Actions	
get card	Move down the gripper and take a card in the position where it is placed the arm.
leave card	Move down the gripper and leave a card in the position where it is placed the arm.

Option	Description
get barcode	Activate the barcode reader, returning the read of the barcode. Previously a card must be taken with the get card button.
Open	Open gripper fingers.
Gripper Status	
Card on gripper	Indicates if it is detected a card on the gripper.
Gripper opened	Indicates if gripper fingers are opened.

3.2.5.2 Presence Detector subtab

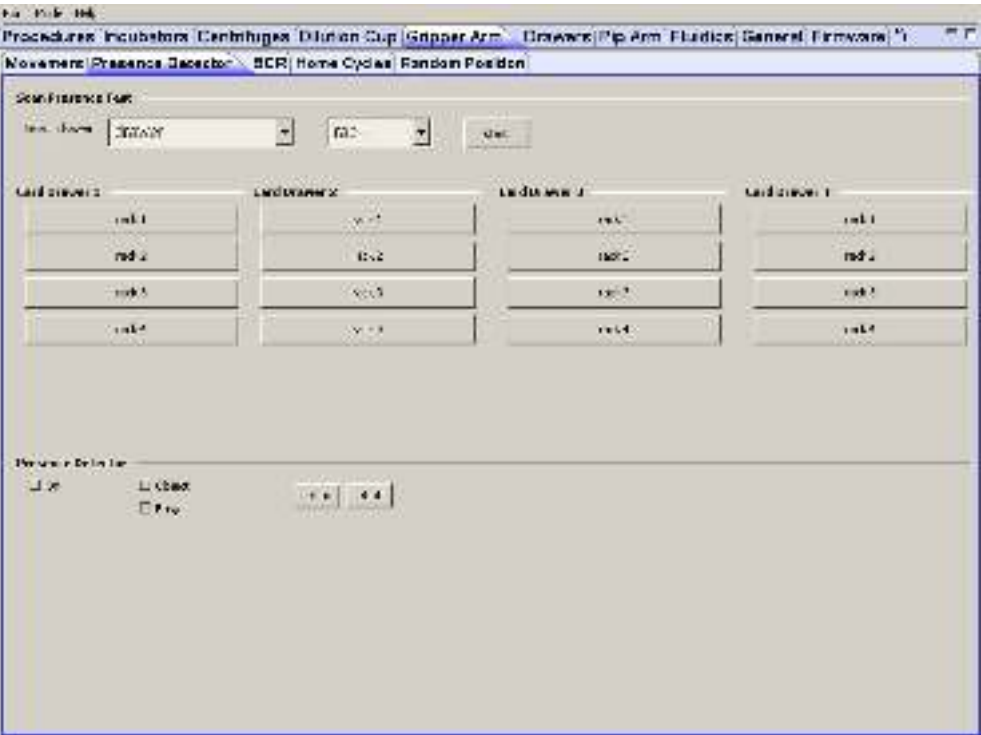


Figure 632 - Presence Detector - Gripper Arm tab in the Technical Service software

Table 76 - List of options in the Presence Detector - Gripper Arm tab

Option	Description										
Scan Presence Test											
Select drawer	List to select a position in order to perform a detection card test. The allowed positions are:										
	<table> <tr> <th>First combo box</th><th>Second combo box</th></tr> <tr> <td>CARD_DRAWER_1</td><td>RACK_1 ... RACK_4</td></tr> <tr> <td>CARD_DRAWER_2</td><td></td></tr> <tr> <td>CARD_DRAWER_3</td><td></td></tr> <tr> <td>CARD_DRAWER_4</td><td></td></tr> </table>	First combo box	Second combo box	CARD_DRAWER_1	RACK_1 ... RACK_4	CARD_DRAWER_2		CARD_DRAWER_3		CARD_DRAWER_4	
First combo box	Second combo box										
CARD_DRAWER_1	RACK_1 ... RACK_4										
CARD_DRAWER_2											
CARD_DRAWER_3											
CARD_DRAWER_4											
start	Move CAT arm to the selected position and it performs a presence detection card test.										
Card Drawer 1, Card Drawer 2, Card Drawer 3 and Card Drawer 4											
rack 1	Show status of the card rack 1 located in the selected card drawer (1, 2, 3 or 4). It indicates the number of detected and undetected cards and the position.										
rack 2	Show status of the card rack 2 located in the selected card drawer (1, 2, 3 or 4). It indicates the number of detected and undetected cards and the position.										
rack 3	Show status of the card rack 3 located in the selected card drawer (1, 2, 3 or 4). It indicates the number of detected and undetected cards and the position.										
rack 4	Show status of the card rack 4 located in the selected card drawer (1, 2, 3 or 4). It indicates the number of detected and undetected cards and the position.										
Presence Detector											

Option	Description
On	Indicate if the presence detector laser beam is on.
Object	Indicates if an object is detected in front of the laser beam.
Busy	Indicates if presence detector is executing a scanning.
set on	Activate the laser beam.
set off	Deactivate the laser beam.

3.2.5.3 BCR subtab

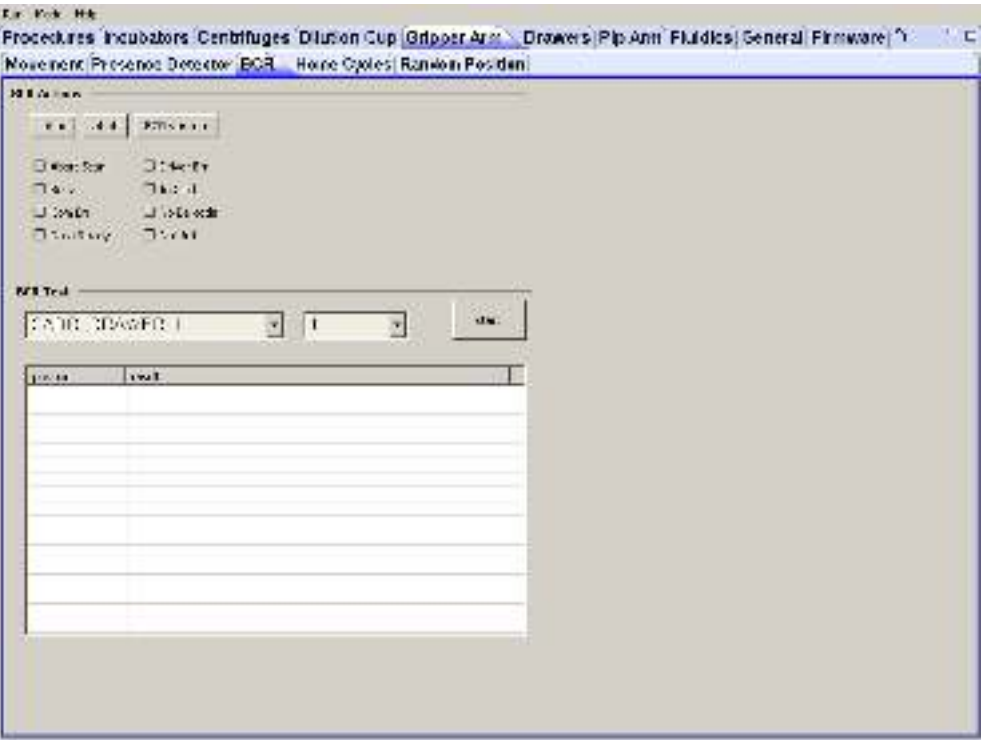


Figure 633 - BCR - Gripper Arm tab in the Technical Service software

Table 77 - List of options in the BCR - Gripper Arm tab

Option	Description
BCR Actions	
set on	Activate barcode reader.
set off	Deactivate barcode reader.
BCR version	Returns barcode reader version.
Abort scan	Indicates that a scanning operation has been aborted.
Busy	Barcode reader is on.
Com Err	Indicates a barcode reader communication error.
Data Ready	Indicates that information obtained by the barcode reader is ready for being recovered by software.
Driver Err	Indicates a barcode reader communication error.
In Cmd	Indicates that there is communication between microcontroller and barcode reader.
No Barcode	No barcode is read.
Not Init	Barcode reader is not initialized.
BCR Test	

Option	Description
start	Button to initiate the process of barcode reading. Previously, a detection test is performed to check where cards are placed. Allowed positions are:
text box	Results of barcode reading process are displayed. In the first column (position) is shown the position of the card detected and in the second column (result) is shown the barcode.

3.2.5.4 Home Cycles subtab



Figure 634 - Home Cycles - Gripper Arm tab in the Technical Service software

Table 78 - List of options in the Home Cycles - Gripper Arm tab

Option	Description
Transport Arm	
	Perform continuous home movements of the CAT arm.
	Stop continuous home movements of the CAT arm.

3.2.5.5 Random Position subtab



Figure 635 - Random Position - Gripper Arm tab in the Technical Service software

Table 79 - List of options in the Random Position - Gripper Arm tab

Option	Description
Transport Arm	
	Perform random movements of the CAT arm.
	Stop random movements of the CAT arm.

3.2.6 Drawers tab

The **Drawers** tab includes all the parameters and functions related to the drawers. It contains five subtabs: **Card Drawers**, **Sample Drawers**, **Reagent Drawers**, **Fluidic Drawers** and **Home Cycles**.

3.2.6.1 Card Drawers subtab

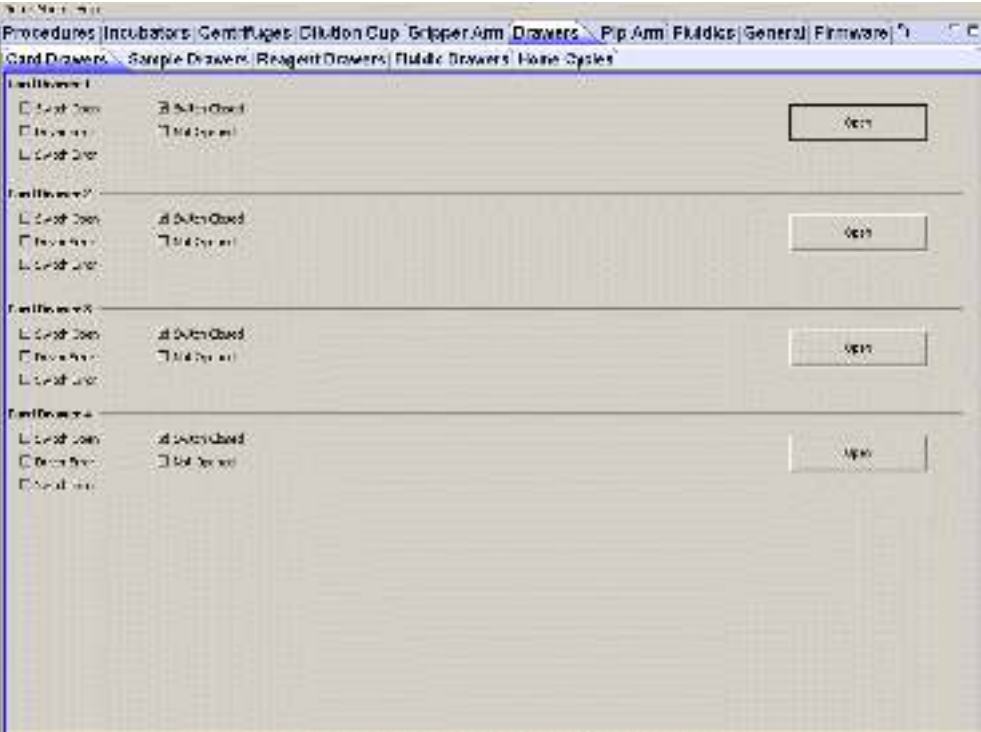


Figure 636 - Card Drawers - Drawers tab in the Technical Service software

Table 80 - List of options in the Card Drawers - Drawers tab

Option	Description
Card Drawer 1, Card Drawer 2, Card Drawer 3 and Card Drawer 4	
Switch Open	Indicates that the drawer is open.
Switch Closed	Indicates that the drawer is closed.
Switch Error	Indicates that an error condition (is open and closed at the same time or not opened nor closed at the same time).
Driver Error	Indicates that an error happened when communicating with the lock auxiliary peripheral, through I2C bus.
Not opened	Indicates that the lock has not opened within one second after an open command has been executed.
Open	Open the selected drawer.

3.2.6.2 Sample Drawers subtab



Figure 637 - Sample Drawers - Drawers tab in the Technical Service software

Table 81 - List of options in the Sample Drawers - Drawers tab

Option	Description
Sample Drawer 1, Sample Drawer 2, Sample Drawer 3 and Sample Drawer 4	
Switch Open	Indicates that the drawer is open.
Switch Closed	Indicates that the drawer is closed.
Switch Error	Indicates that an error condition (is open and closed at the same time or not opened nor closed at the same time).
Driver Error	Indicates that an error happened when communicating with the lock auxiliary peripheral, through I2C bus.
Not opened	Indicates that the lock has not opened within one second after an open command has been executed.
Open	Open the selected drawer.

3.2.6.3 Reagent Drawers subtab

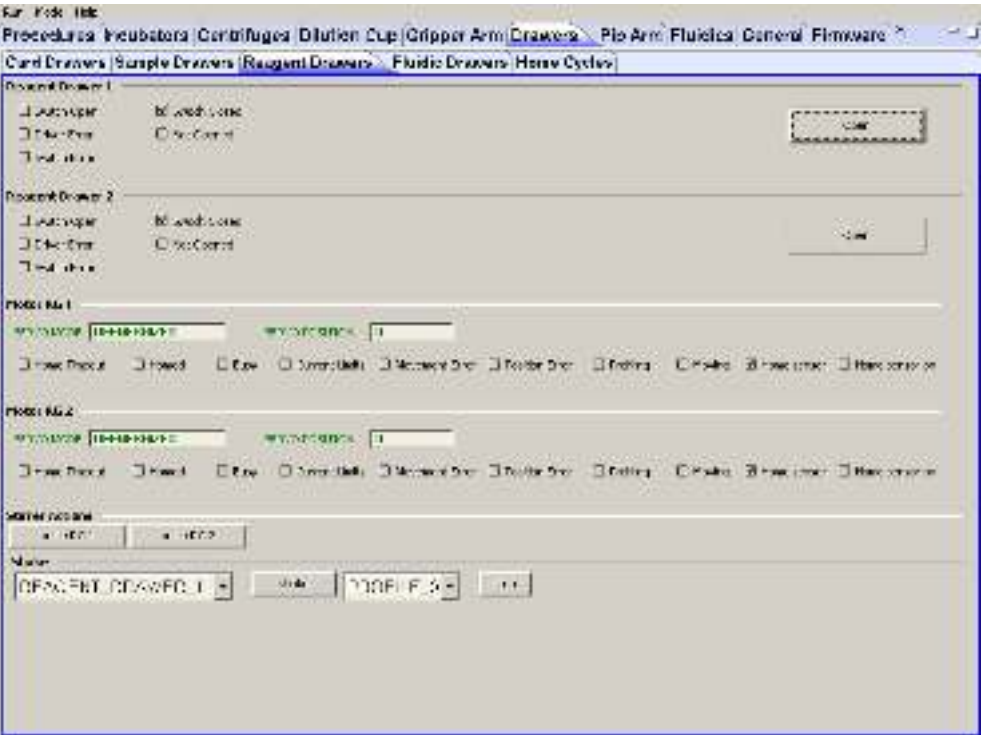


Figure 638 - Reagent Drawers - Drawers tab in the Technical Service software

Table 82 - List of options in the Reagent Drawers - Drawers tab

Option	Description
Reagent Drawer 1 and Reagent Drawer 2	
Switch Open	Indicates that the drawer is open.
Switch Closed	Indicates that the drawer is closed.
Switch Error	Indicates that an error condition (is open and closed at the same time or not opened nor closed at the same time).
Driver Error	Indicates that an error happened when communicating with the lock auxiliary peripheral, through I2C bus.
Not opened	Indicates that the lock has not opened within one second after an open command has been executed.
Open	Button to open the selected drawer.
Motor RG 1 and Motor RG 2	
SERVO MODE	Describes motor movement. Position Control: motor energized, maintaining the actual position. • Deenergized: motor deenergized. • Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. • Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates the encoder counts of the motor.
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.
Homed	Home position has been detected.

Option	Description
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.
Profiling	Servo is executing trapezoidal movement or integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.
Stirrer Actions	
home RG1	Perform a home of the rack 2 of reagent drawer 1.
home RG2	Perform a home of the rack 2 of reagent drawer 2.
Shake	

Option	Description				
shake	Initiate the stirring process with the selected profile: • PROFILE_0: Increases speed during 5 seconds reaching 600 rpm, then decelerates during five more seconds until it stops. • PROFILE_1: Not used • PROFILE_2: Not used • PROFILE_3: Not used <table> <tr> <th>First combo box</th><th>Second combo box</th></tr> <tr> <td>REAGENT_DRAWER_1 REAGENT_DRAWER_2</td><td>PROFILE_0 ... PROFILE_4</td></tr> </table>	First combo box	Second combo box	REAGENT_DRAWER_1 REAGENT_DRAWER_2	PROFILE_0 ... PROFILE_4
First combo box	Second combo box				
REAGENT_DRAWER_1 REAGENT_DRAWER_2	PROFILE_0 ... PROFILE_4				
stop	Cancel / stop the stirring process.				

3.2.6.4 Fluidic Drawers subtab



Figure 639 - Fluidic Drawers - Drawers tab in the Technical Service software

Table 83 - List of options in the Fluidics Drawers - Drawers tab

Option	Description
sol A, sol B, waste tray, waste 1 and waste 2	
Switch Open	Indicates that the drawer is open.
Switch Closed	Indicates that the drawer is closed.
Switch Error	Indicates that an error condition (is opened and closed at the same time or not opened nor closed at the same time).
Driver Error	Indicates that an error happened when communicating with the lock auxiliary peripheral, through I2C bus.
Not Opened	Indicates that the lock has not opened within one second after an open command has been executed.
Open	Open the selected drawer.

3.2.6.5 Home Cycles subtab

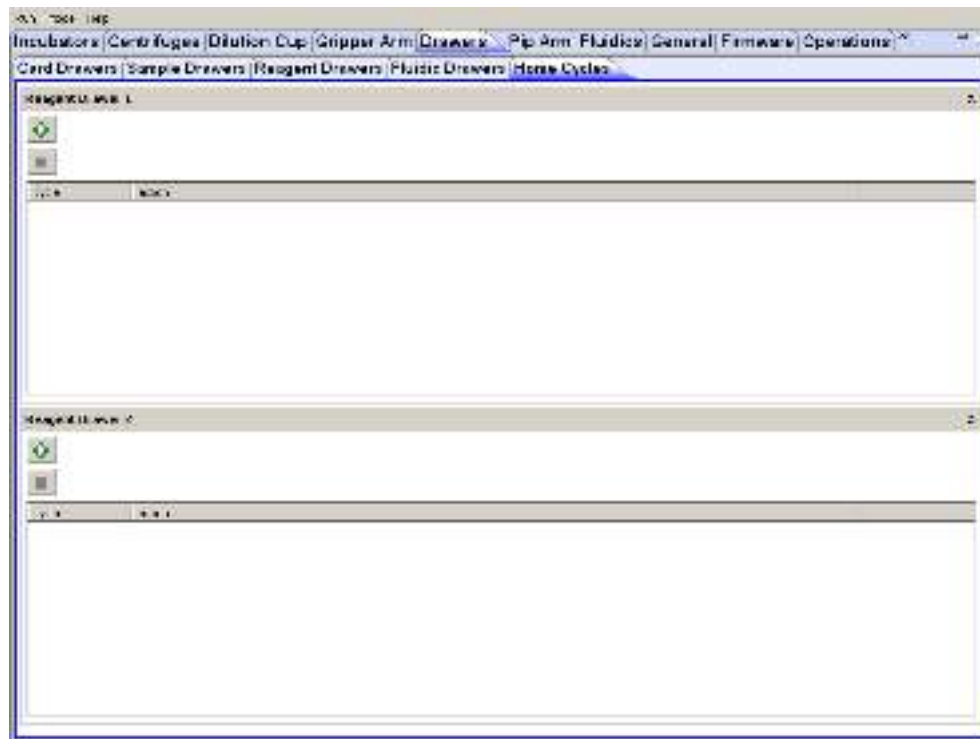


Figure 640 - Home Cycles - Drawers tab in the Technical Service software

Table 84 - List of options in the Home Cycles - Drawers tab

Option	Description
Reagent Drawer 1 and Reagent Drawer 2	
	Perform home cycles of the reagent drawer selected (rack 2).
	Stop the home cycles of the reagent drawer selected (rack 2).

3.2.7 Pip Arm tab

The **Pip Arm** tab includes all the parameters and functions related to the pipetting arm. It contains seven subtabs: **Movement**, **Presence Detector**, **Level Detector**, **BCR**, **CPD**, **Home Cycles** and **Random Position**.

3.2.7.1 Movement subtab



Figure 641 - Movement - the PIP Arm tab in the Technical Service software

Table 85 - List of options in Movement - the PIP Arm tab

Option	Description
X axis (Motor), Y axis (Motor), Z1 axis (Motor), Z2 axis (Motor) and Horns (Motor)	
SERVO MODE	Describes motor movement. There are four possible statuses: • Position Control: motor energized, maintaining the actual position. • Deenergized: motor deenergized. • Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. • Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Move	The selected axis moves to the number of counts written in the Move servo to encoder count text box.
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.
Homed	Home position has been detected.
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.

Option	Description
Profiling	Servo executing trapezoidal movement / integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.
Position Actions	
lock	Lock a servo motor (X, Y, Z1, Z2, Horns or all of them).
unlock/stop	Unlock a servo motor (X, Y, Z1, Z2, Horns or all of them) or to stop the current movement.
home	Perform a home movement. It is possible to select X, Y, Z1, Z2, horns or all of them.
Home sensor on	Activate home optical barriers.
horns up	Move the horns to the upper position.
horns down	Move the horns to the lower position.
up probe 1	Move up probe 1.
down probe 1	Move down probe 1 until adjustment position (depending on where the PIP module is).
up probe 2	Move up probe 2.
down probe 2	Move down probe 2 until adjustment position (depending on where the PIP module is).

Option	Description			
Move Actions				
move to	Button to move the PIP arm to any of the following allowed positions:			
	First combo box	Second combo box	Third combo box	Fourth combo box
	SAMPLE DRAWER_1 SAMPLE DRAWER_2 SAMPLE DRAWER_3 SAMPLE DRAWER_4	RACK_1, RACK_2	POS_1 ... POS_12	PROBE_1, PROBE_2
	REAGENT_DRAWER_1 REAGENT_DRAWER_2		POS_1 ... POS_9	
	INCUBATOR_1 INCUBATOR_2 INCUBATOR_3 INCUBATOR_4	SLOT_1... SLOT_12	WELL_1... WELL_8	
	DILUTION_CUPS	PROBE_1, PROBE_2	ASPIRE DISPENSE	INACTIVE

3.2.7.2 Presence Detector subtab

There are three subtabs inside the Presence Detector subtab: Reagent drawers, Sample drawers and Presence Detector.

3.2.7.2.1 Reagent drawers subtab

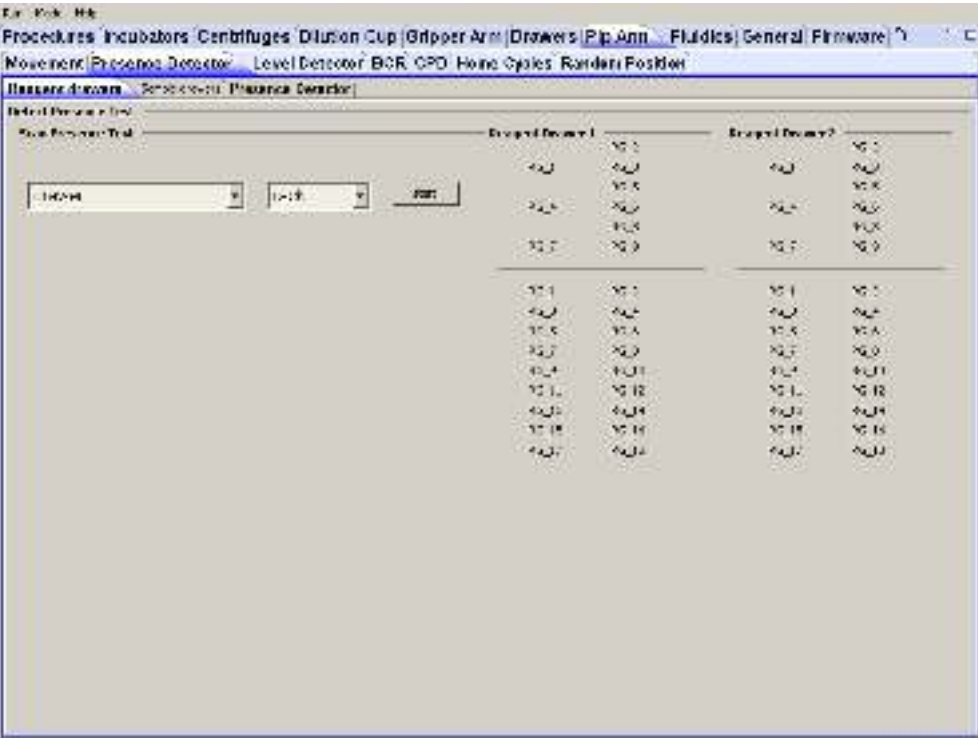


Figure 642 - Reagent drawers - Presence Detector - Pip Arm tab in the Technical Service software

Table 86 - List of options in the Reagent drawers - Presence Detector - Pip Arm tab

Option	Description						
Detect Presence Test							
Scan Presence Test							
start	Initiate the detection process in the reagent rack selected. Allowed positions are:						
	<table> <tr> <th>First combo box</th><th>Second combo box</th></tr> <tr> <td>REAGENT_DRAWER_1</td><td>RACK_1</td></tr> <tr> <td>REAGENT_DRAWER_2</td><td>RACK_2</td></tr> </table>	First combo box	Second combo box	REAGENT_DRAWER_1	RACK_1	REAGENT_DRAWER_2	RACK_2
First combo box	Second combo box						
REAGENT_DRAWER_1	RACK_1						
REAGENT_DRAWER_2	RACK_2						
Reagent Drawer 1 and Reagent Drawer 2	Indicates the presence of a bottle / vial in a position of each drawer of the reagents positions.						

3.2.7.2.2 Sample drawers subtab

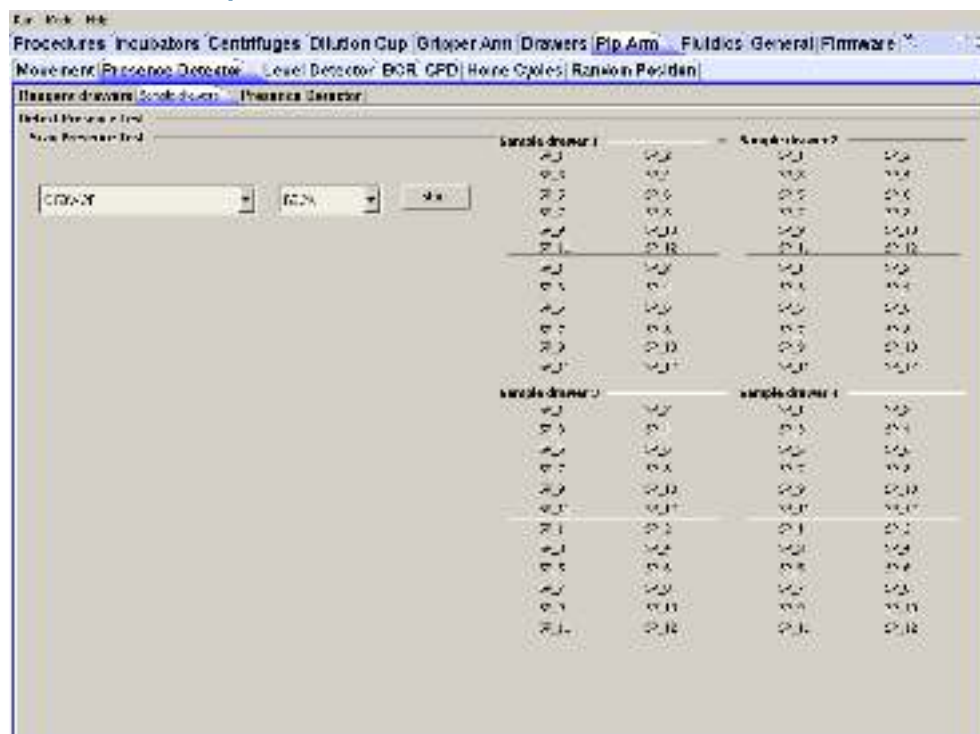


Figure 643 - Sample drawers - Presence Detector - Pip Arm tab in the Technical Service software

Table 87 - List of options in the Sample drawers - Presence Detector - Pip Arm tab

Option	Description										
Detect Presence Test											
Scan Presence Test											
start	Initiate the detection process in the sample rack selected. Allowed positions are:										
	<table> <tr> <th>First combo box</th><th>Second combo box</th></tr> <tr> <td>SAMPLE_DRAWER_1</td><td>RACK_1 ... RACK_2</td></tr> <tr> <td>SAMPLE_DRAWER_2</td><td></td></tr> <tr> <td>SAMPLE_DRAWER_3</td><td></td></tr> <tr> <td>SAMPLE_DRAWER_4</td><td></td></tr> </table>	First combo box	Second combo box	SAMPLE_DRAWER_1	RACK_1 ... RACK_2	SAMPLE_DRAWER_2		SAMPLE_DRAWER_3		SAMPLE_DRAWER_4	
First combo box	Second combo box										
SAMPLE_DRAWER_1	RACK_1 ... RACK_2										
SAMPLE_DRAWER_2											
SAMPLE_DRAWER_3											
SAMPLE_DRAWER_4											
Sample drawer 1, Sample drawer 2, Sample drawer 3 and Sample drawer 4	Indicates the presence of a tube in a position of each drawer of the samples positions. If detected, it returns the diameter of the tube (-1 if diameter can not be calculated).										

3.2.7.2.3 Presence Detector subtab

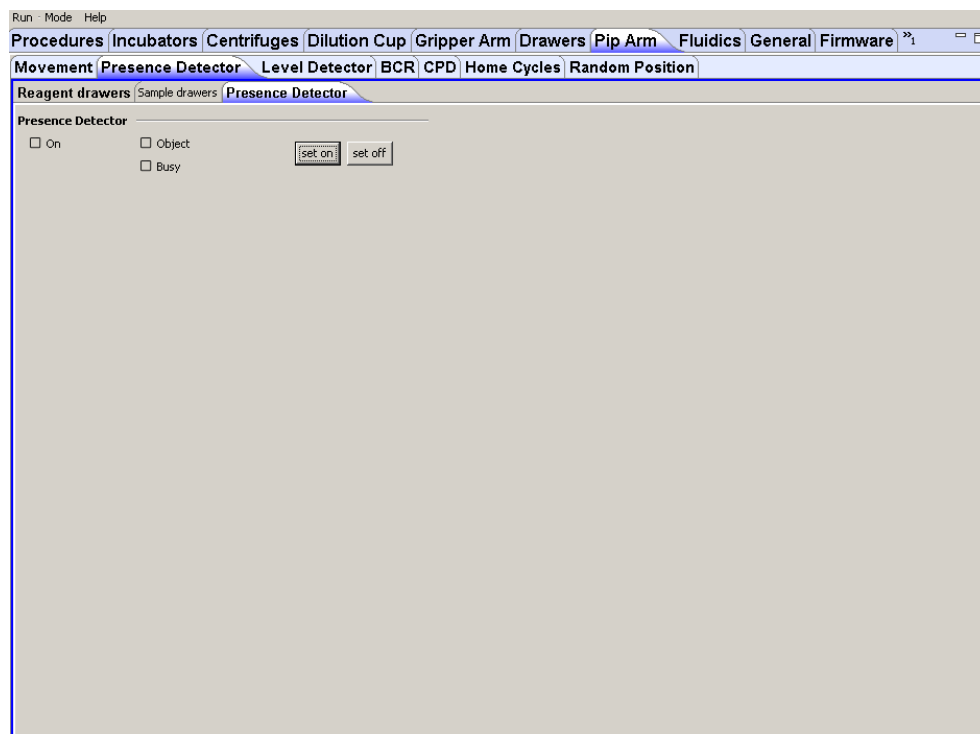


Figure 644 - Presence Detector - Presence Detector - Pip Arm tab in the Technical Service software

Table 88 - List of options in the Presence Detector - Presence Detector - Pip Arm tab

Option	Description
On	Indicates that the presence detector laser beam is on.
Object	Indicates that an object is detected in front of the laser beam.
Busy	Indicates that presence detector is executing a scanning.
set on	Activate the laser beam.
set off	Deactivate the laser beam.

3.2.7.3 Level Detector subtab

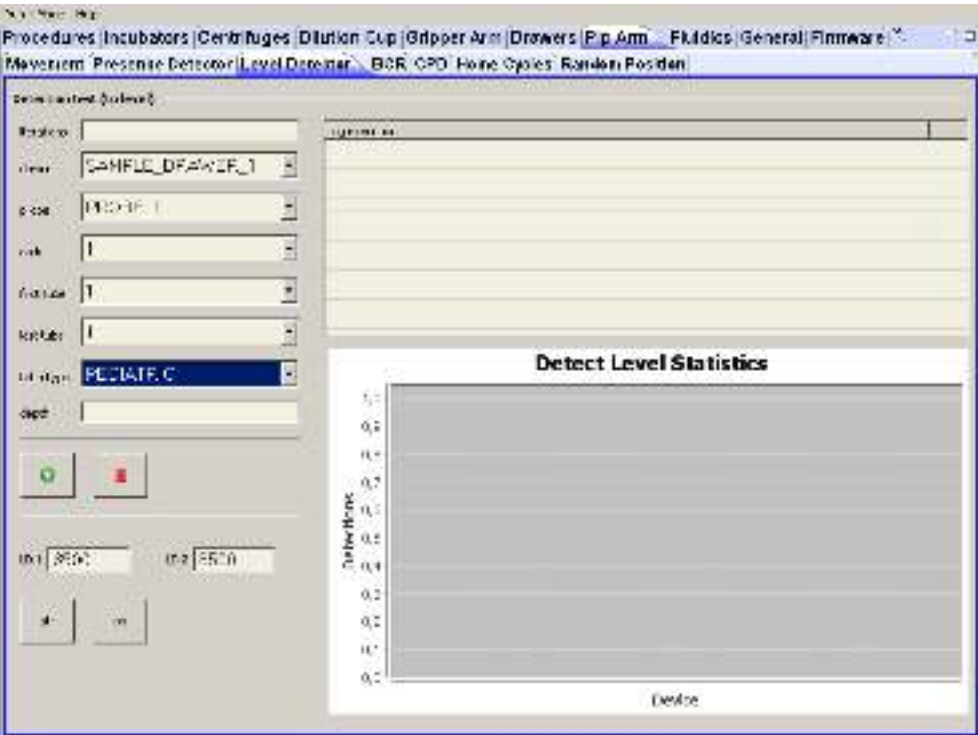


Figure 645 - Level Detector - Pip Arm tab in the Technical Service software

Table 89 - List of options in the Level Detector - Pip Arm tab

Option	Description																																																		
Detection test (to level)																																																			
iterations	Text box to enter the number of times to execute the level detection test.																																																		
device, probe, rack/position, first tube, last tube	Combo box to select the position where to perform the level detection test. Allowed positions are: <table><tr><th>device</th><th>probe</th><th>rack/position</th><th>first tube</th><th>last tube</th></tr><tr><td>REAGENT_DRAWER_1</td><td>PROBE_1</td><td>RACK_1</td><td>1..9</td><td>1..9</td></tr><tr><td></td><td>PROBE_2</td><td>RACK_2</td><td>1..18</td><td>1..18</td></tr><tr><td>REAGENT_DRAWER_2</td><td></td><td>RACK_1</td><td>1..9</td><td>1..9</td></tr><tr><td></td><td></td><td>RACK_2</td><td>1..18</td><td>1..18</td></tr><tr><td>SAMPLE_DRAWER_1</td><td></td><td>RACK_1</td><td>1..12</td><td>1..12</td></tr><tr><td>SAMPLE_DRAWER_2</td><td></td><td>RACK_2</td><td></td><td></td></tr><tr><td>SAMPLE_DRAWER_3</td><td></td><td></td><td></td><td></td></tr><tr><td>SAMPLE_DRAWER_4</td><td></td><td></td><td></td><td></td></tr><tr><td>DILUTION_CUPS</td><td></td><td>1,2</td><td>INACTIVE</td><td></td></tr></table>	device	probe	rack/position	first tube	last tube	REAGENT_DRAWER_1	PROBE_1	RACK_1	1..9	1..9		PROBE_2	RACK_2	1..18	1..18	REAGENT_DRAWER_2		RACK_1	1..9	1..9			RACK_2	1..18	1..18	SAMPLE_DRAWER_1		RACK_1	1..12	1..12	SAMPLE_DRAWER_2		RACK_2			SAMPLE_DRAWER_3					SAMPLE_DRAWER_4					DILUTION_CUPS		1,2	INACTIVE	
device	probe	rack/position	first tube	last tube																																															
REAGENT_DRAWER_1	PROBE_1	RACK_1	1..9	1..9																																															
	PROBE_2	RACK_2	1..18	1..18																																															
REAGENT_DRAWER_2		RACK_1	1..9	1..9																																															
		RACK_2	1..18	1..18																																															
SAMPLE_DRAWER_1		RACK_1	1..12	1..12																																															
SAMPLE_DRAWER_2		RACK_2																																																	
SAMPLE_DRAWER_3																																																			
SAMPLE_DRAWER_4																																																			
DILUTION_CUPS		1,2	INACTIVE																																																
tube type	Select level detection sensitivity: NORMAL or PEDIATRIC .																																																		

Option	Description
depth	This box only appears if PEDIATRIC tube type is selected. Indicates the maximum depth that the probe can move down when entering into the tube (the value is the distance in mm from the standard adjustment Z position).
	Initiate xx times (indicated in iterations field) a test of level detection.
	Stop detection level test.
LD 1	Box where the level detection counts of the probe 1 are displayed.
LD 2	Box where the level detection counts of the probe 2 are displayed.
off	Deactivate the level detection system.
on	Activate the level detection system.
log execution	It details the steps of the level detection test.
Detect Level Statistics	Graph where the statistics of the level detections are shown. Possible results are: • Success: it has been detected level. • Fail: it has not been detected level. • Error: error while detecting level. • Impact: it has been detected an impact.

3.2.7.4 BCR subtab

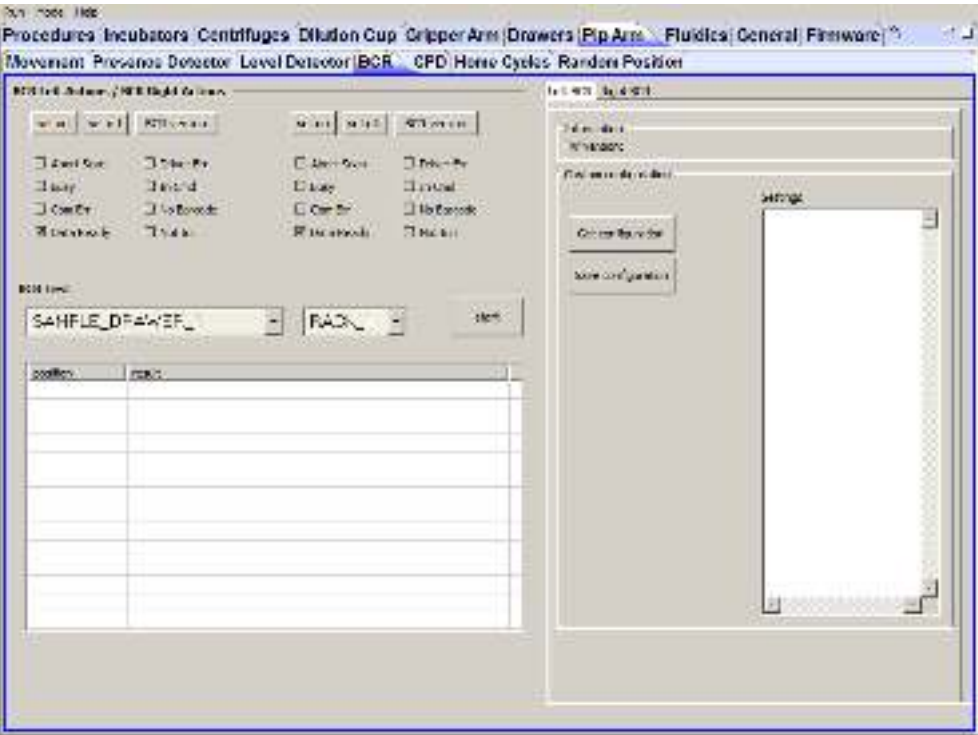


Figure 646 - BCR - Pip Arm tab in the Technical Service software

Table 90 - List of options in the BCR - Pip Arm tab

Option	Description
BCR Left Actions / BCR Right Actions	
set on	Activate barcode reader.
set off	Deactivate the barcode reader.
BCR version	Returns the barcode reader version.
Abort Scan	Indicates that a scanning operation has been aborted.
Busy	Barcode reader is on.
Com Err	Indicates a barcode reader communication error.
Data Ready	Indicates that information obtained by the barcode reader is ready for being recovered by software.
Driver Err	Indicates a barcode reader communication error.
In Cmd	Indicates that there is communication between microcontroller and barcode reader.
No Barcode	No barcode is read.
Not Init	Barcode reader is not initialized.
BCR Test	

Option	Description														
start	Initiate the process of barcode reading. Previously, a detection test is performed to check where tubes, vials or bottles are placed. Allowed positions are: <table> <tr> <th>First combo box</th><th>Second combo box</th></tr> <tr> <td>SAMPLE_DRAWER_1</td><td>RACK_1</td></tr> <tr> <td>SAMPLE_DRAWER_2</td><td>RACK_2</td></tr> <tr> <td>SAMPLE_DRAWER_3</td><td></td></tr> <tr> <td>SAMPLE_DRAWER_4</td><td></td></tr> <tr> <td>REAGENT_DRAWER_1</td><td></td></tr> <tr> <td>REAGENT_DRAWER_2</td><td></td></tr> </table>	First combo box	Second combo box	SAMPLE_DRAWER_1	RACK_1	SAMPLE_DRAWER_2	RACK_2	SAMPLE_DRAWER_3		SAMPLE_DRAWER_4		REAGENT_DRAWER_1		REAGENT_DRAWER_2	
First combo box	Second combo box														
SAMPLE_DRAWER_1	RACK_1														
SAMPLE_DRAWER_2	RACK_2														
SAMPLE_DRAWER_3															
SAMPLE_DRAWER_4															
REAGENT_DRAWER_1															
REAGENT_DRAWER_2															
text box	Results of barcode reading process are displayed. In the first column (position) is shown the position of the vial/bottle/tube detected and in the second column (result) is shown the barcode.														
Settings for BCR Left / BCR Right - Information															
FW version	Firmware version of each barcode reader device.														
Settings for BCR Left / BCR Right - Custom configuration															
Settings	Text box where the customized left / right BCR configuration is shown. Modifications related to this configuration should be directly written on the textbox. The resulting customized configuration will be used to modify the default BCR configuration.														
Get configuration	Reads and shows the saved parameters related to the customized configuration of the selected BCR device.														
Save configuration	Writes and saves the parameters detailed in the Settings text box as the customized configuration for the selected BCR device.														

3.2.7.5 CPD subtab

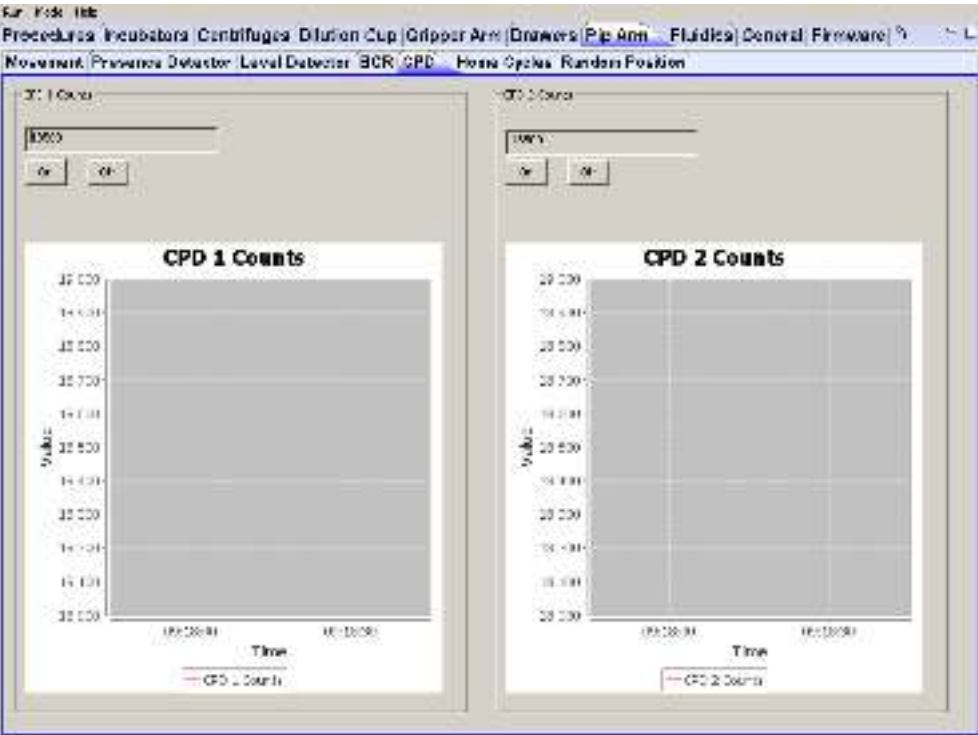


Figure 647 - CPD - Pip Arm tab in the Technical Service software

Table 91 - List of options in the CPD - Pip Arm tab

Option	Description
CPD 1 Counts and CPD 2 Counts	
On	Activates CPD sensor, showing counts in the text box and in the graphs at the bottom of the screen.
Off	Deactivates reading counts process of CPD sensor.

3.2.7.6 Home Cycles subtab



Figure 648 - Home Cycles - Pip Arm tab in the Technical Service software

Table 92 - List of options in the Home Cycles - Pip Arm tab

Option	Description
Pipetting Arm	
	Perform home cycles of the pipetting robot.
	Stop the home cycles of the pipetting robot.

3.2.7.7 Random Position subtab



Figure 649 - Random Position - Pip Arm tab in the Technical Service software

Table 93 - List of options in the Random Position - Pip Arm tab

Option	Description
Pipetting Arm	
	Perform random position movements of the pipetting robot.
	Stop random position movements of the pipetting robot.

Table 94 - List of options in the Fluidics tab

Option	Description
Fluidic Actions	
fluidic empty	Empty all the fluidic circuit.
fluidic prime	Prime fluidic circuit between manifold, probes washer and dilution cup.
fluidic rinse	Wash with solution B fluidic circuit between manifold, probes washer and dilution cup.
initializing fluidic	Initialize the fluidics (home of the PIP arm and dilutors initialization).
stop	Stop the process that is being performed.
ext. end probe 1	External wash of the end of probe 1 (fluidic circuit must be primed).
ext. fast probe 1	External fast wash of probe 1 (fluidic circuit must be primed).
ext. slow probe 1	External slow wash of probe 1 (fluidic circuit must be primed).
ext. end probe 2	External wash of the end of probe 2 (fluidic circuit must be primed).
ext. fast probe 2	External fast wash of probe 2 (fluidic circuit must be primed).
ext. slow probe 2	External slow wash of probe 2 (fluidic circuit must be primed).
prime probe 1	Prime with solution A the fluidic circuit of probe 1 (fluidic circuit must be primed).
prime probe 2	Prime with solution A the fluidic circuit of probe 2 (fluidic circuit must be primed).
rinse probe 1	Rinse with solution B the fluidic circuit of probe 1 (fluidic circuit must be primed).

Option	Description
rinse probe 2	Rinse with solution B the fluidic circuit of probe 2 (fluidic circuit must be primed).
wake up probe 1	Empty and prime probe 1 circuit (same procedure as the one performed during routine after an inactivity wash).
wake up probe 2	Empty and prime probe 2 circuit (same procedure as the one performed during routine after an inactivity wash).
Empty Clot Probe 1	Routine to try to remove a clot from probe 1.
Empty Clot Probe 2	Routine to try to remove a clot from probe 2.
Fast Prime Bottle 1	Fill with liquid fluidic path between bottle 1 and waste bottle/external drain.
Fast Prime Bottle 2	Fill with liquid fluidic path between bottle 2 and waste bottle/external drain.
Fast Prime Bottle 3	Fill with liquid fluidic path between bottle 3 and external drain (present if dual mode is activated).
Fast Prime Bottle 4	Fill with liquid fluidic path between bottle 4 and external drain (present if dual mode is activated).
wash dilution cup	Wash with solution A the dilution cup (fluidic circuit must be primed).
empty dilution cup	Empty the dilution cup (fluidic circuit should be primed previously).

Option	Description
send - Dilutor Controller 1 and send - Dilutor Controller 2	Send the command selected in the text box to the dilutor. INIT: initialize the dilutor. PRIME_PROBE_HALF: five cycles of aspiration and dispensation of 1 ml from from the bottle A to the probe. VENT_PROBE_HALF: five cycles of aspiration and dispensation of 1 ml from the probe to the bottle A. ASPIRATE_1ml_FROM_PROBE: perform an aspiration of 1 ml from the probe, rotating properly the valve. ASPIRATE_1ml_FROM_SOL_A: perform an aspiration of 1 ml from the solution A bottle, rotating properly the valve. DISPENSE_TO_PROBE: perform a dispensation of the liquid that there is in the syringe to the probe, rotating properly the valve. ASPIRATE_GAP: perform an aspiration of a gap. ABORT: abort the movement of the dilutor.
Weight Sensor 1 (BOT A)	Return the value (in ml) of the weight of the load cell placed in the solution A drawer.
Weight Sensor 2 (BOT B)	Return the value (in ml) of the weight of the load cell placed in the solution B drawer.
Weight Sensor 3 (CARDS)	Return the value (cards number) of the weight of the load cell placed in the cards disposal tray drawer.
Weight Sensor 4 (WASTE 1)	Return the value (in ml) of the weight of the load cell placed in the waste 1 drawer.
Weight Sensor 5 (WASTE 2)	Return the value (in ml) of the weight of the load cell placed in the waste 2 drawer.
Pressure Sensor 1 (S3)	Return the value (in mbar) of pressure sensor S3.
Pressure Sensor 2 (S4)	Return the value (in mbar) of pressure sensor S4.
Vacuum Sensor 1 (S5)	Return the value (in mbar) of vacuum sensor S5.
Vacuum Sensor 2 (S6)	Return the value (in mbar) of vacuum sensor S6.

Option	Description
PM1 - SOLA CIRCUIT PUMP	Activates PM1 dispensation pump (solution A). There is a text box where the working pressure can be modified. Allowed values: 160 mbar (default) 180 mbar
EV13 - SOLB TO CIRCUIT A SWITCH	Activates EV13.
EV14 - A1/A2 BOTTLE SELECTOR	Activates EV14.
EV1 - SOLB WASH STATION	Activates EV1.
EV3 - SOLA DILUTION CUP	Activates EV3.
EV4 - VACUUM DILUTION CUP	Activates EV4.
EV5 - VACUUM WASH STATION	Activates EV5.
EV6 - SOLB WASH STATION	Activates EV6.
EV8 - SOLA DILUTION CUP	Activates EV8.
EV9 - VACUUM DILUTION CUP	Activates EV9.
EV10 - VACUUM WASH STATION	Activates EV10.
PM3 - SOLB CIRCUIT PUMP	Activates PM3 dispensation pump (solution B). There is a text box where the working pressure can be modified. Allowed values: 160 mbar (default) 180 mbar
EV15 - EMPTY VALVE	Activates EV15.

Option	Description
EV16 - B1/B2 BOTTLE SELECTOR	Activates EV16.
PM2 - VACUUM PUMP	Activates PM2 aspiration pump.
EV17 - WASTE BOTTLE SELECTOR	Activates EV17.
EV18 - WASTE BOTTLE SELECTOR	Activates EV18.
EV19 - FAST PRIME B CIRCUIT	Activates EV19.
EV20 - FAST PRIME A CIRCUIT	Activates EV20.
EV2 - DILUTION CUP LINES EMPTY	Activates EV2.
EV7 - WASH STATION LINES EMPTY	Activates EV7.

3.2.9 General tab

The **General Module** tab includes all the parameters and functions related to the general module, for example the general LED, drawer lights and temperatures.

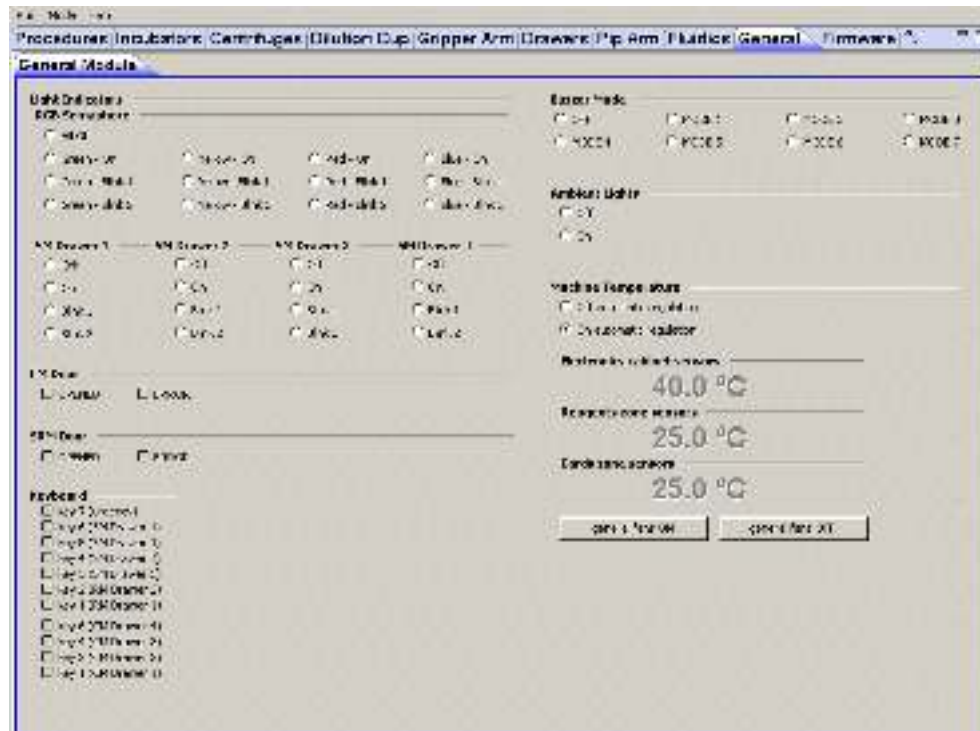


Figure 651 - General Module tab in the Technical Service software

Table 95 - List of options in the General Module tab

Option	Description
Light Indicators	
RBG Semaphore	
All Off	Deactivates any light in the general LED.
Green - On	Activates the green light in the general LED.
Green - Blink 1	Activates the blink frequency 1 with the green light in the general LED.
Green - Blink 2	Activates the blink frequency 2 with the green light in the general LED.
Yellow - On	Activates the yellow light in the status LED.
Yellow - Blink 1	Activates the blink frequency 1 with the yellow light in the general LED.
Yellow - Blink 2	Activates the blink frequency 2 with the yellow light in the general LED.
Red - On	Activates the red light in the general LED.
Red - Blink 1	Activates the blink frequency 1 with the red light in the general LED.
Red - Blink 2	Activates the blink frequency 2 with the red light in the general LED.
Blue - On	Activates the blue light in the general LED.
Blue - Blink 1	Activates the blink frequency 1 with the blue light in the general LED.
Blue - Blink 2	Activates the blink frequency 2 with the blue light in the general LED.

Option	Description
SM Drawer 1, SM Drawer 2, SM Drawer 3 and SM Drawer 4	
Off	Switch off the light in the selected drawer indicator.
On	Switch on the light in the selected drawer indicator.
Blink 1	Activates the blink frequency 1 light in the selected drawer indicator.
Blink 2	Activates the blink frequency 2 light in the selected drawer indicator.
CM Door	
OPENED	Inform that intermediate door is opened.
ERROR	Inform that an error is detected in the intermediate door detector.
SRM Door	
OPENED	Inform that upper door is opened.
ERROR	Inform that an error is detected in the upper door detector.
Keyboard	
key 7 (urgency)	Inform that urgency button is pressed.
key 6 (SM Drawer 4)	Inform that sample drawer 4 button is pressed.
key 5 (SM Drawer 3)	Inform that sample drawer 3 button is pressed.
key 4 (SM Drawer 2)	Inform that sample drawer 2 button is pressed.

Option	Description
key 3 (SM Drawer 1)	Inform that sample drawer 1 button is pressed.
key 2 (RM Drawer 2)	Inform that reagent drawer 2 button is pressed.
key 1 (RM Drawer 1)	Inform that reagent drawer 1 button is pressed.
key 6 (CM Drawer 4)	Inform that card drawer 4 button is pressed.
key 4 (CM Drawer 3)	Inform that card drawer 3 button is pressed.
key 2 (CM Drawer 2)	Inform that card drawer 2 button is pressed.
key 1 (CM Drawer 1)	Inform that card drawer 1 button is pressed.
Buzzer Mode	
OFF	Deactivate the buzzer.
MODE 1	Activate buzzer with mode 1 sound.
MODE 2	Activate buzzer with mode 2 sound.
MODE 3	Activate buzzer with mode 3 sound.
MODE 4	Activate buzzer with mode 4 sound.
MODE 5	Activate buzzer with mode 5 sound.
MODE 6	Activate buzzer with mode 6 sound.
MODE 7	Activate buzzer with mode 7 sound.

Option	Description
Ambient Lights	
Off	Switch off ambient lights (PIP and CAT areas).
On	Switch on ambient lights.
Machine Temperature	
Off automatic regulation	Deactivate temperature regulation that switches on automatically the fans of the electronic cabinet when reaching 30 °C.
On automatic regulation	Activate temperature regulation that switches on automatically the fans of the electronic cabinet when reaching 30 °C.
Electronics cabinet sensors	Information about temperature in real time of the upper part of the rear part, where it is located the power supply, computer, PCA GEN-BCKPL59 and PCAs Motorcontrol.
Reagents zone sensors	Information about temperature in real time of the upper area of the instrument.
Cards zone sensors	Information about temperature in real time of the intermediate area of the instrument.
general fans ON	Activate the fans located in the backpanel.
general fans OFF	Deactivate the fans located in the backpanel.

3.2.10 Firmware tab

The **Firmware** tab includes options related to the firmwares of the PCAs. It contains two subtabs: **PCA Control** and **EPROM control**.

3.2.10.1 PCA control subtab

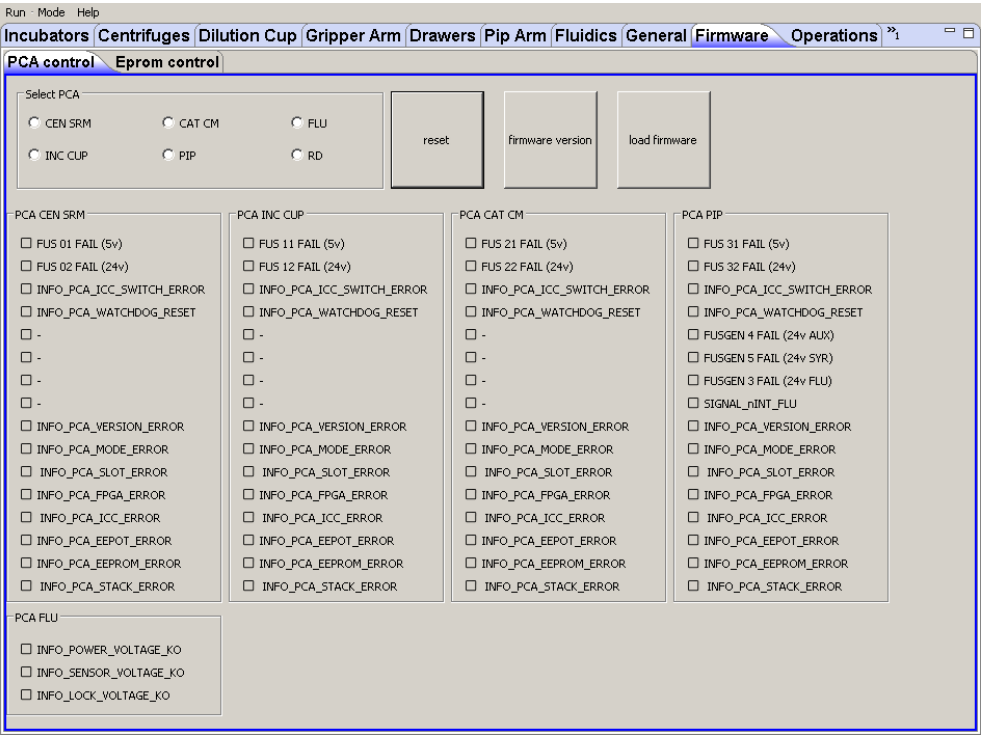


Figure 652 - PCA control - Firmware tab in the Technical Service software

Table 96 - List of options in the PCA control - Firmware tab

Option	Description
Select PCA	
CEN SRM	Select PCA CEN-SRM Control.
CAT CM	Select PCA CAT-CM Control.
FLU	Select PCA FLU-CTRL59.
INC CUP	Select PCA INC-CUP Control.
PIP	Select PCA PIP Control.
RD	Select PCA RD-IL421.
reset	Perform a reset of the selected PCA.
firmware version	Return firmware version of the selected PCA.
load firmware	Load a firmware. It opens a new window.
PCA CEN SRM	
FUS 01 FAIL (5v)	Fus 01 (5 V) of the PCA GEN-BCKPL59 fails.
FUS 02 FAIL (24v)	Fus 02 (24 V) of the PCA GEN-BCKPL59 fails.
INFO_PCA_ICC_SWITCH_ERROR	PCA CEN-SRM Control is not communicating or not working properly with the external IIC switch.
INFO_PCA_WATCHDOG_RESET	Last PCA CEN-SRM Control reset was a watchdog reset.

Option	Description
INFO_PCA_VERSION_ERROR	PCA HW version is erroneous and not compatible with the FW present on it.
INFO_PCA_MODE_ERROR	PCA operating mode is erroneous.
INFO_PCA_SLOT_ERROR	PCA CEN-SRM Control is inserted in an erroneous PCA GEN-BCKPL59 slot.
INFO_PCA_FPGA_ERROR	PCA FPGA programming procedure did not complete successfully.
INFO_PCA_ICC_ERROR	ICC error detected on the Control PCA ICC devices.
INFO_PCA_EEPOT_ERROR	PCA ICC non-volatile potentiometer is not working properly.
INFO_PCA_EEPROM_ERROR	Error during a PCA EEPROM read / write operation.
INFO_PCA_STACK_ERROR	PCA stack has grown over the security margin.
PCA INC CUP	
FUS 11 FAIL (5v)	Fus 11 (5 V) of the PCA GEN-BCKPL59 fails.
FUS 12 FAIL (24v)	Fus 12 (24 V) of the PCA GEN-BCKPL59 fails.
INFO_PCA_ICC_SWITCH_ERROR	PCA INC-CUP Control is not communicating or working properly with the external IIC switch.
INFO_PCA_WATCHDOG_RESET	Last PCA INC-CUP Control reset was a Watchdog reset.
INFO_PCA_VERSION_ERROR	PCA HW version is erroneous and not compatible with the FW present on it.
INFO_PCA_MODE_ERROR	PCA operating mode is erroneous.
INFO_PCA_SLOT_ERROR	PCA INC-CUP Control is inserted in an erroneous PCA GEN-BCKPL59 slot.

Option	Description
INFO_PCA_FPGA_ERROR	PCA FPGA programming procedure did not complete successfully.
INFO_PCA_ICC_ERROR	ICC error detected on the Control PCA ICC devices.
INFO_PCA_EEPOT_ERROR	PCA ICC non-volatile potentiometer is not working properly.
INFO_PCA_EEPROM_ERROR	Error during a PCA EEPROM read / write operation.
INFO_PCA_STACK_ERROR	PCA stack has grown over the security margin.
PCA CAT CM	
FUS 21 FAIL (5v)	Fus 21 (5 V) of the PCA GEN-BCKPL59 fails.
FUS 22 FAIL (24v)	Fus 22 (24 V) of the PCA GEN-BCKPL59 fails.
INFO_PCA_ICC_SWITCH_ERROR	PCA CAT-CM Control is not communicating or working properly with the external IIC switch.
INFO_PCA_WATCHDOG_RESET	Last PCA CAT-CM Control reset was a Watchdog reset.
INFO_PCA_VERSION_ERROR	PCA HW version is erroneous and not compatible with the FW present on it.
INFO_PCA_MODE_ERROR	PCA operating mode is erroneous.
INFO_PCA_SLOT_ERROR	PCA CAT-CM Control is inserted in an erroneous PCA GEN-BCKPL59 slot.
INFO_PCA_FPGA_ERROR	PCA FPGA programming procedure did not complete successfully.
INFO_PCA_ICC_ERROR	ICC error detected on the Control PCA ICC devices.
INFO_PCA_EEPOT_ERROR	PCA ICC non-volatile potentiometer is not working properly.

Option	Description
INFO_PCA_EEPROM_ERROR	Error during a PCA EEPROM read / write operation.
INFO_PCA_STACK_ERROR	PCA stack has grown over the security margin.
PCA PIP	
FUS 31 FAIL (5v)	Fus 01 (5 V) of the PCA GEN-BCKPL59 fails.
FUS 32 FAIL (24v)	Fus 02 (24 V) of the PCA GEN-BCKPL59 fails.
INFO_PCA_ICC_SWITCH_ERROR	Control_PCA is not communicating or working properly with the external IIC switch.
INFO_PCA_WATCHDOG_RESET	Last Control_PCA reset was a Watchdog reset.
FUSGEN 4 FAIL (24v AUX)	FUSGEN4 (24 V) of the PCA GEN-BCKPL59 fails.
FUSGEN 5 FAIL (24v SYR)	FUSGEN5 (24 V) of the PCA GEN-BCKPL59 fails.
FUSGEN 3 FAIL (24v FLU)	FUSGEN3 (24 V) of the PCA GEN-BCKPL59 fails.
SIGNAL_nINT_FLU	Signal nINT_FLU (auxiliary signal coming from the fluidics module) is active.
INFO_PCA_VERSION_ERROR	PCA HW version is erroneous and not compatible with the FW present on it.
INFO_PCA_MODE_ERROR	PCA operating mode is erroneous.
INFO_PCA_SLOT_ERROR	PCA PIP Control is inserted in an erroneous PCA GEN-BCKPL59 slot.
INFO_PCA_FPGA_ERROR	PCA FPGA programming procedure did not complete successfully.
INFO_PCA_ICC_ERROR	ICC error detected on the Control PCA ICC devices.

Option	Description
INFO_PCA_EEPOT_ERROR	PCA ICC non-volatile potentiometer is not working properly.
INFO_PCA_EEPROM_ERROR	Error during a PCA EEPROM read / write operation.
INFO_PCA_STACK_ERROR	PCA stack has grown over the security margin.
PCA FLU	
INFO_POWER_VOLTAGE_KO	No power supply at PCA FLU-CTRL59.
INFO_SENSOR_VOLTAGE_KO	Fluidic sensors voltage error.
INFO_LOCK_VOLTAGE_KO	Fluidic lockers voltage error.

3.2.10.2 Eprom control subtab

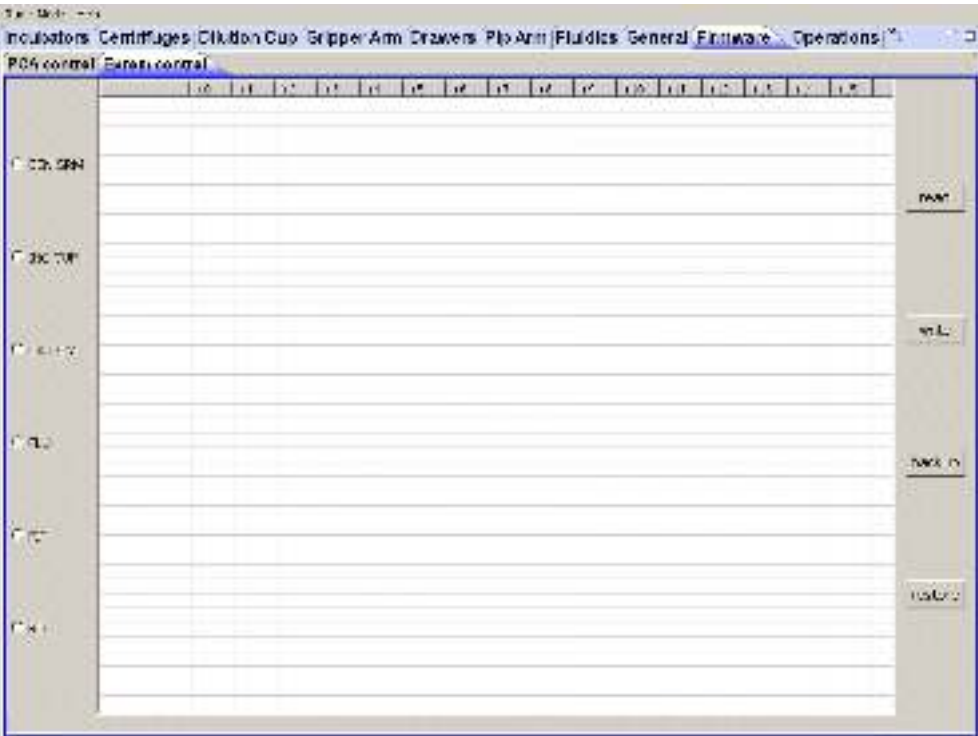


Figure 653 - Eprom control - Firmware tab in the Technical Service software

Table 97 - List of options in the Eprom control - Firmware tab

Option	Description
CEN SRM	Select CEN SRM MotorControl PCA.
INC CUP	Select INC CUP MotorControl PCA.
CAT CM	Select CAT-CM MotorControl PCA.
FLU	Select FLU-CTRL59 PCA.
PIP	Select PIP MotorControl PCA.
RD	Select RD-IL421 PCA.
read	Read EEPROM values of the selected PCA.
write	Write the values shown on the screen to the EEPROM of the selected PCA.
backup	Save into a file the contents of the EEPROM of the selected PCA.
restore	Restore from a file the contents of the EEPROM of the selected PCA.

3.2.11 Operations tab

The **Operations** tab contains two subtabs: **Dilution Cup Operations** and **Aspire and Dispense Operations**.

3.2.11.1 Dilution Cup Operations subtab

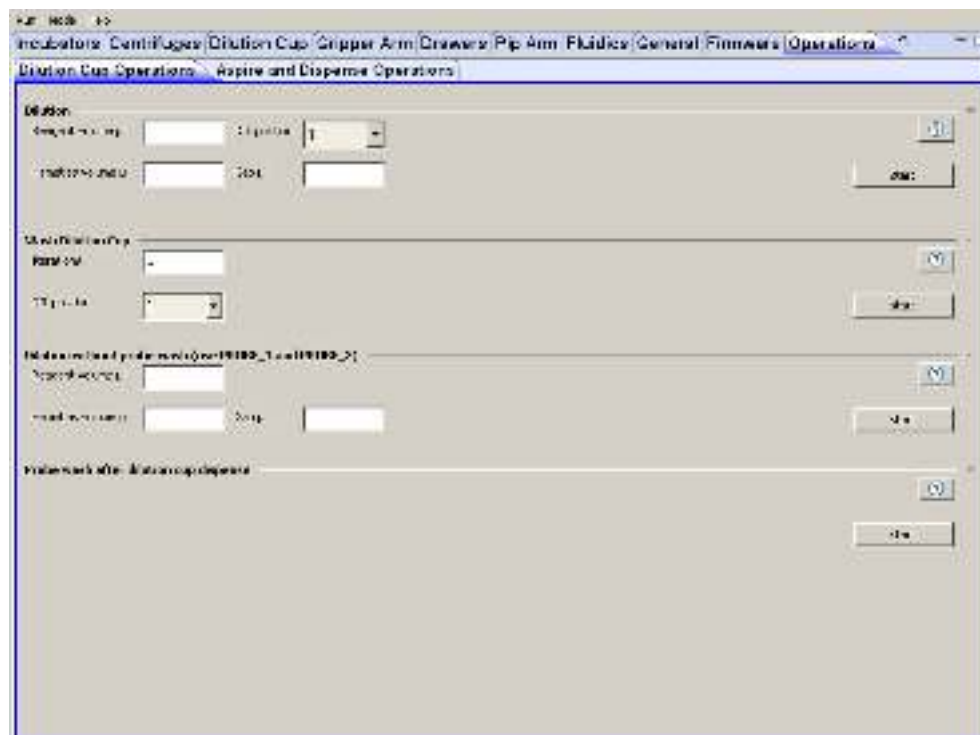


Figure 654 - Dilution Cup Operations in the Technical Service software

Table 98 - List of options in the Dilution Cup Operations tab

Option	Description
Dilution	
Reagent volume μl	Text box to enter the volume [μ l] of the reagent to perform the dilution. Suggested value = 500.
DIL position	Dilution cup well (1 or 2) where to perform the dilution.
Hematies volume μl	Text box to enter the volume [μ l] of the hematies to perform the dilution. Suggested value = 5.
Gap μl	Text box to enter the air gap [μ l] to perform the dilution. Suggested value = 5.
	Information about the prerequisites of the process: • Sample tube in sample drawer 1, rack 1, position 1. • Diluent bottle in reagent drawer 1, rack 1, position 1.
Start	Initiates the dilution process.
Wash Dilution Cup	
Iterations	Text box to enter the number of times to repeat the wash.
DIL position	Dilution cup well (1 or 2) where to perform the wash.
	Information about the process.
Start	Initiates the washing process.
Dilution without probe wash (use PROBE_1 and PROBE_2)	
Reagent volumen μl	Text box to enter the volume [μ l] of the reagent to perform the dilution. Suggested value = 500.

Option	Description
Hematies volume μl	Text box to enter the volume [μl] of the hematies to perform the dilution. Suggested value = 5.
Gap μl	Text box to enter the air gap [μl] to perform the dilution. Suggested value = 5.
	Information about the prerequisites of the process: • Sample tube in sample drawer 1, rack 1, position 1. • Sample tube in sample drawer 1, rack 1, position 2. • Diluent bottle in reagent drawer 1, rack 1, position 1.
start	Initiates the dilution process.
Probe wash after dilution cup dispense	
	Information about the process. It is performed only in probe 1.
start	Initiates the washing process.

3.2.11.2 Aspire and Dispense Operations subtab

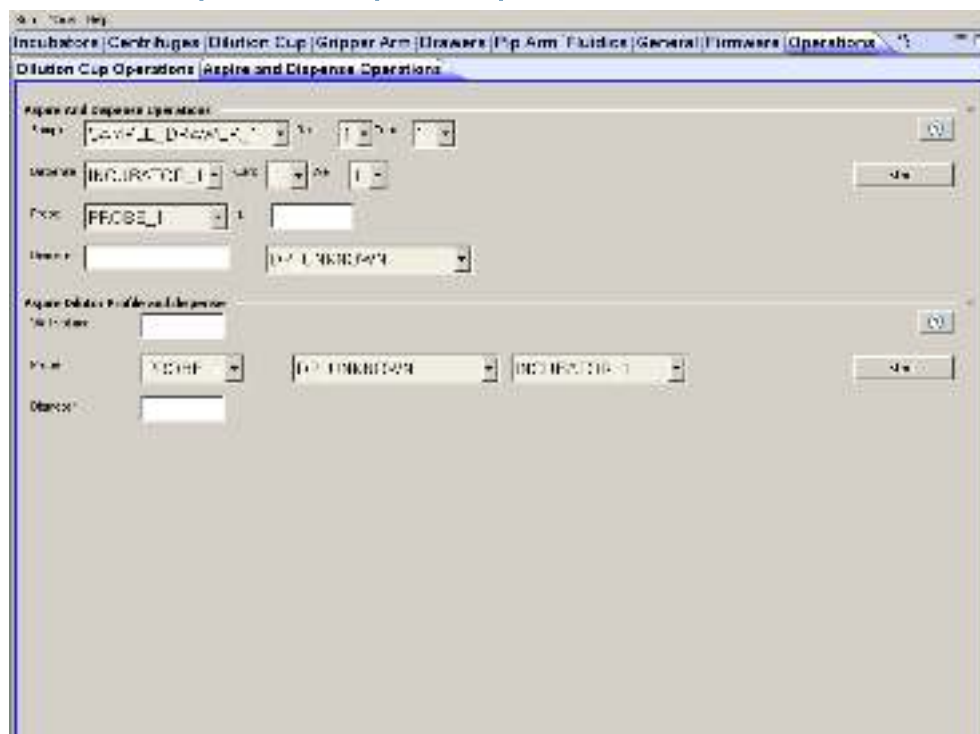


Figure 655 - Aspire and Dispense Operations in the Technical Service software

Table 99 - List of options in the Aspire and Dispense Operations tab

Option	Description
Aspire And Dispense Operations	
Sample	Select sample drawer: 1 to 4.
Rack	Select rack of the drawer: 1 or 2.
Tube	Select position of the tube: 1 to 12.
Dispense	Select incubator where to dispense: 1 to 4.
Card	Select incubator slot where to dispense: 1 to 12.
Well	Select well position in the card where to dispense: 1 to 8.
Probe	Select probe used: 1 or 2.
μl	Text box to enter the serum aspirated [μl] to perform the operation. Suggested value = 25.
Diameter	Text box to enter the diameter of the sample tube.
Selection box	Select dilutor profile.
	Information about the process. It is needed: • sample tube in the selected position. • incubator selected with a card in the selected position in the upper floor.
start	Initiates the process.
Aspire Dilutor Profile and dispense	

Option	Description
Well volume	Text box to enter the volume [μl] of the serum to dispense into the card well. Suggested value = 25.
Probe	Probe (1 or 2) used in the process.
First selection box	Select the dilutor profile.
Second selection box	Select the incubator to dispense.
Diameter	Text box to enter the diameter of the sample tube.
	Information about the process. It is needed: • sample tube in sample drawer 1, rack 1, position 1. • incubator selected filled with cards in the upper floor.
start	Initiates the process.

4 Diagnostic application

The Diagnostic software allows a qualified technician, supervisor, and operator to run self-tests.

4.1 Introduction

This chapter contains an overview of the Diagnostic software that allows a qualified technician / supervisor / operator to run a self-test.

The Diagnostic application can help to diagnose a problem in the instrument or just to check the proper functioning as a usual maintenance. Tubes, vials and a diluent bottle required to execute the Diagnostic application tests are included in P/N 732944 IH DIAGNOSTIC KIT.

In order for a qualified technician to access the Diagnostic application, the boot menu must be accessed. For that, switch on the Erytra, wait for the Erytra logo to appear on screen and then press the grid () button. Log in as `adminUser` (case sensitive, admin password) and the Boot Application options will be displayed. Once in there, press the arrow in the right side of the screen to switch to page 2/2 of the menu, then select the **Diagnostic** icon among the Boot Application options.



Figure 656 - Diagnostic icon

4.2 Overview

Once the Diagnostic application is opened, the following window appears on screen.



Figure 657 - Diagnostic application

Available tests are put into groups depending on the module that are related.

- Initialization.
- Cards Management.
- Pipetting.
- Fluidics.
- Reader.
- Incubators.
- Centrifuges.
- Samples drawers.
- Reagents drawers.
- Cards drawers.
- Dilution cup.
- Miscellaneous.

The Diagnostic application can check completely the equipment or only some parts. This can be chosen in the main screen of the software, by selecting or unselecting the modules to be checked.

At the bottom of the main screen there are four buttons.

Table 100 - Available buttons in the Diagnostic application

Button	Description
	Run the selected tests.
Start button	
	Stop the test. Enabled when the test is started.
Stop button	
	Print the test results. Enabled when the test has finished.
Printer button	
	Exit the software.
Exit button	

4.3 Tests

In the main screen of the software, it is possible to select the tests that are required to be performed. Once tests are selected, the **Start** button must be pressed in the main screen of the software in order to run them.

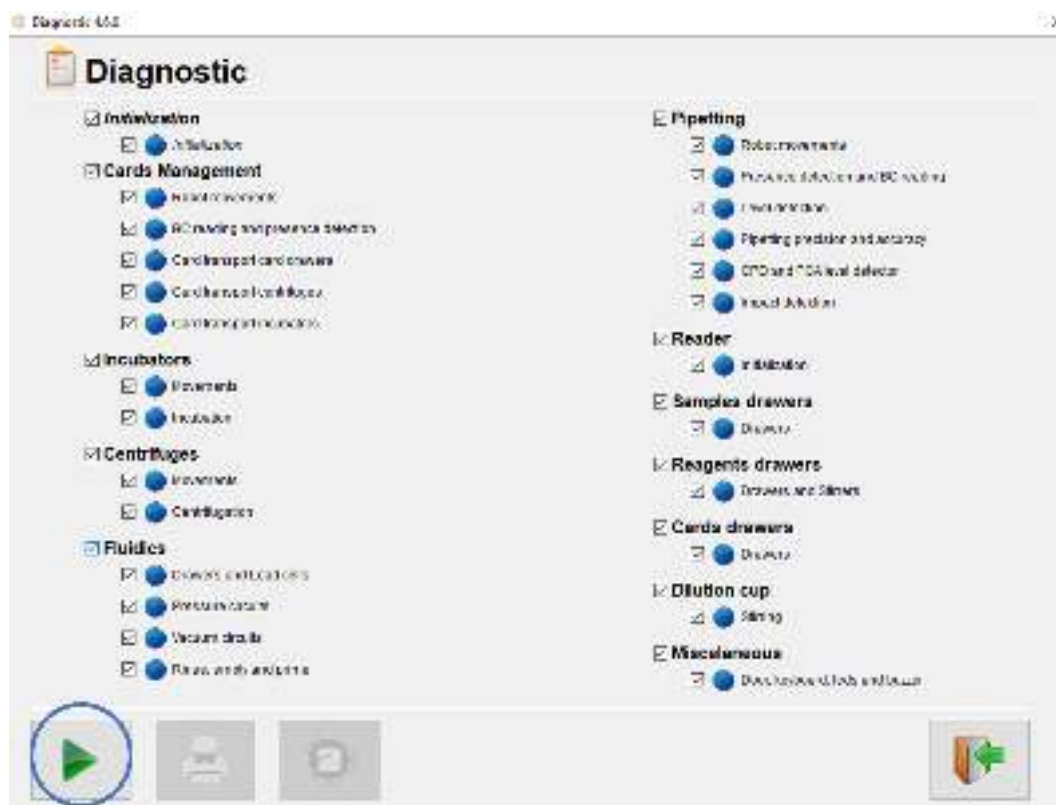


Figure 658 - Start button in Diagnostic application

Once the button is pressed, if any prerequisite is necessary to perform the selected tests, a window message on screen appears with them.

4.3.1 Initialization

This test is executed in each run. If the test fails, no further tests will be executed.



Figure 659 - Initialization option in the Diagnostic application

4.3.1.1 Initialization

Prerequisites:

- None.

Actions:

- Reset the 4 PCA MotorControl (CEN-SRM, CAT-CM, PIP and INC-CUP) and the PCA FLU-CTRL59.
- Home CAT (Z, Y and X axis).
- Home PIP (Z1, Z2, Horns, X and Y axis).

- Home of the incubators.
- Home of the centrifuges.
- Home of the stirring racks of the reagents drawers.

**WARNING**

Performing motor homes implies movement of the robotic arms. Make sure that the door is closed before running the test.

Expected results:

- PCAs can be reset without any error.
- All homes are performed without errors.

4.3.2 Cards Management

It includes the **Robot movements**, **BC reading and presence detection** and **Card Transport** tests.



Figure 660 - Cards Management options in the Diagnostic application

4.3.2.1 Robot movements

Prerequisites:

- No card in gripper.
- No cards in drawers 2 to 4.
- No cards in incubators .
- No cards in centrifuges.
- Card drawers and lower door closed.

Actions:

- Checking that all card drawers and lower door are closed.
- Checking that there is no card in gripper.
- Performing presence detection in card drawer 2 to 4, incubators and centrifuges.
- Perform 50 random movements.
- After each random movement, execute a **Take card** command (ignore no card error).

Expected results:

- Card drawers and lower door are closed.
- No card is detected in gripper.
- No cards are detected in card drawer 2 to 4, incubators and centrifuges.
- No error is detected during the execution.

4.3.2.2 BC reading and presence detection

Prerequisites:

- Card holder with first 12 lower cards in card drawer 1 - rack 4.
- Card drawers and lower door closed.

Actions:

- Checking that all card drawers and lower door are closed.
- Detect presence of placed holder.
- Take all cards and read their barcode .

Expected results:

- Card drawers and lower door are closed.
- No cards are detected in holders' upper floor.
- At least 12 cards are detected in the holders' lower floor.
- All cards are read and no error is reported by the BC reader.

4.3.2.3 Card Transport card drawers

Prerequisites:

- Card drawers and lower door closed.
- Card holder with first 12 lower cards in card drawer 1 - rack 4.
- Empty card holders in card drawer 1 - rack 1 to 3, and drawer 2, 3, and 4, rack 1 to 4.
- No card in gripper.

Actions:

- Checking that all card drawers and lower door are closed.
- Checking there is no card in gripper.
- Detect presence of drawer 1, rack 1 to 3, and drawer 2, 3 and 4, rack 1 to 4.
- Detect presence of placed holder.

- Cards are moved:
 - One by one, move 12 cards to drawer 1, rack 3.
 - One by one, move 12 cards to drawer 1, rack 2.
 - One by one, move 12 cards to drawer 1, rack 1.
 - One by one, move 12 cards to drawer 2, rack 4.
 - One by one, move 12 cards to drawer 2, rack 3.
 - One by one, move 12 cards to drawer 2, rack 2.
 - One by one, move 12 cards to drawer 2, rack 1.
 - One by one, move 12 cards to drawer 3, rack 4.
 - One by one, move 12 cards to drawer 3, rack 3.
 - One by one, move 12 cards to drawer 3, rack 2.
 - One by one, move 12 cards to drawer 3, rack 1.
 - One by one, move 12 cards to drawer 4, rack 4.
 - One by one, move 12 cards to drawer 4, rack 3.
 - One by one, move 12 cards to drawer 4, rack 2.
 - One by one, move 12 cards to drawer 4, rack 1.
 - One by one, move 12 cards to drawer 1, rack 4.

Expected results:

- Card drawers and lower door are closed.
- No card is detected in gripper.
- No cards are detected in holders' upper floor.
- At least 12 cards are detected in the holders' lower floor.
- No cards are detected in drawer 1, rack 1 to 3 and drawer 2, 3 and 4, rack 1 to 4.
- All cards are transported without errors.

4.3.2.4 Card transport centrifuges

Prerequisites:

- Card drawers and lower door closed.
- Card holder with first 12 lower cards in card drawer 1 - rack 4.
- Empty card holders in card drawer 1 - rack 1 to 3, and drawer 2, 3, and 4, rack 1 to 4.
- No card in gripper.
- No cards in centrifuges.

Actions:

- Checking that all card drawers and lower door are closed.
- Checking there is no card in Gripper.
- Detect presence of drawer 1, rack 1 to 3, and drawer 2, 3 and 4, rack 1 to 4.
- Detect presence of placed holder.
- Detect presence of centrifuges.
- Cards are moved:
 - One by one, move 12 cards to Centrifuge 1.
 - One by one, move 12 cards to Centrifuge 2.
 - One by one, move 12 cards to Cards drawer 1, rack 4..

Expected results:

- Card drawers and lower door are closed.
- No card is detected in gripper.
- No cards are detected in holders' upper floor.
- At least 12 cards are detected in the holders' lower floor.
- No cards are detected in drawer 1, rack 1.
- No cards are detected in centrifuges.
- All cards are transported without errors.

4.3.2.5 Card transport incubators

Prerequisites:

- Card drawers and lower door closed.
- Card holder with first 12 lower cards in card drawer 1 - rack 4.
- No card in gripper.
- No cards in incubators.

Actions:

- Checking that all card drawers and lower door are closed.
- Checking there is no card in Gripper.
- Detect presence of placed holder.
- Detect presence of incubators.

- Cards are moved:
 - One by one, move 12 cards to Incubator 1.
 - One by one, move 12 cards to Incubator 2.
 - One by one, move 12 cards to Incubator 3.
 - One by one, move 12 cards to Incubator 4.
 - One by one, move 12 cards to Cards drawer 1, rack 4.

Expected results:

- Card drawers and lower door are closed.
- No card is detected in gripper.
- No cards are detected in holders' upper floor.
- At least 12 cards are detected in the holders' lower floor.
- No cards are detected in incubators.
- All cards are transported without errors.

4.3.3 Pipetting

It includes the **Robot movements**, **Presence detection and BC reading**, **Level detection**, **Pipetting precision and accuracy**, **CPD and PCA level detector** and **Impact detection** tests.



Figure 661 - Pipetting options in the Diagnostic application

4.3.3.1 Robot movements

Prerequisites:

- Sample drawers and upper door closed.
- No samples tubes in samples drawers 4, 5 and 6.

Actions:

- Checking that sample drawers and upper door are closed.
- Performing presence detection in drawers 4, 5 and 6.
- 50 random movements in those positions where there are not tubes in drawers 4 to 6, odd positions (1, 3, ...) with probe 1.

- After each movement executing a Down command with both probes (ignore level error).
- Performing a barcode reading of tubes in drawer 4 rack 2.
- Moving Horns 50 times up/down.

Expected results:

- Sample drawers and upper door are closed.
- No sample tubes are detected in drawers 4, 5 and 6.
- No movement error is detected.

4.3.3.2 Presence detection and BC reading

Prerequisites:

- Reagents and samples drawers and upper door closed.
- No samples in drawers 4, 5 and 6.
- Drawer 3, rack 2 full of 10 mm (positions 1 to 4), 12 mm (positions 5 to 8) and 15 mm (positions 9 to 12) tubes with BC label. Tubes filled with saline solution 0.9%.
- Reagent drawer 1, rack 1 with 1 diluent in position 4 and 2 vials in positions 2 and 9 (with BC label).
- Reagent drawer 1, rack 2 with 4 vials in positions 1, 6, 11 and 18 (with BC label).

Actions:

- Checking that reagents and samples drawers and upper door are closed.
- Detect presence of sample tubes.
- Read barcodes of the tubes.
- Detect presence of vials/bottles.
- Read barcodes of vials/bottles.

Expected results:

- Reagents and samples drawers and upper door are closed.
- Tubes or vials are detected in all required positions of drawer 3, rack 2 and drawer 1.
- No tubes are detected in drawers 4, 5 and 6.
- All diameters match expected setting (+0 mm, -2 mm).
- BCs are read for all tubes.
- Vials/bottles are detected in all rack positions.
- BCs are read for all vials/bottles.



NOTE

If one or many diameters do not match with the expected setting (+0 mm, -2 mm) the test shall be considered as Warning.

4.3.3.3 Level detection

Prerequisites:

- Samples drawers and upper door closed.
- No samples in drawers 4, 5 and 6.
- Drawer 3, rack 2 full of 10 mm (positions 1 to 4), 12 mm (positions 5 to 8) and 15 mm (positions 9 to 12) tubes with BC label. Tubes filled with saline solution 0.9%.
- Circuit primed (if not primed, prime is performed).

Actions:

- Checking that samples drawers and upper door are closed.
- Detect presence of all the sample tubes.
- Check that the fluidic system is primed. If not primed, prime is performed.
- Perform 50 level detections with each probe (probe 1 and 2 in parallel) in all samples tubes (positions 1 and 2, 5 and 6, 9 and 10).



NOTE

If one or many diameters do not match with the expected setting (+0 mm, -2 mm) the test shall be considered as Warning.



NOTE

If any of the probes encounters an error that prevents from continuing, the system shall continue with the other probe.

Expected results:

- Samples drawers and upper door are closed.
- Tubes are detected in all required positions of drawer 3, rack 2.
- No tubes are detected in drawers 4, 5 and 6.
- All diameters match expected setting (+0 mm, -2 mm).
- All level detections are performed without error.
- No error is reported during process.

4.3.3.4 Pipetting precision and accuracy

Prerequisites:

- Reagents and samples drawers, card drawers and doors closed.
- Reagent drawer 1 - rack 1 with 1 diluent bottle in position 4 and 2 vials in positions 2 and 9. All of them must include barcode labels.

- Reagent drawer 1 - rack 2 with 4 vials in positions 1, 6, 11 and 18. All of them must include barcode labels. Tubes filled with saline solution 0.9%. and vial in position 1 must also be colored with tartrazine.
- Cards holder in card drawer 1 - rack 4 with 4 cards in lower floor (minimum).
- Circuit primed (if not primed, prime is performed).
- No cards in Incubators, centrifuges and card drawers 2, 3 and 4.
- No card in gripper.

Actions:

- Checking that reagents and samples drawers, card drawers and doors are closed.
- Checking there is no card in Gripper.
- Detect presence of cards.
- Detect presence of vials.
- If fluidic system is not primed, prime is performed.
- Move four cards from card drawer 1 - rack 4 to incubator 1, performing an integrity check per each card.
- Move incubator 1 to PIP floor.
- Aspirate 720 µl of colouring dilution with each probe.
- Dispense 75 µl of colouring dilution in each well. Probe 1 dispenses into card 1 and probe 2 dispenses into card 3.
- Aspirate 920 µl of colouring dilution with each probe.
- Dispense 100 µl of colouring dilution in each well. Probe 1 dispenses into cards 2 and probe 2 dispenses into card 4.
- Move incubator 1 to the CAT floor.
- Move the four cards from the incubator 1 to centrifuge 1.
- Perform a centrifugation of 2 min.
- Read all cards and calculate dispensed volume.
- Move all cards to the card drawer 1 - rack 4.

Expected results:

- Reagents and samples drawers, card drawers and doors are closed.
- No card detected in gripper.
- Vials are detected in all required positions.
- No cards are detected in holders' upper floor.
- At least four cards are detected in the holders' lower floor.
- No cards are detected in not required positions of card drawers, incubators and centrifuges.
- Cards integrity check gives no error.

- Card read gives no error. Dispensed volume lies within margins ($\pm 12.5 \mu\text{l}$).
- No error is reported during process.

4.3.3.5 CPD and PCA level detector

Prerequisites:

- Circuit primed (if not primed, prime is performed).

Actions:

- Set on CPD of probe 1.
- Wait 5 s.
- Check that value of the CPD sensor is between 17000 and 18400 counts.
- Set off CPD of probe 1.
- Set on CPD of probe 2.
- Wait 5 s.
- Check that value of the CPD sensor is between 17000 and 18400 counts.
- Set off CPD of probe 2.
- Set on level detector of probe 1.
- Wait 5 s.
- Check that value of the level detector is between 7300 and 7900 counts.
- Set off level detector of probe 1.
- Set on level detector of probe 2.
- Wait 5 s.
- Check that value of the level detector is between 9900 and 10900 counts.
- Set off level detector of probe 2.

Expected results:

- Value of the CPD sensors is within margins.
- Value of the level detector sensors is within margins.
- The test is executed without errors.

4.3.3.6 Impact detection

Prerequisites:

- Samples drawers, reagent drawers and upper door closed.

Actions:

- Checking that reagents and samples drawers and upper door are closed.
- Ask operator to insert one diluent bottle with its cap in reagent drawer 2, rack 1, position 4, remove the holder from Samples Drawer 2, Rack 2 and close drawers.
- Performing presence detection in reagent drawer 2, rack 1.
- Move pipetting arm, probe 1, to reagent drawer 2, rack 1, position 4.
- Perform 3 level detections with probe 1, with depth 0 mm, offset maximum depth 0 mm and vial profile level.
- Move the probe up between level detections.
- Move pipetting arm, probe 1, to Samples Drawer 2, Rack 2, Position 6.
- Move probe 1 down.
- If one or more level detections have been performed with probe 1, ask operator for confirmation of three impacts and if the probe 1 can continue executing tests.
- If operator answers No, and there are pending tests related with PIP arm to execute in current execution, stop tests related with PIP zone.
- If operator answered that the probe is not bent, move probe 1 up.
- Move pipetting arm, probe 2, to reagent drawer 2, rack 1, position 4.
- Perform 3 level detections with probe 2, with depth 0 mm, offset maximum depth 0 mm and vial profile level.
- Move the probe up between level detections.
- Move pipetting arm, probe 2, to Samples Drawer 2, Rack 2, Position 6.
- Move probe 2 down.
- If one or more level detections have been performed with probe 2, ask operator for confirmation of three impacts and if the probe 2 can continue executing tests.
- If operator answers No, and there are pending tests related with PIP arm to execute in current execution, stop tests related with PIP zone.
- If operator answered that the probe is not bent, move probe 2 up.
- If operator answered that the probe is not bent, ask the operator to remove the cap and place the diluent bottle to reagent drawer 1, rack 1, position 4, insert the holder in Samples Drawer 2, Rack 2 and close drawers.
- Check reagent drawer 1, reagent drawer 2 and Samples Drawer 2 are closed and if at least one of them is open, show an information message indicating to close them.

Expected results:

- Reagents and samples drawers and upper door are closed.
- A vial is detected in reagent drawer 2, rack 1, position 4.
- The impact or movement error has been detected three times for each probe.

- The operator has confirmed that both probes are not bent.
- The test is executed without errors.

4.3.4 Fluidics

It includes the **Drawers and Load cells**, **Pressure circuits**, **Vacuum circuits** and **Prime, rinse and empty** tests.



Figure 662 - Fluidics options in the Diagnostic application

4.3.4.1 Drawers and Load cells

Prerequisites:

- It is necessary an empty bottle, a full bottle (8 l) and a known 2 kg weight (2 litres of water recommended).
- Fluidic drawers closed.

Actions:

- Checking that all fluidic drawers are closed.
- Solution A drawer is opened. Place an empty bottle, close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct.
- Solution A drawer is opened. Place a full (8 l) bottle, close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct.
- Solution B drawer is opened. Place an empty bottle, close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct.
- Solution B drawer is opened. Place a full (8 l) bottle, close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct.
- Waste bottle 1 drawer is opened. Place a full bottle (8 l), close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct.
- Waste bottle 1 drawer is opened. Place an empty bottle, close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct. Close the drawer

- Waste bottle 2 drawer is opened. Place a full (8 l) bottle, close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct.
- Waste bottle 2 drawer is opened. Place an empty bottle, close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct. Close the drawer
- Card waste drawer is opened. Empty the container, close the drawer, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct.
- Card waste drawer is opened. Place a 2 kg weight into the container by either pouring 2 litres of water (recommended) or placing a known 2 kg weight in the center of the container, wait 2 seconds and press **OK** on the screen message. The system measures the weight and checks whether values are correct.
- Card waste drawer is opened in order to allow the possibility of placing an empty container. Close the drawer.

Expected results:

- All obtained values are correct.
 - Empty bottle 0 ± 0.5 L.
 - Full bottle 8 ± 0.5 L.
 - Empty card waste container: 0 ± 50 cards.
 - Full card waste container: 410 ± 50 cards.
- All drawers are closed and close detection is performed correctly.

4.3.4.2 Prime, rinse and empty

Prerequisites:

- In mono mode: at least 8 l capacity in waste bottles and 4 l available in both solution A and solution B bottles.
- In dual mode: at least 2 L available in a bottle of Sol A and 2 L in a bottle of Sol B.
- All fluidic drawers are closed.

Actions:

- In mono mode, checking that there is at least 1,5 L +/- 200 ml of capacity in a waste bottle and 2 L +/- 200 ml available in both Sol A and Sol B bottles.
- In dual mode, checking that there is at least 2 L +/- 200 ml available in a bottle of Sol A and 2 L +/- 200 ml in a bottle of Sol B.
- Checking that fluidic drawers are closed.
- Move probes to dilution cups.
- Initialize probes
- Perform an empty of the fluidic circuit.

- Perform a rinse of the fluidic circuit.
- Perform an empty of the fluidic circuit.
- Perform a prime of the fluidic circuit.

Expected results:

- Waste and solution bottles capacity are the expected ones.
- Fluidic drawers are closed.
- All processes performed without errors.

4.3.4.3 Pressure circuits

Prerequisites:

- In mono mode: at least 8 l capacity in waste bottles and 4 l available in both solution A and solution B bottles.
- In dual mode: at least 2 L available in a bottle of Sol A and 2L in a bottle of Sol B.
- All fluidic drawers are closed.

Actions:

- a. Checking that fluidic drawers are closed.
- b. In mono mode, checking that there is at least 1,5 L +/- 200 ml of capacity in a waste bottle and 2 L +/- 200 ml available in both Sol A and Sol B bottles. Related electrovalves shall be opened or closed in order to select the bottle with the needed available capacity or volume.
- c. In dual mode, checking that there is at least 2 L +/- 200 ml available in a bottle of Sol A and 2 L +/- 200 ml in a bottle of Sol B. Related electrovalves shall be opened or closed in order to select the bottle with the needed volume.
- d. Close all pumps and electrovalves, except electrovalves related with containers switching.
- e. Open EV20, EV19, EV5 and EV10.
- f. Wait 10 seconds.
- g. Close EV10, EV5, EV19 and EV20.
- h. Switch on Pressure pump A (flowrate to 160).
- i. Wait 500 ms.
- j. Check that once the sensor 3 is stabilized (5 seconds with a variability of +/- 20 mbar), pressure value between 170 and 150 mbar is obtained in sensor 3.
- k. Open EV20.
- l. Wait 40 seconds.
- m. Check that once the sensor 3 is stabilized (5 seconds with a variability of +/- 20 mbar), pressure value between 80 and 280 mbar is obtained in sensor 3.
- n. Switch off Pressure pump A.
- o. Close EV20.

- p. Switch on Pressure pump B.
- q. Configure flowrate of Pressure pump B to 160.
- r. Wait 500 ms.
- s. Check that once the sensor 4 is stabilized (5 seconds with a variability of +/- 20 mbar), pressure value between 170 and 150 mbar is obtained in sensor 4.
- t. Open EV19.
- u. Wait 40 seconds.
- v. Check that once the sensor 4 is stabilized (5 seconds with a variability of +/- 20 mbar), pressure value between 80 and 280 mbar is obtained in sensor 4.
- w. Close EV19.
- x. Switch off Pressure pump B.
- y. If the system is configured in dual mode/external drain, steps **d** to **x** will be repeated with bottle 3 as solution A and bottle 4 as a solution B.

Expected results:

- All fluidic drawers are closed.
- Waste and solution bottles capacity are the expected ones.
- All pressure values obtained are correct.
- Sensors are stabilized in less than 50 seconds.
- The test is executed without errors.

4.3.4.4 Vacuum circuits

Prerequisites:

- In mono mode: at least 8 l capacity in waste bottles and 4 l available in both solution A and solution B bottles.
- In dual mode: at least 2 L available in a bottle of Sol A and 2 L in a bottle of Sol B.
- All fluidic drawers are closed.

Actions:

- Checking that fluidic drawers are closed.
- In mono mode, checking that there is at least 1,5 L +/- 200 ml of capacity in a waste bottle and 2 L +/- 200 ml available in both Sol A and Sol B bottles. Related electrovalves shall be opened or closed in order to select the bottle with the needed available capacity or volume.
- In dual mode, checking that there is at least 2 L +/- 200 ml available in a bottle of Sol A and 2 L +/- 200 ml in a bottle of Sol B. Related electrovalves shall be opened or closed in order to select the bottle with the needed volume.
- Close all pumps and electrovalves, except electrovalves related with containers switching.
- Open EV20, EV19, EV5 and EV10.
- Wait 10 seconds.

- Close EV10, EV5, EV19 and EV20.
- Switch on Vacuum pump.
- Wait 500 ms.
- Check that once the sensor 5 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 5 is greater than or equal to 600 mbar.
- Check that once the sensor 6 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 6 is greater than or equal to 600 mbar.
- Open EV5 and EV10.
- Wait 500 ms.
- Check that once the sensor 5 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 5 is greater or equal than 100 mbar.
- Check that once the sensor 6 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 6 is greater or equal than 100 mbar.
- Close EV5 and EV10.
- Open EV4 and EV9.
- Wait 500 ms.
- Check that once the sensor 5 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 5 is greater or equal than 100 mbar.
- Check that once the sensor 6 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 6 is greater or equal than 100 mbar.
- Close EV4 and EV9.
- Switch off Vacuum pump.

Expected results:

- All fluidic drawers are closed.
- Waste and solution bottles capacity are the expected ones.
- All vacuum values obtained are correct.
- Sensors are stabilized in less than 50 seconds.
- The test is executed without errors.



NOTE

If sensor 5 or sensor 6 are between -450 mbar and -599 mbar in the first check, the test shall be considered as Warning.

4.3.5 Reader

It includes the **Initialization** test.



Figure 663 - Reader option in the Diagnostic application

4.3.5.1 Initialization

Prerequisites:

- None

Actions:

- Perform an initialization of the reader.

Expected results:

- No errors are detected during the execution.

4.3.6 Incubators

It includes the **Movements** and **Incubation** tests.



Figure 664 - Incubators options in the Diagnostic application

4.3.6.1 Movements

Prerequisites:

- Doors closed.

Actions:

- Checking that doors are closed.
- Move CAT robot arm to centrifuge 2.
- Perform 25 movements up and down with incubators 1 and 2.
- Move CAT robot arm to centrifuge 1.
- Perform 25 movements up and down with incubators 3 and 4.

Expected results:

- Doors are closed.
- No error is detected during the execution.

4.3.6.2 Incubation

Prerequisites:

- None.

Actions:

- Start temperature regulation 37 °C in all incubators.
- Check that temperature is reached within 20 min.
- Start temperature regulation 24 °C in all incubators.
- Check that temperature is reached within 20 min.

Expected results:

- No error is detected during the execution.
- Temperatures are reached within expected time.

4.3.7 Centrifuges

It includes the **Movements** and **Centrifugation** tests.



Figure 665 - Centrifuges options in the Diagnostic application

4.3.7.1 Movements

Prerequisites:

- Lower door closed.

Actions:

- Checking that lower door is closed.
- Perform 50 random movements per each centrifuge.

Expected results:

- Lower door is closed.
- No error is detected during the execution.

4.3.7.2 Centrifugation

Prerequisites:

- Lower door closed.

Actions:

- Checking that lower door is closed.
- Perform a centrifugation (PROFILE_1) in each centrifuge.
- Check that average speed is within limits (1206 rpm $\pm$ 3%).
- When centrifugation finishes, check that centrifugation duration longed as expected (120 s to 144 s, it includes centrifuge home action after centrifugation).

Expected results:

- Lower door is closed.
- No error is detected during the execution.
- Speed and time lie within expected limits.

4.3.8 Samples drawers

It includes the **Drawers** test.



Figure 666 - Samples drawers option in the Diagnostic application

4.3.8.1 Drawers

Prerequisites:

- Samples drawers closed.

Actions:

- Check that all samples drawers are closed.
- Software opens automatically, one by one, all samples drawers.
- Close sample drawer 1 (drawer 3) and click **Accept**.
- Close sample drawer 2 (drawer 4) and click **Accept**.
- Close sample drawer 3 (drawer 5) and click **Accept**.
- Close sample drawer 4 (drawer 6) and click **Accept**.

Expected results:

- All drawers are closed.
- All drawers are opened without error.
- All drawers are closed and close detection is performed correctly and in the correct order.

4.3.9 Reagents drawers

It includes the **Drawers and Stirrers** test.



Figure 667 - Reagents drawers option in the Diagnostic application

4.3.9.1 Drawers and stirrers

Prerequisites:

- Reagents drawers closed.

Actions:

- Check that all reagents drawers are closed.
- Software opens automatically, one by one, all reagents drawers.
- Close reagent drawer 1 and click **Ok**.
- Close reagent drawer 2 and click **Ok**.
- Perform 10 stirrings with each reagent drawer.

Expected results:

- All drawers are closed.
- All drawers are opened without error.
- All drawers are closed and close detection is performed correctly and in the correct order.
- Stirrings are performed without errors.

4.3.10 Cards drawers

It includes the **Drawers** test.



Figure 668 - Cards drawers option in the Diagnostic application

4.3.10.1 Drawers

Prerequisites:

- Cards drawers closed.

Actions:

- Check that all cards drawers are closed.
- Software opens automatically, one by one, all cards drawers.
- Close card drawer 1 and click **Accept**.
- Close card drawer 2 and click **Accept**.

- Close card drawer 3 and click **Accept**.
- Close card drawer 4 and click **Accept**.

Expected results:

- All drawers are closed.
- All drawers are opened without error.
- All drawers are closed and close detection is performed correctly and in the correct order.

4.3.11 Dilution cup

It includes the **Stirring** test.



Figure 669 - Dilution cup option in the Diagnostic application

4.3.11.1 Stirring

Prerequisites:

- None.

Actions:

- Perform 25 dilution cup stirrings.

Expected results:

- Stirrings are performed without errors.

4.3.12 Miscellaneous

It includes the **Door, keyboard, leds and buzzer** test.

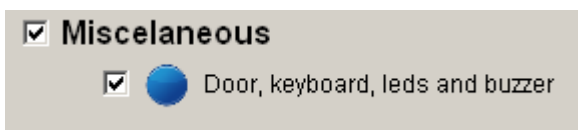


Figure 670 - Miscellaneous option in the Diagnostic application

4.3.12.1 Door, keyboard, LEDs and buzzer

Prerequisites:

- Remove intermediate door screw.

Actions:

- Switch on ambient lights and general LED in red color.
- Check if ambient lights and general LED in red color are switched on. Press **Yes/No** in the software.
- Switch off ambient lights.
- Switch on samples drawers LEDs and general LED in yellow color.
- Check if samples drawers LEDs and general LED in yellow color are switched on. Press **Yes/No** in the software.
- Switch off samples drawers lights.
- Switch on general LED in green color and buzzer.
- Check if general LED is in green color and the buzzer is on. Press **Yes/No** in the software.
- Press all the upper keyboard buttons, one by one, from reagents drawers to stat button.
- Press all the intermediate keyboard buttons, one by one, from card drawer 1 to card drawer 4.
- Open the upper door and press **Ok**.
- Close upper door and press **Ok**.
- Open intermediate door and press **Ok**.
- Close intermediate door and press **Ok**.

Expected results:

- All LEDs are confirmed to work correctly.
- Buzzer is confirmed to work correctly.
- All buttons are pressed.
- Doors open/close is detected.

4.4 Report

Once the selected tests are finished, a report can be obtained by pressing the printer button in the main screen of the software.

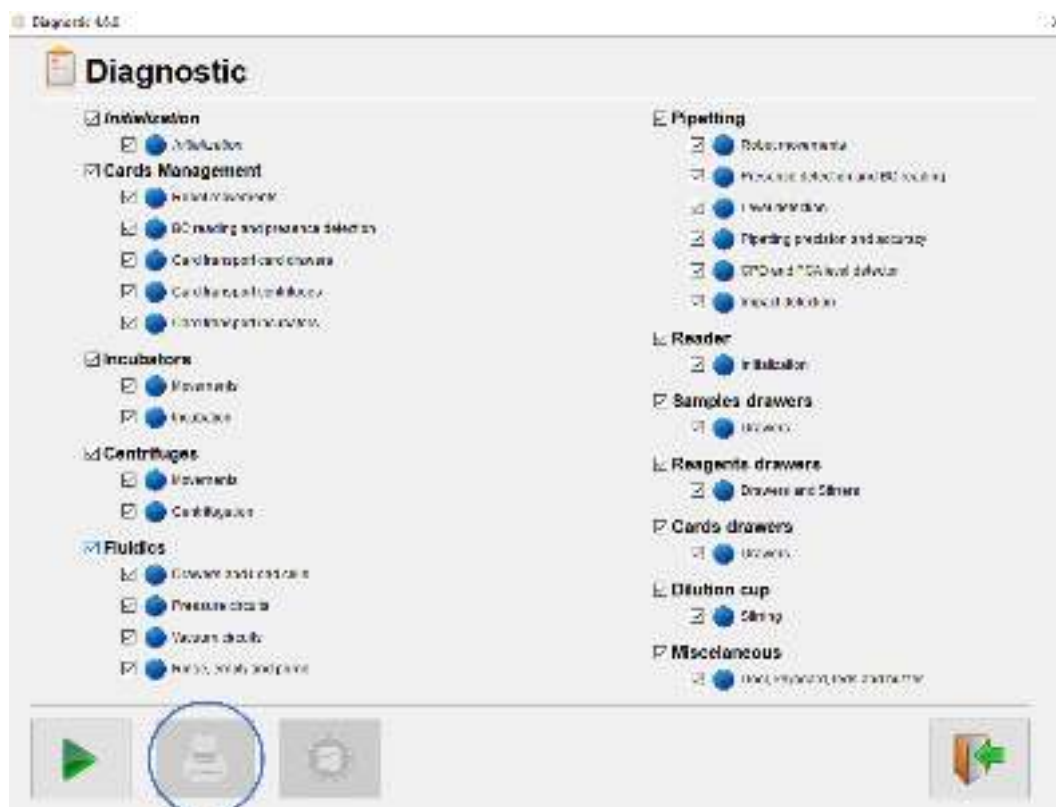


Figure 671 - Printer button in Diagnostic application

The report document is opened in a different window software, including information of instrument serial number, final status of tests, date and time, name and signature.

Possible status of tests in the report are the following ones:

- Pass: test has finished without errors.
- Fail: during the running of the test some errors have been detected.
- Warning: during the running of the test some errors have been detected but they are not critical.
- Not executed: test has not been selected.
- Stopped: running of the test has been stopped during its execution.

Report Example

Vehicle Information:

Vehicle ID	0000
Vehicle	0000

Fault Management:

System monitoring	not executed
ECU monitoring (communication)	not started
ECU monitoring	not started

Planning:

System monitoring	not executed
ECU monitoring (communication)	not started
ECU monitoring	not started
ECU monitoring	not started
ECU monitoring	not started

Results:

System monitoring	not executed
ECU monitoring (communication)	not started
ECU monitoring	not started
ECU monitoring	not started
ECU monitoring	not started

Figure 672 - Report example in Diagnostic application

A list of possible causes that provoke a warning or fail status of the test can be checked in the following table. In order to verify exactly the cause of the problem, **diagnostic.log** must be checked (it can be found in **E:\Program Files\Erytra Diagnostic**).

Table 101 - Warning and Fail results in Diagnostic application

TEST		WARNING	FAIL
Initialization	Initialization	None.	- Identify which PCA could not be reset. - Identify which servo failed to home and the error. - Identify if CAT arm or an incubator can not be homed because they are disturbing each other.
Cards Management	Robot movements	None.	- Home and no home positions are not correctly detected. - Identify if card is detected in gripper. - Identify which card drawer is open or if lower door is open. - Identify if card drawer 2 to 4, which incubator or which centrifuge has present cards. - Identify which servo failed in any movement and the error, and which movement is performed.
Cards Management	BC reading and presence detection	None.	- Identify which card drawer is open or if lower door is open. - Cards have been detected in upper floor. - Number of cards detected is lower than 12. - Card BC has not been read. Identify the position of the card in the rack. - Identify which servo failed a movement and the error.
Cards Management	Card transport drawers	None.	- Identify which card drawer is open or if lower door is open. - Card is detected in gripper. - Cards have been detected in upper floor. - Number of cards detected is lower than 12. - Cards have been detected in drawer 1 rack 1, or any rack of drawer 2, 3 or 4. - Identify which servo failed a movement and the error, and which movement is performed.

	TEST	WARNING	FAIL
Cards Management	Card Transport centrifuges	None.	- Identify which card drawer is open or if lower door is open. - Card detected in gripper. - Cards have been detected in upper floor. - Number of cards detected is lower than 12. - Cards have been detected in a centrifuge. - Identify which servo failed a movement and the error, and which movement is performed.
Cards Management	Card Transport incubators	None.	- Identify which card drawer is open or if lower door is open. - Card detected in gripper. - Cards have been detected in upper floor. - Number of cards detected is lower than 12. - Cards have been detected in an incubator. - Identify which servo failed a movement and the error, and which movement is performed.
Pipetting	Robot movements	None.	- Home and no home positions are not correctly detected - Identify which sample drawer is open or if upper door is open. - Samples detected in drawers 4, 5 or 6. - Identify which servo failed a movement and the error, and which movement is performed.
Pipetting	Presence detection and BC reading	Tubes diameters do not match with the expected ones (0/-2 mm).	- Identify which reagent or sample drawer is open or if upper door is open. - Not all BC have been read. - Tube/vial has not been detected in a required position (position). - Tube detected in a not required position (position). - Diameters different from expected (position, obtained diameter and expected diameter). - Identify which servo failed a movement and the error, and which movement is performed.

	TEST	WARNING	FAIL
Pipetting	Level detection	Tubes diameters do not match with the expected ones (0/-2 mm).	- Identify which sample drawer is open or if upper door is open. - Tube has not been detected in a required position. - Tube detected in a not required position. - Diameters different from expected +0 mm, -2 mm (position, obtained diameter and expected diameter). - Probe and encoder position if level detection fails. - Identify which servo failed a movement and the error, and which movement is performed. - Identify fluidic error performing prime.
Pipetting	Pipetting precision and accuracy	None.	- Identify which reagent or sample drawer or card drawer is open or which door is open. - Card detected in gripper. - Number of detected cards smaller than four or in holder upper floor in card drawer 1, rack 4. - Identify if card drawer 2, 3 or 4, which incubator or which centrifuge has present cards. - Card integrity fail. - Vials not detected in drawer 1. - Dispensed volume is outside limits (detected volume, expected volume). - Identify which servo failed in any movement and the error, and which movement is performed. - Identify fluidic error performing prime.
Pipetting	CPD and PCA level detector	None.	- Identify any fluidic error with process, device and error. - Sensor value does not lie within margins (sensor, expected value, obtained value).
Pipetting	Impact detection	None.	- Identify which reagent or sample drawer is open or if upper door is open. - Vial not detected in reagent drawer 2, rack 1, position 4. - Impact or movement error has not been detected three times for each probe. - Operator notes that a probe is bent. - Identify which servo failed in any movement and the error, and which movement is performed.

	TEST	WARNING	FAIL
Fluidics	Drawers and Load cells	None.	- Measured values are not between margins (± 0.5 l for containers or ± 25 cards for waste tray). - Drawers are open before starting test. - Drawer is not closed.
Fluidics	Prime, rinse and empty	None.	- Waste and solution bottles capacity are not the expected ones. - Identify which fluidic drawer is opened. - Identify which servo failed in any movement and the error, and which movement is performed. - Identify any fluidic error with process, device and error.
Fluidics	Pressure circuits	None.	- A fluidic drawer is open. - Waste or solution bottles capacity are not the expected ones. - P sensor value does not lie within margins. - Sensor not stabilized in less than 50 seconds. - Identify any fluidic error with process, device and error.
Fluidics	Vacuum circuits	None.	- A fluidic drawer is open. - Waste and solution bottles capacity are not the expected ones. - V sensor value does not lie within margins. - Sensor not stabilized in less than 50 seconds. - Identify any fluidic error with process, device and error.
Reader	Initialization	None.	- The reader could not be initialised. - Start-up and check returned an error.
Incubators	Movements	None.	- Identify which door is open. - Identify which servo failed a movement and the error, and which movement is performed.
Incubators	Incubation	None.	Identify what incubator failed to reach temperature in the specified time (less than 20 min).

	TEST	WARNING	FAIL
Centrifuges	Movements	None.	- Lower door is open. - Identify which centrifuge failed in any movement and the error, and which movement is performed.
Centrifuges	Centrifugation	None.	- Lower door is open. - Average speed does not lie within limits (1206 rpm $\pm$ 3%). - Centrifugation time does not correspond to expected value (120 s to 144 s, it includes centrifuge home action after centrifugation).
Samples drawers	Drawers	None.	- Drawers are not closed at the beginning of the test. - Drawer is not opened. - Drawer is not closed and close detection is not performed correctly or in the correct order.
Reagents drawers	Drawers and Stirrers	None.	- Drawers are not closed at the beginning of the test. - Drawer is not opened. - Drawer is not closed. - Close detection is not performed correctly or in the correct order. - Stirring errors, identifying the servo and the error.
Cards drawers	Drawers	None.	- Drawers are not closed at the beginning of the test. - Drawer is not opened (drawer). - Drawer is not closed and close detection is not performed correctly or in the correct order.
Dilution cup	Stirring	None.	Identify any servo error with the stirrer.
Miscellaneous	Door, keyboard, LEDs and buzzer	None.	- LEDs and Buzzer: user answers NO to a question. - Buttons: - User pressed Cancel. - A button pressed was not detected or is not detected in the specified order. - Door open/close has not been detected. LED does not work properly.

5 Importer application

5.1 Introduction

The Erytra Importer application allows a qualified operator to import new reagent packs, activate profiles, and configure settings.

In order for a qualified operator to access the Importer application, the boot menu must be accessed. For that, switch on the Erytra, wait for the Erytra logo to appear on screen and then press the grid () button. Log in as `adminUser` (case sensitive, admin password) and the Boot Application options will be displayed. Once open, select the **Importer** icon among the Boot Application options.

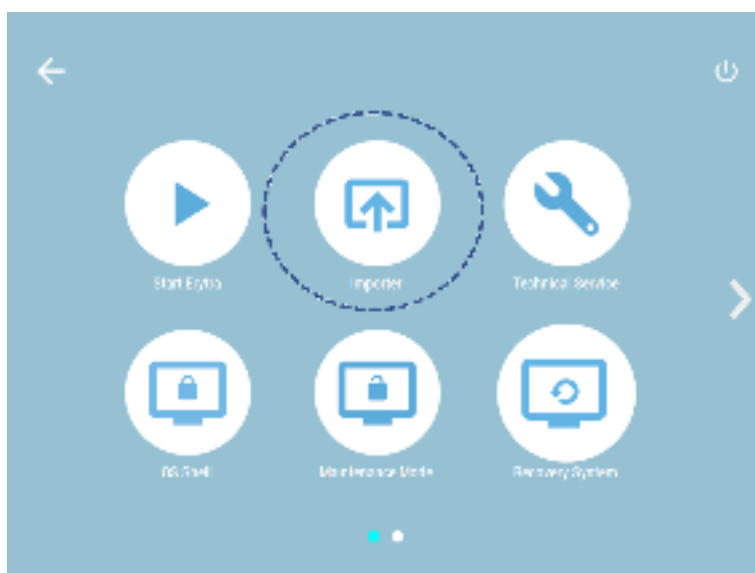


Figure 673 - Importer icon

5.2 Overview

Once the Importer application is opened, the following window appears on screen:



Figure 674 - Importer application

The Importer application has five tabs:

- **Importer:** import reagent pack and language pack files.
- **Profiles:** enable and edit profiles and create not unitary profiles.
- **QC Kits:** view and enable QC kits.
- **Special Results:** configure special results.
- **Setup:** configure setup settings.

5.3 Importer

The **Importer** tab includes functions for importing reagent pack and language pack files.



Figure 675 - Importer tab

5.3.1 Importing a reagent pack file

Reagent packs contain all the available profiles for the Erytra.

Before importing a reagent pack, do the following:

1. Open the reagent pack zip file (e.g. **IHReagentsPack_CE-9.2.0.2.zip**).
2. Double-click the **Erytra** folder.
3. Extract the Erytra reagent pack file (e.g. **ReagentPack_Erytra_CE-9.2.0.2.zip**).
4. Copy the extracted Erytra reagent pack file to **DATA (E:) > Program Files > Packs**.

To import reagent packs, do the following:

1. Click **Import Reagent Pack file**.
2. Double-click **This PC**.
3. From the **Devices and drives (1)** drop-down, double-click **DATA (E:)**.
4. Double-click the **Program Files** folder.
5. Double-click the **Packs** folder.
6. Select the reagent pack zip folder and click **Open**.

If successful, a confirmation window is displayed and the reagent pack file is imported. If unsuccessful, an error message is displayed.

5.3.2 Importing a reagent language pack file

Reagent language packs contain all text related to results, translated to the languages supported by the Importer application.

Before importing a reagent language pack, do the following:

1. Open the reagent pack zip file (e.g. **IHReagentsPack_CE-9.2.0.2.zip**).
2. Double-click the **Erytra** folder.
3. Extract the Erytra reagent language pack file (e.g. **reagentlanguagepack-1.7.0.zip**).
4. Copy the extracted Erytra reagent language pack file to **DATA (E:) > Program Files**.

To import a reagent language pack file, do the following:

1. Click **Import Reagent language Pack file**.
2. Double-click **This PC**.
3. From the **Devices and drives (1)** drop-down, double-click **DATA (E:)**.
4. Double-click the **Program Files** folder.
5. Select the reagent language pack zip folder and click **Open**.

If successful, a confirmation window is displayed and the reagent language pack file is imported. If unsuccessful, an error message is displayed.

5.3.3 Importing an Erytra language pack file

Erytra language packs contain all UI text translated to the languages supported by the Importer application.

To import an Erytra language pack file, do the following:

1. Click **Import Erytra language Pack file**.
2. Double-click **This PC**.
3. From the **Devices and drives (1)** drop-down, double-click **DATA (E:)**.
4. Double-click the **Program Files** folder.
5. Select the Erytra language pack zip folder and click **Open**.

If successful, a confirmation window is displayed and the Erytra language pack file is imported.
If unsuccessful, an error message is displayed.

5.4 Profiles

The **Profiles** tab includes functions for enabling and disabling profiles, editing profiles, and creating not unitary profiles. When the **Profiles** tab is initially selected, no profiles are visible.

The **Profiles** tab displays data in four columns:

- **Enabled:** check box to enable and disable profiles.
- **Name:** the name of the profile.
- **Displayed Name:** the profile name that is displayed in the interface. It can be edited by selecting the cell and entering a name.
- **Version:** the version of the profile that is activated.

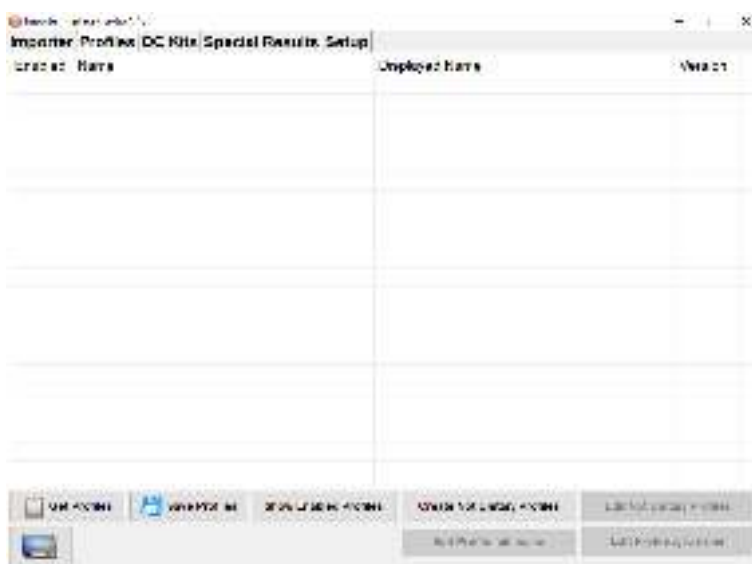


Figure 676 - Profiles tab

When **Get Profiles** is selected, available profiles for the imported reagent pack are displayed. See "Enabling profiles from an imported reagent pack" on page 887.

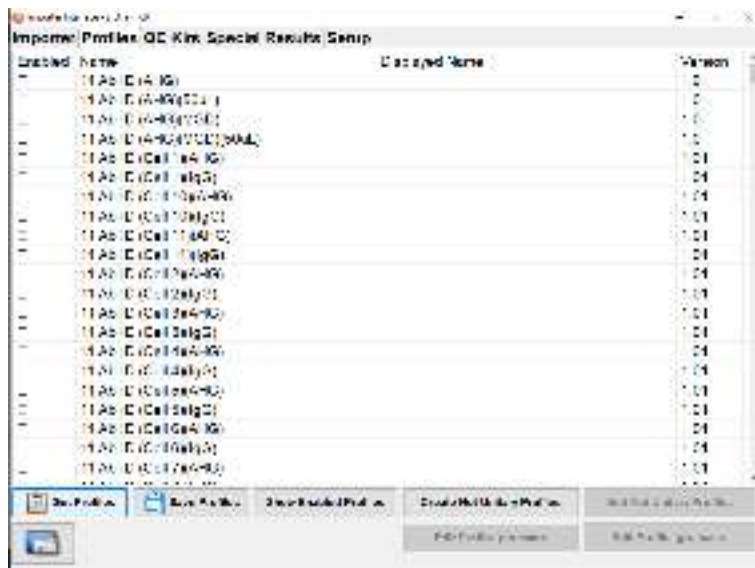


Figure 677 - Profiles tab with all profiles visible

5.4.1 Enabling profiles from an imported reagent pack

To enable profiles, do the following:

1. Import a reagent pack. See "Importing a reagent pack file" on page 885.
2. To view all available profiles, click **Get Profiles**.

After a few seconds, all available profiles in the imported reagent pack are displayed.

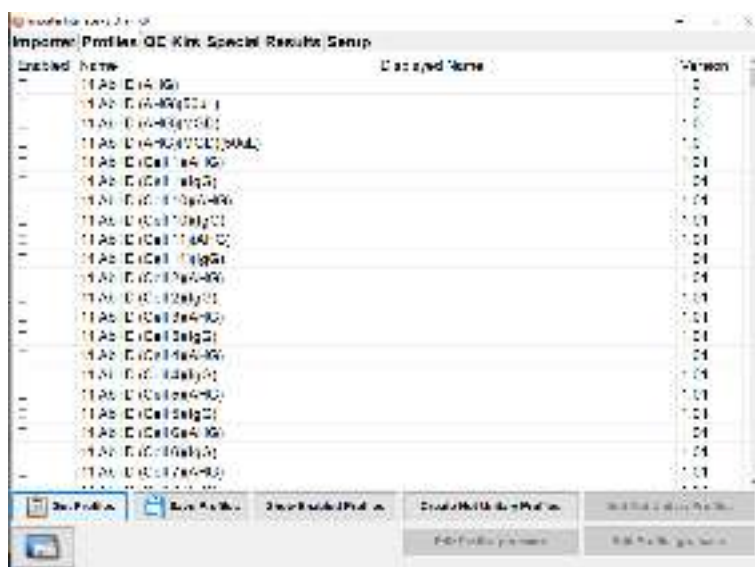


Figure 678 - List of profiles

3. Select the **Enabled** check box for one or more profiles.
4. Click **Save Profiles**.
A confirmation window is displayed.
5. Click **Yes**.
After a few seconds, the **Profiles Updated** window is displayed.
6. Click **Ok**.
7. To check if all required profiles have been correctly enabled, click **Show Enabled Profiles**.
A window with all enabled profiles is displayed.

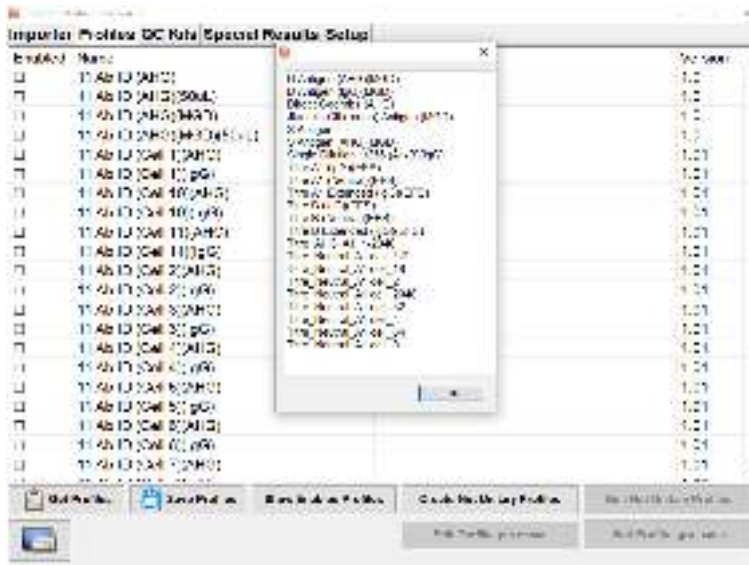


Figure 679 - Show Enabled Profiles window

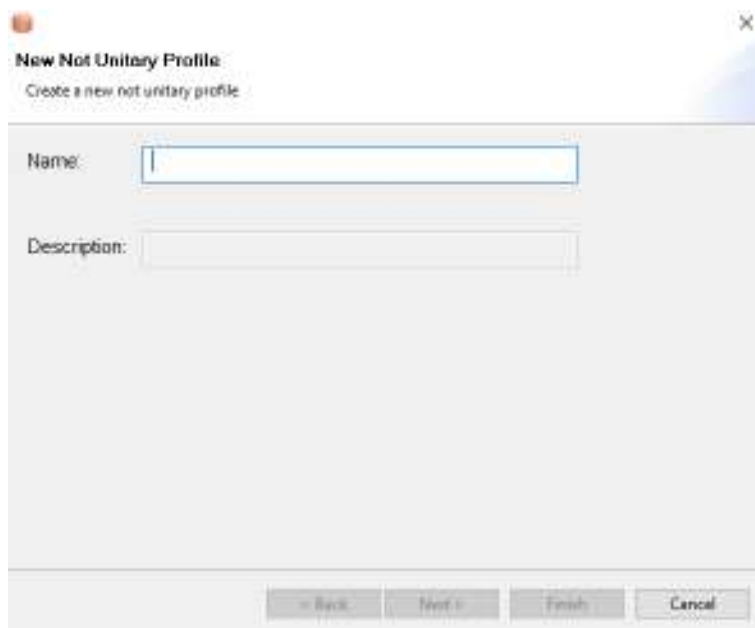
5.4.2 Creating a not unitary profile

The **Create Not Unitary Profiles** function can be used to group several profiles into a single unique profile, performing them simultaneously.

To create a new not unitary profile, do the following:

1. Click **Create Not Unitary Profiles**.

The following window is displayed:



New Not Unitary Profile
Create a new not unitary profile

Name:

Description:

< Back Next > Finish Cancel

Figure 680 - New Not Unitary Profile - name

2. In the **Name** field, enter a name for the profile.
The **Description** field is populated automatically.

3. Click **Next**.

The following window is displayed:

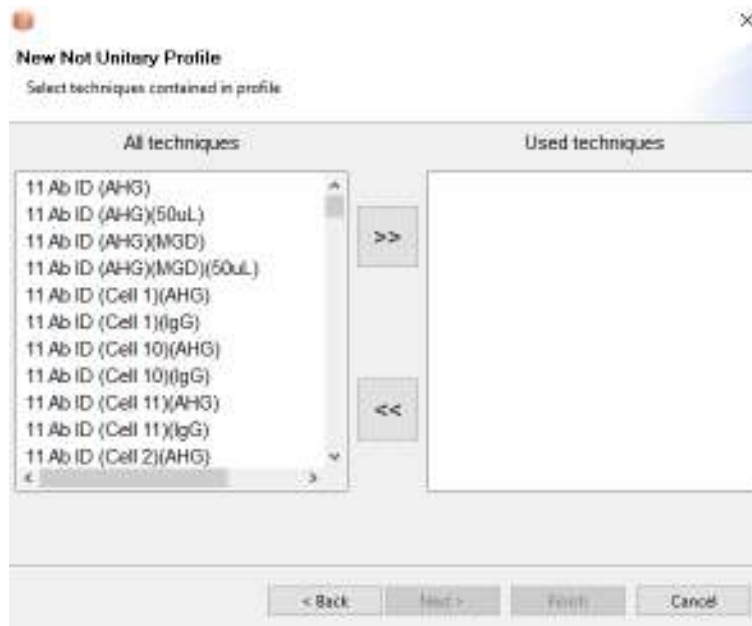


Figure 681 - New Not Unitary Profile - techniques

All available techniques are listed in the **All techniques** column.

4. Select a technique from the **All techniques** column and click the double right arrow () button.

The selected technique appears in the **Used techniques** column.

5. Click **Next**.

The following window is displayed:

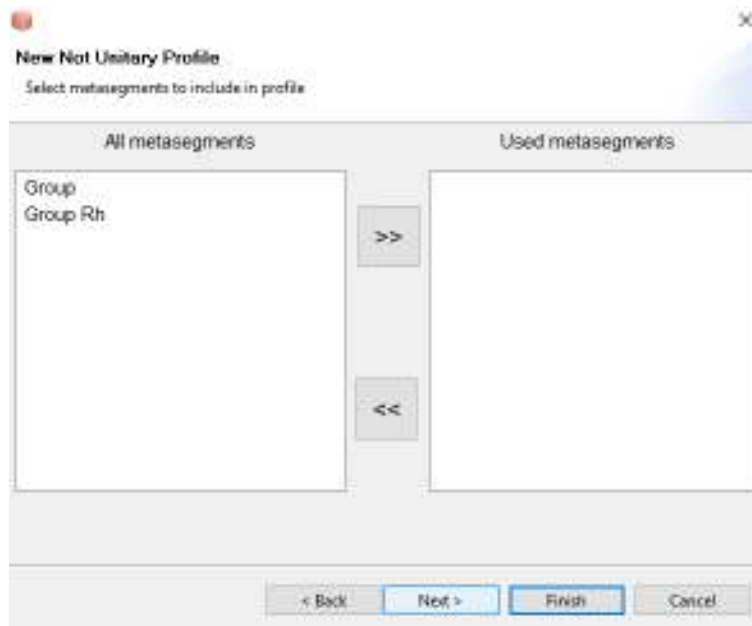


Figure 682 - New Not Unitary Profile - metasegments

All available metasegments are listed in the **All metasegments** column. Metasegments determine the way in which results are displayed in the software.

6. Select a metasegment from the **All metasegments** column and click the double right arrow () button.

The selected metasegment appears in the **Used metasegments** column.



NOTE

The Erytra already has some metasegments enabled by default. If no metasegments are selected, the results are displayed according to these default metasegments.

7. Click **Next**.

The following window is displayed:

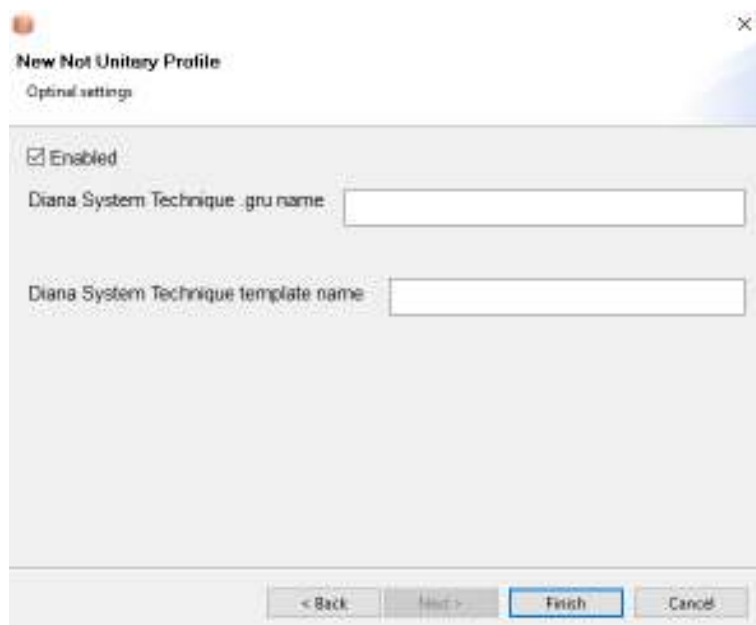


Figure 683 - New Not Unitary Profile - optional settings

8. If the instrument is connected to the LIS, do the following:

- a. In the **Diana System Technique .gru name** field, enter a name.
- b. In the **Diana System Technique template name** field, enter a name.

**NOTE**

The **Diana System Technique .gru name** field is used to set the name of the not unitary profile to be used between the Erytra and the LIS communications.

The following are examples of names that can be used for a not unitary profile, and follow the same format used for profiles:

- DGW99X.gru.
- MDW99X.gru.

**NOTE**

The **Diana System Technique template name** field is used to set the default template to be used to send results to the LIS of this not unitary profile when it has not received an order previously from the LIS (unidirectional communication). It is also used when resending results to the LIS. The field can be left empty, in which case the Erytra.pln template is used.

9. Click **Finish**.

A confirmation window is displayed.

10. Click **Yes**.
11. Click **Save Profiles**.
A confirmation window is displayed.
12. Click **Yes**.
After a few seconds, the **Profiles Updated** window is displayed.
13. Click **Ok**.
14. To check if the new not unitary profile is enabled, click **Show Enabled Profiles**.
A window with all enabled profiles, including the new not unitary profile, is displayed.

5.4.3 Editing not unitary profiles

The **Edit Not Unitary Profiles** function can be used to edit the configuration of not unitary profiles.

To edit the configuration of not unitary profiles, do the following:

1. Click **Edit Not Unitary Profiles**.

The following options can be edited:

- Description.
- Techniques.
- Metasegments.
- Diana System Technique .gru name.
- Diana System Technique template name.

5.5 QC kits

The **QC Kits** tab includes functions for enabling and disabling QC kits. When the **QC Kits** tab is initially selected, no QC kits are visible.

The **QC Kits** tab displays data in three columns:

- **Enabled**: check box to enable and disable QC kits, and the identification number of the QC kit.
- **Name**: the name of the QC kit.
- **Description**: the description of the QC kit.

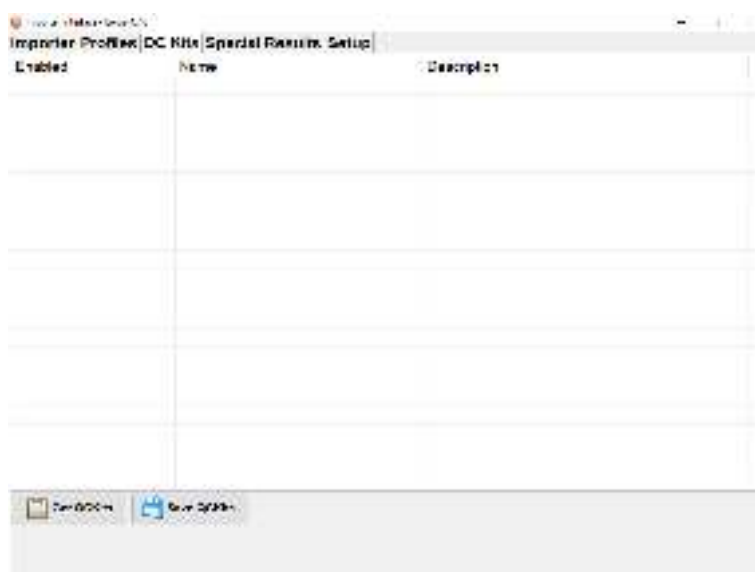


Figure 684 - QC Kits tab

When **Get QCKits** is selected, available QC kits for the imported reagent pack are displayed. See "Enabling QC kits from an imported reagent pack" on page 894.

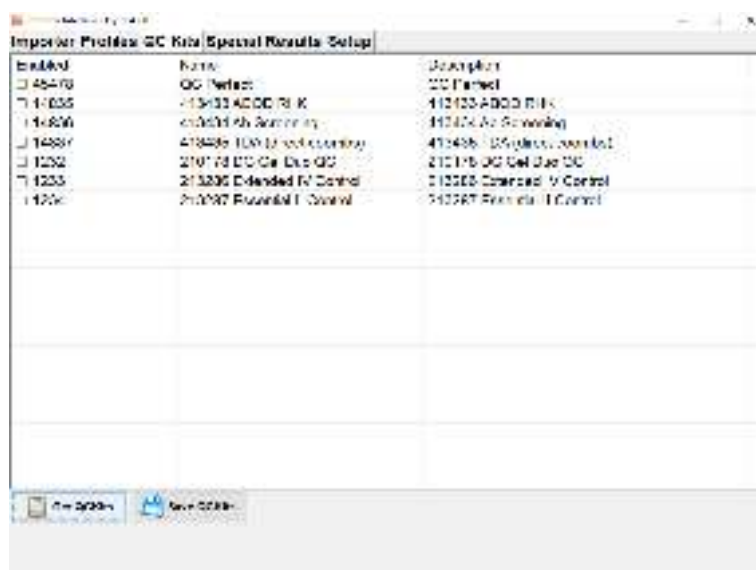


Figure 685 - QC Kits tab with all QC kits visible

5.5.1 Enabling QC kits from an imported reagent pack

To enable QC kits, do the following:

1. Import a reagent pack. See "Importing a reagent pack file" on page 885.
2. To view all available QC kits, click **Get QCKits**.

All available QC kits in the imported reagent pack are displayed.

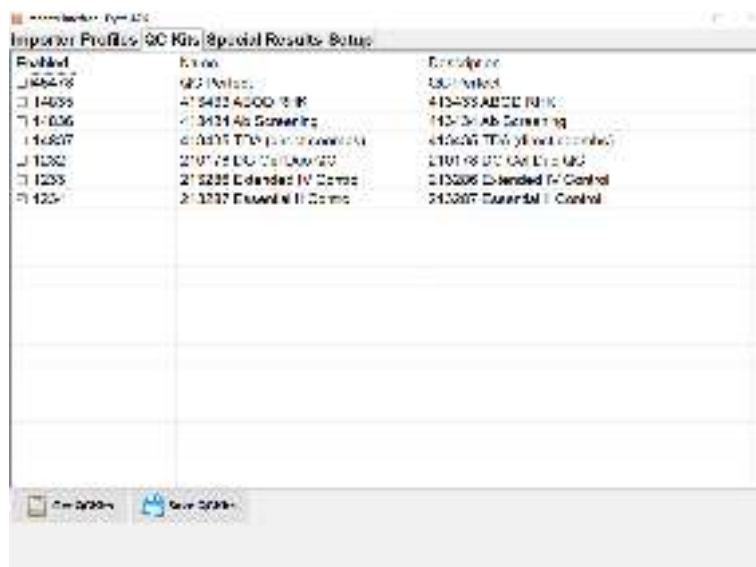


Figure 686 - List of QC kits with an enabled kit

3. Select the **Enabled** check box for one or more QC kits.
4. Click **Save QCKits**.
A confirmation window is displayed.
5. Click **Yes**.
After a few seconds, the **QC Kits Updated** window is displayed.
6. Click **Ok**.

5.6 Special results

The **Special Results** tab includes functions for configuring results of specific profiles as special results. By default, the system has different results that are highlighted as special results, such as double population and doubtful. The options in the **Special Results** tab can be configured so that the system highlights results of a specific profile depending on the configuration of the **Results** section. It is not possible to validate the results until the special result has been accepted. When the **Special Results** tab is initially selected, all sections are blank.



Figure 687 - Special Results tab

When **Get Profiles** is selected, available profiles that can be used to configure special results are displayed. See "Configuring special results that are not set by default" on page 896.

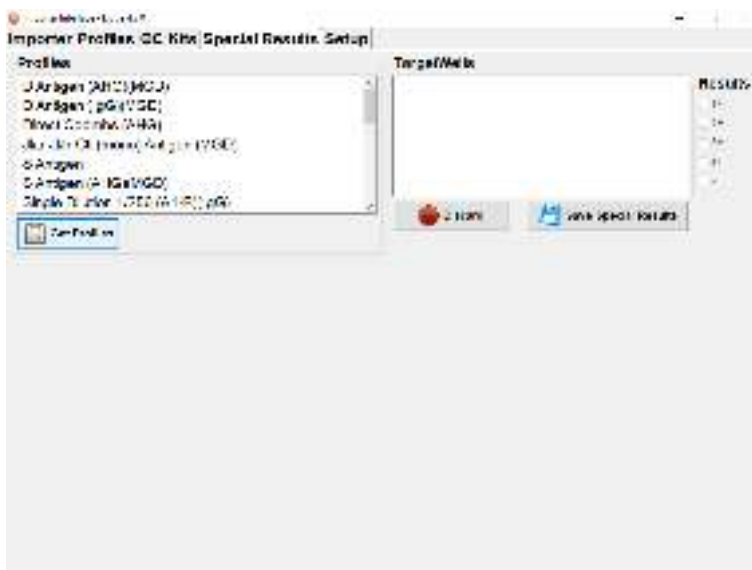


Figure 688 - Special Results tab with all profiles visible

5.6.1 Configuring special results that are not set by default

To configure special results for a specific profile, do the following:

1. Import a reagent pack. See "Importing a reagent pack file" on page 885.
2. To view all available profiles, click **Get Profiles**.

All available profiles that can be used to configure special results are displayed.

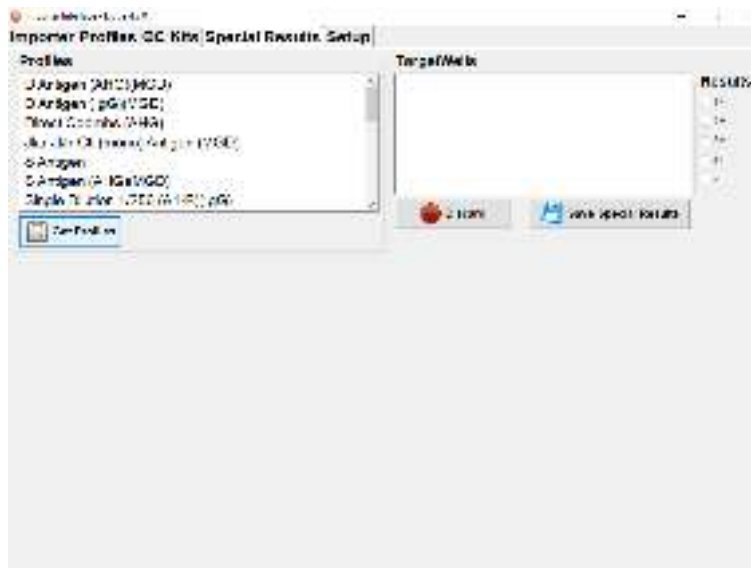


Figure 689 - List of usable profiles

3. From the **Profiles** section, select a profile.

The **TargetWells** section is populated.

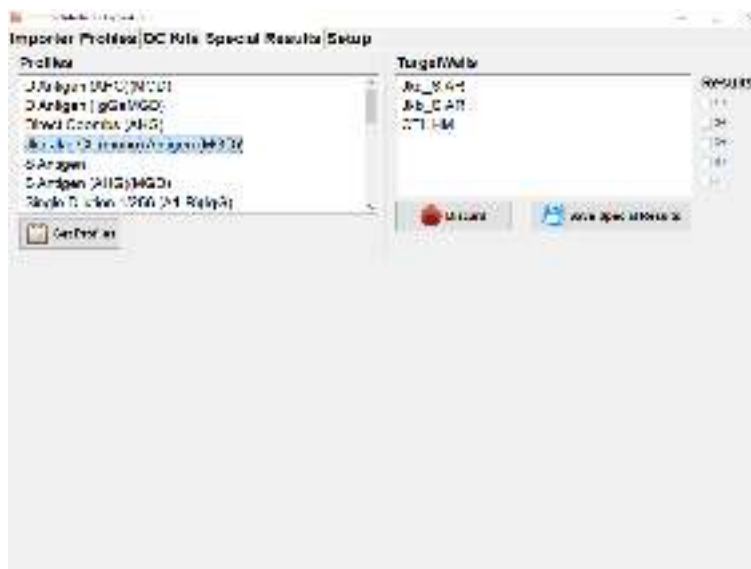


Figure 690 - Profile selected

4. In the **TargetWells** section, select an entry.

- From the **Results** section, select the reading results to be shown as special results for the selected profile.

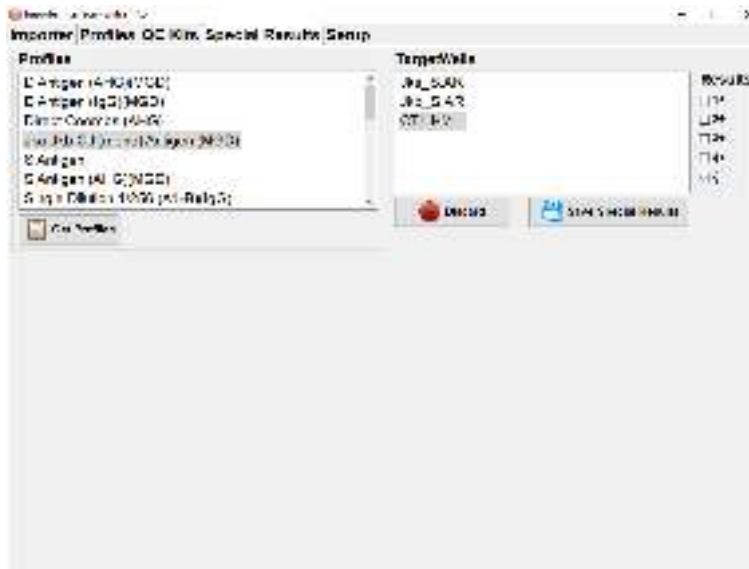


Figure 691 - Results configured

- Click **Save Special Results**.

A confirmation window is displayed.

- Click **Yes**.

After a few seconds, the **Special Results Updated** window is displayed.

- Click **Ok**.

After saving, an icon is displayed next to the related profile in the **Profiles** section and the entry in the **TargetWells** section indicating this configuration.

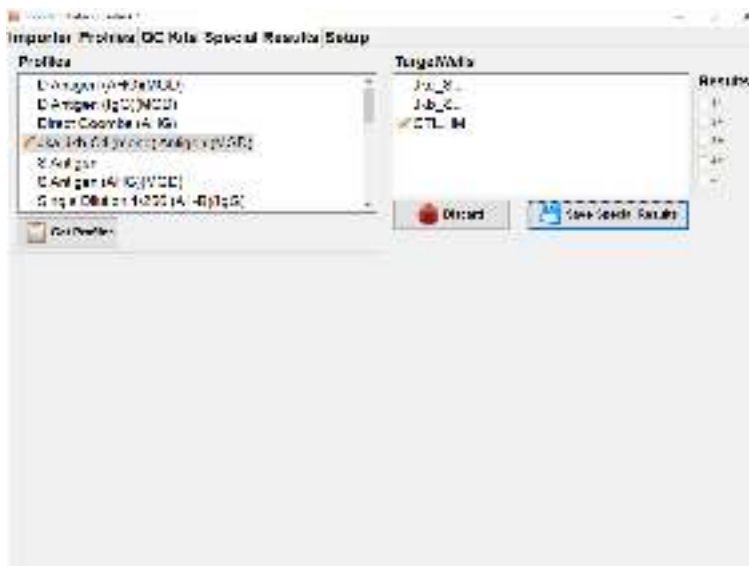


Figure 692 - Special results icon

5.7 Setup

The **Setup** tab includes functions for configuring the Erytra settings.

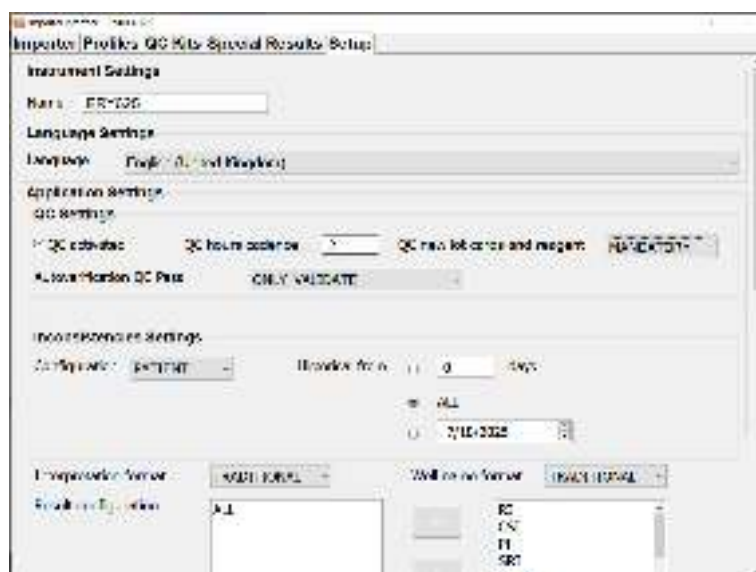


Figure 693 - Setup tab

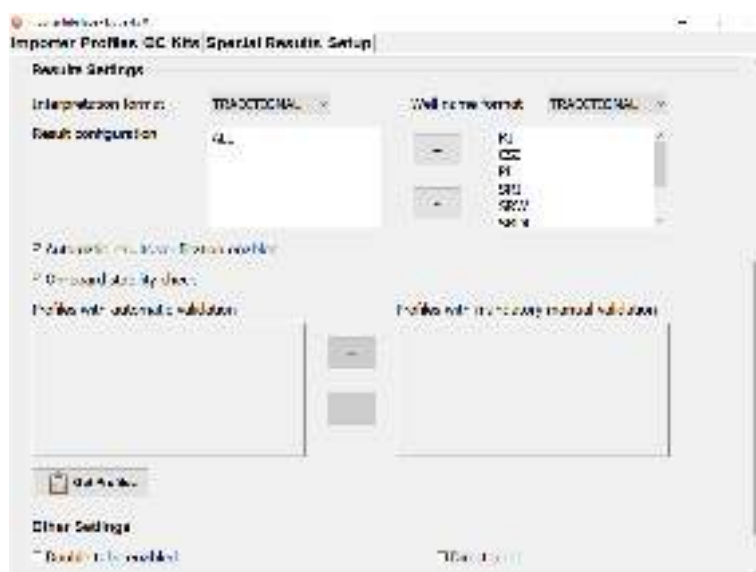


Figure 694 - Setup tab continued

5.7.1 Configuring instrument settings

The **Instrument Settings** section includes one text field:

- **Name:** name of the instrument, displayed on the upper left side of the UI.

To configure the instrument settings, do the following:

1. To change the name of the instrument, click the **Name** field and enter a name.
2. Click **Save Setup**.
A confirmation window is displayed.
3. Click **OK**.
After a few seconds, the **Setup Updated** window is displayed.
4. Click **OK**.

5.7.2 Changing the language

To change the instrument language, do the following:

1. Under **Language Settings**, click the **Language** drop-down.
A list of available languages is displayed.
2. Select a language.
3. Click **Save Setup**.
A confirmation window is displayed.
4. Click **OK**.
After a few seconds, the **Setup Updated** window is displayed.
5. Click **OK**.

5.7.3 Configuring QC settings

The **QC Settings** section includes the following options:

- **QC activated**: check box to activate or deactivate QC.
- **QC hours cadence**: how often the QC must be processed in hours.
- **Autoverification QC Pass**: type of QC pass auto-verification:
 - **NONE**: QC must be validated and exported manually by the operator.
 - **ONLY_VALIDATE**: if QC passes then QC results are validated automatically but must still be exported manually by the operator.
 - **VALIDATE_AND_EXPORT**: if QC passes then the QC results are validated and exported automatically.

- **QC new lot cards and reagent:** Cards and Reagents QC management options:
 - **False**(default option): if this option is configured, the system will not check if a QC has been passed.
 - **Warning:** the system will check if a QC has been passed and show a warning in case it has not.
 - **Mandatory:** the system will check if a QC has been passed. In case it has not, the system will show a warning and will not use that lot of cards or reagents until it has been used in a QC test with a pass result.

To activate QC, do the following:

1. Under **QC Settings**, click the **QC activated** check box.
2. In the **QC hours cadence** field, enter a value.
3. From the **Autoverification QC Pass** drop-down, select an option.
4. From the **QC new lot cards and reagent** drop-dwn, select an option.
5. Click **Save Setup**.
A confirmation window is displayed.
6. Click **OK**.
After a few seconds, the **Setup Updated** window is displayed.
7. Click **OK**.

5.7.4 Configuring inconsistencies settings

The **Inconsistences Settings** section includes options to enable and configure the delta check in order to compare results with historical data by using either Patient ID or Sample ID.

The **Inconsistences Settings** section includes the following options:

- **Configuration:** the variable used to make the comparison:
 - **NONE:** no comparison is made.
 - **PATIENT:** comparison made using Patient ID.
 - **SAMPLE:** comparison made using Sample ID.
- **Historical from:** the time frame of the comparison from which the results are taken. The following three time formats are available:
 - An operator defined number of days.
 - **ALL:** all database results.
 - A specific date.

To enable the delta check, do the following:

1. From the **Configuration** drop-down, select an option.
2. In the **Historical from** section, select an option and configure the settings.

3. Click **Save Setup**.

A confirmation window is displayed.

4. Click **OK**.

After a few seconds, the **Setup Updated** window is displayed.

5. Click **OK**.

5.7.5 Configuring results settings

The **Results Settings** section includes the following options:

- **Interpretation format:** the format used for the interpretation:
 - **TRADITIONAL.**
 - **ISBT.**



NOTE

The ISBT format can be selected, however not all interpretations have it defined in the system. If this format is not defined for a specific interpretation, the TRADITIONAL format is shown.

- **EXTENDED.**



NOTE

The EXTENDED format is not currently defined in the system. If selected, the TRADITIONAL format is shown.

- **Well name format:** the format used for well names:
 - **TRADITIONAL.**
 - **ISBT.**



NOTE

The ISBT format can be selected, however not all well names have it defined in the system. If this format is not defined for a specific well, the TRADITIONAL format is shown.

- **Result configuration:** option to configure that every time a special result appears, the card containing the result can either be kept in the service rack or not.

To configure which special results are taken in to consideration to place the related card in the service rack, the following acronyms are used:

- SRW: special results that are **weak** (+/-).
- SRDL: special results that are **doubtful** (?).
- SRDP: special results that are **double population** (mixed field).
- SROD: special results **defined by the operator**. See "Special results" on page 895.
- SRI: **not interpretable** results and **discrepant interpretation**.
- RI: results with **reading incidences**.
- PI: results with **process incidences**.
- CSI: **card or sample incidences**.
- All: **all the processed cards** are placed in the service rack.
- **Automatic results verification enabled:** check box to enable or disable automatic results verification. Except for any profiles selected for mandatory manual validation, if the results for the routine samples do not include any errors or require any special results to be accepted, the results are validated and exported automatically.
- **On-board stability check:** check box to enable or disable the time tracking of reagents in the instrument.
- **Profiles with automatic validation / Profiles with mandatory manual validation:** if the **Automatic results verification enabled** check box is selected, this section allows the operator to select any configured profiles to be validated manually despite this.

To configure the results settings, do the following:

1. From the **Interpretation format** drop-down, select an option.
2. From the **Well name format** drop-down, select an option.
3. In the **Result configuration** section, do one of the following:
 - To add a special result, select it in the left list and click the plus () button.
 - To remove a special result, select it in the right list and click the minus () button.
4. To enable automatic results verification, click the **Automatic results verification enabled** check box.
5. To enable the time tracking of reagents in the instrument, click the **On-board stability check** check box.
6. Under the **Profiles with automatic validation** section, click **Get Profiles**.
All configured profiles are displayed.

7. To configure the validation behaviour of a profile, do one of the following:
 - If a profile requires manual validation, select it in the **Profiles with automatic validation** list and click the plus () button.
 - If a profile does not require manual validation but it is in the **Profiles with mandatory manual validation** list, select it and click the minus () button.
8. Click **Save Setup**.
A confirmation window is displayed.
9. Click **OK**.
After a few seconds, the **Setup Updated** window is displayed.
10. Click **OK**.

5.7.6 Configuring other settings

The **Other Settings** section includes the following options:

- **Double tube enabled**: check box to enable or disable double tube. Double tube must be configured when plasma and red cells are placed in different tubes.
- **Direct print**: check box to enable or disable direct print. When the print button is selected anywhere in the Erytra software application, the system directly prints the report without displaying a preview.

5.7.7 Password policies

The Password policies section includes the following parameters that will be stored in erytra.properties file:

- Minimum password length (PASSWORD_MINIMUM_LENGTH): password minimum length could from 4 to 20 characters.
- Minimum uppercase characters included (PASSWORD_MINIMUM_UPPERCASE): minimum upper-case characters from 0 to 5.
- Minimum lowercase characters included (PASSWORD_MINIMUM_LOWERCASE): minimum lower-case characters from 0 to 5.
- Minimum special characters included (PASSWORD_MINIMUM_SPECIAL_CHARACTERS): Minimum special characters from 0 to 5.
- Minimum numbers included (PASSWORD_MINIMUM_NUMBERS): minimum number of numbers from 0 to 5.
- Avoid previous passwords (PASSWORD_AVOID_PREVIOUS): number of previous passwords to be remembered in order to check that the new password is not repeated within the previous ones from 0 to 5.

- Password expiration days (PASSWORD_EXPIRATION_DAYS): number of days before a password will expire and a new one has to be set.

6 Mechanical drawings

This chapter describes the available options when viewing the digital drawings representing all the modules of the Erytra instrument, including 3D views and animated assembly procedures that are performed in factory for each of them. As detailed in this section, a free software is required to view the drawings and assembly procedures, which are contained in **.pvz** files. These **.pvz** files are located in "Related documents" on page 19.

6.1 Requirements

- 3D drawings (**.pvz** files). See "Related documents" on page 19.

The file name format of the 3D drawings is **MxxxxxxxVn_myy.pvz**, where xxxxxxx is the module identification number, n is a number indicating the drawing version, and yy is the modification number for the corresponding module.

- 2D drawings. See "Related documents" on page 19.
- Spare parts list. See "Spare parts list" on page 1207.

The raw material codes composing a factory module can be found in the 3D drawings. From there, they can be matched to the codes composing an existing spare part in order to know the corresponding spare part reference.

- 3D viewer.

Creo View Express is the software used to view the available 3D drawings and the corresponding assembly procedures. It can be downloaded and installed (see "Creo View Express installation" on page 906) and configured (see "Creo View configuration" on page 907). Version 8.0 or higher is needed to view the drawings.

6.2 Drawings reference table

For a list of drawing references, see "Related documents" on page 19.

6.3 Creo View Express installation

Information about how you can install Creo View Express within Grifols and from external distributors.

6.3.1 Within Grifols company

Install Creo View Express from the Grifols Software Center.

6.3.2 External distributors

Look for Creo View Express in <https://www.ptc.com> (free viewer) and follow the downloading and installation instructions.

6.4 Creo View configuration

Once a drawing (.pvz file) is opened using Creo View Express, the main structure of that module can be displayed by selecting the **Structure** tab on the left panel of the screen, then double-clicking the main structure code contained in the **Ilustración** field.



NOTE

Some drawings may contain more than one main structure, so they should be opened individually.

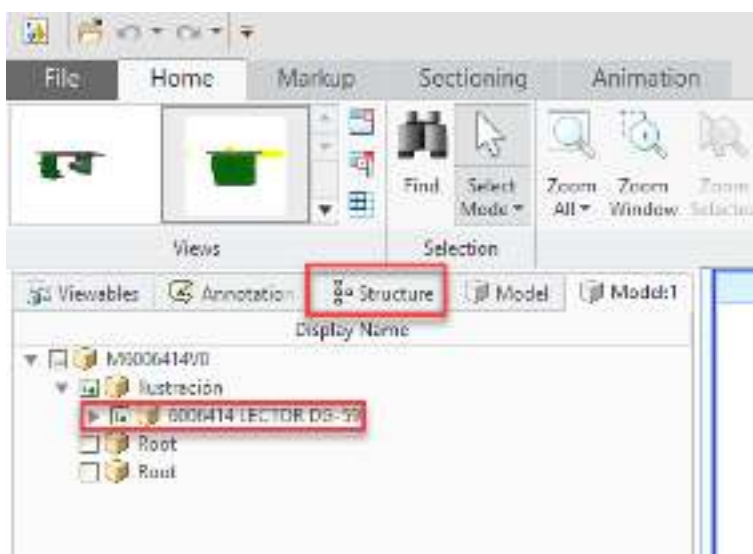


Figure 695 - Main structure code

A new tab named **Model** will be created, and the 3D drawing of the selected module will be shown on screen. At this point, several drawing techniques can be used to represent the model by selecting the **Render Mode** button in the Creo View Express toolbar (the shaded view is selected by default).

When selecting any of the parts composing the drawing, the corresponding component will be highlighted in the navigation tree on the left side of the screen. In order to see the raw material description and its code (factory reference) instead of the default values, the following configuration should be applied:

**NOTE**

This configuration is required if raw material codes need to be searched within the module.

1. Go to **File > Options**.
2. Go to **General > Navigation** window.
3. Enable the **Use alternate path names** checkbox and click the **Configure** button.
4. Click the **Insert Attribute** button.
5. Select **CODIGO** and click the **Insert value** button.
6. Select **DENOMINACION** and click the **Insert value** button.
7. Consider applying a whitespace between both values in the **Configure Alternate Part Names** window.

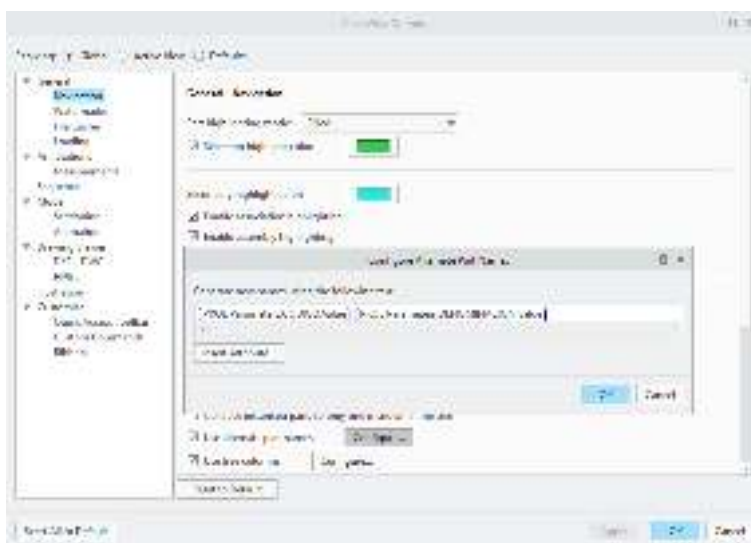


Figure 696 - Creo View options

8. Click the **Close** button, then **OK**.
9. Click the **Apply** button, then **OK**.

10. Now the raw material description and its factory reference can be seen in the navigation tree for each part composing the drawing. This reference can also be found when searching for the **CODIGO** field. See "Spare parts searching" on page 913.

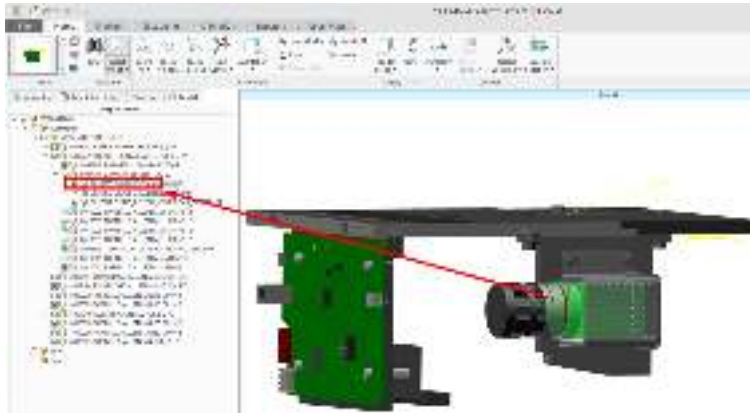


Figure 697 - Highlighted part description after selecting a drawing part

6.5 Models structure and navigation

Information about tree structure and models navigation.

6.5.1 Tree structure

1. Click the **Structure** tab and double-click the main structure code contained in the **Ilustración** field.

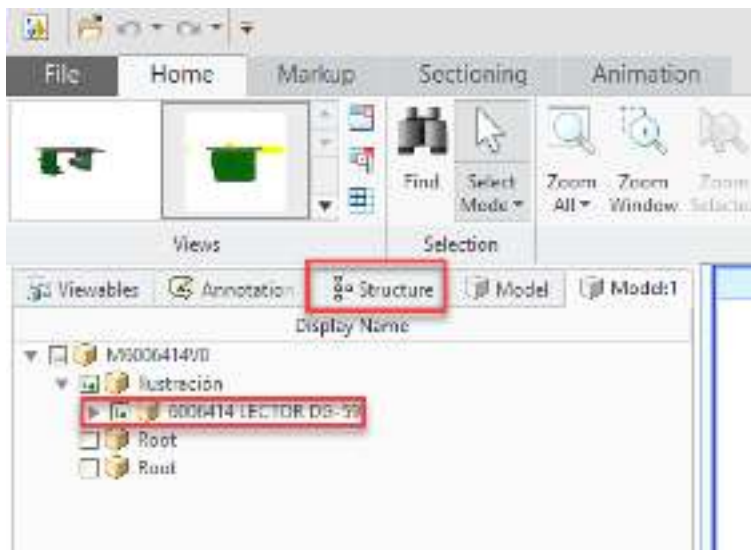


Figure 698 - Main structure code

2. A new tab named **Model** will be created, and the main model of the selected module will be shown on screen. The tree structure (starting on the main structure code) can be expanded or collapsed by clicking on the triangles next to each field.
3. When double-clicking in any field of the tree, the corresponding component in the drawing is zoomed in and changes its colour to blue (subassembly containing raw materials) or green (raw materials).

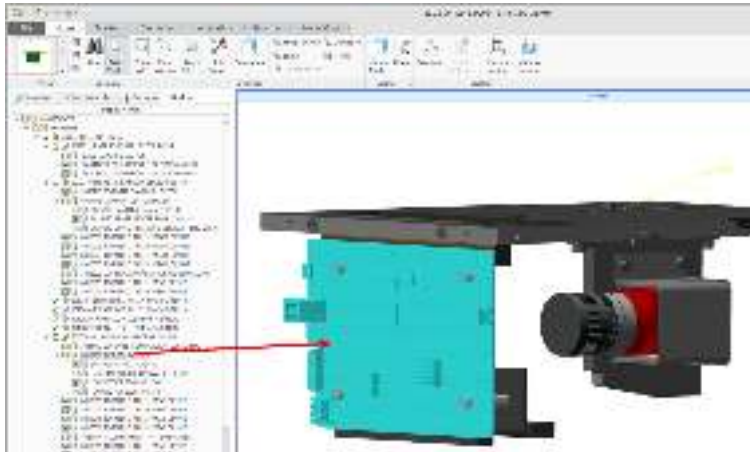


Figure 699 - Tree structure component selection

4. Parts can be hidden or isolated by right-clicking them and selecting the desired action, and then re-established by right-clicking anywhere within the drawing area and selecting the **Unhide All** button. They can also be hidden or unhidden by enabling or disabling the check square near each component in the tree structure.

6.5.2 Models navigation

Once the drawing is displayed on screen, it can be rotated, dragged, zoomed in or out, etc. by using the below listed functionalities for Creo View Express:

- Zoom in/out (pointer area taken as reference point):
 - Mouse: rotate the mouse wheel.
 - Pad: put two fingers on the pad and move them up or down.
- Rotate model:
 - Mouse: press and hold the right button and drag mouse.
 - Pad:
 - Select the **Free Rotation** button in the Creo View Express toolbar.
 - Select the **Ilustración** field in the tree structure, or just select a part of the drawing.
 - Hold left click on the sphere while dragging on the pad.

- Drag model:
 - Mouse: press the wheel and drag mouse.
 - Pad:
 - Select the **Transform** button in the Creo View Express toolbar.
 - Select the **Ilustración** field in the tree structure, or just select a part of the drawing.
 - Hold left click on a sphere's center grey circle while dragging on the pad.

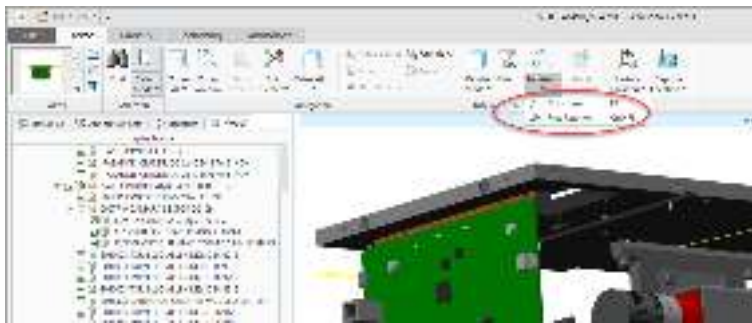


Figure 700 - Transform options in the Creo View Express toolbar

- Hide/Unhide part:
 - Select the part by right-clicking it and select **Hide** or **Unhide All**.
 - Select the part by clicking it and enable or disable it in the tree structure.
- Center model on screen: double-click the background.
- Restore location: click the **Restore location** button in the Creo View Express toolbar. With this, any part that has been moved or rotated from its default position within the drawing area will restore its original position.

6.6 Animated assembly procedures

The **Viewables** tab contains a list of animated assembly procedures detailing the assembly steps as done in factory for each of the modules composing the Erytra instrument. Each component being assembled in an animation is identified with its raw material code.

1. When opening a .pvz file for a specific module, the **Viewables** tab is selected by default, showing the corresponding animated assembly procedures (the **Figures** area).

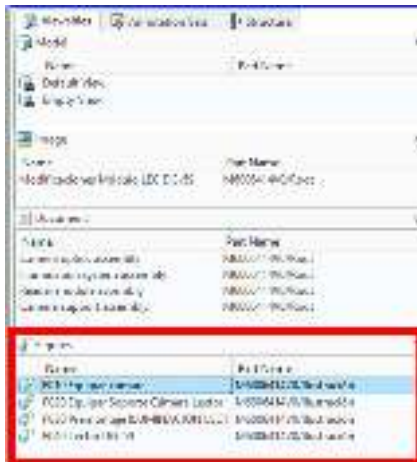


Figure 701 - Figures area

2. Double-click the desired assembly and a new tab will be created, showing the tree structure with the parts involved along with the corresponding animated assembly.
3. The animation can be played and paused for each of the steps composing it by using the **Playback** control buttons at the bottom left corner of the animation screen, or those at the **Play** section in the Creo View Express toolbar.

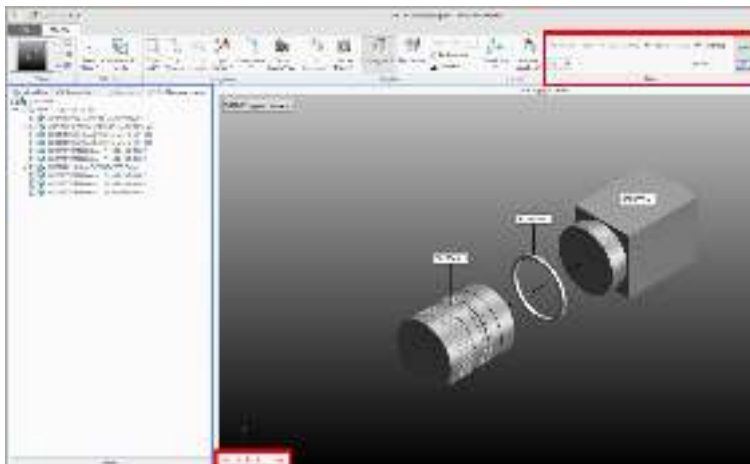


Figure 702 - Playback control buttons for an animated assembly procedure

4. Parts can be hidden or unhidden during the animation by enabling or disabling the square near each component in the tree structure or selecting the corresponding option after right-clicking on it.
5. The background colours can also be modified as desired by editing the **Background** and **Gradient** options of the **Display** section in the Creo View Express toolbar.

6.7 Spare parts searching

The raw codes composing a factory module can be found in the 3D drawings. From there, they can be matched to the codes composing an existing spare part in order to know the corresponding spare part reference.



NOTE

Before searching for a specific raw material code, make sure that the configuration has been applied. See "Creo View configuration" on page 907.

6.7.1 Finding a drawing part for a spare part list's reference code

1. Copy the desired drawing reference code from the Spare Parts List **Drawing reference** column. See "Spare parts list" on page 1207.
2. Click the **Find** button in the Creo View Express toolbar.
3. Paste the previously copied code in the new window and click **Find next**.
4. The raw material corresponding to the selected drawing reference will be highlighted both in the tree structure and in the model drawing.



Figure 703 - Finding a drawing part from a known code

6.7.2 Finding the spare part list's reference code for a drawing part

1. Select the part of the drawing for which the raw material code needs to be known.
2. The part will be selected in the tree structure. If the Creo view configuration has been applied (see "Creo View configuration" on page 907), the raw material code (**CODIGO** field) will be listed as part of the material description. The raw material code (**CODIGO** field) can also be seen by enabling the lower data panel option on the lower left side of the screen.

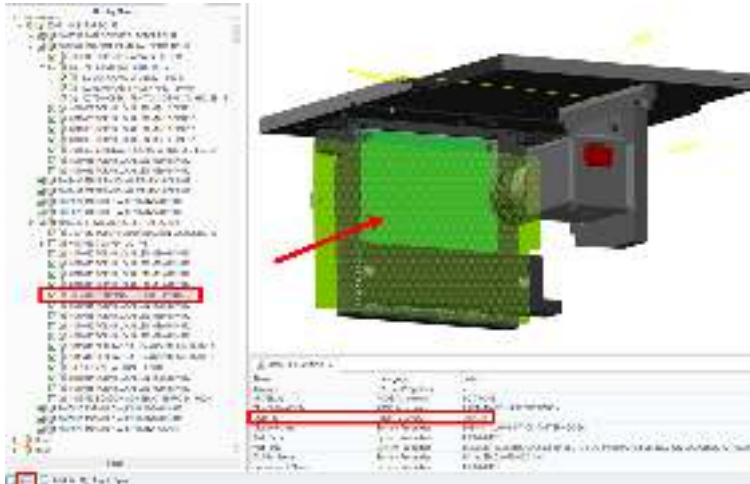


Figure 704 - Find a code by selecting a drawing part

3. Search for the obtained raw material code in the Erytra spare parts list (see "Spare parts list" on page 1207). The code will be listed in the **Drawing Reference** column for one or more spare part codes, so the desired reference can be obtained. If the raw material code cannot be found in the spare parts list:
 - a. If the raw material code is part of a set of codes included in a parent code on top of it in the tree structure, try to look for the parent code in the spare parts list. The parent code of a specific part can be automatically selected by selecting it and then clicking the **Select Parent** button that can be found in the **Structure** tab in the Creo View Express toolbar.
 - b. If the raw material code is not part of a set of codes included in a parent code, or if the parent code cannot be found in the spare parts list, it means that no part number reference has been assigned to the desired raw material code.

7 Software information

This chapter provides information on several procedures such as the Operating System and software installation, update the firmware of the PCAs, touchscreen drivers installation, LIS configuration, a brief description on how to program macros and a description of the log files.

7.1 Software version changes and components versions

This chapter describes the changes that have been made to new Erytra Software versions releases.

7.1.1 Erytra Software v4.5.1

7.1.1.1 Software changes

This chapter describes Erytra Software version 4.5.1 improvements and modifications compared to version 4.5.0

General

Resolved anomalies

1. A default interpretation of results was assigned to an execution that had been automatically cancelled and had not been processed (results were empty), but was not erased from the system due to a concurrency of events when the corresponding automatic re-queueing of the technique took place at the same time . Now a 'NI' interpretation will be assigned to any execution when interpreting empty results.

Complaint 800020269

Alternative browser

The alternative browser has not been modified and remains the same as the version delivered with software version 4.5.0. It is not required for use with the Erytra Database Manager. However, remote connections require a web browser with Flash Player. In cases where remote access is not possible due to the version or configuration of the external computer browser, the alternative browser should be installed to ensure compatibility.

Manager, Controller and User databases and Erytra Database Manager

The Erytra Database Manager has not been modified and remains the same as the version delivered with software version 4.5.1.

This version cannot be installed if a previous version of the Erytra Database Manager is already present on the system. In such cases, the existing version must be uninstalled beforehand, and a database backup should be created if necessary.

Databases created with software version 4.5.1 must be restored using Erytra Database Manager version 4.5.1. Compatibility with earlier versions of the Erytra Database Manager or software is not guaranteed.

Databases from previous software versions must be updated before use. This can be done by running the **updatedb.bat** file located in **E:\Program Files\Liquibase** when restoring in an Erytra instrument, or by executing **updatedb_edm.bat** located in **installation_directory\Grifols\Manager\Program Files\Liquibase** **updatedb_edm.bat** when restoring in the Erytra Database Manager.

7.1.1.2 Application and firmware versions

Table 102 - Erytra v4.5.1 components versions

Name	Version	Description
Apache Tomcat	8.5.86	HTTP web server
Boot Application	1.2.0	Grifols Applications launcher and Database Backup/Restore for Technical Service purposes
DG-59-Backup	1.3.0	Backup tool
DG-59-Restore	1.3.0	Restore tool
Diana System CE	4.6.0.38	Diana System build from v4.6.0 DG-57 project. (Configuration= CE)
Diana System US	2.4.0.19	Diana System build from v2.4.0 DG-517 project. (Configuration= US)
Director.dll	5.10.0	Image processing library
DSC.dll (CE)	6.2.0	Diana System Connectivity. (Configuration= CE)
DSC.dll (US)	6.2.0	Diana System Connectivity. (Configuration= US)
Erytra-databasemanager-4.5.1	4.5.1	Erytra Database Manager
Erytra Diagnostic	4.7.0	Instrument check tool
ErytraSupplyOnOffSwitcher.exe	1.1.0	ErytraSupplyOnOffSwitcher

Name	Version	Description
Importer	4.9.1	Languages and techniques import tool
languagepack.zip	1.27.0	Language pack file
PostgreSQL	10.23	Object-relational database management system
reagentlanguagepack	1.0.0.11	Reagent language pack file
Technical Service	4.9.1	Tool for instrument adjusting

Table 103 - Erytra v4.5.1 FW versions

PCA	Firmware version
CEN_SRM_CTRL	4.5.0
INC_CUP_CTRL	4.5.0
FLUCTRL59	4.6.2
CAT_CM_CTRL	4.5.0
PIP_CTRL	4.10.0
RD_IL421	5.5.0

7.1.2 Erytra Software v4.6.0

7.1.2.1 Software changes

This chapter describes Erytra .Software version 4.6.0 improvements and modifications compared to version 4.5.0

General

New functionalities

- Language pack has been updated (v1.30.0) to include:
 - Turkish (Turkey) is now available.
 - Several messages have been added, reviewed, and improved.

2. The information included in testcounter.txt when restoring an application database with IHImport/Restore application has been improved to accurately compute the number of tests executed by the instrument and determine the Preventive Maintenance category.

Now the offset between the number of tests in previous database and the restored database, and the restoration time stamp will be included:

```
325
restored db. 2024-03-06 08:36:25
Offset: 225
100
```

If testcounter.txt file is not found when restoring the database, it will be created.

The number of tests of the current application database will be searched in testcounter.txt file. If it is not found, it will be searched in the current database and, if no database is found, it will be assumed to be 0.

Databases restoration must be performed with IHImport/Restore application. The old restore application Restore_DataBase.exe (and the folder containing this file, E:\Program Files\ErytraDB Apps) has been removed. The software installer will not copy this file nor create this directory, and delete the directory and its contents if found.

Procedures:

- See "Restoring from Boot application" section (modified)

3. The Support Pack will now include:

- Erytra Diagnostic logs.
- Technical Service logs.
- AppLocker rules.
- The logs in the archive path belonging to the selected time frame.
- Windows users, services, and Application and System events logs with the events belonging to the selected time frame.

Complaint 800010590.

Procedures:

- See "Generating Support Pack from user software" section (modified)

Erytra User Interface

New functionalities

1. The titration profiles with serial serum/plasma dilution have been improved:

- The homogenization of patient serum/plasma dilution has been improved in titration profiles with serial serum/plasma dilution. The system, in this case, will now aspirate and dispense dilution cup contents to improve homogenization.
- A new specific probe internal wash has been created (new DP_WASH_ST in Layout.xml file) to reduce dilutor speed (top_speed from 5000 to 3500) and assure proper dripping detection during washings.
- The steps and volumes used to obtain the dilution have been improved, especially for single point profiles.
- The time elapsed between the reagent red blood cells and the serum/plasma dilution dispensation will now be checked when the profile executed is completed (process check). The message posted in the erytraDebug log files in this step has been modified from

ProcessValidatorChecks - Checking serum dispensation step time duration.
to

ProcessValidatorChecks - Checking serum or serum dilution dispensation step time duration.

2. Air aspiration detection has been improved. The system will execute the air aspiration check in all sample/plasma aspiration but the result of that checking is only considered in sample serum/plasma aspirations of a serial titration profiles. Air aspired detection threshold has been adjusted when a serial titration technique is executed.
3. The process executed when the instrument switches from one waste container to the other that checks that the new selected waste container is properly connected and can be used has been improved.

The dilution cup aspiration line will not be opened, and the system will check that solution B can reach the waste container as in the fast prime of the solution B container process.

4. Priming process has been improved at the end of prime process associated to Dilution Cup. During Prime High the following action is performed:

PRIME_HIGH_DC_9 (this action keeps the EV2 and EV7 4 seconds opened)

This action will improve the priming of the instrument and prevent the presence of bubbles in the syringes.

5. It's been detected that AdobeAir logs could contain sensitive patient demographic data. All logs relative to channelMessage events have been deleted from the logs.

Complaint: 800013755

6. The user password requirements have been improved to increase the password security level of the password set and to allow the configuration of the requirements.

The new requirements will be applied when the current password is changed. The configured password requirements will be shown at the Change Password window.

When the new password does not meet the configured password requirement, the message Password requirements not met will be displayed:

CHANGE PASSWORD

Username ADMIN

Current password *****

New password

Confirm Password

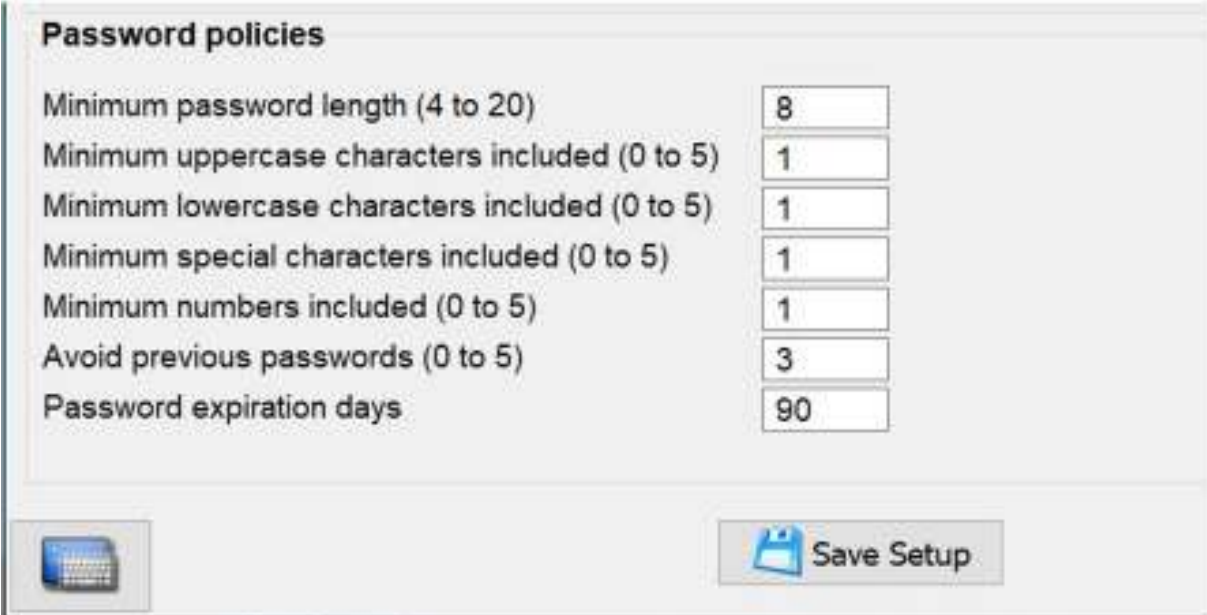
Password requirements not met

The password must include at least:

- 8 characters
- 1 upper case
- 1 lower case
- 1 numbers
- 1 special characters

Figure 705 - Password change window in Erytra software

These new requirements and the password expiration days can be configured from Importer application (new Password policies section at the bottom of setup tab):



Password policies	
Minimum password length (4 to 20)	8
Minimum uppercase characters included (0 to 5)	1
Minimum lowercase characters included (0 to 5)	1
Minimum special characters included (0 to 5)	1
Minimum numbers included (0 to 5)	1
Avoid previous passwords (0 to 5)	3
Password expiration days	90

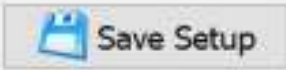
 

Figure 706 - Password configuration in Importer application.

The new requirements are stored in erytra.properties file with following parameters that will be automatically added when the application starts and set to the following default values:

```
PASSWORD_MINIMUM_LENGTH=8
PASSWORD_MINIMUM_UPPERCASE=1
PASSWORD_MINIMUM_LOWERCASE=1
PASSWORD_MINIMUM_NUMBERS=1
PASSWORD_MINIMUM_SPECIAL_CHARACTERS=1
PASSWORD_AVOID_PREVIOUS=3
```

Procedures

- See "Password policies" section (new)
7. The manually requested backup process (MAINTENANCE - ANALYZER - BACKUP) will now include a check that the created backup can be restored successfully. The system will restore the created backup in a temporary database and check if any restoration error occurs (and then delete this temporary database).

Complaint 800013401.

8. The management of the request to enable the feature that allows qualified Field Service Engineers to access Windows File Explorer has been improved to avoid the possibility of creating additional requests while executing the initial request, when the window to enter credentials has not yet been displayed. A new credentials window will be displayed:

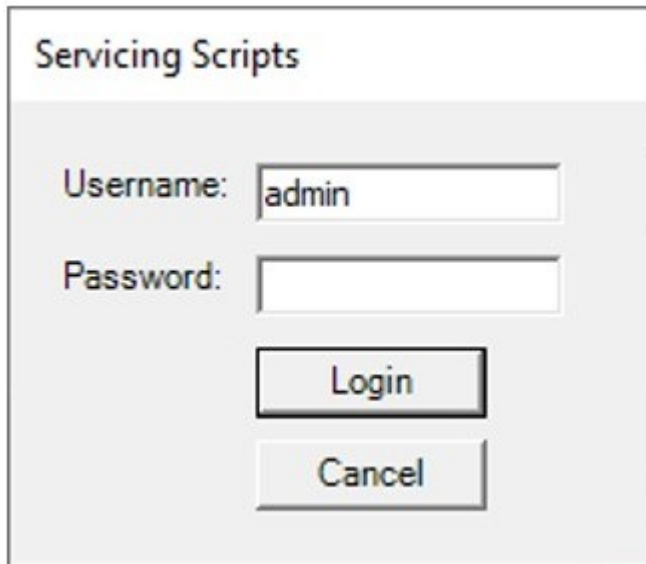
A screenshot of a Windows-style dialog box titled "Servicing Scripts". The dialog has a light gray background and a thin border. Inside, there are two labels: "Username:" and "Password:". The "Username:" label is followed by a text input field containing the text "admin". The "Password:" label is followed by an empty password input field. Below these fields are two buttons: "Login" and "Cancel". The "Login" button is slightly larger and has a darker border than the "Cancel" button.

Figure 707 - File explorer access credentials window

And the information window indicating Starting Servicing Scripts. Please Wait. will not be displayed anymore.

9. The information of "About" screen has been updated. It has been implemented the following information at "About" screen:

- Unique Device Identification (UDI) symbol with the followings 3 fields. The UDI is defined as a series of numeric or alphanumeric characters that is created through a globally accepted device identification and coding standard to unambiguously identify a specific device on the market.
 - Global Trade Item Number (GTIN). Format: (GS1 Application Identifier) GTIN.
 - GS1 Application Identifier will have the value "01" and should be displayed between brackets
 - The Global Trade Item Number (GTIN) will have the value "08436583731604" to Erytra v4.6.0 (major version 4).
 - For example: (01)GTIN number -> (01)08436583731604
- Release Date. Format: (GS1 Application Identifier) Release Date (in "YYMMDD" format).
 - GS1 Application Identifier will have the value "4326" and should be displayed between brackets.
 - Released Date: the date field should be the exact day when the version has been released (compile date of Release version installer). This date must be the same as the one that appears in the Manufacturing Date.
 - For example: (4326)YYMMDD -> (4326)240711
- Software Version. Format: (GS1 Application Identifier) Software Version (in "X.Z.Y" format).
 - GS1 Application Identifier will have the value "8012" and should be displayed between brackets.
 - The Software Version should be the Release version installed in the instrument.
 - For example: (8012)X.Y.Z -> (8012)4.6.0
- Manufacturing Date. Manufacturing year symbol with the compile date (in "YYYY-MM-DD" format):
 - Compile date: the date field should be the exact day when the Release version installer has been compiled. This date must be the same as the one that appears in the Release Date of UDI.
 - For example: YYYY-MM-DD -> 2024-07-11

This is an example of the information that should be added



Figure 708 - Example of added information

Resolved anomalies

1. When a serial titration technique was canceled, diluent was left in the dilution cup because a wash step was not performed. A specific wash of the Dilution Cup will be performed in this situation:
FLU_CMD_WASH_DCUP_HALF(FLUIDIC_SYSTEM,UNKNOWN_DEVICE ,1,2) finished with result: DCUP1_FLU_SERVICE.
2. When the user cancels a SerialTitration profile during aspiration of diluent before starts the homogenization process sometimes a significant error was displayed.
3. When several titration techniques are queued, the blocked volume of the diluent was overestimated. This caused some Profiles to stop despite having enough volume to run them.
4. When a serial titration technique is canceled mid-execution, the diluent volume the instrument locked for the execution was not released.
5. When executing a technique that needs only one lot dependent reagent and this reagent is finished and replaced by another one with a different lot in another drawer, the original reagent lot was asked. Moreover, if the used technique was queued for other samples, this reagent lot was asked from then on. Now, the instrument admits any lot, since only one reagent is used.

6. The screen could turn black when the operator interacted with the toolbar of the reports PDF viewer, usually when clicking the print button. Adobe Acrobat Reader: has been upgraded to version 23.8.20533 and memory management (report unload from memory) has been improved.

Complaint 800013755

7. When there were missing dependent reagents, non dependent reagents were also taken into account in dependent reagents missing calculations.

Even this error affected to all non dependent reagents, this error was only reproduced with diluent reagents (i.e. DG Gel Sol), other kind of reagents were taken into account but the missing was correctly calculated.

The system incorrectly reserved 2147483647 mcl of diluent and then it warned that there was not enough diluent volume.

8. When trying to cancel a profile imported from LIS, an error arose. It was reproducible while the Profile was in 'QUEUED' status. Moreover, related '[ERROR]' traces were only logged in 'erytraError.log' and not in 'erytraDebug.log'.
9. An error message could have been displayed when identifying a reagent whose commercial code or description had been updated when importing a Reagents Pack.

Complaint 800015121

10. The Backup.bat file was modified to not consider the backup process exit values 1, 2 and 3 as an error, as those robocopy exit values do not mean any error. However, the Erytra code is still considering any exit value greater than 0 as an error, and receiving a 1, 2 or 3 will result in an "Error on backup process" error window being shown to the user, despite the backup having been generated successfully.

Complaint 800014238

11. When generating a Support Pack from the instrument application, not all traces related to the Support Pack generation were logged in 'erytraDebug.log' file. Only initial and final traces were logged (START/END Support Pack creation).
12. When a Profile was queued more than once, an error arose. If the queuing process was slow, it is possible to queue the Profile more than once. This creates several 'current' Profiles in the database, which leads to an error 'query did not return a unique result'.

Complaint 800005920

13. When the level detection error is displayed during parallel sample aspiration and the "OPEN" option is selected for the sample rack, probe 1 does not move to the home position. After that, when the user attempts to retry the aspiration, a significant error was displayed.

14. When both probes are disabled and a reagent drawer is queued to be scanned, the same trace was logged infinitely.

Trace:

```
[DEBUG] Scan Drawer  
com.dg.dg59.scheduler.executor.impl.ScanReagentDrawerIncidenceOperation@a99b64  
queued.
```

LIS

Resolved anomalies

1. When LIS communication is set to TCP/IP or serial, DianaServer failed to remove demographic data included in a frame when this frame first characters is the P of the record type.

This would happen if the LIS was configured to build a different package for each line of the message. On the other hand, it would have also been reproducible if the message exceeded the maximum length of the package, which was different depending on Diana Server's configuration:

Mode	size (characters)
Serial port	240
TCP/IP	64.000

Complaint 800014790

Importer

New functionalities

1. The system imported all the profiles found in a Reagents Pack generated from a IH Templates Pack when all the profiles included in that Reagents Pack could not be executed in Erytra. This case has been filtered and not executable techniques will not be imported.

Software installer

New functionalities

1. Software installer will not include/copy Reagent Language Pack file. The updated version of this file is always included in the IH Reagents Pack and the IH Templates Pack.

2. Newer versions of following third parties will be installed:

- PostgreSQL: 16.3 (from 10.23)
- Apache Tomcat: 9.0.85 (from 8.5.31)
- Notepad++: 8.6.2 (from 8.4.9)
- Liquibase: 4.26.0 (from v3.5.5)
- Adobe Acrobat Reader: 23.8.20533 (from v22.3.20322).

Resolved anomalies

1. When running the technical service, importer, backup, restore, diagnostic or service scripts and then exit, the E drive was displayed in application mode.

E: unit is hidden for the following applications: technical service, importer, backup, restore, diagnostic or service scripts.

2. When a boot application (Diagnostic, Technical Service, importer, backup, restore) is not closed correctly, the boot menu was displayed but the options were locked.

The Open/close explorer.exe code before and after launching SetDrivesRestrictions methods has been removed.

It is not needed to Restart and close explorer to make drives restrictions changes to take effect.

Operating System and Boot Application

New functionalities

1. OS image includes Windows Security Updates up to March 2024 (Windows build 10.0.17763.5576):

ID Security Package	Description
KB5005112	Servicing stack update for Windows 10, version 1809: August 10, 2021 This update makes quality improvements to the servicing stack, which is the component that installs Windows updates. Servicing stack updates (SSU) makes sure that you have a robust and reliable servicing stack so that your devices can receive and install Microsoft updates.
KB5033904	Cumulative Update for .NET Framework 3.5 and 4.7.2 for Windows 10, version 1809: January 9, 2024.
KB5035849	OS Build 17763.5576: March 12, 2024. This security update includes improvements. When you install this KB: The update affects Active Directory domains that host mobile device management (MDM) providers. They can transition from "Compatibility mode" to the strong certificate mapping "Enforcement mode." To do this, they can allow an Active Directory Key Distribution Center (KDC) to read user security identifiers (SID) from the Subject Alternative Name (SAN). Then, the providers can populate those values. To learn more, see: • KB5014754: Certificate-based authentication changes on Windows domain controllers. • Preview of SAN URI for Certificate Strong Mapping for KB5014754 • CVE-2022-34691, CVE-2022-26931, and CVE-2022-26923

2. Operating System image (zip file and installation process) has been modified to manage different devices (motherboards,...).

PC Drivers have been removed from Operating System image and have been included in specific drivers files:

PC ASRock IMB-150D drivers file name is imb-150d-drivers-1.1.0.jar.

PC ASRock IMB-1003J drivers file name is imb-1003j-drivers-1.2.0.jar.

This drivers file matching instrument PC shall be copied in the Driver folder of the OS image to restore. This file will be expanded and the integrity of the files to install will be checked.

OS installation process, to install appropriate drivers, will now request to indicate the hard drive type/PC model:

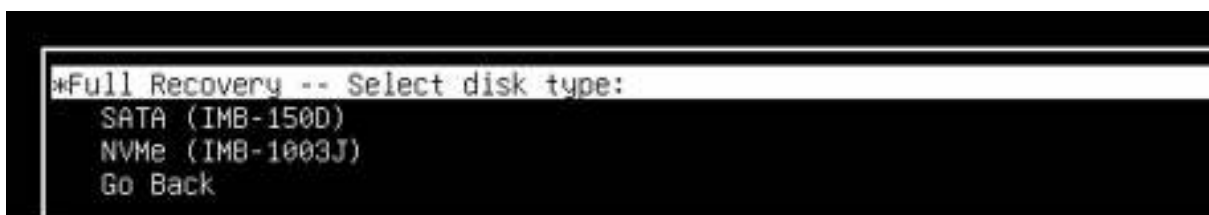


Figure 709 - Installer motherboard selection dialog

Procedures:

- See "Restoring from Boot Application menu" section (modified)
- See "Restoring from USB pen drive" section (modified)



NOTE

Erytra will only use IMB-150D PC.

3.

A Boot Application entry can now be found in Start to allow returning to Boot Application from OS Shell mode without the need to restart the system. To return to the point where the is displayed (and to continue user application start up or allowing to enter Boot Application again).

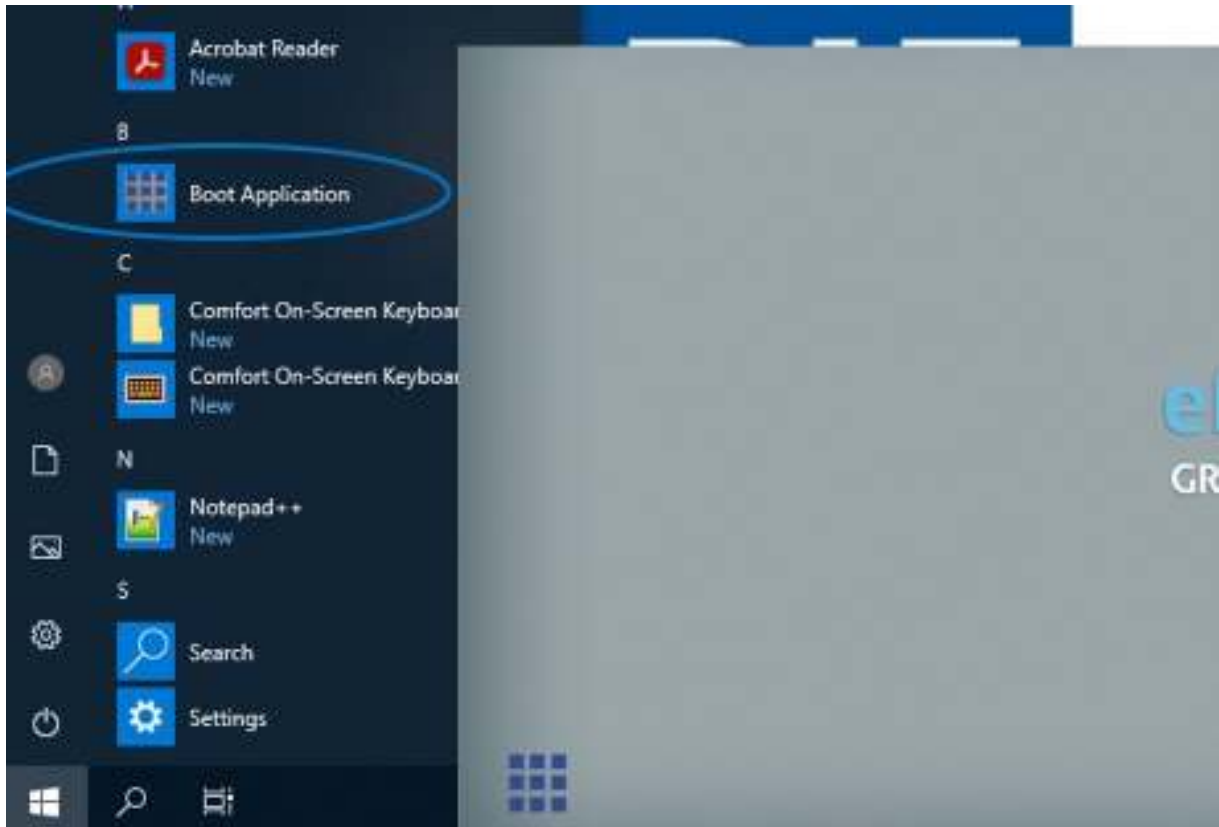


Figure 710 - Return to boot application entry point

Procedures:

- See "Return to Boot Application" section (modified)

4. The system will now display a warning message to the operator at start up when a potential unexpected time change has occurred in the previous session.



Figure 711 - Potential unexpected time change warning

5. The Boot Application and Operating System information will now be registered in a new log named system (E:\log). The startup.log will not be used anymore and installation.log will just include the traces related to software installer

Procedures:

- See "Description of log files" section (modified)
6. The touch screen calibration will now be preserved when requesting OS Full Recovery.

Resolved anomalies

1. The disk/files protection (Windows Unified-Write-Filter) was not automatically enabled in the next reboot when it was (unexpectedly) found disabled.
2. The Boot Application could not correctly manage that a service application was used (Importer, Technical Service,...) and could turn unresponsive or fail to start user application when requested.
3. The File Explorer context menu (mouse right button) was not available when entering Maintenance Mode just after closing the Technical Service software that had been launched from the Boot Application.
4. The Grifols Secure Remote Connectivity tool agent was not removed from recovery partition folder (R:\BeyondTrust) once it had been executed, once OS Full Restore or OS Recovery from Boot Application had been executed.
5. The E drive was available in the user interface after the feature to allow qualified Field Service Engineers to access Windows File Explorer had been used (and closed) or service applications (Technical Service, Importer) had been previously used.

6. The installer did not check the presence of the PC model drivers files in USB:/Drivers. A control of the presence of Drivers artifact under usb:/Drivers has been added.

Instrument backup and restore

New functionalities

1. Testcounter.txt file has been included when performing an export (backup) using the Import application.

Procedures:

- See "Backing up databases and relevant data with boot menu" section (modified)

Resolved anomalies

1. The backup generated with Backup/IHExport tool when requesting to include just Reagents Pack and profiles configuration did not include all the necessary information. This led to incorrect performances if restored: the configured profiles displayed names and special results were not restored, it was not possible to create Not Unitary Profiles, it may not be able to import new tests,...

Technical Service

Resolved anomalies

1. When a macro that performs aspiration with an specific profile was executed in TechnicalService, then the used profile was incorrect.

Erytra Diagnostic

New functionalities

1. Incubator movements test in Diagnostic application have been improved in order to mitigate "Current limit" errors reported during test execution.

In previous versions, the test was executing 25 movements up/down with incubators 1 and 2 (with a delay of 5 seconds between each other) and then the same 25 movements with incubators 3 and 4 in the same way. To avoid impacts of the incubators with the CAT robot arm, it was moved on top of the centrifuges.

Incubator movements test should execute 25 movements up/down with incubators 1 and 3 (with a delay of 5 seconds between each incubator) and then the same 25 movements with incubators 2 and 4 in the same way (with a delay of 5 seconds between each incubator). The positioning of the CAT robot arm should be changed to the "Card Dawer 4, Rack 1 and Card 12" position when making the movements of incubators 1 and 3, and to the position "Card Dawer 1, Rack 1 and Card 1" to prevent the incubators from impacting the CAT robot arm.

Start and End of each block of movements (Incubator 1/3 and Incubator 2/4) should be logged

Resolved anomalies

1. If the Erytra Diagnostic application was opened in French language, the "Incubators Movement" and "Centrifuges Movements" tests name were the same.
 - The "Incubators Movement" test name was displayed as "Mouvements des centrifugeuses". This error is because reference was made to the centrifuges instead of incubators.

"Incubators Movement" and "Centrifuges Movements" tests name have been modified to "Mouvements" to maintain consistency with the tests name in all languages. In the rest of the languages the tests name is "Movements" in their respective languages.

Alternative browser

Alternative browser has not been modified and it is the same as the one delivered with software version 4.5.0.

Alternative browser is not required for Erytra Database Manager.

Remote connection requires a web browser with Flash Player. The alternative browser should be installed on the computers for which the remote connection is not available, due to the external computer browser's version or configuration.

Manager, Controller and user databases and Erytra Database Manager

Erytra Database Manager has been updated to software version 4.6.0.

Erytra Database Manager version 4.6.0 cannot be installed if a previous version of Erytra Database Manager is already installed. The previous version shall be uninstalled (and a database backup created, if necessary, before uninstalling the previous version).

Software version 4.6.0 databases shall be restored in Erytra Database Manager version 4.6.0. The compatibility with previous Erytra Database Manager or software versions is not guaranteed.

Previous software versions databases shall be updated when they are restored in an Erytra instrument (by running the **E:\Program Files\liquibase\updatedb.bat**) or Erytra Database Manager (by running the **installation_directory\Grifols\Manager\Program Files\Liquibase\updatedb_edm.bat**).

7.1.3 Erytra Software v4.6.1

7.1.3.1 Software changes

This chapter describes Erytra Software version 4.6.1 improvements and modifications compared to version 4.6.0

General

New functionalities

1. Pre-analytical barcode reading process.

When scanning a tube barcode from a sample rack, if the system identifies the rack as pre-analytical or detects that the rack label lacks a barcode during the initial scan, it will automatically initiate a retry. This retry will proceed in the reverse direction, operating at a reduced speed and with a narrower reading window. During this process, all tube positions within the rack will be scanned to ensure complete barcode detection.

In case there is a mismatch between the two readings, a 'Barcode not Identified' incidence will be assigned to the tube. This tube will be able to be manually identified.

New adjustment and verification procedure has been implemented. In case the 'horns: pre-analytical' adjustment has not been performed, the identification of the rack will be stopped and the following message will be shown:

- The barcodes of samples identified in the pre-analytical compatible samples holder could not be read automatically.

The barcode reader adjustment in this type of holder is missing.

Please contact your local Grifols service representative.

Complaint 800017189

2. A new check based on level detection to detect dripping has been implemented. This check will be performed immediately after a probe wash following the completion of the below-listed actions:

- A level detection in a reagent vial.
- A level detection on dilution cup during the first step of the dilution.
- An aspiration of sample, reagent, diluent and cleaner from a tube or vial.
- A dilution aspiration from the dilution cup.
- A fluidic inactivity period.
- A pipetting final wash.

The new check basically compares level detection counts taken before the wash and a short time after solution B has been used to clean the probe.

To avoid false positive detections, the check will be repeated up to two additional times. If the difference in counts remains below the threshold on the third attempt, an error will be displayed.

If this new check detects dripping above a tube or vial, the related test will be canceled, the probe will be disabled, the affected tube or vial will be marked with an incidence. Two error window messages will be displayed with only "OK" option to accept the message informing the user that a possible dripping has been detected and that the probe has been disabled.

If this new check detects dripping above the dilution cup, the corresponding test will be canceled, the probe will be disabled, and the dilution cup will be washed. Two error messages will be displayed, each with only an 'OK' button to acknowledge the alert. These messages will inform the user that possible dripping has been detected and that the probe has been disabled.

Complaint 800016541.

3. Language pack has been updated from v1.30 to v1.33 implementing the following:
- All messages related to 'Bleach', 'NaClO', 'sodium hypochlorite' have been replaced by 'decontaminant' due to the possibility to use alternative decontaminants.
 - Change in Portuguese Portugal language:
 - "Próxima intervenção do utilizador **cerca das** 12h54" instead of "**em cerca de**"
 - Change in Finish language:
 - "**ALKUPERÄINEN**" will be used instead of "**ALKUPERÄIN**"
 - Russian language has been removed.

4. The **Adaptive Scanning** feature of the barcode reader has been **disabled**. This feature could interfere with the barcode reader performance in certain situations, as the engine may switch to a narrow scan angle if the scan line does not touch a barcode at a shorter distance. Disabling Adaptive Scanning prevents the scan angle from transitioning to a narrow angle.

**NOTE**

Erytra instruments could be equipped with one of the following barcode reader models:

- SE-955 model.
- SE-965 model.

The **SE-965** barcode reader model features Adaptive Scanning, which automatically optimizes parameters to enhance decode performance based on the barcode's distance and ambient light conditions. When the barcode reaches a certain distance from the engine, the scan angle is automatically reduced to a narrow 10°. See **DG-0059#0078 'NEW BARCODE READER v2'** on the "Bulletin index" on page 1183.

Resolved anomalies

1. If a newly introduced reagent or card shares the same lot number and reagent ID or card ID as one previously used—despite being from a different year—Erytra was unable to distinguish between them. As a result, it incorrectly assumed that quality control had already been performed and did not prompt for a new QC check. This issue has now been addressed: Erytra will distinguish reagents or cards based on their expiration dates and will accurately detect when quality control (QC) is required for newly introduced lots.

Complaint 800003071

2. After a the incubation of a test ends, the system verifies the incubation time to ensure it was completed correctly. Normally, incubation starts right after the last dispensation into a microtube, even though some pipetting-related tasks (like probe cleaning) may still be ongoing. The system typically uses the end of pipetting as the reference point for verification.

However, if there is a delay—such as an interaction with the Laboratory Information System (LIS)—the pipetting end time is recorded much later than the actual incubation start. This causes the system to mistakenly think incubation started late, leading it to incorrectly flag the incubation time as insufficient, even though it was actually correct.

After the incubation of a test ends, the amount of time spent incubating is verified to ensure that the incubation has been properly finished. However, it was possible for some interactions with LIS or some other process to affect this verification.

The system now calculates incubation time starting from the exact moment it is pipetted. This ensures a more accurate incubation time calculation.

Complaint 800014689

3. After a decontamination process, the instrument begins operations without priming the full circuit, preventing the correct performance of the probe and leading to the dispensation of air until enough probe washes are performed to restore the fluidic circuit to its proper state. Consequently, this ends in volume-related test incidents and NR results.

Now the system will be identified as 'not primed' after a decontamination process and it will be primed when needed.

Complaint 800018903

4. The following improvements have been implemented to manage 'Channel Fault' events:

- To prevent 'Channel Fault' errors during validation and export processes caused by multiple instances, the graphical interface will receive execution feedback to avoid triggering redundant instance submissions.

Complaint 800014446

- Report generation process has been improved to reduce the required time and avoid 'Chanel fault' errors.

Complaint 800015183

- More information will be logged in order to properly investigate 'Channel Fault' events.
 - "Verifying connection after reconnection..."
 - "Ping message sent successfully. Connection appears to be stable."
 - "Reconnection successful. All systems operational."
 - "Failed to send ping message after reconnection. Additional issues may be present."
 - "The system reconnected but encountered further communication issues. Please verify."
 - "Checking if critical subscriptions are active after reconnection.."
 - "Checking if critical subscriptions are active after reconnection.."
 - "All critical subscriptions are active."
 - "Known Error 2032 detected. No immediate action required."
 - "Channel fault detected, failed attempts count:"
 - "Channel recovery failed after " + _failedAttempts + " attempts. Disconnecting and alerting user."
 - "The system has made multiple failed attempts to reconnect to the server. Contact technical support."
 - "Attempting automatic reconnection (Attempt " + _failedAttempts + "/" + 3)

Complaint 800015183.

5. When using the Support Pack generator application, the process that obfuscates patient demographic data could fail in a scenario where the communication with LIS is configured to serial port or TCP/IP and the patient demographic data is divided in two frames. This led to an error and an improper completion of the Support Pack creation.

Complaint 800015344.

6. During a dilution process, a level detection has been added to check that the dilution cup is empty the first time the dilution cup is accessed.
 - If liquid is detected at this stage, the system manage the existing liquid before proceeding with any dispensation. A second level detection will then be attempted.
 - If the level is still detected, the probe will be externally washed, and an automatic dilution cup wash will be performed. A third level detection will follow.
 - If the cup is still not empty after the wash, the probe will be washed again, and the following error message will be displayed:

“An error has occurred while executing a [X] profile. The dilution cup is not empty. The related profiles have been canceled.”

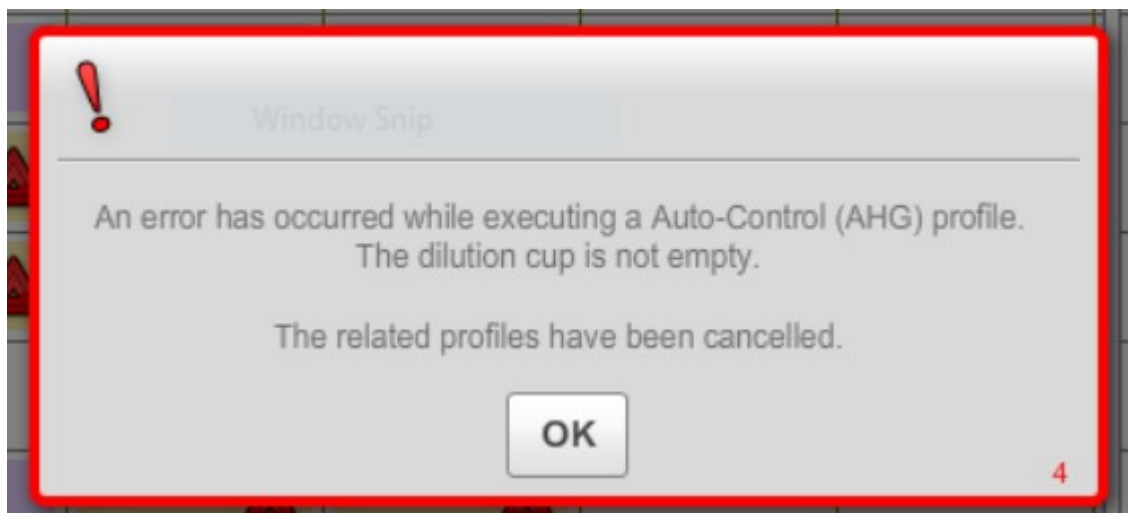


Figure 712 - Dilution cup not empty message

This message will include an “OK” button and a 45-second countdown timer. Regardless of whether the operator clicks “OK” or the timer expires, the system will cancel all techniques with pending dilution steps.

7. During startup, if the system experiences unexpected slowness, the compression of the erytraAir.log file may have failed with the following trace:

```
YYYY-MM-DD HH:MM:SS,000 {hiveStateTimerF} [KEY-ERROR] END
ArchiveTrigger with errors: Architecture must be bootstrapped using
Bootstrap.init(), or service with name
dg.architecture.filemanagement.Service is been searched before
architecture are fully initialized.
```

No error message was displayed to the operator, so the failure went unnoticed.

This failure has been solved by scheduling the compression of erytraAir.log to occur only after the system has fully started.

8. The contaminant reagent wash may use a lower volume of cleaning solution than expected if the system has just aspirated contaminant reagent for several profiles, and some of those profiles are later canceled.
9. During the inactivity wash, Solution B was not pumped as expected. Now dispensation of Solution B during inactivity wash cycles has been adjusted pumping the required Solution B from inactivity wash cycles.

Erytra User Interface

New functionalities

1. The following icons of the information of "About" screen have been updated to the current official ones.

- SN: Serial number
- REF: Catalogue number
- Consult instructions for use
- Manufacturer (black)
- Date of manufacture (white)
- IVD: In vitro diagnostic medical device



2. The system allows store and display demographic data in US market in the same way it does for CE market. The RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA parameter appear in the properties file with default value false fo US configuration.

3. "Fluid System Decontamination" icon has been updated by removing NaClO due to the terminology changes.



Figure 713 - New decontamination icon.

Resolved anomalies

1. When a test was canceled and automatically restored, the canceled instance remained and both were interpreted, sending incorrect results (only when auto-validation is active).

The interpretation algorithm has been improved to prevent unexpected results being generated and marking the segments related to the cancelled (and unexpectedly interpreted) test as "Not Interpretable (NI)".

Complaint 800020269

2. The generation of reports from the Maintenance tab of the user software could cause an unexpected spike of memory usage that led the user interface software to freeze at the moment of closing the report viewer, resulting in black screen. To improve this situation, Adobe Air has been updated to version 51.1.3.4.

Complaint 800016371

3. Short channels disconnections and Outofmemory error caused some unexpected behavior such as:

- graphical interface icons remain activated when they should not
- repeated NetConnection.cAll.Failed: HTTP: failed errors during sample identification.
- Worksheet tab is not updated.
- Reports not being shown and unresponsive screens

Communications between front end and back end and thread management had been improved to reduce occurrence of Out of memory and channel fault errors.

Complaint 800015739

Card Reading

New functionalities

1. After a detailed analysis and volumetry evaluation has been carried out, the final volume conversion factor updated. This improvement will decrease the possibility of an "Incorrect volume" warning in situations where the detected level was slightly below the expected margins.
2. New **DG421_IntegrityCheckPaths.log** in **E:\logs\DG-421** in order to better trace the source path of the microtube images.

QC

New functionalities

1. Cards and Reagents QC management has been improved. The system can now be configured to not use cards or reagents that have not been successfully pass QC test set.

The behavior is now configured by means of Importer.

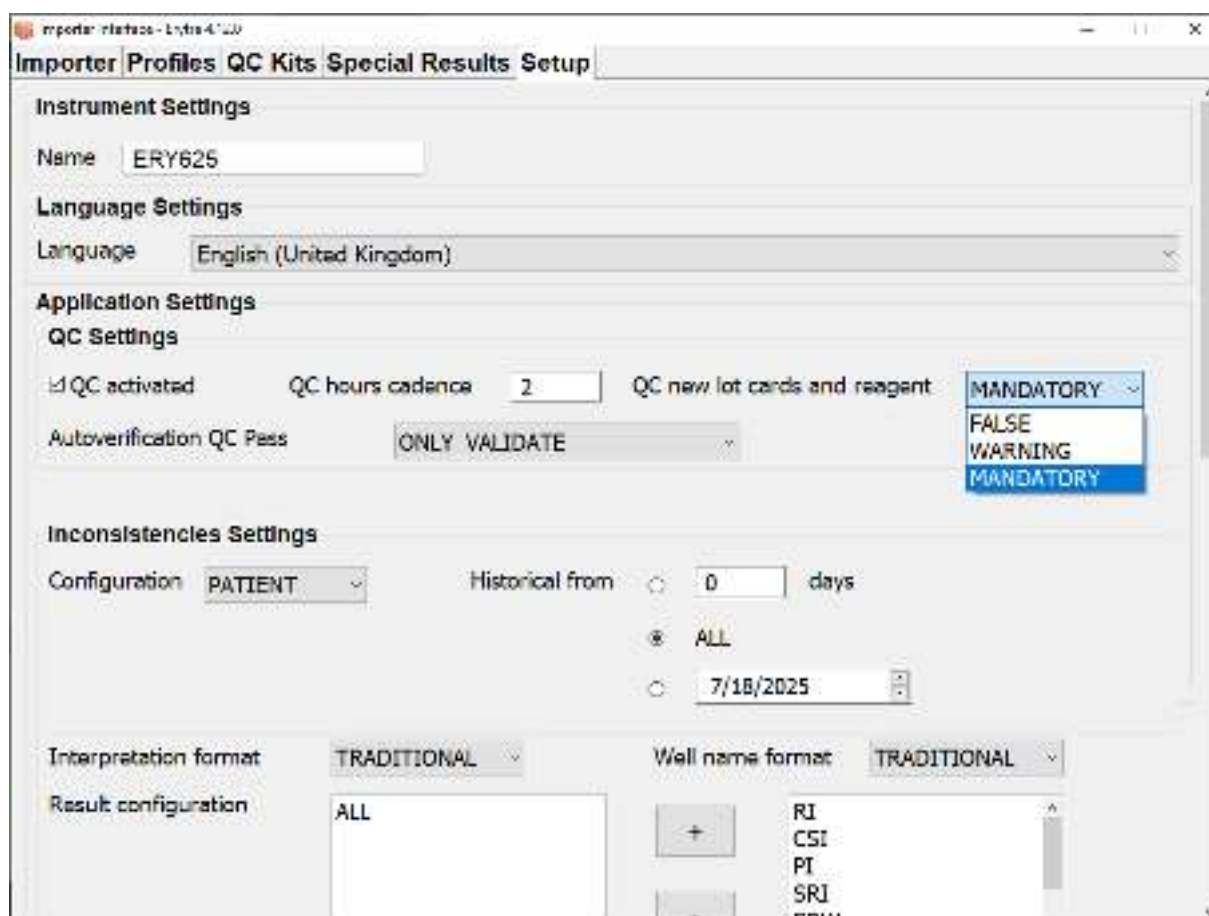


Figure 714 - 'QC new lot cards and reagents' configuration list on Importer Setup tab

- **False** (default option): if this option is configured, the system will not check if a QC has been passed.

- **Warning:** if this option is configured, when the 'QC_ENABLED' parameter is set to true and the card or reagent is used in a technique that has QC, the system will check if a QC test set with passed results has been completed and show a warning in case it has not.



Figure 715 - Warning configuration message

- **Mandatory:** if this option is configured, when the 'QC_ENABLED' parameter is set to true and the card or reagent is used in a technique that has QC, the system will check if a QC test set with passed results has been completed. In case it has not, the system will show a warning and will not use that lot of cards or reagents until this requirement is met.



Figure 716 - Mandatory configuration message

Affected cards or reagents will not be considered during the system planning, regardless of technique. The system will stop the queued techniques and activate the corresponding missing product warning icons.

This setting will be stored in erytra.properties new parameter '**QC_NEW_LOT_CARDS_AND_REAGENTS**' which will replace '**QC_NEW_LOT_WARNING_ENABLED**' parameter.

**NOTE**

The new parameter 'QC_NEW_LOT_CARDS_AND_REAGENTS' has replaced the previous 'QC_NEW_LOT_WARNING_ENABLED' parameter to manage the Cards and Reagents QC lot warning.

When the user application starts, it will check if the parameter 'QC_NEW_LOT_WARNING_ENABLED' exists setting in the Software:

- if the parameter 'QC_NEW_LOT_WARNING_ENABLED' exists and is set to false, the system will set the new parameter 'QC_NEW_LOT_CARDS_AND_REAGENTS' to false and remove the old 'QC_NEW_LOT_WARNING_ENABLED' parameter from the configuration.
- if the parameter 'QC_NEW_LOT_WARNING_ENABLED' exists and is set to true, the system will set the new parameter 'QC_NEW_LOT_CARDS_AND_REAGENTS' to Warning and remove the old 'QC_NEW_LOT_WARNING_ENABLED' parameter from the configuration.
- if the parameter 'QC_NEW_LOT_WARNING_ENABLED' does not exist, the system will set the new parameter 'QC_NEW_LOT_CARDS_AND_REAGENTS' to false (default value).

Importer**Resolved anomalies**

1. When a newer version of a previously installed technique is imported, and the database contains another technique with the same name but different identifier, an error is raised at a certain point of the import process, aborting it, and making it impossible to successfully install the IH Reagents Pack. The error displayed indicated 'object references an unsaved transient instance'. The importation process has been improved to avoid this error.

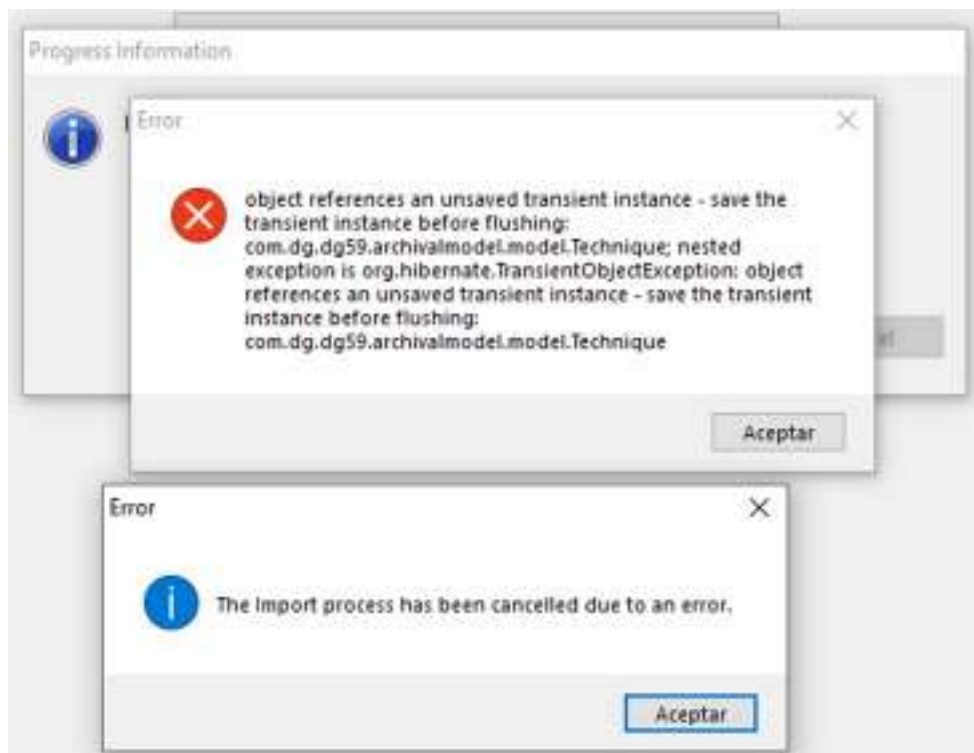


Figure 717 - Error message during importation

Complaints 800010740, 800016943

Firmware

New functionalities

1. CEN-SRM Firmware has been updated from 4.5.0 to v4.6.0 increasing the electronic cabinet fans PWM reference in order to improve fans start-up. this FW is being installed from factory since S/N $\geq$ 059-0001847
2. PIP Firmware has been updated from 4.12.0 to v4.13.0 implementing an improvement on the air aspiration detection for titration techniques: air aspiration detection thresholds have been adjusted to more accurately reflect the actual conditions of the machine.

LIS

Resolved anomalies

1. When importing work from the LIS, a "Host format error #10" occurred, and no test was executed. This error is triggered when the Diana Server attempts to process a file that is empty or contains no lines. To prevent this, a retry mechanism has been implemented to ensure the file is read instead of generating an error. A timeout error will arise if it is not still possible to read the file after a defined time.

Complaint 800014747

2. Diana server could be started a second time and the configuration screen was displayed until the operator accepted it, allowing changes to the configuration. Now the configuration screen will be close automatically.

Complaint 800017236

3. When a LIS message is split at beginning of PatientID field, some demographic data fields were not obfuscated.

Technical Service

New functionalities

1. New adjustment and verification procedures have been implemented to improve the barcode reading of the pre-analytical racks.
 - A new adjustment will determine the center position of the barcode window by performing two barcode reads in opposite directions. The window width will then be verified to ensure the adjustment was successful. This procedure has to be performed for each rack of each samples drawer. The new adjustment can be found in TSS path: 'Procedures tab → Adjustment → Pip floor Modules → horns: pre-analytical'. See "Procedures tab" on page 756.

**NOTE**

Check the barcode reader's laser beam alignment (See "Verifying barcode reader" on page 369) as if the barcode reader's laser beam is misaligned, the adjustment procedure may not complete successfully.

**NOTE**

For instruments S/N < 059-0001887 the barcode reader's laser beam alignment might not be possible to adjust properly due to HW limitations. See "Adjusting barcode reader inclination position" on page 348

In this cases it should be necessary to upgrade the instrument according to **INR-0059#0155 NEW PIP HORNS WALLS**. See "Bulletin index" on page 1183.

- A new verification procedure for the pre-analytical racks adjustment has been implemented to ensure the correct alignment of the horns barcode reader laser beam for each rack, in every drawer, one at a time. During this process, the system automatically positions and activates the barcode reader laser beam at the center of the reading window. This enables the technician to visually verify that the beam is both properly centered and fully contained within the reading window, ensuring accurate and reliable scanning performance. The new verification procedure can be found in TSS path: 'Procedures tab → Checks → PIP Module → PIP module Pre-analytical Rack - Laser alignment check'. See "Procedures tab" on page 756

Complaint 800017189

2. Technical Service logs size have been increased from 2MB and 1 backup to 100MB and 5 backups.

Erytra Diagnostic

New functionalities

1. Diagnostic application limits for 'Pipetting test - CPD and PCA level detector' test have been modified to homogenize them with factory tests. CPD sensor value limits changed from [17000-18400] to [15700-19700].

Others

Resolved anomalies

1. When Erytra executes archive process, the ErytraDebug.log was not archived. This failure has been solved by scheduling the compression of erytraAir.log to occur only after the system has fully started.

Alternative browser

The alternative browser has not been modified and remains the same as the version delivered with software version 4.5.0. It is not required for use with the Erytra Database Manager. However, remote connections require a web browser with Flash Player. In cases where remote access is not possible due to the version or configuration of the external computer browser, the alternative browser should be installed to ensure compatibility.

Manager, Controller and User databases and Erytra Database Manager

The Erytra Database Manager has not been modified and remains the same as the version delivered with software version 4.6.1.

This version cannot be installed if a previous version of the Erytra Database Manager is already present on the system. In such cases, the existing version must be uninstalled beforehand, and a database backup should be created if necessary.

Databases created with software version 4.6.1 must be restored using Erytra Database Manager version 4.6.1. Compatibility with earlier versions of the Erytra Database Manager or software is not guaranteed.

Databases from previous software versions must be updated before use. This can be done by running the **updatedb.bat** file located in **E:\Program Files\liquibase** when restoring in an Erytra instrument, or by executing **iupdatedb_edm.bat** located in **installation_directory\Grifols\Manager\Program Files\Liquibase \updatedb_edm.bat** when restoring in the Erytra Database Manager.

7.1.3.2 Application and firmware versions

Table 104 - Erytra v4.6.1 components versions

Name	Version	Description
Apache Tomcat	9.0.85	HTTP web server
Bootapplication	1.2.0	Grifols Applications launcher and Database Backup/Restore for Technical Service purposes
Diana System CE	4.6.0.38	Diana System build from v4.6.0 DG-57 project. (Configuration= CE)

Name	Version	Description
Diana System US	2.4.0.19	Diana System build from v2.4.0 DG-517 project. (Configuration= US)
Director.dll	5.10.1	Image processing library
DSC.dll (CE)	6.2.0	Diana System Connectivity. (Configuration= CE)
DSC.dll (US)	6.2.0	Diana System Connectivity. (Configuration= US)
Erytra-databasemanager-4.6.1	4.6.1	Erytra Database Manager
Erytra Diagnostic	4.15.0	Instrument check tool
ErytraSupplyOnOffSwitcher.exe	1.1.0	ErytraSupplyOnOffSwitcher
IHExport	1.9.0	It allows performing backup of instrument data as: databases, configuration files, operating system data, diana system data and logs.
IHImport	1.9.0	It allows restoring of instrument data as: databases, configuration files, operating system data, diana system data and logs.
IH-supportpack	1.7.1	Tool to generate a support pack.
Importer	4.15.0	Languages and techniques import tool
languagepack.zip	1.33.0	Language pack file
PostgresSQL	16.3	Object-relational database management system
Technical Service	4.15.0	Tool for instrument adjusting

Table 105 - Erytra v4.6.1 FW versions

PCA	Firmware version
CEN_SRM_CTRL	4.6.0
INC_CUP_CTRL	4.5.0
FLUCTRL59	4.6.2
CAT_CM_CTRL	4.5.0
PIP_CTRL	4.13.0
RD_IL421	5.5.0

7.1.4 Erytra Software v5.2.0

7.1.4.1 Software changes

This chapter describes Erytra Software version 5.2.0 improvements and modifications compared to version 5.1.0

General

New functionalities

1. The Language Pack has been updated (from version 1.25. to 1.30):
 - a. Estonian (Estonia), Hungarian (Hungary), Lithuanian (Lithuania), Latvian (Latvia) and Turkish (Turkey) languages are now available.
 - b. Messages related to new functionalities have been added.
 - c. CONTINUE option has been replaced by OK in significant error messages.
 - d. Several translations in several languages have been added.
 - e. The incorrect display of some words in several languages has been corrected.
 - f. The words that appear in the Analyzer tab have been harmonized (some were displayed in uppercase and others in lowercase).
2. Diana System programs versions delivered with Erytra have been updated. Current versions are:
 - a. CE configuration: Config v4.6.0.12, DianaServer v10.2.0.12, HostDemo v3.0.0.21, Diana O. v4.6.0.17.
 - b. US configuration: Config v2.4.0.6, DianaServer v10.2.0.12, HostDemo v3.0.0.21, Diana O. v2.4.0.7.

The modifications implemented are described in section LIS.

3. The card waste tray warning will be activated with 330 cards (instead of 350) to take into account that card weight decreases as they dry in the card waste tray.
4. Red blood cell dispensation speed (top speed) has changed from 500 to 125 (and now is the same as Erytra Eflexis) allowing lower CPD values and pressure in the circuit.
5. Sample tube diameter detection has been improved. The tube label offset applied (by the Driver) has been modified from 1 mm to 1.5 mm.

The value applied will now be included in the Driver Debug traces related to tube diameter detection.

[DEBUG] PIP Tube selected - Circle center X: 210135 Y: 24156 - Distance: 132.000000 - Max Distance allowed: 1429.000000 - Diameter safety offset: 1.500000 - Compensated Diameter: 11.8316000

6. The system will now post in a log the CPD values measured during aspirations. The new log is named PipettingProfile.log and it will be created in E:\log folder.

Once it reaches 25MB it will be moved to backup folder, compressed, and renamed to PipettingProfile_YYYYMMDD_HHmmss.zip where YYYY=Year, MM=Month, DD=Day, HH=hour in 24hour format, mm=minutes and ss=seconds.

The data logged for each aspiration is: aspirated volume, time required to complete the aspiration, the aspiration profile data and the set of DCP values measured.

7. The system has been prepared for new Pre-analytical Compatible Samples holder (PN 210716, refer to INR-0059#0157) that allows more compatibility with pre-analytical instruments.:

- a. A new holder type has been defined and added to erytra.properties (when application starts): BC0=NORMAL,GN,0,0

- b. Sample drawer identification has been modified:

The system will not detect if a holder is present (will not search for holder handle using present detector) and will proceed to perform tube detection and identification step.

Once the identification is completed, if the following preconditions are accomplished:

- i. A holder with barcode GN is detected or the parameter RACK_BC_ENABLED in Erytra.properties is FALSE.
- ii. Two tubes are detected in two contiguous positions in X axis.
- iii. Only one barcode of these two tubes is read when performing the barcode reading process.

The system will discard the tube with the read barcode and shall acknowledge both tubes as "not read".

This situation will be logged in erytra and erytraDebug log files:

{Probes } [WARN] A tube barcode was not read in a rack that could be preanalytic. The adjacent tube has been marked as 'not read' as well

TUBE NOT READ:

Location: Drawer SAMPLE_DRAWER_4, Rack 2, Position 2

ADJACENT TUBE MARKED AS NOT READ:

Location: Drawer SAMPLE_DRAWER_4, Rack 2, Position 1 Barcode: S11

8. The Importer log files will now be included in the Support Pack (new folder named Importer in Support Pack Erytra\Log folder)

9. The information included in testcounter.txt when restoring an application database with IHImport/Restore application has been improved to accurately compute the number of tests executed by the instrument and determine the Preventive Maintenance category.

Now the offset between the number of tests in previous database and the restored database, and the restoration time stamp will be included:

```
325
restored db. 2024-03-06 08:36:25
Offset: 225
100
```

If testcounter.txt file is not found when restoring the database, it will be created.

The number of tests of the current application database will be searched in testcounter.txt file. If it is not found, it will be searched in the current database and, if no database is found, it will be assumed to be 0.

Databases restoration must be performed with IHImport/Restore application. The old restore application Restore_DataBase.exe (and the folder containing this file, E:\Program Files\ErytraDB Apps) has been removed. The software installer will not copy this file nor create this directory, and delete the directory and its contents if found.

Procedures:

- See "Restoring from Boot application" section (modified)

10. The Support Pack will now include:

- Erytra Diagnostic logs.
- Technical Service logs.
- AppLocker rules.
- The logs in the archive path belonging to the selected time frame.
- Windows users, services, and Application and System events logs with the events belonging to the selected time frame.

Complaint 800010590.

Procedures:

- See "Generating Support Pack from user software" section (modified)

Erytra User Interface

New functionalities

1. The system can now perform serum dilutions. A new set of titration techniques with serum dilutions will be available.

These techniques will be executed alone, without sharing dilutions and without grouping them with other techniques, and will be executed after the techniques that require patient RBC dilutions.

If the volume of serum required to perform a serum dilution is $< 100\mu\text{l}$, the system will aspirate $100\mu\text{l}$ and then dispose of the volume not needed (the difference between $100\mu\text{l}$ and the required volume).

The system, when the required volume of patient serum equals to $100\mu\text{l}$ (in serum dilution techniques), will check if air is aspirated and, should this be the case, the following error will be posted in Driver log file:

```
...{0000003156} [ERROR] ERROR Error Code: (0817031000)
```

Device error

Description: Dilutor's air aspired detected

The analyzer will then:

- Return the aspirated volume to the sample tube using a new specific dilutor profile defined in DilutorsConf.txt.
- Raise and clean the probe externally.
- Detect sample tube liquid level.
- Retry the aspiration.

If the analyzer detects that air has been aspirated in the third attempt, the aspirated volume will be returned to the tube, the probe will be raised and cleaned, and an error message will be displayed informing the operator about the error:



Figure 718 - Air aspired error message

with the following options:

- OPEN: The analyzer shall open the samples drawer where the affected sample is placed.
- RETRY: retry the aspiration

- STOP works: The analyzer should stop the affected test, activate the Sample incidences warning and the Missing samples warning.

The analyzer will begin a 45 second countdown. If no option has been selected before the countdown has elapsed, the analyzer will automatically select STOP works.

Serial Titration techniques including all the dilutions from 1/1 to 1/2048 do not require a serum volume < 100ul to perform a serum dilution.

If an error occurs during the serum dilution step that may affect the dilution, the following error message will be displayed:



The system, when grading the reactions (process validator check step), will verify the volume dispensed in the microtubes as serum has been dispensed.

One new dilutor profile has been implemented.

- The fluidic system can now be enabled from Maintenance tab.

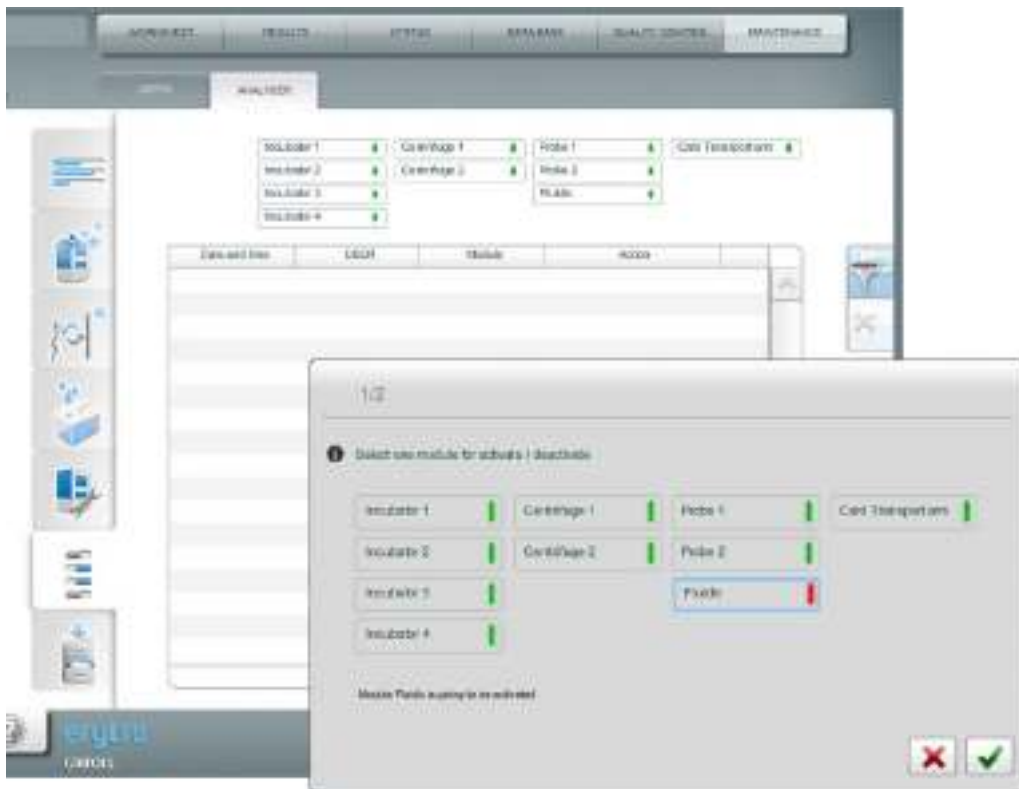


Figure 719 - Erytra software enabling modules from Maintenance tab.

The system will execute a fluidic circuit empty and prime process when the operator requests to enable the fluidic system.

- Patient demographic data stored in instrument database is stored now encrypted.
- The maximum allowed time between patient RBCs and serum dispensation in a given microtube has been reduced from 30 minutes to 20 minutes (RBC to serum timeout incidence) to assure that there is no effect on the reaction.

5. The enabled profiles and their characteristics will now be logged in erytraDebug log file during instrument initialization. The characteristics that will be logged are the profile identifier, the displayed name, profile version and if they are a Unitary Profile (true/false). In case of Not Unitary Profile (Unitary Profile field value is false), its composition will be logged (indented). For example:

...{ost-startStop-2} [DEBUG] Enabled Profiles (Profile identifier, Displayed name, version, is unitary profile):

E7161ABO-Rh (2D), ABO-Rh (2D), 3.00, true
E7115Confirm, Confirm, 2.00, true
E7141Xmatch (IgG), Xmatch (IgG), 2.00, true
E7135Xmatch (Neutral), Xmatch (Neutral), 2.00, true
BG, BG, 1.0, false
E7111ABO-Rh, ABO-Rh, 2.00, true
E7183Confirm P, Confirm P, 2.00, true
E71302 Screen (AHG), 2SCoombs, 1.01, true

6. The information of "About" screen has been updated. It has been implemented the following information at "About" screen:
- Unique Device Identification (UDI) symbol with the followings 3 fields. The UDI is defined as a series of numeric or alphanumeric characters that is created through a globally accepted device identification and coding standard to unambiguously identify a specific device on the market.
 - Global Trade Item Number (GTIN). Format: (GS1 Application Identifier) GTIN.
 - GS1 Application Identifier will have the value "01" and should be displayed between brackets
 - The Global Trade Item Number (GTIN) will have the value "08436583731604" to Erytra v4.6.0 (major version 4).
 - For example: (01)GTIN number -> (01)08436583731604
 - Release Date. Format: (GS1 Application Identifier) Release Date (in "YYMMDD" format).
 - GS1 Application Identifier will have the value "4326" and should be displayed between brackets.
 - Released Date: the date field should be the exact day when the version has been released (compile date of Release version installer). This date must be the same as the one that appears in the Manufacturing Date.
 - For example: (4326)YYMMDD -> (4326)240711
 - Software Version. Format: (GS1 Application Identifier) Software Version (in "X.Z.Y" format).
 - GS1 Application Identifier will have the value "8012" and should be displayed between brackets.
 - The Software Version should be the Release version installed in the instrument.
 - For example: (8012)X.Y.Z -> (8012)4.6.0
 - Manufacturing Date. Manufacturing year symbol with the compile date (in "YYYY-MM-DD" format):
 - Compile date: the date field should be the exact day when the Release version installer has been compiled. This date must be the same as the one that appears in the Release Date of UDI.
 - For example: YYYY-MM-DD -> 2024-07-11

This is an example of the information that should be added

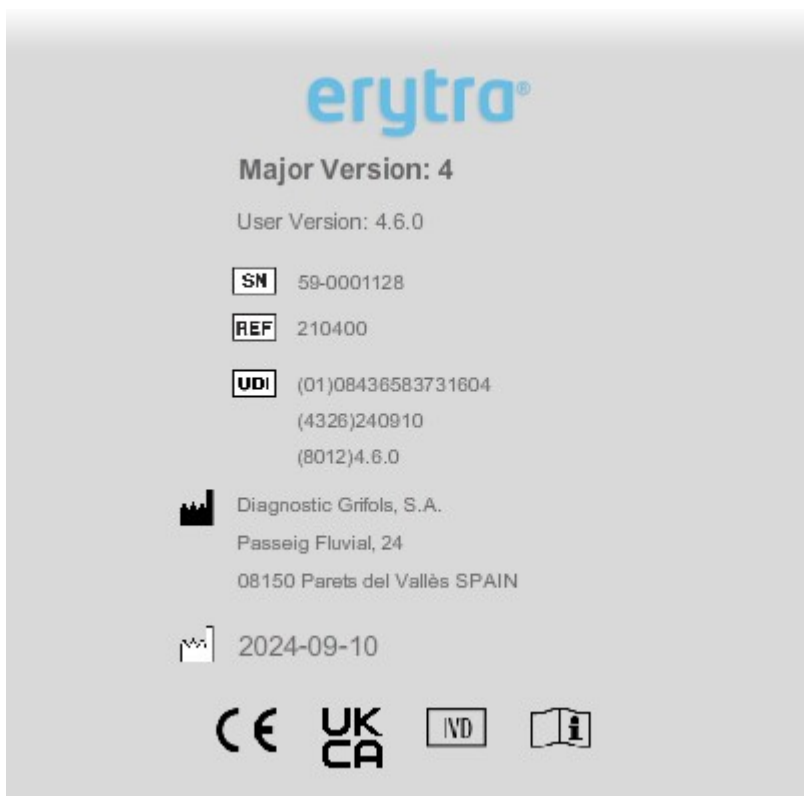


Figure 720 - Example of added information

7. The titration profiles with serial serum/plasma dilution have been improved:

- The homogenization of patient serum/plasma dilution has been improved in titration profiles with serial serum/plasma dilution. The system, in this case, will now aspirate and dispense dilution cup contents to improve homogenization.
- A new specific probe internal wash has been created (new DP_WASH_ST in Layout.xml file) to reduce dilutor speed (top_speed from 5000 to 3500) and assure proper dripping detection during washings.
- The steps and volumes used to obtain the dilution have been improved, especially for single point profiles.
- The time elapsed between the reagent red blood cells and the serum/plasma dilution dispensation will now be checked when the profile executed is completed (process check). The message posted in the erytraDebug log files in this step has been modified from

ProcessValidatorChecks - Checking serum dispensation step time duration.
to

ProcessValidatorChecks - Checking serum or serum dilution dispensation step time duration.

8. Air aspiration detection has been improved. The system will execute the air aspiration check in all sample/plasma aspiration but the result of that checking is only considered in sample serum/plasma aspirations of a serial titration profiles. Air aspired detection threshold has been adjusted when a serial titration technique is executed.

9. The process executed when the instrument switches from one waste container to the other that checks that the new selected waste container is properly connected and can be used has been improved.

The dilution cup aspiration line will not be opened, and the system will check that solution B can reach the waste container as in the fast prime of the solution B container process.

10. Priming process has been improved at the end of prime process associated to Dilution Cup. During Prime High the following action is performed:

PRIME_HIGH_DC_9 (this action keeps the EV2 and EV7 4 seconds opened)

This action will improve the priming of the instrument and prevent the presence of bubbles in the syringes.

11. It's been detected that AdobeAir logs could contain sensitive patient demographic data. All logs relative to channelMessage events have been deleted from the logs.

Complaint: 800013755

12. The user password requirements have been improved to increase the password security level of the password set and to allow the configuration of the requirements.

The new requirements will be applied when the current password is changed. The configured password requirements will be shown at the Change Password window.

When the new password does not meet the configured password requirement, the message Password requirements not met will be displayed:

CHANGE PASSWORD

Username ADMIN

Current password *****

New password

Confirm Password

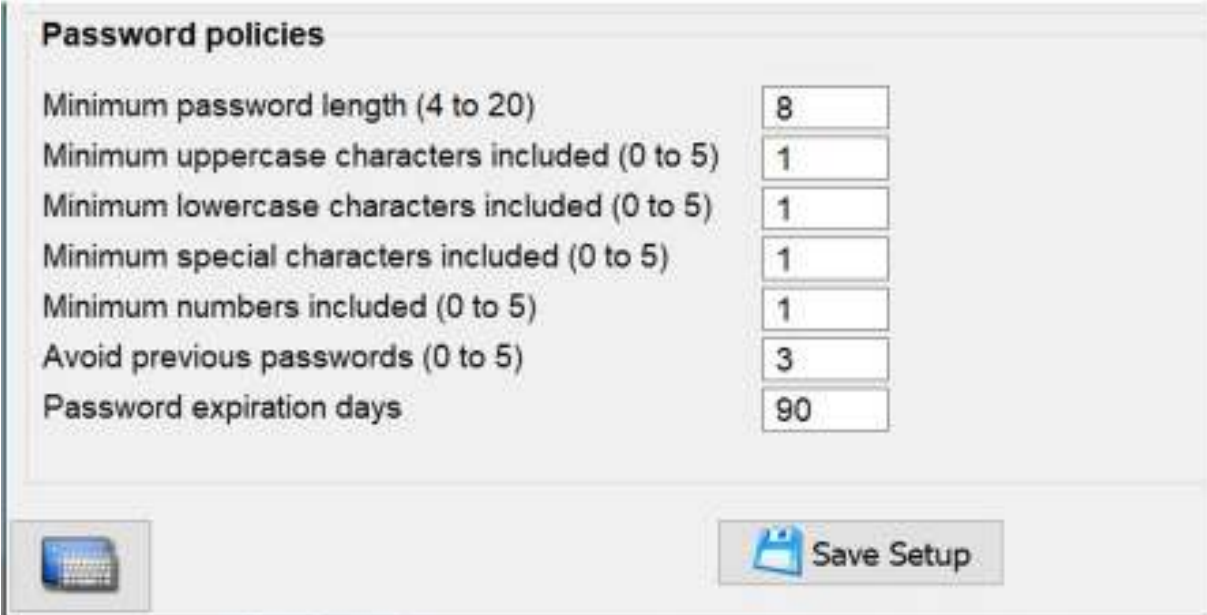
Password requirements not met

The password must include at least:

- 8 characters
- 1 upper case
- 1 lower case
- 1 numbers
- 1 special characters

Figure 721 - Password change window in Erytra software

These new requirements and the password expiration days can be configured from Importer application (new Password policies section at the bottom of setup tab):



Password policies	
Minimum password length (4 to 20)	8
Minimum uppercase characters included (0 to 5)	1
Minimum lowercase characters included (0 to 5)	1
Minimum special characters included (0 to 5)	1
Minimum numbers included (0 to 5)	1
Avoid previous passwords (0 to 5)	3
Password expiration days	90

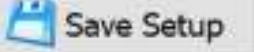
 

Figure 722 - Password configuration in Importer application.

The new requirements are stored in erytra.properties file with following parameters that will be automatically added when the application starts and set to the following default values:

```
PASSWORD_MINIMUM_LENGTH=8
PASSWORD_MINIMUM_UPPERCASE=1
PASSWORD_MINIMUM_LOWERCASE=1
PASSWORD_MINIMUM_NUMBERS=1
PASSWORD_MINIMUM_SPECIAL_CHARACTERS=1
PASSWORD_AVOID_PREVIOUS=3
```

Procedures

- See "Password policies" section (new)

13. The manually requested backup process (MAINTENANCE - ANALYZER - BACKUP) will now include a check that the created backup can be restored successfully. The system will restore the created backup in a temporary database and check if any restoration error occurs (and then delete this temporary database).

Complaint 800013401.

14. The management of the request to enable the feature that allows qualified Field Service Engineers to access Windows File Explorer has been improved to avoid the possibility of creating additional requests while executing the initial request, when the window to enter credentials has not yet been displayed. A new credentials window will be displayed:

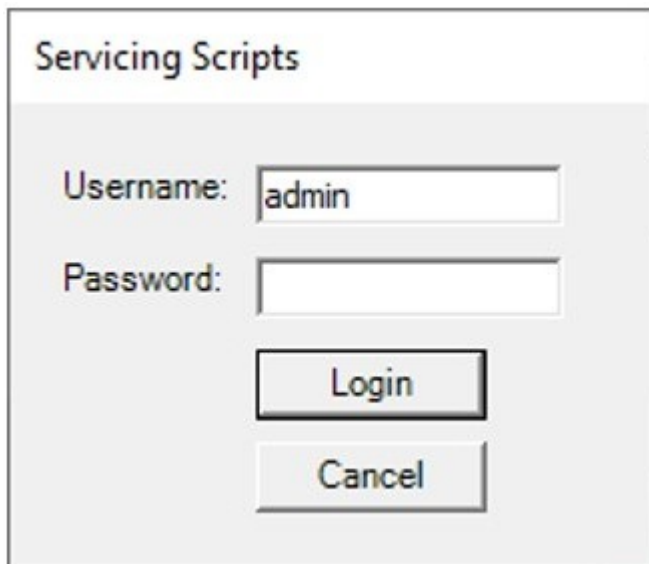


Figure 723 - File explorer access credentials window

And the information window indicating Starting Servicing Scripts. Please Wait. will not be displayed anymore.

Resolved anomalies

1. A desynchronization in the drawers status management occurred when multiple sample drawers were opened and closed in a very short period of time or when a sample drawer was opened and closed during a communication with LIS.

This desynchronization led the instrument to not identify the contents of one drawer. As a result, as one drawer was not identified but the system was waiting for that drawer to be identified, all queued tests remained in queued status (until the affected drawer was opened and closed again and the drawer contents identification was then performed).

Complaints 800011552, 800011641, 800011748 and 800011899.

2. The instrument occasionally failed to calculate the remaining DG Clean volume when there was a low volume in the vial, reporting an insufficient level for subsequent cleaning steps and disabling the probe (as it is considered that there was not enough DG Clean solution to perform the required cleaning steps).

Complaint 800004095.

3. Support Pack was successfully created but took a long time and an error was displayed during the process indicating that an error has occurred in the server when the raw images had to be included and there were many images to include.

Support Pack will now use a java library (instead of 7zip) to compress the data.

Complaint 800008934.

4. Direct Print option did not work. The software, when a report is generated, adds a timestamp to the name of the resulting PDFfile before saving it into the file system to differentiate it from previous reports and to prevent the override of previously generated reports. When the direct print was activated, the software generated the file to print without that timestamp, leading to the reported event because the system did not recognize the file name.

Complaint 800005159.

5. An error message was occasionally displayed when a backup process was completed successfully. The system did not properly manage some of the informational messages related to the copy of files (file skipped,...).

Complaint 800012723.

6. User application buttons turned unresponsive when using remote connection and a PDF report was requested and then closed.

Complaint 800011857.

7. The screen occasionally turned black when the operator interacted with the toolbar of the reports PDF viewer, usually when clicking the print button. Adobe Acrobat Reader: has been upgraded to version 22.3.20322.

Complaint 800012638.

8. Over-incubation incidences were not reported for the techniques that do not include an incubation step.
9. Remaining incubation time was not displayed nor updated during incubation step when the last pipetting step of the technique was a DG Clean step. "00" was displayed instead, during all the incubation time. The issue just affected the graphical display of the incubation time. Incubation was executed properly, and the issue had no impact on technique results.
10. The disabling of the fluidic circuit was not completed successfully during initialization or decontamination processes.

The fluidic circuit was disabled but then the pipetting arm turned unresponsive, and it was not possible to activate/deactivate any module in maintenance tab. The message [ERROR] Monitor Status Thread: thread with name Probes IS DIED was found in erytra and erytraDebug log files.

11. The system could use a probe that had just aspirated a contaminant reagent to identify the contents (vial volume) of a Reagents Drawer without having completed all the probe washing steps for contaminant reagents.

12. Decontamination process has been reviewed and improved:
 - a. A significant error was generated when an obstruction in the probe circuit was detected during decontamination. Now an error window with a message indicating that an obstruction has been detected in the probe, and with OK button, will be displayed. The decontamination process will be stopped when the OK button is pressed.
 - b. The decontamination process could continue after the fluidic circuit had been disabled due to an error that occurred during the decontamination. Once completed, the system considered that the decontamination had not been completed successfully.
 - c. The probe circuit decontamination step will now be executed in parallel for both probes. In previous software versions, it was executed sequentially (one probe at a time)
 - d. The dispensation pump pressure during the Dilution Cup filling with decontamination solution step has changed from 180 mbar to 160 mbar to avoid the possibility of dilution cup overflows.
13. The modification of password re-authentication option configuration in the erytra.properties file was not taken into account in the first user interface application start up after setting the new value. It was successfully applied the next sessions.
14. The system displayed messages indicating that a container drawer was open when the fluidic system had been disabled.
15. On screen keyboard could be not automatically hidden after login.
16. The Result tab list displayed after starting the instrument applied the filter set before the instrument was shut down.
17. The information regarding user interface keyboard open request action has been improved. The erytraAIR log will now include a new trace when the keyboard shall be opened:
...Native application event called for: D:\\startup\\helpers\\show-virtual-keyboard.bat
And a new trace with the results of the opening request.
... Native application event returned OK
Or
... Native application event FAILED
Complaint 800010794.
18. The log traces posted in the eytra and erytraDebug log files dilution have been reviewed to correctly describe all the steps:
 - a. The traces related to a step that was not executed when performing a dilution (dispensing all the dilution in incubator) have been removed.
 - b. The traces related to serum dispensation when two techniques share serum dispensation. The Start and End traces were missing.

19. When a serial titration technique was cancelled, diluent was left in the dilution cup because a wash step was not performed. A specific wash of the Dilution Cup will be performed in this situation:
FLU_CMD_WASH_DCUP_HALF(FLUIDIC_SYSTEM,UNKNOWN_DEVICE ,1,2) finished with result: DCUP1_FLU_SERVICE.
20. When the user cancels a SerialTitration profile during aspiration of diluent before starts the homogenization process sometimes a significant error was displayed.
21. When several titration techniques are queued, the blocked volume of the diluent was overestimated. This caused some SampleProfiles to stop despite having enough volume to run them.
22. When a serial titration technique is cancelled mid-execution, the diluent volume the instrument locked for the execution was not released.
23. When executing a technique that needs only one lot dependant reagent and this reagent is finished and replaced by another one with a different lot in another drawer, the original reagent lot was asked. Moreover, if the used technique was queued for other samples, this reagent lot was asked from then on. Now, the instrument admits any lot, since only one reagent is used.
24. The screen could turn black when the operator interacted with the toolbar of the reports PDF viewer, usually when clicking the print button. Adobe Acrobat Reader: has been upgraded to version 23.8.20533 and memory management (report unload from memory) has been improved.
Complaint 800013755
25. When there were missing dependant reagents, non dependant reagents were also taken into account in dependant reagents missings calculations.
Even this error affected to all non dependant reagents, this error was only reproduced with diluent reagents (i.e. DG Gel Sol), other kind of reagents were taken into account but the missing was correctly calculated.
The system incorrectly reserved 2147483647 mcl of diluent and then it warned that there was not enough diluent volume.
26. When trying to cancel a profile imported from LIS, an error arose. It was reproducible while the profile was in 'QUEUED' status. Moreover, related '[ERROR]' traces were only logged in 'erytraError.log' and not in 'erytraDebug.log'.
27. An error message could have been displayed when identifying a reagent whose commercial code or description had been updated when importing a Reagents Pack.
Complaint 800015121
28. The Backup.bat file was modified to not consider the backup process exit values 1, 2 and 3 as an error, as those robocopy exit values do not mean any error. However, the Erytra code is still considering any exit value greater than 0 as an error, and receiving a 1, 2 or 3 will result in an "Error on backup process" error window being shown to the user, despite the backup having been generated successfully.
Complaint 800014238

29. When generating a Support Pack from the instrument application, not all traces related to the Support Pack generation were logged in 'erytraDebug.log' file. Only initial and final traces were logged (START/END Support Pack creation).
30. When a sampleProfile was queued more than once, an error arised. If the queueing process was slow, it is possible to queue the sampleProfile more than once. This creates several 'current' sampleProfiles in the database, which leads to an error 'query did not return a unique result'.

Complaint 800005920

31. When the level detection error is displayed during parallel sample aspiration and the "OPEN" option is selected for the sample rack, probe 1 does not move to the home position. After that, when the user attempts to retry the aspiration, a significant error was displayed.
32. When both probes are disabled and a reagent drawer is queued to be scanned, the same trace was logged infinitely.

Trace:

```
[DEBUG] Scan Drawer
com.dg.dg59.scheduler.executor.impl.ScanReagentDrawerIncidenceOperation@a99b64
queued.
```

LIS

New functionalities

1. Demographic data will not be posted in DianaServer (DianaASTM) and DSC log files. Those fields will be always filled with *** in these log files.

```
...{0000003564} Receive: H|\^&||| WADIANA^00000735^ERY|||||P|1|20230525124927
...{0000003564} Receive: P|1|P00000002|||***|***|***|
...{0000003564} Receive: O|1|00000002||^ ^^DGW01J.gru^Erytra.pIn|R|||||||||
...{0000003564} Receive: L|1|N
```

Support Pack generation process continues executing the process, like in previous software versions, to anonymize these log files (to assure that the Support Pack will always contain anonymized data. To anonymize log files from previous software versions).

Resolved anomalies

1. The connection with LIS could not be successfully established on some occasion when the instrument was switched on and it was configured to use serial port (Host serial error #8 was found in the log files). The DianaServer or the instrument had to be restarted to reestablish the connection.

System startup has been improved to assure that, when the connection is set to serial port, the serial port had been initialized and are available once the DianaServer starts (system will wait up to 30 seconds, up to 15 attempts).

New traces related to this process will be posted in startup log:

```
...[DIANASERVER] DIANASERVER configured to use COM2 connection
...[DIANASERVER] DIANASERVER.START file exists.
...[COMPORT] Attempt 1 failed. COM port COM2 is not available. Retrying in 2 seconds...
...[DIANASERVER] COM port COM2 is available.
...[DIANASERVER] Starting DianaServer on COM2 - PASS
```

or

```
...[COMPORT] Attempt 15 failed. COM port COM2 is not available. Retrying in 2 seconds...
...[DIANASERVER] Failed to connect to COM port COM2 within the given time frame.
```

Or

```
...[DIANASERVER] Starting DianaServer with TCP/IP connection - PASS
```

Complaint 800012720 and 800012721.

2. System stopped transmitting results to the LIS (until system was rebooted) when the results of a test with a Not Readable (NR, no image available) microtube was requested to be transmitted to LIS.

Now the NR image displayed in the reports will be transmitted to LIS, if microtube images transmission is enabled.

Complaint 800008726.

3. On some occasions, when initializing the instrument, several identical error windows opened indicating that DianaServer Cannot create system shell notification icon.

Now no error will be displayed on screen when the DianaServer cannot display its icon in the Windows task bar.

4. When LIS communication is set to TCP/IP or serial, DianaServer failed to remove demographic data included in a frame when this frame first characters is the P of the record type.

This would happen if the LIS was configured to build a different package for each line of the message. On the other hand, it would have also been reproducible if the message exceeded the maximum length of the package, which was different depending on Diana Server's configuration:

Mode	size (characters)
Serial port	240
TCP/IP	64.000

Complaint 800014790

Card Reading

Director.dll file has been updated to version 5.10.0 (from version 5.9.0).

New functionalities

1. Conversion factor will be estimated and compared with the conversion factor determined during the adjustment in each integrity check and reaction grading image acquisition.

These values will be posted in the DioneExpert log file.

```
...{0000002848} [WARN ] DIONE Acquiring image 713002240850144298_00000362_12_GG_DGG_R.PGM...
```

```
...{0000002848} [WARN ] DIONE 713002240850144298_00000362_12_GG_DGG_R:
Performing reading to image file: 713002240850144298_00000362_12_GG_DGG_R.PGM
```

```
...{0000002848} [WARN ] DIONE 713002240850144298_00000362_12_GG_DGG_R:
Conversion factor: 20.8
```

```
...{0000002848} [WARN ] DIONE 713002240850144298_00000362_12_GG_DGG_R: Red
factor: 1.75
```

```
...{0000002848} [WARN ] DIONE 713002240850144298_00000362_12_GG_DGG_R: Blue
factor: 1.30597
```

```
...{0000002848} [WARN ] DIONE 713002240850144298_00000362_12_GG_DGG_R:
LOCALIZATION: W1(369,577) W2(183,580) m=-0.6161°
```

```
...{0000002848} [WARN ] DIONE 713002240850144298_00000362_12_GG_DGG_R:
Conversion Factor (Current: 20.80 px/mm; Estimated: 20.67 px/mm; Estimated_Error: 0.13
px/mm)
```

```
...{0000002848} [INFO ] DIONE Performing reading of microtube 1
```

```
...{0000002848} [WARN ] DIONE 713002240850144298_00000362_12_GG_DGG_R (1):
FINAL VOLUME: 404 pixels
```

```
...{0000002848} [WARN ] DIONE 713002240850144298_00000362_12_GG_DGG_R (1):
FINAL VOLUME: 19.42 mm
```

Complaints 800002531 and 800003425.

2. Microtubes with a small pellet that were near the threshold of No RBC classification and that were graded as negative now will be classified as microtubes with No RBC.

Complaint 800004568.

3. The commands sent to the reader will now be posted in the DioneExpert log file. For example:

```
...[INFO ] DIONE Sending command <CMD_VERSION> (237). Waiting for reply...
```

```
...[INFO ] DIONE Sending command <CMD_EEREAD> (165). Waiting for reply...
```

```
...[ INFO ] DIONE Sending command <CMD_BACKLIGHT_SETVALUE> (83). Waiting for
reply...
```

```
...[ [INFO ] DIONE Sending command <CMD_BACKLIGHT_READVALUES> (85). Waiting for
reply...
```

```
...[ [INFO ] DIONE Sending command <CMD_QUERY> (161). Waiting for reply...
```

```
...[ [INFO ] DIONE Sending command <CMD_BACKLIGHT_READTIME> (95). Waiting for
reply...
```

```
...[ [INFO ] DIONE Sending command <CMD_EEREAD> (165). Waiting for reply...
```

And in case there is an exception, more information will be posted in the log file

4. The illumination/backlight usage time, in minutes, will now be posted in the log file each time the reader is initialized:
...[INFO] DIONE Sending command <CMD_BACKLIGHT_READTIME> (95). Waiting for reply...
...[INFO] DIONE Response received.
...[WARN] DIONE PCA Illumination Time Use (minutes): 4407
5. The maximum allowed tolerance for the distance between microtube bottom and the card reference line (which permits the ability to check that the microtube bottom has been found successfully) has changed from 0.4 to 0.5mm.

Resolved anomalies

1. The current/illumination and white balance adjusted values during the reader initialization were kept when the initialization checks failed. Now, the initial values (the values stored/set in previous successful adjustment) will be restored.
2. The Microtube localization process (microtube bottom) has been improved to successfully detect it (and to not generate an error) when:
 - a. There is an artifact outside the microtube but in contact the microtube wall.
 - b. The microtube wall is particularly thick in the image.

Complaint 800008726.

3. The logs files including tables (DG421_IntegrityCheck and DG421_Reading) have been modified so all the data remains aligned and can be easily read, when they contain cards with 18 and 26 digit barcodes.

Importer

New functionalities

1. Importer will rollback the changes made in the databases when a Reagent Pack importation error occurs.

A backup of the database section containing Reagent Pack data and profile configurations will be automatically created when importing an Reagent Pack and it will be automatically restored if an importation error occurs.

This temporary backup is created in a temporary folder named tmp in E:\Program Files\Importer. The directory and the backup are deleted once the importation ends.

Complaint 800006764.

2. Importer has been prepared to show in the Not Unitary Profiles creation wizard only the interpretations that are allowed, according to the profiles selected and the IH Reagent Pack information.

3. Importer will not allow the creation of a Not Unitary Profile including just one technique. The Next > button (and Finish button, if applies) will now remain or be disabled when only one technique is currently selected.
4. Importer will not allow the assignment of a Displayed Name or a Not Unitary Profile name matching an already existing Displayed Name or Not Unitary Profile name (so two different profiles will not be displayed with the same name in the worksheet). An error message will be displayed. Both lists of names will be checked when requesting to assign a new name.
5. Importer log file management has been improved. Maximum file size has been increased from 2MB to 100MB. Once the log reaches this size, it will be compressed, moved to a subfolder named Backup, and renamed to:

Importer.log.YYYY-MM-DD-HH-mm-ss.log where YYYY=Year, MM=Month, DD=Day, HH=hour (in 24 hours format), mm=minutes and ss=seconds.

The last (more recent) 10 compressed logs will be maintained.

The Importer log files will now be included in the Support Pack (new folder named Importer in Support Pack Erytra\Log folder)
6. The time required to import the Reagent Pack has been reduced. The interpretations importation step has been improved.
7. The system imported all the profiles found in a Reagents Pack generated from a IH Templates Pack when all the profiles included in that Reagents Pack could not be executed in Erytra. This case has been filtered and not executable techniques will not be imported.

Resolved anomalies

1. Language Pack was not successfully imported. No error was displayed during importation, but new or modified messages were not displayed in user interface.
2. A message indicating that the list of profiles had not changed was displayed, and the list being displayed was not updated, when the Get Profiles button was pressed after creating a Not Unitary Profile.
3. The system did not trace in the log file when the list of techniques that can be executed in the instrument was not found in the Reagent Pack.
4. If a profile had a different name (translation) in a given language, it was displayed in the Profiles tab - Displayed Name field.
5. Profile Displayed Name, once set, could be modified but not removed (set to empty/deleted). An error was displayed when attempting to remove the displayed name.

Software installer

New functionalities

1. System configuration will be actively set to avoid the possibility that user application can be executed in OS Shell or Maintenance Mode.
2. Database engine (PostgreSQL) logs have been enabled. These new logs will be found in a new folder named pg_log in main log folder E:\log.

The log file name is postgresql-YYYY-MM-DD_HHmm.log, where YYYY=year, MM=Month, DD=Day, HH=hour and mm=minutes when PostgreSQL started.

The log files of previous days are compressed in a zip file at system start up in files named postgresSQL_YYY-MM-DD.zip, where YYY=year, MM=Month and DD=Day.

The logs older than 14 days are deleted.

The management of these log files is posted in startup.log (E:\Log folder):

...[LOGROTATE] Compressing log files.

...[LOGROTATE] Log files compressed and deleted successfully.

...[LOGROTATE] Deleting old zip files.

...[LOGROTATE] Old zip files deleted successfully.

...[LOGROTATE] Log rotation completed successfully.

3. 7zip installation has been removed from the installer as it is no longer used to compress files. A java library will be used instead.
4. Software installer will not include/copy Reagent Language Pack file. The updated version of this file is always included in the IH Reagents Pack and the IH Templates Pack.
5. Newer versions of following third parties will be installed:
 - PostgreSQL: 16.3 (from 10.5)
 - Java RunTime environment (from 1.8.0.171): 1.8.0.211.
 - Apache Tomcat: 9.0.85 (from 8.4.4)
 - Notepad++: 8.6.2 (from 8.4.9)
 - AutoHotkey (from v1.1.34): 2.0.2
 - Liquibase: 4.26.0 (from v3.5.5)
 - Adobe Acrobat Reader: 23.8.20533 (from v11.0.2017).
 - Apache Tomcat (from 8.5.31): 8.5.86.

Resolved anomalies

1. User interface (Comfort) keyboard response might lag, and a key may get stuck (the character was displayed as continuous key pressing) when the key was pressed very fast.

Installer will now:

- set to enabled the option Type characters when keys are released (On-Screen keyboard tab)
- set Repeat delay (msec) to 10 (instead of 500, On-Screen keyboard - Keys - Expert settings... - Keys subtab).

Complaint 800011977

Operating System and Boot Application**New functionalities**

1. OS image includes Windows Security Updates up to March 2024 (Windows version 10.0.17763.5576):

Name	Description
KB5027131	Cumulative Update for .NET Framework 3.5 and 4.7.2 for Windows 10, version 1809: June 13, 2023. Security and Quality Improvements.
KB5027222	OS Build 17763.4499: June 13, 2023. This security update includes following improvements: • This update addresses an issue that affects the Storage Spaces Direct (S2D) cluster. It might not come online. This occurs after a periodic password rollover. The error code is 1326. • This update addresses an issue that affects the Appx State Repository. When you remove a user profile, the cleanup is incomplete. Because of this, its database grows as time passes. This growth might cause delays when users sign in to multiuser environments like FSLogix. • This update addresses an issue that affects the Windows Remote Management (WinRM) client. The client returns an HTTP server error status (500). This error occurs when it runs a transfer job in the Storage Migration Service. • This update addresses an issue that affects signed Windows Defender Application Control (WDAC) policies. They are not applied to the Secure Kernel. This occurs when you enable Secure Boot. • This update addresses an issue that might affect the Local Security Authority Subsystem Service (LSASS). It might close sporadically. The system logs the exception 0xc0000710 in the Application Error event 1000. Because of this, the domain controller restarts unexpectedly. This issue affects read-only DCs (RODC) that also run Microsoft Defender Advanced Threat Protection (ATP). • This update addresses an issue that affects the Windows Kernel. This issue is related to CVE-2023-32019. To learn more, see KB5028407.
KB5005112	Servicing stack update for Windows 10, version 1809: August 10, 2021 This update makes quality improvements to the servicing stack, which is the component that installs Windows updates. Servicing stack updates (SSU) makes sure that you have a robust and reliable servicing stack so that your devices can receive and install Microsoft updates.

Name	Description			
KB5033904	Cumulative Update for .NET Framework 3.5 and 4.7.2 for Windows 10, version 1809: January 9, 2024.			
KB5035849	OS Build 17763.5576: March 12, 2024. This security update includes improvements. When you install this KB: The update affects Active Directory domains that host mobile device management (MDM) providers. They can transition from "Compatibility mode" to the strong certificate mapping "Enforcement mode." To do this, they can allow an Active Directory Key Distribution Center (KDC) to read user security identifiers (SID) from the Subject Alternative Name (SAN). Then, the providers can populate those values. To learn more, see: - KB5014754: Certificate-based authentication changes on Windows domain controllers. - Preview of SAN URI for Certificate Strong Mapping for KB5014754 - CVE-2022-34691, CVE-2022-26931, and CVE-2022-26923			

2. Operating System image (zip file and installation process) has been modified to manage different devices (motherboards,...).

PC Drivers have been removed from Operating System image and have been included in specific drivers files:

PC ASRock IMB-150D drivers file name is imb-150d-drivers-1.1.0.jar.

PC ASRock IMB-1003J drivers file name is imb-1003j-drivers-1.2.0.jar.

This drivers file matching instrument PC shall be copied in the Driver folder of the OS image to restore. This file will be expanded and the integrity of the files to install will be checked.

OS installation process, to install appropriate drivers, will now request to indicate the hard drive type/PC model:

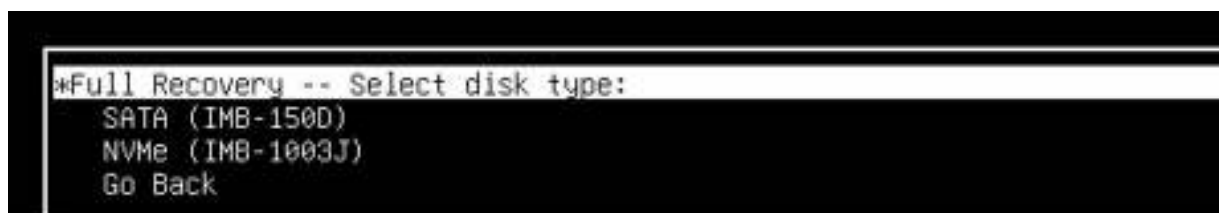


Figure 724 - Installer motherboards selection dialog

Procedures:

- See "Restoring from Boot Application menu" section (modified)
- See "Restoring from USB pen drive" section (modified)



NOTE

Erytra will only use IMB-150D PC.

3. A Boot Application entry can now be found in Start to allow returning to Boot Application from OS Shell mode without the need to restart the system. To return to the point where the is displayed (and to continue user application start up or allowing to enter Boot Application again).

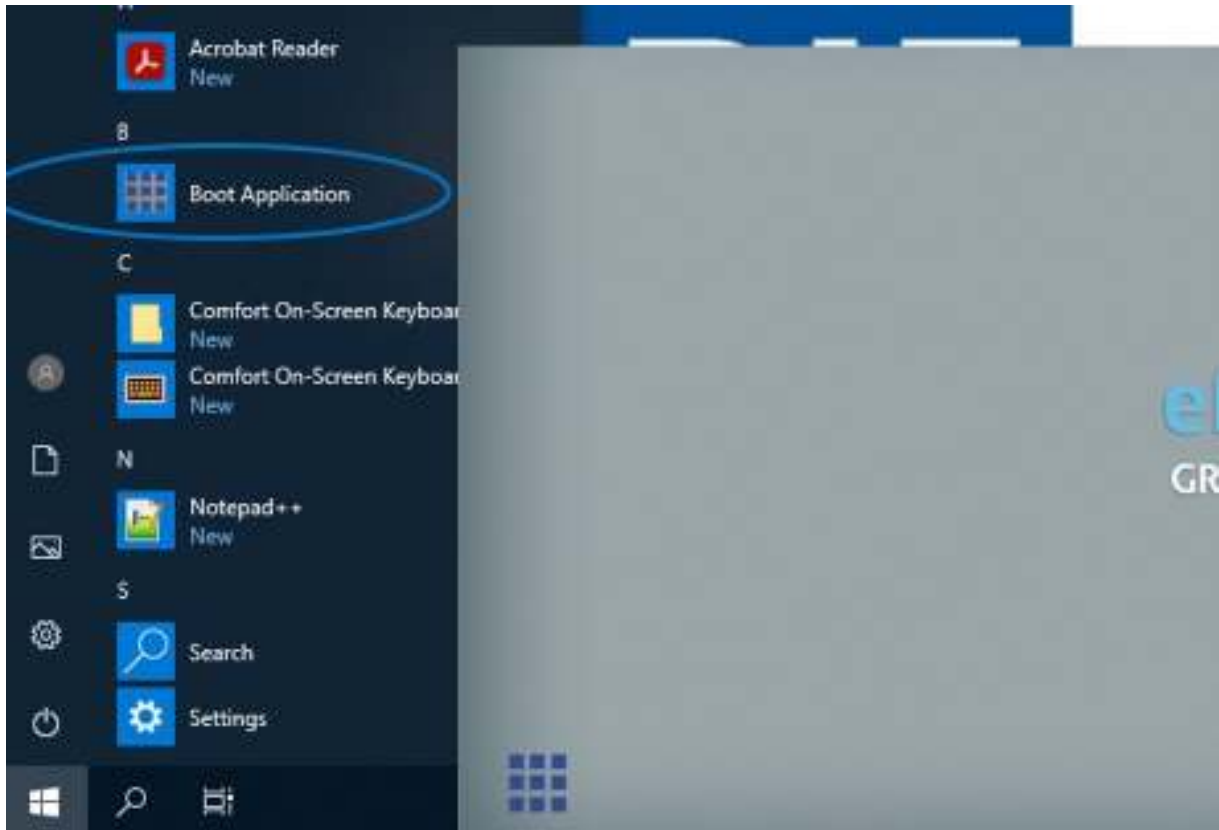


Figure 725 - Return to boot application entry point

Procedures:

- See "Return to Boot Application" section (modified)

4. The system will now display a warning message to the operator at start up when a potential unexpected time change has occurred in the previous session.

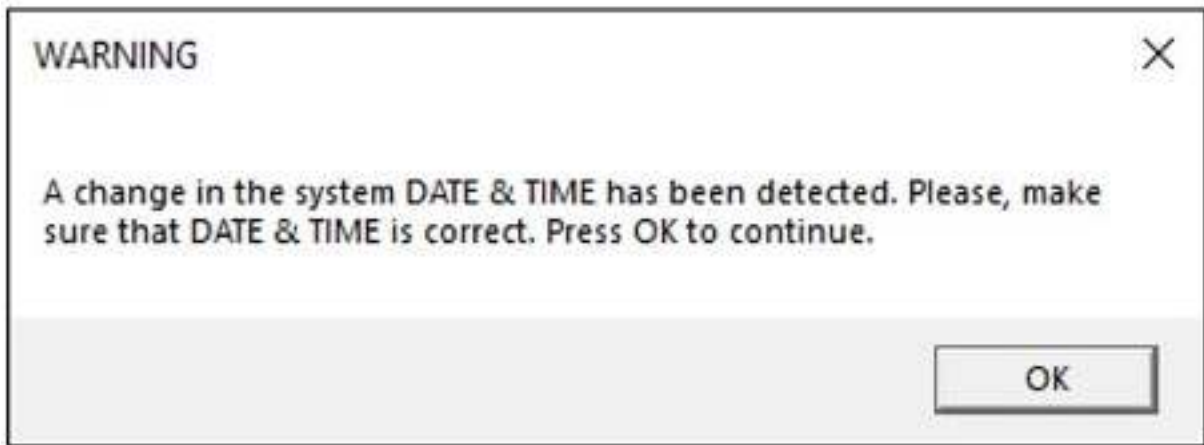


Figure 726 - Potential unexpected time change warning

5. The Boot Application and Operating System information will now be registered in a new log named system (E:\log). The startup.log will not be used anymore and installation.log will just include the traces related to software installer

Procedures:

- See "Description of log files" section (modified)
6. The touch screen calibration will now be preserved when requesting OS Full Recovery.

Resolved anomalies

1. The system failed to map a network drive using the mapping scripts provided with the Operating System and instrument software when the password defined in the script included the character "=". Character "=" is now allowed in any of the MappedDrive file variables (Path, username or password).
Complaint 800010508.
2. The disk/files protection (Windows Unified-Write-Filter) was not automatically enabled in the next reboot when it was (unexpectedly) found disabled.
3. The Boot Application could not correctly manage that a service application was used (Importer, Technical Service,...) and could turn unresponsive or fail to start user application when requested.
4. The File Explorer context menu (mouse right button) was not available when entering Maintenance Mode just after closing the Technical Service software that had been launched from the Boot Application.

5. The Grifols Secure Remote Connectivity tool agent was not removed from recovery partition folder (R:\BeyondTrust) once it had been executed, once OS Full Restore or OS Recovery from Boot Application had been executed.
6. The E drive was available in the user interface after the feature to allow qualified Field Service Engineers to access Windows File Explorer had been used (and closed) or service applications (Technical Service, Importer) had been previously used.
7. The installer did not check the presence of the PC model drivers files in USB:/Drivers. A control of the presence of Drivers artifact under usb:/Drivers has been added.
8. When attempted mapped drives configuration including '=', the network unit was not configured.

Instrument backup and restore

New functionalities

1. The file containing the destination path of the instrument backups (backup.conf file in E:\Program Files\System Tools directory) will now be included in the list of application configuration files that are considered part of the instrument configuration.

This file will be (always) included in Full and Customized instrument configuration backups.

The Information window of the backup application has been updated.

The restore application will now display a new option to select the restoration of this file:



The Information window of the backup application/Restore instructions have been updated.

2. Testcounter.txt file has been included when performing an export (backup) using the Import application.

Procedures:

- See "Backing up databases and relevant data with boot menu" section (modified)

Resolved anomalies

1. The description of a given backup indicated that it included barcode readers configuration files even when these files did not exist, and they were not included in the backup.
2. The backup generated with Backup/IHExport tool when requesting to include just Reagents Pack and profiles configuration did not include all the necessary information. This led to incorrect performances if restored: the configured profiles displayed names and special results were not restored, it was not possible to create Not Unitary Profiles, it may not be able to import new tests,...

Technical Service**New functionalities**

1. Leak tightness test has been improved. CPD values before creating pressure or vacuum will now be checked and the allowed margins have been reviewed and take into account the effects of instrument location (altitude). Now it is more similar to Eflexis leak tightness test.

The leak tightness now:

- a. Empty the fluidic circuit (in Technical Service software → Fluidics tab → initializing fluidic option and then fluidic empty option).
- b. Remove racks of sample drawer 1.
- c. Run the verification in Technical Service software, Procedures tab → Checks section → Others subsection → Leak-tightness button. A wizard window will be opened:

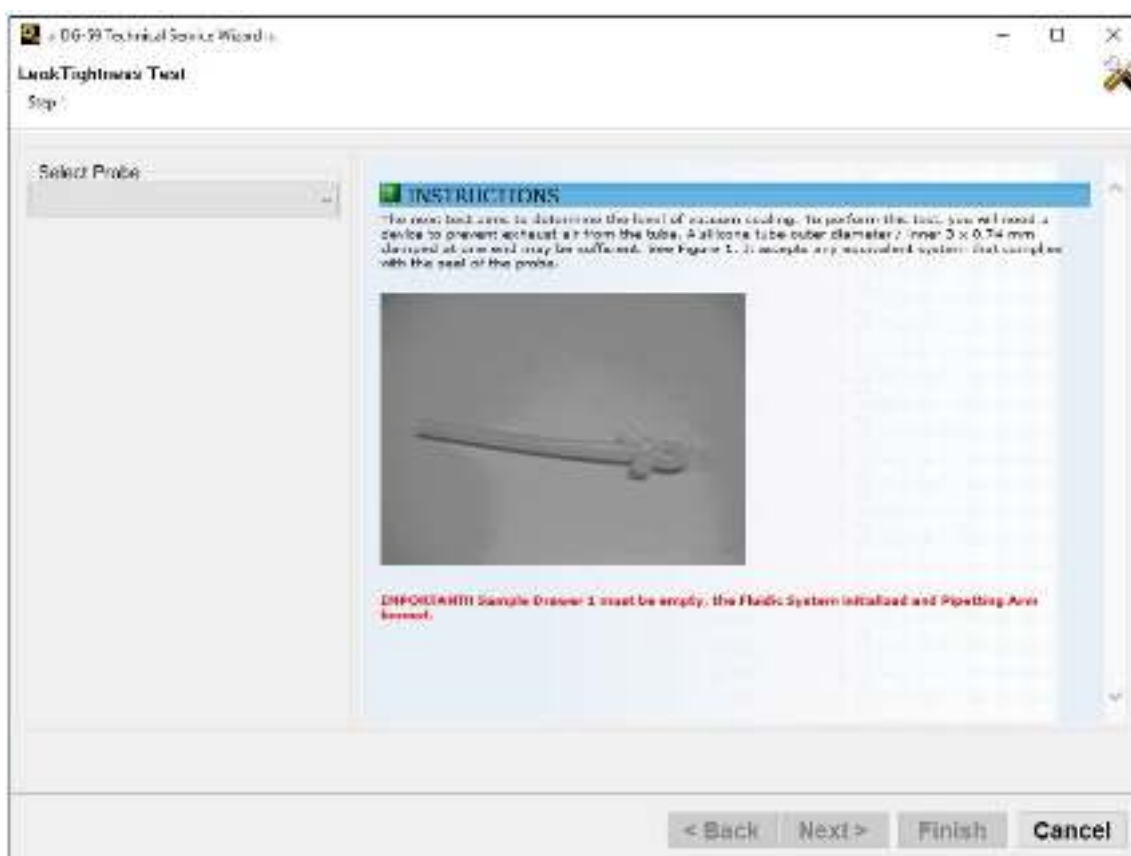


Figure 727 - Leak Tightness step 1 window

- d. Select the probe of the circuit to be checked and press Next> button.
- e. Press Next> button and then OK in the warning message informing that the instrument will perform movements/settings to empty the fluidic circuit between the dilutor and the probe selected. PIP robot will move to rack 2 position of sample drawer 1, it will move down the probe and it will empty the dilutor. Then it will check if CPD value is between 17000 and 21000 counts.

- f. If the above condition is met, the system requests to block the end of the probe so that air cannot enter into the probe. It is recommended to use a piece of silicone tube of 3x0.74 mm (external / internal diameter) tied by one of its sides for this purpose. Place the tube inside the probe and make sure that there is no play between the probe and the tube.
- g. Press Next> button to start the verification. The instrument will perform an aspiration of 1ml from the probe, so that vacuum is created inside the circuit. The system will check whether CPD value is between 6000 and 13000 counts and that the first CPD value minus the current one is less than -6000 counts. Otherwise an error will be shown on screen.
- h. If CPD counts are within margins, the system will check the CPD value on that moment and 30 seconds later. Then it will calculate the difference between these 2 values, which should be smaller than 220 counts.
 - If the differences is more than 220 counts but less than 376 counts, it shall be considered a warning.
 - If the previous test fails, fluidic circuit between dilutor and the probe should be checked.
 - If the test is passed, a new step in the adjustment wizard appears on the screen requesting to remove the blockage from the probe. Remove it and press Next> button.
- i. The instrument will then perform movements/settings corresponding to an empty procedure of the fluidic circuit between the dilutor and the probe. At this point, CPD value should be between 17000 and 21000 counts.
- j. If the above condition is met, an aspiration of 1ml (valve input position) will be made and the software will ask for the silicon tube to be placed again in the probe end so that air cannot enter the probe. Place the tube inside the probe and make sure that there is no play between the probe and the tube.
- k. Press Next> button in order to start the verification. The instrument will perform a dispensation of 1ml (valve output position), so a pressure is created inside the circuit.
- l. The system will check whether CPD value is between 26000 and 38000 counts and that the value of the first CPD value minus the current one is greater than 5700 counts. Otherwise an error will be shown on screen.

- m. If CPD counts are within margins, the system will check the CPD value on that moment and 30 seconds later. Then it will calculate the difference between these 2 values, which should be smaller than 220 counts.
 - If the differences is more than 220 counts but less than 376 counts, it shall be considered a warning.
 - If the previous test fails, fluidic circuit between dilutor and the probe should be checked.
 - If the test is passed, a new step in the adjustment wizard appears on the screen requesting to remove the blockage from the probe. Remove it and press Next> button.
- n. A screen appears showing the parameters obtained during the test. Press the Finish button and OK on the information window (probe will be homed).
- o. Press OK in the confirmation window to finish the verification procedure.

2. Incubator temperature adjustment has been improved. The wizard will now display a message indicating if the incubator is adjusting the temperature still or if it is already stabilized, and the reading of both incubator temperature sensors will be displayed:



Figure 728 - Adjusting incubator step 2 window while regulating temperature

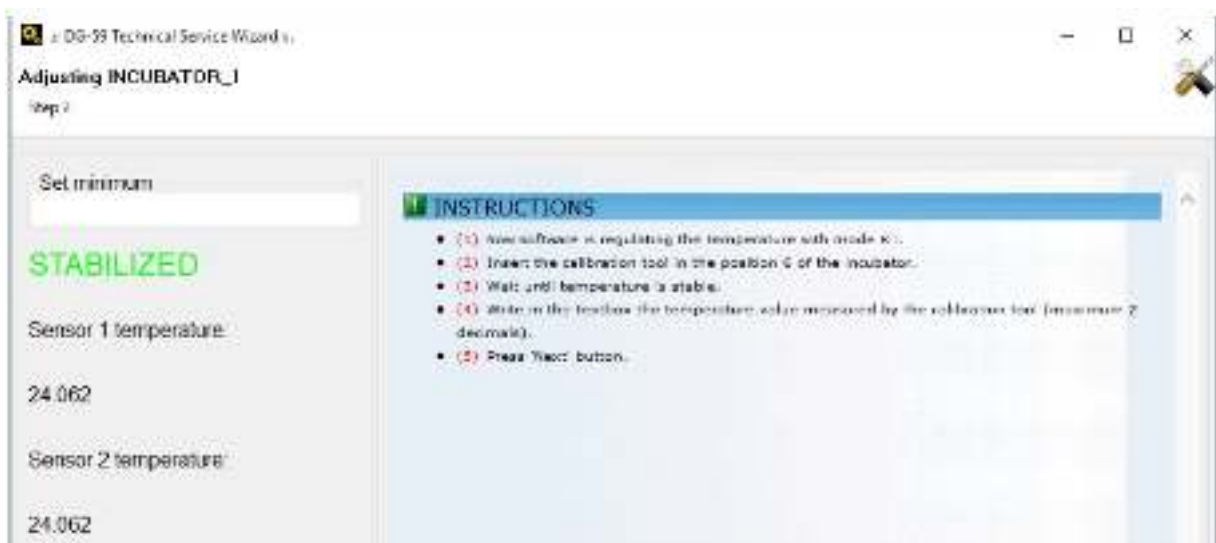


Figure 729 - Adjusting incubator step 2 window while temperature stabilized

The temperature of the incubator is considered as stabilized:

- in mode RT when it does not vary more than 0.4°C in 3 minutes.
- in mode 37°C when its temperature does not vary more than 0.2°C in 5 minutes.

If the temperature is introduced and the Next > button is selected to save it when the temperature is not considered stabilized, a warning message will be displayed:

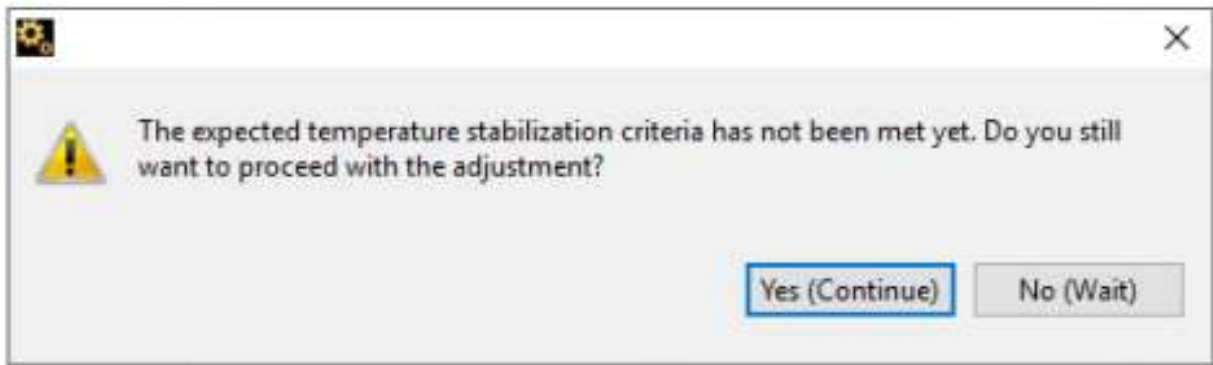


Figure 730 - Temperature stabilization criteria not met message

- When Yes (Continue) option is selected, the action will be logged in DEBUG.technical_service log file with the temperature values that caused the incubator's temperature not to be stabilized.
...INFO com.dg.dg59.ts.model.TemperatureStabilityMonitor [ModalContext] The expected temperature stabilization criteria has not been met yet. User answered YES.
...INFO com.dg.dg59.ts.model.TemperatureStabilityMonitor [ModalContext] The elapsed time regulating the incubator in RT mode is less than 3 minutes
- When No (Wait) option is selected, the warning window will close, and current step temperature regulation will continue. The temperature values that were previously entered in the textbox will be erased.

Resolved anomalies

1. When a macro that performs aspiration with an specific profile was executed in TechnicalService, then the used profile was incorrect.

Firmware

PIP PCA has been modified.

New functionalities

1. New PIP PCA firmware version 4.10.0.

New firmware allows:

- a. Air aspiration detection using CPD values has been implemented. Refer to point 1 of Erytra user interface new features section 1.2.1 for more details.
- b. New firmware has been prepared to register level detection values (level detector and motor encoder counts). Actually, this function is not used, the obtained values are not posted in any log file.

Erytra Diagnostic

New functionalities

1. Pressure circuits test only checked the fluidic section of Solution A container in drawer 1 and Solution B container in drawer 2. If the instrument was configured to use an external drain, the fluidic section of Solution A container in drawer 4 and Solution B container in drawer 5 were not tested, as the selected container did not change from the first one to the second one.
2. The test will now:
 - a. Check solution and waste containers levels.
 - b. Deactivate all electrovalves except the ones related to container switching.
 - c. Switch on PM1.
 - d. Wait 500 ms.
 - e. Check that sensor S3 pressure value is within the range 170-150 mbar.
 - f. Open EV20.
 - g. Wait 30 s.
 - h. Check that sensor S3 pressure value is within the range 170-150 mbar.
 - i. Switch off PM1.
 - j. Close EV20
 - k. Switch on PM3.
 - l. Wait 500 ms
 - m. Check that sensor S4 pressure values is within the range 170-150 mbar.
 - n. Open EV19.
 - o. Wait 30 s.
 - p. Check that Check that S4 pressure value is within the range 170-150 mbar.
 - q. Close EV19.
 - r. Switch off PM3.
 - s. If the system is configured in dual mode/external drain, select Drawer 4 for Solution A and Drawer 5 for Solution B and execute again the steps c to q.

Every sensor value is checked when its value is stabilized (5 seconds with a variability of +/- 20 mbar). An error is generated if sensors values are not stabilized in less than 50 seconds.

3. Incubator movements test in Diagnostic application have been improved in order to mitigate "Current limit" errors reported during test execution.

In previous versions, the test was executing 25 movements up/down with incubators 1 and 2 (with a delay of 5 seconds between each other) and then the same 25 movements with incubators 3 and 4 in the same way. To avoid impacts of the incubators with the CAT robot arm, it was moved on top of the centrifuges.

Incubator movements test should execute 25 movements up/down with incubators 1 and 3 (with a delay of 5 seconds between each incubator) and then the same 25 movements with incubators 2 and 4 in the same way (with a delay of 5 seconds between each incubator). The positioning of the CAT robot arm should be changed to the "Card Dawer 4, Rack 1 and Card 12" position when making the movements of incubators 1 and 3, and to the position "Card Dawer 1, Rack 1 and Card 1" to prevent the incubators from impacting the CAT robot arm.

Start and End of each block of movements (Incubator 1/3 and Incubator 2/4) should be logged

Resolved anomalies

1. If the Erytra Diagnostic application was opened in French language, the "Incubators Movement" and "Centrifuges Movements" tests name were the same.
 - The "Incubators Movement" test name was displayed as "Mouvements des centrifugeuses". This error is because reference was made to the centrifuges instead of incubators.

"Incubators Movement" and "Centrifuges Movements" tests name have been modified to "Mouvements" to maintain consistency with the tests name in all languages. In the rest of the languages the tests name is "Movements" in their respective languages.

Manager, Controller and user databases and Erytra Database Manager

New functionalities

1. Erytra Database Manager has been updated to software version 5.2.0.
2. Erytra Database Manager will check if a Blood Typing Manager is already installed in that computer. Should this be the case, following error will be displayed and the installation will be aborted.

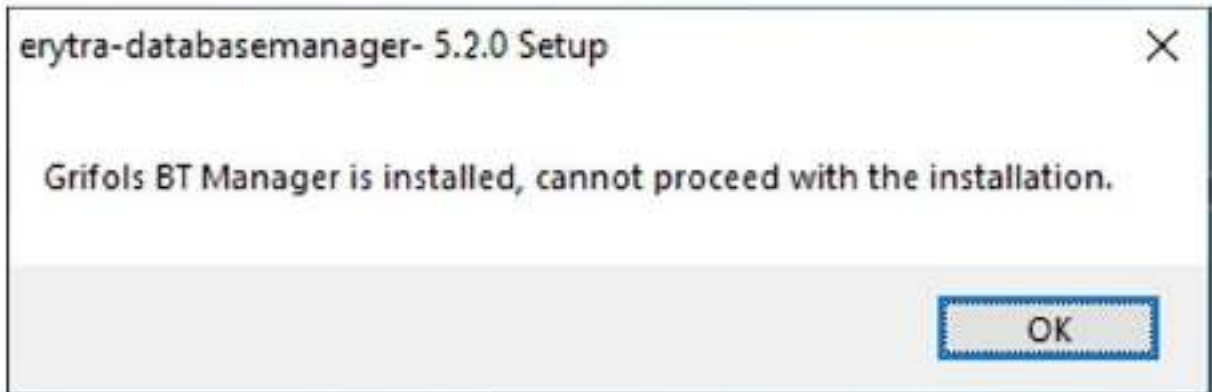


Figure 731 - BTM is already installed message

Nowadays the PostgreSQL version and configuration of both software are not compatible.

Resolved anomalies

1. Erytra Database Manager was not able to update databases using the updated.bat provided. Now updatedb_edm.bat and createdb_edm.bat have been created similarly to Erytra Database Manager.
2. Web application directories were not reset (deleted) during web application server (Apache Tomcat) installation step.

Alternative browser

Alternative browser has not been modified and it is the same as the one delivered with software version 5.1.0.

Alternative browser is not required for Erytra Database Manager.

Remote connection requires a web browser with Flash Player. The alternative browser should be installed on the computers for which the remote connection is not available, due to the external computer browser's version or configuration.

Manager, Controller and user databases and Erytra Database Manager

Erytra Database Manager has been updated to software version 5.2.0.

Erytra Database Manager version 5.2.0 cannot be installed if a previous version of Erytra Database Manager is already installed. The previous version shall be uninstalled (and a database backup created, if necessary, before uninstalling the previous version).

Software version 5.2.0 databases shall be restored in Erytra Database Manager version 5.2.0. The compatibility with previous Erytra Database Manager or software versions are not guaranteed.

Previous software versions databases shall be updated when they are restored in an Erytra instrument (by running the **E:\Program Files\liquibase\updatedb.bat**) or Erytra Database Manager (by running the **installation_directory\Grifols\Manager\Program Files\Liquibase\updatedb_edm.bat**).

7.1.5 Erytra Software v5.2.1

7.1.5.1 Software changes

This chapter describes Erytra Software version 5.2.1 improvements and modifications compared to version 5.2.0

General

New functionalities

1. Pre-analytical barcode reading process.

When scanning a tube barcode from a sample rack, if the system identifies the rack as pre-analytical or detects that the rack label lacks a barcode during the initial scan, it will automatically initiate a retry. This retry will proceed in the reverse direction, operating at a reduced speed and with a narrower reading window. During this process, all tube positions within the rack will be scanned to ensure complete barcode detection.

In case there is a mismatch between the two readings, a 'Barcode not Identified' incidence will be assigned to the tube. This tube will be able to be manually identified.

New adjustment and verification procedure has been implemented. In case the 'horns: pre-analytical' adjustment has not been performed, the identification of the rack will be stopped and the following message will be shown:

- The barcodes of samples identified in the pre-analytical compatible samples holder could not be read automatically.

The barcode reader adjustment in this type of holder is missing.

Please contact your local Grifols service representative.

Complaint 800017189

2. A new check based on level detection to detect dripping has been implemented. This check will be performed immediately after a probe wash following the completion of the below-listed actions:

- A level detection in a reagent vial.
- A level detection on dilution cup during the first step of the dilution.
- An aspiration of sample, reagent, diluent and cleaner from a tube or vial.
- A dilution aspiration from the dilution cup.
- A fluidic inactivity period.
- A pipetting final wash.

The new check basically compares level detection counts taken before the wash and a short time after solution B has been used to clean the probe.

To avoid false positive detections, the check will be repeated up to two additional times. If the difference in counts remains below the threshold on the third attempt, an error will be displayed.

If this new check detects dripping above a tube or vial, the related test will be canceled, the probe will be disabled, the affected tube or vial will be marked with an incidence. Two error window messages will be displayed with only "OK" option to accept the message informing the user that a possible dripping has been detected and that the probe has been disabled.

If this new check detects dripping above the dilution cup, the corresponding test will be canceled, the probe will be disabled, and the dilution cup will be washed. Two error messages will be displayed, each with only an 'OK' button to acknowledge the alert. These messages will inform the user that possible dripping has been detected and that the probe has been disabled.

Complaint 800016541.

3. Language pack has been updated from v1.30 to v1.34, implementing the following:
- All messages related to 'Bleach', 'NaClO', 'sodium hypochlorite' have been replaced by 'decontaminant' due to the possibility to use alternative decontaminants.
 - Change in Portuguese Portugal language:
 - "Próxima intervenção do utilizador **cerca das** 12h54" instead of "**em cerca de**"
 - Change in Finish language:
 - "**ALKUPERÄINEN**" will be used instead of "**ALKUPERÄIN**"
 - Russian language has been removed.

4. The **Adaptive Scanning** feature of the barcode reader has been **disabled**. This feature could interfere with the barcode reader performance in certain situations, as the engine may switch to a narrow scan angle if the scan line does not touch a barcode at a shorter distance. Disabling Adaptive Scanning prevents the scan angle from transitioning to a narrow angle.

**NOTE**

Erytra instruments could be equipped with one of the following barcode reader models:

- SE-955 model.
- SE-965 model.

The **SE-965** barcode reader model features Adaptive Scanning, which automatically optimizes parameters to enhance decode performance based on the barcode's distance and ambient light conditions. When the barcode reaches a certain distance from the engine, the scan angle is automatically reduced to a narrow 10°. See **DG-0059#0078 'NEW BARCODE READER v2'** on the "Bulletin index" on page 1183.

Resolved anomalies

1. If a newly introduced reagent or card shares the same lot number and reagent ID or card ID as one previously used—despite being from a different year—Erytra was unable to distinguish between them. As a result, it incorrectly assumed that quality control had already been performed and did not prompt for a new QC check. This issue has now been addressed: Erytra will distinguish reagents or cards based on their expiration dates and will accurately detect when quality control (QC) is required for newly introduced lots.

Complaint 800003071

2. After a the incubation of a test ends, the system verifies the incubation time to ensure it was completed correctly. Normally, incubation starts right after the last dispensation into a microtube, even though some pipetting-related tasks (like probe cleaning) may still be ongoing. The system typically uses the end of pipetting as the reference point for verification.

However, if there is a delay—such as an interaction with the Laboratory Information System (LIS)—the pipetting end time is recorded much later than the actual incubation start. This causes the system to mistakenly think incubation started late, leading it to incorrectly flag the incubation time as insufficient, even though it was actually correct.

After the incubation of a test ends, the amount of time spent incubating is verified to ensure that the incubation has been properly finished. However, it was possible for some interactions with LIS or some other process to affect this verification.

The system now calculates incubation time starting from the exact moment it is pipetted. This ensures a more accurate incubation time calculation.

Complaint 800014689

3. After a decontamination process, the instrument begins operations without priming the full circuit, preventing the correct performance of the probe and leading to the dispensation of air until enough probe washes are performed to restore the fluidic circuit to its proper state. Consequently, this ends in volume-related test incidents and NR results.

Now the system will be identified as 'not primed' after a decontamination process and it will be primed when needed.

Complaint 800018903

4. The following improvements have been implemented to manage 'Channel Fault' events:

- To prevent 'Channel Fault' errors during validation and export processes caused by multiple instances, the graphical interface will receive execution feedback to avoid triggering redundant instance submissions.

Complaint 800014446

- Report generation process has been improved to reduce the required time and avoid 'Chanel fault' errors.

Complaint 800015183

- More information will be logged in order to properly investigate 'Channel Fault' events.
 - "Verifying connection after reconnection..."
 - "Ping message sent successfully. Connection appears to be stable."
 - "Reconnection successful. All systems operational."
 - "Failed to send ping message after reconnection. Additional issues may be present."
 - "The system reconnected but encountered further communication issues. Please verify."
 - "Checking if critical subscriptions are active after reconnection.."
 - "Checking if critical subscriptions are active after reconnection.."
 - "All critical subscriptions are active."
 - "Known Error 2032 detected. No immediate action required."
 - "Channel fault detected, failed attempts count:"
 - "Channel recovery failed after " + _failedAttempts + " attempts. Disconnecting and alerting user."
 - "The system has made multiple failed attempts to reconnect to the server. Contact technical support."
 - "Attempting automatic reconnection (Attempt " + _failedAttempts + "/"3)

Complaint 800015183.

5. When using the Support Pack generator application, the process that obfuscates patient demographic data could fail in a scenario where the communication with LIS is configured to serial port or TCP/IP and the patient demographic data is divided in two frames. This led to an error and an improper completion of the Support Pack creation.

Complaint 800015344.

6. During a dilution process, a level detection has been added to check that the dilution cup is empty the first time the dilution cup is accessed.
 - If liquid is detected at this stage, the system manage the existing liquid before proceeding with any dispensation. A second level detection will then be attempted.
 - If the level is still detected, the probe will be externally washed, and an automatic dilution cup wash will be performed. A third level detection will follow.
 - If the cup is still not empty after the wash, the probe will be washed again, and the following error message will be displayed:

“An error has occurred while executing a [X] profile. The dilution cup is not empty. The related profiles have been canceled.”

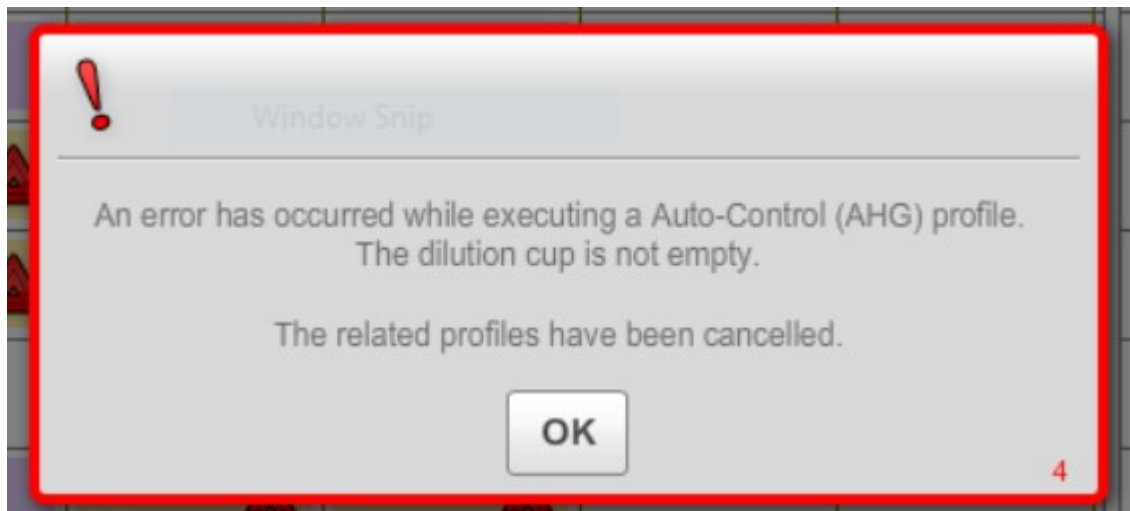


Figure 732 - Dilution cup not empty message

This message will include an “OK” button and a 45-second countdown timer. Regardless of whether the operator clicks “OK” or the timer expires, the system will cancel all techniques with pending dilution steps.

7. During startup, if the system experiences unexpected slowness, the compression of the erytraAir.log file may have failed with the following trace:

```
YYYY-MM-DD HH:MM:SS,000 {hiveStateTimerF} [KEY-ERROR] END
ArchiveTrigger with errors: Architecture must be bootstrapped using
Bootstrap.init(), or service with name
dg.architecture.filemanagement.Service is been searched before
architecture are fully initialized.
```

No error message was displayed to the operator, so the failure went unnoticed.

This failure has been solved by scheduling the compression of erytraAir.log to occur only after the system has fully started.

8. The contaminant reagent wash may use a lower volume of cleaning solution than expected if the system has just aspirated contaminant reagent for several profiles, and some of those profiles are later canceled.
9. During the inactivity wash, Solution B was not pumped as expected. Now dispensation of Solution B during inactivity wash cycles has been adjusted pumping the required Solution B from inactivity wash cycles.

Erytra User Interface

New functionalities

1. The following icons of the information of "About" screen have been updated to the current official ones.

- SN: Serial number
- REF: Catalogue number
- Consult instructions for use
- Manufacturer (black)
- Date of manufacture (white)
- IVD: In vitro diagnostic medical device



2. The system allows store and display demographic data in US market in the same way it does for CE market. The RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA parameter appear in the properties file with default value false fo US configuration.

3. "Fluid System Decontamination" icon has been updated by removing NaClO due to the terminology changes.



Figure 733 - New decontamination icon.

Resolved anomalies

1. When a test was canceled and automatically restored, the canceled instance remained and both were interpreted, sending incorrect results (only when auto-validation is active).

The interpretation algorithm has been improved to prevent unexpected results being generated and marking the segments related to the cancelled (and unexpectedly interpreted) test as "Not Interpretable (NI)".

Complaint 800020269

2. The generation of reports from the Maintenance tab of the user software could cause an unexpected spike of memory usage that led the user interface software to freeze at the moment of closing the report viewer, resulting in black screen. To improve this situation, Adobe Air has been updated to version 51.1.3.4.

Complaint 800016371

3. Short channels disconnections and Outofmemory error caused some unexpected behavior such as:

- graphical interface icons remain activated when they should not
- repeated NetConnection.cAll.Failed: HTTP: failed errors during sample identification.
- Worksheet tab is not updated.
- Reports not being shown and unresponsive screens

Communications between front end and back end and thread management had been improved to reduce occurrence of Out of memory and channel fault errors.

Complaint 800015739

Card Reading

New functionalities

1. After a detailed analysis and volumetry evaluation has been carried out, the final volume conversion factor updated. This improvement will decrease the possibility of an "Incorrect volume" warning in situations where the detected level was slightly below the expected margins.
2. New **DG421_IntegrityCheckPaths.log** in **E:\logs\DG-421** in order to better trace the source path of the microtube images.

QC

New functionalities

1. Cards and Reagents QC management has been improved. The system can now be configured to not use cards or reagents that have not been successfully pass QC test set.

The behavior is now configured by means of Importer.

The screenshot shows the 'Importer Setup' window with the 'QC Kits' tab selected. Under 'QC Settings', the 'QC new lot cards and reagent' dropdown menu is open, displaying a list of options: MANDATORY, FALSE, WARNING, and MANDATORY. The 'FALSE' option is highlighted. Other visible settings include 'Name' set to 'ERY625', 'Language' set to 'English (United Kingdom)', 'QC hours cadence' set to '2', and 'Autoverification QC Pass' set to 'ONLY VALIDATE'.

Figure 734 - 'QC new lot cards and reagents' configuration list on Importer Setup tab

- **False** (default option): if this option is configured, the system will not check if a QC has been passed.

- **Warning:** if this option is configured, when the 'QC_ENABLED' parameter is set to true and the card or reagent is used in a technique that has QC, the system will check if a QC test set with passed results has been completed and show a warning in case it has not.



Figure 735 - Warning configuration message

- **Mandatory:** if this option is configured, when the 'QC_ENABLED' parameter is set to true and the card or reagent is used in a technique that has QC, the system will check if a QC test set with passed results has been completed. In case it has not, the system will show a warning and will not use that lot of cards or reagents until this requirement is met.



Figure 736 - Mandatory configuration message

Affected cards or reagents will not be considered during the system planning, regardless of technique. The system will stop the queued techniques and activate the corresponding missing product warning icons.

This setting will be stored in erytra.properties new parameter '**QC_NEW_LOT_CARDS_AND_REAGENTS**' which will replace 'QC_NEW_LOT_WARNING_ENABLED' parameter.

**NOTE**

The new parameter 'QC_NEW_LOT_CARDS_AND_REAGENTS' has replaced the previous 'QC_NEW_LOT_WARNING_ENABLED' parameter to manage the Cards and Reagents QC lot warning.

When the user application starts, it will check if the parameter 'QC_NEW_LOT_WARNING_ENABLED' exists setting in the Software:

- if the parameter 'QC_NEW_LOT_WARNING_ENABLED' exists and is set to false, the system will set the new parameter 'QC_NEW_LOT_CARDS_AND_REAGENTS' to false and remove the old 'QC_NEW_LOT_WARNING_ENABLED' parameter from the configuration.
- if the parameter 'QC_NEW_LOT_WARNING_ENABLED' exists and is set to true, the system will set the new parameter 'QC_NEW_LOT_CARDS_AND_REAGENTS' to Warning and remove the old 'QC_NEW_LOT_WARNING_ENABLED' parameter from the configuration.
- if the parameter 'QC_NEW_LOT_WARNING_ENABLED' does not exist, the system will set the new parameter 'QC_NEW_LOT_CARDS_AND_REAGENTS' to false (default value).

Importer**Resolved anomalies**

1. When a newer version of a previously installed technique is imported, and the database contains another technique with the same name but different identifier, an error is raised at a certain point of the import process, aborting it, and making it impossible to successfully install the IH Reagents Pack. The error displayed indicated 'object references an unsaved transient instance'. The importation process has been improved to avoid this error.

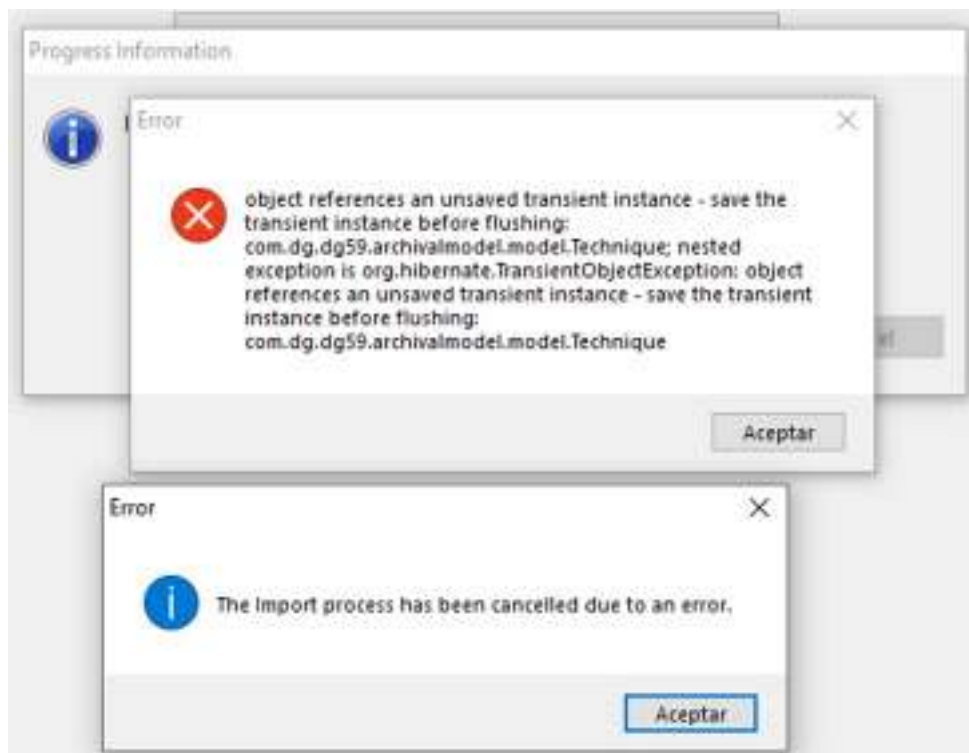


Figure 737 - Error message during importation

Complaints 800010740, 800016943

Firmware

New functionalities

1. CEN-SRM Firmware has been updated from 4.5.0 to v4.6.0 increasing the electronic cabinet fans PWM reference in order to improve fans start-up. this FW is being installed from factory since S/N $\geq$ 059-0001847
2. PIP Firmware has been updated from 4.12.0 to v4.13.0 implementing an improvement on the air aspiration detection for titration techniques: air aspiration detection thresholds have been adjusted to more accurately reflect the actual conditions of the machine.

LIS

Resolved anomalies

1. When importing work from the LIS, a "Host format error #10" occurred, and no test was executed. This error is triggered when the Diana Server attempts to process a file that is empty or contains no lines. To prevent this, a retry mechanism has been implemented to ensure the file is read instead of generating an error. A timeout error will arise if it is not still possible to read the file after a defined time.

Complaint 800014747

2. Diana server could be started a second time and the configuration screen was displayed until the operator accepted it, allowing changes to the configuration. Now the configuration screen will be close automatically.

Complaint 800017236

3. When a LIS message is split at beginning of PatientID field, some demographic data fields were not obfuscated.

Technical Service

New functionalities

1. New adjustment and verification procedures have been implemented to improve the barcode reading of the pre-analytical racks.
 - A new adjustment will determine the center position of the barcode window by performing two barcode reads in opposite directions. The window width will then be verified to ensure the adjustment was successful. This procedure has to be performed for each rack of each samples drawer. The new adjustment can be found in TSS path: 'Procedures tab → Adjustment → Pip floor Modules → horns: pre-analytical'. See "Procedures tab" on page 756.

**NOTE**

Check the barcode reader's laser beam alignment (See "Verifying barcode reader" on page 369) as if the barcode reader's laser beam is misaligned, the adjustment procedure may not complete successfully.

**NOTE**

For instruments S/N < 059-0001887 the barcode reader's laser beam alignment might not be possible to adjust properly due to HW limitations. See "Adjusting barcode reader inclination position" on page 348

In this cases it should be necessary to upgrade the instrument according to **INR-0059#0155 NEW PIP HORNS WALLS**. See "Bulletin index" on page 1183.

- A new verification procedure for the pre-analytical racks adjustment has been implemented to ensure the correct alignment of the horns barcode reader laser beam for each rack, in every drawer, one at a time. During this process, the system automatically positions and activates the barcode reader laser beam at the center of the reading window. This enables the technician to visually verify that the beam is both properly centered and fully contained within the reading window, ensuring accurate and reliable scanning performance. The new verification procedure can be found in TSS path: 'Procedures tab → Checks → PIP Module → PIP module Pre-analytical Rack - Laser alignment check'. See "Procedures tab" on page 756

Complaint 800017189

2. Technical Service logs size have been increased from 2MB and 1 backup to 100MB and 5 backups.

Erytra Diagnostic

New functionalities

1. Diagnostic application limits for 'Pipetting test - CPD and PCA level detector' test have been modified to homogenize them with factory tests. CPD sensor value limits changed from [17000-18400] to [15700-19700].

Others

Resolved anomalies

1. When Erytra executes archive process, the ErytraDebug.log was not archived. This failure has been solved by scheduling the compression of erytraAir.log to occur only after the system has fully started.

Alternative browser

The alternative browser has not been modified and remains the same as the version delivered with software version 5.1.0. It is not required for use with the Erytra Database Manager. However, remote connections require a web browser with Flash Player. In cases where remote access is not possible due to the version or configuration of the external computer browser, the alternative browser should be installed to ensure compatibility.

Manager, Controller and User databases and Erytra Database Manager

The Erytra Database Manager has not been modified and remains the same as the version delivered with software version 5.2.1.

This version cannot be installed if a previous version of the Erytra Database Manager is already present on the system. In such cases, the existing version must be uninstalled beforehand, and a database backup should be created if necessary.

Databases created with software version 5.2.1 must be restored using Erytra Database Manager version 5.2.1. Compatibility with earlier versions of the Erytra Database Manager or software is not guaranteed.

Databases from previous software versions must be updated before use. This can be done by running the **updatedb.bat** file located in **E:\Program Files\liquibase** when restoring in an Erytra instrument, or by executing **iupdatedb_edm.bat** located in **installation_directory\Grifols\Manager\Program Files\Liquibase \updatedb_edm.bat** when restoring in the Erytra Database Manager.

7.1.5.2 Application and firmware versions

Table 106 - Erytra v5.2.1 components versions

Name	Version	Description
Apache Tomcat	9.0.85	HTTP web server
Bootapplication	1.2.0	Grifols Applications launcher and Database Backup/Restore for Technical Service purposes
Diana System CE	4.6.0.38	Diana System build from v4.6.0 DG-57 project. (Configuration= CE)

Name	Version	Description
Diana System US	2.4.0.19	Diana System build from v2.4.0 DG-517 project. (Configuration= US)
Director.dll	5.10.1	Image processing library
DSC.dll (CE)	6.2.0	Diana System Connectivity. (Configuration= CE)
DSC.dll (US)	6.2.0	Diana System Connectivity. (Configuration= US)
Erytra-databasemanager-5.2.1	5.2.1	Erytra Database Manager
Erytra Diagnostic	4.15.0	Instrument check tool
ErytraSupplyOnOffSwitcher.exe	1.1.0	ErytraSupplyOnOffSwitcher
IHExport	1.9.0	It allows performing backup of instrument data as: databases, configuration files, operating system data, diana system data and logs.
IHImport	1.9.0	It allows restoring of instrument data as: databases, configuration files, operating system data, diana system data and logs.
IH-supportpack	1.7.1	Tool to generate a support pack.
Importer	4.15.0	Languages and techniques import tool
languagepack.zip	1.34.0	Language pack file
PostgresSQL	16.3	Object-relational database management system
Technical Service	4.15.0	Tool for instrument adjusting

Table 107 - Erytra v5.2.1 FW versions

PCA	Firmware version
CEN_SRM_CTRL	4.6.0
INC_CUP_CTRL	4.5.0
FLUCTRL59	4.6.2
CAT_CM_CTRL	4.5.0
PIP_CTRL	4.13.0
RD_IL421	5.5.0

7.2 Installation procedures

This section contains installation procedures of the different components of Instrument software.

7.2.1 Downloading Erytra files

Erytra software version and Operating System are distributed in electronic format only.

Download the files needed to install the required Erytra software version. Check the correspondent FSB/INR for each X.Y.Z version.

Before performing any installation it is necessary to verify the Hash of the downloaded files. The concret SHA is provided in the INR/FSB corresponding to the downloaded version. This hash has been calculated using HashMyFiles v2.36 (NirSoft) program.



NOTE

Check the correspondent FSB or INR in "Modifications" on page 22 or MTB in "Bulletin index" on page 1183.

To check downloaded file SHA-256 with HashMyFiles v2.36 (NirSoft) program proceed as follows:

1. Open HashMyFiles v2.36 program.
2. Select **Options** Menu - **Hash Types** sub-menu and check that **SHA-256** option is enabled (ticked) and, if necessary, enable it.



Figure 738 - My Hash files application

3. Select **File** menu - **Add Files** option. Select one or more file names to add window will open.
4. Select downloaded files and press **Open** button.
5. Selected files will be loaded and the enabled hash types will be displayed (resize SHA-256 column to see the whole hash if necessary).
6. Check that just obtained downloaded file hashes are the same as the ones included above.

7.2.2 Installing or restoring Operating System

When restoring the Operating System the following options can be executed:

- **Full Recovery:** This procedure formats **C:**, **D:** and **E:** drives. Instrument adjustments from **E:** unit are not deleted.
- **Os Recovery:** This procedure formats **C:** and **D:** drives. **E:** drive is not affected.

**NOTE**

Full Recovery is mandatory when installing an OS version $\geq 2.2.0$ or higher over an OS version $< v2.2.0$.

**NOTE**

When executed from an USB drive, the **R:** drive will be formatted whenever it exists

**NOTE**

Before performing an SSDD image restore it is necessary to make a backup of the database and the relevant data. See "Database and relevant data" on page 1116.

**NOTE**

The Windows license must be activated after any restoration.

The following methods can be used to install or restore an Operating System:

- OS installation from an USB pen drive. See "Restoring from USB pen drive" on page 1007
- OS installation from Boot Application. See "Restoring from Boot Application menu" on page 1016
- OS installation from Grifols Secure Remote Connectivity tool. See "Restoring from Grifols Secure Remote Connectivity tool" on page 1019

7.2.2.1 Restoring from USB pen drive

7.2.2.1.1 Create USB bootable device

**NOTE**

To create a recovery USB bootable pen drive, the following is required:

- An external computer with:
 - Windows 10 Operating System and USB ports.
 - Access to instrument OS and software programs (DxNet, other file repositories).
 - '7.zip' program installed.
- A USB pen drive with the following specifications:
 - 16 GB minimum recommended.
 - USB 3.0 recommended with FAT32 format.

**CAUTION**

Scan the USB drive for viruses before connecting it to Erytra.

1. Download the OS files in the external computer.

**NOTE**

Current OS version files:

- OS files: w10-1809-2.3.3.zip
- Drivers files: imb-150d-drivers-1.1.0.jar

2. Connect the USB pen drive to the external PC.
3. Open the zip file using 7.zip program.
4. Copy the content of the zip file in the following directories:
 - OS files into **USB:**
 - Drivers files into **USB:\Drivers**

7.2.2.1.2 Grifols Secure Remote Connectivity agent setup

If desired, the operating System can manage the installation of the Grifols Secure Remote Connectivity agent. Download Grifols Secure Remote Connectivity agent file and copy it to: **USB:\BeyondTrust**.

**NOTE**

This procedure shall only be applied if is required to install Grifols Secure Remote Connectivity agent.

**NOTE**

The copy of the installation file will allow the OS to manage the installation the Grifols Secure Remote Connectivity agent.

**CAUTION**

Ensure that you have the latest version of the Grifols Secure Remote Connectivity agent.

7.2.2.1.3 Configure Erytra to boot from USB

Before using the bootable USB drive created, BIOS of the computer must be configured to allow that the system boots from the USB drive.

**NOTE**

It is necessary to connect a USB keyboard and a USB mouse to the instrument.

1. Connect the USB bootable drive to Erytra. It is recommended to use USB 3.0 ports (rear part of the instrument).

2. Switch on the instrument and press the **F11** key during the initialization of the computer to enter the UEFI boot device selection option. Enter password, if required.



Figure 739 - Boot mode window



CAUTION

Before restarting the instrument to boot from the USB pendrive, disconnect any other USB pendrive or external drive (HDD or SSD) from the instrument. Also disconnect the instrument from the network (if connected).

3. Select the USB drive as **Boot option #1** among the available boot device priorities. If two options are found for the USB drive (one starting with USB: and another one with UEFI:) select the **UEFI: USB** option.

**NOTE**

If the USB drive is not displayed or detected, perform one of the following actions and go back to step 1:

- Enter OS Shell mode (or another mode) and check Windows can detect the USB drive.
- Disconnect the instrument from the mains during 5 - 10 seconds.
- Connect the USB drive to another USB port of the instrument.

System will boot from the USB drive and the Operating System installer application is displayed showing the recovery option:

- "OS recovery from USB drive" on page 1010
- "Full recovery from USB drive" on page 1013



Figure 740 - Recovery options

7.2.2.1.4 OS recovery from USB drive

The following steps describe the full restoring procedure of the SSD. This procedure formats **C:**, **D:** and **E:** drives. Instrument adjustments from **E:** unit are not deleted. The **R:** drive will be formatted whenever it exists.

**NOTE**

Full Recovery is mandatory when installed OS v2.2.x over OS v2.1.x.

**NOTE**

The PC shall be configured to run from USB before running this procedure. See "Restoring from USB pen drive" on page 1007.

1. Select **OS RECOVERY** option and press **Enter** in the keyboard.

**NOTE**

With this option the SSD image is partially restored.
Previous information in the **C:** and **D:** units is deleted.
Previous information in the **E:** unit is not deleted

"Select Disk type" window is displayed.



Figure 741 - Select Disk type window

2. Select SSD disk type and Select **Enter** in the keyboard

- SATA (IMB-150D)
- NVMe (IMB-1003J)



NOTE

Erytra is only compatible with the following computer:

- PC IMB-150D

The installation process starts:

- i. System reboots.
- ii. The integrity check of Operating System files in the R recovery unit that are not inside a compressed file is performed. It will last around 2 minutes.
- iii. System reboots automatically.
- iv. Operating System installation menu is displayed briefly indicating that OS Recovery option is selected.
- v. The image of the hard disk partitions related to Operating System (sda4 and sda6) are restored.
- vi. Copied files integrity check is performed.

The PC reboots.

3. Remove the USB pen drive.

4. Once PC reboots, OS preparation is performed.

The computer will reboot automatically.

5. Once PC reboots, OS installation and configuration continues.

The computer will reboot automatically.

6. Once PC reboots, system configuration continues and, AppLocker application is configured.

The computer will reboot automatically.

7. Once PC reboots, system configuration continues to allow Grifols Secure Remote Connectivity Agent installation.

The computer will reboot automatically.



NOTE

This step is only performed if Grifols Secure Remote Connectivity tool agent installation has been setup.

8. Once PC reboots, system configuration continues and Grifols Secure Remote Connectivity Agent application is configured.

**NOTE**

This step is only performed if Grifols Secure Remote Connectivity tool agent installation has been setup.

9. Once PC reboots, configuration is completed, and Boot Application initial window is displayed.
10. Enter Boot application and verify the installation by selecting the **information** option. See "Verifying OS version" on page 1028

**NOTE**

The Windows license must be activated after running this procedure. See "Activating Windows License" on page 1029.

7.2.2.1.5 Full recovery from USB drive

The following steps describe the full restoring procedure of the SSD. This procedure formats **C:**, **D:** and **E:** drives. Instrument adjustments from **E:** unit are not deleted. The **R:** drive will be formatted whenever it exists.

**NOTE**

Full Recovery is mandatory when installed OS v2.2.x over OS v2.1.x.

**NOTE**

The PC shall be configured to run from USB before running this procedure. See "Restoring from USB pen drive" on page 1007.

**NOTE**

The Windows license must be activated after running this procedure. See "Activating Windows License" on page 1029

1. Select **FULL RECOVERY** option and press **Enter** in the keyboard.

**NOTE**

With this option the SSD image is fully restored.

Previous information in the **C:**, **D:**, **E:** and **R:** units is deleted.

Adjustment files will be preserved in **E:** unit.

Files will be temporarily copied to the USB drive and, will be copied back to its location once drive image is restored.

"Select Disk type" window is displayed.



Figure 742 - Select Disk type window

2. Select SSD disk type and Select **Enter** in the keyboard

- SATA (IMB-150D)
- NVMe (IMB-1003J)



NOTE

Erytra is only compatible with the following computer:

- PC IMB-150D

The installation process starts:

- i. The selection menu is removed and GRIFOLS will remain on screen around two minutes.
- ii. The integrity check of files in the USB that are not inside a compressed file is performed. It will last around 2 minutes.
- iii. The relevant data files backup (instrument adjustment files, configuration files, system tools) is created.
- iv. The contents of each partition are copied to the instrument SSD.
- v. The relevant data files backup (instrument adjustment files, configuration files, system tools) is restored to instrument SSD.
- vi. The OS image is copied to Recovery partition and the integrity check of the files that are not inside a compressed file copied to the SSD recovery partitions is performed.

The PC reboots

3. Remove the USB pen drive.
4. Once PC reboots, OS preparation is performed.

The computer will reboot automatically.

5. Once PC reboots, OS installation and configuration continues.

The computer will reboot automatically.

Once PC reboots, system configuration continues and, AppLocker application is configured.

6. The computer will reboot automatically.
7. Once PC reboots, system configuration continues to allow Grifols Secure Remote Connectivity Agent installation.

The computer will reboot automatically.



NOTE

This step is only performed if Grifols Secure Remote Connectivity tool agent installation has been setup.

8. Once PC reboots, system configuration continues and Grifols Secure Remote Connectivity Agent application is configured.

**NOTE**

This step is only performed if Grifols Secure Remote Connectivity tool agent installation has been setup.

9. Once PC reboots, configuration is completed, and Boot Application initial window is displayed.
10. Enter Boot application and verify the installation by selecting the **information** option. See "Verifying OS version" on page 1028

**NOTE**

The Windows license must be activated after running this procedure. See "Activating Windows License" on page 1029.

7.2.2.2 Restoring from Boot Application menu

This option allows to perform an Full Recovery or OS Recovery from the instrument's SSD from the Boot Application, without the need of connecting a USB pen drive.

This procedure restores the OS image files contained in the **R** drive. These files can be manually copied into the drive to allow a newer OS image version to be restored. Initially, the OS restoration files were copied during the Full Recovery from a USB pen drive.

**NOTE**

This procedure can be done remotely. See "Restoring from Grifols Secure Remote Connectivity tool" on page 1019.

**NOTE**

To copy a new OS image version manually into the **R:** drive:

1. Enter **Maintenance Mode** option from the boot application menu. See "Working with Boot Application" on page 88.

The PC loads the OS desktop in 'Maintenance Mode'.

2. Copy the OS image content in the following path:

OS files into **R:**

Drivers files into **R:\Drivers**

3. Copy the Grifols Secure Remote Connectivity agent installation file in the following path:

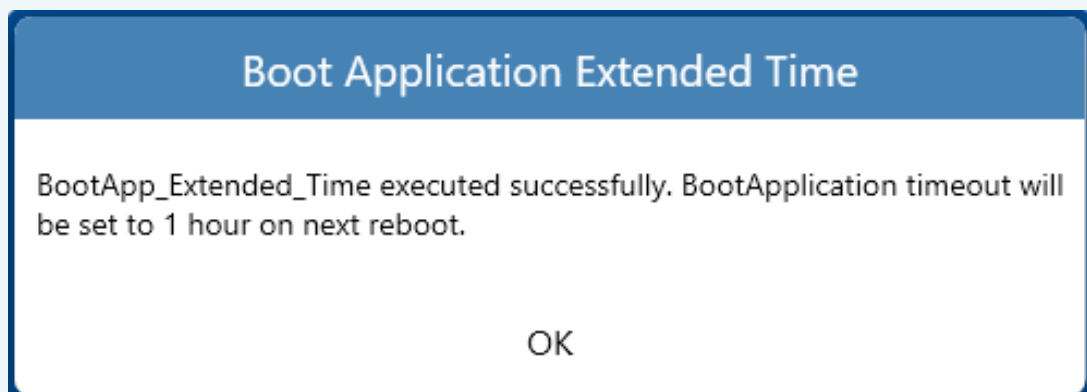
Remote connectivity files into **R:\BeyondTrust**

4. Run the following:

File: **BootApp_Extended_Time**

Path: **Program Files\System Tools**

5. Wait until confirmation message appears and select **OK**.



6. Reboot the instrument PC (This step shall be executed using Windows options of the instrument PC. Do not ever use the Reboot option from Grifols Secure Remote Connectivity Tool.)

**NOTE**

Current OS version 2.3.3 files:

- OS files: w10-1809-2.3.3.zip
- Drivers files: imb-150d-drivers-1.1.0.jar

1. Select the **Recovery System** option from the Boot application menu (see "Working with Boot Application" on page 88). Select the desired option from the pop-up window:



Figure 743 - Recovery system options

2. A warning and confirmation window will be displayed:

OS recovery message

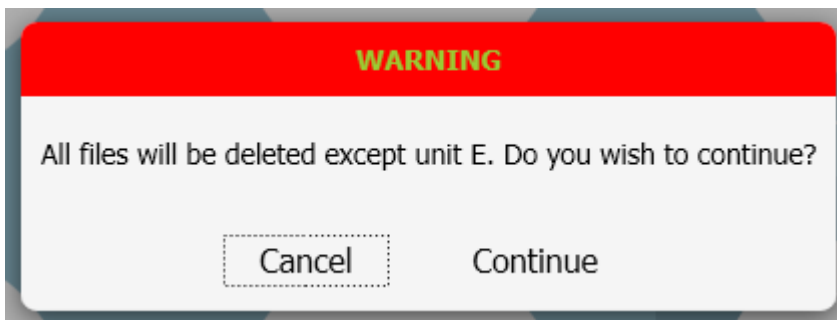


Figure 744 - Recovery system warning for OS recovery option

Full recovery message

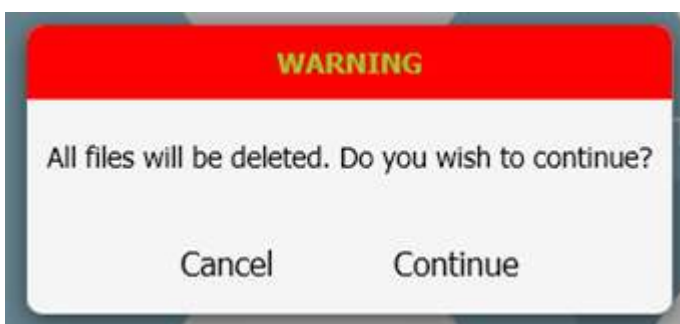


Figure 745 - Recovery system warning for Full recovery option

3. Select **Continue** option.
4. After some minutes, the process finishes and, the computer will reboot automatically.

5. Once PC reboots, OS preparation is performed.
The computer will reboot automatically.
6. Once PC reboots, OS installation and configuration continues.
The computer will be reboot automatically.
7. Once PC reboots, system configuration continues and AppLocker application is configured.
The computer will reboot automatically.
8. Once PC reboots, system configuration continues to allow Grifols Secure Remote Connectivity Agent installation.
The computer will reboot automatically.

**NOTE**

This step is only performed if Grifols Secure Remote Connectivity tool agent installation has been setup.

9. Once PC reboots, system configuration continues and Grifols Secure Remote Connectivity Agent application is configured.

**NOTE**

This step is only performed if Grifols Secure Remote Connectivity tool agent installation has been setup.

10. Once PC reboots, configuration is completed, and Boot Application initial window is displayed.
11. Enter Boot application and verify the installation by selecting the **information** option. See "Verifying OS version" on page 1028

**NOTE**

The Windows license must be activated after running this procedure. See "Activating Windows License" on page 1029.

7.2.2.3 Restoring from Grifols Secure Remote Connectivity tool

**NOTE**

This procedure is only compatible with Operating System v2.2.0 and above.

**NOTE**

This procedure requires that Grifols Secure Remote Connectivity Agent is already installed in the instrument.

**NOTE**

Ensure that you have the latest version of the Grifols Secure Remote Connectivity tool.

**NOTE**

Make sure that the only network connected to the instrument is the network used by the Grifols Secure Remote Connectivity tool.

1. Enter the **Boot Application**.
2. Select **Maintenance Mode** option.
The PC loads the **Maintenance Mode** desktop of the operating system.
3. Connect the PC instrument via **Grifols Secure Remote Connectivity Agent**.
4. Run the following file:
 - File: **BootApp_Extended_Time**
 - Path: **Program Files\System Tools**
5. Wait until confirmation message appears and select **OK**.

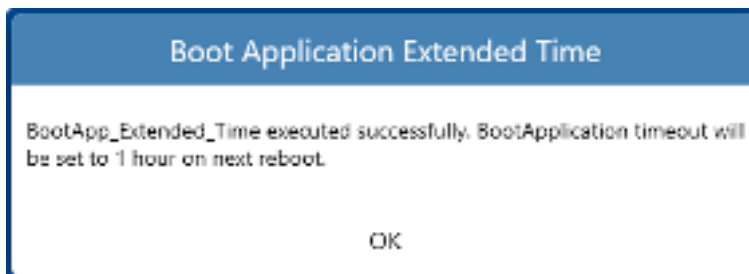


Figure 746 - Boot application extended time

6. Copy the OS image content if required in the following path:
- R:

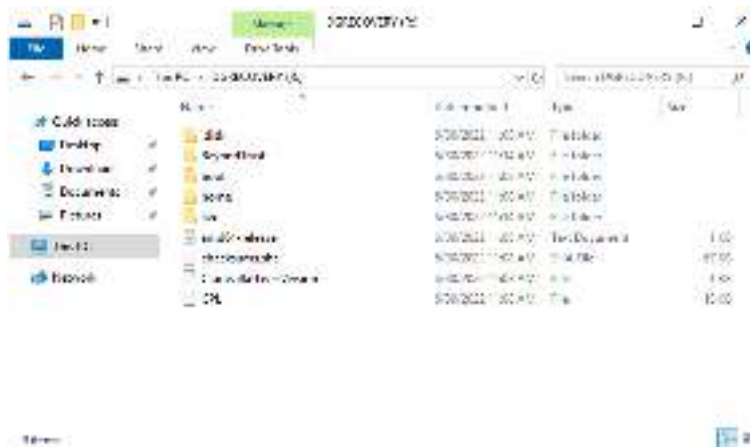


Figure 747 - DGRECOVERY Path



NOTE

By default, the R drive contains the image of the Operating system that is installed on the instrument. To install a different operating system than the current one, the desired version must be copied to this unit.

- Copy the **Grifols Secure Remote Connectivity Agent** installation file in the following path of DGRECOVERY (R:).

R:\BeyondTrust



NOTE

Ensure that you have the latest version of the BGrifols Secure Remote Connectivity Agent agent.

8. Reboot the instrument PC.



CAUTION

This step shall be executed using Windows options of the instrument PC.

Do not ever use the Reboot option from Grifols Secure Remote Connectivity Agent.

9. Wait until Boot Application initial window is displayed. Click the grid () button on the Boot Application

The Boot Application login window will be displayed.

10. Perform a **OS recovery** or **Full recovery** from the boot application.
 - OS recovery from Boot Application

11. Wait until PC instrument is reconnected and displayed with Status **Active[ON]** in Grifols Secure Remote Connectivity Agent Remote console.

Grifols Secure Remote Connectivity Agent Remote Customer client is launched and access is requested.

**CAUTION**

Access request will be required after Operating System installation.



Figure 748 - Requesting access to computer

12. Select **Allow** option.
13. Log in as `adminUser` (case sensitive, admin password). The Boot Application options will be displayed. Select the **Maintenance Mode** option.
14. Restart the instrument.

7.2.2.3.1 OS recovery from Grifols Secure Remote Connectivity tool

**NOTE**

This procedure is only compatible with Operating System v2.2.x and above.

**NOTE**

This procedure requires pre-installation of Grifols Secure Remote Connectivity Agent in the target PC.

**NOTE**

Ensure that you have the latest version of the Grifols Secure Remote Connectivity Agent.

**NOTE**

Make sure that the only network connected to the instrument is the network used by the Grifols Secure Remote Connectivity Agent.

**NOTE**

The Windows license must be activated after running this procedure. See "Activating Windows License" on page 1029.

1. Enter the **Boot Application**.
2. Select **Maintenance Mode** option.
The PC loads the **Maintenance Mode** desktop of the operating system.
3. Connect the PC instrument via **Grifols Secure Remote Connectivity Agent**.
4. Run the following file:
 - File: **BootApp_Extended_Time**
 - Path: **Program Files\System Tools**
5. Wait until confirmation message appears and select **OK**.

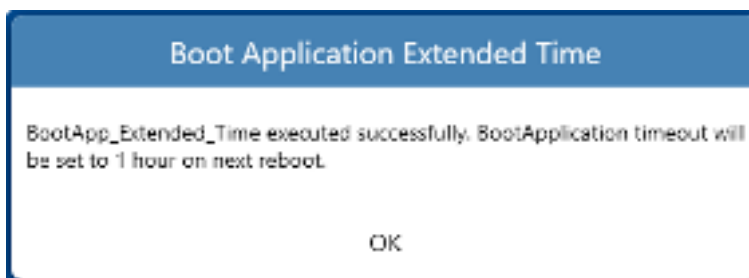


Figure 749 - Boot application extended time

6. Copy the OS image content if required in the following path:
 - R:

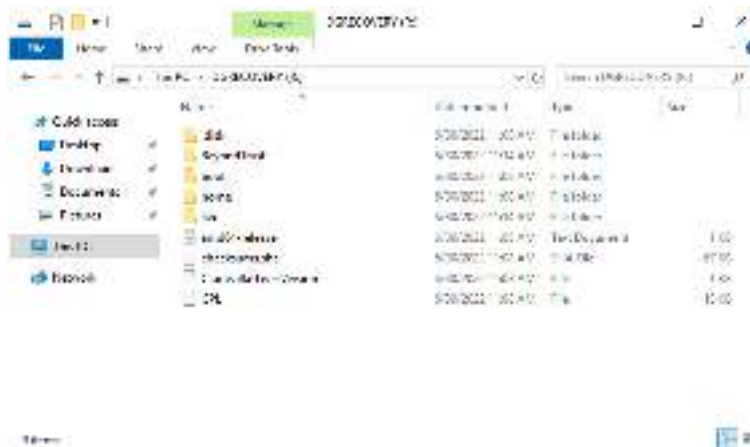


Figure 750 - DGRECOVERY Path



NOTE

By default, the R drive contains the image of the Operating system that is installed on the instrument. To install a different operating system than the current one, the desired version must be copied to this unit.

- Copy the **Grifols Secure Remote Connectivity Agent** installation file in the following path of DGRECOVERY (R:).

R:\BeyondTrust



NOTE

Ensure that you have the latest version of the BGrifols Secure Remote Connectivity Agent agent.

8. Reboot the instrument PC.



CAUTION

This step shall be executed using Windows options of the instrument PC.

Do not ever use the Reboot option from Grifols Secure Remote Connectivity Agent.

9. Wait until Boot Application initial window is displayed. Click the grid () button on the Boot Application

The Boot Application login window will be displayed.

10. Perform a Operating system recovery from the boot application. See "Operating system recovery from Boot Application".

11. Wait until PC instrument is reconnected and displayed with Status **Active[ON]** in Grifols Secure Remote Connectivity Agent Remote console.

Grifols Secure Remote Connectivity Agent Remote Customer client is launched and access is requested.

**CAUTION**

Access request will be required after Operating System installation.



Figure 751 - Requesting access to computer

12. Select **Allow** option.
13. Log in as `adminUser` (case sensitive, admin password). The Boot Application options will be displayed. Select the **Maintenance Mode** option.
14. Restart the instrument.

7.2.2.3.2 Full recovery from Grifols Secure Remote Connectivity tool

The following steps describe the full restoring procedure of the SSD. This procedure formats **C:**, **D:** and **E:** drives. Instrument adjustments from **E:** unit are not deleted. The **R:** drive will be formatted whenever it exists.

**NOTE**

This procedure is only compatible with Operating System v2.2.x and above.

**NOTE**

This procedure requires pre-installation of Grifols Secure Remote Connectivity Agent in the target PC.

**NOTE**

Ensure that you have the latest version of the Grifols Secure Remote Connectivity Agent.

**NOTE**

Make sure that the only network connected to the instrument is the network used by the Grifols Secure Remote Connectivity Agent.

**NOTE**

The Windows license must be activated after running this procedure. See "Activating Windows License" on page 1029.

1. Enter the **Boot Application**.
2. Select **Maintenance Mode** option.
The PC loads the **Maintenance Mode** desktop of the operating system.
3. Connect the PC instrument via **Grifols Secure Remote Connectivity Agent**.
4. Run the following file:
 - File: **BootApp_Extended_Time**
 - Path: **Program Files\System Tools**
5. Wait until confirmation message appears and select **OK**.

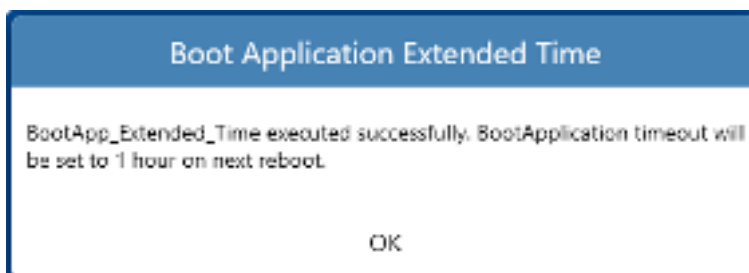


Figure 752 - Boot application extended time

6. Copy the OS image content if required in the following path:



NOTE

By default, the R drive contains the image of the Operating system that is installed on the instrument. To install a different operating system than the current one, the desired version must be copied to this unit.

- R:

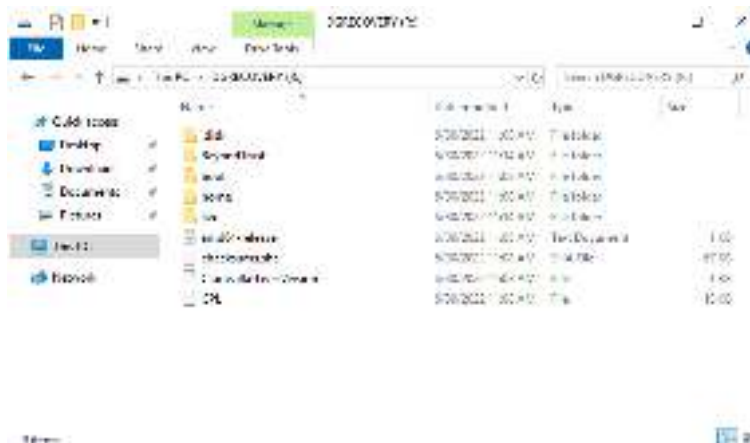


Figure 753 - DGRECOVERY Path

- Copy the **Grifols Secure Remote Connectivity Agent** installation file in the following path of DGRECOVERY (R:).

R:\BeyondTrust



NOTE

Ensure that you have the latest version of the Grifols Secure Remote Connectivity Agent.

8. Reboot the instrument PC.



CAUTION

This step shall be executed using Windows options of the instrument PC.

Do not ever use the Reboot option from Grifols Secure Remote Connectivity Agent.

9. Wait until Boot Application initial window is displayed. Click the grid () button on the Boot Application.

The Boot Application login window will be displayed.

10. Perform a Full Recovery Operating system from the boot application. See "Full recovery from boot application".

11. Wait until PC instrument is reconnected and displayed with Status **Active[ON]** in Grifols Secure Remote Connectivity Agent Remote console.

Grifols Secure Remote Connectivity Agent Remote Customer client is launched and access is requested.

**CAUTION**

Access request will be required after Operating System installation.



Figure 754 - Requesting access to computer

12. Select **Allow** option.
13. Log in as `adminUser` (case sensitive, admin password). The Boot Application options will be displayed. Select the **Maintenance Mode** option.
14. Restart the instrument.

7.2.2.4 Verifying OS version

1. Select the **Information** option from the Boot application menu. See "Working with Boot Application" on page 88

2. Information window about OS version is displayed.

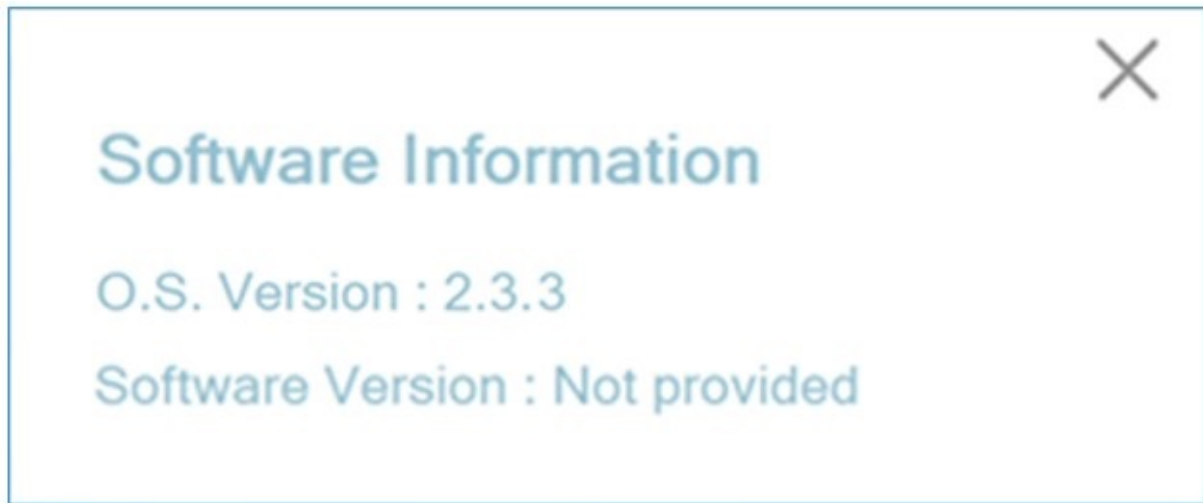


Figure 755 - Information window

3. Verify the OS installed version.
4. Close information window.

7.2.2.5 Activating Windows License

Activate Windows license shall be performed every time the Operating System is installed (OS Recovery or Full Recovery), or the computer has some hardware components modifications, such as SSD, CPU or motherboard and when the instrument is connected to a network.



NOTE

Windows license activation must be performed if the instrument is connected to a network with no access to Microsoft license servers. Otherwise, the Windows license watermark will be displayed.



NOTE

The Windows license watermark does not imply that the license distributed by Diagnostic Grifols is not valid. It simply states that the license could not be activated in this instrument.

This section provides two available methods to activate Windows License. Active Windows License using the method that matches installation characteristics.

- "Activation by accessing Microsoft license servers" on page 1030
- "Activation by phone" on page 1031

7.2.2.5.1 Activation by accessing Microsoft license servers



NOTE

'Confirmation ID' will change on each license activation.
The screenshot is just an example.



NOTE

This procedure requires coordination with the IT department to enable this connection.
These ports that are required for Windows Activation: TCP port 80, TCP port 443.

1. Open the **Maintenance Mode** () option.
2. Connect the Erytra PC to the network with Internet connection.
3. Open CMD as Administrator.

How to: **Start > Windows Administrative Tools.**

Click the right mouse button on **Command Prompt** and select **more > Run as administrator.**

Enter admin user password.



Figure 756 - Run command prompt as administrator

4. Type following command: `slmgr /dlv.`

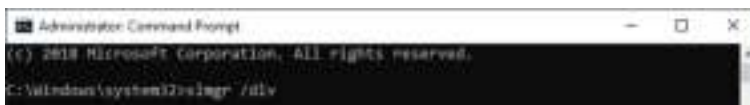


Figure 757 - slmgr /dlv command

This command verifies that the activation of the Windows licenses is performed correctly.

5. The following window is displayed.

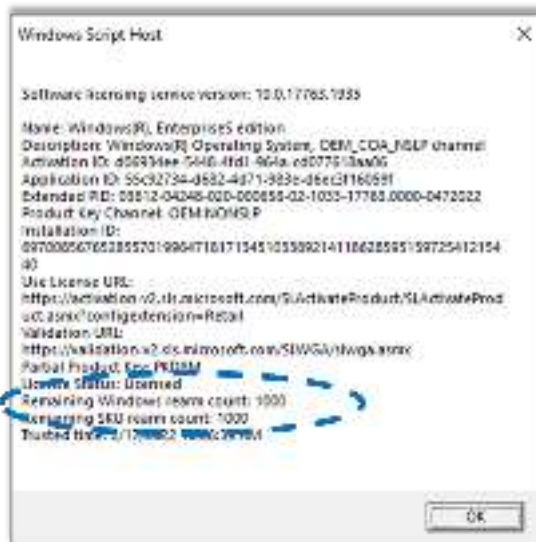


Figure 758 - Windows script host

6. Check that **License Status** field status is **Licensed**. Should this not be the case, wait 15 minutes and check the status again.

If the status does not change to **Licensed** execute the following command to prompt Windows to try online activation (at that time): `slmgr /ato`.

The following window should be displayed:

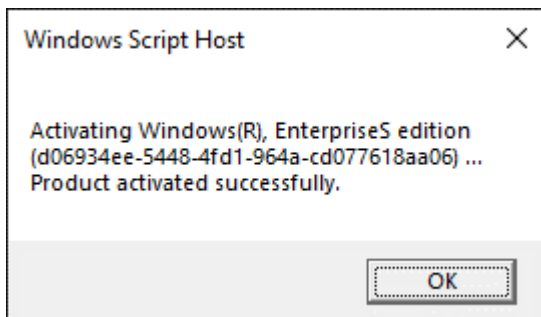


Figure 759 - Activating Windows

Press **OK** and then check again the **License Status** field.

If an activation error displayed and if the status does not change to **Licensed**, activate Windows by phone. See "Activate Windows License by phone" on page 1.

7. Shut down Erytra.

7.2.2.5.2 Activation by phone



NOTE

It is not necessary to be connected to a network.

**NOTE**

This activation procedure is detailed for the 'United States' region.

The region selection may be changed based on language preferences. In this case, the detail of the instructions may change.

1. Open the **Maintenance Mode** () option.
2. Open CMD as Administrator.

How to: **Start > Windows Administrative Tools.**

Click the right mouse button on **Command Prompt** and select **more > Run as administrator.**

Enter admin user password.



Figure 760 - Run command prompt as administrator

3. Type the following command: `slui 4`.

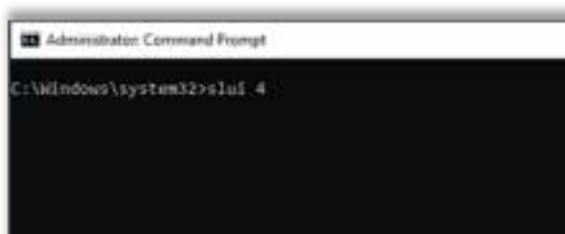


Figure 761 - slui 4 command

4. Select region: <instrument region>.

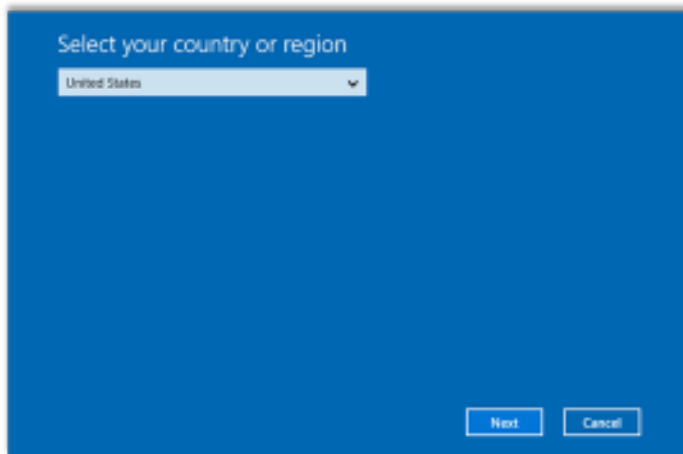


Figure 762 - Instrument region

5. Call the number and follow instructions to activate Windows 10.

Action	Voice recorded message
Call the number.	'Product activation center' welcome message. Consent about the call recording. Press '1' to consent. Press '2' to do not consent.
Press phone key 2.	Introduce phone code for security purposes. Random security code of 3 digits is provided.
Enter the provided random security code.	Welcome to 'Microsoft Product Activation'. If you are trying to activate Windows 10, press '1'. Otherwise press '2'.
Press phone key 1.	Request about product upgrade. If you previously upgraded to Windows 10, press '1'. Otherwise press '2'.
Press phone key 2.	If you got an error code, press '1'. Otherwise press '2'.
Press phone key 2.	Confirmation about you are with the activation screen open. Say 'Yes' or 'No'.
Say "Yes".	Explanation about the 'Installation ID' format. Long number split in groups. It will request one group at a time.
Enter <code>Installation ID</code> by groups with the phone keyboard.	

**NOTE**

Installation ID will change on each license activation.

**NOTE**

Introduce one group at a time and wait for the license activation assistant before introducing a new one.

Action	Voice recorded message
Listen to the message.	Request about waiting to look up the provided 'Installation ID'. Confirmation of verified status.



Figure 763 - Installation ID number

6. Select the **Enter confirmation ID** option.

The **Enter your confirmation ID** window is displayed.

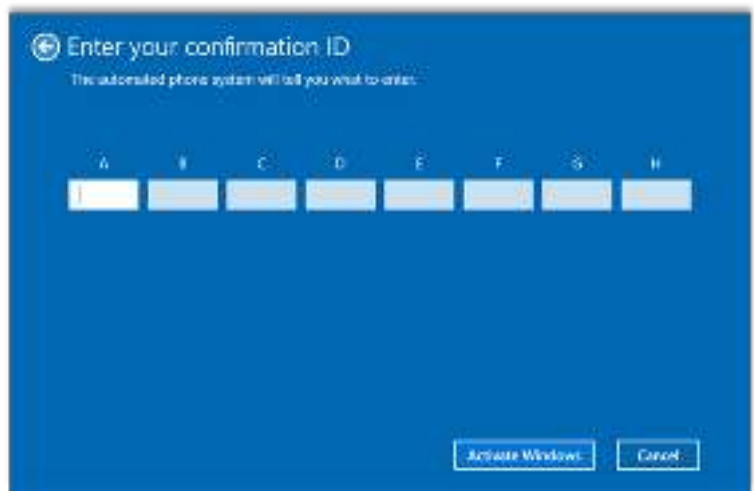


Figure 764 - Confirmation ID window

Action	Voice recorded message
Listen to the message.	Assistant will read the 'Confirmation ID'.
Use your PC keyboard to introduce the provided ID in the empty blocks. You will have the opportunity to listen again the block at the end. Say 'Repeat' at the end, if needed. Say 'Yes' if you are ready to listen the first block.	Assistant will read one block of six digits at a time.
Enter the confirmation numbers provided by phone.	



NOTE

'Confirmation ID' will change on each license activation.
The screenshot is just an example.

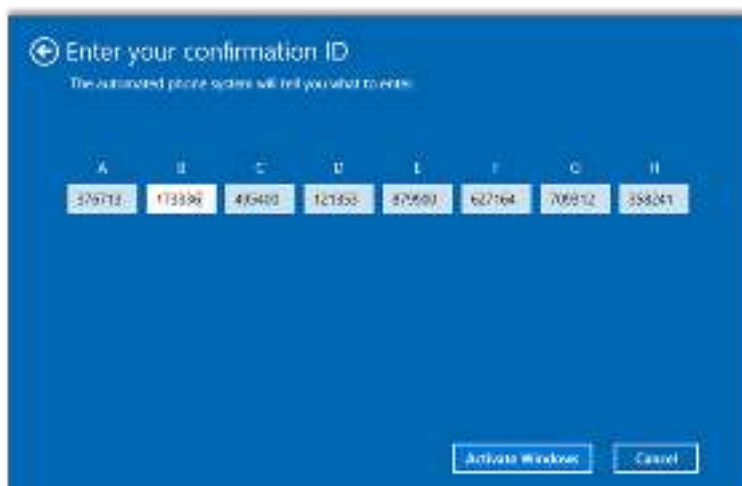


Figure 765 - Populated Confirmation ID window

7. Select the **Activate Windows** option.

The **Thanks, you're all done** window is displayed.

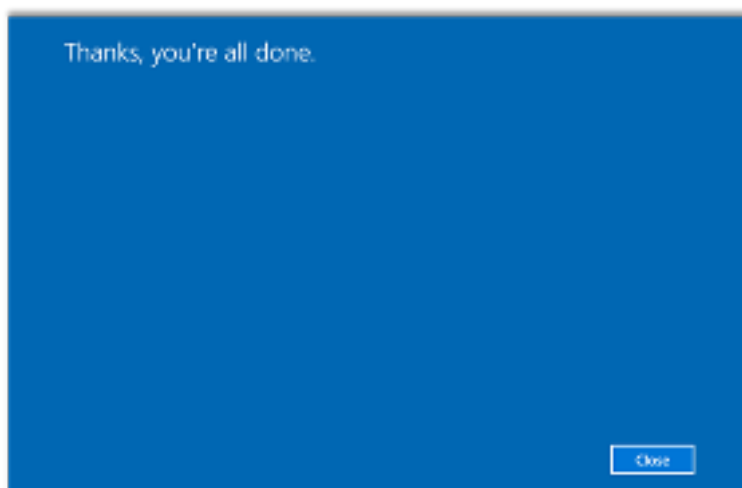


Figure 766 - Thanks, you're all done window

8. Select the **Close** option.
Operating System license activation is performed.
9. Shut down the Erytra instrument.

7.2.2.6 Verifying Licence activation

1. Enter **OS Shell** option (See "Working with Boot Application" section).
2. Open Command Prompt as administrator and enter administrator credentials.



NOTE

- Start - Windows Administrative Tools
- click right mouse option on Command Prompt and select More | Run as administrator
- Enter admin user password

3. Tyep the following command:

```
slmgr /dlv
```

The following will be displayed

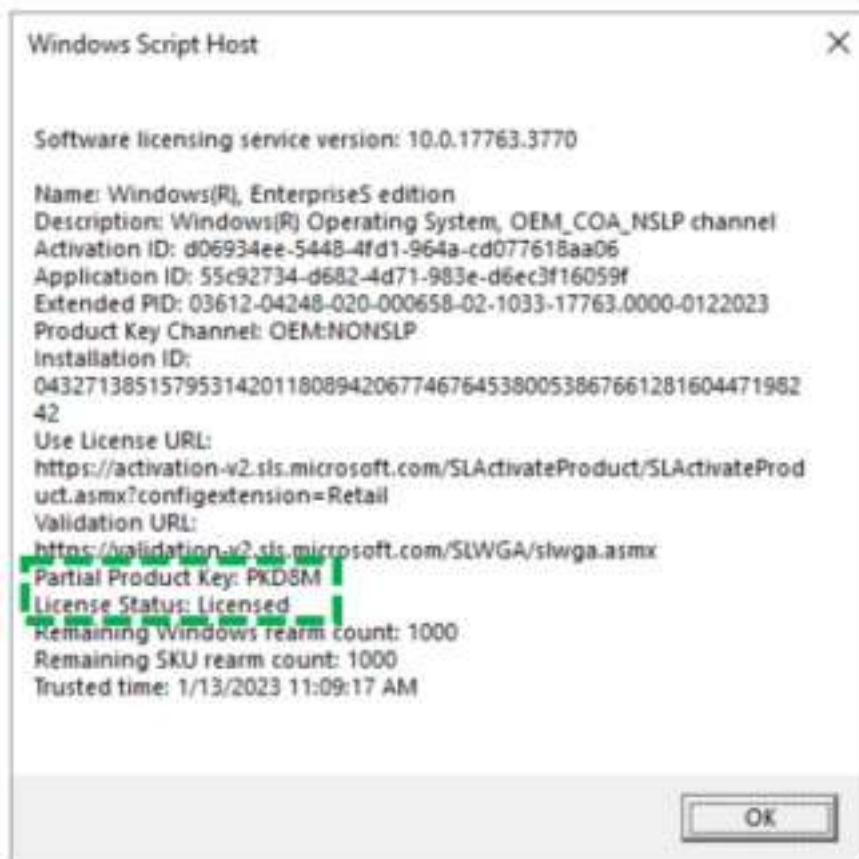


Figure 767 - License status line

4. Check that License Status field status is Licensed.
Should this not be the case, activate the license (See "Activating Windows License" section).

7.2.3 Installing Erytra software v4.6.1

**NOTE**

In this section the general steps for software installation are described. Depending on the version previously installed and the version that it is going to be installed, some additional steps may be needed (check relevant MTB, FSB and INR, see "Bulletin index" on page 1183).

1. Download the required OS and SW. See "Downloading Erytra files" on page 1006.
2. Check the OS version installed on the instrument. See "Verifying OS and SW versions" on page 1097.

**NOTE**

OS version 2.3.3 is equivalent to OS versions 2.3.1 and 2.3.2 with SU v1.7 (March 2025)

3. Select the appropriate path:

**NOTE**

If OS version is 2.3.1 or 2.3.2 make sure to update OS with **FSB-0059#0045 Windows security updates March 2025**. See MOD0045 in "Modifications" on page 22.

OS version 2.3.3 or 2.3.2+SU v1.7 or 2.3.1+ SU v1.7

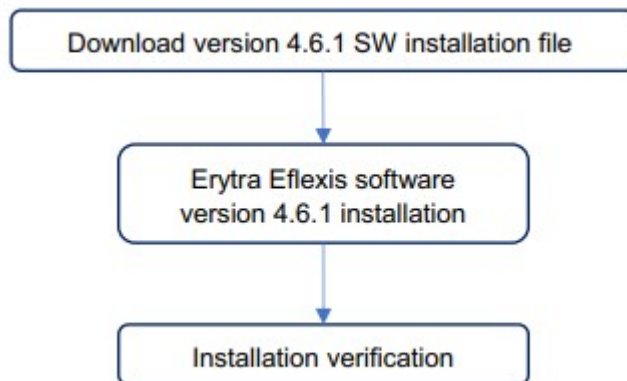


Figure 768 - Installation procedure diagram from OS v2.3.3 and Software v4.6.1

Once the required files have been downloaded and verified:

- "Installing steps in units with OS version v2.3.3 (or v2.3.1+ SU v1.7 or v2.3.2+SU v1.7)" on page 1042.

OS version < 2.3.1 or 3.2.1 (without SU v1.7) or 2.3.2 (without SU v1.7))

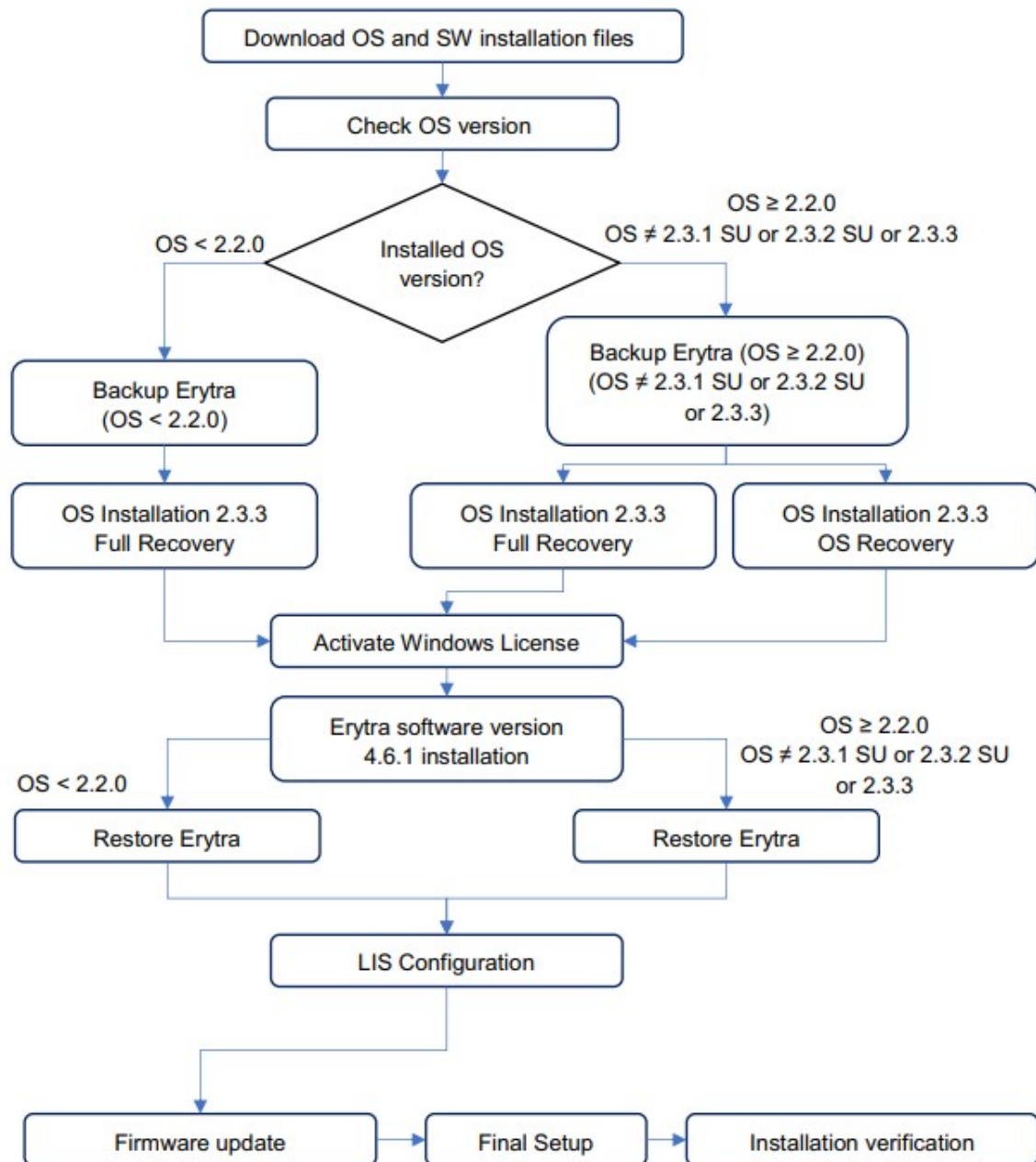


Figure 769 - Installation procedure diagram

Once the required files have been downloaded and verified:

- "Installing steps in units with OS version $\geq 2.2.0$ and OS $\neq 2.3.3$ (or 2.3.1+SU, or 2.3.2+SU)" on page 1045.
- "Installing steps in units with OS version < 2.2.0" on page 1054.

7.2.3.1 Installing steps in units with OS version v2.3.3 (or v2.3.1+ SU v1.7 or v2.3.2+SU v1.7)

To update a properly installed Erytra software version proceed as follows:



CAUTION

All samples results must be validated and exported before starting the installation.



NOTE

This procedure can be executed using Grifols Secure Remote Connectivity tool.

7.2.3.1.1 Erytra Software version 4.6.1 installation

Installation can be performed on side or remotely through Grifols Remote Connectivity tool.

Start software installation from Boot application on-site

1. Copy **erytraW10-INSTALLER-X.Y.Z.exe** file to an USB drive using an external computer to a folder named **installers** on the USB root.
2. Connect the USB drive to Erytra PC.



NOTE

It is recommended to use USB 3.0 ports (rear part of the instrument).

3. Switch on the instrument. Select **Update Software** option from Boot Application menu. See "Working with Boot Application" on page 88.



Figure 770 - Software installer welcome window

- Click on **Next >**. The installer will be copied to E:\update directory and, once copied, an information window requesting to disconnect the USB drive is displayed.

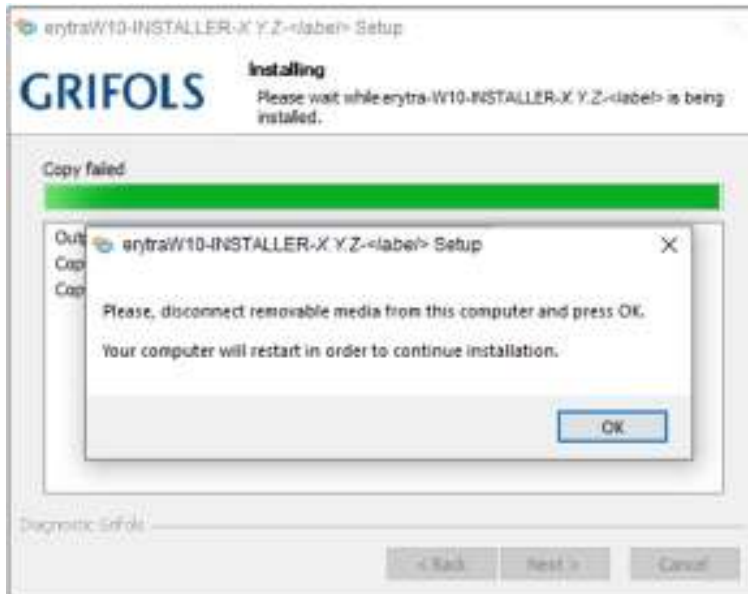


Figure 771 - Remove USB drive from the computer

- Disconnect the USB drive and press **OK** button. Following window will be displayed:

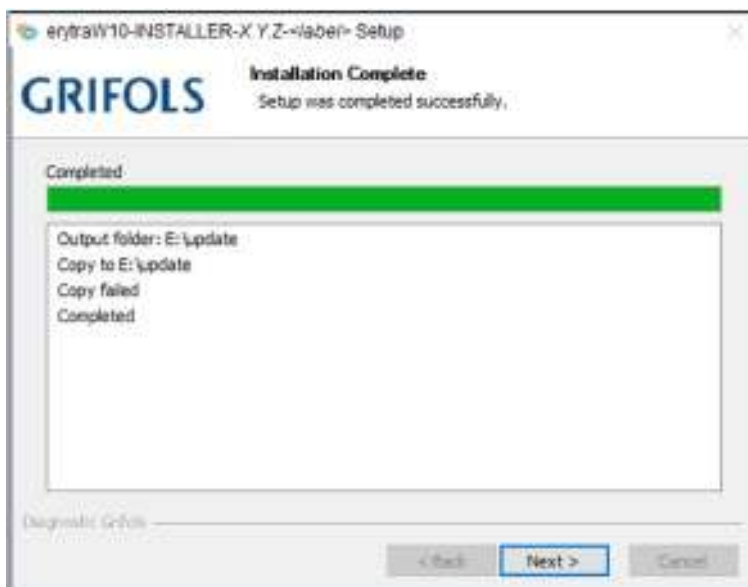


Figure 772 - Installer copied successfully

6. Select **Next >** button. The following window is displayed:



Figure 773 - Installer setup complete

7. Select **Reboot now** option and click on **Finish** button.

After restart, System will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer from E drive. Welcome screen will be displayed. Follow the installer instructions. See "Software installer steps" on page 1065

8. Once the installer is finished. The system will close the AppLocker audit session and apply the new rules, protect the volumes and restart. Enter Boot Application menu. See "Working with Boot Application" on page 88.



NOTE

If Installation has been completed with errors was displayed, or FAILED was displayed for a given installation step, retry software installation from See " Switch on the instrument. Select Update Software option from Boot Application menu. See "Working with Boot Application" on page 88." on page 1042).

Start software installation from Boot application remotely

1. Connect to the instrument through Grifols Secure Remote Connectivity tool in **Maintenance mode**.
2. Run **BootApp_Extended_Time** from **E:\Program Files\System Tools**.
3. Wait until confirmation message is displayed and select **OK**.
4. Restart from Windows taskbar.

5. Enter in **OS Shell** mode from Boot Application. See "Working with Boot Application" on page 88.
6. Copy **erytraW10-INSTALLER-X.Y.Z.exe** file to **E:\update** directory using Grifols Secure Remote Connectivity tool File Transfer.
7. Restart from Windows taskbar.

After restart, System will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer from E drive. Welcome screen will be displayed. Follow the installer instructions. See "Software installer steps" on page 1065

8. Once the installer is finished. The system will close the AppLocker audit session and apply the new rules, protect the volumes and restart. Enter Boot Application menu. See "Working with Boot Application" on page 88.

**NOTE**

If Installation has been completed with errors was displayed, or FAILED was displayed for a given installation step, retry software installation from See " Enter in OS Shell mode from Boot Application. See "Working with Boot Application" on page 88." on page 1045).

7.2.3.1.2 Verify the installation

In order to verify the installation, check the following:

1. Verify OS and SW installation. See "Verifying OS and SW versions" on page 1097.
2. Verify OS licence activation. See "Verifying Licence activation" on page 1038
3. Verify FW versions according to the table. Check firmware version

7.2.3.2 Installing steps in units with OS version $\geq 2.2.0$ and OS $\neq 2.3.3$ (or 2.3.1+SU, or 2.3.2+SU)

To update a properly installed Erytra software version proceed as follows:

**CAUTION**

Before backup perform the following actions:

- Validate and export all samples results and shut down Erytra.
- Disconnect any USB pen drive connected to Erytra

**NOTE**

This procedure can be executed using Grifols Secure Remote Connectivity tool.

**NOTE**

This procedure shall be performed if the Erytra is with OS $\geq 2.2.0$ and OS $\neq 2.3.1$ + Security Updates or 2.3.2. This is SW $\geq 3.0.0$ and SW $\neq 3.2.0$.

**NOTE**

This procedure backups SW Data and OS Settings.

**CAUTION**

If necessary, perform the manual backup of the following directories:

- E:\data\Erytra\Microtubes.
- E:\data\Erytra\Raw.
- E:\data\SistemaDiana.
- E:\data\logs.

**CAUTION**

If destination folder is on a USB drive, check that USB drive does not contain any virus before using it.

1. Select the **Backup** option from the boot application menu. See "Working with Boot Application" on page 88.

Backup application is opened.

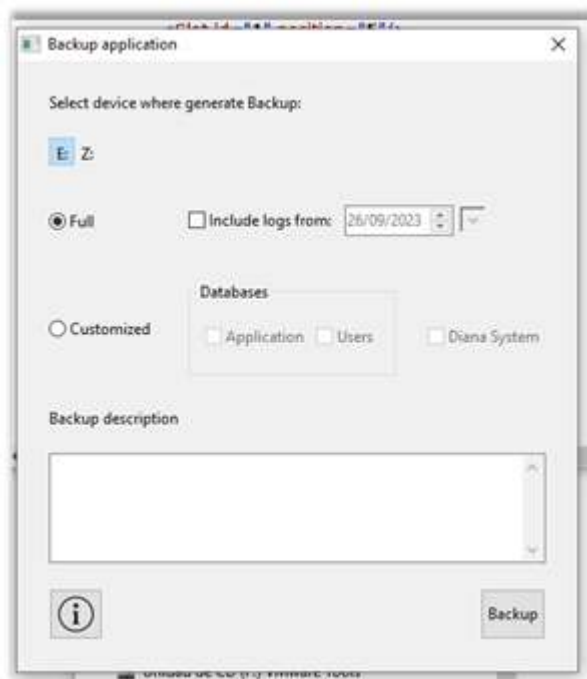


Figure 774 - Backup description and log files choice

2. Select the device where Backup will be generated.
3. Select **Full** backup option and add a description to ease the identification of the backup file, if desired.

4. Press **Backup** button to export instrument configuration. Enter `admin user` password when requested.
5. A message window will be displayed indicating the backup has been completed.

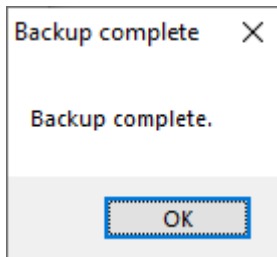


Table 108 - Backup completed

6. Press **OK** button and close Backup application pressing the top-right cross.



NOTE

Backup will be saved on the device that you have previously defined.



NOTE

You can copy the previous backup folder (without modify anything) to save it.

7. Reboot Erytra.

7.2.3.2.1 Operating system installation

See "Installing or restoring Operating System" on page 1006

7.2.3.2.2 Operating System license activation

See "Activating Windows License" on page 1029.

7.2.3.2.3 Erytra Software version 4.6.1 installation

Installation can be performed on side or remotely through Grifols Remote Connectivity tool.

Start software installation from Boot application on-site

1. Copy **erytraW10-INSTALLER-X.Y.Z.exe** file to an USB drive using an external computer to a folder named **installers** on the USB root.
2. Connect the USB drive to Erytra PC.



NOTE

It is recommended to use USB 3.0 ports (rear part of the instrument).

3. Switch on the instrument. Select **Update Software** option from Boot Application menu. See "Working with Boot Application" on page 88.



Figure 775 - Software installer welcome window

4. Click on **Next >**. The installer will be copied to E:\update directory and, once copied, an information window requesting to disconnect the USB drive is displayed.

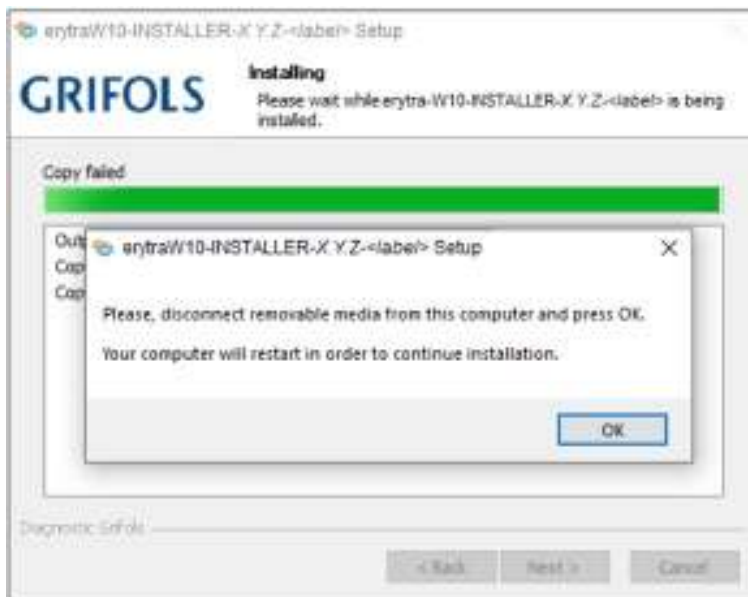


Figure 776 - Remove USB drive from the computer

5. Disconnect the USB drive and press **OK** button. Following window will be displayed:

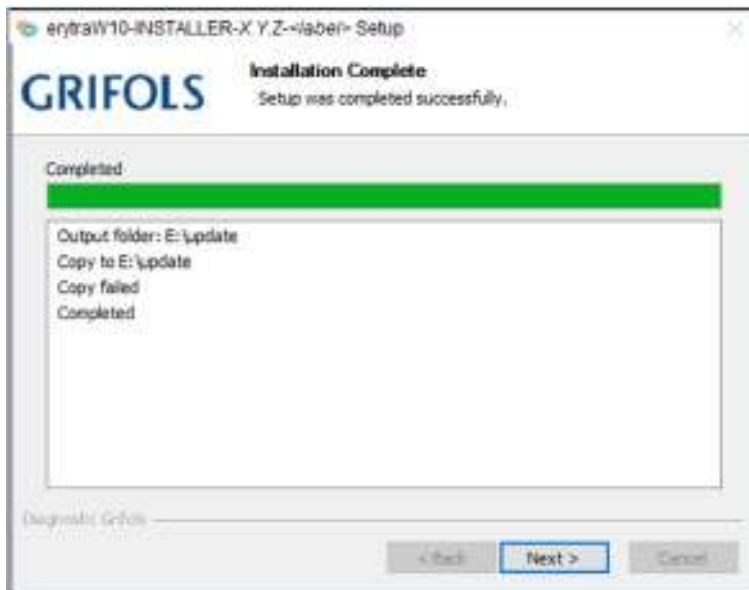


Figure 777 - Installer copied successfully

6. Select **Next >** button. The following window is displayed:



Figure 778 - Installer setup complete

7. Select **Reboot now** option and click on **Finish** button.

After restart, System will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer from E drive. Welcome screen will be displayed. Follow the installer instructions. See "Software installer steps" on page 1065

8. Once the installer is finished. The system will close the AppLocker audit session and apply the new rules, protect the volumes and restart. Enter Boot Application menu. See "Working with Boot Application" on page 88.

**NOTE**

If Installation has been completed with errors was displayed, or FAILED was displayed for a given installation step, retry software installation from See " Switch on the instrument. Select Update Software option from Boot Application menu. See "Working with Boot Application" on page 88." on page 1048).

Start software installation from Boot application remotely

1. Connect to the instrument through Grifols Secure Remote Connectivity tool in **Maintenance mode**.
2. Run **BootApp_Extended_Time** from **E:\Program Files\System Tools**.
3. Wait until confirmation message is displayed and select **OK**.
4. Restart from Windows taskbar.
5. Enter in **OS Shell** mode from Boot Application. See "Working with Boot Application" on page 88.
6. Copy **erytraW10-INSTALLER-X.Y.Z.exe** file to **E:\update** directory using Grifols Secure Remote Connectivity tool File Transfer.
7. Restart from Windows taskbar.

After restart, System will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer from E drive. Welcome screen will be displayed. Follow the installer instructions. See "Software installer steps" on page 1065

8. Once the installer is finished. The system will close the AppLocker audit session and apply the new rules, protect the volumes and restart. Enter Boot Application menu. See "Working with Boot Application" on page 88.

**NOTE**

If Installation has been completed with errors was displayed, or FAILED was displayed for a given installation step, retry software installation from See " Enter in OS Shell mode from Boot Application. See "Working with Boot Application" on page 88." on page 1050).

7.2.3.2.4 Restore relevant data backup



NOTE

If necessary, perform the manual restore of the Microtubes and Raw directories:

- E:\data\Erytra\Microtubes.
- E:\data\Erytra\Raw.
- E:\logs

1. Select **OS shell** option from the boot application. See "Working with Boot Application" on page 88.
2. Update the databases. Run **E:\Program Files\liquibase\updatedb.bat**.
3. Select **Restore** option from the boot application. See "Working with Boot Application" on page 88.

Restore application is opened.



NOTE

Alternatively, Restore application can be launched from **E:\Program Files\Restore\run(.bat)** from OS shell.

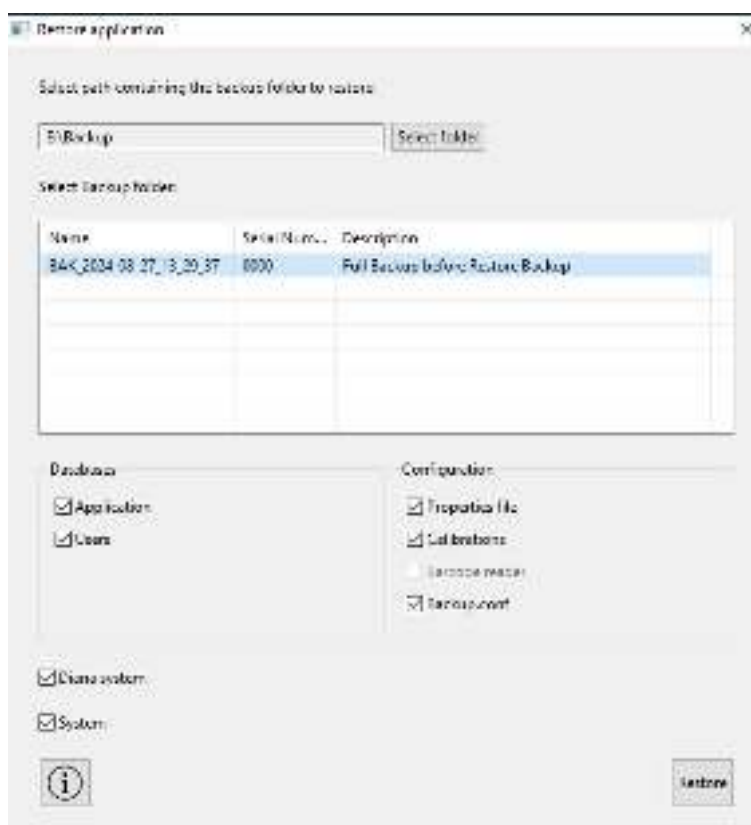


Figure 779 - Restore application

4. Select the backup file to restore. Do not modify any current options selection.
5. Press **Restore** button to import instrument configuration. Enter `admin` user password when requested. A progress windows will be displayed and the system will be rebooted several times.
6. A message informing that System settings have been successfully imported will be displayed. Press **OK** button.



Figure 780 - Import settings PASS



NOTE

In case an empty Database has been restored, the reagent pack and the reagent language pack should be imported again from Importer Application.



NOTE

This procedure shall be only performed if LIS is enabled

1. Enter **Maintenance mode** from the boot application menu. See "Working with Boot Application" on page 88
Wait until OS is loaded
2. If LIS communication mode is set to serial port, review and configure, if necessary, that the current COM port selected is the one connected to LIS (that COM port name is the same after installing new operating system).
3. If LIS communication mode is set to TCP/IP, review and configure, if necessary, that the current TCP/IP IPv4 MS Windows configuration is the one required to connect to LIS. If necessary, indicate the Host IP address and Port in the corresponding DianaServer textboxes, and create a new outbound firewall rule to open the selected port. See "TCP-IP communication mode" on page 1159.
4. Configure LIS encoding to match current LIS encoding.



NOTE

HostDemo program uses (only) ANSI encoding.

- If characters different from the Latin alphabet ones are used, set the suitable language in the Language for non-Unicode programs option

**NOTE**

Start → Control Panel → Clock and Region → Region → Change date, time or number formats → Administrative tab. Administrator password will be requested.

**NOTE**

Language for non-Unicode programs configuration of DianaServer PC, when not running in instrument PC, shall match Erytra configuration.

- If necessary for backup feature or LIS connection, configure Erytra to share or access a shared folder. See "Sharing and accessing a shared folder or printer" on page 1109.

**NOTE**

R and V drives now belong to instrument SSD.

- Restore the PC name and network configuration.

7.2.3.2.5 Firmware upgrade

- Update the PCA firmwares according to the table. See "Updating firmware in a valid PCA" on page 1129.

Table 109 - Erytra v4.6.1 FW versions

PCA	Firmware version
CEN_SRM_CTRL	4.6.0
INC_CUP_CTRL	4.5.0
FLUCTRL59	4.6.2
CAT_CM_CTRL	4.5.0
PIP_CTRL	4.13.0
RD_IL421	5.5.0

7.2.3.2.6 Final setup

See "Software installation Final Setup" on page 1070

7.2.3.2.7 Verify the installation

In order to verify the installation, check the following:

- Verify OS and SW installation. See "Verifying OS and SW versions" on page 1097.
- Verify OS licence activation. See "Verifying Licence activation" on page 1038

3. Verify FW versions according to the table. Check firmware version

7.2.3.3 Installing steps in units with OS version < 2.2.0



NOTE

This procedure cannot be executed using Grifols Secure Remote Connectivity tool as it involves Operating System Full Recovery process from an Operating System version < 2.2.0.

To update a properly installed Erytra software version proceed as follows:



CAUTION

All samples results must be validated and exported before starting the installation. After that, the content of the service racks should be identified before switching off the instrument.

7.2.3.3.1 Instrument backup (OS < 2.2.0)

See "Backing up relevant data manually" on page 1120.

7.2.3.3.2 Operating System image Installation

See "Full recovery from USB drive" on page 1013.

7.2.3.3.3 Operating System license activation

See "Activating Windows License" on page 1029.

7.2.3.3.4 Erytra Software version 4.6.1 installation

Start software installation from Boot application on-site

1. Copy **erytraW10-INSTALLER-X.Y.Z.exe** file to an USB drive using an external computer to a folder named **installers** on the USB root.
2. Connect the USB drive to Erytra PC.



NOTE

It is recommended to use USB 3.0 ports (rear part of the instrument).

3. Switch on the instrument. Select **Update Software** option from Boot Application menu. See "Working with Boot Application" on page 88.



Figure 781 - Software installer welcome window

4. Click on **Next >**. The installer will be copied to E:\update directory and, once copied, an information window requesting to disconnect the USB drive is displayed.

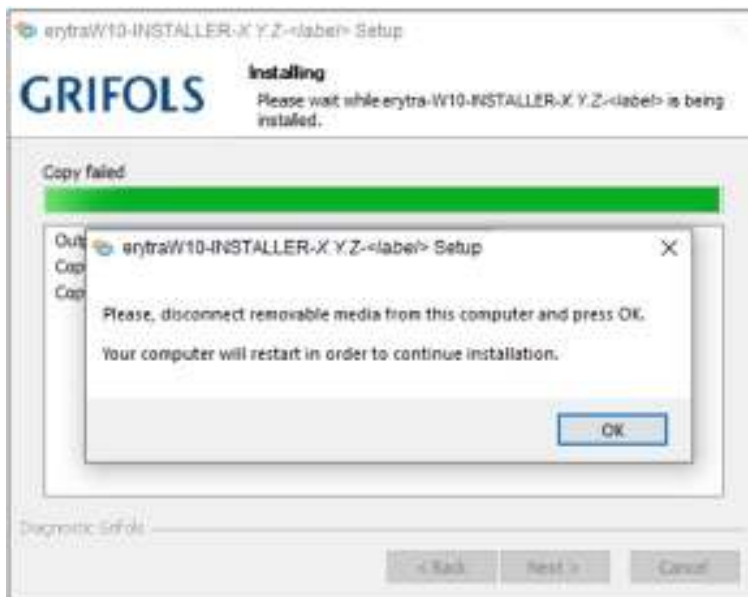


Figure 782 - Remove USB drive from the computer

5. Disconnect the USB drive and press **OK** button. Following window will be displayed:

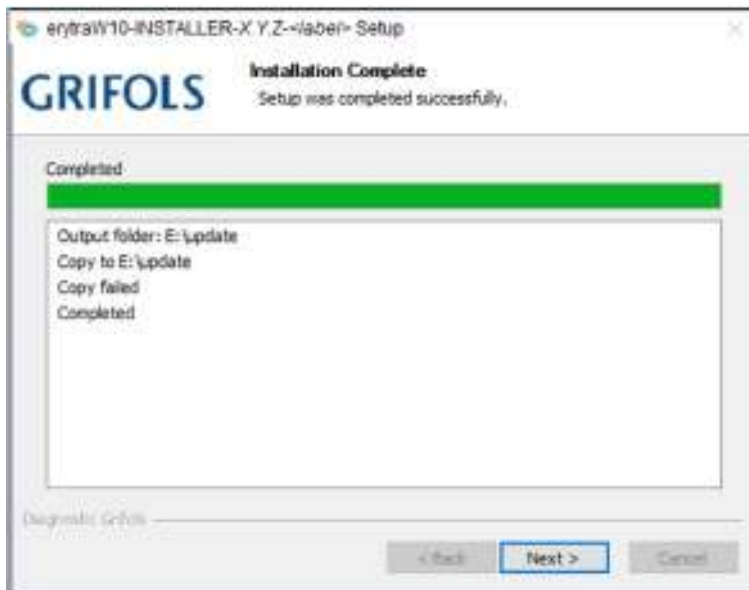


Figure 783 - Installer copied successfully

6. Select **Next >** button. The following window is displayed:



Figure 784 - Installer setup complete

7. Select **Reboot now** option and click on **Finish** button.

After restart, System will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer from E drive. Welcome screen will be displayed. Follow the installer instructions. See "Software installer steps" on page 1065

8. Once the installer is finished. The system will close the AppLocker audit session and apply the new rules, protect the volumes and restart. Enter Boot Application menu. See "Working with Boot Application" on page 88.

**NOTE**

If Installation has been completed with errors was displayed, or FAILED was displayed for a given installation step, retry software installation from See " Switch on the instrument. Select Update Software option from Boot Application menu. See "Working with Boot Application" on page 88." on page 1055).

Start software installation from Boot application remotely

1. Connect to the instrument through Grifols Secure Remote Connectivity tool in **Maintenance mode**.
2. Run **BootApp_Extended_Time** from **E:\Program Files\System Tools**.
3. Wait until confirmation message is displayed and select **OK**.
4. Restart from Windows taskbar.
5. Enter in **OS Shell** mode from Boot Application. See "Working with Boot Application" on page 88.
6. Copy **erytraW10-INSTALLER-X.Y.Z.exe** file to **E:\update** directory using Grifols Secure Remote Connectivity tool File Transfer.
7. Restart from Windows taskbar.

After restart, System will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer from E drive. Welcome screen will be displayed. Follow the installer instructions. See "Software installer steps" on page 1065

8. Once the installer is finished. The system will close the AppLocker audit session and apply the new rules, protect the volumes and restart. Enter Boot Application menu. See "Working with Boot Application" on page 88.

**NOTE**

If Installation has been completed with errors was displayed, or FAILED was displayed for a given installation step, retry software installation from See " Enter in OS Shell mode from Boot Application. See "Working with Boot Application" on page 88." on page 1057).

7.2.3.3.5 Restore Erytra

7.2.4 OS Settings

1. Reboot the instrument.
2. Enter **Maintenance mode** from Boot application. See "Working with Boot Application" on page 88.
Wait until OS is loaded.
3. Check the calibration of the touch screen and, if necessary, calibrate the touch screen. See "Adjusting touchscreen" on page 171.
4. Configure date and time. See "Setting Time Zone, Date and Time" on page 105.

7.2.5 Software data

1. Enter **OS Shell** from Boot application. See "Working with Boot Application" on page 88.
2. Check that adjustment files **PIPCallibrations.xml** and **CATCallibrations.xml** can be found in **E:\data\Erytra\conf** directory. Should this not be the case:
 - a. If a backup of these files exists: restore backup files of adjustments PIPCallibrations.xml and CATCallibrations.xml to **E:\data\Erytra\conf** directory,
 - b. If no backup exists:
 - i. Open **E:\data\Erytra\conf** folder.
 - ii. Rename PIPCallibrations_BASE.xml into PIPCallibrations.xml.
 - iii. Rename CATCallibrations_BASE.xml into CATCallibrations.xml.
 - iv. Perform the adjustments stored in these files after software installation.
CAT: See "Functional adjustments" on page 471.
PIP: See "Functional adjustments" on page 336.
3. Check that **testcounter.txt** can be found in **E:\data\Erytra\conf** directory. Should this not be the case, restore the backup of this file.
4. Restore the backup instrument configuration file **erytra.properties** to **E:\conf**
5. Restore the backup of restore path configuration file **backup.conf** to **E:\program files\System Tools**.



NOTE

R and V drives now belongs to instrument SSD. Check review backup path.

6. Restore the backup of the Microtubes jpg images to **E:\data\Erytra\Microtubes**
7. Restore the backup of the Raw directory to **E:\data\Erytra\Raw**

8. Restore the backup of Diana System definition files, if necessary, to **E:\data\SistemaDiana\Defs**
9. Restore the backup of the Diana System **Tandas** and **Traceability** folders, if necessary.
10. Restore the backup of the templates backup, **Erytra.pln**, if necessary, to **E:\data\SistemaDiana\Plantillas**
11. Restore the backup of the DSC backup files, **DSC.def**, **DSC_Custom.def**, if necessary, to **E:\data\Erytra\Wadiana**.
12. Restore the backup of the Technical Service software Macros, if necessary.
13. Restore the backup of the Erytra log files, if necessary, to **E:\Log** directory and its subdirectories
14. Restore the backup of the Diana System log files, if necessary, to **E:\data\SistemaDiana\log**

15. Restore the backup of the databases.

**NOTE**

Restore_DataBase.exe was removed in this version.

**CAUTION**

To perform this restore it is required to use the **IH Restore** application.

This application requires **IH Manager** and **IH Executor** backups with the following format and placed in the specified location.

File path: **E:\Backup\BAK_YYYY_MM_DD_hh_mm_ss**

Content: **IHExecutor.backup, IHManager.backup.**

- a. Rename and place the database backups s specified.
- b. Select **Restore** option from the boot application. See "Working with Boot Application" on page 88.

Restore application is opened.



NOTE

Alternatively, Restore application can be launched from **E:\Program Files\Restore\run(.bat)** from OS shell.

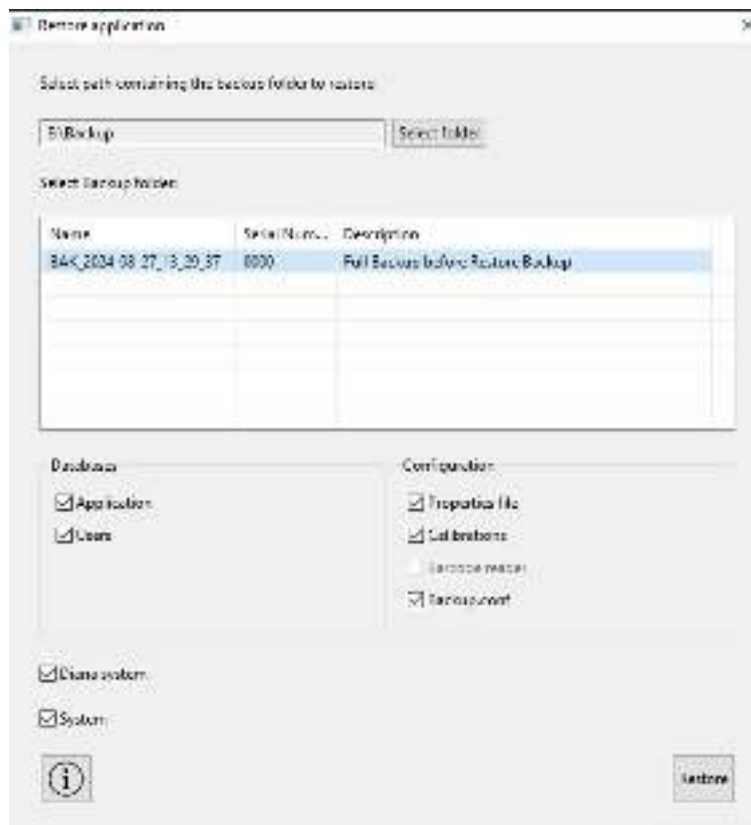


Figure 785 - Restore application

- c. Select the backup file to restore. Do not modify any current options selection.
- d. Press **Restore** button to import instrument configuration. Enter `admin` user password when requested. A progress windows will be displayed and the system will be rebooted several times.

- e. A message informing that System settings have been successfully imported will be displayed. Press **OK** button.



Figure 786 - Import settings PASS



NOTE

In case an empty Database has been restored, the reagent pack and the reagent language pack should be imported again from Importer Application.

16. Update the databases. Run **E:\Program Files\liquibase\updatedb.bat**.

17. Restore the Diana System configuration in Windows registry (Import the backup of Windows 10 HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols , if necessary).

**NOTE**

This step only applies when imported the registry from Windows 7.

- a. Edit registry backup file as follows:
 - All D:\ paths to E:\ paths.
 - All HKEY_LOCAL_MACHINE\SOFTWARE\Grifols paths to HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols paths.
 - Save the modifications.
- b. Open a **Command Prompt** as administrator and enter administrator credentials
 - **Start** → **Windows System** → **Windows Administrative Tools**
 - Click right mouse button on **Command Prompt** and select **more** > **Run as administrator**
 - Enter `admin` user password.



Figure 787 - Opening command prompt

- c. Enter following command (replacing `E:\config.reg` with the location and name of the registry backup file to restore):


```
C:\Windows\SysWOW64\regedit.exe E:\config.reg
```
- d. Confirmation window will be displayed. Press **Yes** button (window will close).
A message window informing that the values have been successfully added to the registry will be displayed.
- e. Press **OK** (window will close).
- f. Close the **Command Prompt** window.

**NOTE**

This procedure shall be only performed if LIS is enabled

1. Enter **Maintenance mode** from the boot application menu. See "Working with Boot Application" on page 88
Wait until OS is loaded

2. If LIS communication mode is set to serial port, review and configure, if necessary, that the current COM port selected is the one connected to LIS (that COM port name is the same after installing new operating system).
3. If LIS communication mode is set to TCP/IP, review and configure, if necessary, that the current TCP/IP IPv4 MS Windows configuration is the one required to connect to LIS. If necessary, indicate the Host IP address and Port in the corresponding DianaServer textboxes, and create a new outbound firewall rule to open the selected port. See "TCP-IP communication mode" on page 1159.
4. Configure LIS encoding to match current LIS encoding.

**NOTE**

HostDemo program uses (only) ANSI encoding.

5. If characters different from the Latin alphabet ones are used, set the suitable language in the Language for non-Unicode programs option

**NOTE**

Start → Control Panel → Clock and Region → Region → Change date, time or number formats → Administrative tab. Administrator password will be requested.

**NOTE**

Language for non-Unicode programs configuration of DianaServer PC, when not running in instrument PC, shall match Erytra configuration.

6. If necessary for backup feature or LIS connection, configure Erytra to share or access a shared folder. See "Sharing and accessing a shared folder or printer" on page 1109.

**NOTE**

R and V drives now belong to instrument SSD.

7. Restore the PC name and network configuration.

7.2.5.0.1 Firmware upgrade

1. Update the PCA firmwares according to the table. See "Updating firmware in a valid PCA" on page 1129.

Table 110 - Erytra v4.6.1 FW versions

PCA	Firmware version
CEN_SRM_CTRL	4.6.0
INC_CUP_CTRL	4.5.0
FLUCTRL59	4.6.2
CAT_CM_CTRL	4.5.0
PIP_CTRL	4.13.0
RD_IL421	5.5.0

7.2.5.0.2 Final setup

See "Software installation Final Setup" on page 1070

7.2.5.0.3 Verify the installation

In order to verify the installation, check the following:

1. Verify OS and SW installation. See "Verifying OS and SW versions" on page 1097.
2. Verify OS licence activation. See "Verifying Licence activation" on page 1038
3. Verify FW versions according to the table. Check firmware version

7.2.5.1 Software installer steps

Once the Installer Welcome screen is displayed.



Figure 788 - Software installer welcome window

1. Press **Next >**.

The installation options window is displayed.

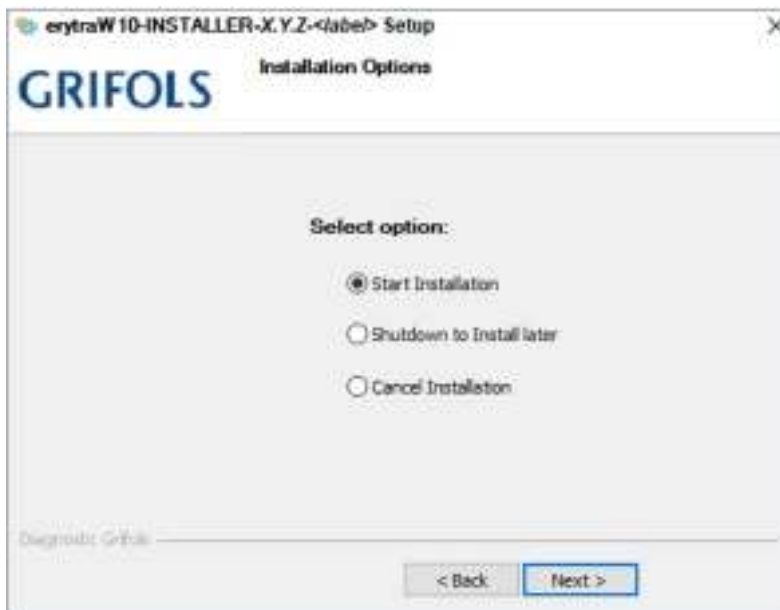


Figure 789 - Software installer options window



CAUTION

If **Hardware Error Found** window is displayed, check PC BIOS version (see **FSB-0059#0032** in "Bulletin index" on page 1183), check firewire PCA and camera and retry software installation.



Figure 790 - Hardware error found window

2. Keep **Start Installation** option selected and click on **Next >**.

The **Instrument Configuration (CE / US)** window will be displayed.



Figure 791 - Instrument Configuration window

3. Select the desired configuration: **CE** or **US** (Default option: **CE**) and press **Install**.

A window informing about installation progress will be displayed:

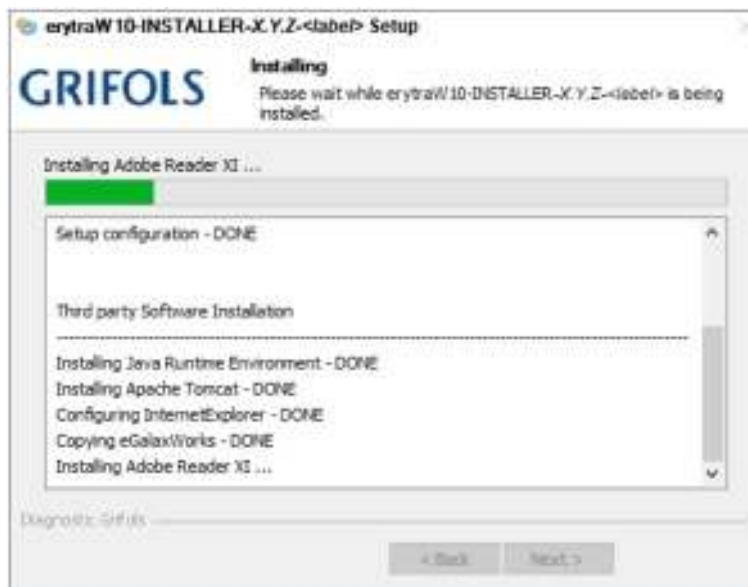


Figure 792 - Installation progress

4. Once complete, review the result and proceed accordingly.
 - DONE is displayed for all the steps. When all steps are DONE and the message **Installation has been completed with errors** is not displayed, click the **Next** button.
 - FAILED is displayed in any step. When a step has FAILED and the message **Installation has been completed with errors** is displayed, click the **Next** button, continue until the system is restarted, then enter boot application and retry software installation.

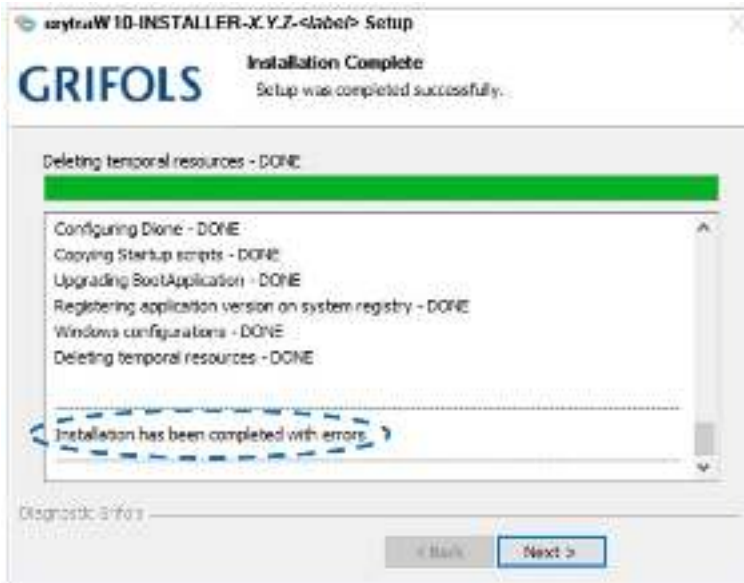


Figure 793 - Installation completed with errors window

5. Following window indicating that installation will be completed will be displayed.



Figure 794 - Completing software installation

6. Click on **Finish**. The system will close the AppLocker audit session and apply the new rules, protect the volumes and restart.

7.2.5.2 Software installation Final Setup

7.2.5.2.1 Close ports

If the installed Operating System is not version 2.3.3, close unnecessary open ports. See "OS Firewall: close a port".

7.2.5.2.2 Importer setup

1. Select **Importer** option in the **Boot Application** menu. See "Working with Boot Application" on page 88.
2. Import Reagents Pack file. See "Importing a reagent pack file" on page 885.



NOTE

Select the Reagents Pack the instrument is using to load all the information necessary for the interpretations. If no Reagent Pack has never been imported, import the desired Reagent Pack.



NOTE

The improvement implemented in Importer regarding interpretations importation requires that this information is imported again after software installation.

3. Import Reagents Language Pack file. See "Importing a reagent language pack file" on page 885.



NOTE

If all relevant data has been restored it is not necessary to re-install Reagents language Pack. Should this not be the case, proceed to install Reagents Language Pack.

4. Close **Importer** application. **Boot Application** menu options will be displayed.

7.2.5.2.3 Adjustments

1. Check that the instrument is properly leveled and perform instrument leveling, if necessary.
2. Perform the following adjustments depending on the previous installed version:

Previous installed version was lower to v4.4.0



NOTE

Below mentioned adjustments require (as pre-condition) that the instrument is properly levelled.

- a. Adjust the Reader. See "Functional adjustments" on page 698
- b. Adjust Gripper in Card Reader area. See "Adjusting gripper in the card reader area" on page 487
- c. If pre-analytical racks are used, perform Horns: pre-analytical adjustment (see "horns: pre-analytical adjustment") and Pre-analytical laser alignment check (See "Pre-analytical laser alignment check").

Previous installed version between v4.4.0 and v4.6.1



NOTE

Below mentioned adjustments require (as pre-condition) that the instrument is properly levelled.

- a. If pre-analytical racks are used, perform Horns: pre-analytical adjustment (see "horns: pre-analytical adjustment") and Pre-analytical laser alignment check (See "Pre-analytical laser alignment check").

7.2.6 Installing Erytra software (v5.2.0)

1. Download the software files listed in the corresponding FSB/INR. See "Downloading Erytra files" on page 1006
2. Check the OS version installed on the instrument. See "Verifying software version" on page 1096.
3. Select the appropriate option:

- See "Installing steps in units with OS version $\geq 2.2.0$ (v5.2.0)" on page 1073
- See "Installing steps in units with OS version $< 2.2.0$ (v5.2.0)" on page 1077

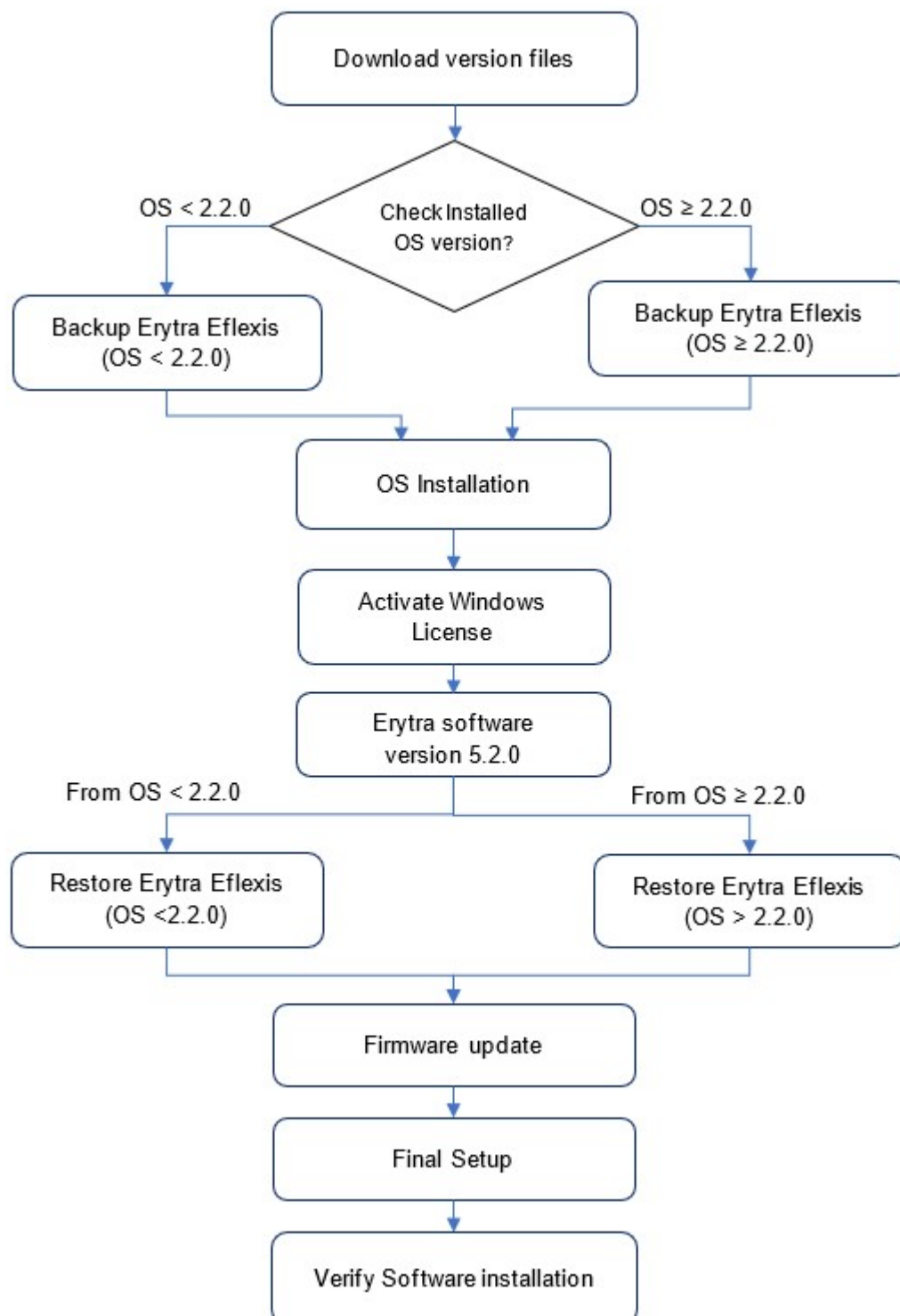


Figure 795 - Software installation process diagram

7.2.6.1 Installing steps in units with OS version $\geq$ 2.2.0 (v5.2.0)

**NOTE**

This procedure can be executed using Grifols Secure Remote Connectivity tool.

To update a properly installed Erytra software version proceed as follows:

**CAUTION**

All samples results must be validated and exported before starting the installation.

7.2.6.1.1 Export instrument configuration

**CAUTION**

If destination folder is on a USB drive, check that USB drive does not contain any virus before using it.

**NOTE**

If necessary, perform the manual backup of the following directories:

- E:\data\Erytra\Microtubes.
- E:\data\Erytra\Raw.
- E:\data\SistemaDiana.
- E:\logs.

1. Select the **Backup** option on the Boot Application (See "Working with Boot Application" section)

Backup application is opened.

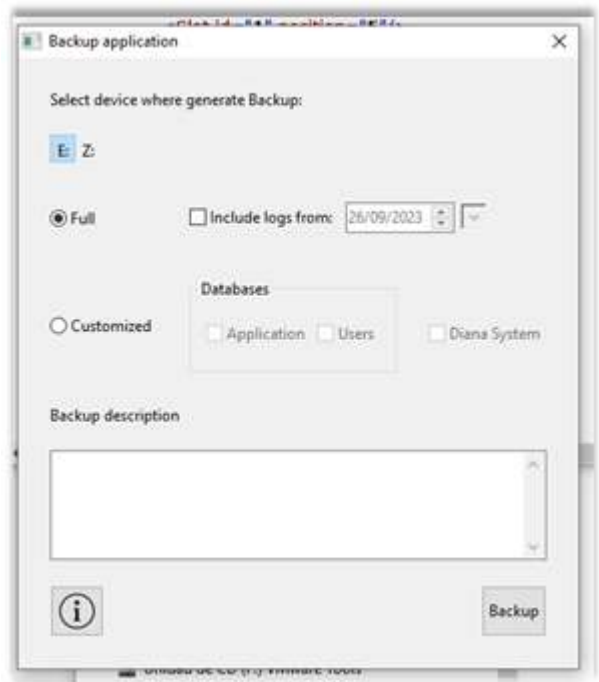


Figure 796 - Backup description and log files choice

2. Keep E drive and Full option selected.
3. Enter a Backup description, if desired.
4. Press **Backup** button to export instrument configuration. Enter `admin user password` when requested.
5. A message window will be displayed indicating the backup has been completed.

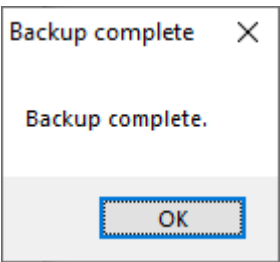


Table 111 - Backup completed

6. Press **OK** button.
7. Close Backup application pressing the top-right cross.
8. Backup will be saved on the device that you have defined.
9. Reboot Erytra.

**NOTE**

You can copy the previous backup folder (without modify anything) to save it.

7.2.6.1.2 Operating system installation

1. Install the Operating system:

- For on-site installation: See "Restoring from Boot Application menu" on page 1016
For remote installation: See "Restoring from Grifols Secure Remote Connectivity tool" on page 1019

7.2.6.1.3 Operating System license activation

1. Activate the Windows License (See "Activating Windows License" on page 1029).

7.2.6.1.4 Erytra Software version v5..0 installation

1. Install the Erytra Software (See "Software installer wizard (v5.2.0)" section)

7.2.6.1.5 Import instrument configuration

**NOTE**

If necessary, perform the manual restore of the Microtubes and Raw directories:

- E:\data\Erytra\Microtubes.
- E:\data\Erytra\Raw.
- E:\logs.

1. Enter **Restore** application (See "Working with Boot Application" section)

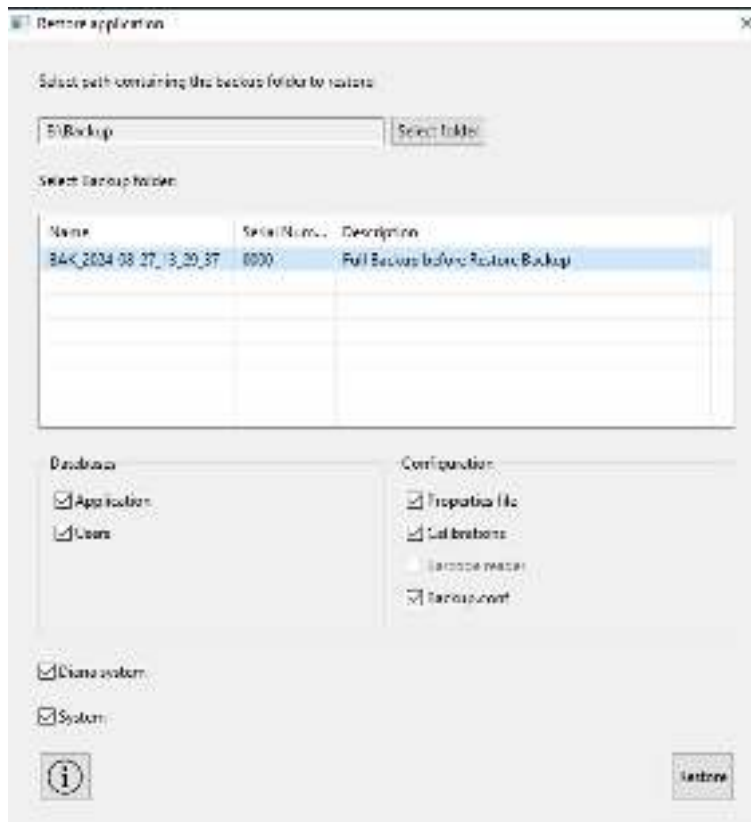


Figure 797 - Restore application

2. Select the backup containing the instrument configuration.



NOTE

Do not modify any current options selection.

3. Press **Restore** button to import instrument configuration. Enter `admin` user when requested. A progress windows will be displayed and the system will be rebooted several times.

4. A message informing that System settings have been successfully imported will be displayed. Press **OK** button.



Figure 798 - Import settings PASS

7.2.6.1.6 Firmware upgrade

Check and update the instrument Firmware (See "Updating firmware" chapter) according to the following table:

PCA	Firmware version
CEN-SRM	4.5.0
INC-CUP	4.5.0
CAT-CM	4.5.0
PIP	4.10.0
FLU	4.6.2
RD	5.5.0

7.2.6.1.7 Final setup

Perform Final Setup operations (See "Final Setup (v5.2.0)" chapter).

7.2.6.1.8 Verify the installation

Verify the software installation (See "Verifying software installation (v5.2.0)" chapter).

7.2.6.2 Installing steps in units with OS version < 2.2.0 (v5.2.0)



NOTE

This procedure cannot be executed using Grifols Secure Remote Connectivity tool as it involves Operating System Full Recovery process from an Operating System version < 2.2.0 (this is SW 5.0.0).

To update a properly installed Erytra software version proceed as follows:

**CAUTION**

All samples results must be validated and exported before starting the installation.

7.2.6.2.1 Relevant data backup

**CAUTION**

If destination folder is on a USB drive, check that USB drive does not contain any virus before using it.

1. Perform a Backup of relevant data (See "Backing up relevant data manually" on page 1120).

7.2.6.2.2 Restore SSD from USB device

1. Perform a **Full Recovery** from a USB pen drive (See "Restoring from USB pen drive" on page 1007.)

7.2.6.2.3 Operating System license activation

1. Activate the Windows License (See "Activating Windows License" on page 1029).

7.2.6.2.4 Erytra Software version v5.2.0 installation

1. Install the Erytra Software (See "Software installer wizard (v5.2.0)" section)

7.2.6.2.5 Initial configuration

1. Enter **Maintenance Mode** (See "Working with Boot Application" section)
2. Configure time zone, date and time. See "Setting Time Zone, Date and Time" on page 105.

3. Restore Diana System configuration in Windows registry (Import the backup of Windows 10 HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols , if necessary. To do so:

- Run **Command Prompt** as Administrator.

How to: **Start** → **Windows System**

- Click right mouse button on **Command Prompt** and select **more** > **Run as administrator**.
- Enter `admin` user password.



Figure 799 - Opening command prompt

- Enter following command (replacing `E:\config.reg` with the location and name of the registry backup file to restore):
C:\Windows\SysWOW64\regedit.exe E:\config.reg
 - Confirmation window will be displayed. Press **Yes** button (window will close).
 - A message window informing that the values have been successfully added to the registry will be displayed. Press **OK** (window will close).
 - Close the **Command Prompt** window.
4. If LIS connection is enabled and communication mode is set to serial port, review/check (and configure, if necessary) that the current COM port selected is the one connected to LIS (that COM port name is the same after installing new operating system).
 5. If LIS connection is enabled and communication mode is set to TCP/IP, indicate the Host IP address and Port in the corresponding DianaServer textboxes, and create a new outbound firewall rule to open the selected port.
 6. If LIS connection is enabled, review/configure LIS encoding to match current LIS encoding.



NOTE

HostDemo program uses (only) ANSI encoding.

7. If LIS connection is enabled and characters different from Latin alphabet ones are used, enter Maintenance mode and set the suitable language in the Language for non-Unicode programs option (**Start → Control Panel → Clock and Region → Region → Change date, time or number formats → Administrative tab**. Administrator password will be requested):

**NOTE**

Language for non-Unicode programs configuration of DianaServer PC, when not running in instrument PC, shall match Erytra configuration.

8. If necessary for backup feature or LIS connection, configure Erytra to share or access a shared folder. See "Sharing and accessing a shared folder or printer" on page 1109.

**NOTE**

R and V drives now belong to instrument SSD.

9. Restore the backup/set the PC name and network configuration.
10. Restart Erytra.
11. Once PC reboots, click on bottom-left corner grid (🖱️) button so **Boot Application** login window is displayed.

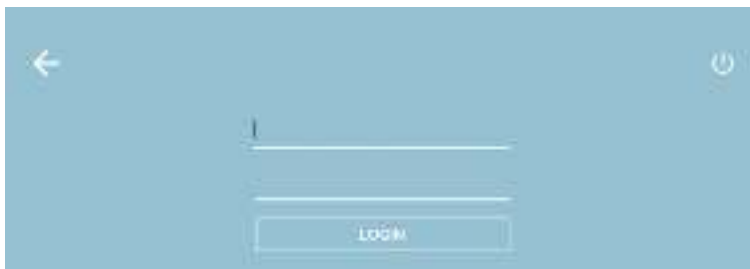


Figure 800 - Boot Application login window

12. Log in as `adminUser` (it is case sensitive). **Boot Application** options will be displayed.

7.2.6.2.6 Restore relevant data backup

1. Enter **OS Shell** (See "Working with Boot Application" section)
2. Check calibration of touchscreen/Calibrate touchscreen (See "Adjusting touchscreen" section).

3. Check that adjustment files PIPCalibrations.xml and CATCalibrations.xml can be found in **E:\data\Erytra\conf** directory. Should this not be the case:
 - a. If a backup of these files exists: restore backup files of adjustments PIPCalibrations.xml and CATCalibrations.xml to **E:\data\Erytra\conf** directory,
 - b. If no backup exists:
 - i. Open **E:\data\Erytra\conf** folder.
 - ii. Rename PIPCalibrations_BASE.xml into PIPCalibrations.xml.
 - iii. Rename CATCalibrations_BASE.xml into CATCalibrations.xml.
 - iv. Perform the adjustments stored in these files after software installation.

CAT: See "Functional adjustments" on page 471.

PIP: See "Functional adjustments" on page 336.
4. Check that **testcounter.txt** can be found in **E:\data\Erytra\conf** directory. Should this not be the case, restore the backup of this file.
5. Check that instrument configuration file (**erytra.properties**) can be found in **E:\conf**. Should this not be the case, restore the backup of this file.
6. Restore Backup path configuration file (**E:\program files\System Tools\backup.conf**).

**NOTE**

R and **V** drives now belongs to instrument SSD. Check review backup path.

7. Restore the backup of Microtubes directory to **E:\data\Erytra\Microtubes**.
8. Restore the backup of Raw directory to **E:\data\Erytra\Raw**.
9. Restore backup of Diana System definition files, if necessary.
10. Restore the backup of Diana System Tandas and Traceability folders, if necessary.
11. Restore, if necessary, the backup of the templates (**Erytra.pln**. Default directory **E:\data\SistemaDiana\Plantillas**).
12. Restore, if necessary, the backup of **DSC.def** and **DSC_Custom.def** files (**E:\data\Erytra\Wadiana**).
13. Restore the backup of the Technical Service software Macros, if necessary.
14. Restore the backup of Erytra log files (**E:\Log** directory and its subdirectories) and Diana System log files (default location **E:\data\SistemaDiana\log**), if necessary.

15. Restore databases backups.

**NOTE**

Restore_DataBase.exe was removed in Erytra SW v5.2.0

**NOTE**

To perform this restore it is required to use the **IH Restore** application.

This application requires IH Manager and IH Executor backups with the following format and placed in the specified location.

File path: E:\Backup\BAK_YYYY_MM_DD_hh_mm_ss\

Content:

- **IHExecutor.backup**
- **IHManager.backup**.

- Rename and place the database backups as specified.
- Enter **Restore** application (See "Working with Boot Application" section)

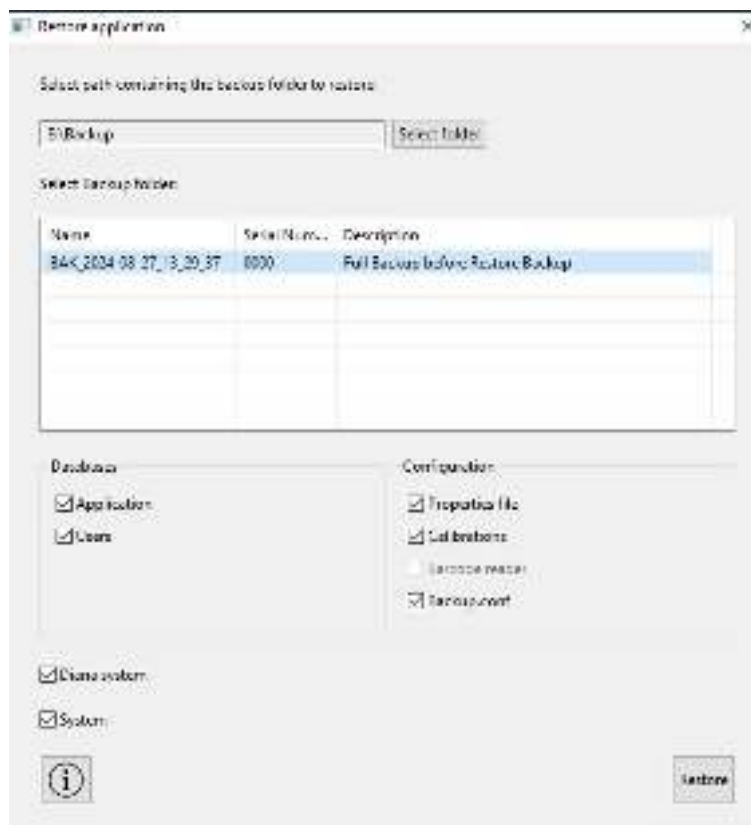


Figure 801 - Restore application

- iii. Select the backup containing the instrument configuration.

**NOTE**

Do not modify any current options selection.

- iv. Press **Restore** button to import instrument configuration. Enter `admin user` when requested. A progress windows will be displayed and the system will be rebooted several times.
- v. A message informing that System settings have been successfully imported will be displayed. Press **OK** button.



Figure 802 - Import settings PASS

16. Update the databases.

- Run `E:\Program Files\liquibase\updatedb.bat`.

17. Restore Diana System configuration in Windows registry, if necessary.

**NOTE**

This step only applies when imported the registry from Windows 7.

- Registry name: HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols
- i. Edit registry backup file as follows:
 - All 'D:\' paths to 'E:\' paths.
 - All 'HKEY_LOCAL_MACHINE\SOFTWARE\Grifols' paths to 'HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols' paths.

Save the modifications.

- ii. Open Command Prompt as administrator and enter administrator credentials.

**NOTE****Start - Windows System - Windows Administrative Tools**

Click right mouse option on Command Prompt and select **More | Run as administrator**.

Enter admin user password: dgroot.

- iii. Enter following command (replacing E:\config.reg with the location and name of the registry backup file to restore):
 - C:\Windows\SysWOW64\regedit.exe E:\config.reg
- iv. Confirmation window is displayed. Select **Yes** option.

An information window is displayed as follows:
Content: Values have been successfully added to the registry.
- v. Select **OK** option.
- vi. Close Command Prompt window.

7.2.6.2.7 Firmware upgrade

Check and update the instrument Firmware (See "Updating firmware" chapter) according to the following table:

PCA	Firmware version
CEN-SRM	4.5.0
INC-CUP	4.5.0
CAT-CM	4.5.0
PIP	4.10.0

PCA	Firmware version
FLU	4.6.2
RD	5.5.0

7.2.6.2.8 Final setup

Perform Final Setup operations (See "Final Setup (v5.2.0)" chapter).

7.2.6.2.9 Verify the installation

Verify the software installation (See "Verifying software installation (v5.2.0)" chapter).

7.2.6.3 Software installer wizard (v5.2.0)

Start installation from USB pen-drive

1. Copy the file as follows using an external computer:
 - File name: ErytraW10-INSTALLER-X.Y.Z.exe (X.Y.Z is 5.2.0)
 - Location: USB ROOT:\installers
2. Connect the USB drive to Erytra PC.



NOTE

It is recommended to use USB 3.0 ports located in the rear part of the instrument.

3. Select 'Update Software' option from the Boot Application (See "Working with Boot Application" section)

Erytra installer will be run and the following window is displayed.



Figure 803 - Software installer welcome window

4. Select 'Next' option.

The installer is copied to 'E:\update' directory.

An information window requesting to disconnect the USB drive is displayed.

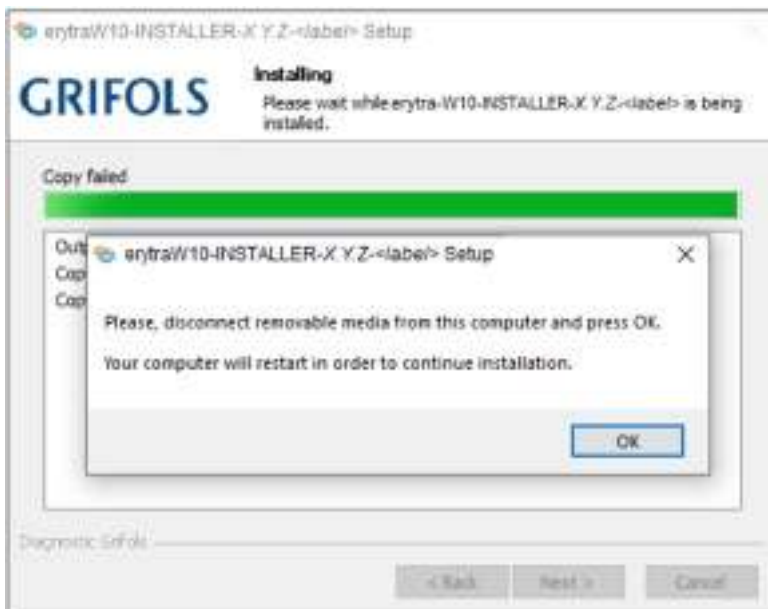


Figure 804 - Disconnect removable media message

5. Disconnect the USB drive and select 'OK' option.

The following window is displayed.

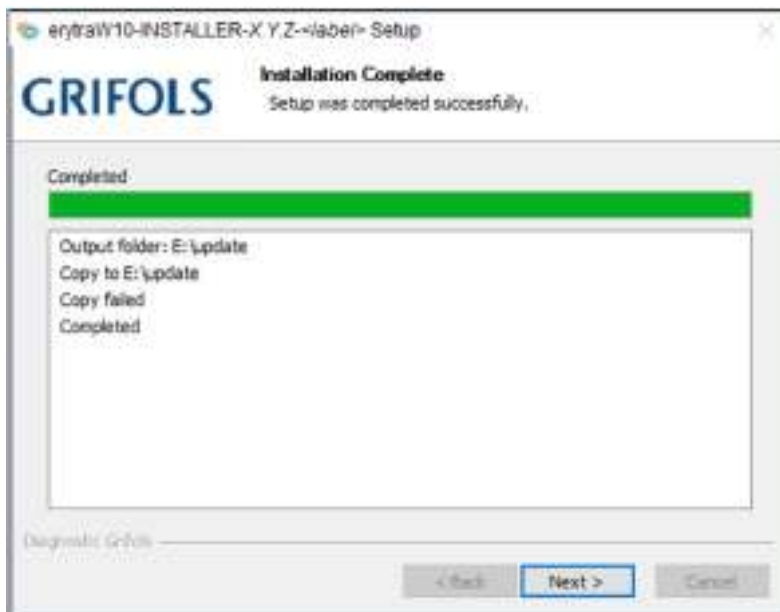


Figure 805 - Installer files copy progress window

6. Select 'Next' option.

The following window is displayed.



Figure 806 - Installer Restart window

7. Select 'Reboot now' and 'Finish' option.

System will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer from E: drive.

Start installation from Boot application on-site



NOTE

This process can be executed with Grifols Remote Connectivity tool.

1. Switch on the instrument and press the grid () button on the Boot Application initial window. The Boot Application login window will be displayed.

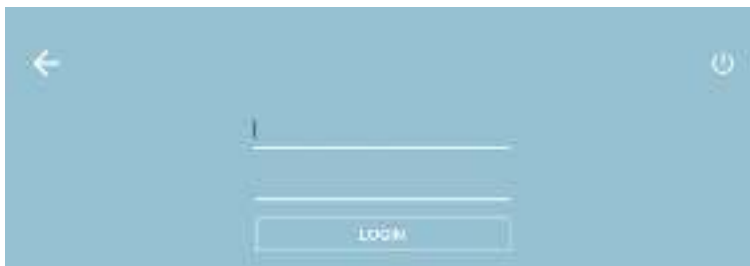


Figure 807 - Boot Application login window

2. Log in as `adminUser` (it is case sensitive). Boot Application options will be displayed. Select the **OS Shell** () option. Copy **erytraW10-INSTALLER-5.2.0.exe** to a USB drive using an external computer.
3. Connect the USB drive to the Erytra PC. It is recommended to use USB 3.0 ports (rear part of the instrument).
4. Copy the **erytraW10-INSTALLER-5.2.0.exe** file to the instrument **E:\update** directory.
5. Restart Erytra.
6. The system will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer (from the **E:** drive).

Start installation from Boot application remotely



NOTE

This process can be executed with Grifols Remote Connectivity tool.

1. Connect to the instrument through Grifols Secure Remote Connectivity tool in Maintenance mode.

2. Run the following file:
 - File: BootApp_Extended_Time
 - Path: Program Files\System Tools
3. Wait until confirmation message is displayed and select 'OK'.
4. Restart from Windows taskbar and enter in 'OS Shell' mode from Boot Application.
5. Copy the file as follows:
 - File name: ErytraW10-INSTALLER-X.Y.Z.exe (X.Y.Z is 5.2.0)
 - Location: E:\update
6. Restart from Windows taskbar

Erytra Software installer wizard

1. The Welcome screen will be displayed:



Figure 808 - Welcome window - software Installation

2. Click on **Next >**.

3. Keep the **Start Installation** option selected and click **Next >**.

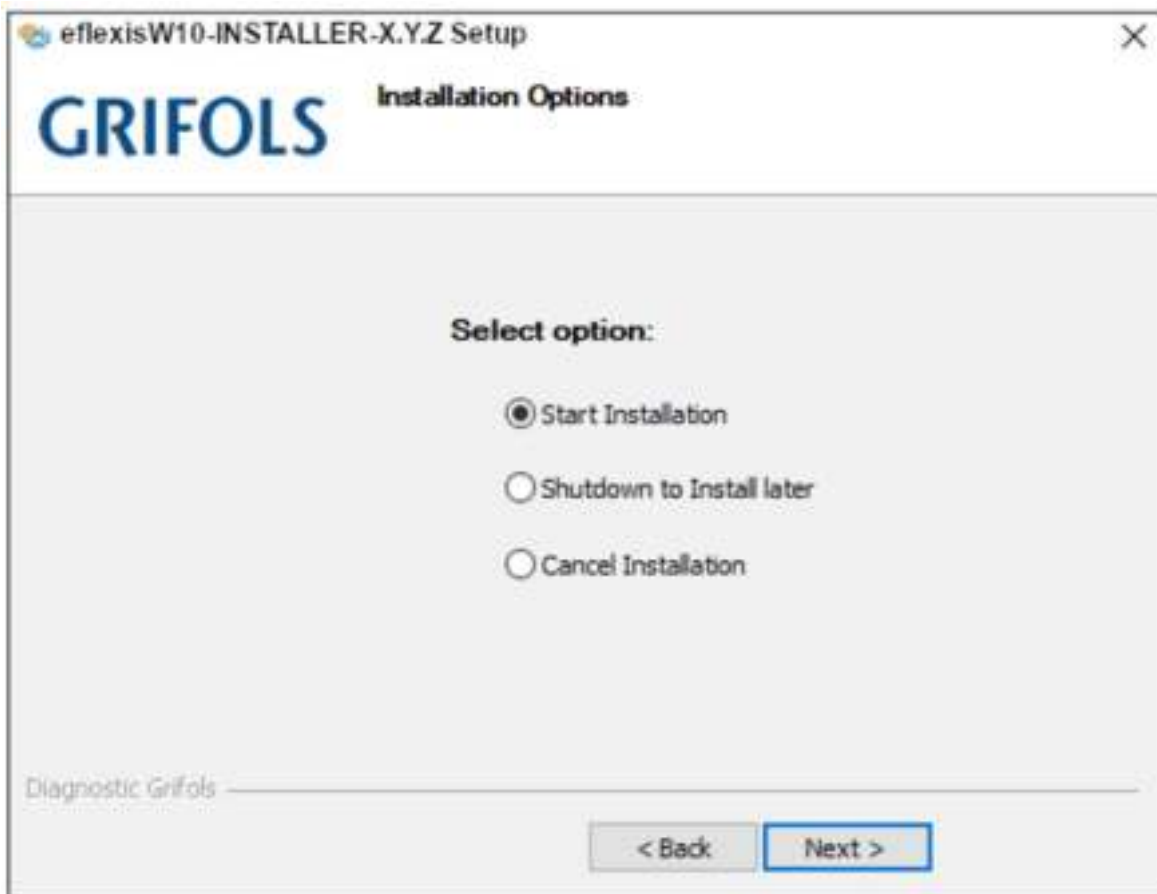


Figure 809 - Installation options window

4. If the **Hardware Error Found** window is displayed, check the PC BIOS version (refer to FSB-0059#0032, see "Bulletin index" on page 1183), firewire PCA and camera and retry software installation.



Figure 810 - Hardware error found window

5. Select the desired configuration (CE or US) and press **Install**.



Figure 811 - Instrument Configuration window

6. A window informing about installation progress will be displayed:

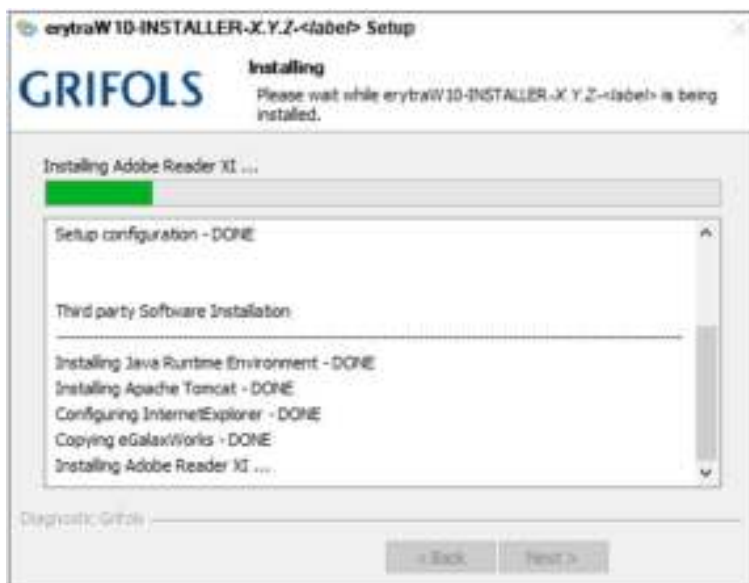


Figure 812 - Installation progress window

7. Once completed, review the result and proceed accordingly:
- DONE** is displayed for all the steps. When all steps are **DONE** and the message **Installation has been completed with errors** is not displayed, click **Next >**.
 - FAILED** is displayed in any step. When a step is **FAILED** and the message **Installation has been completed with errors** is displayed, click **Next >**, continue until step 16 and then go back to step 3 (retry software installation).

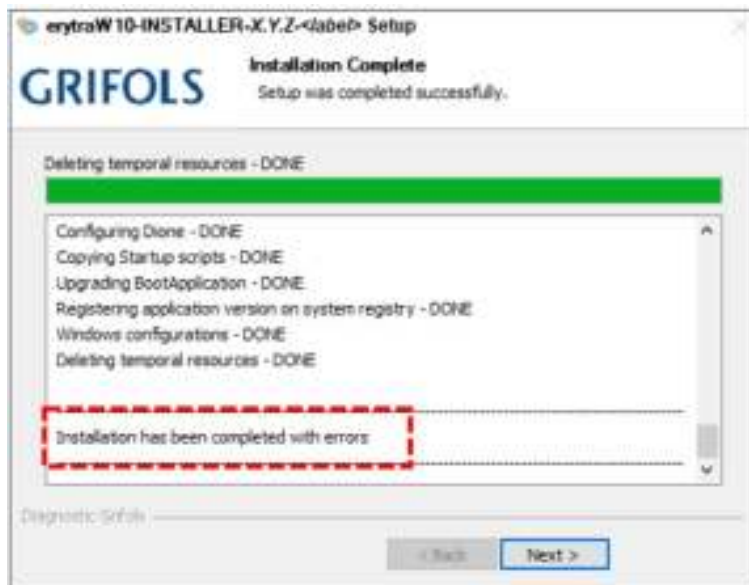


Figure 813 - Installation complete window

8. The following window indicating that installation will be completed will be displayed.



Figure 814 - Installation completed window

9. Click **Finish**. The system will close the AppLocker audit session and apply new rules, protect the volumes and restart.
10. Once the PC reboots, enter the Boot Application options (See "Working with Boot Application" section).

**NOTE**

If in step 7 the message **Installation has been completed with errors** was displayed and/or **FAILED** was displayed for a given installation step, go back to and retry software installation.

7.2.6.4 Final Setup (v5.2.0)

Importer

1. Enter **Importer**(See "Working with Boot Application" section)
2. Import the Reagent Packs that the instrument is using to load all the information necessary for the interpretations. See "Importing a reagent pack file" on page 885.

**NOTE**

Select the Reagents Pack the instrument is using to load all the information necessary for the interpretations. If no Reagent Pack has never been imported, import the desired Reagent Pack.

**NOTE**

The improvement implemented in Importer regarding interpretations importation requires that this information is imported again after software installation.

3. Import Reagents Language Pack file (See "Importing a reagent language pack file" on page 885)

**NOTE**

If all relevant data has been restored it is not necessary to re-install Reagents Pack. Should this not be the case, proceed to install Reagents Pack and Reagents Language Pack.

4. Close **Importer** application.

Adjustments

1. Enter **Technical Service Software** (See "Working with Boot Application" section)
2. Check the instrument is properly levelled and perform instrument levelling, if necessary.

3. Perform the following adjustments depending on the previous installed version:

If previous installed version is lower than v5.0.0

**NOTE**

Below mentioned adjustments require (as pre-condition) that the instrument is properly levelled.

- Reader adjustments - Camera (See "Functional adjustments" on page 698).
- Gripper adjustment in Card Reader area (See "Adjusting gripper in the card reader area" on page 487).

If previous installed version is v5.1.0

**NOTE**

Below mentioned adjustments require (as pre-condition) that the instrument is properly levelled.

- Reader adjustments - Camera (See "Functional adjustments" on page 698).
- Gripper adjustment in Card Reader area (See "Adjusting gripper in the card reader area" on page 487)..

If previous installed version is v5.2.0

- No adjustments are needed

**NOTE**

Centrifuges automatic adjustment is available, but it is not mandatory.

7.2.6.5 Verifying software installation (v5.2.0)

1. Enter **Information** option (See "Working with Boot Application" section).



Figure 815 - Software information

2. Check that the O.S. Version is 2.3.2. See "Installing or restoring Operating System" on page 1006
3. Check that Erytra software version is v5.2.0 Should this not be the case, see "Installing Erytra software (v5.2.0)" on page 1071.
4. Close the information window.
5. Enter **OS Shell** option (See "Working with Boot Application" section).
6. Open Command Prompt as administrator and enter administrator credentials.

**NOTE**

- Start - Windows Administrative Tools
- click right mouse option on Command Prompt and select More | Run as administrator
- Enter admin user password

7. Typ the following command:

```
slmgr /dlv
```

The following will be displayed

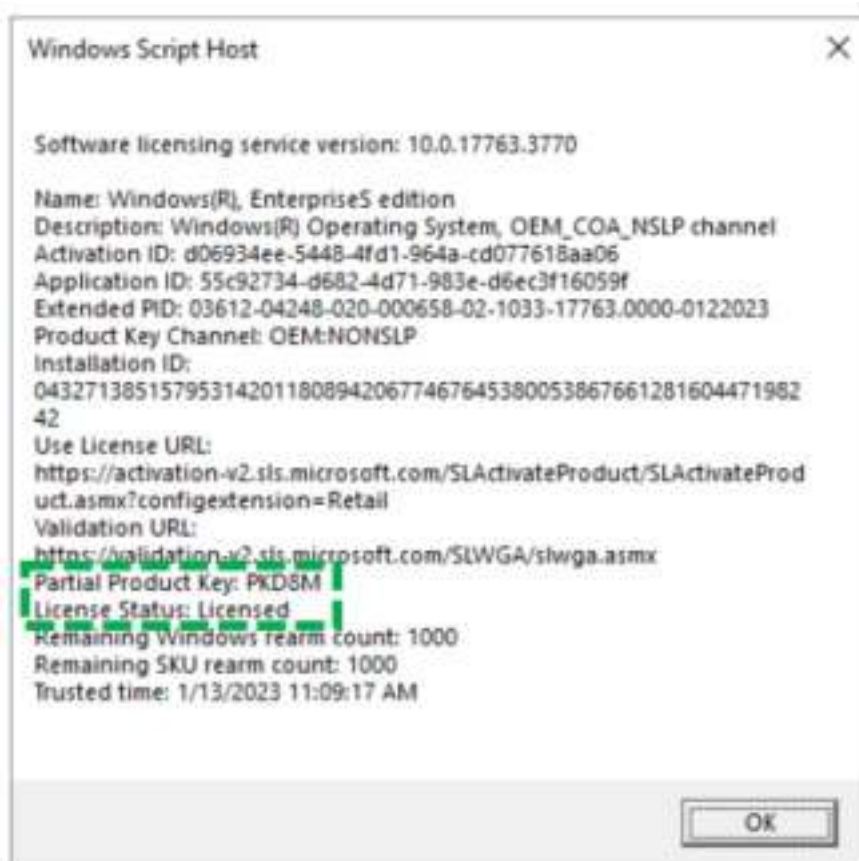


Figure 816 - License status line

8. Check that License Status field status is Licensed.
Should this not be the case, go back to section 2.4.4 Activate Windows License.
9. Enter **Technical Service Software** option (See "Working with Boot Application" section).
10. Check currently installed firmware version.

PCA	Firmware version
CEN-SRM	4.5.0
INC-CUP	4.5.0
CAT-CM	4.5.0
PIP	4.10.0
FLU	4.6.2
RD	5.5.0

Should this not be the case, go back to section 2.8 Firmware Installation.

11. Close Technica Service Software
12. Shut down the instrument

7.2.6.6 Verifying software version

7.2.6.6.1 From the Boot Application

Software version can also be checked from the Boot Application:

1. Switch on the instrument.
2. Click the bottom-left corner grid () button . The Boot Application login window will be displayed.



Figure 817 - Boot application login window

3. Log in as `adminUser` (it is case sensitive). Boot Application options will be displayed.

4. Select the **Information** (i) option.



Figure 818 - Software information

5. Close the information window.

7.2.6.6.2 From ErytraApplication



The software version is displayed when clicking on upper-left area:

7.2.7 Verifying OS and SW versions

7.2.7.1 From Boot Application



NOTE

This procedure is only available fro Erytra v4.3.0 or above

Installed Operating System and Software versions are available from the Boot application:

1. Switch on the instrument.
2. Select the **Information** (i) option from the boobt application. See "Working with Boot Application" on page 88



Figure 819 - Software information

3. Close the information window.

7.2.7.2 From ErytraApplication

Installed Software versions are available from the user application:



The software version is displayed when clicking on upper-left area:

7.3 Operating system

Information related to the operating system (OS) is provided in this section.

7.3.1 Return to Boot Application

A Boot Application entry can be found in Start menu to allow returning to Boot Application from OS Shell mode without the need to restart the system. To return to the point where the Boot Application icon is displayed (and to continue user application start up or allowing to enter Boot Applications again).

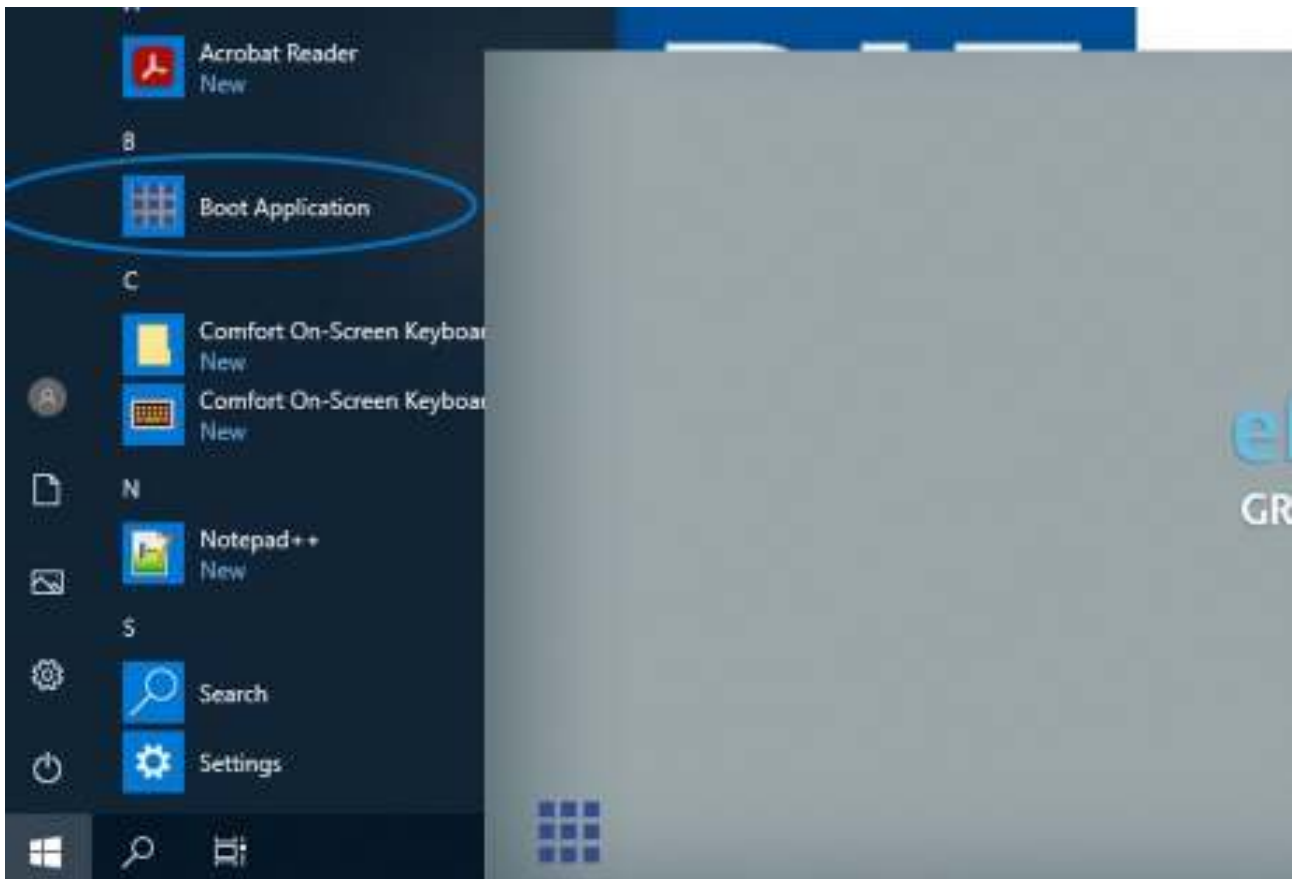


Figure 820 - Return application icon in start menu and Boot Application entry icon while system start.

7.3.2 Windows security packages

The security packages up to September 2024 (Windows version 10.0.17763.6293) are included in:

- OS image v2.3.2
- OS images v2.3.1 with Windows security update v1.6.0.

Table 112 - Security Updates packages

ID Sec. Package	Description
KB5039879	July 9, 2024-KB5039879 Cumulative Update for .NET Framework 3.5 and 4.7.2 for Windows 10, version 1809 and Windows Server 2019. This security update addresses a remote code execution vulnerability detailed in CVE-2024-38081 (.NET Framework Elevation of Privilege Vulnerability).
KB5043050	September 10, 2024-KB5043050 (OS Build 17763.6293) This security update includes improvements. Below is a summary of the key issues that this update addresses when you install this KB. If there are new features, it lists them as well. The bold text within the brackets indicates the item or area of the change we are documenting. • [AppContainer] Printers do not work as you expect when you use them in a restricted setting, like AppContainer. • [BitLocker] You might not be able to decrypt a BitLocker data drive. This occurs when you move that drive from a newer version of Windows to an older version. • [Windows Installer] When it repairs an application, the User Account Control (UAC) does not prompt for your credentials. After you install this update, the UAC will prompt for them. Because of this, you must update your automation scripts. Application owners must add the Shield icon. It indicates that the process requires full administrator access. To turn off the UAC prompt, set the HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\Installer\DisableLUAInRepair registry value to 1. • [Windows Server 2019 (known issue)] Some devices slow down, stop responding, and have high CPU usage. This is most likely to occur with Cryptographic Services. A few companixes report that the issue occurs when antivirus software scans the %systemroot%\system32\catroot2 folder for Windows updates.

Perform the following steps to check which updates are installed:

1. Open the Command Prompt (CMD) with admin rights and execute the following command:

```
wmic qfe get hotfixid
```

The installed security packages are displayed.

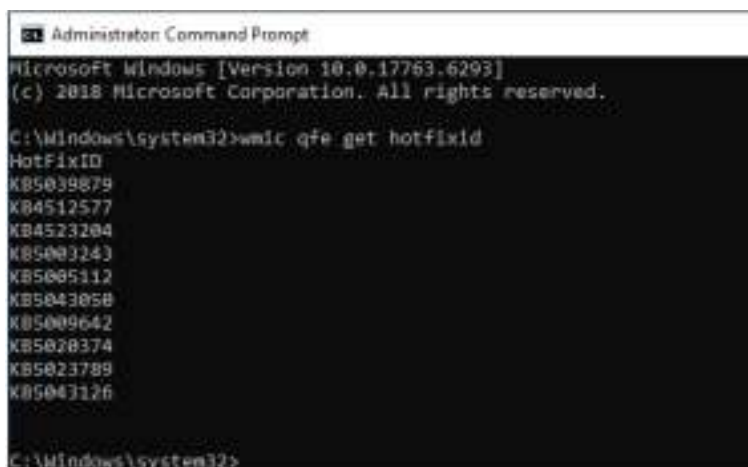


Figure 821 - Installed security packages

2. Check that the packages corresponding to the SO version 2.3.2 or security package update v1.6.0 are included in the list.

**NOTE**

Only the inclusion of the packages included in SO version 2.3.2 or security package update v1.6.0 must be checked. Lists can differ between instruments depending on packages previously installed.

7.3.2.1 Windows security update installation

**NOTE**

This procedure can be executed using Grifols Secure Remote Connectivity tool.

1. Switch on the instrument and enter **Maintenance mode** (See "Working with Boot Application" on page 88).
2. Check that:
 - OS version is 2.3.1 or 2.3.1 + updates v1.6.0 2024/09 or 2.3.2.
 - The Security Updates v1.7.0 are not installed.

How to: Start > Windows System > Control Panel - System and Security > System

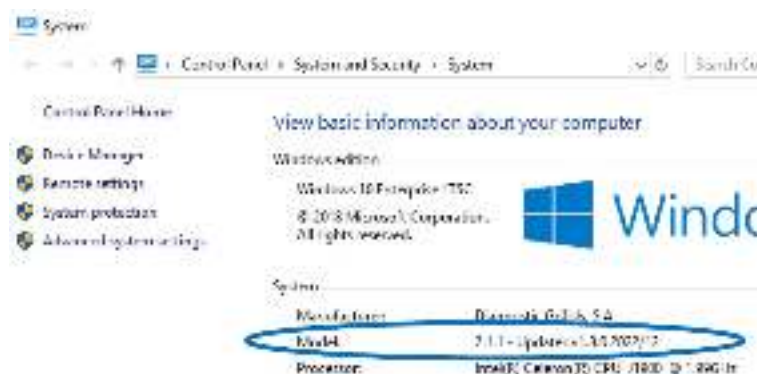


Figure 822 - System model field in Control panel to check OS versions and updates.

Should this not be the case:

- If OS version is not 2.3.1 or 2.3.2, update OS to the latest version: OS 2.3.3
 - If OS version is 2.3.1 - Updates v1.7.0 2025/03 or 2.3.2 - Updates v1.7.0 2025/03, the security updates v1.7.0 are already installed and no further action is required (skip all the remaining steps).
3. Reboot the instrument and enter **OS Shell**(See "Working with Boot Application" on page 88).
 4. Copy **w10-1809-ar-secupdates-installer-1.7.0.exe** file to instrument **E:\update** directory.
 5. Restart the instrument.

System will unprotect the volumes, reboot, open AppLocker audit session and automatically run the installer (from E drive).

6. Welcome screen will be displayed. Click on **Next >**.



Figure 823 - Security update installer window: welcome page

7. keep **Start Installation** option selected and click on **Install**.

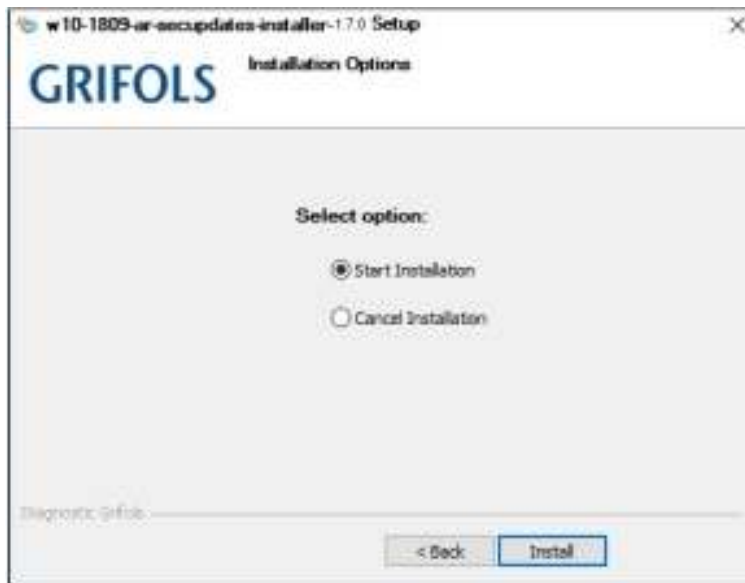


Figure 824 - Security Updates installer window: options selection

The installer will check the compatibility with the system and if all the patches are already installed.

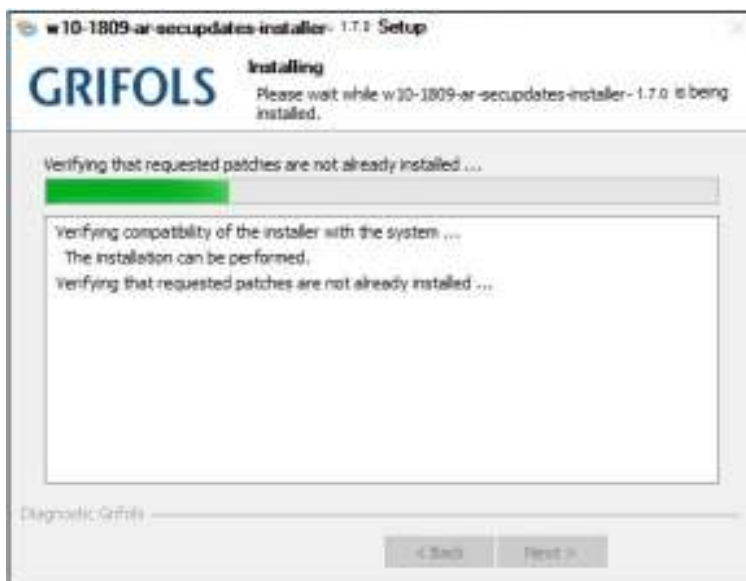


Figure 825 - Security updates installer window: verifying patches

**NOTE**

if the patches cannot be installed, the process will not continue and a window informing the reason why the installation cannot be performed will be displayed:

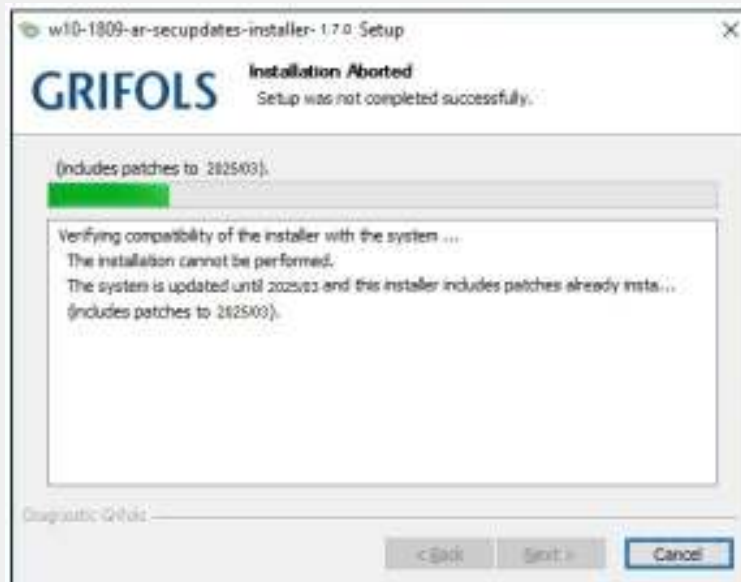


Figure 826 - Security updates installer window: installation aborted.

Close the window using top-right **X** button or the **Cancel** button to exit.

If some or all the packages can be installed, installation will start. A window informing if any package is already installed and informing about the packages installation progress will be displayed:

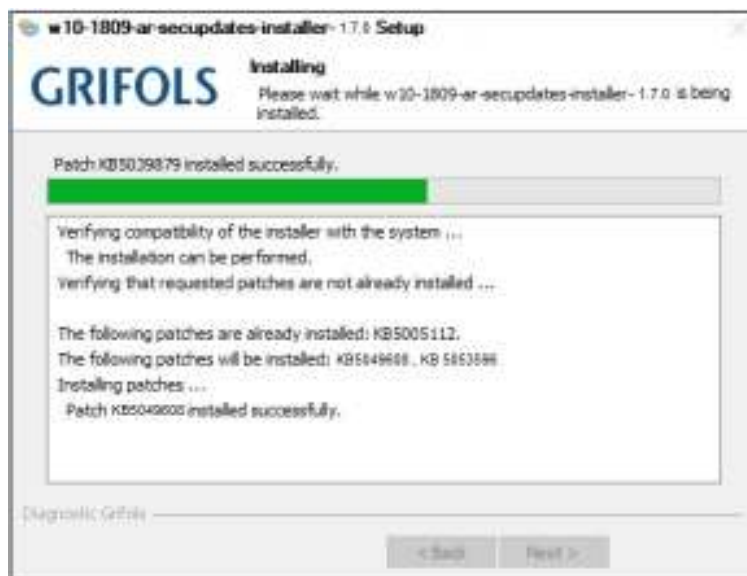


Figure 827 - Security updates installer window: installation successful message

The installation of the complete set of packages takes 40 minutes approximately.

**NOTE**

Operating System Model will be updated to:

- 2.3.1 - Updates v1.7.0 2025/03, for Operating System v2.3.1.
- 2.3.2 - Updates v1.7.0 2025/03, for Operating System v2.3.2

Once completed, the installer shows the security updates that have been applied and informs that the system will restart automatically after 60 seconds.



Figure 828 - Security updates installer window: automatic reboot message

8. Once the system reboots, shut down the instrument.
9. Verify the version installation as in step 2.

7.3.3 Accessing Windows 10 file explorer

This option allows qualified Field Service Engineers to access Windows File Explorer and corresponding taskbar, as well as additional functionalities, from Technical Service, Diagnostic application and User application modes.

**NOTE**

EI SW 4.4.0 includes this functionality.

The executable file launched through INR-0059#0125 is incompatible with this SW version. Should both have been installed, the application won't work. See "Bulletin index" on page 1183.

The details on all the functionalities available are the following:

7.3.3.1 Technical service mode

- Permanent access to Windows 10 File Explorer (taskbar always present at the bottom of the screen).
- Enabled right-click functionality when using the mouse on File Explorer.
- Enabled access to **E:** drive (DATA).
- Enabled shortcut keys, except for **CTRL** and **Fx** keys (where x is a number between 1 and 12).
- Enabled Windows () and left **ALT** keys.



Figure 829 - View of the Technical Service software with access to Windows 10 File Explorer

When closing the Technical Service software, Boot application will be launched again as in a usual start of the instrument (the grid () button should be pressed if no access to user application is intended for the session).

7.3.3.2 Diagnostic application mode



NOTE

A physical keyboard and admin user password is needed to enable the new functionalities in Diagnostic application mode.

- Permanent access to Windows 10 file explorer (taskbar always present at the bottom of the screen).
- Enabled right-click functionality when using the mouse on File Explorer.
- Enabled access to **E:** drive (DATA).

- Enabled shortcut keys, except for **CTRL** and **Fx** keys (where x is a number between 1 and 12).
- Enabled Windows (■) and left **ALT** keys.

In order to enable the above listed functionalities, left **SHIFT** and **↑** keys should be pressed simultaneously using a physical keyboard. Then, a window will be prompted asking for the Administrator user of the system (same as the one used for OS actions requiring administration privileges).



Figure 830 - Request for admin user credentials in Diagnostic application mode

Once credentials have been properly introduced, the new functionalities will be enabled during the current session.



Figure 831 - View of Diagnostic application with access to Windows 10 File Explorer

Once the troubleshooting actions have been performed by the Field Service Engineer, access to Windows 10 File Explorer in the Diagnostic application must be disabled by pressing left **SHIFT** and **↑** keys simultaneously. At that point, the Diagnostic application will not be closed and can be executed as usual.

When closing the Diagnostic application after having accessed Windows 10 File Explorer, the Boot application will be launched again as in a usual start of the instrument (the grid  button should be pressed if no access to user application is intended for the session).

7.3.3.3 User application mode



NOTE

A physical keyboard and admin user password is needed to enable the new functionalities in Diagnostic application mode.

- Permanent access to Windows 10 file explorer (taskbar appearing when required at the bottom of the screen).
- Enabled right-click functionality when using the mouse.
- Enabled access to **E:** drive (DATA).
- Enabled shortcut keys, except for **CTRL** and **Fx** keys (where x is a number between 1 and 12).
- Enabled Windows () and left **ALT** keys.

In order to enable the above listed functionalities, left **SHIFT** and **↑** keys should be pressed simultaneously using a physical keyboard. Then, a window will be prompted asking for the Administrator user of the system (same as the one used for OS actions requiring administration privileges).

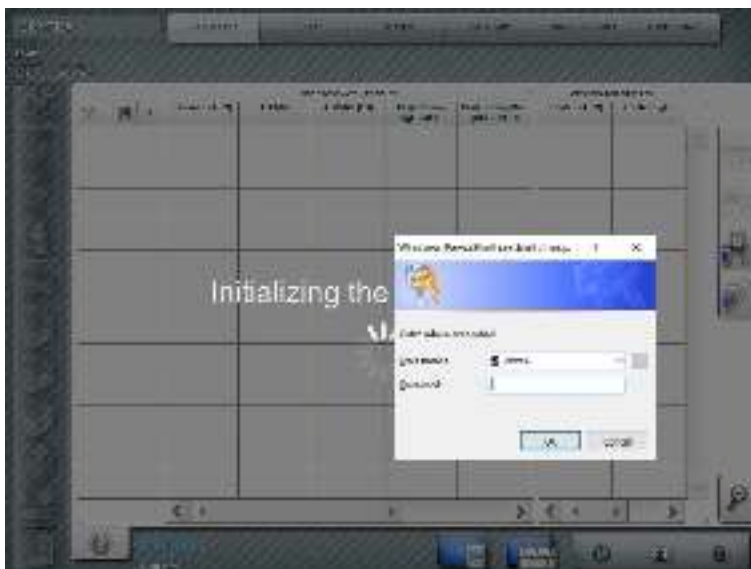


Figure 832 - Request for admin user credentials in User application mode

Once credentials have been properly introduced, the new functionalities will be enabled during the current session.

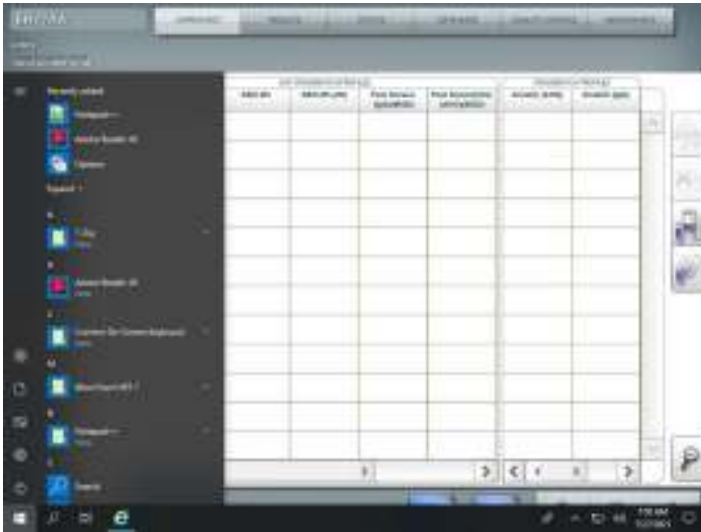


Figure 833 - View of User application with access to Windows 10 File Explorer

Once the troubleshooting actions have been performed by the Field Service Engineer, access to Windows 10 File Explorer in the User application must be disabled by pressing the left **SHIFT** and **↑** keys simultaneously. At that point, User application will not be closed and the user will be able to keep working with the instrument.

7.3.3.4 OS shell mode

- When modifying LIS connection settings (use of DianaServer program, **Listas.exe** or **Config.exe** files) the applied changes will be stored persistently in the corresponding Windows registry keys (no need to do it through Maintenance mode).
- Enabled right-click functionality when using the mouse.
- Enabled shortcut keys, except for **CTRL** and **Fx** keys (where x is a number between 1 and 12).
- Enabled Windows () and left **ALT** keys.

7.3.4 Sharing and accessing a shared folder or printer

The Operating System image version 2.2.x has been configured to be able to share folders without the need to modify any configuration.

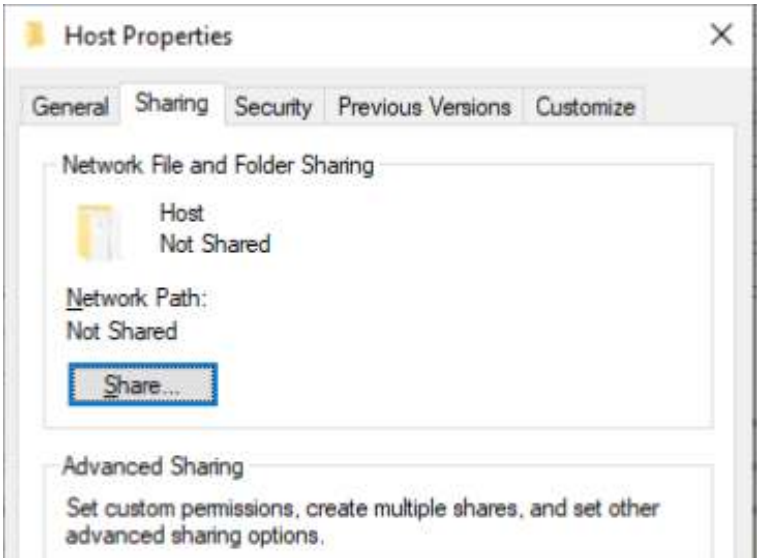


Figure 834 - Sharing

Required Server service and TCP/IP NetBIOS Helper service Startup type are already running (they are set to automatic start up. In previous Operating System versions the Startup type was set to disabled).

The minimum required firewall rules to share or access a share folder have been created and are enabled for all network profiles (Private, Public and Domain).

The automatically enabled inbound firewall rules to share files are:

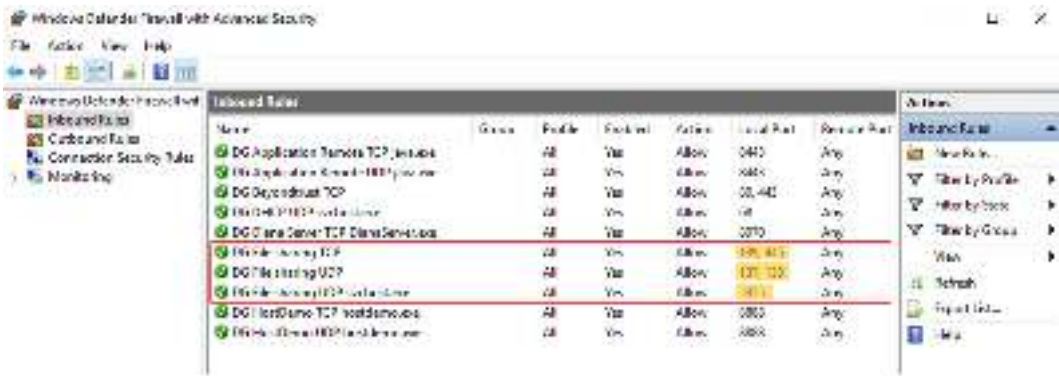


Figure 835 - Inbound rules

The automatically enabled outbound firewall rules to share files are:

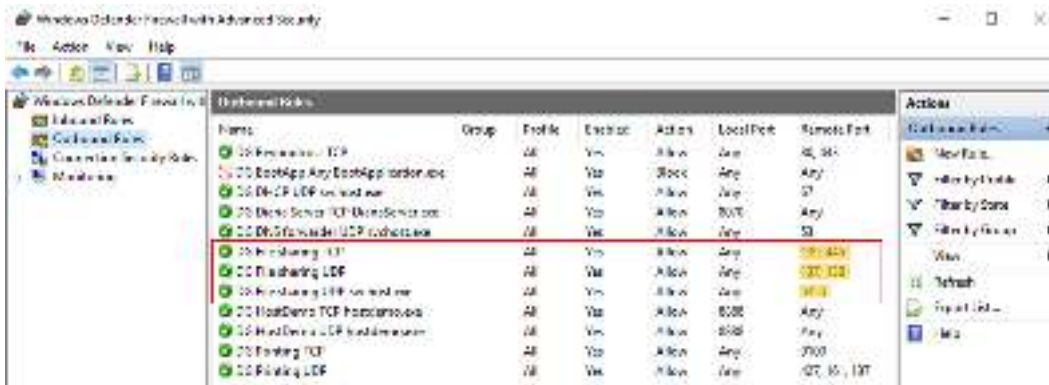


Figure 836 - Outbound rules

Then, no additional action is required to be able to access or share a folder in Operating System version 2.2.x and Erytra software version 4.6.1.

In the event of not being able to share the desired folder with the default configuration, it is possible to create new rules manually as follows:

1. Connect an external mouse to the instrument.
2. Switch on the instrument.
3. Press the grid () button to enter the boot application and log in as `adminUser`.
4. Select the **Maintenance Mode** () option.
5. Open Windows Defender Firewall advanced settings.
 - How to: Select **Start > Windows System > Control Panel**.
 - Once Control panel is opened, select **System and Security > Windows Defender Firewall > Advanced settings**.

6. Select **Inbound Rules** and **New Rule** option in the **Actions** section.

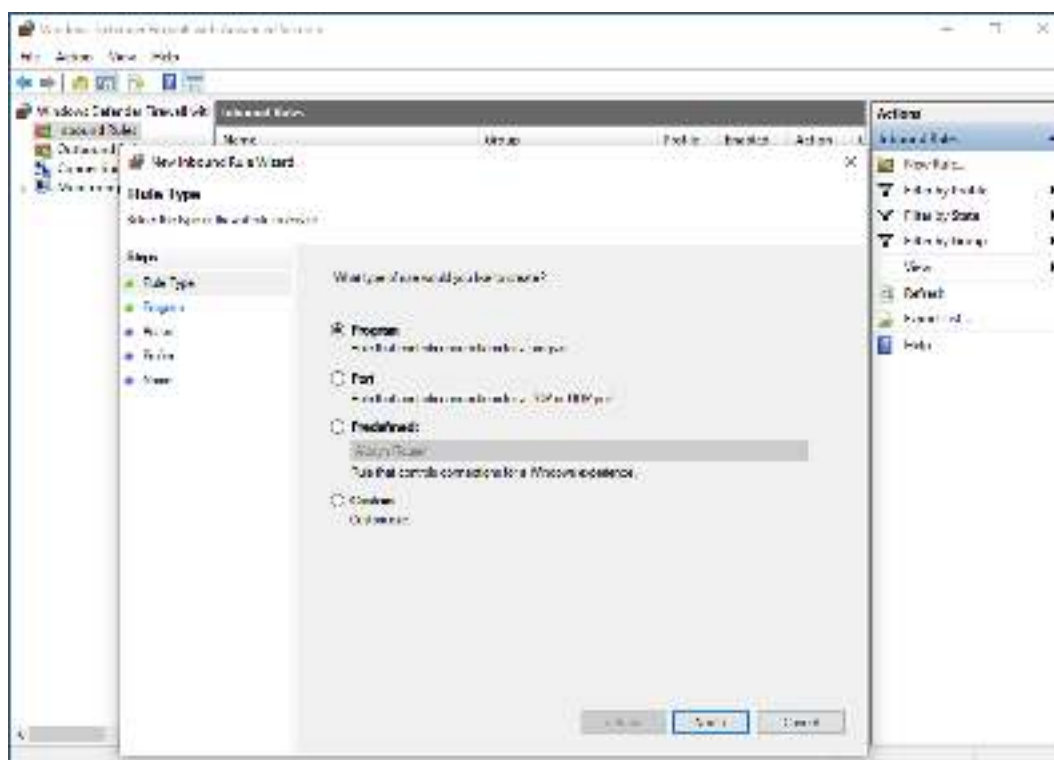


Figure 837 - New inbound rule wizard

7. Select the **Predefined** option and **File Printer Sharing** option in the drop-down menu. Press **Next**.

8. Review the proposed rules, select the desired ones and then select **Next**.

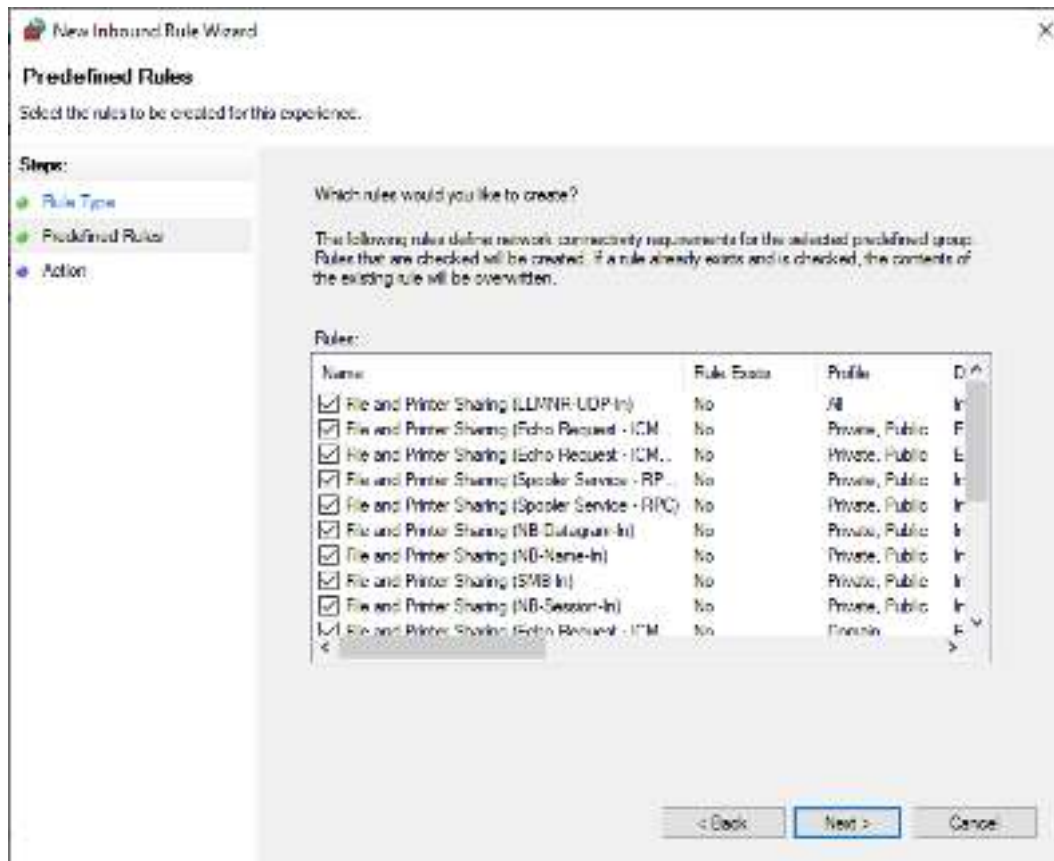


Figure 838 - Predefined rules

9. Select **Allow the connection** or **Allow the connection if it is secure** and select **Finish**. The selected rules will be created and enabled.
10. Select the **Outbound Rules** and **New Rule** options in the **Actions** section.
11. Select the **Predefined** option and **File Printer Sharing** option in the drop-down menu. Press **Next**.
12. Review the proposed rules, select the desired ones and then select **Next**.
13. Select **Allow the connection** or **Allow the connection if it is secure** and select **Finish**. The selected rules will be created and enabled.
14. Close Windows Defender Firewall with the **Advanced Security** window.

To keep access to a shared folder, see "Mapping a network drive" on page 1113.

7.3.5 Mapping a network drive

To access a shared folder of a remote computer, the operating system executes a template script named **mappedDrive.template** that can be found in the folder **E:\Program Files\System Tools**.

The system will execute all files named **MappedDrive** (except the template) at startup.

Each unit to be mapped requires an individual MappedDrive file, properly filled.

Follow the next steps:

1. Enter the Boot Application.
2. Select the **Maintenance Mode** () option.
The PC loads the **Maintenance Mode** desktop of the operating system.
3. Go to the following path:
 - **E:\Program Files\System Tools\.**
4. Open the **mappedDrive.template** file.
5. Edit the file as follows:

Template: 'mappedDrive.template'	Example: 'mappedDrive.M'
localPath=X:	localPath=M:
remotePath=\\server\folder	remotePath=\\server01\folderA
user=username	user=dan
password=password	password=1234

6. Save the changes.

Mapping attempts and result will be logged in installation log (**E:\log**).



NOTE

Duplicate the template as many mapped drive units are required.
Replace filename and 'localPath' with a different unit name for each mapped drive.
C: D: E: and R: units are system units and cannot be used.



NOTE

Ensure there are not USB drives connected with the assigned drive units that are configured.

7.3.6 Collecting Windows event logs

Windows logs of events are not included in the Support Pack and must be collected separately. Both Application and System logs of events should be saved. These logs are needed to investigate software issues such as software freezes.

To collect the Windows logs of events:

1. Switch on the instrument and enter the boot menu pressing the grid () button.
2. Use the credentials: user: `adminUser`, password: `admin` to log in.

3. Access the OS Shell.

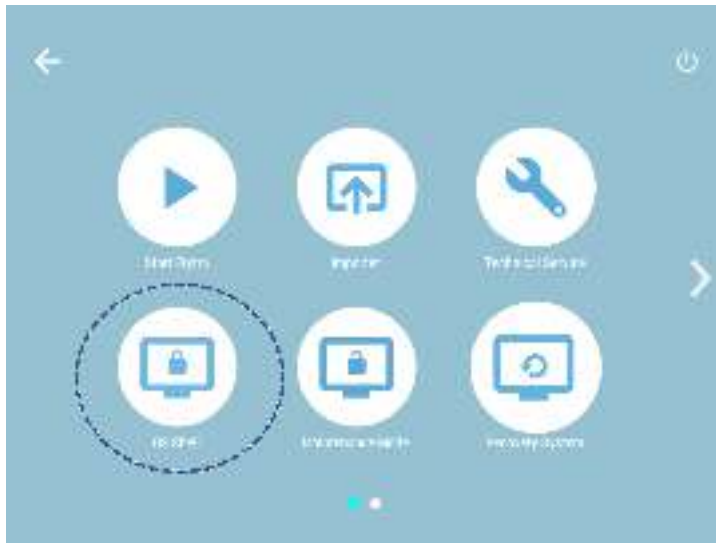


Figure 839 - Access the OS Shell

4. Press the Windows Start button and type `Event Viewer`. Right click the top result and select **Run as administrator**. Then type the administrator password `dgroot`.
5. Select and expand the folder **Windows Logs**: the **Application**, **Security**, **Setup**, **System** and **Forwarded Events** logs can be found.

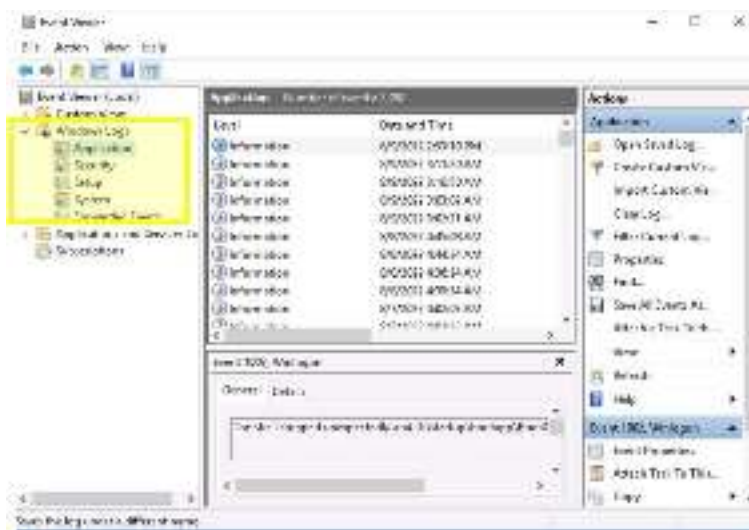


Figure 840 - Windows logs of events

6. Select the application log by clicking **Application**. The log is loaded in the center tab.

7. Select Action > Save All Events As....

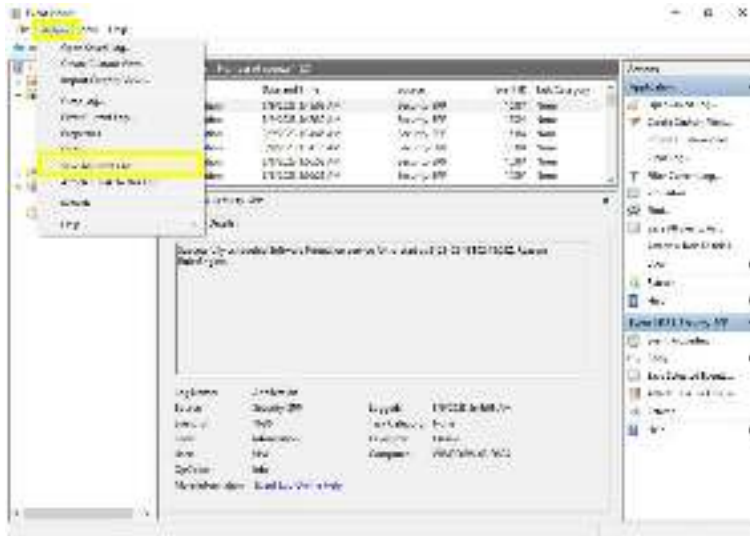


Figure 841 - Saving the Windows logs of events

8. Select Display information for these languages and select English (United States).

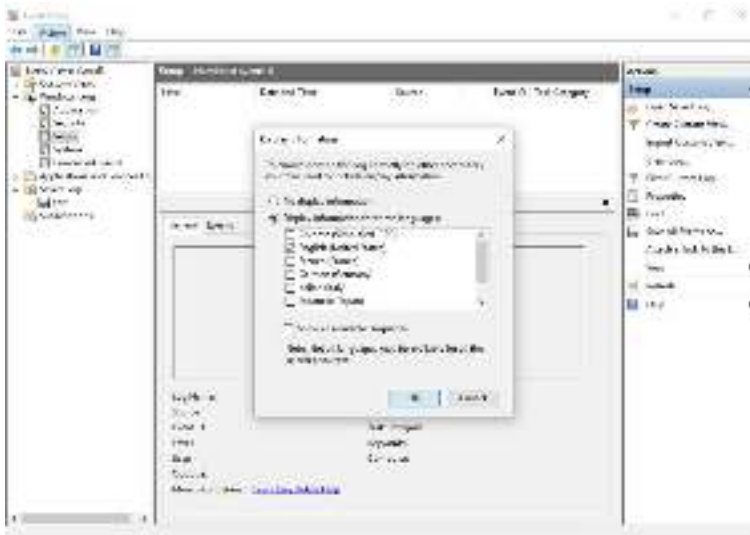


Figure 842 - Display Information

The application logs of events is now saved.

9. Repeat the process to save the **System** logs.

7.4 Database and relevant data

Information related to the database is provided in this section.

7.4.1 Backing up with user interface or OS Shell backup option

It is also possible to perform the Backup from the path **E:\Program Files\Backup** (entering the **Operating System Shell** option from the Boot application) if the current version is higher than 4.3.2 (Operating System is Windows 10). To perform the Backup from this route, execute the **run.bat** and press the **Backup** button and the file will be stored in **E:\Backup\BAK_YYYY-MM-DD_HH_MM_SS** folder.

7.4.2 Backing up databases and relevant data with boot menu

In order to perform manual databases backup from the Boot Application, switch on the PC, wait for the Erytra logo to appear on screen and then press the grid (■) button. Log in as `adminUser` (case sensitive, `admin` password) and the Boot Application options will be displayed. Once in there, press the arrow in the right side of the screen to switch to page 2/2 of the menu, then select the **Backup** icon among the Boot Application options and follow the steps listed below.



Figure 843 - Backup icon

There are two backup options, full backup or custom backup. The two options are listed below:

7.4.2.1 Full backup

This option creates a backup with following data:

- All the databases (IHManager, IHExecutor and UserAccess)
- Instrument configuration files:
 - **PIPCalibrations.xml** and **CATCalibrations.xml**
 - testcounter.txt
 - PIP barcode readers (left and right) customized configuration
 - **erytra.properties** and **erytra.properties.backup**
 - **backup.conf** .
 - **DSC.def** and **DSC_custom.def** .
- Operating System:
 - Installed Drivers (not including the OS image default ones)
 - Local configured printers (not network printers)
 - Active networks configurations
 - Time zone
 - Language for non-unicode programs configuration.
 - Start menu
 - Touchscreen calibration
 - Language
 - Scheduled Tasks
 - Firewall rules
 - Diana System section of Windows registry
- Diana System data:
 - **Log** and **Backups** folder
 - **Def** folder
 - **Plantillas** folder
 - **Tandas** folder
- Logs, if selected, from the selected date to current one.

7.4.2.2 Customized backup

This option creates a backup with following data:

- Databases sections that contain IH Reagents Pack data and profiles configuration (IHManager and IHExecutor).

- Instrument configuration files:
 - **PIPCalibrations.xml** and **CATCalibrations.xml**
 - testcounter.txt
 - PIP barcode readers (left and right) customized configuration
 - **erytra.properties** and **erytra.properties.backup**
 - **backup.conf** .
 - **DSC.def** and **DSC_custom.def** .
- Operating System:
 - Installed Drivers (not including the OS image default ones)
 - Local configured printers (not network printers)
 - Active networks configurations
 - Time zone
 - Language for non-unicode programs configuration.
 - Start menu
 - Touchscreen calibration
 - Language
 - Scheduled Tasks
 - Firewall rules
 - Diana System section of Windows registry
- If Diana System option is selected (checkbox is ticked), Diana System data:
 - **Log** and **Backups** folder
 - **Def** folder
 - **Plantillas** folder
 - **Tandas** folder
- If Users Database option is selected (checkbox is ticked), UserAccess database.

**NOTE**

Admin password will be requested to proceed with OS system configuration backup.

**NOTE**

DSC.def and DSC_Custom.def files are no included in the backup.

The application will copy temporally all the data in **E:\tmp** directory, generate a hash file to check backup integrity and then compress all the data in a 7zip file and move it to destination folder.

A log named **ihexport.log** is created in **E:\Program Files\Restore\logs** folder. Once this file reaches 2 MB it compressed (gnu zip) and renamed to **ihExport-backup-YYYY-MM-DD.log**, where YYYY is the year, MM the month and DD the day.

Follow the steps listed below:

1. Choose:
 - The backup destination
 - Whether log files should be included in the backup or not. If log files should be included, indicate the desired starting date.
 - The type of backup (full or customized)
2. Enter a backup description and press **Backup**. Once the **Backup** button is pressed, a progress window will be displayed describing current step and the total amount of steps.

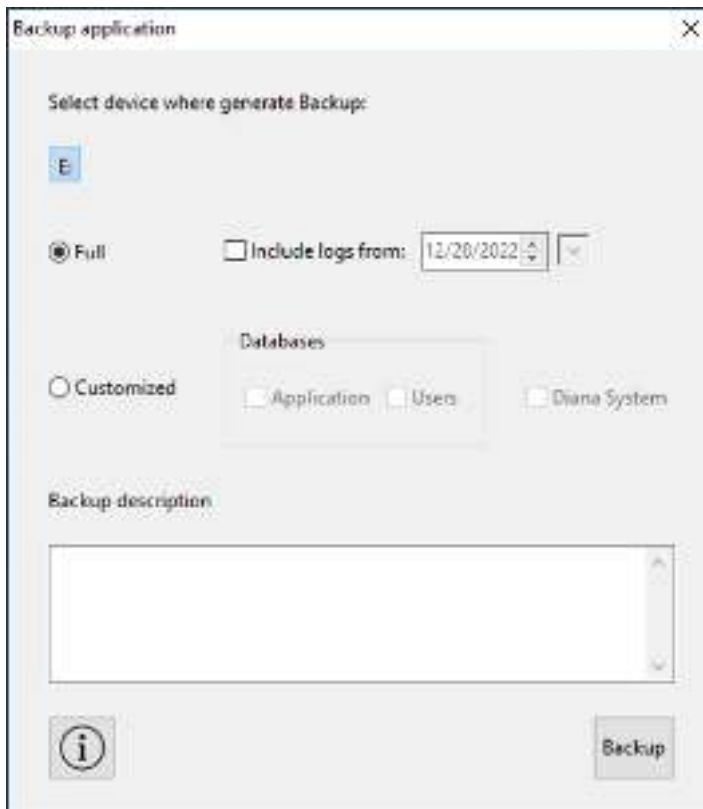


Figure 844 - Backup description and log files choice



NOTE

The information (i) button displays a new window with the backup instructions. It describes the available options, and which will be the content of the backup.

3. A subfolder named **BAK_YYYY-dd-MM_hh_mm_ss** (where yyyy is year, dd is day, MM is month, hh is hour, mm is minute and ss is second when the backup was performed) will be created in the selected backup destination

7.4.3 Backing up relevant data manually

Copy/save in an external device the following relevant data in case of restoring the system in the backing up process:

**CAUTION**

Scan the USB drive for viruses before connecting it to the instrument.

1. Backup Databases and configuration files



CAUTION

To avoid problems with the user list, Databases should be updated before backing them up:

Run **E:\Program Files\liquibase\updatedb.bat**.

Software ≥ 4.3.0 (OS W10)

From boot application main window:

1. Select **Backup** option from boot application. See "Working with Boot Application" on page 88

E:\Backup\BAK_YYYY-MM-DD_HH_MM_SS folder is created with the following content:

- Databases:
 - IH_BBDD_Manager
 - IH_BBDD_Executor
 - DG-userAccessBBDD
- Files:
 - Erytra.properties
 - serialPort.xml
 - testcounter.txt
 - CATCalibrations.xml
 - PIPCalibrations.sml

Software < 4.3.0 (OS W7)

1. Go to: **E:\Program Files\ErytraDB Apps\BRDB**
2. Execute the 'BRDB.exe' selecting only the 'Backup Databases' option.

'**E:\Program Files\Database Backups**' folder is created with the following content:

- Databases:
 - IH_BBDD_Manager
 - IH_BBDD_Executor
 - DG-userAccessBBDD

3. Backup the following configurations files:

From **E:\conf**

- erytra.properties

From **E:\conf\Erytra\conf**

- serialPort.xml
- testcounter.txt
- CATCalibrations.xml
- PIPCalibrations.xml

2. Reboot and enter **OS Shell**(See "Working with Boot Application" section)
3. Backup Microtubes jpg images (**E:\Data\Erytra\Microtubes** subfolders).
4. Backup Raw folder(**E:\Data\Erytra\Raw**).
5. Backup path configuration file (**E:\Program Files\System Tools\backup.conf**).
6. Backup Diana System definition files (**E:\data\SistemaDiana\defs**), if necessary.
7. Backup customized Diana System templates (default directory **E:\Data\SistemaDiana\Plantillas**), if necessary.
8. Backup **DSC.def** file and, if profiles gru names or templates have been customized, **DSC_Custom.def** file (**E:\Data\Erytra\Wadiana**).
9. Backup Technical Service software Macros, if necessary.
10. Backup Erytra log files (**E:\Log** directory and its subdirectories) and Diana System log files (default location **E:\Data\SistemaDiana\log**), if necessary.
11. Backup Diana System **Tandas** and **Traceability** folders, if necessary.
12. Backup Diana System configuration in Windows registry, if necessary.

Software ≥ 4.3.0 (OS W10)

1. run regedit and export

- **HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols**

Software < 4.3.0 (OS W7)

1. run regedit and export

- **HKEY_LOCAL_MACHINE\SOFTWARE\Grifols**

13. Backup the current TCP/IP IPv4 MS Windows configuration and computer name if LIS connection is enabled and communication mode is set to TCP/IP.

7.4.4 Restoring databases and relevant data with boot menu

In order to restore databases backups from the Boot Application, switch on the PC, wait for the Erytra logo to appear on screen and then press the grid () button. Log in as `adminUser` (case sensitive, admin password) and the Boot Application options will be displayed. Once in there, press the arrow in the right side of the screen to switch to page 2/2 of the menu, then select the **Restore** icon among the Boot Application options and follow the steps listed below.



Figure 845 - Restore icon

The optional data to restore is:

- Application Databases: IHManager and IHExecutor databases.
- User Database: UserAccess database.
- Erytra properties: **erytra.properties** and **erytra.properties.backup**.
- Calibrations: **PIPCalibrations.xml** and **CATCalibrations.xml**.
- **backup.conf** file.
- **DSC.def** and **DSC_custom.def** files.
- Barcode reader: PIP left and right customized barcode reader configuration.
- Diana System: Diana System folders **Log**, **Backups**, **Def**, **Plantillas** and **Tandas**.
- System: Operating System configuration: Drivers, Local configured printers, networks configuration, Time zone, Language for non-unicode programs configuration, Start menu, Touchscreen calibration, Language, Scheduled Tasks and Firewall rules.

A log named **ihImport.log** is created in the **E:\Program Files\Restore\logs** folder. Once this file reaches 2 MB it compressed (gnu zip) and renamed to **ihImport-backup-YYYY-MM-DD.log**, where YYYY is the year, MM the month and DD the day.

1. Select the backup to restore.

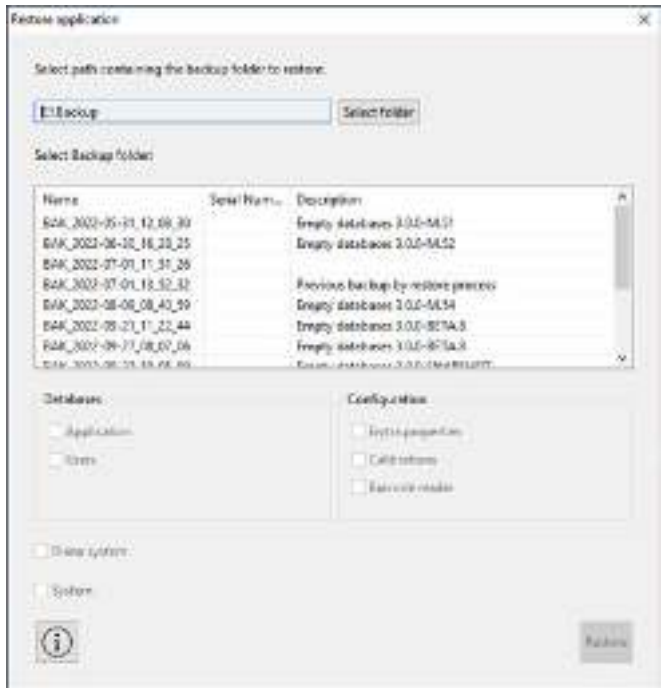


Figure 846 - Restore backup window



NOTE

The information () button displays a new window with the backup instructions. It describes the available options, and which will be the content of the backup.

2. Select the components to be restored and press **Restore**. A progress window will be displayed describing current step and the total amount of steps. Once the **OK** button is pressed, the process is resumed.



Figure 847 - Restore backup window



NOTE

When restoring an empty application database, the relevant reagent, reagent language and erytra language packs must be re-imported in the Importer Application.



NOTE

Admin password will be requested to proceed with OS system configuration backup.

When the backup has been created the data will be uncompressed, and the hash will be checked. If it is ok, the process will continue and a Full backup without log files will be created before starting restore process. Its description will be Full Backup before Restore Backup.

Once restored, the program will update the file registering the number of executed tests in the current database (**testcounter.txt** file in **E:\data\Erytra\conf**), adding a line indicating that the database has been restored. From there, a new entry will be created and executed tests will be registered starting a new count sequence. For instance, when a database has been restored using the **Restore** option from Boot Application in an instrument where 10.000 tests were previously executed, and 2.500 tests have been executed since the database restoring procedure, the following lines will be registered in **testcounter.txt** file:

10.000

restored db

2.500

**NOTE**

It must be taken into account that in case of an operation that requires restoring the previous database, such as a full recovery and this is restored from this option, a new line will be created in the file **testcounter.txt** and once the user application is started, the last line will be updated with the existing database data.

If the **testcounter.txt** file before the restore is:

1.000

In the event the same database is being restored using this backup option:

1.000

restoredb

And once the user applications is started, it will be updated with the database available in the instrument:

1.000

restoredb

1.000

The preventive maintenance replacements are calculated taking into account the processed samples reflected in this file.

7.4.5 Erytra database manager application

Required specifications for any computer where Erytra Database Manager Application will be installed are listed below:

- Operating System: Windows 7 (32 bits), Windows 10 (32 and 64 bits).
- 2 GB RAM minimum.
- **erytra-databasemanager-x.x.x.exe** installer is available.
- Minimum free space on SSD: 922.6 MB.
- Adobe Reader 9.4 or higher.

**NOTE**

Erytra Database Manager software should be installed in a computer where neither Postgres application, Tomcat server or BTM are installed.

**NOTE**

The user to install/run the software must have an administrator role in the used computer.

7.4.5.1 Installation procedure

In order to install Erytra Database Manager in an external computer, follow next steps:

1. Execute installer **Erytra-DATABASEMANAGER-x.y.z.exe**.

**NOTE**

Erytra Database Manager will check if a Blood Typing Manager is already installed in that computer. Should this be the case, following error will be displayed and the installation will be aborted.

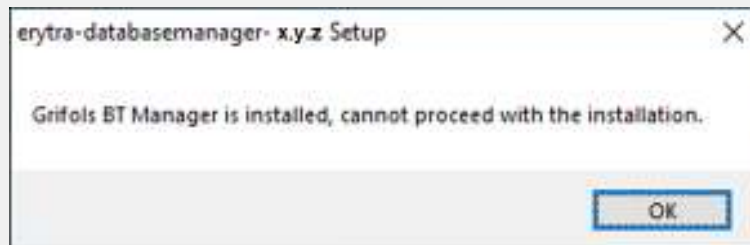


Figure 848 - Database Manager error

- Select the **Browse** button to choose an install location. Navigate to an existing folder or create a new one and then select the **Install** button.

**NOTE**

When the installation begins it opens the Erytra application installer.

- When the installation completes, select the **Finish** button.

1. Load Erytra v4.6.1 database backups.

**NOTE**

This procedure is intended to be performed on a PC where PostgreSQL is not running.

- Open the folder where the application has been installed.
- Execute **RESTORE_DB.vbs** and in the application:
 - Select the **Next** button. Check the **I accept the terms of the License Agreement** checkbox. Select the **Install** button.
 - Choose the **dg59_erytra** database backup file.
 - Choose the **userAccess** database backup file.
 - Select the **Finish** button.

**NOTE**

If the application shows the message error: **Error trying to delete database.** then the file **Data\Database\pg_hba.conf** has to be modified: Comment the last line of the file adding the # symbol at the beginning of the line:

```
#host all all ::1/128 md5
```

7.5 BIOS, Firmware, EEPROM and Drivers procedures

This section contains information about BIOS, firmwares, EEPROM and drivers.

7.5.1 Updating firmware



CAUTION

Do not operate with the instrument while a firmware loading operation is being performed.

There are three possible scenarios: update the firmware in a valid PCA, installing the firmware in a spare part PCA and installing the firmware in a PCA with a non valid firmware, in which a previous firmware loading process failed.

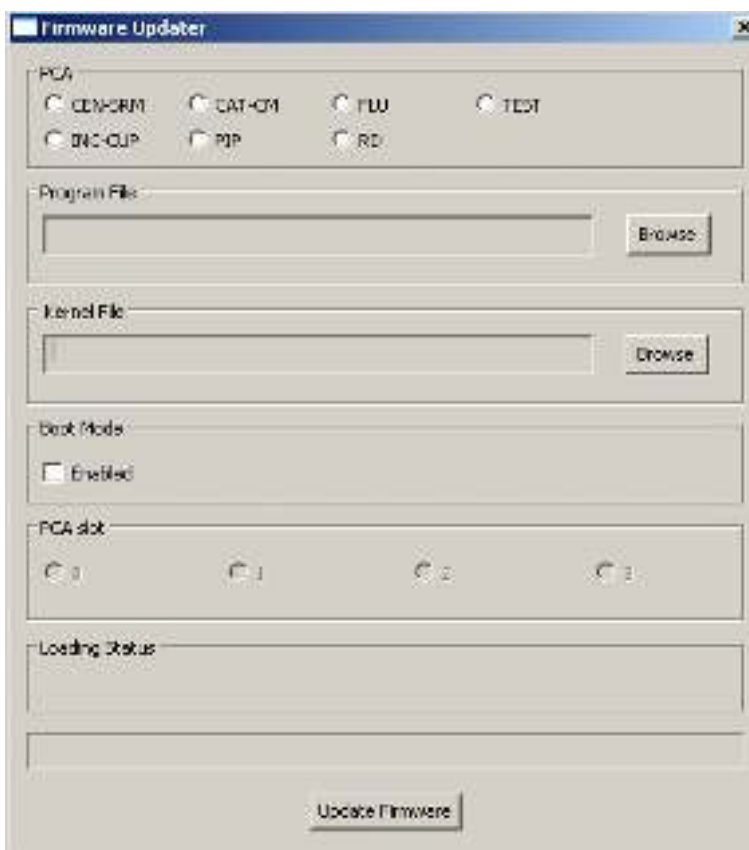


Figure 849 - Firmware update window in the Technical Service software

7.5.1.1 Updating firmware in a valid PCA



CAUTION

Make sure that only one firmware of the resultant type will be present at the Erytra slots after loading process. Otherwise, first remove the necessary cards in order to achieve this.

1. Switch off the instrument and disconnect it from the mains.
2. Connect the instrument to the mains and switch on the instrument.
3. Press the grid () button during system startup (login credentials will be required), then select the **Technical Service** icon among the Boot Application options.
4. Without starting the Driver, go to the **Firmware** tab > **PCA control** section. Press the **load firmware** button.
5. Select the PCA to update the firmware in the **PCA** section.
6. Press the **Browse** button in **Program File** section and select the proper program file for the selected PCA (only the ones related to the previous PCA selected are shown). Program files can be found in **E:\Program Files\Firmwares DG-59**.
7. Press the **Browse** button in Kernel File section and select (only the one related to the previous PCA selected is shown) the proper kernel file (For CEN-SRM, INC-CUP, CAT-CM and PIP select the **H8S2329**; for the FLU select the **Texas**). If the PCA selected is the RD, this option is not activated because is not necessary to indicate any kernel file. Kernels can be found in **E:\Program Files\Firmwares DG-59**.
8. Press the **Update Firmware** button and wait until the process finishes.

7.5.1.2 Updating firmware in a spare part PCA (CEN-SRM, INC-CUP, CAT-CM or PIP MotorControl)



CAUTION

Make sure that only one firmware of the resultant type will be present at the Erytra slots after loading process. Otherwise, first remove the necessary cards in order to achieve this.

1. Switch off the instrument and disconnect it from the mains.
2. Connect the instrument to the mains and switch on the instrument.
3. Press the grid () button during system startup (login credentials will be required), then select the **Technical Service** icon among the Boot Application options.
4. Without starting the Driver, go to the **Firmware** tab > **PCA control** section. Press the **load firmware** button.
5. Select the PCA to update the firmware in the **PCA** section and then select the **Test** option in the PCA section.
6. Press the **Browse** button in the **Program File** section and select the proper program file for the selected PCA (only the ones related to the previous PCA selected are shown). Program files can be found in **E:\Program Files\Firmwares DG-59**.
7. Press the **Browse** button in Kernel File section and select (only the one related to the previous PCA selected is shown) the proper kernel file (For CEN-SRM, INC-CUP, CAT-CM and PIP select the **H8S2329**; for the FLU select the **Texas**). If the PCA selected is the RD, this option is not activated because is not necessary to indicate any kernel file. Kernels can be found in **E:\Program Files\Firmwares DG-59**.

8. Select the proper slot of the PCA in Selected PCA slot section:
 - a. Slot 0: CEN-SRM
 - b. Slot 1: INC-CUP
 - c. Slot 2: CAT-CM
 - d. Slot 3: PIP
9. Press the **Update Firmware** button and wait until process finishes.

7.5.1.3 Updating firmware in a PCA with non valid firmware (CEN-SRM, INC-CUP, CAT-CM or PIP MotorControl)



CAUTION

Only one PCA with non valid firmware must be present at the Erytra slots. Never execute this process if two or more PCAs with no valid firmware are connected at the same time because the process will fail.



CAUTION

Make sure that only one firmware of the resultant type will be present at the Erytra slots after loading process. Otherwise, first remove the necessary cards in order to achieve this.

1. Switch off the instrument and disconnect it from the mains.
2. Connect the instrument to the mains and switch on the instrument.
3. Press the grid () button during system startup (login credentials will be required), then select the **Technical Service** icon among the Boot Application options.
4. Without starting the Driver, go to the **Firmware** tab > **PCA control** section. Press the **load firmware** button.
5. Select the PCA to update the firmware in the **PCA** section.
6. Press the **Browse** button in **Program File** section and select the proper program file for the selected PCA (only the ones related to the previous PCA selected are shown). Program files can be found in **E:\Program Files\Firmwares DG-59**.
7. Press the **Browse** button in the **Kernel File** section and select (only the one related to the previous PCA selected is shown) the proper kernel file (For CEN-SRM, INC-CUP, CAT-CM and PIP select **H8S2329**; for the FLU select **Texas**). If the PCA selected is the RD, this option is not activated because is not necessary to indicate any kernel file. Kernels can be found in **E:\Program Files\Firmwares DG-59**.
8. Activate the **Enabled** box in the **Boot Mode** section.
9. Press the **Update Firmware** button and wait until the process finishes.

7.5.2 BIOS settings

This section explains how to check the BIOS configuration.

**NOTE**

When installing a new BIOS firmware version, the previously configured BIOS settings will be erased.

7.5.2.1 Checking BIOS version

1. Connect a physical USB keyboard. Boot up the system and press **[Del]** when the boot logo page is displayed.

**NOTE**

A password to access the BIOS has been set.

When requested, enter password: TSDGIH.



Figure 850 - Boot logo page

2. In the **Main** tab, check the UEFI Version.

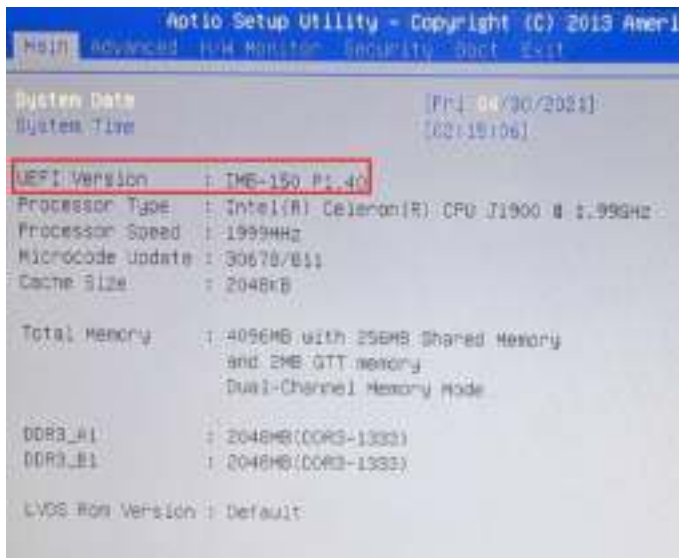


Figure 851 - UEFI version



NOTE

Different BIOS versions may apply:

- P2.32W: s/n ≥ 059-0002099
- P2.20L: s/n ≥ 059-0001775 (see INR-0059#0140 in see "Bulletin index" on page 1183)
- P2.105: s/n ≥ 059-0001435 (see FSB-0059#0032 in see "Bulletin index" on page 1183)
- P1.40: s/n < 059-001435



NOTE

BIOS version P2.20L and later versions require password access.

When requested, enter password: TSDGIH.

7.5.2.2 Configuring BIOS

1. Enter the BIOS configuration. Boot up the system and press **[Del]** when the new boot logo page is displayed.



NOTE

A password to access the BIOS has been set.

When requested, enter password: TSDGIH.



Figure 852 - New boot logo page

2. Go to the **Advanced** tab and select the option **Super IO Configuration**.

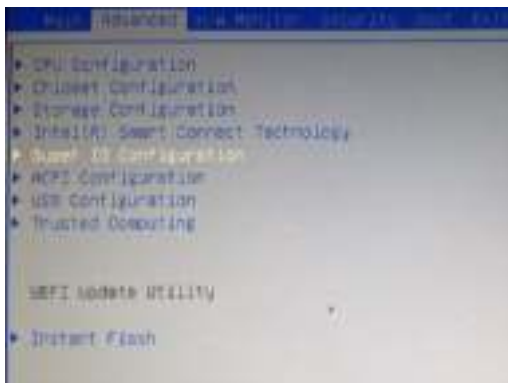


Figure 853 - Super IO configuration selection

3. Check **COM4 Configuration** is set to **Disabled**.



Figure 854 - COM 4 port disabling

4. Check **COM6 Configuration** is set to **Disabled**.

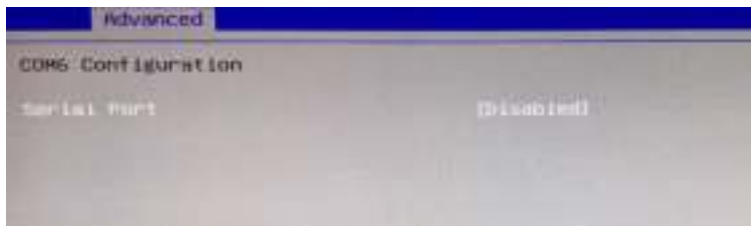


Figure 855 - COM 6 port disabling

5. Check **LPT1 Port Configuration** is set to **Disabled**.

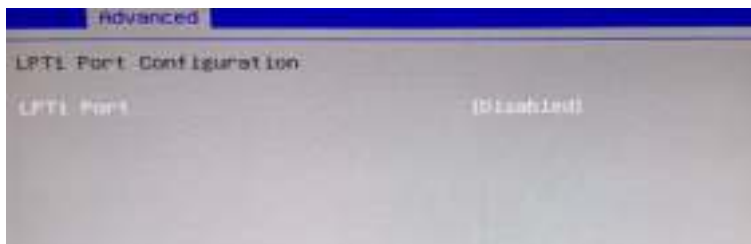


Figure 856 - COM 6 port disabling

6. Go to the **Exit** tab and select **Save Changes and Exit**. Click **Yes**.



Figure 857 - BIOS exit tab

7.5.2.3 Checking assignment of Windows COM ports

COM ports configuration can be checked using Windows Device Manager.

7.5.2.3.1 Windows 10

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required). Then select the **Maintenance Mode** option.

2. Once in Maintenance Mode, type `devmgmt.msc` in the Windows Search Box. **devmgmt.msc** appears as a Command option. Select **Run as administrator** (password for admin user will be required).

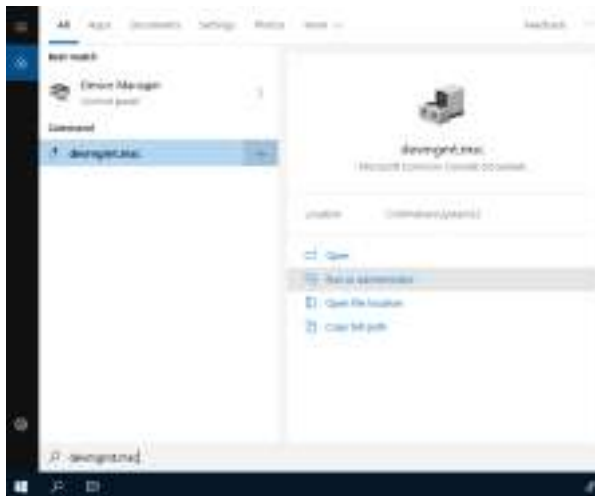


Figure 858 - Running devmgmt.msc as administrator

3. Check the Ports (COM & LPT) assignment:

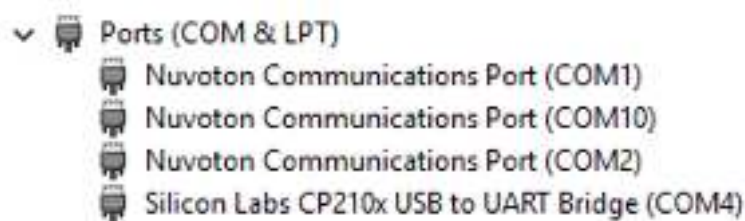


Figure 859 - Ports assignment

Silicon Labs CP210x USB to USART Bridge should be assigned to **COM4**. Should that not be the case, follow the steps listed in "Modify COM ports assignment" on page 1137.

7.5.2.3.2 Modify COM ports assignment

1. Go to the **View** tab in Device Manager and select **Show hidden devices**.
2. If **Nuvoton Communications Port (COM4)** is present, right click and uninstall it.

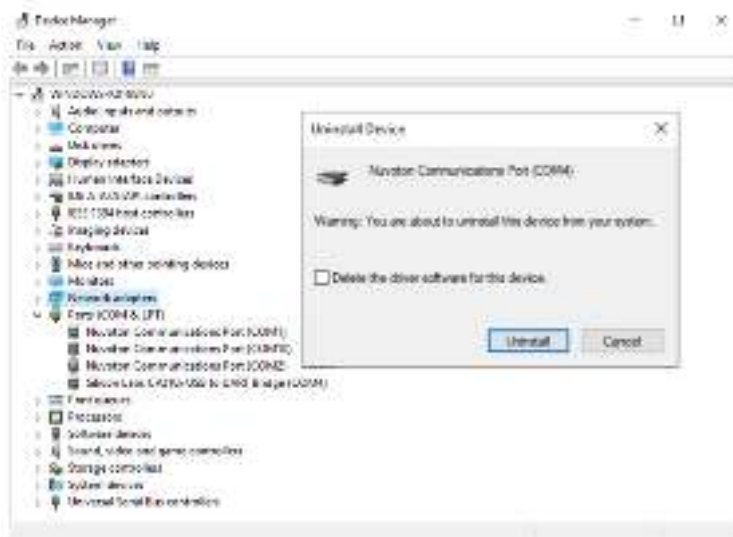


Figure 860 - Uninstalling Nuvoton Communications port (COM4)

3. If **Nuvoton Communications Port (COM6)** is present, right click and uninstall it.

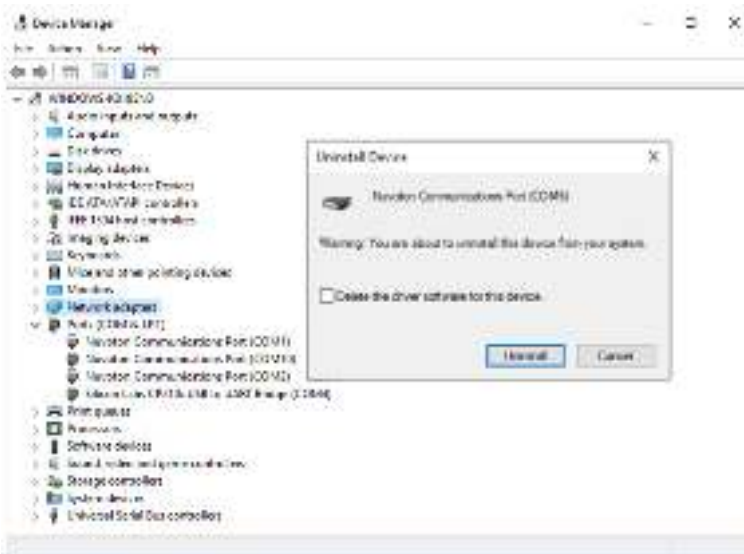


Figure 861 - Uninstalling Nuvoton Communications port (COM6)

4. If **ECP Printer Port (LPT1)** is present, right click and uninstall it.

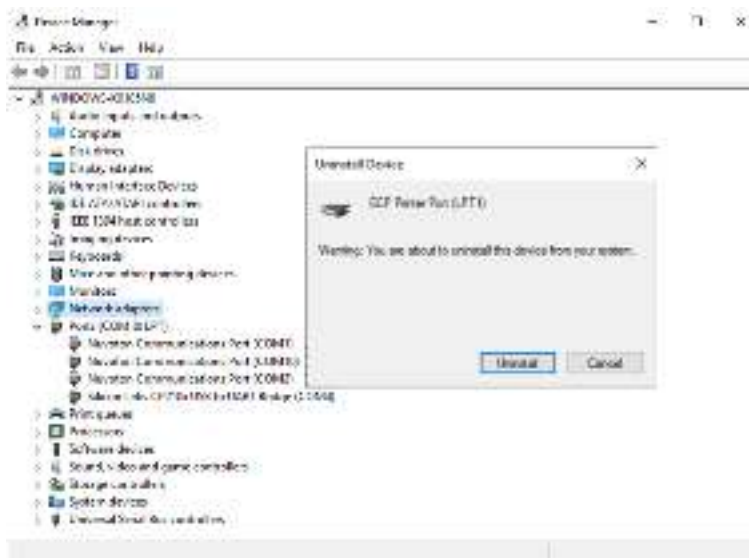


Figure 862 - Uninstalling ECP Printer Port (LPT1)

5. Right-click the COM port assigned to the device **Silicon Labs CP210x USB to UART Bridge** (COM5 in the example below) and click **Properties**.

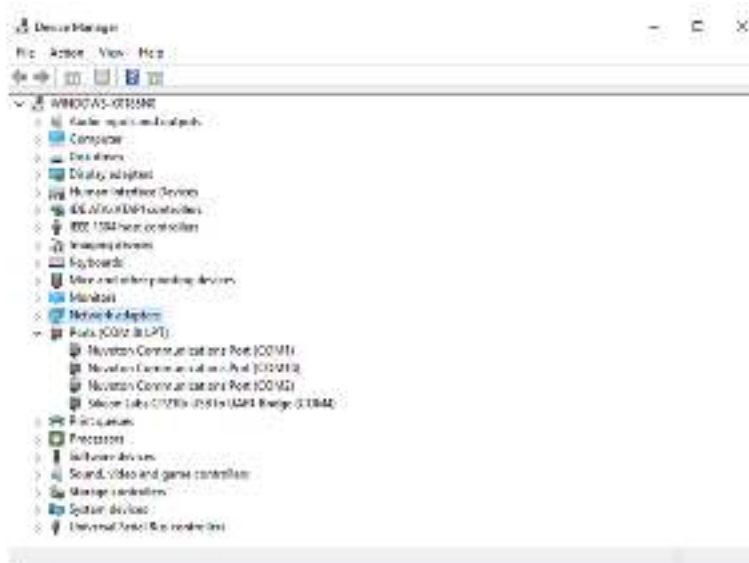


Figure 863 - Silicon Labs CP210x USB to UART Bridge selection

6. Go to the **Port Settings** tab and click **Advanced**.

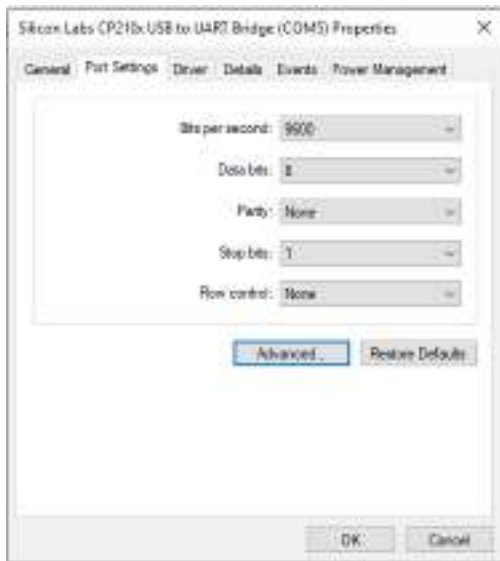


Figure 864 - Port settings tab

7. Select **COM4** in the **COM Port Number** combo box. Click **OK**.



Figure 865 - COM 4 selection in COM port number combo

8. If it exists, delete the file **E:\data\Erytra\conf\serialPort.xml**.
9. Make sure that the **Ports (COM & LPT)** assignment is now the expected one:

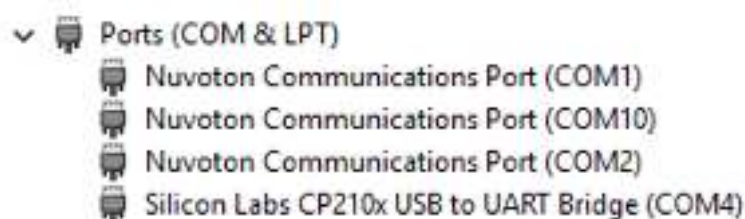


Figure 866 - Expected port assignment in Windows 10

7.5.3 Backing up and restoring EEPROM

In Erytra most of the adjustments are stored in two files (**PIPCalibrations.txt** and **CATCalibrations.txt**) located in **E:\data\Erytra\conf**, except for the following ones:

- Probes offset adjustment: it is stored in EEPROM of PCA PIP-Control.
- Incubator temperature: it is stored in EEPROM of PCA INC-CUP Control.
- Load cells of the fluidic drawers adjustment: it is stored in EEPROM of PCA FLU-CTRL59.
- Fluidic sensors adjustment: it is stored in EEPROM of PCA FLU-CTRL59.
- Reader/camera adjustment: it is stored in EEPROM of PCA RD-IL421.

When replacing any of these PCAs:

- PCA PIP-Control and PCA INC-CUP Control: the stored adjustment must be performed again or restored from a file (before replacing the PCA, a backup of the adjustment must be performed).
- PCA FLU-CTRL59: adjustment backup can be also restored but is recommended to perform the adjustments.
- PCA RD-IL421 it is recommended to perform the adjustments (not recommended restoring them).

7.5.3.1 EEPROM backup

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the **Technical Service** icon among the Boot Application options.
2. Start the Driver and go to the **Firmware** tab > **EPROM control** section.



Figure 867 - EPROM control window in the Technical Service software

3. Select **PCA** to save the EEPROM content.
4. Press the **backup** button. A new window appears in order to select the folder and the name of the file to be saved.

5. After the generation of the EEPROM file (**filename.txt** file), another file is also created with the electronic signature (**filename_signature.txt** file).

**NOTE**

Do not rename the name of the files generated by the backup procedure.

7.5.3.2 EEPROM restore

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the **Technical Service** icon among the Boot Application options.
2. Start the Driver and go to the **Firmware** tab > **EPROM control** section.
3. Select **PCA** to restore the EEPROM content.
4. Press the **restore** button. A new window appears in order to select the folder and the file to be restored.

**NOTE**

EEPROM content file and electronic signature file must be in the same folder.

7.5.4 Installing and configuring touchscreen driver

7.5.4.1 Installing touchscreen driver

During the Operating System installation process, the touchscreen model in use (version 2 or version 3) is automatically detected by the system and the corresponding touchscreen drivers are used.

7.5.4.2 Configuring touchscreen v2 drivers

Settings for touchscreen v2 (SN < 059-0001515) can be modified by means of its Control Panel utility:

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the Maintenance Mode icon among the Boot Application options.

2. Once the PC reboots and the desktop is available, press Windows Start button and select **MicroTouch MT 7** program.

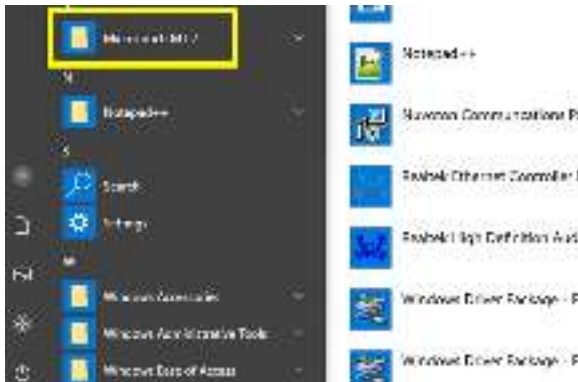


Figure 868 - MicroTouch MT 7 program

3. Click **Control Panel** among the drop down options.

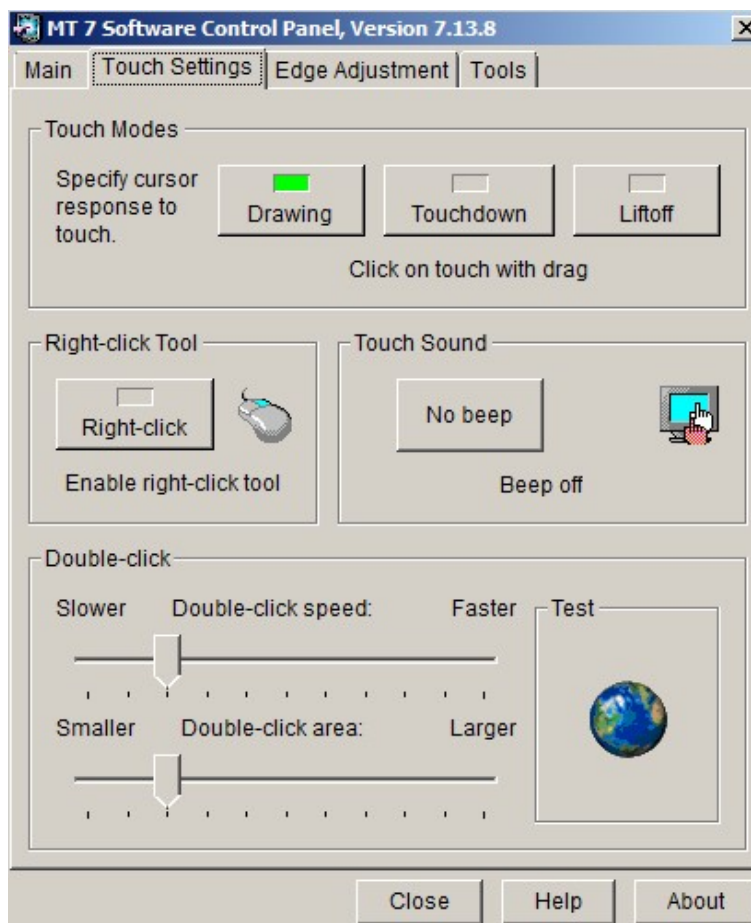


Figure 869 - MT7 Control panel

4. Select **Drawing** touch mode in the **Touch Settings** tab.

5. Make sure that the **Right-click** option in **Right-click Tool** area is disabled, and **Touch Sound** is set to **Beep off**.
6. Set **Double-click** to:
Speed: level 3
Area: level 3
7. Switch off the instrument.

7.6 Applications

The following applications are included or related to Erytrasoftware:

- Boot application
- Erytra user application
- Support pack
- Technical Service
- Macro executor
- Importer application
- Applocker application
- Beyondtrust.

7.6.1 Applocker application

AppLocker is an application whitelisting technology included in Windows operating systems, used to block unauthorized executables on running on Erytra. It uses a dynamic trust model, preventing attacks from unknown malware by allowing only known-good applications to run, registering them on an approved list of entities known as a whitelist.

7.6.1.1 Updating the whitelist

In order to perform new installations or updates of authorized applications or drivers, the whitelist can be dynamically updated by executing the following procedure:

1. Disconnect the instrument from the network (if connected).
2. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the **Maintenance Mode** icon among the Boot Application options.
3. The PC will be rebooted and Maintenance Mode will be accessed in the subsequent startup.



CAUTION

Once the computer has been restarted, drives are not protected.

9. Install/update the desired application or device driver per usual procedure. Once finished, the installed software will be automatically added to the whitelist.
10. Type `eu` (End Update) in the AppLocker Manager graphical interface window.
11. Wait for 10 seconds so that the change to protected mode is completed.



Figure 871 - AppLocker application exiting update mode

12. Restart the system to exit Maintenance mode.
13. Connect (if applies) the instrument to the network.

7.6.1.2 Blocked executables

When the execution of an unauthorized application file has been blocked by AppLocker application, a warning dialog informing of the blocked application is prompted on screen.



Figure 872 - Example of blocked application (!)

If using Windows Command Prompt in order to execute a script or application which is not included in the whitelist, the following messages are shown:



Figure 873 - Example of blocked application (II)

When an application or script has been blocked by AppLocker, the corresponding event is captured to Application event log. In order to see them, go to **E:\log\appLocker.log**. It is also registered in Windows events log.

Additionally, the information of each update session is stored in **E:\AppLockerWorkspace**. It contains the log, the original rules and the final status of the rules.

7.6.2 Erytra Macro executor

The Technical Service software incorporates an option to run test files in order to automate movements and functions of the Erytra for testing purposes.



CAUTION

Be careful with the options programmed in order to not damage the instrument.

In order to run the Macro Executor, see "Technical Service software overview" on page 754.



Figure 874 - Erytra Macro Executor main screen

The test files are **.xml** files. In order to run a test program, follow the next steps:

1. Load the test file by pressing **1.browse test**.
2. Once it is loaded, press **2.validate test**.
3. If no errors appear in the **Macro Executor Log** window, press **start test** to execute the test file.
4. It is possible to stop at any time the test pressing the **stop test** button.

The list of options to include in the test file are detailed in the file **manual.txt**, that it is possible to select in the combo box. Also in the combo box there are detailed examples of all available commands.

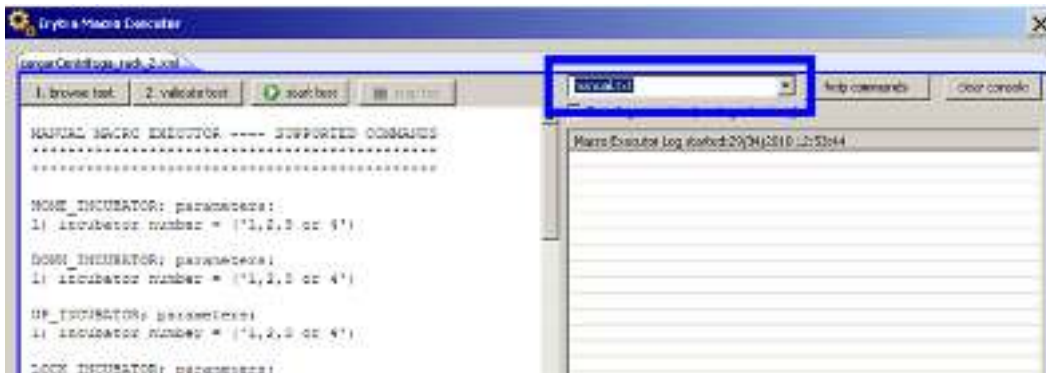


Figure 875 - Combo box in Erytra Macro Executor main screen

Template of the test file is as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<erytraTest xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="./xsd/ErytraTestXMLSchema.xsd">

  <!-- // section where the possible initializations for the test are introduced// -->
  <init>
    <commands>
      .
      .
      .
    </commands>
  </init>

  <!-- // // section where the commands are introduced // -->
  <commands executions="xxxx" where xxxx are the times to be executed

    <command>
      <operations>
        .
        .
        .
      </operations>
    </command>
  </commands>

</erytraTest>
```

7.6.3 Support pack

The Support Pack feature allows the creation of a zipped file with all the data required for support, which will not contain patient data in order to meet the General Data Protection Regulation (GDPR) compliance. This data is required to be stored in a USB drive, using 7-Zip software for its compression.

7.6.3.1 Generating Support Pack from user software

The Support Pack can be created from the user software (**Maintenance > Analyser > Technical Service Action** tab) and can be requested by Technical Service, Operator plus, Specialist, Supervisor and Administrator users.



Figure 876 - Support Pack button in the Technical Service Action tab (user software)

Once the **Support Pack** option is selected and a USB drive has been connected to the instrument, the following menu will be displayed, allowing to choose the below listed options:



Figure 877 - Support Pack menu

- Destination USB drive.
- Time frame for the required information (Date FROM/TO).
- Checkbox allowing the inclusion of RAW images (if available) for the selected time frame.
- Checkbox allowing the inclusion of tan files (if available) for the selected time frame.

When pressing the **Support Pack** button, a progress dialog with the text **Generating Support Pack** will be displayed and the system will collect all the data for the required time frame, erasing patient data in the created database and replacing it by asterisk characters ("****") in the created log files related to communication with LIS.

Data will be temporarily stored in the **E:** drive, naming the temporary folder as **ERYTRA_ssnn_YYMMDDhhmm_SupportPack**, where ssnn is the instrument serial number extracted from **erytra.properties** and YYMMDDhhmm is the date and time when the pack has been created. Then, a folder named **ERYTRA_ssnn_SupportPack** will be created in the selected USB drive, and a 7-Zip file will be created inside with the contents of the temporary folder, being named **ERYTRA_ssnn_YYMMDDhhmm_SupportPack.7z**. Once created, the temporary folder will be deleted from the **E:** drive.

The 7-Zip file created by the Support Pack feature will contain:

- **SupportPack.txt** file indicating pack creation date, selected date range, extended date range and the inclusion (YES or No) of the RAW images and tan files in the pack.
- AppLockerWorkspace folder with all the Applocker rules.

- **Erytra** folder with all the files in **E:\conf** directory, **dg59_erytra** database backup (without demographic data) and the following subfolders:
 - **Conf**: all the files in **E:\data\Erytra\conf**.
 - **CONFIG_DG421**: all the files in **E:\data\Config_DG421**.
 - **Log**: all the logs in **E:\Log** folder and its subfolders (keeping the subfolders structure) belonging to extended date range. It also includes the following subfolders:
 - ArchivedLogs with all logs copied in the ARCHIVE_PATH parameter of erytraproperties file belonging to the extended date range.
 - Erytra Diagnostic with Erytra Diagnostic log (**E:\Program Files\Erytra Diagnostic**) belonging to the extended date range.
 - Technical Service with all Technical Service logs (**E:\Program Files\Technical Service\Resources\logs**) belonging to the extended date range.
 - Tomcat with all Tomcat logs (**E:\Program Files\Apache Software Foundation\Tomcat 6.0\logs**) belonging to the extended date range
 - Importer log (**E:\Program Files\Importer\resources\logs**).
 - **RAW**: all the raw images from **E:\data\Erytra\Raw** belonging to the selected date range, as long as the RAW images have been selected.
 - **Wadiana**: all the files in **E:\data\Erytra\Wadiana**.
- **SistemaDiana** folder with the following subfolders:
 - **defs**: all the files in **E:\data\SistemaDiana\defs**.
 - **Host**: all the files in **E:\data\SistemaDiana\Host** except **LisTest.def** and **LisTest.bak** files.
 - **log**: all the files in **E:\data\SistemaDiana\log** included in the extended date range.
 - **Plantillas**: all the files in **E:\data\SistemaDiana\Plantillas**.
 - **Registry**: **config.reg** file with the Windows registry information from **HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols**.
 - **Tandas**: all tan files from **E:\data\SistemaDiana\Tandas** belonging to the extended date range, as long as the TAN files have been selected.
- Windows folder with windows users, services and application an System event logs

Once completed, a message indicating that the Support Pack was generated successfully will be displayed, and a Technical Service Action record will be created with the comment Support Pack. The Support Pack creation (and errors, if any) are logged in **erytra** and **erytraDebug** log files. The beginning, end and options selected are logged as **WARN** traces and the detail of all the steps of the process are logged as **DEBUG** traces.

7.6.3.2 Generating Support Pack from OS shell

The Support pack generator tool, installed with the software version, allows to perform a Support Pack, same as the User Application but with no need to start the User Application, neither the need for a USB drive or the presence of technical service personnel.

The use of the Grifols Secure Remote Connectivity remote desktop software and the possibility of remotely accessing the instruments allows a faster extraction of the Support Pack with no disturbance to the user work. Additionally, this action can be performed with no need to close the User Application by accessing Windows 10 file explorer using the corresponding scripts if installed, both through local access to the instrument or via the Grifols Secure Remote Connectivity software.

7.6.3.2.1 Support Pack creation

In order to create a Support Pack using the new Support Pack generator follow the next steps:

1. Run the **E:\Program Files\SupportPack\run.bat** file.

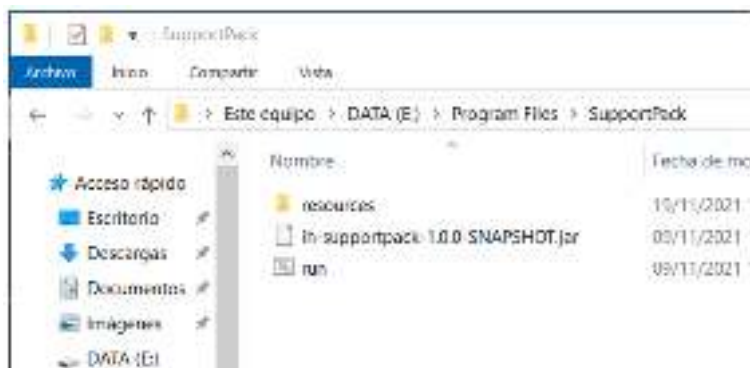


Figure 878 - Support Pack folder

2. A command window will be shown and the **Start date** for the desired time frame is requested. This field must be entered with the format YYYY-MM-DD where:
 - a. YYYY: year (Example: 2021).
 - b. MM: month (Example: 10).
 - c. DD: day (Example: 21).

As long as these requirements are not met, the **Start date** will still be requested.



Figure 879 - Support Pack start date introduction

3. After entering a correct **Start date**, the **End date** for the desired time frame is requested, expecting the same format as when introducing the **Start date**. Additionally, the **End date** must be higher than the **Start date**. As long as these requirements are not met, the **End date** will still be requested.

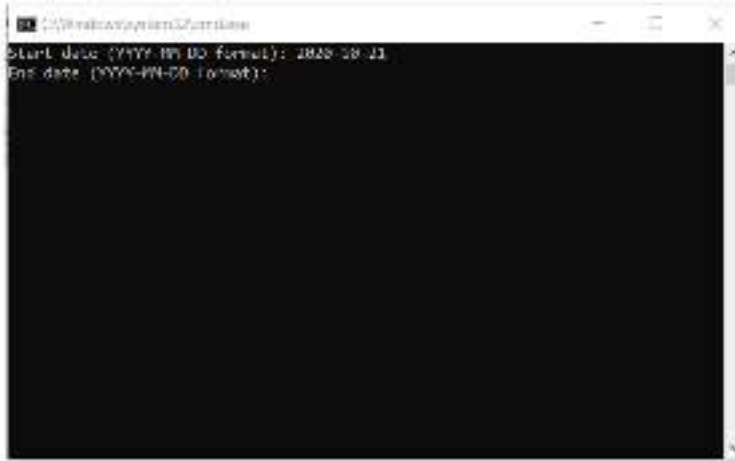


Figure 880 - Support Pack end date introduction

4. After entering a correct **End date**, the application will ask whether RAW images should be included in the Support Pack or not. In case of an affirmative answer by **Y** or **y** keys, the available RAW images in the instrument will be packed into the Support Pack. Otherwise, keying **N** or **n** means that the RAW images will not be included in the Support Pack.

Only the keys **Y**, **y**, **N** and **n** are allowed as an answer. As long as these requirements are not met, the Include RAW image files question will still be requested.

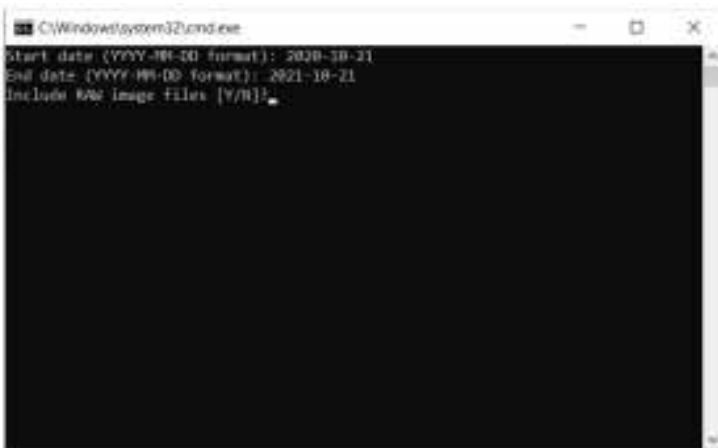


Figure 881 - Support Pack "Include RAW image files" question

5. After entering a valid parameter for the Include RAW image files question, the application will ask whether TAN files should be included in the Support Pack or not. In case of a affirmative answer by Y or y keys, the available TAN files in the instrument will be packed into the Support Pack. Otherwise, keying N or n means that the RAW images will not be included in the Support Pack.

Only the keys Y, y, N and n are allowed as an answer. As long as these requirements are not met, the Include TAN files question will still be requested.

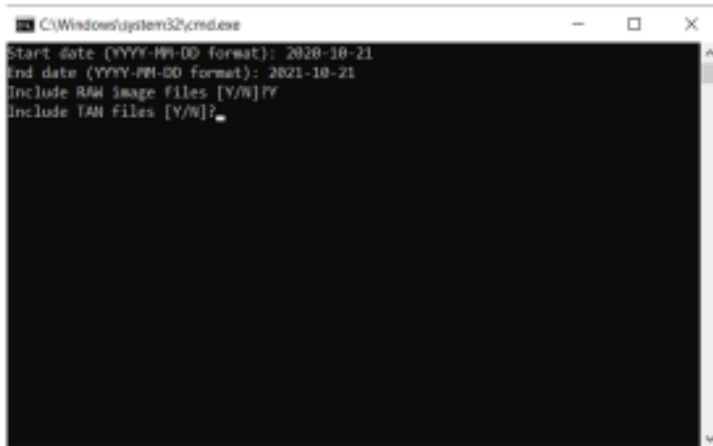


Figure 882 - Support Pack "Include TAN files" question

6. After entering a valid parameter for the "Include TAN files" question, the Support Pack generator will start to generate the pack.

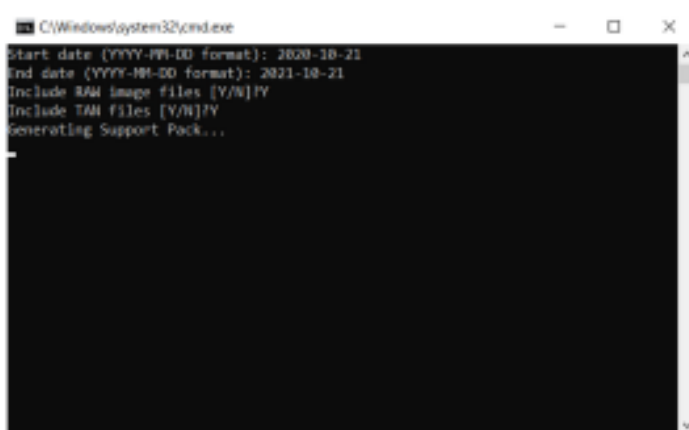


Figure 883 - Support Pack generation

7. At this moment, the Support Pack generator will perform the following actions:
 - a. Determine the extended date range. That is, determine the date range to include the beginning of the session related to the selected **Start date** and the end of the session related to the selected **End date** parameters.
 - b. A temporary folder with name **E:\<INSTRUMENT>_<SERIALNUMBER>_<YYMMDDHHMM>_SupportPack** is then created to generate the pack, where YYMMDDHHMM is the YEAR (YY), month (MM), day (DD), hour (HH) and minutes (MM) at which the pack has been created.
 Additionally, <INSTRUMENT> is ERYTRA for Erytra instruments. For instance:
E:\ERYTRA_0001123_2110211515_SupportPack.
 - c. If **E:\SupportPack** folder does not exist, it will be created.
 - d. The temporary folder is zipped with the name **<INSTRUMENT>_<SERIALNUMBER>_<YYMMDDHHMM>_SupportPack.7z** and copied into **E:\SupportPack\<INSTRUMENT>_<SERIALNUMBER>_SupportPack** folder. For instance: **E:\SupportPack\ERYTRA_0001123_SupportPack\ERYTRA_0001123_2110211515_SupportPack.7z.**
 - e. The temporary folder is deleted.
8. When these actions are finished, the Support Pack generator will inform the user that the SupportPack has been generated successfully.
9. Press any key to close the cmd window.

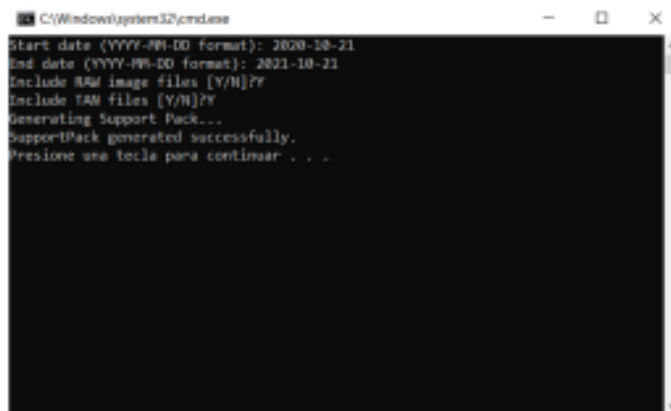


Figure 884 - Support Pack generated successfully

At this point, the zipped Support Pack is ready to be copied or transferred, and subsequent packs will be created in the **E:\SupportPack** folder following the above-mentioned criteria.

Additionally, the Support Pack generator will create a separate log file, located at **E:\log** folder and named **SupportPack.log** where all the steps related to the Support Pack generation process will be traced.

7.7 Additional information

Additional information related to software is provided in this section.

7.7.1 Guidelines on error codes location

User error messages include an error code on the top right corner of the error window.



Figure 885 - Example of user error message

This identification is generated as **MMDDDEESSS**, where **MM** is the module, **DD** is the device and **DEESSS** is the error. Reader module errors are generated as a five digit error code and LIS errors are generated as a seven digit error code.

In order to match the error code shown to the user with the error codes included in this Servicing center, we recommend performing a search with the last six digits in this document. In case of fluidic errors, it is recommended first to check the last three digits to check in which step of the fluidic procedures appears the error.

7.8 Erytra-LIS connection

The connection of Erytra to the LIS is established through Diana System (DianaServer program) and Diana System names (tests, microtubes,...) will be used.

Erytra/Diana System to LIS communication has been implemented in accordance with the CLSI LIS01-A2 and CLSI LIS02-A2 standards (formerly ASTM E1381-02 and ASTM E1394-97, respectively). CLSI LIS02-A2 standard establishes a format for the messages that must be sent from the LIS to the instrument and vice versa, but does not specify the way in which they must be sent. CLSI LIS01-A2 standard describes the way in which messages should be physically sent to the LIS, but does not specify a format.

Erytra can communicate with the LIS in three ways:

- Following CLSI LIS01-A2 standard, where the computers serial channel is used.
- By saving messages in files on local or remote disk units that can be accessed by both LIS and Erytra.
- By using TCP/IP protocol in a network domain.

Diana System Connectivity (DSC) is the Erytra software section that translates/links Erytra and Diana System names.

DianaServer will temporarily store the requests sent by the LIS in the **LisTests.def** file. When a result has to be transmitted to the LIS:

- bi-directional transmission: the existing record in the list is updated
- uni-directional/re-send transmission: a new record is created in the list with the information to be transmitted and, once the transmission has been completed successfully, DianaServer deletes the transmitted record.

7.8.1 DianaServer program

DianaServer is the Diana System program that communicates with the LIS. In order to use it, it should be previously enabled by ticking the corresponding checkbox in the Diana O. program (**Listas.exe**) located in the **E:\data\SistemaDiana > Program configuration > Others** tab. The software installer will set it as enabled when no previous configuration is found. If a previous configuration is found, it will be kept. As the communication between DianaServer and the Erytra software uses IP protocol, the **IP Address** and **Port** fields of the PC where DianaServer is going to be used should also be configured in this window (default values are `localhost` and `8070`, respectively).

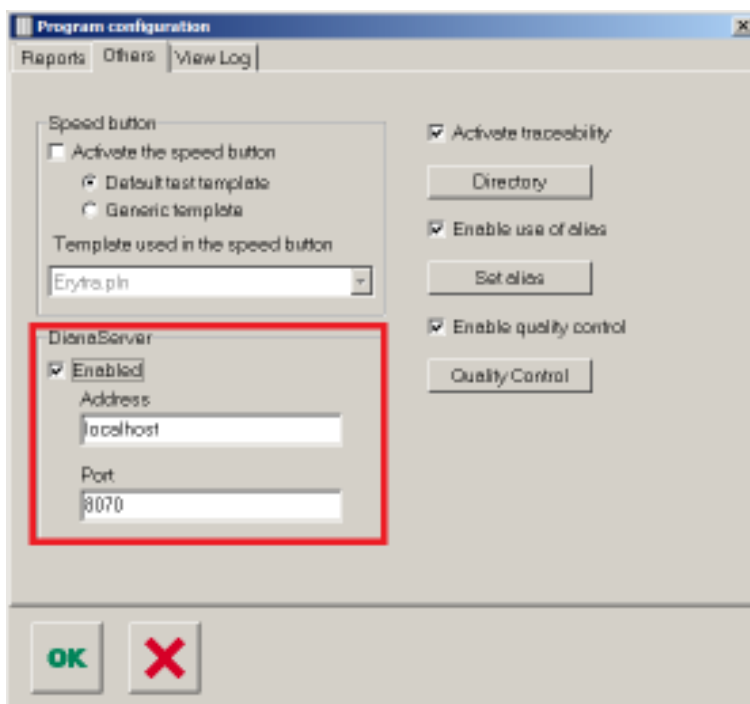


Figure 886 - DianaServer configuration options in Diana O.

When executing DianaServer, the DianaServer () icon appears in the bottom-right edge of the Windows taskbar.

Right clicking with the mouse brings up a context menu with the options **Configuration** and **Close**. Clicking on the **Configuration** option opens following window:

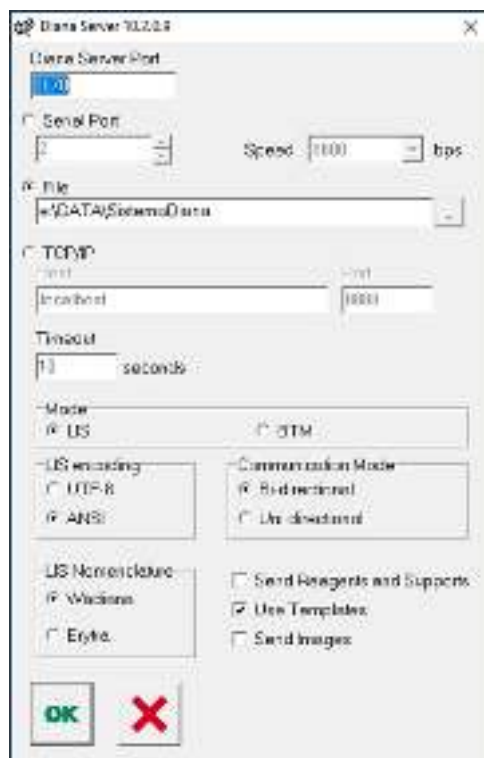


Figure 887 - DianaServer configuration window

The different options to configure are:

- **DianaServer port:** must match with the **Port** number indicated in Diana O. configuration (default value is 8070).
- **Communication mode (Serial Port, File or TCP/IP):** see below for a detailed explanation of each mode.
- **Timeout:** This setting is the maximum time in seconds that the DianaServer will wait for an answer from the LIS once a message has been sent. This timeout is also applied in the communication between Erytra and DianaServer. The transmission of the microtubes images significantly increases the size of the message to transmit and then the time required to complete the transmission is increased. The timeout setting should be reviewed when enabling the microtubes image transmission to LIS. See "Files communication mode" on page 1158.
- **Mode:** Connection mode selection, LIS or BTM. See below for detailed explanation.
- **LIS encoding:** character encoding system. Default option is **ANSI**, but **UTF-8** can also be selected, making sure that the selected system matches the corresponding language in Windows, Region and Language settings.
- **LIS Nomenclature:** nomenclature of interpretations, microtubes, results,... transmitted to LIS. Options are **Diana System** (nomenclature used in previous software versions) or **Erytra system** (the names displayed in user interface will be the ones transmitted to LIS).

- **Communication mode (Bi-directional / Unidirectional):** it does not apply/affect Erytra - LIS communication.
- **Send reagents and supports:** if enabled the barcodes of the cards and reagents used to process each sample are transmitted to LIS. This new option will modify the already existing Windows registry parameter SendBCReacSupp.
- **Use templates:** when enabled, LIS must indicate/send a template and this template will be used to transmit the results to LIS. When disabled, LIS must not indicate/send a template and all the interpretations and microtubes results displayed in user interface results detailed view will be transmitted to LIS.

The default status is enabled.

- **Send images:** option to include microtubes images in the ASTM message to LIS. The default status is disabled.



NOTE

When **Send images** option is activated, Baud rate should be checked and configured to the maximum allowed rate.

Serial communication mode

The following settings may be selected:

- Serial port of Erytra PC that is connected to the LIS (**COM2** expected for PC EMBEDDED v3, v4 and v5, **COM3** expected for PC EMBEDDED v2 / PC EMBEDDED - USA).
- Transmission speed in bauds.



NOTE

When enabling the microtubes image transmission to LIS, baud rate configuration should be set to the maximum supported value.

The rest of the settings are fixed (and set when capturing the port):

- Parity not used.
- 1 start bit.
- 1 stop bit.
- 8 data bits.

File communication mode

It allows setting the files exchange directory.



NOTE

The exchange directory shall be unique. That is, if several systems should be connected to host and more than one DianaServer is used, each DianaServer must be configured with a different exchange directory.

TCP/IP communication mode

It allows setting the host IP address and the related port.

Timeout

This setting is the time in seconds that the DianaServer should wait before receiving an answer once a message has been sent to the LIS.

LIS or BTM connection mode

LIS: when this option is selected, LIS encoding, LIS Nomenclature, Send Reagents and Supports, Use Templates and Send images can be configured.

- When **LIS** and **Erytra** options are selected, the names of the segment, etc will be transmitted in the instrument configured application language.
- When **LIS** and **Wadiana** options are selected, the names of the segment, etc will be transmitted in the Diana System configured applications language.
- Default LIS mode is: LIS.

BTM: when this option is selected, the following parameters are set to required values:

- Encoding: UTF-8.
- Nomenclature: Erytra.
- Send Reagents and Supports: Yes.
- Use Templates: No.
- Send Images: Yes.

And they are greyed out (non-modifiable).

The names of the segment, etc will be transmitted in English and the **Comments** field will be transmitted as 'Remarks'.

7.8.2 Serial communication mode

Using the serial channel system, messages are sent to the Host using the CLSI LIS01-A2 standard.

Serial communication is made through serial port/cable directly to the LIS. The message is chopped into frames.

After each frame is sent, the receiver has to reply with an **OK** message in order to continue the transmission.

Each frame is numbered so a flow control can be established.

7.8.3 Files communication mode

File exchange mode is as follows: When Erytra has to do a request/send a message to the LIS, DianaServer writes the request/message (following the CLSI LIS02-A2 standard) in a file named **TX_TMP.DAT** in the configured exchange directory. Once the complete message is written, this file is renamed to **OUT.DAT**.

LIS shall read this **OUT.DAT** file and then delete it (to indicate that it has been read) within the allowed time set in time out parameter (default time 10 seconds). If the file is not deleted within the allowed timeout, an error is generated.

The LIS shall write the answer message in a file named **IN.DAT** located in the configured exchange directory. DianaServer will read this message and delete it to inform the LIS that the message has been read.

The LIS can not initiate the communication with Erytra.

7.8.4 TCP-IP communication mode

TCP/IP communication mode allows using TCP/IP protocol when exchanging information with the host. In order to use it, the host IP address and port must be indicated in the corresponding textboxes.

Data is encapsulated according to the TCP/IP protocol and transmitted through the network once the latter has been configured accordingly.

In order to allow TCP/IP communication, it is necessary to follow the next steps to create a firewall rule:

1. Connect an external mouse to the instrument.
2. Switch on the instrument.
3. Press the grid () button to enter the boot application and log in as `adminUser`.
4. Select the **Maintenance Mode** () option.
5. Open Windows Defender Firewall advanced settings.
 - How to: Select **Start > Windows System > Control Panel**.
 - Once Control Panel is opened, select **System and Security > Windows Defender Firewall > Advanced settings**.
6. Select **Outbound Rules** and **New Rule** options in the **Actions** section.

7. Select **Port** in the drop-down menu and press the **Next** button.

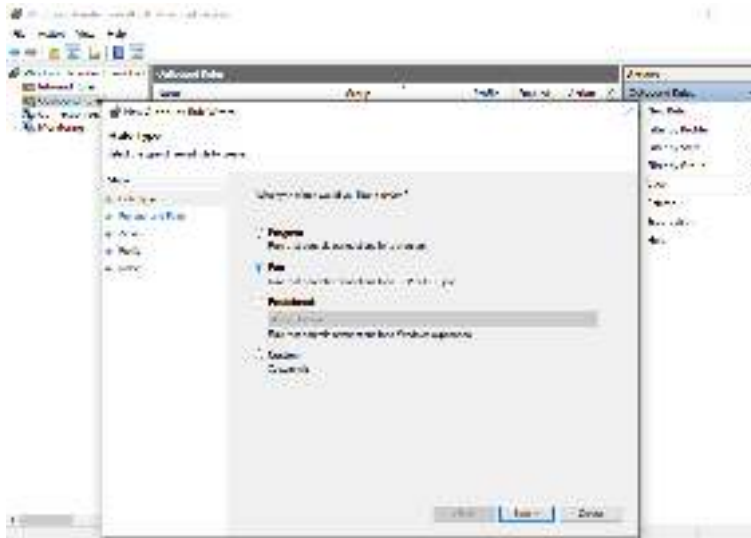


Figure 888 - Rule type: Port

8. Select the **TCP** option and fill the field with the specific remote port. Press the **Next** button.

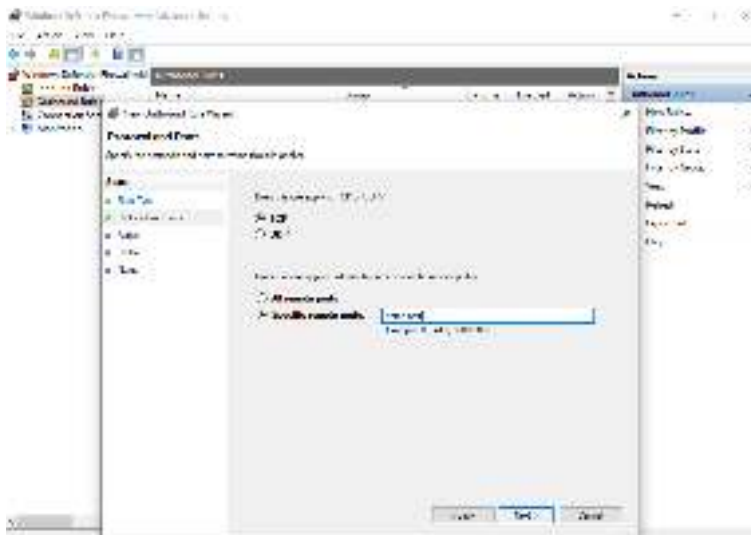


Figure 889 - Protocol and Ports: TCP

9. Select **Allow the connection** or **Allow the connection if it is secure** and select the **Next** button.

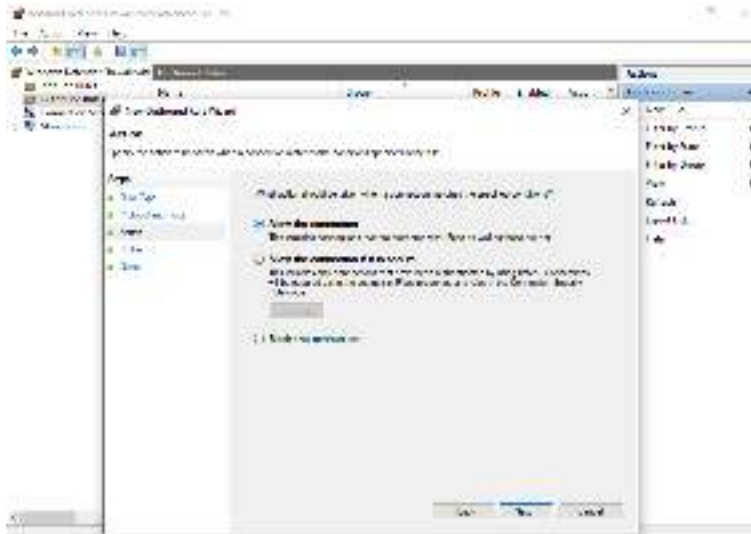


Figure 890 - Action: Allow the connection

10. Do not change the default settings and click the **Next** button.

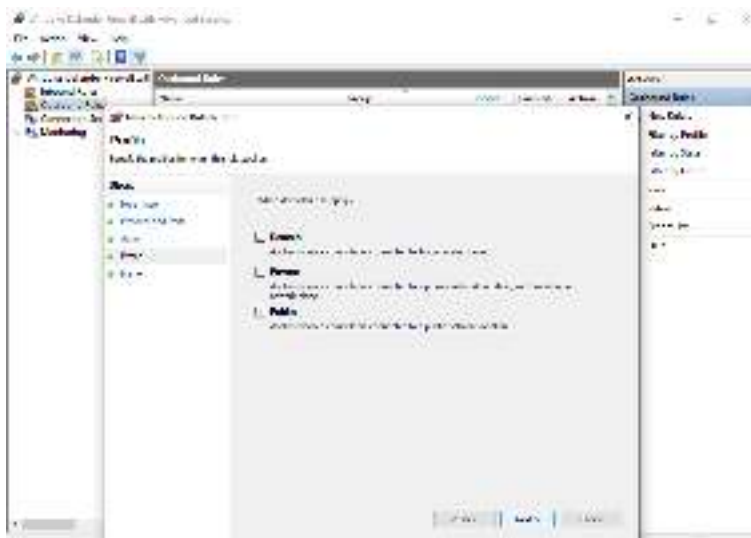


Figure 891 - Profile: default settings

11. Enter the name of the exception and press the **Finish** button. The selected rules will be created and enabled.

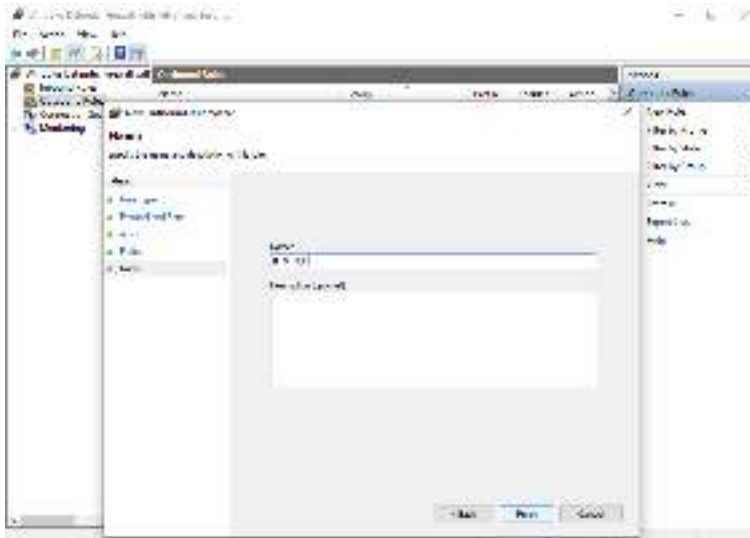


Figure 892 - Name: name of the rule

12. Close the Windows Defender Firewall with the **Advanced Security** window.

7.8.5 HostDemo program

HostDemo.exe is a technical support tool, the aim of which is to check that the LIS connection is properly configured and information can be exchanged between Diana System and LIS.



NOTE

While the character encoding system can be configured in DianaServer program (ANSI/UTF-8), HostDemo can only use ANSI system.

The **HostDemo.exe** file is automatically installed when installing/upgrading Erytra software in the **E:\data\SistemaDiana** directory. Upon opening the program, the software expects the following credentials:

- **User:** ST
- **Password:** d+g_diana

Once the user name and password are introduced, the following window will appear:

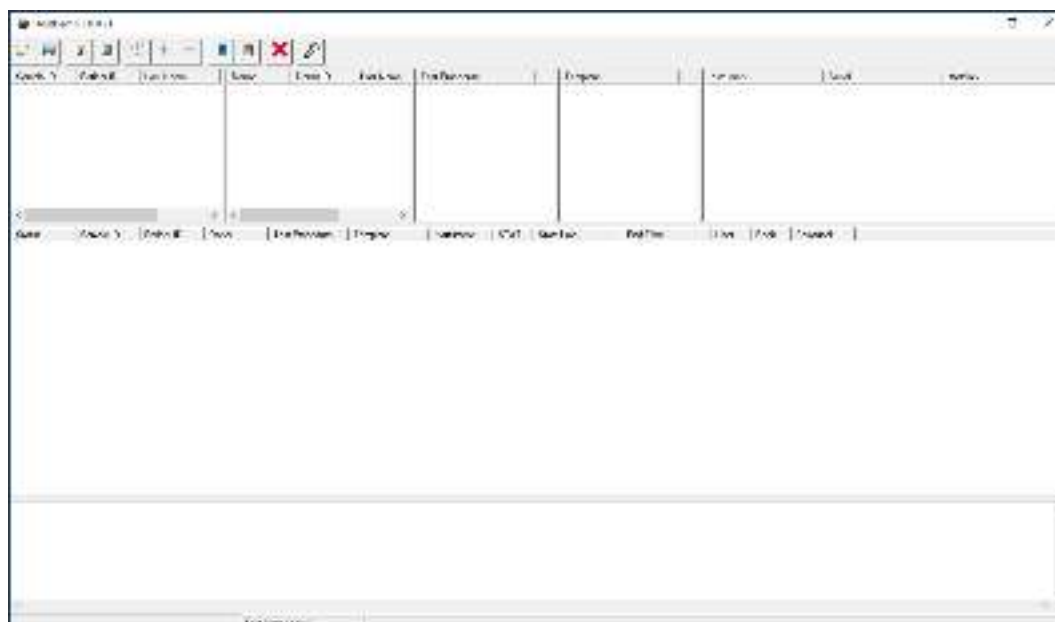


Figure 893 - HostDemo program

The HostDemo user interface includes six different informative sections:

- List of available samples.
- List of available donors.
- List of available test procedures.
- List of available templates.
- Historical list of connected instruments.
- List of test request.

.hst or .dat file extension are allowed.

Option to create samples and donors.

Option to create automatically a random list of samples and donors.

Option to change the STAT flag of a test requests.

Option to import test procedures and templates from a Diana System installation.

Option to delete test procedures.

Option to delete templates.

Option to ignore requests from REPORT (Diana O.) instruments.

Option to automatically change the status of Cancelled tests to New.

Option to view the detailed results of finished test:

- Wells name.
- Segments name.

- Well images.
- Comments.
- Wells results.
- Segments result.
- Reagent and card barcodes.

New **HostDemo.log** in the (installation folder) subdirectory named log.

The HostDemo does not allow to assign two different sample barcodes to a given patient.

An **Access violation** error may be displayed when the loading test procedures and template from Diana System directory has been completed successfully.

The New HostDemo program main window now has the following sections:

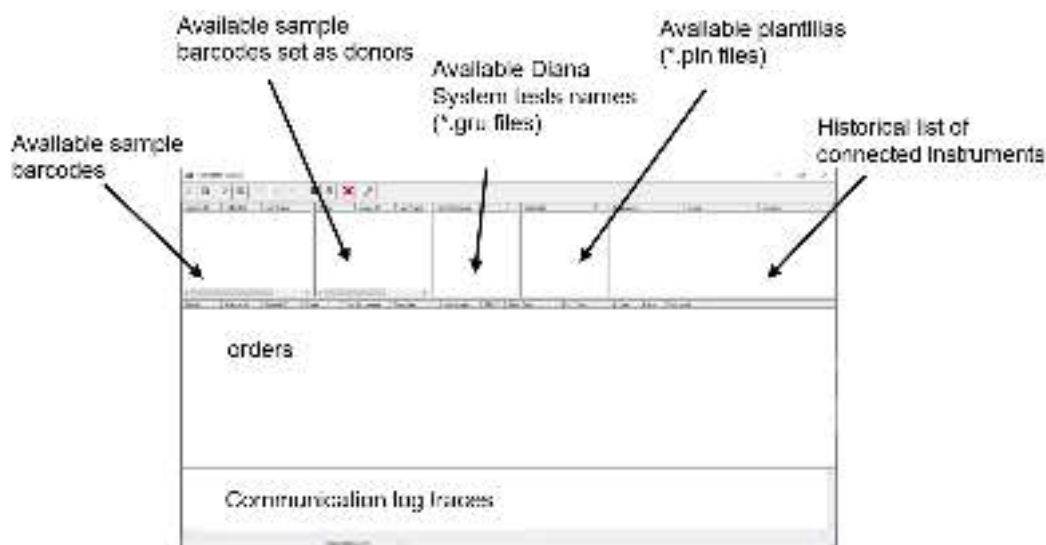


Figure 894 - HostDemo sections

And the following main menu buttons:

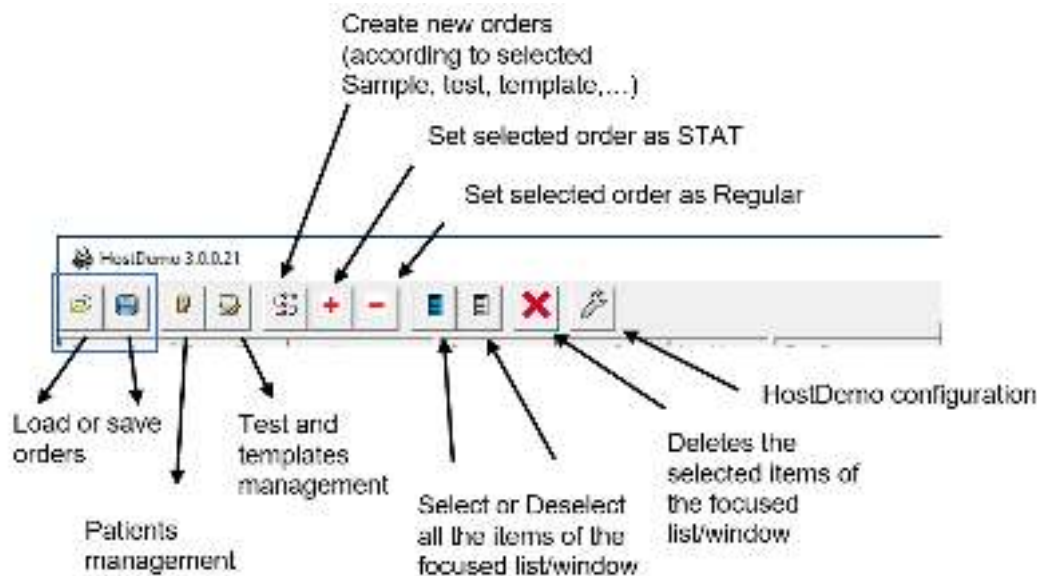


Figure 895 - HostDemo menu buttons

7.8.5.1 Patients Management

The Patients management window is opened when the **Patients Management** button is selected:

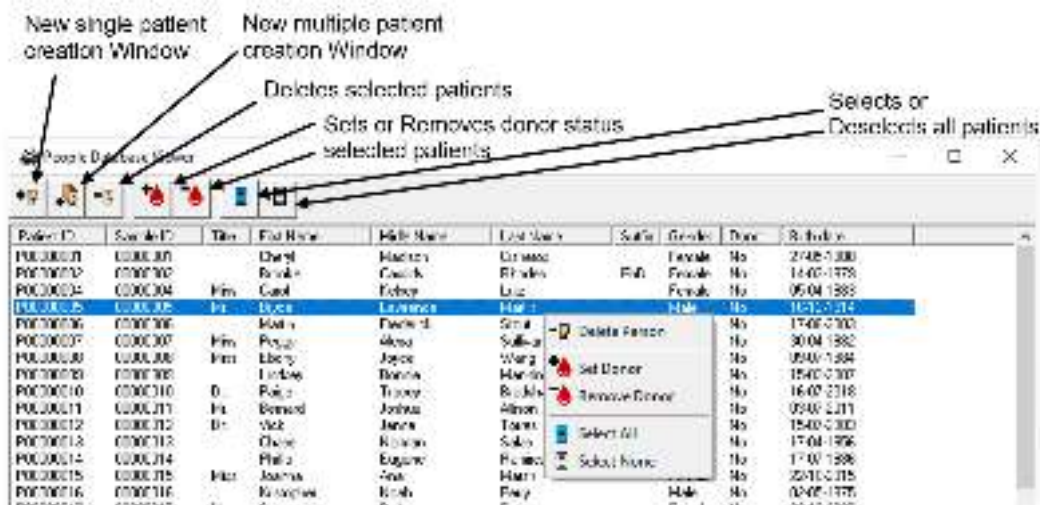


Figure 896 - Patients Management

The mouse right button will show a contextual menu with several options when a patient is selected.

Double-clicking the left mouse button opens in this window or in the main one. Patient data can be edited.



Figure 897 - Edit Person

And the **Create Multiple Patients** window is opened when selecting this option, allowing to define starting patient, how many new patients will be created and patient and sample pattern:

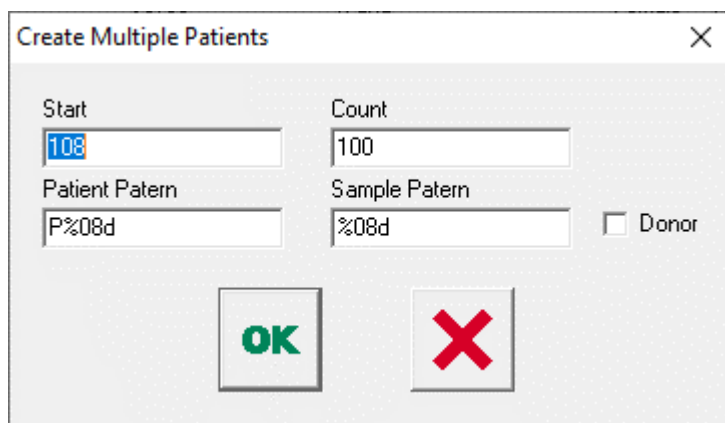


Figure 898 - Create Multiple Patients

7.8.5.2 Test and Templates management

The Test and Templates management window will open with the following options:

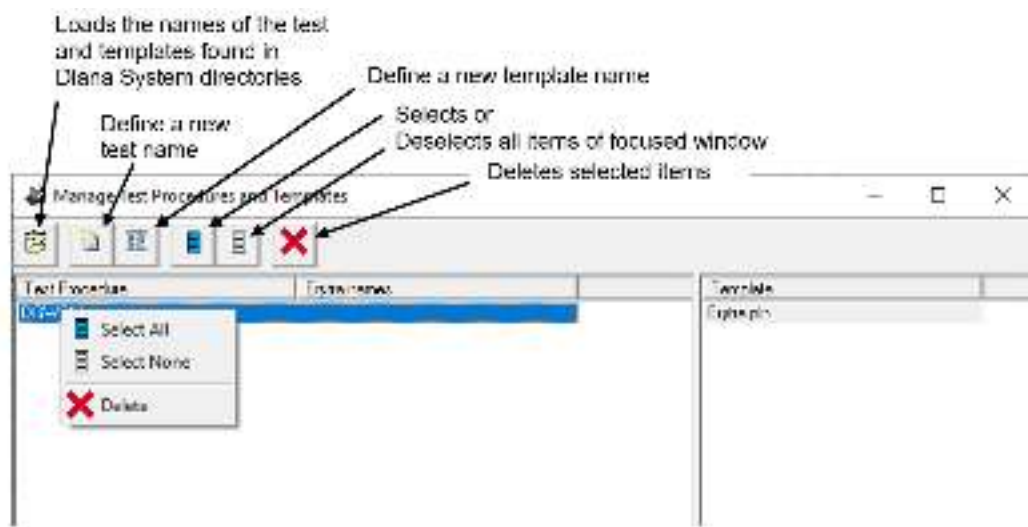


Figure 899 - Test and Templates management

The mouse right button will show a contextual menu with several options when a patient is selected.

7.8.5.3 Configuration

The configuration window includes two tabs, the one related to general options and the one related to Host Settings:

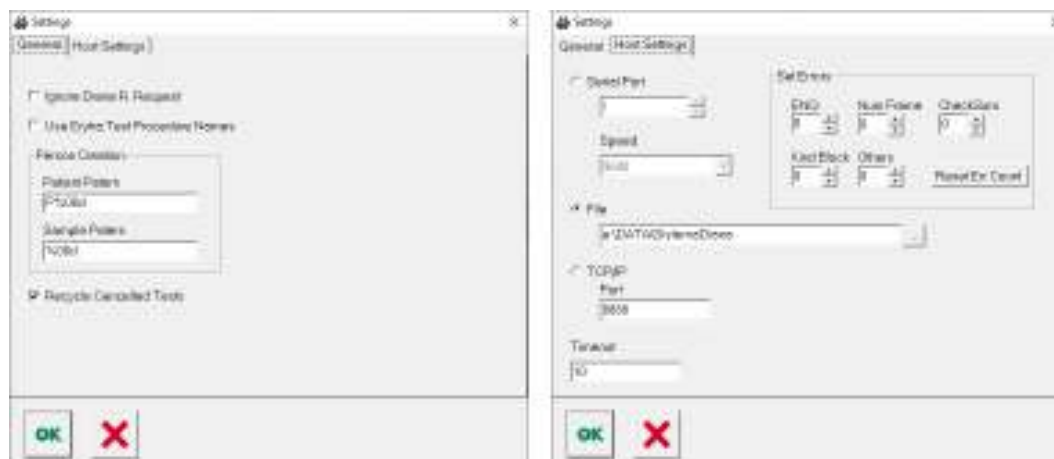


Figure 900 - Configuration

7.8.5.4 Orders section

The orders are displayed in this section of the main window.

Order color indicates order status.

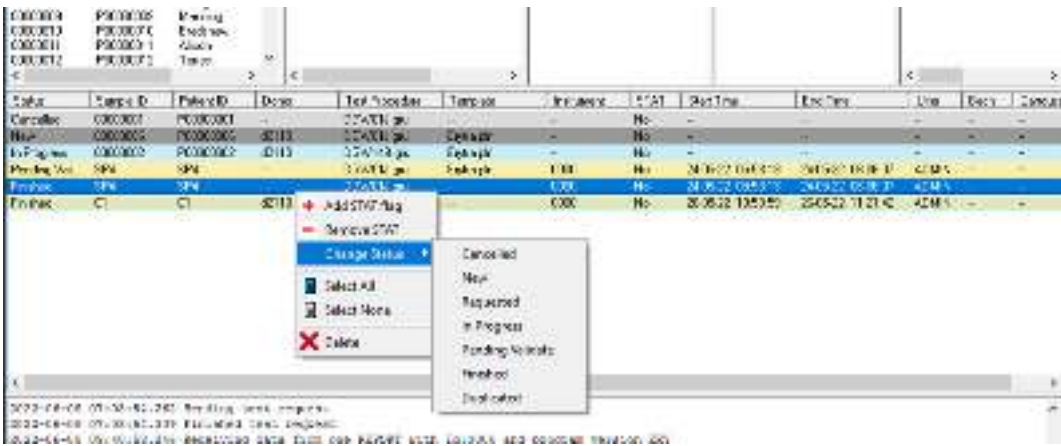


Figure 901 - Orders

The mouse right button will show a contextual menu with several options when an order is selected. One of these option is **Change Status**, which allows you to manually modify order status.

Double-clicking the left mouse button on the selected order will open the **Results Viewer** window, that allows reviewing microtubes results, images, reagents,...

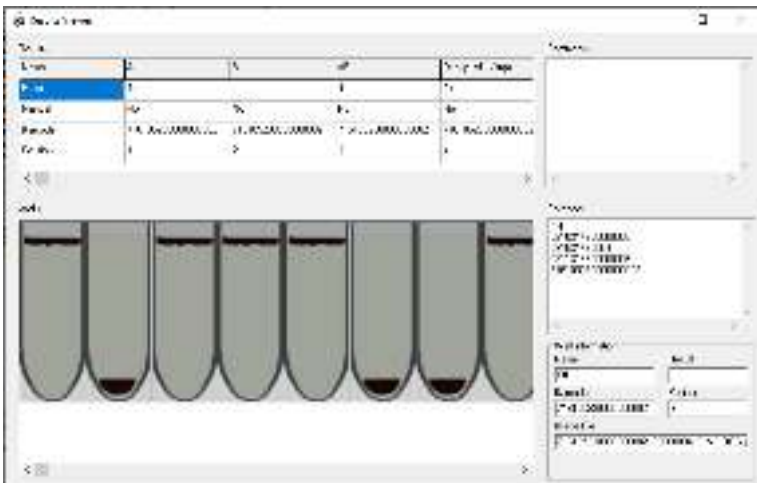


Figure 902 - Test results

Hostdemo created the **DianaASTM.log** file in the **E:\data\SistemaDiana\log\log** folder.

7.8.6 Enabling and configuring LIS connection

To enable and configure the Erytra-LIS connection, proceed as listed below. If connected, it is also recommended to disconnect the instrument from the network during the configuration procedure.

Erytra software configuration

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the **Maintenance Mode** () icon among the Boot Application options.
2. The PC will be rebooted and Maintenance Mode will be accessed in the subsequent startup.
3. Once the desktop is available, press the Windows **Start** button and type `Command Prompt`. Right click at the top result and select **Run as administrator**, then type administrator password `dgroot`.
4. Enter the command: `applocker bu` (Begin Update). The AppLocker Manager graphical interface will be opened in a separate window.
5. Wait for 10 seconds so that the change to update mode is completed.
6. Edit the `E:\conflerytra.properties` file using Notepad++.
7. Set **LIS_ACTIVATED** parameter to **true**.
8. Check/set **LIS_SHARED_MODE** parameter:
 - **false**: sample requests where the sample is missing will be held waiting for the missing sample to be identified (default value).
 - **true**: sample request where the sample is missing will be sent back to LIS as cancelled.
9. Check/set the **RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA** parameter (only for CE instrument configuration):
 - **false**: do not store in Erytra database and do not show demographic data in Erytra application (default value).
 - **true**: store in Erytra database and show demographic data in Erytra application.
10. Check/set **LIS_AUTO_QUERY_BY_SAMPLE** parameter:
 - **false**: no automatic request to LIS will be performed after a sample drawer is identified or a sample tube is manually identified (default value).
 - **true**: once a sample drawer is identified or a sample tube is manually identified, automatically request to LIS all pending work for every identified sample inside the instrument.
11. **AUTO_LIS_QUERY_ENABLED**: it allows enabling and disabling the periodic automatic query to LIS.
 - **false**: Auto LIS query feature is disabled (default value).
 - **true**: Auto LIS query feature is enabled.
12. **AUTO_LIS_QUERY_MODE**: it allows to set the query mode to Query All or to Query by Sample (when **AUTO_LIS_QUERY_ENABLED=true**).
 - **QALL**: Query All will be sent to LIS (when **AUTO_LIS_QUERY_ENABLED=true**)
 - **QBSAMPLE**: Query by Sample will be sent to LIS. (when **AUTO_LIS_QUERY_ENABLED=true**) (default value).

13. **AUTO_LIS_PERIOD_UPDATE**: it allows defining the amount of time in minutes that will elapse between each periodic automatic query (when **AUTO_LIS_QUERY_ENABLED=true**).
 - Values allowed: integers from 1 to 60.
 - Default value: 5.
 (Default value will be applied if a value out of the allowed range is found)
14. Save the file and close the Notepad++ .

DianaServer configuration

1. Enable DianaServer and set the desired IP address and port number in the **Program Configuration** options in Diana O. Then start DianaServer and configure all the parameters. See "DianaServer program" on page 1155 for details on each of the expected parameters.



NOTE

When connecting two or more Erytra computers to the same network domain and workgroup, it is recommended to rename the name of the computers to avoid conflicts between them.

Configure DianaServer to automatically start at PC startup

Software installer will configure the system to start DianaServer automatically when the user interface starts. In case this configuration is disabled, please follow the steps below to enable it:

2. Go to **E:\Program files\Startup** and create an empty text file called **DIANASERVER.START**. Make sure that the original file extension '.txt' is also removed from the file name (in order to delete it, make the extension visible by clicking on the **Organize** button in the current explorer window, select **Folder and search options > View tab** and uncheck the **Hide extensions for known file types** option. Once the '.txt' extension is deleted, enable the option again so that file extensions remain hidden).
3. Type **eu** (End Update) in the AppLocker Manager graphical interface window.
4. Wait for 10 seconds so that the change to protected mode is completed.
5. Reboot the system to exit Maintenance Mode.

7.8.7 Erytra-LIS errors

In this section, the Erytra and LIS communication errors are listed. The following tables include Diana System errors and Erytra software errors related to LIS communication.

7.8.7.1 Diana System errors related to LIS communication

Table 113 - Diana System errors related to LIS communication

Module	Error code	Cause
Message contents errors (ASTM-1394 standard)	Host format error #1	Heading error: wrong separators.
	Host format error #2	Heading error: first line of the message does not begin with 'H'.
	Host format error #3	Error Reading Patient No patient ID.
	Host format error #4	Test: Not enough fields.
	Host format error #5	Test: No specimen field.
	Host format error #6	Test: No test name fields.
	Host format error #7	Error, an O field before a P field.
	Host format error #8	Error, an R field before an O field.
	Host format error #9	Results: Not enough fields.
	Host format error #10	File received has no lines, it is empty.

Module	Error code	Cause
File communication errors	Host file error #1	Exchange file TX_TMP.DAT/ OUT.DAT can not be written in the working directory.
	Host file error #2	Exchange file TX_TMP.DAT/ IN.DAT has been created but it is not possible to fill it.
	Host file error #3	IN.DAT file can not be deleted.
	Host file error #4	OUT.DAT file can not be deleted.
TCP/IP communication errors	Reomote host not ready	LIS server is not ready when trying to establish communication with DianaServer.
Serial or TCP/IP communication errors	Host serial error #1	Time-out when receiving frame.
	Host serial error #2	Incorrect frame received.
	Host serial error #3	Incorrect frame received, STX fault.
	Host serial error #4	Neither ETB or ETX being received in the frame.
	Host serial error #5	Frame received without checksum.
	Host serial error #6	Frame received with wrong checksum.
	Host serial error #7	Frame received with wrong ID.

Module	Error code	Cause
Serial or TCP/IP communication errors	Host serial error #8	Communication error in establishing the connection while awaiting reception.
	Host serial error #9	Undetermined error.
	Host serial error #10	Frame sent six times and returned by LIS as faulty (NAK).
	Host serial error #11	ENQ received while establishing communication.
	Host serial error #12	Strange character received while establishing communication.
	Host serial error #13	EOT missing.

7.8.7.2 List of errors related to Erytra - LIS communication module

Table 114 - List of errors related to Erytra - LIS communication module

Module	Error code	Error	Cause	Text	Options
LIS communication	5000002	ERROR_POCILLOS_DEF	Unable to access Pocillos.def file.	LIS error: Unable to initialize file Pocillos.def . Please contact Technical Service.	OK
	5000003	ERROR_INTERP_DEF	Unable to access INTERP.def file.	LIS error: Unable to initialize file Interp.def . Please contact Technical Service.	
	5000004	ERROR_ALIAS_DEF	Unable to access alias.def file.	LIS error: Unable to initialize file Alias.def . Please contact Technical Service.	
	5000005	ERROR_IDIOMA_L_DEF	Unable to access Idioma_l.def file.	LIS error: Unable to initialize file Idioma_l.def . Please contact Technical Service.	
	5000006	ERROR_DSC_DEF	Unable to access DSC.def file.	LIS error: Unable to initialize file DSC.def . Please contact Technical Service.	

Module	Error code	Error	Cause	Text	Options
LIS communication	5000007	ERROR_ERYTRA_DATA	Unable to find erytra_data environment OS variable.	LIS error: Unable to find erytra_data environment variable. Please contact Technical Service.	OK
	5000008	ERROR_TIMEOUT_REQUEST	Timeout exceeded during a request to DianaServer.	LIS error: Timeout exceeded during a request to DianaServer. Please contact Technical Service.	
	5000009	ERROR_REQUEST	DianaServer request error.	LIS error. DianaServer request error. Please contact Technical Service.	
	5000010	ERROR_NO_INIT	Unable to initialize dsc.dll file.	LIS error: DLL has not been initialized. Please contact Technical Service.	
	5000011	ERROR_GRU_NOT_FOUND	Unable to find gru in DSC.def file.	LIS error: Gru x not found in DSC.def . Please contact Technical Service.	

Module	Error code	Error	Cause	Text	Options
LIS communication	5000012	ERROR_TEMPLATE	Unable to load template.	LIS error: Error loading template x. Please contact Technical Service.	OK
	5000013	ERROR_XML	Error when dsc.dll and java try to communicate.	LIS error: Error loading XML: x. Please contact Technical Service.	
	5000014	ERROR_RESULT_SEGMENT_NOT_FOUND	Unable to find result or segment in DSC.def file.	LIS error: Result of segment x not found in DSC.def . Please contact Technical Service.	
	5000015	ERROR_WELL_NOT_FOUND	Unable to find well in DSC.def file.	LIS error: Well x not found in DSC.def . Please contact Technical Service.	
	5000016	ERROR_RESULT_WELL_NOT_FOUND	Unable to find result in DSC.def file.	LIS error: Result of well x not found in DSC.def. Please contact Technical Service.	
	5000017	ERROR_INCIDENCE_WELL_NOT_FOUND	Unable to find incidence well.	Program error: Well incidence not found in DSC.def . Please contact Technical Service.	
	5000018	ERROR_IMAGE_WELL_NOT_FOUND	Unable to find image well.	LIS error: Well x not found in DSC.def . Please contact Technical Service.	
	5000019	ERROR_DELETING_BATCH	Error when deleting batch.	Program error: Error deleting batch. Please contact Technical Service.	

Module	Error code	Error	Cause	Text	Options
LIS communication	5000020	ERROR_OPEN_DATABASE	Error when accessing database.	Program error: Error open database. Please contact Technical Service.	OK
	5000021	ERROR_ERYTRA_LOG	Error when accessing log file.	Program error: Error in Erytra log. Please contact Technical Service.	
	5000022	ERROR_UNKNOWN	Unknown error.	Program error: Unknown LIS error. Please contact Technical Service.	
	5000023	ERROR_DSC_CUSTOM_DEF	Error when accessing DSC_custom.def file.	Program error: Error in DSC_Custom.def . Please contact Technical Service.	
	5000024	ERROR_COMMUNICATION_DIANASERVER	Error in communication with DianaServer.	Program error: Error in communication with DianaServer. Please contact your local Grifols service representative.	
	5000025	ERROR_INTERPRETATION_NOT_FOUND	Interpretation not found in DSC.def.	Program error: Interpretation xxx not found in DSC.def . Please contact your local Grifols service representative.	
			Interpretation not found in INTERP.DEF.	Program error: Interpretation xxx not found in INTERP.DEF . Please contact your local Grifols service representative.	

Table 115 - LIS troubleshooting

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Error LIS 5000008 (ERROR_TIMEOUT_REQUEST)	Erytra does not communicate with DianaServer	Check if DianaServer is running. Check that Erytra has access to the computer where the DianaServer is running

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS	
Error LIS 5000009 (ERROR_REQUEST)	DianaServer does not communicate with LIS	Communication by files	Check that LIS is running and it has access to the working directory.
			Check that DianaServer is running and it has access to the working directory.
			Increase DianaServer timeout.
		Communication by serial port	Check that serial cable is connected.
			Configure serial ports buffer to minimum.
			Check/replace serial cable.
			Check that LIS is running.
			Increase DianaServer timeout.

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Error LIS 5000012 (ERROR_TEMPLATE)	Error when trying to use a template to send the results	Check that template sent by LIS exists in Plantillas folder of the working directory defined in the Erytra (default in E:\data\SistemaDiana\Plantillas)
		Check that defined template by means of Importer application for the related tests exists in the Plantillas folder of the working directory defined in Erytra (default in E:\data\SistemaDiana\Plantillas).
		Check that default template defined in the OS registry (HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols\SistemaDiana\Diana O.\PlantillaDefecte) exists in the Plantillas folder of the working directory defined in Erytra (default in E:\data\SistemaDiana\Plantillas).
No results are sent to LIS	Incorrect definition of the template	Define correctly the wells/segments to send in the template being used.
	Template is empty	No wells/segments are defined in the template being used.

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Results sent are truncated	Incorrect template definition	Check template definition Results field in ASTM message allocates 200 characters maximum including separators. Template field in ASTM message allocates 255 characters maximum including separators.
Well/segments are sent with another name that defined one	An alias has been defined	Check/configure alias.def located in defs folder of the working directory (default location in E:\data\SistemaDiana\defs).
Segments names are sent in another language that desired one	Diana System language is not configured properly	Check/configure Diana System language by means of Config.exe located in the working directory (default location in E:\data\SistemaDiana).

7.8.8 LIS interface specification

The LIS interface specification describes the message content for transferring information between the Erytra and the laboratory information system (LIS).

To view the LIS interface specification, see "Related documents" on page 19 or refer to the Erytra.

Bulletin index



NOTE

All MTBs are store in the **Relateddocuments > Erytra > MTBs** folder supplied with this document.

All FSBs are stored in the **Relateddocuments > Erytra > FSBs** folder supplied with this document.

All INRs are stored in the **Relateddocuments > Erytra > INRs** folder supplied with this document.



NOTE

IH Reagent Pack (that applies to Erytra, WADiana Compact, Diana/DG Reader and Diana Processor) INRs and FSBs will be sent from 28/JUL/2014 onwards under DG-0423 reference. Refer to DG-0423 index of FSBs and INRs.

Table 116 - Manufacturer Technical Bulletin

Document ID	Title	Communication	Ver.	Date	Applied to
MTB-0059#0001	ERYTRA SOFTWARE VERSION 4.5.1	DXBU-PDS-TB-000012	01	08 SEP 2025	All units not under NMPA
MTB-0059#0002	READER MODULE FOAM REMOVAL	DXBU-PDS-TB-000004	01	30 JUL 2025	s/n ≥ 059-0002085.
MTB-0059#0003	ERYTRA SOFTWARE VERSION 4.6.1	DXBU-PDS-TB-000002	01	29 JUL 2025	All units not under NMPA
MTB-0059#0004	ERYTRA SOFTWARE VERSION 5.2.1	DXBU-PDS-TB-000003	01	DATA PENDING	All units under NMPA

Table 117 - Field Service Bulletin

Document	Ver.	Title	Date	Applied to
DG-0059#0001	NA	SCREWS REPLACEMENT IN CARD TRANSPORT ARM MODULE	21/DEC/10	S/N ≤ 059-0000506 S/N 059-0000508
DG-0059#0002	NA	ERYTRA SYSTEM TOOLS DVD VERSION 1.01	23/DEC/10	S/N ≥ 059-0000510
DG-0059#0003	NA	ERYTRA SYSTEM TOOLS DVD VERSION 1.01 PATCH 1	23/DEC/10	S/N ≥ 059-0000509
DG-0059#0004	NA	ERYTRA NEW PIP PRESENCE DETECTOR HOLDER PLATE	31/MAR/11	S/N ≥ 059-0000510
DG-0059#0005	NA	ERYTRA SOFTWARE VERSION 1.01 PATCH 2 AND PATCH 3	31/MAR/11	S/N ≥ 059-0000527 S/N 059-0000524 S/N 059-0000525
DG-0059#0006	NA	ERYTRA NEW SAMPLES RACK	30/MAY/11	S/N ≥ 059-0000531
DG-0059#0007	NA	ERYTRA SOFTWARE VERSION 1.01 PATCH 4	03/JUN/11	S/N ≥ 059-0000535
DG-0059#0008	NA	ERYTRA SOFTWARE VERSION 2.00	07/JUL/11	S/N ≥ 059-0000551
DG-0059#0009	NA	ERYTRA SOFTWARE VERSION 2.00C PATCH 1 AND PATCH 2	24/FEB/12	S/N ≥ 059-0000575
DG-0059#0010	NA	ERYTRA SOFTWARE VERSION 2.00D	13/APR/12	S/N ≥ 059-0000583
DG-0059#0011	NA	ERYTRA SOFTWARE VERSION 3.00	19/SEP/12	S/N ≥ 059-0000611 S/N 059-0000601
DG-0059#0012	NA	ERYTRA NEW BOTTLES CONFIGURATOR V2	19/SEP/12	S/N ≥ 059-0000611 S/N 059-0000601
DG-0059#0013	NA	ERYTRA SOFTWARE VERSION 3.01	04/MAR/13	S/N ≥ 059-0000633 S/N 059-0000624, S/N 059-0000627 to S/N 059-0000630

Document	Ver.	Title	Date	Applied to
DG-0059#0014	NA	ERYTRA PUMPS REGULATION FLUIDIC SYSTEM	04/MAR/13	S/N ≥ 059-0000633 S/N 059-0000624 S/N 059-0000627 to S/N 059-0000630
DG-0059#0015	NA	ERYTRA PROBES WASHER VER. 2	25/MAR/13	S/N ≥ 059-0000651
DG-0059#0016	NA	ERYTRA VACUUM FILTERS	01/OCT/13	S/N ≥ 059-0000699
DG-0059#0017	NA	ERYTRA AC COVER EDGE PROTECTOR	01/OCT/13	S/N ≥ 059-0000651
DG-0059#0018	NA	ERYTRA SOFTWARE VERSION 3.01 PATCH 3	16/OCT/13	S/N ≥ 059-0000719
DG-0059#0019	NA	ERYTRA SOFTWARE VERSION 3.2	26/MAR/14	S/N ≥ 059-0000777
DG-0059#0020	NA	ERYTRA SOFTWARE US VERSION 1.2 (US)	27/FEB/15	SW (US) 1.2 S/N ≥ 059-0000599
DG-0059#0021	NA	ERYTRA SOFTWARE VERSION 3.4.0	31/JUL/15	S/N ≥ 059-0000975
DG-0059#0022	NA	ERYTRA SOFTWARE VERSION 3.4.2	30/MAR/16	S/N ≥ 059-0001103
DG-0059#0023	NA	ERYTRA SOFTWARE VERSION 3.4.3	05/APR/17	S/N ≥ 059-0001255
DG-0059#0024	NA	ERYTRA SOFTWARE VERSION 4.0.0 (US)	29/NOV/17	SW 4.0.0 S/N ≥ 059-0001360
DG-0059#0025	NA	ERYTRA SOFTWARE VERSION 4.0.3 (US)	16/APR/18	SW 4.0.3
DG-0059#0026	NA	ERYTRA SOFTWARE VERSION 4.0.3	13/APR/18	SW v ≥ 4.0.0 Not in units under CFDA
DG-0059#0027	NA	ERYTRA SOFTWARE VERSION 4.0.4	14/JUN/18	SW v ≥ 4.0.0 Not in units under CFDA
DG-0059#0028	NA	ERYTRA SOFTWARE VERSION 4.1.1	14/JUN/18	SW v ≥ 4.1.0 Not in units under CFDA S/N ≥ 0590001391
DG-0059#0029	NA	ERYTRA SOFTWARE VERSION 4.1.1(US)	14/JUN/18	SW v4.1.1

Document	Ver.	Title	Date	Applied to
FSB-0059#0030	02	NEW SYRINGE LOCK	21/SEP/20	S/N ≥ 0590001583
FSB-0059#0031	01	ERYTRA SOFTWARE VERSION 4.2.0	23/JUN/20	SW v 4.2.0 Not in units under CFDA S/N ≥ 059-0001551
FSB-0059#0032	02	NEW BIOS VERSION FOR ASROCK COMPUTER	25/OCT/21	S/N ≥ 059-0001747
FSB-0059#0033	03	HEMMING OF EDGES IN METALLIC PLATES	19/SEP/22	S/N ≥ 059-0001719
FSB-0059#0034	01	NEW FLU FIRMWARE v4.6.0	19/APR/22	S/N ≥ 059-0001783
FSB-0059#0035	01	PRESSURE SENSORS BATCH VERIFICATION AND REPLACEMENT	23/MAI/22	All
FSB-0059#0036	03	NEW MK VALVES AND SYRINGES	28/APR/23	S/N ≥ 059-0001819
FSB-0059#0037	02	ERYTRA SOFTWARE VERSION 4.4.0	07/JUN/23	All Not in units under NMPA
FSB-0059#0038	02	ERYTRA SOFTWARE VERSION 5.1.0	07/JUN/23	All units under NMPA
FSB-0059#0039	01	NEW FILTERS FOR SOLA and SOLB CIRCUIT	27/JUN/23	S/N ≥ 059-0001879
FSB-0059#0040	03	ERYTRA SOFTWARE VERSION 4.5.0	21/DEC/23	All Not in units under NMPA S/N ≥ 059-0001903 manufactured with SW 4.4.1 & OS 2.2.1
FSB-0059#0041	01	ERYTRA SOFTWARE VERSION 4.6.0	16/SEP/24	All units not under NMPA
FSB-0059#0042	01	ERYTRA SOFTWARE VERSION 5.2.0	16/SEP/24	All units under NMPA
FSB-0059#0043	01	NEW ILLUMINATION PATTERN	27/NOV/24	S/N ≥ 059-0001985
FSB-0059#0044	01	WINDOWS SECURITY UPDATES SEPTEMBER 2024	27/NOV/24	W10 E2019v2.3.1
FSB-0059#0045	01	WINDOWS SECURITY UPDATES MARCH 2025	13/MAI/25	W10 E2019 v2.3.1 and v2.3.2

Table 118 - Information Note Release

Document	Ver.	Title	Date	Applied to
DG-0059#0001	NA	ERYTRA HARDWARE MODIFICATIONS 2	11/NOV/10	S/N ≥ 059-0000506
DG-0059#0002	NA	ERYTRA NEW TOUCHSCREEN Version 2	11/NOV/10	S/N ≥ 059-0000506
DG-0059#0003	NA	ERYTRA NEW SAMPLES RACK AND PROBE ADJUSTMENT TOOL	31/MAR/11	S/N ≥ 059-0000510
DG-0059#0004	NA	ERYTRA SERVICE MANUAL REVISION JULY 2011	07/JUL/11	SW v ≥ 2.00
DG-0059#0005	NA	NEW ERYTRA ADJUSTMENT TOOLS FOR SOFTWARE VERSION ≥ 2.00	07/JUL/11	SW v ≥ 2.00
DG-0059#0006	NA	ERYTRA SOFTWARE VERSION 2.00 PATCH 1	02/SEP/11	SW v 2.00 patch 1 S/N ≥ 059-0000551
DG-0059#0007	NA	ERYTRA SOFTWARE VERSION 2.00 PATCH 2	26/SEP/11	SW v 2.00 patch 2 S/N ≥ 059-0000551
DG-0059#0008	NA	ERYTRA SOFTWARE VERSION 2.00B	24/NOV/11	SW v 2.00B S/N ≥ 059-0000563
DG-0059#0009	NA	ERYTRA SOFTWARE VERSION 2.00C	16/DEC/11	SW v 2.00C S/N ≥ 059-0000567
DG-0059#0010	NA	ERYTRA SAMPLES RACK IMPROVEMENT	10/FEB/12	S/N ≥ 059-0000567 S/N 059-0000565
DG-0059#0011	NA	ERYTRA MIXED REAGENTS' RACK ASSEMBLY IMPROVEMENT	14/MAI/12	S/N ≥ 059-0000580
DG-0059#0012	NA	ERYTRA USER TRAY IMPROVEMENT	30/MAI/12	S/N ≥ 059-0000591 S/N ≥ 059-0000603 top cover.
DG-0059#0013	NA	ERYTRA SOFTWARE VERSION 2.00E	04/JUN/12	SW v 2.00E S/N ≥ 059-0000595

Document	Ver.	Title	Date	Applied to
DG-0059#0014	NA	ERYTRA WEB SERVER CONFIGURATION	25/JUN/12	S/N ≥ 059-0000593 S/N 059-0000585 S/N 059-0000586 S/N 059-0000585 S/N 059-0000588 S/N 059-0000589 S/N 059-0000590 S/N 059-0000591
DG-0059#0015	NA	ERYTRA SERVICE MANUAL REVISION SEPTEMBER 2012	19/SEP/12	SW v ≥ 3.00
DG-0059#0016	NA	ERYTRA NEW PROBES/WAGON ADJUSTMENT TOOL	19/SEP/12	ALL
DG-0059#0017	NA	ERYTRA IMPACT DETECTOR	19/SEP/12	S/N ≥ 059-0000611 S/N 059-0000601 SW v ≥ 3.00
DG-0059#0018	NA	ERYTRA EXTERNAL DRAIN	19/SEP/12	S/N ≥ 059-0000611 S/N 059-0000601 SW v ≥ 3.00
DG-0059#0019	NA	ERYTRA NEW PELTIER MODULE	30/NOV/12	S/N ≥ 059-0000611 S/N 059-0000598
DG-0059#0020	NA	ERYTRA NEW SOLUTION A/B CONTAINERS INTERNAL TUBE	07/FEB/13	S/N ≥ 059-0000639
DG-0059#0021	NA	ERYTRA SERVICE MANUAL REVISION FEBRUARY 2013	05/MAR/13	SW v ≥ 3.01
DG-0059#0022	NA	ERYTRA DISTRIBUTION AND CONNECTION BLOCKS VER. 2	25/MAR/13	S/N ≥ 059-0000651
DG-0059#0023	NA	ERYTRA CAT MIRROR WEDGE	25/MAR/13	S/N ≥ 059-0000651
DG-0059#0024	NA	ERYTRA DILUTION CUP BATH VER.2	25/MAR/13	S/N ≥ 059-0000651
DG-0059#0025	NA	REAGENTS PACK V2.0	06/MAR/13	SW ≥ 3.01 S/N ≥ 059-0000667
DG-0059#0026	NA	ERYTRA SOFTWARE VERSION 3.01 PATCH 2	09/JUL/13	SW 3.01 patch 2 S/N ≥ 059-0000695

Document	Ver.	Title	Date	Applied to
DG-0059#0027	NA	ERYTRA NEW PCA GEN-BCKPL59	01/OCT/13	S/N ≥ 059-0000687
DG-0059#0028	NA	ERYTRA NEW DILUTION CUP ALIGNMENT TOOL	01/OCT/13	ALL
DG-0059#0029	NA	ERYTRA DILUTION CUP ASSEMBLY IMPROVEMENT	27/NOV/13	S/N ≥ 059-0000711
DG-0059#0030	NA	ERYTRA PIP AND CAT X MOTOR MANAGEMENT IMPROVEMENT	27/NOV/13	S/N ≥ 059-0000723 SW ≥ 3.01 CAT-CM FW 1.17 PIP FW 1.22
DG-0059#0031	NA	ERYTRA SERVICE MANUAL REVISION MARCH 2014	26/MAR/14	SW ≥ 3.2
DG-0059#0032	NA	ERYTRA NEW PC EMBEDDED	04/JUN/14	S/N ≥ 059-0000767
DG-0059#0033	NA	ERYTRA NEW PC EMBEDDED (US)	04/JUN/14	S/N ≥ 059-0000767
DG-0059#0034	NA	ERYTRA SOFTWARE VERSION 3.3	20/JUN/14	SW 3.3 S/N ≥ 059-0000807
DG-0059#0035	NA	ERYTRA NEW HAMILTON VALVE AND SYRINGE	23/JUL/14	S/N ≥ 059-0000807
DG-0059#0036	NA	ERYTRA DILUENT RACK IMPROVEMENT	23/JUL/14	S/N ≥ 059-0000815
DG-0059#0037	NA	ERYTRA SOFTWARE VERSION 3.3.1	29/JUL/14	SW 3.3.1 S/N ≥ 059-0000807
DG-0059#0038	NA	ERYTRA WASTE PUMP (LIQUID) (PM2) VER. 2	03/NOV/14	S/N ≥ 059-0000851 S/N 059-0000804
DG-0059#0039	NA	ERYTRA RACK FOR MIXED REAGENTS ASSEMBLY IMPROVEMENT	03/NOV/14	S/N ≥ 059-0000832 S/N ≥ 059-0000843
DG-0059#0040	NA	ERYTRA SOFTWARE US VERSION 1.1	25/NOV/14	SW (US) 1.1 S/N ≥ 059-0000804
DG-0059#0041	NA	ERYTRA SOFTWARE VERSION 3.3.2	11/DEC/14	SW 3.3.2 S/N ≥ 059-0000867
DG-0059#0042	NA	ERYTRA NEW HAMILTON DILUTOR	14/JAN/15	S/N ≥ 059-0000867

Document	Ver.	Title	Date	Applied to
DG-0059#0043	NA	ERYTRA NEW PCA FLU-CTRL59	18/FEB/15	S/N ≥ 059-0000883
DG-0059#0044	NA	ERYTRA 210401 SERVICE MANUAL REVISION FEBRUARY 2025	27/FEB/15	SW (US) ≥ 1.2
DG-0059#0045	NA	ERYTRA NEW DILUTORSCONF.TXT FILE	27/MAR/15	S/N ≥ 059-0000907
DG-0059#0046	NA	ERYTRA NEW DILUTORSCONF.TXT FILE (US)	27/MAR/15	S/N ≥ 059-0000907
DG-0059#0047	NA	ERYTRA NEW FLUCALLIBRATIONS.TXT FILE	27/MAR/15	S/N ≥ 059-0000907
DG-0059#0048	NA	ERYTRA NEW FLUCALLIBRATIONS.TXT FILE (US)	27/MAR/15	S/N ≥ 059-0000907
DG-0059#0049	NA	ERYTRA PROBES HOLDER (IMPACT DETECTOR) (Ver. 2)	27/MAR/15	S/N ≥ 059-0000907
DG-0059#0050	NA	ERYTRA CARD HOLDER RETAINER	29/JUN/15	S/N ≥ 059-0000951
DG-0059#0051	NA	ERYTRA SERVICE MANUAL REVISION JULY 2015	31/JUL/15	SW ≥ 3.4.0
DG-0059#0052	NA	ERYTRA PCB RD IL421 NEW USB CABLE	11/NOV/15	S/N ≥ 059-0000987
DG-0059#0053	NA	ERYTRA NEW PC EMBEDDED (VER.3)	11/NOV/15	S/N ≥ 059-0000987
DG-0059#0054	NA	ERYTRA NEW PC EMBEDDED (VER.3)(US)	1/NOV/15	S/N ≥ 059-0000987
DG-0059#0055	NA	ERYTRA PC EMBEDDED VER. 1 AND VER. 2 RAM MEMORY	11/NOV/15	Erytra PC Embedded Ver. 1 and Ver. 2.
DG-0059#0056	NA	ERYTRA PC EMBEDDED VER. 1 RAM MEMORY (US)	11/NOV/15	Erytra PC Embedded Ver. 1.
DG-0059#0057	NA	ERYTRA NEW HANDHELD BARCODE READER HERON HD3130	22/DEC/2015	S/N ≥ 059-0000995 S/N 059-0000989

Document	Ver.	Title	Date	Applied to
DG-0059#0058	NA	ERYTRA SERVICE MANUAL REVISION MARCH 2016	30/MAR/2016	SW $\geq$ 3.4.2
DG-0059#0059	NA	ERYTRA210401SERVICE MANUAL REVISION MARCH 2016 (US)	06/APR/16	W (US) $\geq$ 1.2
DG-0059#0060	NA	ERYTRA NEW DATABASE CONFIGURATION	13/JUL/2016	S/N $\geq$ 059-0001159
DG-0059#0061	NA	SHOCKWATCH LABEL REMOVED	14/JUL/2016	S/N $\geq$ 059-0001123
DG-0059#0062	NA	ERYTRA NEW REAGENTS STIRRING PROFILE	29/JUL/2016	CEN-SR FW 4.0.0.6, SW 3.4.2, SW (US) 1.2
DG-0059#0063	NA	ERYTRA NEW MIXED REAGENTS RACK HOLDER BEARINGS	23/SEP/2016	S/N $\geq$ 059-0001135
DG-0059#0064	NA	NEW RAM MEMORY FOR ERYTRA PC EMBEDDED VER. 1 AND VER. 2	23/SEP/2016	Erytra PC Embedded Ver. 1 and Ver. 2.
DG-0059#0065	NA	NEW RAM MEMORY FOR ERYTRA PC EMBEDDED VER.1 (US)	23/SEP/16	Erytra PC Embedded Ver. 1.
DG-0059#0066	NA	ERYTRA NEW CAT-CM FIRMWARE	09/DEC/2016	CAT-CM FW 4.0.0.9, SW 3.4.2, SW (US) 1.2
DG-0059#0067	NA	ERYTRA NEW DATABASE CONFIGURATION	22/DEC/2016	SW 3.4.2, S/N $\geq$ 059-0001207
DG-0059#0068	NA	ERYTRA NEW PROBES ADJUSTMENT TOOL	09/MAR/2017	ALL
DG-0059#0069	NA	ERYTRA NEW READER ADJUSTMENT TOOL SUPPORT	09/MAR/2017	ALL
DG-0059#0070	NA	ERYTRA SOFTWARE US VERSION 1.3(US)	09/MAR/17	SW (US) 1.3 S/N $\geq$ 059-0001255
DG-0059#0071	NA	ERYTRA OPERATING SYSTEM SECURITY UPDATE (KB4012212)	17/MAI/2017	sWES7 3.0.0 or 3.1.0 SW 3.4.X

Document	Ver.	Title	Date	Applied to
DG-0059#0072	NA	ERYTRA OPERATING SYSTEM SECURITY UPDATE (KB4012212) (US)	28/MAR/17	WES7 3.0.0 or 3.1.0 SW (US) 1.2
DG-0059#0073	NA	NEW FIREWIRE PCIe v2	29/MAY/2017	Erytra PC Embedded Ver. 3, S/N ≥ 059-0001267
DG-0059#0074	NA	NEW FIREWIRE PCIe v2 (US)	17/MAY/17	Erytra PC Embedded Ver.3
DG-0059#0075	NA	Cancelled. Substituted by DG-0059#0078.	28/JUN/2017	NA
DG-0059#0076	NA	Cancelled. Substituted by DG-0059#0081.	17/JUL/2017	NA
DG-0059#0077	NA	ERYTRA SERVICE MANUAL REVISION JUNE 2017	17/JUL/2017	SW ≥ 4.0.0
DG-0059#0078	NA	NEW BARCODE READER v2	20/SEP/2017	S/N ≥ 059-0001295 (CAT module)
DG-0059#0079	NA	NEW BARCODE READER v2 (US)	NA	S/N ≥ 059-0001295 (CAT module)
DG-0059#0080	NA	NEW DYE FOR ERYTRA DIAGNOSTIC APPLICATION	26/SEP/2017	ALL
DG-0059#0081	NA	Cancelled. Substituted by DG-0059#0085.	26/SEP/2017	NA
DG-0059#0082	NA	ERYTRA NEW FLUCALLIBRATIONS.TXT FILE	06/OCT/2017	SW 4.0.0
DG-0059#0083	NA	ERYTRA NEW FLUCALLIBRATIONS.TXT FILE (US)	26/SEP/17	SW 4.0.0 S/N ≥ 059-0001360
DG-0059#0084	NA	ERYTRA 210401 SERVICE MANUAL REVISION NOVEMBER (US)	NA	SW ≥ 4.0.0
DG-0059#0085	NA	ERYTRA SOFTWARE VERSION 4.0.0	20/DEC/2017	ALL except units under CFDA
DG-0059#0086	NA	ERYTRA NEW USER TRAY V3	23/MAR/2018	S/N ≥ 059-0001363

Document	Ver.	Title	Date	Applied to
DG-0059#0087	NA	ERYTRA SERVICE MANUAL REVISION MARCH 2018	23/MAR/2018	SW ≥ 4.1.0
DG-0059#0088	NA	ERYTRA SOFTWARE VERSION 4.1.0	23/MAR/2018	SW v 4.1.0, Not allowed in units under CFDA, S/N ≥ 059-0001391
DG-0059#0089	NA	ERYTRA SOFTWARE VERSION 4.1.0(US)	NA	SW 4.1.0 S/N >059-0001358
DG-0059#0090	NA	ERYTRA SERVICE MANUAL REVISION MAY 2018	10/MAY/2018	SW ≥ 4.1.0
DG-0059#0091	NA	ERYTRA SERVICE MANUAL REVISION MAY 2018 (US)	27/APR/18	SW ≥4.1.0
DG-0059#0092	NA	NEW FIREWIRE PCIe v3	27/JUL/2018	S/N ≥ 059-0001435
DG-0059#0093	NA	NEW RAM MEMORY FOR ERYTRA PC EMBEDDED VER. 3, VER. 4 AND VER. 5	27/JUL/2018	S/N ≥ 059-0001431
DG-0059#0094	1	HARDWARE MODIFICATIONS IN READER MODULE	12/NOV/2018	S/N ≥ 059-0001451
DG-0059#0095	1	FLUIDIC VERIFICATIONS UPDATE	11/DEC/2018	S/N ≥ 059-0001455
DG-0059#0096	1	ERYTRA NEW READER ADJUSTMENT TOOL - PATTERN	11/MAR/2019	ALL
DG-0059#0097	1	ERYTRA NEW CAT-CM FIRMWARE VERSION 4.2.0.1	07/JUN/2019	CAT-CM FW 4.2.0.1
INR-0059#0098	1	LASER PRODUCT LABEL MODIFICATION	06/SEP/2019	S/N ≥ 059-0001483
INR-0059#0099	1	NEW RAM MEMORY 4GB V2 FOR ERYTRA PC KONTRON	06/SEP/2019	ErytraPC Embedded Ver. 1 and Ver. 2.
INR-0059#0100	1	NEW DIANASERVER V10.1.0.9	04/OCT/2019	SW 4.1.0 or 4.1.1
INR-0059#0101	1	ERYTRA NEW TOUCHSCREEN V3	22/OCT/2019	S/N ≥ 059-0001515
INR-0059#0102	1	NEW IH DIAGNOSTIC KIT	22/JAN/2020	ALL

Document	Ver.	Title	Date	Applied to
INR-0059#0103	1	ERYTRA NEW PCA RD-IL421 V3	10/FEB/2020	S/N ≥ 059-0001537
INR-0059#0104	NA	Cancelled. Substituted by FSB-0059#0031.	06/MAR/2020	NA
INR-0059#0105	1	ERYTRA SERVICE MANUAL REVISION FEBRUARY 2020	19/MAR/2020	SW ≥ 4.2.0
INR-0059#0106	1	NEW CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL	31/MAR/2020	ALL
INR-0059#0107	1	NEW FIREWIRE CABLE ASSEMBLY PROCEDURE	15/APR/2020	SN ≥ 059-0001547
INR-0059#0108	1	NEW CENTRIFUGE CARD HOLDERS	21/MAY/2020	SN ≥ 059-0001559
INR-0059#0109	2	NEW HANDHELD BARCODE READER QUICKSCANTM LITE DATALOGIC QW2420	20/JUL/2020	SN ≥ 059-0001575
INR-0059#0110	3	ERYTRA SOFTWARE VERSION 4.3.0	18/SEP/2020	SN ≥ 059-0001603
INR-0059#0111	1	NEW ELECTRONICS CABINET	18/SEP/2020	SN ≥ 059-0001583
INR-0059#0112	1	NEW DISTRIBUTION BLOCKS BK2 AND BK3	09/OCT/2020	SN ≥ 059-0001595
INR-0059#0113	1	CAMERA COVER AND READER SUPPORT MODIFICATIONS	09/OCT/2020	SN ≥ 059-0001605
INR-0059#0114	1	ERYTRA SERVICE MANUAL REVISION NOVEMBER 2020	19/NOV/2020	SW ≥ 4.3.0
INR-0059#0115	2	ALTERNATIVE BROWSER FOR EDM AND REMOTE CONNECTION	01/SEP/2021	SW ≥ 4.2.0
INR-0059#0116	1	INTERNET ACCESS LIMITATION FOR INTERNET EXPLORER BROWSER	21/DEC/2020	SW 4.3.0
INR-0059#0117	2	ERYTRA SOFTWARE VERSION 4.3.1	22/FEB/2021	SN ≥ 059-0001679, SW 4.3.1 Not allowed in units under CFDA

Document	Ver.	Title	Date	Applied to
INR-0059#0118	1	NEW DILUTION CUP STRUCTURE	22/FEB/2021	SN ≥ 059-0001671
INR-0059#0119	1	NEW USER TRAY V4	01/MAR/2021	SN ≥ 059-0001675
INR-0059#0120	1	NEW INC-CUP FIRMWARE v4.4.0	08/MAR/2021	SN ≥ 059-0001683
INR-0059#0121	2	NEW SCRIPT FOR INCREMENTAL BACKUP CONFIGURATION	15/MAR/2021	SW 4.3.0 / 4.3.1
INR-0059#0122	1	NEW PEDIATRIC SAMPLES HOLDER	15/MAR/2021	ALL
INR-0059#0123	1	NEW SAMPLES HOLDER FOR TUBES > 16 mm	18/MAR/2021	ALL
INR-0059#0124	1	NEW EXTERNAL BARCODE READER HOLDER	18/JUN/2021	SN ≥ 059-0001711
INR-0059#0125	2	ACCESS TO WINDOWS 10 FILE EXPLORER	02/SEP/2021	SW ≥ 4.3.0
INR-0059#0126	1	INTERNAL AC POWER CABLING COLOR CODING STANDARDIZATION	28/SEP/2021	SN ≥ 059-0001731
INR-0059#0127	1	NEW CLAMP FOR FIREWIRE CABLE IN PC BOX	28/SEP/2021	SN ≥ 059-0001731
INR-0059#0128	1	MODIFIED ROUTING FOR TUBE C22	29/OCT/2021	SN ≥ 059-0001735
INR-0059#0129	1	NEW CENTRIFUGE ROTOR	29/OCT/2021	SN ≥ 059-0001739
INR-0059#0130	2	NEW ELECTRIC LOCK v2	09/JAN/2023	SN ≥ 059-0001740
INR-0059#0131	1	NEW RAM MEMORY (ASROCK PC)	12/NOV/2021	SN ≥ 059-0001747 Spare part 733902 lot ≥ 21001 Spare part 733903 lot ≥ 21003
INR-0059#0132	1	SECURITY UPDATES V1.2.0 FOR WINDOWS 10 ENTERPRISE 2019 LTSC v2.1.0	09/DEC/2021	NA

Document	Ver.	Title	Date	Applied to
INR-0059#0133	4	ERYTRA SOFTWARE VERSION 4.3.2	25/MAR/2022	SN ≥ 059-0001783
INR-0059#0134	1	NEW DILUTION STATION CUP v2	03/JAN/2022	SN ≥ 059-0001751
INR-0059#0135	1	NEW SUPPORT PACK GENERATOR	21/JAN/2022	SW ≥ 4.3.1
INR-0059#0136	2	NEW PRESSURE SENSORS	23/SEP/2022	SN 059-0001955 to 059-0001957 and SN ≥ 059-0001959
INR-0059#0137	1	NEW SCREEN DRAWER ASSEMBLY	13/JAN/2022	SN ≥ 059-0001767
INR-0059#0138	1	FLUIDIC DRAWER REDESIGN	16/FEB/2022	SN ≥ 059-0001767
INR-0059#0139	1	NEW CEN-SRM FIRMWARE v4.5.0	23/FEB/2022	SN ≥ 059-0001759
INR-0059#0140	1	NEW PC MOTHERBOARD v3	21/MAR/2022	SN ≥ 059-0001775
INR-0059#0141	1	NEW GEAR DOUBLE WHEEL	21/MAR/2022	SN ≥ 059-0001775
INR-0059#0142	1	NEW CAMERA SUPPORT	27/APR/2022	SN ≥ 059-0001785
INR-0059#0143	1	ERYTRA NEW IQ REVISION MARCH 2022	24/MAR/2022	All new Erytra units' installation
INR-0059#0144	2	ERYTRA SOFTWARE VERSION 4.3.3	16/SEP/2022	All not under NMPA
INR-0059#0145	1	NEW HANDHELD BARCODE READER QUICKSCAN DATALOGIC QW2520 BKK1S	07/SEP/2022	S/N ≥ 059-0001809
INR-0059#0146	1	NEW LASER LABELLING (NMPA)	12/SEP/2022	S/N ≥ 059-0001809
INR-0059#0147	1	ERYTRA SERVICE MANUAL REVISION JULY 2022	16/SEP/2022	SW ≥ 4.3.2
INR-0059#0148	2	NEW GRIPPER PLATE	01/FEB/2023	S/N ≥ 059-0001835

Document	Ver.	Title	Date	Applied to
INR-0059#0149	1	NEW FLUIDIC PROTECTOR PLATES	21/MAR/2023	S/N ≥ 059-0001851
INR-0059#0150	1	NEW SPACER FOR LOAD CELL	21/MAR/2023	S/N ≥ 059-0001855
INR-0059#0151	1	UPPER DOOR SENSOR ASSEMBLY IMPROVEMENT	28/MAR/2023	S/N ≥ 059-0001859
INR-0059#0152	1	SERVICE DOCUMENTATION UPDATE, REVISION APRIL 2023	18/MAI/2023	SW ≥ 4.4.0
INR-0059#0153	1	SYRINGE LOCK REMOVED	14/JUN/2023	S/N ≥ 059-0001887
INR-0059#0154	2	ERYTRA SOFTWARE VERSION 4.4.1	27/JUL/2023	All not under NMPA S/N ≥ 059-0001891
INR-0059#0155	1	NEW PIP HORNS WALLS	31/OCT/2023	All not under NMPA S/N ≥ 059-0001887
INR-0059#0156	1	SOFTWARE NOT PRE-INSTALLED FROM FACTORY	03/NOV/2023	All not under NMPA S/N ≥ 059-0001903
INR-0059#0157	1	NEW PRE-ANALYTICAL COMPATIBLE SAMPLES HOLDER	15/NOV/23	SW ≥ 4.5.0 (except SW 5.1.0).
INR-0059#0158	1	MK VALVE v2	12/DEC/2023	S/N ≥ 059-0001911
INR-0059#0159	1	NEW CAPACITIVE TOUCHSCREEN	28/FEB/2024	S/N ≥ 059-0001911
INR-0059#0160	1	ERYTRA SERVICE MANUAL REVISION MARCH 2024	12/MAR/2024	SW ≥ 4.5.0
INR-0059#0161	1	NEW PREVENTIVE MAINTENANCE PLAN	03/MAI/2024	All Erytra units
INR-0059#0162	1	DDR3 2GB SO DIMM KINGMAX RAM obsolescence	11/MAR/2025	S/N ≥ 059-0002047
INR-0059#0163	1	NEW PROBE MATERIAL	16/MAI/2025	S/N ≥ 059-0002055

7.9 MTB-0059#0001 Erytra Software version 4.5.1 (Patch 1 for version 4.5.0)

TYPE	MANUFACTURER TECHNICAL BULLETIN
COMMUNICATION	DXBU-PDS-TB-000012 Located in RelatedDocuments > Erytra > MTBs.
REASON	This is a patch for Erytra software version 4.5.0 fixing complaint 800020269 "Incorrect result interpretation sent"
SCOPE	All Erytra instruments running under software version 4.5.0 that are not under National Medical Products Administration (NMPA) regulation (China).
CHARACTER	MANDATORY WITH URGENT DEPLOYMENT: The installation of the new software version 4.5.1 is mandatory for all Erytra running under v4.5.0.
REGULATORY IMPACT	Erytra software version 4.5.1 can be installed in all the countries where Erytra is registered except China.

REQUIREMENTS

Erytra software version 4.5.1 requires the following Reagents Pack versions or higher or a Reagents Pack generated from IH Templates Pack (these packs contain stability on-board information):

- **CE configuration with stability on-board feature enabled:** IH Reagents Pack v9.0 or higher, or IH Reagents Pack AU v2.0 or higher (these packs contain stability on-board information for the CE region).
- **CE configuration with stability on-board feature disabled:** IH Reagents Pack v9.0 or higher, or IH Reagents pack AU v2.4.
- **US configuration with stability on-board feature enabled:** IH Reagents Pack US v2.4 and IH Templates pack v12 (these packs contain stability on-board information for the US region).
- **US configuration with stability on-board feature disabled:** IH Reagents pack US v2.4 or higher (this pack and the IH Templates Pack v11 do not contain stability on-board information for the US region).

**NOTE**

Only Serial Titration techniques using the 1/1-1/2048 dilution range (i.e., starting with a 1/1 dilution) are can be used with Erytra software version 4.5.1.

Other Serial Titration ranges, as well as single-point titration techniques, are not allowed with this software version.

**NOTE**

Operating System updates or re-installation of OS version 2.2.1 or higher can be executed remotely and without the need of a USB device.

PROCEDURE/ DOCUMENT

- See "Erytra v4.5.1 installing steps in units running v4.5.0" (new).

7.9.1 CHANGE DESCRIPTION

Erytra Software v4.5.1 located in **Software changes section**.

7.9.2 VERSION INSTALLATION

Table 119 - Version files

Erytra software version v4.5.1		
Erytra software installer	Installer	ERYTRA-4.5.1.zip
		SHA-256: 8697c2bc48d45d9879c1627e4cc15f79fd7c9ec4666be77030889346b3994a3d

Download the files needed to install Erytra software version 4.5.1. See Downloading Erytra software files chapter.

7.9.3 OS & SW INSTALLATION

See "Installing Erytra software version 4.5.1"

7.9.4 VERSION VERIFICATION

In order to verify Erytra software version 4.5.1 installation

- 1. Switch on the instrument.
- 2. Select the **Information** (i) option from the boot application. See "Working with Boot Application" on page 88



Figure 903 - Software information

Check that Erytra software version is 4.5.1. Should this not be the case, reinstall the software from "MTB-0059#0001 Erytra Software version 4.5.1 (Patch 1 for version 4.5.0)" on page 1198

- 3. Close the information window.

7.10 MTB-0059#0002 Reader module foam removal

TYPE	MANUFACTURER TECHNICAL BULLETIN
COMMUNICATION	DXBU-PDS-TB-000004 Located in RelatedDocuments > Erytra > MTBs .
REASON	Improvement to prevent a card from getting stuck on the foam of the reader module visor while being read, causing a 'movement error' in DG Reader Net instruments.
SCOPE	Implemented in factory from s/n ≥ 059-0002085.

CHARACTER	INFORMATIVE
REGULATORY IMPACT	NA
REQUIREMENTS	NA
P/N UPDATE	CREATED: • NA MODIFIED: • 735851 READER ILLUMINATION SYSTEM v4 CANCELED: • 233943 READER FOAM COMPATIBILITY: • New P/N are compatible with all Erytra units.
PROCEDURE	NA

7.10.1 CHANGE DESCRIPTION

The foam stripe has been removed and its support plate has been simplified by removing the horizontal bend. Also the support screws have been changed.

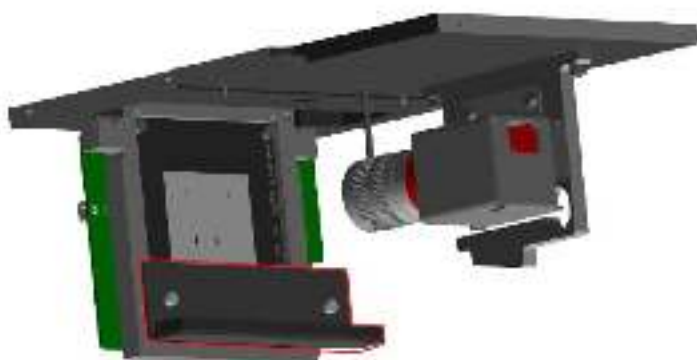


Figure 904 - Foam and support in previous version

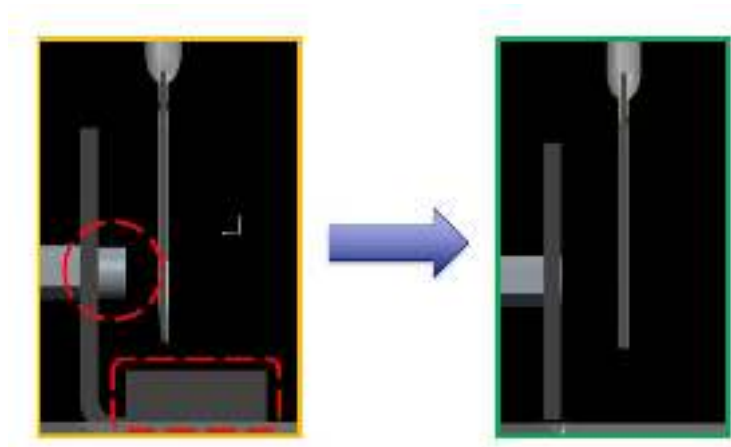


Figure 905 - Foam removal and support plate modification

7.11 MTB-0059#0003 Erytra Software version 4.6.1 (Patch 1 for version 4.6.0)

TYPE	MANUFACTURER TECHNICAL BULLETIN
COMMUNICATION	DXBU-PDS-TB-000002 Located in RelatedDocuments > Erytra > MTBs .
REASON	This is a patch for Erytra software version 4.6.0 fixing several complaints including 800020269 "Incorrect result interpretation sent", 800016371 "Black screen when printing" and 800016541 "Undetected leakage in Erytra".
SCOPE	All Erytra instruments that are not under National Medical Products Administration (NMPA) regulation (China).
CHARACTER	MANDATORY with URGENT DEPLOYMENT: The installation of the new software version v4.6.1 is mandatory with urgent deployment for all Erytra running under software version 4.6.0. It is mandatory for new installations, upgrades, and new customers. Also, it is highly recommended for existing installations and customers to transition from current versions to v4.6.1.
REGULATORY IMPACT	Erytra software version 4.6.1 can be installed in all the countries where Erytra is registered except China.

REQUIREMENTS

Erytra software version 4.6.1 requires the following Reagents Pack versions or higher or a Reagents Pack generated from IH Templates Pack (these packs contain stability on-board information):

- **CE configuration:** IH Reagents Pack v9.0 or higher, or IH Reagents Pack AU v2.0 or higher
- **US configuration:** IH Reagents Pack US v2.4 or higher, or IH Reagents Pack v12.0 or higher (that is, a IH Reagents Pack generated from IH Templates pack v12.0 or higher)

Erytra software version 4.6.1 installation in instrument running under software versions < 4.5.0 cannot be executed using Grifols Secure Remote Connectivity tool as it involves Operating System Full Recovery process from an Operating System version < 2.2.1.

**NOTE**

Erytra units with S/N ≥ 059-0001903 are delivered with Operating System version 2.2.1 or higher and with no software version pre-installed.

**NOTE**

Operating System updates or re-installation of OS version 2.2.1 or higher can be executed remotely and without the need of a USB device.

The update of the alternative browser is required if remote connection to Erytra user interface was already established, and previous software version was < 4.5.0.

PROCEDURE/ DOCUMENT

- See "Installing steps in units with OS version v2.3.3 (or v2.3.1+ SU v1.7 or v2.3.2+SU v1.7)" on page 1042 (new).
- See "Installing steps in units with OS version ≥ 2.2.0 and OS ≠ 2.3.3 (or 2.3.1+SU, or 2.3.2+SU)" on page 1045 (modified)
- See "Installing steps in units with OS version < 2.2.0" on page 1054 (modified)
- See "OS Firewall: close a port" (new)
- See "horns: pre-analytical adjustment" (new)
- See "Pre-analytical laser alignment check" (new)

7.11.1 CHANGE DESCRIPTION

Erytra Software v4.6.1 located in **Software changes section**.

7.11.2 VERSION INSTALLATION

Table 120 - Version files

Erytra software version v4.6.1		
Erytra software installer	Installer	ERYTRA-4.6.1.zip
		SHA-256: 1b535438fe41c57d200a7f50c1a8e343ba2c4cad0e59b4b8402b9abfeaa36f50
Windows 10 Enterprise 2019 LTSC version 2.3.3 (SU march 2025)	Installer	w10-1809-2.3.3.zip
		SHA-256: 24e2989955bd0ad059e1919bbc064183760782a0efd61494e4854b0dbf4f2ac4
		imb-150d-drivers-1.1.0.jar
Remote Connection	Installer	RemoteConnection.zip
		SHA-256: 34072a1035d55d67e2cefdda2afa3930237ed1eb6c282c1f7a5a3375f4d1740f

Download the files needed to install Erytra software version 4.6.1 and Windows 10 Enterprise 2019 LTSC version 2.3.2. See Downloading Erytra software files chapter.

7.11.3 OS & SW INSTALLATION

See "Installing Erytra software v4.6.1"

7.11.4 VERSION VERIFICATION

In order to verify the installation, check the following:

1. Verify OS and SW installation. See "Verifying OS and SW versions" on page 1097.
2. Verify OS licence activation. See "Verifying Licence activation" on page 1038
3. Verify FW versions according to the table. Check firmware version

7.12 MTB-0059#0004 Erytra Software version 5.2.1 (Patch 1 for version 5.2.0)

TYPE	MANUFACTURER TECHNICAL BULLETIN
COMMUNICATION	DXBU-PDS-TB-000003 Located in RelatedDocuments > Erytra > MTBs .
REASON	This is a patch for Erytra software version 5.2.0 fixing several complaints including 800020269 "Incorrect result interpretation sent", 800016371 "Black screen when printing" and 800016541 "Undetected leakage in Erytra".
SCOPE	All Erytra instruments under National Medical Products Administration (NMPA) regulation (China).
CHARACTER	MANDATORY with URGENT DEPLOYMENT: The installation of the new software version v5.2.1 is mandatory with urgent deployment for all Erytra running under software version 5.2.0. It is mandatory for new installations, upgrades, and new customers. Also, it is highly recommended for existing installations and customers to transition from current versions to v5.2.1.
REGULATORY IMPACT	Erytra software version 5.2.1 can only be installed in China.
REQUIREMENTS	Erytra software version 5.2.1 requires the following Reagents Pack versions or higher or a Reagents Pack generated from IH Templates Pack: • CE configuration: IH Reagents Pack v9.0 or higher  NOTE Erytra units with S/N ≥ 059-0001903 are delivered with Operating System version 2.2.1 or higher and with no software version pre-installed.  NOTE Operating System updates or re-installation of OS version 2.2.1 or higher can be executed remotely and without the need of a USB device. The update of the alternative browser is required if remote connection to Erytra user interface was already established, and previous software version was < 5.2.0.
PROCEDURE/ DOCUMENT	• See "Installing steps in units with OS version v2.3.3 (or v2.3.1+ SU v1.7 or v2.3.2+SU v1.7)" (new) • See "Installing steps in units with OS version ≥ 2.2.0 and OS ≠ 2.3.3 (or 2.3.1+SU, or 2.3.2+SU)" (modified) • See "Installing steps in units with OS version < 2.2.0" (modified)

7.12.1 CHANGE DESCRIPTION

Erytra Software v5.2.1 located in **Software changes section**.

7.12.2 VERSION INSTALLATION

Table 121 - Version files

Erytra software version v5.2.1		
Erytra software installer	Installer	ERYTRA-5.2.1.zip
		SHA-256: 82805b9013cd8ca831a7cc80a3f2583cde304b9d6e12a4fb185c8a4081a3a5dc
Windows 10 Enterprise 2019 LTSC version 2.3.3 (SU march 2025)	Installer	w10-1809-2.3.3.zip
		SHA-256: 24e2989955bd0ad059e1919bbc064183760782a0efd61494e4854b0dbf4f2ac4
		imb-150d-drivers-1.1.0.jar
Remote Connection	Installer	SHA-256: 5d42eba4d66ab8ce65c197383f2d4098b26ff0dc6d909bd024fe376462219624
		RemoteConnection.zip
		SHA-256: 34072a1035d55d67e2cefd2a2afa3930237ed1eb6c282c1f7a5a3375f4d1740f

Download the files needed to install Erytra software version 5.2.1 and Windows 10 Enterprise 2019 LTSC version 2.3.3. See Downloading Erytra software version chapter.

OS & SW INSTALLATION: See "Installing Erytra software v5.2.1"

VERSION VERIFICATION: See "Verifying SW version 5.2.1".

8 Spare parts list

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Tools	735569	BOTTLE HOLDER ADJUSTMENT TOOL	BOTTLE HOLDER ADJUSTMENT TOOL	3066630							X				
Computer	732539	WIFI DONGLE	WIFI DONGLE	3056782									X		
Wiring	735719	WIRING KIT MINI-ITX ASROCK IMB-1003J	WIRING KIT MINI-ITX ASROCK IMB-1003J	3065747 3017611 3065748 3065749 8010016									X		
Wiring	735720	GENERAL CABLING V4	GENERAL CABLING V4	6020663 3064048 3065749 3016266 3017611 8010016					YES				X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	735911	ASPIRATION PUMP A (VPA)	ASPIRATION PUMP A (VPA)	6020329	1	1	1	1	YES		X				
Fluidics	735910	ASPIRATION PUMP B (VPB)	ASPIRATION PUMP B (VPB)	6020328	1	1	1	1	YES		X				
Fluidics	735097	POLYURETANE TUBE 6x4 (1m.)	POLYURETANE TUBE 6x4 (1m.)	7002830							X				
Mechanics	734715	CAT ROBOTIC HEAD LEFT COVER	CAT ROBOTIC HEAD LEFT COVER	6012146 8010015x4 8016810x3							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Complementary Material	213796	DG CLEAN RACK ADAPTOR	DG CLEAN ERYTRA RACK ADAPTOR	3039971							X	X			
Complementary Material	210603	ERYTRA EFLEXIS® CARD WASTE CONTAINER	ERYTRA EFLEXIS® CARD WASTE CONTAINER	3042702x5 3024023x5							X				
Complementary Material	210602	ERYTRA EFLEXIS® RACK FOR REAGENTS	ERYTRA EFLEXIS® RACK FOR REAGENTS	6012377							X				
Complementary Material	210601	ERYTRA EFLEXIS® RACK FOR SAMPLES	ERYTRA EFLEXIS® RACK FOR SAMPLES	6012379							X				
Complementary Material	213713	SAMPLES HOLDER	SAMPLES HOLDER	6008854 3034449x2							X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle x is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Compleme ntary Material	210715	SAMPLES HOLDER >16 mm	SAMPLES HOLDER >16 mm	6019139x2 3034449x4							X	X			
Compleme ntary Material	210714	PEDIATRIC SAMPLES HOLDER	PEDIATRIC SAMPLES HOLDER	6019090x2 3033207							X	X			
Compleme ntary Material	210403	WASTE CONTAINER	WASTE CONTAINER	3026421 3031280x2							X	X			
Computer	233771	TOUCHSCREE N COVER	TOUCHSCREEN COVER	3028821 3034004								X			
Computer	732916	TOUCHSCREE N MODULE v2	TOUCHSCREEN MODULE v2	6020080 3056906 8010015x4					YES				X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle x is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Computer	735521	TOUCHSCREE N V3	TOUCHSCREEN V3	3065294 3034004 7006478					YES		X				
Computer	734235	MOTHERBOAR D v3 (MINI-ITX ASROCK IBM-150)	MOTHERBOARD v3 (MINI-ITX ASROCK IBM-150)	3061224					YES			X			
Computer	233990	PC EMBEDDED (Ver. 3) INTERNAL CABLE SET	PC EMBEDDED V3 INTERNAL CABLE SET	6012459					YES			X			
Computer	735899	RAM DDR3L 8GB SO-DIMM	RAM DDR3L 8GB SO-DIMM	3068143					YES		X	X	X		
Computer	736169	RAM DDR4 8GB SO-DIMM KINGMAX (2 units)v2	RAM DDR4 8GB SO-DIMM KINGMAX (2 units)v2	3068991 x2							X		X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Computer	//734234	//DG READERNET PC MOTHERBOARD v2	//DG READERNET PC MOTHERBOARD v2	3061224					YES						
Computer	231370	FUSE T2A (10 units)	DG READER FUSE T2A (10 units)	7006371x10									X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Computer	734233	PC EMBEDDED (Ver. 7, W10 ENT. 2019 LTSC)	PC EMBEDDED (Ver. 7, W10 ENT. 2019 LTSC)	3061224					YES			X			
				3057820											
				3061225x											
				2											
				3043071											
				3052990											
				6012459											
				6012464											
				3043273											
				3016326x											
				2											
				8013500x											
				2											
				8013395x											
				6											
				3053208											
				8016810x											
				4											
				7006347x											
				4											
				7006734											

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Computer	735636	MOTHERBOARD MINI-ITX ASROCK IMB-1003J	MOTHERBOARD MINI-ITX ASROCK IMB-1003J	3065754							X		X		
Computer	735520	TOUCHSCREEN V4	TOUCHSCREEN V4	3065294					YES			X			
Computer	230263	WIFI ANTENNA	WIFI ANTENNA	3047364					YES				X		
Computer	732912	WIFI MODULE v2	WIFI MODULE v2	3056511					YES				X		
Computer	233987	SSD 256 GB mSATA	SSD 256GB mSATA	3043071					YES		X	X	X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Computer	735634	SSD 256GB M.2	SSD 256GB M.2	3065751					YES	IMAGE	X		X		
Computer	735634	SSD 256GB M.2	SSD 256GB M.2	3065751					YES		X		X		
Computer	731695	FIREWIRE PCIe v3	FIREWIRE PCIe v3	3052990					YES		X	X	X		
Computer	731705	FIREWIRE TO PC ADAPTOR 511 V2	FIREWIRE TO PC ADAPTOR 511 V2	3053207 8010016x2 8013500							X				
Computer	731704	FIREWIRE TO PC ADAPTOR 59 v2	FIREWIRE TO PC ADAPTOR 59 V2	3053208 7006347x4								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Computer	//730561	//FIREWIRE TO PC ADAPTOR 59	//FIREWIRE TO PC ADAPTOR 59	3047416 8016810x2 8013395x2											
Electronics	234139	PCA PIP-TOWB511	PCA PIP-TOWB511	6010702					YES		X				
Electronics	234162	PCA CAT-TOW511	PCA CAT-TOW511	6009997		0			YES		X				
Electronics	734459	PCA ELE-BCKPL511 v2	PCA ELE-BCKPL511 v2	6019695					YES		X				
Electronics	234138	PCA PIP-TOWA511	PCA PIP-TOWA511	6010668					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	234126	PCA PIP-OSC511	PCA PIP-OSC511	6010670	1	1	1		YES		X				
Electronics	734998	PCA PIP-BCR511 v2	PCA PIP-BCR511 v2	6019814 3062849 8010000 8010015					YES		X				
Electronics	234170	PCA ELE-CON511	PCA ELE-CON511	6010819					YES		X				
Electronics	234169	PCA GEN-STDBY511	PCA GEN-STDBY511	6012345					YES		X	X			
Electronics	234196	PCA ELE-FRNT511	PCA ELE-FRNT511	6010920					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	235286	PCA ENM-LCK510	PCA ENM-LCK510	6009161 7006352x2					YES		X				
Electronics	234197	PCA FLU-CON511	PCA FLU-CON511	6010666					YES		X				
Electronics	233956	PCA FLU-CTRL59 (Ver. 2)	PCA FLU-CTRL59 (Ver. 2)	6005456					YES			X			
Electronics	234186	PCA FLU-CTRL511	PCA FLU-CTRL511	6009920					YES		X				
Electronics	234102	PCA INC-CTRL511	PCA INC-CTRL511	6009921					YES		X				
Electronics	731379	PCA HOME512	PCA HOME512	6013158					YES				X		
Electronics	231883	PCA GRIPPER	PCA GRIPPER	3015656					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	233937	PCA GEN-BCKPL59 (Ver. 2)	PCA GEN-BCKPL59 (Ver. 2)	6018157					YES			X			
Electronics	234117	PCA GEN-HOME511	PCA GEN-HOME511	6010145					YES		X				
Electronics	234112	PCA PIP-BCKPL511	PCA PIP-BCKPL511	6010667					YES		X				
Electronics	233964	KEYBOARD REAG/SAMPLE DRAWER VER.2	KEYBOARD REAG/SAMPLE DRAWER VER.2	3040141 8013501x4					YES			X			
Electronics	234194	MICRO SWITCH WASTE TRAY SENSOR	MICRO SWITCH WASTE TRAY SENSOR	3042691					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eyt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Electronics	234156	PCA CAT-CON511	PCA CAT-CON511	6009996					YES		X				
Electronics	233806	FUSE MSF 0,8 A (10 units)	FUSE MSF 0,8A (10 units)	3030684x10								X			
Electronics	233805	FUSE MST 1 A (10 units)	FUSE MST 1A (10 units)	3030683x10								X			
Electronics	233803	FUSE MST 4 A (10 units)	FUSE MST 4A (10 units)	3030681x10								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Electronics	233804	FUSE MST 3,15 A (10 units)	FUSE MST 3,15A (10 units)	3030682x10								X			
Electronics	233367	FUSE 6.3 A 5x20 (10 units)	FUSE SCHURTER MSF 6.3A (10 UNITS)	3021843x10								X			
Electronics	233930	FAN ELECTRONICS CABINET	FAN ELECTRONICS CABINET	3026535					YES			X			
Electronics	234166	FAN	FAN	6012353					YES		X		X		
Electronics	234103	FAN (INC)	FAN	6012285					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle x is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Electronics	233941	JUMPER 100Ohm for PCA GEN-STDBY JP3 (5u)	JUMPER 100Ohm for PCA GEN-STDBY JP3 (5u)	3037767x5								X			
Electronics	234202	KEYBOARD	KEYBOARD	6012138					YES		X				
Electronics	234173	ELECTRICAL FILTER	ELECTRICAL FILTER	3043113							X		X		
Electronics	734238	ELECTRIC LOCK v2	ELECTRIC LOCK v2	3061252 3018918	2	2		2	YES		X	X			
Electronics	233965	KEYBOARD CARDS DRAWER VER.2	KEYBOARD CARDS DRAWER VER.2	3040142 8013501x4					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	731393	BARCODE READER	LINEAR AND 2D BARCODE READER	3048640					YES				X		
Electronics	235274	LOAD CELL	LOAD CELL	3025931					YES		X				
Electronics	233736	LOAD CELL	LOAD CELL	3025931 3024571 8010055x2					YES			X			
Electronics	233807	DOOR SENSOR	DOOR SENSOR	3026354 3026636 8010016x2								X			

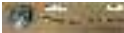
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	731981	CONTAINER PRESENCE DETECTION SYSTEM (1 u)	CONTAINER PRESENCE DETECTION SYSTEM (1u)	6016690			1		YES		X				
Electronics	234111	PCA CAT-BCKPL511	PCA CAT-BCKPL511	6009995					YES		X				
Electronics	234159	PCA CAT-GRP511	PCA CAT-GRP511	6009998					YES		X				
Electronics	234133	PCA PIP-TREAD511	PCA PIP-TREAD511	6012595					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	234118	PCA PIP-CON511	PCA PIP-CON511	6010703					YES		X				
Electronics	234149	PCA SRE-BCKPL511	PCA SRE-BCKPL511	6009896					YES		X				
Electronics	//233751	//PCA PIP-P2OSC59	PCB PIP-P20SC59	3026177					YES			X			
Electronics	233753	PCA PIP-TOW59	PCB PIP-TOW59	6005526					YES			X			
Electronics	233747	PCA RM-CON59	PCB RM-CON59	6007820					YES			X			
Electronics	233713	PCA RM-ST59	PCB RM-ST59	6005770					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	233907	PELTIER MODULE (Ver. 2)	PELTIER MODULE (Ver.2)	3034423					YES		X	X			
Electronics	233756	POWER SUPPLY	POWER SUPPLY	3026607					YES			X			
Electronics	234167	POWER SUPPLY 24V	POWER SUPPLY 24V	3043111					YES		X	X			
Electronics	231806	SOLENOID GRIPPER	SOLENOID GRIPPER LUCAS	7006355x2 3015342 3015814	1	1		1	YES			X			
Electronics	233745	PRESENCE DETECTOR	PRESENCE DETECTOR	3025702	1	1			YES		X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	734608	SOLENOID GRIPPER EFLEXIS	SOLENOID GRIPPER EFLEXIS	7006355x2 3015342 3015814				1	YES		X				
Electronics	731389	BARCODE READER	BARCODE READER	6014092					YES				X		
Electronics	732911	PCA RD-IL421 v3	PCA RD-IL421 v3	6017969			1		YES		X	X	X		
Electronics	734657	BARCODE READER QW2520-BKK1S (W.USB CABLE)	BARCODE READER QW2520- BKK1S(W.USB CABLE)	3062589 3057519					YES		X	X			

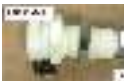
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	730563	BARCODE READER v2	BARCODE READER v2	3046899	1	1	1		YES		X	X			
Electronics	233809	PUSHBUTTON	PUSHBUTTON	3020129					YES		X	X			
Electronics	233895	PCA PIP-P2OSC59 (Impact detector)	PCB PIP-P2OSC59 (Impact detector)	6009119			1		YES			X			
Electronics	233894	PCA PIP-P1OSC59 (Impact detector)	PCB PIP-P1OSC59 (Impact detector)	6009118	1		1		YES			X			
Electronics	234168	POWER SUPPLY 12V	POWER SUPPLY 12V	3043112					YES		X	X	X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	233754	PCA PIP-BCR59	PCB PIP-BCR59	6005590					YES			X			
Electronics	234148	PCA SRE-MOT511	PCA SRE-MOT511	6009895 7008711x2					YES		X				
Electronics	234150	PCA SRE-RCKID511	PCA SRE-RCKID511	6011008					YES		X				
Electronics	733472	PCA VOLTAGE CONVERTER 24V TO 5V	PCA VOLTAGE CONVERTER 24V TO 5V	6012460					YES			X			
Electronics	233742	PCA CAT-CONY59	PCB CAT-CONY59	6005632					YES			X			
Electronics	235206	STEPPER MOTOR	STEPPER MOTOR	6008789x1 7002720x2	1	1	1	1	YES				X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Electronics	233743	PCA CAT-TOW59	PCB CAT-TOW59	6005530					YES			X			
Electronics	233750	PCA PIP-CONY59	PCB PIP-CONY59	6005631					YES			X			
Electronics	233728	PCA CEN-CON59	PCB CENTRIFUGE	6005455					YES		X	X			
Electronics	733698	PCA CUP-ST59 v2	PCB CUP-ST59 v2	6018708 8010027x2			1		YES			X			
Electronics	233772	PCA GEN-CONX59	PCB GEN-CONX59	6005630					YES			X			

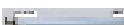
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eyt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Electronics	734038	S/R DRAWER LOCK SYSTEM V3	S/R DRAWER LOCK SYSTEM V3	6018666	1	1	1				X				
Electronics	233808	PCA GEN-FRNT59	PCB GEN-FRNT59	6006500					YES			X			
Electronics	233758	PCA MOTORCONTR OL	PCB MOTORCONTROL	6006090			1		YES		X	X			
Electronics	233702	PCA GEN-LCK59	PCB GEN-LCK59	6005510					YES		X	X			
Electronics	233717	PCA INC-CON59	PCB INC-CON59	6005474					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	233718	PCA INC-BL59	PCB INC-BL59	6005473 3025653					YES		X	X			
Electronics	233810	PCA GEN-TEMP59	PCB GEN-TEMP59	6005914					YES		X	X			
Electronics	233811	PCA GEN-LED59	PCB GEN-LED59	3024684					YES		X	X			
Electronics	233755	PCA GEN-STDBY	PCB GEN-STDBY	3026179					YES			X			
Fluidics	233827	SOLUTION DECONT. BOTTLE (8 litres)	SOLUTION DECONT. BOTTLE (8 Litres)	6006428 3030477 3030478								X			
Fluidics	736237	SAMPLES PROBE (Impact detector) v2	SAMPLES PROBE (Impact detector) v2	3068221	2	2	2	2				X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eyt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Fluidics	733673	TUBE CONNECTION 6x4 tube v2 (25 units)	TUBE CONNECTION 6x4 tube v2 (25 units)	3059488x25							X				
Fluidics	233418	TUBE CONNECTION ADAPTOR PVC 6X4 to 3x1,5	TUBE CONNECTION ADAPTOR PVC 6X4 to 3x1,5	3023502x3				1			X				
Fluidics	233849	TUBE CONNECTION CPC F (bottles)	TUBE CONNECTION CPC F (bottles)	3025572	1	1		1	YES			X			
Fluidics	233848	TUBE CONNECTION CPC F (external waste)	TUBE CONNECTION CPC F (external waste)	3026417				1	YES		X	X			
Fluidics	735501	TUBE CONNECTION CPC GS320 (w.o-ring, 10)	TUBE CONNECTION CPC GS320 (w.o-ring, 10)	3065300x10							X				

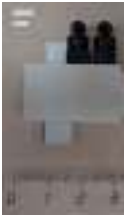
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	735500	TUBE CONNECTION CPC GS520 (w.o-ring, 10)	TUBE CONNECTION CPC GS520 (W.O-RING, 10)	3065299x10							X				
Fluidics	233951	HAMILTON DILUTOR (Ver. 2)	HAMILTON DILUTOR (Ver. 2)	3040300 3040301 3026134 3026161 8016782x2			1		YES		X	X			
Fluidics	233738	SOLUTIONS BOTTLE (8 LITERS)	SOLUTIONS BOTTLE (8 litres)	6006428 3036792x2 3036793x2 3036787 3036788			1					X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	232806	TUBE CONNECTION CPC O-RING (20 units)	TUBE CONNECTION CPC O-RING (20 units)	3030914x20			1					X			
Fluidics	234184	TUBE CONNECTION CPC SMPTD02	TUBE CONNECTION CPC SMPTD02	3033333				1	YES		X				
Fluidics	233876	SILICONE TUBING 3x5 (1 m)	SILICONE TUBING 3x5 (1m)	3034326								X			
Fluidics	233740	SOLUTIONS BOTTLE CAP	SOLUTIONS BOTTLE CAP	3025573 3036787 3036788								X			
Fluidics	733836	SINGLE/DUAL MODE CONNECTOR (10 units)	SINGLE/DUAL MODE CONNECTOR (10 units)	3059492x10							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	734356	CONTAINER COUPLING SYSTEM V3	SOL. CONTAINER COUPLING SYSTEM V3	6017425 3032457 3033333 3042664 3033804 3031559 3032036 8010015 3018725 3018726 3045606 X3 3061110							X				
Fluidics	233920	SOL. CONTAINERS INTERNAL TUBE (w. filter)	SOL. CONTAINER INTERNAL TUBE (w. filter)	3035385 3018726 3018725 3013548 3028747	1	1		1	YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	234190	TUBE CONNECTION (for 2,3x3,6 tube)	TUBE CONNECTION (for 2,3x3,6 tube)	3042635				1			X				
Fluidics	233900	TEFLON TUBE CONNECTION (seal nut+ferrule) (Ver. 2)	TEFLON TUBE CONNECTION (seal nut+fer) V2	3034353 3029872				1	YES		X	X			
Fluidics	735515	SINGLE/DUAL MODE CONN. (MANIFOLD, 10 u.)	SINGLE/DUAL MODE CONN. (MANIFOLD, 10 u.)	3065322 x10					YES		X				
Fluidics	233975	PROBES HOLDER (Impact detector) (Ver. 2)	PROBES HOLDER (Impact detector) (Ver. 2)	6012021 3032861 3042351				1	YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle x is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Fluidics	233845	TUBE CONNECT. STRAIGHT (for 2,3x3,6 tube)	TUBE CONNECT.STRAIGHT (for 2,3x3,6 tube)	3028964				1			X	X			
Fluidics	234185	TUBE CONNECTION CPC SMPTD02BLK	TUBE CONNECTION CPC SMPTD02BLK	3036129				1	YES		X				
Fluidics	734357	POKA-YOKE LOCK (10 units)	POKA-YOKE LOCK (10 units)	3061110x10							X				
Fluidics	733519	O-RING 5.28x1.78 (10 units)	O-RING 5.28x1.78 (10 units)	3008999x10							X	X			
Fluidics	230144	O-RING 3x1 mm (1 unit)	O-RING 3x1 mm	3012942					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Fluidics	233798	MULTILUMEN TUBING 2,3x3,6 (10 m)	MULTILUMEN TUBING 2,3x3,6 (10m)	3024993							X	X			
Fluidics	234210	PRESSURE SENSOR (S3,S4,S5,S6)	PRESSURE SENSOR (S3,S4,S5,S6)	6009378 3008999	2	2	2	1	YES		X	X			
Fluidics	234134	TREADER / WASHING STATION	TREADER / WASHING STATION	3044339 3042739x2	1	1	2	1			X				
Fluidics	734000	PRESSURE SENSOR CABLE	PRESSURE SENSOR CABLE	3036658	1	1		1			X	X			
Fluidics	736238	PROBE v2	PROBE v2	3068221	1	1	2	1			X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	735365	MK VALVE (w. o-ring) v2	MK VALVE (W. O-RING) V2	3065386 3012942	2	2	6	2	YES		X	X			
Fluidics	734835	MK SYRINGE 1ml	MK SYRINGE 1ml	3062972	2	2	6	2			X	X			
Fluidics	735169	PROBE WASHER PIN	PROBE WASHER PIN	3044336							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	735499	MANIFOLD v5	MANIFOLD v5	6020530 3065299 x18 3065300 x6 3021596 x7 3065322 x4 3013548 x1,7m 7006932 x18 8010024					YES		X				
Fluidics	735514	MANIFOLD SPRING CLIPS D7 (20 units)	MANIFOLD SPRING CLIPS D7 (20 units)	7006932x20							X				
Fluidics	234120	MANIFOLD DCP	MANIFOLD DCP	3035386							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	233870	MANIFOLD ASSEMBLY	MANIFOLD ASSEMBLY	6006462 3021596x4 3033787x4 3013548					YES			X			
Fluidics	233924	PROBES WASHER (Ver. 2)	PROBES WASHER (Ver. 2)	3035547 3026804x4	1	1	1	1	YES			X			
Fluidics	233766	LIQUID PUMP (PM1,PM3)	LIQUID PUMP (manifold)	3026301 3031476x2	1	1	1	1	YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	233915	PUMPS REGULATION UPGRADE KIT	PUMPS REGULATION UPGRADE KIT	3035113x2					YES			X			
				3035114x2											
				3035228x2											
				3018660											
				3035225x4											
				3008999x2											
				3035227											
				7004867											
				3011679											
				3024993											
				3013548											
				8010002x4											
				3026301x2											
				3031476x4											

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	231894	TUBE CONNECTION CPC M (right angle)	TUBE CON. CPC PMCD-M RIGHT	3015883			2	1	YES			X			
Fluidics	233417	TUBE CONNECTION RIGHT ANGLE (for 6x4 tube)	TUBE CONECTION RIGHT ANGLE 6X4	3025399				1				X			
Fluidics	233484	TEFLON TUBE 2 WAY ADAPTOR V2	TEFLON TUBE 2 WAY ADAPTOR V2	3035618			1	1			X				
Fluidics	233844	TUBE CONNECT. DISTRIB. (for 2,3x3,6 tube)	TUBE CONNECT.FLUID.DISTR. (2,3x3,6 tube)	3026804				1			X	X			
Fluidics	231915	TUBE PVC 6x4 (50 m.)	PVC TUBE 6x4 mm	3013548			1				X	X			

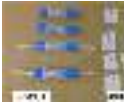
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	234010	FLUIDICS MODULE (complete)	FLUIDICS MODULE (Complete)	6006272					YES			X			
Fluidics	733835	T RACORD HT-5 CPC (20 units)	T RACORD HT-5 CPC	3051690x20							X				
Fluidics	733677	FLUIDIC FITTINGS UPGRADE KIT	FLUIDIC FITTINGS UPGRADE KIT	3059492x2 3051690x4 7006846x8							X				
Fluidics	233847	TUBE CONNECTION MANIFOLD (for 6x4 tube)	TUBE CONNECTION MANIFOLD (for 6x4 tube)	3009000				1				X			
Fluidics	233225	ELECTROVALVE 3 WAYS	ELECTROVALVE 3 WAYS	3021841 8013390x2	2	2	3	1	YES		X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflexis	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	233226	ELECTROVALVE 2 WAYS	ELECTROVALVE 2 WAYS	3021830 8013390x2	2	2	4	1	YES		X	X			
Fluidics	233846	TUBE CONNECTION STRAIGHT (for 6x4 tube)	TUBE CONNECTION STRAIGHT (for 6x4 tube)	3028965				1				X			
Fluidics	735237	EFLEXIS SOLUTION LINES FILTERS SET	EFLEXIS SOLUTION LINES FILTERS SET	3021596x4 3064890x1 3059492x2 3033787x2 3013548 3024993							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	734351	CONTAINER SOLUTIONS V3	CONTAINER SOLUTIONS v3	6017331 3036784 3036786 3044373 3030477 3036787 3036788 3032036 3061110 8010015							X				
Fluidics	233960	WASTE PUMP (Liquid) (Ver. 2) (PM2)	WASTE PUMP (Liquid) (Ver. 2)	3040941x1 3025764x1 8010015x2 8013527x2 8013530x2	1	1	1	1	YES			X			
Fluidics	233910	BOTTLES CONFIGURATOR (with seal) V2	BOTTLES CONFIGURATOR (with seal) V2	3026309 3031014					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	232641	TUBE SILICON 2x4 mm (5 m.)	TUBE SILICON 2x4mm (5 m.)	3033759							X				
Fluidics	734355	WASTE CONTAINER COUPLING SYSTEM V3	WASTE CONTAINER COUPLING SYSTEM V3	6017425 3032457 3033333 3021981 3041791 3031559 3032036 8010015 3061110							X				
Fluidics	733465	DISTRIBUTION BLOCK BK3 (Ver. 3)	DISTRIBUTION BLOCK BK3 (Ver. 3)	3057999 3026804x8					YES			X			
Fluidics	234181	CAP FOR CONTAINERS	CAP FOR CONTAINERS	3025573 3036787 3036788 3030477 3044376							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle xis	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Fluidics	735206	SOLUTION LINES FILTERS SET	SOLUTION LINES FILTERS SET	3021596x4			1		YES		X	X			
Fluidics	733466	DISTRIBUTION BLOCK BK2 (Ver. 3)	DISTRIBUTION BLOCK BK2 (Ver. 3)	3057999 3021981x2 3026804x9					YES			X			
Fluidics	733966	DILUTION STATION CUP V2	DILUTION STATION CUP V2	6019635 3026804x4 3044716 8013453x2			1	1	YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflexis	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	734999	DILUTION STATION CUP EFLEXIS v2	DILUTION STATION CUP EFLEXIS v2	3063373 3026804x2	1	1	2	1	YES		X				
Fluidics	233934	VACUUM FILTERS SET	VACUUM FILTERS SET	3021596x4 3033787x4 3013548x4	1	1	1	1			X	X			
Fluidics	233944	DECONTAMINATION TOOL	DECONTAMINATION TOOL	3013548 3015540x2 3015883 3015587								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	734985	DAMPER SYSTEM v2	DAMPER SYSTEM v2	3063219x2 3033759x2					YES		X				
Fluidics	234140	CPD SENSOR 2K (with PCA connector)	CPD SENSOR 2K (with PCA connector)	6012848 3012942	1	1	2	1			X				
Fluidics	233800	CPD SENSOR (S1,S2)	CPD SENSOR (S1,S2)	3026611 3033759	1	1	2	2	YES			X			
Fluidics	233741	WASTE BOTTLE CAP	WASTE BOTTLE CAP	3028012 3030891 3031281 3031015								X			
Fluidics	233739	WASTE BOTTLE (8 litres)	WASTE BOTTLE (8 litres)	6006780			1					X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	233921	CONNECTION BLOCK BK6 (Ver. 2)	CONNECTION BLOCK BK6 (Ver. 2)	3035548					YES			X			
Fluidics	233799	CONNECTION BLOCK CPD SENSOR (BK4 - BK5)	CONNECTION BLOCK CPD SENSOR	3029808x2 8010024x2			2	1	YES			X			
Fluidics	232655	FILTER FOR CONTAINERS (SOL.)	CONTAINER FILTER (SOLUTIONS)	3018725 3018726			1	1	YES		X	X			
Fluidics	732961	DISPENSING PUMP V2	DISPENSING PUMP V2	3056778	1	1	1	1	YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	233916	EXPANSION VESSEL	EXPANSION VESSEL	3035113 3035114 3035225x2 3008999					YES			X			
Fluidics	734350	CONTAINER WASTE V3	CONTAINER WASTE v3	6019822 3032036							X				
Fluidics	233843	BARBED ADAPTOR OMNIFIT (with O- Ring)	BARBED ADAPTOR OMNIFIT (with O- Ring)	3012942 3026510				1			X	X			
Fluidics	733675	ASPIRATION PUMP SPRING CLIP (40 units)	ASPIRATION PUMP SPRING CLIP (40 units)	7006846x40							X				
Fluidics	232199	TUBE CONNECTION FLUIDICS DISTRIBUTOR (for 6x4 tube)	TUBE CONNECTION FLUIDICS DISTRIBUTOR	3021981				1				X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Fluidics	234182	EXPANSION VESSEL	EXPANSION VESSEL	3042626 3042627 3042629 3008999							X				
Fluidics	735494	FLU FRONT COVER	FLU FRONT COVER	3064797 7009959 x4 8013494 x4							X				
Fluidics	232605	ONE WAY VALVE	1 WAY VALVE 3/16	3016381				1			X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Fluidics	734354	FLUIDIC CONNECTION BLOCK (4 units)	FLUIDIC CONNECTION BLOCK (4 units)	6019672x4 3056897x4 8013463x16 3032456x4 6016690x4 3061069x4 3061110x4 7004681x4 8010016x4 8010027x8 8013519x8	1	1					X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	731980	SCREW M5X30 DIN 912 (10 units)	SCREW M5X30 DIN 912 (10 units)	8010043x10							X				
Mechanics	234002	BALL BEARING 8X19X6 W619/8-2Z STAINLESS	BALL BEARING 8X19X6 W619/8-2Z STAINLESS	3044421				1	YES			X			
Mechanics	233712	TOOTHED BELT T2,5/380	TOOTHED BELT T2,5_380	3025782				1	YES			X			
Mechanics	233978	CARD HOLDER RETAINER KIT	CARD HOLDER RETAINER KIT	3042352 3041153x4 3041152x16							X	X			
Mechanics	733370	PLASTIC BALL BEARING BB-626- B180-30-ES	PLASTIC BALL BEARING BB-626-B180- 30-ES	3045401 8010339					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	234153	PLASTIC BALL BEARING BB-6001- B180-30-ES	PLASTIC BALL BEARING BB-6001-B180-30-ES	3045402 7006730 7006363					YES		X				
Mechanics	734948	TOOTHED GUIDE v2 (PIP BCR)	TOOTHED GUIDE v2 (PIP BCR)	3062840 8010008x3					YES		X				
Mechanics	233841	PIP PRESENCE DETECTOR HOLDER PLATE	PIP PRESENCE DETECTOR HOLDER PLATE	3025272					YES			X			
Mechanics	732573	PIP HEAD LATERAL COVER	PIP HEAD LATERAL COVER	3025271 8010018 7006368 3026152x2 8010015x3								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	231869	TOOTHED GUIDE (PIP BCR)	TOOTHED GUIDE GRIPPER	3015579					YES		X				
Mechanics	731378	CARD HOLDER	CARD HOLDER	3048150									X		
Mechanics	734424	CAMERA SUPPORT PLATE	CAMERA SUPPORT PLATE	3060072 8010029x2 8010024							X	X	X		
Mechanics	735615	PIP FITTING SUPPORT PLATE	PIP FITTING SUPPORT PLATE	3065422 8010015x2							X				
Mechanics	232541	BALL BEARING FAG 135	BALL BEARING FAG 135	3015474					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	733837	CAMERA COVER	CAMERA COVER	3058393 8010015x2							X	X	X		
Mechanics	233867	TOOTHED BELT T2,5/160	TOOTHED BELT T2,5_160	3026422	2	2	1	1	YES		X	X			
Mechanics	731391	CABLE WALL ROUTING	CABLE WALL ROUTING	3034610									X		
Mechanics	735280	PROBE GUIDE V2	PROBE GUIDE V2	3063027 8010010 x2 8013514 x2					YES		X				

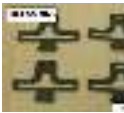
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	735571	PROBE GUIDE V3	PROBE GUIDE V3	3065544 3065550 8010007 x4 8010020 8010009 8013515 8013514 x2							X				
Mechanics	233773	TOOTHED BELT T2,5/145	TOOTHED BELT T2,5/145	3025389				1	YES			X			
Mechanics	233865	TOOTHED BELT T5/2000	TOOTHED BELT T5_2000	3015721					YES			X			
Mechanics	234137	PROBE WAGON	PROBE WAGON	3042588					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	730582	CABLE GROUNDING CLAMP 3.5 mm (10 units)	CABLE GROUNDING CLAMP 3.5mm (10 units)	3017611x10							X	X			
Mechanics	233970	PROBE1 WAGON COVER	PROBE1 WAGON COVER	3025258					YES			X			
Mechanics	234154	BALL BEARING 626 2Z SKF	BALL BEARING 626 2Z SKF	3042505					YES		X				
Mechanics	232049	GRIPPER PIN	PIN (GRIPPER) (VER. 2)	3005586			1	1	YES		X	X			

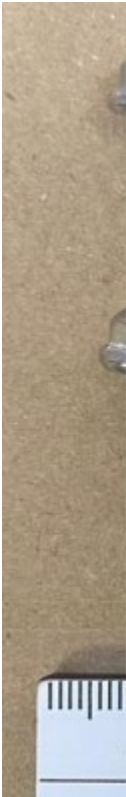
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	731392	CARD STOPPER	CARD STOPPER	3048145									X		
Mechanics	233971	PROBE2 WAGON COVER	PROBE2 WAGON COVER	3025257					YES			X			
Mechanics	233737	BOTTLES TRAY	BOTTLES TRAY	6006291								X			
Mechanics	735247	CONTAINERS DOOR HINGES	CONTAINERS DOOR HINGES	6012159x1 6012161x1 8010026x2 x1 8010037							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	730596	CIRCLIP D 8 DIN 471 (50 units)	CIRCLIP D 8 DIN 471 (50 units)	3017256x50								X			
Mechanics	232761	CIRCLIP BALL BEARING (50 UNITS)	CIRCLIP BALL BEARING (50 UNITS).	3017259x50							X	X			
Mechanics	233729	CENTRIFUGE SHOCK ABSORBERS (3 units)	CENTRIFUGE SHOCK ABSORBERS (3 units)	3025575x3 8016977x3 8013502x3 8010037x3					YES		X	X			
Mechanics	733965	CENTRIFUGE ROTOR V2	CENTRIFUGE ROTOR V2	6019112 3024613					YES		X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	731388	SPRING (10 units)	SPRING (10 units)	3050708x10					YES				X		
Mechanics	233863	CENTRIFUGE POT	CENTRIFUGE POT	3025574					YES		X	X			
Mechanics	233730	CENTRIFUGE LATCH	CENTRIFUGE LATCH	3025155 7002701 3025672 3025156					YES		X	X			
Mechanics	232746	SPACING PLASTIC WASHER (100 UNITS)	SPACING PLASTIC WASHER (100 UNITS)	3016016x100								X			
Mechanics	233883	12 CARDS RACK ADAPTER	12 CARDS RACK ADAPTER	3029365								X			
Mechanics	232043	SCREW ALLEN M3x5 DIN 912 (100 units)	ALLEN SCREW M3X5 DIN 912 (100 units)	8016810x100								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233979	CARD HOLDER RIGHT RETAINER (4 units)	CARD HOLDER RIGHT RETAINER (4 units)	3041152x4							X	X			
Mechanics	731567	CAT ROBOTIC HEAD RIGHT PLATE	CAT ROBOTIC HEAD RIGHT PLATE	6012145							X				
Mechanics	734717	CAT ROBOTIC HEAD REAR COVER	CAT ROBOTIC HEAD REAR COVER	6012144 8016810x4							X				
Mechanics	734745	CAT ROBOTIC HEAD LEFT COVER v2	CAT ROBOTIC HEAD LEFT COVER v2	6019741 8010015x4 8016810x3							X				
Mechanics	233928	CAT ROBOTIC HEAD COVER	CAT ROBOTIC HEAD COVER	3029227					YES			X			

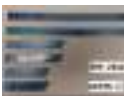
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	234199	WASTE CARDS BASE HOLDER	WASTE CARDS BASE HOLDER	6013290					YES		X				
Mechanics	233931	CAT MIRROR WEDGE	CAT MIRROR WEDGE	3036100								X			
Mechanics	734811	CAT MIRROR COVER	CAT MIRROR COVER	3042651 8016810x3							X				
Mechanics	732517	CAT BARCODE READER COVER v2	CAT BARCODE READER COVER v2	3029228 3052536 3029242 8016810x2 7003718x2 8013452x4					YES			X			

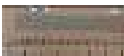
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Therm	DG Spin
					AU S	EME A	LATA M	N A							
Mechanics	735616	CARDS WASTE LOAD SCREWS (10 units)	CARDS WASTE LOAD CELL SCREWS (10 units)	7010143 x10							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233705	CARDS RACK HOLDER	CARDS RACK HOLDER	3025784				1	YES		X	X			
Mechanics	731344	CARDS HOLDER (CENTR.) 12 UNITS (VER.3)	CARDS HOLDER (CENTR.) 12 UNITS (VER.3)	3048672x12					YES		X	X			
Mechanics	233993	CARDS DRAWER LOCK COVER	CARDS DRAWER LOCK COVER	3024132							X	X			
Mechanics	234176	CARDS DRAWER DAMPER	CARDS DRAWER DAMPER	3042690					YES		X				
Mechanics	731625	CARDS DRAWER COSMETIC FRONT COVER	CARDS DRAWER COSMETIC FRONT COVER	3042699 3042697 8010016x2 8010015x4 3024023					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	734716	CAT ROBOTIC HEAD RIGHT COVER	CAT ROBOTIC HEAD RIGHT COVER	6012145 8010015x4 8016810x3							X				
Mechanics	233866	TOOTHED BELT T5/1200	TOOTHED BELT T5_1200	3015721					YES			X			
Mechanics	234108	TOOTHED BELT T5/1460 (X-CAT)	TOOTHED BELT T5/1460 (X-CAT)	3015721					YES		X				
Mechanics	735528	BUTTERFLY NUTS FOR FLU COVER (10 units)	BUTTERFLY NUTS FOR FLU COVER (10 units)	7009959x10							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	731624	REAR PANEL (UPPER)	REAR PANEL (UPPER)	3042678					YES		X				
Mechanics	733467	REAR PANEL (UPPER) v2	REAR PANEL (UPPER) v2	3057513								X			
Mechanics	234007	REAR PANELS SPACERS SET (8 units)	REAR PANELS SPACERS SET (8 units)	3025484x8					YES			X			
Mechanics	730754	REUSABLE CARDS RACK COVER (1 UNIT)	REUSABLE CARDS RACK COVER (1 UNIT)	3042698							X				
Mechanics	730700	RUBBER FOOT V2	RUBBER FOOT V2	3048210					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	234151	S/R DRAWER DAMPER	S/R DRAWER DAMPER	3043091					YES		X				
Mechanics	233859	WHEEL & FOOT ASSEMBLY	WHEEL AND FOOT ASSEMBLY	3027550					YES			X			
Mechanics	234155	S/R MAGNETIC GUIDES SET (SRE)	S/R MAGNETIC GUIDES SET (SRE)	3042486x2 3042498x2 3042499x2							X				
Mechanics	730612	SAMPLE DRAWER HOLDER (2 units)	SAMPLE DRAWER HOLDER (2 units)	3024403x2								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	734220	BCR MIRROR HOLDER SCREWS (40 units)	BCR MIRROR HOLDER SCREWS (40 units)	7008900x40							X				
Mechanics	234130	BCR MIRROR (EFL PIP)	BCR MIRROR	3031711	1	1	1		YES		X				
Mechanics	233777	BCR MIRROR (ERY PIP&CAT / EFL CAT)	BCR MIRROR	3029189	1	1	1	1	YES		X	X			
Mechanics	232760	SCREW ALLEN M2X10 DIN 912 (100 UNITS)	SCREW ALLEN M2X10 DIN 912 (100 UNITS)	8010003x100							X	X			
Mechanics	233864	SCREW FOR LATERAL COVER	SCREW FOR LATERAL COVER	3024851					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eyt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	233855	REAR PANEL (UPPER)	REAR PANEL (UPPER)	3024878								X			
Mechanics	234207	SCREW FOR LATERAL PANEL	SCREW FOR LATERAL PANEL	3032019					YES		X				
Mechanics	730602	SCREW M2x6 DIN 912 (100 units)	SCREW M2x6 DIN 912 (100 units)	8010001x100							X	X			
Mechanics	734739	SCREW M3x4 DIN7985 (100 units)	SCREW M3x4 DIN7985 (100 units)	7002720x100							X				
Mechanics	730603	SCREW M3x6 DIN 912 (100 units)	SCREW M3x6 DIN 912 (100 units)	8010015x100								X	X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	730608	SCREW M3x6 DIN 965H (100 units)	SCREW M3x6 DIN 965H (100 units)	8013452x100								X			
Mechanics	730588	SCREW M3x6 HEX.A2 (100 units)	SCREW M3x6 HEX.A2 (100 units)	7004681x100							X		X		
Mechanics	733912	SCREW M3x10 A2 ISO 7380 FL (20 units)	SCREW M3x10 A2 ISO 7380 FL (20 units)	7006893x20							X				
Mechanics	730609	SCREW M3x12 DIN 965H (100 units)	SCREW M3x12 DIN 965H (100 units)	8013454x100								X			
Mechanics	730604	SCREW M4x6 DIN 912 (100 units)	SCREW M4x6 DIN 912 (100 units)	8010025x100								X			
Mechanics	730599	SCREW M4x6 HEX.A2 (100 units)	SCREW M4x6 HEX.A2 (100 units)	7002712x100								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	730605	SCREW M4x8 DIN 912 (100 units)	SCREW M4x8 DIN 912 (100 units)	8010026x100							X	X			
Mechanics	730606	SCREW M4x10 DIN 912 (100 units)	SCREW M4x10 DIN 912 (100 units)	8010027x100								X			
Mechanics	730600	SCREW M4x10 HEX.A2 (100 units)	SCREW M4x10 HEX.A2 (100 units)	7004402x100								X			
Mechanics	730610	SET SCREW M4X16 DIN 913 (100 units)	SET SCREW M4X16 DIN 913 (100 units)	8013494x100								X			
Mechanics	730607	SCREW M2,5x5 DIN 7985H (100 units)	SCREW M2,5x5 DIN 7985H (100 units)	8013392x100								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	735042	SCREW M2x4 DIN 912 (10 units)	SCREW M2x4 DIN 912 (10 units)	8018118x10							X				
Mechanics	732451	SCREW M5x70 DIN 912 (10 units)	SCREW M5x70 DIN 912 (10 units)	8010049x10								X			
Mechanics	731623	REAR PANEL (LOWER)	REAR PANEL (LOWER)	6012156					YES		X				
Mechanics	233856	REAR PANEL (INTERMEDIATE)	REAR PANEL (INTERMEDIATE)	3024824								X			
Mechanics	233969	PROBES WAGON (Impact detector)	PROBES WAGON (Impact detector)	6009067					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	234122	TOOTHED BELT T5/1260 (Y- PIP/CAT)	TOOTHED BELT T5/1260 (Y-PIP/CAT)	3015721					YES		X				
Mechanics	730668	BARCODE READER COVER	BARCODE READER COVER	6013202							X				
Mechanics	735052	PROTECTOR PLATES (FLU)	PROTECTOR PLATES (FLU)	3063025x3 3063026x2								X			
Mechanics	731380	PULLEY	PULLEY	3048139					YES				X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	735568	BOTTLE HOLDER (FULL) V2	BOTTLE HOLDER (FULL) V2	3012574 x2							X				
				3017260 x2											
				3032456											
				3056897											
				3061069											
				3061110											
				3064075											
				6016690											
				6019672											
				7004596											
				x2											
				7004681											
				7004696											
				x2											
				7006360											
				8010016											
				8013463											
				x2											
				8010027											
				x2											
				8013519											
				x2											

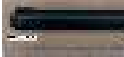
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	733560	BOTTLE HOLDER	BOTTLE HOLDER	3027474 7006352x2								X			
Mechanics	233902	BLUE LATERAL (Right)	BLUE LATERAL (Right)	3024853								X			
Mechanics	233903	BLUE LATERAL (Left)	BLUE LATERAL (Left)	3024854								X			
Mechanics	731390	BLUE BASE	BLUE BASE	3048149					YES				X		
Mechanics	231866	SHOCK ABSORBER PROBE 1/PROBE 2/GRIPPER MOTOR	SILENTBLOCK SHAKER	3015563					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	733086	SHOULDER SCREW 6M5x30 ISO 7379 (10 u.)	SHOULDER SCREW 6M5x30 ISO 7379 (10 u.)	10x7002887								X			
Mechanics	233857	REAR PANEL (LOWER)	REAR PANEL (LOWER)	3024879								X			
Mechanics	231889	TOOTHED BELT T2,5/245	TOOTHED BELT T2.5/245 (XY)	3015729	1	1	1	2	YES		X	X			
Mechanics	731794	BELT PLATE	BELT PLATE	3048141									X		
Mechanics	733350	READER MODULE SUPPORT PLATE	READER SUPPORT PLATE	6018753							X	X	X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	231887	TOOTHED BELT T2,5/120	TOOTHED BELT T2.5/120 (Z)	3015727			1	1	YES		X				
Mechanics	233726	SHOCK ABSORBERS X MOTOR (3 units)	SHOCK ABSORBERS X MOTOR (3 units)	3004968x3 8016977x3 8013507x3					YES			X			
Mechanics	234114	SHOCK ABSORBERS X-Y MOTOR (3 units)	SHOCK ABSORBERS X-Y MOTOR (3 units)	3025575x3 8016977x3 8013507x3					YES		X				
Mechanics	234192	SHOCK ABSORBERS (2 units)	SHOCK ABSORBERS (2 units)	3025575x2					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	733700	SHOCK ABSORBER DILUTION CUP (Ver. 3)	SHOCK ABSORBER DILUTION CUP (Ver. 3)	3004968 3058992 3059009 3045606 8013507 8016977 8013502 8013520					YES			X			
Mechanics	233709	STIRRER BELT TIGHTENING	SHAKER BELT TIGHTENING	3024615 3024616x2 3008828x2 8025918x2 3008951x2 8010055 8013503 7007037x2					YES			X			
Mechanics	731377	BELT	BELT	3050642					YES				X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	235234	THREADED BUMPER	THREADED BUMPER	3025660					YES		X		X		
Mechanics	234003	REAGENTS RACK HOLDER (SST Bearing)	REAGENTS RACK HOLDER (SST Bearing)	3024672 3044421x2 8010339x2					YES			X			
Mechanics	234004	REAGENTS RACK HOLDER (Ver. 2, SST Bear.)	REAGENTS RACK HOLDER (Ver. 2, SST Bear.)	3040278 3044421x2 8010339x2					YES			X			
Mechanics	234011	REAGENTS SPRING CLIP SET (1 unit)	REAGENTS SPRING CLIP SET (1 unit)	3024676								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	735530	READER BOX	READER BOX	3026399 8010027 x2 8013453 x8							X	X			
Mechanics	234113	TOOTHED BELT T5/1195 (X-PIP)	TOOTHED BELT T5/1195 (X-PIP)	3015721					YES		X				
Mechanics	733016	COSMETIC COVER OF VERTICAL GUIDE RACKS	COSMETIC COVER OF VERTICAL GUIDE RACKS	3054412							X				
Mechanics	232671	TOUCHSCREEN / USER TRAY LOCK	CONTAINERS DOOR LATCH	3017373				1	YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Therm	DG Spin
					AU S	EME A	LATA M	N A							
Mechanics	732509	HORNS LATERAL v2 (LEFT)	HORNS LATERAL v2 (LEFT)	3029306 3052536 3029242 8016810x2 7003718x2 8010017x2 8013454x2 8013500x4								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	732508	HORNS LATERAL v2 (RIGHT)	HORNS LATERAL v2 (RIGHT)	3029305								X			
				3052536											
				3029242											
				8016810x2											
				7003718x2											
				8010017x2											
				8013454x2											
				8013500x4											

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eyt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	735327	HORNS LATERAL v3 (LEFT)	HORNS LATERAL v3 (LEFT)	3064621 3052536 3064094 3064619 7003718x2 7004573x2 7007623 7009693x4								X			
Mechanics	735329	HORNS LATERAL v3 (RIGHT)	HORNS LATERAL v3 (RIGHT)	3064620 3052536 3064619 7003718x2 7004573x2 7009693x4								X			

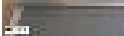
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	733913	HORNS TRANSMISSION GEAR ASSEMBLY	HORNS TRANSMISSION GEAR ASSEMBLY	3059490 3059487 7004681 3059489 3042603 8013487 3015569 8013492					YES		X				
Mechanics	734748	HORNS TRANSMISSION GEAR CROWN	HORNS TRANSMISSION GEAR CROWN	3042603 8013487							X				
Mechanics	734747	HORNS TRANSMISSION GEAR PINION	HORNS TRANSMISSION GEAR PINION	3015569 8013492							X				
Mechanics	734826	INC UPPER COVER	INC UPPER COVER	3025443 3024774x4 7002712x8								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233720	INCUBATOR COUPLING SYSTEM	INCUBATOR COUPLING SYSTEM	3029356 3029362x2								X			
Mechanics	234101	INCUBATOR HOLDER (4 units)	INCUBATOR HOLDER (4 units)	3042630x4					YES		X				
Mechanics	234008	INCUBATOR ISOLATION SET	INCUBATOR ISOLATION SET	6008460								X			
Mechanics	733255	INCUBATOR LATERAL	INCUBATOR LATERAL	3024845 8010040x2								X			
Mechanics	734701	INCUBATOR PLATE EDGE PROTECTORS v2	INCUBATOR PLATE EDGE PROTECTORS v2	7009315x13 7009315x5								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	735331	HORNS LATERAL RIGHT	HORNS LATERAL RIGHT	3064093 3052536 3064091 7003718x2 7004573x2 7009693x4 8013452x2							X				
Mechanics	731850	INCUBATORS COVER	INCUBATORS COVER	3042631							X				
Mechanics	233854	INTERMEDIATE DOOR DAMPER	INTERMEDIATE DOOR DAMPER	3026412					YES			X			
Mechanics	735334	INTERMEDIATE PLATE DC	INTERMEDIATE PLATE DC	3064985							X				

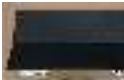
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	731514	ERYTRA CARDS WASTE NOZZLE	ERYTRA CARDS WASTE NOZZLE	3029187								X			
Mechanics	234104	ISOLATOR (4 units)	ISOLATOR (4 units)	3033658							X				
Mechanics	734984	LATERAL COVER v3	LATERAL COVER v3	3064049 8013396					YES				X		
Mechanics	734982	LATERAL FAN UPGRADE KIT	LATERAL FAN UPGRADE KIT	3064048 3064049 6012353 8010020x2 8013396									X		
Mechanics	234205	LATERAL PANEL	LATERAL PANEL	3042684					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	732891	LATERAL PANEL (metacrilate)	LATERAL PANEL (metacrilate)	3042683					YES		X				
Mechanics	234161	LEAD SCREW (GRIPPER)	LEAD SCREW (GRIPPER)	3042653 3042644 3042599					YES		X				
Mechanics	234206	CONTAINERS DOOR	CONTAINERS DOOR	3042681 3033369							X				
Mechanics	233749	LEAD SCREW (Samples)	LEAD SCREW (Samples)	3026117 8013490 X2					YES			X			
Mechanics	233774	LEAD SCREW HOLDER	LEAD SCREW HOLDER	3029226 8016837 8013490x2					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	735102	EFLEXIS PIP GROUNDING PLATE v2	EFLEXIS PIP GROUNDING PLATE v2	6020346 3064798 3062831 8016810x2							X				
Mechanics	734915	TREADER SUPPORT (GUIDE)	TREADER SUPPORT (GUIDE)	3044338 3044341 8010116x4 8013444							X				
Mechanics	233826	INTERMEDIATE DOOR	INTERMEDIATE DOOR	6006586 3023133					YES			X			
Mechanics	233704	LINEAR GUIDE (Cards_Bottles drawer) 500 mm	LINEAR GUIDE (Cards_Bottles drawer)500mm	3025642					YES		X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	735332	HORNS LATERAL LEFT	HORNS LATERAL LEFT	3064092 3052536 3064091 3064094 7003718x2 7004573x2 7009693x4 8013452x2							X				
Mechanics	735328	HORNS-FRONTAL v2	HORNS - FRONTAL v2	3064622 3017508 7009693x8								X			
Mechanics	233959	FLUIDIC DRAWER LOCK COVERS (5units)	FLUIDIC DRAWER LOCK COVERS (5units)	3025825x5								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	240024	RUBBER FOOT (Grey)	RUBBER FOOT (GREY)	3012574									X	X	X
Mechanics	233861	FREE PULLEY (X,Y)	FREE PULLEY (X,Y)	3024094 3025858 3025565x2					YES		X	X			
Mechanics	234204	FRONT PANEL (Big)	FRONT PANEL (Big)	3042685							X				
Mechanics	234203	FRONT PANEL (Small)	FRONT PANEL (Small)	3042686							X				
Mechanics	730627	FRONTAL GUIDE FOR RACKS	FRONTAL GUIDE FOR RACKS	3035327							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	731337	FUNNEL EXTERNAL WASTE CARDS (1 UNIT)	FUNNEL EXTERNAL WASTE CARDS (1 UNIT)	3048946 8010018x2					YES		X				
Mechanics	234193	FUNNEL WASTE CARDS	FUNNEL WASTE CARDS	3042706							X				
Mechanics	733761	FLUIDIC DRAWER EDGE PROTECTOR (10 units)	FLUIDIC DRAWER EDGE PROTECTOR	3033657x10								X			
Mechanics	730601	FLAT CABLE CLAMP	FLAT CABLE CLAMP	7006734				1				X			
Mechanics	733840	GEAR DOUBLE WHEEL (X,Y) v2	GEAR DOUBLE WHEEL (X,Y) v2	3024102 6018663 3025565x2					YES		X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233862	GEAR WHEEL (lead screws)	GEAR WHEEL (lead screws)	3026869 8013488					YES		X	X			
Mechanics	233708	GEAR WHEEL REAGENTS STIRRER	GEAR WHEEL REAGENTS SHAKER	3024617 3019261 3024618 8010016 3029262x2 7002712x2 3029261 7006363					YES			X			
Mechanics	233721	GRIPPER	GRIPPER	6006130 3015580				1	YES			X			
Mechanics	233836	HORNS - HOLDER PLATE	HORNS - HOLDER PLATE	3029311								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	232226	GRIPPER BOTTOM ROUND LABEL (10 units)	GRIPPER BOTTOM ROUND LABEL	3015580x10							X	X			
Mechanics	232242	GRIPPER FINGER SET (w/springs & bolts)	GRIPPER FINGERS SET (W/SPRINGS&BOLTS)	3006342 3006343 3014191x2 3015558x2					YES		X	X			
Mechanics	232127	GRIPPER INTERNAL PARTS SET (ver 2)	GRIPPER INTERNAL PARTS SET (VER. 2)	3006342 3006343 3006345 3006346 3014191x2 3015557 3015576 3016761 3016310					YES		X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233722	LEAD SCREW (gripper)	GRIPPER LEAD SCREW	3026021					YES			X			
Mechanics	233998	GRIPPER MOTOR HOLDER	GRIPPER MOTOR HOLDER	3023933					YES			X			
Mechanics	232010	GRIPPER PIN CIRCLIP (100 units)	GRIPPER PIN'S CIRCLIPS (100 UNITS)	8010301x100					YES		X	X			
Mechanics	232132	GRIPPER PIN SPRING (4 units)	GRIPPER PIN SPRING (4 UNITS)	3015556x4					YES		X	X			
Mechanics	734836	GRIPPER PLATE	GRIPPER PLATE	6020079 8010015x3								X			
Mechanics	231994	GRIPPER FINGERSPRINGS (10 units)	GRIPPER SPRING (10 UNITS)	3015558x10					YES		X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	734759	GRIPPER v2	GRPPER v2	6019935 3015580				1	YES		X				
Mechanics	234119	GUIDE (PIP BCR)	GUIDE (PIP BCR)	3042616					YES		X				
Mechanics	734740	HEXAGONAL SPACER 4 mm M3 (50 units)	HEXAGONAL SPACER 4mm M3 (50 units)	7009073x 50							X				
Mechanics	234175	HINGE	HINGE	3042489					YES		X				
Mechanics	731373	HINGE	HINGE	3048935					YES				X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233835	HORNS - FRONTAL	HORNS - FRONTAL	3029312 3017508								X			
Mechanics	233871	GRIPPER CENTERING SYSTEM	GRIPPER CENTERING SYSTEM	3025489 3015557					YES			X			
Mechanics	234160	LINEAR GUIDE (CAT-GRIPPER)	LINEAR GUIDE (CAT-GRIPPER)	3042740					YES		X		X		
Mechanics	234123	LEAD SCREW (Probe)	LEAD SCREW (Probe)	3042598 3042599					YES		X				
Mechanics	233801	LINEAR GUIDE GRIPPER (CAT)	LINEAR GUIDE GRIPPER (CAT)	3029185					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233977	UPPER DOOR HINGE BAR ASSEMBLY	UPPER DOOR HINGE BAR ASSEMBLY	6005814 3025703x4 3024956x2 3025704x4 3025777x2 8010311x2 7006474x2					YES			X			
Mechanics	234106	TOUCHSCREEN FRONT COVER	TOUCHSCREEN FRONT COVER	6012239					YES		X				
Mechanics	233912	DILUTION CUP BATH	DILUTION CUP BATH	3029054								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233911	DILUTION CUP BATH (Ver. 2)	DILUTION CUP BATH (Ver. 2)	3034575 3035513 3016771 7003270 8013567 7006352x2								X			
Mechanics	730611	NUT M3 DIN 934 (100 units)	NUT M3 DIN 934 (100 units)	8013500x100								X			
Mechanics	234200	MULTILUMEN TUBING HOLDER	MULTILUMEN TUBING HOLDER	3042744					YES		X				
Mechanics	731854	TOP COVER	TOP COVER	6012155							X				
Mechanics	731639	DILUTION CUP COVER	DILUTION CUP COVER	3052912							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	735001	DILUTION CUP HOLDER v2	DILUTION CUP HOLDER v2	3063374 8010018x4							X				
Mechanics	234107	TOUCHSCREEN REAR COVER	TOUCHSCREEN REAR COVER	3042796 3032145 3034610 8010026x4					YES		X				
Mechanics	233703	LINEAR GUIDE (Samp._Reag. Drawer) 450 mm	LINEAR GUIDE (Samp._Reag. drawer) 450mm	3025569					YES		X	X			
Mechanics	233994	TOUCHSCREEN TILTING LOCK SET	TOUCHSCREEN TILTING LOCK SET	3027010 3027017 8010221 8010204								X			
Mechanics	731802	DILUTION CUP STIRRER TUBE	DILUTION CUP STIRRER TUBE	3023706 8x8013452								X			

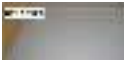
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	735335	DILUTION CUP STRUCTURE (WHOLE) v4	DILUTION CUP STRUCTURE (WHOLE) v4	3042495 7006363 7006730 3045402 3059488 6012096 3063374 8010339 3045401 8013487 3064985 3042496x2 8010018x4							X				
Mechanics	732915	TOUCHSCREEN HOUSING v2	TOUCHSCREEN HOUSING v2	3056906					YES				X		
Mechanics	232559	MOTOR PINION	TOOTHED PULLEY READER / WASHER	3015641									X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	735084	UPPER DOOR v2	UPPER DOOR v2	6020174 8013396x2					YES			X			
Mechanics	233707	DILUENTS RACK HOLDER	DILUENTS RACK HOLDER	3025395 3024671 3025394 8013452x4								X			
Mechanics	234201	CONTAINERS DOOR LATCH	CONTAINERS DOOR LATCH	3032832 3032454					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle x is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	734039	FLUIDICS BOX ASSEMBLY (PLATES)	FLUIDICS BOX ASSEMBLY (PLATES)	3042628x2 8010024x2 6012125x2 6009920x2 6012126x2 8013415x4 3025575x4 7006932x4 6012124x2					YES		X				
Mechanics	234009	COSMETIC FRONT-LEFT PANEL OF UPPER FLOOR	COSMETIC FRONT-LEFT PANEL OF UPPER FLOOR	3024848								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflexis	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233954	COSMETIC FRONT PANEL FOR USER TRAY VER.2	COSMETIC FRONT PANEL FOR USER TRAY VER.2	3040140								X			
Mechanics	233779	COSMETIC FRONT PANEL OF CARDS	COSMETIC FRONT PANEL OF CARDS	3024806 8016810x4 8013454x2								X			
Mechanics	233782	COSMETIC FRONT PANEL OF CONTAINERS	COSMETIC FRONT PANEL OF CONTAINERS	3025491 8019878x3								X			
Mechanics	233780	COSMETIC FRONT PANEL OF REAGENTS	COSMETIC FRONT PANEL OF REAGENTS	3024807 8016810x4 8013454x4								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233781	COSMETIC FRONT PANEL OF SAMPLES	COSMETIC FRONT PANEL OF SAMPLES	3025851 8016810x4 8013454x4								X			
Mechanics	234001	COSMETIC FRONT PANEL OF PUSHBUT. AND USB	COSMETIC FRONT PANEL OF PUSHBUT. AND USB	3024849								X			
Mechanics	731381	WAGON PLATE (POSITION SENSORS)	WAGON PLATE (POSITION SENSORS)	3048140									X		
Mechanics	234174	COVER FOR POWER SUPPLY 12V	COVER FOR POWER SUPPLY 12V	3042794							X				
Mechanics	733620	DG-59 INCUBATOR MODULE	DG-59 INCUBATOR MODULE	6006413								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	733671	USER TRAY (Ver. 4)	USER TRAY (Ver. 4)	6018839					YES			X			
Mechanics	233858	UPPER PANEL	UPPER PANEL	3025745								X			
Mechanics	730795	DILUTION CUP & REAGENTS RACK SHOULDER SCREWS	DIL. CUP&REAG. RACK SHOULDER SCREW (10u)	3042496x10							X				
Mechanics	234136	TREADER GUIDE	TREADER GUIDE	3044337					YES		X				
Mechanics	732420	MONITOR SUPPORT GUIDES	MONITOR SUPPORT GUIDES	3027002 3027004 3027003 3027005 3027007x2								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233733	CENTRIFUGE MOTOR BUSHING	CENTRIFUGE MOTOR BUSHING	3025652					YES		X	X			
Mechanics	234135	TREADER SPRING (10 units)	TREADER SPRING (10 units)	3045403x10					YES		X				
Mechanics	233933	EDGE PROTECTOR AC COVER	EDGE PROTECTOR AC COVER	3033657								X			
Mechanics	233882	LOCK CONNECTOR BCR CABLE (10 units)	LOCK CONNECTOR BCR CABLE (10 units)	3033078x10								X			
Mechanics	233828	LINEAR GUIDE Y (PIP/CAT)	LINEAR GUIDE Y (PIP_ CAT)	30291848013490x2					YES			X			

[illegible]

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	233926	LOWER LEFT LATERAL PANEL	LOWER LEFT LATERAL PANEL	3026651 8013494x6 8013506x6								X			
Mechanics	234121	LINEAR GUIDE Y	LINEAR GUIDE Y	3029184					YES		X				
Mechanics	233925	LOWER RIGHT LATERAL PANEL	LOWER RIGHT LATERAL PANEL	3026652 8013494x6 8013506x6								X			
Mechanics	733559	MAGNETIZED ECCENTRIC WHEEL	MAGNETIZED ECCENTRIC WHEEL	6012096 7006363							X				
Mechanics	233725	LINEAR GUIDE X (PIP/CAT)	LINEAR GUIDE X (PIP/CAT)	3029183 8013490x2					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	732914	MAIN COVER v2	MAIN COVER v2	6017971					YES				X		
Mechanics	233999	UPPER & INTERMED. DOOR HANDLE (w.screws)	UPPER & INTERMED. DOOR HANDLE (w.screws)	3026783 8017718x6								X			
Mechanics	234115	LINEAR GUIDE X	LINEAR GUIDE X	3042749					YES		X				
Mechanics	734279	MONITOR ARM BASE	MONITOR ARM BASE	3031987 8010043x4							X				
Mechanics	234125	LINEAR GUIDE PIP (Probe)	LINEAR GUIDE PIP (Probe)	6012111					YES		X				
Mechanics	233853	UPPER DOOR DAMPER	UPPER DOOR DAMPER	3026413 8010311x2					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eyt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	733055	MONITOR ARM	MONITOR ARM	3009032 8010055 8013463x2 8010123x4 8010026x4							X				
Mechanics	234208	DOOR (only metacrilate)	DOOR (only metacrilate)	6012174					YES		X				
Mechanics	233939	LINEAR GUIDE SET SCREWS (50 units)	LINEAR GUIDE SET SCREWS (50 units)	8013490x50								X			
Mechanics	233892	MIXED REAGENTS RACK CAP (10 units)	MIXED REAGENTS RACK CAP (10units)	3033204x10								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	734760	MIRROR HOLDER v2	MIRRROR HOLDER v2	3062311 3052536 8010016x4					YES		X				
Mechanics	735570	PROBE WAGON V2	PROBE WAGON V2	3065543 8013452 x2 7004681 x2 8013396 8010003 x4							X				
Miscellany	733153	WINDOWS 10 ENTERPRISE 2019 LTSC LICENSE	WINDOWS 10 ENTERPRISE 2019 LTSC LICENSE								X	X			
Miscellany	730572	CD REAGENTS PACK US V1.5 (Build 1.5.0.6)	CD REAGENTS PACK US V1.5 (Build 1.5.0.6)	6014420					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Miscellany	733838	EFLEXIS EXTERNAL BARCODE READER HOLDER	EFLEXIS EXTERNAL BARCODE READER HOLDER	3059867 7008849 8016853							X				
Miscellany	732356	WASTE CONTAINER LABELS V2	WASTE CONTAINER LABELS V2	3044374x2 3044376x2 3055275x2							X				
Miscellany	731805	SOLUTION CONTAINER LABELS	SOLUTION CONTAINER LABELS	3036784 3036786 3044373 3036787 3036788 3030477 3044377x2							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Therm	DG Spin
					AU S	EME A	LATA M	N A							
Miscellany	732355	SOLUTION CONTAINER LABELS V2	SOLUTION CONTAINER LABELS V2	3036784 3036786 3044373 3036787 3036788 3030477 3055275x3							X				
Miscellany	732494	EFLEXIS EARTHQUAKE BRACKET (1 UNIT)	EFLEXIS EARTHQUAKE BRACKET (1 UNIT)	3056039							X				
Miscellany	731803	BOTTLES DOOR LABELS	BOTTLES DOOR LABELS	3044375							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle x is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Miscellany	736209	EFLEXIS PACKAGING v2	EFLEXIS PACKAGING v2	6014022 6021254 3046390 3046388 6021253 8010043 3046391 7006957x2 3018746x3 8010041x4 3046389x26							X				
Miscellany	731483	GRIFOLS BENCH DOOR	GRIFOLS BENCH DOOR								X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Miscellany	233942	PIP&CAT ARM SHIPPING CLAMPS SET (2units)	PIP&CAT ARM SHIPPING CLAMPS SET (2units)	3025575x2								X			
				3026871x2											
				3026872x2											

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eyt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Miscellany	233778	ERYTRA PACKING ASSEMBLY	ERYTRA PACKING ASSEMBLY	6006459 3025575x2 3026871x2 3026872x2 3018746x4 8016824x32 6008463 3031151 3031150								X			
Miscellany	//732907	//DVD W10 IoT v1.1.0 (DG READER NET)	//DVD W10 IoT v1.1.0 (DG READER NET)												

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Therm	DG Spin
					AU S	EME A	LATA M	N A							
Miscellany	//733296	//ERYTRA SOFT v4.3.0 AND WINDOWS 10 ENTERPRISE 2019 LTSC v2.1.0 DVDs	//ERYTRA SOFT v4.3.0 AND WINDOWS 10 ENTERPRISE 2019 LTSC v2.1.0 DVDs												
Miscellany	733839	ERYTRA EXTERNAL BARCODE READER HOLDER	ERYTRA EXTERNAL BARCODE READER HOLDER	3059868 8010041 8016757								X			
Miscellany	731374	PACKAGING	PACKAGING	7004522 6016694 3024771 3018746 7007692									X		
Miscellany	734937	SELF-ADHESIVE TIE (5 units)	SELF-ADHESIVE TIE (5 units)	8013565x5 8013567x5							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Miscellany	730549	SHIPPING BOX CLAMPS (30 UNITS)	SHIPPING BOX CLAMPS (30 UNITS)	3046389x30							X				
Miscellany	731482	GRIFOLS BENCH DOOR HINGES	GRIFOLS BENCH DOOR HINGES								X				
Miscellany	731804	WASTE CONTAINER LABELS	WASTE CONTAINER LABELS	3044374x2 3044376x2 3044377x2							X				
Motors	733951	TREADER MOTOR v2	TREADER MOTOR v2	6019250			1		YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Motors	233723	MOTOR GRIPPER/ PROBE	MOTOR GRIPPER PROBE BCR REAG. STIRRER	3026031	1	1	1	1	YES		X	X			
Motors	233727	MOTOR CENTRIFUGE	MOTOR CENTRIFUGE	6006145 3025282 8013453x4					YES		X	X			
Motors	735099	MOTOR STIRRER v2	MOTOR STIRRER v2	3042519 3042517 6012105 7006480 8016810x3 7008711x2	1	1	1		YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Motors	733699	MOTOR STIRRER DILUTION (Ver. 2)	MOTOR STIRRER DILUTION (Ver. 2)	3025506 3059008 3012942x3 8016810x3 3023731 8013453			1		YES			X			
Motors	233945	MOTOR INCUBATOR (Ver. 2)	MOTOR INCUBATOR (Ver. 2)	3025476 3029362					YES			X			
Motors	733946	MOTOR BCR V3	MOTOR BCR V3	3042519 3045048 7006686 8010108					YES		X				
Motors	233724	MOTOR X-Y	MOTOR X-Y	3009444 3015646 8013486	1	1	1	1			X	X			
Optics	233735	CAMERA (w. optics)	CAMERA (w. optics)	3027549 3026270 3025705					YES		X	X	X		

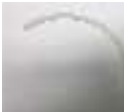
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Tools	734761	COTTON SWAB (100 units)	COTTON SWAB (100 units)								X				
Tools	730672	MICROSPOON	MICROSPOON								X	X			
Tools	733068	LOWER CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL v3	LOWER CENTR._INCUB. ADJUSTMENT TOOL v3								X	X			
Tools	734425	INTERNAL BCR ADJUSTMENT TOOL	INTERNAL BCR ADJUSTMENT TOOL										X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle x is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Tools	732944	IH DIAGNOSTIC KIT	IH DIAGNOSTIC KIT	3057510x4 3066186x4 3066187x4 3030715x1 3044789x6 3003935x12 3003933x1 3055243x6 3013953x1							X	X			
Tools	731586	SWIVEL EYE BOLT (4 units)	SWIVEL EYE BOLT (4 units)	3052452x4							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Tools	233822	FIREWIRE READER CABLE TOOL	FIREWIRE READER CABLE TOOL								X	X			
Tools	233824	DRAWER FRONT PANEL ALIGNMENT TOOL	DRAWER FRONT PANEL ALIGNMENT TOOL									X			
Tools	233938	DILUTION CUP ALIGNMENT TOOL (Ver. 2)	DILUTION CUP ALIGNMENT TOOL (Ver. 2)									X			
Tools	234178	DILUTION CUP/PROBE ADJUSTMENT TOOL	DILUTION CUP ADJUSTMENT TOOL								X				
Tools	230903	DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V	DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V								X	X		X	

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Tools	233878	CARDS DRAWER ADJUSTMENT TOOL (Ver. 2)	CARDS DRAWER ADJUSTMENT TOOL (Ver. 2)								X	X			
Tools	247032	DIGIT. GAUGE DIGITRON MOD.2023	DIGIT. GAUGE DIGITRON MOD.2023								X	X			
Tools	735529	DG GEL CARD HOLDERS (15 units)	DG GEL CARD HOLDERS (15 units)	3047642 x 15							X	X			
Tools	733056	CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL (Ver. 3)	CENTR. INCUB. ADJUSTMENT TOOL (Ver.3)								X	X			
Tools	233819	HORNS ADJUSTMENT TOOL (BCR)	HORNS ADJUSTMENT TOOL (BCR)									X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Tools	233898	OMNIFIT TOOL (for tube connection Ver 2)	OMNIFIT TOOL (for tube connection Ver 2)								X	X			
Tools	232696	SUPER LUBE OIL LUBRICANT 7 ml (5 units)	SUPER LUBE OIL LUBRICANT 7 ml (5 UNITS)								X	X			
Tools	234213	PROBES ADJUSTMENT TOOL (Ver. 4)	PROBES ADJUSTMENT TOOL (Ver. 4)								X	X			
Tools	730671	TARTRAZINE	TARTRAZINE								X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Tools	232697	SUPER LUBE GREASE LUBRICANT 85g (5 units)	SUPER LUBE GREASE LUBRICANT 85g(5 UNITS)								X	X			
Tools	232572	SILICONE TUBE 4x0,76 mm (15 cm)	SILICONE TUBE 4x0,76mm (15cm)				1				X	X			
Tools	230905	TEMP. PROBE PT-100 FOR CARDS - MiniDIN 4v	TEMP. PROBE PT100 FOR CARDS - MINIDIN 4V								X	X		X	
Tools	230902	THERMAL SILICON PASTE (90 ml)	THERMAL SILICONE PASTE (90ml.)								X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Tools	732540	THREAD LOCKER (50 ml)	THREAD LOCKER								X				
Tools	730689	READER SUPPORT PLATE	READER SUPPORT PLATE									X			
Tools	234212	READER ADJUSTMENT TOOL- SUPPORT (Ver. 2)	READER ADJUSTMENT TOOL- SUPPORT (Ver. 2)								X	X			
Tools	731398	READER ADJUSTMENT TOOL- PATTERN v2	READER ADJUSTMENT TOOL- PATTERN v2								X	X	X		

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Tools	233901	PROBES_WAGON ADJUSTMENT TOOL (Ver. 2)	PROBES WAGON ADJUSTMENT TOOL (Ver. 2)									X			
Tools	233908	PAINT SEALANT (12 ml)	PAINT SEALANT (12ml)								X	X			
Tools	733701	PROBES ADJUSTMENT TOOL (holder 16 mm)	PROBES ADJUSTMENT TOOL (holder 16 mm)								X	X			
Tools	233821	CENTRIFUGE ROTOR ADJUSTMENT TOOL	CENTRIFUGE ROTOR ADJUSTMENT TOOL								X	X			
Tools	232596	PROBE UNBLOCKER TOOL	PROBE UNBLOCKER TOOL								X	X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eyt ra	DG Reader Net	DG Therm	DG Spin
					AU S	EME A	LATA M	N A							
Tools	234216	PROBE POSITIONING VERIFICATION TOOL	PROBE POSITIONING VERIFICATION TOOL								X				
Tools	233817	PRESSURE SENSOR ADJUSTMENT TOOL	PRESSURE SENSOR ADJUSTMENT TOOL								X	X			
Wiring	234189	WIRING SET FLUIDICS (FLU)	WIRING SET FLUIDICS (FLU)	3036658 x4 3042813					YES		X				
Wiring	233793	X WIRING/CHAIN (CAT)	X WIRING_CHAIN (CAT)	6006597			1		YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Wiring	233972	CAT RIBBON CABLES (B-CAT_XY/TOW/Z)	CAT RIBBON CABLES (B-CAT_XY/TOW/Z)	3026573x3					YES			X			
Wiring	234110	X WIRING/CHAIN (CAT)	X WIRING/CHAIN (CAT)	6012253					YES		X				
Wiring	234109	X WIRING/CHAIN (PIP)	X WIRING/CHAIN (PIP)	6012248			1		YES		X				
Wiring	234152	WIRING SET S/R DRAWER (SRE)	WIRING SET S/R DRAWER (SRE)	3020845x4 3033524x2 3042815					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Wiring	734938	WIRING SET PRE- BASE (PBSE)	WIRING SET PRE-BASE (PBSE)	3042811 x4 3042965 3042805 3020129 3042808 3042806					YES		X				
Wiring	234131	WIRING SET PIP	WIRING SET PIP	3042819 x2 3036658 x3 3020845					YES		X				
Wiring	234157	Y WIRING/CHAIN (CAT)	Y WIRING/CHAIN (CAT)	6012255					YES		X				
Wiring	233794	Y WIRING/CHAIN (CAT)	Y WIRING _CHAIN (CAT)	6006594			1		YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Wiring	733266	CAMERA FIREWIRE CABLE (READ_FW)	CAMERA FIREWIRE CABLE (READ_FW)	3026115							X	X			
Wiring	233906	WIRING SET LOWER MODULE	WIRING SET LOWER MODULE	3028761 x 5 3029016					YES			X			
Wiring	234006	BKPL-CARD DRAWERS FLAT CABLE (B-CM_LOCK)	BKPL-CARD DRAWERS FLAT CABLE (B-CM_LOCK)	3026113					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Wiring	234198	WIRING SET GEN	WIRING SET GEN	3042810					YES		X				
				3042891											
				3042812											
				3042818											
				3042809											
				3042831											
				3042837											
				3033782											
				3042838											
				3042832											
				3042833											
				3042839											
				3042840											
				3008789											
				x2											
				3020852											
				3042893											
				3042894											
				3042841											
				3042842											
				3042895											
				3042896											
				3042897											
				3042834											

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Wiring	233788	X WIRING/CHAIN (PIP)	X WIRING_CHAIN (PIP)	6006596			1		YES			X			
Wiring	233791	BCR WIRING/CHAIN (PIP)	BCR WIRING_CHAIN (PIP)	6006591			1		YES			X			
Wiring	233792	BCR FLAT CABLE (PIP)	BCR FLAT CABLE (PIP)	3008824			1	1	YES			X			
Wiring	233796	BCR FLAT CABLE (CAT)	BCR FLAT CABLE (CAT)	3026403				1	YES			X			

[illegible]

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Wiring	735639	WIRING SET ELE V4	WIRING SET ELE V4	3017611 3062831 8010015 3042843 3042844 3044600 3042846 3042847 3042848 3042849 3042835 3042836 3042850 3033673 3042898 3033687 3033686 3060548 3065750					YES		X				
Wiring	234211	WIRING SET BRX	WIRING SET BRX	3036658 x2 3042816 x2 3020846 x2 3037747					YES		X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Therm	DG Spin
					AU S	EME A	LATA M	N A							
Wiring	234105	WIRING SET INCUBATOR (INC)	WIRING SET INCUBATOR (INC)	3042807 x3 3020845 x3 3042814 3042966 x3 6012285					YES		X				
Wiring	234164	WIRING SET CAT	WIRING SET CAT	3042819 3036658 3020845					YES		X				
Wiring	234209	WIRING SET DOOR (PTA)	WIRING SET DOOR (PTA)	3033780 3044628					YES		X				
Wiring	733914	BOTTLE DRAWER CABLES	BOTTLE DRAWER CABLES	3026347 3026348								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Wiring	235281	WIRING SET TOUCHSCREEN (PAN)	WIRING SET TOUCHSCREEN (PAN)	3033775 3033863 3033862 3043073 3034254					YES		X				
Wiring	234163	GRIPPER WIRING/CHAIN (CAT)	GRIPPER WIRING/CHAIN (CAT)	6012254			1	1	YES		X				
Wiring	233875	VGA CABLE ADAPTER M-M	VGA CABLE ADAPTER M-M	3026981 3027782					YES			X			
Wiring	233980	REAGENT DRAWER RIBBON CABLES (B-STIR1/2)	REAGENT DRAWER RIBBON CABLES (B-STIR1/2)	3027614x2					YES			X			
Wiring	233802	REAG. STIRRER FLAT CABLE (RM)	REAG. STIRRER FLAT CABLE (RM)	3026141					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Wiring	230909	POWER CORD with lock	POWER CORD with lock v2	3048137							X	X	X	X	X
Wiring	230910	POWER CORD US with lock	POWER CORD U.S.A. with lock v2	3048136							X	X	X	X	X
Wiring	730670	PIP Z CHAIN ADAPTOR PLATE	PIP Z CHAIN ADAPTOR PLATE	7006445							X				
Wiring	//734962	//PC - ELE-BCKPL511 AND READER USB CABLE	PC - ELE-BCKPL511 AND READER USB CABLE	3060531 3062831 3017611 8010015											

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Efle x is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Wiring	735638	PC-ELE-BCKPL511 AND READER USB CABLE V2	PC-ELE-BCKPL511 AND READER USB CABLE v2	3017611 3062831 3065750 8010015							X				
Wiring	233443	USB 2.0 CABLE 1.8 M (TOUCHSCREEN, ELECTRONICS CONNECTORS COVER AND PCA RD IL421)	TOUCHSCREEN USB CABLE	3026455					YES		x	X			
Wiring	233961	TOUCHSCREEN USB EXTENSION CABLE M/F 1.5 m	TOUCHSCREEN USB EXTENSION CABLE M/F 1.5m	3029399					YES			X			
Wiring	233831	TOUCHSCREEN v2&v3 POWER CABLE	TOUCHSCREEN v2 POWER CABLE	3026523 3029407					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Wiring	234129	Y WIRING/CHAIN (PIP)	Y WIRING/CHAIN (PIP)	6012250			1		YES		X				
Wiring	233897	LEVEL DETECTOR WIRING/CHAIN (PIP) (Impact detector)	LEVEL DETECTOR WIRING/CHAIN (PIP)(i. d)	6009226	1	1	1	1	YES			X			
Wiring	234127	LEVEL DETECTOR WIRING/CHAIN (PIP)	LEVEL DETECTOR WIRING/CHAIN (PIP)	6012251	1	1		1	YES		X				
Wiring	233838	KEYBOARD R/S DRAWER FLAT RIBBON CABLE	KEYBOARD R/S DRAWER FLAT RIBBON CABLE	3026673					YES			X			
Wiring	233837	KEYBOARD CARDS DRAWER FLAT RIBBON CABLE	KEYBOARD CARDS DRAWER FLAT RIBBON CABLE	3026672					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Wiring	233716	INCUBATOR WIRING/CHAIN	INCUBATOR WIRING CHAIN	3026142					YES			X			
Wiring	233917	INCUBATOR (2/3) CABLE (B_INC2/B_INC3)	INCUBATOR (2/3) CABLE (B_INC2/B_INC3)	3008749					YES			X			
Wiring	233918	INCUBATOR (1/4) CABLE (B_INC1/B_INC4)	INCUBATOR (1/4) CABLE (B_INC1/B_INC4)	3026743					YES			X			
Wiring	233795	GRIPPER WIRING_CHAIN	GRIPPER WIRING_CHAIN	6006592					YES			X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Therm	DG Spin
					AU S	EME A	LATA M	N A							
Wiring	//734983	//GENERAL CABLING V3	//GENERAL CABLING V3	3042848 3048203 3048204 3048639 3048209 3048208 3048206 3048205 3048207 3052453 3064048 3016266					YES						
Wiring	233919	FLUIDICS/CENTRIF CABLE (B-FLU B- CEN1/2)	FLUIDICS/CENTRIF CABLE (B-FLU B- CEN1/2)	3026744					YES			X			
Wiring	233957	ENCODER CABLE MOTOR X PIP/CAT	ENCODER CABLE JST XA 1496-5-039	3020846					YES			X			

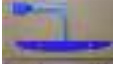
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Therm	DG Spin
					AU S	EME A	LATA M	N A							
Wiring	733259	ELE MODULE WIRING SET V1	ELE MODULE WIRING SET V1	3026527 3026528 3026529 3026530 x3 3026531 3026533 3026534 x2 3026535x2 3026555 3026575 3026576					YES			X			
Wiring	730598	EARTHING STRAP 150x13 (10 units)	EARTHING STRAP 150x13 (10 units)	3028967x10								X			
Wiring	730597	EARTHING STRAP 75x13 (10 units)	EARTHING STRAP 75x13 (10 units)	3025764x10								X			

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Wiring	233991	USB 3.0 CABLES AND CONNECTORS SET	USB 3.0 CABLES AND CONNECTORS SET	3043150x4 3043149x4					YES			X			
Wiring	233874	VGA CABLE M-F	VGA CABLE M-F	3031001					YES			X			
Wiring	233789	Y WIRING/CHAIN (PIP)	Y WIRING_CHAIN (PIP)	6006595			1		YES			X			
Mechanics	735844	INCUBATOR BLOCK (1 unit)	INCUBATOR BLOCK (1 unit)	3042632							X				

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	735851	READER ILLUMINATION SYSTEM v4	READER ILLUMINATION SYSTEM v4	6017969 3025301 3068091 3024413 3017417 x2 8013560 8010015 x10					YES		X	X	X		
Mechanics	735852	ILLUMINATION PATTERN DGGEL v2	ILLUMINATION PATTERN DGGEL v2	3068091					YES		X	X	X		
Electronics	230665	KEYBOARD DG-SPIN	KEYBOARD DG-SPIN	3022945					YES						X
Electronics	230668	PCA DC MOTOR CONTROL (with screws)	PCA DC MOTOR CONTROL (with screws)	3022341 8010008x2 8010010x2					YES						X

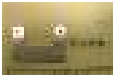
Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	230670	PCA UNBALANCE DETECTOR (with screws)	PCA UNBALANCE DETECTOR (with screws)	3022347 8010015x2 8013529x2					YES						X
Mechanics	230672	CARD HOLDERS (24 units)	CARD HOLDERS (24 units)	3023148x24											X
Mechanics	230674	HOUSING LATERAL PANEL (W/SCREWS)	HOUSING LATERAL PANEL (W/SCREWS)	"3023156 8019878x2"					YES						X
Mechanics	230676	LID JOINT (with screws)	LID JOINT (with screws)	3023155 8019878x8											X
Mechanics	230677	LID SPRING	LID SPRING	3023151											X
Mechanics	230678	MOTOR COVER (with screws)	MOTOR COVER (with screws)	3023145 8010015x3											X

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	230679	SHOCK ABSORBERS (3 units with nuts)	SHOCK ABSORBERS (3 units with nuts)	3023132x3 8010122x3 8013507x3 8016977x3											X
Motors	230680	MOTOR (without unbalance sensor)	MOTOR (without unbalance sensor)	3023144 3023213 8010122x3 6009638					YES						X
Electronics	230683	MAINS SOCKET WITH FILTER (with screws)	MAINS SOCKET WITH FILTER (with screws)	3026961 8013505x2 8016794x2					YES						X
Mechanics	230686	LID (w_joint, latch & screws) (Ver.2)	LID (w_joint, latch & screws) (Ver.2)	6009639 3018918											X

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	736207	HOUSING DG SPIN (w/screws) v3	HOUSING DG SPIN (w/screws) v3	3040144 8019878x6 6021184 3023205											X
Mechanics	230688	HINGE AXIS	HINGE AXIS	3023150											X
Electronics	230818	FUSE 2A 250V MAINS SOCKET	FUSE 2A	7006371										X	X
Tools	230881	TEMPERATURE PROBE PT-100 FOR CARDS	TEMPERATURE PROBE PT-100 FOR CARDS	3003859					YES					X	
Electronics	230886	KEYBOARD DG THERM	KEYBOARD DG THERM	3022944					YES					X	
Electronics	230887	PCA DISPLAY (with screws and plates)	PCA DISPLAY (with screws and plates)	3022943 3023085x2 8019878x2					YES					X	X

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	230889	THERMAL BLANKET (w/ wire & plug)	THERMAL BLANKET (w/ wire & plug)	3023086					YES					X	
Mechanics	230890	DISPLAY WINDOW	DISPLAY WINDOW	3023087					YES					X	X
Mechanics	230891	ELEVATION CARDS PLATE PUSHER	ELEVATION CARDS PLATE PUSHER	3023083										X	
Mechanics	230892	GROUNDING PLATE (w/ screws)	GROUNDING PLATE (w/ screws)	6005691 8010027x2 8013527 8013530x2										X	
Mechanics	230893	HEATING BLOCK DG THERM (w/ screws and th. blanket)	HEATING BLOCK DG THERM (w/ screws and th. blanket)	3023086 3023092x2 6005240 8010027x2 8013530x2										X	

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	230894	HEATING BLOCK DG THERM SEALING JOINT	HEATING BLOCK DG THERM SEALING JOINT	3023091										X	
Mechanics	230896	LID DG THERM	LID DG THERM	3023082										X	
Mechanics	230897	RS-232 PROTECTIVE COVER	RS-232 PROTECTIVE COVER	3018688										X	X
Mechanics	230898	SCREWS & WASHERS SET DG THERM	SCREWS & WASHERS SET DG THERM	8010027x4 8013527 8013530x4 8019878x6										X	

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Mechanics	230899	THERMAL SILICON ADHESIVE TAPE (2 units)	THERMAL SILICON ADHESIVE TAPE (2 units)	8010027x4 8013527 8013530x4 8019878x6										X	
Wiring	230900	CABLE PCA DISPLAY to PCA INCUBATOR	CABLE PCA DISPLAY to PCA INCUBATOR	3023111										X	
Tools	230904	TEMP. PROBE PT-100 FOR TUBES - MiniDIN 4v	TEMP. PROBE PT-100 FOR TUBES - MiniDIN 4v	3029393										X	
Mechanics	230906	MAINS CONNECTOR FIXING PLATE (w screws)	MAINS CONNECTOR FIXING PLATE (w screws)	3023082										X	

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Mechanics	230907	HOUSING DG THERM (w/screws) VER. 2	HOUSING DG THERM (w/screws) VER. 2	3040145 8019878x4 3021122 3023112										X	
Electronics	230908	PCA INCUBATOR DG THERM (w/screws) v2	PCA INCUBATOR DG THERM (w/ screws) v2	6014618 3023092x2 8010027x3 8013527 8013530x3					YES					X	
Wiring	231943	RS-232 COMMUNICATION CABLE	RS-232 CONNECTION CABLE	3015143										X	X
Electronics	240917	CLOCK LITHIUM BATTERY	CLOCK LITHIUM BATTERY	3024518					YES					X	X

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Read er Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Miscellany	732417	PACKAGING (DG THERM)	PACKAGING (DG THERM)	3023112 3021122 3018746 3024772										X	
Miscellany	736208	PACKAGING (DG SPIN) v2	PACKAGING (DG SPIN)	6021184 3023205 3018746 3024771											X
Wiring	734239	CABLES KIT v3 (WITH GROUND SCREW)	CABLES KIT v3 (WITH GROUND SCREW)	6019696 8013501 8013527 8013530 8010027					YES						X
Electronics	734810	LID ELECTRIC LOCK SWITCH v3 (WITH SCREWS/WIRE)	LID ELECTRIC LOCK SWITCH v3 (WITH SCREWS/WIRE)	"3061252 3018918 8013464x2 3039445 3063397 8018618					YES						X

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warra nty covered parts	Picture	Eflex is	Eryt ra	DG Reader Net	DG Ther m	DG Spi n
					AU S	EME A	LATA M	N A							
Electronics	735094	POWER SUPPLY v2 (W/SCREWS)	POWER SUPPLY v2 (W/SCREWS)	3062530 8013515x2 8016810x2					YES						X
Wiring	735095	CABLES KIT v4 (WITH GROUND SCREW)	CABLES KIT v4 (WITH GROUND SCREW)	6020232 8013501 8013527 8013530 8010027					YES						X
Electronics	735096	POWER SUPPLY UPGRADE KIT	POWER SUPPLY UPGRADE KIT	3062530 3062759 3062760 6020257 7002361 8010015x2 8013515x2 8016810x2											X

Module	Part number	Description	Logistics description	Drawing reference	Recomm. stock				Warranty covered parts	Picture	Eflex is	Erytra	DG Reader Net	DG Therm	DG Spin
					AUS	EMEA	LATAM	NA							
Electronics	735344	PCA DISTRIBUTION (w/screws) (Ver. 3)	PCA DISTRIBUTION (w/screws) (Ver. 3)	6020272 8010008x2 8010010x2					YES						X
Mechanics	735760	DG SPIN HEAD v2	DG SPIN HEAD v2	6008821 3066962 3024613 3023148 8010029											X
Mechanics	735916	RACK ENTRY TOP PLATE	RACK ENTRY TOP PLATE	6012102 x1 7004681 x2 8013452 x5							X				

Eflexis Revision history table

Revision	Changes with respect to previous version.
July 2025	Cancelled codes 233943 READER FOAM
May 2025	Added codes 736238 PROBE v2 Cancelled codes 732346 EFLEXIS PACKAGING 234142 PROBE
March 2025	Added codes 736169 RAM DDR4 8GB SO-DIMM KINGMAX (2 units)v2 735916 RACK ENTRY TOP PLATE Cancelled codes 735635 RAM DDR4 8GB SO-DIMM KINGMAX (2 units)
January 2025	Added codes 735899 RAM DDR3L 8GB SO-DIMM 735910 ASPIRATION PUMP B (VPB) 735911 ASPIRATION PUMP A (VPA) Modified codes 734985 DAMPER SYSTEM v2: new drawing references and picture. Cancelled codes 734048 RAM DDR3 2GB SO-DIMM KINGMAX (2 units) 735098 ASPRIATION PUMP V4

Revision	Changes with respect to previous version.
November 2024	Added codes 735852 ILLUMINATION PATTERN DGCEL v2 735851 READER ILLUMINATION SYSTEM v4 735844 INCUBATOR BLOCK (1 unit) Cancelled codes 733042 READER ILLUMINATION SYSTEM V3 233710 ILLUMINATION PATTERN DGCEL 733043 CD IH REAGENTS PACK US V2.1
September 2024	Added recommended stock columns for the different regions.
August 2024	Added codes 735097 POLYURETHANE TUBE 6x4mm (1m) 735098 ASPIRATION PUMP V4 Modified codes 233960 WASTE PUMP (Liquid)(Ver. 2)(PM2): spare part references and picture changed 235206 STEPPER MOTOR: spare part references changed. 732944 IH DIAGNOSTIC KIT: spare part references changed. 734354 FLUIDIC CONNECTION BLOCK (4 units): spare part references and picture changed Cancelled codes 733676 ASPIRATION PUMP V3

Revision	Changes with respect to previous version.
May 2024	Added codes 735280 PROBE GUIDE V2 735568 BOTTLE HOLDER (FULL) V2 735570 PROBE WAGON V2 735571 POBE GUIDE V3 735615 PIP FITTING SUPPORT PLATE 735616 CARDS WASTE LOAD CELL SCREWS (10 units) 735634 SSD 256 GB M.2 735635 RAM DDR4 8GB SO-DIMM KINGMAX (2 units) 735636 MOTHERBOARD MINI-ITX ASROCK IMB-1003J 735638 PC_ELE_BCKPL511 AND READER USBCABLE V2 735639 WIRING SET ELE V4 735529 DG GEL CARD HOLDERS (15 units) 735569 BOTTLE HOLDER ADJUSTMENT TOOL Cancelled codes 734962 PC-ELE-BCKPL511 AND READER USB CABLE 734352 BOTTLE HOLDER (FULL) 734963 WIRING SET ELE V3 730669 PROBE GUIDE 734235 MOTHERBOARD V3 (MINI_ITX ASROCK IBM_150 734607 FLUIDIC CONNECTION BLOCK (1 unit) 732861 EFLEXIS SOFTWARE v1.2.1 & WES7 EF v1.3.0

Revision	Changes with respect to previous version.
February (II) 2024	Added codes 735206 ERYTRA SOLUTION LINES FILTERS SET 735521 TOUCHSCREEN V3 735528 BUTTERFLY NUTS FOR FLU COVER (10 units) 735530 READER BOX 735515 SINGLE/DUAL MODE CONN: (MANIFOLD, 10 u.) Deleted codes 732295 TOUCHSCREEN V2 734834 MK VALVE (w. o-ring)

Revision	Changes with respect to previous version.		
February 2024	Added codes 735365 MK VALVE (w. o-ring) v2 735494 FLU FRONT COVER 735499 MANIFOLD v5 735501 TUBE CONNECTION CPC GS320 (w.o-ring, 10) 735500 TUBE CONNECTION CPC GS520 (w.o-ring, 10) 735514 MANIFOLD SPRING CLIPS D7 (20 units) Other Changed drawing references and picture of item 734039 Changed drawing references for the following items:		
	Codes	Deleted reference	Added reference
	233906	6007924	3028761 x 5 3029016
	234105	6012263	3042807 x3 3020845 x3 3042814 3042966 x3 6012285
	234131	234131	3042819 x2 3036658 x3 3020845
000166 - 1.8.0	234152	6012264	3020845 x4 3033524 x2 3042815

Revision	Changes with respect to previous version.
October (23/10) 2023	Added codes 735331 HORNS LATERAL RIGHT 735332 HORNS LATERAL LEFT 735334 INTERMEDIATE PLATE DC 735335 DILUTION CUP STRUCTURE (WHOLE) v4 Deleted codes 735000 DILUTION CUP STRUCTURE (WHOLE) v3 Other Changed picture for P/N 233798 MULTILUMEN TUBING 2,3x3,6 (10m): added labels to identify the fluidic circuit related to the new tubing being installed. Modified P/N 234161 LEAD SCREW (GRIPPER), adding two washers with code 3016901. Changed picture for P/N 732961 DISPENSING PUMP V2: added labels to identify the fluidic circuit related to the new pump being installed. Changed picture for P/N 733676 ASPIRATION PUMP V3: added labels to identify the fluidic circuit related to the new pump being installed. Changed picture for P/N 734000 PRESSURE SENSOR CABLE: added labels to identify the sensor related to the new cable being installed.
September (18/09) 2023	Added codes 735237 EFLEXIS SOLUTION LINES FILTERS SET 735247 CONTAINERS DOOR HINGES Deleted codes Changed picture for P/N 230903 DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V (added instructions for use in English language). Correction of minor typographic errors

Revision	Changes with respect to previous version.
June (27/06) 2023	Added codes 230903 DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V 247032 DIGIT. GAUGE DIGITRON MOD.2023 734459 PCA ELE-BCKPL511_V2 734999 DILUTION STATION CUP EFLEXIS v2 735000 DILUTION CUP STRUCTURE (WHOLE) v3 735001 DILUTION CUP HOLDER v2 735169 PROBE WASHER PIN Deleted codes 234171 PCA ELE-BCKPL511 730925 DILUTION CUP HOLDER 732031 DILUTION STATION CUP EFLEXIS 733168 SYRINGE LOCK 733672 DILUTION CUP STRUCTURE (WHOLE) v2
March (22/03) 2023	Added codes 232572 SILICONE TUBE 4x0,76mm (15cm) 735042 SCREW M2x4 DIN 912 (10 units) 735099 MOTOR STIRRER v2 735102 EFLEXIS PIP GROUNDING PLATE v2 Deleted codes 234147 MOTOR STIRRER 734837 EFLEXIS PIP GROUNDING PLATE

Revision	Changes with respect to previous version.
December (09/12) 2022	Added codes 734962 PC - ELE-BCKPL511 AND READER USB CABLE 734963 WIRING SET ELE V3 734985 DAMPER SYSTEM v2 734986 MANIFOLD v4 734998 PCA PIP-BCR511 v2 Deleted codes 234128 BCR WIRING/CHAIN (PIP) 234217 PC-PCA ELE-BCKPL511 USB CABLE SET 731840 DAMPER SYSTEM 733674 MANIFOLD v3 234124 PCA PIP-BCR511 733945 WIRING SET ELE V2 Other Changed picture for P/N 734747 HORNS TRANSMISSION GEAR PINION and 734748 HORNS TRANSMISSION GEAR CROWN (setscrew is now partially screwed so that it is not lost when opening the box).
November (07/11) 2022	Added codes 734937 SELF-ADHESIVE TIE (5 units) 734938 WIRING SET PRE-BASE (PBSE) 734948 TOOTHED GUIDE v2 (PIP BCR) Other 734352 BOTTLE HOLDER (FULL): Code 3017260 replaced by 7009680

Revision	Changes with respect to previous version.
October (07/10) 2022	Added codes 734834 MK VALVE (w. o-ring) 734835 MK SYRINGE 1ml 734915 TREADER SUPPORT (GUIDE) Deleted codes 233952 HAMILTON SYRINGE 1ml (Ver. 2) 233953 HAMILTON VALVE (Ver. 2) (w. o-ring) Other Changed picture for P/N 734238 DOOR ELECTRIC LOCK v2 (cord removed)
September (07/09) 2022	Added codes 734759 GRIPPER v2 734760 MIRROR HOLDER v2 734761 COTTON SWAB (100 units) 734811 CAT MIRROR COVER 734837 EFLEXIS PIP GROUNDING PLATE Deleted codes 234158 GRIPPER 234165 MIRROR HOLDER Other Added cap for the diluent bottle in diagnostic kit 732944 and updated its picture

Revision	Changes with respect to previous version.
May (23/05) 2022	Added codes 730588 SCREW M3x6 HEX.A2 (100 units) 734657 BARCODE READER QW2520-BKK1S(W.USB CABLE) 734715 CAT ROBOTIC HEAD LEFT COVER 734716 CAT ROBOTIC HEAD RIGHT COVER 734717 CAT ROBOTIC HEAD REAR COVER 734739 SCREW M3x4 DIN7985 (100 units) 734740 HEXAGONAL SPACER 4mm M3 (50 units) 734745 CAT ROBOTIC HEAD LEFT COVER v2 734747 HORNS TRANSMISSION GEAR PINION 734748 HORNS TRANSMISSION GEAR CROWN Deleted codes 733154 BARCODE READER QW2420-BKK1S (W.USB CABLE)
April (19/04) 2022	Added codes 733840 GEAR DOUBLE WHEEL (X,Y) v2 734424 CAMERA SUPPORT PLATE Deleted codes 233860 GEAR DOUBLE WHEEL (X,Y) 732098 LOCTITE SEALANT 5331 (100ml) 733954 ANTI-STICK LUBRICANT SPRAY Other P/Ns 732098 LOCTITE SEALANT 5331 (100ml) and 733954 ANTI-STICK LUBRICANT SPRAY have been deleted because of being considered as dangerous goods by the logistics company in charge of transportation. 734355 WASTE CONTAINER COUPLING SYSTEM V3: Code 3039481 replaced by 3021981.

Revision	Changes with respect to previous version.
March (09/03) 2022	Added codes 734235 MOTHERBOARD v3 (MINI-ITX ASROCK IBM-150) 734607 FLUIDIC CONNECTION BLOCK (1 unit) 734608 SOLENOID GRIPPER EFLEXIS Deleted codes 231806 SOLENOID GRIPPER 733901 MOTHERBOARD v2 (MINI-ITX ASROCK IBM-150) Other Added code 3042599 and updated picture for P/N 234161 LEAD SCREW (GRIPPER)

Revision	Changes with respect to previous version.
December (20/12) 2021	Added codes 232760 SCREW ALLEN M2X10 DIN 912 (100 UNITS) 730602 SCREW M2x6 DIN 912 (100 units) 734279 MONITOR ARM BASE 734350 CONTAINER WASTE v3 734351 CONTAINER SOLUTIONS v3 734352 BOTTLE HOLDER (FULL) 734354 FLUIDIC CONNECTION BLOCK (4 units) 734355 WASTE CONTAINER COUPLING SYSTEM V3 734356 SOL. CONTAINER COUPLING SYSTEM V3 734357 POKA-YOKE LOCK (10 units) Deleted codes 235275 QUICK COUPLING (Female) 731983 CONTAINER HOLDER PIVOTING PLATE 732357 CONTAINER WASTE V2 732358 CONTAINER SOLUTIONS V2 732366 SOL. CONTAINER COUPLING SYSTEM V2 732439 WASTE CONTAINER COUPLING SYSTEM V2 732987 CONTAINER HOLDER PIVOTING PLATE v2 Other 733055 MONITOR ARM: Code 8010056 replaced by 8010055.

Revision	Changes with respect to previous version.
October (11/10) 2021	Added codes 733945 WIRING SET ELE V2 733954 ANTI-STICK LUBRICANT SPRAY (400 ml) 733965 CENTRIFUGE ROTOR V2 734038 S/R DRAWER LOCK SYSTEM V3 734039 FLUIDICS BOX ASSEMBLY (PLATES) 734048 RAM DDR3 2GB SO-DIMM KINGMAX (2 units) 734220 BCR MIRROR HOLDER SCREWS (40 units) 734238 ELECTRIC LOCK v2 Deleted codes 233701 DRAWER ELECTRIC LOCK 233732 CENTRIFUGE ROTOR 234172 WIRING SET ELE 731346 S/R DRAWER LOCK SYSTEM V2 731399 RAM DDR3L 2GB SO-DIMM Innodisk (2 units)
August (02/08) 2021	Added codes 733838 EFLEXIS EXTERNAL BARCODE READER HOLDER 733913 HORNS TRANSMISSION GEAR ASSEMBLY 733946 MOTOR BCR V3 733951 TREADER MOTOR v2 734000 PRESSURE SENSOR CABLE Deleted codes 732037 MOTOR BCR V2 234132 TREADER MOTOR Other Updated picture and changed code 3036865 by 3034254 for P/N 235281 WIRING SET TOUCHSCREEN (PAN) Changed references to DG Technical Support to DIG PDS

Revision	Changes with respect to previous version.
July (02/07) 2021	Added codes 733901 MOTHERBOARD v2 (MINI-ITX ASROCK IBM-150) 733912 SCREW M3x10 A2 ISO 7380 FL (20 units) Deleted codes 233989 MOTHERBOARD MINI-ITX ASROCK IMB-150 Other Updated picture for P/N 234208 DOOR (only metacrilate) Added codes 3012574 (x2), 7006360 and 7004696 (x2) and updated picture for P/N 732987 CONTAINER HOLDER PIVOTING PLATE v2 (change applies from lot 21005 onwards)
April (19/04) 2021	Added codes 733672 DILUTION CUP STRUCTURE (WHOLE) v2 733673 TUBE CONNECTION 6x4 tube v2 (25 units) 733674 MANIFOLD v3 733675 ASPIRATION PUMP SPRING CLIP (40 units) 733676 ASPIRATION PUMP V3 733677 FLUIDIC FITTINGS UPGRADE KIT 733835 T RACORD HT-5 CPC (20 units) 733836 SINGLE/DUAL MODE CONNECTOR (10 units) 733837 CAMERA COVER Deleted codes 234191 TUBE CONNECTION (for 6x4 tube) 731839 MANIFOLD v2 732125 ASPIRATION PUMP V2 733525 DILUTION CUP STRUCTURE (WHOLE) Other Updated picture for P/N 233817 PRESSURE SENSOR ADJUSTMENT TOOL

Revision	Changes with respect to previous version.
March (19/03) 2021	Added codes 733701 PROBES ADJUSTMENT TOOL (holder 16mm)
November (27/11) 2020	Added codes 733519 O-RING 5.28x1.78 (10 units) 733525 DILUTION CUP STRUCTURE 733559 MAGNETIZED ECCENTRIC WHEEL Other Added screws for the assembly of S/P 234147 MOTOR STIRRER and 234148 PCA SRE-MOT511.
October (08/10) 2020	Added codes 733370 PLASTIC BALL BEARING BB-626-B180-30-ES 733350 READER MODULE SUPPORT PLATE Other Added circlips for the assembly of S/P 234153 PLASTIC BALL BEARING BB-6001-B180-30-ES on dilution cup structure
July (10/07) 2020	Added codes 732987 CONTAINER HOLDER PIVOTING PLATE v2 Other Updated Logistics (SAP) description for P/N 233987 HARD DISK SSD 256GB mSATA
June (08/06) 2020	Added codes 733168 SYRINGE LOCK
May (28/05) 2020	Added codes 733154 BARCODE READER QW2420-BKK1S (W.USB CABLE)

Revision	Changes with respect to previous version.
May (21/05) 2020	Added codes 731344 CARDS HOLDER (CENTR.) 12 UNITS (VER.3) 733043 CD IH REAGENTS PACK US V2.1 733266 CAMERA FIREWIRE CABLE (READ_FW) Deleted codes 232048 CARDS HOLDER (CENTR.) 12 UNITS (VER. 2) 732036 CD IH REAGENTS PACK US V2.0 Other Correction of minor typographic errors
May (08/05) 2020	Added codes 733153 WINDOWS 10 ENTERPRISE 2019 LTSC LICENSE 733159 EFLEXIS SOFTWARE v1.2.2 AND WINDOWS 10 ENTERPRISE 2019 LTSC v2.0.1 DVDs
March (31/03) 2020	Added codes 733055 MONITOR ARM 733056 CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL (Ver. 3) 733068 LOWER CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL v3 730582 CABLE GROUNDING CLAMP 3.5mm (10 units) Deleted codes 233879 CENTRIFUGE/INCUBATOR ADJUSTMENT TOOL (Ver. 2) Other Updated picture for P/N 233811 PCB GEN-LED59

Revision	Changes with respect to previous version.
February (10/02) 2020	Added codes 732494 EFLEXIS EARTHQUAKE BRACKET (1 UNIT) 732911 PCA RD-IL421 v3 733042 READER ILLUMINATION SYSTEM V3 Deleted codes 731680 PCA RD-IL421 v2 731679 READER ILLUMINATION SYSTEM v2 Other Updated templates in 731337 FUNNEL EXTERNAL WASTE CARDS (1 UNIT)
January (23/01) 2020	Added codes 732944 IH DIAGNOSTIC KIT 730627 FRONTAL GUIDE FOR RACKS 733016 COSMETIC COVER OF VERTICAL GUIDE RACKS 730605 SCREW M4x8 DIN 912 (100 units)
November (12/11) 2019	Added codes 732891 LATERAL PANEL (metacrilate) 732961 DISPENSING PUMP V2 Deleted codes 234187 DISPENSING PUMP Other 234217 PC-PCA ELE-BCKPL511 USB CABLE SET: Code 3017610 replaced by 3017611. 234172 WIRING SET ELE: Added code 3017611.

Revision	Changes with respect to previous version.
September (27/09) 2019	Added codes 732295 TOUCHSCREEN V2 732861 EFLEXIS SOFTWARE v1.2.1 & WES7 EF v1.3.0 Deleted codes 235278 TOUCHSCREEN 732294 EFLEXIS SOFTWARE v1.2.1 & WES7 EF v1.2.0
August (28/08) 2019	Added codes 233809 PUSHBUTTON 732540 THREAD LOCKER (50ml)
July (03/07) 2019	Added codes 732346 EFLEXIS PACKAGING 730700 RUBBER FOOT V2 Deleted codes 231892 RUBBER FOOT
June (28/06) 2019	Added codes 232596 PROBE UNBLOCKER TOOL 732355 SOLUTION CONTAINER LABELS V2 732356 WASTE CONTAINER LABELS V2 732366 SOL. CONTAINER COUPLING SYSTEM V2 732357 CONTAINER WASTE V2 732358 CONTAINER SOLUTIONS V2 732439 WASTE CONTAINER COUPLING SYSTEM V2 Deleted codes 234177 SOL. CONTAINER COUPLING SYSTEM 234179 CONTAINER SOLUTIONS 234180 CONTAINER WASTE 235277 WASTE CONTAINER COUPLING SYSTEM

Revision	Changes with respect to previous version.
April (26/04) 2019	Added codes 731981 CONTAINER PRESENCE DETECTION SYSTEM (1u) Deleted codes 234195 PCA CON-DET511
April (12/04) 2019	Added codes 732294 EFLEXIS SOFTWARE v1.2.1 & WES7 EF v1.2.0 Deleted codes 732029 EFLEXIS SOFTWARE v1.2.0 & WES7 EF v1.2.0
March (21/03) 2019	Added codes 732125 ASPIRATION PUMP V2 Deleted codes 234188 ASPIRATION PUMP
March (12/03) 2019	Added codes 731398 READER ADJUSTMENT TOOL-PATTERN v2 Deleted codes 233812 READER ADJUSTMENT TOOL-PATTERN 210403 GRIFOLS BENCH WASTE CONTAINER (as it is an accessory)

Revision	Changes with respect to previous version.
February 2019	Added codes 230909 POWER CORD with lock 230910 POWER CORD US with lock 732098 LOCTITE SEALANT 5331 (100ml) 732036 CD IH REAGENTS PACK US V2.0 Other Changed picture for code 731586 SWIVEL EYE BOLT (4 units) Removed laser label code 3033934 from 233745 PRESENCE DETECTOR. Added laser label code 3052536 to 234165 MIRROR HOLDER. Correction of minor typographic errors
January 2019	Added codes 731980 SCREW M5X30 DIN 912 (10 units) 731983 CONTAINER HOLDER PIVOTING PLATE 732029 EFLEXIS SOFTWARE v1.2.0 & WES7 EF v1.2.0 732031 DILUTION STATION CUP EFLEXIS 732037 MOTOR BCR V2 Deleted codes 232213 DILUTION STATION CUP (Ver. 2) 234116 MOTOR BCR 731568 ERYTRA EFLEXIS SOFT V1.2.0 AND REAGENTS PACK EF V1.7 CDs AND WES7 EFLEXIS V1.2.0 DVD
November 2018	Added codes 731679 READER ILLUMINATION SYSTEM V2 731680 PCA RD IL421 V2 Deleted codes 233734 READER ILLUMINATION SYSTEM 233830 PCA RD IL421 730557 FIREWIRE TO PC ADAPTOR 511

Revision	Changes with respect to previous version.
October 2018	Added codes 731639 DILUTION CUP COVER 731850 INCUBATORS COVER 731803 BOTTLES DOOR LABELS 731804 WASTE CONTAINER LABELS 731805 SOLUTION CONTAINER LABELS 210403 GRIFOLS BENCH WASTE CONTAINER 731854 TOP COVER 731839 MANIFOLD v2 731840 DAMPER SYSTEM Deleted codes 234183 MANIFOLD Other Added revision history table Correction of minor typographic errors

Erytra Revision history table

Revision	Changes with respect to previous version.
July 2025	Cancelled codes 233943 READER FOAM
May 2025	Added codes 736237 SAMPLES PROBE (Impact detector) v2 Cancelled codes 233832 PROBE NUT 233751 PCB PIP-P2OSC59S 233899 SAMPLES PROBE (Impact detector)
January 2025	Added codes 735899 RAM DDR3L 8GB SO-DIMM Cancelled codes 233748 SAMPLES PROBE 233752 PCB PIP-P1OSC59 233790 LEVEL DETECTOR WIRING_CHAIN (PIP) 233880 HARD DISK 233913 PROBES WAGON (No Impact detector) 233914 PROBES HOLDER (No impact detector) 233974 WINDOWS 7 EMBEDDED LICENSE 732489 RAM DDR2 "GB v3 (2 units) 734048 RAM DDR3 2GB SO-DIMM KINGMAX (2 units) 734232 PC EMBEDDED (Ver. 7, WES7, Erytra v4.2.0)

Revision	Changes with respect to previous version.
November 2024	Added codes 735852 ILLUMINATION PATTERN DGGEL v2 735851 READER ILLUMINATION SYSTEM v4 Cancelled codes 733042 READER ILLUMINATION SYSTEM V3 233710 ILLUMINATION PATTERN DGGEL 733043 CD IH REAGENTS PACK US V2.1 731516 ERYTRA SOFT v4.0.3&WES7v4.2.0&IHRPUSv1.5 731603 ERYTRA SOFT v4.1.1&WES7v4.3.0&IHRPUSv1.5 731145 ERYTRA SOFT v3.4.3&WES7v3.1.1&IHRPCEv7.4 732529 ERYTRA SOFT v4.2.0 & WES7 ERYTRA v4.4.0 233976 IMPACT DETECTOR KIT (Ver. 2)
September 2024	Added Recommended Stock columns for the different regions
May 2024	Added codes 735529 DG GEL CARD HOLDERS (15 units) Cancelled codes 733296 ERYTRA SOFT v4.3.0 AND WINDOWS 10 ENTERPRISE 2019 LTSC v2.1.0 730561 FIREWIRE TO PC ADAPTOR 59
February (II) 2024	Added codes 735520 TOUCHSCREEN V4 735530 READER BOX Deleted codes 732288 TOUCHSCREEN V3 734834 MK VALVE (w. o-ring)
February 2024	Added codes 735365 MK VALVE (w. o-ring) v2

Revision	Changes with respect to previous version.
October (24/10) 2023	Added codes 735327 HORNS LATERAL v3 (LEFT) 735328 HORNS-FRONTAL v2 735329 HORNS LATERAL v3 (RIGHT) Deleted codes 233842 TEFLON TUBE CONNECTION (seal nut + ferrule). P/N 233900 TEFLON TUBE CONNECTION (seal nut+ferrule) (Ver. 2) should be used. Other Changed picture for P/N 230903 DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V (added instructions for use in English language).
June (26/06) 2023	Added codes 230903 DIGITAL THERMOMETER HIBOK-15 MINIDIN 4V 247032 DIGIT. GAUGE DIGITRON MOD.2023 735206 ERYTRA SOLUTION LINES FILTERS SET Deleted codes 733168 SYRINGE LOCK
March (20/03) 2023	Added codes 232572 SILICONE TUBE 4x0,76mm (15cm) 735052 PROTECTOR PLATES (FLU) 735084 UPPER DOOR v2 Deleted codes 233763 PRESSURE SENSOR ERYTRA (S3,S4,S5,S6) 233825 UPPER DOOR Other Changed pictures for S/P 233798 MULTILUMEN TUBING 2,3x3,6 (10m) and 734000 PRESSURE SENSOR CABLE(addition of identification labels) Correction of minor typographic errors

Revision	Changes with respect to previous version.
January (13/01) 2023	Added codes 734826 INC UPPER COVER 734836 GRIPPER PLATE
October (07/10) 2022	Added codes 734834 MK VALVE (w. o-ring) 734835 MK SYRINGE 1ml Deleted codes 233952 HAMILTON SYRINGE 1ml (Ver. 2) 233953 HAMILTON VALVE (Ver. 2) (w. o-ring) Other Changed picture for P/N 734238 DOOR ELECTRIC LOCK v2 (cord removed) Added cap for the diluent bottle in diagnostic kit 732944 and updated its picture
June (16/06) 2022	Added codes 734657 BARCODE READER QW2520-BKK1S(W.USB CABLE) 734701 INCUBATOR PLATE EDGE PROTECTORS v2 Deleted codes 733154 BARCODE READER QW2420-BKK1S (W.USB CABLE) 733999 INCUBATOR PLATE EDGE PROTECTORS

Revision	Changes with respect to previous version.
April (19/04) 2022	Added codes 733840 GEAR DOUBLE WHEEL (X,Y) v2 734232 PC EMBEDDED (Ver. 7, WES7, Erytra v4.2.0) 734233 PC EMBEDDED (Ver. 7, W10 ENT. 2019 LTSC) 734235 MOTHERBOARD v3 (MINI-ITX ASROCK IBM-150) 734424 CAMERA SUPPORT PLATE Deleted codes 233860 GEAR DOUBLE WHEEL (X,Y) 732098 LOCTITE SEALANT 5331 (100ml) 733901 MOTHERBOARD v2 (MINI-ITX ASROCK IBM-150) 733902 PC EMBEDDED (Ver. 6, WES7, Erytra v4.2.0) 733903 PC EMBEDDED (Ver. 6, W10 ENT. 2019 LTSC) Other P/N 732098 LOCTITE SEALANT 5331 (100ml) has been deleted because of being considered as dangerous goods by the logistics company in charge of transportation.
January (26/01) 2022	Added codes 234210 PRESSURE SENSOR (S3,S4,S5,S6) 734000 PRESSURE SENSOR CABLE 733519 O-RING 5.28x1.78 (10 units) Other Changed description for P/N 233763 from PRESSURE SENSOR (S3,S4,S5,S6) to PRESSURE SENSOR ERYTRA (S3,S4,S5,S6).

Revision	Changes with respect to previous version.
January (03/01) 2022	Added codes 733966 DILUTION STATION CUP V2 232760 SCREW M2X10 DIN 912 (100 units) 232746 SPACING PLASTIC WASHER (100 units) Deleted codes 233768 DILUTION STATION CUP Other New RAM memory with P/N 734048 implemented from factory in spare parts: • 733902 PC EMBEDDED (Ver. 6, WES7, Erytra v4.2.0 lot ≥ 21001. • 733903 PC EMBEDDED (Ver. 6, W10 ENT. 2019 LTSC) lot ≥ 21003.
October (29/10) 2021	Added codes 733965 CENTRIFUGE ROTOR V2 734048 RAM DDR3 2GB SO-DIMM KINGMAX (2 units) 734238 ELECTRIC LOCK v2 Deleted codes 233701 DRAWER ELECTRIC LOCK 233732 CENTRIFUGE ROTOR 731399 RAM DDR3L 2GB SO-DIMM Innodisk (2 units)
August (02/08) 2021	Added codes 733839 ERYTRA EXTERNAL BARCODE READER HOLDER Other Changed references to DG Technical Support by DIG PDS

Revision	Changes with respect to previous version.
July (02/07) 2021	Added codes 733837 CAMERA COVER 733901 MOTHERBOARD v2 (MINI-ITX ASROCK IBM-150) 733902 PC EMBEDDED (Ver. 6, WES7, Erytra v4.2.0) 733903 PC EMBEDDED (Ver. 6, W10 ENT. 2019 LTSC) 733914 BOTTLE DRAWER CABLES 733999 INCUBATOR PLATE EDGE PROTECTORS Deleted codes 731711 PC EMBEDDED (Ver. 5, WES7, Erytra v4.2.0) 733517 PC EMBEDDED (Ver. 5, W10 ENT.2019 LTSC)
March (19/03) 2021	Added codes 733701 PROBES ADJUSTMENT TOOL (holder 16mm)
March (01/03) 2021	Added codes 733671 USER TRAY (Ver. 4) 733761 FLUIDIC DRAWER EDGE PROTECTOR Deleted codes 731176 USER TRAY (Ver. 3) Other Added picture for P/N 733472 PCA VOLTAGE CONVERTER 24V TO 5V

Revision	Changes with respect to previous version.
February (19/02) 2021	Added codes 733698 PCB CUP-ST59 v2 733699 MOTOR STIRRER DILUTION (Ver. 2) 733700 SHOCK ABSORBER DILUTION CUP (Ver. 3) Deleted codes 233767 MOTOR STIRRER DILUTION 233769 PCB CUP-ST59 233948 SHOCK ABSORBER DILUTOR CUP (Ver. 2)
November (27/11) 2020	Added codes 234001 COSMETIC FRONT PANEL OF PUSHBUT. AND USB 733517 PC EMBEDDED (Ver. 5, W10 ENT.2019 LTSC) 733560 BOTTLE HOLDER 733620 DG-59 INCUBATOR MODULE Other Changed description for P/N 731711 from PC EMBEDDED (Ver. 5) to PC EMBEDDED (Ver. 5, WES7, Erytra v4.2.0).
October (08/10) 2020	Added codes 733350 READER MODULE SUPPORT PLATE 733465 DISTRIBUTION BLOCK BK3 (Ver. 3) 733466 DISTRIBUTION BLOCK BK2 (Ver. 3) Deleted codes 233922 DISTRIBUTION BLOCK BK2 (Ver. 2) 233923 DISTRIBUTION BLOCK BK3 (Ver. 2)

Revision	Changes with respect to previous version.
	Added codes 230909 POWER CORD with lock 230910 POWER CORD US with lock 234167 POWER SUPPLY 24V 234168 POWER SUPPLY 12V 234169 PCA GEN-STDBY511 733259 ELE MODULE WIRING SET V1 733260 BASE MODULE WIRING SET V1 733467 REAR PANEL (UPPER) v2 733472 PCA VOLTAGE CONVERTER 24V TO 5V Deleted codes 230501 POWER CORD 2mts 230836 POWER CORD U.S.A. Other Spare parts list has been unified for CE and US instruments (changes for CE and US areas are listed below). Changed drawing reference for code 233937 PCA GEN-BCKPL59 (Ver. 2) Updated Logistics (SAP) description for P/N 233987 PC EMBEDDED (Ver. 3) HARD DISK SSD 256GB mSATA Added picture for code 733296 ERYTRA SOFT v4.3.0 AND WINDOWS 10 ENTERPRISE 2019 LTSC v2.1.0 DVDs Unified list: Codes only applying to CE instruments 233748 SAMPLES PROBE 233751 PCB PIP-P2OSC59 233752 PCB PIP-P1OSC59 233790 LEVEL DETECTOR WIRING/CHAIN (PIP) 233832 PROBES NUT 233876 SILICONE TUBING 3x5 (1m) 233883 12 CARDS RACK ADAPTER 233912 DILUTION CUP BATH 233913 PROBES WAGON (No Impact detector) 233914 PROBES HOLDER (No impact detector) 731145 ERYTRA SOFT v3.4.3&WES7v3.1.1&IHRPCEv7.4 Unified list: Codes only applying to US instruments 730572 CD REAGENTS PACK US V1.5 (Build 1.5.0.6) 731516 ERYTRA SOFT V4.0.3 AND REAGENTS PACK US V1.5 CDs AND WES7 V4.2.0 DVD 731603 ERYTRA SOFT v4.1.1 AND REAGENTS PACK US v1.5 CDs AND WES7 v4.3.0 DVD 733043 CD IH REAGENTS PACK US V2.1

Revision	Changes with respect to previous version.
June (08/06) 2020	Added codes 733153 WINDOWS 10 ENTERPRISE 2019 LTSC LICENSE 733168 SYRINGE LOCK 733296 ERYTRA SOFT v4.3.0 AND WINDOWS 10 ENTERPRISE 2019 LTSC v2.1.0 DVDs
May (28/05) 2020	Added codes 733154 BARCODE READER QW2420-BKK1S (W.USB CABLE) Deleted codes 232342 BARCODE READER HERON HD3130 W. USB CABLE
May (21/05) 2020	Added codes 731145 ERYTRA SOFT v3.4.3&WES7v3.1.1&IHRPCEv7.4 731344 CARDS HOLDER (CENTR.) 12 UNITS (VER.3) 732489 PC KONTRON RAM DDR2 2GB v3 (2 units) 733255 INCUBATOR LATERAL 733266 CAMERA FIREWIRE CABLE (READ_FW) Deleted codes 232048 CARDS HOLDER (CENTR.) 12 UNITS (VER. 2) 234005 RAM DDR2 2GB v2 (2 units)
March (31/03) 2020	Added codes 733056 CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL (Ver. 3) 733068 LOWER CENTRIFUGE / INCUBATOR ADJUSTMENT TOOL v3 Deleted codes 233879 CENTRIFUGE/INCUBATOR ADJUSTMENT TOOL (Ver. 2) Other Updated picture for P/N 233811 PCB GEN-LED59

Revision	Changes with respect to previous version.
March (10/03) 2020	Added codes 732529 ERYTRA SOFT v4.2.0 & WES7 ERYTRA v4.4.0 733086 SHOULDER SCREW 6M5x30 ISO 7379 (10 u.) Deleted codes 731602 ERYTRA SOFT v4.0.4 AND REAGENTS PACK v8.0 CDs AND WES7 v4.2.0 DVD 731488 ERYTRA SOFT v4.1.0 AND REAGENTS PACK v8.0 CDs AND WES7 v4.3.0 DVD 731601 ERYTRA SOFT v4.1.1 AND REAGENTS PACK v8.0 CDs AND WES7 v4.3.0 DVD
February (10/02) 2020	Added codes 732911 PCA RD-IL421 v3 733042 READER ILLUMINATION SYSTEM V3 Deleted codes 731680 PCA RD-IL421 v2 731679 READER ILLUMINATION SYSTEM v2
January (23/01) 2020	Added codes 732944 IH DIAGNOSTIC KIT
October (22/10) 2019	Added codes 732573 PIP HEAD LATERAL COVER 732288 TOUCHSCREEN V3 Deleted codes 233746 TOUCHSCREEN V2 Other TS description of P/N 233831 TOUCHSCREEN v2 POWER CABLE renamed to TOUCHSCREEN v2&v3 POWER CABLE

Revision	Changes with respect to previous version.
September (06/09) 2019	Added codes 232596 PROBE UNBLOCKER TOOL 732419 MONITOR GUIDES (HINGES) 732420 MONITOR SUPPORT GUIDES 732451 SCREW M5x70 DIN 912 (10 units) 732508 HORNS LATERAL v2 (RIGHT) 732509 HORNS LATERAL v2 (LEFT) 732517 CAT BARCODE READER COVER v2 Deleted codes 233833 HORNS - LATERAL (RIGHT) 233834 HORNS - LATERAL (LEFT) 233927 CAT BARCODE READER COVER Other 233948 SHOCK ABSORBER DILUTOR CUP (Ver. 2) structure and picture have been updated, including 3023707. 731802 DILUTION CUP STIRRER TUBE structure and picture have been updated, including 8x8013452.
March 2019	Added codes 732098 LOCTITE SEALANT 5331 (100ml) 731398 READER ADJUSTMENT TOOL-PATTERN v2 Deleted codes 233812 READER ADJUSTMENT TOOL-PATTERN Other Added page headers. 233745 PRESENCE DETECTOR has been updated (label has been removed). 231806 SOLENOID GRIPPER drawing reference sections has been updated (3014205 updated to 7006355).

Revision	Changes with respect to previous version.
November 2018	Added codes 731679 READER ILLUMINATION SYSTEM V2 731680 PCA RD IL421 V2 731802 DILUTION CUP STIRRER TUBE Deleted codes 233734 READER ILLUMINATION SYSTEM 233830 PCB RD IL421 Other Added revision history table Correction of minor typographic errors

DG Reader Net Revision history table

Revision	Changes with respect to previous version.
July 2025	Cancelled codes 233943 READER FOAM
March 2025	Added codes 736169 RAM DDR4 8GB SO-DIMM KINGMAX (2 units)v2 Cancelled codes 735635 RAM DDR4 8GB SO-DIMM KINGMAX (2 units)
January 2025	Added codes 735899 RAM DDR3L 8GB SO-DIMM Cancelled codes 734048 RAM DDR3 2GB SO-DIMM KINGMAX (2 units)

Revision	Changes with respect to previous version.
November 2024	Added codes 735852 ILLUMINATION PATTERN DGCEL v2 735851 READER ILLUMINATION SYSTEM v4 Cancelled codes 733042 READER ILLUMINATION SYSTEM V3 233710 ILLUMINATION PATTERN DGCEL
September 2024	Added Recommended Stock columns for the different regions
August 2024	Added codes 732539 WIFI DONGLE 735719 WIRING KIT MINI_ITX ASROCK IMB-1003J 735720 GENERAL CABLING V4 Cancelled codes 734234 DG READERNET PC MOTHERBOARD V2 734983 GENERAL CABLING V3
May 2024	Added codes 735634 SSD 256GB M.2 735635 RAM DDR4 8GB SO-DIMM KINGMAX(2 units) 735636 MOTHERBOARD MINI-ITX ASROCK IMB-1003J Cancelled codes 732907 DVD W20 IoT v1.1.0 732908 CD SW v1.1.0 (DG READER NET)

Revision	Changes with respect to previous version.
January (13/01) 2023	Added codes 234166 FAN 734982 LATERAL FAN UPGRADE KIT 734983 GENERAL CABLING v3 734984 LATERAL COVER v3 Deleted codes 732913 LATERAL COVER v3 733944 GENERAL CABLING v2
October (03/10) 2022	Other 732916 TOUCHSCREEN MODULE v2: Code 6017968 replaced by 6020080. 732916 TOUCHSCREEN MODULE v2 with Lot reference ≥ 22006 will have implemented the improvements described in INR-00512#0016
April (19/04) 2022	Added codes 734234 DG READERNET PC MOTHERBOARD v2 734424 CAMERA SUPPORT PLATE 734425 INTERNAL BCR ADJUSTMENT TOOL Deleted codes 732035 DG READERNET PC MOTHERBOARD
October (11/10) 2021	Added codes 733837 CAMERA COVER 733944 GENERAL CABLING V2 734048 RAM DDR3 2GB SO-DIMM KINGMAX (2 units) Deleted codes 731394 GENERAL CABLING 731399 RAM DDR3L 2GB SO-DIMM Innodisk (2 units) Other Document format has been changed according to latest changes in IH instruments spare part lists. Changed references to DG Technical Support by DIG PDS.

Revision	Changes with respect to previous version.
07/2020	Updated description 233987. Added 733350.
01/2020	Removed: 731397, 731395, 731376, 731375, 241187, 731680, 732039, 732173, 731679. - Added: 732907, 732908, 732911, 732912, 732913, 732914, 732915, 732916, 733042.
03/2019	Added picture to 732039. Removed 732040. Added 732173.
12/2018	Removed 233989. Added 732035, 732039, 732040.

DG Spin Revision history table

Revision	Changes with respect to previous version.
May 2025	Added codes 736208 PACKAGING (DG SPIN) v2 736207 HOUSING DG SPIN (w/screws) v3 Cancelled codes 732418 PACKAGING (DG SPIN) 230687 HOUSING DG SPIN VER. 2 (w/screws)
March 2025	Modified codes 735094 POWER SUPPLY v2 (W/SCREWS): power supply image updated. 735096 POWER SUPPLY UPGRADE KIT: power supply image updated.

Revision	Changes with respect to previous version.
January 2025	Added codes 230665 KEYBOARD DG-SPIN 230668 PCA DC MOTOR CONTROL (with screws) 230670 PCA UNBALANCE DETECTOR (with screws) 230672 CARD HOLDERS (24 units) 230674 HOUSING LATERAL PANEL (W/SCREWS) 230676 LID JOINT (with screws) 230677 LID SPRING 230678 MOTOR COVER (with screws) 230679 SHOCK ABSORBERS (3 units with nuts) 230680 MOTOR (without unbalance sensor) 230683 MAINS SOCKET WITH FILTER (with screws) 230686 LID (w_joint, latch & screws)(Ver.2) 230687 HOUSING DG SPIN (w/screws) VER.2 230688 HINGE AXIS 230818 FUSE 2A 250V MAINS SOCKET 230887 PCA DISPLAY (with screws and palates) 230890 DISPLAY WINDOW 230897 RS-232 PROTECTIVE COVER 230909 POWER CORD with lock v2 230910 POWER CORD U.S.A with lock v2 231943 RS-232 COMMUNICATION CABLE 240024 RUBBER FOOT (Grey) 230917 CLOCK LITHIUM BATTERY 732418 PACKAGING (DG SPIN) 734239 CABLES KIT v3 (WITH GROUND SCREW) 734810 LID ELECTRIC LOCK SWITCH v3 (WITH SCREWS/WIRE) 735094 POWER SUPPLY v2 (w/SCREWS) 735095 CABLES KIT v4 (WITH GORUND SCREW) 735096 POWER SUPPLY UPGRADE KIT 735344 PCA DISTRIBUTION (w/screws)(Ver. 3) 735760 DG SPIN HEAD v2

Revision	Changes with respect to previous version.
September 2024	Added codes 735760 DG SPIN HEAD v2 Other Updated picture for P/N 230668 PCB DC MOTOR CONTROL (W-SCREWS)
October 2023	Added codes 735344 PCA DISTRIBUTION (w/screws) (Ver. 3) Cancelled codes 230685 PCA DISTRIBUTION (w/screws) (Ver. 2) Other Changed picture for P/N 734810 LID ELEC.LOCK SWITCH v3 (WITH SCREWS/WIRE) (cord removed)
March 2023	Added codes 735094 POWER SUPPLY v2 (W/SCREWS) 735095 CABLES KIT v4 (WITH GROUND SCREW) 735096 POWER SUPPLY UPGRADE KIT Cancelled codes 230671 POWER SUPPLY (W/SCREWS)
August 2022	Added codes 734810 LID ELECTRIC LOCK SWITCH v3 (WITH SCREWS/WIRE) Cancelled codes 734240 LID ELECTRIC LOCK SWITCH v2 (WITH SCREWS)
February 2022	Other Added code 8018618 in P/N 734240 LID ELEC.LOCK SWITCH v2 (label recommending Firmware v1.3.1 for DG Spin units with s/n < 320-0009551 where this P/N is installed)

Revision	Changes with respect to previous version.
November 2021	Added codes 734239 CABLES KIT v3 (WITH GROUND SCREW) 734240 LID ELECTRIC LOCK SWITCH v2 (WITH SCREWS/WIRE) Cancelled codes 230666 LID ELECTRIC LOCK SWITCH (WITH SCREWS) 733606 CABLES KIT (WITH GROUND SCREW) v2 Other Changed references to DG Technical Support to DIG PDS.
April 2021	Added codes 733606 CABLES KIT (WITH GROUND SCREW) v2 Cancelled codes 230681 CABLES KIT (with ground screw) Other Changed picture for P/N 230686 LID (w_joint, latch & screws)(Ver.2), as code 3018918 (latch) is now delivered separately from the lid
June 2019	Added codes 732418 PACKAGING (DG SPIN)

DG Therm Revision history table

Revision	Changes with respect to previous version.
February 2025	Added codes 230818 FUSE 2A 250V MAINS SOCKET 230881 TEMPERATURE PROBE PT-100 FOR CARDS 230886 KEYBOARD DG THERM 230887 PCA DISPLAY (with screws and palates) 230889 THERMAL BLANKET (w/ wire & plug) 230890 DISPLAY WINDOW 230891 ELEVATION CARDS PLATE PUSHER 230892 GROUNDING PLATE (w/ screws) 230893 HEATING BLOCK DG THERM (w/ screws and th. blanket) 230894 HEATING BLOCK DG THERM SEALING JOINT 230896 LID DG THERM 230897 RS-232 PROTECTIVE COVER 230898 SCREWS & WASHERS SET DG THERM 230899 THERMAL SILICON ADHESIVE TAPE (2 units) 230900 CABLE PCA DISPLAY to PCA INCUBATOR 230903 DIGITAL THERMOMETER HIBOK-15 MiniDIN 4v 230904 TEMP. PROBE PT-100 FOR TUBES - MiniDIN 4v 230905 TEMP. PROBE PT-100 FOR CARDS - MiniDIN 4v 230906 MAINS CONNECTOR FIXING PLATE (w screws) 230907 HOUSING DG THERM (w/screws) v2 230908 PCA INCUBATOR DG THERM (w/screws) v2 230909 POWER CORD with lock v2 230910 POWER CORD U.S.A with lock v2 231943 RS-232 COMMUNICATION CABLE 240024 RUBBER FOOT (Grey) 230917 CLOCK LITHIUM BATTERY 732417 PACKAGING (DG THERM)